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Natural Language Processing

Neural Networks and Large Language Models

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Preface

Natural language processing (NLP) is one of the core subfields of artificial intelligence (AI). For a long time, research in NLP primarily focused on solving specific problems in language understanding and generation, such as parsing and machine translation. This task-driven research approach dominated the development of NLP for several decades. However, with the rise of deep learning, the research paradigm of NLP has undergone a fundamental transformation. The application of deep neural networks has enabled us to tackle increasingly complex tasks. More importantly, researchers have discovered that by conducting large-scale pretraining on a base model with massive datasets, and then fine-tuning it with a small amount of task-specific data and knowledge, it is possible to construct general-purpose models capable of handling multiple tasks simultaneously. This new paradigm is greatly changing the research landscape of NLP, and even the broader field of AI.

This book focuses on modern NLP methods centered around neural networks and foundation models. It aims to provide a practical guide to understanding, building, and applying these powerful models. Unlike traditional NLP textbooks that organize chapters based on specific tasks, this book is structured from the perspective of constructing neural NLP models and is divided into three main parts:

- **Foundations of Machine Learning and Neural Networks** (Chapters 1-2): This part introduces the core concepts and methods of machine learning and neural networks, laying a foundation for the subsequent chapters. It is relatively self-contained and can be studied independently or used as background material when needed.
- **Basic Neural Models for Natural Language Processing** (Chapters 3-6): This part explains the neural networks used in NLP tasks, including word representation models, sequence models, and sequence-to-sequence models. In addition, Transformers are introduced in a dedicated chapter. These models are not limited to individual NLP tasks; rather, they serve as general-purpose tools across many applications.
- **Large Language Models** (Chapters 7-11): This part focuses on large language models (LLMs), covering topics such as pretraining, generative models, prompt engineering, alignment, and inference.

This book is intended for senior undergraduates, graduate students, researchers in related fields, and anyone interested in NLP. We strive for clear and accessible writing, aiming to introduce core concepts and fundamental methods rather than providing an in-depth exploration of all cutting-edge techniques. Therefore, this book can serve both as an introductory text for newcomers and as a reference manual for key concepts and methods in NLP.

The content of this book has gradually taken shape through our years of teaching and research experience. Initially, we only planned to write the first two parts. However, the rapid rise and growing importance of LLMs led us to include this topic as a key part of the book. At the same time, we are delighted to witness the rapid development of NLP, and the writing of this book is also our response to this exciting trend.

Some chapters of this book have been previously published online, such as *Introduction to Transformers: An NLP Perspective* (Chapter 6) and *Foundations of Large Language Models* (Chapters 7-11), and we are grateful for the valuable feedback from many readers, which has greatly contributed to the refinement of the book. Furthermore, during the writing process, we drew significant inspiration from classic works, including *Machine Learning* by Mitchell [1997], *Foundations of Statistical Natural Language Processing* by Manning and Schütze [1999], *Pattern Recognition and Machine Learning* by Bishop [2006], and *Speech and Language Processing* by Jurafsky and Martin [2008]. Many insights from these works profoundly influenced the writing approach of this book.

Lastly, we would like to express our heartfelt thanks to all those who provided suggestions and revisions to the content of this book. They are: Weiqiao Shan, Yongyu Mu, Chenglong Wang, Kaiyan Chang, Yuchun Fan, Hang Zhou, Chuanhao Lv, Xinyu Liu, Tao Zhou, Huiwen Bao, Tong Zheng, Junhao Ruan, Yingfeng Luo, Yuzhang Wu, and Yifu Huo.

Tong Xiao and Jingbo Zhu
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This book is dedicated to our families.

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I Preliminaries

Chapter 1

Foundations of Machine Learning

The goal of machine learning is to develop methods that can automatically detect patterns in data, and then to use the uncovered patterns to predict future data or other outcomes of interest.

– Murphy [2012]

Machine learning can be broadly defined as computational methods using experience to improve performance or to make accurate predictions.

– Mohri et al. [2018]

Data-driven NLP fits the above definitions[1]. It teaches computers to learn language *experience* from corpora, and to understand and utilize language based on that *experience*. Connecting machine learning (ML) with natural language processing is much more than a means that makes computers mimic human language intelligence from data. It is leading a revolution in both areas: natural language processing evolves by using a powerful tool of deriving meaning from corpora, and machine learning evolves by addressing the NLP challenges and testing on real-world data.

In this chapter, we present several basic concepts and models in machine learning. There are no tough bits but some preliminaries for the subsequent chapters. Here we focus on how to apply machine learning to NLP problems, in particular how to define an NLP problem as a statistical learning problem. To do this, we start with classification — one of the most widely-used examples in most introductory books. We then present several fundamental issues of machine learning. They are followed by a discussion on NLP problems from the machine learning perspective.

[1]We drop the term *data-driven* from now on and assume that all NLP models are data-driven in the remainder of this book.

1.1 Math Basics

In the remainder of this chapter and the following chapters, we will talk about machine learning problems using the tool of applied mathematics. Here are the math basics. If you find the details trivial, you can skip to Section 1.2 directly.

1.1.1 Linear Algebra

1. Vectors and Matrices

A **scalar** may or may not be the simplest concept in linear algebra, but is surely the most common concept that one learns in high school or in university. A scalar is a number. It is a quantity that has a magnitude but has no direction. For example, height, weight, distance, temperature are all examples of scalars. Here we use an italic number to denote a scalar, for example, a, b, x, A, and so on.

Vector and **matrix** are defined in terms of scalar. A vector is an array of scalars, or simply a number list. A matrix is a rectangular array of scalars. In this book, we follow the convention of using bold letters to denote vectors and matrices. For example, an n-dimensional vector can be written as

$$\mathbf{a} = \begin{bmatrix} a_1 & a_2 & ... & a_n \end{bmatrix} \tag{1.1}$$

where $\{a_1, a_2, ..., a_n\}$ are the elements (or entries) of the vector. Each element corresponds to a dimension. For convenience of notation, we write a_i as $a(i)$ sometimes. A vector is a real-valued vector only if all the elements are real numbers (i.e., $a_i \in \mathbb{R}$ for each i), denoted as $\mathbf{a} \in \mathbb{R}^n$. Likewise, we can write an $m \times n$ matrix as

$$\mathbf{A} = \begin{bmatrix} A_{11} & A_{12} & ... & A_{1n} \\ A_{21} & A_{22} & ... & A_{2n} \\ ... & ... & ... & ... \\ A_{m1} & A_{m2} & ... & A_{mn} \end{bmatrix} \tag{1.2}$$

where m is the number of rows and n is the number of columns. A_{ij} is the entry (i,j) of the matrix. A real-valued matrix is denoted as $\mathbf{A} \in \mathbb{R}^{m \times n}$. Occasionally, we use $\mathbf{A}_{m \times n}$ to emphasize that the shape of the matrix is $m \times n$.

There are a few special matrices. For example, a matrix whose elements are all zeros is a **zero matrix**, denoted as $\mathbf{0}$. Another example is the **identity matrix**, denoted as $\mathbf{I}$. It is a square matrix whose diagonal elements are all 1, and other elements are 0. Vectors can be treated as a special sort of matrices, too. For example, the vector in Eq. (1.1) is a matrix with only one row.

2. Matrix Transpose

The **transpose** of a matrix $A_{m\times n}$ is a matrix $B_{n\times m}$ subject to $A_{ij} = B_{ji}$ for each pair of i and j. Often, $\mathbf{A}$'s transpose is denoted as $\mathbf{A}^\top$. For example, for a matrix

$$\mathbf{A} = \begin{bmatrix} 8 & 0 & 0 \\ 2 & 9 & 7 \end{bmatrix} \tag{1.3}$$

the transpose is

$$\mathbf{A}^\top = \begin{bmatrix} 8 & 2 \\ 0 & 9 \\ 0 & 7 \end{bmatrix} \tag{1.4}$$

One can transpose a vector as well. For a vector

$$\mathbf{a} = \begin{bmatrix} 1 & 9 & 7 & 3 \end{bmatrix} \tag{1.5}$$

the transpose is

$$\mathbf{a}^\top = \begin{bmatrix} 1 \\ 9 \\ 7 \\ 3 \end{bmatrix} \tag{1.6}$$

In general, a vector with only one row is called a **row vector** (as in Eq. (1.5)), and a vector with only one column is called a **column vector** (as in Eq. (1.6)). In this book, all vectors are row vectors by default.

3. Element-wise Operations on Matrices

Suppose $\mathbf{A}$ and $\mathbf{B}$ are two matrices of the same shape, say $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m\times n}$. The **matrix addition** of $\mathbf{A}$ and $\mathbf{B}$ is written as $\mathbf{A}+\mathbf{B}$. $\mathbf{A}+\mathbf{B}$ is a matrix in $\mathbb{R}^{m\times n}$ such that each element is the sum of the corresponding elements of $\mathbf{A}$ and $\mathbf{B}$. Here is an example.

$$\begin{aligned} \mathbf{A}+\mathbf{B} &= \begin{bmatrix} 8 & 0 & 0 \\ 2 & 9 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 1 & 1 \\ 3 & 9 & 11 \end{bmatrix} \end{aligned} \tag{1.7}$$

In a similar way, we can define element-wise subtraction ($\mathbf{A}-\mathbf{B}$), product ($\mathbf{A} \odot \mathbf{B}$), division ($\mathbf{A} \oslash \mathbf{B}$) and other operations. A special case of element-wise operations is **scalar multiplication**, where we multiply a matrix $\mathbf{A}$ with a scalar k, denoted as $k \times \mathbf{A}$ or $k\mathbf{A}$. See below for an example for $k=2$ and $\mathbf{A} = \begin{bmatrix} 8 & 0 & 0 \\ 2 & 9 & 7 \end{bmatrix}$.

$$\begin{aligned} k\mathbf{A} &= 2\begin{bmatrix} 8 & 0 & 0 \\ 2 & 9 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 16 & 0 & 0 \\ 4 & 18 & 14 \end{bmatrix} \end{aligned} \tag{1.8}$$

Let $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ be matrices in $\mathbb{R}^{m\times n}$, and k and l be scalars in $\mathbb{R}$. Some properties of the matrix operations are:

- Property of the zero matrix:

$$\begin{aligned} \mathbf{A} &= \mathbf{A}+\mathbf{0} \\ &= \mathbf{0}+\mathbf{A} \end{aligned} \tag{1.9}$$

- **Commutativity**:

$$\mathbf{A}+\mathbf{B} = \mathbf{B}+\mathbf{A} \tag{1.10}$$
$$kl\mathbf{A} = lk\mathbf{A} \tag{1.11}$$

- **Associativity**:

$$(\mathbf{A}+\mathbf{B})+\mathbf{C} = \mathbf{A}+(\mathbf{B}+\mathbf{C}) \tag{1.12}$$
$$(kl)\mathbf{A} = l(k\mathbf{A}) \tag{1.13}$$

- **Distributivity**:

$$k(\mathbf{A}+\mathbf{B}) = k\mathbf{A}+k\mathbf{B} \tag{1.14}$$
$$(k+l)\mathbf{A} = k\mathbf{A}+l\mathbf{A} \tag{1.15}$$

4. Dot Product

The **dot product** of two vectors of the same dimensions $\mathbf{a} = \begin{bmatrix} a_1 & a_2 & ... & a_n \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 & b_2 & ... & b_n \end{bmatrix}$ is defined to be

$$\begin{aligned} \mathbf{a}\cdot\mathbf{b} &= \sum_{i=1}^{n} a_i b_i \\ &= a_1b_1 + a_2b_2 + ... + a_nb_n \end{aligned} \tag{1.16}$$

In geometry, a real-valued vector $\mathbf{a}$ can be seen as a geometric object having both magnitude (denoted as $|\mathbf{a}|$) and direction. The dot product of $\mathbf{a}$ and $\mathbf{b}$ can also be defined as

$$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}|\times|\mathbf{b}|\times\cos(\theta) \tag{1.17}$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.

5. Matrix Product

The **matrix product** (or **matrix-matrix product**) operates on two matrices. Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times p}$ and a matrix $\mathbf{B} \in \mathbb{R}^{p \times n}$, the matrix product of $\mathbf{A}$ and $\mathbf{B}$ produces a matrix $\mathbf{C} \in \mathbb{R}^{m \times n}$ whose elements are defined as

$$\begin{aligned} C_{ij} &= \sum_{k=1}^{p} A_{ik} \times B_{kj} \\ &= A_{i1} \times B_{1j} + A_{i2} \times B_{2j} + ... + A_{ip} \times B_{pj} \end{aligned} \tag{1.18}$$

Matrix product requires that the number of columns in $\mathbf{A}$ is exactly the same as the number of rows in $\mathbf{B}$. In this book we use $\mathbf{AB}$ to denote the matrix product of $\mathbf{A}$ and $\mathbf{B}$. Here are a few properties of matrix product.

- **Distributivity**. For $\mathbf{A} \in \mathbb{R}^{m \times p}$ and $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{p \times n}$, the left distributivity is defined as

$$\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC} \tag{1.19}$$

For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times p}$ and $\mathbf{C} \in \mathbb{R}^{p \times n}$, the right distributivity is defined as

$$(\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC}+\mathbf{BC} \tag{1.20}$$

- **Associativity**. For $\mathbf{A} \in \mathbb{R}^{m \times p}$, $\mathbf{B} \in \mathbb{R}^{p \times q}$ and $\mathbf{C} \in \mathbb{R}^{q \times n}$, the associativity defines

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}) \tag{1.21}$$

- **Transpose**. For $\mathbf{A} \in \mathbb{R}^{m \times p}$ and $\mathbf{B} \in \mathbb{R}^{p \times n}$, we have

$$(\mathbf{AB})^{\top} = \mathbf{B}^{\top}\mathbf{A}^{\top} \tag{1.22}$$

Matrix product is not commutative, i.e., we do not have $\mathbf{AB} = \mathbf{BA}$ for all $\mathbf{A}$ and $\mathbf{B}$ even if $\mathbf{A}$ and $\mathbf{B}$ are square matrices with the same shape. Based on matrix-matrix product, we can define vector-matrix product and matrix-vector product accordingly. This is trivial because all we need is to see a vector as a matrix in multiplication.

6. Norm

In a vector space, a **norm** is a function used to measure the "length" of a vector. For a vector $\mathbf{a} \in \mathbb{R}^n$, the p-norm (or l_p norm) is defined as:

$$\|\mathbf{a}\|_p = \left(\sum_{i=1}^{n} |a_i|^p \right)^{1/p} \tag{1.23}$$

The most commonly used p-norms are for $p = 1, 2$, and ∞:

$$\|\mathbf{a}\|_1 = \sum_{i=1}^{n} |a_i| \tag{1.24}$$

$$\|\mathbf{a}\|_2 = \sqrt{\sum_{i=1}^{n} |a_i|^2} \tag{1.25}$$

$$\|\mathbf{a}\|_\infty = \max\{|a_1|, |a_2|, ..., |a_n|\} \tag{1.26}$$

Norms can also measure the distance between two points. Let $\mathbf{b} \in \mathbb{R}^n$ be another vector. The p-norm distance between $\mathbf{a}$ and $\mathbf{b}$ is $\|\mathbf{a} - \mathbf{b}\|_p$.

1.1.2 Probability and Statistics

1. What is Probability

Probability is a measure of uncertainty. It is a quantity that describes how likely an event is to happen. For example, if the event is certain to happen, we will say that the probability is 1; if the event will never happen, we will say that the probability is 0.

An event is the outcome of a random experiment. For example, in a coin-tossing experiment, an event could be "the coin lands heads". A set of related events is described by a **random variable** or **variable** for short. For example, we can define the outcome of tossing a coin as a variable x. As there are two outcomes (heads or tails), we have two choices for the value of x. We can define that $x = 1$ when the coin lands heads, and $x = 0$ otherwise. Hence, x is a **binary variable** or more precisely a 0-1 variable. A random variable taking a value means that an event happens. For example, $x = 1$ means the event of the coin landing heads.

In mathematics, probability is defined as a **probability measure** over a sample space. This measure must satisfy certain axioms, such as countable additivity. While we skip the measure-theoretic details here, we treat probability as a function that maps events to real numbers in the range $[0, 1]$.

Usually, a probability measure is denoted as a function $\Pr(\cdot)$, called a **probability function**. When the input of $\Pr(\cdot)$ is defined on a discrete set of events, the output of the function is the probability that an event happens. For example, $\Pr(x = 1)$ means the probability of x equalling 1. Note that $\Pr(x)$ is a function that varies its output by choosing different values of x. Suppose x_1 is a value that x can take. When we write $\Pr(x_1)$, it means $\Pr(x = x_1)$. Because the probability over all events should be 1, any probability function should be subject to:

$$\sum_{x_i \in X} \Pr(x_i) = 1 \tag{1.27}$$

where X is the set of all events. A probability function can be defined on two variables or more. Here are a few widely-used cases.

- **Joint Probability**. It is the probability that two events x and y happen at the same time, denoted as $\Pr(x,y)$. If x and y are independent, their joint probability is the product of their individual probabilities

$$\Pr(x,y) = \Pr(x)\Pr(y) \tag{1.28}$$

- **Conditional Probability**. It is the probability that x happens in the presence of y happening, denoted as $\Pr(x|y)$. $\Pr(x|y)$ can be defined as:

$$\Pr(x|y) = \frac{\Pr(x,y)}{\Pr(y)} \tag{1.29}$$

- **Marginal Probability**. It is another way to define the probability of a single variable. Given the joint probability on two variables, the marginal probability is defined as

$$\Pr(x) = \sum_{y_j \in Y} \Pr(x, y_j) \tag{1.30}$$

where Y is the event space of y_j. Eq (1.30) says that $\Pr(x)$ is marginalized over Y.

Another note on joint probability. In some cases, one would like to use conditional probabilities to represent a joint probability. To this end, one can rewrite the joint probability by the **product rule** or the **chain rule**, like this

$$\Pr(x,y) = \Pr(x|y)\Pr(y) \tag{1.31}$$

For continuous variables, the probability of the variable taking any exact point value is zero. Instead, we have a density for that point. More formally, given a continuous variable x, a **probability density** of x is written as $\Pr(x)$. Suppose $x \in \mathbb{R}$. The probability of x lying in the interval $[a,b]$ is defined via an integral:

$$\Pr(x \in [a,b]) = \int_a^b \Pr(x)dx \tag{1.32}$$

Obviously, we have

$$\int_{-\infty}^{+\infty} \Pr(x)dx = 1 \tag{1.33}$$

For other properties, such as joint probability and conditional probability, the forms for continuous variables are almost the same as those for discrete variables. We just need to replace the sums in the formulas with the integrals.

2. Distribution and Expectation

A **probability distribution** (or simply **distribution**) describes how probabilities are assigned to the possible values of a random variable. It is defined by a **probability mass function** (**PMF**) for discrete variables or a **probability density function** (**PDF**) for continuous variables.

For example, a uniform distribution on a discrete variable x that takes values from $\{x_1, x_2\}$ can be described as $\Pr(x_i) = 1/2$ because $\Pr(x_1) = \Pr(x_2)$ and $\Pr(x_1) + \Pr(x_2) = 1$; A uniform distribution on a continuous variable in $[-2, 2]$ can be described as $\Pr(x) = 1/4$ because probability density function $\Pr(x)$ is a constant for any $x \in [-2, 2]$ and $\int_{-2}^{2} \Pr(x)dx = 1$. Statisticians have developed many distributions for describing the world we are living in, such as binomial distribution, Bernoulli distribution and Gaussian/normal distribution. One can find details of these distributions in most textbooks on statistics [McClave and Sincich, 2006; Freedman et al., 2007].

For describing properties of a variable, a popular means is to compute the **expected value** or **expectation** of the variable. Let x be a discrete variable that takes values from $\{x_1, ..., x_n\}$, and $\Pr(x)$ be a distribution on x. The expected value of x is defined to be

$$\mathbb{E}_{x \sim \Pr(x)}(x) = \sum_{i=1}^{n} x_i \cdot \Pr(x_i) \tag{1.34}$$

where the subscript $x \sim \Pr(x)$ indicates that x follows the distribution $\Pr(x)$. In many cases, we can drop the subscript and rewrite it as $\mathbb{E}(x)$. $\mathbb{E}(x)$ is essentially the weighted average value of x under the distribution $\Pr(x)$. It is a measure of central tendency, and is sometimes called the **mean** of a variable (denoted as μ).

Then, we can define the **variance** of a variable as the expected squared deviation from its mean

$$\mathrm{Var}(x) = \mathbb{E}[(x - \mathbb{E}(x))^2] \tag{1.35}$$

Informally, it describes how far the values are from the mean. $\mathrm{Var}(x)$ is usually written as σ^2, where σ is called **standard deviation**.

For a continuous variable $x \in \mathbb{R}$, the expected value is defined as:

$$\mathbb{E}(x) = \int_{-\infty}^{+\infty} x \cdot \Pr(x)dx \tag{1.36}$$

where $\Pr(x)$ is a probability density function. For computing the variance of x, we just reuse Eq. (1.35).

3. Entropy

Entropy is one of the most important tools of describing random variables and processes [Shannon, 1948b]. It is a measure of **expected surprisal**. The more deterministic the outcomes are, the less surprise and less information there will be. For simplicity, we restrict the discussion to discrete variables here[2]. Let x be a variable and $\Pr(x)$ be a distribution on x. The entropy is

[2]For continuous variables, we have similar calculations.

written as:

$$H(x) = -\sum_{i=1}^{n} \Pr(x_i) \cdot \log_b \Pr(x_i) \tag{1.37}$$

where b is the base of the logarithm function. The value of b is typically set to 2, 10 and e.

Beyond measuring the uncertainty of a single distribution, we can quantify the dissimilarity between two distributions. Suppose $p(x)$ and $q(x)$ are distributions on x. The **relative entropy** of p with respect to q is defined to be:

$$D_b(p||q) = \sum_{i=1}^{n} p(x_i) \cdot \log_b \frac{p(x_i)}{q(x_i)} \tag{1.38}$$

We can treat $p(x_i)$ as a weight to the log likelihood ratio $\log_b \frac{p(x_i)}{q(x_i)}$. Hence, $D_b(p||q)$ is a weighted sum of the likelihood ratios over all possible values. A smaller value $D_b(p||q)$ indicates that distributions p and q are closer. For example, p and q will be identical if $D_b(p||q) = 0$. The relative entropy is also called the **Kullback-Leibler (KL) divergence**. Note that the relative entropy is asymmetric, i.e., we cannot guarantee $D_b(p||q) = D_b(q||p)$.

Another concept that is popular in machine learning is **cross-entropy**. It is a measure of the information (in terms of the total number of bits) that we need to transmit the events from a source in one distribution with another distribution. More formally, we write the cross-entropy of the distribution p with the distribution q as $H_{\text{cross}}(p,q)$. It is defined to be:

$$H_{\text{cross}}(p,q) = -\sum_{i=1}^{n} p(x_i) \cdot \log_2 q(x_i) \tag{1.39}$$

Like relative entropy, cross-entropy is asymmetric. In the context of optimization, such as training NLP systems, both are frequently used as objective functions. The relationship between them is given by $H(p,q) = H(p) + D_{\text{KL}}(p\|q)$. While relative entropy measures the **additional** bits required due to using q instead of p, cross-entropy measures the total bits required.

1.2 Designing a Text Classifier

Classification is one of the most common problems in machine learning. It aims at automatically categorizing objects into a set of classes. These classes are called **labels**, or **tags**, or **categories**. In general, a program that performs classification is called a **classifier** or **classification system**. There are a vast number of practical applications of classifiers. A simple example is spam filtering where one needs to label an email as “spam” or “not-spam”. More challenging examples include classifying computed tomography images of organs into “normal” or “not-normal”, determining whether a piece of Chinese text is written by native speakers or not, labeling a patent application with a set of the IPC codes it belongs to, and so on.

Many machine learning theories and algorithms are modeled and tested on classification tasks. Following this convention, we consider text classification as an example to get started. Assume that we have a corpus like this.

Text	Label
The game was wonderful.	Not-food
I've tried my best to recreate it in my kitchen. It tastes heavenly.	Food
For centuries seaweed was considered a food for normal people.	Food
Have you finished your coding work today?	Not-Food
I was wondering how you could miss the bus.	Not-Food
I like fruit because it is good for health.	Food
Natural language processing research is amazing.	Not-Food
...	...

Each line of the corpus is a tuple consisting of a piece of text (hereafter referred to as a document) and a label that indicates whether the text is about food or not. We call such tuples **samples**, or more precisely **labeled samples**. Labeled samples are essentially question-answer pairs although they do not strictly follow the standard forms of questions and answers. For example, in the samples presented above, one can take a document as a question and take its label as the answer. In the next few chapters, we will show that such a formulation is general and fits most problems in NLP.

Next, let us assume that we have a classifier that learns how to label documents from these samples. The classifier is then used to label new, unseen documents as "Food" or "Not-Food". For example, for the text

Fruit is not my favorite but I can enjoy it.

the classifier would categorize it as "Food".

However, text classification, though it seems simple on the surface, is much more than merely sorting unlabeled samples into classes. It presents a wide variety of issues, especially when considering the ambiguities and richness of language. Modern classifiers are not systems comprising a set of hand-crafted rules. Instead, they model the classification problem in a probabilistic manner, making it possible to learn to classify from large-scale labeled data.

1.2.1 Problem Formulation

Let x be a document and c be a label. Here we assume a probabilistic classifier which would estimate how likely we choose c as the label of x, denoted as $\Pr(c|x)$. $\Pr(c|x)$ is in general a **classification model**. It describes a distribution over the set of all possible labels, satisfying

$$\sum_{c \in C} \Pr(c|x) \quad = \quad 1 \tag{1.40}$$

where C is the label set. For any document, we choose the most probable label as output via the classification model, like this

$$\hat{c} = \arg\max_{c \in C} \Pr(c|x) \tag{1.41}$$

where $\hat{c}$ is the "best" label predicted by the model. $\arg\max$ is the abbreviation of the arguments of the maxima. It returns the value of the argument that maximizes some function.

Eq. (1.41) is the fundamental equation of classification. It implies three problems

- **The modeling problem**. $\Pr(c|x)$ is a computational challenge because it is not obvious how to obtain the value of $\Pr(c|x)$ for each pair of x and c. To make an adequate model, one may need to represent x and c in a computationally tractable way and define a mathematical relationship connecting them.
- **The learning problem**. From a statistical learning point of view, the general form of $\Pr(c|x)$ represents a range of models configured with different variables or parameters. These models are essentially of the same form but would behave differently if we choose different values of those parameters. Thus, we need to choose a "good" model among them. This is typically addressed by optimizing the parameters on labeled data by some criterion.
- **The prediction problem**. We are addressing a **binary classification** problem here. Predicting document class is thus trivial as we just need to determine which class is more probable than the other. However, one can hardly imagine how difficult the prediction problem is in the real world, especially when predicting tree or graph-like structures and other non-linear structures[3]. For many NLP problems, prediction needs effective and efficient search algorithms.

These problems are general and cover many machine learning and natural language processing tasks. Binary classification, although one of the simplest cases, is sufficient to illustrate the fundamental concepts of machine learning. Classification also has several variants, such as:

- **Multi-class classification**. It is a generalization of binary classification. In multi-class classification, one needs to classify samples into one of three or more classes.
- **Multi-label classification**. This might be confusing because multi-class classification and multi-label classification seem to be the same thing. By conventional use of the terms, multi-label classification is referred to as assigning multiple labels to a sample. By contrast, the problem presented in Eq. (1.41) is a **single-label classification** problem.

Classification would be more interesting and challenging if we extend it to the case of dealing with hierarchical data. For example, for biological and patent data, some classes can be grouped into a super-class. This makes a hierarchy of the classes and requires a hierarchical classification schema.

[3]Predicting trees or other structures is not recognized as a standard sub-problem of classification. It is typically referred to as **structure prediction**. We will show in the later sections that both classification and structure prediction can share a similar machine learning paradigm.

1.2.2 Documents as Feature Vectors

The first problem we confront in designing text classification models is how to represent a document. Treating x as a string is simply not a good solution. One may want a representation by which a human being can understand the text. For example, we can parse each sentence in a document into a syntax tree and use trees as a text representation. This, however, requires efforts for developing additional NLP tools (such as syntactic parsers).

Representation, of course, is a fundamental issue in NLP. We omit complex, state-of-the-art models here to present a simple yet effective approach — **the bag-of-words (BOW) model**. The bag-of-words model is a feature-based model of representing documents. In machine learning, a **feature** is a property of a sample. It can be defined not only as a categorical attribute (e.g., name or gender) but also as a numerical value, such as a real number.

In the bag-of-words model, a feature corresponds to the frequency (or count) of a word. Let V be a vocabulary. A document can be represented as a $|V|$-dimensional feature vector. Each dimension corresponds to a word count feature, representing the number of times the i-th word in V appears in the document. More formally, let $\mathbf{x}$ be a feature vector. The i-th entry of $\mathbf{x}$ is defined as:

$$x_i = \mathrm{count}(V_i) \tag{1.42}$$

where V_i is the i-th word in V and $\mathrm{count}(\cdot)$ is a counting function. Consider, for example, the following lines of text[4].

As I went to Bonner

I met a pig

Without a wig,

Upon my word and honor.

We then have a vocabulary extracted from the corpus[5], like this

$$\begin{aligned} V = \{&\text{“a”, “and”, “as”, “Bonner”, “honor”,} \\ &\text{“I”, “met”, “word”, “my”, “pig”,} \\ &\text{“to”, “upon”, “went”, “wig”, “without”,} \\ &\text{“,”, “.”}\} \end{aligned}$$

Each line of the text can be seen as a document and represented as a feature vector. See below for the feature vectors generated by using the bag-of-words model.

[4] The text is from *Mother Goose rhymes*.

[5] We convert all words to lowercase at the beginning of each line.

	a	and	as	Bonner	honor	I	met	word	my	pig	to	upon	went	wig	without	,	.
As I went to Bonner	0	0	1	1	0	1	0	0	0	0	1	0	1	0	0	0	0
I met a pig	1	0	0	0	0	1	1	0	0	1	0	0	0	0	0	0	0
Without a wig,	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0
Upon my word and honor.	0	1	0	0	1	0	0	1	1	0	0	1	0	0	0	0	1

The bag-of-words model defines a vector space[6]. In this space, the similarity between two vectors is measured using metrics such as the dot product. This facilitates establishing relationships between documents: two documents with more overlapping words are considered more similar. This intuitive concept underpins many text classification systems.

The beauty of the bag-of-words model comes from its simple form in that any word is independent of other words. The independence assumption makes it possible to encode a document with almost infinite word relations as a countable, feasibly sized vector. On the other hand, representing a document as a feature vector of word counts is not the only option. As an improvement, one might take more context into account, and/or design more powerful features. This leads to an active line of research on text representation methods, ranging from heuristics-based methods to representation learning methods. We will see a few of them in the subsequent chapters.

1.2.3 Linear Classifiers

The **linear classifier** represents one of the most fundamental models in machine learning. Let $\mathbf{x} \in \mathbb{R}^n$ denote an input feature vector, and let $\mathbf{w} \in \mathbb{R}^n$ and $b \in \mathbb{R}$ denote the weight vector and bias parameter, respectively. A linear scoring function is defined as:

$$\begin{aligned} s(\mathbf{x}, \mathbf{w}, b) &= \mathbf{w} \cdot \mathbf{x} + b \\ &= \sum_{i=1}^{n} w_i x_i + b \end{aligned} \tag{1.43}$$

To simplify notation, we often employ the **bias trick**, defining an augmented input vector $\mathbf{x}' = [\mathbf{x}; 1]$ and weight vector $\mathbf{w}' = [\mathbf{w}; b]$. This allows Eq. (1.43) to be expressed compactly as a dot product:

$$s(\mathbf{x}', \mathbf{w}') = \mathbf{w}' \cdot \mathbf{x}' \tag{1.44}$$

Henceforth, we will omit the explicit bias term and refer to the parameters simply as $\mathbf{w}$.

In the context of classification, the linear function quantifies the affinity between an input and a class. For binary classification with classes $C = \{c_a, c_b\}$, we associate a weight vector with each class, denoted $\mathbf{w}_a$ and $\mathbf{w}_b$.

To perform prediction, we map the continuous scores to discrete class labels using a decision function (often termed an **activation function** or **threshold function**), $\psi(\cdot)$. For

[6] A vector space should be closed under vector addition and scalar multiplication.

binary classification, the decision rule is typically:

$$\begin{aligned} \hat{c} &= \psi(s(\mathbf{x}, \mathbf{w}_a) - s(\mathbf{x}, \mathbf{w}_b)) \\ &= \begin{cases} c_a & s(\mathbf{x}, \mathbf{w}_a) > s(\mathbf{x}, \mathbf{w}_b) \\ c_b & \text{otherwise} \end{cases} \end{aligned} \tag{1.45}$$

By linearity, $s(\mathbf{x}, \mathbf{w}_a) - s(\mathbf{x}, \mathbf{w}_b) = s(\mathbf{x}, \mathbf{w}_a - \mathbf{w}_b)$. The term $f(\mathbf{x}) = s(\mathbf{x}, \mathbf{w}_a - \mathbf{w}_b)$ serves as the **linear discriminant function**.

While discriminant functions directly predict class labels, they do not inherently provide probability estimates. To obtain a probabilistic output (i.e., $\Pr(c|x)$), we must map the scores to a valid probability distribution using a normalization function $\psi(\cdot)$, such as the Softmax function:

$$\begin{bmatrix} \Pr(c_a|\mathbf{x}) & \Pr(c_b|\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \frac{\exp(s(\mathbf{x},\mathbf{w}_a))}{\exp(s(\mathbf{x},\mathbf{w}_a))+\exp(s(\mathbf{x},\mathbf{w}_b))} & \frac{\exp(s(\mathbf{x},\mathbf{w}_b))}{\exp(s(\mathbf{x},\mathbf{w}_a))+\exp(s(\mathbf{x},\mathbf{w}_b))} \end{bmatrix} \tag{1.46}$$

Under this formulation, predicting the class with the highest probability remains equivalent to predicting the class with the highest score, aligning with Eq. (1.45).

Geometrically, a linear classifier defines a **separating hyperplane** (or **decision boundary**) in the feature space. Figure 1.1 depicts this in two dimensions. The objective of learning is to identify a hyperplane (like Hyperplanes 1 and 2) that correctly segregates the data points by class, avoiding the errors made by Hyperplane 3.

It is worthy of noting that linearity is the basis of many classifiers although most of them do not exist in the same form as Eqs. (1.43 - 1.44). A linear model is ideal for **linearly separable** problems. For non-linearly separable data, linear models are often combined with non-linear components, such as non-linear feature transformations or activation functions in neural networks, to capture more complex decision boundaries.

1.2.4 Generative vs Discriminative

Classification approaches are generally categorized into two paradigms: **generative models** and **discriminative models**. While both approaches often result in linear decision boundaries, they differ fundamentally in how they model the underlying data distribution.

1. Generative Models

A goal of classification is to learn $\Pr(c|\mathbf{x})$. Generative models do not explicitly model this conditional probability. Instead, they model the joint probability $\Pr(\mathbf{x}, c)$, and use Bayes' rule to compute $\Pr(c|\mathbf{x})$. This is given by the following equation.

$$\begin{aligned} \Pr(c|\mathbf{x}) &= \frac{\Pr(\mathbf{x}, c)}{\Pr(\mathbf{x})} \\ &= \frac{\Pr(c)\Pr(\mathbf{x}|c)}{\Pr(\mathbf{x})} \end{aligned} \tag{1.47}$$

where $\Pr(\mathbf{x}, c)$ is rewritten as $\Pr(\mathbf{x}|c)\Pr(c)$. For an optimal class $\hat{c}$, we choose a class c by

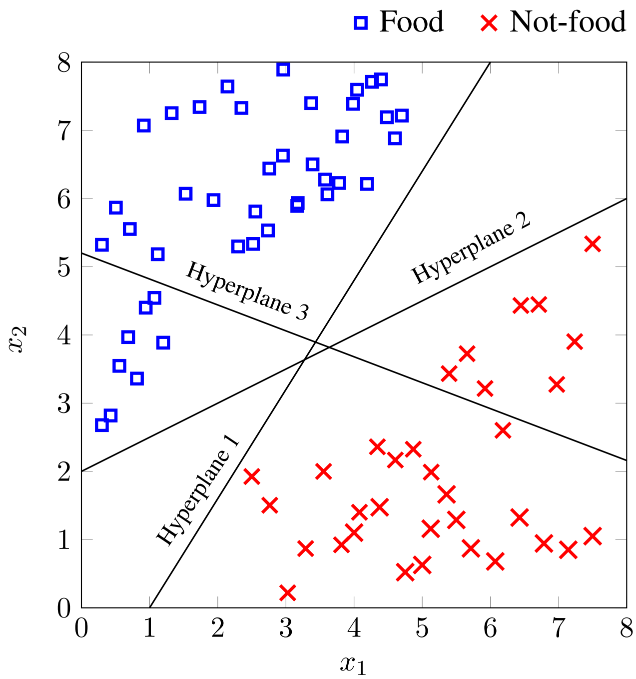


Figure 1.1: Data points and separating hyperplanes in two dimensions. There are two classes of data points (food and non-food). Both hyperplanes 1 and 2 separate the space into two sub-spaces where the two classes of data points are isolated. In this sense, the problem here is linearly separable. On the other hand, hyperplane 3 fails to separate the two classes, that is, the data points in the same class are classified into two different classes.

maximizing $\Pr(c|\mathbf{x})$ (see Eq.(1.41)). Since the denominator $\Pr(\mathbf{x})$ is a constant for the class c, we just need to maximize the numerator. Then, we rewrite Eq.(1.41) in the form

$$\begin{aligned} \hat{c} &= \arg\max_{c \in C} \Pr(c|\mathbf{x}) \\ &= \arg\max_{c \in C} \frac{\Pr(c)\Pr(\mathbf{x}|c)}{\Pr(\mathbf{x})} \\ &= \arg\max_{c \in C} \Pr(c)\Pr(\mathbf{x}|c) \end{aligned} \tag{1.48}$$

where $\Pr(c)$ is the prior of c, and $\Pr(\mathbf{x}|c)$ is the conditional probability of the input document vector $\mathbf{x}$ given c. Computing $\Pr(c)$ is straightforward. For example, the **maximum likelihood estimation** (**MLE**) defines $\Pr(c)$ as a relative frequency:

$$\Pr(c) = \frac{\text{count}(c)}{\sum_{c' \in C} \text{count}(c')} \tag{1.49}$$

where $\text{count}(c)$ counts the occurrences of c in a corpus.

However, estimating $\Pr(\mathbf{x}|c)$ is challenging because data sparsity prevents accurate estimation for high-dimensional vectors. Recall that the bag-of-words model defines x_i as the frequency of word V_i in the document. We assume here that the feature vector $\mathbf{x} = [x_1, ..., x_n]$ is generated by a multinomial distribution parameterized by $(p_1(c), ..., p_n(c))$, where $p_i(c)$ is the probability of word V_i occurring in class c. Based on MLE, we estimate $p_i(c)$ as:

$$p_i(c) = \frac{\text{count}(V_i, c)}{\sum_{1 \leq i' \leq n} \text{count}(V_{i'}, c)} \tag{1.50}$$

where $\text{count}(V_i, c)$ is the number of occurrences of V_i in all documents labeled as c. Then, $\Pr(\mathbf{x}|c)$ is given by

$$\Pr(\mathbf{x}|c) = \frac{\left(\sum_{i=1}^{n} x_i\right)!}{x_1! \cdot x_2! \cdot ... \cdot x_n!} \cdot \prod_{i=1}^{n} p_i(c)^{x_i} \tag{1.51}$$

Substituting Eq. (1.51) into Eq. (1.48), we have

$$\hat{c} = \arg\max_{c \in C} \Pr(c) \cdot \frac{\left(\sum_{i=1}^{n} x_i\right)!}{x_1! \cdot x_2! \cdot ... \cdot x_n!} \cdot \prod_{i=1}^{n} p_i(c)^{x_i} \tag{1.52}$$

Note that $\frac{\left(\sum_{i=1}^{n} x_i\right)!}{x_1! \cdot x_2! \cdot ... \cdot x_n!}$ is constant with respect to the class variable c. We drop it in $\arg\max$, and rewrite the right-hand side of the equation in log scale:

$$\hat{c} = \arg\max_{c \in C} \log(\Pr(c)) + \sum_{i=1}^{n} x_i \cdot \log(p_i(c)) \tag{1.53}$$

Clearly, this is a linear model. It defines the feature vector and weight vector as below

$$\mathbf{x} = \begin{bmatrix} 1 & x_1 & ... & x_n \end{bmatrix} \tag{1.54}$$

$$\mathbf{w} = \begin{bmatrix} \log(\Pr(c)) & \log(p_1(c)) & ... & \log(p_n(c)) \end{bmatrix} \tag{1.55}$$

Such a form is sometimes called a **log-linear** model, as the linearity comes from transforming the original problem via a logarithmic transformation.

In general, Eq (1.53) is called the **multinomial naive Bayes** approach. There are, of course, the naive Bayes variants for other types of feature vectors. For example, for binary-valued feature vectors, one can assume a Bernoulli distribution on each entry of a vector and design a **Bernoulli naive Bayes** classifier; for vectors with continuous features, one can assume a Gaussian distribution over continuous data and design a **Gaussian naive Bayes** classifier.

2. Discriminative Models

The model defined by $\Pr(c|\mathbf{x}) = \frac{\Pr(\mathbf{x},c)}{\Pr(\mathbf{x})} = \frac{\Pr(c)\Pr(\mathbf{x}|c)}{\Pr(\mathbf{x})}$ (see Eq. (1.47)) is called generative because it assumes a specific process for generating data $\mathbf{x}$ given label c. The idea is to use the joint probability $\Pr(\mathbf{x}, c)$ as an intermediate step to compute $\Pr(c|\mathbf{x})$. Alternatively, **discriminative models** do not attempt to model the distribution of $\mathbf{x}$. Instead, they estimate

$\Pr(c|\mathbf{x})$ directly.

An example is **logistic regression**. Recall that for binary text classification, a naive Bayes classifier predicts label c_a for a given document $\mathbf{x}$ only if the log-odds function is positive:

$$f_a(\mathbf{x}) = \log \frac{\Pr(c_a|\mathbf{x})}{\Pr(c_b|\mathbf{x})} \tag{1.56}$$

We can assume that this quantity follows a linear model:

$$\log \frac{\Pr(c_a|\mathbf{x})}{\Pr(c_b|\mathbf{x})} = \mathbf{w} \cdot \mathbf{x} \tag{1.57}$$

Since $\Pr(c_b|\mathbf{x}) = 1 - \Pr(c_a|\mathbf{x})$, solving for $\Pr(c_a|\mathbf{x})$ yields:

$$\Pr(c_a|\mathbf{x}) = \frac{1}{1+\exp(-\mathbf{w} \cdot \mathbf{x})} \tag{1.58}$$

This represents a sigmoid function, specifically the logistic function. Under this model, we predict c_a only if $\mathbf{w} \cdot \mathbf{x} > 0$. Eq. (1.58) defines the logistic regression classifier, which functions as a discriminative analog to the naive Bayes classifier.

An advantage of discriminative models is the flexibility they offer in framing classification tasks. While generative models attempt to estimate the data distribution of $\mathbf{x}$, discriminative models focus on establishing a robust boundary between classes. Discriminative models, therefore, care more about which class is prioritized over another given $\mathbf{x}$, or in probabilistic terms, which class is more likely to appear, instead of making strong assumptions on individual data points. This enables model learning by minimizing a specific error metric, rather than strictly maximizing the likelihood of the input data points. This approach falls under the framework of **error-driven learning**.

There are many ways to define these errors. Similar to generative models, one can train a discriminative model by optimizing parameters $\mathbf{w}$ to maximize the conditional likelihood of the training data. Let $\{(\mathbf{x}^{(1)}, c^{(1)}), ..., (\mathbf{x}^{(K)}, c^{(K)})\}$ be a set of labeled documents. The optimal parameter vector $\hat{\mathbf{w}}$ is given by:

$$\hat{\mathbf{w}} = \arg\max_{\mathbf{w}} \sum_{k=1}^{K} \log \Pr(c^{(k)}|\mathbf{x}^{(k)}) \tag{1.59}$$

Applying this to Eq. (1.58), the process corresponds to maximizing the log-likelihood:

$$\begin{aligned}
\hat{\mathbf{w}} &= \arg\max_{\mathbf{w}} \sum_{k=1}^{K} \left[\delta(c_a, c^{(k)}) \log \Pr(c_a|\mathbf{x}^{(k)}) + \delta(c_b, c^{(k)}) \log \Pr(c_b|\mathbf{x}^{(k)})\right] \\
&= \arg\max_{\mathbf{w}} \sum_{k=1}^{K} \Bigg[\delta(c_a, c^{(k)}) \log\left(\frac{1}{1+\exp(-\mathbf{w} \cdot \mathbf{x}^{(k)})}\right) + \\
&\qquad \delta(c_b, c^{(k)}) \log\left(1 - \frac{1}{1+\exp(-\mathbf{w} \cdot \mathbf{x}^{(k)})}\right)\Bigg]
\end{aligned} \tag{1.60}$$

where $\delta(\cdot,\cdot)$ is an **indicator function** that returns 1 if the two arguments are equal, and 0 otherwise.

Alternatively, we can train the model by minimizing the 0-1 loss, that is, we count an error whenever the predicted label does not match the true label:

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \sum_{k=1}^{K} \left(1 - \delta(\hat{c}^{(k)}, c^{(k)})\right) \tag{1.61}$$

where $\hat{c}^{(k)}$ is the prediction. While mathematically sound, optimizing 0-1 loss directly is difficult because it is non-differentiable, leading to the use of surrogate losses like the log-likelihood mentioned above.

The choice of training objective is crucial in discriminative models. This topic is broad and will be discussed further in Section 1.3.4. Furthermore, discriminative models do not require features to strictly adhere to a probabilistic generative process. $\mathbf{x}$ can be any feature vector engineered for the task. For example, let $g(\mathbf{x})$ be the output score of another system. Following Eq. (1.58), a new binary classification model can be constructed:

$$\Pr(c_a|\mathbf{x}) = \frac{1}{1+\exp(-\mathbf{w}\cdot g(\mathbf{x}))} \tag{1.62}$$

To learn $g(\mathbf{x})$, one can either pre-train it on some additional data, or train it jointly with $\mathbf{w}$ (i.e., the parameters of the upper-level model $\Pr(c_a|\mathbf{x})$).

1.2.5 OOV Words and Smoothing

The **out-of-vocabulary** (**OOV**) problem occurs when some of the words of a document are not found in the vocabulary. OOV words are common in NLP because new words are always there no matter how much text we have seen. Figure 1.2 gives two curves to illustrate this problem. As shown in Figure 1.2 (a), new words continuously appear when more data is available. When we fix the data that is used for testing the coverage of a vocabulary, OOV words remain even if we have an extremely large vocabulary (Figure 1.2 (b)).

For practical systems, OOV words are common because the vocabulary is often restricted to a "small" number of entries. A standard method is to keep the top-n most frequent words and discard the rest. In this case, OOV words are treated as *unknown* words. For example, a new symbol <unk> is introduced into the vocabulary so that all OOV words are denoted as <unk>. A more aggressive idea is to build an **open-vocabulary** system that accepts every possible word, but it would require more sophisticated algorithms and probably a task-specific design of data structures. The use of the <unk> is still the de facto standard for the development of current NLP systems.

For words that are already in the vocabulary, there are also **unseen words** that are absent in the parameter estimation phase but appear in a new document. A naive implementation of the model described in the previous section would be tough when dealing with unseen words. For example, Eq. (1.50) will simply give a zero probability if the word V_i does not occur in any training example labeled with c. In consequence, for a new example containing V_i, Eq.

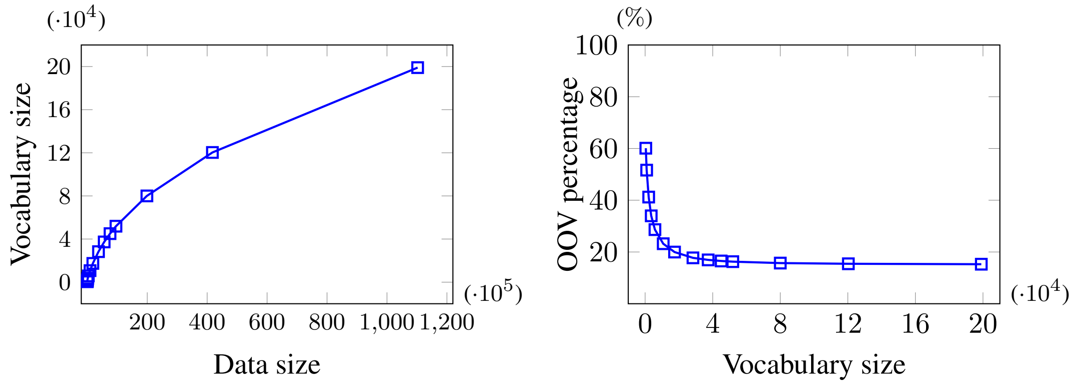


Figure 1.2: Data size (in number of words), vocabulary size and percentage of OOV. The statistics are collected on the English data provided in the WMT21 Zh-En translation task. The more data we use, the larger vocabulary we have. The increase in vocabulary size continues even if we build the vocabulary on data of more than 100 million words. However, the slope of the curve tends to be smaller as more data is involved. Interestingly, the OOV percentage converges to a certain level as the vocabulary size increases, indicating that new words always occur no matter how many words we have observed.

(1.51) will assign it a zero probability. This result is obviously unreasonable. However, we should not simply attribute it to the model design itself. Instead, a primary reason for this is the insufficient data used for parameter estimation. This is also explained by **Zipf's Law**: a small number of words occur quite often, while a large number of words occur rarely.

However, we cannot suppose that we always have access to some data where every word occurs sufficiently. Alternatively, one could adopt smoothing techniques to redistribute the probability over the vocabulary. Consider Eq. (1.50) as an example. We can add a small number to each V_i, and rewrite the equation as:

$$p_i(c) = \frac{\mathrm{count}(V_i, c) + \alpha}{\sum_{1 \le i' \le n} \mathrm{count}(V_{i'}, c) + n \cdot \alpha} \tag{1.63}$$

where α is the default count that we assign to each word. In this way, $p_i(c)$ gives a non-zero probability even if $\mathrm{count}(V_i, c) = 0$. Eq. (1.63) shifts probability mass from high-frequency words to low-frequency words. This method is called **additive smoothing** or **add-α smoothing**. It is one of the simplest smoothing methods. For other methods, we refer the reader to language modeling papers where smoothing is in heavy use [Chen and Goodman, 1999].

It is worth noting that in modern NLP, the traditional OOV problem is less frequently discussed. As we will introduce in later chapters, recent NLP systems often use **subword** tokenization methods. In these systems, the processing granularity is no longer traditional linguistic words, but rather statistically learned "units". Since the vast majority of unseen words can be assembled from these subword units (in the extreme case, words can be treated simply as sequences of characters), using a relatively compact subword vocabulary can significantly alleviate the OOV problem.

1.3 General Problems

Building a simple text classifier is a good start but not enough for solving complicated, diverse real-world problems. For a more general picture of how modern machine learning systems work, we now discuss some problems that are important when designing such systems.

1.3.1 Supervised and Unsupervised Models

Supervised learning deals with labeled samples, by which we mean that an input x is associated with an output y. Given a set of input-output pairs $\{(x^{(1)}, y^{(1)}), ..., (x^{(K)}, y^{(K)})\}$, the task here is to learn a function $f(\cdot)$ that maps each $x^{(k)}$ to $y^{(k)}$:

$$y^{(k)} \quad = \quad f(x^{(k)}) \tag{1.64}$$

This process is called supervised because learning $f(\cdot)$ is guided by the manually annotated answer $y^{(k)}$ for $x^{(k)}$. In general, $y^{(k)}$ is referred to as the ground-truth or gold-standard label of $x^{(k)}$. After that, when a new input x_{new} comes, we use the learned function $f(\cdot)$ to predict the output. We will say that the supervised learning succeeds if the prediction $f(x_{\text{new}})$ matches the ground-truth y_{new}.

The vast majority of NLP tasks can be framed as supervised learning problems. Assigning a class to a document is undoubtedly one of the simplest cases. Other NLP tasks include, but are not limited to, generating a sequence of labels, a piece of text, a syntax tree, or a graph.

In contrast to supervised learning, **unsupervised learning** deals with unlabeled samples, in other words, for each sample, we have an input x in the absence of the correct output y. In this case, we need an algorithm that learns from the unlabeled data $\{x^{(k)}\}$ the mapping function from $x^{(k)}$ to some output. Since there is no human intervention on the output of the function, algorithms of this kind need to discover patterns in $\{x^{(k)}\}$ and optimize the way we represent $\{x^{(k)}\}$ and function outputs by some criteria. These criteria are typically inspired by human prior knowledge so that the resulting function could output something that we expect.

A common example in unsupervised learning is **word clustering**. It groups a set of words by assigning similar words into the same cluster[7]. Based on such a criterion, we can bias $f(\cdot)$ towards outputting the same cluster for similar words. A more difficult case is **unsupervised bilingual dictionary induction**. It learns a word-level mapping between two languages without the need of parallel data. The problem is usually addressed, in part, by making use of the isomorphism of word representation spaces of different languages.

Falling halfway between supervised and unsupervised learning is **semi-supervised learning**. It deals with scenarios where only a small portion of the input data is labeled, while the vast majority remains unlabeled. Thus, it leverages the benefits of both paradigms. For example, in machine translation, we may have a certain amount of parallel data (labeled) and orders of magnitude more monolingual data (unlabeled). Often, a base system is trained on the parallel data in a supervised fashion. Improvements can then be made by training components of this base system on the large-scale monolingual data. This is implemented either

[7] It might be difficult and ambiguous to determine if two words are similar or not. We leave this issue to Chapter 3.

by combining a translation model learned on bilingual data with a language model learned on target-language monolingual data, or by pre-training parts of the model on monolingual data and fine-tuning the entire model on the bilingual data.

Broadly speaking, all learning algorithms need supervision. This sounds counterintuitive because unsupervised learning seems to not be signaled by any ground-truth data. However, from a general learning perspective, supervision signals should not be restricted to human-provided labels. Even in an unsupervised learning problem, the learning process is supervised by prior knowledge and the hidden distributional patterns within the input data $\{x^{(k)}\}$. In this sense, unsupervised learning is not strictly "learning without supervision".

Taking "supervision" in this broader sense, additional paradigms emerge as instances of machine learning, though they do not fit neatly into traditional supervised or unsupervised categories. An example is **reinforcement learning**. It models how a system makes a sequence of decisions by operating an agent within an environment. The agent receives feedback (or a reward) from the environment when taking an action. The goal of reinforcement learning is to learn a decision policy that maximizes the cumulative reward over the steps the agent takes. Often, the true reward is only available when the agent reaches a terminal state, such as winning or losing a game. As such, reinforcement learning excels at describing problems characterized by **delayed reward** (or sparse reward). This significantly differentiates it from standard supervised learning, which relies on instant, step-by-step supervision signals encoded in advance. This delayed-reward characteristic fits many NLP problems. For example, a text generation system outputs a sequence of words from left to right, but it is often difficult to determine if a single word was properly chosen until the entire text has been generated and evaluated.

Another example is **self-supervised learning**. It addresses unsupervised learning problems in a supervised learning manner. A general idea is to frame the unsupervised learning task as a pretext task that can be used in solving the original problem. In the pretext task, ground truth can be generated from input data. The learning therefore receives supervision from self-made signals instead of manual labels.

Self-supervised learning has been very successful in NLP. **Pre-training** is perhaps the most prominent application driving recent progress. In pre-training, a base model (such as a language model) is trained via self-supervised learning on massive corpora, and then its learned parameters are applied to a downstream system (such as a sentiment analysis model). This offers two major advantages. First, self-supervised models can be scaled to practically limitless amounts of text because they require no manual labeling. Second, the representations learned during pre-training are highly generalizable and can be applied to a wide range of downstream tasks. In Chapter 7, we will explore several examples of applying self-supervised learning to modern NLP models.

1.3.2 Inductive Bias

We informally describe (supervised) machine learning as a process of **inductive reasoning** (or **inductive inference**): we use specific observations (such as labeled documents) to construct a generalized model (such as a classifier). For example, if we observe that the word *cooking* frequently occurs in documents about food, inductive reasoning leads us to conclude that *cooking* is a strong indicator for all food-related documents. This "specific-to-general" method is simply called **induction**. Once we have an induced model, applying it to predict new, unseen observations is a process of **deduction**.

Induction is the most widely-used principle in designing learners of modern machine learning systems[8]. Imagine that there is a hypothesis space (or model space, or learnable function space) consisting of all possible models that we could make. Learning a model is thus the same as selecting a model in the hypothesis space by inducing from the given training samples. However, we cannot simply assume an oracle model that works well on all unseen samples. A reason for this is that searching for the "best" model in a huge hypothesis space is computationally infeasible. There would be an infinite number of dimensions along which we can design models if the hypothesis space is unconstrained. This is fundamentally related to **the curse of dimensionality**[9]. Another reason is that many models in the hypothesis space can fit training samples well, but only some of them can fit unseen samples. There is a risk of selecting a model that perfectly memorizes the training data but fails to generalize to unseen data. This is relevant to **overfitting**, a concept that we will discuss later.

A natural solution is to define priors on the hypothesis space in a way that allows some models to be more preferable than others. A simple example is that we restrict classifiers to linear models (see Section 1.2.3). This is equivalent to imposing a prior that excludes all non-linear models from the hypothesis space.

Such a prior is known as an **inductive bias**. In a nutshell, an inductive bias is a set of assumptions about the learning problem[10]. For instance, one can assume a specific mathematical form for the model (model bias), choose a particular optimization algorithm (algorithmic bias), or make assumptions about how the data was generated (sample bias)[11].

Inductive biases try to tell in what way we should describe a problem. Better results are generally favorable when the inductive bias aligns well with the true underlying data distribution. This explains why solutions to some problems prefer certain model architectures (or model biases). Of course, more and stronger inductive biases could make it easier to solve a problem. However, inductive biases are not always helpful, especially when they are not close to the reality.

Let us consider a dice-rolling game to illustrate this. Suppose you have a 6-sided die. Before rolling, you guess a number from 1 to 6, aiming to win as many times as possible. Based on common sense, a random guess is your only logical choice because all sides should have an equal chance of appearing ($1/6$). This holds true for a fair die. However, imagine one day you are playing with a weighted die, and winning becomes difficult. You notice the outcomes no

[8] Not all machine learning methods strictly follow induction. Other paradigms include deductive reasoning, abductive reasoning, and analogical reasoning (or transduction) [Hurley, 2011].

[9] The curse of dimensionality refers to the problems that generally appear as the dimensionality of the hypothesis space increases. For example, data sparseness is a common problem that arises when processing high-dimensional data, and is thus a kind of the curse of dimensionality.

[10] A more formal definition can be found in standard machine learning textbooks [Mitchell, 1997].

[11] It is worth noting that the term *bias* has different meanings in other contexts (e.g., bias in neural networks or cognitive bias). We will clarify when a different meaning is intended.

longer follow a uniform distribution. Then, you assume the outcomes follow a multinomial distribution (because you had the experience of developing naive Bayes classifiers). Before betting, you observe 100 practice rolls, choose the most frequent side for future games, and start winning again.

In this example, your initial assumption (i.e., all six sides are equally likely) is a very strong inductive bias because your mental model had zero degrees of freedom. It seemed obvious but failed for the weighted die. Your second inductive bias (the multinomial distribution), though mathematically more complex, is actually a *weaker* assumption. It defines a broader family of models, giving you the flexibility to find the specific probabilities that match the weighted die.

Ultimately, all machine learning models rely on inductive biases. Many of these assumptions are implicit; sometimes, we are not even aware we are making them because they feel entirely "logical". Yet, if an assumption is fundamentally flawed, it is harmful to the entire learning process. Therefore, practical experience remains essential to recognize and avoid these easily overlooked pitfalls.

1.3.3 Non-linearity

Most real-world problems are inherently non-linear, making them difficult to solve with simple linear models. Figure 1.3 provides examples of varying classification difficulty. In Figure 1.3 (a), the two classes can be perfectly separated by a straight line (a hyperplane). In this case, the problem is **linearly separable** because the decision boundary can be represented as a linear function. In contrast, in Figure 1.3 (b), we cannot draw a single hyperplane to perfectly separate the classes. Instead, we require a non-linear decision boundary, such as a hypersphere (a circle in 2D). An even more difficult case is shown in Figure 1.3 (c), where the classes are intertwined and the decision boundary is highly complex.

While modeling non-linear systems introduces computational complexities, several established methods exist to incorporate non-linearity into machine learning systems.

- **Feature mapping** and **kernel methods**. Recall that a linear classifier can be formulated as a function $f(\mathbf{w}\cdot\mathbf{x})$, where $\mathbf{w}$ is the weight vector, $\mathbf{x}$ is the feature vector, and $f(\cdot)$ is the activation function that returns one class (say c_a) if its argument is greater than zero and the other class (say c_b) otherwise. The idea of feature mapping is to map the feature vector $\mathbf{x}$ into a higher-dimensional space so that the problem is linearly separable in the new space. For example, let $\phi(\cdot):\mathbb{R}^2\rightarrow\mathbb{R}^4$ be a mapping function and $\mathbf{x}=\begin{bmatrix}x_1 & x_2\end{bmatrix}$ be a 2-dimensional vector. We assume that

$$\phi(\begin{bmatrix}x_1 & x_2\end{bmatrix}) = \begin{bmatrix}x_1^2+x_2^2 & x_1 & x_2 & 1\end{bmatrix} \tag{1.65}$$

  By choosing $\mathbf{w}=\begin{bmatrix}1 & -8 & -8 & 28\end{bmatrix}$, we get a new classifier:

$$\begin{aligned} f(\mathbf{w}\cdot\phi(\mathbf{x})) &= f\big(\begin{bmatrix}1 & -8 & -8 & 28\end{bmatrix}\cdot\begin{bmatrix}x_1^2+x_2^2 & x_1 & x_2 & 1\end{bmatrix}\big) \\ &= f\big((x_1-4)^2+(x_2-4)^2-4\big) \end{aligned} \tag{1.66}$$

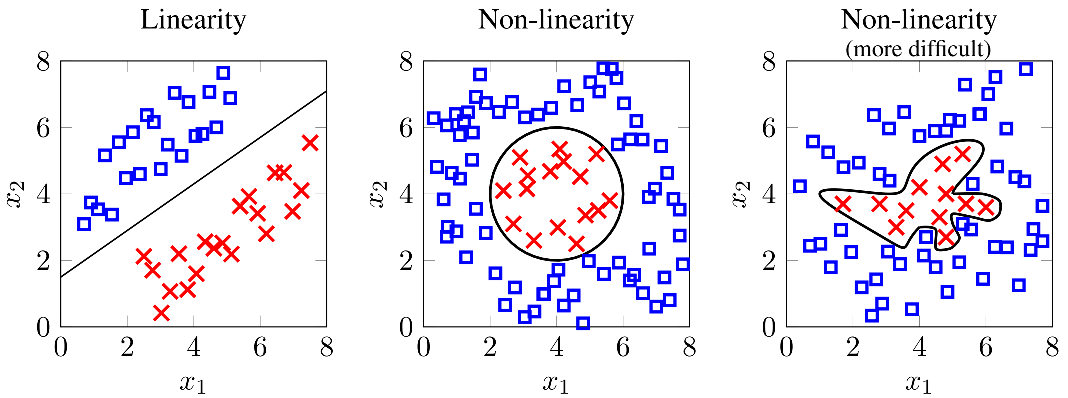


Figure 1.3: Linearity and non-linearity in binary classification. The first problem (left) is linearly separable because there exists (at least) a hyperplane that perfectly separates the data points in the two classes. The property of linear separability does not hold in the second problem (middle). Rather, we need a circle-like decision boundary. The decision boundary would be more complex if there are areas where the two classes of the data points are mixed and more or less indistinguishable (right).

It defines a decision boundary (i.e., a hypersphere $(x_1-4)^2+(x_2-4)^2=4$) that perfectly classifies the samples in Figure 1.3 (b). In other words, we use a linear model on the mapped feature space to create a non-linear model. However, computing the mapping function might be inefficient. This is typically addressed by using kernel methods. In kernel methods, the calculation of vector dot-product in the new space is performed efficiently by using a kernel function in the old space. This method is called **the kernel trick**. It has been successfully adopted in classification and other machine learning models, such as **support vector machines** [Cortes and Vapnik, 1995].

- **Non-linear activation functions**. Another standard approach to introducing non-linearity is through activation functions. A common paradigm is to apply a non-linear activation function to the output of a linear transformation. For instance, the function $f(\cdot)$ used in the previous example acts as a simple non-linear activation. Depending on the desired output format, various activation functions can be selected. In sophisticated models, multiple non-linear activation functions are embedded within intermediate computational steps to drastically increase the model's expressive power. A **deep neural network**, for example, is essentially a stack of sub-models (commonly referred to as layers), where each layer applies a linear transformation followed by a non-linear activation.

- **Non-parametric methods**. The term *non-parametric* originates from statistics In non-parametric statistics, statistical inferences are made without any assumption on underlying distributions of data. In machine learning, non-parametric methods follow the same idea. They do not assume any mapping function from input to output as Eq. (1.66). This differentiates them from **parametric methods** that explicitly learn a mathematical form of variables (or parameters) to describe the problem. An example of non-parametric methods is k**-nearest neighbors**, which classifies a new sample based on the majority vote of its k closest training samples. Note that non-parametric does not mean parameter-free. Rather, it means the number of parameters is not fixed in advance. The complexity of the model grows dynamically as more training data becomes available. As a reward, they can handle highly non-linear problems when training samples are sufficient.

Still, non-linear methods do not work alone. Linearity is an important component of most practical machine learning systems. This has two flavors. First, more non-linearity is not always better. If a linear model sufficiently resolves the problem, introducing non-linear complexity is unnecessary and risks overfitting. In accordance with **Occam's Razor** — *the simplest solution is almost always the best* — most state-of-the-art models maintain a balance of linear and non-linear components. Second, linear models serve as excellent approximations for non-linear behaviors. If the non-linearity of a problem is mild, using linear functions to approximate the solution is often significantly more efficient and interpretable for practical engineering purposes.

1.3.4 Training and Loss Functions

Almost all machine learning algorithms involve a training step. Training typically refers to the process of estimating the mapping function and its associated parameters from data. Here, we follow a conventional definition of the training problem: given a model or mapping function, we optimize a specific objective by evaluating the model through some training experience [Mitchell, 1997]. For example, training a naive Bayes text classifier requires maximizing a likelihood function (the objective) over a set of labeled documents (the training experience).

Often, the training problem can be framed as an **optimization** task. As such, we optimize an **objective function** via a training algorithm. Although the ideal objective is the actual performance measure on test samples, we cannot use it directly for optimization, as test samples and their corresponding labels are inaccessible during the training phase. Instead, practical objective functions are defined as computable surrogates for the test-time metric. These objective functions are not necessarily exact performance measures, but rather some metrics that are assumed to correlate with the performance on test data.

Let us consider a general case. Suppose $\mathbf{y}_\theta = f_\theta(\mathbf{x})$ is a model that takes a feature vector $\mathbf{x}$ as input and produces an n-dimensional output vector $\mathbf{y}_\theta$. For example, in text classification, $\mathbf{x}$ is the bag-of-words representation of a document, and $\mathbf{y}_\theta$ is a probability distribution over a set of classes. Here, θ represents the parameters of the model, and the subscript emphasizes that the behavior of the model is determined by θ. We further suppose that $\mathbf{y}_{\mathrm{gold}}$ is the ground-truth (or gold-standard) vector. We then define a **loss function**, denoted as $L(\mathbf{y}_\theta, \mathbf{y}_{\mathrm{gold}})$, which quantifies the discrepancy or error between the prediction $\mathbf{y}_\theta$ and the true label $\mathbf{y}_{\mathrm{gold}}$. It penalizes the model for inaccurate predictions. Given a model architecture, the training problem is described as finding the "best" parameters $\hat{\theta}$ such that the loss $L(\mathbf{y}_\theta, \mathbf{y}_{\mathrm{gold}})$ is minimized:

$$\hat{\theta} \;=\; \arg\min_{\theta} L(\mathbf{y}_{\theta}, \mathbf{y}_{\text{gold}}) \tag{1.67}$$

This formulation can be easily extended to the case of K training samples:

$$\hat{\theta} \;=\; \arg\min_{\theta} \frac{1}{K} \sum_{k=1}^{K} L(\mathbf{y}_{\theta}^{(k)}, \mathbf{y}_{\text{gold}}^{(k)}) \tag{1.68}$$

Once we obtain $\hat{\theta}$, we can use $f_{\hat{\theta}}(\mathbf{x})$ as a fixed model for further predictions.

$L(\mathbf{y}_{\theta}, \mathbf{y}_{\text{gold}})$ is typically referred to as the **loss function** (computed per sample), while its average over the dataset, $\frac{1}{K}\sum_{k=1}^{K} L(\mathbf{y}_{\theta}^{(k)}, \mathbf{y}_{\text{gold}}^{(k)})$, is often called the **cost function** or **empirical risk**. A loss function can be defined in many ways, depending on the problem we address and the priors we want to impose upon training. Here, we first consider the case in which $\mathbf{y}_{\theta}$ is a probability distribution. This is quite common in NLP, for example, $\mathbf{y}_{\theta}$ could be a distribution over a vocabulary, a list of documents, or a set of syntactic labels. For this type of model output, the most commonly used loss functions are measures of divergence:

- **Divergence-based Loss**. Divergence-based loss functions quantify the difference between the two distributions $\mathbf{y}_{\theta}$ and $\mathbf{y}_{\text{gold}}$. For example, cross-entropy (see Section 1.1.2) is one of the most popular loss functions used in NLP. One can also choose other divergence-based measures, such as the KL divergence and the Jensen-Shannon (JS) divergence, which can be found in most statistics textbooks. Note that the objective of **maximum likelihood estimation** (**MLE**) is fundamentally tied to divergence-based objectives. Specifically, minimizing the negative log-likelihood (the standard MLE objective) is mathematically equivalent to minimizing the cross-entropy loss when $\mathbf{y}_{\text{gold}}$ is a one-hot vector (where the entry of the correct label is 1 and all other entries are 0).

However, machine learning systems are not always restricted to distribution-like outputs. Rather, $\mathbf{y}_{\theta}$ could simply be a vector in $\mathbb{R}^n$. An example is the discriminant functions used in classification (see Section 1.2.3). These functions assign a score to each class, indicating how strongly the model believes a specific class is the correct answer. One way to define loss functions for real-valued vectors is to transform them into distribution-like forms[12] and resort to divergence-based losses. However, normalization is not always necessary, especially when we require scores outside the range of $[0,1]$. It is more common to compute losses directly on the raw outputs of these models. Below are some examples.

- **Distance-based Loss**. It is natural to formulate the loss as a geometric distance. A general example is the p-norm distance (see Section 1.1.1):

$$L(\mathbf{y}_{\theta}, \mathbf{y}_{\text{gold}}) \;=\; \Big(\sum_{i=1}^{n} |y_{\theta}(i) - y_{\text{gold}}(i)|^p\Big)^{1/p} \tag{1.69}$$

[12] For example, by normalizing the entries of a vector by the sum of all its entries.

For example, setting $p=2$ yields the standard Euclidean distance. Distance-based losses intrinsically describe a **curve fitting** problem: we learn a function $\mathbf{y}_\theta = f_\theta(\mathbf{x})$ to fit a set of target points $\{(\mathbf{x}^{(k)}, \mathbf{y}_{\text{gold}}^{(k)})\}$. This approach is typical in **regression** tasks[13]. A concrete example is the quality estimation of machine translation[14]. The model learns to predict a continuous translation quality score (y_θ) for a source-target sentence pair ($\mathbf{x}$). The prediction is deemed accurate if it closely matches human evaluation. Building upon Eq. (1.69), **mean squared error** (**MSE**) is a highly popular regression loss. It drops the $1/p$ root and simply calculates the sum of squared Euclidean distances:

$$L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) = \sum_{i=1}^{n} |y_\theta(i) - y_{\text{gold}}(i)|^2 \tag{1.70}$$

Another example is **mean absolute error** (**MAE**). It is precisely the form of Eq. (1.69) when $p=1$.

- **0-1 Loss**. The 0-1 loss is widely used in classification problems. It assigns a penalty of 1 for an incorrect prediction and 0 for a correct one. Let $c_\theta = \arg\max_c y_\theta(c)$ be the predicted label, selected by finding the highest value in $\mathbf{y}_\theta$. Likewise, let $c_{\text{gold}} = \arg\max_c y_{\text{gold}}(c)$ be the ground-truth label. The 0-1 loss is formally defined as:

$$\begin{aligned} L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) &= L_{\text{0-1}}(c_\theta, c_{\text{gold}}) \\ &= \begin{cases} 1 & c_\theta \neq c_{\text{gold}} \\ 0 & c_\theta = c_{\text{gold}} \end{cases} \end{aligned} \tag{1.71}$$

- **Margin-based Loss**. A **margin** represents the difference in predicted scores between the correct label c_{gold} and an incorrect label c:

$$\text{margin}(c, c_{\text{gold}}) = y_\theta(c_{\text{gold}}) - y_\theta(c) \tag{1.72}$$

It indicates a distinction between c_{gold} and c. So, a natural idea is to ensure that the margin is sufficiently large, or at least exceeds a minimum. This paradigm is known as **large-margin training**. Let $\triangle(c, c_{\text{gold}})$ be a predefined penalty cost for misclassifying c_{gold} as c, satisfying $\triangle(c, c_{\text{gold}}) \geq 0$, and $\triangle(c, c_{\text{gold}}) = 0$ if and only if $c = c_{\text{gold}}$. Our goal is to maximize the expression $\text{margin}(c, c_{\text{gold}}) - \triangle(c, c_{\text{gold}})$; equivalently, the loss should penalize instances where this value is small or negative. The margin-based loss

[13] When the model output is a vector with two or more dimensions, the problem is known as **multivariate regression**.

[14] In machine translation, quality estimation comprises several sub-tasks. Here, we use the term to refer to predicting an evaluation score directly.

is given by:

$$L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) = \max\Big(0, \max_c -\big(\text{margin}(c, c_{\text{gold}}) - \triangle(c, c_{\text{gold}})\big)\Big)$$
$$= \max\Big(0, \max_c \big(y_\theta(c) - y_\theta(c_{\text{gold}}) + \triangle(c, c_{\text{gold}})\big)\Big)$$
$$= \max_c \Big(0, y_\theta(c) - y_\theta(c_{\text{gold}}) + \triangle(c, c_{\text{gold}})\Big) \quad (1.73)$$

Designing $\triangle(c, c_{\text{gold}})$ depends on the target problem. A simple choice is $\triangle(c, c_{\text{gold}}) = 1$ for $c \neq c_{\text{gold}}$. This formulation makes Eq. (1.73) a variation of the **hinge loss**. An alternative is to use a sum instead of a maximum over all classes c:

$$L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) = \sum_c \max\Big(0, y_\theta(c) - y_\theta(c_{\text{gold}}) + \triangle(c, c_{\text{gold}})\Big) \quad (1.74)$$

- **Ranking-based Loss**. Ranking-based loss (or ranking loss) is used in several different areas, such as information retrieval, classification and metric learning. It deals with the problem where we want to order a set of scored items. Suppose the model output $\mathbf{y}_\theta$ evaluates n items, $\{c_i\}$. We define $\{\psi_\theta(c_i)\}$ as the predicted ranks of $\{c_i\}$ sorted by their scores $\{y_\theta(i)\}$. For example, given scores $\{y_\theta(i)\} = \{0.3, -2, 1\}$, the ranks would be $\psi_\theta(c_1) = 2$, $\psi_\theta(c_2) = 3$, and $\psi_\theta(c_3) = 1$. We analogously define $\{\psi_{\text{gold}}(c_i)\}$ as the gold-standard ranks. Note that if a task only requires relative ordering rather than absolute scores, $\{\psi_{\text{gold}}(c_i)\}$ can be defined without target values $\mathbf{y}_{\text{gold}}$. The idea is to penalize ranking inversions in $\{\psi_\theta(c_i)\}$ compared to $\{\psi_{\text{gold}}(c_i)\}$. A straightforward method is to penalize incorrectly ordered item pairs, sharing the idea of binary classification. Let Ω be the set of correctly ordered ground-truth pairs:

$$\Omega = \{(i,j) \mid \psi_{\text{gold}}(i) < \psi_{\text{gold}}(j)\} \quad (1.75)$$

The loss function aggregates penalties over these pairs:

$$L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) = \sum_{(i,j)\in\Omega} L_{\text{pair}}(y_\theta(i), y_\theta(j)) \quad (1.76)$$

where L_{pair} is a pairwise classification loss, such as the hinge loss used in [Collobert and Weston, 2008]:

$$L_{\text{pair}}(y_\theta(i), y_\theta(j)) = \max\big(0, y_\theta(i) - y_\theta(j) + 1\big) \quad (1.77)$$

This approach is known as the **pairwise method**. Ranking losses can also be defined in pointwise or listwise manners, which are widely used in modern learning-to-rank systems.

- **Contrastive Loss**. Contrastive loss is typically used in **contrastive learning**. It assumes that, given a sample, there is a similar sample that is labeled as “positive”, and there are a number of dissimilar samples that are labeled as “negative”. A natural idea is to minimize the distance between similar samples and simultaneously maximize the distance between dissimilar samples. Returning to our formulation, let $\mathbf{y}^{+}$ be the positive output and $\mathbf{Y}^{-} = \{\mathbf{y}^{-}\}$ be the set of negative outputs. Also, we use $\mathbf{y}_{\text{gold}}$ to denote the tuple of $\mathbf{y}^{+}$ and $\mathbf{Y}^{-}$ instead of a single gold-standard vector. A form of the contrastive loss function is given by the equation:

$$\begin{aligned} L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) &= L(\mathbf{y}_\theta, \mathbf{y}^{+}, \mathbf{Y}^{-}) \\ &= \log D(\mathbf{y}_\theta, \mathbf{y}^{+}) - \log \sum_{\mathbf{y}^{-} \in \mathbf{Y}^{-}} D(\mathbf{y}_\theta, \mathbf{y}^{-}) \\ &= \log \frac{D(\mathbf{y}_\theta, \mathbf{y}^{+})}{\sum_{\mathbf{y}^{-} \in \mathbf{Y}^{-}} D(\mathbf{y}_\theta, \mathbf{y}^{-})} \end{aligned} \tag{1.78}$$

where $D(\alpha, \beta)$ represents a distance metric (e.g., Euclidean distance) between α and β. A design challenge is generating these positive and negative outputs. In supervised setups, the ground-truth vector naturally serves as the positive target, while negatives can be drawn from other classes. In unsupervised setups, $\mathbf{y}^{+}$ and $\mathbf{Y}^{-}$ are often defined heuristically. For example, in an **auto-encoding** framework, $\mathbf{y}^{+}$ could simply be the input $\mathbf{x}$ itself (or an augmented view of it), while $\mathbf{Y}^{-}$ is a set of randomly generated samples.

- **Error-based Loss**. Evaluation metrics, generally used to quantify system errors, can also be used as loss functions. For example, in machine translation, BLEU is a commonly-used metric (where higher is better, bounded between 0 and 1). One could define the objective as minimizing $1 - \text{BLEU}$. Let $\text{Error}(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}})$ denote the calculated metric error. The error-based loss is simply this value:

$$L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) = \text{Error}(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) \tag{1.79}$$

So far we have presented several loss functions for a wide variety of problems, such as classification, regression, and ranking. As we will see in this book, different loss functions have different effects on model behavior. However, testing all possible loss functions is computationally impractical. Instead, practitioners must carefully select or design the most suitable objective function for their specific task. This process requires time and domain expertise but is essential.

On another hand, there are general methods to improve the design of loss functions. For example, rather than treating the model output $\mathbf{y}_\theta$ as a deterministic vector, we can model it as a **random variable** drawn from a probability distribution. As a result, the individual loss $L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}})$ also becomes a random variable. Under this framework, we redefine the objective as the **expected loss** (or expected risk) over the distribution of $\mathbf{y}_\theta$:

$$\begin{aligned} L(\mathbf{x}, \mathbf{y}_{\text{gold}}) &= \mathbb{E}_{\mathbf{y}_\theta \sim \Pr(\cdot|\mathbf{x})} \big[L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) \big] \\ &= \sum_{\mathbf{y}_\theta} L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) \cdot \Pr(\mathbf{y}_\theta | \mathbf{x}) \end{aligned} \tag{1.80}$$

By integrating over the entire space of possible predictions $\mathbf{y}_\theta$, this method offers a more robust estimation of the loss. Such a paradigm is rooted in **the Bayesian approach**, and $L(\cdot)$ (also denoted as $\mathbb{L}(\cdot)$) is commonly referred to as the **Bayesian risk** or simply **expected risk**.

Another way to improve training is introducing priors into the objective. A typical method is to add a regularization term R to the objective, as follows

$$\hat{\theta} = \arg\min_{\theta} L(\mathbf{y}_\theta, \mathbf{y}_{\text{gold}}) + \alpha \cdot R \tag{1.81}$$

where $R(\theta)$ is a function that penalizes certain aspects of the model complexity, such as the magnitude or the number of non-zero parameters. The coefficient α is a hyperparameter that dictates the strength of the regularization, which balances the trade-off between fitting the training data and maintaining model simplicity. The design of R is itself a critical area of research in practical machine learning systems. We will explore several specific regularization techniques later in this book.

Once the objective function is defined, an optimization algorithm is required to find the optimal parameters. Optimization is a vast field within machine learning. Hence, a detailed discussion of specific algorithms is beyond the scope of this chapter. Ultimately, there is no single universal algorithm capable of efficiently solving all training problems (a concept related to the "No Free Lunch" theorem). Instead, specific algorithms are chosen based on the mathematical properties of the objective function. For example, one can use *gradient descent* to train a neural language model with a cross-entropy loss [Bengio et al., 2003a], use *quadratic programming* to train an SVM with a hinge loss [Cortes and Vapnik, 1995], or use **minimum error-rate training** (**MERT**) to optimize a statistical machine translation model using a non-differentiable $1 - \text{BLEU}$ loss [Och and Ney, 2002].

1.3.5 Overfitting and Underfitting

The standard process of (supervised) machine learning comprises a training step and a test step. While one may try to minimize the loss on training samples, the learned model is used to deal with new samples that are never seen before. This is analogous to a common life experience: a student studying for final exams. While *studying hard* = *good grades* generally holds true, memorizing textbook questions and answers is a poor strategy for an exam featuring novel questions. Exam questions inevitably differ from the exact examples studied. We therefore need the ability to generalize.

In machine learning, generalization is used to describe how well a model learned through experience predicts on new data. A system is thought to have excellent generalization performance if, guided by appropriate inductive biases, it successfully extracts the underlying patterns from the training data without memorizing the noise.However, achieving good generalization does not mean less training. Instead, practitioners would like to train a machine learning model on more training data to prevent it from memorizing all the things. Generalization is a very complex issue determined by several factors, including problem complexity, model architecture, amount of training data, training algorithm and so on. While there are no standard rules to ensure good generalization, researchers always try to address it somehow.

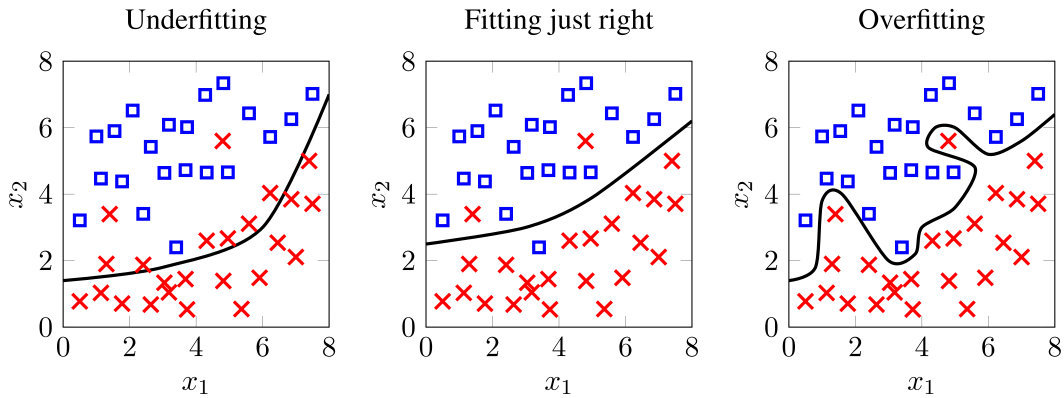


Figure 1.4: Decision boundaries of a binary classification problem. left = underfitting, right = overfitting, and middle = fitting just right. In the underfitting case, there are several obvious mistakes that are made in separating the two classes of data points. By shifting the decision boundary up a bit (middle), we obtain a satisfactory separation result, where most of the data points belonging to the same class are placed on the same side of the decision boundary. By contrast, a perfect separation requires a highly complex decision boundary instead (right).

To describe how well a model generalizes to new data, there are two important concepts, **underfitting** and overfitting. Underfitting occurs when a model fails to learn the underlying patterns in the training data sufficiently, resulting in poor performance on both the training and test datasets. This typically happens if the training process is prematurely terminated or if the capacity of the model is simply too low for the task. If a model underfits, one can improve it by increasing the training duration, engineering more expressive features, or using a model with a more complex architecture.

Conversely, overfitting refers to the phenomenon where a model fits the training data exceptionally well but generalizes poorly to test data (see Figure 1.4). In NLP, a basic manifestation of overfitting occurs when a text classifier rigidly memorizes the exact vocabulary seen during training, causing it to fail completely when OOV words appear (see Section 1.2.5).

The causes of overfitting are diverse. A primary cause is training an overly complex model on a small dataset. The model complexity often matters when we design a machine learning model. If the model is complex and has many parameters, then it would be much easier to overfit a small number of samples (see Figure 1.4). The problem would be more difficult if there is noisy data, because of the errors of "garbage in, garbage out" in training. In addition, excessive training is another cause of overfitting. For example, we can heavily tune a system to enforce it to model the data with no errors. The system would be fragile for new samples, even when there are small fluctuations in input.

Overfitting can be alleviated in many ways. Here are some commonly-used techniques.

- **Using more (high-quality) training data**. Large-scale training helps the model capture the true patterns in data. However, adding noisy data would do this in a negative way.
- **Using validation data**. Validation data is a disjoint subset of the available data used to

evaluate the model during the training process, without directly updating the parameters. For example, a dataset can be split into a training set and a **held-out validation set**. Practitioners can monitor performance on the validation set and apply **early stopping**, for example, halting the training process as soon as the validation performance begins to degrade.

- **Using simpler model architectures**. As noted previously, Occam's Razor is a principle we can follow in model design. If a problem is inherently simple, reducing the model's capacity (e.g., fewer layers or parameters) forces it to focus on the dominant, generalizable patterns in the data.
- **Regularization**. Regularization is another way to control the model complexity. Typically, it regularizes model parameters by priors. An example is smoothing (see Section 1.2.5). It re-estimates the distribution of words after training. A more general method is regularized training (see Eq. (1.81)). For example, we can define the regularization factor as the l_1 norm of the parameters, and bias the model to those whose parameters are not in large absolute values.
- **Combining multiple models**. A better prediction can also be made by ensembling multiple models. These models (call them **component models**) are in general of different parameters or architectures, and/or are trained with different portions of the data. The variance in models can reduce the risk that all these models overfit the data in exactly the same manner. These models are, therefore, less likely to make similar mistakes in prediction.

1.3.6 Prediction and Inference

Although we restricted our discussion to classification in previous sections, (supervised) machine learning is not just a task of predicting a label for an input object. There are many types of machine learning problems, depending on what form of the prediction is defined.

- **Classification**. Classification is perhaps one of the most common machine learning problems. A classification system is required to assign one or more classes to an input object.
- **Regression**. In statistics, regression studies the relationship between a **dependent variable** (or an outcome) and an **independent variable**. While regression has many applications, it is often framed as score prediction in NLP. For example, taking a movie review as input (i.e., an independent variable), the regression model learns to predict a recommendation score (i.e., a dependent variable).
- **Ranking**. A ranking model predicts the order of a set of input objects. For example, a model ranks a number of translations in terms of translation quality.
- **Structure prediction**. Many machine learning models are required to output not only a real value or a class but a tree or a sequence. The task of predicting structured outputs is called structure prediction. For example, a syntactic parser is a structure prediction system, as its output is a tree structure.

Despite being a fundamental aspect of machine learning, prediction is conventionally assumed to be trivial, given that many models and methods are tested on standard classification and regression tasks. On the other hand, prediction is non-trivial in structure prediction tasks, such as parsing and machine translation, which are very common in NLP. Essentially, predicting a tree or a sequence is a **search problem**. For example, there exists a theoretically infinite number of valid translations for a given source-language sentence. Even with a perfect model to evaluate every translation, finding the optimal translation within this search space presents a severe computational challenge.

In such cases, we must employ efficient ways to perform the search. This is implemented by either resorting to general search algorithms from artificial intelligence or developing new heuristics for specific problems. Furthermore, analyzing this search problem offers an important perspective on the mistakes made by a machine learning model: errors can be categorized either as **model errors** (due to inaccurate probability modeling) or as **search errors** (due to the search algorithm failing to find the true optimum). While eliminating search errors is a primary goal, doing so often requires a prohibitively large amount of computational effort. Therefore, deploying a machine learning model for practical purposes invariably requires a trade-off between efficiency and accuracy. We will see examples of this trade-off in Chapter 5.

1.4 Model Selection and Evaluation

For most machine learning problems, the goal is to find a model that would perform the best on new data. Two problems can be separated out from this goal [Hastie et al., 2009]:

- **Model selection**. Selecting the best model on training data by some criteria.
- **Model evaluation**. Estimating the performance of a given model on new data.

As noted in Section 1.3.4, loss functions (or error functions) are common ways of measuring errors in a prediction $\mathbf{y}_\theta = f_\theta(\mathbf{x})$ with respect to a gold-standard $\mathbf{y}_{\text{gold}}$. Given K labeled training samples $\{(\mathbf{x}^{(1)}, \mathbf{y}_{\text{gold}}^{(1)}), ..., (\mathbf{x}^{(K)}, \mathbf{y}_{\text{gold}}^{(K)})\}$, the **training error** is given by

$$\mathrm{Err}_{\text{train}} = L(\{\mathbf{y}_\theta^{(k)}\}, \{\mathbf{y}_{\text{gold}}^{(k)}\}) \tag{1.82}$$

where $\{\mathbf{y}_\theta^{(k)}\}$ are the predictions over the training dataset, and $\{\mathbf{y}_{\text{gold}}^{(k)}\}$ are the corresponding gold-standards. $L(\{\mathbf{y}_\theta^{(k)}\}, \{\mathbf{y}_{\text{gold}}^{(k)}\})$ is in general defined as the averaged loss over all training samples:

$$L(\{\mathbf{y}_\theta^{(k)}\}, \{\mathbf{y}_{\text{gold}}^{(k)}\}) = \frac{1}{K} \sum_{k=1}^{K} L(\mathbf{y}_\theta^{(k)}, \mathbf{y}_{\text{gold}}^{(k)}) \tag{1.83}$$

or defined as a single measure on the entire set of training samples. Likewise, we can define the **test error** on the test dataset, denoted as $\mathrm{Err}_{\text{test}}$. The test error serves as an estimate of the **generalization error**, indicating how well a model is expected to perform on new data.

In the preceding sections, we assumed that minimizing $\mathrm{Err}_{\text{train}}$ is the primary objective of

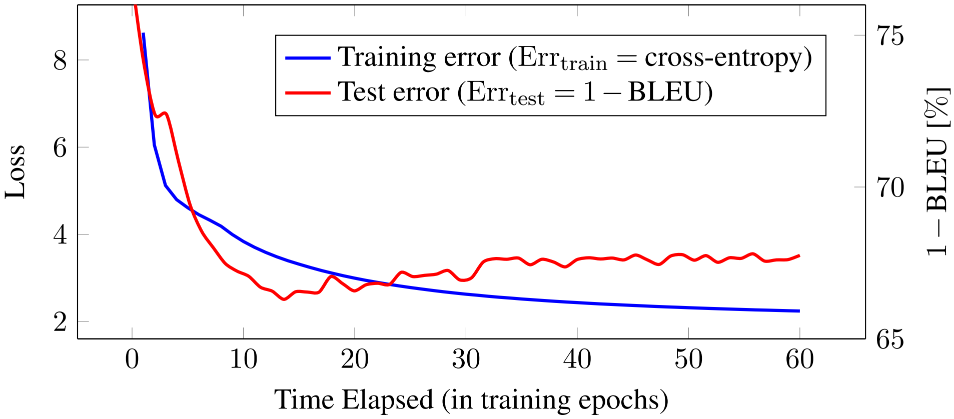


Figure 1.5: Curves of training error and test error for a machine learning system. The training error is measured in terms of the cross-entropy loss, and the test error is measured in terms of $1-\mathrm{BLEU}$. All statistics are collected by running a neural machine translation system on the IWSLT De-En benchmark. The training error continues to drop as more training epochs are involved. The test error, on the other hand, follows a trend of first going down and then going up. When the test error starts to increase, the model is likely to overfit the training data.

training, i.e., $\hat{\theta} = \arg\min_{\theta} \mathrm{Err}_{\mathrm{train}}$. However, we cannot assume that the resulting function $f_{\hat{\theta}}(\cdot)$ will achieve the minimum $\mathrm{Err}_{\mathrm{test}}$ simultaneously. Figure 1.5 illustrates the learning curves of a machine translation system. Clearly, $\mathrm{Err}_{\mathrm{test}}$ does not perfectly correlate with $\mathrm{Err}_{\mathrm{train}}$. While the training error decreases continuously as training proceeds, the test error initially follows this downward trend but eventually begins to increase, indicating model overfitting. This makes the problem a bit more complicated, as we cannot always trust $\mathrm{Err}_{\mathrm{train}}$ although it is and should be the measure of the goodness of training. Surely, we need some way to select a better model, in addition to looking at $\mathrm{Err}_{\mathrm{train}}$ only.

1.4.1 Strategies for Model Selection

Choosing the optimal model based solely on training data is inherently ambitious; we expect a machine learning model to generalize to the entire population from a finite, often small, sample set. From a statistical learning perspective, generalization is only possible because of how we formulate the learning problem. A fundamental assumption in standard machine learning is the i.i.d. assumption — that both the training and test data are independent and identically distributed, generated by the same underlying data-generating distribution.

If this assumption holds and the dataset is sufficiently large, a model that learns from the training data will likely perform well on the test data. By contrast, if the training and test sets are generated in a completely arbitrary, unconstrained manner (e.g., following a uniform distribution over the entire space of all possible functions), the model will fail to generalize. In this case, observing the training data provides absolutely no information about the test data.

This concept is formalized by the **No Free Lunch Theorem** [Wolpert, 1996; Wolpert and Macready, 1997], which proves that if we average performance across all mathematically possible problems, all learning algorithms perform equally well (equivalent to random guessing)[15]. In other words, universal learning is impossible: an algorithm can only succeed if it incorporates inductive biases that align with the specific problem at hand. Fortunately, in real-world applications, the training and test data is generally assumed to at least in part follow some distribution. Therefore, there are indeed some ways to capture this distribution and improve generalization ability of a model. Two scenarios are generally considered in improving machine learning systems:

- Given a fixed model architecture and training algorithm, how can we develop or curate training data to reduce test error?
- Given a fixed training dataset, how can we design or select a model architecture to reduce test error?

The first scenario is complex and involves practical data engineering issues, such as annotation, data cleaning, and data quality estimation. We will leave these topics for subsequent chapters and focus here on the model selection problem in the second scenario.

1. Model Complexity

The simplest method of model selection might be testing the models on validation data. Typically, this data does not overlap with either training or test data, but is assumed to be generated in the same way as the test data. However, such data is not always available. In some cases, we do not even know anything about the test data. So many model selection methods are validation-free.

A common alternative is to rely on **model complexity** (or **model capacity**) as a guiding indicator. In machine learning, model complexity can be interpreted in several different ways. Intuitively, a non-linear model is more complex than a linear one, and a model with millions of parameters is more complex than one with only a few. More formal definitions could be found in computational learning theory, such as the **Vapnik-Chervonenkis (VC) dimension** [Vapnik and Chervonenkis, 1971]. For our purposes, we treat model complexity broadly as a measure of the expressive power of a model: higher complexity equates to a larger hypothesis space.

While highly expressive models are powerful, increased complexity is not strictly beneficial. Complex models are highly susceptible to overfitting, particularly when trained on small datasets. Overly simplistic models, by contrast, are prone to underfitting. The goal of model selection is to identify the "optimal" level of model complexity. Figure 1.6 plots training and test errors against model capacity. Importantly, the optimal complexity is not found where the training error converges to zero, but rather at the point where the validation/test error reaches its global minimum before beginning to rise.

When explicit validation error is unavailable, Occam's Razor is frequently used: when presented with multiple models that perform comparably well on the training data, the simplest model should be preferred. Many criteria are available to quantify model complexity. For example,

[15] The no free lunch theorem was originally presented in a classification scenario [Wolpert, 1996], and was further extended to search and optimization problems [Wolpert and Macready, 1997].

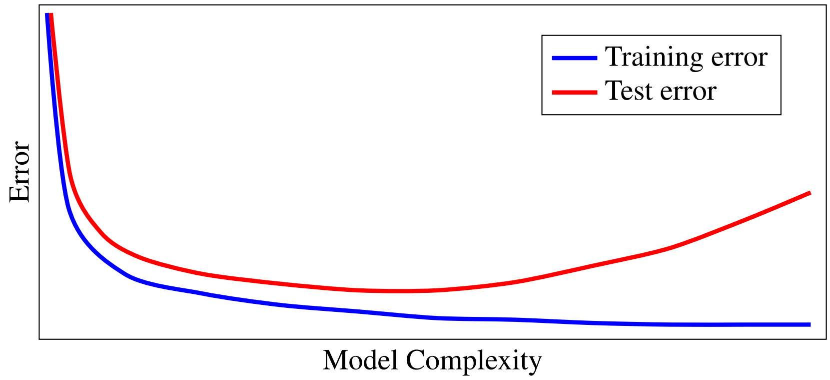


Figure 1.6: Curves of training error and test error under different model complexities. Complex models help in reducing the training error as they can compute complex functions in fitting data points. However, large model complexity is more likely to lead to overfitting and is harmful to the generalization ability of the models. For example, the test error increases as more complexity is added.

- **Number of parameters**. Though very simple, counting the number of free parameters is an intuitive and highly effective heuristic. In linear smoothing models, this is sometimes formalized as the **effective degrees of freedom**, defined as the trace of the projection matrix that maps the ground-truth $\mathbf{y}_{\text{gold}}$ to the predictions $\mathbf{y}_\theta$.
- **p-norm of parameters**. The p-norm of a weight matrix is a strong indicator of complexity (see Section 1.1.1). For example, under an L_1 or L_2 norm constraint, a model with larger absolute parameter values exhibits highly curved decision boundaries and is thus deemed more complex.
- **Description length**. Rooted in information theory and data compression, this evaluates the number of bits required to encode the model and its errors. The **Minimum Description Length** (**MDL**) principle favors the model that yields the highest degree of data compression.
- **The VC dimension**. It is originally from computational learning theory. The VC dimension formally quantifies capacity as the maximum number of data points that the classifier family can perfectly shatter (assign every possible combination of binary labels to).

In addition, there are other choices for defining the criterion, including **the Akaike information criterion** (**AIC**), **the Bayesian information criterion** (**BIC**), **the minimum message length** (**MML**) and so on. They can be found in many textbooks on statistics and/or statistical learning [Burnham and Anderson, 2002; Konishi and Kitagawa, 2007; Hastie et al., 2009].

2. Bias-Variance Tradeoff

Controlling model complexity to avoid overfitting and underfitting is also linked to the tradeoff between bias and variance. Bias (or prediction bias) is the amount that the model prediction differs from the true value. In statistics, bias is a **systematic error** that cannot cancel out even if we run a large number of repeated experiments. In general, bias error results from the wrong assumptions about the problem, such as approximating a non-linear problem via a linear model. This is very interesting! We can establish the connection of the bias error here with the inductive bias used in model design (see Section 1.3.2). For example, given training data, a large bias model is usually due to the fact that there are more assumptions and the model is not complex enough. Simply put, we would say that more (or stronger) inductive biases can result in a lower model complexity and more bias error in prediction. Occasionally, the term *bias* is used as a shorthand for both bias in prediction (from a statistics perspective) and inductive bias (from a model design perspective), although they are considered to have different meanings[16].

Variance, on the other hand, describes how spread out the predictions are when there are variations in the training data. The variance error also correlates with model complexity. For example, a complex model tends to exhibit higher variance.

Both bias and variance contribute to the overall error of a system. A classic illustration is the bias-variance decomposition of the mean squared error. Here we use some notation that differs slightly from previous sections. Let D be a set of K training samples and $f_{\hat{\theta}(D)}(\cdot)$ be a model learned on D. Furthermore, given a new sample $\mathbf{x}$, let $\mathbf{y}_{\hat{\theta}(D)} = f_{\hat{\theta}(D)}(\mathbf{x})$ be the model prediction and $\mathbf{y}_{\text{gold}}$ be the "true" prediction. The bias and variance are defined as:

$$\begin{aligned}
\text{bias} &= \mathbb{E}_D[\mathbf{y}_{\hat{\theta}(D)}] - \mathbf{y}_{\text{gold}} && (1.84) \\
\text{variance} &= \mathbb{E}_D\left[\left\|\mathbb{E}_D[\mathbf{y}_{\hat{\theta}(D)}] - \mathbf{y}_{\hat{\theta}(D)}\right\|^2\right] && (1.85)
\end{aligned}$$

where $\mathbb{E}_D[\mathbf{y}_{\hat{\theta}(D)}]$ is the expected prediction averaged over all possible training datasets of size K. Thus, the bias represents the deviation between the expected prediction and the true value, while the variance represents the expected squared deviation of the individual predictions from their mean[17]. Taking the mean squared error as the error measure, we can write the expected error as:

$$\begin{aligned}
\text{error} &= \mathbb{E}_D\left[\left\|\mathbf{y}_{\hat{\theta}(D)} - \mathbf{y}_{\text{gold}}\right\|^2\right] \\
&= \|\text{bias}\|^2 + \text{variance} && (1.86)
\end{aligned}$$

For lower mean squared error, reducing both bias and variance simultaneously is obviously the ideal goal. However, decreasing one typically increases the other (see Figure 1.7). Practitioners must seek the optimal level of model complexity to balance this tradeoff, thereby preventing overfitting and underfitting. This balance heavily depends on the inherent complexity of the problem. A simple linear model generally exhibits low variance but high bias, which leads to underfitting if the underlying problem is highly non-linear.

[16] Bias is more often used in statistics to describe some aspect of an estimator.

[17] Strictly speaking, bias as defined here is a vector, whereas variance evaluates to a scalar. Furthermore, since $\mathbf{y}$ represents a probability distribution in this chapter, the discrepancy between distributions is typically measured using divergence metrics (such as cross-entropy or Kullback-Leibler divergence) rather than Euclidean distance. The Euclidean-based bias formulation presented here serves as a mathematically intuitive illustration. Conceptually, one can replace the Euclidean difference with a divergence measure when evaluating distributions.

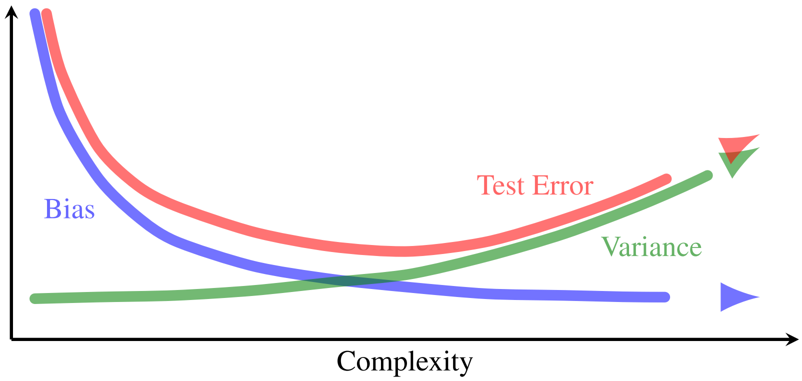


Figure 1.7: Bias and variance against model complexity [Goodfellow et al., 2016]. The curves show a conflict in reducing the bias error and the variance error simultaneously. By varying the model complexity, one can obtain either a low-bias, high-variance model or a high-bias, low-variance model. Both cases exhibit high test error. For example, a high-variance model is often of a larger model complexity. While such a model is able to deal with complex problems, it is more likely to overfit the data. On the other hand, a high-bias model often means a simpler model but tends to underfit the data. To improve generalization on test data, one can seek a tradeoff between bias and variance. For example, there is low test error when a "moderately complex" model is chosen.

Returning to the model selection problem, the bias-variance tradeoff is not a rule for model selection, but a principle guiding model design. Often, one must make compromises to create a model that generalizes well. It is also worth noting that, in many applications, complex models are usually accompanied by computational inefficiency. An appropriate method might be to start with a simple model and only add complexity when it is needed.

3. Model Combination

Selecting a single optimal model is not the only strategy to reduce generalization error. Alternatively, one can combine multiple models to form a single, more robust predictor. This paradigm is known as **ensemble learning** [Seni et al., 2010; Zhou, 2012a]. The core idea of ensemble learning is to train a set of **component models** (often called base learners) and aggregate their outputs to produce a superior prediction. The simplest approach is **model averaging** (or soft voting), which computes the average of the predicted scores from the individual models. More sophisticated methods, such as mixtures of experts, dynamically weight or combine the internal representations of these sub-models.

In general, component models should exhibit predictive diversity. This diversity is typically achieved by training models on different subsets of the data, using different feature subsets, or applying different weight initializations. From a statistical perspective, these methods can systematically reduce generalization error. For example, bagging helps to lower variance [Breiman, 1996], and boosting helps to lower bias [Schapire, 1990]. These are linked back to what we presented in Section 1.4.1: the generalization error can be reduced by either reducing the bias error or reducing the variance error.

But discussing how to combine models is beyond the scope of this chapter. Although it may not be strictly appropriate to categorize model combination as a topic related to model selection, it can be seen as a means of improving the generalization ability. In this sense, both model combination and model selection address problems on a similar theme. In fact, model combination is remarkably effective for many NLP tasks. For example, most state-of-the-art systems in NLP are based on the combination of multiple models.

1.4.2 Training, Validation and Test Data

We turn now to the data problem. As discussed in the previous sections, in the training stage, a training dataset is used to fit the parameters of the model. In the test stage, a test dataset is used to evaluate the learned model. Closely related to, yet distinct from, the test data is the validation dataset. While a validation dataset is also used for evaluation, it is employed during the training stage. Its primary purpose is to guide model selection and hyperparameter tuning (e.g., determining when to apply early stopping).

In an ideal scenario, researchers have access to separate datasets for training, validation, and testing. However, this is rarely realistic in real-world applications. For example, developers cannot access true future user data before a system is deployed. From a scientific point of view, there is no "real" new data for test — when you see new data, it is not new anymore.

A simple method, as in many research papers, is to verify machine learning models on benchmark tasks. In these tasks, all data is prepared in advance, and all you need is to run your models on the data. Such a method makes it easy to compare different systems directly, as all these systems are trained and tested on the same datasets. Occasionally, we are provided with a single dataset without predefined splits. In such cases, the data can be divided into parts each of which is used for some purposes. For example, a split could be 60% for training, 20% for validation, and 20% for test.

While data splitting provides a way to assess the performance of a model, the assessment result is not always stable due to sampling bias. Sometimes, the performance varies greatly across different runs of data splitting. The problem is more obvious when the dataset is too small to perform sufficient training or test.

A commonly-used way to mitigate the effect of this bias is **cross-validation**. Cross-validation is a resampling method where the data is repeatedly partitioned, and the final result is the average of the assessments over all rounds. A simple variant is random subsampling, which repeatedly randomizes the partition of the data and averages the performance across runs. A more systematic method is k-fold cross-validation. It divides the data into k mutually exclusive parts (or folds). In each round of cross-validation, $k-1$ parts are combined to form the training data, while the remaining single part is held out as the test (or validation) data. For example, in 10-fold cross-validation, a model is trained and evaluated 10 times, each time using a different fold as the test dataset.

Another note on the scale of data. For practitioners, one of the most frequent questions is how many samples are enough for learning a good model. There is rarely a universal consensus. While computational learning theory provides formal error bounds given a specific sample size, the prevailing empirical heuristic is simply “the more, the better”. However, modern machine learning increasingly emphasizes **sample efficiency**. A sample-efficient system can achieve a high level of performance using significantly fewer training examples or requiring fewer training epochs. For example, in recent research on NLP, fine-tuning a pre-trained model is highly sample-efficient because it leverages extensive prior knowledge, requiring only modest updates rather than learning representations from scratch. An extreme manifestation of this is **few-shot learning**, which aims to generalize robustly after observing only a handful of task-specific samples.

1.4.3 Performance Measure

As an essential part of every machine learning problem, a performance measure describes how well a system performs given some data. Usually it is used in either designing the training objectives or evaluating the result of the final system. For example, all those loss functions described in Section 1.3.4 can serve as performance measures.

When evaluating performance on test data, metrics are often designed to count actual errors. Thus, re-using the loss functions in training might not be a good choice for reporting the final score. For example, the widely-used measures for classification problems are precision, recall and F_1 score. They are proposed to quantify the ability of a classification system in certain aspects: given a class c, precision computes the fraction of correct predictions among all samples predicted as c, and recall computes the fraction of correct predictions on all samples labeled as c. The F_1 score is a measure that combines precision and recall.

Note that performance measures are not necessarily designed for optimization. In this sense, they may not guarantee some mathematical properties, such as being differentiable or continuous. An example is the **BLEU** metric widely used in machine translation [Papineni et al., 2002]. BLEU computes a modified n-gram precision score combined with a brevity penalty. These discrete counting operations make the metric inherently non-differentiable. NLP features many such task-specific evaluation metrics. This leads to a fundamental training-evaluation discrepancy: the loss function optimized during training differs from the metric used to judge the final model. Practitioners sometimes need to take this gap into account when designing systems.

Another challenge with performance measures in NLP is that a single input often admits multiple valid outputs. For example, there are generally multiple good translations for a source-language sentence. One solution is to take multiple gold-standards into account when designing a performance measure. BLEU is such a case. It counts the maximum number of the correct translation segments over all reference translations. The second solution involves human evaluation. Such a way of evaluation is more accurate but of course is more expensive. When developing practical systems, practitioners usually train and tune the systems using automatic measures, and call for human evaluations for the final test.

1.4.4 Significance Tests

Now, assuming you are improving a system in some way, you might be wondering if the improvement is significant enough or not. All you have is a performance measure. So you can tell the performance difference between any two points in developing the system, but you cannot tell if the difference is real or happens by chance.

In this example, you implicitly try to accept or reject a claim that a system is better than another system (or not). In statistics, **significance tests** provide a formal framework to model this problem. Suppose we evaluate two systems, A and B, across multiple datasets or test splits using a consistent performance measure. We formulate two opposing hypotheses:

H_0: System A performs no better than system B.

H_1: System A performs better than system B.

where H_0 is the **null hypothesis**, and H_1 is the **alternative hypothesis** that is contradictory to the null hypothesis. By testing these hypotheses against empirical data, we aim to determine whether we have sufficient evidence to reject H_0 in favor of H_1, or if we fail to reject H_0. Statistical testing is inherently prone to errors: we might incorrectly reject a true null hypothesis (a **Type I error**, or false positive), or we might fail to reject a false null hypothesis (a **Type II error**, or false negative). These two error types trade off against each other. A standard approach in frequentist statistics is to bound the probability of a Type I error below a predefined threshold α, while seeking to minimize the Type II error. This threshold α is known as the **significance level** of the test. Common practice in machine learning involves choosing a significance level of 1% or 5%.

When conducting the test, we calculate a p**-value**. Note that the p-value is not the probability of making a Type I error. Rather, it is the probability of observing a performance difference as extreme as, or more extreme than, the one actually observed, assuming the null hypothesis is true. If this p-value is less than the significance level α, we reject the null hypothesis. For example, with a significance level of 5%, an observed p-value of 3% indicates that the performance improvement is statistically significant. For a more comprehensive treatment of hypothesis testing, we refer the reader to common statistics textbooks [McClave and Sincich, 2006; Freedman et al., 2007; Freedman, 2009].

Note that the conclusion of significance tests depends on several factors, such as the number of experiments and the variance in the results of experiments. A problem with applying significance tests to NLP tasks is that there are often very few datasets for running the experiments [Dror et al., 2020]. When we can safely assume that the test statistic follows a known probability distribution (such as a Normal distribution), we can employ a parametric test. However, performance of real-world NLP systems rarely adheres to these strict assumptions. As a result, researchers frequently rely on classical **non-parametric tests** (which make no assumptions about the underlying distribution) or empirical resampling methods (such as bootstrapping or approximate randomization). These resampling techniques approximate the underlying distribution by repeatedly sampling the existing dataset or randomly shuffling the system outputs.

Significance tests are important for drawing convincing conclusions in developing machine learning systems, although they are often ignored unintentionally. Figure 1.8 shows evaluation results of three models. Each of them is run for several times with different initial parameters. While system A is superior to system B in terms of the averaged performance, there are large

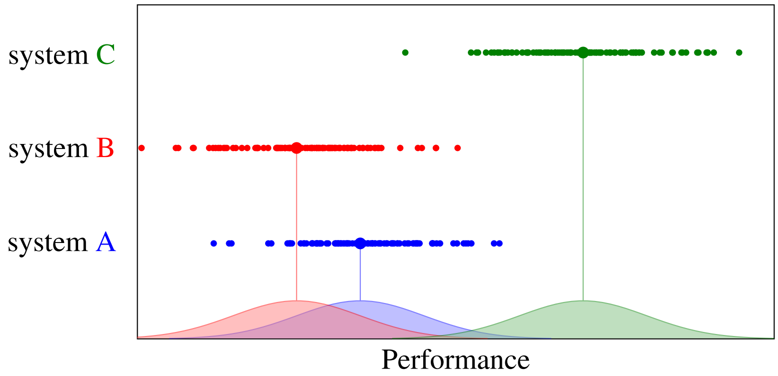


Figure 1.8: Performance of three machine learning systems. For each system, there are many different results because we introduce some randomness into training (e.g., data shuffling, random starting points, etc). Although it seems that System A outperforms System B, there is no real distinction between them, because they overlap a lot in the distributions of the performance (see the bottom of the figure). When comparing System C with System A or B, the difference in performance is significant because we could accept the H_1 hypothesis (i.e., System C outperforms System A or B) given a large number of experiments.

variances in their results. The significance test indicates that the difference is not significant. By contrast, the difference between system A and system C is significant because their performance differs greatly enough in most cases. On the other hand, researchers have found that there are indeed some thresholds of performance gain to indicate significance under certain circumstances. For example, we would say that the significance can be roughly indicated by a certain metric gain if we compare similar systems [Berg-Kirkpatrick et al., 2012].

1.5 NLP Tasks as ML Tasks

While there is a wide variety of NLP tasks, many of them can be formulated in a common machine learning framework. This enables a unified paradigm to solve diverse NLP problems using general machine learning approaches. Typically, an NLP task can be described as learning to map language units to some output. Following the notation used in this chapter, we use $\mathbf{x}$ to denote the input feature vector (or matrix) of an NLP task, and use $f(\mathbf{x})$ to denote the function that is learned to process $\mathbf{x}$. Here are some of the common tasks in NLP.

1.5.1 Classification

Suppose there is a set of classes or labels C. Each class is represented by a distinct integer in $\{1, ..., |C|\}$. A classification model is a function that maps the input $\mathbf{x}$ to a $|C|$-dimensional vector $\mathbf{y}$, i.e., $\mathbf{y} = f(\mathbf{x})$. Each entry of $\mathbf{y}$ is a score corresponding to class i, denoted by $y(i)$. The task here is to assign $\mathbf{x}$ to one or more classes having the highest scores. Consider single-label classification as an example. The prediction is given by the equation

$$\hat{c} \quad = \quad \arg\max_{1 \le i \le |C|} y(i) \tag{1.87}$$

where $\hat{c}$ is the “best” class assigned to $\mathbf{x}$. Sometimes, one needs a probability-like output (see Section 1.2.1). Let $\psi(\cdot)$ be a function that normalizes a vector into a distribution[18]. We then obtain a probabilistic classifier:

$$\mathbf{y} \quad = \quad \psi(f(\mathbf{x})) \tag{1.90}$$

Classification may be the most common problem in NLP. There are many applications in addition to categorizing documents into predefined classes. Among them are choosing a sense for a word [Yarowsky, 1994], determining the polarity of a sentence [Pang et al., 2002], checking whether two entities should be linked [Krebs et al., 2018], classifying the way of associating a semantic argument with a verb [Gildea and Jurafsky, 2002], and so on. When adapting a classification model to these tasks, all you need is to design the form of $\mathbf{x}$ and the set of classes.

1.5.2 Sequence Labeling

An extension to standard classification is to classify a set of samples simultaneously. **Sequence labeling** is an example of such a problem. In sequence labeling, the input is a sequence of n tokens, such as a sequence of n words. A sequence labeling system is required to assign each input token $\mathbf{x}(i)$ a label $l(i)$. Here the boldface in $\mathbf{x}(i)$ is used to emphasize that the token is represented as a feature vector. For convenience, we write $\mathbf{x}(i)$ as $\mathbf{x}_i$ and $l(i)$ as l_i. The function $f(\cdot)$ maps the sequence $\mathbf{x}_1 \ldots \mathbf{x}_n$ into another sequence $\mathbf{y}_1 \ldots \mathbf{y}_n$, where $\mathbf{y}_i$ is the output vector corresponding to $\mathbf{x}_i$. This can be formulated as:

$$\begin{bmatrix} \mathbf{y}_1 & \ldots & \mathbf{y}_n \end{bmatrix} \quad = \quad f(\begin{bmatrix} \mathbf{x}_1 & \ldots & \mathbf{x}_n \end{bmatrix}) \tag{1.91}$$

For vector $\mathbf{y}_i$, each entry $y_i(c)$ corresponds to the prediction score of a class $c \in C$. Note that $f(\cdot)$ allows for the use of a larger context. For example, one can condition the prediction $\mathbf{y}_i$ on the entire input sequence [Lafferty et al., 2001]. The final output of the system can be defined as the “optimal” label sequence induced from $\mathbf{y}_1 \ldots \mathbf{y}_n$. A simple method is to choose

[18] A similar idea can be found in Eq. (1.46). Given a vector $\mathbf{a} = \begin{bmatrix} a(1) & \ldots & a(n) \end{bmatrix}$, the normalization function has the form:

$$\psi(\mathbf{a}) \quad = \quad \begin{bmatrix} \frac{a(1)}{\sum_{i=1}^{n} a(i)} & \ldots & \frac{a(n)}{\sum_{i=1}^{n} a(i)} \end{bmatrix} \tag{1.88}$$

Another way is using the Softmax function:

$$\psi(\mathbf{a}) \quad = \quad \begin{bmatrix} \frac{\exp(a(1))}{\sum_{i=1}^{n} \exp(a(i))} & \ldots & \frac{\exp(a(n))}{\sum_{i=1}^{n} \exp(a(i))} \end{bmatrix} \tag{1.89}$$

Tokens :	Most	are	expected	to	fall	below	previous-month	levels	.
POS tags :	JJS	VBP	VBN	TO	VB	IN	JJ	NNS	.
Chunk tags :	B-NP	B-VP	I-VP	I-VP	I-VP	B-PP	B-NP	I-NP	O
	NP	VP				PP	NP		

Figure 1.9: An example of sequence labeling for POS tagging and chunking. The example is from the training data of the CoNLL 2000 shared task. Each token is labeled with a POS tag and a chunk tag. A chunk tag has an initial character chosen from $\{\mathrm{B}, \mathrm{I}, \mathrm{O}\}$, where B = beginning of a chunk, I = inside a chunk, and O = outside a chunk. So, a chunk always starts with a "B" tag, optionally followed by "I" tags. For example, the VP (verb phrase) chunk in the example spans over the chunk tag sequence "B-VP I-VP I-VP I-VP".

the label sequence that maximizes the sum of the scores over all positions

$$\begin{bmatrix} \hat{l}_1 & \dots & \hat{l}_n \end{bmatrix} = \underset{l_1,\dots,l_n \in C}{\arg\max} \sum_{i=1}^{n} y_i(l_i) \tag{1.92}$$

A straightforward application of sequence labeling to NLP is to tag each token of the input sequence, such as **part-of-speech tagging** (or **POS tagging**). Furthermore, sequence labeling is able to deal with more complex problems by using labels in a clever way. A well-known example is the use of the "IOB" label format in identifying chunks spanning multiple tokens (call it **chunking**). In this method, "I", "O" and "B" stand for a token inside a chunk, a token outside a chunk, and the leftmost token of a chunk [Ramshaw and Marcus, 1995]. As such, a chunk always starts with a "B" and ends just before the next "B" or a new "O". See Figure 1.9 for POS tagging and chunking results on an example sentence. As sequence labeling allows the labeling of both tokens and spans, it has been applied with strong results to many tasks, including POS tagging [Bahl and Mercer, 1976], chunking [Tjong Kim Sang and Buchholz, 2000], **named entity recognition** (**NER**) [Tjong Kim Sang, 2002], and so on.

1.5.3 Language Modeling/Word Prediction

Statistical language modeling (or **language modeling** for short) is the task of assigning a probability $\Pr(w_1, ..., w_n)$ to a sequence of words $w_1...w_n$. This joint probability is generally decomposed into a product of conditional probabilities, by using the chain rule:

$$\begin{aligned} \Pr(w_1, ..., w_n) &= \Pr(w_1) \cdot \Pr(w_2|w_1) \cdots \Pr(w_n|w_1, ..., w_{n-1}) \\ &= \prod_{i=1}^{n} \Pr(w_i|w_1, ..., w_{i-1}) \end{aligned} \tag{1.93}$$

Eq. (1.93) describes a procedure that generates a word sequence from left to right (call it **auto-regressive** generation). Estimating $\Pr(w_i|w_1,...,w_{i-1})$ is essentially a missing word prediction problem: we mask out the last word of a sequence and guide the language model to predict the correct word at that position. See below for a word sequence where the last word is missing.

Pride and prejudice is one of the best known __

We can reuse the idea in classification to model the probability distribution $\Pr(\ \_\ |\ \textit{Pride}, \textit{and}, ..., \textit{known})$. Let $\mathbf{x}_i$ be the vector representation of w_i. We can define a function that reads $\mathbf{x}_1...\mathbf{x}_{i-1}$ and produces a vector $\mathbf{h}_i$:

$$\mathbf{h}_i = f(\mathbf{x}_1,...,\mathbf{x}_{i-1}) \tag{1.94}$$

where $\mathbf{h}_i$ represents the intermediate state of the model at position i. To obtain a valid probability distribution, we normalize $\mathbf{h}_i$ by some normalization function $\psi(\cdot)$. Thus, the distribution at position i would be

$$\begin{aligned} \mathbf{y}_i &= \psi(\mathbf{h}_i) \\ &= \psi(f(\mathbf{x}_1,...,\mathbf{x}_{i-1})) \end{aligned} \tag{1.95}$$

Clearly, $y_i(w_i)$ is the probability of w_i given previous words, i.e., $y_i(w_i) = \Pr(w_i|w_1,...,w_{i-1})$. Note that Eq. (1.93) only considers the left context when predicting a word. A natural extension to this is to condition the prediction on all available context. Consider, for example, a sentence with a masked word in the middle.

Pride and __ is one of the best-known novels

In this example, we can predict the masked word by using both the left context (*Pride and*) and the right context (*is one of the best known novels*):

$$\mathbf{y}_i = \psi(f(\mathbf{x}_1,...,\mathbf{x}_{i-1},\mathbf{x}_{i+1},...,\mathbf{x}_n)) \tag{1.96}$$

This is a bidirectional model, and is commonly used in **auto-encoding** methods for learning sequence representation models [Devlin et al., 2019].

Note that while language modeling has historically been viewed as a basic task in NLP, it is now recognized as the absolute cornerstone of modern NLP systems. In particular, with the development of large language models, word prediction has emerged as the primary mechanism through which models acquire deep linguistic and world knowledge. As we will see in Chapters 7–11, the language modeling paradigm is extensively used to solve a wide variety of downstream tasks.

1.5.4 Sequence Generation

Sequence generation covers a range of NLP problems, including machine translation, summarization, question answering, dialogue systems, and so on. Usually, it refers to mapping some data to a sequence. Here we focus on the **sequence-to-sequence** problem, in that a source-side sequence is transformed to a target-side sequence, although sequence generation is not specialized to work with chain structures on the source-side.

For notational convenience, we use boldface variables to denote sequences from now on. For example, $\mathbf{a}$ is a sequence of size n. It can be written as either $\begin{bmatrix} a_1 & ... & a_n \end{bmatrix}$ or $a_1...a_n$. Let $\mathbf{s} = s_1...s_m$ and $\mathbf{t} = t_1...t_n$ be the source and target sequences, respectively. The sequence-to-sequence problem can be described as finding a target-side sequence that maximizes $\Pr(\mathbf{t}|\mathbf{s})$:

$$\hat{\mathbf{t}} = \arg\max_{\mathbf{t}} \Pr(\mathbf{t}|\mathbf{s}) \tag{1.97}$$

Like language modeling, $\Pr(\mathbf{t}|\mathbf{s})$ can be formalized in an auto-regressive fashion:

$$\begin{aligned}
\Pr(\mathbf{t}|\mathbf{s}) &= \Pr(t_1,...,t_n|\mathbf{s}) \\
&= \Pr(t_1|\mathbf{s}) \cdot \Pr(t_2|\mathbf{s},t_1) \cdots \Pr(t_n|\mathbf{s},t_1,...,t_{n-1}) \\
&= \prod_{i=1}^{n} \Pr(t_i|\mathbf{s},t_1,...,t_{i-1})
\end{aligned} \tag{1.98}$$

Eq. (1.98) differs from Eq. (1.93) only in the additional condition (i.e., $\mathbf{s}$) introduced to these probabilities. In this sense, we can use Eqs. (1.94-1.95) to model $\Pr(t_i|\mathbf{s},t_1,...,t_{i-1})$. However, conditioning on $\mathbf{s}$ makes the problem more complex, as we must model the cross-sequence relationship between $\mathbf{s}$ and t_i. A straightforward approach to sequence generation is to formulate $\Pr(t_i|\mathbf{s},t_1,...,t_{i-1})$ in the **encoder-decoder** paradigm. There are two steps: an encoder is first used to represent $\mathbf{s}$ as some intermediate form (e.g., a vector), and a decoder is then used to model both the target-side words and the correlation between the encoder output and the target-side words. Putting these together, the output of the encoder-decoder model can be defined to be

$$\mathbf{y}_i = \mathrm{Dec}(\mathrm{Enc}(\mathbf{s}),t_1,...,t_{i-1}) \tag{1.99}$$

where $\mathrm{Enc}(\cdot)$ is the encoder, and $\mathrm{Dec}(\cdot)$ is the decoder. $\mathbf{y}_i$ is a probability distribution over the target-side vocabulary at position i, i.e., $y_i(t_i) = \Pr(t_i|\mathbf{s},t_1,...,t_{i-1})$. Chapter 5 will provide a detailed description of the encoder-decoder model.

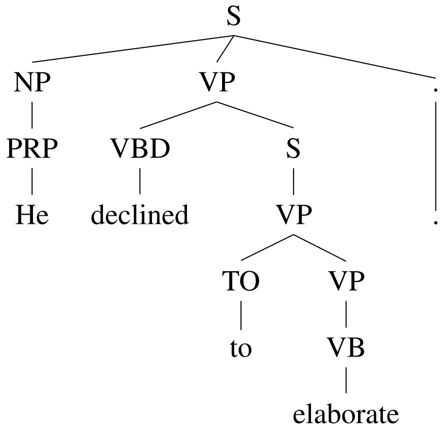
1.5.5 Tree Generation

In NLP, trees are usually used to represent the structures or meanings of sequential data. For example, a **syntactic parser** analyzes a sentence to form a **syntax tree** or **parse tree**. More formally, given a sequence of words $\mathbf{s} = s_1...s_m$, the parsing problem can be defined as:

$$\hat{d} = \arg\max_{d \in D} \Pr(d|\mathbf{s}) \tag{1.100}$$

where d is a parse tree, and D is the set of all parse trees yielding $s_1...s_m$. Computing $\Pr(d|\mathbf{s})$ is

Parse Tree:



CFG Rules:

r_1 : PRP $\rightarrow$ He
r_2 : VBD $\rightarrow$ declined
r_3 : TO $\rightarrow$ to
r_4 : VB $\rightarrow$ elaborate
r_5 : . $\rightarrow$.
r_6 : NP $\rightarrow$ PRP
r_7 : VP $\rightarrow$ VB
r_8 : VP $\rightarrow$ TO VP
r_9 : S $\rightarrow$ VP
r_{10} : VP $\rightarrow$ VBD S
r_{11} : S $\rightarrow$ NP VP .

$$\begin{aligned}\mathrm{P}(d|\mathbf{s}) = {} & \mathrm{P}(\mathrm{PRP} \rightarrow \mathrm{He}) \cdot \mathrm{P}(\mathrm{VBD} \rightarrow \mathrm{declined}) \cdot \mathrm{P}(\mathrm{TO} \rightarrow \mathrm{to}) \cdot \mathrm{P}(\mathrm{VB} \rightarrow \mathrm{elaborate}) \cdot \\ & \mathrm{P}(. \rightarrow .) \cdot \mathrm{P}(\mathrm{NP} \rightarrow \mathrm{PRP}) \cdot \mathrm{P}(\mathrm{VP} \rightarrow \mathrm{VB}) \cdot \mathrm{P}(\mathrm{VP} \rightarrow \mathrm{TO\ VP}) \cdot \mathrm{P}(\mathrm{S} \rightarrow \mathrm{VP}) \cdot \\ & \mathrm{P}(\mathrm{VP} \rightarrow \mathrm{VBDS}) \cdot \mathrm{P}(\mathrm{S} \rightarrow \mathrm{NP\ VP\ .}) \\ = {} & \textstyle\prod_{i=0}^{11} \mathrm{P}(r_i)\end{aligned}$$

Figure 1.10: An example parse tree and CFG rules. The sentence is from the training data of the CoNLL 2000 shared task. The parse tree is represented as a derivation of CFG rules. The probability of the parse tree is defined as the product of rule probabilities.

challenging, as the modeling complexity increases exponentially when moving from sequences to trees. In **statistical parsing**, a solution is to model d as a derivation of syntactic rules. In this way, $\Pr(d|\mathbf{s})$ can be formulated as a product of rule probabilities. Figure 1.10 presents an example of parsing with context-free grammar (CFG) rules. Alternatively, $\Pr(d|\mathbf{s})$ can be modeled in an end-to-end manner. For example, some recent approaches perform parsing by defining a neural network over the parse tree. The probability of a sub-tree rooted at a node is computed by considering the interaction between this node and child nodes.

Another method is to frame parsing as sequence generation. For example, one can linearize a parse tree and represent it as a sequence of words and syntactic labels, or transform the tree generation process into a sequence of actions. This allows the use of sequence-to-sequence techniques in addressing a sequence-to-tree problem.

In linguistics and NLP, tree structures are in heavy use for syntactic analysis. In addition to parsing sentences, they are also applied to represent the internal hierarchical structure of words (morphology), phrases, and entire discourses. However, trees are not the only way to visualize non-linear relationships. A more general data structure is the graph. While trees can be conceptualized as a special class of graphs, there are linguistic phenomena that trees cannot adequately capture [Fellbaum, 2005; Singhal, 2005; Banarescu et al., 2013]. For example, in the semantic representation of a sentence, we often require a graph to connect predicates with their arguments, allowing nodes to possess multiple parents. Although generating general graphs is computationally more demanding than parsing a sentence into a tree, we can successfully adapt many of the sophisticated methods originally developed for sequence and tree generation.

1.5.6 Relevance Modeling

Generally speaking, relevance refers to the degree to which one entity relates to another. The concept of relevance is used in many different sub-fields of NLP and information science. For example, in information retrieval, relevance is used to describe to what extent a retrieved document meets the query. Additional applications of this concept are fundamental to question answering, dialogue systems, semantic analysis, and various other tasks requiring a matching or retrieval mechanism.

Let us consider a more general description. Assume that we have a query and a key that represents something we intend to match with query. Then, we define the feature vectors of query and key as

$$\mathbf{q} = Q(\mathrm{query}) \tag{1.101}$$

$$\mathbf{k} = K(\mathrm{key}) \tag{1.102}$$

where $Q(\cdot)$ and $K(\cdot)$ are feature extractors. The relevance between the query and the key is then computed by a scoring function:

$$r = f(\mathbf{q}, \mathbf{k}) \tag{1.103}$$

The function $f(\cdot)$ can range from a simple distance metric (if $\mathbf{q}$ and $\mathbf{k}$ reside in the same vector space) to a complex parameterized model that performs non-linear transformations. In fact, the way of defining relevance can be adopted in several different scenarios. Sometimes, relevance is also termed similarity or correlation. Let x and y be two samples (e.g., two words). The similarity of x and y is given by

$$r = f(g(x), g(y)) \tag{1.104}$$

where $g(\cdot)$ is a feature extractor, and $f(\cdot)$ is a **similarity function**. Learning both $g(\cdot)$ and $f(\cdot)$ is called **similarity learning**. In one setup of similarity learning, we fix $f(\cdot)$ and learn $g(\cdot)$ in a way that similar samples exhibit similar outputs of $g(\cdot)$. Furthermore, the training of the feature extractor does not necessarily need to be coupled with the downstream similarity function. For example, for obtaining the similarity between words, we can pre-train $g(\cdot)$ via a language modeling objective and subsequently apply it in conjunction with various off-the-shelf similarity functions (such as cosine similarity). This reflects a broader philosophy in modern machine learning: decoupling representation learning from the specific downstream task. Such an idea is widely adopted in pre-training, advancing the recent state-of-the-art on many NLP tasks.

In another setup of similarity learning, we can learn $f(\cdot)$ directly. This can be performed by either jointly learning $f(\cdot)$ and $g(\cdot)$, or learning $f(\cdot)$ on top of fixed $g(\cdot)$. The problem is also related to **metric learning**. Typically, metric learning is framed as a supervised problem [Kulis, 2013]. A desired similarity function could be learned with the supervision regarding some gold-standard similarity. However, in practice there is usually no such supervised information in NLP. In this case, one could take relative distance as a proxy for supervision. For example, the similarity function can be learned by optimizing a contrastive loss (see Section 1.3.4).

Measuring the similarity between objects plays an important role in many machine learning methods, such as clustering and nearest neighbor classification. In NLP, it is useful for exploring the relationship between words, phrases, sentences, and documents, e.g., similarity is a way to examine how word vectors correspond to our understanding of word meanings [Mikolov et al., 2013c; Pennington et al., 2014].

1.5.7 Linguistic Alignment

Linguistic alignment is a set of problems where we establish some correspondence between two sets of linguistic units. In NLP, the sequence-to-sequence and sequence-to-tree problems are typically linguistic alignment problems, as they both connect two linguistic units. However, by convention, the term *alignment* refers to aligning multiple objects simultaneously[19].

As an example, consider the well-known **word alignment** task: we align the words of a sentence to the words of another sentence. We reuse the notation in Section 1.5.4 as both the sequence-to-sequence and word alignment tasks perform on a pair of sequences. Given a source-side word sequence $\mathbf{s} = s_1...s_m$ and a target-side word sequence $\mathbf{t} = t_1...t_n$, the word alignment between the two sequences is denoted as an $m \times n$ matrix $\mathbf{A}$. $A(i,j) = 1$ if there is an **alignment link** between s_i and t_j, and $A(i,j) = 0$ otherwise. The optimal alignment can be defined as:

$$\hat{\mathbf{A}} = \arg\max_{\mathbf{A}} \Pr(\mathbf{A}|\mathbf{s},\mathbf{t}) \tag{1.105}$$

where $\Pr(\mathbf{A}|\mathbf{s},\mathbf{t})$ is the word alignment probability. Like in other machine learning problems, we can model $\Pr(\mathbf{A}|\mathbf{s},\mathbf{t})$ in either a generative or discriminative manner (see Section 1.2.4). For example, in Brown et al. [1993]'s work, the word alignment model is factored into several generative steps, each accounting for some assumptions about the problem[20].

A 0-1 alignment matrix indicates a hard alignment scheme. A problem here is that the hard model may not describe well the highly ambiguous word alignments. We therefore can represent $\mathbf{A}$ as a real-valued matrix (call it a **soft word alignment matrix** or a **word alignment weight matrix**). Assume that the source-side words are represented as a sequence of feature vectors $\mathbf{x} = \begin{bmatrix}\mathbf{x}_1 & ... & \mathbf{x}_m\end{bmatrix}$. Likewise, the target-side words are represented as $\mathbf{y} = \begin{bmatrix}\mathbf{y}_1 & ... & \mathbf{y}_n\end{bmatrix}$. A soft word alignment model is given by:

$$\mathbf{A} = a(\mathbf{s},\mathbf{t}) \tag{1.106}$$

[19]The concept of alignment is wide-ranging. We use the term *linguistic alignment* here to differentiate it from the *alignment* of large language models discussed in subsequent chapters.

[20]More precisely, Brown et al. [1993] model $\Pr(\mathbf{A},\mathbf{s}|\mathbf{t})$ which is a surrogate of $\Pr(\mathbf{A}|\mathbf{s},\mathbf{t})$, as $\Pr(\mathbf{A}|\mathbf{s},\mathbf{t}) = \frac{\Pr(\mathbf{A},\mathbf{s}|\mathbf{t})}{\Pr(\mathbf{s}|\mathbf{t})} = \frac{\Pr(\mathbf{A},\mathbf{s}|\mathbf{t})}{\sum_{\mathbf{A}'}\Pr(\mathbf{A}',\mathbf{s}|\mathbf{t})}$

where $a(\cdot)$ is a word alignment function that computes the alignment weight $A(i, j)$ for each pair of $\mathbf{x}_i$ and $\mathbf{y}_j$. In fact, all the methods discussed in Section 1.5.6 are applicable to the design of $a(\cdot)$. This links the modeling of word alignment with the modeling of similarity, and makes it possible to address different NLP problems by using the same machine learning approach.

Eq. (1.106) offers a very general way to discover the underlying connection over pairs of variables. In addition to aligning words in sequences, it is useful for aligning unordered objects. For example, in bilingual dictionary induction, we can learn such a weight matrix to estimate how strongly a word in one language corresponds to a word in another language.

A final note concerns the role of linguistic alignment models. While linguistic alignment could be thought of as an independent NLP task, it is commonly used in designing sub-models of some downstream systems. Many systems that model word-level relationships involve implicit representation of linguistic alignment. As a consequence, linguistic alignment is treated as some latent states, and is a by-product of these systems. For example, in the early days of statistical machine translation, word alignment was a hidden variable used in modeling the mapping between sequences. The word alignment result can be easily induced from a machine translation model. More recently, neural sequence-to-sequence models — most notably attentional models [Bahdanau et al., 2014] — have attempted to do something similar to word alignment by computing attention weights among words.

1.5.8 Information Extraction

In NLP, *extraction* is not a kind of task but a kind of behavior that a system exhibits. Informally, it denotes a process of gathering and distilling structured information from some information sources. So, the term extraction generally appears together with other terms to form a specific task, such as **keyword extraction**, **event extraction**, and **relation extraction**. Many of these tasks can be categorized into an area — **information extraction**. Information extraction is perhaps the broadest topic in NLP. There is no exhaustive list of information extraction tasks. According to Jurafsky and Martin [2008]'s book, it includes but is not limited to named entity recognition, coreference resolution, relation extraction, event extraction, and template filling.

Because information extraction is a heterogeneous collection of diverse problems, it cannot be formulated in a single machine learning paradigm. Fortunately, most of these problems can be framed as standard machine learning problems, such as classification and sequence labeling, and can be solved by using off-the-shelf tools. In some cases, it may require a slight update of existing models for adaptation to a new task. For example, extracting a specific segment from text may require the system to produce a span that indicates the beginning and ending positions of the extracted segment (call it **span prediction**).

On the practical side, machine learning is not always necessary in extracting information from text. Many problems can be solved by using hand-crafted rules. An example is using regular expressions to identify locations and dates in text. In practice real-world systems are usually combinations of heuristic methods and automatic machine learning methods.

1.5.9 Remarks

Figure 1.11 shows illustrative examples of the above NLP tasks. Note that we only focused on NLP problems in the supervised learning paradigm. Many unsupervised tasks are important for NLP research as well. For example, it is common to cluster unlabeled words or documents to ease the processing in downstream systems. Several methods are directly applicable to this task [Murphy, 2012]. A recent trend in NLP is that it is not necessary to set a strict boundary between the use of supervised learning and the use of unsupervised learning. In many cases, unsupervised methods help supervised tasks, and vice versa. A notable example is that we learn a pre-trained feature extractor on unlabeled data and build a supervised classifier on top of it. This approach is widely known as **self-supervised learning**: it still follows a supervised learning framework, but the supervision signal is intrinsically derived from the unlabeled data itself. This has led to a paradigm shift toward advancing representation models (i.e., feature extractors) independently, without relying on specific downstream supervised tasks.

1.6 Summary

This chapter has given the basic ideas of machine learning and its applications to NLP problems. In particular, we have presented a simple text classification problem to get started with machine learning. Also, we have discussed several general problems on machine learning, e.g., types of machine learning methods, inductive biases, loss functions, overfitting and so on. They are followed by a discussion on model selection and assessment. In addition, we have described how modern NLP problems are framed as machine learning problems.

However, machine learning is a huge research field. Several interesting topics fall outside the scope of this chapter. One such topic that we said little about is reinforcement learning. In general, reinforcement learning is very powerful. It should be and has been considered as an approach to addressing NLP problems, e.g., training a sequence-to-sequence by using a risk-based loss function. A reinforcement learning textbook will offer the general ideas of reinforcement learning [Sutton and Barto, 2018]. Another topic we missed here is Bayesian learning [Gelman et al., 2020; McElreath, 2020; Downey, 2021]. It opens up a notable strand of research in statistical learning, and has been successfully used in NLP tasks. Moreover, there are many other topics that are specialized in certain aspects of machine learning and are of interest to NLP researchers and engineers. Some of them are efficient machine learning [Tay et al., 2020b], multi-task learning [Zhang and Yang, 2021], few-shot/zero-shot learning [Wang et al., 2019b; 2020c], and so on.

A final point to wrap up this chapter. We skip the detailed discussion on certain machine learning models and algorithms, such as classification and regression models, because the reader interested in them can find several excellent, comprehensive introductions [Bishop, 2006; Hastie et al., 2009; Murphy, 2012; Mohri et al., 2018]. In the next chapter we will discuss a bit more about artificial neural networks which are the basis of deep learning and recent state-of-the-art NLP models.

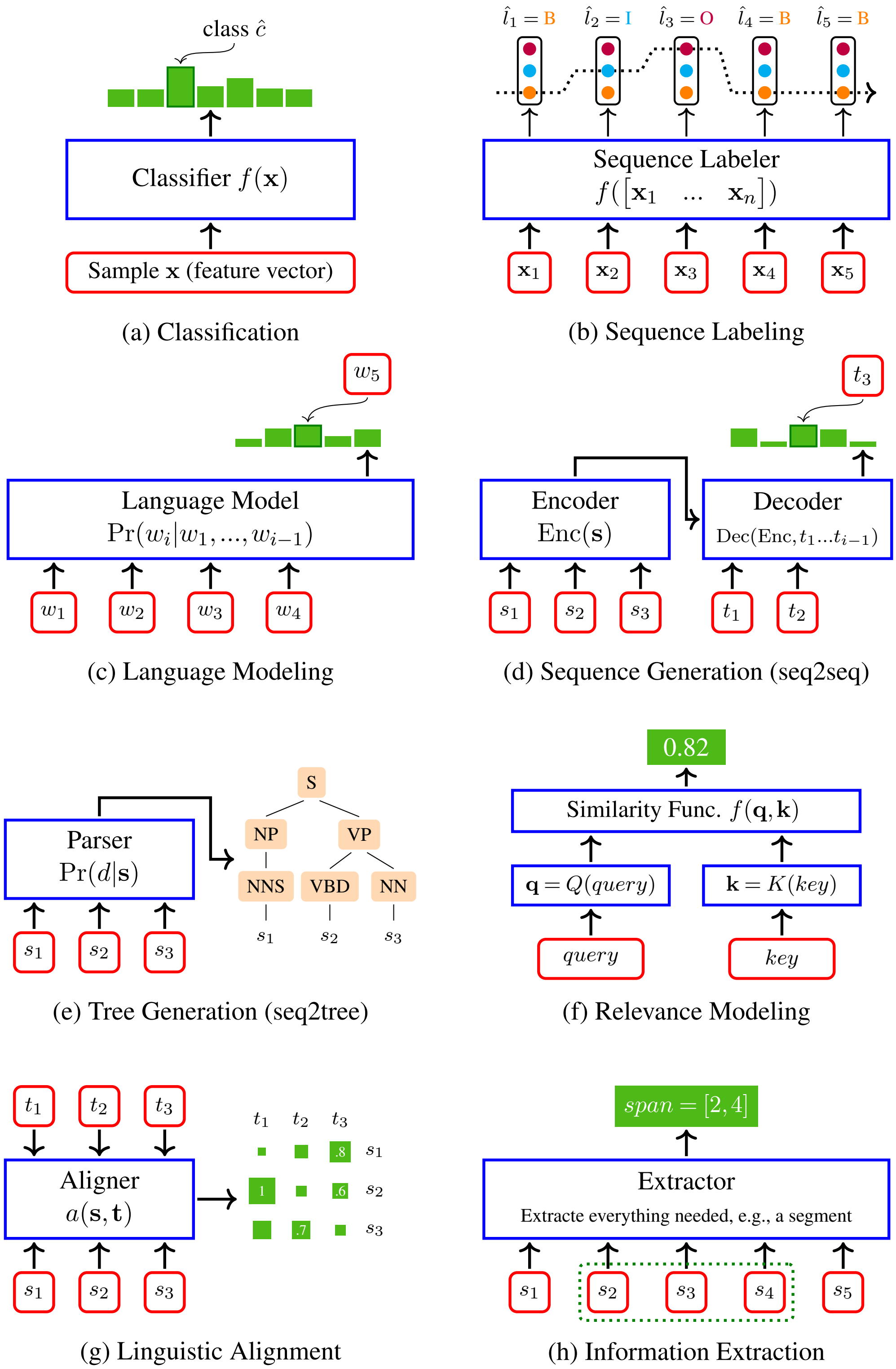


Figure 1.11: Natural language processing tasks from a machine learning perspective.

Chapter 2

Foundations of Neural Networks

Artificial neural networks (or **neural networks**, or **neural nets** for short) are powerful machine learning tools that have advanced the state-of-the-art in NLP in recent years. Although the history of neural networks can be traced back to the 1940s [McCulloch and Pitts, 1943], for a long time, they did not consistently outperform other machine learning counterparts. This changed around 2006 when novel ideas were developed to train **deep neural networks** [Hinton et al., 2006; Hinton, 2007]. These methods have since become known as **deep learning**. To date, deep learning has undoubtedly become one of the most active and influential areas in artificial intelligence, benefiting not only from "deep" model architectures but also from numerous techniques that help to train and deploy such models.

In this chapter, we will present the fundamental concepts of neural networks and deep learning. While it does not focus on the latest cutting-edge research, this chapter covers several important concepts and techniques widely used in implementing neural systems. These include fundamental model architectures, training and regularization methods, unsupervised learning techniques, and auto-encoders. We will also present an example of applying neural networks to the task of language modeling.

2.1 Multi-layer Neural Networks

To get started, we give a brief introduction to **single-layer perceptrons** and extend them to a more general case where multiple layers are stacked to form a more complex network.

2.1.1 Single-layer Perceptrons

Single-layer perceptrons (or **perceptrons** for short) may be the simplest neural networks that have been developed for practical use [Rosenblatt, 1957; Minsky and Papert, 1969]. Often, they are thought of as biologically-inspired models that transform some input to some output. A perceptron comprises a number of **neurons** connecting input and output variables. Figure 2.1 shows a perceptron where there is only one neuron. In this example, there are two real-valued variables x_1 and x_2 for input and a binary variable y for output. The neuron reads the input variables and determines which output value is chosen. This procedure is analogous to that of

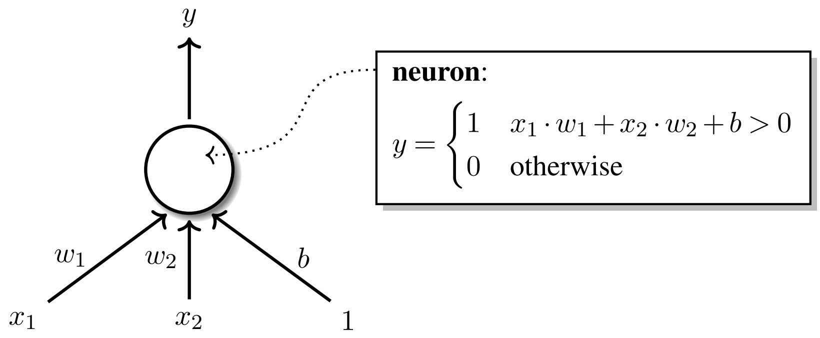


Figure 2.1: A perceptron with two input variables $\{x_1, x_2\}$ and an output variable y. There are two weights $\{w_1, w_2\}$, each corresponding to an input variable. The output depends on the sum of the weighted input variables and the bias term b, e.g., $y = 1$ if $x_1 \cdot w_1 + x_2 \cdot w_2 + b > 0$, and $y = 0$ otherwise.

a biological neuron — it receives electrochemical inputs from other neurons and determines whether the electrochemical signal is passed along.

Mathematically, a perceptron can be described as a mapping function. Let $\mathbf{x}$ be a vector of input variables (i.e., a **feature vector**). An **affine transformation** of $\mathbf{x}$ is given by[1]:

$$\begin{aligned} f(\mathbf{x}) &= \mathbf{x} \cdot \mathbf{w} + b \\ &= \sum_i x_i \cdot w_i + b \end{aligned} \tag{2.1}$$

where $\mathbf{w}$ is a weight vector and b is a bias term. Then, a standard perceptron can be defined as:

$$\begin{aligned} y &= \psi(f(\mathbf{x})) \\ &= \begin{cases} 1 & f(\mathbf{x}) > 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned} \tag{2.2}$$

where $\psi(\cdot)$ is a binary **step function**. The function $\psi(\cdot)$ is also known as an activation function. This links the perceptron to the classification models discussed in Chapter 1. In other words, Eq. (2.2) acts as a classifier itself: $f(\mathbf{x})$ is a discriminant function defined on each input $\mathbf{x}$, followed by an activation function $\psi(\cdot)$ used to produce the desired output[2].

When there are two or more neurons, we can group these neurons into a **layer**. As shown in Figure 2.2, all the neurons in a layer receive signals from the same input feature vector but are weighted in different ways. The output of the layer is a new feature vector, each entry of which corresponds to a neuron. More formally, taking $\psi(\cdot)$ and $f(\cdot)$ as vector functions, the

[1] In mathematics, a **linear transformation** maps each vector $\mathbf{v}$ in a space to $f(\mathbf{v})$ in another space, such that for any vectors $\mathbf{x}$ and $\mathbf{y}$, and scalars α and β, we have $f(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y})$. An affine transformation is a linear transformation followed by a translation, often written in the form $f(\mathbf{x}) + \mathbf{b}$.

[2] Since the decision boundary is linear, the perceptron is considered a linear classifier, even though the step function itself is non-linear.

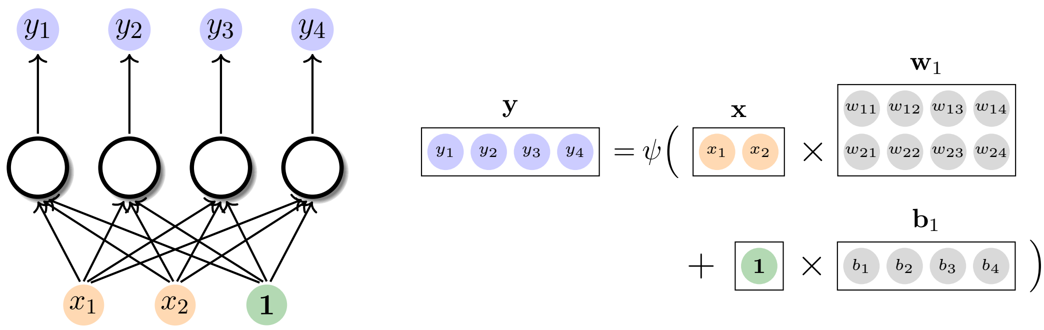


Figure 2.2: A single-layer perceptron involving four neurons. All these neurons receive information from the input variables $\{x_1, x_2\}$. The perceptron describes a process where 1) we first transform the input vector of variables by an affine transformation $f(\mathbf{x}) = \mathbf{x} \cdot \mathbf{w} + \mathbf{b}$; and 2) compute the output by feeding $f(\mathbf{x})$ into the activation function $\psi(\cdot)$.

mathematical form of the single-layer perceptron is given by the equations:

$$\mathbf{y} = \psi(f(\mathbf{x})) \tag{2.3}$$

$$f(\mathbf{x}) = \mathbf{x} \cdot \mathbf{w} + \mathbf{b} \tag{2.4}$$

where $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{w} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^n$.

Another note on the activation function: the step function, though extensively used, is not the only form of the activation function. There are many different ways to perform activation. For example, we can use the Softmax function if we want a probability distribution-like output; we can use the Sigmoid function if we want a monotonic, continuous, easy-to-optimize output; we can use the ReLU function if we want a ramp-shaped output. Table 2.1 shows several commonly used activation functions. Note that, although a layer of neurons equipped with these activations can be loosely called a single-layer perceptron, it can be categorized as a more general concept, called a **single-layer neural network**. If not specified otherwise, we will use the term *single-layer neural network* throughout this book.

2.1.2 Stacking Multiple Layers

A next obvious step is to create a neural network comprising multiple layers. To do this, we simply stack multiple layers to form a **multi-layer neural network**. Figure 2.3 illustrates an example. In this architecture, the output of every neuron in a given layer is connected to all neurons in the subsequent layer. So the network is **fully connected**. Essentially, a multi-layer neural network describes a composition of functions. a multi-layer neural network describes a composition of functions. For example, we can formulate the neural network in Figure 2.3 as a complex function formed by the composition of several simpler functions:

$$\mathbf{y} = \mathrm{Softmax}(\mathrm{Sigmoid}(\mathrm{ReLU}(\mathbf{x} \cdot \mathbf{w}_1) \cdot \mathbf{w}_2) \cdot \mathbf{w}_3 + \mathbf{b}_3) \tag{2.5}$$

where $\mathbf{w}_1 \in \mathbb{R}^{3 \times 4}$, $\mathbf{w}_2 \in \mathbb{R}^{4 \times 3}$, $\mathbf{w}_3 \in \mathbb{R}^{3 \times 3}$, and $\mathbf{b}_3 \in \mathbb{R}^3$ are parameters.

Name	Formula (for entry i of a vector)
Identity	$y_i = s_i$
Binary Step	$y_i = \begin{cases} 1 & s_i > 0 \\ 0 & s_i \leq 0 \end{cases}$
Hyperbolic Tangent	$y_i = \dfrac{\exp(s_i) - \exp(-s_i)}{\exp(s_i) + \exp(-s_i)}$
Hard Tangent	$y_i = \begin{cases} 1 & s_i > 1 \\ s_i & -1 \leq s_i \leq 1 \\ -1 & s_i < -1 \end{cases}$
Sigmoid (Logistic)	$y_i = \dfrac{1}{1 + \exp(-s_i)}$
ReLU (Rectified Linear Unit)	$y_i = \begin{cases} s_i & s_i > 0 \\ 0 & s_i \leq 0 \end{cases}$
Softplus	$y_i = \ln(1 + \exp(s_i))$
Gaussian	$y_i = \exp\left(-\dfrac{1}{2} \cdot \dfrac{(s_i - \mu_i)^2}{\sigma_i^2}\right)$
Softmax	$y_i = \dfrac{\exp(s_i)}{\sum_{i'=1}^{n} \exp(s_{i'})}$
Maxout	$y_i = \max(s_1, ..., s_n)$

Table 2.1: Activation functions ($\mathbf{y} = \psi(\mathbf{s})$, where $\mathbf{s}, \mathbf{y} \in \mathbb{R}^n$). All these functions are vector functions. We show formulas for entry i of the input and output vectors. μ_i and σ_i^2 are the mean and variance respectively.

Typically, the **depth** of a neural network (sometimes referred to as **model depth**) is measured by the number of its layers. For example, treating the input vector as an initial layer, the depth of the example network in Figure 2.3 is 4. A related concept is **model width**, which is defined per layer rather than across the entire network. A common measure for the width of a layer is the number of neurons it contains. For example, the width of the output layer in Figure 2.3 is 3. If all layers of a neural network share the same width n, we can simply state that the overall model width is n. Both model depth and model width have important implications for the representational capacity of the resulting neural network. For example, it has been mathematically proven that even a neural network with just two layers of neurons and a Sigmoid activation function can approximate any continuous function [Cybenko, 1989]. However, for more complex tasks, increasing the depth of the network generally yields significant performance improvements.

Stacking layers results in a very common kind of neural network — **feed-forward neural networks** (**FFNNs**). These networks are called "feed-forward" because there are no cycles in connections between layers and all the data moves in one direction. As we will see throughout this book, many of modern neural networks are feed-forward in nature, though a few notable exceptions will be discussed in Section 2.3.

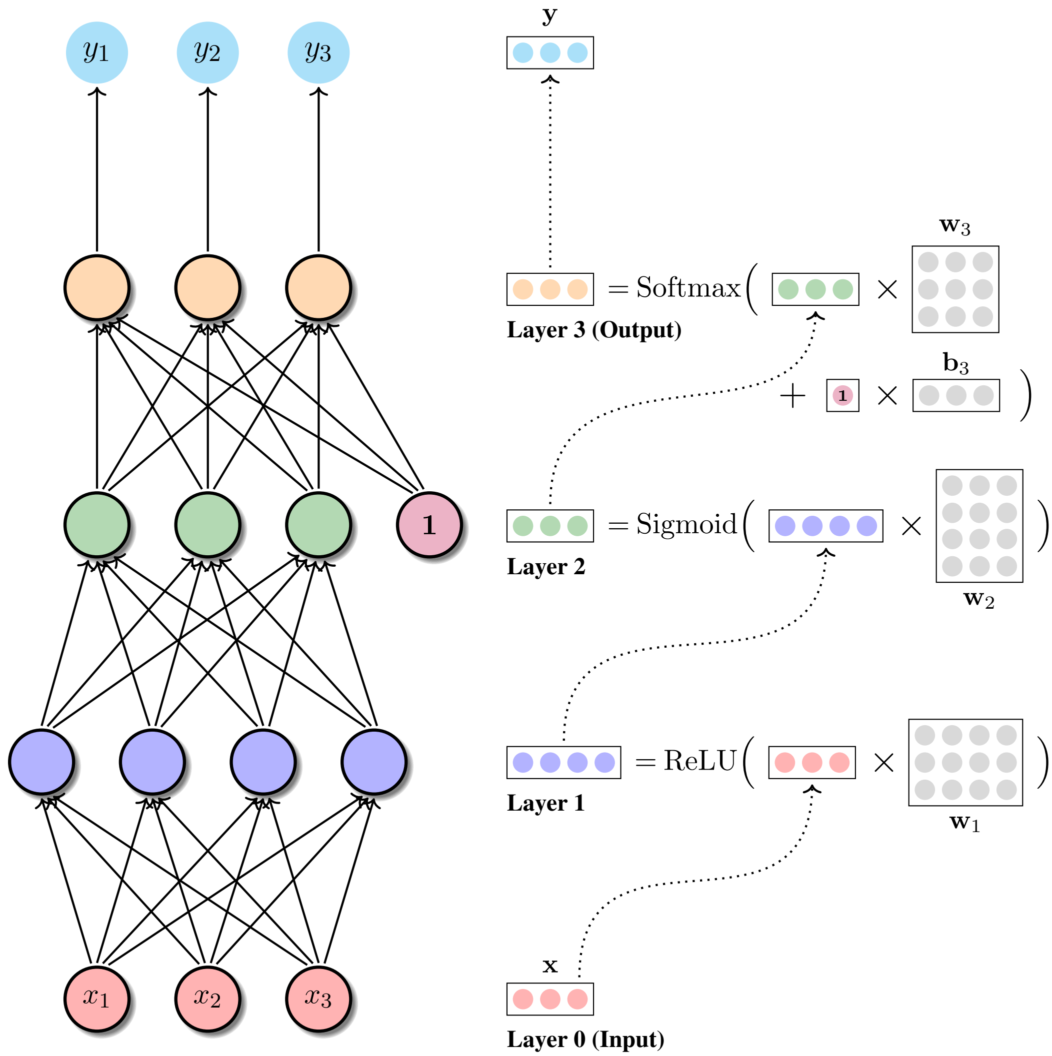


Figure 2.3: A multi-layer neural network. The input layer consists of three variables $\{x_1, x_2, x_3\}$. These variables are fully connected to all neurons of layer 1. The output of layer 1 is a new vector $\mathbf{h}_1 = \text{ReLU}(\mathbf{x} \cdot \mathbf{w}_1)$. It is then fully connected to layer 2, performing the mapping $\mathbf{h}_2 = \text{Sigmoid}(\mathbf{h}_1 \cdot \mathbf{w}_2)$. Its output $\mathbf{h}_2$ is fed into layer 3, which generates the final output $\mathbf{y} = \text{Softmax}(\mathbf{h}_2 \cdot \mathbf{w}_3 + \mathbf{b}_3)$. The parameters of this neural network are $\mathbf{w}_1$, $\mathbf{w}_2$, $\mathbf{w}_3$ and $\mathbf{b}_3$.

2.1.3 Computation Graphs

Computation graphs are a common way of representing neural networks. As graphs in mathematics, a computation graph is made up of nodes and edges between nodes. Each node represents either a mathematical operation or a variable, and each edge represents the data flow from one node to another. Therefore, computation graphs are directed[3]. Consider, for example,

[3]While a number of machine learning models can be represented as undirected computation graphs, they are not the focus of this book.

three functions:

$$\mathbf{y} = \mathbf{x} + \mathbf{w} \tag{2.6}$$

$$\mathbf{y} = \text{Softmax}(\mathbf{x} \cdot \mathbf{w} + \mathbf{b}) \tag{2.7}$$

$$\mathbf{y} = \text{Sigmoid}(\mathbf{x} \cdot \mathbf{w}_1 + \mathbf{b}_1) - \text{ReLU}(\mathbf{x} \cdot \mathbf{w}_2) \tag{2.8}$$

Figure 2.4 shows the computation graphs of these functions. From a parsing perspective, all neural networks can be viewed as mathematical expressions. A computation graph is therefore the representation of the result when parsing a mathematical expression. In this way, each node of the graph yields a sub-expression, and the root node yields the whole expression.

In a computation graph, a node can be connected to multiple nodes beneath it and/or above it. This enables the reuse of sub-graphs in representing complex functions. For example, in Eq. (2.8), the variable $\mathbf{x}$ is used twice and the corresponding node has two outgoing edges. In fact, organizing neural networks into computation graphs resembles the compositional nature of neural networks — typically, a large network is built by composing small networks. Take Eq. (2.8) as an instance. It can be rewritten as a system of three equations:

$$\mathbf{y} = \mathbf{h}_1 - \mathbf{h}_2 \tag{2.9}$$

$$\mathbf{h}_1 = \text{Sigmoid}(\mathbf{x} \cdot \mathbf{w}_1 + \mathbf{b}_1) \tag{2.10}$$

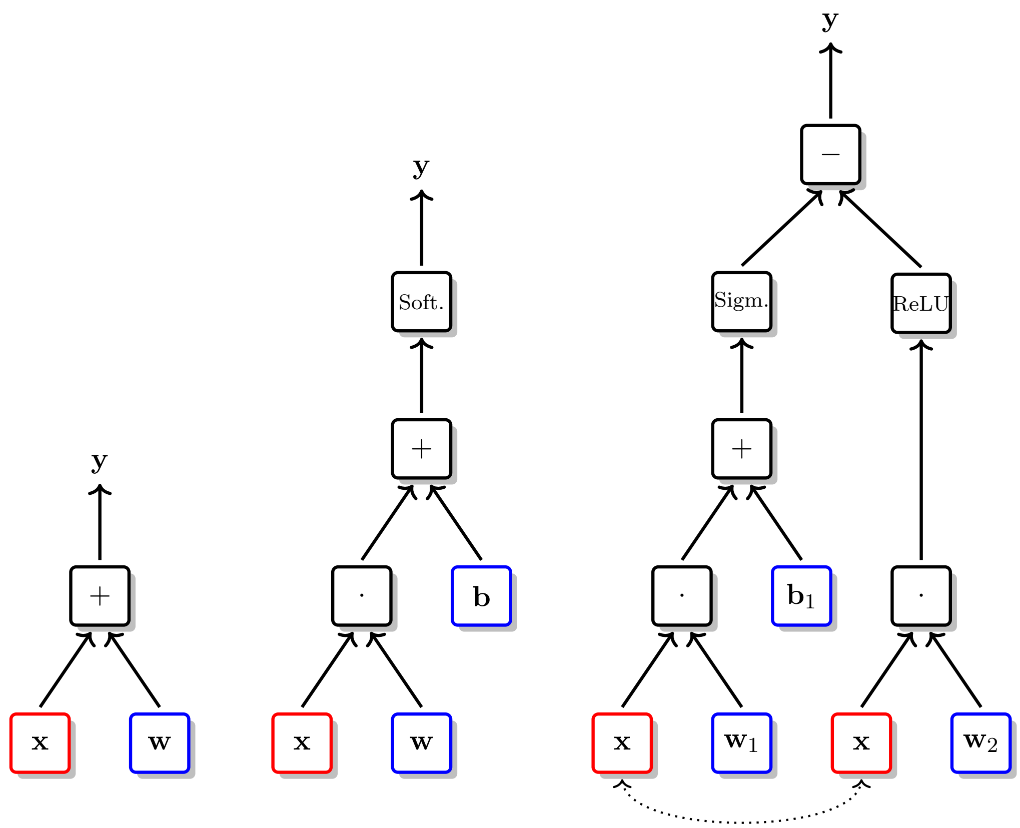
$$\mathbf{h}_2 = \text{ReLU}(\mathbf{x} \cdot \mathbf{w}_2) \tag{2.11}$$

In the composition operation, the nodes of $\mathbf{h}_1$ and $\mathbf{h}_2$ in Eq (2.9) are replaced by the graphs of Eqs. (2.10-2.11).

The main use of computation graphs is in executing the function. This is exactly the same thing as predicting the output of a neural network. The method is quite simple. First, the nodes of the graph are topologically sorted such that they are placed in an order consistent with the information flow. Then, given the values that are fed into the input nodes, the graph is traversed in a way that we compute the output of each node and forward it to its parent nodes. The final result is obtained from the output node. This procedure is typically called a **forward pass**. A forward pass can be efficient, as every node only needs to be visited once and its output can be reused by multiple nodes without the need of recomputing the result. Moreover, a forward pass can be optimized by reconstructing the graph. This can develop the reuse idea a bit more and avoid unnecessary computation and memory consumption.

Another use of computation graphs is to compute gradients automatically. In training neural networks, it is in general required to compute the partial derivatives of the loss function L with respect to every weight matrix ($\mathbf{w}$) and every bias term ($\mathbf{b}$), such as $\frac{\partial L}{\partial \mathbf{w}}$ and $\frac{\partial L}{\partial \mathbf{b}}$. Before seeing how these partial derivatives are used in updating a model (see Section 2.4.1), though, we first give an idea of computing derivatives in a computation graph. For example, consider the function below:

$$\mathbf{y} = \psi(\mathbf{x} \cdot \mathbf{w}_1 + \mathbf{b}_1) \cdot \mathbf{w}_2 \tag{2.12}$$



(a) $\mathbf{y} = \mathbf{x} + \mathbf{w}$ (b) $\mathbf{y} = \mathrm{Softmax}(\mathbf{x} \cdot \mathbf{w} + \mathbf{b})$ (c) $\mathbf{y} = \mathrm{Sigmoid}(\mathbf{x} \cdot \mathbf{w}_1 + \mathbf{b}_1) - \mathrm{ReLU}(\mathbf{x} \cdot \mathbf{w}_2)$

Figure 2.4: Computation graphs of three example neural networks. The black boxes represent the mathematical operations, and the colored boxes represent the variables. A mathematical operation node has incoming edges from other nodes, and each of these nodes can be treated as an argument of the operation. For example, in sub-figure (a), the addition node has two child nodes labeled with $\mathbf{x}$ and $\mathbf{w}$ respectively. This node reads the output of the nodes $\mathbf{x}$ and $\mathbf{w}$, and generates the output $\mathbf{y} = \mathbf{x} + \mathbf{w}$. Larger graphs exhibit more complex data flows. In sub-graph (b), the output of the dot node (i.e., $\mathbf{x} \cdot \mathbf{w}$) is passed along the edge to the addition node. Then, the addition node computes the sum of $\mathbf{x} \cdot \mathbf{w}$ and $\mathbf{b}$ as its output. We can repeat the same process over all the mathematical operation nodes in a bottom-up manner, and get the final result of computing the whole expression out of the top-most node.

To obtain $\frac{\partial L}{\partial \mathbf{w}_1}$, $\frac{\partial L}{\partial \mathbf{b}_1}$ and $\frac{\partial L}{\partial \mathbf{w}_2}$, it is natural to use the chain rule of differentiation. For example, for a composite function $y = p(q(x))$, the formula of the chain rule is given as:

$$\frac{\partial y}{\partial x} = \frac{\partial p}{\partial q} \cdot \frac{\partial q}{\partial x} \tag{2.13}$$

However, the analytic formula of a derivative based on Eq. (2.13) would result in a lengthy equation. Instead, we can decompose a complex function into several functions, each standing for an operation. Then, Eq. (2.12) can be rewritten as:

$$\mathbf{y} = \mathbf{h}_1 \cdot \mathbf{w}_2 \tag{2.14}$$
$$\mathbf{h}_1 = \psi(\mathbf{h_2}) \tag{2.15}$$
$$\mathbf{h}_2 = \mathbf{h}_3 + \mathbf{b}_1 \tag{2.16}$$
$$\mathbf{h}_3 = \mathbf{x} \cdot \mathbf{w}_1 \tag{2.17}$$

All these variables can be better understood intuitively from a computation graph: each variable is a node of the graph, and nodes are connected by algebraic operations and function compositions. Taking Eq. (2.13) and some basic knowledge of calculus, we compute the derivatives of the variables, as follows:

$$\text{node 1:} \quad \frac{\partial L}{\partial \mathbf{y}} = \delta_{\mathbf{y}} \tag{2.18}$$
$$\text{node 2:} \quad \frac{\partial L}{\partial \mathbf{h}_1} = \frac{\partial L}{\partial \mathbf{y}} \cdot \mathbf{w}_2^{\top} \tag{2.19}$$
$$\text{node 3:} \quad \frac{\partial L}{\partial \mathbf{w}_2} = \mathbf{h}_1^{\top} \cdot \frac{\partial L}{\partial \mathbf{y}} \tag{2.20}$$
$$\text{node 4:} \quad \frac{\partial L}{\partial \mathbf{h}_2} = \frac{\partial L}{\partial \mathbf{h}_1} \odot \psi'(\mathbf{h}_2) \tag{2.21}$$
$$\text{node 5:} \quad \frac{\partial L}{\partial \mathbf{h}_3} = \frac{\partial L}{\partial \mathbf{h}_2} \tag{2.22}$$
$$\text{node 6:} \quad \frac{\partial L}{\partial \mathbf{b}_1} = \frac{\partial L}{\partial \mathbf{h}_2} \tag{2.23}$$
$$\text{node 7:} \quad \frac{\partial L}{\partial \mathbf{x}} = \frac{\partial L}{\partial \mathbf{h}_3} \cdot \mathbf{w}_1^{\top} \tag{2.24}$$
$$\text{node 8:} \quad \frac{\partial L}{\partial \mathbf{w}_1} = \mathbf{x}^{\top} \cdot \frac{\partial L}{\partial \mathbf{h}_3} \tag{2.25}$$

where $\delta_{\mathbf{y}}$ is the derivative of the loss with respect to the model output. $\delta_{\mathbf{y}}$ depends on the choice of the loss function, e.g., if we use the squared loss $L = \frac{1}{2}(\mathbf{y} - \mathbf{y}_{\text{gold}})^2$, where $\mathbf{y}_{\text{gold}}$ is the ground-truth, then $\delta_{\mathbf{y}} = \mathbf{y} - \mathbf{y}_{\text{gold}}$. The above process is essentially a **backward pass**, as the gradients are passed in a top-down fashion. Another name for this is **backpropagation**. It has been the de facto standard for training deep neural networks. For a better understanding of how forward and backward passes work, Figure 2.5 shows two running examples.

2.2 Example: Neural Language Modeling

Language modeling is a fundamental NLP task that estimates a probability distribution over sequences of words. Given a sequence of m words $w_1 ... w_m$, the joint probability $\Pr(w_1, ..., w_m)$

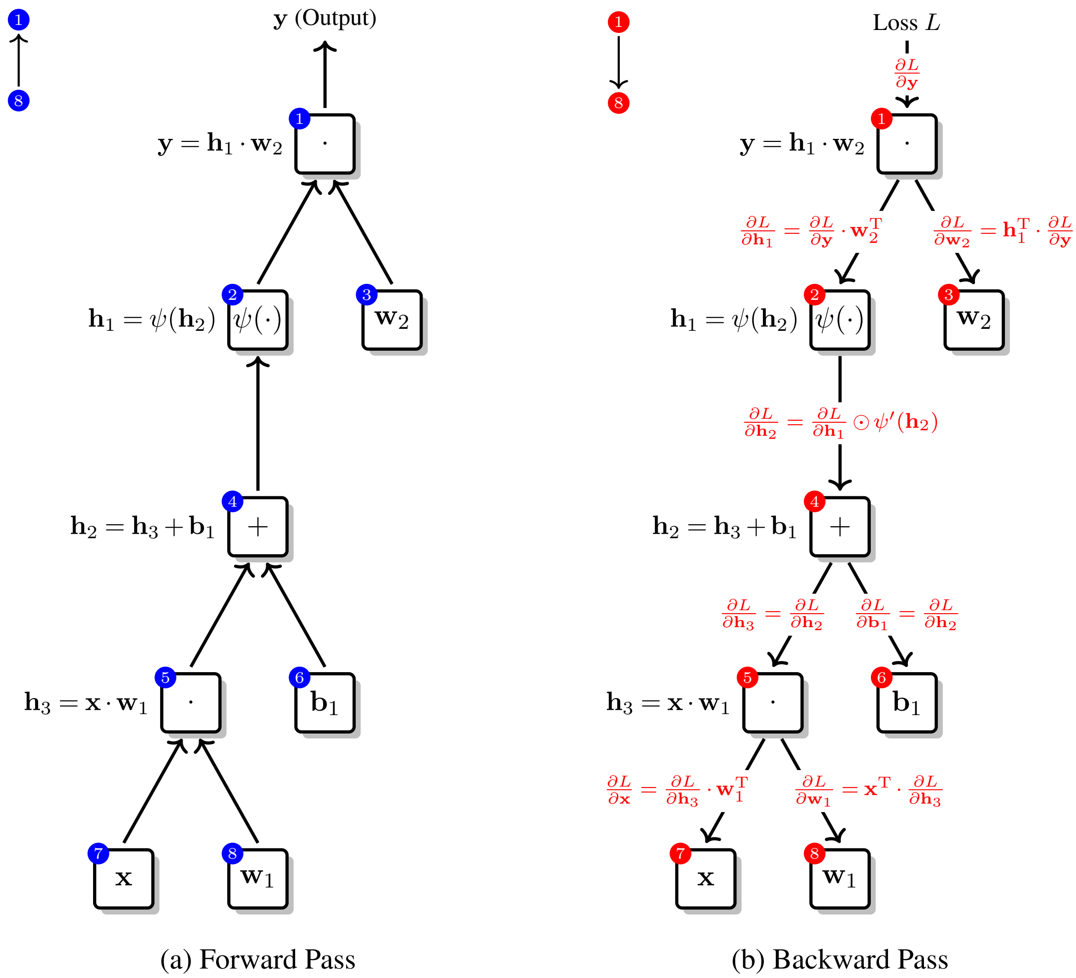


Figure 2.5: The forward pass and backward pass for an example computation graph. In the forward pass (left), the nodes are visited in an order from the input to the output, e.g., from node 8 to 1. On each node, we execute the corresponding function, such as addition, to generate the output, which is then consumed by the subsequent nodes. In contrast, in the backward pass (right), the nodes are visited in the reverse order, e.g., from node 1 to 8. During this process, we pass the gradient of the loss (or error) from the output to the input, that is, for each node, we compute the gradient at the input point of the node by using the chain rule, given the gradient at the output point of the node.

is given by the equation:

$$\Pr(w_1, ..., w_m) = \prod_{i=1}^{m} \Pr(w_i | w_1, ..., w_{i-1}) \tag{2.26}$$

As such, the language modeling problem is framed as predicting the next word given all previous context words. A straightforward method to model $\Pr(w_i | w_1, ..., w_{i-1})$ is to condition the prediction on a context window covering a fixed number of preceding words, as follows:

$$\Pr(w_i|w_1,...,w_{i-1}) \quad \approx \quad \Pr(w_i|w_{i-n+1},...,w_{i-1}) \tag{2.27}$$

where n is the window size. One way to estimate this probability is the **n-gram language modeling** approach: we compute the relative frequency for each n-gram $w_{i-n+1}...w_i$, i.e., $\Pr(w_i|w_{i-n+1},...,w_{i-1}) = \frac{\mathrm{count}(w_{i-n+1}...w_i)}{\mathrm{count}(w_{i-n+1}...w_{i-1})}$. Although n-gram language models dominated the NLP field for decades, they require massive tables to store all n-gram probabilities. In consequence, as the training corpus grows and longer contexts are considered, the data becomes exceedingly sparse. This is a classic instance of the curse of dimensionality.

Here we consider neural networks in addressing the language modeling problem [Bengio et al., 2000; 2003b]. Unlike n-gram models, which operate in a discrete space requiring an exponentially large number of distinct feature vectors, **neural language models** generalize in a continuous space by encoding words as dense, low-dimensional real-valued vectors. In particular, a feed-forward network is utilized here to predict the likelihood of w_i occurring given the context $w_{i-n+1}...w_{i-1}$.

Figure 2.6 presents the architecture of the **feed-forward neural network based language model** (**FFNNLM**). The input is the context words $w_{i-n+1}...w_{i-1}$. Each is a discrete variable taking values from a vocabulary V. Since the neural network operates on vectors, all words are vectorized as **one-hot representations**. In this case, the word $w = V_k$ is a $|V|$-dimensional vector in which entry k is 1 and other entries are all 0. For example, consider a vocabulary $V = \{\text{“I”}, \text{“you”}, \text{“he”}, \text{“she”}, \text{“they”}\}$. The one-hot representation of “you” is expressed as

$$\mathbf{1}_{\text{you}} \quad = \quad \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \end{bmatrix} \tag{2.28}$$

Although one-hot vectors uniquely identify words, they fail to capture semantic relationships (i.e., the expectation that similar words should be situated close to one another in the vector space). By relaxing these indicator-based representations into continuous, real-valued vectors, we can capture word relationships by computing similarities between these vectors. To this end, an effective technique is to transform one-hot representations into **distributed representations**. More formally, let $\mathbf{x}$ be a one-hot vector of a word w. Its distributed representation is a real-valued vector, given by:

$$\mathbf{e} \quad = \quad \mathbf{x} \cdot \mathbf{C} \tag{2.29}$$

where the representation $\mathbf{e}$ is a vector $\in \mathbb{R}^{d_e}$, and d_e is the dimensionality of the representation. Each dimension of $\mathbf{e}$ can be viewed as some countable aspect of the word, though it is not required to be interpreted linguistically. $\mathbf{C}$ is a $|V| \times d_e$ matrix, of which the k-th row corresponds to the vector for V_k. Hence, $\mathbf{x} \cdot \mathbf{C}$ is to “select” a row from $\mathbf{C}$. For example, given

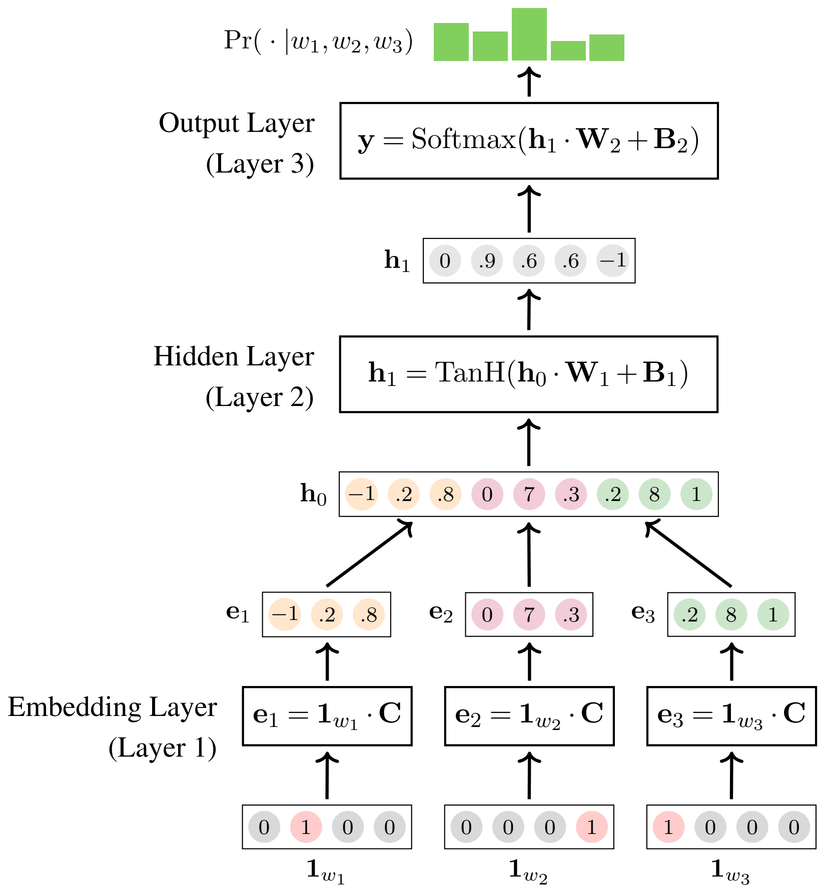


Figure 2.6: A neural language model [Bengio et al., 2003b]. Blue boxes represent the layers of the neural network. The input is three context words in their one-hot representations $\{\mathbf{1}_{w_1}, \mathbf{1}_{w_2}, \mathbf{1}_{w_3}\}$, and the output is the probability distribution of the next word $\Pr(w_4|w_1,w_2,w_3)$. First, an embedding layer is used to map each word into the distributed representation (i.e., the word embedding). The embeddings of these words are concatenated to form a bigger vector $\mathbf{h}_0$ such that the concatenated vector encodes all input information. Then, $\mathbf{h}_0$ is taken as the input to a normal layer that performs the mapping $\mathbf{h}_1 = \mathrm{TanH}(\mathbf{h}_0 \cdot \mathbf{W}_1 + \mathbf{B}_1)$. The final layer reads $\mathbf{h}_1$ and produces a distribution over the vocabulary, i.e., $\mathbf{y} = \mathrm{Softmax}(\mathbf{h}_1 \cdot \mathbf{W}_2 + \mathbf{B}_2)$ where $y_k = \Pr(V_k|w_1,w_2,w_3)$.

$\mathbf{C} \in \mathbb{R}^{5\times 3}$, the distributed representation of "you" is given by:

$$\begin{aligned} e(\text{"you"}) &= \mathbf{1}_{\text{you}} \cdot \mathbf{C} \\ &= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 73 & 12 & 0.1 \\ 12 & 0.5 & 18 \\ 37 & 0.7 & 28 \\ 61 & 0.4 & 23 \\ 62 & 11 & 0.4 \end{bmatrix} \\ &= \begin{bmatrix} 12 & 0.5 & 18 \end{bmatrix} \end{aligned} \tag{2.30}$$

Eq. (2.29) introduces the foundational concept of learning word representations, which has driven massive advancements in NLP. Typically, the vector $\mathbf{e}$ is called a **word embedding**, and the parameter matrix $\mathbf{C}$ is called the **embedding matrix**. While many methods exist for learning word embeddings, we will not focus on them in this chapter. The reader is referred to Chapter 3 for a detailed discussion on this topic.

To encode the context words $\{w_{i-n+1},...,w_{i-1}\}$, a simple method is to concatenate the word embeddings $\{\mathbf{e}_{i-n+1},...,\mathbf{e}_{i-1}\}$ as a new vector $\mathbf{h}_0$:

$$\mathbf{h}_0 = [\mathbf{e}_{i-n+1},...,\mathbf{e}_{i-1}]$$

The next part of the model is a 2-layer feed-forward neural network. The first layer, called a **hidden layer**, is a standard layer of neurons, followed by the hyperbolic tangent activation function. The layer produces a d_h-dimensional vector:

$$\mathbf{h}_1 = \mathrm{TanH}(\mathbf{h}_0 \cdot \mathbf{W}_1 + \mathbf{B}_1) \tag{2.31}$$

The second layer is the output layer. It produces a distribution over V. This can be formulated as:

$$\Pr(\cdot|w_{i-n+1},...,w_{i-1}) = \mathrm{Softmax}(\mathbf{h}_1 \cdot \mathbf{W}_2 + \mathbf{B}_2) \tag{2.32}$$

The parameters of the model are $\mathbf{C} \in \mathbb{R}^{|V| \times d_e}$, $\mathbf{W}_1 \in \mathbb{R}^{(n-1)d_e \times d_h}$, $\mathbf{B}_1 \in \mathbb{R}^{d_h}$, $\mathbf{W}_2 \in \mathbb{R}^{d_h \times |V|}$, and $\mathbf{B}_2 \in \mathbb{R}^{|V|}$. A popular way to optimize these parameters is to minimize the cross-entropy loss via gradient descent. Additionally, training can be improved via regularization. These methods will be discussed in Sections 2.4 and 2.5.

A few remarks on the neural language model: first, by using distributed feature vectors, "senses" can be shared in part by different words. This enables learnable word senses by which the similarity between words is implicitly considered. An advantage of such a model is that a small change in word vectors would not lead to a big change in the result. For example, suppose we have seen "grapes are fruits" many times but have never seen "peaches are fruits". If the embeddings for "grapes" and "peaches" are close, the model can logically infer that the phrase "peaches are fruits" is highly probable. This differentiates neural language models greatly from n-gram language models where different surface forms are treated as completely independent entities.

Second, the dense representation of words results in a smaller model. For example, a common setting of d_e and d_h is less than 1000, keeping the total number of parameters manageable. By contrast, the size of an n-gram language model increases by a factor of $|V|$ as n increases. For example, there will be a huge table of probabilities for a common vocabulary if n is larger than 3.

Third, the neural language model is computationally expensive because of the heavy use of vector and matrix operations, such as matrix multiplication. This is a common problem with most deep neural network-based systems. A common solution is to break the computation problem into independent sub-problems so that these sub-problems can be handled in parallel.

At a lower level, one can use GPUs or other parallel computing devices to speed up linear algebra operators. At a higher level, one can distribute parts of the model or parts of the data to multiple devices for model-level or task-level speed-ups.

2.3 Basic Model Architectures

We now describe, in more detail, several basic building blocks for neural networks. They are widely used in developing state-of-the-art neural models in NLP.

2.3.1 Recurrent Units

Recurrent neural networks (**RNNs**) are a class of neural networks that read and/or produce sequential data or time-series data. As with a feed-forward neural network, an RNN comprises layers of neurons and connections between neurons [Hopfield, 1984; Rumelhart et al., 1986; Williams and Zipser, 1989; Elman, 1990]. Some of the neurons are used as a "memory" that keeps the state of the problem when the processing moves on along a sequence of signals. As a result, it is straightforward to use RNNs to deal with variable-length problems, such as machine translation and speech recognition.

The main idea behind RNNs is to repeatedly utilize a **recurrent unit** (or **recurrent cell**) to compute the output at each position of an input sequence. To be more precise, given a sequence of vectors $\mathbf{x}_1 ... \mathbf{x}_m$, a standard **recurrent unit** can be described as a function $\mathrm{RNN}(\cdot)$ that consumes an input $\mathbf{x}_i$ and a state $\mathbf{s}_{i-1}$ at each time step and generates a new state $\mathbf{s}_i$, given by:

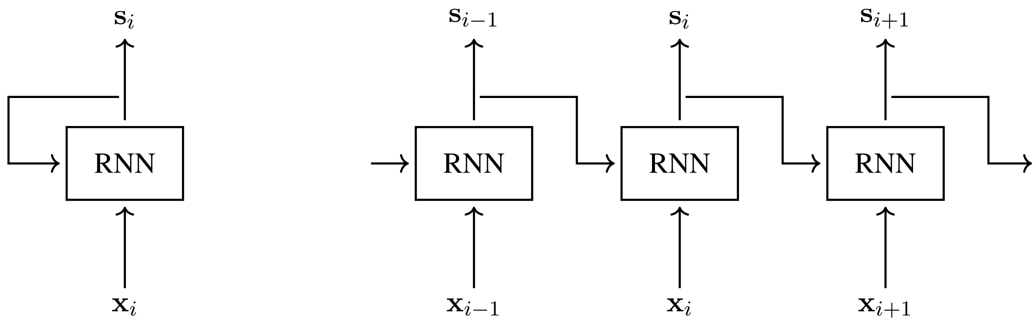
$$\mathbf{s}_i = \mathrm{RNN}(\mathbf{s}_{i-1}, \mathbf{x}_i) \tag{2.33}$$

The state $\mathbf{s}_i$ can be viewed as a "memory" that summarizes past information and is updated as new data arrives. See Figure 2.7 (a) for visualization of Eq. (2.33). The circle here indicates the reuse of the recurrent unit. This can be understood by rewriting Eq. (2.33) in a sequence of calls of the function $\mathrm{RNN}(\cdot)$:

$$\begin{aligned}
\mathrm{RNN}(\mathbf{s}_{i-1}, \mathbf{x}_i) &= \mathrm{RNN}(\mathrm{RNN}(\mathbf{s}_{i-2}, \mathbf{x}_{i-1}), \mathbf{x}_i) \\
&= \mathrm{RNN}(\mathrm{RNN}(\mathrm{RNN}(\mathbf{s}_{i-3}, \mathbf{x}_{i-2}), \mathbf{x}_{i-1}), \mathbf{x}_i) \\
&= ... \\
&= \mathrm{RNN}(\mathrm{RNN}(... (\mathrm{RNN}(\mathbf{s}_0, \mathbf{x}_1), \mathbf{x}_2) ... \mathbf{x}_{i-1}), \mathbf{x}_i)
\end{aligned} \tag{2.34}$$

Figure 2.7 (b) illustrates the structure of this expanded network. This formulation is typically referred to as the **unrolled** (or **unfolded**) structure of an RNN. Fundamentally, Figures 2.7 (a) and (b) represent the exact same mathematical operations. While a rolled RNN offers a compact and elegant mathematical form, an unrolled RNN provides a clearer visualization of the data flow through time. Therefore, we will use the unrolled representation of RNNs throughout this book.

Moreover, it is worth noting that an unrolled RNN can be viewed as a deep feed-forward



(a) Rolled RNN Structure (b) Unrolled RNN Structure

Figure 2.7: Example of RNN (rolled vs unrolled). An RNN unit reads the input at each time step i and the output at the last time step $i-1$, and produces a new output s_i. As such we can reuse the same RNN unit to make predictions over a sequence of inputs (see sub-figure (a)): for each i, the current input $\mathbf{x}_i$ and last output $\mathbf{s}_{i-1}$ are consumed and mapped to the output that is fed into the same RNN unit for the processing at the next time step. A better way to visualize the RNN is to unroll it into a network with no cycles (see sub-figure (b)). The unrolled RNN can be regarded as a deep feed-forward neural network in that all RNN units share the same set of parameters.

neural network, where each application of the recurrent unit acts as a "layer" receiving information from the preceding "layer" (i.e., the previous time step). In this sense, an RNN is effectively a stack of layers extending horizontally across time. A benefit of this perspective is that one can utilize similar optimization techniques to train both architectures. For example, both RNNs and feed-forward networks rely on gradient-based optimization, though RNNs specifically employ a variant of backpropagation known as **backpropagation through time** (**BPTT**) to propagate errors backward through the unrolled computation graph.

There are many RNN variants, differing in how the function $\mathrm{RNN}(\cdot)$ is defined. The simplest of these is to formulate $\mathrm{RNN}(\cdot)$ as a single-layer neural network. Assume that $\mathbf{s}_{i-1}$ and $\mathbf{x}_i$ are in $\mathbb{R}^{d_h}$. The form of $\mathrm{RNN}(\cdot)$ is given by:

$$\mathrm{RNN}(\mathbf{s}_{i-1}, \mathbf{x}_i) = \psi(\mathbf{s}_{i-1} \cdot \mathbf{U} + \mathbf{x}_i \cdot \mathbf{V}) \tag{2.35}$$

where $\mathbf{U} \in \mathbb{R}^{d_h \times d_h}$ and $\mathbf{V} \in \mathbb{R}^{d_h \times d_h}$ are parameters. The common choices for the activation function ψ are $\mathrm{TanH}(\cdot)$, $\mathrm{Sigmoid}(\cdot)$, $\mathrm{ReLU}(\cdot)$, among others. Eq. (2.35) is a single-layer neural network because it has the same form as Eqs. (2.3-2.4):

$$\psi(\mathbf{s}_{i-1} \cdot \mathbf{U} + \mathbf{x}_i \cdot \mathbf{V}) = \psi([\mathbf{s}_{i-1}, \mathbf{x}_i] \cdot \mathbf{W}) \tag{2.36}$$

where $[\mathbf{s}_{i-1}, \mathbf{x}_i]$ is the concatenation of $\mathbf{s}_{i-1}$ and $\mathbf{x}_i$, and $\mathbf{W} \in \mathbb{R}^{2d_h \times d_h}$ is the parameter matrix that is formed by $\begin{bmatrix} \mathbf{U} \\ \mathbf{V} \end{bmatrix}$.

In practice, RNNs are frequently used as sub-components within larger architectures. For example, the input to a recurrent unit could be a direct representation of raw data (such as word embeddings) or the output from another neural network module. Similarly, additional neural network layers are routinely stacked on top of recurrent units. In many real-world NLP systems, an output layer is stacked directly upon the hidden state $\mathbf{s}_i$ to project it into the desired output space for classification or prediction tasks.

2.3.2 Convolutional Units

Convolutional neural networks (**CNNs**) are another well-known class of neural networks [Waibel et al., 1989; LeCun et al., 1989]. In a biological sense, they are inspired by the human visual system: neurons react to stimuli in a restricted region of the visual field (known as the **receptive field**) [Hubel and Wiesel, 1959]. In CNNs, the receptive field describes the region in the input space that is involved in generating the output for a neuron. CNNs are therefore "partially connected" models in which each neuron only considers input features within a restricted region. This differentiates CNNs from fully connected feed-forward neural networks. In general, CNNs naturally capture the hierarchical nature of features describing data and scale more efficiently in terms of complexity.

While CNNs have many applications in processing 2D data, such as image classification, we discuss them here in a sequential data processing scenario to maintain a consistent treatment of the problem throughout this chapter. Typically, a CNN consists of a **convolutional layer**, a **pooling layer**, and, optionally, other types of layers. It begins with the convolutional layer, where the receptive field is defined by a set of **convolution kernels** or **filters**. A filter is a linear mapping function that convolves the input features in the receptive field to form a single output feature. For example, consider a sequence of numbers $x_1 ... x_m$. The filter ranging from position i to position $i+r-1$ is defined to be:

$$
\begin{aligned}
v &= \mathrm{Conv}(\mathbf{x}_{[i,i+r-1]}, \mathbf{W}) \\
&= \mathbf{x}_{[i,i+r-1]} \cdot \mathbf{W} \\
&= \sum_{k=0}^{r-1} x_{i+k} \cdot W_k
\end{aligned}
\tag{2.37}
$$

where r is the size of the receptive field, $\mathbf{x}_{[i,i+r-1]}$ is the sub-sequence $x_i ... x_{i+r-1}$, and $\mathbf{W} \in \mathbb{R}^r$ are the parameters of the filter. Then, a sequence of output features can be generated by moving the filter along the input sequence. Let stride be the distance between consecutive moves. The output for move i is then defined to be:

$$
v_i = \mathrm{Conv}(\mathbf{x}_{[\mathrm{stride}\times i, \mathrm{stride}\times i+r-1]}, \mathbf{W}) \tag{2.38}
$$

In this way, the convolutional layer transforms the input sequence $x_1 ... x_m$ to the output sequence $v_1 ... v_{\lfloor \frac{m}{\mathrm{stride}} \rfloor}$[4]. A remark here is that the parameters $\mathbf{W}$ are shared across positions of the input sequence. This method is known as **parameter sharing** or **weight sharing**. Parameter

[4] $\lfloor \cdot \rfloor$ stands for the floor function.

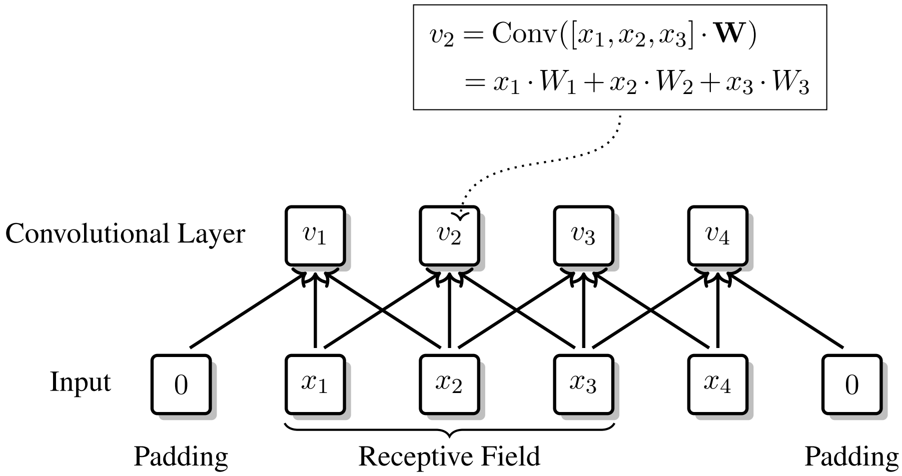


Figure 2.8: Convolution over a sequence of numbers $\{x_1, x_2, x_3, x_4\}$ ($r = 3$ and $\mathrm{stride} = 1$). The receptive field defines the region in the input that is taken in computing the output. Here the receptive field has a size of 3, that is, the convolutional operation covers three consecutive numbers in the input sequence. The filter (or convolutional kernel) outputs a weighted sum of these numbers. Each time we slide the receptive field over the input, the filter generates a new output. As such, the output of the convolutional layer is a vector of numbers. Also, a padding number (i.e. 0) is added to each end of the sequence so that the convolution is feasible when the receptive field is incomplete.

sharing makes a CNN efficient because it requires fewer parameters than a feed-forward neural network given the same number of neurons.

A problem with the above formulation is that the use of the filter may not be tiled to fit the input sequence. For example, when $\mathrm{stride} \times i + r - 1 > m$, the input of the filter is incomplete. A commonly used solution is **padding**. It simply sets the features outside the input sequence to a constant. For example, we can append dummy feature vectors to each end of the sequence so that all convolution operations are feasible. See Figure 2.8 for an example filter computed over a sequence of numbers. Note that the receptive fields of different filter applications may overlap. This is beneficial sometimes because it reduces information loss in feature representation when a low-level feature is used in forming multiple high-level features.

In addition, a convolutional layer can involve an activation function $\psi(\cdot)$ to perform some non-linear mapping on the filter output. Let $m_k = \lfloor \frac{m}{\mathrm{stride}} \rfloor$ be the number of filter applications. The output of a convolutional layer is given by

$$\begin{bmatrix} h_1 & \dots & h_{m_k} \end{bmatrix} = \psi(\begin{bmatrix} v_1 & \dots & v_{m_k} \end{bmatrix}) \tag{2.39}$$

In general, a convolutional layer may not be restricted to a scalar-based input or a single filter. Often, we can take a vector as the representation of a token in the input sequence, and take a set of filters as feature extractors. To this end, we can adopt the same formulation as in Eqs. (2.37-2.39), but replace x_i, v_i and h_i with the vectorized counterparts.

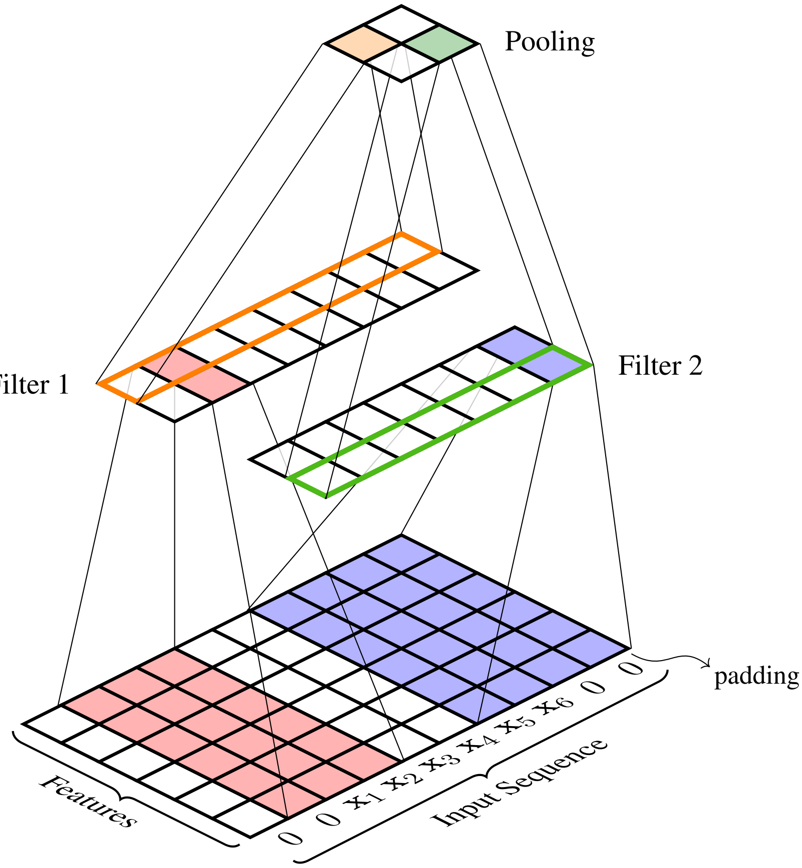


Figure 2.9: Example of CNN. There are two filters for the convolutional layer. The input is a sequence of 6 tokens represented in their feature vectors $\{\mathbf{x}_1, ..., \mathbf{x}_6\}$. To tile the filters to fit the input sequence, two padding vectors are attached to each end of the sequence. When applying a filter, we map the feature vectors in the receptive field to a new feature vector. For example, for filter 1, the receptive field is a 6×3 rectangle in the input, and the output is a 2-dimensional feature vector. By sweeping the filter on the sequence, we obtain a sequence of feature vectors, e.g., a sequence of 8 feature vectors, each having 2 features. The pooling layer fuses features along the sequence. For example, performing the pooling on the output of filter 1 results in 2 fused features. The final output of the CNN is two 2-dimensional feature vectors.

A convolutional layer is typically followed by a pooling layer. Like convolution, pooling is a function that sweeps a filter on a sequence. But the pooling operation does not have any parameters. It can be instead thought of as an aggregation function that performs down-sampling on the input sequence. There are several ways to design a pooling function. One of the most common methods is **max pooling**, which outputs the maximum value in the receptive field. Another method is **average pooling** which outputs the averaged value over the receptive field. For a complete picture of how a CNN works, Figure 2.9 depicts a running example where convolutional and pooling operations are performed on a sequence of feature vectors via two filters.

2.3.3 Gate Units

In neural networks, a **gate** is used to decide how much information is passed along [Hochreiter and Schmidhuber, 1997]. Consider a standard RNN as an example. At each time step i, instead of directly passing the previous state $\mathbf{s}_{i-1} \in \mathbb{R}^{d_h}$ to the recurrent unit, it is often more effective to determine what portion of the information in $\mathbf{s}_{i-1}$ is relevant for the subsequent decision. In this context, we want $\mathbf{s}_{i-1}$ to function more like a true memory: as time progresses, certain information should be retained while other information is discarded.

A common approach to achieving this is to introduce a gating vector to control the scale of the data flow. Here, we reuse the notation from RNNs (see Section 2.3.1), but our description is general and applies to any architecture requiring such a mechanism. Let $\mathbf{z} \in [0,1]^{d_h}$ be a gating vector, where $z_k = 0$ implies complete forgetting for dimension k, and $z_k = 1$ implies complete retention for dimension k. Applying $\mathbf{z}$ as a gate to $\mathbf{s}_{i-1}$ can be formulated as:

$$\mathrm{Gate}(\mathbf{z}, \mathbf{s}_{i-1}) = \mathbf{z} \odot \mathbf{s}_{i-1} \tag{2.40}$$

where $\odot$ denotes the element-wise multiplication of two vectors. The operation $\mathrm{Gate}(\mathbf{z}, \mathbf{s}_{i-1})$ represents a filtered update to $\mathbf{s}_{i-1}$. The vector $\mathbf{z}$ can be referred to as an update gate, a forget gate, or a similarly functioning component. Alternatively, the gating function can be defined as:

$$\mathrm{Gate}(\mathbf{z}, \mathbf{s}_{i-1}) = (1 - \mathbf{z}) \odot \mathbf{s}_{i-1} \tag{2.41}$$

Eqs. (2.40) and (2.41) describe the same underlying mechanism but entail different interpretations for $\mathbf{z}$.

The central question is how to compute $\mathbf{z}$. A standard method is to define $\mathbf{z}$ as the output of an internal neural network layer. For example, in a recurrent unit, $\mathbf{z}$ can be defined as:

$$\mathbf{z} = \mathrm{Sigmoid}(\mathbf{s}_{i-1} \cdot \mathbf{W}_1 + \mathbf{x}_i \cdot \mathbf{W}_2 + \mathbf{b}) \tag{2.42}$$

The use of the Sigmoid activation function ensures that the output values fall strictly within the range $[0,1]$. Note that Eq. (2.42) describes a learnable gate. This incorporates the gating mechanism directly into the model, allowing it to be optimized via standard training procedures. Various gating architectures exist, several of which will be explored in detail in Chapter 4.

2.3.4 Normalization (Standardization) Units

A neural network works by transforming feature vectors layer by layer. While the multi-layer, multi-dimensional nature of neural networks enables the models to compute complex functions, it might lead to very different distributions of output activations across layers or features. This is a problem with deep neural networks because such models have to adapt to different distributions over different layers or different features [Ioffe and Szegedy, 2015]. Sometimes, as model parameters are initialized randomly in all layers and in all feature dimensions, some features are likely to take on disproportionately large values. In this case, the model would become biased toward those features with larger magnitudes.

A way to mitigate this issue is **normalization**, which standardizes an n-dimensional feature vector $\mathbf{s}$ as

$$\text{Normalize}(\mathbf{s}) = \boldsymbol{\alpha} \odot \frac{\mathbf{s} - \boldsymbol{\mu}}{\boldsymbol{\sigma} + \boldsymbol{\epsilon}} + \boldsymbol{\beta} \tag{2.43}$$

where $\boldsymbol{\mu} \in \mathbb{R}^n$ and $\boldsymbol{\sigma} \in \mathbb{R}^n$ are the mean and standard deviation of $\mathbf{s}$, respectively. $\boldsymbol{\epsilon}$ is a small number used for numerical stability [Chiang and Cholak, 2022]. $\boldsymbol{\alpha} \in \mathbb{R}^n$ and $\boldsymbol{\beta} \in \mathbb{R}^n$ are the parameters of the normalization unit. A simple choice is $\alpha = \mathbf{1}$ and $\beta = \mathbf{0}$, whereas a more sophisticated method is to learn α and β together with other parameters.

We may implement Eq. (2.43) in several ways that differ in how to define $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$. Let us consider this for one dimension in $\mathbf{s}$, say s, in a general setting. Suppose that s is drawn from a set of feature values Ω_k. The mean and the standard deviation on Ω_k are then defined to be:

$$\mu_k = \frac{1}{|\Omega_k|} \sum_{s \in \Omega_k} s \tag{2.44}$$

$$\sigma_k = \sqrt{\frac{1}{|\Omega_k|} \sum_{s \in \Omega_k} (s - \mu_k)^2} \tag{2.45}$$

Several methods are available to define Ω_k. For example, one can define Ω_k as features in the same layer [Ba et al., 2016], or as features along the same dimension over different samples or input positions [Ioffe and Szegedy, 2015; Ulyanov et al., 2016], or as a hybrid grouping of these dimensions [Wu and He, 2018].

An advantage of normalization is to put features on the same scale and make them comparable. This has been found to be very helpful for stabilizing the training process and improving the convergence behavior. As we will see in subsequent chapters, normalization plays an important role in many successful systems.

As an aside, while the term *normalization* in deep learning usually refers to a process of subtracting the mean and dividing by the standard deviation, it is in fact a **standardization** process. In other areas, by contrast, normalization is more often referred to as a technique that scales all entries of a vector to the interval $[0,1]$. Standardization has no such requirement. It instead tends to have the input centered around zero. In this sense, the term *normalization* is somewhat of a misnomer in deep learning. Nevertheless, normalization and standardization are used interchangeably in this book when referring to processes like Eq. (2.43).

2.3.5 Residual Units

The success of deep neural networks has been mostly attributed to the increasing number of layers used in forming more complex functions. Although stacking a large number of layers is the simplest way to obtain a deep model, it has been pointed out that such models are difficult to train. There are several reasons for this, such as the vanishing or exploding gradient problem. Furthermore, sequentially feeding a single representation through every layer can be disadvantageous when upper-level feature extractors could actually benefit from direct access to intermediate representations from earlier layers.

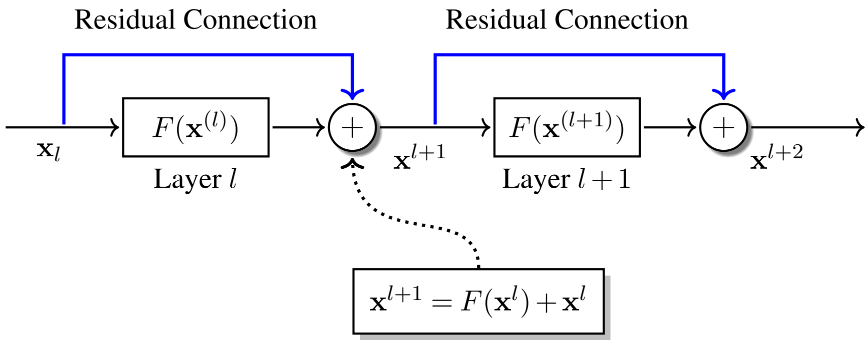


Figure 2.10: A 2-layer residual neural network. For each layer (or block), there is a skip or shortcut connection (in blue) that directly adds the input to the output.

Residual neural networks are one of the most effective approaches to addressing these issues [He et al., 2016a]. They are a special type of neural network that adds **residual connections** (or **skip connections**, or **shortcut connections**) across layers in a layer stack. Let $F(\mathbf{x})$ be a layer or a sequence of layers that maps $\mathbf{x}$ to some output. A residual block built on $F(\mathbf{x})$ is defined by summing the mapping $F(\mathbf{x})$ and the identity mapping $\mathbf{x}$:

$$\mathbf{y} = F(\mathbf{x}) + \mathbf{x} \tag{2.46}$$

A more common use of residual connections is in a neural network consisting of a number of identical layers. Let $\mathbf{x}^l$ and $\mathbf{y}^l$ be the input and output of layer l in a residual multi-layer neural network. The output of layer l can be defined as:

$$\mathbf{x}^{l+1} = F(\mathbf{x}^l) + \mathbf{x}^l \tag{2.47}$$

or

$$\mathbf{y}^l = F(\mathbf{y}^{l-1}) + \mathbf{y}^{l-1} \tag{2.48}$$

Figure 2.10 shows the architecture of a 2-layer residual neural network. Clearly, the residual connections add the inputs of a layer (or block) directly to its outputs. The added identity mapping is generally thought of as one of the most effective ways to simplify the optimization and ease the information flow in a deep model.

2.4 Training Neural Networks

In this section, we turn to the training problem, which is fundamental in developing neural network-based systems. Most of the discussion here is focused on methods in a supervised learning setting. We will discuss unsupervised methods in Section 2.6.

2.4.1 Gradient Descent

The gradient method has been proven to be one of the most successful methods for training neural networks. The basic idea is to iteratively update parameters so that we can minimize a differentiable loss function. In an update, the values of the parameters are adjusted in a way that the loss decreases the fastest. In a mathematical sense, it requires the update to be in the opposite direction of the gradient of the loss. This is known as **gradient descent** or **steepest descent**. Let θ_t be the parameters at step t (call it an **update step** or a **training step**), and $L(\theta_t)$ be the loss computed by the model parameterized by θ_t. The **update rule** (or **delta rule**) of gradient descent is given by the equation:

$$\theta_{t+1} \quad = \quad \theta_t - \eta \cdot \frac{\partial L(\theta_t)}{\partial \theta_t} \tag{2.49}$$

where $\frac{\partial L(\theta_t)}{\partial \theta_t}$ is the gradient of the loss with respect to the parameters at step t. It can be obtained by running the backpropagation algorithm presented in Section 2.1.3. Since θ_t is usually of multiple dimensions, $\frac{\partial L(\theta_t)}{\partial \theta_t}$ could be a vector or matrix that has the same shape as θ_t. η is the **learning rate** that controls how big a step we take in the direction of the minimum. While η can be simply set to a constant value during training, it is more common to adjust its value as the training proceeds (see Section 2.4.4 for a discussion). See Figure 2.11 for an illustration of gradient descent.

Eq. (2.49) gives a very basic definition of gradient descent. There are a number of improvements to the form of Eq. (2.49). Some of them are:

- **Gradient Descent with Momentum**. In physics, **momentum** is a vector quantity that describes the mass in motion. If we think of updating parameters as moving an object in a space, then we need to consider the momentum of the object at a position to determine the direction of the next move. This idea can be implemented by re-defining the update rule as:

$$\theta_{t+1} \quad = \quad \theta_t + v_t \tag{2.50}$$

where v_t is the velocity vector of the momentum. In the classic momentum method [Polyak, 1964], v_t is defined to be:

$$v_t \quad = \quad \lambda \cdot v_{t-1} - \eta \cdot \frac{\partial L(\theta_t)}{\partial \theta_t} \tag{2.51}$$

v_t retains some of the previous momentum (i.e., v_{t-1}), followed by a correction based on the gradient (i.e., $\frac{\partial L(\theta_t)}{\partial \theta_t}$). λ is a scalar for weighting v_t in an update. A well-known improvement to Eq. (2.51) is to take into account the momentum in the gradient, and thus prevent the velocity from becoming excessively large when approaching the minimum [Nesterov, 1983], like this

$$v_t \quad = \quad \lambda \cdot v_{t-1} - \eta \cdot \frac{\partial L(\theta_t + \lambda \cdot v_{t-1})}{\partial \theta_t} \tag{2.52}$$

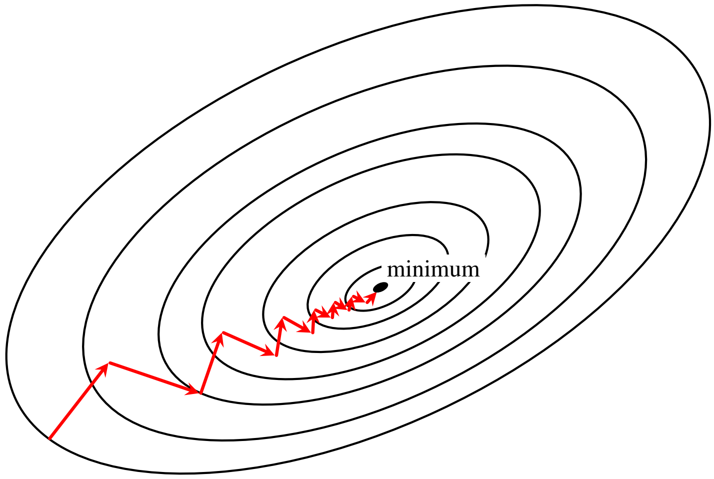


Figure 2.11: Gradient descent in a 2D space (blue lines stand for level sets). The goal is to find the parameters (i.e., values along the two dimensions) that minimize the value of a given loss function. Gradient descent does this by starting at a random point and stepping to the minimum in a number of updates of the parameters. In each update, it adjusts the parameters θ_t in the direction that makes the loss lower. The idea here is that the update chooses the direction of the steepest ascent, that is, the model moves a step in the direction of the negative gradient of the loss (i.e., $-\frac{\partial L(\theta_t)}{\partial \theta_t}$). The size of the move is controlled by a hyper-parameter η, called the learning rate. Thus, the amount of the change to the parameters is $-\eta \cdot \frac{\partial L(\theta_t)}{\partial \theta_t}$. By adding this to θ_t, we obtain the new parameters θ_{t+1}. This process repeats for a number of updates until the value of the loss function is close to the minimum.

A more detailed discussion on the difference between Eq. (2.51) and Eq. (2.52) can be found in Sutskever et al. [2013]'s paper.

- **Adaptive Gradient Descent**. In adaptive methods for gradient descent, the update rule is adapted to every parameter, rather than the whole model. **AdaGrad** is a method of this kind [Duchi et al., 2011]. It scales down the learning rate for frequently updated parameters and scales it up for infrequently updated ones. Assume that θ_t and $\frac{\partial L(\theta_t)}{\partial \theta_t}$ are both d-dimensional vectors. We can define a new variable $G \in \mathbb{R}^{d \times d}$ as the sum of the outer product of the gradient over the past t steps[5]:

$$G_t = \sum_{i=1}^{t} [\frac{\partial L(\theta_i)}{\partial \theta_i}]^{\top} \cdot \frac{\partial L(\theta_i)}{\partial \theta_i} \tag{2.54}$$

In general, $(G_t)^{\frac{1}{2}} \in \mathbb{R}^{d \times d}$ can be viewed as an indicator that describes to what extent a parameter has been updated so far. However, computing $(G_t)^{\frac{1}{2}}$ is extremely expensive. So it is more common to use the diagonal of G_t instead. Then, the update rule of AdaGrad is given by:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\mathrm{diag}(G_t) + \epsilon}} \odot \frac{\partial L(\theta_t)}{\partial \theta_t} \tag{2.55}$$

where $\mathrm{diag}(G_t)$ is the diagonal of G_t, i.e., $\mathrm{diag}(G_t)(k) = G_t(k,k)$. ϵ is a smoothing factor for numerical stability. Instead of summing over those squared gradients in an unweighted manner, another way is to reduce the impact of "old" gradients and make "recent" gradients more important. **AdaDelta** considers this by accumulating squared gradients with a **decay factor** [Zeiler, 2012]:

$$g_t^2 = \sigma \cdot g_{t-1}^2 + (1-\sigma) \cdot \left(\frac{\partial L(\theta_t)}{\partial \theta_t} \odot \frac{\partial L(\theta_t)}{\partial \theta_t} \right) \tag{2.56}$$

where σ is the decay factor of a value < 1. Like Eq. (2.55), the update rule for AdaDelta can be given by replacing $\mathrm{diag}(G_t)$ with g_t^2:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{g_t^2 + \epsilon}} \odot \frac{\partial L(\theta_t)}{\partial \theta_t} \tag{2.57}$$

Since $\sqrt{g_t^2 + \epsilon}$ can be seen as the root mean square (RMS) of the gradient, Eqs. (2.56-2.57) are also known as the **RMSProp** method [Hinton, 2018].

- **Adam** (**Adaptive Moment Estimation**). The Adam optimizer combines the merits of both the adaptive gradient descent and momentum methods [Kingma and Ba, 2014]. It defines an estimate of the mean of the gradient (the first moment) and an estimate of the uncentered variance of the gradient (the second raw moment). Let m_t and v_t be the two

[5] Given two vectors $\mathbf{a} = \begin{bmatrix} a_1 & \cdots & a_d \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 & \cdots & b_d \end{bmatrix}$, the outer product of $\mathbf{a}$ and $\mathbf{b}$ is:

$$\begin{aligned} \mathbf{a} \otimes \mathbf{b} &= \mathbf{a}^{\mathrm{T}} \cdot \mathbf{b} \\ &= \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} \cdot \begin{bmatrix} b_1 & \cdots & b_d \end{bmatrix} \\ &= \begin{bmatrix} a_1 b_1 & \cdots & a_1 b_d \\ \vdots & \ddots & \vdots \\ a_d b_1 & \cdots & a_d b_d \end{bmatrix} \end{aligned} \tag{2.53}$$

moment estimates. They are given by the equations:

$$m_t = \beta_1 \cdot m_{t-1} + (1-\beta_1) \cdot \frac{\partial L(\theta_t)}{\partial \theta_t} \tag{2.58}$$

$$v_t = \beta_2 \cdot v_{t-1} + (1-\beta_2) \cdot \left(\frac{\partial L(\theta_t)}{\partial \theta_t} \odot \frac{\partial L(\theta_t)}{\partial \theta_t} \right) \tag{2.59}$$

where $\beta_1, \beta_2 \in [0,1]$ are hyper-parameters for a trade-off between the previous estimate and the gradient (or squared gradient) at the current step. β_1 and β_2 are also treated as the decay factors of these averages. For example, common choices for β_1 and β_2 are 0.9 and 0.999. As the initial moments are set to 0, these estimates are biased toward zero vectors at the very beginning of the training process. To address this issue, bias corrections are used in Adam, leading to bias-corrected estimates:

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t} \tag{2.60}$$

$$\hat{v}_t = \frac{v_t}{1-\beta_2^t} \tag{2.61}$$

Since $\beta_1, \beta_2 < 1$, the corrections would be sufficiently small if a larger number of updates are performed. The update rule is finally defined to be:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \tag{2.62}$$

Eq. (2.62) resembles the general form of gradient descent, but makes use of both the momentum method (i.e., the moving average of the past gradients) and the adaptive method (i.e., the moving average of the past squared gradients). In practice, Adam has become a popular optimizer for training neural networks.

Improving gradient descent is an active sub-field of deep learning, but a full discussion of all those techniques is beyond the scope of this book. A few related issues will be discussed in the remainder of this section.

On a last note of this subsection, a practical issue that one should consider in utilizing iterative training methods is when to stop training. **Stopping criterion** is a general topic in optimization. For gradient descent and its variants, it is common practice to set a maximum number of training steps or **training epochs**[6], say 20,000 steps, or 100 epochs. As an alternative, we can perform training until convergence. For example, we can say that the training converges if the loss tends to be stable for a number of training steps. When there is some data for validating the model, a better method may be to check the states of the model on validation data. For example, we can stop the training when the prediction error increases on the validation data. This method, known as **early stopping**, is often used as a means of regularization. In Section 2.5.3, we will see more details about how to apply early stopping during training by using a validation dataset. On the algorithmic side, there has been much

[6]A training epoch means that the trainer goes over the whole training dataset exactly once.

interest and work in studying the convergence and error bounds for machine learning methods. We refer the interested reader to a few textbooks for further discussions [Mohri et al., 2018; Kochenderfer and Wheeler, 2019].

2.4.2 Batching

The loss function is an essential aspect of the training of neural networks. While a number of mathematical forms are available to define the loss function (see Chapter 1), we still need to decide the scale of samples over which to compute the loss function. Perhaps the simplest method is **stochastic gradient descent** (**SGD**). In each update of parameters, SGD computes the loss function on a single sample that is randomly selected from the training dataset. Let D be a set of training samples, and $(\mathbf{x}^{(i)}, \mathbf{y}_{\text{gold}}^{(i)})$ be a randomly selected sample from D. Given a neural network

$$\mathbf{y}_{\theta}^{(i)} = F_{\theta}\left(\mathbf{x}^{(i)}\right) \tag{2.63}$$

the loss of SGD is defined to be:

$$L(\theta) = L\left(\mathbf{y}_{\theta}^{(i)}, \mathbf{y}_{\text{gold}}^{(i)}\right) \tag{2.64}$$

where $L(\mathbf{y}_{\theta}^{(i)}, \mathbf{y}_{\text{gold}}^{(i)})$ is a sample-level loss function that quantifies the error in the model output $\mathbf{y}_{\theta}^{(i)}$ with respect to the ground truth $\mathbf{y}_{\text{gold}}^{(i)}$.

SGD has been one of the most important optimization methods in machine learning due to its simplicity. However, SGD exhibits high variance in its updates because a single sample provides only a noisy approximation of the true gradient. To estimate the gradient in a more precise way, we can take into account a set of samples (call it a **batch**) in computing the loss. This method is known as **batching**. Let S be a set of samples from D. The loss function is then defined on S, as follows

$$\begin{aligned} L(\theta) &= \frac{1}{|S|} \sum_{(\mathbf{x}, \mathbf{y}_{\text{gold}}) \in S} L(F_{\theta}(\mathbf{x}), \mathbf{y}_{\text{gold}}) \\ &= \frac{1}{|S|} \sum_{(\mathbf{x}, \mathbf{y}_{\text{gold}}) \in S} L(\mathbf{y}_{\theta}, \mathbf{y}_{\text{gold}}) \end{aligned} \tag{2.65}$$

If $S = D$, then we have the **batch gradient descent** (**BGD**) method, i.e., the gradient is estimated on the entire set of training samples. In general, batch gradient descent is what we would ordinarily call gradient descent. However, calculating the loss on all the training samples simultaneously is time consuming. In practice, it is more common to use a batch much smaller than D. This is known as **mini-batch gradient descent**. It is adopted in learning real-world systems for its good efficiency and strong performance.

As another "bonus", batching is generally used as a way to make dense computation on matrices for system speed-up. Assume that S consists of m samples $\{(\mathbf{x}^{(i_1)}, \mathbf{y}_{\text{gold}}^{(i_1)}), ..., (\mathbf{x}^{(i_m)}, \mathbf{y}_{\text{gold}}^{(i_m)})\}$.

We can batch all input vectors and ground-truth vectors as matrices:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}^{(i_1)} \\ \vdots \\ \mathbf{x}^{(i_m)} \end{bmatrix} \tag{2.66}$$

$$\mathbf{Y}_{\text{gold}} = \begin{bmatrix} \mathbf{y}_{\text{gold}}^{(i_1)} \\ \vdots \\ \mathbf{y}_{\text{gold}}^{(i_m)} \end{bmatrix} \tag{2.67}$$

Then, we can run the neural network on the batched input and output, like this:

$$\mathbf{Y}_\theta = F_\theta(\mathbf{X}) \tag{2.68}$$

Likewise, we can compute the batched loss

$$L(\theta) = \frac{1}{m} \cdot L(\mathbf{Y}_\theta, \mathbf{Y}_{\text{gold}}) \tag{2.69}$$

where $L(\mathbf{Y}_\theta, \mathbf{Y}_{\text{gold}})$ vectorizes the computation of $\sum_{k=1}^{m} L\left(\mathbf{y}_\theta^{(i_k)}, \mathbf{y}_{\text{gold}}^{(i_k)}\right)$. Eqs. (2.68-2.69) prevent the repetitive calls of the forward and backward passes on individual samples. They instead pack everything in a single pass through the network. This makes better use of maximum available compute on modern GPUs which are the majority of the devices for running deep learning systems.

2.4.3 Parameter Initialization

Gradient descent requires the training process to start from some initial parameters. Since the training objective in a practical system is often a highly non-convex function with many local minima, the performance of the resulting model is extremely sensitive to the parameter initialization step. Here, we describe some of the most common methods for parameter initialization.

- **Constant Initialization**. The first method could assign the same value to all parameters (or all entries of a parameter matrix). This method, though straightforward, causes all hidden units to update identically during training. As a result, the network becomes unable to learn complex patterns. It generally performs poorly in practice unless specific architectural randomness is introduced.
- **Initialization with Predefined Distributions**. A more practical approach is to randomly initialize parameters using predefined probability distributions. The simplest method of this kind is to draw parameter values from a uniform or Gaussian distribution, e.g., assigning a random value in the interval $[-0.1, 0.1]$. Interestingly, this heuristic is satisfactory for many basic practical applications.
- **Layer-sensitive Initialization**. An extension to basic random initialization is to use tailored distributions based on the dimensions of different neural network layers. Given

a layer $\mathbf{y} = \psi(\mathbf{x} \cdot \mathbf{W} + \mathbf{B})$, let d_{in} and d_{out} denote the fan-in and fan-out (i.e., the row and column dimensions of $\mathbf{W}$), respectively. The **LeCun initialization** method [LeCun et al., 2012] assigns a random number to every parameter of $\mathbf{W}$ by considering only the input dimension:

$$\mathbf{W} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}} \sim U\left(-\frac{1}{\sqrt{d_{\text{in}}}}, \frac{1}{\sqrt{d_{\text{in}}}}\right) \tag{2.70}$$

where $U(-a, a)$ denotes a uniform distribution over the interval $[-a, a]$. Likewise, the bias term can be initialized in a similar manner. Building upon this, the standard **Xavier initialization** (also known as Glorot initialization) considers both d_{in} and d_{out} to stabilize the variance of activations and gradients across layers [Glorot and Bengio, 2010]:

$$\mathbf{W} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}} \sim U\left(-\sqrt{\frac{6}{d_{\text{in}} + d_{\text{out}}}}, \sqrt{\frac{6}{d_{\text{in}} + d_{\text{out}}}}\right) \tag{2.71}$$

Note that these uniform distributions can be readily replaced by normal distributions with a mean of 0 and a variance of $\frac{1}{d_{\text{in}}}$ (for LeCun initialization) or $\frac{2}{d_{\text{in}} + d_{\text{out}}}$ (for Xavier initialization).

Many parameter initialization methods are designed for certain types of neural networks. For example, Xavier initialization is derived assuming the use of symmetric activation functions, such as the Sigmoid and hyperbolic tangent. For ReLU, one can refer to He et al. [2015]'s work. Another example is initialization for deep neural networks. It has been found that appropriate initialization is critical to the success of extremely deep models in NLP. Considering the model depth as an additional factor in initialization, we can modify Eq. (2.71) to be:

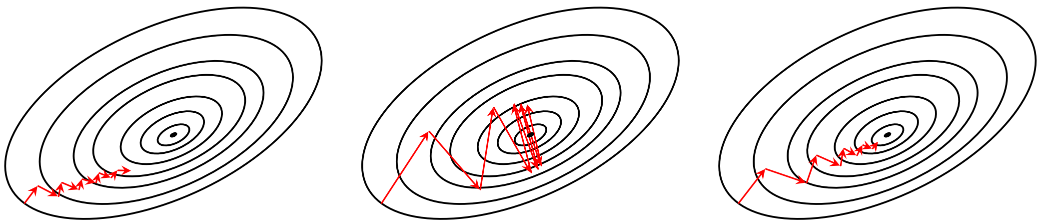
$$\mathbf{W} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}} \sim U\left(-\frac{\alpha_s}{l} \cdot \sqrt{\frac{6}{d_{\text{in}} + d_{\text{out}}}}, \frac{\alpha_s}{l} \cdot \sqrt{\frac{6}{d_{\text{in}} + d_{\text{out}}}}\right) \tag{2.72}$$

where l is the depth of a layer, and α_s is a hyper-parameter. Apart from this, several methods have been proposed to address the initialization of deep neural networks, including the Lipschitz initialization [Xu et al., 2020], the T-Fixup initialization [Huang et al., 2020a], the Admin initialization [Liu et al., 2020c], and so on.

As a final note, in practice, we do not have to restrict training to a single starting point. It is common to perform multiple trials using different initialization methods or random seeds, and then select the best-performing model from these trials. This strategy is particularly helpful when local minima abound in the optimization landscape.

2.4.4 Learning Rate Scheduling

To achieve desirable results, it is essential to carefully configure the learning rate throughout the training process. While some of the update rules, as noted above, have considered scaling the gradient for different parameters, **learning rate scheduling** is conventionally focused more on designing heuristics to adjust η over training steps. In practice, an excessively large learning



(a) A small learning rate (b) A large learning rate (c) A desirable learning rate

Figure 2.12: Learning with different learning rates. Small learning rates (left) help us step toward the minimum in a precise manner, but require significantly more time for convergence. Large learning rates (middle) lead to rapid initial updates, which is highly beneficial when far from the minimum. However, as we approach the minimum, excessively large learning rates cause overshooting. A more desirable strategy (right) is to update the model rapidly when far from convergence, and to slow down the updates as convergence approaches.

rate usually leads to overshooting around the minimum, whereas a learning rate that is too small results in painfully slow convergence (see Figure 2.12). A common strategy is to take large steps at the beginning (i.e., using a large learning rate) and smaller steps as the loss approaches the minimum. Here, we present some of the popular methods for learning rate scheduling.

- **Fixed Learning Rates**. Fixing the learning rate is generally a sub-optimal strategy, but could be used when prototyping systems, e.g., conducting a quick test of a new method by training it for only a few epochs.
- **Learning Rate Decay**. Decay is a commonly used technique for learning rate scheduling. There are many approaches to this idea. For example, one can halve the learning rate after each training epoch. Here we use n_t to represent the current training step, and n_{decay} to represent how often we change the learning rate (e.g., 100 steps). Table 2.2 shows several decay functions for learning rate scheduling.
- **Warmup and Decay**. As noted previously, it is common to initialize model parameters with random values before training a neural network. However, starting from scratch with a large learning rate is usually inadvisable because gradients at the early stages of training are highly noisy, and the state of the model is unstable. Therefore, it is more reasonable to start with a small learning rate and gradually increase it. Then, after the model has trained for some time, the learning rate begins to decay as usual. This intuition motivates the warmup and decay method. A popular formulation of this method was proposed by Vaswani et al. [2017], as follows:

$$\eta_{n_t} = \eta_0 \cdot \min\left(\left(\frac{n_t}{n_{\text{decay}}}\right)^{-0.5}, \frac{n_t}{n_{\text{decay}}} \cdot (n_{\text{warmup}})^{-1.5} \right) \tag{2.73}$$

where η_0 is the initial learning rate, and n_{warmup} is a hyper-parameter that specifies the

Entry	Formula	Hyper-parameters
Piecewise Constant Decay	$\eta_{n_t} = \beta_i$ if $\gamma_i \leq \frac{n_t}{n_{\text{decay}}} < \gamma_{i+1}$	values $\{\beta_1, ..., \beta_m\}$ thresholds $\{\gamma_1, ..., \gamma_m\}$
Exponential Decay	$\eta_{n_t} = \eta_0 \cdot \lambda^{\frac{n_t}{n_{\text{decay}}}}$	decay rate λ, init. rate η_0
(Drop) Exponential Decay	$\eta_{n_t} = \eta_0 \cdot \lambda^{\lfloor \frac{n_t}{n_{\text{decay}}} \rfloor}$	decay rate λ, init. rate η_0
Natural Exponential Decay	$\eta_{n_t} = \eta_0 \cdot \exp(-\lambda \cdot \frac{n_t}{n_{\text{decay}}})$	decay rate λ, init. rate η_0
Inverse Time Decay	$\eta_{n_t} = \eta_0 \cdot \frac{1}{1+\lambda \cdot \frac{n_t}{n_{\text{decay}}}}$	decay rate λ, init. rate η_0
(Drop) Inverse Time Decay	$\eta_{n_t} = \eta_0 \cdot \frac{1}{1+\lambda \cdot \lfloor \frac{n_t}{n_{\text{decay}}} \rfloor}$	decay rate λ
Cosine Decay	$\eta_{n_t} = \eta_0 \cdot \big((1-\alpha) \cdot c_{\text{decay}} + \alpha\big)$ $c_{\text{decay}} = \frac{1}{2} \cdot \big(1 + \cos(\pi \cdot \frac{n_t}{n_{\text{decay}}})\big)$	coefficient α init. rate η_0

Table 2.2: Decay functions. λ = decay rate, η_0 = initial learning rate, and $\{\beta_i\}$, $\{\gamma_i\}$ and α = other hyper-parameters.

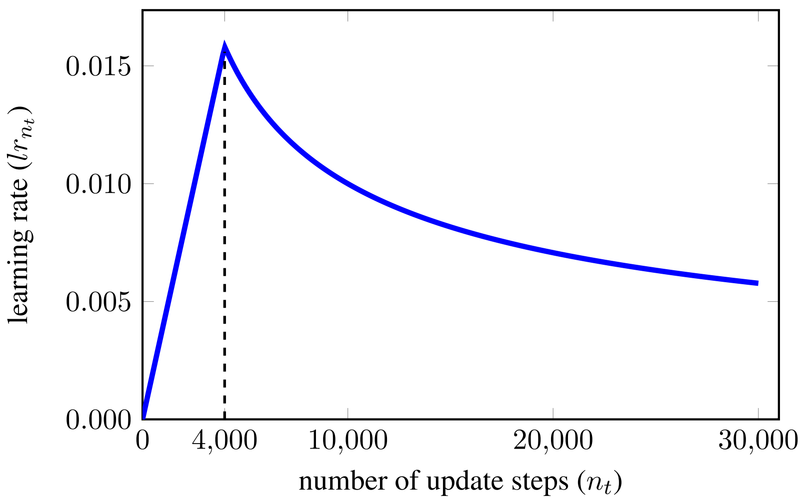


Figure 2.13: Learning rate scheduling: warmup and then decay ($n_{\text{warmup}} = 4,000$, $n_{\text{decay}} = 1$, and $\eta_0 = 1$). The learning rate increases linearly with n_t for the first $4,000$ steps. Then, the learning rate follows an inverse square root function and decays as the training continues. The change in the learning rate will be small if n_t is sufficiently large, indicating the fine adjustment of the parameters when we are approaching the minimum of the loss.

number of steps for the warmup phase. Figure 2.13 plots the curve of Eq. (2.73) where n_{warmup}, n_{decay}, and η_0 are set to $4,000$, 1 and 1, respectively. We see that the learning rate increases linearly in the first n_{warmup} steps and then decays as an inverse square root function.

Choosing an appropriate learning rate scheduling strategy is a highly empirical problem, and there are no universally optimal choices. The problem becomes even more complex when considering the correlation between the learning rate and other training dynamics, even though learning rate scheduling is typically treated as an isolated task. For example, when a larger batch size is used in training, a larger learning rate is generally required to achieve good results [Ott et al., 2018b; Smith et al., 2018]. As a result, selecting an optimal learning rate remains difficult and time-consuming in neural network applications. Occasionally, one must rely on numerous trial-and-error runs to find a desirable learning rate setup for the specific problem at hand.

2.5 Regularization Methods

We now discuss the regularization methods for preventing overfitting. While regularization is a wide-ranging topic in machine learning, we present some of those that are commonly adopted in training neural networks.

2.5.1 Norm-based Penalties

One of the most popular methods involves a regularization term based on the l_p norm. A general form of the regularized objective can be defined as:

$$\hat{\theta} = \arg\min_{\theta} L(\theta) + \alpha \cdot R(\theta) \tag{2.74}$$

where $R(\theta)$ is the regularization term weighted by a coefficient α. In general, $R(\theta)$ serves as an additional loss that penalizes complex models. This is motivated by the fact that complex models are more likely to overfit the data (see Chapter 1). To penalize model complexity, a straightforward approach is to define $R(\theta)$ as the l_1 norm of the parameter vector θ. The l_1 norm-based penalty is given by

$$R(\theta) = \sum_i |\theta_i| \tag{2.75}$$

Eq. (2.75) penalizes models with very large parameter values. This can be intuitively understood by analogy to a polynomial function: large coefficients of variables in a polynomial function lead to a complex curve. Typically, regularization with the l_1 norm is referred to as l_1 regularization or **Lasso regularization**. This method requires no fundamental modifications to the optimization algorithm and can be implemented using standard gradient descent. More interestingly, l_1 regularization typically provides sparse solutions to the original training objective. It forces many parameters to be exactly zero, leaving only a small subset of critical features with non-zero weights. This also implies an inherent ability of feature selection because parameters are forced to be zero for less important features.

An alternative to l_1 regularization is l_2 **regularization**, also widely known as **Ridge regularization**. In l_2 regularization, the regularization term is given by the squared l_2 norm of the parameters

$$R(\theta) = \sum_i \theta_i^2 \tag{2.76}$$

Like the l_1 norm, the l_2 norm penalizes parameter vectors that deviate significantly from the origin. However, it slightly differs from the l_1 norm in that the l_2 norm enforces all parameters to have small values (but not necessarily to be zero) and there are no large value parameters. In this sense, the use of the l_2 norm does not introduce strict sparsity into the solution, but rather performs a "smoothing" effect on the underlying distribution of features. Hence, it is sometimes called **weight decay** to emphasize its ability to continuously shrink parameter values during gradient updates.

In a broader sense of machine learning, Eq. (2.74) offers a generalized framework for injecting prior knowledge into the training of a neural network. There are numerous ways to design the regularization term, and mitigating overfitting is merely one objective of these designs. We will encounter various applications of this approach in NLP, and we will explore a few examples in the remaining chapters of this book.

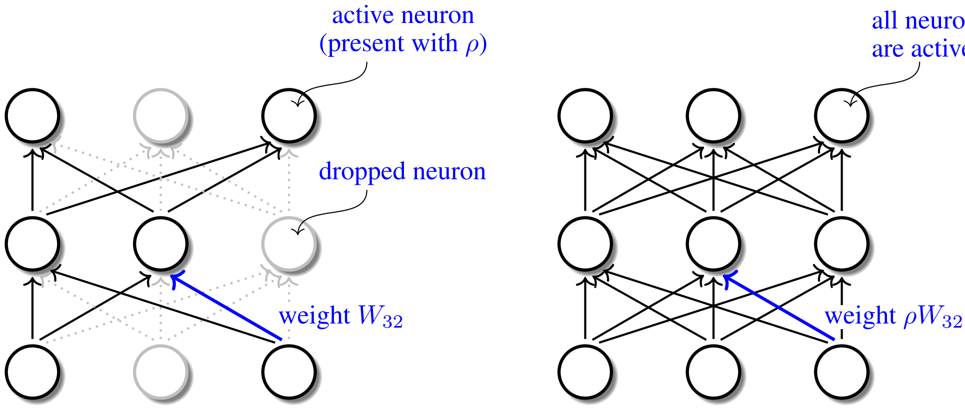
2.5.2 Dropout

In a real-world neural network, a layer typically involves hundreds or thousands of neurons and produces a feature vector accordingly. While each of these features is computed by a single neuron, they work together to form the input to each neuron of the following layer. As a result, a feature is forced to cooperate with other features. It is like a group of people sitting together and making a collective decision. Although a member could have opinions independently, they occasionally try to correct errors made by the rest of the group. In this case, every group member is co-adapted to others in the group [Hinton et al., 2012]. From a feature engineering standpoint, the **co-adaptation** of neurons helps when modeling complex problems, as it implicitly makes some sort of higher order features. Beyond this, the strong supervision information (e.g., propagating errors through layers) could strengthen the co-adaptation in training. This explains more or less why a neural network with a large number of neurons can fit complex curves. At test time, however, the co-adaptation prevents generalization. Since all neurons of a layer are learned to collaborate well on the training data, a small change in the input could affect all these neurons and lead to a big change in the behavior of the neural network.

A way to mitigate or eliminate complex co-adaptations is to learn for each neuron to predict in the absence of other neurons. To this end, one can simply drop some of the neurons in training. This method is known as **dropout** [Srivastava et al., 2014]. Let n be the number of neurons of a layer. Given a probability ρ (call $1-\rho$ the **dropout rate**), we can generate an n-dimensional mask vector $\mathbf{M}_{\text{drop}}$ where every entry is set to 1 with a probability of ρ, and set to 0 with a probability of $1-\rho$. Then, a dropout layer can be defined as

$$\mathbf{y} = \mathbf{M}_{\text{drop}} \odot \psi(\mathbf{x} \cdot \mathbf{W} + \mathbf{B}) \tag{2.77}$$

where $\psi(\mathbf{x} \cdot \mathbf{W} + \mathbf{B})$ is a usual single-layer neural network. Eq. (2.77) only activates the neurons whose masks are 1. For dropped neurons, all connections from/to these neurons are blocked (see Figure 2.14 (a)). During training, $\mathbf{M}_{\text{drop}}$ is randomly generated in a call of the forward and backward passes. A neuron therefore can learn to work with different



(a) Training the dropout network (b) Testing with the dropout network

Figure 2.14: Dropout for a multi-layer neural network (training vs test). At training time, every neuron is randomly dropped with a probability of $1-\rho$, resulting in a slimmed network. In this sense, dropout training is essentially a process of learning an exponentially large number of sub-networks. At test time, the full network is used as usual, which is the result of combining all those sub-networks for prediction. Since all connections between neurons are activated with the probability ρ during training, the weights of the inference network are scaled down with ρ.

neurons each time and would not adapt to the same group of "co-workers". Another way to understand dropout is to view it as learning sub-models of a "big" model. The use of Eq. (2.77) is essentially a sampling process that extracts a sub-network from the original network. So, training with dropout is akin to training an exponentially large ensemble of sub-networks[7]. On the other hand, the training is efficient because these sub-networks share the same parameters for the same neuron and the update of a parameter can benefit exponentially many sub-networks.

At test time, all these sub-networks are combined for prediction. In this case, we do not need to drop any neuron but use the original network as usual. This makes it simple to implement dropout: a neuron is present with some probability on the training data, and all neurons are present and work together on the test data. Since the connections between neurons are involved with a probability of ρ in training, the learned weights are scaled down with ρ in the inference network, i.e., a layer has a form:

$$\mathbf{y} = \psi(\mathbf{x} \cdot \rho\mathbf{W} + \rho\mathbf{B}) \tag{2.78}$$

where the weights $\mathbf{W}$ and $\mathbf{B}$ are scaled by ρ[8].

[7]For a single-layer network having n neurons, there are 2^n possible sub-networks.

[8]Note that placing the scaling factor inside the activation function ψ (e.g., using $\rho\mathbf{W}$) implicitly assumes that ψ is a linear function or possesses positive homogeneity (e.g., the ReLU function, where $\mathrm{ReLU}(cx) = c\mathrm{ReLU}(x)$ for $c > 0$). If a non-linear activation function is used, such as Sigmoid, the scaling factor cannot be distributed inside. In such cases, the scaling is applied directly to the output of the activation function to properly preserve the expected value. The mathematical forms for standard inference scaling and inverted dropout training would then be

See Figure 2.14 for a comparison of training and applying a dropout network. Eq. (2.78) requires an update of the inference system. An alternative is to take into account the scaling issue only in the training process and leave the inference system as it is. For example, we can scale up all the parameters with $\frac{1}{\rho}$ in dropout training, like this

$$\mathbf{y} = \mathbf{M}_{\text{drop}} \odot \psi(\mathbf{x} \cdot \frac{1}{\rho}\mathbf{W} + \frac{1}{\rho}\mathbf{B}) \tag{2.79}$$

Since multiplying $\frac{1}{\rho}\mathbf{W}$ with ρ yields $\mathbf{W}$ (this also holds for the bias term $\mathbf{B}$), we can use $\mathbf{W}$ (and $\mathbf{B}$) as the parameters of the inference system.

2.5.3 Early Stopping

In Chapter 1 we have discussed a bit of how to stop the training by monitoring the performance on the validation data. This can be viewed as a form of model selection that seeks an optimal balance between underfitting and overfitting. Note that early stopping is not merely an empirical heuristic; it is also well-founded in statistical learning theory. For example, researchers have found that, under certain conditions, early stopping has an effect similar to l_2 regularization, restricting the learning to a region of small parameter values [Bishop, 1995a; Goodfellow et al., 2016]. Furthermore, studies show that specific early stopping rules share a close relationship with the bias-variance trade-off and can guarantee desirable convergence properties [Yao et al., 2007].

In practice, however, early stopping requires several heuristics to make it practical and useful. The primary problem lies in determining the exact stopping criterion. Ideally, one might imagine that there is a perfect U-shaped error curve on the validation data, and the training can be halted immediately when the error starts to increase. The truth, however, is that the error curve cannot be simply described as a strictly convex function of the training time. After drops in the error in a certain number of training steps, the performance of the model tends to fluctuate, leading to many local minima. The problem would be more interesting if one wants to save time and stop the training as early as possible. However, we never know whether the current choice or decision is the best one because we have no idea of what happens next. A commonly used method is to decide whether the training should stop by checking the model states for a number of past update steps (or epochs) [Prechelt, 1998]. Some early stopping conditions are:

- The change in the performance is below a threshold for a given number of steps (or epochs).
- The change in the model parameters is below a threshold for a given number of steps (or epochs).
- The average performance over a given number of steps (or epochs) starts to decrease.
- The maximum performance over a given number of steps (or epochs) starts to degrade.

However, using the model at the point that we stop the training is not always a good choice.

$\mathbf{y} = \rho \cdot \psi(\mathbf{x} \cdot \mathbf{W} + \mathbf{B})$ and $\mathbf{y} = \frac{1}{\rho}\left(\mathbf{M}_{\text{drop}} \odot \psi(\mathbf{x} \cdot \mathbf{W} + \mathbf{B})\right)$, respectively.

In practice, a model often has a large variance in generalization error around that point, making model selection more difficult. Instead of “selecting” a model, an alternative way is to combine multiple models. For example, we can save the model for every run of a given number of training steps (call each copy of the model a checkpoint). The final model is obtained by averaging the parameters of the last few checkpoints. For better results, one may use more sophisticated ensemble methods (see Chapter 1).

2.5.4 Smoothing Output Probabilities

In statistics, smoothing refers to the process of reducing the influence of noisy data points (typically extreme values) and reallocating probability mass to normal data points or unobserved events. It is widely used when a distribution is estimated from limited data, where the probabilities of rare events are poorly represented. For example, consider the language modeling problem described in Section 2.2. A language model is trained in a way that enforces the model to output a one-hot distribution, that is, the entire probability mass of 1 is assigned to a single word, leaving other words assigned zero probabilities. It may be more desirable to distribute the probability to all words, even though many of them are not observed to be the answer given the previous words. In this way, the model learns to make a soft prediction of word probabilities so that it can generalize better on unseen data.

Given a distribution $\mathbf{p} = \begin{bmatrix} p_1 & \dots & p_n \end{bmatrix}$, it is the purpose of smoothing that we obtain the new estimate between $\mathbf{p}$ and a uniform distribution $\frac{1}{n}$. A common approach to this idea is to use a **shrinkage estimator** to improve $\mathbf{p}$ by making it closer to $\frac{1}{n}$. For example, the additive smoothing mentioned in Chapter 1 is a simple type of shrinkage estimator. Here we consider, for example, smoothing a multinomial distribution. Let p_k denote the probability of event k and s_k denote a quantity that describes some observed “count” of the event. The probability p_k is given by

$$p_k = \frac{s_k}{\sum_{k=1}^{n} s_k} \tag{2.80}$$

Then, the smoothed version of p_k is defined as

$$\hat{p}_k = \frac{s_k + \alpha}{\sum_{k=1}^{n} (s_k + \alpha)} \tag{2.81}$$

It simply adds a quantity α to each s_k. The value of α controls the smoothness of the resulting estimate. For example, $\hat{p}_k = p_k$ if $\alpha = 0$, and $\hat{p}_k \approx \frac{1}{n}$ as $\alpha \to \infty$.

Apart from additive smoothing, we can smooth a distribution using a Softmax function with temperature scaling, defined as follows:

$$\hat{p}_k = \frac{\exp(s_k/\beta)}{\sum_{k=1}^{n} \exp(s_k/\beta)} \tag{2.82}$$

This form is a classic instance of the **Boltzmann distribution** [Uffink, 2017], where s_k can be viewed as the negative energy of a state, and β represents the temperature that controls the

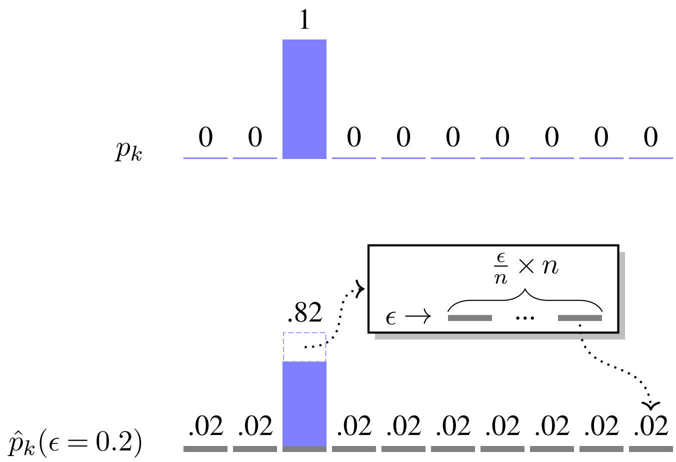


Figure 2.15: Smoothing a distribution by interpolating it with a uniform distribution: $\hat{p}_k = (1-\epsilon) \cdot p_k + \frac{\epsilon}{n}$. Mathematically, this operation scales down each original probability p_k by an amount of ϵp_k, pooling a total probability mass of ϵ. This mass is then redistributed evenly across all n dimensions, ensuring every variable receives an additive smoothing factor of $\frac{\epsilon}{n}$.

degree of smoothing. Note that s_k can be interpreted in various ways. In a neural network, for example, s_k is typically the pre-activation logit of a neuron. Occasionally, s_k can even be a raw probability. This flexibility implies that we can directly apply Eqs. (2.81–2.82) to any distribution $\mathbf{p}$ even without prior knowledge of how $\mathbf{p}$ was originally estimated. We can rewrite these equations by substituting s_k with p_k:

$$\hat{p}_k = \frac{p_k + \alpha}{\sum_{k=1}^{n}(p_k + \alpha)} \tag{2.83}$$

$$\hat{p}_k = \frac{\exp(p_k/\beta)}{\sum_{k=1}^{n}\exp(p_k/\beta)} \tag{2.84}$$

Another straightforward method of smoothing is to linearly interpolate $\mathbf{p}$ with a uniform distribution. A standard form of this interpolation is given by:

$$\hat{p}_k = (1-\epsilon) \cdot p_k + \frac{\epsilon}{n} \tag{2.85}$$

where ϵ is a hyper-parameter indicating the extent to which we rely on the uniform distribution. To illustrate how Eq. (2.85) works, suppose $\mathbf{p}$ is a one-hot vector (e.g., $p_k = 1$ if $k = z$, and $p_k = 0$ otherwise). By applying Eq. (2.85), we effectively scale down the original distribution by $(1-\epsilon)$ and add a uniform probability mass of $\frac{\epsilon}{n}$ to every dimension. This redistribution makes the resulting distribution flatter and smoother. See Figure 2.15 for a visual illustration.

In NLP, since many systems make probability-like predictions, a common application of smoothing is to smooth a system's output. There are two ways. First, we can smooth the ground-truth probability such that the model is guided by the generalized error rather than the error made by hard decisions. For example, the **label smoothing** technique adopts the same form as Eq. (2.85) and improves the ground-truth representation on categorical data [Szegedy et al., 2016]. Second, we can reduce the sharpness and increase the tailedness of a predicted distribution[9]. This method is often used when the posterior probability of the prediction is required, such as minimum Bayesian risk decoding/training [Bickel and Doksum, 2015; Goodman, 1996a; Kumar and Byrne, 2004a].

2.5.5 Training with Noise

Above, we have shown that adding some amount to each observed count of events in predicting a probability can improve generalization. From a **robust statistics** point of view [Olive, 2022], this is equivalent to improving the robustness of an estimator where a skewed distribution often leads to a biased model. The addition of a small perturbation to the estimate can prevent large biases caused by outliers and unexpected observations of rare events. In this sense, smoothing can be regarded as a way of introducing noise into training, that is, we impose a prior of uniform distribution on the estimate though the correct estimate may not be uniform.

Training with noise is based on the idea that a model should learn to operate under non-ideal conditions, thereby avoiding the overfitting of extreme data points. Here, the term *noise* takes on a broad meaning, and there are several distinct ways to regularize training via noise injection. One of the simplest methods is to introduce noise into the training objectives. For example, label smoothing (as discussed previously) can be viewed as injecting noise into the ground-truth annotations. Alternatively, we can inject random noise directly into the inputs, intermediate hidden states, or outputs of a neural network. A ubiquitous choice is **Gaussian noise**. Suppose we have an activation vector $\mathbf{x} \in \mathbb{R}^n$. The addition of Gaussian noise yields a new corrupted vector as follows:

$$\mathbf{x}_{\text{noise}} = \mathbf{x} + \mathbf{g} \tag{2.86}$$

where $\mathbf{g} \in \mathbb{R}^n$ is the noise vector. Each element of $\mathbf{g}$ is drawn independently from a Gaussian distribution:

$$\mathbf{g} \sim \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2)) \tag{2.87}$$

For entry k of $\mathbf{g}$, it defines the probability $\Pr(g_k)$ to be:

$$\Pr(g_k) = \frac{1}{\sigma_k\sqrt{2\pi}} \cdot \exp\left(-\frac{(g_k - \mu_k)^2}{2\sigma_k^2}\right) \tag{2.88}$$

where μ_k is the mean, and σ_k is the standard deviation. Typically, μ_k is set to 0. The standard deviation σ_k is a hyper-parameter controlling the magnitude of the injected noise. A larger σ_k means the random noise spreads over a wider region around the mean, increasing the likelihood of generating severe perturbations.

Eq. (2.86) is highly generic and can be applied almost anywhere within a neural network.

[9] In statistics, this is loosely related to transforming a leptokurtic distribution into a more mesokurtic or platykurtic one to prevent overconfidence.

Given a layer $\mathbf{y} = \psi(\mathbf{x} \cdot \mathbf{W} + \mathbf{B})$, noise (denoted as $\mathbf{g}_{\text{input}}$) can be added directly to the input:

$$\mathbf{y} = \psi\left((\mathbf{x} + \mathbf{g}_{\text{input}}) \cdot \mathbf{W} + \mathbf{B}\right) \tag{2.89}$$

Likewise, the noise (say $\mathbf{g}_{\text{output}}$) can be added to the activation (or output):

$$\mathbf{y} = \psi(\mathbf{x} \cdot \mathbf{W} + \mathbf{B}) + \mathbf{g}_{output} \tag{2.90}$$

For example, one can simply corrupt the raw inputs (or targets) and run the network as usual, or alternatively, inject random noise into the activations of all hidden layers. While it is common practice to add noise to layer inputs and activations [Plaut et al., 1986; Holmström and Koistinen, 1992; Bishop, 1995b], another distinct approach is to add random noise directly to the model parameters or gradients [Graves et al., 2013b; Neelakantan et al., 2015]. For example, perturbing the weight matrix takes the following form:

$$\mathbf{y} \quad = \quad \psi(\mathbf{x} \cdot (\mathbf{W} + \mathbf{g}_{\text{w}}) + \mathbf{B}) \tag{2.91}$$

where $\mathbf{g}_{\text{w}}$ is a noise matrix matching the dimensions of $\mathbf{W}$. Furthermore, we can inject noise (denoted as $\mathbf{g}_{\text{gradient}}$) into the loss gradients with respect to $\mathbf{W}$. Let $\mathbf{s} = \mathbf{x} \cdot \mathbf{W} + \mathbf{B}$ represent the pre-activation logits. The noisy gradient can be written as:

$$\begin{aligned} \frac{\partial L}{\partial \mathbf{W}} &= \mathbf{x}^{\top} \cdot \frac{\partial L}{\partial \mathbf{s}} + \mathbf{g}_{\text{gradient}} \\ &= \mathbf{x}^{\top} \cdot \left(\frac{\partial L}{\partial \mathbf{y}} \odot \frac{\partial \mathbf{y}}{\partial \mathbf{s}}\right) + \mathbf{g}_{\text{gradient}} \\ &= \mathbf{x}^{\top} \cdot \left(\frac{\partial L}{\partial \mathbf{y}} \odot \psi'(\mathbf{s})\right) + \mathbf{g}_{\text{gradient}} \end{aligned} \tag{2.92}$$

The use of noisy gradients has been found to not only be helpful for robust training but also to ease the gradient flow in the network [Gulcehre et al., 2016].

It should be noted that noise is only present during training and the model works without the addition of noise when making predictions on new data. In this sense, many of the regularization methods could fall under the noisy training framework that is used to prevent fitting the training data precisely and enable the inference system to generalize well on the test data. For example, dropout randomly masks out layer activations so that every neuron learns to work in a noisy environment. At test time, all neurons work together as in a usual neural network.

An additional advantage of noisy training is that the use of random noise creates "new" training samples. Even for the same sample, different noise could lead to different training results. In other words, we essentially train the model on an infinite number of samples. This idea is also linked to another line of research on training with synthetic data, called **data augmentation**. In simple terms, data augmentation encompasses a suite of techniques designed to generate new training samples from existing ones. An example is **back-translation** [Sennrich et al., 2016a]. When developing a machine translation system from language A to language

B, we can first train a reverse translation system (say the B→A system) on the bilingual data. Then, we use the B→A system to translate some additional target-language data to source-language data. This results in new bilingual data where the target-language data is real and the source-language data is synthetic. This new data can be used together with other bilingual data to train the A→B system. Beyond back-translation, there are many data augmentation methods in NLP, including replacing words with synonyms, swapping two words, and deleting or inserting words. Furthermore, augmentation can be performed in the continuous latent space, such as substituting a word vector with a geometrically proximate vector. Since data augmentation covers a wide variety of topics, we refer the reader to a few survey papers for further details [Feng et al., 2021; Shorten and Khoshgoftaar, 2019].

One last note on data augmentation: synthetic data can also be crafted for specific objectives. A popular idea is **adversarial machine learning**. It generates **adversarial samples** on which a model would make mistakes (call such processes **attacks**) [Szegedy et al., 2014a; Goodfellow et al., 2015]. The model is trained to make correct predictions on these samples, i.e., it defends against the attacks. For example, in some cases, a machine translation system might output completely erroneous translations if the gender of the subject in the input sentence is merely flipped. For a more robust system, one may train the translation model by using more gender-balanced data, gathered either manually or automatically. But it is not easy to craft samples that look like normal sentences but can fool the model [Zhang et al., 2020b]. This in turn makes it interesting yet challenging to generate adversarial samples in NLP, since a small change in a sentence (such as word replacement) could lead to something with a very different meaning[10]. The challenge also motivates a thread of research on investigating adversarial samples in NLP [Jia and Liang, 2017; Belinkov and Bisk, 2018; Ebrahimi et al., 2018; Alzantot et al., 2018].

2.6 Unsupervised Methods and Auto-encoders

Unsupervised learning is concerned with discovering the underlying patterns in a set of unlabeled data points. A number of problems can be viewed as classical unsupervised learning problems, although we will not discuss them in detail in this chapter. For example, data clustering aims to find groupings in a collection of data objects, without explicit supervision regarding the correct groupings. Another well-known example is association rule mining, which is often framed as a process of establishing relationships among sets of data objects. While these problems are indeed covered by unsupervised learning, we will focus on problems of unsupervised representation learning or **feature learning**, where a model learns to map an object from an input space to a low-dimensional feature vector space[11].

Learning low-dimensional representations has been extensively studied in the context of finding a linear transformation from the original space to a new space. For example, **principal**

[10] By contrast, in computer vision, it is much easier to create adversarial samples by making a small change in the input (e.g., pixels), since the input space is continuous and a small input perturbation has very little effect on the whole image.

[11] In addition to learning to represent data objects, this section also covers some topics on the generation of data objects. We will explore them in Section 2.6.3.

component analysis (**PCA**) and its variants aim to find a linear mapping function such that a (high-dimensional) data object can be represented by its coordinates along the directions of greatest variance [Pearson, 1901; Wold et al., 1987]. Here, we extend the mapping function to its natural non-linear generalization and use neural networks as a solution to the mapping problem.

As with other machine learning models, a neural network is typically trained by optimizing model parameters with respect to some loss function. A considerable challenge with unsupervised learning is that there are no ground-truth labels to guide the learning process. One solution to this issue is to resort to non-parametric methods or heuristics (see Chapter 1). However, such methods themselves are not designed to address the optimization issues of large-scale neural networks, particularly when a network is built with a massive number of parameters. In the unsupervised training of neural networks, therefore, it is more common to derive supervisory signals directly from the input data itself. While there are several ways to achieve this [Hopfield, 1982; Ackley et al., 1985; Dayan et al., 1995; Hinton and Salakhutdinov, 2006], we will focus on **auto-encoders** in this section. We choose auto-encoders for discussion because they resemble the general form of supervised models and can be seamlessly trained via back-propagation.

An auto-encoder is a type of neural network that tries to reconstruct the input data from its representation. It is inspired by the idea of dimensionality reduction:

> *High-dimensional data can be converted to low-dimensional codes by training a multilayer neural network with a small central layer to reconstruct high-dimensional input vectors.*
>
> – Hinton and Salakhutdinov [2006]

This also develops the idea of representation learning in that the information of an object can be sufficiently represented by a low-dimensional real-valued vector. Typically, an auto-encoder involves a (probably non-linear) dimensionality reduction function (call it an **encoder**) to map the input object to its low-dimensional feature vector representation (call it a **code**). Also, it involves a reverse function (call it a **decoder**) that maps the code back to the object. So, although an auto-encoder is called an "encoder", it is not just an encoder but a combination of an encoder and a decoder. More formally, let $\mathbf{x} \in \mathbb{R}^{d_x}$ be the input vector of the model, such as a high-dimensional representation of a word. The encoder outputs a vector describing the code or low-dimensional representation of $\mathbf{x}$, denoted as $\mathbf{h} \in \mathbb{R}^{d_h}$. This vector is defined as:

$$\mathbf{h} = \mathrm{Enc}(\mathbf{x}) \tag{2.93}$$

where $\mathrm{Enc}(\cdot)$ is the encoding network. $\mathrm{Enc}(\cdot)$ is typically a multi-layer neural network and works as a plug-in module for other systems. Thus, $\mathrm{Enc}(\cdot)$ is a general-purpose model. In subsequent chapters, we will see many examples where encoders are trained and applied as parts of "bigger" systems.

Once we obtain the code, we use the decoder to map it back to the input space:

$$\widetilde{\mathbf{x}} = \mathrm{Dec}(\mathbf{h}) \tag{2.94}$$

where $\widetilde{\mathbf{x}} \in \mathbb{R}^{d_x}$ is the reconstruction of the input, and $\mathrm{Dec}(\cdot)$ is the decoding network. Given the original input $\mathbf{x}$ and the reconstructed input $\widetilde{\mathbf{x}}$, the objective of the auto-encoder is to minimize the discrepancy between $\mathbf{x}$ and $\widetilde{\mathbf{x}}$. Suppose that the encoder and the decoder are parameterized by θ and ω, denoted as $\mathrm{Enc}_\theta(\cdot)$ and $\mathrm{Dec}_\omega(\cdot)$, respectively. The training objective over a set of samples $\{\mathbf{x}^{(1)},...,\mathbf{x}^{(m)}\}$ is defined as

$$\begin{aligned}
(\hat{\theta},\hat{\omega}) &= \underset{(\theta,\omega)}{\arg\min} \sum_{i=1}^{m} L\left(\mathbf{x}^{(i)},\widetilde{\mathbf{x}}^{(i)}\right) \\
&= \underset{(\theta,\omega)}{\arg\min} \sum_{i=1}^{m} L\left(\mathbf{x}^{(i)},\mathrm{Dec}_\omega(\mathbf{h}^{(i)})\right) \\
&= \underset{(\theta,\omega)}{\arg\min} \sum_{i=1}^{m} L\left(\mathbf{x}^{(i)},\mathrm{Dec}_\omega(\mathrm{Enc}_\theta(\mathbf{x}^{(i)}))\right)
\end{aligned} \tag{2.95}$$

where $L(\cdot)$ is the loss function that computes the discrepancy between $\mathbf{x}$ and $\widetilde{\mathbf{x}}$. It is sometimes called the **reconstruction loss**. Popular loss functions for reconstruction include mean squared error loss, crossentropy loss, etc.

Putting together the encoder and the decoder, it is tempting to think of a network in which we feed something into the input layer and get back the same thing out of the output layer. The challenge here is that the low-dimensional vector $\mathbf{h}$ serves as a bottleneck in information flow. There is a risk of information loss in transformation either from $\mathbf{x}$ to $\mathbf{h}$ or from $\mathbf{h}$ to $\widetilde{\mathbf{x}}$, making it difficult to "copy" the input to the output. Rather, we need to "squeeze" an object from a high-dimensional space to a dense, low-dimensional space, and then "unsqueeze" it from the new space to the original high-dimensional space. A consequence of this squeeze-and-unsqueeze process is that the encoder is forced to compress the data but retain the information as much as possible. So, the auto-encoder discussed here is also called the **undercomplete auto-encoder**, because $\mathbf{h}$ has a smaller size than $\mathbf{x}$ and $\widetilde{\mathbf{x}}$. Figure 2.16 shows an illustration of the undercomplete auto-encoder structure.

Given the loss function $L(\cdot)$, the encoder $\mathrm{Enc}_\theta(\cdot)$ and the decoder $\mathrm{Dec}_\omega(\cdot)$, the parameters $\hat{\theta}$ and $\hat{\omega}$ can be optimized by using the gradient descent method as in supervised learning (see Section 2.4.1). When applying the auto-encoder, one can simply drop the decoder and use the encoder as a feature extractor, that is, given a new input $\mathbf{x}_{\mathrm{new}}$, we generate a new representation

$$\hat{\mathbf{h}}_{\mathrm{new}} = \mathrm{Enc}_{\hat{\theta}}(\mathbf{x}_{\mathrm{new}}) \tag{2.96}$$

Note that the encoder is not a standalone system but typically works with other models for a complete working system. For example, we can train an auto-encoder on some sentences

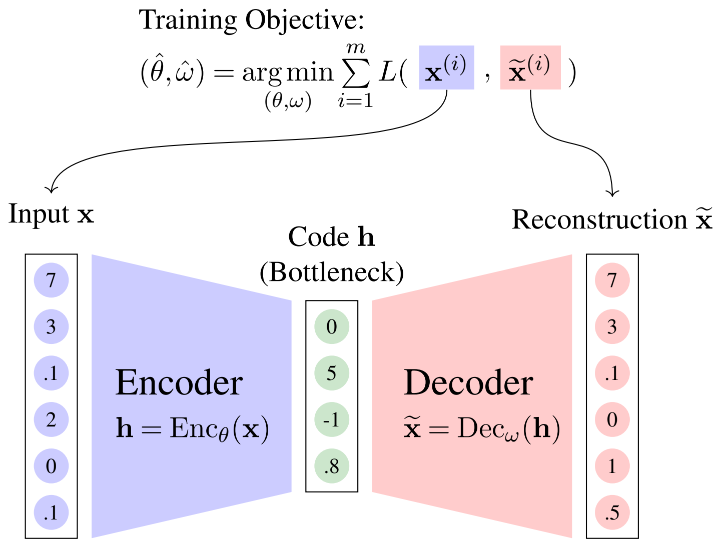


Figure 2.16: An undercomplete auto-encoder with an encoder, a decoder and a bottleneck layer sandwiched between them. An input $\mathbf{x}$ (left) is transformed into a code $\mathbf{h}$ (middle) and then a reconstruction $\widetilde{\mathbf{x}}$ (right). The parameters of both the encoder and decoder are optimized by minimizing the discrepancy between the input $\mathbf{x}$ and the reconstruction $\widetilde{\mathbf{x}}$ on a number of unlabeled samples $\{\mathbf{x}^{(1)},...,\mathbf{x}^{(m)}\}$. On new samples, we throw away the decoder, and use the encoder to generate new codes or representations.

and place a Softmax layer on the output of the encoder to build a sentence classifier. The classifier can be further trained on some task-specific data to solve a new problem, such as tagging a sentence with its sentiment polarity. This also makes the application of auto-encoder fall under the general paradigm of pre-training: a sub-model (i.e., an encoder) is first trained on large-scale, task-agnostic data, and then used as a component of a bigger model on a downstream task.

2.6.1 Auto-encoders with Explicit Regularizers

As more complex neural networks are involved, an auto-encoder tends to learn an identity transformation although the bottleneck makes it a bit harder to pass through without information loss. This is what we would ordinarily expect: we could make $\mathbf{h}$ a surrogate of $\mathbf{x}$ and decode $\mathbf{h}$ to something very similar to $\mathbf{x}$. On the other hand, learning an exact identity transformation requires a highly complicated model and is heavily prone to overfitting. Fortunately, as with other machine learning models, we can regularize the training by using the methods presented in Section 2.5. One of the most popular methods is adding an explicit regularization term to the loss function. Taking together Eq. (2.74) and Eq. (2.95), we can define the training objective to be:

$$(\hat{\theta},\hat{\omega}) = \arg\min_{(\theta,\omega)} \sum_{i=1}^{m} L\left(\mathbf{x}^{(i)}, \mathrm{Dec}_\omega(\mathrm{Enc}_\theta(\mathbf{x}^{(i)}))\right) + \alpha \cdot R \tag{2.97}$$

where R is the regularization term accounting for some prior knowledge we want to impose on the training, and α is its weighting coefficient. A common choice for R in auto-encoders is a **sparsity penalty** (resulting in what is known as a **sparse auto-encoder**). The simplest way to implement such a penalty is to apply the l_1 or l_2 norm on the code, as follows:

$$R_{l_1} = \sum_{i=1}^{m}\sum_{k}\left|h_k^{(i)}\right| \tag{2.98}$$

$$R_{l_2} = \sum_{i=1}^{m}\sqrt{\sum_{k}\left(h_k^{(i)}\right)^2} \tag{2.99}$$

It is worth noting that, unlike those penalties on model parameters (see Section 2.5.1), the sparsity penalty regularizes the code $\mathbf{h}$ (or the output of the encoder) to be sparse. The idea of encouraging sparseness in representations stems from **sparse coding** [Olshausen and Field, 1997]. It states that the information of an object is embedded in complex dependencies among the original attributes (or features) of the object. A desirable representation learning system should extract such dependencies and reform them to be a set of independent features. And there should be a small number of these independent features that are active, while the active features vary when we switch to a new object. Note also that, from a Bayesian learning point of view, other penalties in regularized training could be interpreted as priors over models. The sparsity penalty, however, is not a prior because it does not depend on models (or model parameters) but on the training data [Goodfellow et al., 2016]. In this view, the sparsity penalty should not be treated as a "regularization" term, but simply some distribution over the model's intermediate states. On the other hand, the sparseness of the code, though not well explained by conventional use of regularization terms, is indeed helpful in many applications of auto-encoders, because it directly models the way of representing the input and imposing "priors" on outcomes of encoders. When considered from an empirical point of view, the sparsity penalty is still thought of as a regularizer that biases the training to certain models.

There are other choices for defining the regularization term R in addition to Eqs. (2.98-2.99). For example, a way of forcing sparsity is to penalize the cases where the average value of each entry h_k is far away from a predefined value [Nair and Hinton, 2009]. When h_k chooses values from $[0, 1]$, the regularization can be implemented by defining R as the KL divergence between the average code over a number of samples and the expected code[12]. Let $\bar{\mathbf{h}}$ denote the average code over $\{\mathbf{x}^{(1)}, ..., \mathbf{x}^{(m)}\}$, where the value of $\bar{h}_k$ is the mean of the k-th variable of the code:

$$\bar{h}_k = \frac{1}{m} \cdot \sum_{i=1}^{m} \mathrm{Enc}(\mathbf{x}^{(i)})(k) \tag{2.100}$$

Also, let $\mathbf{q}$ be the expected code, where $q_k = \tau$ for any k. If each entry of the average code is viewed as a Bernoulli random variable with mean $\bar{h}_k$, and each entry of the expected code is

[12] In general, we can set all entries of the expected code to $\tau \in [0, 1]$. Sparse codes will be preferred if τ is close to 0, as features are "inactive" in most cases. By contrast, dense codes will be preferred if τ is close to 1.

viewed as another Bernoulli random variable with mean τ, then the regularization term can be defined as the sum of the KL divergence between $\bar{h}_k$ and q_k over all entries:

$$R = \sum_k D_{\mathrm{KL}}(q_k \| \bar{h}_k) = \sum_k \tau \cdot \log \frac{\tau}{\bar{h}_k} + (1-\tau) \cdot \log \frac{1-\tau}{1-\bar{h}_k} \tag{2.101}$$

In this form, R penalizes the model when $\bar{\mathbf{h}}$ deviates from $\mathbf{q}$.

As another auto-encoder variant, the **contractive auto-encoder** (**CAE**) tries to improve the robustness of representation by introducing a new regularization term into training [Rifai et al., 2011]:

$$R = \sum_{i=1}^{m} \left\| \frac{\partial \mathbf{h}^{(i)}}{\partial \mathbf{x}^{(i)}} \right\|_F^2 = \sum_{i=1}^{m} \left\| \frac{\partial \mathrm{Enc}(\mathbf{x}^{(i)})}{\partial \mathbf{x}^{(i)}} \right\|_F^2 \tag{2.102}$$

where $\frac{\partial \mathrm{Enc}(\mathbf{x}^{(i)})}{\partial \mathbf{x}^{(i)}}$ (or $\frac{\partial \mathbf{h}^{(i)}}{\partial \mathbf{x}^{(i)}}$) is the **Jacobian matrix** of the representation [13], and $\|\cdot\|_F$ is the **Frobenius norm** of a matrix[14]. The contractive penalty helps resist the influence of small perturbations to the input. In the geometric sense, it encourages that the neighborhood relationship holds for output data points if the input data points are close neighbors, in other words, it forces $\mathrm{Enc}(\cdot)$ to behave more like a **contraction mapping**[15], hence the name of contractive auto-encoder.

[13] Suppose that the encoder is a function $\mathrm{Enc}(\cdot)$: $\mathbf{x} \in \mathbb{R}^{d_x} \to \mathbf{h} \in \mathbb{R}^{d_h}$. The Jacobian matrix of $\mathbf{h} = \mathrm{Enc}(\mathbf{x})$ is a $d_h \times d_x$ matrix:

$$\begin{aligned} \text{Jacobian} &= \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \\ &= \begin{bmatrix} \frac{\partial \mathbf{h}}{\partial x_1} & \cdots & \frac{\partial \mathbf{h}}{\partial x_{d_x}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_{d_x}} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_{d_h}}{\partial x_1} & \cdots & \frac{\partial h_{d_h}}{\partial x_{d_x}} \end{bmatrix} \end{aligned} \tag{2.103}$$

[14] For a matrix $\mathbf{A} \in \mathbb{R}^{d_x \times d_h}$, the Frobenius norm is essentially the l_2 norm of its flattened vector representation, given by the equation:

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2} \tag{2.104}$$

Note that in strict linear algebra, the l_2 norm of a matrix refers to its maximum singular value (the spectral norm), which is why the term Frobenius norm is explicitly used here to describe the element-wise magnitude.

[15] Let X be a metric space with a metric d. Given a function $f(\cdot)$ from X to X, $f(\cdot)$ is a *contraction mapping* if there exists a constant $\epsilon \in [0,1)$ such that for any $x_1, x_2 \in X$:

$$d(f(x_1), f(x_2)) \le \epsilon \cdot d(x_1, x_2) \tag{2.105}$$

2.6.2 Denoising Auto-encoders

Another source of inspiration for improving the robustness of a model arises from the **denoising** idea: we add noise to the input and then remove it to recover the clean data. **Denoising auto-encoders** (**DAEs**) are a class of neural networks that combine the principles of auto-encoding and denoising. First, noise is added to the input vector in a stochastic manner. This can be described as a process of generating a noisy input $\mathbf{x}_{\text{noise}} \in \mathbb{R}^{d_x}$ given the original input $\mathbf{x} \in \mathbb{R}^{d_x}$:

$$\mathbf{x}_{\text{noise}} \sim \Pr_{\text{noise}}(\mathbf{x}_{\text{noise}}|\mathbf{x}) \tag{2.106}$$

where $\Pr_{\text{noise}}(\cdot|\mathbf{x})$ is a conditional distribution for sampling $\mathbf{x}_{\text{noise}}$. For example, we can follow the method presented in Section 2.5.5 and draw the noisy input from a multivariate Gaussian distribution:

$$\mathbf{x}_{\text{noise}} \sim \mathcal{N}(\mathbf{x}, \sigma^2\mathbf{I}) \tag{2.107}$$

where $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a Gaussian distribution with mean vector $\boldsymbol{\mu} \in \mathbb{R}^{d_x}$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{d_x \times d_x}$. Equivalently, this additive noise can be explicitly formulated by isolating the zero-mean noise term:

$$\mathbf{x}_{\text{noise}} = \mathbf{x} + \mathbf{g} \tag{2.108}$$

$$\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}) \tag{2.109}$$

where $\sigma^2\mathbf{I}$ is a diagonal covariance matrix, indicating that independent Gaussian noise with variance σ^2 is applied to each dimension of the input, and $\mathbf{I}$ is the $d_x \times d_x$ identity matrix.

Sometimes, this process is called the **corruption** of the input, and $\mathbf{x}_{\text{noise}}$ is called the **corrupted input**. Aside from additive Gaussian noise, there are a few different ways to corrupt the input [Vincent et al., 2010]. One of the popular methods is to zero some of the entries of $\mathbf{x}$. For example, we can set each entry to 0 with a pre-defined probability. This is also called **masking noise**. Another method is to use **salt-and-pepper noise** or **impulse noise** for corruption. It randomly chooses some of the entries, and sets each of them to a minimum or maximum value with a pre-defined probability. Different types of noise are applied to different applications of auto-encoders. For example, the masking noise is popular in training language models, and the salt-and-pepper noise is more commonly used in image processing.

Then, the corrupted input $\mathbf{x}_{\text{noise}}$ is fed into an encoder-decoder network, and the network produces a reconstructed input $\tilde{\mathbf{x}} = \text{Dec}(\text{Enc}(\mathbf{x}_{\text{noise}}))$. The training process is regular. We reuse Eq. (2.95) to minimize the loss of replacing $\mathbf{x}$ with $\tilde{\mathbf{x}}$. Thus, we can rewrite Eq. (2.95) to adapt the objective to the denoising case:

$$(\hat{\theta}, \hat{\omega}) = \underset{(\theta,\omega)}{\arg\min} \sum_{i=1}^{m} L\left(\mathbf{x}^{(i)}, \text{Dec}_{\omega}(\text{Enc}_{\theta}(\mathbf{x}_{\text{noise}}^{(i)}))\right) \tag{2.110}$$

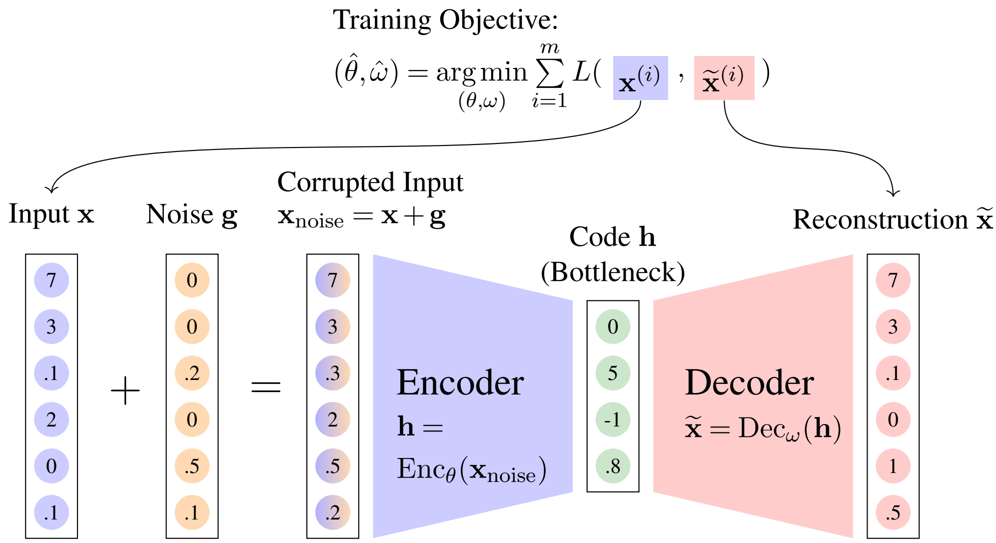


Figure 2.17: The structure of a denoising auto-encoder. An input $\mathbf{x}$ is first corrupted into a noisy or corrupted input $\mathbf{x}_{\text{noise}}$. Then, it is passed through an encoder to form a code $\mathbf{h}$. Then, the code is passed through a decoder to form a reconstructed input $\widetilde{\mathbf{x}}$. The training is performed by minimizing the loss between $\mathbf{x}$ and $\widetilde{\mathbf{x}}$. This process is termed "denoising" because it tries to remove the noise from $\mathbf{x}_{\text{noise}}$ and recover the original input $\mathbf{x}$.

Eq. (2.110) differs from Eq. (2.95) only in that the input of the auto-encoder is $\mathbf{x}_{\text{noise}}$ instead of $\mathbf{x}$. In other words, we denoise the corrupted input to recover the original input. See Figure 2.17 for the structure of denoising auto-encoders.

Note that both contractive/sparse auto-encoders and denoising auto-encoders can be thought of as ways to improve the robustness of auto-encoders. Their difference lies in that they regularize the training at different points of the model. Contractive auto-encoders aim at improving the robustness of encoding, that is, the representation is trained to be not so sensitive to small perturbations to the input. Denoising auto-encoders, on the other hand, aim at improving the robustness of reconstruction. They affect both encoders and decoders simultaneously. In some sense, denoising auto-encoders are direct applications of noisy training to auto-encoders (see Section 2.5.5). It is of course difficult to say which models are better. For example, contractive auto-encoders have more direct guidance on learning the representation, which is often our primary objective in feature learning. The training of denoising auto-encoders, although it has an indirect effect on encoding, receives additional denoising signals from the decoder. This offers a new view of robust training: a robust representation can be learned at both its point of generation (the denoising encoder) and its point of application (the denoising decoder).

2.6.3 Variational Auto-encoders

Despite the name, **variational auto-encoders** (**VAEs**) were not originally proposed to solve standard representation encoding problems. They are fundamentally designed to generate new data similar to observed data, hence possessing very different mathematical formulations from the classical auto-encoders discussed above. In statistics and machine learning, variational auto-encoders are more often viewed as instances of variational Bayesian methods and used to perform efficient statistical inference over latent variables when the posterior probabilities of these variables are intractable [Kingma and Welling, 2014; 2019]. On the other hand, variational auto-encoders, implicitly or explicitly, deal with what we do in inducing the underlying representation of an observed object. We therefore include them in this section for a relatively complete discussion.

We begin with a generative story describing how each data point is generated. Suppose that, for an observed sample $\mathbf{x}$ in our dataset, there is an unobserved latent variable $\mathbf{h}$ that describes $\mathbf{x}$. Now we intend to develop a probabilistic model to describe the generation process of $\mathbf{x}$, specifically, estimating the probability $\Pr(\mathbf{x})$. This can be computed as the marginal distribution:

$$\Pr(\mathbf{x}) = \int \Pr(\mathbf{x}, \mathbf{h}) d\mathbf{h} \tag{2.111}$$

where we explicitly introduce the latent variable $\mathbf{h}$ into the inference of $\mathbf{x}$. To solve Eq. (2.111), we use a model $p_\omega(\mathbf{x}, \mathbf{h})$ to approximate $\Pr(\mathbf{x}, \mathbf{h})$ (i.e., $p_\omega(\mathbf{x}, \mathbf{h}) \approx \Pr(\mathbf{x}, \mathbf{h})$), and we have

$$p_\omega(\mathbf{x}) = \int p_\omega(\mathbf{x}, \mathbf{h}) d\mathbf{h} \tag{2.112}$$

where $p_\omega(\mathbf{x}, \mathbf{h})$ is a probability density function parameterized by ω. We replace the left-hand side of Eq. (2.112) with $p_\omega(\mathbf{x})$ to emphasize that the probability is determined by the model $p_\omega(\cdot)$. There are generally many ways to define $p_\omega(\mathbf{x}, \mathbf{h})$. Here we can simply think of it as a neural network.

Then, we can rewrite Eq. (2.112) by using the chain rule:

$$p_\omega(\mathbf{x}) = \int p_\omega(\mathbf{h}) \cdot p_\omega(\mathbf{x}|\mathbf{h}) d\mathbf{h} \tag{2.113}$$

where $p_\omega(\mathbf{h})$ is the prior over $\mathbf{h}$, e.g., a Gaussian prior. The conditional probability $p_\omega(\mathbf{x}|\mathbf{h})$ describes how likely $\mathbf{x}$ is observed given the latent variable $\mathbf{h}$. To model this generation process, $p_\omega(\mathbf{x}|\mathbf{h})$ is often assumed to be a Gaussian distribution that is parameterized with its mean $\boldsymbol{\mu}_p$ and variance σ_p:

$$p_\omega(\mathbf{x}|\mathbf{h}) = \mathcal{N}(\boldsymbol{\mu}_p, \sigma_p^2 \mathbf{I}) \tag{2.114}$$

where $\boldsymbol{\mu}_p \in \mathbb{R}^{d_x}$ and $\sigma_p \in \mathbb{R}$ are determined by a decoding network $\mathrm{Dec}_\omega(\cdot)$ (we will explain later on why it is called "decoding"):

$$(\boldsymbol{\mu}_p, \sigma_p) = \mathrm{Dec}_\omega(\mathbf{h}) \tag{2.115}$$

However, the marginal likelihood in Eq. (2.112) remains intractable. Although we can easily evaluate the prior $p_\omega(\mathbf{h})$ and the conditional $p_\omega(\mathbf{x}|\mathbf{h})$, performing the exact integration over the entire continuous latent space of $\mathbf{h}$ is computationally prohibitive. This also leads to an intractable posterior:

$$p_\omega(\mathbf{h}|\mathbf{x}) = \frac{p_\omega(\mathbf{h}) \cdot p_\omega(\mathbf{x}|\mathbf{h})}{p_\omega(\mathbf{x})} \tag{2.116}$$

It looks like we are stuck with $p_\omega(\mathbf{x})$ and $p_\omega(\mathbf{h}|\mathbf{x})$! Variational auto-encoders address this issue by approximating $p_\omega(\mathbf{h}|\mathbf{x})$ with a tractable posterior $q_\theta(\mathbf{h}|\mathbf{x})$:

$$q_\theta(\mathbf{h}|\mathbf{x}) \approx p_\omega(\mathbf{h}|\mathbf{x}) \tag{2.117}$$

where θ is the parameter of the new model. Like Eqs. (2.114-2.115), $q_\theta(\mathbf{h}|\mathbf{x})$ is defined as another Gaussian distribution:

$$q_\theta(\mathbf{h}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_q, \sigma_q^2 \mathbf{I}) \tag{2.118}$$

$$(\boldsymbol{\mu}_q, \sigma_q) = \mathrm{Enc}_\theta(\mathbf{x}) \tag{2.119}$$

where $\mathrm{Enc}_\theta(\cdot)$ is the encoding network that reads $\mathbf{x}$ and generates the mean and variance of the distribution $q_\theta(\mathbf{h}|\mathbf{x})$. This is interesting! We now have a feasible path to compute $\Pr(\mathbf{x})$: we can sample a latent variable $\mathbf{h}$ from the variational posterior $q_\theta(\mathbf{h}|\mathbf{x})$, and then evaluate the joint probability $p_\omega(\mathbf{x},\mathbf{h})$. Mathematically, we can reformulate the log-likelihood of the observation as an expectation over the variational posterior:

$$\begin{aligned}
& \log p_\omega(\mathbf{x}) \\
= \; & \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log p_\omega(\mathbf{x})\right] \\
= \; & \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log \frac{p_\omega(\mathbf{h}) \cdot p_\omega(\mathbf{x}|\mathbf{h})}{p_\omega(\mathbf{h}|\mathbf{x})}\right] \\
= \; & \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log \frac{p_\omega(\mathbf{h}) \cdot p_\omega(\mathbf{x}|\mathbf{h})}{p_\omega(\mathbf{h}|\mathbf{x})} \cdot \frac{q_\theta(\mathbf{h}|\mathbf{x})}{q_\theta(\mathbf{h}|\mathbf{x})}\right] \\
= \; & \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log \frac{q_\theta(\mathbf{h}|\mathbf{x})}{p_\omega(\mathbf{h}|\mathbf{x})}\right] + \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log \frac{p_\omega(\mathbf{h}) \cdot p_\omega(\mathbf{x}|\mathbf{h})}{q_\theta(\mathbf{h}|\mathbf{x})}\right]
\end{aligned} \tag{2.120}$$

The first term on the right-hand side of Eq. (2.120) is precisely the KL divergence (or relative entropy) between the approximate posterior $q_\theta(\mathbf{h}|\mathbf{x})$ and the true posterior $p_\omega(\mathbf{h}|\mathbf{x})$:

$$D_{\mathrm{KL}}(q_\theta(\mathbf{h}|\mathbf{x}) \,\|\, p_\omega(\mathbf{h}|\mathbf{x})) = \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log \frac{q_\theta(\mathbf{h}|\mathbf{x})}{p_\omega(\mathbf{h}|\mathbf{x})}\right] \tag{2.121}$$

Since the KL divergence is strictly non-negative ($D_{\mathrm{KL}}(q \,\|\, p) \geq 0$, and equals zero if and only if $q = p$), we can establish a strict lower bound on the log-likelihood:

$$\begin{aligned}\log p_\omega(\mathbf{x}) &\geq \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log \frac{p_\omega(\mathbf{h})\cdot p_\omega(\mathbf{x}|\mathbf{h})}{q_\theta(\mathbf{h}|\mathbf{x})}\right] \quad (2.122)\\ &= \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log p_\omega(\mathbf{x}|\mathbf{h}) + \log \frac{p_\omega(\mathbf{h})}{q_\theta(\mathbf{h}|\mathbf{x})}\right] \quad (2.123)\\ &= \mathbb{E}_{\mathbf{h}\sim q_\theta(\mathbf{h}|\mathbf{x})}\left[\log p_\omega(\mathbf{x}|\mathbf{h})\right] - D_{\mathrm{KL}}(q_\theta(\mathbf{h}|\mathbf{x}) \,\|\, p_\omega(\mathbf{h})) \quad (2.124)\end{aligned}$$

The right-hand side of Eq. (2.124) constitutes the foundational **evidence lower bound (ELBO)** of the data likelihood. The first term of the ELBO (the expected reconstruction log-likelihood) can be approximated using Monte Carlo sampling of $\mathbf{h}$. The second term is computationally straightforward because an analytical solution exists for the KL divergence between two Gaussian distributions. Let $L(\mathbf{x}, \theta, \omega)$ denote the negative ELBO. The training process of a variational auto-encoder is then framed as minimizing $L(\cdot)$ over a dataset of observed samples $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(m)}\}$:

$$(\hat{\theta}, \hat{\omega}) = \underset{(\theta,\omega)}{\arg\min} \sum_{i=1}^{m} L\left(\mathbf{x}^{(i)}, \theta, \omega\right) \quad (2.125)$$

Note that because sampling $\mathbf{h}$ from $q_\theta(\mathbf{h} \mid \mathbf{x})$ is a stochastic, non-differentiable operation, its gradient cannot be computed directly. To fit the training of auto-encoders in standard back-propagation, a common way is to use the so-called **reparameterization trick**. Here we skip the details and refer the reader to a few papers for more information [Kingma and Welling, 2014; Doersch, 2016].

Figures 2.18 illustrates how a variational auto-encoder works. It presents a two-step generation process:

- **Encoding**. For an input sample $\mathbf{x}$, we sample a latent variable $\mathbf{h}$ from $q_\theta(\mathbf{h}|\mathbf{x})$. This involves an encoding network $\mathrm{Enc}_\theta(\mathbf{x})$ that generates the mean $\boldsymbol{\mu}_q$ and variance σ_q^2 of the Gaussian distribution $q_\theta(\mathbf{h}|\mathbf{x})$. The latent variable is then generated by sampling from $\mathcal{N}(\boldsymbol{\mu}_q, \sigma_q^2\mathbf{I})$.
- **Decoding**. For the latent variable $\mathbf{h}$, we sample the original input $\mathbf{x}$ from $p_\omega(\mathbf{x}|\mathbf{h})$. It follows again a Gaussian sampling process: a decoding network $\mathrm{Dec}_\omega(\mathbf{h})$ is used to determine the mean $\boldsymbol{\mu}_p$ and variance σ_p^2 of the distribution. $\mathbf{x}$ is generated by following $\mathcal{N}(\boldsymbol{\mu}_p, \sigma_p^2\mathbf{I})$.

Sometimes, $q_\theta(\mathbf{h}|\mathbf{x})$ and $p_\omega(\mathbf{x}|\mathbf{h})$ are called an "encoder" and a "decoder", as they try to "map" an input to a representation and then "map" it back to the input. However, $q_\theta(\mathbf{h}|\mathbf{x})$ and $p_\omega(\mathbf{x}|\mathbf{h})$ themselves imply some non-deterministic models, that is, they output the probability density functions of the variables rather than point estimates. An important consequence of this result is that variational auto-encoders do not tend to find the "best" representation for the input. At first glance, this might appear counterintuitive, as conventional models typically yield fixed-value representations. This, however, is the case of the Bayesian inference — we only learn a distribution over possible values of a latent variable. From a practical perspective, if a

$$\text{Training Objective: } (\hat{\theta},\hat{\omega}) = \arg\min_{(\theta,\omega)} \sum_{i=1}^{m} L(\mathbf{x}^{(i)},\theta,\omega)$$

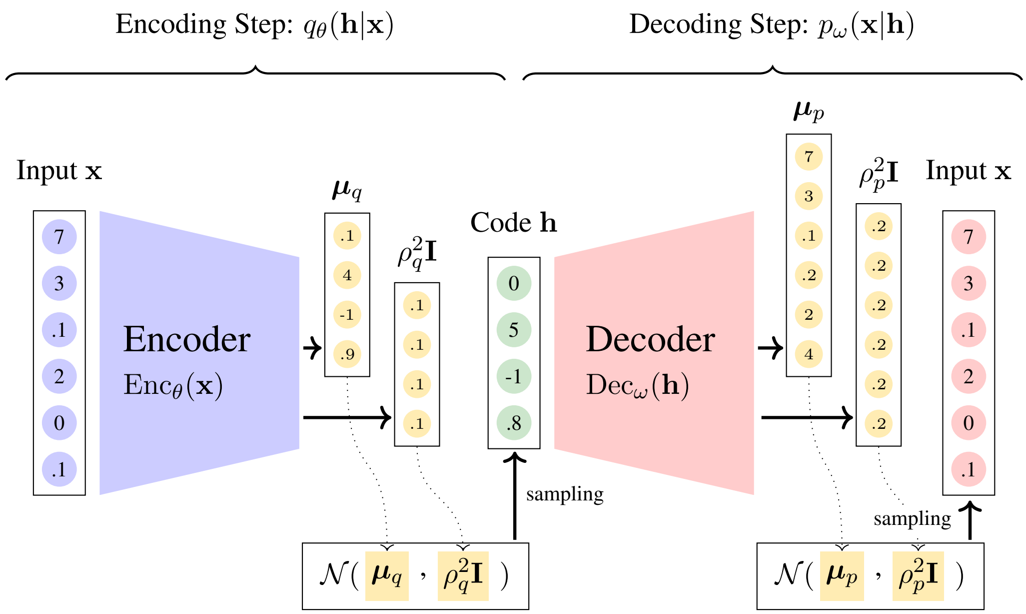


Figure 2.18: The generative story of a variational auto-encoder. For an input sample $\mathbf{x}$, we generate a latent variable $\mathbf{h}$ by using an encoder $q_\theta(\mathbf{h}|\mathbf{x})$. In the encoding step, a neural network $\mathrm{Enc}_\theta(\cdot)$ is first used to produce the mean and variance of a Gaussian distribution, i.e., $\boldsymbol{\mu}_q$ and σ_q^2. The latent variable $\mathbf{h}$ is then drawn according to $\mathcal{N}(\boldsymbol{\mu}_q,\sigma_q^2\mathbf{I})$. After that, we regenerate the original sample $\mathbf{x}$ by using a decoder $p_\omega(\mathbf{x}|\mathbf{h})$. In the decoding step, like the generation process in the encoder, a neural network $\mathrm{Dec}_\omega(\cdot)$ is used to generate the mean $\boldsymbol{\mu}_p$ and variance σ_p^2 of $\mathcal{N}(\boldsymbol{\mu}_p,\sigma_p^2\mathbf{I})$. The same input $\mathbf{x}$ is generated by sampling from $\mathcal{N}(\boldsymbol{\mu}_p,\sigma_p^2\mathbf{I})$.

deterministic representation is required for downstream tasks, one can simply use the expected value (i.e., the mean μ_q) of the inferred distribution, which reliably captures the essential features of the input [Knight, 2009].

In practice, the main use of variational auto-encoders is in generation but not representation. At test time, provided the optimized parameters $\hat{\theta}$ and $\hat{\omega}$, the encoder (i.e., $q_{\hat{\theta}}(\mathbf{h}|\mathbf{x})$) is removed, and the decoder (i.e., $p_{\hat{\omega}}(\mathbf{x}|\mathbf{h})$) works with randomly generated $\mathbf{h}$'s. More precisely, we sample a latent variable $\mathbf{h}_{\text{new}}$ from a Gaussian distribution, and infer a sample $\mathbf{x}_{\text{new}}$ by $p_{\hat{\omega}}(\mathbf{x}|\mathbf{h}_{\text{new}})$ as usual. We will see in the subsequent chapters that many NLP problems can be categorized as generation problems where sequential or hierarchical data objects are generated on the condition of some given data objects or latent variables.

2.7 Summary

In this chapter, we have discussed what a neural network is, as well as a few basic architectures that are commonly used as building blocks for developing powerful deep learning systems. Furthermore, we have examined how to train neural networks, regularize the training process, and apply neural networks to feature learning in an unsupervised manner.

However, neural networks and deep learning are wide-ranging topics, and our discussions merely scratch the surface. For a more comprehensive introduction, Goodfellow et al. [2016]'s book is an excellent resource. It also covers several advanced techniques, such as deep structured models and randomized methods, for developing state-of-the-art systems. As always, there is a significant difference between understanding a technique theoretically and being proficient in applying it to solve real-world problems. Therefore, for practitioners who wish to apply neural networks and deep learning in various scenarios, there are many books detailing the implementation of deep learning systems [Géron, 2019; Zhang et al., 2021; Chollet, 2021], along with open-source projects that provide reference codebases[16].

In the following chapters, we will focus on how to use neural models to address NLP problems. Along the way, we will explore how to learn the representations of words and sentences using the methods discussed so far (Chapters 3-4), and how to model various NLP problems using several advanced neural network-based methods. These include the attention mechanism and Transformers (Chapters 5-6), pre-training (Chapter 7), and large language models (Chapters 8-11).

[16]URLs to a few popular online tutorials: `https://pytorch.org/tutorials`, `https://keras.io/examples/nlp`, and `https://www.tensorflow.org/tutorials`

II Basic Models

Chapter 3

Words and Word Vectors

Words are the basic units of language [Jackendoff, 1992]. Most natural language systems used for expression and communication involve the creation, composition, and combination of words. Before exploring how words form larger linguistic units, it is worth first understanding what a word is. This involves two fundamental questions:

- What is the surface form of a word?
- What is the meaning of a word?

However, answering these questions is challenging because no simple rules can fully describe how words are formed or how their meanings are defined and induced. While various linguistic theories address these questions, NLP researchers focus more on two practical issues:

- **Tokenization**: Given a string, how to segment it into a sequence of words (also called **tokens**) such that these units can be used in downstream NLP tasks?
- **Word Representation Learning**: Given a corpus, how to represent each word in a computable form, and how can NLP models perform computation over this representation?

One goal of this chapter is to demonstrate how sentences are segmented using either linguistic or statistical methods. Specifically, we describe several approaches to tokenizing a string of characters into words or subwords using heuristic rules or statistical models learned from data. Another goal is to show how words can be represented as real-valued vectors. We present modern approaches to learning and evaluating these word vectors. The focus of this section is not on the derivation of formulas and models, but on the core concept of word vector representation, which serves as the foundation for many NLP systems. In the following chapters, we will see the generalization of this idea to modeling larger language units, such as sentences.

3.1 Tokenization

In computer science and related fields, the term *token* can be used in many different ways. Here we simply think of a token as a word in linguistics, although it can be something different (see Section 3.1.4). In NLP, tokenization or segmentation is a task related to **morphological**

Chinese	Input:	一直以来，完美的机器翻译是人类的梦想之一。
	Output:	一直/以来/，/完美/的/机器翻译/是/人类/的/梦想/之一/。
Japanese	Input:	西日本や海はく晴れて、汗ばむ暑さとなる。
	Output:	西日本/や/海/は/く/晴れて/、/汗ばむ/暑さ/と/なる/。
English	Input:	She said, “Deep learning is not the solution to all world’s problems”.
	Output:	She/said/,/“/Deep/learning/is/not/the/solution/to/all/world/ ’s/problems/”/.

Figure 3.1: Tokenization for different languages (slash = word boundary). For Chinese and Japanese where there are no delimiters between words, tokenization is often called **word segmentation**.

analysis [Aronoff and Fudeman, 2011]. While morphological analyzers or parsers are generally used to study the internal structure of words, tokenization is concerned with how sentences are broken down into words. It appears that we need to know how words are composed if we want to know how sentences are formed by words. This complexity is further compounded by linguistic diversity, which makes it difficult to establish a universal system for describing morphology. For example, analytic languages (such as Chinese) have little inflection, and rely on word order to convey meaning. By contrast, synthetic languages (such as French) may have rich inflection and the meaning of a word is highly influenced by morphology.

On the other hand, dividing sentences into smaller linguistic units is important in many NLP tasks, even though many of the world’s languages have little morphology. For example, Chinese is a morphologically simple language that has no explicit word boundaries. While it also makes sense to take characters as units in understanding the meaning of Chinese text, it is more desirable and reasonable to consider larger units in processing the text. Note that, even for languages having delimiters between words, such as English, we still have to tokenize sentences such that they are standardized when serving as the input and/or output of an NLP system.

In this section, we skip the discussion on what exactly a word is in morphology and syntax, but simply view tokenization as a task of adding word or token boundaries to a given string (see Figure 3.1). We will show that a sentence can be broken down into words or tokens in either a heuristic or statistical manner. Note that this process is designed to produce some units that can ease the processing of languages in NLP systems, not necessarily to make strict linguistic sense.

3.1.1 Tokenization via Rules and Heuristics

A common and simple approach to tokenization is to identify every word in a sentence by applying a set of pre-defined rules. In general, these rules are linguistically motivated and reflect our prior knowledge of what the form of a word should be. For example, consider the English example in Figure 3.1. We can define the following rules for tokenizing the sentence:

- Words do not contain spaces. Therefore, we can split the sentence into “word candidates” using spaces.

- Every word candidate that is made up of English letters only (i.e., *a-z* and *A-Z*) is a word.
- Every punctuation mark (e.g., quote, comma, period) should be isolated to form a word.
- *'s* is treated as a distinct token, indicating a possessive noun.

This is perhaps the minimal rule set we can use in English tokenization. Surely, more rules can be added to cover more linguistic phenomena, e.g., words with dashes, words containing non-English letters, and so on. However, there are no standards to define such a set of rules. In practice, and particularly in NLP applications, we want a minimal set of rules to deal with most problems, and the tokenization is usually implemented by a number of **regular expressions**. Here we will not discuss these rules and regular expressions in detail, but refer the reader to a few textbooks for further reading [Lawson, 2003; Friedl, 2006; Jurafsky and Martin, 2008][1].

Also, it is common to normalize the text before tokenization so that the input to the tokenizer is canonical. For example, for English and other alphabetic languages, **normalization** or **canonicalization** refers to a process of lowercasing words, normalizing character representation (e.g., Unicode characters), and so on. In addition, we can map different forms of a word to the same form for further generalization of the tokenization. A simple way to do this is to conflate all inflected forms of a word into its base form. In linguistics, the base form of a word is called **lemma**, and the process of mapping words to lemmas is called **lemmatization**. Here are some examples of lemmatization.

$$\begin{aligned} \mathit{learn} &\rightarrow \mathit{learn} \\ \mathit{learning} &\rightarrow \mathit{learn} \\ \mathit{learns} &\rightarrow \mathit{learn} \\ \mathit{best} &\rightarrow \mathit{good} \end{aligned}$$

There are words that correspond to two or more different lemmas (often with different part-of-speech). In this case, we should select the correct lemma according to the context. In other words, lemmatization is context-dependent.

Closely related to lemmatization is **stemming**, which represents a word as its **stem**. Like lemmas, a stem is some base form of a word. However, unlike lemmas, a stem is not necessarily a valid word, although there are many words whose lemmas and stems are identical. Another difference from lemmatization is that stemming is performed on individual words, without the need for context for disambiguation. So, stemming is context-independent. There are several efficient algorithms for stemming. A popular one is **suffix stripping** [Porter, 1980]. It simply removes the suffixes *ing*, *ed*, *ion*, etc., as shown below:

[1] Tokenization scripts can be found in many open-source projects, such as Moses [Koehn et al., 2007] (`https://github.com/moses-smt/mosesdecoder/blob/master/scripts/tokenizer/tokenizer.perl`) and the tokenizers in SacreBLEU (`https://github.com/mjpost/sacrebleu/tree/master/sacrebleu/tokenizers`).

Original	She said, “Deep learning is not the solution to all world’s problems”.
Normalization	she said, “deep learning is not the solution to all world’s problems”.
Tokenization	she/said/,/“/deep/learning/is/not/the/solution/to/all/world/’s/problems/”/.
Lemmatization	she/say/,/“/deep/learning/be/not/the/solution/to/all/world/’s/problem/”/.
Stemming	she/said/,/“/deep/learn/is/not/the/solut/to/all/world/’s/problem/”/.

Figure 3.2: Normalization, lemmatization, and stemming of an English sentence. In normalization, the whole sentence is lowercased. In lemmatization, every word is lemmatized and rewritten as its lemma. In stemming, the suffixes of some words are removed.

$$\begin{aligned} \textit{remove} &\rightarrow \textit{remov} \\ \textit{removing} &\rightarrow \textit{remov} \\ \textit{removal} &\rightarrow \textit{remov} \\ \textit{best} &\rightarrow \textit{best} \end{aligned}$$

For more examples, Figure 3.2 shows normalization, lemmatization, and stemming results for an English sentence.

It is worth noting that the above methods are typically implemented using regular expressions, dictionary lookups, and additional heuristics. While this brief overview might suggest that tokenization is straightforward, much more work is needed to make it practical. In particular, when dealing with languages with a non-alphabetic writing system, or languages without explicit spacing between words, tokenization becomes a complex problem, and in that case, using simple rules would not be a good strategy. In the following subsections, we will reframe tokenization as a machine learning problem where the way to tokenize or segment sentences is learned from data. These methods are language-independent and can be applied to a wide range of tokenization or segmentation-like problems.

3.1.2 Tokenization as Language Modeling

Now let us move to statistical modeling of the tokenization problem. For ease of discussion, this subsection focuses on writing systems without explicit word boundaries, but these methods can generalize to languages with delimiters. Let $\mathbf{x} = x_1...x_l$ be a string of characters, and $\mathbf{y} = y_1...y_m$ be a sequence of words or tokens. We would say that $\mathbf{y}$ is a tokenization result of $\mathbf{x}$ if $\mathbf{y}$ defines a segmentation on $\mathbf{x}$. Consider the following Chinese sentence:

$$\mathbf{x} \quad = \quad \text{机器翻译是人类的梦想之一。}$$

We can define a segmentation on the sentence, for example[2],

$$
\begin{aligned}
\mathbf{y} &= \text{机器翻译/是/人类/的/梦想/之一/。} \\
&= \begin{bmatrix} \text{“机器翻译”} & \text{“是”} & \text{“人类”} & \text{“的”} & \text{“梦想”} & \text{“之一”} & \text{“。”} \end{bmatrix}
\end{aligned} \tag{3.1}
$$

In this way, tokenization can be framed as a problem of mapping $\mathbf{x}$ to $\mathbf{y}$. Given an input string, the output is the most likely segmentation:

$$
\begin{aligned}
\hat{\mathbf{y}} &= \arg\max_{\mathbf{y}} \Pr(\mathbf{y}|\mathbf{x}) \\
&= \arg\max_{\mathbf{y}} \log \Pr(\mathbf{y}|\mathbf{x})
\end{aligned} \tag{3.2}
$$

Eq. (3.2) describes a prediction model we have been referencing several times in this book. However, the problem we are dealing with is easier because any valid segmentation $\mathbf{y}$ deterministically yields the string $\mathbf{x}$. Thus, finding the most likely segmentation simplifies to maximizing the prior probability $\Pr(\mathbf{y})$:

$$
\begin{aligned}
\hat{\mathbf{y}} &= \arg\max_{\mathbf{y}} \log \Pr(\mathbf{y}) \\
&= \arg\max_{\mathbf{y}} \log \Pr(y_1, ..., y_m)
\end{aligned} \tag{3.3}
$$

It is easy to check that Eq. (3.3) in fact describes a language modeling problem. There are a few different ways to estimate the joint probability $\Pr(y_1, ..., y_m)$. A simple method is to rewrite $\log \Pr(y_1, ..., y_m)$ into a sum of log-scale conditional probabilities:

$$
\log \Pr(y_1, ..., y_m) = \log \Pr(y_1) + \log \Pr(y_2|y_1) + ... + \log \Pr(y_m|y_1, ..., y_{m-1}) \tag{3.4}
$$

Each conditional probability $\Pr(y_i|y_1, ..., y_{i-1})$ can be approximated by

$$
\Pr(y_i|y_1, ..., y_{i-1}) = \Pr(y_i|y_{i-n+1}, ..., y_{i-1}) \tag{3.5}
$$

that is, the generation of y_i only depends on the $n-1$ previous context words. To compute $\Pr(y_i|y_{i-n+1}, ..., y_{i-1})$, we can either use the relative frequency methods or neural networks (see Chapter 2).

Now we can think of tokenization as a supervised learning problem. The process is outlined here:

- Prepare some sentences that are correctly segmented.
- Learn a language model $\Pr(\mathbf{y})$ on these labeled sentences.
- For a new sentence, find the “best” tokenization $\hat{\mathbf{y}}$ that maximizes $\Pr(\hat{\mathbf{y}})$, as in Eq. (3.3).

While this procedure follows a standard pipeline of supervised learning, there are several

[2] Following the notation used previously, we use both $\mathbf{y} = y_1 ... y_m$ and $\mathbf{y} = \begin{bmatrix} y_1 & ... & y_m \end{bmatrix}$ to denote a sequence of variables.

practical issues we have to iron out. First, the language model requires a predefined vocabulary to compute the probabilities for each token y_i, but new words are always around. To handle them, one way is to segment an unknown substring into characters, that is, we treat characters as words if the substring yielding these characters is not contained in the vocabulary. An alternative is to take into account all substrings that are not covered by the vocabulary, and replace them with the <unk> tag. The <unk> trick is widely adopted in state-of-the-art language models and is usually helpful.

Second, the language model described above has a bias towards short sequences because $\Pr(y_1,...,y_m)$ would be large if m is a small number. A general way to mitigate this bias is to introduce a length reward (or length bonus) to the model, for example,

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \log \Pr(\mathbf{y}) + \lambda \cdot m \tag{3.6}$$

or

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \frac{\log \Pr(\mathbf{y})}{m^{\lambda}} \tag{3.7}$$

where $\lambda \cdot m$ and m^{λ} reward long sequences and $\lambda > 0$ is a hyperparameter controlling how much we rely on the reward in assessing the goodness of $\mathbf{y}$. Interestingly, it is found that the length bias is not a big problem with tokenization in practice because the variance in length is small for those "good" tokenization results. For example, using a unigram language model (i.e., $n = 1$) without any length reward works well in many real-world applications. We will see a few examples in Section 3.1.4.

Third, performing $\arg\max$ is difficult because there are exponentially many tokenization candidates. However, the use of language models here enables efficient search algorithms. Consider, for example, applying a unigram language model to tokenization. For the input string $\mathbf{x}$, we keep, at each position j of $\mathbf{x}$, a state that describes the probability of the best tokenization on $x_1...x_j$ (denoted as $p(j)$) as well as the last word of this tokenization. At position $j+1$, we create a new state and compute the probability of the best tokenization on $x_1...x_{j+1}$ by

$$\begin{aligned} p(j+1) &= \max_{1 \le i \le j} p(i) \cdot \Pr(x_{i+1}...x_j) \\ &= \max_{1 \le i \le j} p(i) \cdot \Pr(w_{[i+1,j]}) \end{aligned} \tag{3.8}$$

where $\Pr(w_{[i+1,j]})$ is the probability of the word spanning $x_{i+1}...x_j$. On the algorithmic side, Eq. (3.8) describes a dynamic programming method that has a time complexity of $O(l^2)$ for an input of length l. For the final output, we can trace back from the final state and dump the word sequence along the path of the optimal tokenization.

Note that the methods here are generic and can be applied to tokenization for other languages. For example, when applying it to English, we only need a slight update on the format of the input: the input is not a character sequence but a sequence of the smallest possible pieces separated out by punctuation and spaces. For example, for the sentence *Is this Tom's laptop?*, we have

$$\mathbf{x} \quad = \quad \begin{bmatrix} \text{Is} & \text{this} & \text{Tom} & \text{'} & \text{s} & \text{laptop} & \text{?} \end{bmatrix} \tag{3.9}$$

Then, the tokenization process can proceed as in Eqs. (3.2-3.7).

3.1.3 Tokenization as Sequence Labeling

One common approach to segmenting characters into words is to assign labels from a predefined set to each character in a sequence, a process known as sequence labeling. Although we focus here on the problem of word segmentation, this framework is also applicable to various other NLP tasks, such as part-of-speech tagging and named entity recognition. Since the idea of sequence labeling has been discussed in Chapter 1, we present here how it is adapted to the tokenization task.

The label sets used in tokenization are generally simple. One example is the "IB" set. The "I" label indicates a linguistic piece inside a word, and the "B" label indicates the beginning of a word. The label set can be enriched by adding the "E" label (i.e., the ending of a word) and/or splitting the "B" label into sub-labels (e.g., B_1 and B_2 indicate the first and the second linguistic pieces of a word) [Zhao et al., 2006]. Given an input sequence $\mathbf{x}$ and a tokenization result $\mathbf{y}$, transforming $\mathbf{y}$ to a label sequence is fairly simple. Consider again the example used in the previous subsection. We can label the sequence in different formats:

$\mathbf{x}$:	机	器	翻	译	是	人	类	的	梦	想	之	一	。
$\mathbf{y}$:	$\underline{\text{机 器 翻 译}}$				$\underline{\text{是}}$	$\underline{\text{人 类}}$		$\underline{\text{的}}$	$\underline{\text{梦 想}}$		$\underline{\text{之 一}}$		$\underline{\text{。}}$
$\{\mathrm{I},\mathrm{B}\}$	B	I	I	I	B	B	I	B	B	I	B	I	B
$\{\mathrm{I},\mathrm{B},\mathrm{E}\}$	B	I	I	E	B	B	E	B	B	E	B	E	B
$\{\mathrm{I},\mathrm{B}_1,\mathrm{B}_2,\mathrm{E}\}$	B_1	B_2	I	E	B_1	B_1	B_2	B	B_1	B_2	B_1	B_2	B_1

Since the label sequence can be treated as another form of the tokenization, we can restate the problem as finding the best label sequence given an input:

$$\hat{\mathbf{c}} \quad = \quad \arg\max_{\mathbf{c}} \ \log \Pr(\mathbf{c}|\mathbf{x}) \tag{3.10}$$

where $\mathbf{c} = c_1...c_l$ is a label sequence of length l, and $\hat{\mathbf{c}}$ is the label sequence with the highest probability. Many methods have been proposed to model $\Pr(\mathbf{c}|\mathbf{x})$. A classic way is given by rewriting $\Pr(\mathbf{c}|\mathbf{x})$ using Bayes' rule:

$$\begin{aligned} \hat{\mathbf{c}} &= \arg\max_{\mathbf{c}} \ \log \frac{\Pr(\mathbf{x}|\mathbf{c})\Pr(\mathbf{c})}{\Pr(\mathbf{x})} \\ &= \arg\max_{\mathbf{c}} \ \log \Pr(\mathbf{x}|\mathbf{c}) + \log \Pr(\mathbf{c}) \end{aligned} \tag{3.11}$$

Since $\Pr(\mathbf{x})$ is a constant across all candidate sequences $\mathbf{c}$, it can be disregarded during the search process. $\Pr(\mathbf{x}|\mathbf{c})$ is the probability of generating the input $\mathbf{x}$ (i.e., observations) given the label sequence $\mathbf{c}$ (i.e., latent variables), and $\Pr(\mathbf{c})$ is a language model defined on the label sequence. Simplifications are in general required for a tractable model. For example, we can make a **Markov assumption** that the choice of c_i is dependent only on the choice of c_{i-1}. This leads to the **hidden Markov model** (**HMM**) which is widely used in generative modeling for NLP problems.

An alternative method is discriminative modeling. A common idea is to treat sequence labeling as a series of independent classification problems. For example, we can develop a local classifier that conditions the prediction of c_i on a set of features around position i. In more sophisticated models, such as **conditional random fields** (**CRFs**), the context of the entire sequence can be used in the prediction. A detailed discussion of these sequence labeling models is beyond the scope of this section. The reader is referred to [Kupiec, 1992; McCallum et al., 2000; Lafferty et al., 2001] for thorough discussions of how these models are developed and applied. In addition, for a comparison of generative modeling and discriminative modeling, we refer the reader to Chapter 1.

3.1.4 Learning Subwords

Words are widely regarded as the basic units of language. However, they are not the smallest linguistic elements. Rather, they can be decomposed into smaller meaningful units, such as morphemes. This hierarchical structure, where morphemes form words, and words form phrases and sentences, highlights the central role of words in syntax. Nevertheless, in NLP, treating sentences strictly as word sequences is often suboptimal. A primary issue is data sparsity caused by rare words, which hinders effective model training. For instance, the word *uncopyrightable* rarely appears in corpora. An NLP system might treat it as an out-of-vocabulary token, even though its meaning can be derived by decomposing it into *un*, *copy*, *right*, and *able*. Another issue is that linguistics-based tokenization standards constrain the ability of models to learn optimal segmentation units automatically in a machine learning sense. Therefore, it is often more effective to identify "subword" units that, while not strictly linguistic, are better suited for machine learning systems.

1. Byte Pair Encoding

Byte Pair Encoding (**BPE**) is one of the most successful methods for learning **subword** units from a set of word sequences [Sennrich et al., 2016b]. While the BPE approach stems from data compression [Gage, 1994], it is more often used in NLP as a solution to the open-vocabulary problem. The basic idea of BPE is that we repeatedly replace the most frequent pair of bytes in the data to form a new byte. As a result, common bytes are often involved in merging substrings of bytes, and rare bytes are often isolated and considered unique units. The outcome of BPE is a byte vocabulary that can be used to encode new data.

In NLP, a byte can roughly correspond to a character. Each entry of the vocabulary is a character sequence, called a symbol or subword. BPE begins with splitting a given text into a sequence of characters, for example, we can insert a space after each occurrence of an English letter or a punctuation mark. However, this typically yields an excessively long sequence. While BPE itself has no restrictions on input length, a more common way is to prevent cross-word symbols for efficiency considerations. Thus, we can represent the text as a list of space-separated words, each being associated with the frequency of the word. For example, consider a word list:

$$
\begin{array}{r}
\mathrm{f\ l\ o\ w\ \#} : 2 \\
\mathrm{b\ l\ o\ w\ \#} : 2 \\
\mathrm{f\ l\ a\ t\ \#} : 1 \\
\mathrm{f\ l\ a\ g\ \#} : 4
\end{array}
$$

where # is a special symbol indicating the end of a word[3]. From this word list, we can collect an initial vocabulary:

$$
\begin{array}{rcr}
\mathrm{f} : 7 & \quad & \mathrm{b} : 2 \\
\mathrm{l} : 9 & & \mathrm{a} : 5 \\
\mathrm{o} : 4 & & \mathrm{t} : 1 \\
\mathrm{w} : 4 & & \mathrm{g} : 4 \\
\# : 9 & &
\end{array}
$$

Then, we count the occurrences of each symbol bigram:

$$
\begin{array}{rcr}
\mathrm{f\ \ l} : 7 & \quad & \mathrm{a\ g} : 4 \\
\mathrm{l\ \ a} : 5 & & \mathrm{g\ \#} : 4 \\
\mathrm{l\ \ o} : 4 & & \mathrm{b\ \ l} : 2 \\
\mathrm{o\ w} : 4 & & \mathrm{a\ \ t} : 1 \\
\mathrm{w\ \ \#} : 4 & & \mathrm{t\ \#} : 1
\end{array}
$$

We merge the most frequent symbol bigram “f l” into a new symbol “fl” and replace in the word list each occurrence of “f l” with “fl”:

$$
\begin{array}{r}
\mathrm{fl\ o\ w\ \#} : 2 \\
\mathrm{b\ l\ o\ w\ \#} : 2 \\
\mathrm{fl\ a\ t\ \#} : 1 \\
\mathrm{fl\ a\ g\ \#} : 4
\end{array}
$$

[3] Instead of treating # as a separate symbol, it can be appended to the final character of each word

$$
\begin{array}{r}
\mathrm{f\ l\ o\ w\#} : 2 \\
\mathrm{b\ l\ o\ w\#} : 2 \\
\mathrm{f\ l\ a\ t\#} : 1 \\
\mathrm{f\ l\ a\ g\#} : 4
\end{array}
$$

where “w#”, “t#”, and “g#” represent characters that occur at the end of a word.

Accordingly, the symbol "fl" is added to the vocabulary:

f : 7	b : 2
l : 9	a : 5
o : 4	t : 1
w : 4	g : 4
# : 9	fl : 7

This process is then iterated. This time, we merge the symbol bigram "fl a" and create a new symbol "fla". As such, we have a new word list:

fl o w # : 2
b l o w # : 2
fla t # : 1
fla g # : 4

and a new vocabulary:

f : 7	b : 2	fla : 5
l : 9	a : 5	
o : 4	t : 1	
w : 4	g : 4	
# : 9	fl : 7	

This merging process can be repeated for a predefined number of iterations. The more times we perform the merge process, the larger the vocabulary is. The entries of the final vocabulary are reordered by symbol frequencies. For example, if we set the number of merge operations to 6, we will have a vocabulary, as follows:

l : 9	fla : 5	ow# : 4
# : 9	o : 4	flag : 4
f : 7	w : 4	flag# : 4
fl : 7	g : 4	b : 2
a : 5	ow : 4	t : 1

It corresponds to the word list:

fl ow# : 2
b l ow# : 2
fla t # : 1
flag# : 4

Having obtained a vocabulary like above, we can apply it to tokenize new words. The subword tokenization follows the same procedure of merging symbol bigrams as that used in building the vocabulary. Given a BPE vocabulary, we first segment the input text into character symbols. Then, we examine each symbol bigram in the sequence, and merge the one that has the highest frequency in the vocabulary. We repeat this operation until there are no further merges. Consider, for example, the following text:

tow a flag

It is first transformed into a character sequence:

$$\text{t o w \# a \# f l a g \#}$$

By using the BPE vocabulary we have obtained, we can perform BPE merge operations on this sequence, as follows

$$\begin{aligned}
& \quad \text{t o w \# a \# f l a g \#} \\
\xrightarrow{\text{f l} \Rightarrow \text{fl}} & \quad \text{t o w \# a \# } \textcolor{red}{\text{fl}} \text{ a g \#} \\
\xrightarrow{\text{fl a} \Rightarrow \text{fla}} & \quad \text{t o w \# a \# } \textcolor{red}{\text{fla}} \text{ g \#} \\
\xrightarrow{\text{o w} \Rightarrow \text{ow}} & \quad \text{t } \textcolor{red}{\text{ow}} \text{ \# a \# fla g \#} \\
\xrightarrow{\text{ow \#} \Rightarrow \text{ow\#}} & \quad \text{t } \textcolor{red}{\text{ow\#}} \text{ a \# fla g \#} \\
\dots \qquad & \quad \dots \\
\longrightarrow \quad & \quad \text{t ow\# a \# flag\#}
\end{aligned}$$

This subword sequence can serve as input and/or output for a downstream NLP task, such as machine translation. Sometimes, we want to map subwords back to words. This is simple: we keep the space after each occurrence of the # symbol, and remove all other spaces and #. Also note that the BPE method we describe here requires word-segmented inputs, that is, we need a pre-tokenizer to roughly tokenize the input sequence into some units. This can be done by using the methods presented in Sections 3.1.1-3.1.3.

2. WordPiece

WordPiece is very similar to BPE: it first divides the input text into the smallest symbols and then progressively merges consecutive symbols to form larger symbols [Schuster and Nakajima, 2012]. The difference between them is only in the way of selecting which symbol bigram to merge. At each step, BPE merges the symbol bigram with the highest frequency. Let (x_i, x_{i+1}) be a bigram in the sequence $\mathbf{x}$. The merge rule of BPE can be described as

$$(x_{\hat{i}}, x_{\hat{i}+1}) \quad = \quad \underset{i \in [1, |\mathbf{x}|-1]}{\arg\max} \ \mathrm{count}(x_i, x_{i+1}) \tag{3.12}$$

where the function $\mathrm{count}(x_i, x_{i+1})$ returns the frequency of (x_i, x_{i+1}) in the corpus, and $(x_{\hat{i}}, x_{\hat{i}+1})$ is the bigram with the highest frequency.

The WordPiece method, instead, adopts a maximum likelihood criterion for bigram selection. Specifically, it merges the bigram that yields the greatest increase in the likelihood of the training data. This can be formalized as:

$$\begin{aligned}(x_{\hat{i}}, x_{\hat{i}+1}) &= \underset{i\in[1,|\mathbf{x}|-1]}{\arg\max} \log \frac{\Pr(x_i, x_{i+1})}{\Pr(x_i)\Pr(x_{i+1})} \\ &= \underset{i\in[1,|\mathbf{x}|-1]}{\arg\max} [\log \Pr(x_i, x_{i+1}) - \log(\Pr(x_i)\Pr(x_{i+1}))] \end{aligned} \tag{3.13}$$

The term $\log \Pr(x_i, x_{i+1}) - \log(\Pr(x_i)\Pr(x_{i+1}))$ represents the increase in the log-likelihood of the corpus upon replacing consecutive symbols (x_i, x_{i+1}) with the merged symbol $x_i x_{i+1}$[4]. Thus, applying this merge rule yields a sequence of operations that incrementally increases the overall data likelihood. The outcome of this process is a codebook (i.e., a vocabulary) by which we can define the most likely code sequence for the given text.

3. SentencePiece

Both the BPE and WordPiece methods require that the input text be pre-tokenized into words. This dependency adds complexity to the tokenization pipeline. As an alternative, SentencePiece offers a more general framework that processes raw text directly by treating all characters, including spaces, as basic atomic units [Kudo and Richardson, 2018]. While SentencePiece can be paired with various subword algorithms, it is often associated with unigram language models (or simply the unigram method), which adopts a fundamentally different vocabulary construction strategy: rather than building the vocabulary bottom-up, it begins with a big vocabulary and progressively prunes it to maximize the overall data likelihood [Kudo, 2018] [5].

The unigram method frames subword segmentation as a unigram language modeling problem, resembling the general form of Eqs. (3.3-3.4). Let $\mathbf{x}$ be a sequence of characters and $\mathbf{y}$ be a sequence of symbols or subwords that concatenates to form $\mathbf{x}$. The probability of $\mathbf{y}$ is given by:

$$\Pr(\mathbf{y}) = \prod_{i=1}^{|\mathbf{y}|} \Pr(y_i) \tag{3.14}$$

Then, we can write the likelihood of $\mathbf{x}$ in terms of the joint probability of $\mathbf{x}$ and $\mathbf{y}$:

$$\Pr(\mathbf{x}) = \sum_{\mathbf{y}\in Y(\mathbf{x})} \Pr(\mathbf{x}, \mathbf{y}) \tag{3.15}$$

where the sum is over all possible tokenization results $Y(\mathbf{x})$. Since $\mathbf{x}$ is deterministically derived from $\mathbf{y}$, the joint probability simplifies to $\Pr(\mathbf{x}, \mathbf{y}) = \Pr(\mathbf{y})$. Thus, we can rewrite Eq.

[4] In statistics, $\frac{\Pr(a,b)}{\Pr(a)\Pr(b)}$ is called the **pointwise mutual information** of variables a and b. See more details in Section 3.3.1. This concept is also known as **information gain** and can be interpreted by using the Kullback-Leibler divergence or other measures in information theory (see Chapter 1).

[5] The term *vocabulary size* typically refers to the number of entries in the vocabulary. In some computational contexts, however, it may denote the total number of bytes required to store the vocabulary.

(3.15) as:

$$\begin{aligned}\Pr(\mathbf{x}) &= \sum_{\mathbf{y}\in Y(\mathbf{x})} \Pr(\mathbf{y}) \\ &= \sum_{\mathbf{y}\in Y(\mathbf{x})} \prod_{i=1}^{|\mathbf{y}|} \Pr(y_i) \end{aligned} \tag{3.16}$$

Taking this equation, the log-likelihood of a set of strings X is given by

$$\begin{aligned}\Pr(X) &= \log \prod_{\mathbf{x}\in X} \sum_{\mathbf{y}\in Y(\mathbf{x})} \prod_{i=1}^{|\mathbf{y}|} \Pr(y_i) \\ &= \sum_{\mathbf{x}\in X} \log \left(\sum_{\mathbf{y}\in Y(\mathbf{x})} \prod_{i=1}^{|\mathbf{y}|} \Pr(y_i) \right) \end{aligned} \tag{3.17}$$

If we consider $-\Pr(X)$ as a loss function, then the task here can be stated as finding the best estimate for each unigram probability $\Pr(y)$ so as to make $\Pr(X)$ as large as possible. At first glance this optimization problem looks complicated. Fortunately, there are several powerful tools to solve it. A popular method is to use **the Expectation-Maximization (EM) algorithm** [Dempster et al., 1977], which is commonly used when one tries to find a statistical model that maximizes the likelihood of the data. Note that the EM-based solution to Eq. (3.17) is similar to those for other NLP problems, such as statistical machine translation, and has been well discussed in those contexts. So we refer the reader to [Brown et al., 1993] for details about these methods. In this chapter we use EM as an off-the-shelf tool to estimate $\Pr(y)$ given Eq. (3.17). [6]

SentencePiece is essentially a "pruning" method that removes low probability entries from the vocabulary. It starts with a big initial vocabulary V. For example, we can create the initial

[6] In EM, we can view X as an observation, and $\Pr(X|\theta)$ as a statistical model that describes how likely the observation occurs. Here θ is the model parameters that we intend to determine. EM is based on an objective of maximum likelihood estimation, that is

$$\hat{\theta} = \arg\max_{\theta} \Pr(X|\theta) \tag{3.18}$$

For the model here, we can view $\{\Pr(y)\}$ as model parameters. We skip the derivation details about the EM estimate of $\Pr(y)$ but directly present the result. The EM algorithm involves two steps.

- **The Expectation Step** (or the E-step): Given the current estimate of $\Pr(y)$ (say, $\Pr_t(y)$), we compute the posterior $\Pr_t(\mathbf{y})$ for each $\mathbf{y}$ according to Eq. (3.14). Then, we compute the **fractional count** of each subword y in the vocabulary V, as follows

$$\mathrm{fcount}(y) = \sum_{\mathbf{x}\in X} \sum_{\mathbf{y}\in Y(\mathbf{x})} \left(\Pr_t(\mathbf{y}) \sum_{i=1}^{|\mathbf{y}|} \delta(y, y_i) \right) \tag{3.19}$$

  where $\delta(y, y_i)$ returns 1 if $y = y_i$, and 0 otherwise. $\sum_{i=1}^{|\mathbf{y}|} \delta(y, y_i)$ counts the number of times y occurs in the subword sequence $\mathbf{y}$.

- **The Maximization Step** (or the M-step): Given the fractional counts obtained in the E-step, we re-estimate

subword sequence	unigram probabilities ([subword]:probability)
t/ow_/a/_flag	[t]:0.030 [ow_]:0.002 [a]:0.041 [_flag]:0.001
t/ow/_/a_/f/lag	[t]:0.030 [ow]:0.005 [_]:0.113 [a_]:0.093 [f]:0.041 [lag]:0.002
t/ow/_a_/fla/g	[t]:0.030 [ow]:0.005 [_a_]:0.084 [fla]:0.003 [g]:0.027
tow/_/a_/f/lag	[tow]:0.001 [_]:0.113 [a_]:0.093 [f]:0.041 [lag]:0.002
t/ow_/a_/flag	[t]:0.030 [ow_]:0.002 [a_]:0.093 [flag]:0.001

Figure 3.3: Different tokenization results for *tow a flag*. Every subword is assigned a probability that is estimated through a unigram language model. Every whitespace is replaced with "_" for a clear presentation.

vocabulary by enumerating all strings with a length constraint. Typically, cross-word strings are excluded to reduce the vocabulary size. Then, we run the following steps:

- Estimate the probability for each entry y of V by optimizing Eq. (3.17).
- Compute the loss for each entry y of V via the **remove-one** strategy, that is, the loss is the reduction in the likelihood (see Eq. (3.17)) when y is removed from the vocabulary.
- Remove a certain percentage of entries of V with large losses. For example, we keep 80% of the entries, and discard the rest.

The outcome of this process is a new vocabulary as well as the probability assigned to each subword. We can repeat this process a number of times until the vocabulary size is reduced to a desirable level.

SentencePiece differs from BPE and WordPiece in that it considers all possible subword sequences for a given string (see the sum $\sum_{\mathbf{y}\in Y(\mathbf{x})}$ in Eq. (3.15)). From the machine learning point of view, this can be seen as a way of regularization, that is, we can reduce the risk of overestimating the parameters corresponding to the single-best subword sequence that may have errors. An alternative way is to only consider some of the subword sequences in $Y(\mathbf{x})$ for the sake of efficiency. For example, we can sample k subword sequences according to $\Pr(\mathbf{y})$ to form the candidate set $Y(\mathbf{x})$.

Note that the SentencePiece method does not depend on word-separated input sequences. While the BPE and WordPiece methods can also deal with raw text if updated, the SentencePiece method explicitly takes the space and other delimiters as parts of the subwords. See Figure 3.3 for a few tokenization results for *tow a flag*.

Given a learned vocabulary and the corresponding unigram probabilities, we can apply them to deal with a new text. This is in fact a search problem: we find the most likely subword

the unigram probabilities by the equation:

$$\Pr_{t+1}(y) = \frac{\mathrm{fcount}(y)}{\sum_{y'\in V}\mathrm{fcount}(y')} \tag{3.20}$$

The two steps are iterated for a number of rounds until the parameters converge to some values.

sequence in terms of the unigram probability. As language modeling is a well-studied topic in NLP, many search algorithms are directly applicable to the case here. For example, the methods presented in Section 3.1.2 are straightforwardly applicable here.

3.2 Vector Representation for Words

Words inherently carry meaning[7]. In the broadest sense, the meaning of a word is derived from how it is interpreted. This is something behind the surface form of a word but can be understood by language speakers. For example, consider the following lines of text from a poem [Knight, 2018]:

There was a little sparrow

Who sat on a wheelbarrow,

And tweeted to all her friends around.

A cat with open jaws

And very pointed claws,

Spied her as he raced along the ground.

These words are not merely arbitrary strings of characters; they evoke specific concepts. For example, "little" means *small in size*, "sparrow" means *a kind of bird*, and "friends" means *people who you like and trust*. From an NLP perspective, however, word meaning (or **word sense**) cannot merely remain a cognitive abstraction; it must be formalized into a computable, machine-readable format.

3.2.1 One-hot Representation

A natural way to represent word meanings is to use language to describe them. For example, we can look up the aforementioned words in a dictionary to find their ids and definitions[8]:

[7]While we have so far discussed several linguistic elements used in NLP, such as subwords, we still use *words* as the basic units in our discussion here. The methods we will present in the remaining part of this chapter can generally be applied to other linguistic units used in NLP systems, including characters, subwords, and so on.

[8]All these words and their meanings are found in `https://dictionary.cambridge.org/`.

cat	511	*A small animal with fur, four legs, a tail, and claws, usually kept as a pet or for catching mice*
her	5220	*Used, usually as the object of a verb or preposition, to refer to a woman, girl, or female animal that has just been mentioned or is just about to be mentioned*
jaws	6186	*The mouth, including the teeth*
ground	6402	*The surface of the earth*
sparrow	8331	*A common, small, gray-brown bird*
wheelbarrow	9954	*A large, open container for moving things in with a wheel at the front and two handles at the back, used especially in the garden*

To represent a word, the simplest idea is to replace it with its dictionary id. In this way, each word representation is a unique number. An equivalent form to this is the **one-hot** representation. It is a vector whose dimensionality is equal to the vocabulary size. In this vector, only the element corresponding to the word index is set to 1, while all other elements are exactly 0. For example, the word *sparrow* can be represented as a one-hot vector $\mathbf{1}_{\text{sparrow}}$ based on its id (8331):

$$\begin{array}{ccccccccc} [\ 0 & 0 & \ldots & 0 & 1 & 0 & \ldots & 0 & 0\ ] \\ & & & & \uparrow & & & & \\ & & & & \text{id} = 8331 & & & & \end{array}$$

3.2.2 Distributed Representation

However, it appears that the one-hot representation only provides the "identity" of the word but not the "description" of what the word is. An obvious problem is that every word is orthogonal to other words. Because of this orthogonality, it is impossible to mathematically quantify semantic relationships, even for words that are highly related in natural language. Here, our desire is a model in which words are described as quantifiable attributes and semantic closeness can be naturally captured. A way to do this is to enrich the representation with the word description. Consider again the word *sparrow* for example. In the dictionary, we have its meaning *a common, small, gray-brown bird*. By using the tokenization and normalization methods mentioned in Section 3.1.1, this text can be transformed into a sequence of words

$$\begin{bmatrix} a & common & , & small & , & gray & - & brown & bird \end{bmatrix}$$

Then, we vectorize this sequence using the bag-of-words model (see Chapter 1), leading to a new vector of numbers

$$\begin{array}{ccccccccccccccccccccc} [\ 0 & 0 & \ldots & 1 & \ldots & 1 & \ldots & 1 & \ldots & 1 & \ldots & 1 & \ldots & 1 & \ldots & 1 & \ldots & 1 & \ldots & 0 & 0\ ] \\ & & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \uparrow & & & \\ & & & , & & - & & a & & bird & & brown & & common & & gray & & small & & & \end{array}$$

where the value of an entry is 1 if the corresponding word is present, and 0 otherwise. This way enables the sharing of content among words. We would say that two words are similar if they have overlaps in their word vectors. Consider a new word *cuckoo*. We can find its meaning in a dictionary, e.g., *a grey bird with a two-note call that sounds similar to its name*. It is easy to know that *sparrow* and *cuckoo* are two words that share something similar because they both mark the "bird" dimension as 1 and the vector similarity between the two word vectors is greater than 0[9].

Treating words as vectors of numbers offers a general tool to represent words in various different ways. We do not even have to explain a word vector from the viewpoint of semantics. For example, we can introduce a new dimension into the vector to mark if the word belongs to some syntactic category. In a broad sense, we can define an arbitrary function on each entry of the vector and view the function's output as a feature describing the word. For example, a simple improvement to the above representation is to use a function counting the occurrences of a word instead of the binary-valued function marking the presence of the word. More feature functions can be found in Section 3.3.

Note that it is not necessary to constrain the feature functions to forms that make linguistic sense although linguistically motivated designs of the feature functions are usually of interest to NLP researchers. A more general form for word representation is simply a real-valued, multi-dimensional vector. It is often called the **distributed representation** of a word, or the **word embedding**. For example, the word *sparrow* can be represented as a vector

$$\begin{bmatrix} 1.9 & -7 & 3 & -1.2 & ... & 2.01 & -2.05 \end{bmatrix}$$

From a machine learning perspective, this vector can describe some underlying attributes of a word. These attributes may not be explainable in human understanding but can be learned from data. One of the challenges in learning such a representation is that one can hardly measure the goodness of a vector. In general, it makes no sense to ask whether the distributed representation of a single word is good or not. Rather, we would like to know if the representations of a group of words are well behaved. For example, it is a common belief that similar words should have similar representations. So, the relationship between words is often thought of as some "distance" between the word representations in a vector space. This leads to a number of methods to visualize and evaluate word representations. In Section 3.6, we will give a more detailed discussion about these issues.

Typically, word representations do not work alone in NLP systems but are used as some intermediate states of a model. A standard approach in NLP, to learn distributed representations of words, is to take it as a by-product of training a "big" system. That is, the representation model works as a component of a system, and is optimized together with other components when the system is trained in some way. This inspires a promising paradigm of representation learning: the representation model is learned as a sub-model in an easy-to-train system, and can be used as a plug-in for a completely different system. In neural language modeling, for instance, we can force the model to map each input one-hot word vector into a real-valued,

[9]The similarity of two vectors can be measured by the cosine of the angle between them.

low-dimensional distributed representation. These distributed representations are fed into a neural network that predicts a probability distribution over the vocabulary. The mapping function or embedding function is trained so as to minimize the loss of the language model on some data (see Section 3.4). When applying the learned embedding function, we drop all other parts of the language model and use the function to generate the distributed representation for each word in downstream tasks. An alternative strategy is to specifically tailor the model to the word representation learning problem. Systems of this kind are typically not designed to deal with standard NLP problems, but with an emphasis on specific problems in word representation learning, such as explicitly modeling the relationship between words (see Section 3.5).

3.2.3 Compositionality and Contextuality

While our discussion is restricted to word representation learning in NLP, studying word meaning is a traditional sub-field of linguistics. In **lexical semantics**, for instance, researchers investigate how word meanings are defined and how they compose sentence-level meaning. Historically, computational word representation has not strictly adhered to linguistic semantics, focusing instead on transforming linguistic units into computational formats. However, because semantics is fundamental to language understanding, it remains crucial to incorporate semantic principles into the design of representation models. Consider the following properties:

- **Compositionality**. Compositionality is a common concept in semantics, logic and related fields. It often comes out with **the principle of compositionality**:

  > *The meaning of a complex expression is determined by its structure and the meanings of its constituents.*
  >
  > – Szabó [2020]

  The principle of compositionality is fundamental and exists everywhere in natural language. For instance, the meaning of the phrase *white cat* is easily derived from the individual meanings of *white* and *cat*. Similarly, compound sentences derive their overall meaning from the combination of independent clauses and conjunctions. Note that the principle of compositionality is not merely a simple rule that we use to describe how larger linguistic units are composed of smaller ones, although researchers have attempted to define it formally [Montague, 1974]. There are ongoing debates regarding how this principle should be interpreted and adequately modeled by semantic theories. Still, if we focus on NLP applications and set aside theoretical linguistic debates, compositionality remains a highly valuable property for system design and evaluation. When a problem exhibits compositionality, it implies that complex tasks can be systematically decomposed into more manageable sub-components. For word representation learning, we desire resulting vectors that exhibit compositionality to reflect the combinatorial nature of language. In Section 3.6, we will see several examples of this. For example, representations learned by neural networks often demonstrate meaningful algebraic properties under linear operations, even though the models themselves are non-linear. However, compositionality is not an absolute rule. It frequently breaks down in cases such as collocations and idioms, where natural language exhibits non-compositionality. This explains why NLP problems are so challenging.

- **Contextuality**. Contextuality represents some sort of non-compositionality. It states that a word may have multiple meanings, and its specific interpretation is determined by the surrounding context. For example, consider the following sentences[10]

  They sat round the dinner table, arguing about politics.

  Come to the table everybody - supper's ready.

  He came in with four shopping bags and dumped them on the table.

  The table can help you evaluate the potential risks of investing in the Fund.

  Building societies dominate the best-value tables for mortgages.

  This table represents export sales.

  In these example sentences, *table* is a polysemous word with two meanings:

  Sense 1: *a flat surface used for putting things on.*

  Sense 2: *an arrangement of items in rows, or columns, or blocks.*

  In other words, *table* is an ambiguous word. This ambiguity is typically resolved by examining the adjacent words. For example, when *table* follows *dinner*, it is easy to figure out that it refers to sense 1. The ambiguity also exists when a word stems from a few different forms or lexemes (a phenomenon known as **homonymy**). For example, *bear* can be either a verb or a noun. Disambiguating a word for a given set of word senses has been studied for decades in NLP and is commonly known as **word sense disambiguation** (**WSD**) [Kelly and Stone, 1975]. However, the word representation problem discussed here is more challenging because we usually do not have a pre-defined set of word senses in hand. We instead want a contextual representation model that can generate a word representation dependent on its context. Thus, it is important to recognize that word meaning is not static. This makes the problem somewhat different from what we discussed at the beginning of the section, as we no longer have a lookup table for word representations, but a model that produces different representations of a word in different contexts.

In the remaining sections of this chapter, we focus on learning vector representations of words from their distributions in language use. We leave the discussion on the contextual models for learning dense word representations to Chapters 4-6.

[10] All these sentences are from `https://dictionary.cambridge.org/dictionary/english/table`

3.3 Count-based Models

We have framed the induction of word meanings as a problem of learning word vectors. In this section, we proceed by assuming that the meaning of a word is determined by the environment where the word is used. This is usually stated as the **distributional hypothesis** — words are semantically similar if they appear in similar contexts [Harris, 1954; Firth, 1957]. A word representation learned under this hypothesis is also called the **distributional word representation** or **distributional representation**[11]. To ease the reading, however, we will still use the terms *word vector* and *word representation* throughout this book. Next, we introduce several methods for modeling the distribution of words in texts, and then offer some refinements.

3.3.1 Co-occurrence Matrices

In **distributional semantics**, words are represented with semantic models that consider various aspects of the context. These models differ in how the context of a word is modeled, for example, how large the context is considered, how each occurrence of a word is counted, how the dimensionality of a distribution is defined, and so on. In this section we assume, as in most models used in NLP, that word representations are learned from a collection of documents.

A document can be viewed simply as an unordered collection of words (i.e., the bag-of-words assumption). Then we can think of each occurrence of these words as an independent context indicator. In this way, the distribution of a word in its context can be described as the number of times the word co-occurs with the context words. This is achieved by constructing a co-occurrence matrix, where each cell records the co-occurrence frequency of a row item and a column item. Consider, for example, the following documents[12]:

Doc 1 *A berry is a small, pulpy, and often edible fruit.*

Doc 2 *In botanical terminology, a berry is a simple fruit with seeds and pulp produced from the ovary of a single flower.*

Doc 3 *The term "banana" is also used as the common name for the plants that produce the fruit.*

Doc 4 *Banana seeds are large and hard and spiky and liable to crack teeth.*

Doc 5 *A banana is an elongated, edible fruit - botanically a berry - produced by several kinds of large herbaceous flowering plants in the genus Musa.*

For each pair of words, we collect the total number of times they co-occur in these documents, leading to a matrix, called the **word-word co-occurrence matrix** or **term-term**

[11] It should be noted that *distributional representation* and *distributed representation* are two different concepts. A distributional representation refers to a representation that describes the distribution of language items in language use. A related term is *non-distributional representation* which means something that is obtained from lexical databases, such as the interpretation of a word in a dictionary. On the other hand, a distributed representation refers to a vector of variables corresponding to some underlying attributes of a language item. In contrast to distributed representation, a one-hot representation just describes the word symbol.

[12] The texts are from Wikipedia.

co-occurrence matrix. Here is a subset of the matrix for the above documents.

	flowering	*fruit*	*herbaceous*	...	*often*	*plants*	*seeds*
berry	1	3	1	...	1	1	1
terminology	0	1	0	...	0	0	1
common	0	1	0	...	0	1	0
teeth	0	0	0	...	0	0	1
banana	1	2	1	...	0	2	1
simple	0	1	0	...	0	0	1
and	0	2	0	...	1	0	2

In the matrix, each row word is associated with a word vector of $|V|$ entries. The numbers in the entries describe how often the row word co-occurs with different context words, that is, how a given word is distributed in different "contexts". In a geometric sense, if two words have similar distributions in co-occurring with the same group of context words, then the angle between the word vectors would be small[13]. For example, if we think of these words as vectors in a vector space, *berry* is closer to *banana* than *teeth* (see Figure 3.4). This geometric intuition is the basis of many representation models. More examples will be given in Chapters 4 and 5.

A problem with this method is that the distance between words is not taken into account although the correlation is not that strong when the context word is distant. A simple solution is to constrain context words in a window, called the **context window** or window for short [Lund and Burgess, 1996]. For example, for each word in a document, we only count the -2 and +2 words surrounding it (i.e., a window of size 5).

Note that the word vectors learned by the bag-of-words model in Section 3.2 are a special instance of the co-occurrence matrix. In that example, the entire corpus consists of a single document, and an indicator function marks presence rather than frequency. Beyond simple counts or binary indicators, the entries of a word vector can be populated using various association measures. In practice, the value of an entry of a word vector can be thought of as the degree of the correspondence between words. If two words are correlated with each other in some context, a feature function may assign a score between them in any manner. This score does not necessarily have to be a count, but can be an arbitrary real number. As such, the problem can be stated as measuring the association strength between words. It is common practice to define such a measure on the basis of correlation models. In statistics, correlation describes to what extent two variables are associated, measured by **correlation coefficients**. Common correlation coefficients include the Pearson correlation coefficient (Pearson's r), the Spearman's rank correlation coefficient (Spearman's ρ), and so on[14]. In NLP, a widely used measure is the **pointwise mutual information** (**PMI**) [Church and Hanks, 1990]. Let a and b

[13] The angle between two vectors is independent of their magnitudes. If the vectors are normalized in some way (e.g., by vector norm), similar vectors mean that most entries of the two vectors have similar values.

[14] Some of the correlation coefficients assume certain distributions of the data. For example, the Pearson correlation coefficient is calculated based on two variables following normal distributions.

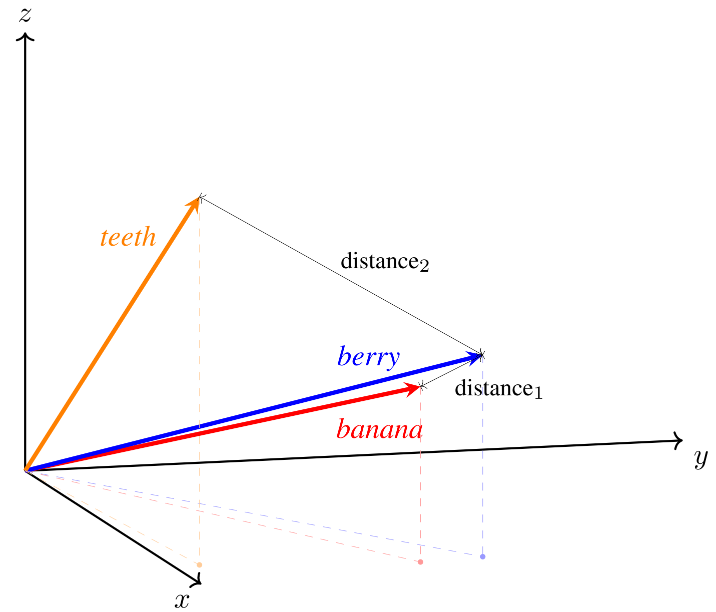


Figure 3.4: Word vectors in a vector space that is built from the word co-occurrence statistics on the English data from WMT 2012. All the vectors are normalized and represented as arrows. For visualization, we project these vectors from a high-dimensional space to a 3-dimensional space via principal component analysis. As expected, *berry* is closer to *banana* than to *teeth*.

be two words. The mathematical form of PMI is given by

$$\mathrm{PMI}(a,b) = \log \frac{\Pr(a,b)}{\Pr(a)\Pr(b)} \tag{3.21}$$

where $\Pr(a,b)$ is the joint probability of a and b co-occurring, and $\Pr(a)$ (or $\Pr(b)$) is the probability of a (or b) occurring. These probabilities can be simply estimated on the texts by the relative frequency method[15]. Given a word a and a vocabulary of context words $\{b_1,...,b_{|V|}\}$, the PMI-based word vector of a is written as

$$e(a) = \begin{bmatrix}\mathrm{PMI}(a,b_1) & ... & \mathrm{PMI}(a,b_{|V|})\end{bmatrix} \tag{3.22}$$

Correlation coefficients are generally used to test whether two variables are (linearly) related. So, an alternative method is to define an entry of the word vector as the outcome of a test. For example, the entry (a,b) chooses a value of 1, if the correlation coefficient between words a and b is larger than a threshold, or the correlation of words a and b is sufficiently supported by hypothesis testing.

However, modeling words as vectors of correlation scores between words somewhat limits the scope of contextual information one may use. An alternative macro-level approach is

[15]A problem with PMI is that the measure becomes unstable when the words are rare. For example, if a very rare word happens to appear in a document, the PMI value of this word and any other word in this document would be unreasonably large.

Entry	Mathematical form
Binary	$\mathrm{tf}(a,d) = \begin{cases} 1 & a \text{ occurs in } d \\ 0 & \text{otherwise} \end{cases}$
Count	$\mathrm{tf}(a,d) = \mathrm{count}(a;d)$
Exponential Count	$\mathrm{tf}(a,d) = \mathrm{count}(a;d)^{\alpha}$
Log-scale Count	$\mathrm{tf}(a,d) = \log(1+\mathrm{count}(a;d))$
Normalized Count (or Frequency)	$\mathrm{tf}(a,d) = \frac{\mathrm{count}(a;d)}{\sum_{a'}\mathrm{count}(a';d)}$

Table 3.1: Functions of the term-frequency weighting scheme. $\mathrm{count}(a;d)$ counts the occurrences of the word a in the document d.

to analyze how words distribute across entire documents, forming a **word-document co-occurrence matrix** or **term-document co-occurrence matrix**. For example, for the above mentioned documents, we can build a matrix

	Doc 1	Doc 2	Doc 3	Doc 4	Doc 5
berry	1	1	0	0	1
terminology	0	1	0	0	0
common	0	0	1	0	0
teeth	0	0	0	1	0
banana	0	0	1	1	1
simple	0	1	0	0	0
and	1	1	0	2	0

In the matrix, the value of entry (a,d) is defined to be the number of times the word a occurs in the document d, quantifying the association strength between a and d. This is commonly called the **term frequency** (**TF**) of a in d (denoted by $\mathrm{tf(a,d)}$). Also, we can use a 0-1 indicator function to mark the presence of the word occurrence (see Section 3.2). See Table 3.1 for a few variations of the TF weighting function.

This structure serves a dual purpose: each row provides a vector representation of a specific word, while each column acts as a vector representation of an entire document. This word-document co-occurrence matrix is conceptually identical to the bag-of-words model introduced for text classification in Chapter 1, where word order is discarded but relative frequencies are preserved. Here we perform document vectorization via this model on a collection of documents.

3.3.2 TF-IDF

Modeling word-document associations is important for many NLP tasks. A significant refinement over raw frequency counts for building these vectors is the **term frequency-inverse**

document frequency (**TF-IDF**) weighting scheme. Given a set of documents D, TF-IDF assigns a score to each word-document pair (a,d) via the following equation:

$$\mathrm{tfidf}(a,d,D) = \mathrm{tf}(a,d) \cdot \mathrm{idf}(a,D) \tag{3.23}$$

where

- $\mathrm{tf}(a,d)$ is the term frequency (see Table 3.1). When $\mathrm{tf}(a,d)$ is large, the word a is a good indicator for the document d. In contrast, when $\mathrm{tf}(a,d)$ is small, the word-document association is not that strong.
- $\mathrm{idf}(a,D)$ is the **inverse document frequency** (**IDF**). It is developed based on the fact that common words across documents are less informative. For example, for a collection of documents on sports, it is likely to see *player* and *players* in most documents. In this case, the words *player* and *players* are less interesting in discriminating different documents or contexts. Let $\mathrm{df}(a,D)$ be the number of documents in D containing the word a. A common form of $\mathrm{idf}(a,D)$ is given by

  $$\mathrm{idf}(a,D) = \log \frac{|D|}{\mathrm{df}(a,D)} \tag{3.24}$$

  This equation penalizes words that appear frequently across the corpus. Similarly, we can have a smoothed version of $\mathrm{idf}(a,D)$

  $$\mathrm{idf}(a,D) = \log \frac{|D|}{\mathrm{df}(a,D)+1} + 1 \tag{3.25}$$

Using the TF-IDF weighting scheme, we can construct a word-document matrix where the value of entry (a,d) is $\mathrm{tfidf}(a,d,D)$. As previously described, each row of this matrix serves as the vector representation of its corresponding word. Traditionally, TF-IDF and term-document matrices are extensively used for document representation. For instance, in information retrieval, both queries and documents can be represented as TF-IDF column vectors, allowing us to compute query-document relevance via vector similarity. While vector space models in information retrieval fall outside the scope of this chapter, readers are referred to standard textbooks for comprehensive coverage [Manning et al., 2008; Buttcher et al., 2016].

3.3.3 Low-Dimensional Models

Co-occurrence matrices are often high dimensional. Suppose, for example, that there is a vocabulary of $20,000$ unique words and a collection of $10,000,000$ documents. Then, a word-document co-occurrence matrix has $20,000 \times 10,000,000 = 2 \times 10^{11}$ entries. Given the computational burden associated with such a model, it is impractical to represent a word as a $10,000,000$-dimensional vector and a document as a $20,000$-dimensional vector. Instead, we expect that the representation of a word (or a document) requires only a reasonably small number of features. In this subsection, we discuss standard approaches for transforming high-dimensional co-occurrence matrices into lower-dimensional representations. Because these techniques have been extensively studied and successfully applied across various disciplines [Barber, 2012; Wright and Ma, 2022], we bypass the underlying mathematical proofs and focus instead on their practical application for deriving word and document vectors.

1. Latent Semantic Analysis

In NLP, **latent semantic analysis** (**LSA**) is a method of seeking the latent semantic structure behind the word-document associations [Deerwester et al., 1990; Landauer et al., 1998][16]. It assumes that both words and documents can be represented as low-dimensional vectors that are distilled from the co-occurrence matrix, preserving the property of the original vector space model, e.g., the angle between vectors is small for similar words.

More specifically, LSA factorizes the co-occurrence matrix into a matrix for word representation, a matrix for document representation, and a third matrix connecting the first two matrices. Mathematically, this can be framed as a **singular value decomposition** (**SVD**) process [Stewart, 1993]. Let $\mathbf{M} \in \mathbb{R}^{|V| \times |D|}$ be a co-occurrence matrix over a vocabulary V and a document set D. The SVD produces a factorization of $\mathbf{M}$

$$\mathbf{M} = \mathbf{P}\mathbf{\Sigma}\mathbf{Q}^{\mathrm{T}} \tag{3.26}$$

where $\mathbf{P} \in \mathbb{R}^{|V| \times r}$, $\mathbf{\Sigma} \in \mathbb{R}^{r \times r}$, and $\mathbf{Q}^{\mathrm{T}} \in \mathbb{R}^{r \times |D|}$. In this factorization, the matrices $\mathbf{P}$ and $\mathbf{Q}^{\mathrm{T}}$ are **semi-unitary** (or **semi-orthogonal**), meaning that the columns of both $\mathbf{P}$ and $\mathbf{Q}$ are mutually **orthonormal vectors** (orthogonal unit vectors). Consequently, the columns of $\mathbf{P}$ and $\mathbf{Q}$ form orthonormal bases for the column space and row space of $\mathbf{M}$, respectively, where r is the rank of $\mathbf{M}$. This guarantees that $\mathbf{M}$ is represented using the minimum necessary number of dimensions. The matrix $\mathbf{\Sigma}$ is diagonal:

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_r \end{bmatrix} \tag{3.27}$$

The diagonal entries $\{\sigma_1, ..., \sigma_r\}$ are all non-negative real numbers, and are called the **singular values** of $\mathbf{M}$. Typically, $\{\sigma_1, ..., \sigma_r\}$ are arranged in descending order (i.e., $\sigma_1 \geq \sigma_2 \geq ... \geq \sigma_r$). This ordering standardizes the decomposition. If we write $\mathbf{P}$ as a sequence of column vectors (call them **left-singular vectors**)

$$\mathbf{P} = \begin{bmatrix} \mathbf{p}_1, ..., \mathbf{p}_r \end{bmatrix} \tag{3.28}$$

[16] Latent semantic analysis is also called **latent semantic indexing** (**LSI**). This term is more often used in information retrieval and related fields.

and $\mathbf{Q}^{\mathrm{T}}$ as a sequence of row vectors (call them **right-singular vectors**)

$$\mathbf{Q}^{\mathrm{T}} = \begin{bmatrix} \mathbf{q}_1^{\mathrm{T}} \\ \vdots \\ \mathbf{q}_r^{\mathrm{T}} \end{bmatrix} \tag{3.29}$$

then we can write $\mathbf{M}$ as

$$\mathbf{M} = \sum_{i=1}^{r} \sigma_i \mathbf{p}_i \mathbf{q}_i^{\mathrm{T}} \tag{3.30}$$

To obtain word representations, we interpret the i-th row of $\mathbf{P}$ as an r-dimensional feature vector $\mathbf{e}_i$ for the i-th word a_i in the vocabulary. In other words, the vector representation of a_i is

$$\mathbf{e}_i = \begin{bmatrix} p_1(i) & \dots & p_r(i) \end{bmatrix} \tag{3.31}$$

Similarly, the vector representation of a document d_j can be written as

$$\mathbf{h}_j = \begin{bmatrix} q_1(j) & \dots & q_r(j) \end{bmatrix} \tag{3.32}$$

In this way, we have two separate representation models for words and documents: $\mathbf{P}$ deals with word representations and $\mathbf{Q}$ deals with document representations. Thus, we can take $\mathbf{M}$ as a product of these representation models

$$\begin{aligned} \mathbf{M} &= \mathbf{P}\boldsymbol{\Sigma}\mathbf{Q}^{\mathrm{T}} \\ &= \begin{array}{cc} \text{words} & \begin{bmatrix} \mathbf{e}_1 \\ \vdots \\ \mathbf{e}_{|V|} \end{bmatrix} \end{array} \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_r \end{bmatrix} \begin{array}{c} \text{documents} \\ \begin{bmatrix} \mathbf{h}_1^{\mathrm{T}} & \dots & \mathbf{h}_{|D|}^{\mathrm{T}} \end{bmatrix} \end{array} \end{aligned} \tag{3.33}$$

In practice, the rank r is usually much smaller than $|V|$ and $|D|$. Thus, we have, for each word (or each document), a new representation whose dimensionality is much smaller than the representation contained in the co-occurrence matrix. A further improvement can make use of the r^* largest singular values (i.e., $\{\sigma_1, ..., \sigma_{r^*}\}$) and throw away the rest. As a consequence, we only keep the first r^* left-singular vectors and right-singular vectors in $\mathbf{P}$ and $\mathbf{Q}$ respectively. Here $r^* < r$ is a hyperparameter specifying the number of vectors in $\mathbf{P}$ and $\mathbf{Q}$, i.e., the number of features used to describe a word or a document. In this way, we have a new factorization of $\mathbf{M}$ as

$$\mathbf{M} \approx \sum_{i}^{r^*} \sigma_i \mathbf{p}_i \mathbf{q}_i^{\mathrm{T}} \tag{3.34}$$

The right hand side of Eq. (3.34) is also known as a **low-rank approximation** of $\mathbf{M}$. By specifying r^*, it can approximate $\mathbf{M}$ with a matrix having an arbitrary rank $< r$.

There are a number of algorithms for implementing the SVD [Cline and Dhillon, 2014]. In fact, most of the modern implementations of the SVD are efficient and scalable. One can use them as off-the-shelf toolkits in NLP applications.

2. Principal Component Analysis

In data analysis, **principal component analysis** (**PCA**) is a widely-used technique for dimensionality reduction. Given a set of data points, PCA finds a sequence of orthogonal directions in the coordinate space so that the variance of the data points along these directions is maximized. These directions are typically represented as unit vectors, called **principal component loadings** or **principal component coefficients**. As a result, they form a new coordinate space to which we can map the given data points by an orthogonal linear transformation.

Consider a word-document co-occurrence matrix $\mathbf{M} \in \mathbb{R}^{|V|\times|D|}$, where each row is a $|D|$-dimensional word vector or feature vector. PCA defines a linear mapping from $\mathbb{R}^{|D|}$ to $\mathbb{R}^p$, that is, we transform each $|D|$-dimensional word vector into a p-dimensional word vector. This is given by

$$\mathbf{N} = \mathbf{MC} \tag{3.35}$$

where $\mathbf{N} \in \mathbb{R}^{|V|\times p}$ represents the matrix of mapped word vectorsover the vocabulary V, and $\mathbf{C} \in \mathbb{R}^{|D|\times p}$ is the matrix of the linear mapping. Then, we can write $\mathbf{C}$ as a sequence of column vectors

$$\mathbf{C} = \begin{bmatrix} \mathbf{c}_1 & ... & \mathbf{c}_p \end{bmatrix} \tag{3.36}$$

Each column vector $\mathbf{c}_i = \begin{bmatrix} c_i(1) \\ \vdots \\ c_i(|D|) \end{bmatrix}$ is a group of principal component coefficients, indicating a linear function that combines the input features into a new feature. For example, if we view $\mathbf{M}$ as the values of a bunch of feature functions (say, column vectors $\{\mathbf{m}_1, ..., \mathbf{m}_{|D|}\}$), we can map $\mathbf{M}$ to a new feature space in terms of $\mathbf{c}_i$:

$$\begin{aligned} \mathbf{Mc}_i &= \begin{bmatrix} \mathbf{m}_1 & ... & \mathbf{m}_{|D|} \end{bmatrix} \begin{bmatrix} c_i(1) \\ \vdots \\ c_i(|D|) \end{bmatrix} \\ &= \sum_{k=1}^{|D|} c_i(k)\mathbf{m}_k \end{aligned} \tag{3.37}$$

The product $\mathbf{Mc}_i$ (which forms the i-th column of $\mathbf{N}$) is a column vector where each entry represents the new feature value for a word in V. In PCA, the vectors $\{\mathbf{c}_1, ..., \mathbf{c}_p\}$ are generated sequentially to maximize the variance of the projected data. Thus, for each $i \in [1, p]$, the optimal principal component coefficients are defined as:

$$\begin{aligned} \hat{\mathbf{c}}_i &= \arg\max_{\mathbf{c}_i} \mathrm{Var}(\mathbf{M}\mathbf{c}_i) \\ &= \arg\max_{\mathbf{c}_i} \mathbf{c}_i^{\mathrm{T}} \mathbf{S} \mathbf{c}_i \end{aligned} \tag{3.38}$$

where $\mathrm{Var}(\mathbf{M}\mathbf{c}_i)$ is the variance of the projected data, and $\mathbf{S}$ is the $|D| \times |D|$ covariance matrix of $\mathbf{M}$. To ensure a well-defined solution, it is standard practice to constrain $\mathbf{c}_i$ to be a unit vector (i.e., $\mathbf{c}_i^{\mathrm{T}} \mathbf{c}_i = 1$). Using a Lagrange multiplier λ_i, this constrained optimization problem can be formulated by defining the Lagrangian:

$$\mathcal{L}(\mathbf{c}_i, \lambda_i) = \mathbf{c}_i^{\mathrm{T}} \mathbf{S} \mathbf{c}_i - \lambda_i (\mathbf{c}_i^{\mathrm{T}} \mathbf{c}_i - 1) \tag{3.39}$$

Solving for the stationary points of this Lagrangian reveals that $\hat{\mathbf{c}}_i$ must be an eigenvector of $\mathbf{S}$, with λ_i as its corresponding eigenvalue [Jolliffe, 2002]. Because $\mathbf{S}$ is a $|D| \times |D|$ symmetric matrix, it possesses exactly $|D|$ eigenvalues and corresponding mutually orthogonal eigenvectors. By ordering these eigenvectors according to their associated eigenvalues in descending order, we select the top p eigenvectors to form $\{\hat{\mathbf{c}}_1, ..., \hat{\mathbf{c}}_p\}$. In other words, $\hat{\mathbf{c}}_1$ is the eigenvector associated with the largest eigenvalue, $\hat{\mathbf{c}}_2$ with the second largest, and so forth. The projection $\mathbf{M}\hat{\mathbf{c}}_i$ is termed the i-th **principal component** of $\mathbf{M}$.

A geometric way to conceptualize PCA is as a coordinate transformation in Euclidean space. For a dataset $\mathbf{M}$, each row represents the coordinates of a data point in a $|D|$-dimensional space A. PCA seeks to represent these data points in a new, p-dimensional coordinate system B. The i-th dimension of this new system is defined by the direction of the unit vector $\mathbf{c}_i$. To find the i-th coordinate of a data point in B, we project its original coordinates onto the line spanned by $\mathbf{c}_i$. The optimal direction $\mathbf{c}_i$ is chosen to maximize the spread (variance) of these projected points. Essentially, we seek the axis along which the data points are most dispersed. Through this process, we generate a sequence of principal component coefficients by successively solving Eq. (3.38). Figure 3.5 illustrates this concept by projecting 2-dimensional data onto a 1-dimensional subspace.

In real-world applications, p is typically chosen to be much smaller than $|D|$, allowing PCA to significantly reduce the dimensionality of the word representations. Note that PCA is a very general method and is found to be useful in many disciplines. More broadly, if $\mathbf{M}$ represents observations over a set of variables, PCA can condense these observations into a smaller set of uncorrelated variables (the principal components).

3. Other Methods

Dimensionality reduction is a fundamental problem in machine learning and has been generalized in several directions. For instance, the neural word embedding models described in Sections 3.4 and 3.5 inherently learn low-dimensional, real-valued word vectors from text. Here we present some of the dimensionality reduction methods one may come across in the NLP and machine learning literature.

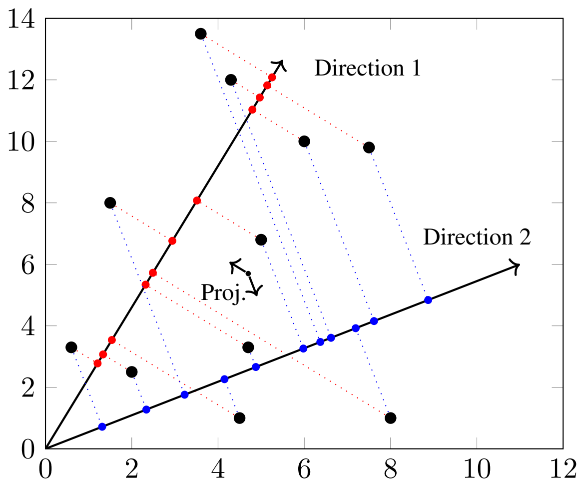


Figure 3.5: Transforming 2-dimensional data to 1-dimensional data via PCA. Consider a set of data points (represented by black circles) on a Euclidean plane. By using PCA, we find a direction (represented by an arrow) such that the variance of the projected data (represented by colored circles) in this direction is maximized. Such a direction can be represented by a unit vector of principal component coefficients. In this example, the principal component coefficients describe a 1-dimensional coordinate space. We can map the data from the 2-dimensional coordinate space to the 1-dimensional coordinate space via linear transformation. The mapped data is called the principal component of the original data points.

- **Topic models**. Technically, topic models are not ways of dimensionality reduction, but tools for describing how documents and words are generated based on distributions over topics [Blei, 2012]. For example, **latent Dirichlet allocation** (**LDA**) models the generation of a document by using document-topic and topic-word distributions [Blei et al., 2003]. As a by-product, we obtain a distribution over words for each topic, indicating the likelihood of a word occurring given a specific topic. If we write all these topic-word distributions as a $|V| \times K$ matrix where $|V|$ is the number of words and K is the number of topics, we obtain word representations conceptually similar to those described in previous sections. Typically, K is a relatively small number (e.g., 200). In this case, we have a low-dimensional model for representing words. Although LDA is not so popular in learning word representations in NLP applications, it offers a way to represent words as distributions over latent thematic structures.
- **Auto-encoders**. Undercomplete auto-encoders are a type of neural model that encodes features into low-dimensional codes such that the input features can be reconstructed from the codes. An advantage of auto-encoders is that they do not make assumptions on the hidden structures of the features. Thus, auto-encoders learn to transform any type of data into low-dimensional representations. For example, in Chapter 7 we will see examples of applying auto-encoders to learn sentence representations. For more details about auto-encoders, the reader can refer to Chapter 2.
- **Supervised dimensionality reduction**. Traditional dimensionality reduction methods

(such as PCA) operate in an unsupervised manner. However, when target task labels are available, it is advantageous to leverage this supervision signal. A classic example is supervised dimensionality reduction for classification. In methods like **Fisher's linear discriminant** and **linear discriminant analysis**, high-dimensional data is mapped into a lower-dimensional space that maximizes the separation between predefined classes. This concept is generalized to multi-dimensional target spaces via Canonical Variates analysis [Barber, 2012].

- **Feature selection**. Feature selection is the process of identifying a subset of the most relevant features to represent an object, thereby explicitly reducing the representation's dimensionality. It is a broad topic in machine learning involving many methods [Guyon and Elisseeff, 2003; Liu and Motoda, 2012]. The simplest is to frame it as a search problem: we search in the space of feature subsets so that the selected features maximize (or minimize) some objective. In general, the design of the objective depends on the task where we apply the features. This tight coupling makes feature selection challenging, as it requires balancing many factors, such as the performance measure of the target task, the search efficiency, and the representation of each feature subset. Note that feature selection is generally discussed in supervised learning, which requires labeled data to compute loss for optimization. The reader is referred to Solorio-Fernández et al. [2020]'s review paper for unsupervised feature selection methods.

In statistics, many methods can fall under the dimensionality reduction framework and are related to what we discussed in this section. For example, **factor analysis** is a method similar to PCA because they both seek a linear mapping from the input variables to a smaller number of new variables. The difference between them is that factor analysis focuses on modeling the common variance of variables, while PCA focuses on maximizing the variance of the projected data. Another example is **independent component analysis** (**ICA**). Unlike PCA, the goal of ICA is to find independent components that are additively separable. More examples can be found in machine learning and statistics textbooks [McClave and Sincich, 2006; Freedman et al., 2007; Barber, 2012].

3.4 Inducing Word Embeddings from NLMs

Counting word-word or word-document occurrences is a straightforward method for representing words based on their distributional properties in a corpus. While this method is effective in many applications, it imposes a strict constraint on word representations: the entries of a word vector must be interpretable as explicit evidence of how the word distributes across different contexts. Ideally, we would like to represent words in a more generalized form — namely, as dense, real-valued vectors (commonly referred to as **word embeddings**) without enforcing predefined constraints or assumptions on the specific meaning of each vector entry.

Learning word vectors without constraints comes at a cost. Unlike the count-based methods presented in Section 3.3, we do not rely on heuristics or prior statistical knowledge to estimate the vector values. Instead, we wish to induce meaningful representations directly from data.

A primary challenge here is the absence of a ground-truth gold standard, as it is impractical to manually annotate high-dimensional, continuous representations for an entire vocabulary. Thus, we frequently frame the induction of word vectors as an auxiliary objective within a well-defined task (often referred to as a **background task**). The learned word vectors emerge as a valuable byproduct of training on this background task.

A common example is the induction of word vectors from neural language models (NLMs). Recall the NLM described in Chapter 2. Its goal is to predict the probability of a word given its preceding words [Bengio et al., 2003a]. More formally, let w_i be the word we want to predict, and $\{w_{i-n+1},...,w_{i-1}\}$ be the context words we have seen. First, the context words $\{w_{i-n+1},...,w_{i-1}\}$ are projected into d_e-dimensional dense vectors $\{\mathbf{e}_{i-n+1},...,\mathbf{e}_{i-1}\}$ via an embedding layer. Assuming $\mathbf{1}_{w_j}$ is the one-hot representation of word j (a row vector of size $|V|$), the dense word vector $\mathbf{e}_j$ is given by:

$$\mathbf{e}_j = \mathbf{1}_{w_j}\mathbf{C} \tag{3.40}$$

where $\mathbf{C} \in \mathbb{R}^{|V|\times d_e}$ is the parameter of the embedding layer. $\mathbf{C}$ is often known as the word embedding table in which the k-th row is the representation of the k-th word in V.

Then, we use a feed-forward neural network to compute the probability distribution of the word at position i. This is given by

$$\Pr(\cdot|w_{i-n+1},...,w_{i-1}) = F_\theta(\mathbf{e}_{i-n+1},...,\mathbf{e}_{i-1}) \tag{3.41}$$

where $F_\theta(\cdot)$ is a feed-forward neural network parameterized by θ. Typically, the embedding layer can be seen as a component of the NLM. Here we use slightly different notation to emphasize that the NLM is a function of both θ and $\mathbf{C}$

$$\Pr_{\theta,\mathbf{C}}(\cdot|w_{i-n+1},...,w_{i-1}) = F_{\theta,\mathbf{C}}(w_{i-n+1},...,w_{i-1}) \tag{3.42}$$

For training, we optimize both θ and $\mathbf{C}$ to minimize a loss function. A popular method is maximum likelihood training which maximizes the sum of log-likelihood over all n-grams in the data. Given a sequence of words $w_1...w_m$, the objective of the training is defined to be[17]

$$(\hat{\theta},\widehat{\mathbf{C}}) = \arg\max_{\theta,\mathbf{C}} \sum_{i=n}^{m} \log \Pr_{\theta,\mathbf{C}}(w_i|w_{i-n+1},...,w_{i-1}) \tag{3.43}$$

Having obtained the optimized parameters $\hat{\theta}$ and $\widehat{\mathbf{C}}$, we can apply $F_{\hat{\theta},\widehat{\mathbf{C}}}(\cdot)$ to deal with unseen n-grams. More importantly, we have some well-trained word vectors (i.e., $\widehat{\mathbf{C}}$) that can be used in systems other than NLMs. This is also known as the pre-training of word vectors. In pre-training, we can define $F_{\theta,\mathbf{C}}(\cdot)$ as any system that makes use of the word vectors $\mathbf{C}$. Thus, the task of learning $\mathbf{C}$ is transformed to the task of optimizing $F_{\theta,\mathbf{C}}(\cdot)$ on the background task (see Figure 3.6 for an illustration). The main advantage of this method is that we can reuse existing NLP tasks to train the word vectors. A risk here is that the "best" word vectors found

[17]This can be generalized to a data set consisting of multiple sequences.

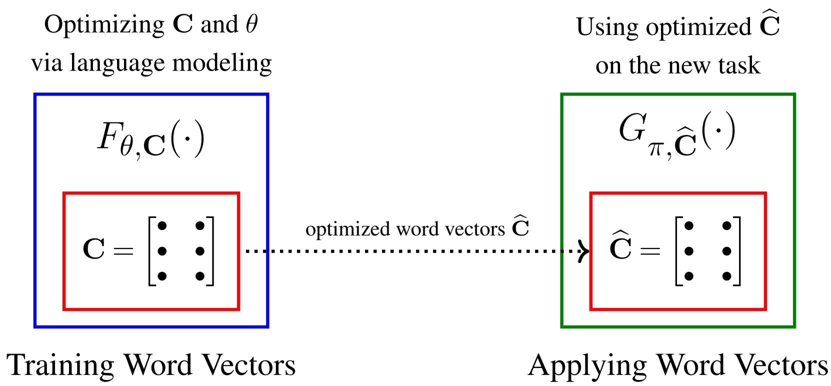


Figure 3.6: Illustration of pre-training word vectors in an NLM. The NLM can be denoted as a function $F_{\theta,\mathbf{C}}(\cdot)$ of the word embedding table (i.e., $\mathbf{C}$) and other parameters of the NLM (i.e., θ). The pre-training of $\mathbf{C}$ is essentially a process of training $F_{\theta,\mathbf{C}}(\cdot)$ on a background task. The outcome is the optimized word vectors $\widehat{\mathbf{C}}$ which are then applied to a new system $G_{\pi,\widehat{\mathbf{C}}}(\cdot)$ that might be different from the NLM. In the new system, $\widehat{\mathbf{C}}$ is the word embedding table learned from the NLM and π represents the parameters specialized to $G(\cdot)$.

in training $F_{\theta,\mathbf{C}}(\cdot)$ might not be well suited for the system where the word vectors are in actual use. Interestingly, in many situations, word vectors that are pre-trained by NLMs are of good quality for downstream tasks, or at least provide a good starting point for further tuning of these word vectors in the target system.

3.5 Word Embedding Models

In principle, word vectors can be learned in any manner. Treating word vectors as components of existing NLP systems is one option, but this approach typically lacks task-specific considerations. Another option is to develop methods specifically tailored to the problem of learning representations. Because the training of such systems does not need to satisfy the strict constraints of standard downstream NLP tasks, inducing high-quality word vectors becomes significantly more tractable.

3.5.1 Word2Vec

Word2Vec refers to a suite of models proposed in [Mikolov et al., 2013a;c]. As with neural language models, Word2Vec models are built upon neural networks. However, rather than resorting to the heavy autoregressive generative modeling of n-grams, Word2Vec frames the learning of word vectors in a computationally efficient log-linear fashion. Consequently, the architectures of these models differ markedly from those used in traditional language modeling. There are two primary architectures in the Word2Vec family:

- **The continuous bag-of-words model** (or **the CBOW model**). The CBOW model is a word prediction model. It is used to predict how likely a word at position i occurs given the $-n$ and $+n$ word windows around it. The structure of the CBOW model is similar to that of the neural language model introduced in Chapter 2 (see Figure 3.7 (a)). First, we use an embedding layer to transform the context words $w_{i-n}...w_{i-1}$ and $w_{i+1}...w_{i+n}$ to corresponding word vectors. This is performed by multiplying the one-hot representation of each input word w_j with the embedding table $\mathbf{C} \in \mathbb{R}^{|V| \times d_e}$, as shown in Eq. (3.40). These word vectors are then averaged to produce a single representation for the input words, giving us

$$\mathbf{h} = \frac{1}{2n}\left(\sum_{j=i-n}^{i-1} \mathbf{1}_{w_j}\mathbf{C} + \sum_{j=i+1}^{i+n} \mathbf{1}_{w_j}\mathbf{C}\right) \tag{3.44}$$

Note that the above defines a model that completely ignores the order of input words because of the use of the sum operation. This explains why the CBOW model is called *bag-of-words*. The output layer of the CBOW model is a standard Softmax layer that projects $\mathbf{h}$ to a probability distribution over the vocabulary

$$\mathbf{y} = \mathrm{Softmax}(\mathbf{hU} + \mathbf{b}) \tag{3.45}$$

where $\mathbf{U} \in \mathbb{R}^{d_e \times |V|}$ is the parameter matrix of the linear mapping and $\mathbf{b} \in \mathbb{R}^{|V|}$ is the bias term. $\mathbf{y}$ is a distribution over the vocabulary, and $\Pr(w_i|w_{i-n},...,w_{i-1},w_{i+1},...,w_{i+n}) = y(w_i)$. Eqs. (3.44-3.45) describe a very simple neural network. An advantage is that the resulting model is small and efficient as compared to NLMs. Training the CBOW model is straightforward. We can frame it as finding the maximum likelihood estimation of the parameters of the model. For simplicity, let θ denote the parameters other than $\mathbf{C}$ (i.e, $\theta = \{\mathbf{U}, \mathbf{b}\}$). We have

$$(\hat{\theta}, \widehat{\mathbf{C}}) = \arg\max_{\theta, \mathbf{C}} \sum_{i=n+1}^{m-n-1} \log \Pr_{\theta,\mathbf{C}}(w_i|w_{i-n},...,w_{i-1},w_{i+1},...,w_{i+n}) \tag{3.46}$$

where m is the length of the word sequence. After training, we can simply drop $\hat{\theta}$ and use $\widehat{\mathbf{C}}$ as a word vector look-up table.

- **The continuous skip-gram model** (or **the skip-gram model**). The skip-gram model is another word prediction model. It models the reverse of the task described in Eqs. (3.44-3.45). To be more precise, our objective is to predict each of the $\pm n$ context words given w_i. This is generally framed as estimating the probability of w_j occurring given w_i ($i-n \le j \le i-1$ or $i+1 \le j \le i+n$). Figure 3.7 (b) shows the structure of the skip-gram model. The embedding layer deals with w_i as usual. The representation of w_i is given by

$$\mathbf{h} = \mathbf{1}_{w_i}\mathbf{C} \tag{3.47}$$

It is then passed to a Softmax layer to predict the probability for each context word w_j

(assuming $j = i + k$)[18]

$$\mathbf{y}_k = \mathrm{Softmax}(\mathbf{h}\mathbf{V}_k + \mathbf{b}_k) \quad (3.48)$$

where $\mathbf{V}_k$ and $\mathbf{b}_k$ are the parameters of the model ($-n \le k \le -1$ and $1 \le k \le n$). We have

$$\begin{aligned} \Pr(w_j|w_i) &= \Pr(w_{i+k}|w_i) \\ &= y_k(w_{i+k}) \end{aligned} \quad (3.49)$$

Let θ be a short representation of $\{\mathbf{V}_k\}$ and $\{\mathbf{b}_k\}$. The training problem can be defined as

$$(\hat{\theta}, \widehat{\mathbf{C}}) = \arg\max_{\theta,\mathbf{C}} \sum_{i=n+1}^{m-n-1} \sum_{\substack{-n \le k \le -1, \\ 1 \le k \le n}} \log \Pr_{\theta,\mathbf{C}}(w_{i+k}|w_i) \quad (3.50)$$

Both of the above models make an analogy to cloze tests by considering only the pairwise dependency between words. Because they ignore word order and complex syntactic relationships, their predictive distributions are less precise compared to language models. Note, however, that the goal of these models is not to precisely predict missing words given their contexts, but to learn word representations from some task that captures word-word relationships. It is therefore not so important to care about the word prediction performance of the learned model.

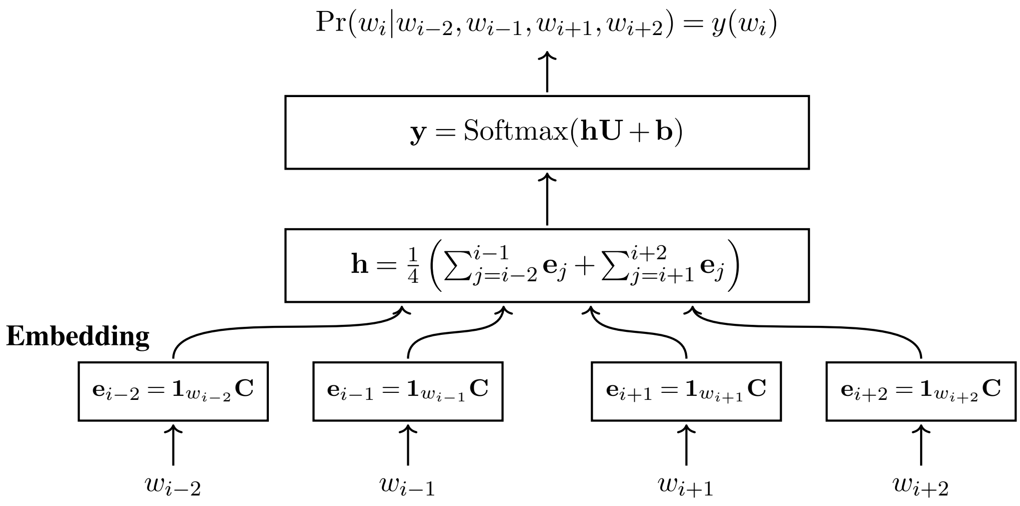
Another merit of these models is that they have very simple, easy-to-train architectures. For example, in both models there are no hidden layers and the embedding layer is directly connected to the output layer. These model structures can be seen as instances of log-linear modeling in machine learning: the input variables are linearly transformed into a dense feature vector (e.g., Eq. (3.44)), followed immediately by a log-linear classification function (e.g., Eq. (3.45)).

3.5.2 GloVe

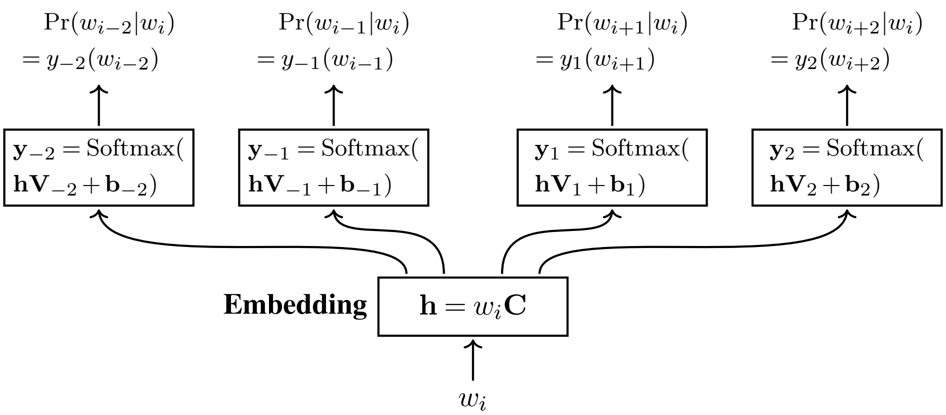
Global vectors, commonly known as **GloVe**, are word embeddings learned by combining the advantages of global matrix factorization and local context window methods [Pennington et al., 2014]. The GloVe method begins by constructing a global word-word co-occurrence matrix (as discussed in Section 3.3) and subsequently formulates a log-bilinear model based on a specific set of theoretical assumptions regarding word meanings.

Given a word-word co-occurrence matrix $\mathbf{M}$, where each cell $M(a,b) = \mathrm{count}(a,b)$ represents the number of times word $a \in V$ co-occurs with word $b \in V$ within a specified

[18]When $k > 0$, w_j is a word in the right context window of w_i; when $k < 0$, w_j is a word in the left context window.



(a) CBOW



(b) Skip-gram

Figure 3.7: The CBOW and skip-gram architectures. The CBOW model computes the probability $\Pr(w_i|w_{i-2},w_{i-1},w_{i+1},w_{i+2})$ where w_i is a word in a sequence and $\{w_{i-2},w_{i-1},w_{i+1},w_{i+2}\}$ are words in the ± 2 context windows. The context representation $\mathbf{h}$ is the mean of the word vectors that are produced through an embedding layer. $\mathbf{h}$ is then fed into a Softmax layer to output a distribution over the vocabulary (i.e., $\mathbf{y}$). The prediction probability of w_i is $\Pr(w_i|w_{i-2},w_{i-1},w_{i+1},w_{i+2}) = y(w_i)$. The skip-gram model is also based on the embedding + Softmax structure. It models the probability of each context word w_j given the word w_i. This is achieved by simply computing the output of a standard Softmax layer that takes the vector representation of w_i as input. Both the CBOW and skip-gram models are trained in a maximum likelihood fashion. The resulting lookup table of the embedding layer is the word vectors (or embeddings) for the words in the vocabulary.

context window, we can compute the conditional probability $\Pr(b|a)$ as follows:

$$\begin{aligned}\Pr(b|a) &= \frac{\text{count}(a,b)}{\sum_{b'}\text{count}(a,b')} \\ &= \frac{\text{count}(a,b)}{\text{count}(a)}\end{aligned} \tag{3.51}$$

where $\text{count}(a)$ represents the total number of times word a occurs in the corpus.

To understand the motivation behind GloVe, consider an example where we wish to distinguish between the thermodynamic concepts of *air* and *water*. Using Eq. (3.51), we can compute the probability of these target words occurring given various context probe words. The following table displays a small fraction of the $\Pr(b|a)$ matrix derived from a 3.8 billion-word corpus in WMT14:

Entry	$w = \textit{fly}$	$w = \textit{drink}$	$w = \textit{breath}$	$w = \textit{live}$	$w = \textit{flow}$
$\Pr(\textit{air}|w)$	1.5×10^{-4}	6.2×10^{-5}	2.2×10^{-4}	1.6×10^{-4}	3.6×10^{-4}
$\Pr(\textit{water}|w)$	1.3×10^{-5}	4.1×10^{-4}	1.8×10^{-5}	1.4×10^{-4}	3.0×10^{-4}
$\Pr(\textit{air}|w)/\Pr(\textit{water}|w)$	11.54	0.15	12.2	1.14	1.2

In this table, $\Pr(\textit{air}|w)$ and $\Pr(\textit{water}|w)$ indicate how strongly *air* and *water* correlate with different probe words w. The crucial insight arises when we examine the probability ratio $\Pr(\textit{air}|w)/\Pr(\textit{water}|w)$, shown in the last row. This ratio effectively isolates the semantic differences between the two target words. When w is highly relevant to *air* but not *water* (e.g., $w = \textit{fly}$ or $w = \textit{breath}$), the ratio is large. Conversely, when w is relevant to *water* but not *air* (e.g., $w = \textit{drink}$), the ratio is small. When w is relevant to both (e.g., $w = \textit{live}$ or $w = \textit{flow}$), or irrelevant to both, the ratio is close to 1.

The foundational premise of GloVe is that word vectors should explicitly encode this ratio of co-occurrence probabilities. A natural starting point is to posit a function F that takes word vectors as input and approximates this ratio:

$$F(\mathbf{e}_a, \mathbf{e}_b, \tilde{\mathbf{e}}_w) = \frac{\Pr(a|w)}{\Pr(b|w)} \tag{3.52}$$

where $\mathbf{e}_a, \mathbf{e}_b \in \mathbb{R}^{d_e}$ are the target vector representations of words a and b, and $\tilde{\mathbf{e}}_w \in \mathbb{R}^{d_e}$ is the context vector representation of the probe word w. Notice the distinction between $\mathbf{e}$ and $\tilde{\mathbf{e}}$: the former is drawn from a target embedding table $\mathbf{C}$, while the latter is drawn from a separate context embedding table $\widetilde{\mathbf{C}}$. Utilizing two distinct embedding tables helps mitigate overfitting during training. Ultimately, the final word representation is typically taken as the sum (or average) of the two: $\mathbf{e}_{\text{final}} = \mathbf{C} + \widetilde{\mathbf{C}}$.

Because vector spaces are inherently linear, the most natural way to encode the relationship between the target words a and b is via vector subtraction: $\mathbf{e}_a - \mathbf{e}_b$. Furthermore, to quantify the relationship between this difference and the context word w, we can employ the vector dot product. This restricts the function F to take a single scalar input:

$$F\left((\mathbf{e}_a - \mathbf{e}_b)\tilde{\mathbf{e}}_w^\top\right) = \frac{\Pr(a|w)}{\Pr(b|w)} \tag{3.53}$$

To enforce a principled algebraic structure, we require F to be a homomorphism between the additive group in the vector space and the multiplicative group in the probability space. This ensures that vector subtraction maps directly to probability division:

$$F\left((\mathbf{e}_a - \mathbf{e}_b)\tilde{\mathbf{e}}_w^\top\right) = \frac{F(\mathbf{e}_a\tilde{\mathbf{e}}_w^\top)}{F(\mathbf{e}_b\tilde{\mathbf{e}}_w^\top)} \tag{3.54}$$

The unique continuous solution to Eq. (3.54) is the exponential function, $F(x) = \exp(x)$. Thus, we have:

$$\begin{aligned} F(\mathbf{e}_a\tilde{\mathbf{e}}_w^\top) &= \exp(\mathbf{e}_a\tilde{\mathbf{e}}_w^\top) \\ &= \Pr(a|w) \\ &= \frac{\mathrm{count}(a,w)}{\mathrm{count}(a)} \end{aligned} \tag{3.55}$$

Taking the logarithm of both sides yields:

$$\mathbf{e}_a\tilde{\mathbf{e}}_w^\top = \log\mathrm{count}(a,w) - \log\mathrm{count}(a) \tag{3.56}$$

An issue with Eq. (3.56) is that the term $\log\mathrm{count}(a)$ breaks the symmetry; exchanging the target word a and context word w (which corresponds to transposing the co-occurrence matrix) should leave the underlying geometry unchanged. To restore this symmetry, the $\log\mathrm{count}(a)$ term is absorbed into two learned bias terms, β_a and $\tilde{\beta}_w$:

$$\mathbf{e}_a\tilde{\mathbf{e}}_w^\top + \beta_a + \tilde{\beta}_w = \log\mathrm{count}(a,w) \tag{3.57}$$

Eq. (3.57) establishes the ideal relationship we want our learned embeddings to satisfy. To enforce this, we formulate a regression problem, aiming to minimize the squared difference between the predicted log co-occurrence and the actual log co-occurrence. The squared loss for a single word pair (a,w) is defined as:

$$L_{a,w} = \left(\mathbf{e}_a\tilde{\mathbf{e}}_w^\top + \beta_a + \tilde{\beta}_w - \log\mathrm{count}(a,w)\right)^2 \tag{3.58}$$

The total cost function over the entire vocabulary is the weighted sum of these individual losses:

$$L_{\mathrm{GloVe}} = \sum_{a,w\in V} f(\mathrm{count}(a,w))\cdot L_{a,w} \tag{3.59}$$

where $f(\mathrm{count}(a,w))$ is a weighting function designed to prevent rare word pairs from dominating the loss, while also capping the influence of overly frequent word pairs. Pennington et al. [2014] propose the following weighting function:

$$f(x) = \begin{cases} \left(\frac{x}{x_{\max}}\right)^{\alpha} & \text{if } x < x_{\max} \\ 1 & \text{otherwise} \end{cases} \tag{3.60}$$

where $x_{\max}$ and α are hyperparameters (typically, $x_{\max} = 100$ and $\alpha = 3/4$). As such, $f(x)$ penalizes rare word-pairs by assigning them smaller weights, while ensuring that the weight for frequent pairs does not grow unbounded.

Eqs. (3.58-3.59) describe a highly efficient framework for learning word vectors using standard neural network operations (e.g., dot products and scalar additions). A defining characteristic of GloVe is its reliance on a simple log-bilinear objective. Because the training avoids the computationally expensive cross-entropy loss and softmax normalization required by full NLMs and Word2Vec, GloVe is remarkably fast to train. The core intuition is that semantic relationships can be modeled via simple regression on global corpus statistics, rather than relying strictly on probability-based divergence over local context windows.

Another note about the use of global data bears repeating. The co-occurrence matrix encodes information aggregated across the entire corpus. By directly factoring this matrix, GloVe explicitly frames learning as a global optimization problem. While Word2Vec also seeks a globally optimal set of parameters (e.g., maximizing the likelihood over the entire input space), its objective is complex and typically optimized via online learning (stochastic gradient descent on local batches). GloVe, by pre-aggregating the data into a co-occurrence matrix, distills the learning process into a more direct and computationally efficient global model.

3.5.3 Remarks

We have seen in the previous sections how word vectors can be induced using several different methods. We now discuss several important considerations when training or applying these word embeddings in practice.

1. Count-based vs Neural Network-based

The simplicity and interpretability of count-based methods have long been appreciated in the NLP community. Both count-based and neural network-based models fundamentally rely on the distributional hypothesis — the idea that words occurring in similar contexts possess similar meanings. However, count-based methods explicitly construct high-dimensional, sparse matrices based on this assumption, which often leads to data sparsity issues due to the curse of dimensionality.

At the other end of the spectrum are neural network-based models. Rather than relying on explicit co-occurrence counting and manual weighting schemes, these methods treat word vectors as low-dimensional, dense intermediate states within a neural network optimized for a specific background task. By removing explicit linguistic constraints on the specific meaning of each vector dimension, neural network-based models effectively learn latent semantic features that are difficult to manually engineer.

The contrast between these two approaches is similar to the broader comparison of two well-known machine learning paradigms: feature engineering vs. end-to-end representation learning. However, we will not delve deeper into this philosophical debate here. It is simply worth noting that there is no absolute superiority of one specific form over the other. In general, the choice of word vector type depends entirely on the target downstream application and the desired level of interpretability. For example, if the goal is to obtain an explicitly interpretable, mathematically transparent computable form directly from a corpus, factorizing a co-occurrence matrix remains a robust choice. Conversely, if the objective is to extract continuous, dense word vectors intended for seamless integration into a deep neural architecture, neural representation learning is typically preferred.

It should be noted that learning continuous, dense word embeddings has become the dominant paradigm in recent years, demonstrating massive advancements in neural models for NLP. Concurrently, there has been significant academic interest in comparing count-based and neural network-based methods, and revealing the profound theoretical relationships between them [Levy and Goldberg, 2014b; Baroni et al., 2014; Levy and Goldberg, 2014c; Schnabel et al., 2015a; Levy et al., 2015; Gladkova et al., 2016].

2. Shallow Models vs. Deep Models

While neural networks have become the standard approach for inducing word vectors, the model structures introduced thus far in this chapter are simple. Technically, they consist of only one or two neural layers and are therefore categorized as shallow models. A notable example is the **vLBL** word embedding model [Mnih and Kavukcuoglu, 2013], which models the interactions among words using a strictly two-layer neural network. This architecture, which omits even a Softmax normalization function, is one of the simplest word embedding models to the best of our knowledge. Despite such simplicity, these models remain highly effective in many scenarios. A primary benefit of shallow models is their computational efficiency and scalability to massive datasets. This efficiency allows them to scale to more complex NLP tasks. A prominent example is the **fastText** system for text classification [Joulin et al., 2017]. It shares a similar architecture with the CBOW model (see Section 3.5.1). In fastText, the input text is represented as a bag of word vectors that are averaged to form a global hidden representation of the text. This is followed immediately by an output layer that maps the hidden representation to a probability distribution over predefined classes. By using this mechanism, the classification objective and the underlying word vectors are trained jointly.

Although shallow models are remarkably effective for learning static word representations, deep neural networks offer significantly greater representational capacity. As with most multi-layer architectures, learning representations via deep neural networks offers several distinct advantages [Telgarsky, 2016]. First, deep models allow us to explore a much larger hypothesis space, potentially discovering superior functional mappings. Second, the injection of multiple non-linear transformations dramatically increases the model's ability to capture highly complex patterns in the data. There are many strategies for leveraging depth in representation learning. The most straightforward approach is to incrementally stack feed-forward, recurrent, or convolutional layers on top of a base embedding layer. With the widespread adoption of large-scale pretrained models in NLP, the paradigm has shifted toward extracting contextualized representations directly from very deep architectures, achieving state-of-the-art performance across a vast array of NLP benchmarks [Radford et al., 2018; Devlin et al., 2019]. However, this increased modeling power introduces severe computational bottlenecks and training instabilities. Consequently, a substantial body of research focuses on mitigating these optimization challenges [Pascanu et al., 2013; Bapna et al., 2018; Wang et al., 2019a; Zhang et al., 2019a; Pham et al., 2019; Li et al., 2020b]. In Chapters 4-6, we will explore several successful NLP models built upon these deep neural architectures.

3. Training Objectives

The idea of treating word vectors as model parameters aligns well with latent-variable modeling: a model is parameterized by learnable word vectors, and their optimal values are inferred by optimizing a global objective function. While this learning paradigm is standard, the specific training objective can vary significantly. A primary difficulty is the absence of an explicit objective dedicated solely to supervising word vector representations. A straightforward solution is to leverage well-defined NLP tasks. For instance, we can use word vectors as the input representations for language modeling or text classification systems. As a result, the word vectors serve as standard model parameters and are optimized end-to-end.

An alternative approach is to design novel training objectives. As is typical in machine learning, the design space for such objectives is vast. A common strategy, therefore, is to adapt existing tasks. For example, the training objective of the CBOW model is essentially a generalized word prediction problem and shares a similar mathematical form with traditional language modeling.

We will explore several novel objectives derived from language modeling in Chapter 7. In a strict sense, these training tasks are not inherently about learning word vectors. Rather, they provide a well-defined surrogate objective for representation learning. Furthermore, we typically cannot assume a purely supervised learning scenario: the learned embeddings will ultimately be deployed across diverse downstream systems unknown during the initial training phase. This frames the induction process as a form of self-supervised learning, lacking direct supervision from the ultimate target application. However, when the target application is accessible and labeled task-specific data is available, we can further fine-tune these pre-trained word embeddings to capture domain-specific semantics.

3.6 Evaluating Word Embeddings

Having obtained vector representations of words, we must assess their quality. Ideally, we would evaluate these word vectors against a gold standard. Unfortunately, such a gold standard rarely exists, as it is impractical for human annotators to directly assign high-dimensional, continuous numerical values to capture the semantics of a word. A practical solution is to evaluate the vectors based on the performance of a downstream working system that utilizes them. Broadly, evaluation approaches fall into two categories [Schnabel et al., 2015b]:

- **Extrinsic Evaluation** (or end-to-end testing). We incorporate the word vectors into a downstream NLP system and measure their impact on the system's overall performance.
- **Intrinsic Evaluation**. We test the ability of the word vectors to capture specific morphological, syntactic, and semantic properties of language.

Below, we briefly describe how these approaches are applied to word vector evaluation.

3.6.1 Extrinsic Evaluation

This approach is widely adopted in practice, as it allows researchers and engineers to quickly examine how a real-world system behaves when its underlying representations are modified. Since many NLP systems take words as input, it is standard practice to replace symbolic word representations with dense word vectors. We have already seen systems of this kind, which typically employ an embedding layer to transform the one-hot representation of each input word into a real-valued vector (e.g., the neural language model in Chapter 2).

Given such a system and a set of pre-trained word vectors, we can use the downstream performance of the system as a proxy measure for the quality of the embeddings. Depending on how the word vectors are utilized, there are two primary training paradigms:

- **Word Vectors as Fixed Parameters**. We freeze the word vectors and only train the remaining parameters of the system as usual.
- **Word Vectors as Initial Parameters** (Fine-Tuning). We initialize the embedding layer with the pre-trained word vectors and update them alongside the rest of the model's parameters during training.

Both methods leverage the concept of pre-training, and can be extended to many scenarios where part of a model is trained extensively before being applied to a downstream task. Freezing word vectors simplifies the training process and isolates the evaluation of the representations themselves. In contrast, treating the word vectors as learnable parameters increases training complexity but allows the model to fine-tune the representations, adapting them specifically to the target working system.

Note that while extrinsic evaluation is highly practical, its results are heavily dependent on the specific downstream architecture and task. Because developing state-of-the-art NLP systems involves complex architectures and extensive hyperparameter tuning, performance variations cannot always be attributed to the quality of the word embeddings alone. This task dependency is common in NLP. For instance, a tokenization method that excels in a machine translation system might perform poorly in an information retrieval system. Therefore, to rigorously test the generalizability of a given set of word vectors, it is standard practice to evaluate them across a diverse suite of downstream NLP tasks.

3.6.2 Intrinsic Evaluation

Although much of word representation research involves end-to-end tests in NLP applications, it also involves examining the ability of the representation to deal with certain problems, such as interpreting the relationship between two words. There are many ways to design intrinsic evaluation, each addressing a specific problem. In the following we describe some of these methods. For a more comprehensive survey of intrinsic evaluation methodologies, we refer the reader to the extensive literature on this subject [Baroni et al., 2014; Bakarov, 2018; Rogers et al., 2018].

1. Semantic Relatedness

Modeling the relatedness between words is perhaps the most prevalent method for evaluating the quality of word vectors in NLP [Reisinger and Mooney, 2010; Huang et al., 2012; Baroni et al., 2014]. Fundamentally, this involves computing a spatial distance between words (referred to as **word semantic distance** or simply **word distance**). The motivation is that the geometric distance in the continuous vector space should correlate with human cognitive judgments of semantic relatedness [Rubenstein and Goodenough, 1965]. For example, we expect the vector for *dog* to be geometrically proximal to *wolf*, while *peach* should be distant from *television*. Mathematically, many metrics can be used to calculate the distance or angle between two vectors. A simple and commonly used measure is the Euclidean distance. Alternatively, computing the cosine similarity between two vectors yields a score in the interval $[-1,1]$[19].

In this evaluation setup, we are provided with a dataset of word pairs, each assigned an expected relatedness score by human annotators. Given a pair of words, we correlate this expected human score with the geometric distance measured in the word vector space. The quality of the word vectors is reflected by the strength of this correlation. A primary challenge, however, is the absence of an absolute gold-standard metric for semantic distance. Even human annotators struggle to quantify semantic proximity with absolute numerical precision. A practical alternative is to map the distance into discrete categories or Likert-style rating scales, such as integers in $[1,5]$ [Reisinger and Mooney, 2010]. This significantly reduces the cognitive load during data annotation. Another evaluation variant requires the model to identify the most similar word to a given target from a small set of candidates (e.g., synonym tests). This avoids predicting absolute values. Instead we only need a mechanism to rank relative similarities [Baroni et al., 2014].

Evaluating relationships between words, however, can be ambiguous due to the fluid nature of language. In general, multiple linguistic factors affect human judgments of relatedness [Faruqui et al., 2016]. For example, *corn* and *cornea* exhibit morphological similarity due to subword overlap, yet they are semantically disjoint. Ambiguity also stems from the very definitions of the tasks. **Semantic relatedness** and **semantic similarity are** often used interchangeably, yet they denote distinct linguistic concepts. For instance, *car* is topically related to *road*, but in a strict taxonomic sense, *car* is similar to *van*.

Furthermore, the meaning of a word is generally context-dependent, and thus confound the evaluation of words with multiple distinct senses (**polysemy**). Broadly speaking, this is an inherent limitation of static word vector models, where every word type is deterministically mapped to a single, fixed vector. To address this, subsequent chapters will introduce contextualized word embeddings, which dynamically generate a distinct computable form for a word

[19] In practice, it is common to convert this into a *cosine distance* (defined as $1-\cos(\theta)$), where a distance of 0 indicates two vectors pointing in the exact same direction, and 1 indicates orthogonal vectors.

based strictly on its surrounding text.

2. Word Analogy

The word analogy task is concerned with modeling analogical relations between pairs of words. This task is motivated by the premise that semantic relationships between words can be captured via simple algebraic operations on their corresponding continuous representations. A canonical example presented by Mikolov et al. [2013d] demonstrates that relational semantics can be described by vector subtraction. Specifically, subtracting the vector for *man* from *king* and adding the vector for *woman* yields a vector remarkably close to that of *queen*:

$$\mathbf{e}_{\text{king}} - \mathbf{e}_{\text{man}} + \mathbf{e}_{\text{woman}} \quad \approx \quad \mathbf{e}_{\text{queen}} \tag{3.61}$$

Formally, word analogy is the task of comparing two word pairs, (a, a^*) and (b, b^*). An analogy holds if the semantic relationship between a and a^* mirrors that between b and b^*. This reflects underlying **linguistic regularities** in the vector space, which can be algebraically formulated as:

$$\mathbf{e}_{a^*} - \mathbf{e}_a \quad \approx \quad \mathbf{e}_{b^*} - \mathbf{e}_b \tag{3.62}$$

Word analogy can thus be framed as an analogical reasoning task: the objective is to predict $\mathbf{e}_{b^*}$ given $\mathbf{e}_a$, $\mathbf{e}_{a^*}$, and $\mathbf{e}_b$. More specifically, we expect the vector $\mathbf{e}_{a^*} - \mathbf{e}_a + \mathbf{e}_b$ to closely approximate $\mathbf{e}_{b^*}$ if (a, a^*) and (b, b^*) share similar linguistic relations. There are many ways to refine this formulation. For example, rather than merely optimizing the spatial distance to predict $\mathbf{e}_{b^*}$, one can evaluate the cosine angle between the difference vectors $\mathbf{e}_{a^*} - \mathbf{e}_a$ and $\mathbf{e}_{b^*} - \mathbf{e}_b$ to capture relational similarity more robustly [Levy and Goldberg, 2014b].

The word analogy task provides a straightforward mechanism to examine the linear algebraic properties of a word vector model. Such characteristics are rarely observed in traditional count-based methods. Remarkably, modern word embedding models exhibit strong linear behavior even though such geometric constraints are not explicitly enforced during training. This emergent property provides researchers with insights into the latent structures learned by these models and their potential downstream applications [Levy and Goldberg, 2014b; Linzen, 2016; Allen and Hospedales, 2019]. However, word analogy accuracy is not a universally indicative metric. In many scenarios, it does not strongly correlate with the performance of downstream NLP systems. As a result, it often serves as an analytical tool for probing the structural and semantic properties of word representations.

3. Word Categorization (or Clustering)

Another method to evaluate whether word vectors align with human semantic understanding is to examine their capacity to be grouped into meaningful categories. This is typically achieved by applying standard clustering algorithms to the word vectors. Ideally, semantically similar words should be assigned to the same cluster, while dissimilar words are separated into distinct clusters. For instance, *apple*, *grape*, *peach*, and *orange* should naturally cluster together under the broader semantic category of fruits. An advantage of this evaluation approach is that many clustering algorithms and established word categorization benchmarks are readily applicable. However, as with most unsupervised clustering tasks, practical challenges must be addressed, such as predetermining the optimal number of clusters.

Because most clustering methods in machine learning rely on computing spatial distances between data points, word clustering is rooted in the concept of semantic relatedness. However, unlike the tasks discussed in Section 3.6.2, clustering evaluates the global structure rather than explicitly judging individual pairwise distance scores. While this highlights the intrinsic connections among different evaluation protocols, it also means that word clustering has the inherent limitations of these related methods. Most notably, it is difficult to establish an absolute gold-standard criterion for clustering quality, as semantic taxonomies are subjective, and words can be categorized in multiple ways depending on the chosen semantic perspective.

4. Subconscious Evaluation

The goal of subconscious (or cognitive) evaluation is to examine the correlation between the use of word vectors and human subconscious behaviors or brain activity during reading. A wide variety of psycholinguistic phenomena can serve as evaluation proxies [Mitchell and Lapata, 2010]. A well-known paradigm is **priming**, which investigates how a person responds to sequential stimuli [Schacter and Buckner, 1998; Tulving and Schacter, 1990; Wiggs and Martin, 1998]. For example, an experiment can be designed to measure the speed at which a participant reads a given word (referred to as the **target word**) immediately following another word (the **prime word**) [Meyer and Schvaneveldt, 1971; Lund, 1995; McNamara, 2005]. If a target word t is processed more rapidly following word a compared to word b, it indicates a stronger cognitive association between t and a. This psychological latency can then serve as a proxy to evaluate the semantic similarity between their corresponding word vectors. To measure reading latency, a popular methodology is the **self-paced reading** task[20]. An alternative is to use eye-tracking devices to automatically record precise information regarding eye movements and fixations. By using these techniques, many datasets and methodologies have been developed to study a variety of psycholinguistic issues [Mitchell and Lapata, 2010; Hutchison et al., 2013; Lapesa and Evert, 2014; Klerke et al., 2015; Søgaard, 2016; Auguste et al., 2017].

In addition to tracking behavioral responses during reading, we can directly monitor brain activity using neuroimaging techniques, such as functional magnetic resonance imaging (fMRI) and electroencephalography (EEG) [Devereux et al., 2010; Søgaard, 2016; Bhattasali et al., 2020]. For example, language comprehension elicits specific neural activation patterns in the brain. Therefore, we can attempt to correlate these physiological activation patterns directly with the computable form of word meaning provided by vector representations. However, a limitation is that our understanding of the underlying cognitive mechanisms driving these processes remains incomplete, making it difficult to sufficiently correlate the results of such psycholinguistic studies with the performance of real-world NLP systems [Baroni et al., 2014; Bakarov, 2018].

[20] In self-paced reading, the text is segmented into words or phrases, and the participant presses a button to advance the display segment by segment.

5. Linguistically Motivated Evaluation

Linguistically motivated evaluation is based on the assumption that data-driven word vectors should reflect the structural and semantic knowledge encoded in linguistic resources. An interesting approach to performing such evaluation is to align the word vectors with representations of dictionary entries [Tsvetkov et al., 2015; Acs and Kornai, 2016]. The quality of the word embeddings is then quantified by the correlation between these continuous vectors and the discrete linguistic representations[21]. Beyond standard dictionaries, word vectors can be evaluated against topological structures within semantic networks, such as WordNet. In this way, evaluation metrics can be enriched by employing graph-based algorithms to measure the structural alignment between the vector space and these semantic resources [Agirre et al., 2009].

3.6.3 Visualization

By treating word vectors as multi-dimensional data points, we can adopt dimensionality reduction techniques to project them onto a two- or three-dimensional space for visualization. Such projections enable us to visually analyze the semantic patterns encoded within these vectors and explore the latent relationships between words. One method is PCA which seeks a linear mapping from a high-dimensional space to a low-dimensional space (see Section 3.3.3). Another well-known method is **t-distributed stochastic neighbor embedding** (**t-SNE**) [Hinton and Roweis, 2002; Van der Maaten and Hinton, 2008]. It is a non-linear dimensionality reduction method, and has been widely used in visualizing high-dimensional data. Apart from these, one can consider the methods presented in Section 3.3.3 as well as those tailored for visualizing word vectors [Zhang et al., 2019b; Liu et al., 2017].

3.7 Summary

In this chapter we discussed two interesting problems in NLP: tokenization and word (or token) representation. First, we introduced models for dividing a sentence into units that are meaningful and/or well suited for downstream tasks. Second, we introduced the idea of word vector models with particular attention to learning both count-based high-dimensional models and real-valued low-dimensional models. While most of these models are simple, they are often used in complex NLP systems and form the basis of many advanced models, as will be shown in the following chapters.

Tokenization (or segmentation) is an important process in NLP, commonly as a preprocessing step for many applications [Webster and Kit, 1992]. However, the use of the term *tokenization* is slightly misleading as it originally refers to the basic computer science process of dividing a string into substrings. In modern NLP, tokenization draws on concepts from multiple sub-fields. Linguistically, it is deeply tied to two core questions: how words are morphologically composed and how they syntactically form sentences. It is therefore natural to use morphological and syntactic theories to define the basic units of a language, leading to many

[21] In this context, a linguistic representation is typically a handcrafted feature vector derived from a formal linguistic resource, such as a lexicon.

rule-based tokenization systems covering a variety of languages. From a machine learning perspective, tokenization has long been cast as a problem of learning token boundaries from data in either a supervised or unsupervised manner [Mielke et al., 2021]. A common approach is to first annotate some tokenized text with human knowledge about what basic language units should be, and then learn to tokenize on this annotated data (see Section 3.1.3). More recently, learning tokenizers without linguistic constraints has been found to be promising (see Section 3.1.4). Since natural languages are themselves sets of characters or byte sequences, it is also possible to segment a sentence into characters or bytes [Ling et al., 2015; Lee et al., 2017]. Tokenization-free methods in general help when one wants a language-independent tokenizer and a simpler pipeline for processing the text.

From a mathematical standpoint, tokenization can be thought of as a mapping from the input data to a sequence of variables. Thus, the concept of tokenization can be generalized by relaxing the assumption that both the input and output variables are constrained to discrete values. Recent image and speech processing systems, for example, transform continuous sensory data (such as pixels and acoustic signals) into sequences of vector-based "tokens" [Schneider et al., 2019; Dosovitskiy et al., 2021]. Some interesting extensions of these ideas are even to transform image and speech data to a sequence of discrete indices, thereby aligning computer vision and speech processing more closely with traditional NLP paradigms [Oord et al., 2017; Baevski et al., 2020; Hsu et al., 2021].

Having divided the input text into smaller segments, the natural next step is to represent these pieces in some way that captures their underlying features. While representing language units as dense numerical vectors has become the de facto standard in modern NLP, the work on vector representation dates back to the very early days of computational linguistics. According to many popular textbooks and papers [Manning and Schütze, 1999; Jurafsky and Martin, 2008], the idea of using a statistical distribution to represent word meaning, known as **distributional semantics**, emerged in the 1950s alongside the rise of empiricism. During that era, much of the research was heavily influenced by Harris's distributionalism [Harris, 1954] and related philosophical frameworks [Firth, 1957; Wittgenstein, 1953]. Concurrently, Osgood [1952] proposed defining the meaning of a concept psychologically, as a point within a multidimensional space. These ideas significantly shaped how linguists and NLP researchers conceptualized word meaning over the following decades.

Modern approaches to distributional semantics appeared in the 1990s, mainly as a result of the revival of empiricism in artificial intelligence [Church, 2011]. Most of these were driven by the distributional hypothesis: words having similar meanings are more likely to occur in similar contexts. In response, a number of methods were developed, differing in the way the contexts are modeled. For example, a context can be the words in a context-window, or the words with a relation to the given word in a syntax tree. Apart from those mentioned in Section 3.3, methods that are not covered in this chapter include hyperspace analogue of language (HAL) [Lund and Burgess, 1996], distributional memory [Baroni and Lenci, 2010], dependency-based semantic space models [Padó and Lapata, 2007], and so on. For comprehensive surveys of distributional semantics, the reader can refer to dedicated literature [Lenci, 2018; Mitchell and Lapata, 2010]. Note that most of the above-mentioned work can be thought of as instances of the vector space model which can deal with problems beyond lexical semantics. In compositional distributional semantics, for example, the meaning of a phrase or sentence is derived by performing systematic algebraic operations on its constituent word vectors [Clark et al., 2008; Mitchell and Lapata, 2010; Blacoe and Lapata, 2012].

While distributional models have attracted attention in the NLP community for many years, word embedding models that learn low-dimensional, real-valued word vectors directly from texts have been a predominant approach recently. As described in Sections 3.4-3.5, rather than explicitly constructing sparse co-occurrence matrices, these models implicitly leverage the distributional hypothesis by optimizing a vector of latent attributes to predict surrounding contexts. The resulting model is an extension of the feature-based semantic model [Markman, 2013]. A recognized difference with traditional feature-based methods is that we do not need to manually define the features. We instead take these features as parameters of the model, and train them in an end-to-end fashion.

Formulating word representation as an end-to-end learning problem brings with it several benefits. A primary advantage is that new features can be found because no constraints are placed on how these features are learned and interpreted. Furthermore, as demonstrated in Section 3.6.2, word vectors obtained in this way indeed show some linguistic properties, though the word embedding models are not trained to achieve this. Another benefit is that the word embedding models also fall within the broader vector space model paradigm, enabling the easy use of word vectors in various applications. There are also many examples of methods that attempt to improve standard word embedding systems. Researchers have successfully infused pre-trained embeddings with external linguistic knowledge [Levy and Goldberg, 2014a; Cotterell and Schütze, 2015; Tissier et al., 2017] and aligned vector spaces to induce universal representations in multiple languages [Klementiev et al., 2012; Mikolov et al., 2013b; Ammar et al., 2020; Smith et al., 2017; Artetxe et al., 2017].

Widely associated with neural models in NLP, the idea of distributed representation has been successfully generalized to structural units far beyond individual words. This includes learning dense representations for sentences [Le and Mikolov, 2014; Kalchbrenner et al., 2014; Kiros et al., 2015; Hill et al., 2016; Arora et al., 2017; Lin et al., 2017; Conneau et al., 2017b] as well as complex tree and graph structures [Socher et al., 2011; Perozzi et al., 2014; Tai et al., 2015; Grover and Leskovec, 2016]. In particular, contextualized representations of words, though not discussed in this chapter, are generally appreciated for modeling sequential data [McCann et al., 2017; Peters et al., 2018; Devlin et al., 2019].

Foundations

Chapter 4

Recurrent and Convolutional Sequence Models

The whole is more than the sum of its parts.

–Aristotle, 384-322 BC [Ross, 1924]

Aristotle might or might not have had linguistic phenomena in mind when he expressed this thought, but it is indeed what we want to convey in this chapter: a sentence or phrase is much more than a mere collection of words. While words carry meaning on their own, when they combine to form a sentence or phrase, the meaning of the whole can be significantly more complex and diverse. This leads to the most beautiful aspect of language: human beings can express an infinite variety of meanings using a finite set of elements (e.g., words or characters).

The infinite and highly context-dependent nature of language makes modeling a sequence of words far more difficult than modeling individual words. One challenge is that a word's sense can change drastically depending on its context. Building upon the concept of word embeddings, where a word is represented as a low-dimensional, real-valued vector, the "meaning" of a language unit can be modeled in a continuous space. It is therefore possible to extend methods of distributed representation from single words to entire sequences. This motivates the exploration of models that process variable-length word sequences in a fundamentally continuous manner.

Here, we explore the general approach to learning distributed representations of word sequences. Specifically, we consider recurrent and convolutional neural networks, which have been extensively used in fields ranging from speech processing to computer vision. For natural language inputs, applying these models yields a sequence-level representation. This representation can be either a single real-valued vector or a sequence of such vectors, each corresponding to a contextualized representation for a specific word within the input sequence. These representation models, often referred to as **encoders**, are commonly used in a variety of systems that process sequential data. We will see several examples of these in this chapter.

4.1 Problem Statement

For many NLP applications, our objective is to make predictions based on an input sequence. Let us consider again the text classification problem mentioned in Chapter 1. Given a text that may or may not discuss food, we want to assign one of two classes to it (e.g., Food or $\mathrm{Not\text{-}food}$). To do this, a common classification method is to represent the text as a bag of features, denoted as $\mathbf{H}$. Then, a probability is assigned to each of the classes using a probabilistic model $\Pr(y|\mathbf{H})$. The predicted class is the one that maximizes this probability $\hat{y} = \arg\max_y \Pr(y|\mathbf{H})$.

While this is a standard procedure for classification, the underlying concept describes a much broader problem. Formally, let $\mathbf{w} = w_1...w_m$ be a sequence of words[1]. A sequence-level NLP system can be formulated as a function that maps the input sequence $\mathbf{w}$ to some output $\mathbf{y}$. This process can be divided into two distinct steps: representation (or encoding) and prediction.

- **Representation** (or **Encoding**). This step transforms the input sequence $\mathbf{w}$ to a feature representation $\mathbf{H}$ using an encoder $\mathrm{Enc}(\cdot)$:

$$\mathbf{H} \quad = \quad \mathrm{Enc}(\mathbf{w}) \tag{4.1}$$

- **Prediction**. A predictor $\mathrm{Predict}(\cdot)$ takes $\mathbf{H}$ and generates the output:

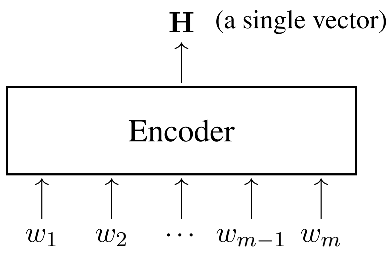
$$\mathbf{y} \quad = \quad \mathrm{Predict}(\mathbf{H}) \tag{4.2}$$

A simple form of $\mathbf{H}$ is a feature vector. For example, $\mathbf{H}$ could be a set of human-designed indicator features extracted from $\mathbf{w}$ (representing a high-dimensional sparse representation), or a set of real numbers indicating latent features (representing a low-dimensional dense representation). In NLP, another common form of $\mathbf{H}$ is a sequence of vectors, where each vector $\mathbf{h}_i$ corresponds to an input word w_i (see Figure 4.1). In this case, $\mathbf{h}_i$ can be viewed as a contextualized representation of both w_i and its surrounding context in $\mathbf{w}$[2]. The correspondence between $\mathbf{h}_i$ and w_i enables the representation to distinguish between different positions in the sequence and, more importantly, to dynamically adapt its modeling power for variable-length inputs.

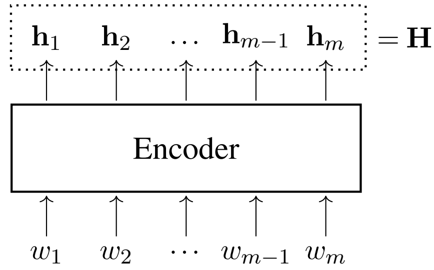
The form of $\mathbf{y}$ depends on the problem we intend to deal with. For example, in classification problems, $\mathbf{y}$ is the index of a class (or a distribution of classes); for regression problems, $\mathbf{y}$ is a real number; and in machine translation, $\mathbf{y}$ is a sequence of words in the target language. Note that, in this model, representation and prediction can be regarded as two separate problems. A major advantage of decoupling them is that the same encoder can be reused across multiple applications with different predictors. This also motivates a highly promising line of research where a general-purpose encoder is pre-trained on large-scale data and then used as a core component in various downstream systems [Peters et al., 2018; Devlin et al., 2019].

[1] Although we restrict our discussion to word sequences, these methods can handle sequences of any language units, such as sub-words or characters.

[2] This architecture can be extended to encoders where the input and output have different lengths. For example, the input is $w_1...w_m$ and the output is $\mathbf{h}_1...\mathbf{h}_n$ ($m \neq n$).



(a) Encoding the sequence as a vector



(b) Encoding the sequence as a vector sequence

Figure 4.1: Representing a word sequence as (a) a vector or (b) a sequence of vectors.

There are many possible implementations for $\mathrm{Enc}(\cdot)$ and $\mathrm{Predict}(\cdot)$. For example, in text classification, one way is to define $\mathrm{Enc}(\cdot)$ as a function that computes a feature vector using hand-crafted templates, while defining $\mathrm{Predict}(\cdot)$ as a traditional statistical classifier (e.g., an SVM or a maximum entropy model). Another way is to define $\mathrm{Enc}(\cdot)$ as a multi-layer neural network that outputs a real-valued vector, and define $\mathrm{Predict}(\cdot)$ as a simple neural network consisting of a single Softmax layer. In this chapter, we focus on neural network-based encoders. Section 4.5 will demonstrate how these encoders can be applied to a variety of NLP tasks.

4.2 Recurrent Models

A comprehensive study of various sequence models is not easy work. It is convenient, however, to first introduce one of the most common and practical neural models, called recurrent neural networks (RNNs). We will see later that RNNs are extensively used in sequence modeling, and the techniques presented here are generic and applicable to many systems.

4.2.1 RNN-based Language Models

Perhaps the most prevalent application of sequence models in NLP is estimating the probability of a word sequence, known as language modeling. Mathematically, language modeling is a specific instance of a fundamental problem in the field of **stochastic processes** (or **random processes**): the modeling of **time series** data [Hamilton, 1994; Chatfield, 2003; Fuller, 2009]. Treated as a time series, a sequence of words constitutes discrete data points occurring at sequential time steps. In this sense, the methods presented here are broadly applicable, although an exhaustive discussion on general time series analysis falls outside the scope of this text.

Given a sequence of words $w_1...w_m$, the goal of language modeling is to compute $\Pr(w_1,...,w_m)$. This joint probability is typically written as a product of conditional probabilities using the chain rule:

$$\Pr(w_1,...,w_m) = \Pr(w_1) \cdot \Pr(w_2|w_1) \cdots \Pr(w_m|w_1,...,w_{m-1}) \tag{4.3}$$

In other words, the problem of generating $w_1...w_m$ is the same as the problem of generating a word w_{i+1} at a time based on the previous words $w_1...w_i$. RNN-based language models represent $w_1...w_i$ via a recurrent unit $\mathrm{RNN}(\cdot)$ [Mikolov et al., 2010]:

$$\mathbf{h}_i = \mathrm{RNN}(\mathbf{h}_{i-1}, \mathbf{x}_i) \tag{4.4}$$

where $\mathbf{x}_i \in \mathbb{R}^{d_e}$ is the word vector (or word embedding) for w_i. Let V be the vocabulary from which we can choose a word. If $\mathbf{1}_{w_i}$ is the one-hot representation of w_i, then $\mathbf{x}_i$ is given by multiplying $\mathbf{1}_{w_i}$ with the word embedding table $\mathbf{C} \in \mathbb{R}^{|V| \times d_e}$:

$$\begin{aligned} \mathbf{x}_i &= \mathrm{Embed}(w_i) \\ &= \mathbf{1}_{w_i}\mathbf{C} \end{aligned} \tag{4.5}$$

As shown in Chapter 1, the use of $\mathbf{C}$ transforms a $|V|$-dimensional (and probably high-dimensional) vector to a d_e-dimensional (and probably low-dimensional) vector. Note that $\mathbf{C}$ is essentially a lookup table, with a distinct table entry (i.e., a row) for each word in V. So, the right-hand side of Eq. (4.5) is a lookup operation which extracts the corresponding row from $\mathbf{C}$.

Now we go back to Eq. (4.4). Intuitively, this equation implies that the state of the context we have seen so far (i.e., $\mathbf{h}_i$) is some representation of the combination of the current input (i.e., $\mathbf{x}_i$) and the state of the earlier context ($\mathbf{h}_{i-1}$). Put another way, it can be thought of as a process of repeatedly adding information of a new word to a cache of "history". An elegant aspect of this process is that it can be easily implemented by repeatedly applying Eq. (4.4) until the end of the sequence.

$\mathrm{RNN}(\cdot)$ can be any function that takes $\mathbf{h}_{i-1}$ and $\mathbf{x}_i$, and produces a new vector $\mathbf{h}_i$. The vanilla RNN has a form

$$\mathrm{RNN}(\mathbf{h}_{i-1}, \mathbf{x}_i) = \psi(\mathbf{h}_{i-1}\mathbf{U} + \mathbf{x}_i\mathbf{V}) \tag{4.6}$$

where $\psi(\cdot)$ is an activation function, such as $\mathrm{TanH}(\cdot)$ and $\mathrm{Sigmoid}(\cdot)$. Together with Eqs. (4.4) and (4.5), we can define $\mathbf{h}_i$ as a function of $\mathbf{h}_{i-1}$ and w_i

$$\mathbf{h}_i = \psi(\mathbf{h}_{i-1}\mathbf{U} + \mathbf{1}_{w_i}\mathbf{C}\mathbf{V}) \tag{4.7}$$

where $\mathbf{U} \in \mathbb{R}^{d_h \times d_h}$, $\mathbf{V} \in \mathbb{R}^{d_e \times d_h}$, and $\mathbf{C} \in \mathbb{R}^{|V| \times d_e}$ are learnable parameters of the model, and d_h is a hyper-parameter indicating the number of dimensions of $\mathbf{h}_i$ and $\mathbf{h}_{i-1}$.

We now have an encoder that represents the word sequence $w_1...w_m$ as a sequence of RNN's outputs $\mathbf{H} = \{\mathbf{h}_1, ..., \mathbf{h}_m\}$. Given that each $\mathbf{h}_i$ encodes the sub-sequence spanning from w_1 to w_i, we can place a Softmax layer on $\mathbf{h}_i$ to obtain a distribution of words:

$$\mathbf{y}_{i+1} = \mathrm{Softmax}(\mathbf{h}_i\mathbf{O} + \mathbf{b}) \tag{4.8}$$

where $\mathbf{O} \in \mathbb{R}^{d_h \times |V|}$ and $\mathbf{b} \in \mathbb{R}^{|V|}$. Taking the word index w_{i+1}, we have

$$\Pr(w_{i+1}|w_1, ..., w_i) = y_{i+1}(w_{i+1}) \tag{4.9}$$

Thus, we have developed a language model that produces a probability $\Pr(w_{i+1}|w_1, ..., w_i)$ at each step. Figure 4.2 shows an illustration of the RNN-based language model for an example sequence. To run this model on a word sequence, we surely wish to start with predicting w_1 but this requires a preceding word w_0 that is taken as the input. A simple and widely applicable method for giving an appropriate starting state to RNNs is to add a beginning symbol $\langle\text{SOS}\rangle$ to the sequence so that all sequences start with the same "word". Likewise, we can attach an end symbol $\langle\text{EOS}\rangle$ to the sequence to model the completeness of the sequence. This leads to a new form of the probability of the sequence

$$\begin{aligned}
\Pr(\langle\text{SOS}\rangle, w_1, ..., w_m, \langle\text{EOS}\rangle) = \; & \Pr(\langle\text{SOS}\rangle) \cdot \\
& \Pr(w_1|\langle\text{SOS}\rangle) \cdot \\
& \Pr(w_2|\langle\text{SOS}\rangle, w_1) \cdot \\
& \cdots \\
& \Pr(w_m|\langle\text{SOS}\rangle, w_1, ..., w_{m-1}) \cdot \\
& \Pr(\langle\text{EOS}\rangle|\langle\text{SOS}\rangle, w_1, ..., w_m)
\end{aligned} \tag{4.10}$$

We can simply assume $\Pr(\langle\text{SOS}\rangle) = 1$. To obtain $\Pr(\langle\text{SOS}\rangle, w_1, ..., w_m, \langle\text{EOS}\rangle)$, we take $\langle\text{SOS}\rangle\ w_1...w_m$ as an input sequence and $w_1...w_m\ \langle\text{EOS}\rangle$ as the output sequence.

4.2.2 Training

As a differentiable neural architecture, the RNN-based language model can be trained using standard optimization techniques. The foundational principles of training neural networks have been thoroughly discussed in Chapter 2. Therefore, rather than providing an exhaustive description here, we outline the fundamental concepts and discuss several refinements to sequence models.

RNN-based language modeling can be framed as a next-step-prediction problem. Suppose we are given a corpus S consisting of word sequences. For each sequence $\mathbf{w} = w_1, \ldots, w_{|\mathbf{w}|}$ in S, we construct a sequence of input-target pairs, structured as follows[3]:

$$\{(w_1, w_2), (w_2, w_3), ..., (w_{|\mathbf{w}|-1}, w_{|\mathbf{w}|})\}$$

The language model processes the input prefix sequence $w_1, ..., w_{|\mathbf{w}|-1}$ and emits a corresponding sequence of predicted probability distributions $\hat{\mathbf{y}}_2, ..., \hat{\mathbf{y}}_{|\mathbf{w}|}$. See the following table for an illustration of the inputs and outputs of the model.

[3] While the $\langle\text{SOS}\rangle$ and $\langle\text{EOS}\rangle$ boundary tokens are essential in real-world implementations, we omit them from the notation here for simplicity.

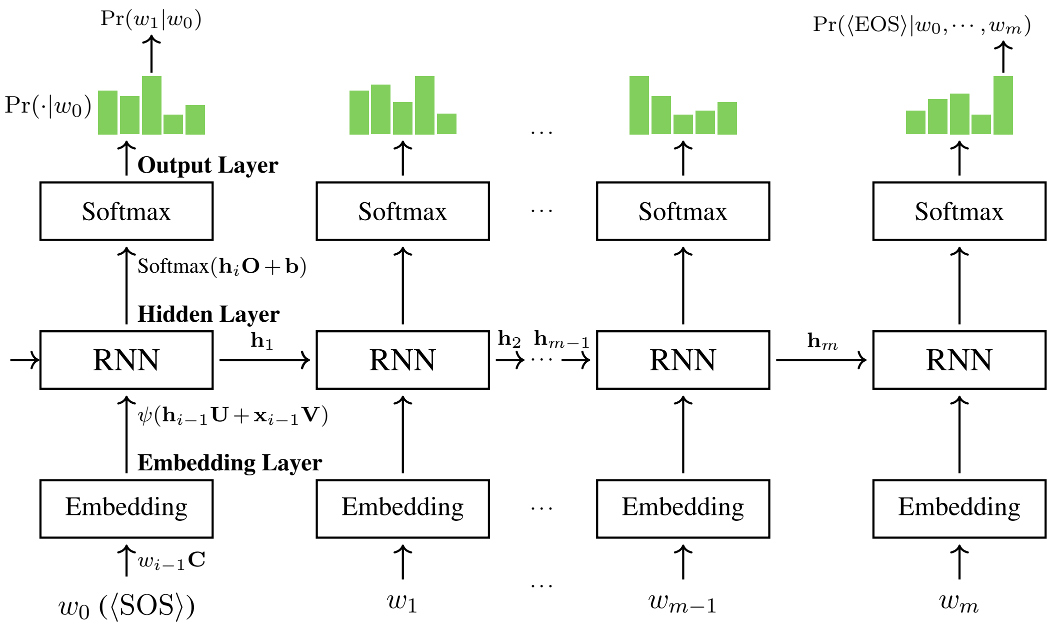


Figure 4.2: Illustration of using an RNN-based language model to calculate $\Pr(\langle\text{SOS}\rangle w_1 ... w_m \langle\text{EOS}\rangle)$. The input is $\langle\text{SOS}\rangle w_1 ... w_m$, and the output is the probability $\Pr(w_1|\langle\text{SOS}\rangle)\Pr(w_2|\langle\text{SOS}\rangle\ w_1)...\Pr(\langle\text{EOS}\rangle|\langle\text{SOS}\rangle\ w_1...w_m)$. As $\Pr(\langle\text{SOS}\rangle) = 1$, the probability of generating the sequence is simply $\Pr(\langle\text{SOS}\rangle w_1...w_m\langle\text{EOS}\rangle) = \Pr(\langle\text{SOS}\rangle)\Pr(w_1|\langle\text{SOS}\rangle)\Pr(w_2|\langle\text{SOS}\rangle\ w_1)...\Pr(\langle\text{EOS}\rangle|\langle\text{SOS}\rangle w_1...w_m)$. For each input w_i, we first represent it as a word vector $\mathbf{x}_i$ via the embedding layer, resulting in a sequence of word vectors $\mathbf{x}_0...\mathbf{x}_m$. The RNN layer maps $\mathbf{x}_0...\mathbf{x}_m$ to a sequence of hidden states $\mathbf{h}_1...\mathbf{h}_{m+1}$. In this process, the operation is applied recursively: an RNN unit takes both $\mathbf{h}_{i-1}$ and $\mathbf{x}_i$ and produces a new state $\mathbf{h}_i$. On top of that, we use the output layer (Softmax) to obtain $\Pr(w_{i+1}|\langle\text{SOS}\rangle\ w_1...w_i)$.

Step	History	Input	Output (Distribution)	Gold-Standard
1		w_1	$\hat{\mathbf{y}}_2$	w_2
2	w_1	w_2	$\hat{\mathbf{y}}_3$	w_3
3	w_1, w_2	w_3	$\hat{\mathbf{y}}_4$	w_4
...	...	...	...	...
$\|\mathbf{w}\|-2$	$w_1, w_2, ..., w_{\|\mathbf{w}\|-3}$	$w_{\|\mathbf{w}\|-2}$	$\hat{\mathbf{y}}_{\|\mathbf{w}\|-1}$	$w_{\|\mathbf{w}\|-1}$
$\|\mathbf{w}\|-1$	$w_1, w_2, ..., w_{\|\mathbf{w}\|-3}, w_{\|\mathbf{w}\|-2}$	$w_{\|\mathbf{w}\|-1}$	$\hat{\mathbf{y}}_{\|\mathbf{w}\|}$	$w_{\|\mathbf{w}\|}$

A loss function $L(\hat{\mathbf{y}}_i, \mathbf{1}_{w_i})$ is defined to quantify the discrepancy between the predicted distribution $\hat{\mathbf{y}}_i$ and the true one-hot target representation $\mathbf{1}_{w_i}$. The standard choice for this task is the cross-entropy loss, which measures the divergence between the two distributions [Mitchell, 1997; Bishop, 2006].

Then, the loss over the entire set is defined to be

$$L = \sum_{\mathbf{w} \in S} \sum_{i=2}^{|\mathbf{w}|} L(\hat{\mathbf{y}}_i, \mathbf{1}_{w_i}) \tag{4.11}$$

Once we know the loss, the training of the RNN-based language model can be achieved by using gradient descent. A simple form of this method is the delta rule

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \cdot \frac{\partial L}{\partial \theta} \tag{4.12}$$

where θ stands for the parameters. For the model described in Section 4.2.1, θ includes $\mathbf{C}$, $\mathbf{U}$, $\mathbf{V}$, $\mathbf{O}$ and $\mathbf{b}$. $\frac{\partial L}{\partial \theta}$ is the derivative of the loss with respect to the parameters, called **error gradient**.

Eq. (4.12) can be understood as a process of moving the current parameters a small step in the steepest downhill direction (i.e., the direction of $-\frac{\partial L}{\partial \theta}$). Here η stands for how far we move in each step of going downhill, also called the learning rate. Obtaining $\frac{\partial L}{\partial \theta}$ often requires a back-propagation algorithm that propagates errors backward through the unrolled time steps. In modern implementations of deep learning systems, in which neural networks are represented as computation graphs, back-propagation is simple since it is just a by-product of graph traversal and there are many automatic differentiation toolkits to do this. Similar algorithms, called **back-propagation through time** (**BPTT**), were also used in earlier systems [Werbos, 1990]. For further information about training neural networks, see Chapter 2 and/or textbooks on this subject [Goodfellow et al., 2016; Zhang et al., 2021].

If the input is a long sequence, the application of RNNs would result in a deep neural network. In this case, the use of the chain rule of ordered derivatives makes large or small loss derivatives accumulate, and the update to the parameters in Eq. (4.12) is consequently very large or small. These are typically known as the **exploding and vanishing gradient problems**. There are several methods to mitigate these problems for RNNs [Sutskever, 2013]. Some of them are

- **Regularization**. Introducing regularization terms (such as the l_1 and l_2 norms on parameter matrices) into training can avoid models in which most of the parameters have large values, and thus help to avoid exploding gradients. Similarly, one can penalize the cases in which the norms of the gradients are too small [Pascanu et al., 2013].
- **Gradient Clipping**. When the norm of the gradients is too large, it is natural to directly scale down their magnitudes. A simple method is to clip the gradient norm in terms of a threshold τ. If the norm $||\frac{\partial L}{\partial \theta}||$ is larger than τ, we can rescale $\frac{\partial L}{\partial \theta}$ accordingly, e.g., $\frac{\partial L}{\partial \theta} = \frac{\tau}{||\frac{\partial L}{\partial \theta}||} \cdot \frac{\partial L}{\partial \theta}$.[4]

[4] It is usually formulated as an equation

$$\frac{\partial L}{\partial \theta} = \frac{\tau}{\max(\tau, ||\frac{\partial L}{\partial \theta}||)} \cdot \frac{\partial L}{\partial \theta} \tag{4.13}$$

- **Truncated Back-propagation**. Another idea is to break a long sequence of input-output pairs into shorter pieces, and train RNNs on these separate sub-sequences [Williams and Peng, 1990; Elman, 1990]. This reduces both the cost of training and the risk of too large or small values in accumulating error gradients.
- **Improved Architectures**. It is also possible to redesign the model to overcome the limits of standard RNNs, usually using the memory mechanism. In Section 4.3, we will see a few examples of redesigning the recurrent unit for addressing the vanishing gradient problem.
- **Initialization and Constraints of Parameters**. Initializing the model parameters to a desirable region is generally helpful for optimization, and, sometimes, helpful for preventing very small gradients. An alternative method is to randomly set the model and only learn the parameters of the output layers [Jaeger and Haas, 2004].
- **Non-saturating Activations**. Many common activation functions have a compact range of outputs, e.g., the Sigmoid function has a range of $[0,1]$. They are also called **saturating activation functions**[5]. The use of saturating activation functions often leads to the decay of gradients over layers, i.e., the vanishing gradient. It is therefore promising to use non-saturating activation functions instead, e.g., the ReLU function.
- **Normalization of Activations**. Saturating activations may also result in getting stuck in a saturated region of outputs, and we need a large learning rate to escape from local optima [Ioffe and Szegedy, 2015]. Thus, the training would be unstable, and subtle changes in inputs and/or model parameters would lead to a big variance in model behavior. A possible solution is to normalize the activations to reduce the variance, e.g., subtracting the mean of the activations in a group of samples (e.g., samples in a mini-batch of training), and dividing by their standard deviation.

4.2.3 Layer Stacking

If we think of the application of a recurrent unit as a function mapping a variable sequence to a new variable sequence of the same length, it is natural to compose this function with another function of the same type, or even with itself. This makes it very easy to extend RNNs to deep neural networks simply by stacking recurrent layers.

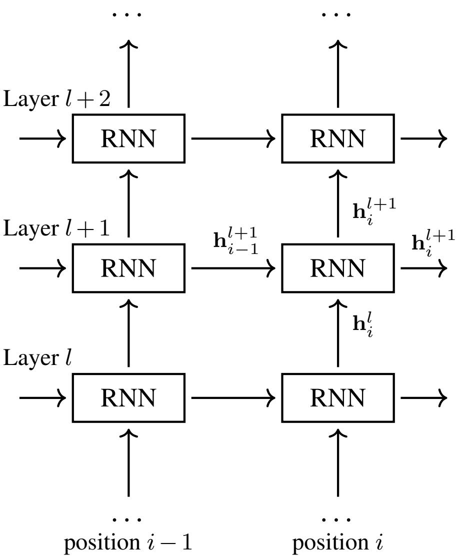
Let $\mathbf{h}_i^l$ be the output of the l-th recurrent unit in the stack at position i. We can apply a new recurrent unit to $\mathbf{h}_i^l$, resulting in a new output at level $l+1$

$$\mathbf{h}_i^{l+1} = \mathrm{RNN}(\mathbf{h}_{i-1}^{l+1}, \mathbf{h}_i^l) \tag{4.14}$$

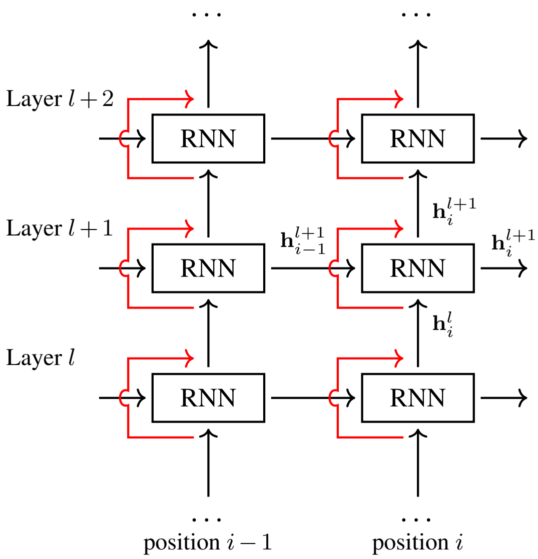
where $\mathbf{h}_{i-1}^{l+1}$ is the output of the previous step at level $l+1$. To make Eq. (4.14) well-formed, we typically define $\mathbf{h}_i^0 = \mathbf{x}_i$. In other words, the stack starts off with the word vector $\mathbf{x}_i$, then a series of RNN outputs (i.e., $\mathbf{h}_i^1$, $\mathbf{h}_i^2$, $\mathbf{h}_i^3$, etc).

To illustrate, Figure 4.3 (a) shows a stacked RNN for language modeling. We see that

[5] An activation function $f(x)$ is non-saturating if and only if when $x \to \infty$ (or $-\infty$), $f(x) \to \infty$. An activation function is saturating if it is not a non-saturating activation function.



(a) A 3-layer RNN.



(b) A 3-layer RNN with residual connections.

Figure 4.3: 3-layer RNNs (with and without residual connections). To stack RNN layers, we feed the output of layer l to layer $l+1$. Thus the output of layer $l+1$ is given by $\mathbf{h}_i^{l+1} = \mathrm{RNN}(\mathbf{h}_{i-1}^{l+1}, \mathbf{h}_i^l)$. Red lines stand for the residual connections which directly add the input of a layer to its output, resulting in $\mathbf{h}_i^{l+1} = \mathrm{RNN}(\mathbf{h}_{i-1}^{l+1}, \mathbf{h}_i^l) + \mathbf{h}_i^l$.

applying a stack of recurrent units is equivalent to creating multiple layers of RNNs simultaneously. However, there would be a risk of confusion if we call an unrolled recurrent network a *layer*, as the term *layer* typically refers to a set of neurons receiving the same inputs in a feed-forward neural network. Here we extend the term *layer* to cover a more general concept: a group of neurons that are topologically placed on the same level. So, we say that the language model in Figure 4.3 has 3 RNN layers.

Stacking multiple layers of RNNs, we build a model which is deeper but more difficult to train. This difficulty arises in part from the barriers of passing information through many-layered RNNs. To make the training easier, a widely-used approach is to introduce **skip connections** or **residual connections** into a multi-layer neural network [He et al., 2016a]. These connections are intended to leverage an additional path to allow information to skip layers. As described in Chapter 2, the form of a residual neural network is given by

$$\mathbf{y}^{l+1} = F(\mathbf{y}^l) + \mathbf{y}^l \tag{4.15}$$

where $\mathbf{y}^l$ is the output of layer l. Extending this formulation to Eq. (4.14) leads to multi-layer RNNs with residual connections, given by

$$\mathbf{h}_i^{l+1} = \mathrm{RNN}(\mathbf{h}_{i-1}^{l+1}, \mathbf{h}_i^l) + \mathbf{h}_i^l \tag{4.16}$$

The only difference from Eq. (4.14) is that we introduce the identity map of $\mathbf{h}_i^l$ to the right-hand side of Eq. (4.16). Thus, the input $\mathbf{h}_i^l$ is directly accessible from layer $l+1$. This greatly simplifies the way that the information flows through the neural network, and allows the system to "skip" layers in propagating errors. Figure 4.3 (b) shows a 3-layer RNN with residual connections.

4.2.4 Bi-directional Models

The use of RNNs enables us to formulate the problem of encoding a word sequence as a problem of left-to-right generation of words. One advantage of this approach is that the modeling of contextual information arises naturally: the output of an RNN unit in some way describes the context words up to that point. This feature makes it very straightforward to model the probability distribution $\Pr(w_{i+1}|w_1,...,w_i)$, as we can use $\mathbf{h}_i$ as a representation of the context $w_1...w_i$, that is, $\Pr(w_{i+1}|w_1,...,w_i) = \Pr(w_{i+1}|\mathbf{h}_i)$.

The left-to-right generation is widely used in sequence generation, such as machine translation. It can be viewed as an instance of **autoregressive processes** (**AR processes**) in which the state of a variable is dependent on the state of the previous variables [Chatfield, 2003; Box et al., 2015][6]. However, such a method is not the only choice for modeling sequences. We do not even necessarily restrict ourselves to language modeling for training a sequence encoder. This gives rise to an interesting question: how can we develop an encoder of word sequences without assumptions regarding the predictor? Answering the question leads us to isolate the learning of the text encoder from a specific NLP task, and to regard it as a separate task whose result can be applied to many other systems. A more detailed discussion is not the focus here and we leave it to subsequent chapters.

We now present a simple extension of the left-to-right sequence model by returning to RNNs. Note that in sequence modeling our desire is some representation of the entire sequence. A problem with usual RNNs is that they are **uni-directional models** in which the context words following w_i are absent. To consider both the left and right contexts of a given word, we can instead use **bi-directional models**. Figure 4.4 shows an example of the bi-directional RNN. There are two sub-models: a left-to-right RNN and a right-to-left RNN. They have the

[6] As a stochastic process, an autoregressive process expresses a variable at time t by relating it to the past values of the process and the current value of an error process [Chatfield, 2003]. Formally, a time series $\{z_1,...,z_T\}$ describes an autoregressive process of order p if for any $t \in \{p+1,...,T\}$

$$z_t = \sum_{i=1}^{p} \alpha_i z_{t-i} + \epsilon_t \tag{4.17}$$

where $\{\alpha_1,...,\alpha_p\}$ are the parameters of the process, and ϵ_t is the error at time t. This process is called *regressive* because it has the same form as the **multiple linear regression model**. The prefix *auto-* comes from the way we regress z_t: z_t is dependent on its past values instead of additional independent variables. One way to interpret language modeling in an autoregressive process perspective is to simply treat $\{z_1,...,z_T\}$ as representations of a sequence of words $\{w_1,...,w_T\}$. Thus, we can gain some idea of predicting w_t using previous words $\{w_1,...,w_{t-1}\}$ by considering the autoregressive property of the problem. However, it should be noted that most of the sequence generation models used in NLP are not mathematically equivalent to Eq. (4.17), although they are often called regressive models. For example, the RNN-based language model discussed here is not a linear model. Rather, it takes layers of non-linearity to describe the complex relationships among words.

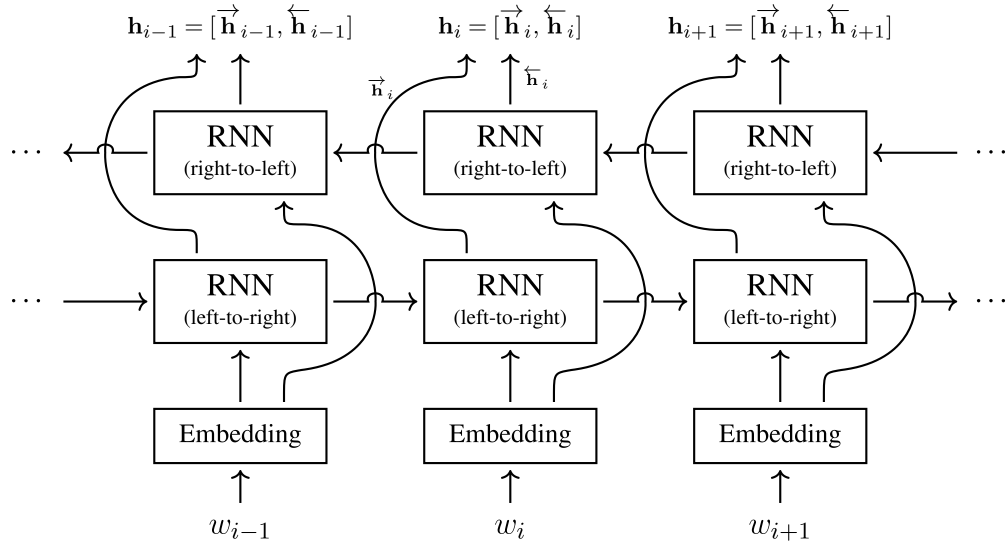


Figure 4.4: A bi-directional RNN model. Given a word sequence, we run an RNN from left to right and another RNN from right to left. Therefore, at each position we obtain a left-to-right representation and a right-to-left representation. The output is the concatenation of the two representations so that it involves both the left and right contexts.

same architecture but work in opposite directions. For each input word w_i, the left-to-right RNN outputs a vector representing the context $\{w_1, ..., w_i\}$ (denoted by $\overrightarrow{\mathbf{h}}_i$), and the right-to-left RNN outputs a vector representing the context $\{w_i, ..., w_m\}$ (denoted by $\overleftarrow{\mathbf{h}}_i$). We can concatenate $\overrightarrow{\mathbf{h}}_i$ and $\overleftarrow{\mathbf{h}}_i$ to obtain a bi-directional representation

$$\mathbf{h}_i = [\overrightarrow{\mathbf{h}}_i, \overleftarrow{\mathbf{h}}_i] \tag{4.18}$$

Thus, the bi-directional RNN has the same form of output as that of the uni-directional RNN, that is, a sequence of vectors $\{\mathbf{h}_1, .., \mathbf{h}_m\}$. Unlike the uni-directional RNN, the representation $\mathbf{h}_i$ here describes the context on both sides.

To increase expressive power, the bi-directional RNN can be extended to a neural network of multiple RNN layers. For example, we can run deep RNNs in two directions and combine their results as in Eq. (4.18). Such model architectures have been extensively used in language and speech processing tasks, including machine translation [Wu et al., 2016], sentiment analysis [Tang et al., 2015], POS tagging [Huang et al., 2015], speech recognition [Graves et al., 2013a;b], and so on.

4.3 Memory

RNNs are appropriate for sequence learning as they effectively summarize past inputs at each step and base subsequent predictions on this historical summary. A key benefit of RNNs is that we can represent a history of arbitrary length as a fixed-size vector and update it dynamically when new information arrives. In other words, they use an implicit **memory** to store contextual information. Next, we demonstrate that this memory mechanism is a general concept and can be used to improve sequence models.

4.3.1 Memory as A System

In psychology, memory is the cognitive ability of the mind to encode, store, and retrieve information. Among the many cognitive models in psychology, a foundational framework is the **multi-store model** [Atkinson and Shiffrin, 1968]. It defines memory as a system comprising three structural components: fleeting **sensory memory**, **short-term memory**, and **long-term memory**. Sensory memory retains raw stimuli that decay rapidly, such as immediate visual and olfactory data. Short-term memory stores information for a slightly longer duration but remains transient. For example, when attempting to memorize a phone number, the digits are held transiently and may be forgotten shortly after. Long-term memory, conversely, is permanent, retaining information indefinitely. For instance, adults can recall specific events from their childhood.

Given this categorization, there appear to be interesting connections between the above model of memory and the neural networks we discuss here. For example, the hidden state of a recurrent unit can be viewed as short-term memory: it maintains active information throughout a sequence but is entirely reset when switching to a new sequence[7]. On the other hand, the entire language model and its associated parameters function akin to a long-term memory: the model encodes global probabilistic word prediction patterns from the training corpus, retaining this knowledge permanently for future inference. Furthermore, standard cognitive metrics perfectly align with computational paradigms: **coding** or encoding is referred to as how the information is stored in a memory, **duration** is referred to as how long the information is stored in a memory, and **capacity** is referred to as how much information is stored in a memory.

In machine learning and NLP, we can gain an understanding of memory by considering it from an information processing point of view. Broadly, memory can be viewed as a system that writes information to a "storage" and reads it when queried. It has the following functions.

- **Encoding**. The input of the system is encoded in a form that is easy to process. For text inputs, this can be simply thought of as the same encoding process as we discuss in both Chapter 3 and this chapter: a word or a sequence of words is represented as a feature vector or a sequence of feature vectors.
- **Update**. Given the encoded information, we store it in the memory. This operation is generally dependent on the organization of the memory. For example, one can treat a group of encoded items as a datastore with an indexing system. In this case, storing an item requires finding the right place to keep it. Alternatively, one can represent the memory as a single vector of numbers.
- **Retrieval**. The stored information can be retrieved. This typically involves matching each item in the memory against an input query. If the memory is represented in a

[7]Alternatively, the hidden state at step i may simply retain very little information about the earliest steps due to information decay.

simpler form, such as a vector, it may not be explicitly retrieved, and we return the entire memory when required.

These functions can be designed in many different ways, leading to a variety of NLP systems. One simple example is information retrieval (IR) [Manning et al., 2008]. A typical IR system indexes a vast collection of documents and allows users to search for relevant information. To enable efficient search, documents are encoded into convenient representations. For example, a bag-of-words model might compute the matching score between a document and a query, while an inverted index efficiently maps query terms to their storage locations. Systems of this type cover a wide range of applications, including translation memory, dialogue, summarization, document classification, and so on.

Another design choice made for memory systems is to consider, either partially or fully, a continuous form for the above components. One method is to encode each input item as a real-valued vector (e.g., a word embedding) but use the same modules of update and retrieval as in usual information retrieval-like systems [Weston et al., 2015; Khandelwal et al., 2019]. An alternative method is to adopt differentiable functions for all the steps in building and accessing the memory. These models are typically implemented using neural networks and trained using gradient descent [Sukhbaatar et al., 2015; Graves et al., 2014; Kumar et al., 2016; Graves et al., 2016; Miller et al., 2016]. This idea motivates work on exploring approaches to coupling neural networks with memories, such as **end-to-end memory networks** and **neural Turing machines**. Note that the above models are sometimes called **external memories**, as they are used as separate modules working with other systems.

Memory can also work as an internal or hidden component of a system. In this case, the memory is typically rebuilt for each input sample, and so it can be regarded as an instance of the short-term memory. There are various ways of using this type of memory to improve sequence models. In the remainder of the section, we will focus on using the memory mechanism in RNNs. In Chapter 5, we will see how the idea of memory is extended to model the correspondence between tokens of two sequences.

4.3.2 Long Short-Term Memory

In the vanilla RNN presented in Section 4.2.1, the summarization of the context words was given by the output of a recurrent unit. It implicitly defines a memory, and thus in principle allows modeling dependencies over arbitrary lengths. The memory simply combines the representations of the earlier history $w_1 ... w_{i-1}$ and the input at the current step w_i, but does not consider how much information from different steps should be compressed into a fixed-length representation. A problem with this model is that, when predictions require long-term dependencies, the memory may retain insufficient relevant information, and it may be hard to learn these dependencies through back-propagation [Bengio et al., 1994; Pascanu et al., 2013]. A more powerful approach, therefore, is to compute what should be retained at each step, and to let the model learn to decide whether to memorize or forget.

Long short-term memory (**LSTM**) is perhaps the best-known variant of RNNs to accomplish the above goal [Hochreiter and Schmidhuber, 1997]. The basic idea of LSTM is that a recurrent unit can learn to memorize useful things and forget useless things by maintaining an explicit memory [Gers et al., 2000]. To this end, the vanilla recurrent unit is replaced with an LSTM unit that is made up of an output vector (call it a **recurrent cell**), a memory vector (call it a **memory cell**), and three gates to control the information flow inside the LSTM unit. As an extension to RNNs, an LSTM network deals with an input sequence as usual: it starts with some initial states, and then repeatedly takes an input and outputs a vector. A key difference between LSTM networks and RNNs is that the LSTM unit of step i takes both the recurrent cell and the memory cell of its previous step. The form of an LSTM unit is given by

$$(\mathbf{h}_i, \mathbf{c}_i) = \mathrm{LSTM}(\mathbf{h}_{i-1}, \mathbf{c}_{i-1}, \mathbf{x}_i) \tag{4.19}$$

where $\mathbf{h}_i \in \mathbb{R}^{d_h}$ is the recurrent cell of step i, $\mathbf{c}_i \in \mathbb{R}^{d_h}$ is the memory cell of step i, and $\mathbf{x}_i \in \mathbb{R}^{d_e}$ is the input of step i. Given $\mathrm{LSTM}(\cdot)$, applying the LSTM model is straightforward. We simply repeat the call of $\mathrm{LSTM}(\cdot)$ for the inputs $\{\mathbf{x}_1, ..., \mathbf{x}_m\}$ and obtain the outputs $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$. This resembles the way we use vanilla RNNs, making it very easy to extend LSTM to multi-layer models (see Section 4.2.3) and bi-directional models (see Section 4.2.4).

We can divide $\mathrm{LSTM}(\cdot)$ into three steps.

- **Step 1: Forget**. Assuming that $\mathbf{c}_{i-1}$ contains the information that the model memorizes at step $i-1$, we need to determine how much information in $\mathbf{c}_{i-1}$ is discarded in building $\mathbf{c}_i$. To do this, a gate is used to control to what extent we forget for each dimension of $\mathbf{c}_{i-1}$. The **forget gate** is defined to be:

$$\mathbf{f}_i = \mathrm{Sigmoid}(\mathbf{h}_{i-1}\mathbf{U}_\mathrm{f} + \mathbf{x}_i\mathbf{V}_\mathrm{f} + \mathbf{b}_\mathrm{f}) \tag{4.20}$$

  where $\mathbf{f}_i \in [0,1]^{d_h}$ is a vector with the same number of dimensions as $\mathbf{c}_{i-1}$. The Sigmoid function maps the input data to the range $[0,1]$. Thus, an entry of $\mathbf{f}_i$ indicates how much is preserved for the same entry of $\mathbf{c}_{i-1}$. Taking this further, $\mathbf{f}_i \odot \mathbf{c}_{i-1}$ represents the memory that is retained after passing through the forget gate. See Figure 4.5 (a) for an illustration of the forget gate in the LSTM unit.

- **Step 2: Update**. Next we update the memory by considering both the previous state of the memory (i.e., $\mathbf{c}_{i-1}$) and the input of the LSTM unit (i.e., $\mathbf{x}_i$ and $\mathbf{h}_{i-1}$). We first combine $\mathbf{x}_i$ and $\mathbf{h}_{i-1}$ using a simple neural network:

$$\hat{\mathbf{c}}_i = \mathrm{TanH}(\mathbf{h}_{i-1}\mathbf{U}_\mathrm{c} + \mathbf{x}_i\mathbf{V}_\mathrm{c} + \mathbf{b}_\mathrm{c}) \tag{4.21}$$

  $\hat{\mathbf{c}}_i$ can be treated as the new information we intend to add to the memory at step i. Again, we need a way to control the amount of information coming into the memory. Hence we define an **input gate** as

$$\mathbf{g}_i = \mathrm{Sigmoid}(\mathbf{h}_{i-1}\mathbf{U}_\mathrm{g} + \mathbf{x}_i\mathbf{V}_\mathrm{g} + \mathbf{b}_\mathrm{g}) \tag{4.22}$$

  This equation is similar to Eq. (4.20) but with different parameters. We then define $\mathbf{g}_i \odot \hat{\mathbf{c}}_i$ to be the actual new information that we are interested in. Taking both $\mathbf{f}_i \odot \mathbf{c}_{i-1}$

and $\mathbf{g}_i \odot \hat{\mathbf{c}}_i$, the memory cell at step i is given by

$$\mathbf{c}_i = \mathbf{f}_i \odot \mathbf{c}_{i-1} + \mathbf{g}_i \odot \hat{\mathbf{c}}_i \tag{4.23}$$

In other words, we forget something old in $\mathbf{c}_{i-1}$ and memorize something new in $\hat{\mathbf{c}}_i$. See Figure 4.5 (b) for an illustration of the update step.

- **Step 3: Output**. In the last step we generate the output $\mathbf{h}_i$ based on the memory $\mathbf{c}_i$. Instead of copying $\mathbf{c}_i$ to $\mathbf{h}_i$, we feed $\mathbf{c}_i$ to the hyperbolic tangent function and multiply its result with the output gate. Like Eqs. (4.20) and (4.22), the output gate is given by

$$\mathbf{o}_i = \mathrm{Sigmoid}(\mathbf{h}_{i-1}\mathbf{U}_\mathrm{o} + \mathbf{x}_i\mathbf{V}_\mathrm{o} + \mathbf{b}_\mathrm{o}) \tag{4.24}$$

Then, the output of the LSTM unit is defined to be

$$\mathbf{h}_i = \mathbf{o}_i \odot \mathrm{TanH}(\mathbf{c}_i) \tag{4.25}$$

See Figure 4.5 (c) for an illustration of the output step.

The LSTM model is parameterized by $\mathbf{U}_\mathrm{f}, \mathbf{U}_\mathrm{c}, \mathbf{U}_\mathrm{g}, \mathbf{U}_\mathrm{o} \in \mathbb{R}^{d_h \times d_h}$, $\mathbf{V}_\mathrm{f}, \mathbf{V}_\mathrm{c}, \mathbf{V}_\mathrm{g}, \mathbf{V}_\mathrm{o} \in \mathbb{R}^{d_e \times d_h}$, and $\mathbf{b}_\mathrm{f}, \mathbf{b}_\mathrm{c}, \mathbf{b}_\mathrm{g}, \mathbf{b}_\mathrm{o} \in \mathbb{R}^{d_h}$. Compared with vanilla RNNs, additional parameters arise directly from this gate-based architecture. In practice one can implement LSTM units in many different ways, e.g., using activation functions other than $\mathrm{Sigmoid}(\cdot)$ and $\mathrm{TanH}(\cdot)$, removing the bias terms $\mathbf{b}_\mathrm{f}$, $\mathbf{b}_\mathrm{c}$, $\mathbf{b}_\mathrm{g}$, and $\mathbf{b}_\mathrm{o}$. Training LSTM models follows the standard paradigm of training RNN-based models. For example, we can build an LSTM-based language model and train it by using the methods presented in Section 4.2.2.

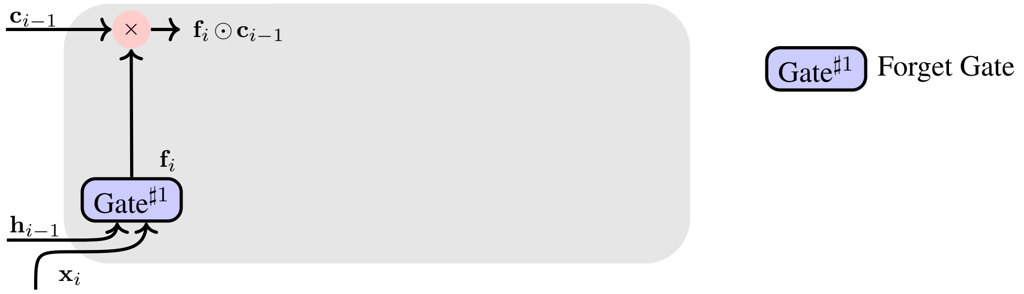
4.3.3 Gated Recurrent Units

Above, we saw the important role played by the gate units and the memory cell. In general the use of these neural networks makes the model computationally more expensive. A computationally efficient alternative to the LSTM is the **gated recurrent unit** (**GRU**), which adopts a simplified model structure with fewer gate functions [Cho et al., 2014; Chung et al., 2014]. Unlike LSTM, a GRU does not have a memory cell so, as an RNN unit, it takes both the previous state vector $\mathbf{h}_{i-1}$ and the current input vector $\mathbf{x}_i$, and produces the current state vector $\mathbf{h}_i$.

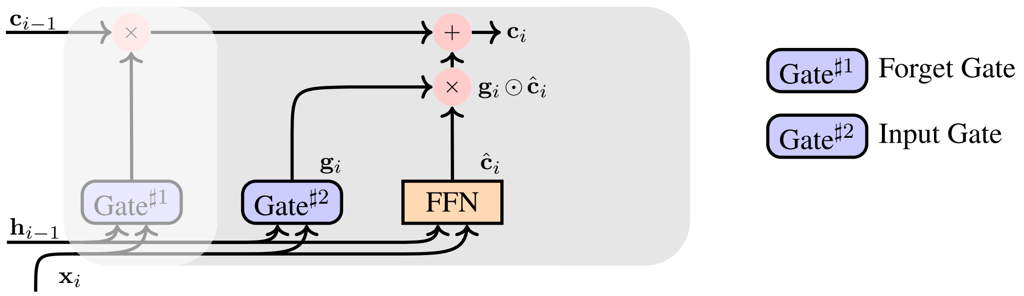
The GRU architecture incorporates two gate units: the **reset gate** and the **update gate**. The reset gate, as the name suggests, is used to reset (or rescale) the state of the GRU (i.e., $\mathbf{h}_{i-1}$). Analogous to the gates in an LSTM, the reset gate is defined to be

$$\mathbf{r}_i = \mathrm{Sigmoid}(\mathbf{h}_{i-1}\mathbf{U}_\mathrm{r} + \mathbf{x}_i\mathbf{V}_\mathrm{r} + \mathbf{b}_\mathrm{r}) \tag{4.26}$$

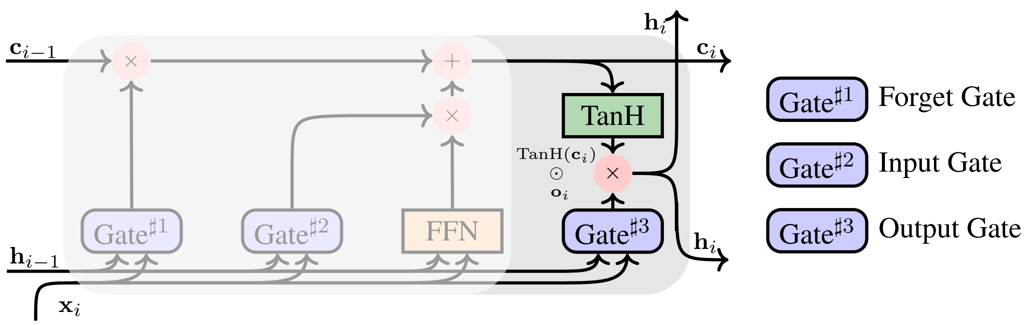
where $\mathbf{r}_i \in [0,1]^{d_h}$ is a vector of scalars, each dimension describing how much information in the corresponding dimension of $\mathbf{h}_{i-1}$ is retained. Thus, we have a representation of retained



(a) The forget gate.



(b) The input gate.



(c) The output gate.

Figure 4.5: The architecture of the LSTM unit. At step i, it takes the input $\mathbf{x}_i$, and then updates both the memory cell ($\mathbf{c}_{i-1} \rightarrow \mathbf{c}_i$) and the recurrent cell ($\mathbf{h}_{i-1} \rightarrow \mathbf{h}_i$). This process involves three gates: the forget gate controls how much information in $\mathbf{c}_{i-1}$ is retained at step i, the input gate controls how much information in $\mathbf{c}_{i-1}$ and $\mathbf{x}_i$ is retained at step i, and the output gate controls how much information in $\mathbf{c}_i$ is used to form $\mathbf{h}_i$.

information

$$v_{i-1} \quad = \quad \mathbf{r}_i \odot \mathbf{h}_{i-1} \tag{4.27}$$

Taking both the retained information v_{i-1} and the current input $\mathbf{x}_i$, a new state vector is defined to be

$$\hat{\mathbf{h}}_i = \mathrm{TanH}(v_{i-1}\mathbf{U}_\mathrm{h} + \mathbf{x}_i\mathbf{V}_\mathrm{h} + \mathbf{b}_\mathrm{h}) \tag{4.28}$$

The update gate is then given by

$$\mathbf{u}_i = \mathrm{Sigmoid}(\mathbf{h}_{i-1}\mathbf{U}_\mathrm{u} + \mathbf{x}_i\mathbf{V}_\mathrm{u} + \mathbf{b}_\mathrm{u}) \tag{4.29}$$

$\mathbf{u}_i$ can be thought of as an interpolation weight vector that balances the preservation of the old state against the integration of the new candidate state. Finally, the output of the GRU is defined as a linear interpolation of $\hat{\mathbf{h}}_i$ and $\mathbf{h}_{i-1}$

$$\mathbf{h}_i = \mathbf{u}_i \odot \hat{\mathbf{h}}_i + (1 - \mathbf{u}_i) \odot \mathbf{h}_{i-1} \tag{4.30}$$

Figure 4.6 illustrates the information flow in a GRU unit. The parameters here are $\mathbf{U}_\mathrm{r}, \mathbf{U}_\mathrm{h}, \mathbf{U}_\mathrm{u} \in \mathbb{R}^{d_h \times d_h}$, $\mathbf{V}_\mathrm{r}, \mathbf{V}_\mathrm{h}, \mathbf{V}_\mathrm{u} \in \mathbb{R}^{d_e \times d_h}$, and $\mathbf{b}_\mathrm{r}, \mathbf{b}_\mathrm{h}, \mathbf{b}_\mathrm{u} \in \mathbb{R}^{d_h}$. Therefore, the GRU model is smaller than the LSTM model because of the use of fewer gate units. Note that removing the memory cell makes GRUs more efficient. In this case, the role of memory is implicitly assumed by the hidden state $\mathbf{h}_i$, which dynamically updates to preserve important information over time.

4.4 Convolutional Models

In this section we describe another type of model for sequence modeling, called convolutional neural networks (CNNs). While our overview covers the basic concepts, it does not serve as an exhaustive introduction to various CNN variants and cutting-edge techniques. Instead, we focus on the application of CNNs to sequential data and present some architectural refinements.

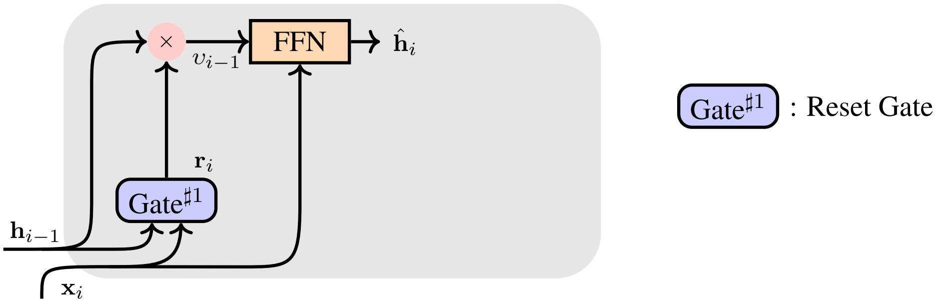
4.4.1 Convolution

A key property of CNNs is their shared-weight architecture by which a kernel or filter slides over the input data and produces a feature map. The idea is that the filter only receives signals from a restricted region of data at a time (call it the **receptive field**), and computes the weighted sum of these input signals. To illustrate this, we consider the common case where a filter in CNNs is generally used to deal with 2D data. Consider a 3×3 data matrix:

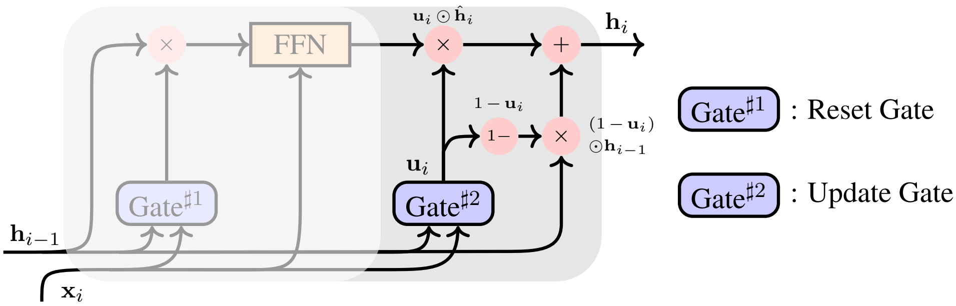
$$\mathbf{A} = \begin{bmatrix} 1 & 9 & 7 \\ 3 & 1 & 2 \\ 0 & 1 & -1 \end{bmatrix} \tag{4.31}$$

and a 2×2 filter with a weight matrix

$$\mathbf{W} = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix} \tag{4.32}$$



(a) The reset gate.



(b) The update gate.

Figure 4.6: The architecture of the GRU. Unlike the LSTM unit, the GRU does not involve a memory cell, and thus follows the same input and output forms of a standard RNN unit. There are two gates in the GRU. The reset gate controls how much information in $\mathbf{h}_{i-1}$ is retained at step i. The retained information is then fused with the input $\mathbf{x}_i$, generating the candidate output $\hat{\mathbf{h}}_i$. The update gate seeks a balance between $\hat{\mathbf{h}}_i$ and $\mathbf{h}_{i-1}$ in computing the final output of the GRU.

We apply this filter to every 2×2 sub-matrix of $\mathbf{A}$ (yielding four such sub-matrices in this example) and compute the sum of the elements weighted by $\mathbf{W}$. For instance, considering the 2×2 sub-matrix in the upper-left corner of $\mathbf{A}$, the output of the filter is calculated as:

$$\begin{aligned}
\mathrm{Conv}(\begin{bmatrix} 1 & 9 \\ 3 & 1 \end{bmatrix}, \mathbf{W}) &= \mathrm{Conv}(\begin{bmatrix} 1 & 9 \\ 3 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}) \\
&= 1 \times 2 + 9 \times 0 + 3 \times 2 + 1 \times 2 \\
&= 10
\end{aligned} \tag{4.33}$$

$\mathrm{Conv}(\cdot)$ defines a **convolution operation** that sums the entries of the element-wise product of the two matrices. The convolution operation can be extended to cover the entire input matrix by sliding the filter over it, as follows

$$
\begin{aligned}
\mathrm{Conv}(\mathbf{A},\mathbf{W}) &= \mathrm{Conv}(\begin{bmatrix} 1 & 9 & 7 \\ 3 & 1 & 2 \\ 0 & 1 & -1 \end{bmatrix}, \mathbf{W}) \\
&= \begin{bmatrix} \mathrm{Conv}(\begin{bmatrix} 1 & 9 \\ 3 & 1 \end{bmatrix}, \mathbf{W}) & \mathrm{Conv}(\begin{bmatrix} 9 & 7 \\ 1 & 2 \end{bmatrix}, \mathbf{W}) \\ \mathrm{Conv}(\begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix}, \mathbf{W}) & \mathrm{Conv}(\begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}, \mathbf{W}) \end{bmatrix} \\
&= \begin{bmatrix} 10 & 24 \\ 8 & 2 \end{bmatrix}
\end{aligned} \tag{4.34}
$$

The output $\begin{bmatrix} 10 & 24 \\ 8 & 2 \end{bmatrix}$ is also called the **feature map** for the filter $\mathbf{W}$ on $\mathbf{A}$. Sometimes, the convolution operation $\mathrm{Conv}(\mathbf{A},\mathbf{W})$ is written as $\mathbf{A} * \mathbf{W}$ where the symbol $*$ stands for the **convolution product**.[8]

Now let us consider a more general description of convolution in CNNs. Suppose that $\mathbf{A}$ is a multi-dimensional data array. A filter defines a window (or receptive field) on $\mathbf{A}$. We can move the window on $\mathbf{A}$ in different directions. This results in a set of data arrays, denoted by Ω. Each data array $\mathbf{a}_p \in \Omega$ is formed by the elements from the corresponding region of $\mathbf{A}$. For example, there are four sub-matrices in Eq. (4.34): $\mathbf{a}_1 = \begin{bmatrix} 1 & 9 \\ 3 & 1 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 9 & 7 \\ 1 & 2 \end{bmatrix}$, $\mathbf{a}_3 = \begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix}$, and $\mathbf{a}_4 = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$. Also, we suppose the filter is parameterized by a weight array $\mathbf{W}$ of the same size as $\mathbf{a}_p$, i.e., $|\mathbf{a}_p| = |\mathbf{W}|$. The result of applying the filter to $\mathbf{A}$ is an array of features

$$
\mathrm{Conv}(\mathbf{A},\mathbf{W}) = \begin{bmatrix} v_1 & \dots & v_{|\Omega|} \end{bmatrix} \tag{4.37}
$$

[8]In mathematical analysis, given two integrable functions $f(\cdot)$ and $g(\cdot)$, **convolution** defines a new integrable function $f * g(\cdot)$ to describe the integral of $f(\cdot)$ weighted by reflected, shifted $g(\cdot)$. More formally, the convolution for continuous functions is defined as

$$
f * g(x) = \int_{\mathbb{R}} f(y) g(x-y) \mathrm{d}y \tag{4.35}
$$

where $f(y)$ is the function that we are concerned with, and $g(x-y)$ is the weight function which is translated by reflecting $g(y)$ along the y-axis and then shifting it by x. A special case is that x and y are both integers. In this case, we can define $f * g(\cdot)$ as

$$
f * g(x) = \sum_{y} f(y) g(x-y) \tag{4.36}
$$

which is the basic form of Eq. (4.33). In CNNs, x, y and $x-y$ can be seen as indices of items in data arrays. $f(y)$ is a data item in the input array, and $g(x-y)$ is the corresponding weight in the filter. By using Eq. (4.36), we calculate the value of the item indexed by x in the output array $f * g(x)$ (i.e., the feature map).

Each feature v_p is given by

$$
\begin{aligned}
v_p &= \mathrm{Conv}(\mathbf{a}_p, \mathbf{W}) \\
&= \mathbf{a}_p \cdot \mathbf{W} \\
&= \sum_{k=1}^{|\mathbf{W}|} a_p(k) \cdot W(k)
\end{aligned} \tag{4.38}
$$

where $a_p(k)$ and $W(k)$ represent the k-th elements of the flattened arrays $\mathbf{a}_p$ and $\mathbf{W}$, respectively. Note that while the array $\begin{bmatrix} v_1 & \dots & v_{|\Omega|} \end{bmatrix}$ can be reshaped into various formats (such as a matrix or a 3D tensor) based on the input dimensions, these representations are fundamentally equivalent from a data storage perspective. For instance, applying a 2D filter to 2D input data yields a 2D feature matrix, as demonstrated in Eq. (4.34).

Furthermore, we need to address two issues to make the model practical. First, we must specify the **stride**, which represents the step size of the filter as it slides over $\mathbf{A}$. In the previous example, we assumed a stride of 1 (i.e., $\mathrm{stride} = 1$). By employing a larger stride, we effectively downsample $\mathbf{A}$, compressing it into a smaller number of features. Second, to control the spatial dimensions of the output feature map, we can introduce dummy elements (or **padding**) around the input data boundaries. Zero-padding, which surrounds the input region with zeros, is the most common approach. For example, consider the 2×2 data matrix:

$$
\mathbf{A} = \begin{bmatrix} 1 & 9 \\ 7 & 3 \end{bmatrix} \tag{4.39}
$$

We can add zero-valued entries around it to obtain a 4×4 matrix

$$
\mathbf{A}_{\mathrm{padding}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 9 & 0 \\ 0 & 7 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{4.40}
$$

Using the same filter as in Eq. (4.33) with $\mathrm{stride} = 1$, we have a 3×3 feature map

$$
\mathrm{Conv}_{\mathrm{stride}=1}(\mathbf{A}_{\mathrm{padding}}, \mathbf{W}) = \begin{bmatrix} 2 & 20 & 18 \\ 14 & 22 & 24 \\ 0 & 14 & 6 \end{bmatrix} \tag{4.41}
$$

If $\mathrm{stride} = 2$, then we would have a feature map with the same size as the input data

$$
\mathrm{Conv}_{\mathrm{stride}=2}(\mathbf{A}_{\mathrm{padding}}, \mathbf{W}) = \begin{bmatrix} 2 & 18 \\ 0 & 6 \end{bmatrix} \tag{4.42}
$$

Note that strictly speaking, the operation implemented in deep learning is technically **cross-correlation** rather than true mathematical convolution. True convolution requires the filter to be reflected (flipped) before sliding it over the input. However, because the filter weights in CNNs are learned from scratch, this flipping step is mathematically redundant. Most deep learning frameworks omit it for computational efficiency, though the field conventionally retains the term "convolution".

4.4.2 CNNs for Sequence Modeling

Following the formulation in the previous sections, we assume that the input of a sequence model is a vector sequence $\mathbf{x}_1...\mathbf{x}_m$ and the output is another vector sequence $\mathbf{h}_1...\mathbf{h}_m$. For example, we can think of $\mathbf{x}_1...\mathbf{x}_m$ as a matrix $\mathbf{X} \in \mathbb{R}^{m \times d_e}$ in which the i-th row vector represents the i-th word of the sequence.

It is straightforward to perform convolution on $\mathbf{X}$. Since $\mathbf{x}_i$ does not impose an ordering across feature dimensions, it is unnecessary to apply convolution across feature dimensions. Hence, we can utilize a receptive field of size $r \times d_e$ and consider all dimensions of $\mathbf{x}_i$ simultaneously in the convolution operation. In practical applications, multiple filters are often employed to capture different aspects of the inputs. For instance, one might use a filter with a large receptive field to incorporate broader context, while using another with a small receptive field to concentrate on highly local features. Figure 4.7 illustrates the application of two distinct filters to an input sequence.

To distribute the extracted features to $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$, we associate each application of a filter with a specific position in the sequence. To ensure the input and output sequences maintain the same length, a padding vector is added to both ends of the input sequence. The following table shows the input and output of a CNN for an example sequence:

Position	Input	Receptive Field	Output
0	$\mathbf{x}_0(=0)$	N/A	N/A
1	$\mathbf{x}_1$	$\{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2\}$	$\mathbf{h}_1$
2	$\mathbf{x}_2$	$\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$	$\mathbf{h}_2$
3	$\mathbf{x}_3$	$\{\mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\}$	$\mathbf{h}_3$
4	$\mathbf{x}_4$	$\{\mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5\}$	$\mathbf{h}_4$
5	$\mathbf{x}_5(=0)$	N/A	N/A

An activation function is typically applied to introduce non-linearity to the final output. In this manner, we construct a standard convolutional layer, which can be viewed as a sequence of fully connected neural networks, each taking inputs from a fixed-size window. For the i-th position, the output of the convolutional layer is given by

$$v_i = \psi(\mathrm{Conv}(\mathbf{a}_i, \mathbf{W})) \tag{4.43}$$

where $\mathbf{a}_i$ represents the inputs in the receptive field[9], and $\mathbf{W}$ represents the parameters of the filter. In situations involving multiple filters (e.g., d_h filters), we have a set of param-

[9] For an $r \times d_e$ receptive field, $\mathbf{a}_i$ is defined to be $\{\mathbf{x}_{\lceil i-\frac{r}{2} \rceil}, ..., \mathbf{x}_{\lceil i+\frac{r}{2}-1 \rceil}\}$ or $\{\mathbf{x}_{\lfloor i-\frac{r}{2} \rfloor}, ..., \mathbf{x}_{\lfloor i+\frac{r}{2}-1 \rfloor}\}$.

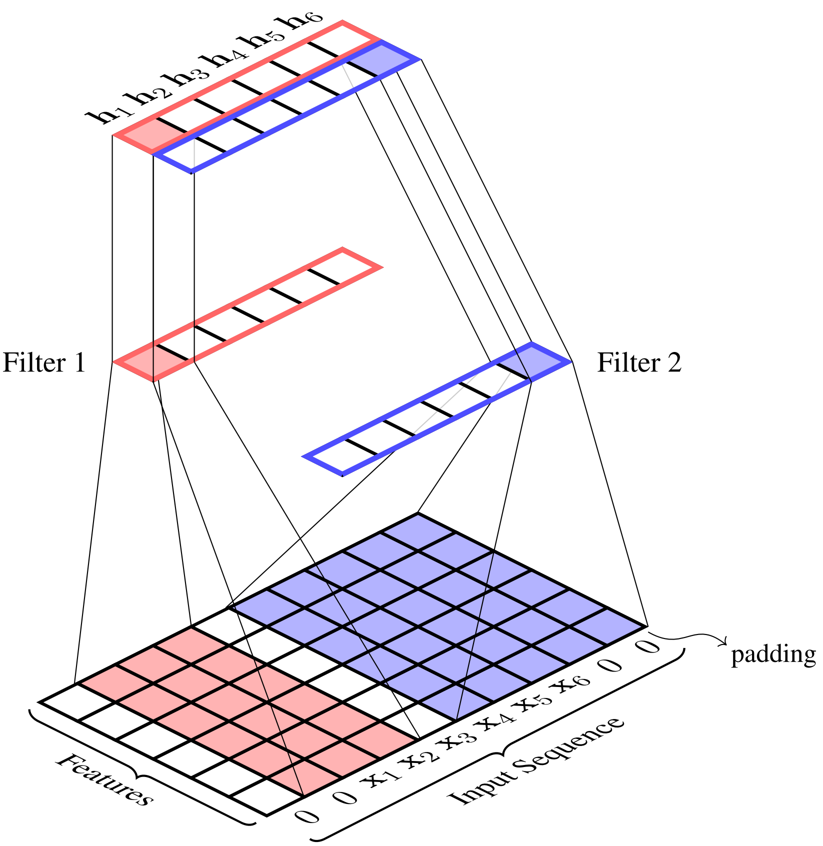


Figure 4.7: Two filters applied to a sequence of word vectors. The input involves ten word vectors (words $\mathbf{x}_1 ... \mathbf{x}_6$ and two padding words on each of the two ends of the sequence). Each word vector has 6 dimensions, and so, the input is a 10×6 matrix. Filter 1 has a receptive field of size 3×6. By sliding it over the input matrix, we obtain a sequence of outputs, each corresponding to a position (i.e., a sequence of 6 outputs). Similarly, we apply filter 2 to the input sequence and obtain another sequence of outputs. The two output sequences are then organized as a 2×6 matrix in which the i-th row vector is $\mathbf{h}_i$.

eters $\{\mathbf{W}^{(1)}, ..., \mathbf{W}^{(d_h)}\}$, a set of activation functions $\{\psi^{(1)}, ..., \psi^{(d_h)}\}$, and a set of inputs $\{\mathbf{a}_i^{(1)}, ..., \mathbf{a}_i^{(d_h)}\}$. Each tuple $(\mathbf{W}^{(k)}, \psi^{(k)}, \mathbf{a}_i^{(k)})$ gives an output by

$$v_i^{(k)} = \psi^{(k)}(\mathrm{Conv}(\mathbf{a}_i^{(k)}, \mathbf{W}^{(k)})) \tag{4.44}$$

Note that $v_i^{(k)}$ is simply an entry of $\mathbf{h}_i$. Thus, $\mathbf{h}_i$ can be written as

$$\mathbf{h}_i = \begin{bmatrix} v_i^{(1)} & ... & v_i^{(d_h)} \end{bmatrix} \tag{4.45}$$

Many CNN-based systems of practical interest comprise two or more convolutional layers. The most straightforward approach to achieve this is through layer stacking, analogous to multi-layer RNNs (see Section 4.2.3). Specifically, the output sequence of a lower convolutional layer

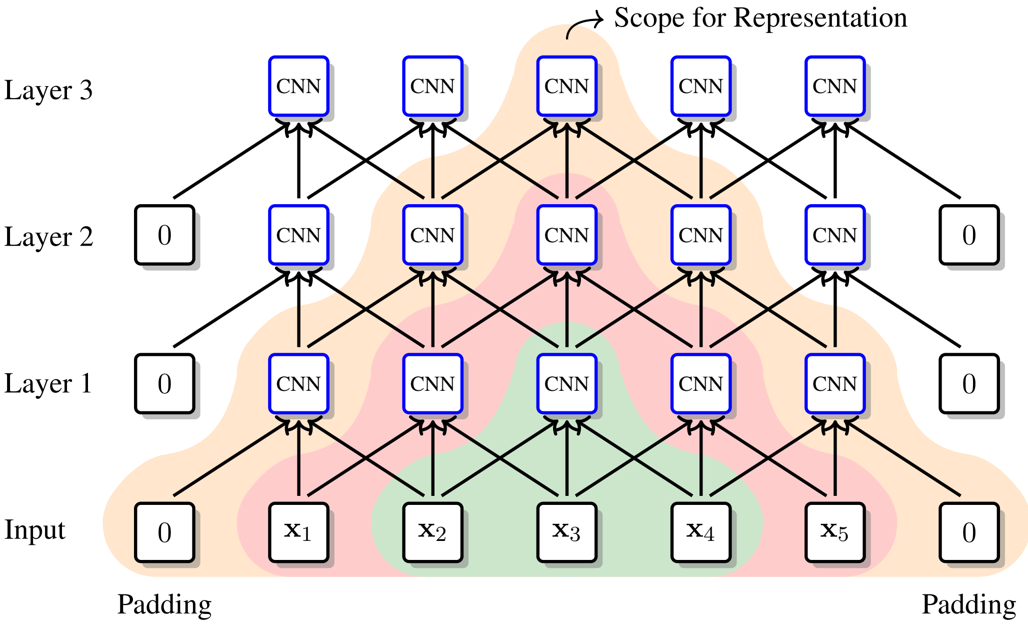


Figure 4.8: A CNN with 3 convolutional layers (stride $= 1$ and $r = 3$). Each layer takes a sequence of vectors and produces another sequence of vectors. In this process a filter moves over the input and performs the convolution operation in each move. In layer 1, the receptive field of the filter is a region of three input items (see green shadows). The higher a layer is, the larger receptive field a filter has. For example, in layer 3, an application of the filter can at most cover the entire input sequence (see orange shadows).

serves directly as the input to the subsequent layer. Figure 4.8 provides an example of a CNN architecture involving three stacked convolutional layers. A primary advantage of multi-layer CNNs is the exponential expansion of their receptive fields, leading to a broader scope for representation. As illustrated in Figure 4.8, while a neuron in Layer 1 aggregates information strictly from three input vectors, a neuron in Layer 3 indirectly aggregates information spanning seven input vectors. Because neurons in higher layers process signals from a much larger contiguous span of the original sequence, they are capable of extracting higher-level semantic representations and effectively capturing long-distance dependencies.

In some applications, we need a fixed-length, low-dimensional representation of the entire sequence. A common way is to add a pooling layer to merge $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$ into a single vector. For example, we can select the maximum value (i.e., max-pooling) or average the values (i.e., average-pooling) along each dimension. This is a generic method in machine learning and is applicable to most of the sequence models discussed in this book.

4.4.3 Handling Positional Information

One compelling property of CNNs is their ability to balance complexity and efficiency by restricting full connectivity to a localized region of the input data. This local connectivity allows us to model a convolutional layer mathematically similarly to a fully-connected layer: the output of a neuron is a non-linear transformation of the weighted sum of its inputs. Despite the computational efficiency of this local connectivity, a limitation of standard CNNs is their translation invariance. While they capture local structures within the receptive field, they lack an awareness of absolute global positions within the sequence.

Interestingly, if we restrict ourselves to sequence modeling, this might seem unproblematic because the output of the model is itself a sequence. It seems reasonable to assume that the output sequence preserves the ordering information of the input sequence. However, applying CNNs to sequential data does not guarantee a strict one-to-one mapping between the input and output items. Technically, $\mathbf{h}_i$ is not simply a representation of $\mathbf{x}_i$. Rather, it encodes a window of inputs centered around $\mathbf{x}_i$. This significantly complicates the problem, as it becomes exceedingly difficult to infer from $\mathbf{h}_i$ exactly how those localized inputs were originally ordered.

Explicitly modeling word orders is very important in NLP, and has been extensively studied in tasks like machine translation [Lopez, 2008; Koehn, 2010]. For neural network-based models, one may address the problem by resorting to order-sensitive model architectures like RNNs. A more popular approach in recent systems is to develop a **positional encoding** sub-model and incorporate it into existing sequence models [Gehring et al., 2017b; Vaswani et al., 2017; Shaw et al., 2018; Dufter et al., 2022]. Formally, we say that the input at position i is a combination of the original input $\mathbf{x}_i$ and the positional encoding of i (denoted by $\mathrm{PE}(\cdot)$):

$$\mathbf{e}_i = \mathrm{Merge}(\mathbf{x}_i, \mathrm{PE}(i)) \tag{4.46}$$

where $\mathrm{Merge}(\cdot)$ defines a combination operation. The use of positional encoding is straightforward: one simply needs to replace $\{\mathbf{x}_1,...,\mathbf{x}_m\}$ with $\{\mathbf{e}_1,...,\mathbf{e}_m\}$ in a sequence model. So, this approach is model-agnostic.

In this subsection, we present several versions of $\mathrm{PE}(\cdot)$ and ways to combine them with $\mathbf{x}_i$. Note that the following discussion is not specific to CNNs. We consider it here because positional encoding is useful for models that are insensitive to the order of inputs, and CNNs are a good example to see how it is used [Gehring et al., 2017a;b]. In Chapter 6, we will see an application of positional encoding in Transformer which is a state-of-the-art neural model in many areas.

1. Offset-based Positional Encoding

The simplest way to represent a position i is to use its absolute index. This can be formulated as the "distance" from a reference point

$$\mathrm{PE}(i) = i - i_0 \tag{4.47}$$

where i_0 is an integer indicating where we start counting. If $i_0 = 0$, $\mathrm{PE}(i) = i$ yields the standard absolute position. Note that $\mathrm{PE}(i)$ could be a negative number if $i_0 > i$. While not a physical distance in the strict sense, it remains a perfectly valid feature for machine learning algorithms. To enforce a non-negative measure, the right-hand side of Eq. (4.47) can be bounded using an absolute value:

$$\mathrm{PE}(i) = |i - i_0| \tag{4.48}$$

Treating positions as simple integers leads to unbounded, discrete positional encoding. A more desirable method might be to use a continuous representation in a range of values, because it allows the system to work within a sample space that is smooth and easy to optimize. A simple way to do this is normalization. For example, dividing $i - i_0$ by some maximum value, we obtain a normalized version of the offset-based encoding

$$\mathrm{PE}(i) = \frac{i - i_0}{i_{\max} - i_0} \tag{4.49}$$

For example, we can set $i_{\max}$ = the maximum possible length of the sequence and $i_0 = 0$, so that $\mathrm{PE}(i)$ chooses its value in $[0,1]$. Another common choice is to set $i_{\max} = m$ (i.e., the length of the input sequence) and define $\mathrm{PE}(i)$ as a ratio whose value varies as m changes.

To make use of these scalar positions, it is straightforward to enrich the original input vectors by adding new dimensions, provided they can be viewed as new features. Thus, $\mathbf{e}_i$ is given by

$$\mathbf{e}_i = [\mathbf{x}_i, \mathrm{PE}(i)] \tag{4.50}$$

where $[\cdot]$ stands for the concatenation operation.

2. Sinusoidal Positional Encoding

The next obvious step is to represent positions as vectors instead of scalars. Although vectorizing the representations of positions sounds complicated, a simple idea is to use a carrying system which describes how a natural number is expressed by a polynomial with respect to a base [Kernes, 2021]. For example, i can be written as

$$i = \sum_{k=0}^{k_{\max}} a(i,k) b^k \tag{4.51}$$

where $a(i,k)$ is the k-th digit, $k_{\max}+1$ is the maximum number of digits, and b is the base of the system. The carrying occurs when $a(i,k)$ reaches b: we increase $a(i,k+1)$ by 1 and roll back $a(i,k)$ to 0. In this way we can change $a(i,k)$ with a period of b^k, that is, $a(i,0)$ changes with a period of b^0, $a(i,1)$ changes with a period of b^1, $a(i,2)$ changes with a period of b^2, and so on.

Using this system, i can be represented as a vector

$$\mathrm{PE}(i) = \begin{bmatrix} a(i,0) & a(i,1) & \dots & a(i,k_{\max}) \end{bmatrix} \tag{4.52}$$

For example, when $b = 2$, $\mathrm{PE}(11) = \begin{bmatrix} 1 & 1 & 0 & 1 \end{bmatrix}$. However, in Eq. (4.52), $\mathrm{PE}(i)$ is still a discrete function. As discussed throughout this book, we may want a continuous vector representation that can describe intermediate states between discrete events. Considering $a(i,k)$ as a periodic function, a common choice is the **sine** function. Thus $a(i,k)$ can be re-defined, as follows

$$a(i,k) = \sin(i \cdot \omega_k) \tag{4.53}$$

This function has an amplitude of 1 and a period of $\frac{2\pi}{\omega_k}$. Using an analogous form of periods to that used in Eq. (4.51), we define ω_k as

$$\omega_k = \frac{1}{(b_{\text{model}})^{k/d_{\text{model}}}} \tag{4.54}$$

where $b_{\text{model}} > 0$ and $d_{\text{model}} > 0$ are hyper-parameters of the model. Clearly, we have $\frac{2\pi}{\omega_0} < \frac{2\pi}{\omega_1} < ... < \frac{2\pi}{\omega_{k_{\max}}}$.

Similarly, we can define $a(i,k)$ via the **cosine** function

$$a(i,k) = \cos(i \cdot \omega_k) \tag{4.55}$$

Taking both Eqs. (4.53) and (4.55), we create a new representation of i, as follows

$$\mathrm{PE}(i) = \begin{bmatrix} \sin(i \cdot \omega_0) & \cos(i \cdot \omega_0) & ... & \sin(i \cdot \omega_{k_{\max}}) & \cos(i \cdot \omega_{k_{\max}}) \end{bmatrix} \tag{4.56}$$

Vaswani et al. [2017] instantiated the above form by setting $b_{\text{model}} = 10,000$. Let $\mathrm{PE}(i,k)$ be the k-th dimension of $\mathrm{PE}(i)$. Vaswani et al. [2017]'s version of positional encoding is written as

$$\mathrm{PE}(i,2k) = \sin(i \cdot \frac{1}{10000^{2k/d_{\text{model}}}}) \tag{4.57}$$

$$\mathrm{PE}(i,2k+1) = \cos(i \cdot \frac{1}{10000^{2k/d_{\text{model}}}}) \tag{4.58}$$

Choosing $b_{\text{model}} = 10,000$ is empirical. One can adjust it for specific tasks. Figure 4.9 plots the positional encoding for different positions. We see that, when k becomes larger, the change of the color follows a larger period.

Note that Eqs. (6.14) and (6.15) have a useful property that $\mathrm{PE}(i+\mu)$ can be easily

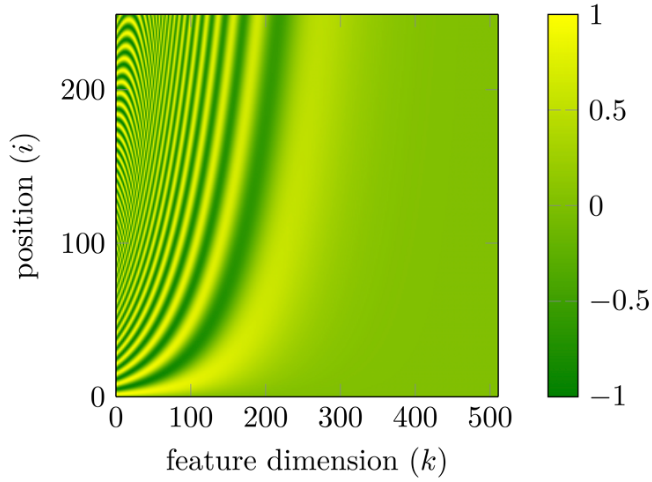


Figure 4.9: A heat map of the positional embedding model of Eqs. (6.14) and (6.15) ($b_{\text{model}} = 10,000$ and $d_{\text{model}} = 512$). Consider a position i (i.e., the i-th row), then move another position j from i upwards or downwards. Intuitively, when i and j are closer, the corresponding row vectors are more similar. By contrast, when j moves away from i, the similarity is not that obvious. This property helps explain the idea behind the positional embedding model: the "distance" between two positions is implicitly modeled by comparing their multi-dimensional representations.

expressed by a linear function of $\text{PE}(i)$ for a given offset μ[10]

$$\begin{aligned}\text{PE}(i+\mu, 2k) &= \text{PE}(i, 2k) \cdot \text{PE}(\mu, 2k+1) + \\ &\quad \text{PE}(i, 2k+1) \cdot \text{PE}(\mu, 2k) \qquad (4.61)\\ \text{PE}(i+\mu, 2k+1) &= \text{PE}(i, 2k+1) \cdot \text{PE}(\mu, 2k+1) - \\ &\quad \text{PE}(i, 2k) \cdot \text{PE}(\mu, 2k) \qquad (4.62)\end{aligned}$$

This property is advantageous because it allows the model to attend to relative positions. That is, the state at position $i+\mu$ can be described by starting with i and then appending it with the offset μ.

When applying the sinusoidal positional encoding, one way is to use Eq. (4.50) to concatenate $\mathbf{x}_i$ and $\text{PE}(i)$. In Vaswani et al. [2017]'s work, they instead assume $\text{PE}(i)$ to be a vector of the same size as $\mathbf{x}_i$ (i.e., $|\text{PE}(i)| = |\mathbf{x}_i| = d_e$), and add $\text{PE}(i)$ to $\mathbf{x}_i$

$$\mathbf{e}_i = \mathbf{x}_i + \text{PE}(i) \qquad (4.63)$$

[10] One can derive these by taking

$$\begin{aligned}\sin(\alpha+\beta) &= \sin(\alpha)\cdot\cos(\beta) + \cos(\alpha)\cdot\sin(\beta) \qquad (4.59)\\ \cos(\alpha+\beta) &= \cos(\alpha)\cdot\cos(\beta) - \sin(\alpha)\cdot\sin(\beta) \qquad (4.60)\end{aligned}$$

This sinusoidal additive model has been the basis of many positional encoding approaches [Dehghani et al., 2018; Likhomanenko et al., 2021; Su et al., 2021].

3. Learnable Positional Encoding

The application of sinusoidal positional encoding yields a lookup table $\mathbf{C}_{\mathrm{PE}} \in \mathbb{R}^{m_{\max} \times d_e}$ (where $m_{\max}$ is the maximum sequence length)

$$\mathbf{C}_{\mathrm{PE}} = \begin{bmatrix} \mathrm{PE}(1) \\ \vdots \\ \mathrm{PE}(m_{\max}) \end{bmatrix} \tag{4.64}$$

In this table, each row vector $\mathrm{PE}(i)$ corresponds to the embedding of a position i. These vectors, as described previoiusly, these sinusoidal vectors are deterministic and derived from fixed mathematical functions. An alternative approach is to treat vectors of positions as parameters of the model and learn them as usual. In this case, both word embeddings and position embeddings can be trained in the same manner. See Chapters 2 and 3 for more information about learning word embeddings in neural language models.

One last note on positional encoding. What we have shown in this section can broadly be characterized as an **absolute positional encoding** paradigm: a position is described by its location in a coordinate system. Another concept that is worth exploring is **relative positional encoding** [Shaw et al., 2018]. For example, we can extend Eq. (4.48) to define the distance between two positions i and j

$$\mathrm{PE}(i,j) = |i-j| \tag{4.65}$$

In this case, the positional encoding is no longer an attribute of the i-th input but some representation of the distance relative to a reference position j. In fact, most of the methods for relative positional encoding are variants on a theme in which positions are described by their pair-wise relationships. This relational inductive bias forms the foundation of several advanced architectures, which will be extensively detailed in Chapters 6 and 8.

4.5 Examples

Both recurrent and convolutional models have been successfully used in numerous applications. Here we discuss a few of the interesting examples. While these models are mostly basic, they form the foundations of many state-of-the-art systems.

4.5.1 Text Classification

To illustrate how sequence models can be used, we first consider the text classification task, where we assign a predefined class to a given text sequence. This framework naturally extends to a broad spectrum of NLP problems, including categorizing news articles, analyzing sentiment, identifying spam emails, detecting deceptive comments, and so forth.

In text classification we are interested in predicting the class label from a set C, given a word sequence $w_1...w_m$:

$$\hat{c} = \arg\max_{c \in C} \mathrm{Score}(c, w_1...w_m) \tag{4.66}$$

Here $\mathrm{Score}(c, w_1...w_m)$ measures how well a class c is predicted for the input sequence $w_1...w_m$. We map the sequence of words to the sequence of word vectors (or word embeddings), that is, $w_1...w_m \to \mathbf{x}_1...\mathbf{x}_m$. Assuming $\mathrm{Score}(\cdot)$ is a probabilistic function that describes the distribution of the classes, we can reformulate the problem as

$$\begin{aligned} \hat{c} &= \arg\max_{c \in C} \Pr(c|w_1...w_m) \\ &= \arg\max_{c \in C} \Pr(c|\mathbf{x}_1...\mathbf{x}_m) \end{aligned} \tag{4.67}$$

The central issue here is the modeling of $\Pr(c|\mathbf{x}_1...\mathbf{x}_m)$. We define $\Pr(c|\mathbf{x}_1...\mathbf{x}_m)$ by following the general encoder-predictor framework, as follows

- The input $\mathbf{x}_1...\mathbf{x}_m$ is represented as a feature vector $\mathbf{H} \in \mathbb{R}^{d_h}$ by using an encoder

$$\mathbf{H} = \mathrm{Encode}(\mathbf{x}_1...\mathbf{x}_m) \tag{4.68}$$

- $\mathbf{H}$ is fed to a standard Softmax layer to predict the class distribution

$$\Pr(\cdot|\mathbf{H}) = \mathrm{Softmax}(\mathbf{H} \cdot \mathbf{U} + \mathbf{b}) \tag{4.69}$$

  where $\mathbf{U} \in \mathbb{R}^{d_h \times |C|}$ and $\mathbf{b} \in \mathbb{R}^{|C|}$ are model parameters.

The $\mathrm{Encode}(\cdot)$ component is the same as the architectures discussed in the preceding sections. As a result, a wide variety of sequence encoding models are directly applicable here. For example, consider the CNN-based encoder presented in Kim [2014]'s work. Kim [2014]'s model is based on a single convolution layer involving d_h filters. The application of a filter produces a set of features, each being associated with a position of the sequence. Since we want a single vector for representing the entire sequence, a pooling layer is added so that the number of features for each filter is reduced to one. Then, for any c, the probability $\Pr(c|\mathbf{H})$ can be computed directly according to Eq. (4.69). See Figure 4.10 for an illustration of this classifier.

Training such a model involves optimizing a predefined loss function, and, as mentioned several times in this book, one of the most common methods is to minimize the cross-entropy loss using gradient descent. Also, we can use regularization to improve the training of CNNs. More details about these techniques can be found in Chapter 2.

It is worth noting that while this baseline model is straightforward, it remains one of the most highly effective architectures for text classification. Many architectural refinements exist. Prominent extensions of this framework include deep CNNs [Conneau et al., 2017c], character-level CNNs [Santos and Gatti, 2014; Zhang et al., 2015], and recurrent-convolutional

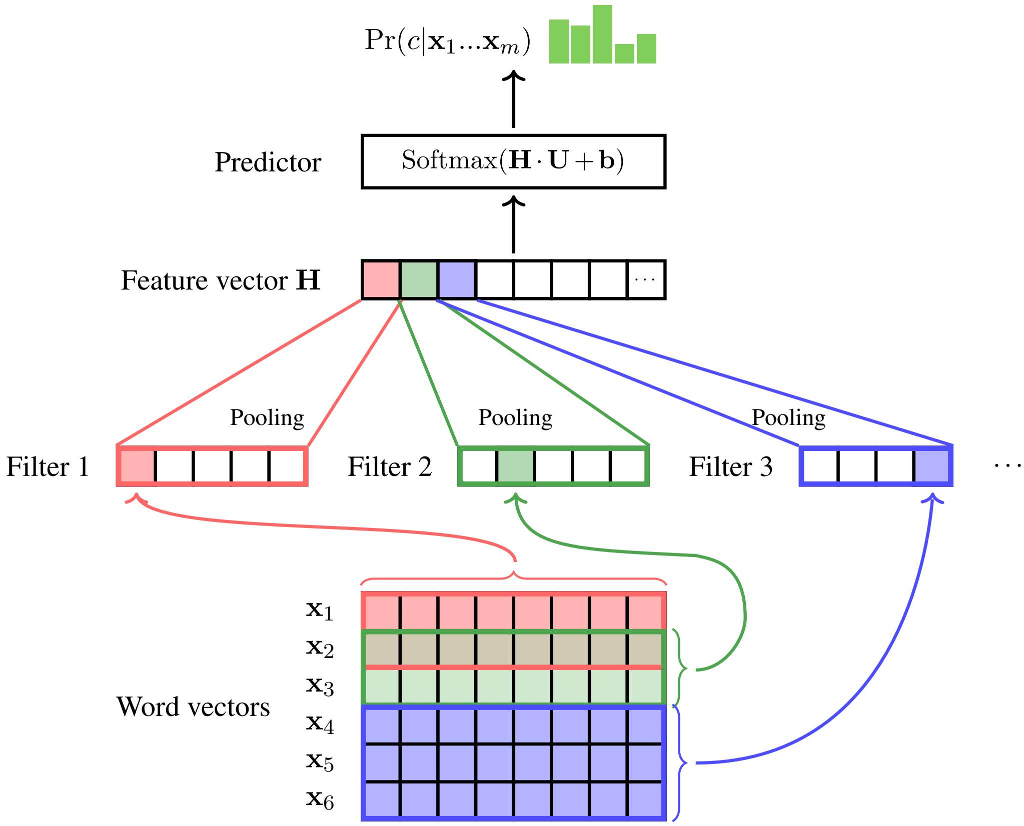


Figure 4.10: A CNN-based text classifier [Kim, 2014]. The input is a sequence of word vectors. A convolutional layer involving multiple filters is used to extract features in different dimensions. A pooling layer is used to reduce the number of features for representing the input text, leading to a low-dimensional feature vector $\mathbf{H}$. The prediction conditions on $\mathbf{H}$ and is made by using a standard Softmax layer.

hybrids [Lai et al., 2015].

4.5.2 End-to-End Speech Recognition

Speech recognition is the task of mapping a sequence of acoustic signals to a sequence of words or characters (called a **transcription**) [Reddy, 1976; Rabiner and Juang, 1993]. Since the original input is an acoustic waveform over the time domain, it is standard practice to transform it into a sequence of waveform fragments (called **frames**). Typically, a frame is represented as a feature vector, denoted by $\mathbf{x}_i$. This is achieved by employing either engineered feature functions from signal processing [Davis and Mermelstein, 1980; Picone, 1993; Campbell, 1997] or learnable neural embeddings [Chorowski et al., 2019; Schneider et al., 2019]. Regarding the output, speech recognition systems generally do not directly produce full words. Instead, they output sequences of **transcription units** (or **transcription labels**), such as phonemes, characters, or sub-words. In this subsection, we assume that the output of the system is a sequence of characters, denoted by $y_1, \ldots, y_n \in V_y^n$. The vocabulary

V_y consists of standard English letters $(\mathrm{a}-\mathrm{z})$, digits $(0-9)$, spaces $(\langle \mathrm{sp} \rangle)$, periods $(\langle \mathrm{pe} \rangle)$, and other punctuation marks. As with most modern speech recognition systems, we add a blank symbol ϵ to the alphabet in order to indicate the null output.

The goal here is to find a string $\hat{y}_1...\hat{y}_n$ for a given input sequence $\mathbf{x}_1...\mathbf{x}_m$, so that

$$\hat{y}_1...\hat{y}_n = \arg\max_{y_1...y_n} \Pr(y_1...y_n|\mathbf{x}_1...\mathbf{x}_m) \tag{4.70}$$

This problem is more challenging than the classification problem described in the previous subsection, as the output can be an arbitrary string, rather than a class in a predefined class set. The string generation problem leads to two difficulties. First, we need some mechanism to model $\Pr(y_1...y_n|\mathbf{x}_1...\mathbf{x}_m)$ for an exponentially large number of pairs of input and output sequences. Second, in practice $y_1...y_n$ is often much shorter than $\mathbf{x}_1...\mathbf{x}_m$ (i.e., $n < m$), and so we need some mechanism to align a long sequence to a short one. However, we do not need to consider these difficulties in the stage of representing the input sequence, and can still encode the input sequence $\mathbf{x}_1...\mathbf{x}_m$ in the same way as other sequence models. Specifically, we represent $\mathbf{x}_1...\mathbf{x}_m$ in the following form

$$\mathbf{h}_1...\mathbf{h}_m = \mathrm{Encode}(\mathbf{x}_1...\mathbf{x}_m) \tag{4.71}$$

The encoder can be RNNs, CNNs, or more advanced models (such as Transformers). Here we consider the encoder architecture used in Graves et al. [2013b] and Graves and Jaitly [2014]'s work. It is a multi-layer bi-directional LSTM model. We skip the details of this model without loss of continuity, as the reader may already be familiar with it in Sections 4.2.3, 4.2.4 and 4.3.2.

Having obtained the sequence representation $\mathbf{H} = \mathbf{h}_1...\mathbf{h}_m$, a softmax layer is used to map each $\mathbf{h}_i \in \mathbb{R}^{d_h}$ to a distribution of transcription labels, given by

$$\Pr(\cdot|\mathbf{h}_i) = \mathrm{Softmax}(\mathbf{h}_i \cdot \mathbf{U} + \mathbf{b})$$

where $\mathbf{U} \in \mathbb{R}^{d_h \times |V_y|}$ and $\mathbf{b} \in \mathbb{R}^{|V_y|}$ are model parameters. $\Pr(\cdot|\mathbf{h}_i) \in \mathbb{R}^{|V_y|}$ is a probability distribution over V_y, and the probability of transcription label l_i at position i is simply $\Pr(l_i|\mathbf{h}_i)$. We can then write the probability of a label sequence in the form

$$\Pr(l_1...l_m|\mathbf{H}) = \prod_{i=1}^{m} \Pr(l_i|\mathbf{h}_i) \tag{4.72}$$

We can apply the $\arg\max$ operation to find the most probable label sequence as usual. However, $l_1...l_m$ cannot be straightforwardly used as a system output, because it often contains many duplicate and blank symbols. To post-process $l_1...l_m$, we first collapse consecutive identical labels into a single label, and then remove the blank symbols. For example, consider a label sequence

$$\mathrm{s} \quad \mathrm{s} \quad \epsilon \quad \mathrm{e} \quad \mathrm{e} \quad \epsilon \quad \epsilon \quad \mathrm{e} \quad \langle \mathrm{sp} \rangle \quad \mathrm{y} \quad \epsilon \quad \mathrm{o} \quad \mathrm{o} \quad \mathrm{u}$$

By merging "s s", "e e", and "o o", we have

$$\mathrm{s}\quad \epsilon\quad \mathrm{e}\quad \epsilon\quad \epsilon\quad \mathrm{e}\quad \langle\mathrm{sp}\rangle\quad \mathrm{y}\quad \epsilon\quad \mathrm{o}\quad \mathrm{u}$$

Then, we remove all ϵ and obtain

$$\mathrm{s}\quad \mathrm{e}\quad \mathrm{e}\quad \langle\mathrm{sp}\rangle\quad \mathrm{y}\quad \mathrm{o}\quad \mathrm{u}$$

The above sequence is what we would call a transcription. Clearly, different label sequences can correspond to the same transcription. Let $B(y_1...y_n)$ be the set of label sequences corresponding to $y_1...y_n$[11]. We now turn to the following form of the transcription probability (see Figure 4.11 for an illustration)

$$\Pr(y_1...y_n|\mathbf{x}_1...\mathbf{x}_m) \quad = \sum_{l_1...l_m \in B(y_1...y_n)} \Pr(l_1...l_m|\mathbf{H}) \tag{4.73}$$

A problem with this model is that the number of sequences in $B(y_1...y_n)$ grows exponentially with n (and m). Fortunately, there are very efficient methods for computing $\sum_{l_1...l_m \in B(y_1...y_n)} \Pr(l_1...l_m|\mathbf{H})$. See [Graves et al., 2006] for a dynamic programming algorithm for solving this problem.

Note that Eq. (4.73) is also known as a form of **connectionist temporal classification** (**CTC**) [Graves et al., 2006]. It is one of the most widely used methods for training end-to-end speech recognition and speech translation systems. One of the merits of CTC is that it allows us to align any label sequence to a transcription in a very simple and efficient way. It is easy to make use of CTC in training a speech recognition system. Suppose there is a set of pairs of input sequence and transcription, denoted by S. A common training objective is to maximize the likelihood of these transcriptions given the corresponding inputs, written as

$$\hat{\theta} \quad = \quad \arg\max_{\theta} \sum_{(y_1...y_n, \mathbf{x}_1...\mathbf{x}_m) \in S} \log \Pr(y_1...y_n|\mathbf{x}_1...\mathbf{x}_m, \theta) \tag{4.74}$$

where $\Pr(y_1...y_n|\mathbf{x}_1...\mathbf{x}_m, \theta)$ denotes the probability computed via Eq. (4.73), and θ denotes the model parameters.

When testing on new data, we search for an optimal transcription as in Eq. (4.70). This process, also known as **decoding**, generally involves optimized search algorithms and pruning techniques. For example, we can use the 1-best label sequence instead of all possible label sequences to obtain an approximation to Eq. (4.73), that is, $\Pr(y_1...y_n|\mathbf{x}_1...\mathbf{x}_m) = \max \Pr(l_1...l_m|\mathbf{H})$. This leads to an efficient decoding method, called **Viterbi decoding**, which has been extensively used in speech recognition and machine translation [Lopez, 2008]. For more details about the decoding of sequence generation, we refer the reader to Chapter 5.

[11] $B(y_1...y_n)$ may contain label sequences of arbitrary lengths. However, if we restrict input to the sequence $\mathbf{x}_1...\mathbf{x}_m$, then each sequence in $B(y_1...y_n)$ is of length m.

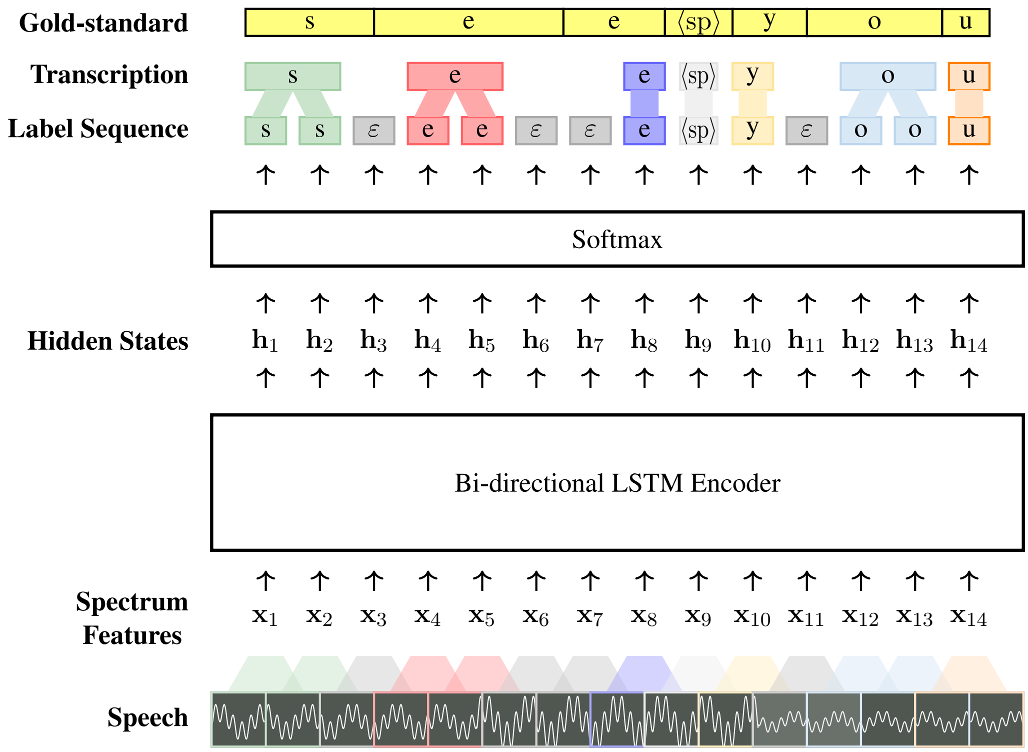


Figure 4.11: An end-to-end speech recognition architecture. The input of the system is a sequence of acoustic signals that are represented as a sequence of feature vectors (i.e.., $\mathbf{x}_1 ... \mathbf{x}_{14}$). These feature vectors are taken by a bi-directional LSTM encoder. The output of the encoder is a sequence of contextualized representations (i.e., $\mathbf{h}_1 ... \mathbf{h}_{14}$) which is then fed into a softmax layer for generating a distribution of labels at each position. We can then generate a sequence of labels from these distributions. We map each label sequence to a form of final output by eliminating duplicate symbols and blank symbols. An output of the system corresponds to multiple label sequences, and the probability of the output is the sum of the probabilities of these label sequences.

4.5.3 Sequence Labeling with LSTM and Graphical Models

Sequence labeling is a conceptually straightforward approach to labeling sequence data. In NLP, it has been widely used in many sub-areas, such as word segmentation, part-of-speech tagging, and chunking. Learning in these models involves predicting a label from a predefined label set V_y at each position of a sequence. Ideally, we wish to predict the optimal label sequence given the entire input, formulated as:

$$\hat{y}_1 ... \hat{y}_m = \underset{y_1 ... y_m}{\arg\max} \Pr(y_1 ... y_m | \mathbf{x}_1 ... \mathbf{x}_m) \tag{4.75}$$

where $\mathbf{x}_1 ... \mathbf{x}_m$ is an input sequence (such as a sequence of word vectors), and $y_1 ... y_m$ is a label sequence in which each label y_i corresponds to an input item $\mathbf{x}_i$.

As we have seen in this chapter, Eq. (4.75) fits well within the formulation of the sequence modeling problem. As a first step, we use an encoder to map $\mathbf{x}_1 ... \mathbf{x}_m$ to a sequence of contextualized representations, as follows

$$\mathbf{h}_1...\mathbf{h}_m = \mathrm{Encode}(\mathbf{x}_1...\mathbf{x}_m) \tag{4.76}$$

We define $\mathrm{Encode}(\cdot)$ as a bi-directional LSTM model because it involves a memory mechanism for modeling long-range dependencies in both left and right contexts. Hence, the architecture of the encoder is the same as that used in the preceding subsection.

$\mathbf{h}_1...\mathbf{h}_m$ can then serve as the input to a standard sequence labeling system (see Figure 4.12). The sequence labeling problem has been discussed in Chapter 1, and many models can be applied to address it. The simplest is a classifier (such as maximum entropy models and SVM-based models) that predicts a label distribution for each $\mathbf{h}_i$. A limitation of these models is that the predictions are made independently of one another. A more powerful approach is to use **graphical models** to consider dependencies among predicted labels. For example, **hidden Markov models** (**HMMs**) describe how a sequence of observations (i.e., $\mathbf{x}_1...\mathbf{x}_m$) is generated given a sequence of variables (i.e., $y_1...y_m$). The key idea is to rewrite $\Pr(y_1...y_m|\mathbf{x}_1...\mathbf{x}_m)$ by making independence assumptions to factorize the distribution, and approximate $\Pr(y_1...y_m|\mathbf{x}_1...\mathbf{x}_m)$ by a product of simple factors. However, these models require probability density functions of continuous variables (e.g., $\Pr(\mathbf{x}_i|y_i)$) which are difficult to estimate. This significantly differentiates the use of HMMs in neural models from that in conventional models where all states and observations are discrete variables[12].

HMMs and their descendants can be viewed as instances of generative models. Another class of models commonly used for sequence labeling is discriminative models. One such model is **conditional random fields** (**CRFs**) [Lafferty et al., 2001]. The CRF model is characterized by its ability to directly model the conditional probability of a label sequence given the input sequence, which facilitates the integration of arbitrary features. Consider, for example, the **linear-chain CRF** [Sutton and McCallum, 2012]. It defines $\Pr(y_1...y_m|\mathbf{h}_1...\mathbf{h}_m)$

[12]In HMMs, a sequence of variables can be viewed as a path of transiting over some states whose values are chosen from a pre-defined set. In each transition from one state to another, something is observed (call it an observation). By making some assumptions, we can approximate $\Pr(y_1...y_m|\mathbf{x}_1...\mathbf{x}_m)$ in the following fashion

$$\begin{aligned}\Pr(y_1...y_m|\mathbf{x}_1...\mathbf{x}_m) &= \frac{\Pr(y_1...y_m)\cdot\Pr(\mathbf{x}_1...\mathbf{x}_m|y_1...y_m)}{\Pr(\mathbf{x}_1...\mathbf{x}_m)} \\ &\approx \frac{\prod_{i=1}^{m}\Pr(y_i|y_{i-1})\cdot\prod_{i=1}^{m}\Pr(\mathbf{x}_i|y_i)}{\Pr(\mathbf{x}_1...\mathbf{x}_m)} \\ &= \frac{\prod_{i=1}^{m}\Pr(y_i|y_{i-1})\Pr(\mathbf{x}_i|y_i)}{\Pr(\mathbf{x}_1...\mathbf{x}_m)}\end{aligned} \tag{4.77}$$

where $\Pr(y_i|y_{i-1})$ is the **transition probability** of moving from y_{i-1} to y_i (when $i=1$, we define $\Pr(y_i|y_{i-1}) = \Pr(y_1|y_0) = \Pr(y_1)$), and $\Pr(\mathbf{x}_i|y_i)$ is the **emission probability** of observing $\mathbf{x}_i$ given y_i. As the denominator $\Pr(\mathbf{x}_1...\mathbf{x}_m)$ is a constant for different $y_1...y_m$, it can be dropped in the argmax operation of Eq. (4.75), as follows

$$\hat{y}_1...\hat{y}_m = \underset{y_1\cdots y_m}{\arg\max}\prod_{i=1}^{m}\Pr(y_i|y_{i-1})\Pr(\mathbf{x}_i|y_i) \tag{4.78}$$

To estimate $\Pr(\mathbf{x}_i|y_i)$, a possible solution is to take $\Pr(\mathbf{x}_i|y_i) = \frac{\Pr(y_i|\mathbf{x}_i)\Pr(\mathbf{x}_i)}{\Pr(y_i)}$, and use a neural network to compute $\Pr(y_i|\mathbf{x}_i)$.

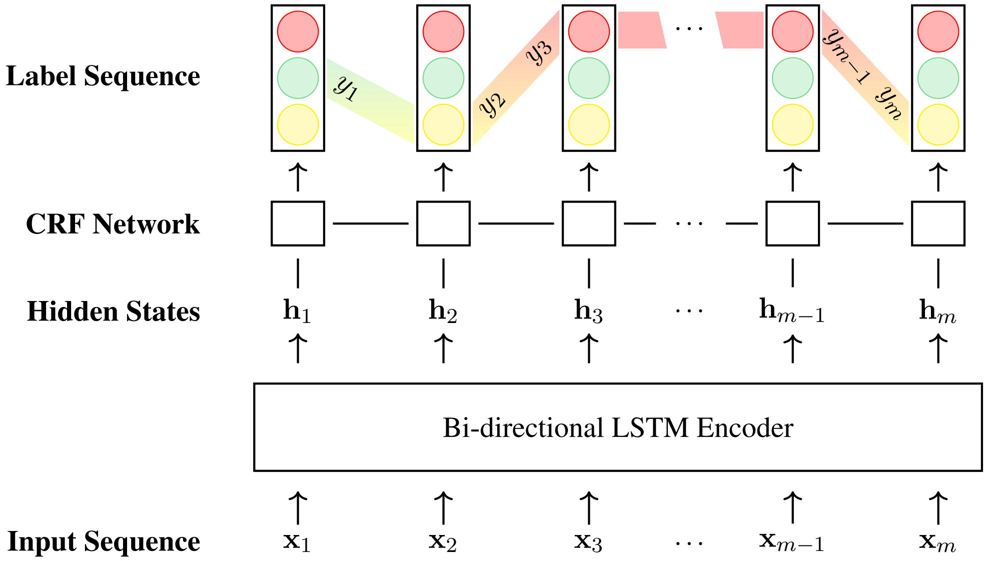


Figure 4.12: The BiLSTM + graphical model architecture for sequence labeling. The encoder is a standard bi-directional LSTM model. Given a sequence of input feature vectors (i.e., $\mathbf{x}_1...\mathbf{x}_m$), it produces a new sequence of feature vectors for mapping the input to contextualized representations (i.e., $\mathbf{h}_1...\mathbf{h}_m$). A CRF network is placed on $\mathbf{h}_1...\mathbf{h}_m$ to predict a distribution of label sequences. The optimal label sequence is the one that has the maximum probability.

in the following form

$$\begin{aligned}\Pr(y_1...y_m|\mathbf{h}_1...\mathbf{h}_m) &= \frac{\Pr(y_1...y_m,\mathbf{h}_1...\mathbf{h}_m)}{\Pr(\mathbf{h}_1...\mathbf{h}_m)} \\ &= \frac{\exp(\mathrm{Score}(y_1...y_m,\mathbf{h}_1...\mathbf{h}_m))}{Z(\mathbf{h}_1...\mathbf{h}_m)} \end{aligned} \tag{4.79}$$

where $Z(\mathbf{h}_1...\mathbf{h}_m)$ is a normalization factor, and has the form

$$Z(\mathbf{h}_1...\mathbf{h}_m) = \sum_{y'_1...y'_m} \exp(\mathrm{Score}(y'_1...y'_m,\mathbf{h}_1...\mathbf{h}_m)) \tag{4.80}$$

$\mathrm{Score}(\cdot)$ is a score for weighting the sequence pair $(y_1...y_m,\mathbf{h}_1...\mathbf{h}_m)$. It is given by summing over the values of a set of feature functions $\{f_1(\cdot),...,f_J(\cdot)\}$

$$\mathrm{Score}(y_1...y_m,\mathbf{h}_1...\mathbf{h}_m) = \sum_{i=1}^{m}\sum_{j=1}^{J} f_j(y_i,y_{i-1},\mathbf{h}_i) \tag{4.81}$$

The outer loop of the summation corresponds to a visit to each position i. Given i, each function $f_j(\cdot)$ takes the current label y_i, the previous label y_{i-1} and the current input vector $\mathbf{h}_i$, and then returns the value of a feature.

This model is called *linear-chain* because it assumes a first-order Markov property: each label y_i is dependent only on its immediate neighbors y_{i-1} and y_{i+1}, conditioned on the observed sequence $\mathbf{H}$[13], as follows

$$\begin{array}{ccccccccccccc} y_1 & — & \cdots & — & y_{i-1} & — & y_i & — & y_{i+1} & — & \cdots & — & y_m \\ | & & & & | & & | & & | & & & & | \\ \mathbf{h}_1 & & \cdots & & \mathbf{h}_{i-1} & & \mathbf{h}_i & & \mathbf{h}_{i+1} & & \cdots & & \mathbf{h}_m \end{array}$$

In CRFs, it is assumed that the variables in the graph depend only on their neighboring variables. Therefore, $f_j(y_i, y_{i-1}, \mathbf{h}_i)$ can be defined according to how y_i is connected. There are generally two types of features.

- **Transition-like Features**. These features model the connection between consecutive labels (y_{i-1}, y_i), given by

$$f_1(y_i, y_{i-1}, \mathbf{h}_i) \quad = \quad u(y_{i-1}, y_i) \tag{4.82}$$

  where $u(y_{i-1}, y_i)$ is an entry of a weight matrix $\mathbf{u}$, indexed by (y_{i-1}, y_i).
- **Emission-like Features**. The second type of feature models the connection between a label y_i and the associated input $\mathbf{x}_i$, given by

$$f_2(y_i, y_{i-1}, \mathbf{h}_i) \quad = \quad g_i(y_i) \tag{4.83}$$

  where $g_i(y_i)$ is the entry y_i of a vector $\mathbf{g}_i \in \mathbb{R}^{|V_y|}$. The vector $\mathbf{g}_i$ represents the weights of associating $\mathbf{h}_i$ with each label in the form

$$\mathbf{g}_i \quad = \quad \mathbf{h}_i \cdot \mathbf{v} \tag{4.84}$$

  where $\mathbf{v} \in \mathbb{R}^{d_h \times |V_y|}$ is a weight matrix.

To simplify notation, we use y_i (or y_{i-1}) to denote the one-hot representation for a label[14]. Then, substituting the above feature functions into Eq. (4.81) allows the scoring function to be written in the form

$$\begin{aligned} \text{Score}(y_1 ... y_m, \mathbf{h}_1 ... \mathbf{h}_m) &= \sum_{i=1}^{m} u(y_{i-1}, y_i) + g_i(y_i) \\ &= \sum_{i=1}^{m} y_{i-1} \cdot \mathbf{u} \cdot y_i^{\top} + \mathbf{h}_i \cdot \mathbf{v} \cdot y_i^{\top} \\ &= \sum_{i=1}^{m} (y_{i-1} \cdot \mathbf{u} + \mathbf{h}_i \cdot \mathbf{v}) \cdot y_i^{\top} \end{aligned} \tag{4.85}$$

The right-hand side of the equation only involves simple algebraic operations on vectors

[13] CRFs can broadly be categorized as a type of **undirected graphical models**. They define a graph over a set of observed variables and a set of unobserved variables. These variables are connected in some way that forms a graph.

[14] In this case, $y_i \in \mathbb{R}^{|V_y|}$, although it is originally used as a scalar.

and matrices, allowing this model to be viewed as a standard neural network. In this way, it is convenient to implement the sequence labeling system with various neural network toolkits. We just need to stack a CRF network on an encoder network and learn the entire network as usual. For example, one can train this system by maximum likelihood, and optimize the loss function by gradient descent. Note that, as with other chain or lattice-based models, the CRF network can be efficient because there are dynamic programming algorithms, called the forward-backward methods, for computing both $\mathrm{Score}(y_1...y_m, \mathbf{h}_1...\mathbf{h}_m)$ and $Z(\mathbf{h}_1...\mathbf{h}_m)$. We refer the interested reader to related papers for more detailed discussions [Lafferty et al., 2001; Sutton and McCallum, 2012].

One primary advantage of integrating distributed representations with sequence labeling is that it eliminates the need for manual feature engineering. Instead, the model autonomously learns features from raw input that are most effective for optimizing the sequence labeling objective. Such an architecture has been used as the backbone model for several state-of-the-art systems [Huang et al., 2015; Lample et al., 2016; Ma and Hovy, 2016; Li et al., 2020c].

4.5.4 Hybrid Models for Language Modeling

As we have already noted, many sequence modeling problems can be handled by either RNN-based or CNN-based models. Each of these two architectures has its own characteristics: RNNs were originally designed to process variable-length temporal data, whereas CNNs are more effective at capturing local information in restricted regions of the input. Here, we consider a hybrid approach to language modeling to leverage the benefits of both.

Recall from Section 4.2.1 that a neural language model is trained to predict a probability distribution over a vocabulary, given some representation of the preceding words. Let $w_1...w_m$ be a word sequence whose probability we want to compute. First, we represent each word w_i as a word vector $\mathbf{x}_i$. Then, an RNN model processes one word vector at a time and outputs the probability:

$$\begin{aligned}\Pr(w_{i+1}|w_1...w_i) &= \Pr(w_{i+1}|\mathbf{x}_1...\mathbf{x}_i) \\ &= \Pr(w_{i+1}|\mathbf{h}_i)\end{aligned} \tag{4.86}$$

where $\mathbf{h}_i$ is the state of the recurrent unit at step i.

The process of converting words from symbols to continuous representations plays an important role in this model. While it is common for practitioners to obtain $\mathbf{x}_i$ from a word embedding table, this approach treats each word as a whole and simply ignores its internal structure. As a result, it might be difficult to learn distinct vectors for rare words in languages with large vocabularies [Bojanowski et al., 2017].

Here we consider a different way of representing words. The idea is simple: an additional neural network is used to embed words [Ling et al., 2015; Kim et al., 2016]. Suppose every word w_i can be expressed as a sequence of characters. We represent these characters as real-valued vectors $\mathbf{z}_{i,1}...\mathbf{z}_{i,\mathrm{len}_i}$ via a character embedding table. Following Kim et al. [2016]'s

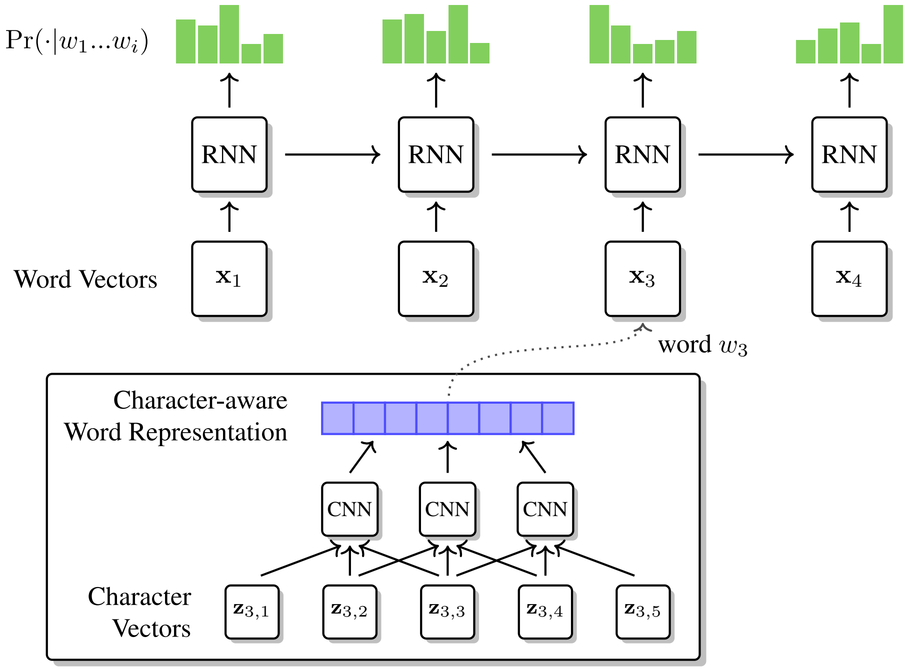


Figure 4.13: A language model with character-aware word representations [Kim et al., 2016]. As a language model, the goal of this model is to compute the probability $\Pr(w_{i+1}|w_1...w_i)$ for each i. We represent each word w_i as a real-valued vector $\mathbf{x}_i$. This vector is the output of a CNN that takes a sequence of characters corresponding to this word. Then, the sequence of the word vectors $\mathbf{x}_1...\mathbf{x}_m$ is used as the input to an RNN + Softmax model. The model outputs at each position i a distribution of words, where the entry corresponding to w_{i+1} gives $\Pr(w_{i+1}|w_1...w_i)$. This hierarchical structure provides a multi-scale approach to language modeling: a sentence is modeled by considering words, and a word is modeled by considering characters.

work, we can use a CNN to represent $\mathbf{z}_{i,1}...\mathbf{z}_{i,\mathrm{len}_i}$ as a word vector in the following form

$$\begin{aligned}\mathbf{x}_i &= \mathrm{CNN}(\mathbf{z}_{i,1}...\mathbf{z}_{i,\mathrm{len}_i}, \mathbf{W}) \\ &= \mathrm{Pooling}(\mathrm{TanH}(\mathrm{Conv}(\mathbf{z}_{i,1}...\mathbf{z}_{i,\mathrm{len}_i}, \mathbf{W}))) \end{aligned} \tag{4.87}$$

where $\mathrm{Conv}(\cdot, \mathbf{W})$ is a convolutional layer with parameters $\mathbf{W}$, $\mathrm{TanH}(\cdot)$ is a hyperbolic tangent function, and $\mathrm{Pooling}(\cdot)$ is a pooling layer.

Figure 4.13 shows the architecture of the model. This model exhibits a hierarchical structure, in which characters form words, and words form sentences or phrases. From a practical perspective, in many NLP tasks it is quite natural to consider the hierarchical nature of language. We will see a few examples of making use of the relationships between different levels of language representations in later chapters.

4.6 Summary

This chapter has introduced the recurrent and convolutional neural network approaches to modeling sequences of words. On one hand, recurrent neural networks are designed for handling sequential data, and have broad applicability in NLP. To improve the modeling power of these models, memory mechanisms are commonly used. In particular, we have introduced LSTM and GRU which are two popular models for long sequences. On the other hand, while convolutional neural networks are commonly used to process visual data, they can be directly applied to sequence modeling. We have seen that all these models can be used in several language and speech processing tasks, including text classification, speech recognition, sequence labeling, and language modeling.

The roots of modeling sequences of language units can be traced back to early work in several different fields. For example, the process of generating a sequence of words can be described as a Markov chain where the prediction of a word only depends on a limited number of previous words [Markov, 1913]. This idea motivates the n-gram methods for sequence modeling [Shannon, 1948a], as well as hidden Markov models which later appeared and became popular in modeling sequences of pairs of observed and unobserved variables [Baum and Petrie, 1966; Baum et al., 1970]. These models and their variants laid the foundations of many successful NLP systems in past decades [Manning and Schütze, 1999; Jurafsky and Martin, 2008].

The idea of using neural networks in sequence modeling has also been investigated for some time. One example illustrating how neural networks have been developed and applied to sequence modeling is speech recognition [Lippmann, 1989]. Most of the studies in the early days of this research area tried to either combine neural networks with existing models [Bourlard and Wellekens, 1990; Bourlard and Morgan, 1993; Trentin and Gori, 2001], or address sub-problems of speech recognition [Tank and Hopfield, 1987; Waibel et al., 1989; Lang et al., 1990; Bengio, 1991]. However, scaling up neural networks was challenging because training deep neural networks requires a lot of computational resources and data. The field had long been dominated by approaches based on hidden Markov models and **Gaussian mixture models** (**GMMs**), with a pipeline of several modules that require careful tuning. On the other hand, while fully neural approaches were not state-of-the-art during that time, researchers were aware of their potential in learning representations of acoustic inputs and freeing them from hand-crafted features [LeCun and Bengio, 1995].

A dramatic shift from conventional pipelined approaches to end-to-end approaches came with the revival of neural networks in the 2000s [Hannun et al., 2014; Graves et al., 2013b]. This shift was so influential that it brought a wide range of fields together like never before. For instance, in both computer vision and speech processing, the past decade has witnessed tremendous performance gains driven by very deep neural networks and end-to-end learning [Hinton et al., 2006; Graves et al., 2013b; He et al., 2016a; Krizhevsky et al., 2017]. In NLP, the paradigm shift started with the work on word embeddings [Mikolov et al., 2013a; Pennington et al., 2014], and continues as more powerful sequence representation models are developed [Vaswani et al., 2017]. Simple approaches to sequence modeling, though not discussed in depth in this chapter, are **compositional models** [Janssen, 2012]. For example, we can use the bag-of-words model to sum or average word vectors of a sequence. Despite the simple architectures of these approaches, they achieve satisfactory results in many tasks, providing strong baselines for further research on more advanced models [Conneau et al., 2018]. As the next step, applying recurrent and convolutional neural models to sequence modeling is straightforward. This is not surprising because these models are fairly well studied in other fields [Lipton et al., 2015; Li et al., 2021d; Khan et al., 2020]. In particular, the LSTM model is well suited to deal with long sequences and thus of great interest to NLP researchers [Sundermeyer et al., 2012; Huang et al., 2015; Wu et al., 2016]. However, research in this area is ongoing. Sequence modeling is one of the most active research fields with no end in sight. There are many models that are based on new architectures and show stronger performance in various tasks. More discussions on some of these models can be found in Chapters 6, 7 and 8.

Note that the term *sequence modeling* is currently used in many different ways, referring to different tasks. In many cases it is more common to use the terms *encoding* and *encoder* to emphasize the process of mapping a sequence of symbols to a continuous representation. As discussed in the previous sections, a benefit of viewing encoding as an individual task is that we can learn a general representation model that is not dependent on where we apply it. It opens the door to a wide range of pre-trained encoders for learning to represent various types of data, such as text [Peters et al., 2018; Devlin et al., 2019], speech [Oord et al., 2018; Hsu et al., 2021; Chen et al., 2022], vision [Chen and He, 2021; Bao et al., 2021; He et al., 2022], and combinations of these modalities [Chuang et al., 2020; Li et al., 2021c; Kim et al., 2021]. A closely related concept to text encoding is **text embedding** or **sentence embedding** [Conneau et al., 2017a; Cer et al., 2018]. These can be broadly considered equivalent. In general, an embedding model in NLP means a process of transforming the input text into a single low-dimensional vector rather than producing sequences of contextualized vectors [Kiros et al., 2015; Hill et al., 2016].

In many NLP problems, systems are not necessarily sequential on their input and/or output. For example, in text classification, a system may take tree-structured input and produce a label [Tai et al., 2015; Yang et al., 2016]. In this case we need some mechanism to encode hierarchical structures. An alternative approach is to convert trees to sequences (or **linearized trees**) so that we can directly make use of sequence models to handle non-sequential data [Vinyals et al., 2015]. This is an appealing idea because it opens up the possibility of developing a universally applicable encoder to represent various types of data if the input of the encoder can be linearized in some way. For example, by representing an image as a sequence of patches, sequence models can be directly applied to image classification, achieving state-of-the-art results on several tasks [Chen et al., 2020a; Dosovitskiy et al., 2021].

Foundations

Chapter 5

Sequence-to-Sequence Models

天下万物之理，无独必有对。

According to the Principle of Heaven and Earth and all things, nothing exists in isolation but everything necessarily has its opposite.

– 《近思录》

Reflections on things at hand

朱熹/Xi Zhu (AD 1130-1200)
吕祖谦/Zuqian Lv (AD 1137-1181)
translated by Chang [1967]

In the world of language, things often come in pairs. If there is a question, there is an answer; if there is a text in Chinese, there is an English translation; if there is a sentence, there is a syntactic parse. Many NLP systems are designed to model the correspondence between such pairs, i.e., one element is taken as input and the other as output. These problems can be expressed in a form that we have encountered several times, as follows:

$$\hat{y} = \arg\max_{y} \Pr(y|x) \tag{5.1}$$

This chapter focuses on a particular family of problems where both x and y are sequences of words, known as **sequence-to-sequence** (or **seq2seq**) problems. Unlike classification problems, where the output $\hat{y}$ is selected from a fixed set of classes, sequence-to-sequence problems require the model to produce an output from an exponentially large output space. Obtaining $\hat{y}$ in this case is considerably more complex than in classification, as it requires more powerful models to represent $\Pr(y|x)$ and more efficient search algorithms to solve Eq. (5.2).

This chapter first introduces the well-known **encoder-decoder architecture** for sequence-to-sequence modeling, and then presents the attention mechanism as an important extension of this architecture. These models lay the foundation for several state-of-the-art approaches discussed in subsequent chapters. In addition, the chapter will discuss the search (decoding) problem, which plays an important role in sequence generation and related tasks.

In the world of language, things often come in pairs. If there is a question, there would be an answer; if there is a Chinese text, there would be an English translation; if there is a sentence, there would be a parse of it according to some syntax. Many NLP systems are designed to model the correspondence between these pairs, i.e., one of the two is taken as input and the other is taken as output. These problems can be expressed in a form that we have encountered several times, like this

$$\hat{y} = \arg\max_{y} \Pr(y|x) \tag{5.2}$$

where x is an input variable, y is an output variable, and $\Pr(y|x)$ is a model that estimates how likely y would be the true output given x.

This chapter is more interested in a particular family of problems where both x and y are sequences of words, called **sequence-to-sequence** (or **seq2seq**) problems. Unlike classification problems where the output $\hat{y}$ is selected from a fixed set of classes, sequence-to-sequence problems require producing an output from an exponentially larger set of sequences. Obtaining $\hat{y}$ in this case turns out to be a much more complex problem than the case of classification, because we need more powerful models to describe $\Pr(y|x)$ and more efficient search algorithms to solve Eq. (5.2).

This chapter will discuss the well-known **encoder-decoder architecture** for sequence-to-sequence modeling. Also, this chapter will discuss the attention mechanism which is an improvement on this architecture. Both of these models lay the foundation of discussions of several state-of-the-art models in the following chapters. Furthermore, this chapter will discuss the search problem which plays an important role in sequence generation and related problems.

5.1 Sequence-to-Sequence Problems

We use machine translation as an illustrative example throughout this chapter, because it is now one of the most popular sequence-to-sequence tasks. We use $\mathbf{x} = x_1...x_m$ to denote a sequence of words in one language (call it a **source-side sequence** or **source sequence**), and use $\mathbf{y} = y_1...y_n$ to denote a sequence of words in another language (call it a **target-side sequence** or **target sequence**). Using this notation, Eq. (5.2) can be written as:

$$\begin{aligned} \hat{\mathbf{y}} &= \arg\max_{\mathbf{y}} \Pr(\mathbf{y}|\mathbf{x}) \\ &= \arg\max_{y_1...y_n} \Pr(y_1...y_n|x_1...x_m) \end{aligned} \tag{5.3}$$

As discussed in Chapter 1 and in [Brown et al., 1993], this formulation implies three fundamental issues.

- **Modeling**. First, we need to define the form of $\Pr(\mathbf{y}|\mathbf{x})$. In this chapter we show that $\Pr(\mathbf{y}|\mathbf{x})$ can be modeled using a single neural network based on the encoder-decoder architecture and the attention mechanism. Note that sometimes we just need a model for

Task	Source	Target
Machine Translation	Text in One Language	Translation in Another Language
Question Answering	Question	Answer
Dialogue Systems	Text/Speech for Conversation	Response
Summarization	Long Text	Summary of the Text
Text Simplification	Text	Simpler Text
Text Style Transfer	Text in One Style	Text with the Same Content in Another Style
Grammar Correction	Text with Errors	Corrected Text
Speech Recognition	Speech	Transcription
Speech Synthesis	Text	Speech
Speech Translation	Speech in One Language	Translation in Another Language

Table 5.1: Examples of sequence-to-sequence problems.

discriminating "good" from "bad" target sequences. In this case, it is not necessary to require the model to be probabilistic, and we can take a discriminant function instead.

- **Training**. Then, we need to learn the parameters of the model $\Pr(\mathbf{y}|\mathbf{x})$ given some training data. As $\Pr(\mathbf{y}|\mathbf{x})$ is implemented as a neural network, we can train it in a standard fashion: we optimize a loss function by gradient descent. See Chapter 3 for common approaches to training neural networks. We will also discuss techniques that are tailored for specific tasks in this and the following chapters.
- **Search** (or **Decoding**). Once we have learned a model, we will obtain $\hat{\mathbf{y}}$ by searching for a target sequence that maximizes $\Pr(\mathbf{y}|\mathbf{x})$. This poses a computational challenge because the number of candidate sequences grows with the maximum sequence length and the size of the vocabulary. In Section 5.4, we will discuss efficient and effective search methods for sequence-to-sequence problems, particularly for machine translation.

Many NLP problems that fit the form of Eq. (5.3) can be formulated as sequence-to-sequence problems. Table 5.1 shows common examples of sequence-to-sequence problems taken from the literature. When the target side is text, such problems can broadly be categorized as **text generation** problems, although a general text generation system does not require the source-side to be sequential. In addition to language and speech processing, sequence-to-sequence problems can be generalized to cases where the input and/or output of a system are not naturally sequential. For example, **image-to-text generation** (or **image captioning**) and **text-to-image generation** systems both involve dealing with images that are typically represented as 2D data. By representing images as sequences in some way (such as sequences of patches), sequence-to-sequence models are directly applicable to these tasks.

Historically, most systems in these tasks were developed somewhat independently, resulting in different architectures, features, and training methods for different tasks. However, as shown in this chapter, when we represent these models as neural networks and train them in an end-to-end fashion, there appears to be a "universal" paradigm for all these problems. This is a big change for the AI community because many research fields come together and systems can be shared across them. We can gain some insight into the common nature of a broad variety of problems, though there are many task-specific considerations in practice. In the following sections, we will discuss some of the common threads among sequence-to-sequence models.

5.2 The Encoder-Decoder Architecture

In this section we discuss the encoder-decoder architecture and a simple neural machine translation model built upon this architecture.

5.2.1 Encoding and Decoding

From a supervised learning viewpoint, we would ideally like to learn a model from a number of sequence pairs such that any source-side sequence can be mapped to the corresponding target-side sequence. However, learning mappings between sequences of discrete variables often requires handling high-dimensional data, which inevitably suffers from the curse of dimensionality.

One approach to learning such a mapping is to divide the problem into simpler sub-problems. We assume that there is a low-dimensional representation shared by $\mathbf{x}$ and $\mathbf{y}$, denoted by $\mathbf{H}$. Then, the mapping $\mathbf{x} \rightarrow \mathbf{y}$ can be achieved by mapping $\mathbf{x}$ to $\mathbf{H}$ and then to $\mathbf{y}$. Formally, given a source-side sequence $\mathbf{x}$, we map it to the representation $\mathbf{H}$ using an **encoding system** (call it an encoder)

$$\mathbf{H} = \mathrm{Encode}(\mathbf{x}) \tag{5.4}$$

Then, we map $\mathbf{H}$ to the target-side sequence $\mathbf{y}$ by using a **decoding system** (call it a decoder)[1]

$$\mathbf{y} = \mathrm{Decode}(\mathbf{H}) \tag{5.5}$$

This architecture, also known as the encoder-decoder architecture, is widely used in modern sequence-to-sequence systems (see Figure 5.1 for an illustration). It is easy to see that the

[1] It is important to distinguish between the concept of *decoding* (or *decoder*) used in conventional sequence-to-sequence systems and that used in the encoder-decoder architecture. The two are often confused, though they are different in nature. In many machine translation or speech recognition systems, *decoding* has the same meaning as *translation* or *transcription*, that is, we recover the optimal $\mathbf{y}$ from $\mathbf{x}$. As pointed out in Eq. (5.3), this process involves a search over all candidate $\mathbf{y}$. Therefore, the conventional use of decoding in these systems is to refer to a search process (i.e., the $\arg\max$ operation in Eq. (5.3)) [Koehn, 2010]. By contrast, in the encoder-decoder architecture *decoding* means a process of recovering the target-side sequence $\mathbf{y}$ from the intermediate representation $\mathbf{H}$. It is all about modeling rather than searching. It is also worth noting that, while the term *decoding* (or *decoder*) is used in different ways, it can be thought of as a process of mapping an encoded message back to the original message in a communication system as defined in information theory [Shannon, 1948b]. In this sense, the *decoding* processes in these systems do the same thing as the word sounds like: convert something to its original form.

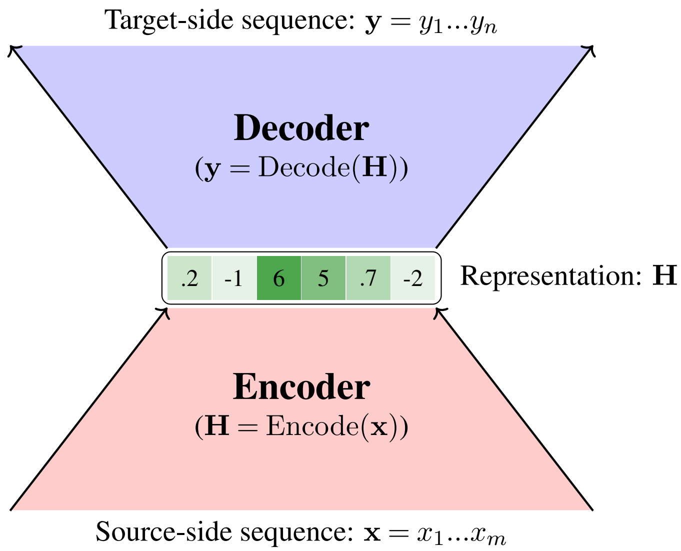


Figure 5.1: The encoder-decoder architecture. In the case of sequence-to-sequence problems, it transforms a source-side sequence $\mathbf{x} = x_1...x_m$ to a target-side sequence $\mathbf{y} = y_1...y_n$. This procedure involves two steps: $\mathbf{x}$ is first encoded as a representation $\mathbf{H}$, and this representation is then decoded to $\mathbf{y}$.

form of Eq. (5.4) is the same as that of the sequence models mentioned in Chapter 4, and so there are many encoding models to choose from, such as bi-directional LSTM. The goal of the decoder is to produce a "best" target-side sequence given the representation of the source-side sequence. Like classification models, the prediction is made by first producing a distribution over all possible sequences, and then selecting the one with the maximum probability. As such, we can re-define $\mathrm{Decode}(\cdot)$ as a probability function

$$\begin{aligned}\Pr(\cdot|\mathbf{H}) &= \mathrm{Decode}(\mathbf{H}) \\ &= \mathrm{Decode}(\mathrm{Encode}(\mathbf{x})) \end{aligned} \tag{5.6}$$

In other words, given a target-side sequence $\mathbf{y}$ and a source-side sequence $\mathbf{x}$, the decoder assigns a probability to $\mathbf{y}$

$$\Pr(\mathbf{y}|\mathbf{x}) = \Pr(\mathbf{y}|\mathbf{H}) \tag{5.7}$$

Then, the optimal sequence $\hat{\mathbf{y}}$ is obtained by performing $\arg\max_{\mathbf{y}} \Pr(\mathbf{y}|\mathbf{x})$ as in Eq. (5.3). In many systems based on the encoder-decoder architecture, both $\mathrm{Encode}(\cdot)$ and $\mathrm{Decode}(\cdot)$ are models constructed from neural networks. Thus, we can treat the sequence-to-sequence model as a single neural network and train it as usual, provided the entire model is some combination of $\mathrm{Encode}(\cdot)$ and $\mathrm{Decode}(\cdot)$.

To apply the encoder-decoder architecture to a real-world task, we need to make a number of design choices, such as the forms of $\mathbf{H}$, $\mathrm{Encode}(\cdot)$ and $\mathrm{Decode}(\cdot)$. As a very simple example, consider the task of reconstructing an input word. We can define $\mathrm{Encode}(\cdot)$ as a feed-forward neural network that takes a word (in one-hot representation) and outputs a word vector. In this way, $\mathbf{H}$ is a distributed representation of the word. Then, we define $\mathrm{Decode}(\cdot)$ as another feed-forward neural network that takes the word vector and generates a distribution over the vocabulary. For training, we wish to learn a system that assigns the largest probability to the input word. As discussed in Chapter 2, we can call this an auto-encoder which is a special instance of the encoder-decoder architecture.

5.2.2 Example: Neural Machine Translation

Next, we illustrate the application of the encoder-decoder architecture using a working example: **neural machine translation** (**NMT**). We consider a well-known NMT model that employs RNNs or their variants for both the encoder and the decoder [Cho et al., 2014; Sutskever et al., 2014]. The encoder of this NMT model is a standard RNN-based sequence model. Since the RNN-based sequence model has been discussed in detail in Chapter 4, we provide only a brief review here. Suppose that the source-side vocabulary is V_x and each source-side word x_j is represented as a one-hot vector in $\mathbb{R}^{|V_x|}$. Then, x_j is transformed into an h_s-dimensional vector (or word embedding)

$$\mathbf{x}_j^e = \mathrm{Embed_s}(x_j) \tag{5.8}$$

where $\mathrm{Embed_s}(\cdot)$ is the word embedding function. More details about word embedding models can be found in Chapter 3.

The RNN model takes the sequence of word embeddings $\mathbf{x}_1^e...\mathbf{x}_m^e$ and produces a sequence of RNN state vectors $\mathbf{h}_1...\mathbf{h}_m$. An RNN state vector $\mathbf{h}_j \in \mathbb{R}^{d_h}$ is defined as

$$\mathbf{h}_j = \mathrm{RNN}(\mathbf{h}_{j-1}, \mathbf{x}_j^e) \tag{5.9}$$

Here, $\mathrm{RNN}(\cdot)$ is an RNN unit that summarizes the information up to position j by combining the previous state $\mathbf{h}_{j-1}$ and the current input $\mathbf{x}_j^e$. Then, the final state $\mathbf{h}_m$ can be treated as a compressed representation of the input sequence $x_1...x_m$, and we use $\mathbf{h}_m$ as the output of the encoder, written as

$$\mathbf{h}_m = \mathrm{Encode}(x_1...x_m) \tag{5.10}$$

Figure 5.2 (a-b) illustrates the encoding process. Note that the model described above involves only a single-layer RNN. In practical systems, this framework can be easily extended to include multiple layers and more powerful recurrent units (such as LSTM units).

The decoder of the NMT model is a standard RNN-based language model, that is, it predicts the next word y_{i+1} given all previous words $y_1...y_i$. To incorporate the source-side information into the translation, a straightforward method is to treat $\mathbf{h}_m$ as the initial state of the target-side RNN. Let $\mathbf{y}_0^e \in \mathbb{R}^{d_s}$ be the word vector of the start symbol $\langle\mathrm{SOS}\rangle$ (denoted by

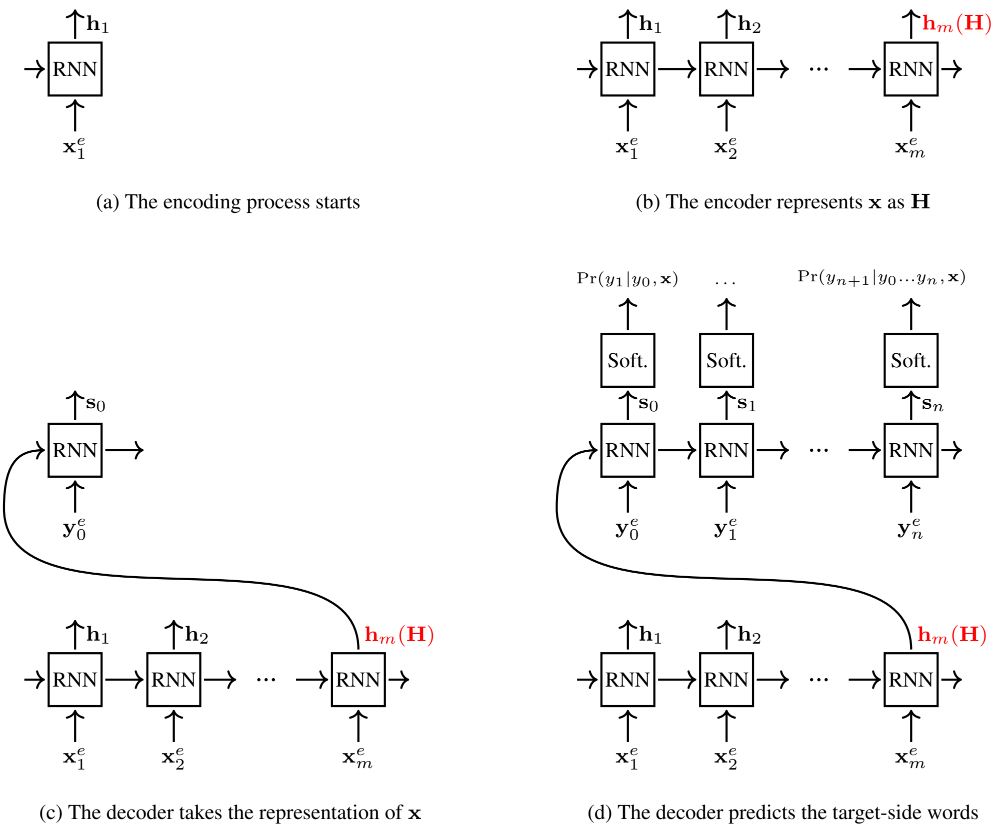


Figure 5.2: The encoding and decoding steps for an RNN-based NMT system. The encoder is a standard RNN. The encoding process starts with the first source-side word and concludes with the last source-side word. The final state of the RNN is taken as the representation of the entire source-side sequence (i.e., $\mathbf{H} = \mathbf{h}_m$). The decoder is another RNN. In the first step, it receives $\mathbf{H}$ from the encoder. After representing $(\mathbf{y}^e_0 ... \mathbf{y}^e_i, \mathbf{H})$ as $\mathbf{s}_i$ at position i, a Softmax layer is applied to predict the next word y_{i+1}.

y_0). The corresponding initial RNN state is given by

$$\mathbf{s}_0 = \mathrm{RNN}(\mathbf{h}_m, \mathbf{y}^e_0) \tag{5.11}$$

Here, $\mathrm{RNN}(\cdot)$ has the same form as the recurrent unit used in the encoder, but with different parameters.

For $i > 0$, the state vector $\mathbf{s}_i \in \mathbb{R}^{d_s}$ is defined as

$$\mathbf{s}_i = \mathrm{RNN}(\mathbf{s}_{i-1}, \mathbf{y}^e_i) \tag{5.12}$$

Then, $\mathbf{s}_i$ is fed into a Softmax layer to produce a distribution over the target-side vocabulary

V_y. The output of the Softmax layer is given by

$$\begin{aligned}\Pr(\cdot|y_1...y_i, x_1...x_m) &= \Pr(\cdot|\mathbf{s}_i) \\ &= \mathrm{Softmax}(\mathbf{s}_i\mathbf{U}_y + \mathbf{b}_y)\end{aligned} \tag{5.13}$$

where $\mathbf{U}_y \in \mathbb{R}^{d_s \times |V_y|}$ and $\mathbf{b}_y \in \mathbb{R}^{|V_y|}$ are the parameters of the Softmax layer. $\Pr(y_{i+1}|y_1...y_i, x_1...x_m)$ can be seen as the probability of predicting word y_{i+1} conditioned on both the translated words $y_1...y_i$ and the source-side sequence $x_1...x_m$. See Figure 5.2 (c-d) for an illustration of the word predictions of a decoder.

Based on this word prediction model, the joint probability frequently used in NMT literature is formulated as follows:

$$\begin{aligned}\Pr(\mathbf{y}|\mathbf{x}) &= \Pr(y_0 y_1...y_n|x_1...x_m) \\ &= \Pr(y_0|x_1...x_m)\Pr(y_1...y_n|y_0, x_1...x_m) \\ &= \prod_{i=0}^{n-1} \Pr(y_{i+1}|y_0...y_i, x_1...x_m)\end{aligned} \tag{5.14}$$

Sometimes, this equation is also written in an equivalent form:

$$\Pr(\mathbf{y}|\mathbf{x}) = \prod_{i=1}^{n} \Pr(y_i|y_0...y_{i-1}, x_1...x_m) \tag{5.15}$$

Here, we assume that $\mathbf{y}$ always starts with y_0 (i.e., $\langle\mathrm{SOS}\rangle$), such that $\Pr(y_0|x_1...x_m) = 1$. In many practical systems, it is also common to assume that $\mathbf{y}$ ends with a special symbol $\langle\mathrm{EOS}\rangle$. Therefore, we can modify this equation to include $\langle\mathrm{SOS}\rangle$ and $\langle\mathrm{EOS}\rangle$ on both the source and target sides, as follows:

$$\begin{aligned}\Pr(y_0\mathbf{y}y_{n+1}|x_0\mathbf{x}x_{m+1}) &= \Pr(y_0 y_1...y_n y_{n+1}|x_0 x_1...x_m x_{m+1}) \\ &= \Pr(y_0|x_0...x_{m+1}) \cdot \\ &\quad\ \Pr(y_1...y_n y_{n+1}|y_0, x_0...x_{m+1}) \\ &= \prod_{i=0}^{n} \Pr(y_{i+1}|y_0...y_i, x_0...x_{m+1})\end{aligned} \tag{5.16}$$

where $x_0 = y_0 = \langle\mathrm{SOS}\rangle$, $x_{m+1} = y_{n+1} = \langle\mathrm{EOS}\rangle$, and $\Pr(y_0|x_0 x_1...x_m x_{m+1}) = 1$.

Since $\Pr(\mathbf{y}|\mathbf{x})$ can be implemented as a neural network, training this model is straightforward. As described in Chapter 4, RNN-based language models are trained using cross-entropy loss and gradient descent. NMT uses this same method for training model parameters. Once we obtain the optimized model, we can use it to translate new sentences. Finding the best translation for any given source-side sentence is a standard search problem, which we will discuss in Section 5.4.

5.3 The Attention Mechanism

The NMT model discussed in the previous section is based on a fixed-length representation of the source-side sequence. While this model is easy to implement, in many practical applications it is often inadequate, as a fixed-length vector is insufficient to represent a variable-length sequence, especially when the sequence is long. We therefore need some mechanism to couple the encoder and the decoder in a fine-grained manner. In this section we discuss the attention mechanism by which the model can learn, for each word in the target-side sequence, an adaptive representation that focuses more on important parts of the source-side sequence.

This idea is closely related to models of attention in psychology because translation is itself a cognitive process [Sternberg, 1996; Neisser, 2014]. The key idea behind this type of model is intuitive: attention is generally concentrated on specific parts of the input. This intuition forms the basis of many state-of-the-art sequence-to-sequence models, and the attention mechanism has been the de facto standard in sequence-to-sequence modeling.

5.3.1 A Basic Model

Recall that in the NMT model of the previous section, the encoder represents a source-side word sequence as $\mathbf{h}_1, ..., \mathbf{h}_m$, and the decoder represents a target-side word sequence as $\mathbf{s}_1, ..., \mathbf{s}_n$. The attention mechanism addresses the question of how a representation can be learned from $\mathbf{h}_1, ..., \mathbf{h}_m$ so that this representation can encode the source-side sequence well for a given query state $\mathbf{s}_i$[2]. From an information processing perspective, so long as we ignore the meanings of $\mathbf{h}_1, ..., \mathbf{h}_m$ and $\mathbf{s}_i$ in NMT, attention can be thought of as a generic mechanism for processing the input information $\mathbf{h}_1, ..., \mathbf{h}_m$ by considering how each $\mathbf{h}_j$ is related to the query state $\mathbf{s}_i$. Figure 5.3 compares NMT architectures with and without the attention mechanism.

More formally, an attention model produces a weighted combination of $\{\mathbf{h}_1,...,\mathbf{h}_m\}$ in the form

$$\mathbf{c}_i = \sum_{j=1}^{m} \alpha_{i,j} \cdot \mathbf{h}_j \tag{5.17}$$

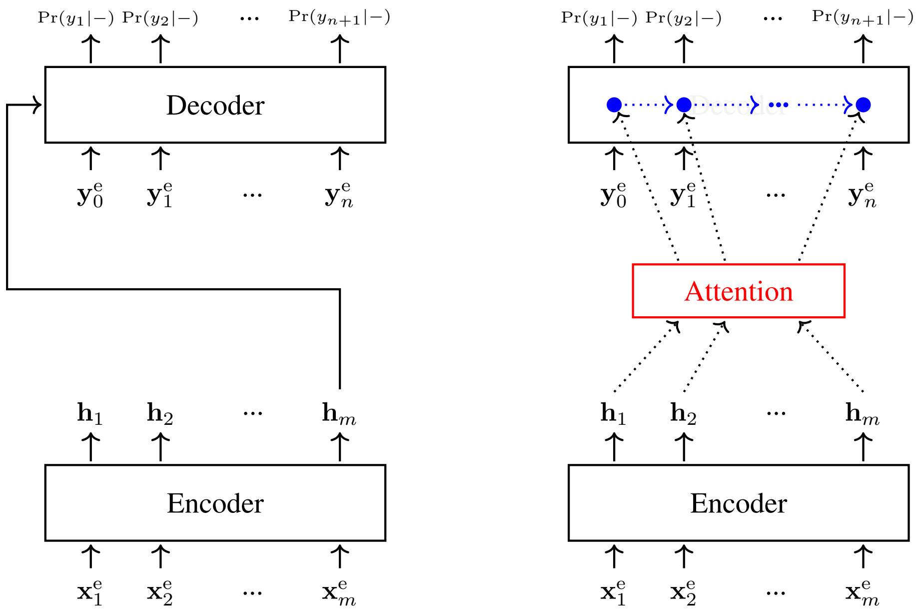
where $\alpha_{i,j}$ is the **attention weight** that describes how much the model should rely on $\mathbf{h}_j$ when computing $\mathbf{c}_i$ for $\mathbf{s}_i$. Sometimes $\mathbf{c}_i$ is also called a **context vector**.

A common approach to computing attention weights is to normalize **alignment scores** in the following form

$$\alpha_{i,j} = \frac{\exp(a(\mathbf{s}_i, \mathbf{h}_j))}{\sum_{j'=1}^{m} \exp(a(\mathbf{s}_i, \mathbf{h}_{j'}))} \tag{5.18}$$

Here the alignment score $a(\mathbf{s}_i, \mathbf{h}_j)$ measures how strongly $\mathbf{h}_j$ is related to $\mathbf{s}_i$. In general, $a(\mathbf{s}_i, \mathbf{h}_j)$ can be defined in several different ways [Graves et al., 2014; Bahdanau et al., 2014; Luong et al., 2015]. A comprehensive list of these functions can be found in survey papers on

[2]Following the convention in machine translation [Brown et al., 1993], we use j to represent a position in the source-side sequence, and use i to represent a position in the target-side sequence.



(a) An NMT system without attention (b) An NMT system with attention

Figure 5.3: NMT architectures without (left) and with (right) the attention model. When the attention model is not involved, a fixed-length representation is considered for generating the entire target-side sequence. By contrast, when the attention model is involved, a new representation is computed specifically for each target-side state so that the decoder can learn to concentrate on different parts of the source-side sequence for predicting a target-side word.

this subject [Chaudhari et al., 2021]. Here we introduce some of the common ones.

- **Dot-product Attention**. One of the simplest methods is to measure the similarity between $\mathbf{h}_j$ and $\mathbf{s}_i$. Thus, we can calculate the dot-product of the two vectors, as follows

$$\begin{aligned} a(\mathbf{s}_i, \mathbf{h}_j) &= \mathbf{s}_i \mathbf{h}_j^{\top} \\ &= \sum_{k=1}^{d_h} s_i(k) \cdot h_j(k) \end{aligned} \tag{5.19}$$

  A variant of this model, called **scaled dot-product attention**, applies a scaling factor of $\frac{1}{\beta}$ to the right-hand side of Eq. (5.19), as follows

$$a(\mathbf{s}_i, \mathbf{h}_j) = \frac{\mathbf{s}_i \mathbf{h}_j^{\top}}{\beta} \tag{5.20}$$

  where β is a hyperparameter. We will see an example of this model later in this section.

- **Cosine Attention**. Another commonly used similarity measure in vector algebra is the cosine of the angle between two vectors, given by

$$\begin{aligned} a(\mathbf{s}_i, \mathbf{h}_j) &= \cos(\mathbf{s}_i, \mathbf{h}_j) \\ &= \frac{\mathbf{s}_i \mathbf{h}_j^{\top}}{\|\mathbf{s}_i\|_2 \cdot \|\mathbf{h}_j\|_2} \end{aligned} \tag{5.21}$$

where $\|\mathbf{a}\|_2 = (\mathbf{a} \cdot \mathbf{a})^{\frac{1}{2}}$ is the Euclidean norm of the vector $\mathbf{a}$.

- **Weighted Dot-product Attention**. This attention model involves a linear mapping of the input vectors before performing the dot-product operation, given by

$$a(\mathbf{s}_i, \mathbf{h}_j) = \mathbf{s}_i \mathbf{W}_{\mathrm{a}} \mathbf{h}_j^{\top} \tag{5.22}$$

  where $\mathbf{W}_{\mathrm{a}} \in \mathbb{R}^{d_h \times d_h}$ is the parameter matrix of the linear mapping. Both this approach and the dot-product attention approach are also called **multiplicative attention** [Ruder, 2017].

- **Additive Attention**. In additive attention, the entries of the two vectors are summed in some way. A widely-used form is given by Bahdanau et al. [2014]

$$a(\mathbf{s}_i, \mathbf{h}_j) = \mathbf{v}_{\mathrm{a}}^{\top} \mathrm{TanH}(\mathbf{s}_i \mathbf{W}_{\mathrm{s}} + \mathbf{h}_j \mathbf{W}_{\mathrm{h}}) \tag{5.23}$$

  where $\mathbf{W}_{\mathrm{h}}, \mathbf{W}_{\mathrm{s}} \in \mathbb{R}^{d_h \times d_a}$ and $\mathbf{v}_{\mathrm{a}} \in \mathbb{R}^{d_a}$ are parameters. $\mathrm{TanH}(\mathbf{s}_i \mathbf{W}_{\mathrm{s}} + \mathbf{h}_j \mathbf{W}_{\mathrm{h}})$ produces a d_a-dimensional vector where each entry is a transformed weighted sum of the entries of $\mathbf{h}_j$ and $\mathbf{s}_i$. It is followed by a dot-product with another weight vector $\mathbf{v}_{\mathrm{a}}$.

Now let us return to Eqs. (5.17-5.18) and rethink the role of attention weights. Eq. (5.18) informally defines a "distribution" over $\mathbf{h}_1 ... \mathbf{h}_m$, written as

$$\mathrm{Pr}(\mathbf{h}_j | \mathbf{s}_i) = \alpha_{i,j} \tag{5.24}$$

If we consider $\mathbf{h}$ as a random variable that takes a value from $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$, then $\alpha_{i,j}$ can be thought of as the probability of $\mathbf{h} = \mathbf{h}_j$, conditioned on $\mathbf{s}_i$, and Eq. (5.17) can be rewritten as

$$\begin{aligned} \mathbf{c}_i &= \sum_{j=1}^{m} \mathrm{Pr}(\mathbf{h}_j | \mathbf{s}_i) \cdot \mathbf{h}_j \\ &= \mathbb{E}_{\mathbf{h} \sim \mathrm{Pr}(\mathbf{h}|\mathbf{s}_i)}(\mathbf{h}) \end{aligned} \tag{5.25}$$

In other words, $\mathbf{c}_i$ can be viewed as an **expected representation** of the source-side sequence given the target-side state $\mathbf{s}_i$, that is, the expectation of $\mathbf{h}$ over the set $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$ under the distribution $\mathrm{Pr}(\mathbf{h}_j | \mathbf{s}_i)$. This provides a general framework for describing the way the decoder receives the information from the encoder: the decoder is a **receiver** that determines how much information is accepted from each **sender**. For example, in the NMT model of the previous section, there is only one sender $\mathbf{h}_m$, and so the receiver receives all the information the sender sends. By contrast, in the NMT model equipped with the attention mechanism, there are m senders $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$ and the receiver receives information according to a distribution of preferences over the senders.

It is straightforward to introduce the attention model into the process of word prediction. To make use of both source-side and target-side information at each decoding step, we slightly modify the definition of $\mathbf{s}_i$ to include the context vector corresponding to the previous state $\mathbf{s}_{i-1}$, as follows:

$$\mathbf{s}_i = \mathrm{RNN}(\mathbf{s}_{i-1}, \mathbf{c}_{i-1}, \mathbf{y}_i^e) \tag{5.26}$$

Compared with the model of Eq. (5.12), the model of Eq. (5.26) takes $\mathbf{c}_{i-1}$ as an additional input. Therefore, this model considers both the representation of the target-side words $y_1...y_{i-1}$ (as encoded in $\mathbf{s}_{i-1}$ and $\mathbf{y}_i^e$) and the representation of the entire source-side sequence $x_1...x_m$ (as encoded in $\mathbf{c}_{i-1}$). Then, the distribution of target words at position i can be conditioned on $\mathbf{s}_i$ as usual:

$$\Pr(\cdot|y_1, ..., y_i, x_1, ..., x_m) = \Pr(\cdot|\mathbf{s}_i) \tag{5.27}$$

where $\Pr(\cdot|\mathbf{s}_i)$ is generally a Softmax layer. This process is illustrated in Figure 5.4.

We now have a model for computing $\Pr(y_{i+1}|y_1...y_i, x_1...x_m)$. A brief outline of the key steps of this model is given by

1. Encode the source-side sequence as $\mathbf{h}_1...\mathbf{h}_m$ where $\mathbf{h}_j = \mathrm{RNN}(\mathbf{h}_{j-1}, \mathbf{x}_j^e)$.
2. Repeat the following procedure from $i = 1$ to $n-1$.
   a. Compute the alignment score $a(\mathbf{s}_{i-1}, \mathbf{h}_j)$ for each j.
   b. Compute the attention weights $\{\alpha_{i-1,1}, ..., \alpha_{i-1,m}\}$ where $\alpha_{i-1,j} = \frac{\exp(a(\mathbf{s}_{i-1}, \mathbf{h}_j))}{\sum_{j'=1}^{m} \exp(a(\mathbf{s}_{i-1}, \mathbf{h}_{j'}))}$.
   c. Compute the context vector $\mathbf{c}_{i-1} = \sum_{j=1}^{m} \alpha_{i-1,j} \cdot \mathbf{h}_j$.
   d. Compute the target-side state $\mathbf{s}_i = \mathrm{RNN}(\mathbf{s}_{i-1}, \mathbf{c}_{i-1}, \mathbf{y}_i^e)$.
   e. Compute the distribution of target-side words $\Pr(\cdot|\mathbf{s}_i)$.
   f. Compute $\Pr(y_{i+1}|y_1...y_i, x_1...x_m) = \Pr(y_{i+1}|\mathbf{s}_i)$ for a given word y_{i+1} (as in training), or select the most likely word $\hat{y}_{i+1} = \arg\max_{y_{i+1}} \Pr(y_{i+1}|y_1...y_i, x_1...x_m)$ (as in testing).

In real-world systems, this basic model can be modified to better predict the target-side words. For example, we can introduce fusion layers to combine $\mathbf{s}_i$, $\mathbf{c}_{i-1}$, and $\mathbf{y}_i^e$ before the Softmax layer so that we have a deeper model for prediction [Bahdanau et al., 2014]. Another commonly used approach is to stack multiple RNN layers on the target-side. In this case, one can perform attention in either each layer of the stack [Wu et al., 2016] or the top-most layer of the stack [Luong et al., 2015]. See Section 5.3.4 for more information about multi-layer approaches to attention.

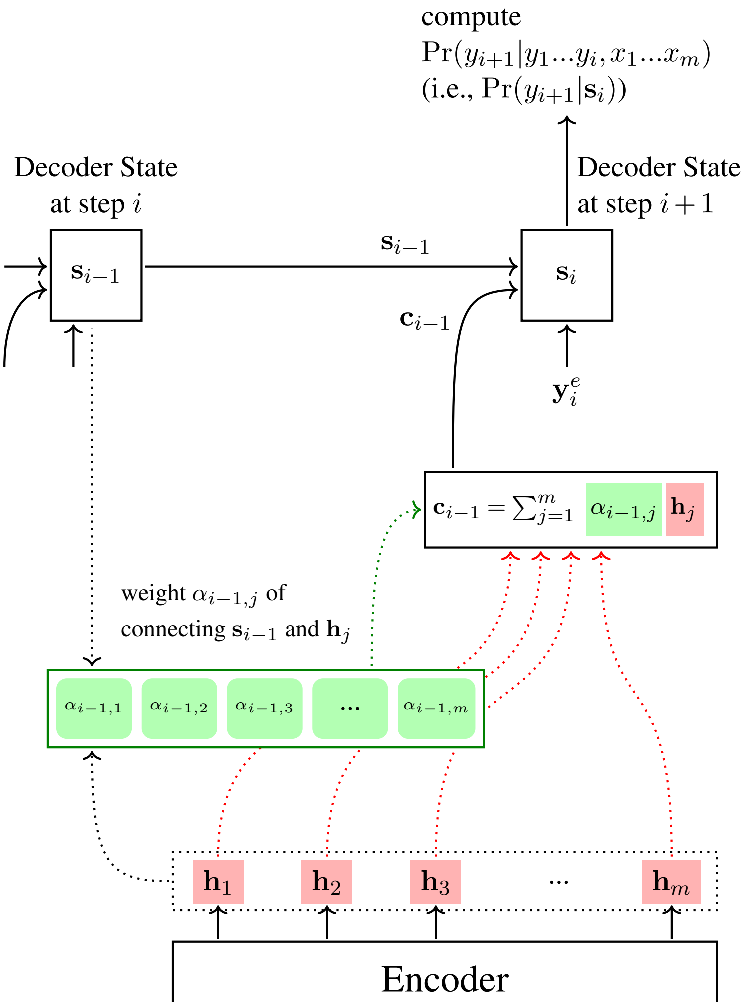


Figure 5.4: An attention model for NMT. Suppose we have obtained the representations $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$ and the decoder state $\mathbf{s}_{i-1}$ up to this point. We wish to obtain the decoder state at the next step. To this end, we first compute attention weights by normalizing the attention scores between $\mathbf{s}_{i-1}$ and $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$, and then compute a context vector $\mathbf{c}_{i-1}$ by summing over $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$ with the attention weights. A new decoder state $\mathbf{s}_i$ is created by taking the context vector $\mathbf{c}_{i-1}$, the previous state $\mathbf{s}_{i-1}$, and the word representation $\mathbf{y}_i^e$. $\mathbf{s}_i$ will be used as a condition for predicting a distribution of words at step $i+1$.

5.3.2 The QKV Attention

Because the attention mechanism is such a powerful approach, many variants have been developed. Perhaps the most widely used approach is to reframe the attention problem as one of matching a query against a set of key-value pairs. It lays the foundation for the well-known sequence model — Transformer [Vaswani et al., 2017].

Here we assume that there are a number of key-value pairs $\{(\mathbf{k}_1, \mathbf{v}_1), ..., (\mathbf{k}_m, \mathbf{v}_m)\}$ and a query $\mathbf{q}$. The goal of the **query-key-value attention** (or **QKV attention**) model is to aggregate values by considering the correspondence between the query and the keys. This is a standard search problem in database systems in which information is returned in its original form or a new form when it matches the query. In the QKV attention, the result of the search is not a single value in $\{\mathbf{v}_1, ..., \mathbf{v}_m\}$ but instead a weighted combination of these values. This is the key difference between this attention model and conventional search models.

Formally, the result of the QKV attention is defined to be

$$\mathbf{c} = \sum_{j=1}^{m} \alpha_j \mathbf{v}_j \tag{5.28}$$

where

$$\alpha_j = \frac{\exp(\mathbf{q}\mathbf{k}_j^{\top}/\beta)}{\sum_{j'=1}^{m} \exp(\mathbf{q}\mathbf{k}_{j'}^{\top}/\beta)} \tag{5.29}$$

is the attention weight. It turns out that the above model has precisely the same general form as the model described in the previous subsection, and $\mathbf{c}$ can be simply viewed as a context vector.

While the basic form of the QKV attention is not something new, it can handle a variety of problems by giving $\mathbf{q}$, $\mathbf{k}_j$ and $\mathbf{v}_j$ appropriate meanings. Here we consider a more general case where there are n queries $\{\mathbf{q}_1, ..., \mathbf{q}_n\}$ and n output vectors $\{\mathbf{c}_1, ..., \mathbf{c}_n\}$. To simplify notation, we use $\mathbf{Q}$ to denote a matrix whose i-th row is $\mathbf{q}_i$:

$$\mathbf{Q} = \begin{bmatrix} \mathbf{q}_1 \\ \vdots \\ \mathbf{q}_n \end{bmatrix} \tag{5.30}$$

Likewise, we can define $\mathbf{K} = \begin{bmatrix} \mathbf{k}_1 \\ \vdots \\ \mathbf{k}_m \end{bmatrix}$, $\mathbf{V} = \begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_m \end{bmatrix}$, and $\mathbf{C} = \begin{bmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_n \end{bmatrix}$. Then, the attention model can be formulated as

$$\mathbf{C} = \mathrm{Softmax}(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\beta})\mathbf{V} \tag{5.31}$$

Figure 5.5 shows an illustration of this equation. Note that $\mathrm{Softmax}(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\beta})$ is the row-wise Softmax function, which produces a matrix of attention weights

$$\mathrm{Softmax}(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\beta}) = \begin{bmatrix} \alpha_{1,1} & ... & \alpha_{1,m} \\ \vdots & & \vdots \\ \alpha_{n,1} & ... & \alpha_{n,m} \end{bmatrix} \tag{5.32}$$

where a row vector $\begin{bmatrix} \alpha_{i,1} & ... & \alpha_{i,m} \end{bmatrix}$ represents a distribution over the indices of the values.

We can then expand Eq. (5.31) to facilitate understanding of the model

$$
\begin{aligned}
\mathbf{C} &= \begin{bmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_n \end{bmatrix} \\
&= \begin{bmatrix} \sum_{j=1}^{m} \alpha_{1,j}\mathbf{v}_j \\ \vdots \\ \sum_{j=1}^{m} \alpha_{n,j}\mathbf{v}_j \end{bmatrix} \\
&= \begin{bmatrix} \alpha_{1,1} & \dots & \alpha_{1,m} \\ \vdots & & \vdots \\ \alpha_{n,1} & \dots & \alpha_{n,m} \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_m \end{bmatrix}
\end{aligned}
\tag{5.33}
$$

In sequence-to-sequence modeling, $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ can be defined in several different ways. To describe the correspondence between the source-side and target-side sequences, one approach, called **encoder-decoder attention**, is to simply assume that

$$
\mathbf{Q} = \begin{bmatrix} \mathbf{s}_1 \\ \vdots \\ \mathbf{s}_n \end{bmatrix} \tag{5.34}
$$

and

$$
\mathbf{K} = \mathbf{V} = \begin{bmatrix} \mathbf{h}_1 \\ \vdots \\ \mathbf{h}_m \end{bmatrix} \tag{5.35}
$$

In this case, $\mathbf{C}$ is a sequence of new representations of the source-side sequence given the representations of the target-side sequence. As with the model described in the previous subsection, each $\mathbf{c}_i \in \mathbf{C}$ can be used to predict the word y_{i+1}.

In addition to applying the model to sequence-to-sequence problems, another type of approach is to regard it as a sequence model, that is, we use the QKV attention to represent a sequence in one language. In this case, the QKV attention is also called **self-attention** which forms the basis of the well-known Transformer model [Vaswani et al., 2017]. Consider, for example, the sequence of states $\mathbf{h}_1 \dots \mathbf{h}_m$. The self-attention model assumes that

$$
\mathbf{Q} = \mathbf{K} = \mathbf{V} = \begin{bmatrix} \mathbf{h}_1 \\ \vdots \\ \mathbf{h}_m \end{bmatrix} \tag{5.36}
$$

Then, the output of the model is a sequence of representations $\mathbf{c}_1 \dots \mathbf{c}_m$. $\mathbf{c}_j$ is a representation which captures the correlations between $\mathbf{h}_j$ and all elements of the input sequence. We will see a more detailed discussion on this model in Chapter 6.

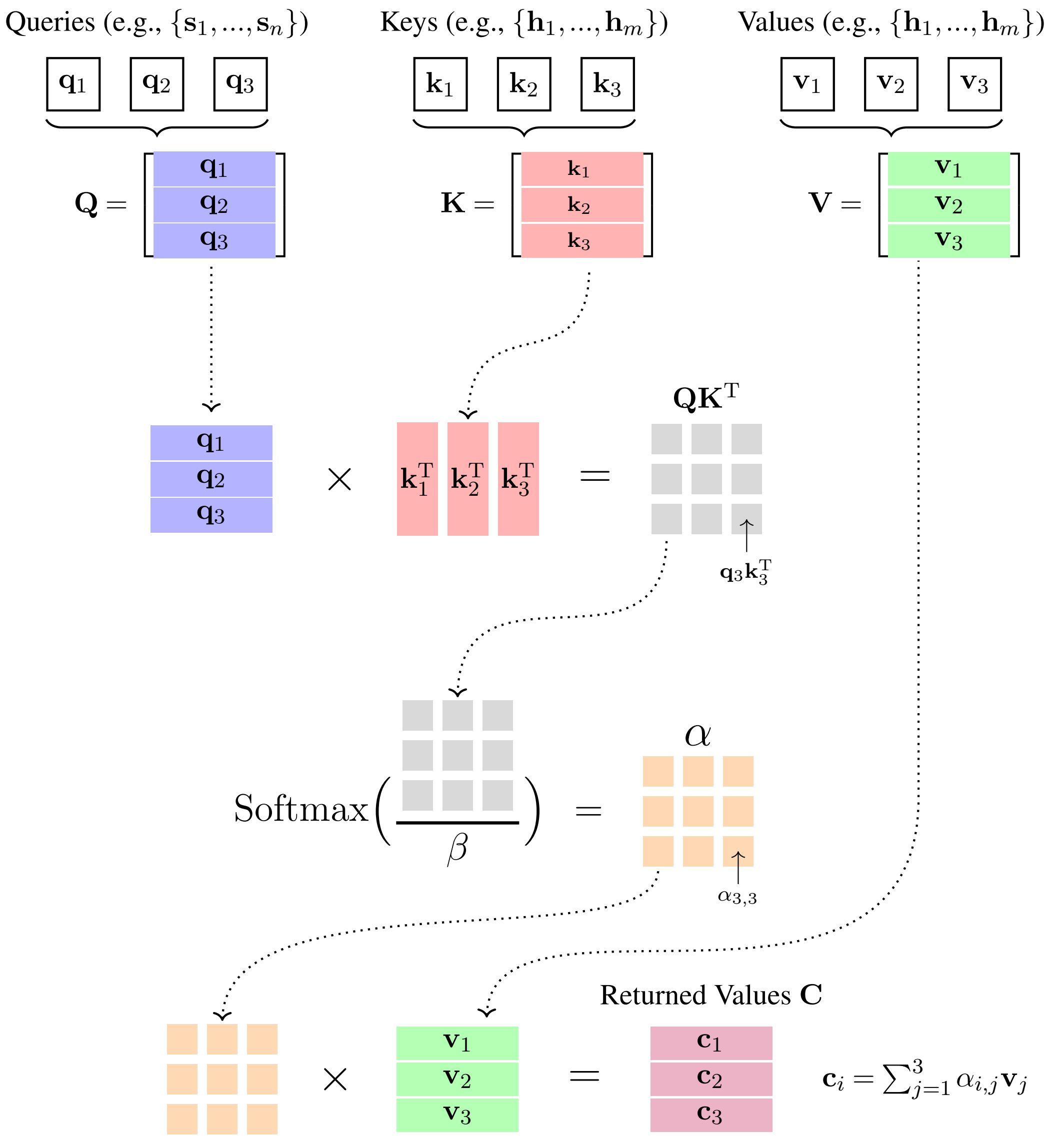


Figure 5.5: The QKV attention model for batches of queries ($\mathbf{Q}$), keys ($\mathbf{K}$), and values ($\mathbf{V}$). The figure shows a direct implementation of the formula $\mathbf{C} = \mathrm{Softmax}(\frac{\mathbf{QK}^{\top}}{\beta})\mathbf{V}$. $\mathrm{Softmax}(\frac{\mathbf{QK}^{\top}}{\beta})$ computes the attention weights by normalizing a scaled dot-product of $\mathbf{Q}$ and $\mathbf{K}^{\top}$. This results in a matrix α in which a row vector represents the weights over different values. By multiplying α with $\mathbf{V}$, we obtain a sequence of new values, each expressing a weighted sum of the original values.

5.3.3 Multi-head Attention

Multi-head attention is a natural extension to the above models. The key idea is to perform attention in different subspaces of the representation space simultaneously rather than in a single representation space. To illustrate, consider a standard attention model that takes sequences of source-side and target-side states and outputs a sequence of new states, written as

$$\mathbf{c}_1 ... \mathbf{c}_n = \mathrm{Att}(\mathbf{h}_1 ... \mathbf{h}_m, \mathbf{s}_1 ... \mathbf{s}_n) \tag{5.37}$$

where $\mathbf{h}_j, \mathbf{s}_i, \mathbf{c}_i \in \mathbb{R}^{d_h}$, and $\mathrm{Att}(\cdot)$ is the attention function. We can map $\mathbf{h}_j$ into τ vectors $\{\mathbf{h}_j^{[1]}, ..., \mathbf{h}_j^{[\tau]}\}$ using the following linear transformations

$$\mathbf{h}_j^{[1]} = \mathbf{h}_j \mathbf{W}_h^{[1]} \tag{5.38}$$

$$\vdots$$

$$\mathbf{h}_j^{[\tau]} = \mathbf{h}_j \mathbf{W}_h^{[\tau]} \tag{5.39}$$

where $\mathbf{h}_j^{[1]}, ..., \mathbf{h}_j^{[\tau]} \in \mathbb{R}^{\frac{d_h}{\tau}}$, and $\mathbf{W}_h^{[1]}, ..., \mathbf{W}_h^{[\tau]} \in \mathbb{R}^{d_h \times \frac{d_h}{\tau}}$.

Similarly, we can map $\mathbf{s}_i$ into τ vectors $\{\mathbf{s}_i^{[1]}, ..., \mathbf{s}_i^{[\tau]}\}$. We then define τ **feature subspaces** in which the attention function is performed independently. For the k-th feature subspace, we have

$$\mathbf{c}_1^{[k]} ... \mathbf{c}_n^{[k]} = \mathrm{Att}(\mathbf{h}_1^{[k]} ... \mathbf{h}_m^{[k]}, \mathbf{s}_1^{[k]} ... \mathbf{s}_n^{[k]}) \tag{5.40}$$

The output of the model is a sequence of d_h-dimensional vectors, each of which is obtained by concatenating the vectors that are produced in all these feature subspaces, followed by a linear transformation. This procedure is given by

$$\mathbf{c}_1 = [\mathbf{c}_1^{[1]}, ..., \mathbf{c}_1^{[\tau]}] \mathbf{W}_c \tag{5.41}$$

$$...$$

$$\mathbf{c}_n = [\mathbf{c}_n^{[1]}, ..., \mathbf{c}_n^{[\tau]}] \mathbf{W}_c \tag{5.42}$$

where $\mathbf{W}_c \in \mathbb{R}^{d_h \times d_h}$.

Following the notation used in the previous subsection, we can express a sequence of vectors as a matrix, e.g., $\mathbf{H} = \begin{bmatrix} \mathbf{h}_1 \\ \vdots \\ \mathbf{h}_m \end{bmatrix} \in \mathbb{R}^{m \times d_h}$, $\mathbf{S} = \begin{bmatrix} \mathbf{s}_1 \\ \vdots \\ \mathbf{s}_n \end{bmatrix} \in \mathbb{R}^{n \times d_h}$, and $\mathbf{C} = \begin{bmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_n \end{bmatrix} \in \mathbb{R}^{n \times d_h}$.
Using this notation, we rewrite Eq. (5.37) as

$$\mathbf{C} = \mathrm{Att}(\mathbf{H}, \mathbf{S}) \tag{5.43}$$

To give a formal definition of multi-head attention, we first introduce the split and merge functions. The split function divides each row vector of a matrix into a number of sub-vectors, resulting in a 3D tensor. For example, splitting an $m \times d_h$ matrix $\mathbf{A}$ with τ produces a $\tau \times m \times \frac{d_h}{\tau}$ tensor[3]

$$\mathbf{A}_{\text{heads}} = \mathrm{Split}(\mathbf{A}, \tau) \tag{5.44}$$

[3] A $a \times b \times c$ tensor can be treated as an array of a matrices whose shapes are $b \times c$.

The merge function performs the inverse operation of the split function. Given a $\tau \times n \times \frac{d_h}{\tau}$ tensor (say $\mathbf{A}_{\text{heads}}$), it merges each group of τ $\frac{d_h}{\tau}$-dimensional sub-arrays in the form

$$\mathbf{A}_{\text{merge}} = \text{Merge}(\mathbf{A}_{\text{heads}}, \tau) \tag{5.45}$$

Thus the form of multi-head attention is given by

$$\begin{aligned} \mathbf{C} &= \mathbf{C}_{\text{merge}} \mathbf{W}_c \\ &= \text{Merge}(\mathbf{C}_{\text{heads}}, \tau) \mathbf{W}_c \\ &= \text{Merge}(\text{Att}(\mathbf{H}_{\text{heads}}, \mathbf{S}_{\text{heads}}), \tau) \mathbf{W}_c & (5.46) \\ \mathbf{H}_{\text{heads}} &= \text{Split}(\mathbf{H}\mathbf{W}_h, \tau) & (5.47) \\ \mathbf{S}_{\text{heads}} &= \text{Split}(\mathbf{S}\mathbf{W}_s, \tau) & (5.48) \end{aligned}$$

where $\mathbf{W}_h, \mathbf{W}_s \in \mathbb{R}^{d_h \times d_h}$ are the parameters. $\text{Split}(\mathbf{H}\mathbf{W}_h, \tau)$ implements the projections of Eqs. (5.38-5.39) for all $\mathbf{h}_j$. Likewise, $\text{Split}(\mathbf{S}\mathbf{W}_s, \tau)$ has an analogous interpretation. Note that here $\text{Att}(\cdot)$ is extended to deal with multi-head inputs. See Figure 5.6 for an illustration of this model.

Multi-head attention is a very general approach that can be extended to many models. As a simple example of this extension, consider the QKV attention model discussed in the previous subsection. Let $\text{Att}_{\text{QKV}}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ be the attention function, and $\mathbf{Q} \in \mathbb{R}^{d_k}, \mathbf{K} \in \mathbb{R}^{d_k}, \mathbf{V} \in \mathbb{R}^{d_v}$ be the inputs. The multi-head QKV attention model is given by

$$\begin{aligned} \mathbf{C} &= \text{Merge}(\text{Att}_{\text{QKV}}(\mathbf{Q}_{\text{heads}}, \mathbf{K}_{\text{heads}}, \mathbf{V}_{\text{heads}})) \mathbf{W}_c & (5.49) \\ \mathbf{Q}_{\text{heads}} &= \text{Split}(\mathbf{Q}\mathbf{W}_q, \tau) & (5.50) \\ \mathbf{K}_{\text{heads}} &= \text{Split}(\mathbf{K}\mathbf{W}_k, \tau) & (5.51) \\ \mathbf{V}_{\text{heads}} &= \text{Split}(\mathbf{V}\mathbf{W}_v, \tau) & (5.52) \end{aligned}$$

where $\mathbf{W}_q \in \mathbb{R}^{d_k \times d_k}, \mathbf{W}_k \in \mathbb{R}^{d_k \times d_k}, \mathbf{W}_v \in \mathbb{R}^{d_v \times d_v}, \mathbf{W}_c \in \mathbb{R}^{d_v \times d_v}$ are the model parameters.

One advantage of multi-head attention is that each feature subspace describes a different perspective of attention (call it an **attention head** or **head** for short). Therefore, the concatenation of the outputs over these heads represents an ensemble of attention models that deal with different parts of the data. This is similar to learning a group of models independently and combining them to form a stronger model. Such approaches have been proven to be useful in many problems [Opitz and Maclin, 1999; Zhou, 2012b]. Note that the multi-head attention models discussed here are parameterized by the linear projections on the input and output spaces. The use of these linear projections is generally helpful as the models become deeper and can describe more complex problems.

From an architecture design perspective, multi-head attention falls into a broad class of neural networks — those involving a number of branches of layer stacks for dealing with the same input (call them **multi-branch neural networks**). However, unlike conventional approaches, which require different model architectures for different branches, the multi-head attention approach is based on a single model for all the heads. As a result, such systems are

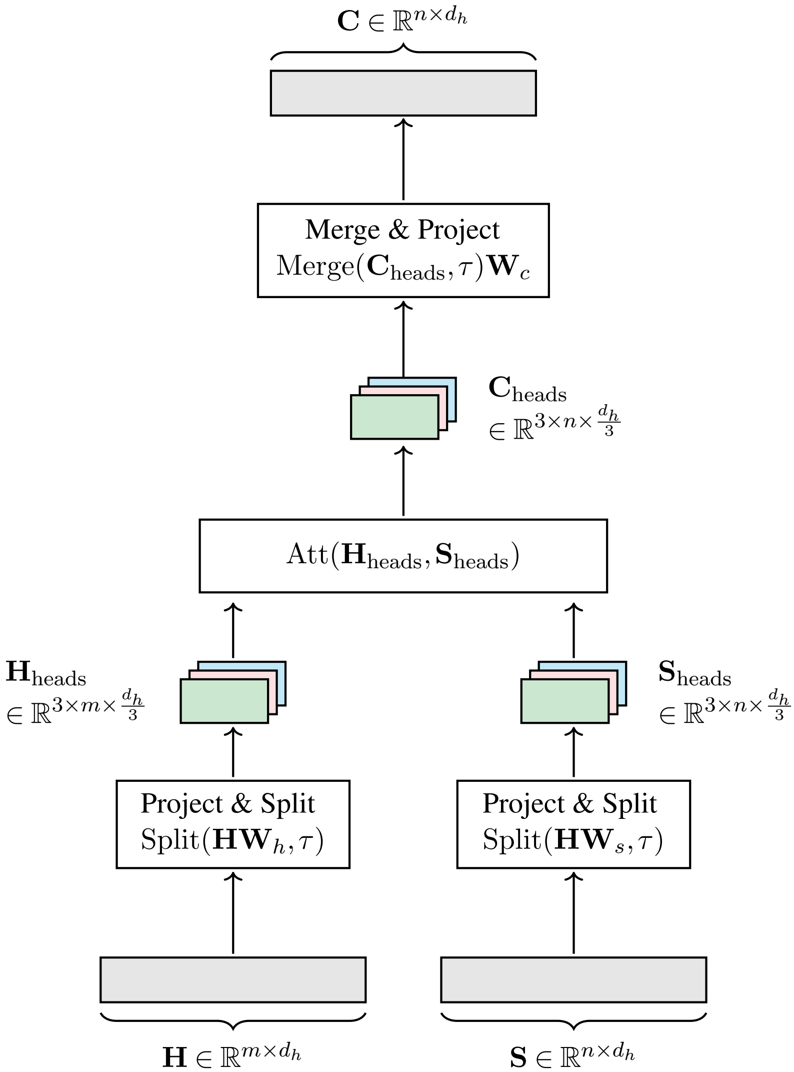


Figure 5.6: An attention model with $\tau = 3$ heads. First, we transform the input matrices into multi-head representations, i.e., 3D tensors $\mathbf{H}_{\mathrm{heads}} \in \mathbb{R}^{3 \times m \times \frac{d_h}{3}}$ and $\mathbf{S}_{\mathrm{heads}} \in \mathbb{R}^{3 \times n \times \frac{d_h}{3}}$. These tensors are then fed into an attention model. The output of this model is a tensor $\mathbf{C}_{\mathrm{heads}} \in \mathbb{R}^{3 \times n \times \frac{d_h}{3}}$. We then merge the heads of $\mathbf{C}_{\mathrm{heads}}$, followed by a linear transformation. Finally, we obtain n vectors of size d_h, represented by an $n \times d_h$ matrix.

very efficient in practice because the attention procedure can run in parallel over these heads.

5.3.4 Multi-layer Attention

So far we have considered the case of **single-layer attention** — the output of the attention models can be written as a linear combination of the source-side representations. Now we extend it in a natural way to **multi-layer attention** in which the single-layer attention procedure runs a number of times for forming a "deeper" attention model.

To do this, a multi-layer neural network is created on the target-side. The model architecture is regular. We stack a number of attention layers, each interacting with the source-side sequence and feeding its output to the next layer. In an attention layer, we perform attention as usual. For the l-th layer in the stack, this step takes the source-side sequence (denoted by $\mathbf{h}_1...\mathbf{h}_m$) as well as the output of the previous layer (denoted by $\mathbf{s}_1^{l-1}...\mathbf{s}_n^{l-1}$), and produces a sequence of vectors by

$$\mathbf{c}_1^l...\mathbf{c}_n^l \quad = \quad \mathrm{Att}(\mathbf{h}_1...\mathbf{h}_m, \mathbf{s}_1^{l-1}...\mathbf{s}_n^{l-1}) \tag{5.53}$$

where $\mathrm{Att}(\cdot)$ could be any attention function described in this chapter.

Then, we create another neural network $f(\cdot)$ to give more modeling power to the model. The output of the attention layer is thus defined to be

$$\mathbf{s}_1^l...\mathbf{s}_n^l \quad = \quad f(\mathbf{c}_1^l...\mathbf{c}_n^l, \mathbf{s}_1^{l-1}...\mathbf{s}_n^{l-1}) \tag{5.54}$$

$f(\cdot)$ can be designed in many ways [Sukhbaatar et al., 2015; Wu et al., 2016; Vaswani et al., 2017]. A popular choice is to define $f(\cdot)$ as a feed-forward neural network with a residual connection, given by

$$f(\mathbf{c}_1^l...\mathbf{c}_n^l, \mathbf{s}_1^{l-1}...\mathbf{s}_n^{l-1}) \quad = \quad \mathrm{FFN}(\mathbf{c}_1^l...\mathbf{c}_n^l) \ + \ \mathbf{s}_1^{l-1}...\mathbf{s}_n^{l-1} \tag{5.55}$$

Substituting for the vectors $\mathbf{c}_1^l...\mathbf{c}_n^l$, using Eq. (5.53), the output of layer i is written in the form

$$\mathbf{s}_1^l...\mathbf{s}_n^l \quad = \quad \mathrm{FFN}(\mathrm{Att}(\mathbf{h}_1...\mathbf{h}_m, \mathbf{s}_1^{l-1}...\mathbf{s}_n^{l-1})) \ + \ \mathbf{s}_1^{l-1}...\mathbf{s}_n^{l-1} \tag{5.56}$$

As with the models in the previous subsections, it is convenient to use a more compact notation by expressing a sequence of vectors as a matrix. Thus this model can be given in another form

$$\mathbf{S}^l \quad = \quad \mathrm{FFN}(\mathrm{Att}(\mathbf{H}, \mathbf{S}^{l-1})) + \mathbf{S}^{l-1} \tag{5.57}$$

Here $\mathrm{FFN}(\cdot)$ is generally a multi-layer neural network with non-linear activation functions. The identity mapping (i.e., $+\mathbf{S}^{l-1}$) creates a direct path from the input to the output of the layer, making it easier to train a deep neural network.

Figure 5.7 shows the architecture of this model. The attention model starts with the initial representation of the target-side sequence, that is, $\mathbf{S}^0 = \mathbf{S} = \begin{bmatrix} \mathbf{s}_1 \\ \vdots \\ \mathbf{s}_n \end{bmatrix}$. If there are L attention layers, then the final output will be $\mathbf{S}^L$.

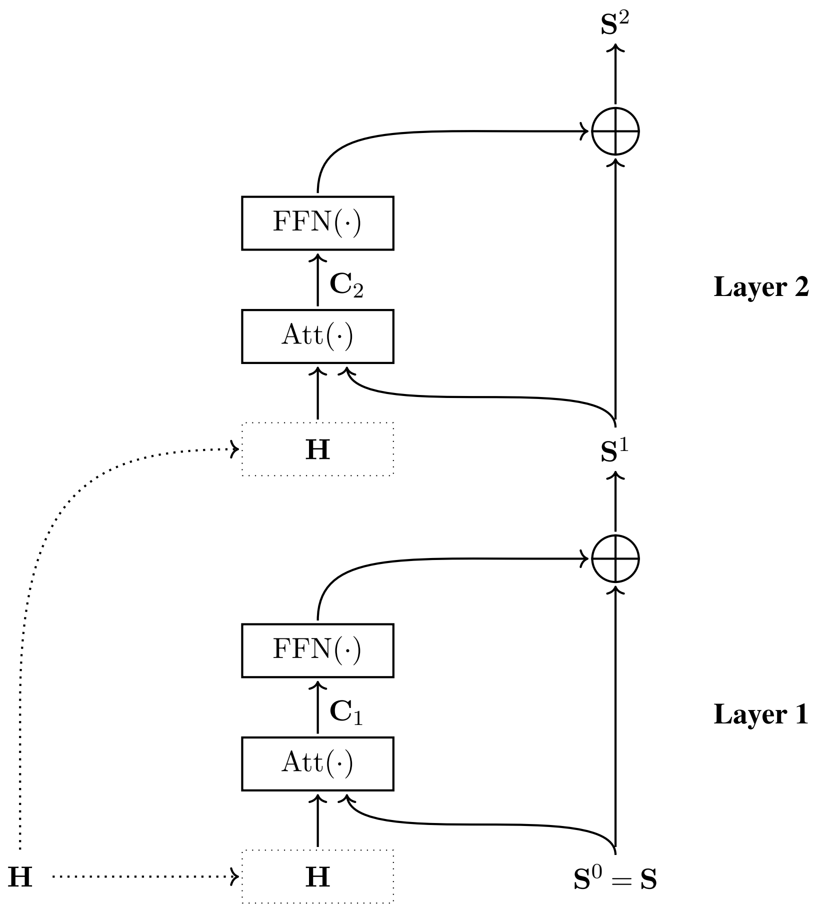


Figure 5.7: A 2-layer attention model. These layers take the same "key-value" pairs (i.e., $\mathbf{H}$) but each takes a different "query" (i.e., $\mathbf{S}^l$). The attention model is standard: context vectors $\mathbf{C}^l$ are generated by taking both $\mathbf{H}$ and $\mathbf{S}^l$. A feed-forward neural network is built to transform $\mathbf{C}^l$, followed by an addition of $\mathbf{S}^l$. So this model can be formulated as $\mathbf{S}^l = \mathrm{FFN}(\mathrm{Att}(\mathbf{H}, \mathbf{S}^{l-1})) + \mathbf{S}^{l-1}$. $\mathbf{S}^l$ is then used in the next layer as the query, that is, layer $l+1$ takes $\mathbf{H}$ and $\mathbf{S}^l$, and repeats the above process. The output of the last layer can be viewed as a deeper representation of $\mathbf{H}$ for $\mathbf{S}$.

5.3.5 Remarks

Above we considered a basic attention model and a series of refinements. The literature on attention and related topics contains a wide range of methods for modeling how a system concentrates on different parts of the input, as well as many applications of such models. This subsection discusses some interesting issues.

1. Word Alignment vs. Attention

Attention is related to a long line of research on (word) alignment which models the correspondence between two groups of language units. In NLP, alignment is a very general concept that is used to refer to several problems. For example, most statistical machine translation systems are trained on bilingual texts with word-to-word alignments [Koehn et al., 2003; Chiang, 2005]. **Word alignment** models are thus developed to generate links between words in two sentences [Vogel et al., 1996; Och and Ney, 2003; Taskar et al., 2005; Dyer et al., 2013]. While the outputs of these systems are discrete variables, the underlying models are typically probabilistic and may involve continuous representations. Therefore, the correspondence between word alignment and the attention models discussed here is apparent because they both assign weights to pairs of words.

Note that despite the similarity between word alignment and attention problems, their goals are different. In most cases word alignment models are used as individual systems to produce alignment results for downstream systems, whereas attention models are typically treated as components of bigger systems and do not work alone (see Figure 5.8 for a comparison of these models). As a result, they are suited to different types of sequence-to-sequence systems in practice: word alignment is one step in cascaded systems, and attention represents intermediate states of a neural network.

Nevertheless, word alignment and attention are two related problems, and can help each other in some cases. For example, one way to see how an attention model behaves is to induce word alignments from it and measure the quality of these word alignments [Tu et al., 2016; Li et al., 2019; Garg et al., 2019]. Also, systems equipped with the attention mechanism can be guided by external word alignment resources [Mi et al., 2016b; Liu et al., 2016b].

2. Introducing Priors

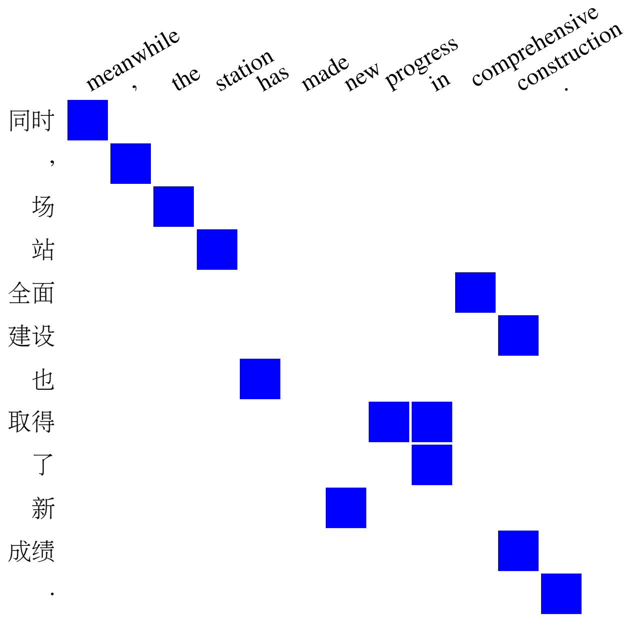
As discussed in Section 5.3.1, the attention models implicitly define an attention distribution over $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$ which is used to compute a weighted sum of these representations. This distribution is expressed in terms of the alignment weights and is learned as part of the model. In addition to learning the attention distribution in an end-to-end fashion, we can also define it based on prior knowledge of how attention is allocated in sequences.

One approach is to directly impose some structure on the distribution. A straightforward example is to restrict attention to a local span of the sequence and discard the rest. Let $[\rho_i - D, \rho_i + D]$ be a $(2D+1)$-word window centered at position ρ_i of the source-side sequence. We can restrict attention from $\mathbf{s}_i$ to $\mathbf{h}_{\rho_i - D} ... \mathbf{h}_{\rho_i + D}$, yielding a local context vector formulated as:

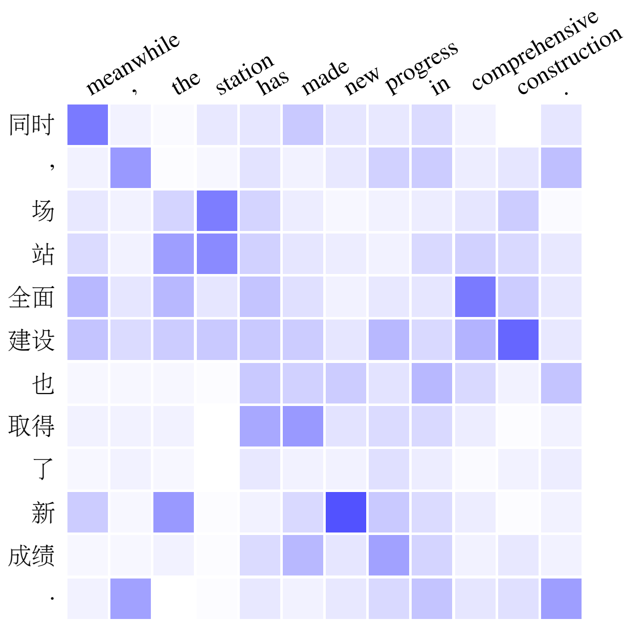
$$\mathbf{c}_i = \mathrm{Att}(\mathbf{h}_{\rho_i - D} ... \mathbf{h}_{\rho_i + D}, \mathbf{s}_i) \tag{5.58}$$

This approach is also called **local attention**. Determining ρ_i is similar to the **reordering problem** in machine translation. For translation between languages with similar word orders, it is common to assume that the translation is monotonic and ρ_i is linear with respect to i [Koehn, 2004], e.g., $\rho_i = \lfloor m \frac{i}{n} \rfloor$ or $\lceil m \frac{i}{n} \rceil$. An alternative method is to use a neural network to predict a relative position in the source-side sequence (denoted by r_i) [Luong et al., 2015]. ρ_i can then be defined to be $\lfloor m r_i \rfloor$ or $\lceil m r_i \rceil$.

In another thread of research, a new distribution is constructed by combining the original



(a) Heat map of word alignments



(b) Heat map of attention weights

Figure 5.8: Heat maps of a word alignment model and an attention model for a pair of Chinese and English sentences. The heat maps are represented as shaded grids in which each cell describes the correspondence of a pair of Chinese and English words. This correspondence is shown on a color scale ranging from white denoting a weight of 0 to pure blue denoting a weight of 1.

attention distribution and some prior distribution. The simplest such distribution takes the form of linear interpolation

$$\widetilde{\Pr}(\mathbf{h}_j|\mathbf{s}_i) = \eta \cdot \Pr(\mathbf{h}_j|\mathbf{s}_i) + (1-\eta) \cdot \mathrm{Prior} \tag{5.59}$$

where Prior denotes the prior distribution, and η denotes the interpolation coefficient. When $\eta = 1$, it is a standard attention model. By contrast, when $\eta = 0$, the attention is completely dependent on the prior [You et al., 2020].

The prior can be chosen in many ways. A simple choice is to assume Prior to be a Gaussian distribution $\mathrm{Gaussian}(\mu, \sigma^2)$. This provides a clear intuitive interpretation: attention is concentrated at a focal point in the sequence and decays as the distance from this point increases. To determine the mean and variance of the Gaussian distribution, we can use the same technique described above, for example, we develop two neural networks to compute the mean and variance, respectively.

The interpolation can also be considered an intermediate step in computing the attention distribution. For example, consider the QKV attention discussed in Section 5.3.2. The interpolation can be applied to the query-key dot-product [Yang et al., 2018a; Guo et al., 2019]. To this end, we can modify Eq.(5.29) in the following form

$$\begin{aligned} \alpha_j &= \mathrm{Softmax}(\frac{\mathbf{q}\mathbf{k}_j^{\top}}{\beta} + \eta \mathrm{Prior}) \\ &= \mathrm{Softmax}(\frac{\mathbf{s}_i\mathbf{h}_j^{\top}}{\beta} + \eta \mathrm{Prior}) \end{aligned} \tag{5.60}$$

As $\frac{\mathbf{q}\mathbf{k}_j^{\top}}{\beta}$ (or $\frac{\mathbf{s}_i\mathbf{h}_j^{\top}}{\beta}$) is not constrained to $[0,1]$, Prior is re-scaled by a hyper-parameter η.

Sometimes, priors arise when knowledge about attention comes from external sources. As discussed above, incorporating word alignments into attention models is one of the simplest ways to do this. The idea can be extended to make use of more structural information, such as fertility and coverage [Cohn et al., 2016; Feng et al., 2016; Tu et al., 2016], or more task-specific constraints, such as monotonic alignments between input and output sequences [Raffel et al., 2017; Chiu and Raffel, 2018]. Also, as with syntactic machine translation systems, parse trees can be used to bias the process of attention as an auxiliary input. For example, dependency trees are a widely used source of information in modeling word correspondence for either sequence-to-sequence [Chen et al., 2018a] or sequence modeling problems [Zhang et al., 2020c; Nguyen et al., 2020; Xu et al., 2021b].

Because standard attention mechanisms can be computationally expensive in large-scale applications, researchers have developed efficient alternatives by introducing stronger inductive biases into the model design [Tay et al., 2020b]. This line of research falls under the broader category of efficient NLP methods. Chapter 6 discusses these methods in detail.

3. Attention in Memory Networks

As well as being of great interest in sequence-to-sequence systems, the attention mechanism is extensively used in memory-based neural models [Sukhbaatar et al., 2015; Graves et al., 2014; Kumar et al., 2016]. As discussed in Chapter 4, a memory system maintains a collection of data items, allowing the model to retain and retrieve information. Given a query, the system computes a match score between the query and the key of each data item. This procedure is also known as **addressing** [Graves et al., 2014]. Such addressing is typically implemented by first representing the query and the data items as real-valued vectors, and then calculating a weight based on a similarity measure between them. The result of the retrieval is a weighted sum of the data items. This formalism aligns perfectly with the QKV attention model discussed in Section 5.3.2.

Given the conceptual link between attention and memory mechanisms, at least from a computational, if not psychological perspective, we can view attention as a process of retrieving information from a memory bank (i.e., $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$) using a given query (i.e., $\mathbf{s}_i$). Thus we can interpret a sequence-to-sequence system with the attention mechanism as follows. On the source-side, we store information in a memory represented as a sequence of vectors $\mathbf{h}_1...\mathbf{h}_m$. Then, we retrieve from this memory to recover step by step the source-side information on the target-side.

4. Beyond Sequence-to-Sequence Problems

While we restrict our discussion in this section to the problem of transforming one sequence into another, the general attention mechanism can be applied to address a broader range of tasks. As mentioned in Section 5.3.2, and as will be discussed in Chapter 6, a well-known variant of this approach is self-attention. In self-attention, the QKV attention model is used as a sequence model. It provides a general framework for modeling the interactions and dependencies between input variables, which makes it applicable to a variety of problems. For example, we can concatenate $\mathbf{h}_1...\mathbf{h}_m$ and $\mathbf{s}_1...\mathbf{s}_n$ into a single combined sequence $\mathbf{h}_1...\mathbf{h}_m\mathbf{s}_1...\mathbf{s}_n$, and apply the model to this new sequence. In this way, self-attention can be naturally extended to a sequence-to-sequence model [Lample and Conneau, 2019; Raffel et al., 2020]. Such an approach is effective even when $\mathbf{h}_1...\mathbf{h}_m$ and $\mathbf{s}_1...\mathbf{s}_n$ represent different modalities. For instance, if we use $\mathbf{h}_1...\mathbf{h}_m$ to represent text and $\mathbf{s}_1...\mathbf{s}_n$ to represent an image, we obtain a multi-modal model that fuses textual and visual representations by performing self-attention across them [Chen et al., 2020c].

Another approach to learning joint representations of sequences is to build multiple attention models, allowing each sequence to attend to the other. An example of this approach is **co-attention**, which has been widely adopted in multi-modal language processing [Lu et al., 2016]. In tasks such as **visual question answering** (**VQA**), we aim to determine which parts of an image correspond to a specific word in the question, and conversely, which words in the question align with a given visual region. To achieve this, we construct two attention models: one for image-to-text attention, and another for text-to-image attention. The outputs of both models serve as joint representations of the image and text, and thus can be used as features for answer prediction.

The attention models discussed in this section are invariant to the order of their inputs. While this permutation invariance poses a challenge for modeling sequential data, it can be addressed by explicitly encoding position information into the inputs themselves (see Chapters 4 and 6). On the other hand, the simplicity of this formulation makes these models broadly applicable: the input data need not be sequential. As a result, attention mechanisms can be directly applied to more complex data structures, such as graphs [Veličković et al., 2018; Lee et al., 2019].

5.4 Search

Search is a fundamental problem in artificial intelligence and plays a crucial role in many NLP systems. The search problem presents a computational challenge here because the number of hypotheses in the search space increases exponentially with the length of the sequence and the size of the target vocabulary. Exhaustive search in this context is computationally prohibitive. Therefore, real-world systems typically employ search algorithms or heuristics to ensure that optimal or near-optimal solutions can be found in acceptable computational time.

For many practical sequence-to-sequence applications, the search problem, often referred to as inference or decoding, can be formulated as follows:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}\in\Omega} \mathrm{Score}(\mathbf{x},\mathbf{y}) \tag{5.61}$$

where $\mathrm{Score}(\mathbf{x},\mathbf{y})$ is a scoring function that measures the quality of $\mathbf{y}$ given $\mathbf{x}$.

This formulation differs slightly from Eq. (5.3). First, we use $\mathrm{Score}(\mathbf{x},\mathbf{y})$ instead of $\Pr(\mathbf{y}|\mathbf{x})$ as the objective function. While a standard approach to training sequence-to-sequence models is to maximize $\Pr(\mathbf{y}|\mathbf{x})$ (or $\Pr(\mathbf{x},\mathbf{y})$), we often need to account for task-specific issues, such as length bias, at inference time. It is therefore common to incorporate other penalty and reward terms alongside $\Pr(\mathbf{y}|\mathbf{x})$ to define the final search objective (see Section 5.4.1). A second difference arises from the search space itself, which is constrained to a subset Ω. In practice, Ω is a pruned search space containing a relatively small number of highly probable hypotheses. This is typically achieved through pruning techniques and advanced search algorithms (see Section 5.4.2). In this section, we explore solutions to these challenges and discuss several refinements. While these methods are largely motivated by developments in machine translation, the concepts are general and apply broadly to most text generation tasks.

5.4.1 The Length Problem

Recall from Section 5.2.2 that the probability of the target-side sequence $\mathbf{y}$ given the source-side sequence $\mathbf{x}$ can be written as the product of the probabilities of each word y_i given both the previously generated words $y_0...y_{i-1}$ and $\mathbf{x}$. Here, we re-express Eq. (5.15) using a simpler notation:

$$\Pr(\mathbf{y}|\mathbf{x}) = \prod_{i=1}^{n} \Pr(y_i|\mathbf{y}_{<i},\mathbf{x}) \tag{5.62}$$

where $\mathbf{y}_{<i}$ denotes the sequence $y_0...y_{i-1}$. This can be equivalently expressed in terms of log probabilities

$$\log\Pr(\mathbf{y}|\mathbf{x}) = \sum_{i=1}^{n} \log\Pr(y_i|\mathbf{y}_{<i},\mathbf{x}) \tag{5.63}$$

While this simple form of modeling offers clear advantages for practical NLP applications, it unfortunately introduces a strong bias toward shorter target sequences. Therefore, directly using $\mathrm{Score}(\mathbf{x},\mathbf{y}) = \Pr(\mathbf{y}|\mathbf{x})$ or $\log\Pr(\mathbf{y}|\mathbf{x})$ as the search objective is problematic, as the output is highly likely to be an excessively short sequence (e.g., a sequence comprising only one or two words). This problem is a direct consequence of the model's inductive bias. From a supervised learning perspective, this issue arises because **teacher forcing** is used during training: the model is exposed to ground-truth target sequences and is forced to predict them. By contrast, during inference on test data, we must explore a large hypothesis space of $\mathbf{y}$ with varying lengths, comparing them using a scoring function that effectively differs from the one optimized during training.

This problem can be addressed through a technique called **length reward**, which gives bonuses to longer sequences by adding a term to $\mathrm{Score}(\mathbf{x},\mathbf{y})$ [He et al., 2016c]. As discussed in Chapter 3, a commonly used form of length reward is given by

$$\mathrm{Score}(\mathbf{x},\mathbf{y}) = \log\Pr(\mathbf{y}|\mathbf{x}) + \lambda \cdot n \tag{5.64}$$

Here the length reward term is the length of $\mathbf{y}$ (i.e., $n = |\mathbf{y}|$), weighted by the parameter $\lambda > 0$. Improvements to this approach involve replacing n with an estimated sequence length by using a length prediction model. For example, we can bound the reward in the following form [Huang et al., 2017b; Yang et al., 2018b]

$$\mathrm{Score}(\mathbf{x},\mathbf{y}) = \log\Pr(\mathbf{y}|\mathbf{x}) + \lambda \cdot \max(l_p, n) \tag{5.65}$$

where l_p is a predicted length, generally defined to be a scaled length of $\mathbf{x}$, that is, $l_p = \mathrm{scalar}_p \cdot m$.

If we substitute the log probability $\log\Pr(\mathbf{y}|\mathbf{x})$ given by Eq. (5.63) into Eq. (5.64), we obtain

$$\begin{aligned}\mathrm{Score}(\mathbf{x},\mathbf{y}) &= \sum_{i=1}^{n} \log\Pr(y_i|\mathbf{y}_{<i},\mathbf{x}) + \lambda \cdot n \\ &= \sum_{i=1}^{n} [\log\Pr(y_i|\mathbf{y}_{<i},\mathbf{x}) + \lambda]\end{aligned} \tag{5.66}$$

Thus, we can interpret the length reward term as a reward to each word y_i. Such a method has been widely used in **statistical machine translation** (**SMT**) systems in which the rewards are treated as features of a log-linear model [Koehn et al., 2003; Chiang, 2007]. To find an appropriate value of λ, we can either use minimum error rate training [Och, 2003], following the convention in SMT, or use gradient-based methods as in neural network-based systems [Murray and Chiang, 2018].

A second approach to biasing search towards longer sequences, called **length normalization**, is to divide $\log \Pr(\mathbf{y}|\mathbf{x})$ by a length correction term, written in the following form

$$\text{Score}(\mathbf{x},\mathbf{y}) = \frac{\log \Pr(\mathbf{y}|\mathbf{x})}{n_{\text{correct}}} \tag{5.67}$$

A simple example of this model is to define the length correction term as the sequence length [Jean et al., 2015], like this

$$\begin{aligned} n_{\text{correct}} &= n \\ &= |\mathbf{y}| \end{aligned} \tag{5.68}$$

In this case, $\frac{\log \Pr(\mathbf{y}|\mathbf{x})}{n} = \frac{\sum_{i=1}^{n} \log \Pr(y_i|\mathbf{y}_{<i},\mathbf{x})}{n}$ can be viewed as the log-scale geometric mean of the probabilities $\{\Pr(y_i|\mathbf{y}_{<i},\mathbf{x})\}$[4].

Another prominent example is the normalization strategy adopted in the GNMT system [Wu et al., 2016]. It employs an exponentially scaled and shifted length penalty:

$$n_{\text{correct}} = \frac{(5+n)^{\alpha}}{(5+1)^{\alpha}} \tag{5.71}$$

where the power α is a hyper-parameter and can be determined empirically on a tuning set. To compare different methods, Table 5.2 shows a list of scoring functions for length reward and length normalization.

In machine translation, the length problem is intrinsically linked to the **coverage** problem, a topic studied extensively in SMT. When translating a source sequence, it is important to monitor how often each word is translated. We consider **over-translation** to occur (yielding an erroneously long translation) if source words are translated multiple times, and **under-translation** to occur (yielding a truncated translation) if source words are insufficiently translated. Traditionally, the coverage of a source sequence is modeled using an m-dimensional **coverage**

[4] Suppose $\{a_1,...,a_n\}$ are n variables. Since

$$\exp\left(\frac{\sum_{i=1}^{n} \log a_i}{n}\right) = \left(\prod_{i=1}^{n} a_i\right)^{\frac{1}{n}} \tag{5.69}$$

we have

$$\frac{\sum_{i=1}^{n} \log a_i}{n} = \log\left(\prod_{i=1}^{n} a_i\right)^{\frac{1}{n}} \tag{5.70}$$

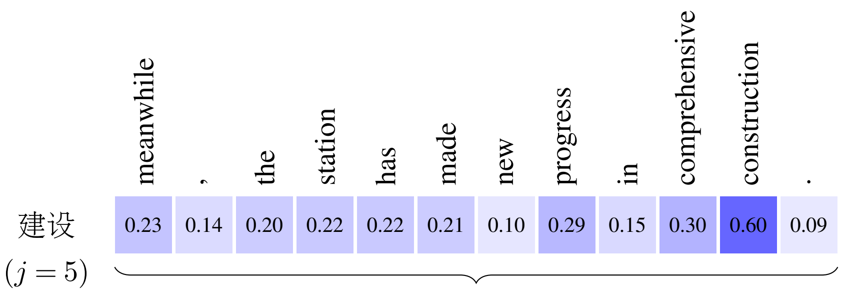
Method	Form of $\mathrm{Score}(\mathbf{x},\mathbf{y})$
No Reward/Normalization	$\mathrm{Score}(\mathbf{x},\mathbf{y}) = \log \Pr(\mathbf{y}\|\mathbf{x})$
Length Reward	$\mathrm{Score}(\mathbf{x},\mathbf{y}) = \log \Pr(\mathbf{y}\|\mathbf{x}) + \lambda \cdot n$
Bounded Length Reward	$\mathrm{Score}(\mathbf{x},\mathbf{y}) = \log \Pr(\mathbf{y}\|\mathbf{x}) + \lambda \cdot \max(l_p, n)$
Length Normalization (Basic)	$\mathrm{Score}(\mathbf{x},\mathbf{y}) = \frac{\log \Pr(\mathbf{y}\|\mathbf{x})}{n}$
Length Normalization (GNMT)	$\mathrm{Score}(\mathbf{x},\mathbf{y}) = \frac{\log \Pr(\mathbf{y}\|\mathbf{x})}{(5+n)^{\alpha}/(5+1)^{\alpha}}$

Table 5.2: Scoring functions for length reward and length normalization. $m = |\mathbf{x}|$, $n = |\mathbf{y}|$, and $l_p = \mathrm{scalar}_p \cdot m$. λ and α are parameters.

vector $\begin{bmatrix} \upsilon_1 & ... & \upsilon_m \end{bmatrix}$. Each element υ_j indicates the degree to which the source word x_j has been translated. In SMT systems, υ_j is typically a binary variable: 0 denotes $\mathrm{untranslated}$, and 1 denotes $\mathrm{translated}$. However, standard NMT systems lack such an explicit symbolic coverage mechanism. Instead, they rely on attention weights distributed across the target words. Therefore, one way to quantify the coverage of a source word x_j is to measure how strongly it is attended to by the generated target words. To achieve this, we redefine υ_j as a continuous variable:

$$\upsilon_j = \sum_{i=1}^{n} \alpha_{i,j} \quad (5.72)$$

υ_j can thus be viewed as the "number of times" x_j is translated, e.g., $\upsilon_j = 0$ means that x_j is not translated at all, and $\upsilon_j = 1$ means that x_j is counted only once in translation. Consider the example in Figure 5.8. For the source-side word 建设, the corresponding attention weights are shown below.



$$\upsilon_5 = \sum_{i=1}^{12} \alpha_{i,5} = 2.75$$

We will say that 建设is translated 2.75 times. It is possible to make use of $\{\upsilon_1, ..., \upsilon_m\}$ to define how much the source-side sequence is covered in translation. A simple way to do this is to develop a coverage score $\mathrm{cp}(\mathbf{x},\mathbf{y})$ by combining $\{\upsilon_1, ..., \upsilon_m\}$. For example, the GNMT system defines $\mathrm{cp}(\mathbf{x},\mathbf{y})$ in the following form

$$\mathrm{cp}(\mathbf{x},\mathbf{y}) = \beta \sum_{j=1}^{m} \log(\min(\upsilon_j, 1)) \quad (5.73)$$

where β dictates the weight of the coverage model. The intuition is that if $\upsilon_j \geq 1$, the word x_j is considered adequately translated; if $\upsilon_j < 1$, x_j lacks sufficient translation. As a result, $\mathrm{cp}(\mathbf{x},\mathbf{y})$ penalizes hypotheses where source words are under-translated. A refined version of this penalty was proposed by Li et al. [2018]:

$$\mathrm{cp}(\mathbf{x},\mathbf{y}) = \beta \sum_{j=1}^{m} \log(\max(\upsilon_j, \gamma)) \tag{5.74}$$

where γ is the hyper-parameter for truncation, giving a tolerance for under-translation. A similar form was proposed in [Chorowski and Jaitly, 2017]

$$\mathrm{cp}(\mathbf{x},\mathbf{y}) = \beta \sum_{j=1}^{m} \mathbb{1}(\upsilon_j > \gamma) \tag{5.75}$$

It counts the number of times that $\upsilon_j > \gamma$.

$\mathrm{cp}(\mathbf{x},\mathbf{y})$ can be easily introduced into search by adding it to $\mathrm{Score}(\mathbf{x},\mathbf{y})$. For example, the GNMT-style scoring function is defined to be

$$\mathrm{Score}(\mathbf{x},\mathbf{y}) = \frac{\log \Pr(\mathbf{y}|\mathbf{x})}{(5+n)^{\alpha}/(5+1)^{\alpha}} + \mathrm{cp}(\mathbf{x},\mathbf{y}) \tag{5.76}$$

In practice, modifying $\mathrm{Score}(\mathbf{x},\mathbf{y})$ is not the only method for mitigating the length and coverage problems during search. An alternative strategy involves introducing architectural modifications to explicitly model these phenomena [Tu et al., 2016; Mi et al., 2016a; Sankaran et al., 2016; See et al., 2017; Malaviya et al., 2018]. Note that in some cases, the length of the target sequence is specified or predicted in advance. In such cases, we can either employ non-autoregressive models [Gu et al., 2018] or design length-controllable text generation systems for specific applications [Rush et al., 2015; Kikuchi et al., 2016].

5.4.2 Pruning and Beam Search

There are many ways to define a search space. As a general concept in computer science, a search space is typically defined as the set of all possible solutions to a problem. For sequence-to-sequence problems, we can view a hypothesis as a mapping from a source-side sequence $\mathbf{x}$ to a target-side sequence $\mathbf{y}$, and the search space as the collection of all such hypotheses[5].

We can implement a search algorithm by organizing hypotheses. Recall that in Eqs. (5.62) and (5.63), we compute the probability of $\mathbf{y}$ given $\mathbf{x}$ using a left-to-right factorization. A typical search system maintains a set of hypotheses (or partial hypotheses) and expands them incrementally[6]. The search procedure begins with an initial hypothesis set Z_0 containing

[5] Here we use $(\mathbf{x},\mathbf{y})$ to denote a hypothesis. When there are multiple mappings from $\mathbf{x}$ to $\mathbf{y}$, a hypothesis can be represented as $(\mathbf{x},\mathbf{y},d)$ where d denotes the specific mapping. For example, if we transform $\mathbf{x}$ to $\mathbf{y}$ using a synchronous grammar, there might be multiple derivations of grammar rules to achieve this.

[6] A hypothesis is considered partial when the corresponding target-side sequence does not end with $\langle\mathrm{EOS}\rangle$, meaning it is an incomplete sequence. In this section, we use the terms *hypothesis* and *partial hypothesis* interchangeably because their forms are identical.

a single hypothesis z_0, whose target side is $y_0 = \langle \text{SOS} \rangle$, representing the start symbol for all target sequences. We then expand this hypothesis set over a series of search steps. Let $Z_0 ... Z_{n_{\max}}$ be a sequence of hypothesis sets, where $n_{\max}$ is the maximum number of search steps. At step i, we extend each hypothesis by appending a new word v_k drawn from the vocabulary V_y. Let $z.src$ and $z.tgt$ denote the source and target sides of a hypothesis z, respectively. Clearly, $z.src = \mathbf{x}$ for any z. Given a hypothesis $z_{\text{cur}} \in Z_{i-1}$, we can extend it to generate $|V_y|$ new hypotheses, denoted as $\{z^1_{\text{next}}, \ldots, z^{|V_y|}_{\text{next}}\}$:

$$\begin{aligned} \{z^1_{\text{next}}, ..., z^{|V_y|}_{\text{next}}\} &= \text{Extend}(z_{\text{cur}}, V_y) \\ &= \bigcup_{v_k \in V_y} \text{Extend}(z_{\text{cur}}, v_k) \end{aligned} \tag{5.77}$$

Here, $\text{Extend}(z_{\text{cur}}, v_k)$ is a function that extends the input hypothesis z_{cur} with a word $v_k \in V_y$. The target side of the resulting hypothesis is the concatenation of $z_{\text{cur}}.tgt$ and v_k, written as[7]:

$$z^k_{\text{next}}.tgt = z_{\text{cur}}.tgt \circ v_k \tag{5.78}$$

These new hypotheses $\{z^1_{\text{next}}, \ldots, z^{|V_y|}_{\text{next}}\}$ are then added to Z_i. Figure 5.9 illustrates the first few steps of this extension process. All hypotheses can be organized into a tree structure, where Z_i corresponds to the set of nodes at level i of the search tree. We thus have:

$$|Z_i| = |V_y| \cdot |Z_{i-1}| \tag{5.79}$$

In other words, the size of Z_i grows exponentially with the number of steps, yielding $|Z_i| = |V_y|^i$.

Each hypothesis z is associated with a log-probability $\log \Pr(z.tgt|z.src)$. Following the form of Eq. (5.63), this probability can be computed recursively:

$$\begin{aligned} \log \Pr(z^k_{\text{next}}.tgt|z^k_{\text{next}}.src) &= \log \Pr(z_{\text{cur}}.tgt|z_{\text{cur}}.src) + \\ &\quad \log \Pr(v_k|z_{\text{cur}}.tgt, z_{\text{cur}}.src) \end{aligned} \tag{5.80}$$

As an example, suppose $z^k_{\text{next}}.tgt = y_0 ... y_{i+1}$. The form of Eq. (5.80) becomes clear from the following expression

$$\begin{aligned} \underbrace{\log \Pr(y_0 ... y_{i+1}|\mathbf{x})}_{\log \Pr(z^k_{\text{next}}.tgt|z^k_{\text{next}}.src)} &= \underbrace{\log \Pr(y_0 ... y_i|\mathbf{x})}_{\log \Pr(z_{\text{cur}}.tgt|z_{\text{cur}}.src)} + \underbrace{\log \Pr(y_{i+1}|y_0 ... y_i, \mathbf{x})}_{\log \Pr(v_k|z_{\text{cur}}.tgt, z_{\text{cur}}.src)} \\ &= \sum_{k=1}^{i} \log \Pr(y_k|\mathbf{y}_{<k}, \mathbf{x}) + \log \Pr(y_{i+1}|y_0 ... y_i, \mathbf{x}) \\ &= \sum_{k=1}^{i+1} \log \Pr(y_k|\mathbf{y}_{<k}, \mathbf{x}) \end{aligned} \tag{5.81}$$

[7]We use $a \circ b$ to denote the concatenation of two strings a and b.

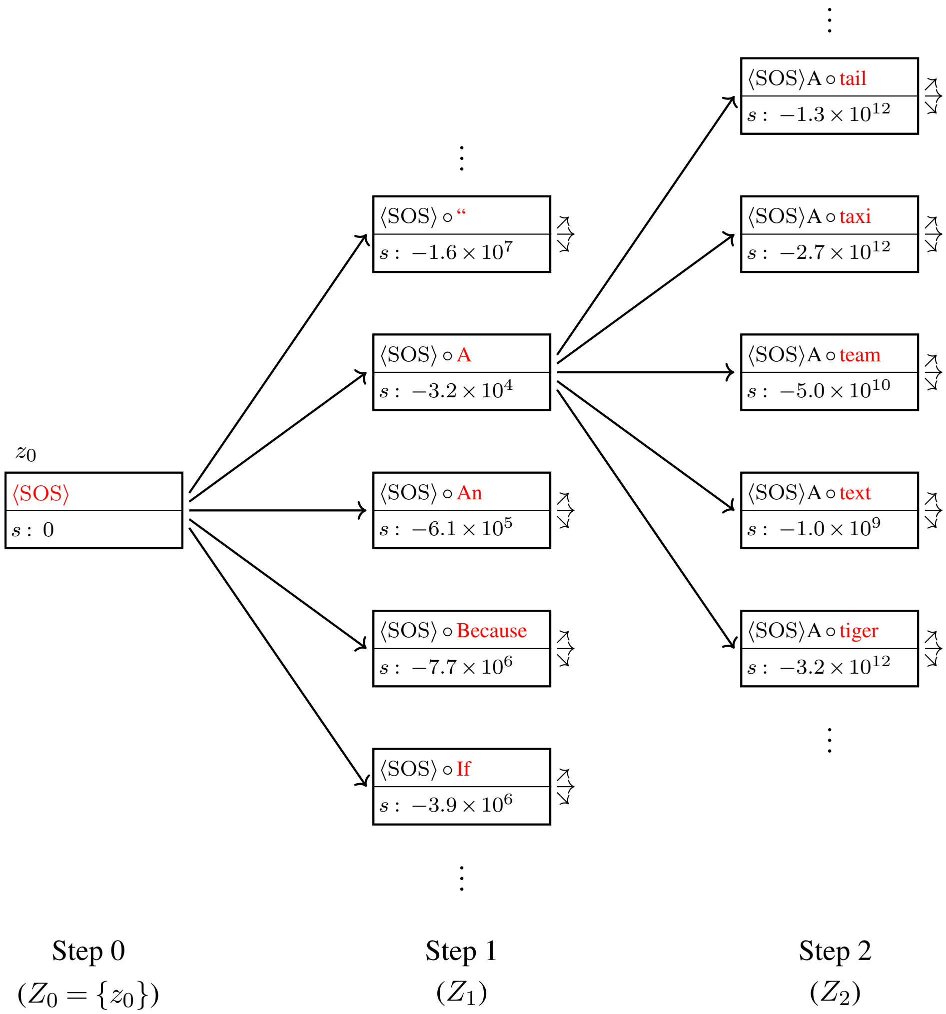


Figure 5.9: Illustration of hypothesis extension in the first 3 steps. Each (partial) hypothesis is represented as a box in which we show the corresponding target-side sequence and model score. Each search step is associated with a hypothesis set Z_i. We start with a hypothesis $z_0 \in Z_0$ denoting the start symbol ⟨SOS⟩. In step i, we extend every hypothesis in Z_{i-1} by trying to append every word from a vocabulary V_y (see words in red). This operation will result in $|V_y|\cdot|Z_{i-1}|$ hypotheses, forming the hypothesis set Z_i. The hypothesis extension procedure represents a breadth-first search algorithm: we create all the nodes (or search states) at depth $i-1$ before moving to depth i. A tree structure is created along with this procedure, and a leaf node of the tree can trace the search path back to the root node.

Given this probability, we can then compute $z.score = \mathrm{Score}(z.src, z.tgt)$, as in Section 5.4.1. This score allows us to rank and compare different hypotheses. If a hypothesis ends with the ⟨EOS⟩ symbol, it is considered complete and is no longer extended. Once complete, it is pushed onto a max-heap[8]. We can then extract the highest-scoring hypotheses from this heap. The search procedure typically terminates when a predefined number of complete hypotheses are found, for instance, when the heap reaches its capacity. The resulting search algorithm is presented below.

Algorithm: A Simple Breadth-first Search Algorithm

SimpleSearch($\mathbf{x}$)

// Search for the best hypothesis given the source-side sequence $\mathbf{x}$

1. Create a Heap with $size_{\text{heap}}$ elements
2. $Z_0 = \{z_0\}$ where $z_0.src = \mathbf{x}$ and $z_0.tgt = y_0$
3. For each step $i = 1$ to $n_{\max}$
4. For each hypothesis $z_{\text{cur}} \in Z_{i-1}$
5. For each word $v_k \in V_y$
6. $z_{\text{next}} = \text{Extend}(z_{\text{cur}}, v_k, \mathbf{x})$
7. If $z_{\text{next}}.tgt$ ends with $\langle\text{EOS}\rangle$, then
8. Add z_{next} to Heap
9. Else
10. Add z_{next} to Z_i
11. If Heap is full and/or other stopping criteria are met, then
12. Break all the loops
13. Return Heap.Pop()

Extend(z_{cur}, v_k, src)

// Create a new hypothesis by appending a new word v_k to the target-side of z_{cur}

1. Create a new hypothesis z_{next}
2. $z_{\text{next}}.src = src$
3. $z_{\text{next}}.tgt = z_{\text{cur}}.tgt \circ v_k$
4. $z_{\text{next}}.prob = z_{\text{cur}}.prob + \log \Pr(v_k | z_{\text{cur}}.tgt, z_{\text{cur}}.src)$ // see Eq. (5.80)
5. $z_{\text{next}}.score = \text{Score}(z_{\text{next}}.src, z_{\text{next}}.tgt)$ // see Section 5.4.1
6. Return z_{next}

If the hypothesis heap has infinite capacity ($size_{\text{heap}} = \infty$), this algorithm performs an exhaustive search over the space of all hypotheses with lengths up to $n_{\max}$, producing at most $1 + |V_y| + |V_y|^2 + \cdots + |V_y|^{n_{\max}} = \frac{|V_y|^{n_{\max}+1}-1}{|V_y|-1}$ complete hypotheses. This search space is extremely large and computationally intractable in practice[9]. As a result, practical systems heavily rely on pruning techniques to render the search process feasible. In the remainder of this section, we will introduce two popular search algorithms that utilize pruning for efficient decoding.

[8] Given a max-heap a, we use a.Pop() to denote a function that removes and returns the top item of a, and a.PopAll() for a function that extracts all items.

[9] Consider, for example, a vocabulary size of $20{,}000$ ($|V_y| = 20{,}000$) and a length limit of 20 ($n_{\max} = 20$). The total number of hypotheses $\frac{|V_y|^{n_{\max}+1}-1}{|V_y|-1}$ would be approximately 1.05×10^{86}.

1. Greedy Search

The **greedy strategy** is a fundamental concept in algorithm design. It is based on the heuristic that a globally optimal solution can be obtained by making locally optimal decisions. Although this strategy only guarantees local optimality, it is sufficient for many practical applications, and its low computational complexity is a clear advantage.

Applying the greedy strategy to our search problem is straightforward. At each extension step i, we only consider the single best hypothesis up to this step. To be more precise, for any set Z_i, we retain only the hypothesis with the highest model score and discard the rest. The best hypothesis at each step of the greedy search is given by

$$z_{\text{best}} = \underset{z_{\text{next}} \in \text{Extend}(Z_{i-1}, V_y)}{\arg\max} z_{\text{next}}.score \tag{5.82}$$

Here the function $\text{Extend}(Z_{i-1}, V_y)$ has the same meaning as that in Eq. (5.77), but operates on a set of hypotheses, that is,

$$\text{Extend}(Z_{i-1}, V_y) = \bigcup_{z \in Z_{i-1}} \text{Extend}(z, V_y) \tag{5.83}$$

Then, Z_i is defined to be

$$Z_i = \{z_{\text{best}}\} \tag{5.84}$$

A greedy search algorithm for sequence-to-sequence problems is given below.

Algorithm: A Greedy Search Algorithm
GreedySearch($\mathbf{x}$)
// Search for the "best" hypothesis in a greedy manner
1. Create a hypothesis z_{best}
2. $Z_0 = \{z_0\}$ where $z_0.src = \mathbf{x}$ and $z_0.tgt = y_0$
3. For each step $i = 1$ to $n_{\max}$
4. $\quad$ $z_{\text{best}}.score = -\infty$
5. $\quad$ For each hypothesis $z_{\text{cur}} \in Z_{i-1}$
6. $\quad\quad$ For each word $v_k \in V_y$
7. $\quad\quad\quad$ $z_{\text{next}} = \text{Extend}(z_{\text{cur}}, v_k, \mathbf{x})$
8. $\quad\quad\quad$ If $z_{\text{best}}.score < z_{\text{next}}.score$, then
9. $\quad\quad\quad\quad$ $z_{\text{best}} = z_{\text{next}}$
10. $\quad$ If $z_{\text{best}}.tgt$ ends with $\langle\text{EOS}\rangle$ and/or other stopping criteria are met, then
11. $\quad\quad$ Break the loop
12. $\quad$ $Z_i = \{z_{\text{best}}\}$
13. Return z_{best}

At each search step, we have only one active hypothesis to extend (i.e., $|Z_{i-1}| = 1$). Thus, we generate $|V_y|$ extensions, from which we select the best one for the subsequent step. The total number of times $\text{Extend}(z_{\text{cur}}, v_k)$ is called is $|V_y| \cdot n_{\max}$. Assuming $\text{Extend}(z_{\text{cur}}, v_k)$ operates in constant time, the overall time complexity of the algorithm scales linearly with respect to $|V_y|$ and $n_{\max}$.

2. Beam Search

Beam search is a natural extension of the above greedy search algorithm. It is based on the greedy heuristic as well, and is thus a type of greedy algorithm. The core idea of beam search is to retain a predetermined number of the most promising hypotheses at each step, rather than just the single best one. A beam is a data structure that stores the best hypotheses we have generated so far. The number of hypotheses in a beam is a predetermined parameter, called **beam width** or **beam size**. Here we can simply view Z_i as a beam, written as

$$Z_i = \{z_{\text{best}}^1, ..., z_{\text{best}}^{size_{\text{beam}}}\} \tag{5.85}$$

where $size_{\text{beam}}$ is the beam size. z_{best}^1 is the best hypothesis in the set $\text{Extend}(Z_{i-1}, V_y)$ (see Eq. (5.82)), z_{best}^2 is the 2nd best hypothesis in $\text{Extend}(Z_{i-1}, V_y)$, and so on.

The following pseudo-code describes a beam search algorithm for sequence-to-sequence problems.

Algorithm: A Beam Search Algorithm
$\text{BeamSearch}(\mathbf{x})$
// Search for the "best" hypothesis by considering a number of best candidates
// in each step
Create a Heap with $size_{\text{heap}}$ elements
$Z_0 = \{z_0\}$ where $z_0.src = \mathbf{x}$ and $z_0.tgt = y_0$
For each step $i = 1$ to $n_{\max}$
    Create a heap Beam with $size_{\text{beam}}$ elements
    For each hypothesis $z_{\text{cur}} \in Z_{i-1}$
        For each word $v_k \in V_y$
            $z_{\text{next}} = \text{Extend}(z_{\text{cur}}, v_k, \mathbf{x})$
            If $z_{\text{next}}.tgt$ ends with $\langle\text{EOS}\rangle$, then
                Add z_{next} to Heap
            Else
                $\text{UpdateBeam}(\text{Beam}, z_{\text{next}})$
            If Heap is full and/or other stopping criteria are met, then
                Break all the loops
    $Z_i = \text{Beam.PopAll}()$
Return Heap.Pop()
$\text{UpdateBeam}(\text{Beam}, z_{\text{next}})$
// Update Beam with a newly-generated hypothesis z_{next}

1. Add z_{next} to Beam [a]

[a] Beam is a bounded max-heap that maintains the top-$size_{\text{beam}}$ hypotheses. So, if $z_{\text{next}}.score$ is lower than the lowest score in the heap, it is discarded. In other words, Beam only stores top-$size_{\text{beam}}$ best hypotheses and ignores the rest.

The function $\text{UpdateBeam}(\text{Beam}, z_{\text{next}})$ can be viewed as an instance of **histogram pruning**. Note that this general-purpose framework provides a simple way to implement other pruning methods, and one can modify $\text{UpdateBeam}(\text{Beam}, z_{\text{next}})$ as needed. For example, an alternative method, called **threshold pruning**, retains the hypotheses whose model scores differ from the best hypothesis in Beam by less than a predefined threshold, e.g., we discard z_{next} in $\text{UpdateBeam}(\text{Beam}, z_{\text{next}})$ if

$$z_{\text{next}}.score \quad < \quad z_{\text{best}}.score - \theta_{\text{beam}} \tag{5.86}$$

where z_{best} is the best hypothesis in Beam. Alternatively, we can consider a relative threshold method [Freitag and Al-Onaizan, 2017], given by

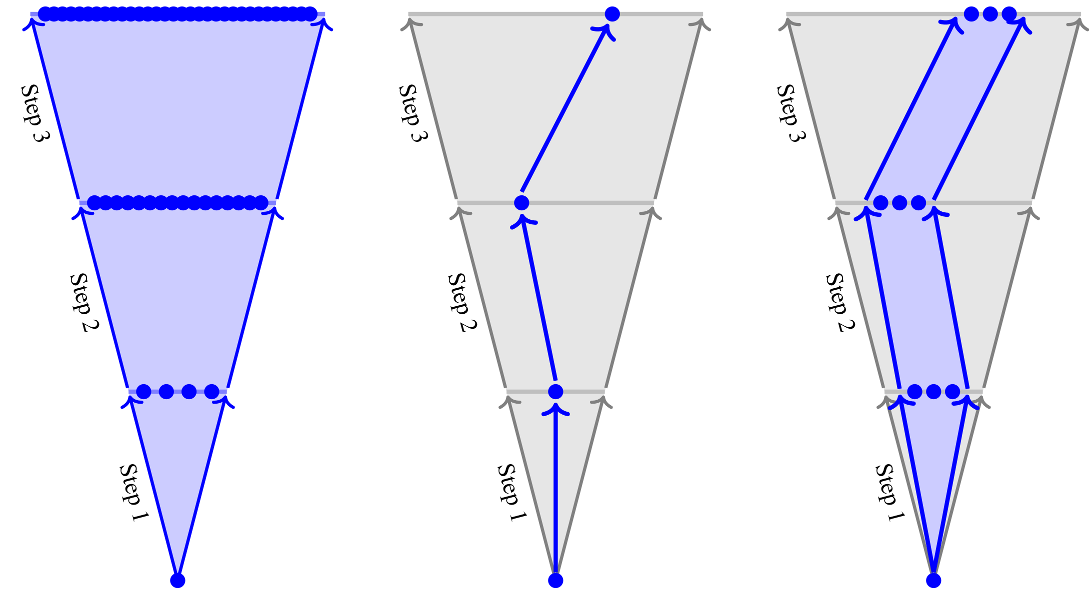
$$z_{\text{next}}.score \quad < \quad z_{\text{best}}.score \cdot \theta_{\text{beam}} \tag{5.87}$$

Figure 5.10 shows a comparison of exhaustive search, (1-best) greedy search and beam search. At one extreme, the optimal solution is guaranteed, but an exponentially large number of search states are visited. At the other extreme, only the minimum number of search states are visited, but the solution is sub-optimal. By contrast, beam search makes a trade-off between the two methods. A larger beam size means more search effort and a greater chance of finding better hypotheses, while a smaller beam size means faster search and a higher risk of missing the optimum. It is also possible to use a variable beam size to make a better trade-off during search [Buckman et al., 2016; Post and Vilar, 2018; Kulikov et al., 2019].

An important problem related to these search algorithms is the problem of **search errors**. In general, search errors can be defined in several different ways. Here we say that a search error occurs if the search result is not the same as that of exhaustive search. Intuitively, minimizing search errors should yield higher-quality results. Thus, one might expect that increasing the beam size would consistently improves the generated sequences. However, this assumption does not always hold true. For instance, in neural machine translation systems, decoding with an excessively large beam size can lead to degraded translation quality [Koehn and Knowles, 2017]. This counterintuitive phenomenon has inspired research into the performance deterioration associated with large beam sizes [Ott et al., 2018b; Yang et al., 2018b; Stahlberg and Byrne, 2019].

3. Stopping Criteria

Although the time complexities of the previously discussed algorithms are bounded by the maximum number of search steps (i.e., n_{max}), it is important to design more effective stopping criteria to terminate the search as early as possible, particularly for latency-sensitive applications. This typically requires heuristics to design additional criteria that halt the search



(a) Exhausted Search (b) (1-best) Greedy Search (c) Beam Search

Figure 5.10: A comparison of exhaustive search, (1-best) greedy search and beam search. Balls represent search states or partial hypotheses. Exhaustive search explores all search states in the search space. By contrast, greedy search keeps only the 1-best path of search states and prunes away the rest. Beam search is a trade-off between them and keeps the most promising search states in a beam in each step.

procedure at an optimal point. Common stopping criteria include:

- If a given number of complete hypotheses are created, then we stop searching. For example, in the beam search algorithm described in this subsection, the search program terminates when we have $size_{\text{heap}}$ complete hypotheses. Another way to implement this idea is to shrink the beam as the number of complete hypotheses increases. In Bahdanau et al. [2014]'s system, once a new complete hypothesis is created, the beam size decreases by 1. Therefore, the search program will terminate if the beam size is reduced to 0.
- If every hypothesis at step i has a score lower than that of the best complete hypothesis in Heap by some margin, then we stop searching. Suppose $z_{\text{bestinall}}$ is the best hypothesis we have generated so far (i.e., $z_{\text{bestinall}} = \mathrm{Heap.Pop}()$). If every hypothesis z_{next} at step i satisfies

$$z_{\text{bestinall}}.score - z_{\text{next}}.score \quad \geq \quad \theta_{\text{all}} \tag{5.88}$$

then the search process concludes. Here, θ_{all} is a hyper-parameter typically determined empirically. A simple choice is $\theta_{\text{all}} = 0$, which is employed in several popular sequence-to-sequence frameworks [Ott et al., 2019]. Under certain conditions, such an **early**

stopping strategy can even guarantee search optimality [Huang et al., 2017b; Yang et al., 2018b].

- If every hypothesis at step i has a score lower than that of the last complete hypothesis in Heap by some margin, then we stop searching. This is a weak condition for early stopping.
- If the top-ranked hypotheses at step i are all complete hypotheses, then we stop searching. This is a more aggressive version of early stopping. For example, in Klein et al. [2017]'s system, the search program terminates at step i if the top-1 hypothesis is a complete hypothesis.
- If the computational budget (e.g., a maximum number of floating-point operations or a strict wall-clock time limit) is exhausted, the search is terminated. In applications where computational resources are highly constrained and latency is a primary concern, this criterion becomes essential.

Occasionally, the search algorithm may fail to find any complete hypothesis before hitting the length limit $n_{\max}$. In practice, a common fallback strategy in such cases is to complete the highest-scoring partial hypothesis by appending the $\langle \mathrm{EOS} \rangle$ token to its end.

Note that selecting appropriate stopping criteria reflects a compromise between computational efficiency and predictive accuracy during inference (often referred to as the **speed-accuracy trade-off**). While allocating more computational time does not guarantee superior outputs for a sequence-to-sequence system, it is important to understand how closely our heuristic results approximate the optimal solution by exploring a broader region of the search space. Exact search algorithms that guarantee such optimality are discussed further in Section 5.4.4.

5.4.3 Online Search

So far in our general discussion of sequence-to-sequence problems, we have assumed that all the source-side words are observed as a complete sequence. However, in some practical applications, the inputs are received sequentially, and we wish to make predictions conditioned on partial observations. An example of this is online automatic speech recognition in which the system continually takes new acoustic signals and simultaneously outputs the corresponding transcription units.

Intuitively, we might think of the generation of the i-th target-side word as a problem of mapping a prefix of the source-side sequence to a target-side word. We can formulate this by introducing a function $g(i)$ which denotes the maximum length of the prefix of $\mathbf{x}$ we use in generating y_i. Thus, the probability of y_i given the entire sequence $\mathbf{x}$ and the previously generated words $\mathbf{y}_{<i}$ can be approximated by

$$\Pr(y_i|\mathbf{y}_{<i}, \mathbf{x}) \quad \approx \quad \Pr(y_i|\mathbf{y}_{<i}, \mathbf{x}_{\leq g(i)}) \tag{5.89}$$

where $\mathbf{x}_{\leq g(i)}$ denotes the sub-sequence $x_1...x_{g(i)}$. Then, the log probability of the target-side sequence $\mathbf{y}$ given the source-side sequence $\mathbf{x}$ is written as

$$\begin{aligned}\log \Pr(\mathbf{y}|\mathbf{x}) &= \sum_{i=1}^{n} \log \Pr(y_i|\mathbf{y}_{<i}, \mathbf{x}) \\ &\approx \sum_{i=1}^{n} \log \Pr(y_i|\mathbf{y}_{<i}, \mathbf{x}_{\le g(i)}) \end{aligned} \tag{5.90}$$

This equation frames a sequence-to-sequence problem as a prefix-to-prefix problem, that is, the prefix $\mathbf{y}_{\le i}$ depends only on the prefix $\mathbf{x}_{\le g(i)}$. Inference for this model is simple. For each i, the search system waits until the first $g(i)$ source-side words have been received, and then extends the hypotheses as usual. This can be done by reusing the algorithms described in the previous subsection. For example, we can modify the beam search algorithm and obtain the following online search algorithm.

Algorithm: An Online Beam Search Algorithm

OnlineBeamSearch$(\mathbf{x}, g(\cdot))$

// Online search executed once a sufficient number of input words are received.

// At each search step, a predefined number of promising candidates are considered.

Create a Heap with capacity $size_{\text{heap}}$

$Z_0 = \{z_0\}$ where $z_0.tgt = y_0$

$j = 1$ // source word index

$i = 1$ // target generation step

$input = \emptyset$

While $i \le n_{\max}$ do

If $j \le g(i)$, then // read a word from the input stream

$input = input \circ x_j$

j++

Else // make a prediction at step i

// when $g(i)$ input words are observed (stored in $input$)

Create a bounded heap Beam of size $size_{\text{beam}}$

For each hypothesis $z_{\text{cur}} \in Z_{i-1}$

For each word $v_k \in V_y$

$z_{\text{next}} = \text{Extend}(z_{\text{cur}}, v_k, input)$

If $input$ equals $\mathbf{x}$ and $z_{\text{next}}.tgt$ ends with $\langle\text{EOS}\rangle$, then

Add z_{next} to Heap

Else

UpdateBeam$(\text{Beam}, z_{\text{next}})$

If Heap is full and/or other stopping criteria are met, then

Break all the loops

OutputPartial(Beam)

$Z_i = \text{Beam.PopAll}()$

24. i++
25. Return Heap.Pop()

OutputPartial(Beam)

// Output a partial result

1. Display the best hypothesis in Beam

An advantage of this system is that the output at step i can be produced immediately once we have seen $\mathbf{x}_{\leq g(i)}$. This results in an **online sequence-to-sequence system** in which input words arrive in a continuous stream and predictions are generated as soon as a sufficient number of input words have been observed.

While the search problem here seems simple, much remains to be done to define $g(i)$. Clearly, $g(i)$ is a monotonically non-decreasing function. As a simple example, we can define $g(i) = m$ for any i. This reduces the above algorithm exactly to the standard beam search algorithm that works with a complete input sequence. By contrast, in online sequence-to-sequence tasks, we want $g(i)$ to be as small as possible, and so we can start computation as early as possible in inference. The simplest case is that the input and output sequences are synchronous in some way. For example, an automatic speech recognition system assigns each spectral frame a transcription unit. In this case, we have a simple correspondence between inputs and outputs: $m = n$ (i.e., $|\mathbf{x}| = |\mathbf{y}|$), and x_i corresponds to y_i. Then, we can simply define $g(i) = i$, in other words, each time a new input arrives, we make a prediction.

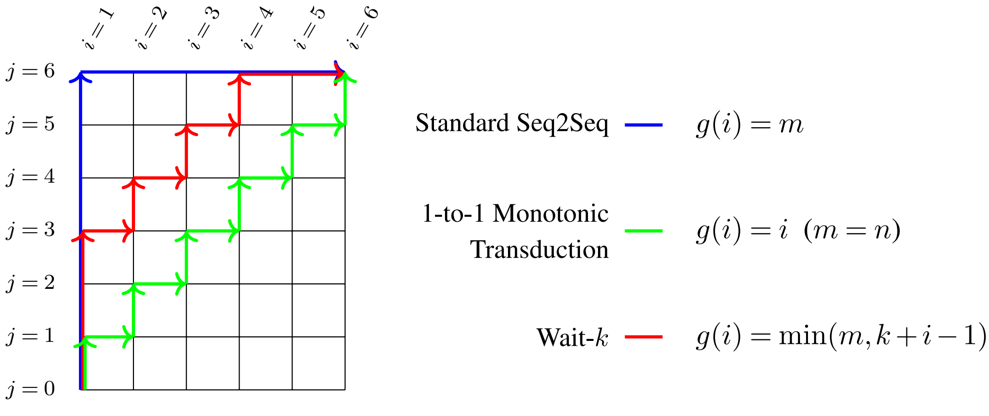
A more complicated case is online sequence-to-sequence problems with reordering, such as **simultaneous translation**, in which a target-side word may depend on source-side words with long-range dependencies. A simple way to address this is to delay the predictions for a number of steps. For example, the wait-k method forces each prediction to lag behind the inputs by k words [Ma et al., 2019]. More formally, the wait-k version of the function $g(i)$ is defined to be

$$g(i) \quad = \quad \min(m, k+i-1) \tag{5.91}$$

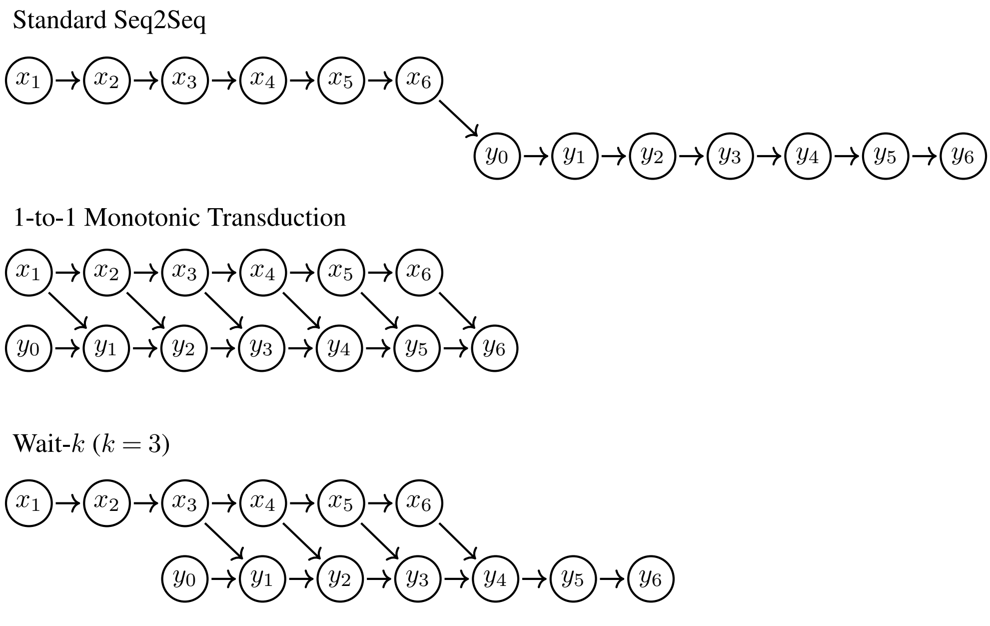
Here k is a hyper-parameter that controls how large a source-side context is considered in predicting target-side words. When $k = \infty$, it is the same as the standard search methods for sequence-to-sequence inference. In simultaneous translation and related tasks, results are in general satisfactory by using a small value of k. A comparison of different $g(i)$ is shown in Figure 5.11.

In some applications of online sequence-to-sequence problems, we may know when to perform search and when to read inputs. For example, in **interactive machine translation** [Casacuberta et al., 2009], the translation of a partial input sequence is triggered directly by user actions (e.g., key presses). In this case, we do not need to define $g(i)$, but view it as an input variable of the model.

Note that while one can directly employ pre-trained sequence-level models for online inference, developing such systems often requires additional training effort. A more principled approach to online sequence-to-sequence modeling is to model the transformation from $\mathbf{x}$ to $\mathbf{y}$ as a sequence of actions [Grissom II et al., 2014; Cho and Esipova, 2016; Gu et al., 2017;



(a) Visualization of $g(i)$.



(b) Action sequences.

Figure 5.11: Visualization (top) and action sequences (bottom) of different $g(i)$ for a pair of sequences $(\mathbf{x}=x_1...x_6, \mathbf{y}=y_1...y_6)$. In an action sequence, a circled x_j stands for the action of reading a source-side word (x_j), and a circled y_i stands for the action of predicting the probability of y_i given $\mathbf{x}_{\leq g(i)}$ and $\mathbf{y}_{<i}$. Arrows here stand for dependencies between words. Because y_0 denotes the start symbol $\langle$SOS$\rangle$, it could be generated without dependencies on any words.

Zheng et al., 2019]. For example, an action can be either a predict operation that performs search at the current step, or a read operation that accepts a new input word. Then, we can frame the task of designing the function $g(i)$ as learning a policy to determine which action is taken given a source-side prefix $\mathbf{x}_{\leq j}$ and a target-side prefix $\mathbf{y}_{<i}$. Thus, sequence-to-sequence models can be trained on the states of these action sequences so that they can make better predictions conditioned on part of the input. However, a discussion of training online sequence-to-sequence models lies outside the scope of this section. We refer the reader to the above papers for more details on these methods.

5.4.4 Exact Search

From a formal point of view, we would ideally like to develop a system with no search errors. Although approximate search algorithms have been used successfully in many applications, it is important to study model errors of these systems, and thus to focus on the problem in principle, not just in practice. So developing exact search algorithms for sequence-to-sequence models has long been an interesting topic in NLP research. However, the search problem for a simple word-based machine translation system with n-gram language models has been found to be an NP-hard problem [Knight, 1999]. Much of the earlier research formulated the search problem as classical **combinatorial optimization problems**, such as the linear programming problem and the traveling salesman problem, and employed the corresponding solvers [Germann et al., 2004; Zaslavskiy et al., 2009]. Additional research efforts explored exact search algorithms for statistical machine translation systems by using the Lagrangian relaxation technique [Chang and Collins, 2011; Rush and Collins, 2012] and finite-state automata [de Gispert et al., 2010; Allauzen et al., 2014].

Unlike these methods, which are more or less dependent on the integration of n-gram language models into sequence-to-sequence models, the models described in this chapter take a simpler form. We begin with a basic model in which the scoring function $\mathrm{Score}(\mathbf{x},\mathbf{y})$ is the log probability $\log\Pr(\mathbf{y}|\mathbf{x})$. Eq. (5.63) tells us that $\log\Pr(\mathbf{y}|\mathbf{x})$ can be written as a sum of word-level log probabilities, and $\log\Pr(\mathbf{y}|\mathbf{x})$ becomes smaller as more target-side words are generated (i.e., a larger n)[10]. In other words, $\log\Pr(\mathbf{y}|\mathbf{x})$ is a monotonically decreasing function with respect to the target-side length n: for any i, we have

$$\begin{aligned} \log\Pr(\mathbf{y}_{\leq i}|\mathbf{x}) &= \log\Pr(\mathbf{y}_{<i}|\mathbf{x}) + \log\Pr(y_i|\mathbf{y}_{<i},\mathbf{x}) \\ &\leq \log\Pr(\mathbf{y}_{<i}|\mathbf{x}) \end{aligned} \tag{5.92}$$

This is also called the **monotonicity** of the scoring function.

Then, by making use of the monotonic nature of model scores, we can develop a heuristic to rule out hypotheses that would never be the best. Let $z_{\mathrm{bestinall}}$ be the globally best complete hypothesis discovered so far. If a new partial hypothesis has a model score strictly lower than $z_{\mathrm{bestinall}}.score$, it cannot possibly be extended to surpass it, and thus we do not need to extend it. This allows us to explore a region that is significantly smaller than the original search space without any loss of optimality. Note that as the search proceeds, $z_{\mathrm{bestinall}}.score$ continually increases. It becomes increasingly difficult to find a better hypothesis, resulting in more aggressive pruning as the search continues. The pseudo-code below outlines an exact

[10] Consider $\log\Pr(\mathbf{y}|\mathbf{x}) = \sum_{i=1}^{n}\log\Pr(y_i|\mathbf{y}_{<i},\mathbf{x})$. Since $\log\Pr(y_i|\mathbf{y}_{<i},\mathbf{x})$ has a non-positive value, $\log\Pr(\mathbf{y}|\mathbf{x})$ will be smaller or unchanged if n grows.

search algorithm for the sequence-to-sequence model defined by Eq. (5.63).

Algorithm: An Exact Search Algorithm
ExactSearch($\mathbf{x}$)
// Search for the "best" hypothesis by making use of the monotonicity of the
// scoring function ($\mathrm{Score}(\mathbf{x}, \mathbf{y}) = \log \Pr(\mathbf{y}|\mathbf{x})$).
Create a priority queue (max-heap) Queue
Create a hypothesis z_{best} with $z_{\text{best}}.score = -\infty$
While Queue is not empty do
    $z_{\text{cur}} = \text{Queue.Pop()}$
    If $|z_{\text{cur}}.tgt| > n_{\max}$, then
        skip z_{cur} and continue the loop
    For each word $v_k \in V_y$
        $z_{\text{next}} = \text{Extend}(z_{\text{cur}}, v_k, \mathbf{x})$
        $bound = z_{\text{best}}.score$ // a lower bound on model scores
        If $bound < z_{\text{next}}.score$, then // admissible pruning
            If $z_{\text{next}}.tgt$ ends with $\langle \text{EOS} \rangle$, then
                $z_{\text{best}} = z_{\text{next}}$
                $bound = z_{\text{next}}.score$
            Else
                Add z_{next} to Queue
Return z_{best}

This is a general algorithm for exact search, and its search efficiency is greatly influenced by the design of the priority queue [Meister et al., 2020]. For example, we can view $\mathrm{Score}(\mathbf{x}, \mathbf{y})$ as the priority of each hypothesis in the priority queue, as in a max-heap[11]. Then, the resulting algorithm performs a procedure of breadth-first-like search, since a hypothesis with a shorter target-side sequence is more likely to have a higher model score and to be a top-ranked item in the priority queue. For efficient search, however, we wish to find complete hypotheses as early as possible, such that more unpromising hypotheses can be thrown away in the early stage of search. To do this, we can bias the priority of a hypothesis towards a longer target-side sequence. This provides a **depth-first search** algorithm which is more likely to find complete hypotheses in a shorter time [Stahlberg and Byrne, 2019].

While the exact search algorithm becomes apparent by considering the monotonicity of $\Pr(\mathbf{y}|\mathbf{x})$, in practical systems, as discussed in Section 5.4.1, $\mathrm{Score}(\mathbf{x}, \mathbf{y})$ often has a more complex form involving length reward or normalization, and so the monotonic property does not hold. Fortunately, the assumption of monotonicity can be dropped at the expense of slightly relaxing the lower bound on model scores for pruning. Here we define *bound* to be the lowest model score that a hypothesis should have so that it can at best be extended to a hypothesis as good as z_{best}. For example, consider a simple word reward model described in Eq. (5.64):

[11] We can implement a priority queue using a max-heap.

$\text{Score}(\mathbf{x},\mathbf{y}) = \log \Pr(\mathbf{y}|\mathbf{x}) + \lambda \cdot n$. For a hypothesis z_{next}, there are at most $n_{\max} - |z_{\text{next}}.tgt|$ words we can predict to obtain a complete hypothesis. Suppose all these $n_{\max} - |z_{\text{next}}.tgt|$ words are predicted with a probability of 1. Then, the model score of the resulting hypothesis (denoted by z_{new}) will be given by

$$\begin{aligned} z_{\text{new}}.score &= z_{\text{next}}.score + \sum_{i=|z_{\text{next}}.tgt|+1}^{n_{\max}} (\log 1 + \lambda) \\ &= z_{\text{next}}.score + \lambda \cdot (n_{\max} - |z_{\text{next}}.tgt|) \end{aligned} \tag{5.93}$$

Using this result, we can define *bound* as

$$bound = z_{\text{best}}.score - \lambda \cdot (n_{\max} - |z_{\text{next}}.tgt|) \tag{5.94}$$

An alternative way to derive the lower bound is to simply subtract the maximum possible word reward, given by

$$bound = z_{\text{best}}.score - \lambda \cdot n_{\max} \tag{5.95}$$

This is a loose lower bound and leads to less pruning.

In the case of length normalization, we can do this in a similar way. For example, consider the length normalization model $\text{Score}(\mathbf{x},\mathbf{y}) = \frac{\log \Pr(\mathbf{y}|\mathbf{x})}{n}$, as in Eqs. (5.67-5.68). A lower bound on admissible model scores is given by

$$bound = \frac{\Pr(z_{\text{next}}.tgt|\mathbf{x})}{n_{\max}} \tag{5.96}$$

In practice, such a lower bound can be defined in several different ways to guarantee the optimality of search, depending on which model and search strategy are used in the sequence-to-sequence systems [Huang et al., 2017b; Stahlberg and Byrne, 2019].

We can easily apply these lower bounds to the above exact search algorithm by replacing line 9 with Eq. (5.94) or (5.96). As a side effect, the search will explore more hypotheses and thus be much slower.

5.4.5 Differentiable Search

We have addressed the search problem through the introduction of heuristic search algorithms in which we try to maximize the scoring function on a set of sequences of discrete variables. An alternative is to relax these discrete variables to continuous variables and formulate the problem as a **continuous optimization** problem [Hoang et al., 2017; Kumar et al., 2021]. While we try to use a consistent notation throughout this book, it is convenient here to introduce notation that slightly differs from that used in previous chapters. We will use a vector $\mathbf{y}_i^w \in \{0,1\}^{|V_y|}$ to denote the one-hot representation of y_i. Suppose the output at step i is a distribution over V_y, denoted by $\Pr(\cdot|\mathbf{y}_{<i},\mathbf{x})$. Then, we can express the log probability of y_i as a dot product of

two vectors:

$$
\begin{aligned}
\log \Pr(y_i|\mathbf{y}_{<i},\mathbf{x}) &= \mathbf{y}_i^w \cdot \log \Pr(\cdot|\mathbf{y}_{<i},\mathbf{x}) \\
&= \mathbf{y}_i^w \cdot \log \Pr(\cdot|\mathbf{y}_0^w ... \mathbf{y}_{i-1}^w,\mathbf{x})
\end{aligned} \tag{5.97}
$$

where $\mathbf{y}_{<i} = y_0...y_{i-1}$ is represented as a sequence of one-hot vectors $\mathbf{y}_0^w...\mathbf{y}_{i-1}^w$. As discussed in Chapter 3, the right-hand side of the above equation corresponds to selecting the entry y_i of the vector $\log \Pr(\cdot|\mathbf{y}_{<i},\mathbf{x})$ (or $\log \Pr(\cdot|\mathbf{y}_0^w...\mathbf{y}_{i-1}^w,\mathbf{x})$).

Using this notation, we can write $\log \Pr(\mathbf{y}|\mathbf{x})$ as

$$
\begin{aligned}
\log \Pr(\mathbf{y}|\mathbf{x}) &= \sum_{i=1}^{n} \log \Pr(y_i|\mathbf{y}_{<i},\mathbf{x}) \\
&= \sum_{i=1}^{n} \mathbf{y}_i^w \cdot \log \Pr(\cdot|\mathbf{y}_0^w ... \mathbf{y}_{i-1}^w,\mathbf{x})
\end{aligned} \tag{5.98}
$$

Provided we use $\log \Pr(\mathbf{y}|\mathbf{x})$ as the objective function (i.e., $\mathrm{Score}(\mathbf{x},\mathbf{y}) = \log \Pr(\mathbf{y}|\mathbf{x})$), the search problem can be formulated as

$$
\hat{\mathbf{y}}_0^w...\hat{\mathbf{y}}_n^w = \underset{\mathbf{y}_1^w...\mathbf{y}_n^w}{\arg\max} \sum_{i=1}^{n} \mathbf{y}_i^w \cdot \log \Pr(\cdot|\mathbf{y}_0^w...\mathbf{y}_{i-1}^w,\mathbf{x}) \tag{5.99}
$$

This is equivalent to the standard form for inference of sequence-to-sequence models, given by

$$
\begin{aligned}
\hat{\mathbf{y}} &= \hat{y}_0...\hat{y}_n \\
&= \underset{y_0...y_n}{\arg\max} \Pr(y_0...y_n|\mathbf{x})
\end{aligned} \tag{5.100}
$$

Given Eq. (5.99), we can now relax each one-hot vector to a real-valued vector with a constraint that the sum of all its entries is equal to 1, that is,

$$
\mathbf{y}_i^w \in \mathbb{R}_+^{|V_y|} \tag{5.101}
$$

$$
\text{s.t. } ||\mathbf{y}_i^w||_1 = 1 \tag{5.102}
$$

In this way, $\mathbf{y}_i^w$ can be informally treated as a $|V_y|$-dimensional representation of y_i, although it possesses many more dimensions than typical dense embeddings used in NLP. Instead of corresponding to a specific word in the vocabulary, $\mathbf{y}_i^w$ now describes a continuous distribution over the vocabulary. As a result, the term $\mathbf{y}_i^w \cdot \log \Pr(\cdot|\mathbf{y}_0^w \dots \mathbf{y}_{i-1}^w,\mathbf{x})$ computes the **expected log-probability** of the prediction at step i. To make the conditional distribution $\Pr(\cdot|\mathbf{y}_0^w \dots \mathbf{y}_{i-1}^w,\mathbf{x})$ differentiable with respect to the continuous prefix, Hoang et al. [2017] replace the standard discrete word embedding lookup at each decoder step with an **expected embedding**, which is a weighted average of the word embedding matrix guided by the probability vector $\mathbf{y}_{i-1}^w$. Interestingly, this formulation implies that Eq. (5.99) defines a "new" task in which we attempt to maximize a continuous objective function (i.e., a sum of n expected log-probabilities).

We can solve Eq. (5.99) using off-the-shelf optimization toolkits. Because $\mathbf{y}_i^w$ is constrained to a **simplex**[12], it is straightforward to apply general **constrained optimization** algorithms. Alternatively, one can use algorithms that are designed to solve the optimization problem with simplex constraints.

A third approach to solving Eq. (5.99) is to explicitly incorporate the constraints into the objective function and to use gradient descent methods to optimize this function. For example, Hoang et al. [2017] modify Eq. (5.99) and obtain a new form for optimization

$$\hat{\mathbf{y}}_0^w ... \hat{\mathbf{y}}_n^w \quad = \quad \underset{\mathbf{y}_1^w ... \mathbf{y}_n^w}{\arg\max} \sum_{i=1}^{n} \mathrm{Softmax}(\mathbf{y}_i^w) \cdot \log \Pr(\cdot | \mathbf{y}_0^w ... \mathbf{y}_{i-1}^w, \mathbf{x}) \tag{5.106}$$

Here we remove the simplex constraint on $\mathbf{y}_i^w$, and instead impose it via the Softmax output.

Once we have obtained the optimal sequence $\hat{\mathbf{y}}_0^w ... \hat{\mathbf{y}}_n^w$, we need to map each $\mathbf{y}_i^w$ to a word. A straightforward method is to select the word corresponding to the entry of $\mathbf{y}_i^w$ with the largest value. However, this may break the optimality of the solution because the condition $\mathbf{y}_0^w ... \mathbf{y}_{i-1}^w$ is changed when these variables are discretized. A more practical method is to perform optimization to predict the next word given a prefix, e.g., we fix $\mathbf{y}_0^w ... \mathbf{y}_{i-1}^w$ to the one-hot representations of the optimal prefix, and maximize $\sum_{k=i}^{n} \mathbf{y}_k^w \cdot \log \Pr(\cdot | \mathbf{y}_0^w ... \mathbf{y}_{k-1}^w, \mathbf{x})$. Then, we select the best word at position i and move on to the next position.

So far, we have assumed that the search objective is derived from the log probability $\log \Pr(\mathbf{y}|\mathbf{x})$ and that the output length is predetermined. To search over sequences of varying lengths, we can repeat the above optimization procedure for every $n \in [1, n_{\max}]$ and select the sequence with the highest score. This approach also facilitates the integration of length normalization and rewards into the search. Specifically, we can ignore length bias during the fixed-n optimization and apply length models post-hoc. That is, we leave the initial search objective unchanged and, in the final step, select the best candidate across all lengths based on a length-normalized $\mathrm{Score}(\mathbf{x}, \mathbf{y})$.

5.4.6 Hypothesis Diversity

Generating multiple outputs is often necessary when one wants to rescore these outputs and/or interact with the system. One of the most widely used methods is to employ beam search to generate top-ranked hypotheses. For example, we can simply view the elements of Heap

[12] A simplex is a concept from geometry. In a Euclidean space, a k-simplex is a k-dimensional polytope described by a set of $k+1$ independent points $\{\mathbf{p}_0, \mathbf{p}_1, ..., \mathbf{p}_k\}$. This polytope is defined as the set of points

$$P_{k\text{-simplex}} \quad = \quad \{a_0 \cdot \mathbf{p}_0 + a_1 \cdot \mathbf{p}_1 + ... + a_k \cdot \mathbf{p}_k\} \tag{5.103}$$

where

$$\sum_{i=0}^{k} a_i \quad = \quad 1 \tag{5.104}$$

$$a_i \quad \geq \quad 0 \quad \text{for all } i \in [0, k] \tag{5.105}$$

Rank	Output
1	Manuela Arbelaez accidentally revealed the correct answer to a guessing game for a new Hyundai Sonata. Host Drew Carey couldn't stop laughing. It's been a busy week for "The Price Is Right" when Bob Barker, 91, showed up to run his old show.
2	Manuela Arbelaez accidentally revealed the correct answer to a guessing game for a new Hyundai Sonata. Host Drew Carey couldn't stop laughing. It's been a busy week for "The Price Is Right" when Bob Barker showed up to run his old show.
3	Manuela Arbelaez accidentally revealed the correct answer to a guessing game for a new Hyundai Sonata. Host Drew Carey couldn't stop laughing. It's been a busy week for "The Price Is Right" when Bob Barker, 91, showed up to run the show.
4	Manuela Arbelaez accidentally revealed the correct answer to a guessing game for a new Hyundai Sonata. Host Drew Carey couldn't stop laughing. It's been a busy week for "The Price Is Right" when Bob Barker, 91, showed up to run his show.

Table 5.3: 4-best outputs of a text summarization system on a sample in the CNN/Daily Mail dataset (beam size $= 4$). We see that these texts differ only by a few words.

as the k-best hypotheses in beam search (see Section 5.4.2). However, this approach often produces highly similar hypotheses, and it is difficult to figure out which one is better even when more options are available. Table 5.3 shows the 4-best outputs of a text summarization system. We see that these texts are fairly similar to each other. One reason for this phenomenon is that diverse hypotheses, even those that might eventually achieve high model scores upon completion, will be pruned if they fall out of the beam at any intermediate stage of the search. From a modeling perspective, we can interpret this as an inherent limitation of the locally normalized models used here: every prediction is made at an intermediate step of the search, and there is no mechanism to correct past decisions once a prefix is committed [Murray and Chiang, 2018].

One approach to improving hypothesis diversity is to introduce penalties when the hypotheses in the beam lack variation [Li and Jurafsky, 2016; Vijayakumar et al., 2018]. A simple objective function is given by

$$\mathrm{Score}_d(\mathbf{x}, \mathbf{y}) = \mathrm{Score}(\mathbf{x}, \mathbf{y}) - \lambda \cdot dp \tag{5.107}$$

It combines the original model score $\mathrm{Score}(\mathbf{x}, \mathbf{y})$ and a diversity penalty dp. dp can be defined in several ways. One approach is to penalize hypotheses that are close in the search tree. For example, one can define dp as the rank of a hypothesis in the set of its siblings that are extended from the same parent hypothesis, and so the beam can spread its members over a larger region of the hypothesis space [Li and Jurafsky, 2016]. Another way to introduce diversity measures is to consider the differences among the target-side sequences of the hypotheses in the beam. For example, we can define dp as the average string similarity between a given hypothesis and other hypotheses in the beam [Xiao et al., 2013].

The above idea can also be expressed as constraints imposed on the search procedure. For example, we can constrain the beam to include only the hypotheses that are rooted at different parents in the last step [Boulanger-Lewandowski et al., 2013]. More precisely, for each hypothesis $z_{\text{cur}} \in Z_{i-1}$, we select the best next-step hypothesis as follows

$$\hat{z}_{\text{next}} = \underset{z_{\text{next}} \in \text{Extend}(z_{\text{cur}}, V_y)}{\arg\max} \Pr(z_{\text{next}}.tgt|\mathbf{x}) \tag{5.108}$$

The hypothesis $\hat{z}_{\text{next}}$ is then added to Z_i. Note that this is essentially a **subspace method** that divides the hypothesis space into subspaces, and collects results over these subspaces. An intuition behind this method is that different subspaces can describe different aspects of the problem, and so we can have diverse solutions.

Another approach to addressing the diversity issue is to perturb beam search by introducing randomly generated hypotheses into the beam [Holtzman et al., 2020a; Wiher et al., 2022]. One common way to do this is to choose some random words for extending a hypothesis, and to add the extended hypotheses to the beam. In general, these words can be sampled from the distribution $\Pr(\cdot|\mathbf{y}_{<i}, \mathbf{x})$ over the entire vocabulary or its subset. Randomness can also be added to the system inputs at test time. For example, an input word can be represented as a linear combination of its original embedding and the embedding of a word sampled from a randomly selected training sequence [Li et al., 2021b]. In problems having many local minima, adding random noise to the search process is generally helpful, as we can explore more diverse hypotheses and prevent the system from getting stuck in certain regions of the search space.

Instead of performing search using a single system, we can use multiple systems to obtain diverse hypotheses. These systems can be built on either different architectures or different hidden structures/configurations [He et al., 2018; Shen et al., 2019; Wu et al., 2020a; Sun et al., 2020a]. Although such methods do not fall under the search framework that we have been discussing, combining the results from multiple systems is generally helpful. The following section will present a discussion on this issue.

5.4.7 Combining Multiple Models

In machine learning, **ensembling** refers to methods aimed at addressing modeling limitations rather than search issues. In this subsection, we discuss these methods because their implementations typically require modifications to the search modules, and we can gain some insight into the resulting system by viewing it from the search perspective.

Ensemble methods aim to improve predictive performance by combining the outputs of multiple **constituent systems** (or **component systems**). The problem of combining multiple models has been extensively discussed since the emergence of statistical models in NLP, where such techniques are often referred to as **system combination methods** to emphasize their practical utility. For the sequence-to-sequence models discussed here, a widely adopted form of system combination is prediction averaging [Sutskever et al., 2014]. Suppose we have an ensemble of K trained sequence-to-sequence models. The log-probability of generating the target-side word y_i, given its left context $\mathbf{y}_{<i}$ and the source-side sequence $\mathbf{x}$, can be defined

using the geometric average:

$$\log \Pr(y_i|\mathbf{y}_{<i}, \mathbf{x}) = \frac{1}{K}\sum_{k=1}^{K} \log \Pr_k(y_i|\mathbf{y}_{<i}, \mathbf{x}) \quad (5.109)$$

or alternatively by using the arithmetic average

$$\log \Pr(y_i|\mathbf{y}_{<i}, \mathbf{x}) = \log \frac{1}{K}\sum_{k=1}^{K} \Pr_k(y_i|\mathbf{y}_{<i}, \mathbf{x}) \quad (5.110)$$

where $\Pr_k(y_i|\mathbf{y}_{<i}, \mathbf{x})$ is the output of the k-th component system. These forms are so simple that one can implement them for any sequence-to-sequence models without significant modifications to existing systems, and they have been used as the basis of many successful systems in various evaluation tasks [Barrault et al., 2020; Akhbardeh et al., 2021].

A limitation of prediction averaging is that all component systems must adhere to the same basic autoregressive formulation (see Eq. (5.63)), and we need direct access to their step-wise output probabilities $\{\Pr_k(y_i|\mathbf{y}_{<i}, \mathbf{x})\}$. When dealing with black-box systems, we must instead perform sequence-level ensembling. A common strategy is to aggregate the complete sequences produced by the component systems. The most straightforward approach is **hypothesis selection** [Hildebrand and Vogel, 2008], where we simply select the single "best" sequence from the generated pool based on a predefined criterion. An alternative is to regenerate a novel sequence that may differ from any individual output in the pool [Matusov et al., 2006; Rosti et al., 2007]. This typically requires compressing the generated sequences into a compact unified structure (such as a word lattice), followed by an additional search pass (e.g., lattice decoding and rescoring) to extract the optimal consensus sequence [Deoras et al., 2011; Stahlberg et al., 2016; Khayrallah et al., 2017].

Note that the ensembling of sequence-to-sequence models is related to the diversity issue discussed in the previous subsection. It is widely recognized that component systems must exhibit sufficient diversity to yield meaningful ensemble gains. Thus, models should be trained or constructed to capture different behavioral patterns [Sutskever et al., 2014; Zhou et al., 2017]. One of the most popular methods is **checkpoint ensembling**. It uses multiple snapshots of a single model at different checkpoints during the training process and combines their predictions via averaging. This method can be useful for alleviating the overfitting problem in practice. Also, different models can be created from a base model under different settings. For example, we can build models with different numbers of parameters based on a common backbone model. A more general approach is to take models based on different architectures, although this is at the expense of more development effort.

Another way to view sequence ensembling is that it provides a two-pass search scheme. In the first pass, multiple independent systems perform inference. Because each system possesses its own inductive biases regarding modeling and decoding, they naturally explore different regions of the search space. A high-quality hypothesis discovered by one system might be entirely missed by another. The outcome of this first pass is a diverse pool of hypotheses, each considered "optimal" from a specific perspective. In the second pass, this diverse pool defines a constrained, high-quality search space, over which a fine-grained rescoring model can navigate to determine the final, globally optimal result.

5.4.8 More Search Objectives

In this subsection, we consider more objective functions that can be applied to the search problem.

1. Search with Future Scores

Most of the algorithms described previously can be viewed as variants of best-first search algorithms [Meister et al., 2020]. A* search is another example of a best-first search algorithm and is widely considered to be a good solution to the general search problem. Vanilla A* search requires that all search states are sorted at every search step, which is computationally intractable in many NLP problems. We therefore continue to focus on beam search and greedy search, but employ an A*-like objective function instead. Specifically, given a search state $(\mathbf{x}, \mathbf{y}_{\leq i})$, the A*-like objective function can be defined as:

$$\mathrm{Score}_{\mathrm{A}*}(\mathbf{x}, \mathbf{y}_{\leq i}) = g(\mathbf{x}, \mathbf{y}_{\leq i}) + h(\mathbf{x}, \mathbf{y}_{\leq i}) \tag{5.111}$$

Here $g(\mathbf{x}, \mathbf{y}_{\leq i})$ is the cumulative reward from the start state to $(\mathbf{x}, \mathbf{y}_{\leq i})$, and $h(\mathbf{x}, \mathbf{y}_{\leq i})$ is the estimated reward of the "optimal" path from $(\mathbf{x}, \mathbf{y}_{\leq i})$ to the final goal. Because $g(\mathbf{x}, \mathbf{y}_{\leq i})$ and $h(\mathbf{x}, \mathbf{y}_{\leq i})$ can take arbitrary forms, this framework is very general. For example, if we define

$$g(\mathbf{x}, \mathbf{y}_{\leq i}) = \mathrm{Score}(\mathbf{x}, \mathbf{y}_{\leq i}) \tag{5.112}$$

$$h(\mathbf{x}, \mathbf{y}_{\leq i}) = 0 \tag{5.113}$$

then $\mathrm{Score}_{\mathrm{A}*}(\mathbf{x}, \mathbf{y})$ reduces to the objective functions discussed previously.

To make full use of this formulation, it is natural to seek a function that estimates future rewards or costs. Ideally, we would want $h(\mathbf{x}, \mathbf{y}_{\leq i})$ to exactly compute how much additional reward can be obtained by extending $(\mathbf{x}, \mathbf{y}_{\leq i})$ to the optimal complete hypothesis. This, however, is intractable because it requires exploring all possible hypotheses extended from $(\mathbf{x}, \mathbf{y}_{\leq i})$ to identify the best one. In practice, it is common to use a computationally cheaper model that approximates the true future reward. Conventional approaches rely on heuristics to define $h(\mathbf{x}, \mathbf{y}_{\leq i})$ [Koehn et al., 2007], such as estimating the contributions of words that may be generated. These heuristics can be adapted based on the architectural design of sequence-to-sequence models [He et al., 2017; Zheng et al., 2018]. A more principled approach is to use a value-based formulation [Ren et al., 2017; Li et al., 2017a; Leblond et al., 2021]. One can develop a policy that predicts the distribution of y_i given $\mathbf{x}$ and $\mathbf{y}_{<i}$, alongside a **value function** that estimates future rewards. Eq. (**??**) can therefore be interpreted as a linear combination of the policy score for $(\mathbf{x}, \mathbf{y}_{\leq i})$ and its corresponding value. Such a treatment of search objectives falls into the framework of **value-based search** and has been successfully

employed in reinforcement learning [Silver et al., 2017].

2. Search with Language Models

For a long time, language models have played a central role in text generation tasks. For example, statistical machine translation and automatic speech recognition systems typically rely on large n-gram language models to produce fluent text. While modern sequence-to-sequence models do not require separate language models, integrating them during sequence-to-sequence search remains intuitively appealing for machine translation and related problems.

Following the convention that a language model can be treated as a feature of a log-linear (or linear) model [Och and Ney, 2002], the language model-augmented objective can be defined as

$$\text{Score}_{\text{lm}}(\mathbf{x},\mathbf{y}) = \log \Pr(\mathbf{y}|\mathbf{x}) + \lambda \cdot \log \Pr(\mathbf{y}) \tag{5.114}$$

This formulation does not involve length reward and normalization terms, but both can be easily incorporated as additional features. In general, the language model $\Pr(\mathbf{y})$ is trained solely on target-side sequences, enabling the use of large-scale monolingual data in sequence-to-sequence models [Gulcehre et al., 2017]. Interestingly, it has been found that current sequence-to-sequence models are themselves strong language models if they are sufficiently trained, and a more effective strategy for leveraging target-side data is to create synthetic corpora, a technique known as **data augmentation**. An example is **back-translation**, where a target-to-source system translates monolingual target sentences to generate synthetic source sentences. This synthetic bilingual data is then used to augment the training set [Sennrich et al., 2016a; Edunov et al., 2018]. In many tasks, such a simple method can achieve significant improvements in translation quality, but these results raise questions about the necessity of using additional language models in neural machine translation.

Note that the model of Eq. (5.114) depends on the choice of the coefficient λ. For machine translation, we are usually interested in a positive value of λ so that our system can produce more fluent outputs. By contrast, a negative value of λ means that we want less frequent outputs. For example, if $\lambda = -1$, then Eq. (5.114) can be written as the point-wise mutual information of $\mathbf{x}$ and $\mathbf{y}$

$$\begin{aligned} \text{Score}_{\text{lm}}(\mathbf{x},\mathbf{y}) &= \log \Pr(\mathbf{y}|\mathbf{x}) - \log \Pr(\mathbf{y}) \\ &= \log \frac{\Pr(\mathbf{x},\mathbf{y})}{\Pr(\mathbf{x}) \cdot \Pr(\mathbf{y})} \end{aligned} \tag{5.115}$$

This scoring function has been shown to be useful for generating more diverse outputs for neural conversation systems [Li et al., 2016].

3. Minimum Bayes Risk Search

So far, our discussion of search objectives has focused on selecting the highest-scoring hypothesis, a process known as **maximum a posteriori** (**MAP**) search[13]. An underlying assumption of this method is that the posterior probability $\Pr(\mathbf{y}|\mathbf{x})$ (or the model score $\mathrm{Score}(\mathbf{x},\mathbf{y})$) correlates with the quality of the outputs. In practice, this assumption offers several useful properties. For instance, the search system is straightforward to implement, and the search objective remains consistent with the training objective. However, MAP search suffers from several shortcomings, which have prompted researchers to explore more powerful alternatives. One issue with MAP search is that its objective does not align with the actual evaluation metrics of the system. The metrics used in the end-to-end evaluation of a system may take forms that differ significantly from $\Pr(\mathbf{y}|\mathbf{x})$. A second problem is that MAP, as a specific point estimate in Bayesian decision theory, provides no measure of uncertainty and is often overconfident. In certain applications, sequence-to-sequence models spread probability mass too thinly across many different hypotheses [Ott et al., 2018a], meaning the MAP estimate may fail to capture the bulk of the true distribution.

Here, we consider **minimum Bayes risk** (**MBR**) search, which provides a principled way to incorporate evaluation metrics into the decoding process and leverage the full distribution over hypotheses. The MBR method defines a risk function over a pair of sequences, denoted by $R(\mathbf{y},\mathbf{y}_r)$. It computes the cost of predicting $\mathbf{y}$ when the true reference is $\mathbf{y}_r$, evaluated according to a specific metric. For example, in machine translation, we could define the risk score as $1-\mathrm{BLEU}$. The total risk for a candidate $\mathbf{y}$ over a set Ω is then defined as the

[13]In statistics, MAP is a method for inferring the parameters of a statistical model. Suppose we have a model that describes the distribution of a variable x, parameterized by θ. MAP seeks the optimal value of θ by maximizing the probability of θ given x, written as:

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \Pr(\theta|x) \tag{5.116}$$

$\hat{\theta}_{\mathrm{MAP}}$ is also called the **mode** of the posterior distribution of θ. For the MAP search problem here, we simply denote θ by $\mathbf{y}$ and seek the mode of $\Pr(\mathbf{y}|\mathbf{x})$.

As a Bayesian method, we can re-express the above equation using Bayes' rule

$$\begin{aligned}\hat{\theta}_{\mathrm{MAP}} &= \arg\max_{\theta} \frac{\Pr(x|\theta)\cdot\Pr(\theta)}{\Pr(x)} \\ &= \arg\max_{\theta} \Pr(x|\theta)\cdot\Pr(\theta) \end{aligned} \tag{5.117}$$

where θ is treated as a random variable with a prior distribution $\Pr(\theta)$.

By contrast, MLE directly maximizes the likelihood function $\Pr(x|\theta)$

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \Pr(x|\theta) \tag{5.118}$$

Thus, the MAP result can be viewed as an estimate of θ that considers both the likelihood of x given θ and the prior distribution of θ. Note that MAP and MLE are equivalent if $\Pr(\theta)$ is a uniform distribution.

expectation of $R(\mathbf{y}, \mathbf{y}_r)$ with respect to the posterior distribution $\Pr(\mathbf{y}_r|\mathbf{x})$:

$$\begin{aligned} \text{Risk}(\mathbf{y}) &= \mathbb{E}_{\mathbf{y}_r \sim \Pr(\mathbf{y}_r|\mathbf{x})} R(\mathbf{y}, \mathbf{y}_r) \\ &= \sum_{\mathbf{y}_r \in \Omega} R(\mathbf{y}, \mathbf{y}_r) \cdot \Pr(\mathbf{y}_r|\mathbf{x}) \end{aligned} \tag{5.119}$$

However, the summation over all possible target-side sequences is computationally infeasible. We therefore restrict Ω to the k-best outputs or sampled outputs of a system [Eikema and Aziz, 2020], denoted by Ω_{system}. Then, we take $\text{Score}(\mathbf{x}, \mathbf{y}) = -\text{Risk}(\mathbf{y})$ and obtain the following objective for MBR search

$$\begin{aligned} \hat{\mathbf{y}} &= \arg\max_{\mathbf{y}} -\text{Risk}(\mathbf{y}) \\ &= \arg\min_{\mathbf{y}} \sum_{\mathbf{y}_r \in \Omega_{\text{system}}} R(\mathbf{y}, \mathbf{y}_r) \cdot \Pr(\mathbf{y}_r|\mathbf{x}) \end{aligned} \tag{5.120}$$

This model is very general and applies to a wide range of NLP problems in which one needs to search for an optimal hypothesis in a large set of candidates [Goodman, 1996b; Goel and Byrne, 2000; Kumar and Byrne, 2004b]. It allows for flexible forms of risk functions, for instance, incorporating various factors in evaluating hypotheses. MBR search has recently been of interest to NLP researchers because it has been found to be effective in eliminating the biases caused by MAP search [Müller and Sennrich, 2021; Freitag et al., 2022]. In addition to providing a formulation of search objectives, MBR methods can be used for training sequence-to-sequence models, and are considered a promising solution to the discrepancy between training and evaluation objectives [Shen et al., 2016].

5.5 Summary

In this chapter, we provided an overview of sequence-to-sequence modeling, which serves as a foundation for many modern NLP systems. Sequence-to-sequence modeling is a very rich area of research, and has been widely discussed in different disciplines, even beyond NLP. This chapter is not a review of all the literature on this subject, but focuses on some of the core methods and ideas. We started with an introduction to sequence-to-sequence problems, as well as the encoder-decoder architecture which lays the foundations for most of the state-of-the-art sequence-to-sequence systems. As an illustration of the application of this architecture, we considered the problem of neural machine translation, and built a simple neural machine translation model using the basic knowledge we have learned so far.

We also presented the attention mechanism and a series of refinements. If we look back to the past few years, we find that exploring attention models is the next natural step in developing sequence-to-sequence models. Although attention models were initially popularized by their success in machine translation, they have since come to dominate the NLP community. There is also great interest in attention models in some other sub-fields of AI, such as computer vision [Borji and Itti, 2012; Xu et al., 2015; Jaderberg et al., 2015] and speech processing [Chorowski et al., 2015; Chan et al., 2016; Bahdanau et al., 2016]. As a result, the past few years have been an exciting time for people in these areas.

Sequence-to-sequence models are so successful that there is a growing trend toward unifying diverse problems under a single framework. Not only have we developed powerful sequence-to-sequence models to deal with very general problems, but current research trends increasingly favor unified approaches. An example is the Transformer, a self-attention-based sequence-to-sequence model that has become a fundamental backbone for tasks spanning various data modalities from text to vision and speech. It can even be extended to deal with multimodal problems which are sometimes more challenging. This makes things more interesting and exciting: an improvement to one model can be used to improve systems in a variety of tasks. And we are seeing a significant change in our research paradigm in which the NLP and machine learning fields are converging, and results in NLP research are becoming more influential. However, on the other side of the coin is that we are allocating more attention to some problems while neglecting others. In recent NLP conferences, there are many papers discussing how to train big sequence-to-sequence models and apply them to different text generation tasks, but there are a relatively small number of papers on parsing. There have always been debates on this over the past few decades, for example, what and how much prior knowledge do we need to build an NLP system? [Church, 2011; See, 2018] A detailed discussion of these debates is beyond the scope of this chapter. Fortunately, NLP research promises to continue to be diverse and active, providing valuable insights from both perspectives. For example, there are interesting findings that the neural sequence models can learn some linguistic properties from data, and linguistic structures can help system design. In Chapter 6, we will see a few examples.

The "bias" of research focus also exists on the machine learning side of problem-solving. For example, for sequence-to-sequence problems discussed here, recent years have witnessed a drastic increase of interest in model design and training methods, but only a relatively small group of people discuss the search problem. While search is a classical problem in AI and plays an important role in practical systems [Russell and Norvig, 2010], it is not even discussed in recent tutorials and surveys in NLP. This motivates us to write a section on this subject so that we can have a more complete picture of the problem. However, our general discussion does not cover all aspects of the search problem. One important topic we left out is efficiency [Birch et al., 2018; Heafield et al., 2021]. While this chapter includes some discussions on the efficiency issue, such as stopping criteria of search algorithms, efficient methods are a wide-ranging topic and are generally dependent on model architectures. A more detailed discussion of them can be found in Chapter 6. Another topic that one may be interested in is **constrained search** in which constraints are imposed on the search process [Hokamp and Liu, 2017; Anderson et al., 2017]. In general, these constraints come from our prior knowledge or interactions with users. For example, constrained search has been used to enforce term translation constraints on machine translation [Hasler et al., 2018; Post and Vilar, 2018].

One last note on limitations of this chapter. The formulation of the general sequence-to-sequence problem described here is based on the left-to-right factorization of $\Pr(\mathbf{y}|\mathbf{x})$, resulting in an autoregressive model. One limitation of this formulation is that each prediction at some step depends only on the preceding words, and so the model cannot access the right context. To make use of the right context of a word, a simple approach is to build another model that performs right-to-left generation. The left-to-right and right-to-left models can then be combined to generate a better output sequence [Liu et al., 2016a; Hoang et al., 2017; Zhang et al., 2018b; 2020a]. An alternative approach is given by **non-autoregressive generation** or **non-autoregressive decoding** in which the constraint of autoregressive generation is removed and each word prediction is conditioned on the global context [Gu et al., 2018; Ghazvininejad et al., 2019; Lee et al., 2020]. A key advantage of non-autoregressive generation is the possibility of system speed-up, since all the words in a sequence can be generated in parallel and we can do this efficiently using GPUs.

Chapter 6

Transformers

So far we have discussed several basic models for solving sequence-to-sequence problems. We now explore a new class of models based on a powerful architecture, called **Transformers**. Transformers differ in several ways from the models presented in Chapters 4 and 5. First, they do not depend on recurrent or convolutional neural networks for modeling sequences of words, but use only attention mechanisms and feed-forward neural networks. Second, the use of self-attention in Transformers makes it easier to deal with global contexts and dependencies among words. Third, Transformers are very flexible architectures and can be easily modified to accommodate different tasks. The past decade has seen the rise of Transformers not only in NLP but also in several other fields. As Transformers and their variants continue to mature, these models are playing an increasingly important role in the research and application of artificial intelligence.

In this chapter, we will discuss the core ideas of Transformers. We will begin our discussion by looking at the standard Transformer architecture. Then we will look at some notable developments, such as improvements to the basic architecture and efficient methods. We will also present several applications in which Transformer models have been extensively used. However, the discussion of Transformers is a wide-ranging topic, and there have been many related papers. This chapter is not intended to provide a comprehensive survey of the literature, but rather to present a collection of selected topics that may be of interest to the NLP community.

6.1 The Basic Model

Here we consider the model presented in Vaswani et al. [2017]'s work. We start by considering the Transformer architecture and then discuss its components in detail.

6.1.1 The Transformer Architecture

Figure 6.1 shows the standard Transformer model which follows the standard encoder-decoder framework. A Transformer encoder comprises a number of stacked **encoding layers** (or **encoding blocks**). Each encoding layer has two different sub-layers (or sub-blocks), called the self-attention sub-layer and the feed-forward neural network (FFN) sub-layer. Suppose we have a source-side sequence $\mathbf{x} = x_1...x_m$ and a target-side sequence $\mathbf{y} = y_1...y_n$. The input of an encoding layer is a sequence of m vectors $\mathbf{h}_1...\mathbf{h}_m$, each having d_{model} dimensions (or d dimensions for simplicity). We follow the notation adopted in the previous chapters, using $\mathbf{H} \in \mathbb{R}^{m \times d}$ to denote these input vectors[1]. The self-attention sub-layer first performs a self-attention operation $\text{Att}_{\text{self}}(\cdot)$ on $\mathbf{H}$ to generate an output $\mathbf{C}$:

$$\mathbf{C} = \text{Att}_{\text{self}}(\mathbf{H}) \tag{6.1}$$

Here $\mathbf{C}$ is of the same size as $\mathbf{H}$, and can thus be viewed as a new representation of the inputs. Then, a residual connection and a layer normalization unit are added to the output so that the resulting model is easier to optimize.

The original Transformer model employs the **post-norm** structure where a residual connection is created before layer normalization is performed, as follows:

$$\mathbf{H}_{\text{self}} = \text{LNorm}(\mathbf{C} + \mathbf{H}) \tag{6.2}$$

where the addition of $\mathbf{H}$ denotes the residual connection, and $\text{LNorm}(\cdot)$ denotes the layer normalization function. Substituting Eq. (6.1) into Eq. (6.2), we obtain the form of the self-attention sub-layer

$$\begin{aligned} \text{Layer}_{\text{self}}(\mathbf{H}) &= \mathbf{H}_{\text{self}} \\ &= \text{LNorm}(\text{Att}_{\text{self}}(\mathbf{H}) + \mathbf{H}) \end{aligned} \tag{6.3}$$

The definitions of $\text{LNorm}(\cdot)$ and $\text{Att}_{\text{self}}(\cdot)$ have been given in Chapters 2 and 5, and we will also discuss them later in the section.

The FFN sub-layer takes $\mathbf{H}_{\text{self}}$ and outputs a new representation $\mathbf{H}_{\text{ffn}} \in \mathbb{R}^{m \times d}$. It has the same form as the self-attention sub-layer, with the attention function replaced by the FFN function, given by

$$\begin{aligned} \text{Layer}_{\text{ffn}}(\mathbf{H}_{\text{self}}) &= \mathbf{H}_{\text{ffn}} \\ &= \text{LNorm}(\text{FFN}(\mathbf{H}_{\text{self}}) + \mathbf{H}_{\text{self}}) \end{aligned} \tag{6.4}$$

Here $\text{FFN}(\cdot)$ can be any feed-forward neural network with non-linear activation functions. The most common structure of $\text{FFN}(\cdot)$ is a two-layer network involving two linear transformations and a ReLU activation function between them.

For deep models, we can stack the above neural networks. Let $\mathbf{H}^l$ be the output of layer l. Then, we can express $\mathbf{H}^l$ as a function of $\mathbf{H}^{l-1}$. We write this as a composition of two

[1]Provided $\mathbf{h}_j \in \mathbb{R}^d$ is a row vector, we have $\mathbf{H} = \begin{bmatrix} \mathbf{h}_1 \\ \vdots \\ \mathbf{h}_m \end{bmatrix}$.

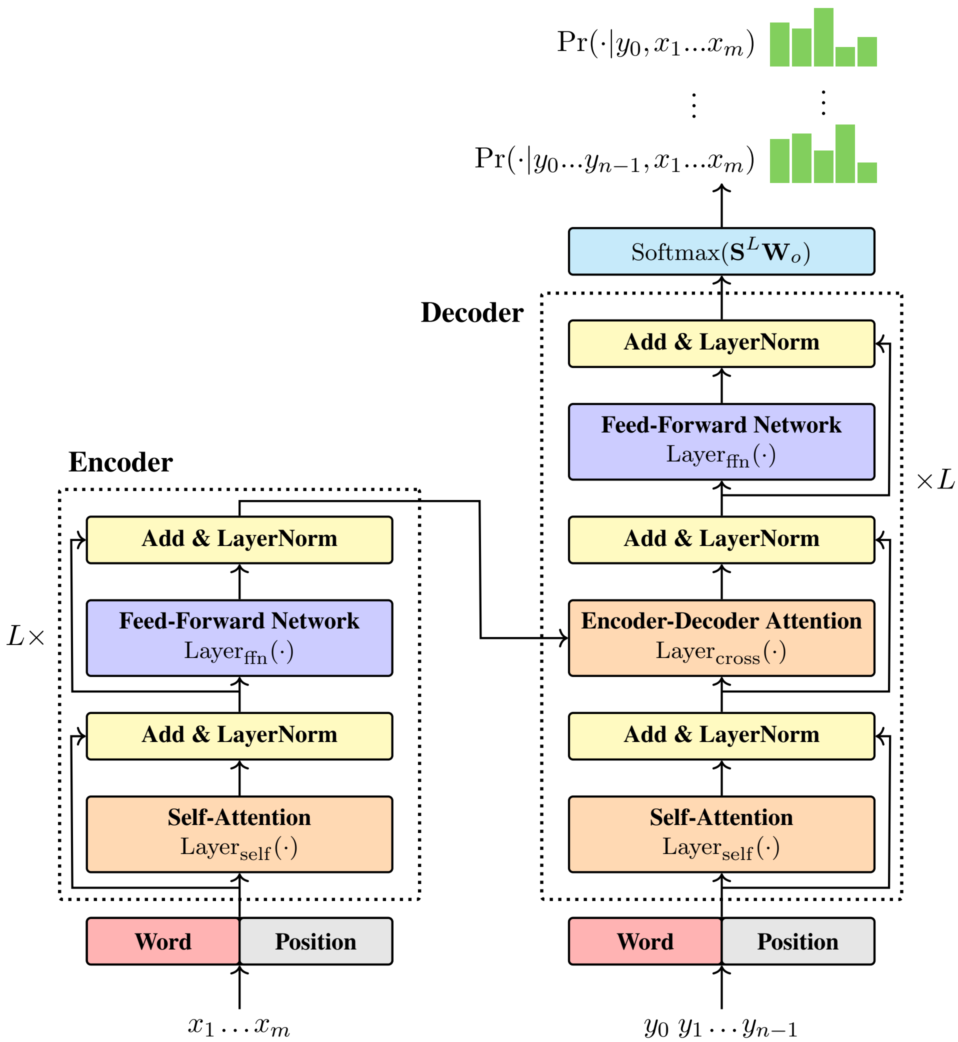


Figure 6.1: The Transformer architecture [Vaswani et al., 2017]. There are L stacked layers on each of the encoder and decoder sides. An encoding layer comprises a self-attention sub-layer and an FFN sub-layer. Both of these sub-layers share the same structure which involves a core function (either $\text{Layer}_{\text{self}}(\cdot)$ or $\text{Layer}_{\text{ffn}}(\cdot)$), followed by a residual connection and a layer normalization unit. Each decoding layer has a similar architecture to the encoding layers, but with an additional encoder-decoder attention sub-layer sandwiched between the self-attention and FFN sub-layers. As with most sequence-to-sequence models, Transformer takes $x_1...x_m$ and $y_0...y_{i-1}$ for predicting y_i. The representation of an input word comprises a sum of a word embedding and a positional embedding. The distributions $\{\Pr(\cdot|y_0...y_{i-1}, x_1...x_m)\}$ are generated in sequence by a Softmax layer, which operates on a linear transformation of the output from the last decoding layer.

sub-layers

$$\mathbf{H}^l = \text{Layer}_{\text{ffn}}(\mathbf{H}^l_{\text{self}}) \tag{6.5}$$

$$\mathbf{H}^l_{\text{self}} = \text{Layer}_{\text{self}}(\mathbf{H}^{l-1}) \tag{6.6}$$

If there are L encoding layers, then $\mathbf{H}^L$ serves as the final output of the encoder. In this case, $\mathbf{H}^L$ can be viewed as a learned contextual representation of the input sequence. $\mathbf{H}^0$ denotes the initial input to the encoder. In recurrent and convolutional models, $\mathbf{H}^0$ typically consists simply of the word embeddings of the input sequence. The Transformer takes a different approach to representing input words and explicitly encodes positional information. In Section 6.1.2, we will discuss the embedding model used in the Transformer.

The Transformer decoder has a similar structure to the Transformer encoder, comprising L stacked **decoding layers** (or **decoding blocks**). Let $\mathbf{S}^l$ be the output of the l-th decoding layer. We can formulate a decoding layer using the following equations:

$$\mathbf{S}^l = \mathrm{Layer}_{\mathrm{ffn}}(\mathbf{S}^l_{\mathrm{cross}}) \tag{6.7}$$

$$\mathbf{S}^l_{\mathrm{cross}} = \mathrm{Layer}_{\mathrm{cross}}(\mathbf{H}^L, \mathbf{S}^{l-1}_{\mathrm{self}}) \tag{6.8}$$

$$\mathbf{S}^l_{\mathrm{self}} = \mathrm{Layer}_{\mathrm{self}}(\mathbf{S}^{l-1}) \tag{6.9}$$

Here there are three decoder sub-layers. The self-attention and FFN sub-layers are the same as those used in the encoder. $\mathrm{Layer}_{\mathrm{cross}}(\cdot)$ denotes a cross attention sub-layer (or encoder-decoder sub-layer) which models the transformation from the source-side to the target-side. In Section 6.1.6 we will see that $\mathrm{Layer}_{\mathrm{cross}}(\cdot)$ can be implemented using the same function as $\mathrm{Layer}_{\mathrm{self}}(\cdot)$.

The Transformer decoder outputs a distribution over a vocabulary V_{y} at each target-side position. This is achieved by using a softmax layer that normalizes a linear transformation of $\mathbf{S}^L$ to distributions of target-side words. To do this, we map $\mathbf{S}^L$ to an $n \times |V_{\mathrm{y}}|$ matrix $\mathbf{O}$ by

$$\mathbf{O} = \mathbf{S}^L \cdot \mathbf{W}_{\mathrm{o}} \tag{6.10}$$

where $\mathbf{W}_{\mathrm{o}} \in \mathbb{R}^{d \times |V_{\mathrm{y}}|}$ is the parameter matrix of the linear transformation.

Then, the output of the Transformer decoder is given in the form

$$\begin{aligned} \begin{bmatrix} \Pr(\cdot|y_0, \mathbf{x}) \\ \vdots \\ \Pr(\cdot|y_0...y_{n-1}, \mathbf{x}) \end{bmatrix} &= \mathrm{Softmax}(\mathbf{O}) \\ &= \begin{bmatrix} \mathrm{Softmax}(\mathbf{o}_1) \\ \vdots \\ \mathrm{Softmax}(\mathbf{o}_n) \end{bmatrix} \end{aligned} \tag{6.11}$$

where $\mathbf{o}_i$ denotes the i-th row vector of $\mathbf{O}$, and y_0 denotes the start symbol $\langle\mathrm{SOS}\rangle$. Under this model, the probability of $\mathbf{y}$ given $\mathbf{x}$ can be defined as usual,

$$\log \Pr(\mathbf{y}|\mathbf{x}) = \sum_{i=1}^{n} \log \Pr(y_i|y_0...y_{i-1}, \mathbf{x}) \tag{6.12}$$

This equation resembles the general form of language modeling: we predict the word at position i given all the words up to position $i-1$. Therefore, the target output sequence is shifted one step forward in time relative to the input sequence, that is, the decoder takes $y_0...y_{n-1}$ as input to predict the output sequence $y_1...y_n$.

The Transformer architecture discussed above has several variants which have been successfully used in different fields of NLP. For example, we can use a Transformer encoder to represent text (referred to as the **encoder-only architecture**), a Transformer decoder to generate text (referred to as the **decoder-only architecture**), or a standard encoder-decoder Transformer model to transform an input sequence to an output sequence. In the rest of this chapter, much of the discussion is independent of the particular choice of application, and will be primarily focused on the encoder-decoder architecture. In Section 6.5, we will see applications of the encoder-only and decoder-only architectures.

6.1.2 Positional Encoding

In their original form, both FFNs and attention models used in Transformers ignore an important aspect of sequence modeling, namely that the order of the words plays a crucial role in expressing the meaning of a sequence. This means that the encoder and decoder are insensitive to the positional information of the input words. A simple approach to overcoming this problem is to add positional encodings to the representation of each word in the sequence. More formally, a word x_j can be represented as a d-dimensional vector

$$\mathbf{e}_j = \mathbf{x}_j + \mathrm{PE}(j) \tag{6.13}$$

Here $\mathbf{x}_j \in \mathbb{R}^d$ is the word embedding, which can be obtained using standard word embedding models, as described in Chapter 3. $\mathrm{PE}(j) \in \mathbb{R}^d$ is the representation of the position j. The vanilla Transformer employs a sinusoidal positional encoding model, which we write in the form

$$\mathrm{PE}(i,2k) = \sin(i \cdot \frac{1}{10000^{2k/d}}) \tag{6.14}$$

$$\mathrm{PE}(i,2k+1) = \cos(i \cdot \frac{1}{10000^{2k/d}}) \tag{6.15}$$

where $\mathrm{PE}(i,k)$ denotes the k-th entry of $\mathrm{PE}(i)$. The idea of positional encoding is to distinguish different positions using continuous functions. Here we use the sine and cosine functions with different frequencies. The interested reader can refer to Chapter 4 to see that such a method can be interpreted as a positional numeral system. Because the encoding is based on individual positions, it is also called **absolute positional encoding**. In Section 6.3.1, we will see an improvement to this method.

Once we obtain these embeddings, the sequence $\mathbf{e}_1...\mathbf{e}_m$ is used as input to the Transformer

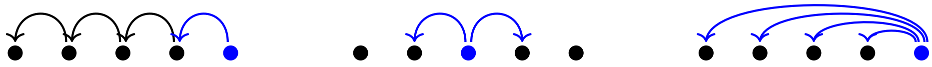


Figure 6.2: Information flows in recurrent, convolutional, and self-attention models, shown as arrow lines between positions.

encoder, that is,

$$\mathbf{H}_0 = \begin{bmatrix} \mathbf{e}_1 \\ \vdots \\ \mathbf{e}_m \end{bmatrix} \tag{6.16}$$

Similarly, we can define the input on the decoder side.

6.1.3 Multi-head Self-attention

The use of self-attention is perhaps one of the most significant advances in sequence-to-sequence models. It attempts to learn and make use of direct interactions between all pairs of inputs. From a representation learning perspective, self-attention models assume that the learned representation at position i (denoted by $\mathbf{c}_i$) is a weighted sum of the inputs over the sequence. The output $\mathbf{c}_i$ is thus given by

$$\mathbf{c}_i = \sum_{j=1}^{m} \alpha_{i,j} \mathbf{h}_j \tag{6.17}$$

where $\alpha_{i,j}$ represents the attention weight assigned to $\mathbf{h}_j$ when computing the representation at position i. We can thus view $\mathbf{c}_i$ as a representation of the global context at position i. $\alpha_{i,j}$ can be defined in different ways depending on the specific attention model considered. Here we use the scaled dot-product attention function to compute $\alpha_{i,j}$, as follows

$$\begin{aligned} \alpha_{i,j} &= \mathrm{Softmax}\left(\mathbf{h}_i \mathbf{h}_j^{\top} / \beta\right) \\ &= \frac{\exp(\mathbf{h}_i \mathbf{h}_j^{\top} / \beta)}{\sum_{k=1}^{m} \exp(\mathbf{h}_i \mathbf{h}_k^{\top} / \beta)} \end{aligned} \tag{6.18}$$

where the scaling factor β is typically set to $\sqrt{d}$.

Compared with conventional recurrent and convolutional models, an advantage of self-attention models is that they shorten the computational "distance" between two inputs. Figure 6.2 illustrates the information flow in these models. We can see that, given the input at position i, self-attention models can directly access any other input. By contrast, recurrent and convolutional models might need two or more jumps to see the entire sequence.

We can adopt a more general perspective on self-attention by using the QKV attention model. Suppose we have a sequence of κ queries $\mathbf{Q} = \begin{bmatrix} \mathbf{q}_1 \\ \vdots \\ \mathbf{q}_\kappa \end{bmatrix}$, and a sequence of ψ key-value pairs ($\mathbf{K} = \begin{bmatrix} \mathbf{k}_1 \\ \vdots \\ \mathbf{k}_\psi \end{bmatrix}, \mathbf{V} = \begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_\psi \end{bmatrix}$). The output of the model is a sequence of vectors, each corresponding to a query. The form of the QKV attention is given by

$$\mathrm{Att}_{\mathrm{qkv}}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Softmax}(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}})\mathbf{V} \tag{6.19}$$

We can write the output of the QKV attention model as a sequence of row vectors

$$\begin{aligned} \mathbf{C} &= \begin{bmatrix} \mathbf{c}_1 \\ \vdots \\ \mathbf{c}_\kappa \end{bmatrix} \\ &= \mathrm{Att}_{\mathrm{qkv}}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \end{aligned} \tag{6.20}$$

To apply this equation to self-attention, we simply have

$$\mathbf{H}^q = \mathbf{H}\mathbf{W}^q \tag{6.21}$$

$$\mathbf{H}^k = \mathbf{H}\mathbf{W}^k \tag{6.22}$$

$$\mathbf{H}^v = \mathbf{H}\mathbf{W}^v \tag{6.23}$$

where $\mathbf{W}^q, \mathbf{W}^k, \mathbf{W}^v \in \mathbb{R}^{d \times d}$ represent linear transformations of $\mathbf{H}$.

By considering Eq. (6.1), we then obtain

$$\begin{aligned} \mathbf{C} &= \mathrm{Att}_{\mathrm{self}}(\mathbf{H}) \\ &= \mathrm{Att}_{\mathrm{qkv}}(\mathbf{H}^q, \mathbf{H}^k, \mathbf{H}^v) \\ &= \mathrm{Softmax}(\frac{\mathbf{H}^q[\mathbf{H}^k]^\top}{\sqrt{d}})\mathbf{H}^v \end{aligned} \tag{6.24}$$

Here $\mathrm{Softmax}(\frac{\mathbf{H}^q[\mathbf{H}^k]^\top}{\sqrt{d}})$ is an $m \times m$ matrix in which each row represents a distribution over all input positions, that is

$$\text{row } i = \begin{bmatrix} \alpha_{i,1} & ... & \alpha_{i,m} \end{bmatrix} \tag{6.25}$$

We can improve the above self-attention model by using a technique called **multi-head attention**. This method can be motivated from the perspective of learning from multiple lower-dimensional feature subspaces. It projects the input into multiple subspaces and learns separate representations in each. Specifically, we project the entire input space into τ subspaces (referred to as **heads**). For example, we transform $\mathbf{H} \in \mathbb{R}^{m \times d}$ into τ matrices of size $m \times \frac{d}{\tau}$, denoted by $\{\mathbf{H}_1^{\text{head}}, \dots, \mathbf{H}_\tau^{\text{head}}\}$. The attention model is then applied τ times, once for each head. Finally, the outputs of these model runs are concatenated and transformed by a linear projection. This procedure can be expressed by

$$\mathbf{C} = \text{Merge}(\mathbf{C}_1^{\text{head}}, ..., \mathbf{C}_\tau^{\text{head}})\mathbf{W}_c \tag{6.26}$$

$$\tag{6.27}$$

For each head h,

$$\mathbf{C}_h^{\text{head}} = \text{Softmax}\left(\frac{\mathbf{H}_h^q[\mathbf{H}_h^k]^\top}{\sqrt{d}}\right)\mathbf{H}_h^v \tag{6.28}$$

$$\mathbf{H}_h^q = \mathbf{H}\mathbf{W}_h^q \tag{6.29}$$

$$\mathbf{H}_h^k = \mathbf{H}\mathbf{W}_h^k \tag{6.30}$$

$$\mathbf{H}_h^v = \mathbf{H}\mathbf{W}_h^v \tag{6.31}$$

Here $\text{Merge}(\cdot)$ is the concatenation function, and $\text{Att}_{\text{qkv}}(\cdot)$ is the attention function described in Eq. (6.20). $\mathbf{W}_h^q, \mathbf{W}_h^k, \mathbf{W}_h^v \in \mathbb{R}^{d \times \frac{d}{\tau}}$ are the parameters of the projections from a d-dimensional space to a $\frac{d}{\tau}$-dimensional space for the queries, keys, and values. Thus, $\mathbf{H}_h^q$, $\mathbf{H}_h^k$, $\mathbf{H}_h^v$, and $\mathbf{C}_h^{\text{head}}$ are all $m \times \frac{d}{\tau}$ matrices. $\text{Merge}(\mathbf{C}_1^{\text{head}}, ..., \mathbf{C}_\tau^{\text{head}})$ produces an $m \times d$ matrix. It is then transformed by a linear mapping $\mathbf{W}_c \in \mathbb{R}^{d \times d}$, leading to the final result $\mathbf{C} \in \mathbb{R}^{m \times d}$.

Although the notation may seem tedious, implementing multi-head models is straightforward using modern deep learning toolkits. A common method in Transformer-based systems is to store inputs from all the heads in data structures called tensors, enabling us to leverage parallel computing resources for greater efficiency. A more general discussion of the QKV attention and multi-head attention models can be found in Chapter 5.

6.1.4 Layer Normalization

Layer normalization provides a simple and effective means to make the training of neural networks more stable by standardizing the activations of the hidden layers in a layer-wise manner. As introduced by Ba et al. [2016], given a layer's output $\mathbf{h} \in \mathbb{R}^d$, the layer normalization method computes a standardized output $\text{LNorm}(\mathbf{h}) \in \mathbb{R}^d$ by

$$\text{LNorm}(\mathbf{h}) = \boldsymbol{\alpha} \odot \frac{\mathbf{h} - \mu}{\sigma + \epsilon} + \boldsymbol{\beta} \tag{6.32}$$

Here $\mu \in \mathbb{R}$ and $\sigma \in \mathbb{R}$ are the scalar mean and standard deviation of the activations, respectively. Let h_k be the k-th dimension of $\mathbf{h}$. The scalars μ and σ are defined as

$$\mu = \frac{1}{d}\sum_{k=1}^{d} h_k \tag{6.33}$$

$$\sigma = \sqrt{\frac{1}{d}\sum_{k=1}^{d}(h_k - \mu)^2} \tag{6.34}$$

Here $\boldsymbol{\alpha} \in \mathbb{R}^d$ and $\boldsymbol{\beta} \in \mathbb{R}^d$ are the scale and shift parameters. They can be treated as parameters of layer normalization, whose values are learned jointly with other parameters of the Transformer model. The addition of ϵ to σ is used for numerical stability. In general, ϵ is chosen to be a very small number.

We illustrate the layer normalization method for the hidden states of an encoder in the following example (assume that $m = 4$, $d = 3$, $\boldsymbol{\alpha} = \mathbf{1}$, $\boldsymbol{\beta} = \mathbf{0}$, and $\epsilon = 0.1$).

$$\begin{array}{c} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \\ \mathbf{h}_4 \end{array} \begin{bmatrix} 1 & 1 & 2 \\ 0.9 & 0.9 & 0 \\ 0.7 & 0.8 & 0 \\ 3 & 1 & 7 \end{bmatrix} \quad \begin{array}{l} \mu = 1.3,\ \sigma = 0.5 \\ \mu = 0.6,\ \sigma = 0.4 \\ \mu = 0.5,\ \sigma = 0.4 \\ \mu = 3.7,\ \sigma = 2.5 \end{array} \quad \Longrightarrow \quad \begin{bmatrix} \frac{1-1.3}{0.5+0.1} & \frac{1-1.3}{0.5+0.1} & \frac{2-1.3}{0.5+0.1} \\ \frac{0.9-0.6}{0.4+0.1} & \frac{0.9-0.6}{0.4+0.1} & \frac{0-0.6}{0.4+0.1} \\ \frac{0.7-0.5}{0.4+0.1} & \frac{0.8-0.5}{0.4+0.1} & \frac{0-0.5}{0.4+0.1} \\ \frac{3-3.7}{2.5+0.1} & \frac{1-3.7}{2.5+0.1} & \frac{7-3.7}{2.5+0.1} \end{bmatrix}$$

As discussed in Section 6.1.1, the layer normalization unit in each sub-layer is used to standardize the output of a residual block. Here we describe a more general formulation for this structure. Suppose that $F(\cdot)$ is a neural network sub-layer. Then, the post-norm structure of $F(\cdot)$ is given by:

$$\mathbf{H}_{\text{out}} = \text{LNorm}(F(\mathbf{H}_{\text{in}}) + \mathbf{H}_{\text{in}}) \tag{6.35}$$

where $\mathbf{H}_{\text{in}}$ and $\mathbf{H}_{\text{out}}$ are the input and output of this sub-layer. Clearly, Eq. (6.3) is an instance of this equation.

An alternative approach to incorporating layer normalization and residual connections into the modeling is to execute the $\text{LNorm}(\cdot)$ function right before the $F(\cdot)$ function, and to establish an identity mapping from the input to the output of the entire sub-layer. This structure, known as the **pre-norm** structure, can be expressed in the form

$$\mathbf{H}_{\text{out}} = F(\text{LNorm}(\mathbf{H}_{\text{in}})) + \mathbf{H}_{\text{in}} \tag{6.36}$$

Both post-norm and pre-norm Transformer models are widely used in NLP systems. See Figure 6.3 for a comparison of these two structures. In general, residual connections are considered an effective means to make the training of multi-layer neural networks easier. In this sense, the pre-norm Transformer seems promising because it follows the convention that a residual connection is created to bypass the whole network and that the identity mapping from the input to the output leads to easier optimization of deep models. However, when considering the expressive power of a model, there may be modeling advantages to using the post-norm Transformer, because it does not rely as heavily on residual connections and enforces more sophisticated modeling for representation learning. In Section 6.3.2, we will present a further discussion on this issue.

6.1.5 Feed-forward Neural Networks

The use of FFNs in Transformers is inspired in part by the fact that complex outputs can be formed by transforming the inputs through non-linearities. While the self-attention model itself has some non-linearity (in $\text{Softmax}(\cdot)$), a more common way to introduce non-linearity

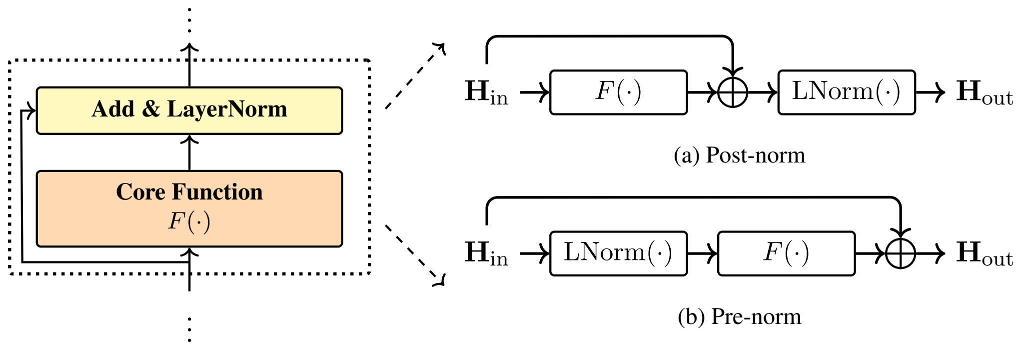


Figure 6.3: The post-norm and pre-norm structures. $F(\cdot) =$ core function, $\text{LNorm}(\cdot) =$ layer normalization, and $\oplus =$ residual connection.

is to consider additional layers with non-linear activation functions and linear transformations. Given an input $\mathbf{H}_{\text{in}} \in \mathbb{R}^{m \times d}$ and an output $\mathbf{H}_{\text{out}} \in \mathbb{R}^{m \times d}$, the $\mathbf{H}_{\text{out}} = \text{FFN}(\mathbf{H}_{\text{in}})$ function in the Transformer has the following form

$$\mathbf{H}_{\text{out}} = \mathbf{H}_{\text{hidden}} \mathbf{W}_f + \mathbf{b}_f \tag{6.37}$$

$$\mathbf{H}_{\text{hidden}} = \text{ReLU}(\mathbf{H}_{\text{in}} \mathbf{W}_h + \mathbf{b}_h) \tag{6.38}$$

where $\mathbf{H}_{\text{hidden}} \in \mathbb{R}^{m \times d_{\text{ffn}}}$ represents the hidden states, and $\mathbf{W}_h \in \mathbb{R}^{d \times d_{\text{ffn}}}$, $\mathbf{b}_h \in \mathbb{R}^{d_{\text{ffn}}}$, $\mathbf{W}_f \in \mathbb{R}^{d_{\text{ffn}} \times d}$ and $\mathbf{b}_f \in \mathbb{R}^d$ are the parameters. This is a two-layer FFN in which the first layer (or hidden layer) introduces a non-linearity through $\text{ReLU}(\cdot)$[2] and the second layer involves only a linear transformation. It is common practice in the Transformer to use a larger hidden layer size. For example, a common choice is $d_{\text{ffn}} = 4d$, that is, the size of each hidden representation is 4 times that of the input.

Note that using a wide FFN sub-layer has proven to be of great practical value in many state-of-the-art systems. However, a consequence of this is that the model's parameter count is dominated by the FFN. Table 6.1 shows the number of parameters and time complexities for different modules of a standard Transformer-based system. We see that FFNs dominate the model size when d_{ffn} is large, though they are not the most time-consuming components. In the case of very large Transformer models, we therefore wish to address this problem to build more efficient systems.

6.1.6 Decoder-side Attention

A decoder layer involves two attention sub-layers: the first is a self-attention sub-layer, and the second is a cross-attention sub-layer. These sub-layers are based on either the post-norm or the pre-norm structure, but differ in how the attention functions are defined. Consider, for example, the post-norm structure described in Eq. (6.35). We can define the cross-attention

[2] $\text{ReLU}(x) = \max\{0, x\}$.

Sub-model		# of Parameters	Time Complexity	$\times$
Encoder	Multi-head Self-attention	$4d^2$	$O(m^2 \cdot d)$	L
	Feed-forward Network	$2d \cdot d_{\mathrm{ffn}} + d + d_{\mathrm{ffn}}$	$O(m \cdot d \cdot d_{\mathrm{ffn}})$	L
	Layer Normalization	$2d$	$O(d)$	$2L$
Decoder	Multi-head Self-attention	$4d^2$	$O(n^2 \cdot d)$	L
	Multi-head Cross-attention	$4d^2$	$O(m \cdot n \cdot d)$	L
	Feed-forward Network	$2d \cdot d_{\mathrm{ffn}} + d + d_{\mathrm{ffn}}$	$O(n \cdot d \cdot d_{\mathrm{ffn}})$	L
	Layer Normalization	$2d$	$O(d)$	$3L$

Table 6.1: Number of parameters and time complexities of different Transformer modules under various setups. $m =$ source-sequence length, $n =$ target-sequence length, $d =$ default number of dimensions of a hidden layer, $d_{\mathrm{ffn}} =$ number of dimensions of the FFN hidden layer, $\tau =$ number of heads in the attention models, and $L =$ number of encoding or decoding layers. The column $\times$ indicates the number of times a sub-model is applied on the encoder or decoder side. The time complexities are estimated by counting the number of floating-point multiplications.

and self-attention sub-layers for a decoding layer to be

$$\begin{aligned} \mathbf{S}_{\mathrm{cross}} &= \mathrm{Layer}_{\mathrm{cross}}(\mathbf{H}_{\mathrm{enc}}, \mathbf{S}_{\mathrm{self}}) \\ &= \mathrm{LNorm}(\mathrm{Att}_{\mathrm{cross}}(\mathbf{H}_{\mathrm{enc}}, \mathbf{S}_{\mathrm{self}}) + \mathbf{S}_{\mathrm{self}}) \end{aligned} \tag{6.39}$$

$$\begin{aligned} \mathbf{S}_{\mathrm{self}} &= \mathrm{Layer}_{\mathrm{self}}(\mathbf{S}) \\ &= \mathrm{LNorm}(\mathrm{Att}_{\mathrm{self}}(\mathbf{S}) + \mathbf{S}) \end{aligned} \tag{6.40}$$

where $\mathbf{S} \in \mathbb{R}^{n \times d}$ is the input to the self-attention sub-layer, $\mathbf{S}_{\mathrm{cross}} \in \mathbb{R}^{n \times d}$ and $\mathbf{S}_{\mathrm{self}} \in \mathbb{R}^{n \times d}$ are the outputs of the sub-layers, and $\mathbf{H}_{\mathrm{enc}} \in \mathbb{R}^{m \times d}$ is the output of the encoder [3].

As with conventional attention models, cross-attention is primarily used to model the correspondence between the source-side and target-side sequences. The $\mathrm{Att}_{\mathrm{cross}}(\cdot)$ function is based on the QKV attention model, which generates an output by querying a collection of key-value pairs. More specifically, we define the queries, keys, and values as linear mappings of $\mathbf{S}_{\mathrm{self}}$ and $\mathbf{H}_{\mathrm{enc}}$, as follows

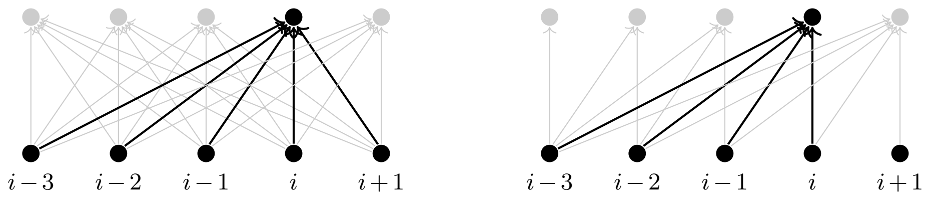
$$\mathbf{S}^q_{\mathrm{self}} = \mathbf{S}_{\mathrm{self}} \mathbf{W}^q_{\mathrm{cross}} \tag{6.41}$$

$$\mathbf{H}^k_{\mathrm{enc}} = \mathbf{H}_{\mathrm{enc}} \mathbf{W}^k_{\mathrm{enc}} \tag{6.42}$$

$$\mathbf{H}^v_{\mathrm{enc}} = \mathbf{H}_{\mathrm{enc}} \mathbf{W}^v_{\mathrm{enc}} \tag{6.43}$$

where $\mathbf{W}^q_{\mathrm{cross}}, \mathbf{W}^k_{\mathrm{enc}}, \mathbf{W}^v_{\mathrm{enc}} \in \mathbb{R}^{d \times d}$ are the parameters of the mappings. In other words, the queries are defined based on $\mathbf{S}_{\mathrm{self}}$, while the keys and values are defined based on $\mathbf{H}_{\mathrm{enc}}$.

[3] For an encoder having L encoder layers, $\mathbf{H}_{\mathrm{enc}} = \mathbf{H}^L$.



(a) Encoder-side Self-attention (b) Decoder-side Self-attention

Figure 6.4: Self-attention on the encoder and decoder sides. Each line connects an input and an output of the self-attention model, indicating a dependency of an output state on an input state. For encoder self-attention, the output at any position is computed by having access to the entire sequence. By contrast, for decoder self-attention, the output at position i is computed by seeing only inputs at positions up to i.

$\mathrm{Att}_{\mathrm{cross}}(\cdot)$ is then defined as

$$\begin{aligned}\mathrm{Att}_{\mathrm{cross}}(\mathbf{H}_{\mathrm{enc}}, \mathbf{S}_{\mathrm{self}}) &= \mathrm{Att}_{\mathrm{qkv}}(\mathbf{S}^q_{\mathrm{self}}, \mathbf{H}^k_{\mathrm{enc}}, \mathbf{H}^v_{\mathrm{enc}}) \\ &= \mathrm{Softmax}\left(\frac{\mathbf{S}^q_{\mathrm{self}}[\mathbf{H}^k_{\mathrm{enc}}]^\top}{\sqrt{d}}\right)\mathbf{H}^v_{\mathrm{enc}} \end{aligned} \tag{6.44}$$

The $\mathrm{Att}_{\mathrm{self}}(\cdot)$ function has a similar form to $\mathrm{Att}_{\mathrm{cross}}(\cdot)$, with linear mappings of $\mathbf{S}$ taken as the queries, keys, and values, like this

$$\begin{aligned}\mathrm{Att}_{\mathrm{self}}(\mathbf{S}) &= \mathrm{Att}_{\mathrm{qkv}}(\mathbf{S}^q, \mathbf{S}^k, \mathbf{S}^v) \\ &= \mathrm{Softmax}(\frac{\mathbf{S}^q[\mathbf{S}^k]^\top}{\sqrt{d}} + \mathbf{M})\mathbf{S}^v \end{aligned} \tag{6.45}$$

where $\mathbf{S}^q = \mathbf{S}\mathbf{W}^q_{\mathrm{dec}}$, $\mathbf{S}^k = \mathbf{S}\mathbf{W}^k_{\mathrm{dec}}$, and $\mathbf{S}^v = \mathbf{S}\mathbf{W}^v_{\mathrm{dec}}$ are linear mappings of $\mathbf{S}$ with parameters $\mathbf{W}^q_{\mathrm{dec}}, \mathbf{W}^k_{\mathrm{dec}}, \mathbf{W}^v_{\mathrm{dec}} \in \mathbb{R}^{d\times d}$.

This form is similar to that of Eq. (6.20). However, a key difference compared to encoder self-attention is that the model here must follow the rule of left-to-right generation (see Figure 6.4). That is, given a target-side word at position i, the model can only see the target-side words in the left context $y_1, \ldots, y_{i-1}$. To enforce this, we add a masking variable $\mathbf{M}$ to the unnormalized weight matrix $\frac{\mathbf{S}^q[\mathbf{S}^k]^\top}{\sqrt{d}}$. Both $\mathbf{M}$ and the weight matrix are of size $n \times n$, so large negative values in $\mathbf{M}$ suppress the corresponding attention scores. To prevent attention to the right context (future words) at step i, $\mathbf{M}$ is defined as:

$$M(i,j) = \begin{cases} 0 & i \geq j \\ -\infty & i < j \end{cases} \tag{6.46}$$

where $M(i,j)$ indicates a bias term for the alignment score between positions i and j. Below we show an example of how the masking variable is applied (assume $n = 4$).

$$
\begin{aligned}
& \mathrm{Softmax}(\frac{\mathbf{S}^q[\mathbf{S}^k]^\top}{\sqrt{d}} + \mathbf{M}) \\
= \quad & \mathrm{Softmax}(\begin{bmatrix} 2 & 0.1 & 1 & 1 \\ 0 & 0.9 & 0.9 & 0.9 \\ 0.2 & 0.8 & 0.7 & 2 \\ 0.3 & 1 & 0.3 & 3 \end{bmatrix} + \begin{bmatrix} 0 & -\infty & -\infty & -\infty \\ 0 & 0 & -\infty & -\infty \\ 0 & 0 & 0 & -\infty \\ 0 & 0 & 0 & 0 \end{bmatrix}) \\
= \quad & \mathrm{Softmax}(\begin{bmatrix} 2 & -\infty & -\infty & -\infty \\ 0 & 0.9 & -\infty & -\infty \\ 0.2 & 0.8 & 0.7 & -\infty \\ 0.3 & 1 & 0.3 & 3 \end{bmatrix}) \\
= \quad & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 \\ 0.2 & 0.4 & 0.4 & 0 \\ 0.05 & 0.1 & 0.05 & 0.8 \end{bmatrix}
\end{aligned}
\tag{6.47}
$$

As noted in Section 6.1.3, it is straightforward to improve these models by using the multi-head attention mechanism. In addition, because decoders are typically the most time-consuming components in practical applications, the overall computational efficiency of these systems depends heavily on optimizing decoder-side attention.

6.1.7 Training and Inference

Transformers can be trained and used in a standard manner. For example, we can train a Transformer model by performing gradient descent to minimize a loss function on the training data (see Chapter 2), and test the trained model by performing beam search on unseen data (see Chapter 5). Below we present some of the techniques that are typically used in the training and inference of Transformer models.

- **Learning Rate Scheduling**. Like standard neural networks, Transformers can be directly trained using back-propagation. The training process is generally iterated many times to make the model fit the training data well. In each training step, we update the weights of the neural networks by moving them a small step in the direction of the negative gradient of the loss function. There are many ways to design the update rule for training. A popular choice is the Adam optimization method [Kingma and Ba, 2014]. To adjust the learning rate during training, Vaswani et al. [2017] present a learning rate scheduling strategy that increases the learning rate linearly for a number of steps and then decays it gradually. They design a learning rate of the form

$$
\eta \quad = \quad \eta_0 \cdot \min\left\{n_{\mathrm{step}}^{-0.5},\ n_{\mathrm{step}} \cdot (n_{\mathrm{warmup}})^{-1.5}\right\} \tag{6.48}
$$

  where η_0 denotes the initial learning rate, n_{step} denotes the number of training steps we have executed, and n_{warmup} denotes the number of warmup steps. In the first n_{warmup}

steps, the learning rate η grows larger as training proceeds. It reaches its highest value at the point where $n_{\text{step}} = n_{\text{warmup}}$, and then decreases as an inverse square root function (i.e., $\eta_0 \cdot n_{\text{step}}^{-0.5}$).

- **Batching and Padding**. To make a trade-off between global optimization and training convergence, it is common practice to update the weights using a relatively small collection of samples, called a **minibatch**. Therefore, we can consider a batch version of the forward and backward computation processes in which the entire minibatch is used together to obtain the gradient information. One advantage of batching is that it allows the system to leverage efficient tensor operations to process multiple sequences in a single run. This requires all input sequences in a minibatch to be stored in a single memory block so they can be read and processed simultaneously. To illustrate this idea, consider a minibatch containing four samples whose source-sides are:

$$\begin{array}{cccccc} \text{A} & \text{B} & \text{C} & \text{D} & \text{E} & \text{F} \\ \text{M} & \text{N} & & & & \\ \text{R} & \text{S} & \text{T} & & & \\ \text{W} & \text{X} & \text{Y} & \text{Z} & & \end{array}$$

We can store these sequences in a 4×6 continuous block where each "row" represents a sequence, like this

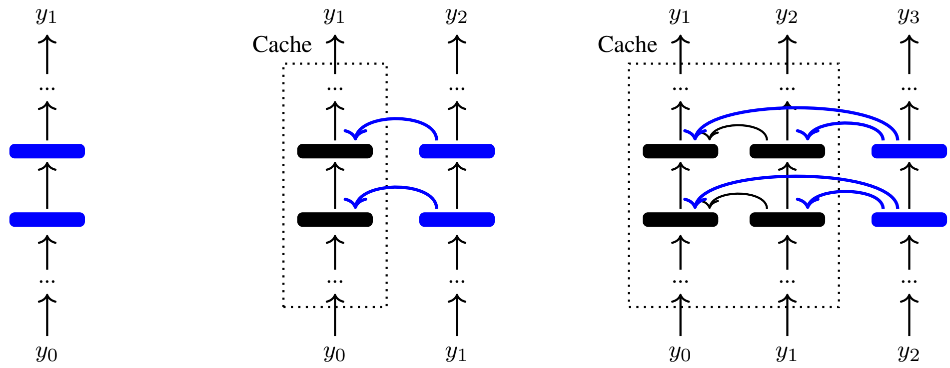
$$\begin{array}{cccccc} \text{A} & \text{B} & \text{C} & \text{D} & \text{E} & \text{F} \\ \text{M} & \text{N} & \square & \square & \square & \square \\ \text{R} & \text{S} & \text{T} & \square & \square & \square \\ \text{W} & \text{X} & \text{Y} & \text{Z} & \square & \square \end{array}$$

Here, padding words $\square$ are appended to the shorter sequences so that all sequences are properly aligned in memory. Typically, we do not want padding to affect the operation of the system, so we can simply define $\square$ as a zero vector (referred to as **zero padding**). On the other hand, in some cases we are interested in using padding to describe something that is not covered by the input sequences. For example, we can replace padding words with context words from the left (or right) of a sequence, though this may require modifications to the system to ensure that the newly added context words do not cause unintended content to appear in the output.

- **Search and Caching**. At test time, we need to search the space of candidate hypotheses (or candidate target-side sequences) to identify the hypothesis with the highest score

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \text{score}(\mathbf{x}, \mathbf{y}) \tag{6.49}$$

where $\text{score}(\mathbf{x}, \mathbf{y})$ is the model score of the target-side sequence $\mathbf{y}$ given the source-side sequence $\mathbf{x}$. While there are many search algorithms designed to achieve this, most share a similar structure: the search program operates by extending a pool of candidate target-side sequences one token at a time. In this way, the resulting algorithm can be viewed as



(a) decoding step 1 (b) decoding step 2 (c) decoding step 3

Figure 6.5: Illustration of the caching mechanism in Transformer decoders. Rectangles indicate the states of decoding layers or sub-layers. At step i, all the states from previous steps are stored in a cache (see dotted boxes), and we only need to compute the states for the current step (see blue rectangles and arrows). We then add the newly generated states to the cache and move on to step $i+1$.

a left-to-right generation procedure. For a more detailed discussion of search algorithms and model scores of general sequence-to-sequence models, see Chapter 5. Note that all designs for $\mathrm{score}(\mathbf{x},\mathbf{y})$, no matter how complex, are based on computing $\Pr(\mathbf{y}|\mathbf{x})$. Because the attention models used in Transformers require computing the dot-product of each pair of input vectors in a layer, the time complexity of the search algorithm is a quadratic function of the length of $\mathbf{y}$. It is therefore inefficient to repeatedly compute the outputs of the attention models for previously processed positions. This problem can be addressed by caching the states of each layer for the words we have already seen. Figure 6.5 illustrates the use of the caching mechanism in a search step. All states for positions $< i$ are maintained and easily accessed in a cache. At position i, all we need is to compute the states for the newly added word and then update the cache.

6.2 Syntax-aware Models

Although Transformers are deep learning models that do not explicitly make use of linguistic structures or assumptions, it may be necessary to incorporate our prior knowledge into such systems. This is in part because NLP researchers have long believed that a higher level of abstraction in data representation is needed to develop ideal NLP systems, and there have been many systems that use structural priors. However, structure is a broad topic. For a discussion of the various types of structure one might consider, see the work of See [2018]. For example, the inductive biases used in our model design can be thought of as some structural prior, and NLP models can also learn the underlying structure of problems by themselves. In this subsection we will discuss some of these issues. We will focus on the methods of introducing linguistic structure into Transformer models. As Transformers can be applied to many NLP tasks, which differ much in their input and output formats, we will primarily discuss modifications to Transformer encoders (call them **syntax-aware Transformer encoders**). Our discussion, however, is general, and these methods can easily be extended to Transformer decoders.

6.2.1 Syntax-aware Input and Output

One of the simplest methods of incorporating structure into NLP systems is to modify the input sequence, leaving the system unchanged. As a simple example, consider a sentence where each word x_j is assigned a set of κ syntactic labels $\{\mathrm{tag}_j^1, ..., \mathrm{tag}_j^\kappa\}$ (e.g., POS labels and dependency labels). We can concatenate these symbols to define a new "word"

$$x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa$$

Then, the embedding of this word is given by

$$\mathbf{e}_j = e(x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa) + \mathrm{PE}(j) \tag{6.50}$$

where $e(x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa) \in \mathbb{R}^d$ is the embedding of $x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa$. Since $x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa$ is a complex symbol, we decompose the learning problem of $e(x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa)$ into easier problems. For example, we can develop κ embedding models, each producing an embedding for a given tag. Then, we write $e(x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa)$ as a sum of the word embedding and tag embeddings

$$e(x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa) = \mathbf{x}_j + e(\mathrm{tag}_j^1) + ... + e(\mathrm{tag}_j^\kappa) \tag{6.51}$$

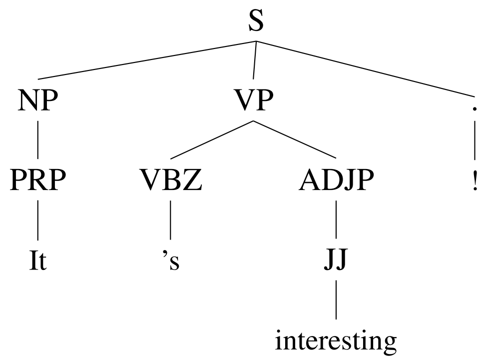
where $\{e(\mathrm{tag}_j^1), ..., e(\mathrm{tag}_j^\kappa)\}$ are the embeddings of the tags. Alternatively, we can combine these embeddings via a neural network in the form

$$e(x_j/\mathrm{tag}_j^1/.../\mathrm{tag}_j^\kappa) = \mathrm{FFN}_{\mathrm{embed}}(\mathbf{x}_j, e(\mathrm{tag}_j^1), ..., e(\mathrm{tag}_j^\kappa)) \tag{6.52}$$

where $\mathrm{FFN}_{\mathrm{embed}}(\cdot)$ is a feed-forward neural network that has one or two layers.

We can apply a similar technique on the decoder side, treating $y_i/\mathrm{tag}_i^1/\cdots/\mathrm{tag}_i^\kappa$ as a syntax-augmented target word. However, doing so may drastically increase the size of the target-side vocabulary, posing a computational challenge for both training and inference.

Another common method for representing a sentence is via a syntax tree. In linguistics, the syntax of a sentence can be interpreted in many different ways, resulting in various grammars and corresponding tree-based (or graph-based) representations. While these representations differ in their exact syntactic forms, a general approach for utilizing them in sequence modeling is **tree linearization**. Consider the following sentence annotated with a constituency-based parse tree:



We can write this tree structure as a sequence of words, syntactic labels and brackets via a tree traversal algorithm, as follows

(S (NP (PRP It)$_{\text{PRP}}$)$_{\text{NP}}$ (VP (VBZ 's)$_{\text{VBZ}}$ (ADJP (JJ interesting)$_{\text{JJ}}$)$_{\text{ADJP}}$)$_{\text{VP}}$ (. !)$_{.}$)$_{\text{S}}$

This sequence of syntactic tokens can be used directly as input to the encoder, that is, each token is represented by the sum of its word and positional embeddings, and this combined embedding is treated as a standard input to the encoder. One application of linearized trees is tree-to-string machine translation, in which a syntax tree in one language is translated into a string in another language [Li et al., 2017b; Currey and Heafield, 2018]. Linearized trees can also be utilized for tree generation. For example, parsing tasks can be framed as sequence-to-sequence problems, mapping an input text to a sequential representation of its corresponding syntax tree [Vinyals et al., 2015; Choe and Charniak, 2016]. Figure 6.6 illustrates these models. It should be noted that the methods described here are not specific to the Transformer and can be applied to many sequence-based architectures, such as RNN-based models.

6.2.2 Syntax-aware Attention Models

For Transformer models, it is also natural to make use of syntax trees to guide the process of learning sequence representations. In the previous section we saw how representations of a sequence can be computed by relating different positions within that sequence. This allows us to impose structure on the attention distributions over positions. To do this, we use the encoder self-attention with an additive mask

$$\mathrm{AttSyn}_{\text{self}}(\mathbf{H}) = \mathrm{Softmax}(\frac{\mathbf{H}^q[\mathbf{H}^k]^\top}{\sqrt{d}} + \mathbf{M})\mathbf{H}^v \tag{6.53}$$

or alternatively with a multiplicative mask

$$\mathrm{AttSyn}_{\text{self}}(\mathbf{H}) = \mathrm{Softmax}(\frac{\mathbf{H}^q[\mathbf{H}^k]^\top}{\sqrt{d}} \odot \mathbf{M})\mathbf{H}^v \tag{6.54}$$

where $\mathbf{M} \in \mathbb{R}^{m \times m}$ is a matrix of masking variables in which a larger value of $M(i,j)$ indicates a stronger syntactic correlation between positions i and j. In the following description we

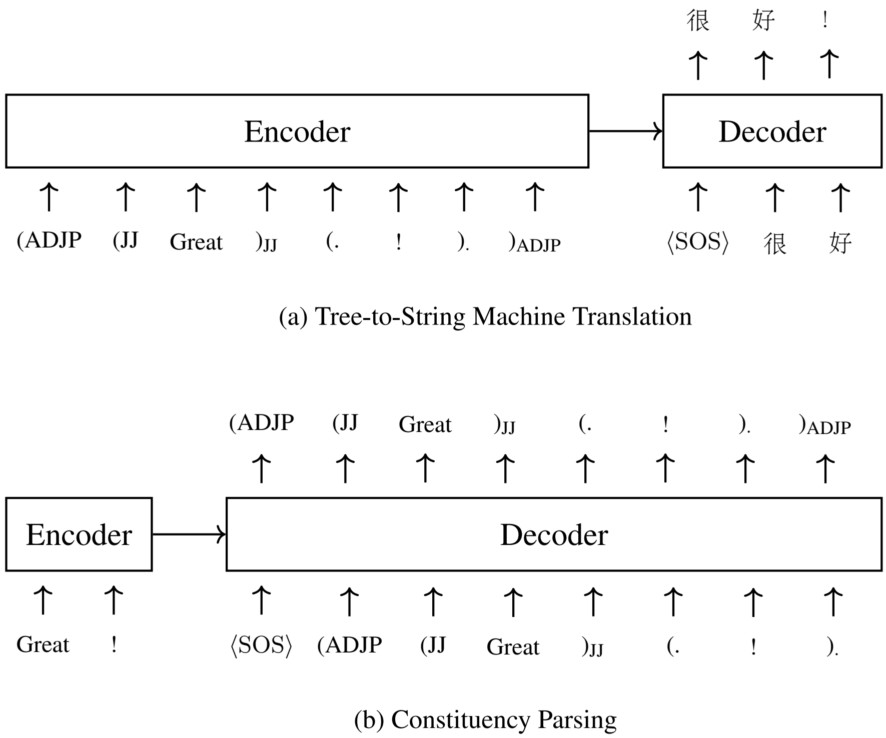


Figure 6.6: Illustration of tree linearization on either the encoder or decoder side. For tree-to-string machine translation, the encoder takes a sequential representation of an input parse tree, and the decoder outputs the corresponding translation. For parsing, the encoder takes a sentence, and the decoder outputs the corresponding syntax tree.

choose Eq. (6.54) as the basic form.

One common way to design $\mathbf{M}$ is to project syntactic relations of the input tree structure into constraints over the sequence. Here we consider constituency parse trees and dependency parse trees for illustration. Generally, two types of masking methods are employed.

- **0-1 Masking**. This method assigns $M(i,j) = 1$ if the words at positions i and j are considered syntactically correlated and $M(i,j) = 0$ otherwise [Zhang et al., 2020c; Bai et al., 2021]. To model the relation between two words in a syntax tree, we can consider the distance between their corresponding nodes. One of the simplest forms is given by

$$M(i,j) = \begin{cases} 1 & \omega(i,j) \leq \omega_{\max} \\ 0 & \text{otherwise} \end{cases} \tag{6.55}$$

  where $\omega(i,j)$ is the shortest path length between the nodes corresponding to the words at positions i and j. For example, given a dependency parse tree, $\omega(i,j)$ is the number of dependency edges in the path between the two words. For a constituency parse tree, all words are leaf nodes. Thus, $\omega(i,j)$ represents the tree distance between the two leaves within the same branch of the tree. $\omega_{\max}$ is a parameter used to control the maximum distance between two nodes that can still be considered syntactically correlated. For example, assuming a dependency parse tree with $\omega_{\max} = 1$, Eq. (6.55) enforces the constraint that the attention score between positions i and j is computed only if they have a parent-dependent relation[4].

- **Soft Masking**. Instead of treating $\mathbf{M}$ as a hard constraint, we can use it as a soft constraint that scales the attention weight between positions i and j based on the degree to which the corresponding words are correlated. One approach is to reduce the attention weight as $\omega(i,j)$ becomes larger. A simple method is to transform $\omega(i,j)$ such that $M(i,j)$ is negatively correlated with $\omega(i,j)$ and falls within the interval $[0,1]$:

$$M(i,j) \quad = \quad \mathrm{DNorm}(\omega(i,j)) \tag{6.56}$$

There are several alternative designs for $\mathrm{DNorm}(\cdot)$. For example, one can compute a standardized score of $-\omega(i,j)$ by subtracting its mean and dividing by its standard deviation [Chen et al., 2018a], or normalize $1/\omega(i,j)$ over all possible j in the sequence [Xu et al., 2021b]. In cases where parsers can output a score between positions i and j, this score can also be used to compute $M(i,j)$. For example, a dependency parser can produce the probability of the word at position i being the parent of the word at position j [Strubell et al., 2018]. We can then write $M(i,j)$ as

$$M(i,j) \quad = \quad \mathrm{Pr}_{\mathrm{parent}}(i|j) \tag{6.57}$$

or alternatively

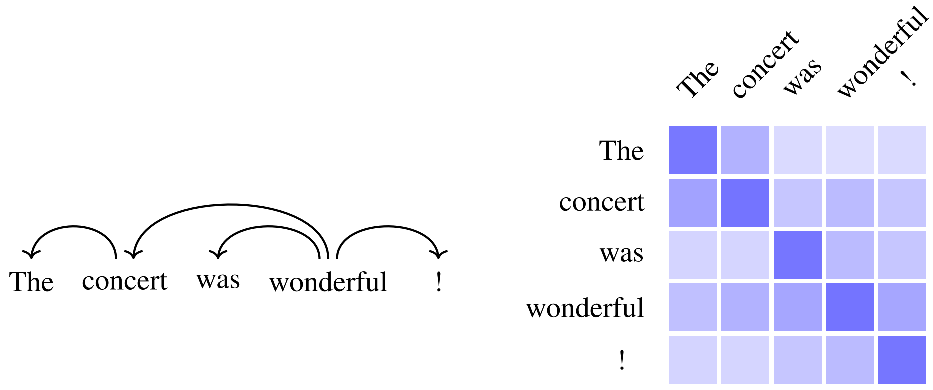
$$M(i,j) \quad = \quad \max\{\mathrm{Pr}_{\mathrm{parent}}(i|j), \mathrm{Pr}_{\mathrm{parent}}(j|i)\} \tag{6.58}$$

where $\mathrm{Pr}_{\mathrm{parent}}(i|j)$ and $\mathrm{Pr}_{\mathrm{parent}}(j|i)$ are the probabilities given by the parser. See Figure 6.7 for an example of inducing a soft masking variable from a dependency parse tree.

6.2.3 Multi-branch Models

Integrating syntax into NLP systems presents several challenges. This is partly because automatic parse trees may contain errors, and partly because the use of syntax may lead to strong assumptions about the underlying structure of a sentence. Rather than combining syntactic and word information into a single monolithic model, it may be more flexible and effective to build one model to encode syntactic information and another to encode word sequences. One way to achieve this is through the use of multiple neural networks (referred to as **branches** or **paths**), each dealing with one type of input. The outputs of these branches are

[4]For multiplicative masks, $M(i,j) = 0$ does not mean that the final attention weight between j and i is zero, because an input of zero to the Softmax function does not yield a zero probability. A method to strictly "mask out" an entry of Softmax$\left(\frac{\mathbf{H}^q[\mathbf{H}^k]^\top}{\sqrt{d}}\right)$ is to use an additive mask and set $M(i,j) = -\infty$ if $\omega(i,j) > \omega_{\max}$.



(a) Dependency Parse Tree (b) Mask $\mathbf{M}$ (darker color means larger value)

Figure 6.7: Priors induced from a dependency parse tree. The row i of the matrix $\mathbf{M}$ represents a distribution that describes how much weight we can give to $M(i, j)$ in terms of the syntactic distance between i and j.

then combined to produce a final output [Xie et al., 2017; Fan et al., 2020; Lin et al., 2022b]. Various methods have therefore been used to combine different types of input for neural models like Transformers.

One commonly used approach is to build two separate encoders, in which one model is trained to encode the syntactic input (denoted by $\mathbf{t}$), and the other is trained to encode the standard textual input (denoted by $\mathbf{x}$). Figure 6.8(a) illustrates this multi-encoder architecture. The syntactic encoder $\mathrm{Encode}_{\mathrm{syn}}(\mathbf{t})$ is based on models presented in Sections 6.2.1 and 6.2.2, and the text encoder $\mathrm{Encode}_{\mathrm{text}}(\mathbf{x})$ is a standard Transformer encoder. The representations generated by these encoders are then fed into the combination model as input and combined into a hybrid representation, given by

$$
\begin{aligned}
\mathbf{H}_{\mathrm{hybrid}} &= \mathrm{Combine}(\mathbf{H}_{\mathrm{syn}}, \mathbf{H}_{\mathrm{text}}) \\
&= \mathrm{Combine}(\mathrm{Encode}_{\mathrm{syn}}(\mathbf{t}), \mathrm{Encode}_{\mathrm{text}}(\mathbf{x})) \qquad (6.59)
\end{aligned}
$$

There are several designs for $\mathrm{Combine}(\cdot)$, depending on the tasks to which the encoders are applied. For example, if we want to develop a text classifier, $\mathrm{Combine}(\cdot)$ can be a simple pooling network. For more complicated tasks, such as machine translation, $\mathrm{Combine}(\cdot)$ can also be a Transformer encoder, and we can fuse information from different sources by performing self-attention on $[\mathbf{H}_{\mathrm{syn}}, \mathbf{H}_{\mathrm{text}}]$.

Although we focus primarily on syntactic models in this section, the general multi-encoder architecture can be used in many tasks where inputs from additional sources are required. For example, one can use one encoder to represent a sentence and another to represent the previous sentence in the same document. Combining the two encoders thus yields a context-aware model [Voita et al., 2018; Li et al., 2020a]. Furthermore, the architectures of the encoders do not need to be restricted to Transformers. We can choose different models for different

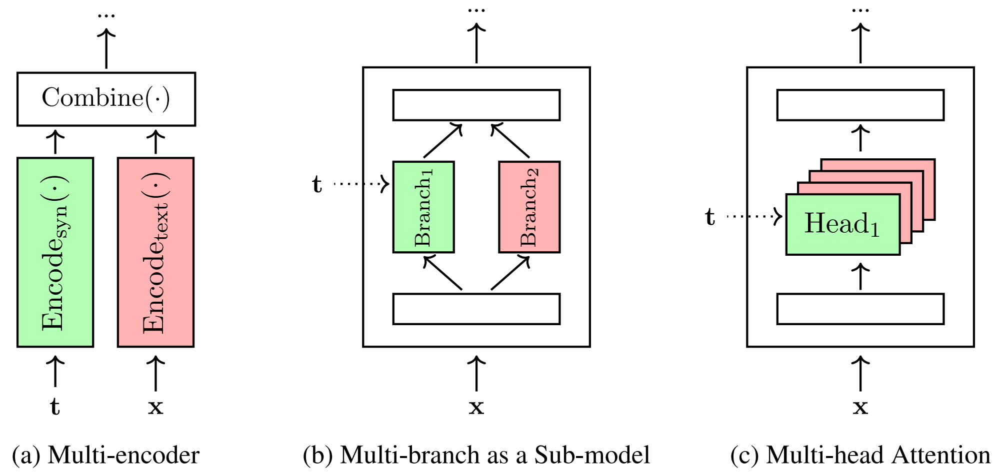


Figure 6.8: Multi-branch architectures. There are two inputs: a sentence (denoted by $\mathbf{x}$) and the syntax tree of the sentence (denoted by $\mathbf{t}$). In the multi-encoder architecture (see sub-figure (a)), two encoders are constructed to encode $\mathbf{x}$ and $\mathbf{t}$, respectively. A combination model then takes the outputs of the encoders and produces a combined representation of $\mathbf{x}$ and $\mathbf{t}$. The idea of multi-branch networks can be used for designing sub-models of the encoder. A simple example is that we create multiple paths in parallel for some layers of the encoder (see sub-figure (b)). Another example is multi-head attention (see sub-figure (c)) where we use different heads to learn different representations.

branches. For instance, in a widely used two-branch encoding architecture, we can employ a CNN-based encoder to model local context and a Transformer encoder to model global context [Wu et al., 2020b].

Sub-models of Transformers can also be multi-branch neural networks. See Figure 6.8 (b) for an example involving two self-attention branches. One is the standard self-attention network $\mathrm{Att}_{\mathrm{self}}(\mathbf{H})$, and the other is the syntax-aware self-attention network $\mathrm{AttSyn}_{\mathrm{self}}(\mathbf{H})$. The output of the self-attention model is a linear combination of the outputs of these two branches [Xu et al., 2021b], given by

$$\mathbf{H}_{\mathrm{self}} = \alpha \cdot \mathrm{Att}_{\mathrm{self}}(\mathbf{H}) + (1-\alpha) \cdot \mathrm{AttSyn}_{\mathrm{self}}(\mathbf{H}) \tag{6.60}$$

where α is a combination coefficient. $\mathbf{H}_{\mathrm{self}}$ can be used as usual by applying a layer normalization function and adding a residual connection, thus keeping the overall architecture identical to standard Transformer models.

Multi-head attention networks can also be viewed as a form of multi-branch models. Therefore, we can provide guidance from syntax to only some of the heads while keeping the rest unchanged [Strubell et al., 2018]. This approach is illustrated in Figure 6.8 (c) where only one head of the self-attention sub-layer makes use of syntax trees for computing attention weights.

6.2.4 Multi-scale Models

In linguistics, syntax studies how sentences are built from smaller constituents. Different levels of these constituents are generally organized in a hierarchical structure, called a **syntactic hierarchy**. It is therefore possible to use multiple levels of syntactic constituents to explain the same sentence. For example, words capture how the sentence is constructed from small meaningful units, and phrases capture how the sentence is constructed from larger linguistic units.

Multi-scale Transformers leverage different levels of abstraction in data to represent a sentence using features at multiple scales. A common approach is to write a sentence in multiple different forms and then combine them using a multi-branch network [Hao et al., 2019]. For example, consider the following sentence:

The oldest beer-making facility was discovered in China.

We can tokenize it into a sequence of words, denoted by

$$\mathbf{x}_{\text{words}} = \text{The oldest beer-making facility was discovered in China .}$$

Alternatively, we can write it as a sequence of phrases by using a parser, denoted by

$$\mathbf{x}_{\text{phrases}} = \text{[The oldest beer-making facility]}_{\text{NP}} \text{ [was discovered in China]}_{\text{VP}} \text{ [.]}.$$

The simplest way to build a multi-scale model is to encode $\mathbf{x}_{\text{words}}$ and $\mathbf{x}_{\text{phrases}}$ using two separate Transformer encoders. The outputs of these encoders are then combined in some way. This leads to the same form as Eq. (6.59), and we can view this model as an instance of the general multi-encoder architecture.

Both $\mathbf{x}_{\text{words}}$ and $\mathbf{x}_{\text{phrases}}$ can be viewed as sequences of tokens. For example, $\mathbf{x}_{\text{words}}$ has nine word-based tokens, and $\mathbf{x}_{\text{phrases}}$ has three phrase-based tokens[5]. However, enumerating all possible phrases would result in a huge vocabulary. We therefore need a method to represent each phrase as an embedding in a computationally efficient manner. By treating phrase embedding as a sequence modeling problem, it becomes straightforward to learn sub-sequence representations by employing the sequence models described in previous chapters. We now have a two-stage learning process. In the first stage, we learn the embeddings of input units at different scales using separate models. In the second stage, we learn to encode sequences at different scales using a multi-branch model.

More generally, we do not need to restrict ourselves to linguistically meaningful units in multi-scale representation learning. For example, we can learn sub-word segmentations from data and represent an input sentence as a sequence of sub-words. This results in a hierarchical representation of the sentence (e.g., sub-words $\rightarrow$ words $\rightarrow$ phrases). While the learned sub-words may not have linguistic meanings, they provide new insights into modeling words and

[5] $\mathbf{x}_{\text{phrases}}$ comprises three tokens: *The oldest beer-making facility*, *was discovered in China*, and .

phrases, as well as a new scale of features. Also, we do not strictly need to develop multiple encoders for multi-scale modeling. An alternative approach is to incorporate representations at different scales directly into the multi-head self-attention modules, which makes it easier to model the interactions among different scales [Guo et al., 2020; Li et al., 2022b].

A problem with the approaches described above, however, is that the representations (or attention weight matrices) learned at different scales are of different sizes. For example, in the scenarios above, the representation learned from $\mathbf{x}_{\text{words}}$ is a $9 \times d$ matrix, while the representation learned from $\mathbf{x}_{\text{phrases}}$ is a $3 \times d$ matrix. A simple solution to this problem is to perform upsampling on the phrase-based representation to expand it to a $9 \times d$ matrix. Likewise, we can perform downsampling on the word-based representation to shrink it to a $3 \times d$ matrix. The combination model $\text{Combine}(\cdot)$ can then be implemented as described in Section 6.2.3.

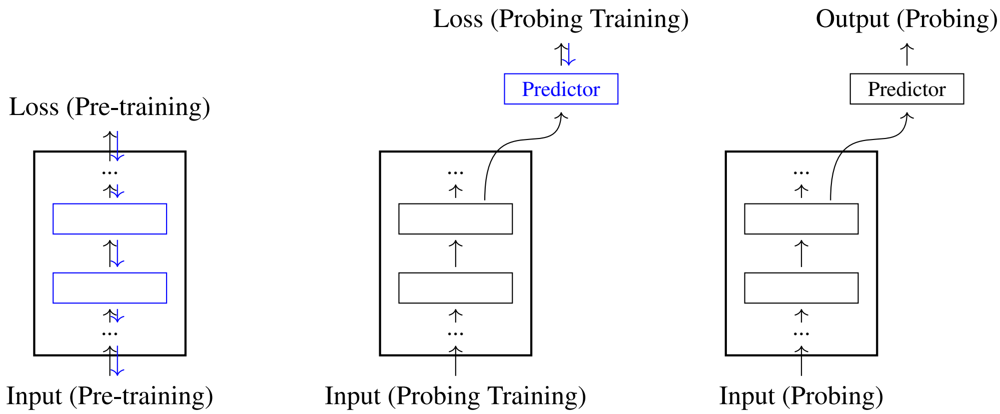
It is worth noting that multi-scale modeling is widely discussed across several fields. In computer vision, for example, multi-scale modeling is often referred to as the process of learning a series of feature maps from the input image [Fan et al., 2021; Li et al., 2022f]. Unlike the multi-branch models presented here, multi-scale vision Transformer models make use of the hierarchical nature of features when representing images. Systems of this kind are typically based on a stack of layers in which each layer learns features at a larger scale (e.g., a higher channel capacity) from the features produced by the previous layer.

6.2.5 Transformers as Syntax Learners

So far we have discussed syntax trees as constraints or priors on the encoding process so that we can make use of linguistic representations in learning neural networks. It is natural to wonder whether these neural models can learn some knowledge of linguistic structure directly from data, without human-designed linguistic annotations. This reflects a fundamental goal in NLP: to learn linguistic knowledge from data and encode it implicitly within models.

To explore the linguistic properties learned by NLP systems, a straightforward method is to examine the syntactic behaviors of the systems' outputs. For example, we can examine whether the outputs of language generation systems contain grammatical errors. Another approach is to ask these systems to accomplish tasks that are linguistically meaningful, even though they were not explicitly trained to do so [Brown et al., 2020]. However, merely examining how model predictions exhibit syntactic abilities is not sufficient to fully answer the question. It is also possible that neural networks learn linguistic knowledge but do not utilize it during prediction [Clark et al., 2019a]. Therefore, we need to investigate what is modeled and learned inside the neural networks themselves.

One approach to examining the latent linguistic structure in Transformer models is to develop **probes** to determine whether and to what extent these models capture linguistic notions such as dependency relations and parts-of-speech. A general probing approach involves extracting the internal representations of the models and probing them for linguistic phenomena. For Transformers, this is usually achieved by examining the attention maps and/or the outputs of attention layers. We then construct a **probing predictor** (or **probing classifier**) that takes these internal representations as input and outputs the corresponding linguistic



(a) Training the Transformer Model (b) Training the Probing Predictor (c) Probing on New Data

Figure 6.9: An overview of probing for Transformer-based models. Given a Transformer model (e.g., a Transformer-based language model), we first optimize the model parameters on some unlabeled data. Then, we develop a predictor which takes the states of a hidden layer of the Transformer model and generates outputs for a probing task (see sub-figure (a)). The predictor can be trained as usual in which only the parameters of the predictor are optimized and the parameters of the Transformer model are fixed (see sub-figure (b)). The Transformer model and the predictor are used together to make predictions on new data for probing (see sub-figure (c)).

notions [Belinkov, 2022]. The probing predictor can be based on either simple heuristics or parameterized models optimized specifically for the probing task. Recent work has shown that large-scale Transformer-based language models exhibit strong capabilities, referred to as **emergent abilities**, across various probing tasks. We will not discuss the details of these language modeling systems in this chapter, but rather leave them for the following chapters. Nevertheless, we assume here that we have a Transformer encoder that has been fully trained on unlabeled data and is available for probing. Figure 6.9 illustrates this probing process.

Many probing methods have been used in recent work on analyzing and understanding what is learned in neural encoders. Here we describe some of the popular ones.

- **Trees**. Given a trained Transformer encoder, it is easy to determine how likely it is that two words in a sentence share a linguistic relationship by computing the attention weight between them. We can use this quantity to define a metric measuring the syntactic distance between two words at positions i and j

$$d_s(i,j) = 1 - \alpha(i,j) \tag{6.61}$$

By using this metric, it is straightforward to construct the **minimum-spanning tree** for the sentence. That is, we connect all the words to form a tree structure with the minimum total distance. This tree structure can be viewed as a latent tree representation of the sentence induced from the neural network. While this dependency-tree-like structure can serve as a source of learned syntactic information for downstream tasks, it says little about the model's actual knowledge of syntax. An approach to aligning the representations in the encoder with linguistic structure is to learn to produce syntax trees that are consistent with human annotations. To achieve this, we need to develop a probing predictor trained on tree-annotated data. Suppose we have a human-annotated dependency tree for a given sentence. For each pair of words, we can obtain a distance $\omega(i,j)$ by counting the number of edges between them. Then, we can learn a distance metric based on the internal representations of the encoder to approximate $\omega(i,j)$. A simple form of such a metric relies on the Euclidean distance [Manning et al., 2020]. Let $\mathbf{A} \in \mathbb{R}^{d \times k_s}$ be a parameter matrix. The squared Euclidean distance is given by

$$d_s^2(i,j) \quad = \quad \|(\mathbf{h}_i - \mathbf{h}_j)\mathbf{A}\|_2^2 \tag{6.62}$$

where $\mathbf{h}_i$ and $\mathbf{h}_j$ are the representations produced by an encoding layer at positions i and j[6]. Given a set of tree-annotated sentences S, we can optimize the model by

$$\hat{\mathbf{A}} \quad = \quad \arg\max_{\mathbf{A}} \sum_{s \in S} \frac{1}{|s|^2} \sum_{i \in s, j \in s} |\omega(i,j) - d_s^2(i,j)| \tag{6.63}$$

where $|s|$ is the length of the sentence s, and (i,j) indicates a pair of words in s. The optimized model is then used to parse test sentences via the minimum-spanning tree algorithm, and we can compare the parse trees against the human-annotated trees. To obtain directed trees, which are standard forms of dependency syntax, one can update the above model by considering the relative distance of a word to the root. More details can be found in Manning et al. [2020]'s work. Here the probing predictor functions similarly to a neural parser, trained to predict a syntax tree based on a representation of the input sentence. This idea can be extended to other forms of syntactic structure, such as phrase structure trees [Shi et al., 2016].

- **Syntactic and Semantic Labels**. Many syntactic and semantic parsing tasks can be framed as problems of predicting linguistic labels given a sentence or its segments. A simple example is part-of-speech tagging in which each word of a sentence is labeled with a word class. A probe for part-of-speech tagging can be a classifier that takes a representation $\mathbf{h}_j$ each time and outputs the corresponding word class. One general probing approach to these problems is **edge probing** [Tenney et al., 2019b;a]. Given a sentence, a labeled edge is defined as a tuple

$$(\mathrm{span}_1, \mathrm{span}_2, \mathrm{label})$$

where span_1 is a span $[i_1, j_1]$, and span_2 is an optional second span $[i_2, j_2]$, and label is the corresponding label. Our goal is to learn a probe to predict label given span_1 and span_2. For example, for part-of-speech tagging, span_1 is a unit span $[j, j]$ for each

[6] In general, $\mathbf{h}_i$ and $\mathbf{h}_j$ are the outputs of the last layer of the encoder. Alternatively, they can be weighted sums of the outputs of all the layers.

position j, span_2 is an empty span, and label is the part-of-speech tag corresponding to the j-th word of the sentence. For dependency parsing and coreference resolution, span_1 and span_2 are two words or entities, and label is the relationship between them. For constituency parsing, span_1 is a span of words, span_2 is an empty span, and label is the syntactic category of the tree node yielding span_1. In simple cases, the probing model can be a multi-layer feed-forward neural network with a Softmax output layer. As usual, this model is trained on labeled data, and then tested on new data.

- **Surface Forms of Words and Sentences**. Probing tasks can also be designed to examine whether the representations encode the surface information of sentences or words [Adi et al., 2016; Conneau et al., 2018]. A simple sentence-level probing task is **sentence length prediction**. To do this, we first represent the sentence as a single vector $\mathbf{h}$[7], and then build a classifier to categorize $\mathbf{h}$ into the corresponding length bin. Similarly, probes can be built to predict whether two words at positions i and j are reordered in the sentence given $\mathbf{h}_i$ and $\mathbf{h}_j$. Also, we can develop probes to address conventional problems in morphology. For example, we reconstruct the word at position j or predict its sense with the representation $\mathbf{h}_j$. In addition, probing tasks can be focused on particular linguistic problems, for example, numeracy [Wallace et al., 2019] and function words [Kim et al., 2019].

- **Cloze**. Of course, we can probe neural models for problems beyond syntax and morphology. One perspective on large-scale pre-trained Transformer models is to view them as knowledge bases containing facts about the world. It is therefore tempting to see if we can apply them to test factual knowledge. A simple method is to ask a probe to recover the missing item of a sentence [Petroni et al., 2019]. For example, if we have a cloze test

  Shiji was written by ____.

  we wish the probe to give an answer *Sima Qian* because there is a subject-relation-object fact (Shiji, written-by, Sima Qian). This probe can simply be a **masked language model** that is widely used in self-supervised learning of Transformer encoders.

In NLP, probing is closely related to the pre-training of large language models (see Chapters 7 and 8). In general, we can view probing tasks as downstream applications of these pre-trained models, though probing is ordinarily used as an analytical tool to quickly test the capabilities of the models. Ideally, we want to develop a probe that makes the best use of the internal representations to solve a given task. However, when a probe is complex and sufficiently well-trained, it becomes difficult to determine whether the problem was solved by the rich information in the representations or by the modeling capacity of the probe itself. A common way to isolate the contribution of the representations is to compare the probes against reasonable baselines or to conduct comparisons using control tasks [Hewitt and Liang, 2019; Belinkov, 2022].

[7] $\mathbf{h}$ can be computed by performing a pooling operation on $\{\mathbf{h}_1, ..., \mathbf{h}_m\}$

6.3 Improved Architectures

In this section we present several improvements to the vanilla Transformer model. Unlike the previous section, most of the improvements are from the perspective of machine learning, rather than linguistics.

6.3.1 Locally Attentive Models

The self-attention methods discussed in Section 6.1.3 can be viewed as learning representations over the entire input sequence. While this global attention mechanism provides a strong ability to capture long-distance dependencies, it does not explicitly model local information. Here, we consider a few techniques that attempt to explicitly capture locality in representations.

1. Priors for Local Modeling

One of the simplest ways to introduce local context modeling into Transformers is to add a penalty term to the attention function, thereby discouraging large attention weights between distant positions. On the encoder side, this leads to a form that we have already encountered several times in this chapter:

$$\mathrm{AttLocal}_{\mathrm{self}}(\mathbf{H}) = \mathrm{Softmax}\left(\frac{\mathbf{H}^q[\mathbf{H}^k]^\top}{\sqrt{d}} - \gamma \cdot \mathbf{G}\right)\mathbf{H}^v \tag{6.64}$$

where γ is the weight (or temperature) of the penalty term, and $\mathbf{G} \in \mathbb{R}^{m \times m}$ is the penalty matrix. Each entry $G(i,j)$ indicates how heavily we penalize the model for attending to position j from position i. A simple choice for $G(i,j)$ is the absolute distance between i and j, for example

$$G(i,j) = |i-j| \tag{6.65}$$

Alternatively, $G(i,j)$ can be defined using a Gaussian penalty function [Yang et al., 2018a]

$$G(i,j) = \frac{(i-j)^2}{2\sigma_i^2} \tag{6.66}$$

where σ_i is the standard deviation of the Gaussian distribution. For a given i, both penalty terms increase (either linearly or exponentially) as the distance $|i-j|$ grows.

This method can be extended to the cross-attention model, as follows

$$\mathrm{AttLocal}_{\mathrm{cross}}(\mathbf{H},\mathbf{S}) = \mathrm{Softmax}\left(\frac{\mathbf{S}^q[\mathbf{H}^k]^\top}{\sqrt{d}} - \gamma \cdot \mathbf{G}\right)\mathbf{H}^v \tag{6.67}$$

where $\mathbf{G}$ is an $n \times m$ matrix. Each entry of $\mathbf{G}$ can be defined as

$$G(i,j) = \frac{(\mu_i-j)^2}{2\sigma_i^2} \tag{6.68}$$

where μ_i is the mean of the Gaussian distribution over the source-side positions. Both μ_i

and σ_i can be determined using heuristics. Alternatively, we can develop additional neural networks to model them, learning their parameters jointly with the rest of the model. For example, we could use a feed-forward neural network to predict μ_i given $\mathbf{s}_i$.

An alternative to Eq. (6.64) (or Eq. (6.67)) is to treat the penalty term as a separate distribution and linearly combine it with the original attention model. For example, we can define the self-attention model as

$$\mathrm{AttLocal}_{\mathrm{self}}(\mathbf{H}) = \Big((1-\beta)\cdot\mathrm{Softmax}\Big(\frac{\mathbf{H}^q[\mathbf{H}^k]^\top}{\sqrt{d}}\Big)+\beta\cdot\mathrm{Softmax}(-\gamma\cdot\mathbf{G})\Big)\mathbf{H}^v \tag{6.69}$$

where $\beta \in [0,1]$ is the linear combination coefficient. Note that, to avoid manually tuning the hyperparameter β, we can use a gating network to dynamically predict β and train it end-to-end.

Another alternative is to use a multiplicative mask to incorporate the prior into the modeling, similar to Eq. (6.54). This is given by

$$\mathrm{AttLocal}_{\mathrm{self}}(\mathbf{H}) = \mathrm{Softmax}\Big(\frac{\mathbf{H}^q[\mathbf{H}^k]^\top}{\sqrt{d}}\odot\mathbf{G}'\Big)\mathbf{H}^v \tag{6.70}$$

Here, $\mathbf{G}' \in [0,1]^{m\times m}$ is a matrix with entries in $[0,1]$. The scalar $G'(i,j)$ takes a maximum value of 1 when $i=j$ and decays as j moves further away from i. $G'(i,j)$ can be obtained by normalizing $-G(i,j)$ over all j, or by using other decaying functions.

2. Local Attention

The term *local attention* has been used broadly to describe a wide range of problems and to refer to many different models in the NLP literature. The methods discussed above are those that impose soft constraints on attention models. In fact, local attention has its origins in attempts to restrict the scope of attention models for modeling and computational considerations [Luong et al., 2015]. Research in this area often focuses on introducing hard constraints, so that the resulting models can focus on parts of the input and ignore the rest. For example, we can predict a span of source-side positions to attend to, given a target-side position [Sperber et al., 2018; Yang et al., 2018a; Sukhbaatar et al., 2019]. Also, attention spans can be induced from syntax trees, for example, syntactic subtree structures can help narrow the attention scope. Thus, many syntax-constrained models can be viewed as instances of local attention (see Section 6.2.4). In addition, the concept of local attention can be generalized to a rich set of models, such as **sparse attention models**, although these models are often discussed in the context of efficient machine learning methods. We will see a few examples in Section 6.4.

In deep learning, one of the most widely used architectures for learning features from a restricted region of the input is CNNs. It is thus interesting to consider methods of combining CNNs and Transformer models to obtain the benefits of both approaches, for example, CNNs deal with short-term dependencies, and self-attention models deal with long-term dependencies. One approach is to build a two-branch sequence model where one branch is based on CNNs and the other is based on self-attention models [Wu et al., 2020b]. Another approach is to incorporate CNN layers into Transformer blocks in a way that enables the model to learn both local and global representations [Wu et al., 2019; Gulati et al., 2020].

3. Relative Positional Embedding

Relative positional embedding, also known as **relative positional representation** (**RPR**), is an improvement to the absolute positional embedding method used in standard Transformers[Shaw et al., 2018; Huang et al., 2018]. The idea of RPR is that we model the distance between two positions in a sequence rather than giving each position a fixed representation. As a result, we have a pair-wise representation $\mathrm{PE}(i,j)$ for two positions i and j. One simple way to define $\mathrm{PE}(i,j)$ is to consider it as a lookup table for all pairs of i and j. More specifically, let $\mathbf{u}_{\pi}$ be a d-dimensional representation for a given distance π. The form of $\mathrm{PE}(i,j)$ in the vanilla RPR method is given by

$$\mathrm{PE}(i,j) \quad = \quad \mathbf{u}_{\mathrm{clip}(j-i,k_{\mathrm{rpr}})} \tag{6.71}$$

where $\mathrm{clip}(x,k_{\mathrm{rpr}})$ is a function that clips x in the interval $[-k_{\mathrm{rpr}},k_{\mathrm{rpr}}]$

$$\mathrm{clip}(x,k_{\mathrm{rpr}}) \quad = \quad \max\{-k_{\mathrm{rpr}},\min\{x,k_{\mathrm{rpr}}\}\} \tag{6.72}$$

Thus, we have a model with parameters

$$\mathbf{U}_{\mathrm{rpr}} \quad = \quad \begin{bmatrix} \mathbf{u}_{-k_{\mathrm{rpr}}} \\ \vdots \\ \mathbf{u}_0 \\ \vdots \\ \mathbf{u}_{k_{\mathrm{rpr}}} \end{bmatrix} \tag{6.73}$$

While this notation is somewhat informal, we can view $\mathbf{U}_{\mathrm{rpr}}$ as a matrix $\in \mathbb{R}^{(2k_{\mathrm{rpr}}+1)\times d}$, and select a row corresponding to $\mathrm{clip}(j-i,k_{\mathrm{rpr}})$ when computing RPR for given i and j.

Using the above method, we can define three RPR models $\mathrm{PE}^q(i,j)$, $\mathrm{PE}^k(i,j)$ and $\mathrm{PE}^v(i,j)$ for queries, keys, and values, respectively. Then, following the form of Eq. (6.17), the output of the self-attention model at position i can be written as

$$\begin{aligned} \mathbf{c}_i \quad &= \quad \sum_{j=1}^{m} \alpha_{i,j} \left[\mathbf{h}_j^v + \mathrm{PE}^v(i,j)\right] \\ &= \quad \sum_{j=1}^{m} \alpha_{i,j}\mathbf{h}_j^v + \sum_{j=1}^{m} \alpha_{i,j}\mathrm{PE}^v(i,j) \end{aligned} \tag{6.74}$$

where $\mathbf{h}_j^v$ is the j-th row vector of $\mathbf{H}^v$. This representation comprises two components: $\sum_{j=1}^{m} \alpha_{i,j}\mathbf{h}_j^v$ is the basic representation, and $\sum_{j=1}^{m} \alpha_{i,j}\mathrm{PE}^v(i,j)$ is the positional representation.

The attention weight $\alpha_{i,j}$ is computed in a regular way, but with additional terms $\mathrm{PE}^q(i,j)$

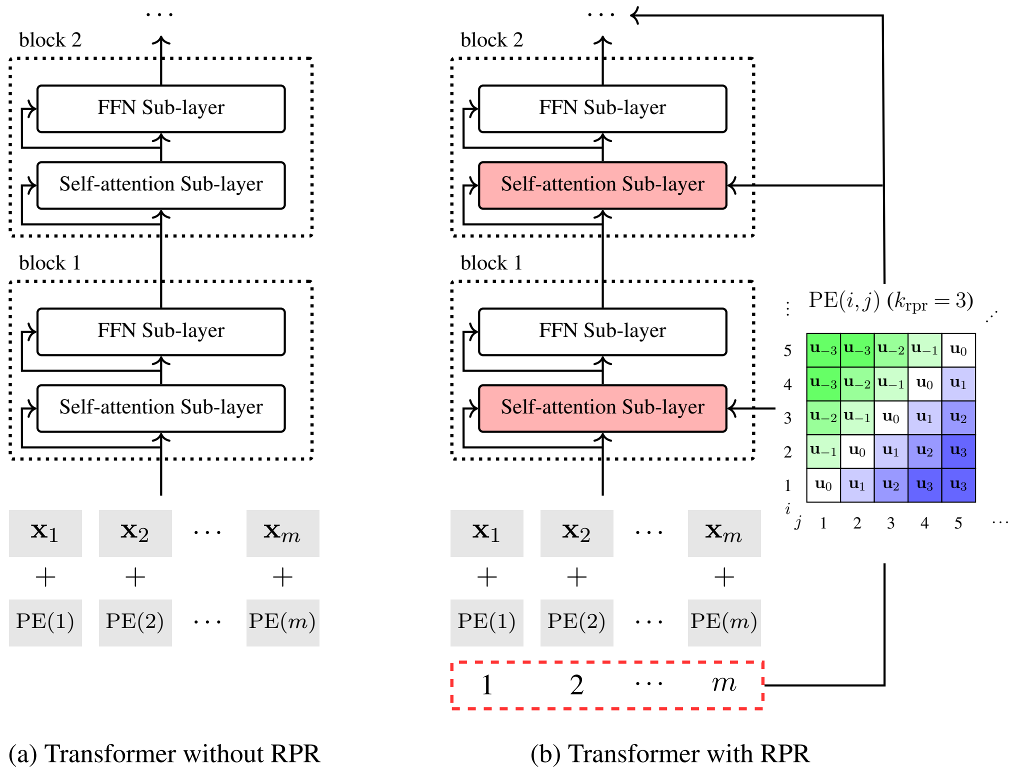


Figure 6.10: Transformer encoders without and with relative positional representation (RPR). In RPR, each pair of positions is represented as a vector $\mathrm{PE}(i,j)$ using a model parameterized by $\mathbf{U}_{\text{rpr}}$. $\mathrm{PE}(i,j)$ is fed into each self-attention sub-layer so that we can make use of the positional information in intermediate steps of learning representations.

and $\mathrm{PE}^k(i,j)$ added to each query and key.

$$\alpha_{i,j} = \mathrm{Softmax}(\frac{[\mathbf{h}_i^q + \mathrm{PE}^q(i,j)][\mathbf{h}_j^k + \mathrm{PE}^k(i,j)]^\top}{\sqrt{d}}) \tag{6.75}$$

Figure 6.10 shows the Transformer encoder architectures with and without RPR. When RPR is adopted, $\mathrm{PE}^q(i,j)$, $\mathrm{PE}^k(i,j)$, $\mathrm{PE}^v(i,j)$ are directly fed to each self-attention sub-layer, and so we can make better use of positional information for sequence modeling. Note that the use of the clipping function (see Eq. (6.72)) makes the modeling simple because we do not need to distinguish the relative distances for cases $|j-i| \geq k_{\text{rpr}}$. This clipped distance-based model can lead to better modeling in local context windows.

Eqs. (6.74) and (6.75) provide a general approach to position-sensitive sequence modeling. There are many variants of this model. In the early work on RPR by Shaw et al. [2018], the positional representations for queries are removed, and the model works only with $\mathrm{PE}^k(i,j)$

and $\mathrm{PE}^v(i,j)$, like this

$$\alpha_{i,j} = \mathrm{Softmax}(\frac{\mathbf{h}_i^q[\mathbf{h}_j^k + \mathrm{PE}^k(i,j)]^\top}{\sqrt{d}}) \tag{6.76}$$

By contrast, some variants attempt to improve the RPR model when computing attention weights but ignore $\mathrm{PE}^v(i,j)$ when computing value representations [Dai et al., 2019; He et al., 2021]. Instead of treating RPR as an additive term to each representation, researchers have also explored other ways of introducing RPR into the Transformer [Huang et al., 2020b; Raffel et al., 2020]. We refer interested readers to these papers for more details.

6.3.2 Deep Models

Many state-of-the-art NLP systems are based on deep Transformer models. For example, recent large language models generally comprise tens of Transformer layers (or more precisely, hundreds of layers), demonstrating strong performance on many tasks [Ouyang et al., 2022; Touvron et al., 2023a]. By stacking Transformer layers, it is straightforward to construct a deep model. However, as is often the case, training very deep neural networks is challenging. One difficulty arises from the fact that the error surfaces of deep neural networks are highly non-convex and contain many local optima, making the training process likely to get stuck. While there are optimization algorithms that can help alleviate this problem, most practical approaches rely on gradient-based methods for optimizing deep neural networks. As a result, training a model with many Transformer layers becomes challenging due to vanishing and exploding gradients during back-propagation. Here, we consider several techniques for training deep Transformer models.

1. Re-thinking the Pre-Norm and Post-Norm Architectures

As introduced previously, a Transformer sub-layer is a residual network where a shortcut is created to add the input of the network directly to the output of the sub-layer. This allows gradients to flow more directly from the output back to the input, mitigating the vanishing gradient problem. In general, a residual connection in a Transformer is used together with a layer normalization unit to form a sub-layer. This leads to two types of architectures, called post-norm and pre-norm. Specifically, recall from Section 6.1.4 that the post-norm architecture can be expressed as:

$$\mathbf{z}^l = \mathrm{LNorm}(F^l(\mathbf{z}^{l-1}) + \mathbf{z}^{l-1}) \tag{6.77}$$

where $\mathbf{z}^l$ and $\mathbf{z}^{l-1}$ are the output and input of the sub-layer l, respectively, and $F^l(\cdot)$ is the core function of this sub-layer. The pre-norm architecture takes the identity mapping $\mathbf{z}^{l-1}$ outside the layer normalization function, given in the form

$$\mathbf{z}^l = F^l(\mathrm{LNorm}(\mathbf{z}^{l-1})) + \mathbf{z}^{l-1} \tag{6.78}$$

Consider the difference between the information flow in these two architectures:

- The post-norm architecture prevents the identity mapping of the input from being directly added to the final output of the sub-layer. This is not a true residual network because all the information is passed through a non-linear function (i.e., the layer normalization unit). Thus, the post-norm architecture is less effective for back-propagation. Wang et al. [2019a] show that the gradient of the loss of an L-sub-layer Transformer network with respect to $\mathbf{z}^l$ is given by

$$\frac{\partial E}{\partial \mathbf{z}^l} = \frac{\partial E}{\partial \mathbf{z}^L} \cdot \prod_{k=l}^{L-1} \frac{\partial \mathrm{LNorm}(\mathbf{v}^k)}{\partial \mathbf{v}^k} \cdot \prod_{k=l}^{L-1} \left(1 + \frac{\partial F^k(\mathbf{z}^k)}{\partial \mathbf{z}^k}\right) \tag{6.79}$$

  where $\mathbf{z}^L$ is the output of the last layer, $\mathbf{v}^k$ is short for $F^k(\mathbf{z}^{k-1})$, and E is the error measured by some loss function. $\frac{\partial \mathrm{LNorm}(\mathbf{v}^k)}{\partial \mathbf{v}^k}$ and $\frac{\partial F^k(\mathbf{z}^k)}{\partial \mathbf{z}^k}$ are the gradients of the layer normalization function and the core function, respectively. Although the equation here appears complex, we see that $\prod_{k=l}^{L-1} \frac{\partial \mathrm{LNorm}(\mathbf{v}^k)}{\partial \mathbf{v}^k}$ is simply a product of $L - l$ factors. This means that the error gradient will be rescaled repeatedly as L becomes larger, leading to a higher risk of vanishing or exploding gradients in deeper models.

- The pre-norm architecture describes a standard residual neural network where the input of the entire network is added directly to its output. We can write the gradient of the error at $\mathbf{z}^l$ as:

$$\begin{aligned} \frac{\partial E}{\partial \mathbf{z}^l} &= \frac{\partial E}{\partial \mathbf{z}^L} \cdot \left(1 + \sum_{k=l}^{L-1} \frac{\partial F^k(\mathrm{LNorm}(\mathbf{z}^k))}{\partial \mathbf{z}^k}\right) \\ &= \frac{\partial E}{\partial \mathbf{z}^L} + \frac{\partial E}{\partial \mathbf{z}^L} \cdot \sum_{k=l}^{L-1} \frac{\partial F^k(\mathrm{LNorm}(\mathbf{z}^k))}{\partial \mathbf{z}^k} \end{aligned} \tag{6.80}$$

  It is easy to see that $\frac{\partial E}{\partial \mathbf{z}^l}$ receives direct feedback regarding the error, because the first term on the right-hand side (i.e., $\frac{\partial E}{\partial \mathbf{z}^L}$) is the gradient of the model output, which is independent of the network depth.

The use of the pre-norm architecture also helps optimization during early gradient descent steps. For example, it has been found that pre-norm Transformer models can be trained using a larger learning rate in the early stages of training, instead of requiring a gradual learning rate warmup from a small value [Xiong et al., 2020].

While the pre-norm architecture facilitates easier optimization of deep Transformer models, we cannot simply conclude that it is a universally better choice compared to the post-norm architecture. In fact, both post-norm and pre-norm Transformer models have been successfully used in many applications. For example, the post-norm architecture is widely used in BERT-like models, while the pre-norm architecture is the more popular choice for recent generative large language models. Broadly, these two architectures provide different paradigms for designing deep Transformer models, each with distinct advantages and disadvantages. The post-norm architecture forces the representation to be learned through more non-linear functions, resulting in a highly expressive but complicated model that is relatively hard to train. By contrast, the pre-norm architecture makes the training of Transformer models easier, but it may be less expressive than its post-norm counterpart if the learned model becomes overly dependent on the shortcut paths.

An improvement to these architectures is to control the extent to which we want to "skip" a sub-layer. A simple way to do this is to weight different paths rather than treating them equally. For example, a scalar factor of a residual connection can be introduced to determine how heavily we weight this residual connection relative to the path of the core function [He et al., 2016b; Liu et al., 2020a;b]. A more general form of this model is given by

$$\mathbf{z}^l = \mathrm{LNorm}(F^l(\mathbf{z}^{l-1}) + \beta \cdot \mathbf{z}^{l-1}) + \gamma \cdot \mathbf{z}^{l-1} \tag{6.81}$$

where β is the weight of the identity mapping inside the layer normalization function, and γ is the weight of the identity mapping outside the layer normalization function. Clearly, both the post-norm and pre-norm architectures can be seen as special cases of this equation. That is, if $\beta = 1$ and $\gamma = 0$, then it will become Eq. (6.77); if $\beta = 0$ and $\gamma = 1$, it will become Eq. (6.78). This model provides a multi-branch view of building residual blocks. The input to this block can be computed through multiple paths with different modeling complexities. When β and γ are small, the representation is forced to be learned through a "deep" model with multiple layers of cascaded non-linear units. In contrast, when β and γ are large, the representation is more likely to be learned using a "shallow" model with fewer layers. To determine the optimal choices of β and γ, one can give them fixed values by considering some theoretical properties or system performance on validation sets, or compute these values by using additional functions that can be trained to do so [Srivastava et al., 2015]. It should be emphasized that many other types of architecture can be considered in the design of a Transformer sub-layer. It is possible, for instance, to introduce more layer normalization units into a sub-layer [Ding et al., 2021; Wang et al., 2022b], or, conversely, to simply remove them from a sub-layer [Bachlechner et al., 2021].

2. Parameter Initialization

As with other deep neural networks, there is considerable interest in developing parameter initialization methods for deep Transformer models in order to facilitate optimization in a more favorable region of the parameter space. However, initialization is a broad topic in the optimization of machine learning models, and a general discussion of this topic lies beyond the scope of this section. Here, we will focus on specific parameter initialization methods used in Transformer-based systems rather than general optimization problems.

While the parameters of a neural network can be initialized in various ways, most practical systems adopt simple techniques to provide appropriate initial values. Consider, for example, the Xavier initialization for a parameter matrix $\mathbf{W} \in \mathbb{R}^{d_{\mathrm{in}} \times d_{\mathrm{out}}}$ [Glorot and Bengio, 2010]. We define a variable η as follows

$$\eta = \mathrm{gain} \cdot \sqrt{\frac{6}{d_{\mathrm{in}} + d_{\mathrm{out}}}} \tag{6.82}$$

where gain is a hyper-parameter which equals 1 by default. Then, each entry of $\mathbf{W}$ can be initialized by using a uniform distribution

$$W \sim U(-\eta, \eta) \tag{6.83}$$

or, alternatively, using a Gaussian distribution

$$W \sim \mathcal{N}(0, \eta^2) \tag{6.84}$$

This method can be easily adapted to initialize Transformer models with a large number of layers. One common approach is to find a more suitable value for gain by taking into account the fact that the optimal initial conditions for optimization might differ across neural networks of varying depths. For example, one can increase the value of gain as the depth of the model grows. Thus, gain can be defined as a function of the network depth in the following form

$$\mathrm{gain} = a \cdot L^b \tag{6.85}$$

where a is a scalar, and L^b is the network depth raised to the power of b. Typically, both a and b are positive numbers, indicating a preference for larger initial parameter values in deeper models. For example, Wang et al. [2022a] demonstrate that by choosing appropriate values for a and b, a very deep Transformer model can be successfully trained.

Eq. (6.85) assigns gain the same value for all of the sub-layers. However, it is found that the norm of gradients becomes smaller as a sub-layer is located farther from the output layer. This consistent application of gain across the entire model could result in under-training of the lower layers due to the gradient vanishing problem. For this reason, one can develop methods that are sensitive to the position of a sub-layer in the neural network. The general form of such methods is given by

$$\mathrm{gain} = \frac{a}{l^b} \tag{6.86}$$

Here l denotes the depth of a sub-layer. If l is larger (i.e., the sub-layer is closer to the output), gain will be smaller and the corresponding parameters will be set to smaller values. An example of this method can be found in the work of Zhang et al. [2019a].

It is also, of course, straightforward to apply general methods of initializing deep multi-layer neural networks to Transformer models. An example is to consider the **Lipschitz constant** in parameter initialization, which has been shown to help improve the stability of training deep models [Szegedy et al., 2014b; Xu et al., 2020]. Another approach is to use second-order methods to estimate the proper values of the parameters. For example, one can compute the Hessian of each parameter matrix to model its curvature [Skorski et al., 2021].

For models with a large number of layers, it is also possible to pre-train some of the layers via smaller models and use their trained parameters to initialize bigger models [Chen et al., 2015]. That is, we first obtain a rough estimation of the parameters in a cheap way, and then continue the training process on the whole model as usual. These methods fall into a class of training methods, called **model growth** or **depth growth**.

As a simple example, consider a Transformer model (e.g., a Transformer encoder) of $2L$ sub-layers. We can train this model by using the **shallow-to-deep training** method [Li et al., 2020b]. First, we train an L-sub-layer model (call it the shallow model) in a regular way. Then, we create a $2L$-sub-layer model (call it the deep model) by stacking the shallow model twice, and further train this deep model. To construct deeper models, this procedure can be repeated multiple times, say, we start with a model of L sub-layers, and obtain a model of $L \cdot 2^I$ after I iterations. Note that many pre-trained models are used in a similar manner. For example, for BERT-like methods, a Transformer encoder is trained on large-scale data, and the optimized parameters are then used to initialize downstream systems.

3. Layer Fusion

Another problem with training a deep Transformer model is that the final prediction is conditioned on the last layer of the neural network. While the use of residual connections enables direct access to lower-level layers from a higher-level layer, there is still a "long" path for passing information from the bottom to the top. One simple way to address this is to create residual connections that skip more layers. For example, consider a group of L Transformer sub-layers. For the sub-layer at depth l, we can introduce $l-1$ residual connections, each connecting this sub-layer to a previous sub-layer. In this way, we develop a densely connected network where each sub-layer takes as input the outputs of all previous sub-layers [Huang et al., 2017a]. The output of the last sub-layer can be seen as a combination of representations at different levels.

Following the notation used in the previous subsections, we denote the output of the sub-layer at depth l by $\mathbf{z}^l$, and denote the function of the sub-layer by $\mathrm{Layer}^l(\cdot)$. Then, $\mathbf{z}^l$ can be expressed as

$$\mathbf{z}^l = \mathrm{Layer}^l(\mathbf{z}^1, ..., \mathbf{z}^{l-1}) \tag{6.87}$$

We can simply view $\mathrm{Layer}^l(\cdot)$ as a function that fuses the information from $\{\mathbf{z}^1, ..., \mathbf{z}^{l-1}\}$. There are many possible choices for $\mathrm{Layer}^l(\cdot)$. For example, a simple form of $\mathrm{Layer}^l(\cdot)$ is given by

$$\mathrm{Layer}^l(\mathbf{z}^1, ..., \mathbf{z}^{l-1}) = \mathrm{LNorm}(F^l(\mathbf{Z}^l)) \tag{6.88}$$

$$\mathbf{Z}^l = \phi(\mathbf{z}^1, ..., \mathbf{z}^{l-1}) \tag{6.89}$$

Here $\phi(\cdot)$ takes the layer outputs $\{\mathbf{z}^1, ..., \mathbf{z}^{l-1}\}$ and fuses them into a single representation $\mathbf{Z}^l$. A simple instance of $\phi(\cdot)$ is average pooling which computes the sum of $\{\mathbf{z}^1, ..., \mathbf{z}^{l-1}\}$ divided by $l-1$. See Table 6.2 for more examples of $\phi(\cdot)$.

Adopting a similar architecture for a Transformer sub-layer, we can also consider a post-

Entry	Function
Average Pooling	$\phi(\mathbf{z}^1,...,\mathbf{z}^{l-1}) = \frac{1}{l-1}\sum_{k=1}^{l-1}\mathbf{z}^k$
Weighted Sum	$\phi(\mathbf{z}^1,...,\mathbf{z}^{l-1}) = \sum_{k=1}^{l-1}\mathrm{weight}_k \cdot \mathbf{z}^k$
Feedforward Network	$\phi(\mathbf{z}^1,...,\mathbf{z}^{l-1}) = \mathrm{FFN}([\mathbf{z}^1,...,\mathbf{z}^{l-1}])$
Self-attention	$\phi(\mathbf{z}^1,...,\mathbf{z}^{l-1}) = \mathrm{FFN}([\mathrm{Att}_{\mathrm{self}}(\mathbf{z}^1,...,\mathbf{z}^{l-1})])$

Table 6.2: Fusion functions. $\mathrm{FFN}(\cdot)$ = feedforward neural network, $[\cdot]$ = concatenating the input vectors, and $\mathrm{Att}_{\mathrm{self}}(\cdot)$ = self-attention function. All fusion functions can be followed by a layer normalization function, for example, we can write the weighted sum of $\{\mathbf{z}^1,...,\mathbf{z}^{l-1}\}$ as $\phi(\mathbf{z}^1,...,\mathbf{z}^{l-1}) = \mathrm{LNorm}(\sum_{k=1}^{l-1}\mathrm{weight}_k \cdot \mathbf{z}^k)$.

norm form

$$\begin{aligned} \mathrm{Layer}^l(\mathbf{z}^1,...,\mathbf{z}^{l-1}) &= \mathrm{LNorm}(\mathbf{Z}^l) & (6.90) \\ \mathbf{Z}^l &= \phi(F^l(\mathbf{z}^{l-1}),\mathbf{z}^1,...,\mathbf{z}^{l-1}) & (6.91) \end{aligned}$$

or a pre-norm form

$$\begin{aligned} \mathrm{Layer}^l(\mathbf{z}^1,...,\mathbf{z}^{l-1}) &= \mathbf{Z}^l & (6.92) \\ \mathbf{Z}^l &= \phi(\mathrm{LNorm}(F^l(\mathbf{z}^{l-1})),\mathbf{z}^1,...,\mathbf{z}^{l-1}) & (6.93) \end{aligned}$$

These models are quite general. For example, a standard post-norm encoder sub-layer can be recovered as a special case of Eqs. (6.90-6.91), if we remove the dependencies on sub-layers 1 to $l-2$, and define $\phi(\cdot)$ to be

$$\phi(F^l(\mathbf{z}^{l-1}),\mathbf{z}^1,...,\mathbf{z}^{l-1}) = F^l(\mathbf{z}^{l-1}) + \mathbf{z}^{l-1} \quad (6.94)$$

Densely connected networks make it easier for information to flow through direct connections between sub-layers, but the resulting models are slightly more complex, especially when we use parameterized fusion functions. In practice, we typically add dense connections only to a subset of the sub-layers, so the overall network is not overly dense. For example, we might only add connections from lower sub-layers to the final few sub-layers. Thus, predictions can be made with direct access to different levels of representation [Wang et al., 2018a].

4. Regularization

In machine learning, regularization is used to mitigate overfitting when training deep neural networks. It is therefore straightforward to apply regularization techniques to Transformer models. Since the topic of regularization has already been discussed in Chapter 2, here we consider several methods that have not yet been covered in this book but are particularly useful for training deep Transformer models.

One approach to regularizing deep Transformer models is to randomly skip sub-layers or layers during training [Huang et al., 2016; Pham et al., 2019]. In each run of the model, we retain each sub-layer with a probability ρ and stack the selected sub-layers to form a new model.

Thus, we essentially train an ensemble of different neural networks with shared architectures and parameters on the same dataset. In this way, each sub-layer learns to operate somewhat independently, and overfitting is reduced by preventing the co-adaptation of sub-layers. In fact, dropping out sub-layers (or layers) and dropping out neurons are essentially variations on the same theme. Sometimes, the method described here is referred to as **sub-layer dropout** or **layer dropout**.

At test time, we need to combine all possible networks to make predictions. A simple method to achieve this is to rescale the outputs of the stochastic components of the model [Li et al., 2021a]. As an example, suppose each sub-layer has a pre-norm architecture. Then, the output of the sub-layer at depth l is given by

$$\mathbf{z}^l = \rho \cdot \mathrm{LNorm}(F^l(\mathbf{z}^{l-1})) + \mathbf{z}^{l-1} \tag{6.95}$$

Another idea is to share parameters across sub-layers. One of the simplest methods is to use the exact same parameters for all corresponding sub-layers [Dehghani et al., 2018]. For example, all the FFN sub-layers could be based on the same feed-forward network. This approach has a regularizing effect similar to methods that add the norms of parameter matrices to the loss function to penalize model complexity. For practical systems, there can be significant benefits to adopting a shared architecture, as it allows us to reuse the same sub-model to build a deep neural network, thereby substantially reducing the overall memory footprint. We will see further discussion regarding efficiency issues in Section 6.4.4.

6.3.3 Numerical Method-Inspired Models

A residual network computes its output as the sum of the identity mapping and a transformation of the input. Such a model can be interpreted as an Euler discretization of **ordinary differential equations** (**ODEs**) [Ee, 2017; Haber and Ruthotto, 2017]. To illustrate this idea, we consider a general form of residual networks

$$\mathbf{z}^l = f^l\big(\mathbf{z}^{l-1}\big) + \mathbf{z}^{l-1} \tag{6.96}$$

where $f^l(\mathbf{z}^{l-1})$ denotes a function that takes an input variable $\mathbf{z}^{l-1}$ and produces an output variable in the same space. Clearly, a Transformer sub-layer is a special case of this equation. For example, for a pre-norm Transformer, we have $f^l(\cdot) = \mathrm{LNorm}(F^l(\cdot))$.

For notational simplicity, we rewrite the above equation in an equivalent form

$$\mathbf{z}(l) = f\big(\mathbf{z}(l-1), l\big) + \mathbf{z}(l-1) \tag{6.97}$$

We use the notations $\mathbf{z}(l)$ and $f(\mathbf{z}(\cdot, l))$ to emphasize that $\mathbf{z}(\cdot)$ and $f(\cdot)$ are functions of l. Here we assume that l is a discrete variable. If we relax l to a continuous variable and $\mathbf{z}(l)$ to a continuous function of l, then we can express Eq. (6.97) as

$$\mathbf{z}(l) = \triangle l \cdot f\big(\mathbf{z}(l - \triangle l), l\big) + \mathbf{z}(l - \triangle l) \tag{6.98}$$

This can be further written as

$$\frac{\mathbf{z}(l)-\mathbf{z}(l-\triangle l)}{\triangle l} \quad = \quad f\big(\mathbf{z}(l-\triangle l), l\big) \tag{6.99}$$

Taking the limit $\triangle l \to 0$, we have an ODE

$$\frac{\mathrm{d}\mathbf{z}(l)}{\mathrm{d}l} \quad = \quad f\big(\mathbf{z}(l), l\big) \tag{6.100}$$

We can therefore say that a pre-norm Transformer sub-layer (i.e., Eqs. (6.97) and (6.96)) is an Euler discretization of the above ODE. This is an interesting result! A sub-layer essentially serves as an ODE solver.

Eqs. (6.97) and (6.96) represent standard forms of the **Euler method**. This method computes a new estimate of the solution by moving one step forward along l from the previous estimation. In general, two aspects can be considered in the design of numerical methods for ODEs.

- **Linear Multi-step Methods**. A linear multi-step method computes the current estimate of the solution by taking the estimations and derivative information from multiple previous steps. A general formulation of p-step methods can be expressed as

$$\mathbf{z}(l) \quad = \quad \sum_{i=1}^{p} a_i \cdot \mathbf{z}(l-i) + h \sum_{i=1}^{p+1} b_i \cdot f\big(\mathbf{z}(l-i),\ l-i+1\big) \tag{6.101}$$

  where h is the size of the step we move each time[8], that is, $\triangle l$ in Eqs. (6.98) and (6.99). $\{a_i\}$ and $\{b_i\}$ are coefficients of the solution points and derivatives in the linear combination. Given this definition, we can think of the Euler method as a single-step, low-order method of solving ODEs[9].

- **(Higher-order) Runge-Kutta Methods**. Runge-Kutta (RK) methods and their variants provide ways to compute the next step solution by taking intermediate results in solving an ODE. As a result, we obtain higher-order methods but still follow the form of single-step methods, that is, the estimated solution is dependent only on $\mathbf{z}(l-1)$ rather than on the outputs at multiple previous steps.

In fact, linear multi-step methods, though not explicitly mentioned, have been used in the layer fusion methods discussed in Section 6.3.2. For example, taking Eqs. (6.92) and (6.93) and a linear fusion function, a pre-norm sub-layer with dense connections to all previous sub-layers can be expressed as

$$\mathrm{Layer}^l(\mathbf{z}^1,...,\mathbf{z}^{l-1}) \quad = \quad a_1 \cdot \mathbf{z}^{l-1} + ... + a_{l-1} \cdot \mathbf{z}^1 + b_1 \cdot \mathrm{LNorm}(F^l(\mathbf{z}^{l-1})) \tag{6.102}$$

[8] Let $\{t_0,...,t_i\}$ denote the values of the variable l at steps $\{0,...,i\}$. In linear multi-step methods, it is assumed that $t_i = t_0 + ih$.

[9] In numerical analysis, the **local truncation error** of a method of solving ODEs at a step is defined to be the difference between the approximated solution computed by the method and the true solution. The method is called order p if it has a local truncation error $O(h^{p+1})$.

This equation is an instance of Eq. (6.101) where we set $h = 1$ and remove some of the terms on the right-hand side.

It is also straightforward to apply Runge-Kutta methods to Transformer models [Li et al., 2022a]. Given an ODE as described in Eq. (6.100), an explicit p-th order Runge-Kutta solution is given by

$$\mathbf{z}(l) = \mathbf{z}(l-1) + \sum_{i=1}^{p} \gamma_i \cdot \mathbf{g}_i \tag{6.103}$$

$$\mathbf{g}_i = h \cdot f\big(\mathbf{z}(l-1) + \sum_{j=1}^{i-1} \beta_{i,j} \cdot \mathbf{g}_j,\ l-1+\lambda_i \cdot h\big) \tag{6.104}$$

Here $\mathbf{g}_i$ represents an intermediate step which is present only during the above process. $\{\gamma_i\}$, $\{\beta_{i,j}\}$ and $\{\lambda_i\}$ are coefficients that are determined by using the Taylor series of $\mathbf{z}(l)$. To simplify the model, we assume that the same function f is used for all $\{\mathbf{g}_i\}$. Then, we remove the dependency on the term $l-1+\lambda_i \cdot h$ in f, and rewrite Eq. (6.104) as

$$\mathbf{g}_i = h \cdot f\big(\mathbf{z}(l-1) + \sum_{j=1}^{i-1} \beta_{i,j} \cdot \mathbf{g}_j\big) \tag{6.105}$$

where $f(\cdot)$ is a function that is independent of i.

As an example, consider the 4th-order Runge-Kutta (RK4) solution

$$\mathbf{z}(l) = \mathbf{z}(l-1) + \frac{1}{6}(\mathbf{g}_1 + 2\mathbf{g}_2 + 2\mathbf{g}_3 + \mathbf{g}_4) \tag{6.106}$$

$$\mathbf{g}_1 = h \cdot f(\mathbf{z}(l-1)) \tag{6.107}$$

$$\mathbf{g}_2 = h \cdot f(\mathbf{z}(l-1) + \frac{1}{2}\mathbf{g}_1) \tag{6.108}$$

$$\mathbf{g}_3 = h \cdot f(\mathbf{z}(l-1) + \frac{1}{2}\mathbf{g}_2) \tag{6.109}$$

$$\mathbf{g}_4 = h \cdot f(\mathbf{z}(l-1) + \mathbf{g}_3) \tag{6.110}$$

These equations define a new architecture of sub-layer. For example, by setting $h = 1$ and $f(\cdot) = \mathrm{LNorm}(F^l(\cdot))$, we obtain an RK4 Transformer sub-layer, as shown in Figure 6.11. This method leads to a deep model because each sub-layer involves four evaluations of $f(\cdot)$ in sequence. On the other hand, the resulting model is parameter efficient because we reuse the same function $f(\cdot)$ within the sub-layer, without introducing additional parameters.

So far in this subsection, our discussion has focused on designing Transformer architectures inspired by dynamical systems. While the basic ODE model is continuous with respect to the depth l, these methods still follow the general framework of neural networks in which l is treated as a discrete variable, and the representational power of the models is largely determined by this hyper-parameter. An alternative approach is to use neural ODEs to relax the "depth" into a continuous variable. In this way, we can obtain a continuous-depth model for computing the solution to ODEs. However, as a detailed discussion of neural ODEs lies

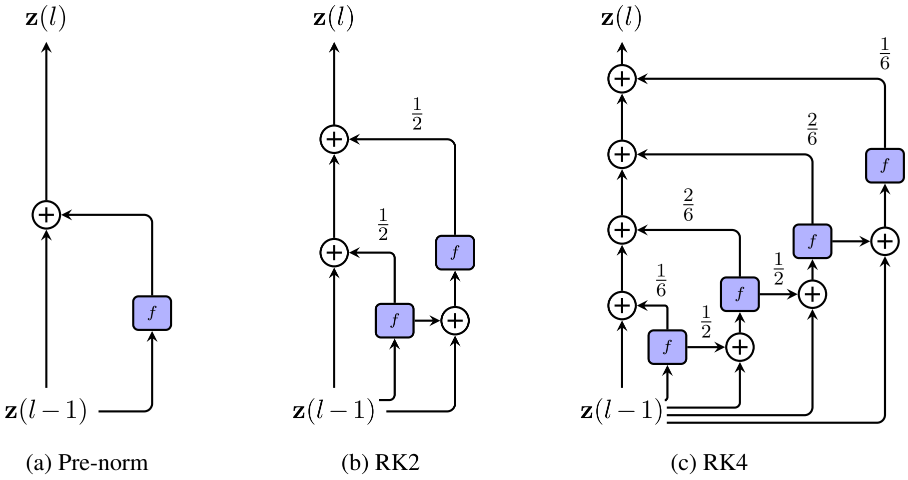


Figure 6.11: Pre-norm (a) and Runge-Kutta (b and c) sub-layer architectures. $\mathbf{z}(l-1)$ denotes the input of a sub-layer at depth l, $\mathbf{z}(l)$ denotes the output of the sub-layer, and f (in blue boxes) denotes the function $f(\cdot) = \text{LNorm}(F^l(\cdot))$.

beyond the scope of this chapter, we refer interested readers to related papers for more details [Chen et al., 2018c; Kidger, 2022; Xiao et al., 2026].

6.3.4 Wide Models

Most of the methods that we have studied so far in this section are examples of learning and using deep models. Another design choice we generally face is determining the width of a neural network. Typically, the **width** of a Transformer model is defined as the dimensionality of a representation at some position in the input sequence, that is, the parameter d. Increasing this width is a common method for obtaining a more complex and powerful model. For example, in the work of Vaswani et al. [2017], a wide model (referred to as Transformer-Big) leads to significant improvements in translation quality for machine translation systems. Such wide models have been widely used to boost system performance on large-scale tasks [Lepikhin et al., 2021; Fedus et al., 2022b].

However, developing very wide Transformer models is difficult. One major challenge is that training such systems is computationally expensive. While the number of model parameters (the model size) grows linearly with d, the time complexity of the model grows quadratically with d (see Table 6.1). In some NLP tasks, it has been found empirically that the training effort required to obtain satisfactory performance follows a scaling law [Kaplan et al., 2020]. These results emphasize the need for ways to improve training efficiency when enlarging d.

One simple method is to incrementally grow the model along the dimension of d, rather than training the model from scratch. Suppose we have an initial model with a $d_1 \times d_1$ parameter matrix $\mathbf{W}_1$, for example, the linear transformation of each query or key in some layer. We can train this model to obtain optimized $\mathbf{W}_1$ in a regular way. Then, we want to extend this model to a wider model where $\mathbf{W}_1$ is replaced by a $d_2 \times d_2$ parameter matrix $\mathbf{W}_2$.

Let us assume for simplicity that $d_2 = kd_1$. There are several ways to expand a $d_1 \times d_1$ matrix to a $kd_1 \times kd_1$ matrix. The simplest of these may be to use $\mathbf{W}_1$ to fill $\mathbf{W}_2$. We can write $\mathbf{W}_2$ in the form

$$\mathbf{W}_2 = \begin{array}{cc} \text{k times} & \\ \begin{bmatrix} \frac{\mathbf{W}_1}{\rho} & \dots & \frac{\mathbf{W}_1}{\rho} \\ \vdots & & \vdots \\ \frac{\mathbf{W}_1}{\rho} & \dots & \frac{\mathbf{W}_1}{\rho} \end{bmatrix} & \text{k times} \end{array} \tag{6.111}$$

where ρ is a hyper-parameter that is used to control the norm of $\mathbf{W}_2$. For example, if $\rho = k$, $\mathbf{W}_2$ will have the same l_1 norm as $\mathbf{W}_1$. The above equation provides a good starting point for training the wide model, and we can train $\mathbf{W}_2$ as usual after initialization. The procedure can be repeated a number of times for constructing a model with arbitrary width. Both this method and the depth growth method described in Section 6.3.2 are instances of the general method of model growth. In other words, we can obtain a larger model by extending a small model either vertically or horizontally, or both. Alternative methods for transforming $\mathbf{W}_1$ to $\mathbf{W}_2$ involve those considering other mathematical properties of the transformation [Chen et al., 2015]. These models can fall under the reusable neural networks where we are concerned with models and algorithms for transferring parameters from small models to (significantly) larger models [Wang et al., 2023b].

A second difficulty in building a wide Transformer model is the large memory requirement. Since the feedforward network generally has a larger hidden layer than other parts of the model, it demands relatively more memory as the model becomes wider. Consider the feedforward network described in Section 6.1.5

$$\begin{aligned} \mathbf{H}_{\text{out}} &= \text{FFN}(\mathbf{H}_{\text{in}}) \\ &= \text{ReLU}(\mathbf{H}_{\text{in}} \cdot \mathbf{W}_h + \mathbf{b}_h) \cdot \mathbf{W}_f + \mathbf{b}_f \end{aligned} \tag{6.112}$$

where $\mathbf{W}_h \in \mathbb{R}^{d \times d_{\text{ffn}}}$ and $\mathbf{W}_f \in \mathbb{R}^{d_{\text{ffn}} \times d}$ are the parameters of the linear transformations. d_{ffn} is typically several times larger than d. Therefore, $\mathbf{W}_h$ and $\mathbf{W}_f$ will dominate the model size if d and d_{ffn} have very large values.

In some cases, the size of the feedforward network may exceed the memory capacity of a single device. This problem can be addressed by using the **mixture-of-experts** (**MoE**) models [Shazeer et al., 2017]. An MoE model consists of M expert models $\{e_1(\cdot), ..., e_M(\cdot)\}$. Given an input $\mathbf{h}_{\text{in}} \in \mathbb{R}^d$, each expert model produces an output $e_k(\mathbf{h}_{\text{in}})$. The output of the MoE model is a linear combination of $\{e_1(\mathbf{h}_{\text{in}}), ..., e_M(\mathbf{h}_{\text{in}})\}$, given by

$$\mathbf{h}_{\text{out}} = \sum_{i=1}^{M} g_i(\mathbf{h}_{\text{in}}) \cdot e_i(\mathbf{h}_{\text{in}}) \tag{6.113}$$

where $g(\cdot)$ is a gating model (also called **routing model**). Its output is a vector $g(\mathbf{h}_{\text{in}}) = \begin{bmatrix} g_1(\mathbf{h}_{\text{in}}) & \dots & g_M(\mathbf{h}_{\text{in}}) \end{bmatrix}$ in which each entry $g_i(\mathbf{h}_{\text{in}})$ indicates the weight of the corresponding expert model. In many applications, it is assumed that $g(\mathbf{h}_{\text{in}})$ is a sparse vector. This means that only a small number of expert models are involved in computing the output. A widely-used form of $g(\mathbf{h}_{\text{in}})$ is given by using the Softmax layer

$$g(\mathbf{h}_{\text{in}}) = \text{Softmax}(\mathbf{h}_{\text{in}} \cdot \mathbf{W}_g) \tag{6.114}$$

where $\mathbf{W}_g \in \mathbb{R}^{d \times M}$ is the parameter matrix of the layer. To enforce sparsity on $g(\mathbf{h}_{\text{in}})$, we can select the top-$k$ entries of $g(\mathbf{h}_{\text{in}})$, that is, we set non-top-k entries to 0. An alternative method is to first perform top-k selection on $\mathbf{h}_{\text{in}} \cdot \mathbf{W}_g$ and then normalize the top-k entries using the Softmax function.

Let π be the set of the indices of the top-k expert models. The MoE model with top-k routing has the following form

$$\mathbf{h}_{\text{out}} = \sum_{i \in \pi} g_i(\mathbf{h}_{\text{in}}) \cdot e_i(\mathbf{h}_{\text{in}}) \tag{6.115}$$

An advantage of this approach is that we can distribute different expert models to different processors, making it possible to execute these models on parallel computing machines. In each run of the MoE model, either during training or inference, we only need to activate and use k expert models rather than all of the expert models. In this way, the MoE approach is automatically learning a sparse model by limiting the number of active expert models each time in training and inference. The sparsity is determined by the hyperparameter k, e.g., a small value of k leads to a sparse model, and a large value of k leads to a dense model.

Let us return to the discussion of Eq. (6.112). It is straightforward to apply the MoE approach to feedforward neural networks. To simplify the discussion, consider the linear transformation of the first layer as shown in Eq. (6.112), that is, $\mathbf{H}_{\text{in}} \cdot \mathbf{W}_h$. We can approximate $\mathbf{H}_{\text{in}} \cdot \mathbf{W}_h$ in an MoE form

$$\begin{aligned} \mathbf{H}_{\text{in}} \cdot \mathbf{W}_h &\approx \sum_{i \in \pi} g_i(\mathbf{H}_{\text{in}}) \cdot e_i(\mathbf{H}_{\text{in}}) \\ &= \sum_{i \in \pi} g_i(\mathbf{H}_{\text{in}}) \cdot [\mathbf{H}_{\text{in}} \cdot \mathbf{W}_h^i] \end{aligned} \tag{6.116}$$

Here $\mathbf{W}_h$ is divided into M slices (or sub-matrices) $\{\mathbf{W}_h^1, ..., \mathbf{W}_h^M\}$, written as

$$\mathbf{W}_h = \begin{bmatrix} \mathbf{W}_h^1 & ... & \mathbf{W}_h^M \end{bmatrix} \tag{6.117}$$

Hence each expert model $e_i(\mathbf{H}_{\text{in}}) = \mathbf{H}_{\text{in}} \cdot \mathbf{W}_h^i$ solves a sub-problem of the original linear mapping, and Eq. (6.116) can be thought of as a divide-and-conquer solution to the matrix multiplication problem.

We can, of course, treat any feedforward neural network as an expert model, resulting in

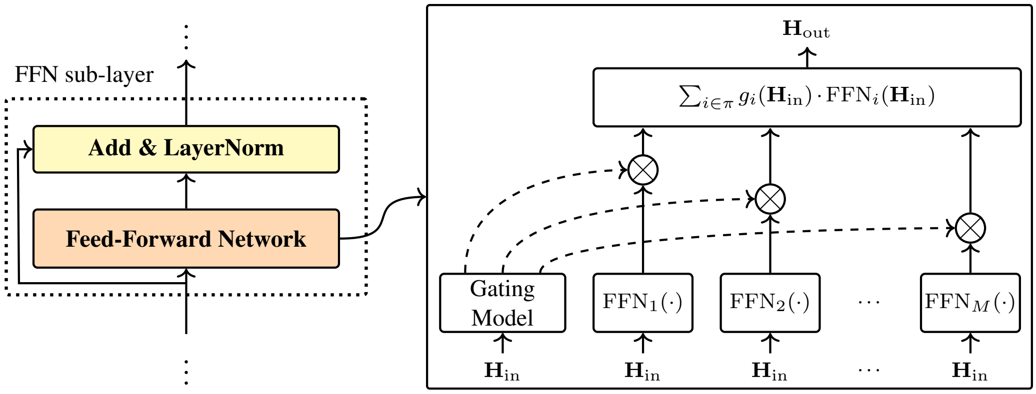


Figure 6.12: An illustration of the MoE model applied to an FFN sub-layer. There are M FFNs (call them expert models) and a gating model. Each FFN is weighted by the gating model. The output of the model is the sum of the weighted outputs of the top-k FFNs (denoted by π). Because these FFNs work independently and can be placed on different computing devices, the model can be easily scaled up as M is larger.

the following model

$$\mathbf{H}_{\mathrm{out}} = \sum_{i\in\pi} g_i(\mathbf{H}_{\mathrm{in}}) \cdot \mathrm{FFN}_i(\mathbf{H}_{\mathrm{in}}) \tag{6.118}$$

where $\mathrm{FFN}_i(\cdot)$ is a "small" feedforward neural network that has the same form as Eq. (6.112). This model is illustrated with an example in Figure 6.12. In practical implementations, all these expert models can be run in parallel on different devices, and so the resulting system is efficient.

Note that, from a perspective of machine learning, MoE is a general approach to combining different neural networks, each of which is developed to address a different aspect of the problem [Yuksel et al., 2012; Masoudnia and Ebrahimpour, 2014]. The application here is just a special instance of the general framework of MoE. The approach is also often used to improve the overall performance of predictors, which can be discussed in the field of ensemble learning [Zhou, 2012a].

Another difficulty in developing massive Transformer models is training instability. As with many other exceptionally large neural networks, the straightforward optimization of a Transformer model with billions of parameters may lead to the model getting trapped in local minima and, occasionally, to large spikes in the loss during training [Lepikhin et al., 2021; Fedus et al., 2022b; Chowdhery et al., 2022]. Even with careful tuning of hyperparameters, training strategies, and initial parameter values, it is common to encounter situations where the training must be restarted from an earlier checkpoint to escape unstable optimization regions. One of the reasons for this training difficulty is that the standard implementations of linear algebra operations, such as matrix multiplication, can become numerically unstable when they operate on extremely large vectors and matrices. It is therefore crucial to improve training stability by employing numerically robust methods and mixed-precision techniques.

6.4 Efficient Models

Efficiency is an important consideration for many practical applications of Transformer models. For example, we may wish to run and/or train a Transformer model given memory and time constraints. Efficiency is not a single issue, but covers a wide range of problems. While these problems can be categorized in several different ways, there are two fundamental aspects one may consider in an efficiency problem.

- **Time and Space Efficiencies**. For a given problem, we wish the model to be small and fast, while remaining as accurate as possible in solving the problem. For example, in some machine translation applications, we may develop a model with a small number of parameters to fit into limited memory, and may develop a fast search algorithm to achieve low-latency translation. A practical difficulty here is that improving efficiency often leads to worse predictions. In many cases, we need to seek a trade-off between efficiency and accuracy.
- **Scalability**. When the problem is scaled up, we wish that the additional computational overhead required is minimized. For example, the training of a neural network is called efficient if it takes a reasonably short time to optimize it as the volume of training data increases. Another metric of efficiency is the amount of resources consumed in processing larger inputs. For instance, a machine translation system is inefficient in translating long sentences if the memory footprint and latency grow exponentially with the number of input words.

In this section, we will not discuss all the issues related to efficiency, which is a very broad topic. We instead consider the widely-used efficient approaches to Transformer-based sequence modeling and generation, some of which are refinements of model architectures, and some of which are architecture-agnostic approaches and could be used in other systems as well. Most of the discussions here are focused on developing lightweight and fast Transformer models that are relatively robust to long input and output sequences.

In general, the same optimization method can be applied to different modules of a Transformer-based system. To simplify the discussion, we will mostly consider self-attention sub-layers and FFN sub-layers in this section. Our discussion, however, is general and the methods presented here can be applied to other parts of a Transformer model, for example, cross-attention sub-layers.

6.4.1 Sparse Attention

In practice, the attention mechanisms used in Transformers are time-consuming, especially when the input sequences are long. To illustrate, consider a Transformer decoder that predicts a distribution over the vocabulary at each time step given the previous words. Suppose the sequence generated by the decoder has length n and the input to a self-attention sub-layer is an $n \times d$ matrix $\mathbf{S}$. First, $\mathbf{S}$ is linearly transformed to obtain the queries $\mathbf{S}^q \in \mathbb{R}^{n \times d}$, keys $\mathbf{S}^k \in \mathbb{R}^{n \times d}$, and values $\mathbf{S}^v \in \mathbb{R}^{n \times d}$. To simplify the notation in this subsection, we use $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ to represent $\mathbf{S}^q$, $\mathbf{S}^k$, and $\mathbf{S}^v$, respectively.

The output of the self-attention sub-layer can then be computed using

$$\mathrm{Att}_{\mathrm{self}}(\mathbf{S}) = \mathbf{AV} \tag{6.119}$$

where $\mathbf{A}$ is an $n \times n$ attention matrix or attention map

$$\mathbf{A} = \mathrm{Softmax}(\frac{\mathbf{QK}^{\top}}{\sqrt{d}} + \mathbf{M}) \tag{6.120}$$

Here, $\mathbf{M}$ is a masking matrix used to prevent the model from attending to future context at each position. Specifically, for a position i, $M(i,j) = 0$ if $j \le i$, and $M(i,j) = -\infty$ otherwise. Both the time and space complexity of the self-attention sub-layer scale quadratically with n[10]. As a result, standard self-attention becomes computationally prohibitive for large values of n.

The standard implementation of the above model relies on dense matrix computations, such as the multiplications in Eqs. (6.119) and (6.120). One common approach to reducing both memory footprint and floating-point operations is sparsification. By assuming $\mathbf{A}$ is a sparse matrix, only a fraction $\varrho \cdot n^2$ of its entries remain non-zero, where ϱ is the **sparsity ratio**. Using sparse matrix representations drastically reduces memory requirements. Furthermore, computing $\frac{\mathbf{QK}^{\top}}{\sqrt{d}}$ and $\mathbf{AV}$ becomes more efficient, as the model only processes a small subset of relevant positions.

Given a position i, we define the **attention field** π_i to be the set of positions that are considered in computing the representation at this position. We therefore only need to compute the dot-product attention between the given position i and each position $j \in \pi_i$. This results in a sparse attention matrix $\mathbf{A}'$ where

$$A'(i,j) = \begin{cases} a_{i,j} & j \in \pi_i \text{ and } j \le i \\ 0 & \text{otherwise} \end{cases} \tag{6.121}$$

where $a_{i,j}$ is a non-zero weight. A simple implementation of this model involves a slight modification to $\mathbf{M}$, leading to a new masking variable $\mathbf{M}'$

$$M'(i,j) = \begin{cases} 0 & j \in \pi_i \text{ and } j \le i \\ -\infty & \text{otherwise} \end{cases} \tag{6.122}$$

In practical implementations, a more efficient approach is to employ sparse operations for $\mathbf{QK}^{\top}$ and $\mathbf{A}'\mathbf{V}$ by considering $\mathbf{M}'$ and $\mathbf{A}'$, respectively. That is, we compute only for pairs with non-zero attention weights and skip the rest.

Several approaches to sparse self-attention can be considered.

- **Span-based Attention/Local Attention**. As discussed in Section 6.3.1, the use of context in sequence modeling is local in many cases. The basic idea of local attention is

[10] More precisely, the memory footprint is $n^2 + n \cdot d$, which is dominated by the n^2 term when $n \gg d$.

to restrict attention to a local region of the input sequence. We can then write π_i as

$$\pi_i = [a_i^l, a_i^r] \tag{6.123}$$

where a_i^l and a_i^r are the left and right ends of π_i. $a_i^r - a_i^l + 1$ determines the size of the region, and so we can use it to control the sparsity of the attention model, for example, if $a_i^r - a_i^l + 1 \ll n$, the model would be very sparse. a_i^l and a_i^r can be obtained by using either heuristics or machine learning methods. The reader may refer to related papers for more details [Luong et al., 2015; Sperber et al., 2018; Yang et al., 2018a; Sukhbaatar et al., 2019]. See Figure 6.13 (b) for an illustration of local attention.

- **Chunked Attention**. In this method, we segment a sequence into chunks and run the attention model on each of them [Parmar et al., 2018; Qiu et al., 2020a]. Given a sequence $\{1,...,n\}$, we define $\{\mathrm{chunk}_1,...,\mathrm{chunk}_q\}$ to be a segmentation of the sequence. A chunk can be expressed as a span

$$\mathrm{chunk}_k = [c_k^l, c_k^r] \tag{6.124}$$

In the attention step, we treat each chunk as a sequence and perform self-attention on it as usual. In other words, the representation at position i is computed by using only the context in the chunk that i belongs to. In this sense, this model can be thought of as a variant of local attention. Figure 6.13 (c) shows an illustration of this model. A remaining issue is how to segment the sequence. There are several ways to do this. For example, as discussed in Section 6.2.4, we can do segmentation from a linguistic perspective, and segment the sequence into linguistically motivated units. In practical systems, it is sometimes more convenient to segment the sequence into chunks that are of equal length. Thus, the sparsity of the model is controlled by the size of these chunks, for example, the use of smaller chunks would lead to a more sparse attention model.

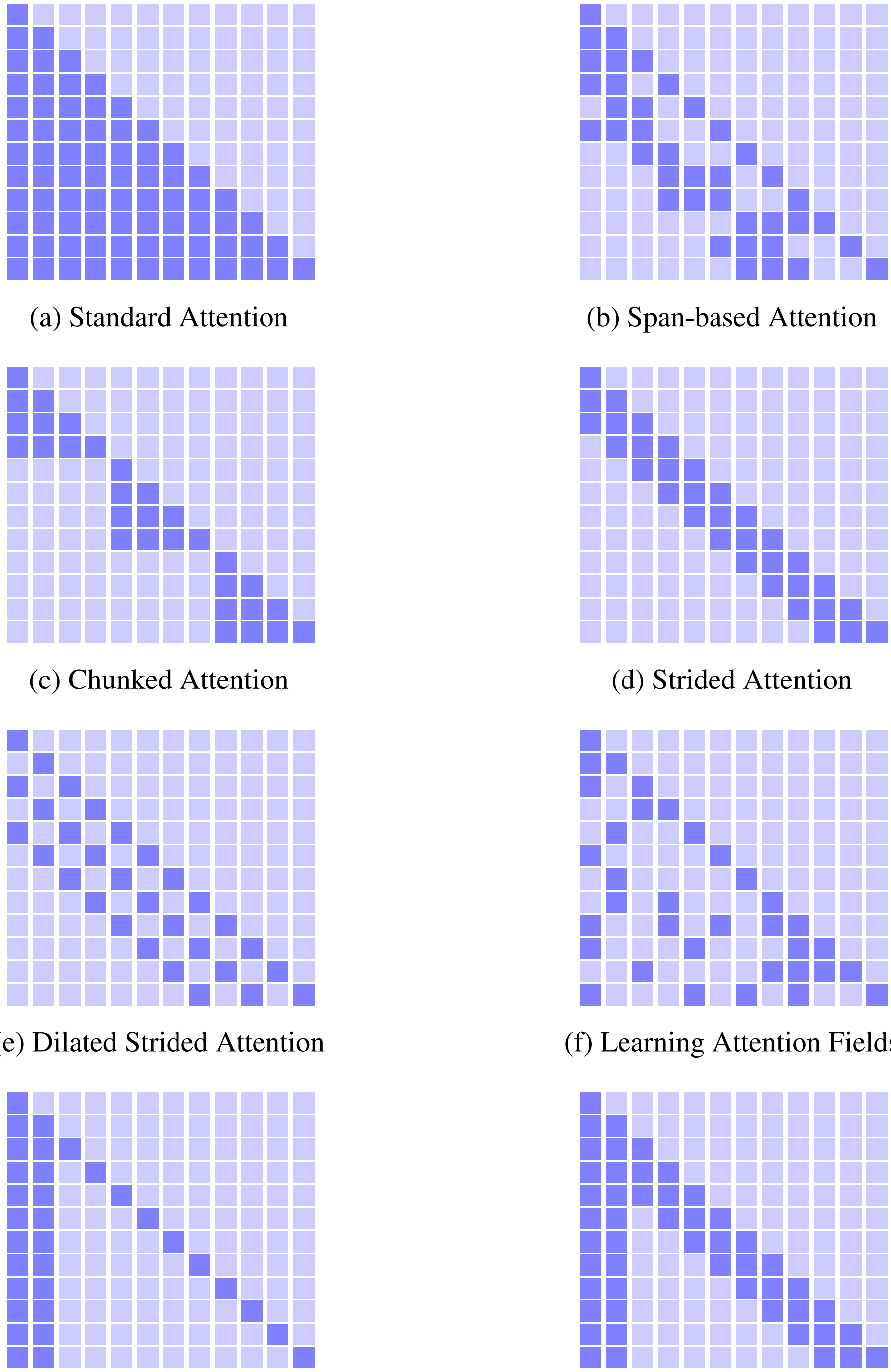
- **Strided Attention**. Since the chunked attention approach enforces a hard segmentation on the input sequence, it may lose the ability to learn representations from inputs in different chunks. An alternative way to achieve chunk-wise attention is to allow overlap between chunks [Child et al., 2019; Beltagy et al., 2020; Ainslie et al., 2020]. This approach is analogous to the family of approaches that are commonly used to apply a local model to 1D or 2D data to generate outputs of the same shape. Like CNNs, we use a context window to represent the field of input of the attention model. The context window slides along the sequence, each time moving forward a step of size stride. As a special case, if stride equals the size of the context window, this model reduces to the chunked attention model mentioned above. If stride is smaller than the size of the context window, the attention model will become denser. Figure 6.13 (d) shows the case of $\mathrm{stride} = 1$ where the chunk overlapping is maximized. A way to achieve relatively sparser attention is to use a **dilated context window**. Figure 6.13 (e) shows an example of the dilated strided attention model, where the context window is discontinuous, with gaps of size 1.

- **Learning Attention Fields**. Because the attention field π_i can be any subset of $\{1, ..., n\}$, we can develop more general sparse attention models by considering attention maps beyond chunk-based patterns. The only question is how to determine which positions the model attends to for a given position. One simple approach is to use a computationally cheaper model to estimate the "importance" of each position. Then, attention weights are computed only for some of the positions which are thought to be most important [Zhou et al., 2021]. A second approach is grouping: positions are grouped, and then the attention weights are computed only for positions in the same group. This can be achieved by clustering keys and queries. For example, we can cluster keys and queries via k-means clustering. The centroids of the clusters can be treated as additional parameters of the attention model, and so can be learned during optimization [Roy et al., 2021]. One benefit of learning attention fields is that the model can spread its attention more broadly over the sequence. This is a useful property for many NLP problems because word dependencies are sometimes long-range, not restricted to a local context window. See Figure 6.13 (f) for an example of the attention map learned through this model. Alternative approaches to learning to attend are to use sorting or hashing functions to group similar key and query vectors [Kitaev et al., 2020; Tay et al., 2020a]. These functions can be either heuristically designed functions or neural networks with learnable parameters. By using these functions, we can reorder the sequence so that the inputs in the same group are adjacent in the reordered sequence. In this way, the resulting attention map follows a chunk-wise pattern, and the model is computationally efficient through the use of the chunked attention approach.
- **Hybrid Methods**. Above, we have discussed a range of different sparse attention models. It is natural to explore methods that combine multiple models together to make use of their benefits in some way. A simple way to do this is to combine the attention fields of different models. For example, in Zaheer et al. [2020]'s system, the attention map is generated by considering three different sparse models, including local attention (chunked attention), global attention, and random attention[11]. The resulting model is still a sparse model, but is somewhat more robust as it involves multiple patterns from different perspectives of attention modeling. Another way of combining multiple attention models is to use different models for different heads in multi-head attention [Child et al., 2019; Beltagy et al., 2020]. For example, one can use one head as a local attention model, and use another head as a global attention model (see Figure 6.13 (g-h)).

One disadvantage of sparse models compared to dense models is that they are often less efficient on GPUs/CPUs due to hardware constraints. While sparse models can ideally reduce both the memory and computation requirements, the computational throughput of sparse models is much slower than that of dense models. In practice, it is difficult for sparse models to approach the peak FLOPS of a GPU or CPU[12]. Therefore, they are primarily used to improve

[11] Here the global attention model attends each word only to a special word which accounts for the entire sequence and is often placed at the beginning of the sequence. The random attention model attends each word to a random set of the words of the sequence.

[12] FLOPS = floating point operations per second.



(a) Standard Attention

(b) Span-based Attention

(c) Chunked Attention

(d) Strided Attention

(e) Dilated Strided Attention

(f) Learning Attention Fields

(g) Global Attention

(h) Hybrid Methods

Figure 6.13: Illustration of the attention maps of different models (self-attention on the decoder side). Dark cells mean $A'(i,j) \neq 0$ (i.e., i attends to j), and light cells mean $A'(i,j) = 0$ (i.e., i does not attend to j). In all these attention maps, we assume that every position attends to itself by default (see diagonals).

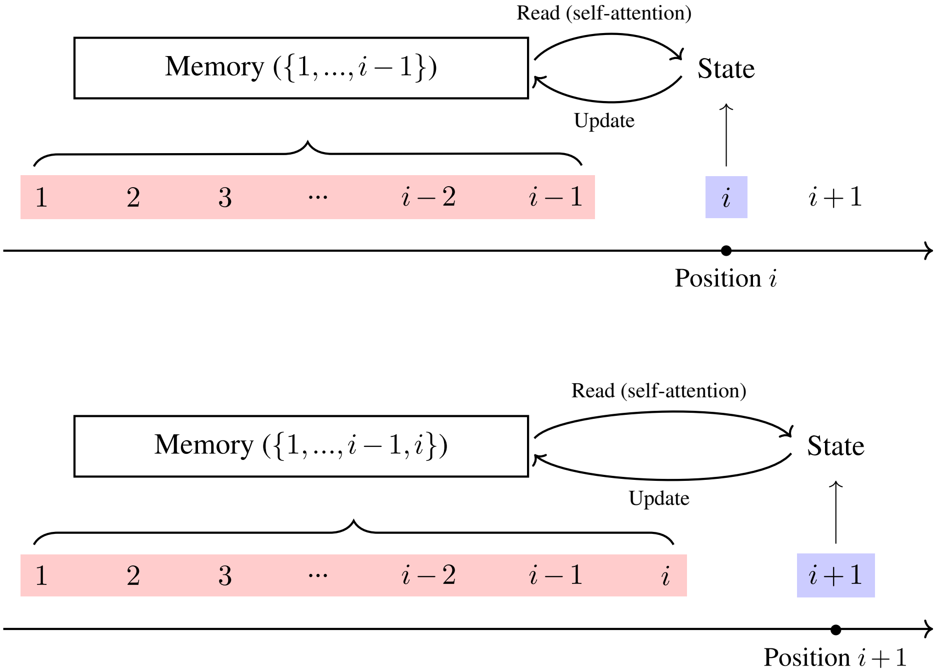


Figure 6.14: The Transformer as a memory system. At position i, the collection of key-value pairs from positions $\{1,...,i-1\}$ serves as a memory of past context. The model accesses this memory to generate an output, and subsequently adds the key-value pair of position i to the memory. Moving to the next position, this cycle of memory access and update is repeated.

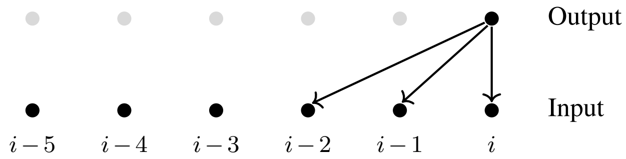
memory efficiency rather than achieving raw computational speedups on current hardware. On the other hand, sparse models are still of great use to NLP practitioners in the context of memory-efficient Transformer, especially when Transformers are used to deal with extremely long sequences.

6.4.2 Recurrent and Memory Models

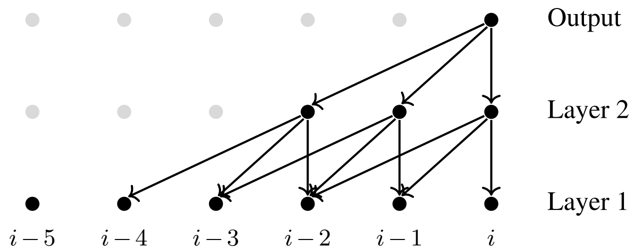
For sequence generation problems, Transformers can also be thought of as memory systems. Consider again the general autoregressive setting, in which we are given the states of the previous $i-1$ positions and wish to predict the next state. In self-attention, this is achieved by using the query at position i (i.e., $\mathbf{q}_i$) to access the key-value pairs of the previous positions (i.e., $\{(\mathbf{k}_1,\mathbf{v}_1),...,(\mathbf{k}_{i-1},\mathbf{v}_{i-1})\}$). Then, we move to position $i+1$ and append $(\mathbf{k}_i,\mathbf{v}_i)$ to the collection of key-value pairs. This procedure can be intuitively interpreted using memory mechanisms (see Chapter 4). A Transformer model maintains a memory bank that retains past information. As the model processes the sequence, it repeats the same operation: generating outputs by reading from the memory, and then updating the memory to incorporate new information. This is illustrated in Figure **??**.

1. Cache-based Memory

The memory here can be viewed as a datastore of vectors. From a machine learning perspective, this is a non-parametric model, and the cost of accessing the model grows as the sequence becomes longer. Clearly, such a variable-length memory will generally be infeasible if the model deals with extremely long sequences. For modeling sequences of arbitrary length, it is common to use a fixed-length memory instead. As in many NLP problems, one of the simplest ways to do this is to maintain a cache of recent information, that is, we restrict the modeling to a context window. Let n_c be the size of the context window. The model keeps track of the $n_c - 1$ latest states preceding the current position, so that its closest predecessors can be considered at each step. This means that, for each position, a self-attention sub-layer attends to $n_c - 1$ previous positions, as follows

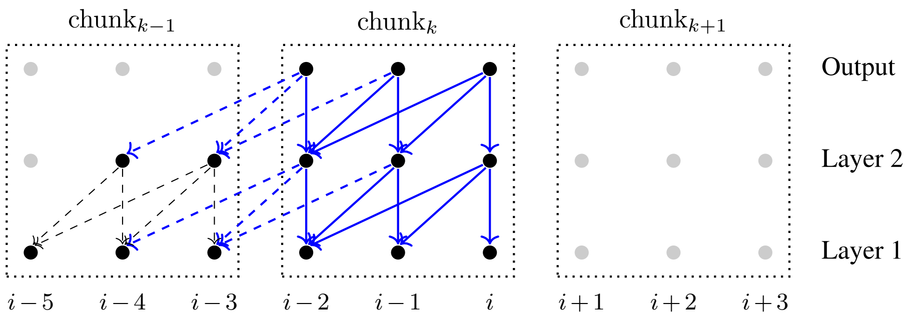


If we stack multiple self-attention sub-layers, a larger context window would be considered. For example, a model involving two self-attention sub-layers has a context window of size $2n_c - 1$, as follows

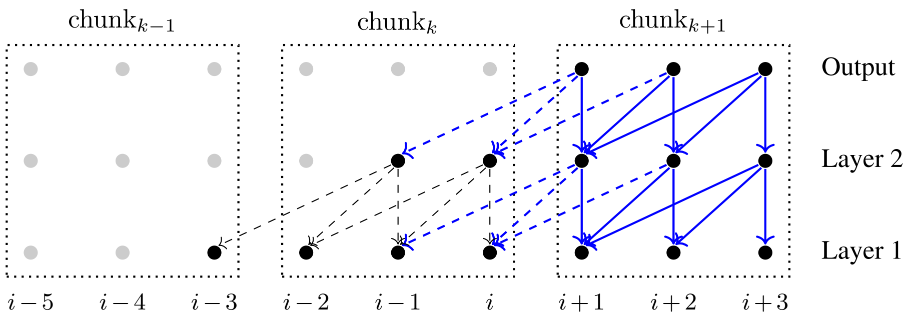


Therefore, we can capture a sufficiently large context by using a multi-layer Transformer model. Note that the context window model here is essentially the same as the strided attention model presented in the preceding section. Systems of this type are often easy to implement: we slide a window along the sequence, and, in each move, we make predictions at the last position of the window (for inference), or back-propagate errors (for training).

An alternative strategy for training a context window model is chunked attention. The sequence is explicitly partitioned into sub-sequences (or chunks) of a fixed length n_c. These chunks are then treated as independent training samples. This approach, however, completely ignores the relationship between inputs in different chunks. One way to address this issue is to introduce dependence between chunks. For example, the **Transformer-XL** model allows every chunk to access one or more preceding chunks [Dai et al., 2019]. In the simplest case, consider an example in which chunk_k can see its predecessor chunk_{k-1}. Each position in chunk_k can attend to all its preceding positions in both chunk_k and chunk_{k-1}.



(a) Step k of chunk-wise training



(b) Step $k+1$ of chunk-wise training

Figure 6.15: Illustration of chunk-wise training in Transformer-XL [Dai et al., 2019]. The input sequence is divided into non-overlapping chunks of length n_c. Training proceeds one chunk at a time. Within chunk_k, the attention span for every token is restricted to a left-context window of size n_c. This structure permits cross-chunk attention. For example, position $i-2$ in chunk_k can attend to positions $i-3$ and $i-4$ residing in chunk_{k-1} (see sub-figure (a)). During back-propagation, gradients are computed for the current sub-network (chunk_k), treating cached activations from previous chunks as fixed constants. Dashed lines denote forward-pass information flows that do not receive gradients. After completing the training step for chunk_k, the process moves to the next chunk and repeats.

In Transformer-XL, this approach is implemented in a simplified form. First, each position is constrained to attend to $n_c - 1$ previous positions so that the size of the attention field of a position is the same in the training and inference stages. Such a method turns the problem back to strided attention, making the implementation of the attention model straightforward. However, unlike standard strided attention, Transformer-XL performs training in a chunk-wise manner. Once we finish the training on a chunk, we directly move to the next chunk, rather than sliding the context window a small step forward. Second, while this approach allows for connections between chunks, the parameters of the sub-network corresponding to chunk_{k-1} are fixed, and we only update the parameters of the sub-network on chunk_k in the k-th step. See Figure 6.15 for an illustration.

The above model is similar in spirit to recurrent models because both of them require the computation in one step to depend on the states of the preceding steps. However, it is not in the standard form of a recurrent model, in which the output of a recurrent unit in one step is the input in the next step. Instead, the "recurrence" is expressed by involving connections across layers and chunks, that is, the output of one layer in chunk_{k-1} is used as the input of a higher-level layer in chunk_k.

2. Encoding Long-term Memory

Another idea for representing the states of a sequence is to frame the task as an encoding problem. Instead of storing all key-value vectors during left-to-right generation, we compress the memory of the entire "history" into a fixed number of encoded key-value vectors. These encoded vectors can be either a small subset of the original sequence $\{(\mathbf{k}_1,\mathbf{v}_1),...,(\mathbf{k}_{i-1},\mathbf{v}_{i-1})\}$, or a small set of newly generated vectors that encode this entire history.

One way to achieve this encoding is to apply a pooling operation over $\{(\mathbf{k}_1,\mathbf{v}_1),...,(\mathbf{k}_{i-1},\mathbf{v}_{i-1})\}$ [Rae et al., 2019a]. For example, by using average pooling, the memory is reduced to a single key-value pair $(\bar{\mathbf{k}},\bar{\mathbf{v}})$:

$$\bar{\mathbf{k}} = \frac{1}{i-1}\sum_{j=1}^{i-1}\mathbf{k}_j \tag{6.125}$$

$$\bar{\mathbf{v}} = \frac{1}{i-1}\sum_{j=1}^{i-1}\mathbf{v}_j \tag{6.126}$$

This leads to a very efficient model, as we only need to update the aggregated vectors $(\bar{\mathbf{k}},\bar{\mathbf{v}})$ incrementally [Zhang et al., 2018a]. Let $(\bar{\mathbf{k}}[i],\bar{\mathbf{v}}[i])$ represent the state of the memory at position i. A more general formulation is given recursively:

$$\bar{\mathbf{k}}[i] = \mathrm{KMem}(\bar{\mathbf{k}}[i-1],\mathbf{k}_{i-1}) \tag{6.127}$$

$$\bar{\mathbf{v}}[i] = \mathrm{VMem}(\bar{\mathbf{v}}[i-1],\mathbf{v}_{i-1}) \tag{6.128}$$

where $\mathrm{KMem}(\cdot)$ and $\mathrm{VMem}(\cdot)$ are functions that update the memory by taking both the states of the memory at the previous position (i.e., $\bar{\mathbf{k}}[i-1]$ and $\bar{\mathbf{v}}[i-1]$) and the new states (i.e., $\mathbf{k}_{i-1}$ and $\mathbf{v}_{i-1}$). There are many possible choices for functions $\mathrm{KMem}(\cdot)$ and $\mathrm{VMem}(\cdot)$. For example, if $\mathrm{KMem}(\cdot)$ and $\mathrm{VMem}(\cdot)$ are weighted sum functions, we can derive the same forms as Eqs. (6.125) and (6.126). If $\mathrm{KMem}(\cdot)$ and $\mathrm{VMem}(\cdot)$ are recurrent cells (e.g., standard RNN or LSTM cells), we obtain a recurrent model of memory.

Extending the above concept to memories with more than one key-value pair is straightforward. One approach is chunk-level representation. Let $\{(\bar{\mathbf{k}}_1,\bar{\mathbf{v}}_1),...,(\bar{\mathbf{k}}_\kappa,\bar{\mathbf{v}}_\kappa)\}$ be a memory of size κ, where each $(\bar{\mathbf{k}}_j,\bar{\mathbf{v}}_j)$ is a representation of a chunk of length n_c. This memory can thus encode a sequence of maximum length $\kappa \cdot n_c$, computing each $(\bar{\mathbf{k}}_j,\bar{\mathbf{v}}_j)$ via Eqs. (6.127) and (6.128) over its corresponding chunk. A second approach is to organize the memory into a priority queue. A learned scoring function, which is often an auxiliary neural network, evaluates and assigns a priority score to each generated key-value pair. High-scoring pairs are inserted into the queue via a push operation. In this way, the memory actively maintains a dynamic collection of only the most valuable key-value pairs across the input sequence.

Although representing memory as a set of vectors is an intuitive design choice for Transformers, its capacity is bounded by the number of vectors. An alternative paradigm is **continuous memory**. This approach leverages function approximation, treating the sets of keys $\{\mathbf{k}_1, ..., \mathbf{k}_{i-1}\}$ or values $\{\mathbf{v}_1, ..., \mathbf{v}_{i-1}\}$ as a series of data points to be fitted by a continuous function. Instead of explicitly storing these vectors, the memory is parameterized by the functions themselves. A common method is to use a linear combination of simple basis functions to approximate the complex function defined by the discrete key or value vectors [Martins et al., 2022].

It is also straightforward to use a short-term memory and a long-term memory simultaneously so that we can combine the merits of both. For example, we use a cache-based memory to capture local context, and use an efficient long-term memory that encodes the entire history to model long-range dependencies. This idea is also similar to that used in combining different sparse attention models as discussed in the previous subsection.

3. Retrieval-based Methods

So far in this subsection, we have discussed approaches based on fixed-length models. It is also possible to develop efficient memory models by improving the efficiency of accessing memory, instead of strictly reducing memory capacity. One way to achieve this is to store the past key-value pairs in a database (call it a **vector database**), and to find the most similar key-value pairs when querying the database. To be more precise, given a query $\mathbf{q}$, we use the database to find a set of top-p relevant key-value pairs (denoted by Ω_p) by performing a similarity search based on the dot-product between the query and key vectors. The query $\mathbf{q}$ then attends to Ω_p as in standard self-attention models. The idea behind this method is to consider only a small number of elements that contribute most to the attention result. Therefore, the model is essentially a sparse attention model which is computationally efficient. Another advantage of this method is that it allows for fast similarity search over a very large set of vectors because of the highly optimized implementation of vector databases. Framing memory as a retrieval system falls under the broader paradigm of **retrieval-augmented generation** (**RAG**) or retrieval-augmented approaches. It provides a scalable framework for incorporating external memory into neural architectures like Transformers [Guu et al., 2020; Lewis et al., 2020b; Wu et al., 2021].

6.4.3 Low-dimensional Models

In many practical applications, Transformers are "high-dimensional" models. This is not only because the input and/or output data is in high-dimensional spaces, but also because some of the intermediate representations of the data in the model are high-dimensional. As discussed in Section 6.4.1, this high dimensionality arises in part from computing the weighted sum of value vectors, as in Eq. (6.129) (repeated here for convenience)

$$\mathrm{Att}_{\mathrm{self}}(\mathbf{S}) = \mathbf{AV} \tag{6.129}$$

and computing the attention matrix itself, as in Eq. (6.130)

$$\mathbf{A} = \mathrm{Softmax}(\frac{\mathbf{QK}^{\top}}{\sqrt{d}} + \mathbf{M}) \tag{6.130}$$

which involves large matrix multiplications $\mathbf{QK}^{\top}$ and $\mathbf{AV}$ when the length n and the hidden dimensionality d are large.

The $\mathbf{AV}$ and $\mathbf{QK}^{\top}$ operations scale with a time complexity of $O(n^2 \cdot d)$ and a space complexity of $O(n^2 + n \cdot d)$. While previous approaches have mitigated this complexity using sparse models, this subsection focuses on methods that approximate these operations via dense computation. One straightforward idea is to project $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ into smaller matrices, thereby reducing the computational burden of matrix multiplication. Since $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ all reside in $\mathbb{R}^{n \times d}$, this can be achieved by reducing either the length dimension n, the hidden dimension d, or both.

1. Reducing n

Note that the output $\mathrm{Att}_{\mathrm{self}}(\mathbf{S})$ is required to be an $n \times d$ matrix, and so we cannot reduce the number of queries. We instead consider reducing the number of keys and values. Suppose n' is an integer with $n' < n$, and $\mathbf{K}$ and $\mathbf{V}$ can be transformed into $n' \times d$ matrices $\mathbf{K}'$ and $\mathbf{V}'$ in some way. We can obtain a "smaller" model simply by replacing $\mathbf{K}$ and $\mathbf{V}$ with $\mathbf{K}'$ and $\mathbf{V}'$, giving

$$\mathrm{Att}_{\mathrm{self}}(\mathbf{S}) = \mathbf{A}\mathbf{V}' \tag{6.131}$$

$$\mathbf{A} = \mathrm{Softmax}(\frac{\mathbf{Q}[\mathbf{K}']^{\top}}{\sqrt{d}} + \mathbf{M}) \tag{6.132}$$

This model is in the standard form of self-attention, but has lower time and space complexities, that is, $O(n' \cdot n \cdot d) < O(n^2 \cdot d)$ and $O(n' \cdot n + n' \cdot d) < O(n^2 + n \cdot d)$. If $n' \ll n$, the resulting model will be linear with n.

The key problem here is how to obtain $\mathbf{K}'$ and $\mathbf{V}'$ in a way that retains much of the information in $\mathbf{K}$ and $\mathbf{V}$. There are several ways to do so. One simple method is to select the keys and values that are thought to be important. The importance of a key (or value) can be computed in terms of some computationally inexpensive measure. For example, we can sample a small number of query-key dot-products and estimate the importance of a key by collecting these dot-product results.

Although straightforward, the above method still relies on sparse operations like sampling and aggregation. Alternatively, dense computation can be used to project $\mathbf{K}$ and $\mathbf{V}$ down to $\mathbf{K}'$ and $\mathbf{V}'$. A common approach employs CNNs [Liu et al., 2018]. Let $\mathrm{Conv}(\cdot)$ denote a 1D convolution operation sliding along the n dimension. $\mathbf{K}'$ is then computed as:

$$\mathbf{K}' = \mathrm{Conv}(\mathbf{K}, \mathbf{W}_c, \mathrm{size}_r, \mathrm{stride}) \tag{6.133}$$

where $\mathbf{W}_c$ is the parameter matrix of the filters, size_r is the size of the receptive field, and stride is the step size of the convolution. In general, we can achieve a high compression rate by choosing large values for size_r and stride. Likewise, we can compute $\mathbf{V}'$ using another convolutional function. It is worth noting that, if the parameter n' is fixed for all samples, compression of $\mathbf{K}$ and $\mathbf{V}$ along the length dimension is essentially the same as the fixed-length memory model as described in the preceding subsection. The methods presented here are more general and could be applied to variable-length memories.

We might also be tempted to model the attention function by considering the attention matrix $\mathbf{A}$ as a high-dimensional representation of data and then applying conventional dimensionality reduction methods. Empirically, in many problems, $\mathbf{A}$ (or more precisely $\mathbf{Q}\mathbf{K}^\top$) is found to be a low-rank matrix. In this case, we can compress $\mathbf{A}$ while retaining as much information as possible. There are many ways to do so. For example, we might use a product of smaller matrices as an approximation to $\mathbf{A}$ via the SVD technique (see Chapter 3). However, this introduces computational overhead in using SVD compared with the standard attention model. A simpler idea is to directly transform $\mathbf{K}$ and $\mathbf{V}$ into smaller-sized matrices via linear mappings, given by

$$\mathbf{K}' = \mathbf{U}^k\mathbf{K} \tag{6.134}$$

$$\mathbf{V}' = \mathbf{U}^v\mathbf{V} \tag{6.135}$$

where $\mathbf{U}^k \in \mathbb{R}^{n' \times n}$ and $\mathbf{U}^v \in \mathbb{R}^{n' \times n}$ are parameter matrices. Clearly, this leads to a model which is equivalent to that described in Eqs. (6.131) and (6.132). While highly intuitive, this approach has been proven to yield a sufficiently small approximation error ϵ, provided that n' scales linearly with d/ϵ^2 [Wang et al., 2020b].

2. Reducing d

Another approach to reducing computational complexity is to reduce the dimensionality d. One of the simplest methods is to project all queries and keys onto a d'-dimensional space ($d' < d$), and to compute the dot-product between query-key pairs in the new space. For modeling, we only need to replace $\mathbf{Q} \in \mathbb{R}^{n \times d}$ and $\mathbf{K} \in \mathbb{R}^{n \times d}$ with new representations $\mathbf{Q}' \in \mathbb{R}^{n \times d'}$ and $\mathbf{K}' \in \mathbb{R}^{n \times d'}$. We can easily modify Eq. (6.130) to use $\mathbf{Q}'$ and $\mathbf{K}'$ in computing the attention matrix

$$\mathbf{A} = \mathrm{Softmax}(\frac{\mathbf{Q}'[\mathbf{K}']^\top}{\sqrt{d'}} + \mathbf{M}) \tag{6.136}$$

$\mathbf{Q}'$ and $\mathbf{K}'$ are given by

$$\mathbf{Q}' = \mathbf{Q}\mathbf{U}^q \tag{6.137}$$
$$\mathbf{K}' = \mathbf{K}\mathbf{U}^k \tag{6.138}$$

where $\mathbf{U}^q \in \mathbb{R}^{d \times d'}$ and $\mathbf{U}^k \in \mathbb{R}^{d \times d'}$ are parameter matrices of linear transformations.

It is also possible to exploit **kernel methods** to obtain an efficient dot-product attention model. The basic idea is to map data points (represented as vectors) from one space to another, transforming a problem that is difficult in the original space into one that is more tractable in the new space. The "kernel trick" allows us to compute such inner products implicitly via a kernel function, without explicitly constructing the mapping $\phi(\cdot)$[13]. Such inner products in feature space are commonly represented by kernel functions, denoted as $K(\cdot,\cdot)$.

It is interesting to approximate $\mathbf{A}$ in a fashion analogous to $K(\cdot,\cdot)$ in kernel methods. To illustrate, note in Eq. (6.130) $\mathbf{A}$ represents normalized attention weights. The numerator can be written in the form

$$\widetilde{\mathbf{A}} = \mathrm{Mask}(\exp(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}})) \tag{6.140}$$

Here $\mathrm{Mask}(\cdot)$ is a function which has the same effect as using the additive masking variable $\mathbf{M}$. Then, $\mathbf{A}$ can be expressed as

$$\mathbf{A} = \mathbf{D}^{-1}\widetilde{\mathbf{A}} \tag{6.141}$$

where $\mathbf{D}$ is an $n \times n$ diagonal matrix. Each entry of the main diagonal is the sum of the entries of the corresponding row in $\widetilde{\mathbf{A}}$, denoting the normalization factor of Softmax. Substituting this equation into Eq. (6.130), we have

$$\mathrm{Att}_{\mathrm{self}}(\mathbf{S}) = \mathbf{D}^{-1}\widetilde{\mathbf{A}}\mathbf{V} \tag{6.142}$$

In this model, $\widetilde{A}(i,j)$ can be viewed as a similarity function over all query-key pairs in a d-dimensional space. Here we assume that this function, which is in the form of the dot-product of vectors, can be approximated by a kernel function

$$\begin{aligned} \widetilde{A}(i,j) &= K(\mathbf{q}_i, \mathbf{k}_j) \\ &= \langle \phi(\mathbf{q}_i), \phi(\mathbf{k}_j) \rangle \end{aligned}$$

[13] In mathematical analysis, the inner product is a generalized notion of the dot-product, typically denoted by $\langle\cdot,\cdot\rangle$. A formal definition requires that $\langle\cdot,\cdot\rangle$ satisfies several properties in a vector space. Although the inner product takes different forms in different contexts, in Euclidean space $\mathbb{R}^d$, it is equivalent to the dot-product. That is, given two vectors $\mathbf{a} \in \mathbb{R}^d$ and $\mathbf{b} \in \mathbb{R}^d$, we have

$$\begin{aligned} \langle \mathbf{a}, \mathbf{b} \rangle &= \mathbf{a} \cdot \mathbf{b} \\ &= \sum_{i=1}^{d} a_i \cdot b_i \end{aligned} \tag{6.139}$$

$\phi(\cdot)$ is a mapping from $\mathbb{R}^d$ to $\mathbb{R}^{d'}$. We can represent the queries and keys in the following form

$$\begin{aligned} \mathbf{Q}' &= \phi(\mathbf{Q}) \\ &= \begin{bmatrix} \phi(\mathbf{q}_1) \\ \vdots \\ \phi(\mathbf{q}_n) \end{bmatrix} \end{aligned} \tag{6.143}$$

$$\begin{aligned} \mathbf{K}' &= \phi(\mathbf{K}) \\ &= \begin{bmatrix} \phi(\mathbf{k}_1) \\ \vdots \\ \phi(\mathbf{k}_n) \end{bmatrix} \end{aligned} \tag{6.144}$$

Then, we develop a kernelized attention model by approximating the attention weight $\alpha_{i,j}$ in the form

$$\alpha_{i,j} \approx \frac{\phi(\mathbf{q}_i)\phi(\mathbf{k}_j)^\top}{\sum_{j'=1}^{n} \phi(\mathbf{q}_i)\phi(\mathbf{k}_{j'})^\top} \tag{6.145}$$

The key idea behind this kernelized attention model is that we can remove the Softmax function if the queries and keys are mapped to a new space. Using this approximation, the i-th output vector of the attention model (i.e., the i-th row vector of $\mathrm{Att}_{\mathrm{self}}(\mathbf{S})$) is given by

$$\begin{aligned} \mathbf{c}_i &= \sum_{j=1}^{n} \alpha_{i,j} \cdot \mathbf{v}_j \\ &\approx \sum_{j=1}^{n} \Big(\frac{\phi(\mathbf{q}_i)\phi(\mathbf{k}_j)^\top}{\sum_{j'=1}^{n} \phi(\mathbf{q}_i)\phi(\mathbf{k}_{j'})^\top} \cdot \mathbf{v}_j \Big) \\ &= \frac{\sum_{j=1}^{n} \phi(\mathbf{q}_i)\phi(\mathbf{k}_j)^\top \mathbf{v}_j}{\sum_{j'=1}^{n} \phi(\mathbf{q}_i)\phi(\mathbf{k}_{j'})^\top} \\ &= \frac{\phi(\mathbf{q}_i)\big(\sum_{j=1}^{n} \phi(\mathbf{k}_j)^\top \mathbf{v}_j\big)}{\phi(\mathbf{q}_i)\big(\sum_{j'=1}^{n} \phi(\mathbf{k}_{j'})^\top\big)} \end{aligned} \tag{6.146}$$

Here $\phi(\mathbf{q}_i) \in \mathbb{R}^{1 \times d'}$, while $\phi(\mathbf{k}_j)^\top \mathbf{v}_j \in \mathbb{R}^{d' \times d}$. Therefore, the inner term $\sum_{j=1}^{n} \phi(\mathbf{k}_j)^\top \mathbf{v}_j$ is a $d' \times d$ matrix, and the final product yields a vector in $\mathbb{R}^{1 \times d}$.

Although the equation appears a bit complicated, the idea is simple: instead of attending the query to all keys to obtain the attention weight $\alpha_{i,j}$, we can compute the sum of the multiplications $\sum_{j=1}^{n} \phi(\mathbf{k}_j)^\top \mathbf{v}_j \in \mathbb{R}^{d' \times d}$ and then multiply it with the kernelized query $\phi(\mathbf{q}_i)$. Returning to the notation used in Eq. (6.142), we define the i-th diagonal entry of $\mathbf{D}$ to be $\phi(\mathbf{q}_i) \sum_{j'=1}^{n} \phi(\mathbf{k}_{j'})^\top$. Then, the attention model can be re-expressed in the form

$$\begin{aligned} \mathrm{Att}_{\mathrm{self}}(\mathbf{S}) &= \mathbf{D}^{-1} \phi(\mathbf{Q}) \phi(\mathbf{K})^\top \mathbf{V} \\ &= \mathbf{D}^{-1} \mathbf{Q}' \mathbf{K}'^\top \mathbf{V} \\ &= \mathbf{D}^{-1} \big(\mathbf{Q}' (\mathbf{K}'^\top \mathbf{V})\big) \end{aligned} \tag{6.147}$$

Here we change the order of computation from left-to-right to right-to-left using parentheses. Given that $\mathbf{Q}' \in \mathbb{R}^{n\times d'}$ and $\mathbf{K}' \in \mathbb{R}^{n\times d'}$, this model has time and space complexities of $O(n\cdot d\cdot d')$ and $O(n\cdot d + n\cdot d' + d\cdot d')$, respectively. Therefore, the model is linear with respect to the sequence length n, and is sometimes called the **linear attention model**. One computational advantage of this model is that we need only compute the multiplication $\mathbf{K}'^{\top}\mathbf{V}$ (i.e., $\sum_{j=1}^{n}\phi(\mathbf{k}_j)^{\top}\mathbf{v}_j$) and the corresponding normalization factor (i.e., $\sum_{j'=1}^{n}\phi(\mathbf{k}_{j'})^{\top}$) once. The results can then be used for any query [Katharopoulos et al., 2020]. The model needs to maintain $\sum_{j=1}^{n}\phi(\mathbf{k}_j)^{\top}\mathbf{v}_j$ and $\sum_{j'=1}^{n}\phi(\mathbf{k}_{j'})^{\top}$ and update them when new key and value vectors arrive.

Still, there are several limitations regarding this kernelized model, for example, how to develop the feature map $\phi(\cdot)$ to obtain a good approximation to the standard attention model. Interested readers may refer to Choromanski et al. [2020]'s work for more details.

A second idea for reducing d is to take sub-space models, in which a problem in a d-dimensional space is transformed into sub-problems in lower-dimensional spaces, and the solution to the original problem is approximated by some combination of the solutions to these sub-problems. In a general sub-space model, a d-dimensional key vector $\mathbf{k}$ can be mapped into a set of d'-dimensional vectors $\{\mathbf{K}'_1, ..., \mathbf{K}'_\eta\}$. To simplify modeling, we can do this by vector segmentation, that is, we segment $\mathbf{k}$ into η sub-vectors, each having $d' = \frac{d}{\eta}$ dimensions. We can transform all query and value vectors in the same way. Then, the attention model is applied in each of these sub-spaces.

This method, however, does not reduce the total amount of computation. As presented in Lample et al. [2019]'s work, we can instead approximate the dot-product attention over a set of key-value pairs by considering top-p candidates in each sub-space. More precisely, we find p-best key-value pairs in each sub-space, which is computationally cheaper. The Cartesian product of these p-best key sets consists of p^η product keys. Likewise, we obtain p^η product values. The remaining work is simple: the d-dimensional queries attend to these d-dimensional product keys and values. An interesting difference between this sub-space model and the d-dimensional space model is that the generated product keys and values may be different from any of the original key-values $\{(\mathbf{k}_1, \mathbf{v}_1), ..., (\mathbf{k}_{i-1}, \mathbf{v}_{i-1})\}$. This provides a way for learning new representations of the past information.

So far we have discussed approaches to dimensionality reduction along either the n or d dimension. It is straightforward to combine them to develop a "lower-dimensional" model. As an example, suppose that we have the $n \to n'$ reduction for keys and values, and the $d \to d'$ reduction for queries and keys. The model takes the form

$$\begin{aligned} \mathrm{Att}_{\mathrm{self}}(\mathbf{S}) &= \mathbf{A}\mathbf{V}' \\ \mathbf{A} &= \mathrm{Softmax}(\frac{\mathbf{Q}'\mathbf{K}'^{\top}}{\sqrt{d'}} + \mathbf{M}) \end{aligned} \tag{6.148}$$

where $\mathbf{Q}' \in \mathbb{R}^{n\times d'}$, $\mathbf{K}' \in \mathbb{R}^{n'\times d'}$, and $\mathbf{V}' \in \mathbb{R}^{n'\times d'}$ are low-dimensional representations for queries, keys and values. As usual, we can easily obtain these representations through the linear mappings of $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$. The time and space complexities of this model are $O(n'\cdot n\cdot d')$ and

$O(n' \cdot n + n' \cdot d')$.

6.4.4 Parameter and Activation Sharing

Redundancy is common to most large-scale neural networks. As a result, many of these models are over-parameterized, making training and inference less efficient. One common approach to reducing redundancy is to simplify the model by removing unnecessary components. For example, we can either prune a complex model or share sub-modules across different parts of the model to obtain a more compact model. In this subsection, we discuss methods for sharing parameters and intermediate states in Transformer models. We leave the discussion of model transfer and pruning to Section 6.4.7.

Shared-parameter architectures are widely used in neural network-based systems. Well-known examples include CNNs and RNNs, where the same set of parameters (or layers) is applied across different regions of the input. This produces a large neural network, parts of which have the same architecture and the same shared parameters. For Transformers as well as other sequence models, the sharing mechanism can be applied to different levels of modeling. A simple example is embedding sharing. In machine translation, a typical strategy for dealing with words in two languages is to develop two separate embedding models. Alternatively, one can use a single embedding model for both languages. The parameters of the model are then learned during the training of both the source-side and target-side networks. Such a strategy is also often adopted in multilingual sequence models, such as language models that are able to deal with text in many different languages.

For multi-layer neural networks, a popular method is layer-wise sharing. Suppose there is a stack of layers, all of which have the same form

$$\mathbf{S}^l = \mathrm{Layer}(\mathbf{S}^{l-1}; \theta^l) \tag{6.149}$$

We can tie the parameters for some or all of these layers. For example, given a set of layers $\{l_1, l_2, ..., l_n\}$, we enforce the constraint $\theta^{l_1} = \theta^{l_2} = ... = \theta^{l_n}$, so that we can obtain a smaller model and the optimization of the model can be easier. In practice, this shared-layer model is highly advantageous if many layers are involved, because we can repeat the same process many times to construct a very deep neural network [Dehghani et al., 2018]. For example, sharing a single FFN sub-layer across all encoder layers is found to be effective in reducing the redundancy in machine translation systems [Pires et al., 2023].

For Transformers, sharing can also be performed in multi-head attention. An example of this is **multi-query attention** [Shazeer, 2019]. Recall from Section 6.1.3 that the output of a head h in standard multi-head self-attention can be written as

$$\begin{aligned} \mathbf{C}_h^{\mathrm{head}} &= \mathrm{Att}_{\mathrm{qkv}}(\mathbf{S}_h^q, \mathbf{S}_h^k, \mathbf{S}_h^v) \\ &= \mathrm{Att}_{\mathrm{qkv}}(\mathbf{S}\mathbf{W}_h^q, \mathbf{S}\mathbf{W}_h^k, \mathbf{S}\mathbf{W}_h^v) \end{aligned} \tag{6.150}$$

Here $\mathbf{S}_h^q = \mathbf{S}\mathbf{W}_h^q$, $\mathbf{S}_h^k = \mathbf{S}\mathbf{W}_h^k$, and $\mathbf{S}_h^v = \mathbf{S}\mathbf{W}_h^v$ are the queries, keys, and values, which are obtained by linearly transforming the input $\mathbf{S}$ with distinct parameter matrices $\mathbf{W}_h^q$, $\mathbf{W}_h^k$, and

$\mathbf{W}_h^v$. In multi-query attention, we share the same keys and values across all the heads, but use different queries for different heads. The form of this model is given by

$$\mathbf{C}_h^{\text{head}} = \mathrm{Att}_{\text{qkv}}(\mathbf{S}\mathbf{W}_h^q, \mathbf{S}\mathbf{W}_0^k, \mathbf{S}\mathbf{W}_0^v) \tag{6.151}$$

Because the keys $\mathbf{S}\mathbf{W}_0^k$ and values $\mathbf{S}\mathbf{W}_0^v$ are independent of h, they are computed and stored only once, rather than redundantly for each head. This yields significant computational and memory bandwidth savings during inference, especially when the number of attention heads is large. Due to its efficiency, multi-query attention (and its variant, grouped-query attention) has been successfully integrated into modern large language models, including Llama 2 [Touvron et al., 2023b] and Falcon[14].

By extending the idea of sharing to more general situations, intermediate states can, in principle, be shared across a neural network. Reusing neuron activations, for instance, allows sub-components to be evaluated multiple times efficiently. In Transformers, this method can be applied directly to the self-attention matrices. Studies have empirically shown that attention maps across adjacent layers are often highly correlated in certain NLP tasks [Xiao et al., 2019]. As a result, a computationally efficient strategy is to compute the dense attention map once and reuse it across subsequent layers.

More broadly, the sharing mechanism can be viewed as a process by which we reuse previously computed results rather than recomputing them on the fly. It is thus possible to reuse the information across different runs of a neural network. A related example is **reversible residual networks**, in which activations of one layer can be recovered from the activations of the following layer [Gomez et al., 2017]. Hence we only keep the output of the latest layer in the forward pass. Then, in the backward pass of training, we reconstruct the output of each layer from its successor. One advantage of this reversible design is that the information produced in the forward pass is shared implicitly, and the model is memory-efficient [Kitaev et al., 2020].

6.4.5 Alternatives to Self-Attention

We have seen that the use of self-attention is one of the main sources of high computational and memory costs in Transformers. It is natural to wonder if there are efficient alternatives to self-attention models. Here, we briefly present some Transformer variants in which self-attention sub-layers are replaced with other types of neural architectures.

1. CNNs as a Replacement for Self-Attention

CNNs are simple and widely used neural networks, and are considered potential alternatives to self-attention models. To apply CNNs to Transformers, all we need is to construct a convolutional sub-layer to replace the self-attention sub-layer in a Transformer block. While a CNN filter has a restricted receptive field and thus takes inputs from a "local" context window, large contexts can be easily modeled by stacking multiple convolutional sub-layers. One key advantage of CNNs is that their computational complexity scales linearly with the

[14]https://falconllm.tii.ae/index.html

sequence length n, compared to the quadratic scaling of self-attention models. Furthermore, the availability of highly optimized CNN implementations makes them easy to apply to sequence modeling in practical systems. To further improve memory efficiency, we can utilize lightweight CNN variants, such as depthwise CNNs [Wu et al., 2018a] [15].

2. Linear Attention

As with many practical approaches to sequence modeling, there is also considerable interest in developing linear models in order to speed up the processing of long sequences. While there are many ways to define a linear model, one general form that is commonly used in sequence models is

$$\mathbf{z}_i = f(a \cdot \mathbf{z}_{i-1} + b \cdot \mathbf{s}_i) \tag{6.153}$$

Here $\mathbf{s}_i$ represents some intermediate states of the model at step i, and $\mathbf{z}_i$ represents the summary of the past states up to step i. It is easy to see that this is a recurrent model: the output at step i depends only on the input at the current step and the output at the previous step. As with the popular design choices in neural network-based systems, the linear part is followed by a transformation $f(\cdot)$ which can be either an activation function or a feedforward neural network. Note that, Eq. (6.153) defines a standard linear model only if $f(\cdot)$ is a linear function. The use of $f(\cdot)$ gives greater flexibility in modeling the problem, although the term *linear model* may not be applied if $f(\cdot)$ chooses a non-linear form.

The above formula describes a linearly structured model which can be seen as an instance of a general family of mathematical models. Typically, it can be represented as a chain structure, or an ordered set of nodes. The model repeats the same computation process from the first node to the last, each time taking the information from the current and previous steps and producing an output vector that is used in the following time steps. As a result, the space and time cost of the model scales linearly with the length of the chain.

We can extend Eq. (6.153) to a standard RNN model by simply applying a linear transformation of the current input and the previous state, that is, $\mathbf{z}_i = f(\mathbf{z}_{i-1} \cdot \mathbf{W}_z + \mathbf{s}_i \cdot \mathbf{W}_s)$. It is thus straightforward to apply RNNs and their variants to Transformer to obtain a hybrid model. For example, we can use LSTM and GRUs in building some of the Transformer layers

[15] Recall from Chapter 2 that in CNNs, a filter (or a set of filters) maps the input variables within its receptive field into an output variable (or a set of output variables) via a linear mapping. Suppose the input and output of a problem are represented as sequences of feature vectors. Given a standard filter with a $k \times d$ receptive field, we slide it along the sequence. At each step, the filter takes $k \times d$ input features and produces a single output feature. This procedure is typically expressed as:

$$y = \mathrm{ReduceSum}(\mathbf{x} \odot \mathbf{W}) \tag{6.152}$$

where $\mathbf{x} \in \mathbb{R}^{k \times d}$ is the matrix representation of the input, $y \in \mathbb{R}$ is the output feature, and $\mathbf{W} \in \mathbb{R}^{k \times d}$ is the weight matrix. The function $\mathrm{ReduceSum}(\cdot)$ computes the sum of all element-wise products between $\mathbf{x}$ and $\mathbf{W}$. If we want the input and output to have the same number of features d, we must apply d distinct filters, resulting in a total of $d \cdot (k \cdot d) = d^2 \cdot k$ parameters. In depthwise CNNs, we decouple the spatial convolution from the feature dimensions. Instead of applying full $k \times d$ filters, each of the d input feature channels is convolved independently with its own 1D spatial filter of size k. Thus, the number of unique parameters is drastically reduced to $d \cdot k$ (with each of the d channels corresponding to a dedicated filter containing k unique parameters).

to combine the merits of both recurrent models and self-attentive models [Chen et al., 2018b]. As the conventional recurrent models have been discussed at length in Chapter 2, we skip the discussion of them here.

In fact, we may be more interested in developing linear attention models, so that we can obtain an efficient system, while still retaining the benefit of globally attentive sequence modeling. Part of the difficulty in doing this is that the form of self-attention is not linear. Let us take a moment to see how this difficulty arises. Recall that the result of self-attention can be written in the following form

$$\begin{aligned} \mathrm{Att}_{\mathrm{self}} &= \mathbf{A} \cdot \mathbf{V} \\ &= \psi(\mathbf{Q} \cdot \mathbf{K}^{\top}) \cdot \mathbf{V} \end{aligned} \tag{6.154}$$

Here $\psi(\cdot)$ is a function that is composed of scaling, exponentiation, masking, and normalization operations (i.e., $\psi(\mathbf{a}) = \mathrm{Normalize}(\mathrm{Mask}(\exp(\frac{\mathbf{a}}{\sqrt{d}})))$). Because $\psi(\cdot)$ is a complex non-linear function, there is no straightforward way to simplify the computation directly, and we typically need to calculate the two matrix multiplications separately (one inside $\psi(\cdot)$ and one outside $\psi(\cdot)$). As a consequence, we need to store all the key-value pairs explicitly, and visit each of them given a query. Not surprisingly, this leads to a model whose computational cost grows quadratically with the sequence length n.

Although in self-attention keys and values are coupled, they are used in separate steps. An elegant solution decouples the query from the key-value interaction so that the context information can be encoded independently of any specific query. The key insight is to bypass the non-linear Softmax by applying a feature map $\phi(\cdot)$ to the queries and keys. As introduced in Section 6.4.3, we transform $\mathbf{Q}$ and $\mathbf{K}$ into $\mathbf{Q}' = \phi(\mathbf{Q}) \in \mathbb{R}^{n \times d'}$ and $\mathbf{K}' = \phi(\mathbf{K}) \in \mathbb{R}^{n \times d'}$. The attention mechanism can then be reformulated as:

$$\begin{aligned} \mathrm{Att}_{\mathrm{self}} &\equiv \psi'(\mathbf{Q}' \cdot \mathbf{K}'^{\top}) \cdot \mathbf{V} \\ &= \mathbf{D}^{-1}(\mathbf{Q}' \cdot \mathbf{K}'^{\top}) \cdot \mathbf{V} \\ &= \mathbf{D}^{-1}\Big(\mathbf{Q}' \cdot (\mathbf{K}'^{\top} \cdot \mathbf{V})\Big) \end{aligned} \tag{6.155}$$

where $\mathbf{D}$ is the diagonal normalization matrix, and $\psi'(\mathbf{Z}) = \mathbf{D}^{-1}\mathbf{Z}$[16]. In this transformed space, the query-key dot product does not require exponential Softmax normalization. It corresponds to a normalization by multiplying with $\mathbf{D}^{-1}$. Because the core operation is now pure matrix multiplication, we can exploit matrix associativity to change the order of computation.

This yields a powerful autoregressive formulation: keys and values are first aggregated via $\mathbf{K}'^{\top} \cdot \mathbf{V}$, and then queries attend to this global representation. Given that $\mathbf{K}'^{\top} \cdot \mathbf{V} = \sum_{j=1}^{n} \mathbf{k}'^{\top}_j \cdot \mathbf{v}_j$, we can write $\mathbf{K}'^{\top} \cdot \mathbf{V}$ in the form of Eq. (6.153), as follows

$$\mu_j = \mu_{j-1} + \mathbf{k}'^{\top}_j \cdot \mathbf{v}_j \tag{6.156}$$

Here $\mu_j \in \mathbb{R}^{d' \times d}$ is a variable that adds $\mathbf{k}'^{\top}_j \cdot \mathbf{v}_j$ at a time. Likewise, we can define another

[16] If $\mathbf{D}$ is a diagonal matrix, we occasionally denote $\mathbf{D}^{-1}\mathbf{A}$ as $\frac{\mathbf{A}}{\mathbf{D}}$.

variable $\nu_j \in \mathbb{R}^{d'}$

$$\nu_j = \nu_{j-1} + \mathbf{k'}_j^{\top} \tag{6.157}$$

Then, the output of self-attention for the j-th query can be written as (see also Eq. (6.146))

$$\mathrm{Att}_{\mathrm{self},j} = \frac{\mathbf{q}'_j \cdot \mu_n}{\mathbf{q}'_j \cdot \nu_n} \tag{6.158}$$

Clearly, this yields a linear-time model, because μ_n and ν_n are linear with respect to n. In simple implementations of this model, only μ_j and ν_j are kept. Each time a new query is encountered, we update μ_j and ν_j using Eqs. (6.156) and (6.157), and then compute $\mathrm{Att}_{\mathrm{self},j} = \frac{\mathbf{q}'_j \cdot \mu_j}{\mathbf{q}'_j \cdot \nu_j}$[17].

One straightforward extension to the linear attention model is to allow Eqs. (6.156) and (6.157) to combine different terms with different weights. For example, we can redefine μ_j and ν_j as

$$\mu_j = a \cdot \mu_{j-1} + (1-a) \cdot \mathbf{k'}_j^{\top} \cdot \mathbf{v}_j \tag{6.159}$$

$$\nu_j = a \cdot \nu_{j-1} + (1-a) \cdot \mathbf{k'}_j^{\top} \tag{6.160}$$

and train the parameter a as usual. Also, we can treat a as a gate and use another neural network to compute a [Peng et al., 2021]. Another model design is to add more terms to Eqs. (6.156) and (6.157) in order to provide a more expressive formulation of the linear attention approach [Bello, 2020; Schlag et al., 2021].

We have seen a general idea of designing linear models for the attention mechanism. The key design choice of such models is to remove the Softmax-based normalization, thereby yielding linear forms of representation based on various intermediate states of the models. This motivates several recently developed alternatives to self-attention in which efficient inference systems are developed on the basis of recurrent models of sequence modeling [Peng et al., 2023; Sun et al., 2023]. While these systems have different architectures, the underlying models have a similar form, as described in Eq. (6.153). Note that, by using the general formulation of recurrent models, we are not restricted to standard QKV attention. Instead we may give new meanings and forms to the queries, keys, and values.

The discussion here is also related to the memory models discussed in Section 6.4.2. From a memory perspective, the keys and values can be treated as encodings of the context. Therefore, in the linear attention model above we have a memory system in which two simple variables μ_j and ν_j are used to represent all the context information up to position j. This results in a fixed-length memory which is very useful in practice. There are also other linear approaches to encoding long sequences. For example, we can view the **moving average** model as an instance of Eq. (6.153), and average a series of state vectors of a Transformer model, either weighted or unweighted.

[17] In autoregressive generation, we generate a sequence from left to right. In this case, we need not consider the keys and values for positions $> j$.

3. State-Space Models

In control systems, **state-space models** (**SSMs**) are representations of a system whose input and output are related by some **state variables** (or states for short), and whose dynamics are described by first-order differential equations of these states. As a simple example, we consider a continuous-time linear time-invariant system which is given in the form of the state-space representation. We adopt a row-vector convention throughout this section, where all variables are represented as row vectors and linear transformations are applied on the right:

$$\frac{d\mathbf{z}(t)}{dt} = \mathbf{z}(t) \cdot \mathbf{A} + \mathbf{s}(t) \cdot \mathbf{B} \tag{6.161}$$

$$\mathbf{o}(t) = \mathbf{z}(t) \cdot \mathbf{C} + \mathbf{s}(t) \cdot \mathbf{D} \tag{6.162}$$

Here, $\mathbf{s}(t)$, $\mathbf{o}(t)$, and $\mathbf{z}(t)$ represent the values of the input variable, output variable, and state variable at time t, respectively[18]. In a general setting, these variables may possess different dimensionalities. For simplicity, we assume $\mathbf{s}(t), \mathbf{o}(t) \in \mathbb{R}^d$ and $\mathbf{z}(t) \in \mathbb{R}^{d_z}$[19]. Eq. (6.161) is called the **state equation**, where $\mathbf{A} \in \mathbb{R}^{d_z \times d_z}$ is the state matrix and $\mathbf{B} \in \mathbb{R}^{d \times d_z}$ is the input matrix. Eq. (6.162) is called the **output equation**, where $\mathbf{C} \in \mathbb{R}^{d_z \times d}$ is the output matrix and $\mathbf{D} \in \mathbb{R}^{d \times d}$ is the feedforward matrix.

These equations describe a continuous mapping from the input $\mathbf{s}(t)$ to the output $\mathbf{o}(t)$ over time. They are, therefore, often used to deal with continuous time series data. To apply this model to sequence modeling, we need to modify the above equations to obtain a discrete form of the state-space representation. Suppose that $\{\mathbf{s}_0, \mathbf{s}_1, ..., \mathbf{s}_n\}$ is a sequence of input data points sampled from $\mathbf{s}(t)$ with time step Δt. Similarly, we define $\{\mathbf{z}_0, \mathbf{z}_1, ..., \mathbf{z}_n\}$ and $\{\mathbf{o}_0, \mathbf{o}_1, ..., \mathbf{o}_n\}$ as sequences of the state and output vectors. Given this notation, we now have a discretized version of the SSM, written as

$$\mathbf{z}_t = \mathbf{z}_{t-1} \cdot \overline{\mathbf{A}} + \mathbf{s}_t \cdot \overline{\mathbf{B}} \tag{6.163}$$

$$\mathbf{o}_t = \mathbf{z}_t \cdot \overline{\mathbf{C}} + \mathbf{s}_t \cdot \overline{\mathbf{D}} \tag{6.164}$$

This formulation of the SSM defines an RNN with a residual connection. To be more precise, Eq. (6.163) describes a recurrent unit that reads the input at step t and the state at step $t-1$, without using any activation function. Eq. (6.164) describes an output layer that sums the linear transformations of the state $\mathbf{z}_t$ and the residual mapping $\mathbf{s}_t$.

The parameters $\overline{\mathbf{A}}$, $\overline{\mathbf{B}}$, $\overline{\mathbf{C}}$, and $\overline{\mathbf{D}}$ can be induced from $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ in several different ways, depending on how Eq. (6.161) is approximated by Eq. (6.163)[20]. One approach to time discretization is the **bilinear transform** (also known as Tustin's method). An alternative

[18] We use boldface letters to emphasize that these variables are vectors.

[19] In general state-space theory, these variables and matrices are defined over the field of complex numbers. Because models defined over the complex field generalize to real-valued state spaces, we restrict our discussion to the multi-dimensional real number field for simplicity.

[20] The discretization process can be interpreted as numerically solving the underlying differential equation. Note that Eq. (6.161) is an ODE:

$$\frac{d\mathbf{z}(t)}{dt} = g(\mathbf{z}(t), t) \tag{6.165}$$

approach is to use the **Zero-Order-Hold** (**ZOH**) discretization. A detailed discussion of these approaches lies beyond the scope of this book, and we refer the interested reader to standard textbooks on control theory for further details [Åström and Wittenmark, 2013].

The recurrent form of Eq. (6.163) makes it easy to compute the states and outputs over a sequence of discrete time steps. We can unroll $\mathbf{z}_t$ and $\mathbf{o}_t$ in a feedforward fashion

$$
\begin{array}{l|l}
\mathbf{z}_0 = \mathbf{s}_0 \cdot \overline{\mathbf{B}} & \mathbf{o}_0 = \mathbf{s}_0 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{C}} + \mathbf{s}_0 \cdot \overline{\mathbf{D}} \\
\mathbf{z}_1 = \mathbf{s}_0 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}} + \mathbf{s}_1 \cdot \overline{\mathbf{B}} & \mathbf{o}_1 = \mathbf{s}_0 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}} \cdot \overline{\mathbf{C}} + \mathbf{s}_1 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{C}} + \mathbf{s}_1 \cdot \overline{\mathbf{D}} \\
\mathbf{z}_2 = \mathbf{s}_0 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^2 + \mathbf{s}_1 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}} + \mathbf{s}_2 \cdot \overline{\mathbf{B}} & \mathbf{o}_2 = \mathbf{s}_0 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^2 \cdot \overline{\mathbf{C}} + \mathbf{s}_1 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}} \cdot \overline{\mathbf{C}} + \\
 & \qquad \mathbf{s}_2 \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{C}} + \mathbf{s}_2 \cdot \overline{\mathbf{D}} \\
\ldots\ldots & \ldots\ldots
\end{array}
$$

It is easy to write

$$
\mathbf{z}_t = \sum_{i=0}^{t} \mathbf{s}_i \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^{t-i} \tag{6.171}
$$

$$
\mathbf{o}_t = \sum_{i=0}^{t} \mathbf{s}_i \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^{t-i} \cdot \overline{\mathbf{C}} + \mathbf{s}_t \cdot \overline{\mathbf{D}} \tag{6.172}
$$

Clearly, the right-hand side of Eq. (6.172) can be interpreted as the sum of a convolutional term and a linear term. Given that

$$
\begin{aligned}
\sum_{i=0}^{t} \mathbf{s}_i \cdot \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^{t-i} \cdot \overline{\mathbf{C}} = & \begin{bmatrix} \mathbf{s}_0 & \mathbf{s}_1 & \ldots & \mathbf{s}_t \end{bmatrix} \cdot \\
& \begin{bmatrix} \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^{t} \cdot \overline{\mathbf{C}} & \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^{t-1} \cdot \overline{\mathbf{C}} & \ldots & \overline{\mathbf{B}} \cdot \overline{\mathbf{C}} \end{bmatrix}^{\top}
\end{aligned} \tag{6.173}
$$

---

where

$$
g(\mathbf{z}(t), t) = \mathbf{z}(t) \cdot \mathbf{A} + \mathbf{s}(t) \cdot \mathbf{B} \tag{6.166}
$$

There are many numerical methods to approximate the solution to an ODE. For example, the Euler method for solving ODEs can be expressed as (see Section 6.3.3)

$$
\mathbf{z}_t = \mathbf{z}_{t-1} + \Delta t \cdot g(\mathbf{z}_{t-1}, t) \tag{6.167}
$$

Substituting Eq. (6.166) into Eq. (6.167) yields

$$
\begin{aligned}
\mathbf{z}_t &= \mathbf{z}_{t-1} + \Delta t(\mathbf{z}_{t-1} \cdot \mathbf{A} + \mathbf{s}_t \cdot \mathbf{B}) \\
&= \mathbf{z}_{t-1} \cdot (\mathbf{I} + \Delta t \cdot \mathbf{A}) + \mathbf{s}_t \cdot (\Delta t \cdot \mathbf{B})
\end{aligned} \tag{6.168}
$$

This gives one of the simplest forms of the discretized state equations [Gu et al., 2022b], that is,

$$
\overline{\mathbf{A}} = \mathbf{I} + \Delta t \cdot \mathbf{A} \tag{6.169}
$$

$$
\overline{\mathbf{B}} = \Delta t \cdot \mathbf{B} \tag{6.170}
$$

we define a convolutional filter with parameters

$$\mathbf{W}_{\text{ssm}} = \begin{bmatrix} \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^{n_{\max}} \cdot \overline{\mathbf{C}} & \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^{n_{\max}-1} \cdot \overline{\mathbf{C}} & \ldots & \overline{\mathbf{B}} \cdot \overline{\mathbf{C}} \end{bmatrix} \tag{6.174}$$

where $n_{\max}$ is the maximum length of the sequence[21]. Then, the output of the state-space model for a sequence $\mathbf{S} = \begin{bmatrix} \mathbf{s}_0 \\ \vdots \\ \mathbf{s}_n \end{bmatrix}$ can be expressed as

$$\mathbf{O} = \text{Conv}(\mathbf{S}, \mathbf{W}_{\text{ssm}}) + \text{Linear}(\mathbf{S}, \overline{\mathbf{D}}) \tag{6.175}$$

where $\text{Conv}(\cdot)$ is the convolution operation, and $\text{Linear}(\cdot)$ is the linear transformation operation. Such a treatment of the state-space model enables the system to be efficiently implemented using fast parallel convolution algorithms.

However, naive implementations of this model often perform poorly. As with many deep neural architectures, careful parameter initialization is crucial. Specifically, restricting the state matrix $\mathbf{A}$ to particular structures (e.g., HiPPO matrices) has proven essential for memorizing and generalizing over long sequences [Gu et al., 2022a].

Another computational bottleneck is the repeated matrix multiplication required to compute $\overline{\mathbf{A}}^n$. For large n, this operation is computationally expensive and numerically unstable. A common approach in modern SSMs is **diagonalization**. The idea is to apply a basis transformation to the state space such that $\mathbf{A}$ (or $\overline{\mathbf{A}}$) becomes diagonal. Given an SSM parameterized by $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$, we can define an equivalent model $(\mathbf{U}\mathbf{A}\mathbf{U}^{-1}, \mathbf{B}\mathbf{U}^{-1}, \mathbf{U}\mathbf{C}, \mathbf{D})$ via an invertible transformation matrix $\mathbf{U}$. It can be shown that both representations are mathematically equivalent under a change of basis[22].

By applying this transformation, and assuming $\overline{\mathbf{A}}$ can be diagonalized into the canonical form $\mathbf{P}^{-1}\Lambda\mathbf{P}$ (where Λ is diagonal), we enforce the constraint that the state matrix is diagonal, giving rise to **diagonal state-space models**. To illustrate, consider the filter used in the convolutional representation (Eq. (6.173)). Substituting $\overline{\mathbf{A}} = \mathbf{P}^{-1}\Lambda\mathbf{P}$, we can rewrite the term $\overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^t \cdot \overline{\mathbf{C}}$ as:

$$\begin{aligned} \overline{\mathbf{B}} \cdot \overline{\mathbf{A}}^t \cdot \overline{\mathbf{C}} &= \overline{\mathbf{B}} \cdot (\mathbf{P}^{-1}\Lambda\mathbf{P})^t \cdot \overline{\mathbf{C}} \\ &= \overline{\mathbf{B}} \cdot (\mathbf{P}^{-1}\Lambda\mathbf{P}) \cdot (\mathbf{P}^{-1}\Lambda\mathbf{P}) \cdots (\mathbf{P}^{-1}\Lambda\mathbf{P}) \cdot \overline{\mathbf{C}} \\ &= (\overline{\mathbf{B}} \cdot \mathbf{P}^{-1}) \cdot \Lambda^t \cdot (\mathbf{P} \cdot \overline{\mathbf{C}}) \end{aligned} \tag{6.176}$$

Since Λ is a diagonal matrix, we can efficiently compute Λ^t by simply raising all the entries

[21] Here $\mathbf{W}_{\text{ssm}}$ can be represented as an $n_{\max} \times d \times d$ tensor.

[22] A state-space transformation maps the state variables into a new coordinate system: $\mathbf{z}'(t) = \mathbf{z}(t) \cdot \mathbf{U}^{-1}$.

of Λ to the t-th power. We then have a computationally cheaper model, in which

$$\overline{\mathbf{A}}' = \Lambda \tag{6.177}$$

$$\overline{\mathbf{B}}' = \overline{\mathbf{B}} \cdot \mathbf{P}^{-1} \tag{6.178}$$

$$\overline{\mathbf{C}}' = \mathbf{P} \cdot \overline{\mathbf{C}} \tag{6.179}$$

$$\overline{\mathbf{D}}' = \overline{\mathbf{D}} \tag{6.180}$$

More detailed discussions of diagonal state-space models in sequence modeling can be found in Gu et al. [2021]'s work.

The application of state-space models to Transformers is simple. Each self-attention sub-layer is replaced in this case by an SSM sub-layer as described in Eqs. (6.163) and (6.164). As we have seen there is a close relationship between state-space models and both CNNs and RNNs. For sequence modeling, we can deal with a sequence of tokens either sequentially as in RNNs, or in parallel as in CNNs. This leads to a new paradigm that takes both the sequential view and the parallel view of the sequence modeling problem — for training, the system operates like CNNs to make use of fast parallel training algorithms; for prediction, the problem is re-cast as a sequential update problem which can be efficiently solved by using RNN-like models.

While the formalism of state-space models is different from that we discussed in this chapter, it provides a unifying framework for sequence modeling. By viewing the problem through complementary perspectives (parallel and sequential), we can optimize for different hardware and latency constraints. Several recent architectures are motivated by this duality, yielding systems that effectively combine the parallelizability of Transformers with the inference efficiency of RNNs [Orvieto et al., 2023; Sun et al., 2023].

6.4.6 Conditional Computation

So far in our discussion of efficient Transformer models, we have assumed that the model architecture is determined before training and remains fixed throughout. We now turn to the case of learning efficient model architectures. We can write a model in the form

$$\mathbf{y} = \mathrm{Model}(\mathbf{x}, g(\mathbf{x})) \tag{6.181}$$

where $\mathbf{x}$ and $\mathbf{y}$ are the input and output of the model. $g(\mathbf{x})$ is a function that returns the model architecture and corresponding parameters for the given input $\mathbf{x}$. In general, we adopt the convention prevalent in learning problems of using a fixed model architecture and learning only the parameters, say, $g(\mathbf{x}) = \theta$. In this case, the goal is to learn optimal parameter values given the architecture and training data. On test data, we make predictions using the same model architecture along with the optimized parameters.

A natural extension of this approach is to consider the learning of both the model architecture and parameters. In architecture learning, we would like to find a function $\hat{g}(\mathbf{x})$ that produces the optimal model architecture and parameter values given the input $\mathbf{x}$. However, searching the entire hypothesis space of all possible combinations of architectures and parameters is intractable, and so practical methods are required. Two classes of methods can be applied:

- **Neural Architecture Search** (**NAS**). In **automated machine learning** (**AutoML**), neural architecture search is the process of exploring a space of neural networks to find one that best fits some criteria [Zoph and Le, 2016; Elsken et al., 2019b]. Once the optimal neural network is determined, its parameters are trained as usual, and then applied to new data. In order to make search tractable, several additional techniques, such as search space pruning and fast search algorithms, are typically used. Applying neural architecture search to the development of efficient neural networks is straightforward [Howard et al., 2019; Tan and Le, 2019]. We need only incorporate efficiency measures into the performance estimation of neural networks, for example, the search can be guided by a criterion that penalizes neural networks with high latency or excessive memory requirements.
- **Dynamic Neural Networks**. The key idea of dynamic neural networks is to adapt a neural network to various inputs dynamically [Gupta et al., 2004; Han et al., 2021b]. Ideally, we would like to learn $\hat{g}(\cdot)$, and then, for any input $\mathbf{x}_{\mathrm{new}}$, we apply the model $\mathrm{Model}(\mathbf{x}_{\mathrm{new}}, \hat{g}(\mathbf{x}_{\mathrm{new}}))$. As a result, at test time we may have different model structures and/or different parameters for different inputs. However, it is infeasible to develop a function $\hat{g}(\cdot)$ that can model arbitrary neural networks. In practice, $\hat{g}(\cdot)$ is often considered to represent a family of sub-networks of a super-network. The problem is therefore reframed as a simpler problem to learn to choose which sub-network is used for a given input.

From a machine learning perspective, the approaches to neural architecture search are general and can be applied to any neural network. On the other hand, from a practical perspective, it is still difficult to find an efficient neural network that is sufficiently powerful and generalizes well. While neural architecture search provides interesting ideas for developing efficient Transformer models, we do not discuss it in depth here. Instead, the reader can refer to the above papers to have a general idea of it, and refer to So et al. [2019], Wang et al. [2020a], and Hu et al. [2021]'s work for its application to Transformers.

In this subsection, we focus on a particular family of approaches to dynamic neural networks, called **conditional computation**. This concept was originally motivated by the dynamic selection of neurons of a neural network [Bengio et al., 2013; 2015]. More recently, it has often been used to refer to the process of dynamically selecting parts of a neural network. A narrow view of conditional computation is to see $g(\cdot)$ as an adaptive neural network which dynamically reduces or grows the number of computation units (such as neurons and layers). As a result, computation can adapt to changing conditions, and we can seek a good accuracy-latency trade-off by this adaptation mechanism.

A common way to achieve this is to learn how to skip some computation steps so that we can work with a subset of the network [Xu and Mcauley, 2023]. One of the simplest methods, sometimes called **early exiting**, is to stop the computation at some point during encoding or decoding a sequence. This technique is often used in practical sequence generation applications where a low latency is required. Suppose $y_1 \ldots y_{n_{\max}}$ is the longest sequence that the system can generate, and $\mathbf{s}_1 \ldots \mathbf{s}_{n_{\max}}$ is the corresponding sequence of the hidden states of the top Transformer layer. Then we develop a model $f_{\text{stop}}(\cdot)$ that takes one hidden state $\mathbf{s}_i$ at a time and produces a distribution of a binary variable $c \in \{\text{stop}, \text{nonstop}\}$:

$$\Pr(c|\mathbf{s}_i) = f_{\text{stop}}(\mathbf{s}_i) \tag{6.182}$$

The generation process terminates if $\Pr(\text{stop}|\mathbf{s}_i)$ is sufficiently large, for example

$$\Pr(\text{stop}|\mathbf{s}_i) \geq \Pr(\text{nonstop}|\mathbf{s}_i) + \theta_{\text{stop}} \tag{6.183}$$

where θ_{stop} denotes the minimal margin for distinguishing the two actions[23]. This formulation is also related to the stopping criterion problem that is frequently discussed in search algorithms for sequence generation (see Chapter 5). $f_{\text{stop}}(\cdot)$ can be designed in several different ways. For example, in many practical applications, the stopping criterion is based on simple heuristics. Alternatively, we can define the function $f_{\text{stop}}(\cdot)$ as a neural network and train it using labeled data.

The above approach can be easily extended to handle situations in which some of the tokens are skipped. This learning-to-skip approach is typically used in the encoding stage in which all input tokens are given in advance. Let $\mathbf{h}_1 \ldots \mathbf{h}_m$ be low-level representations of a sequence $x_1 \ldots x_m$. Like Eq. (6.182), we can develop a model $\Pr(c|\mathbf{s}_i)$ ($c \in \{\text{skip}, \text{nonskip}\}$) to determine whether the token x_i can be skipped. Figure 6.16 (a) and (b) show illustrations of early exiting and skipping. Note that the learning-to-skip method overlaps with other lines of research on training neural networks. For example, erasing some of the input tokens in training is found to be useful for achieving higher generalization of Transformer models [Shen et al., 2020a; Kim and Cho, 2021]. This method is also related to the downsampling methods which will be discussed in Section 6.4.8.

A second approach to conditional computation is to resort to **sparse expert models**, or its popular instance — MoE [Yuksel et al., 2012]. In deep learning, a model of this kind is typically built from a number of experts which are neural networks having the same structure but with different parameters. In this way, we can construct a large model by simply increasing the number of experts. When running this model, during either training or prediction, we activate only a small number of the experts by some routing algorithms (see Figure 6.16 (c)). An MoE model is an adaptive network since each time we have a new input, the model routes it to different experts. In Section 6.3.4, we presented the basic form of MoE, and showed how Transformer models can be scaled up by this sparse method. For a comprehensive review of the recent advances in MoE, we refer the interested reader to Fedus et al. [2022a]'s work.

A third approach that can be used to adapt a Transformer model to changing input is to dynamically shrink the number of layers. Several methods have been proposed to do this in an attempt to improve inference efficiency. The simplest of these is to exit at some hidden layers by which we can still make accurate predictions for the input (see Figure 6.16 (d) and

[23] An equivalent form of Eq. (6.183) is $\Pr(\text{stop}|\mathbf{s}_i) \geq \frac{1+\theta_{\text{stop}}}{2}$.

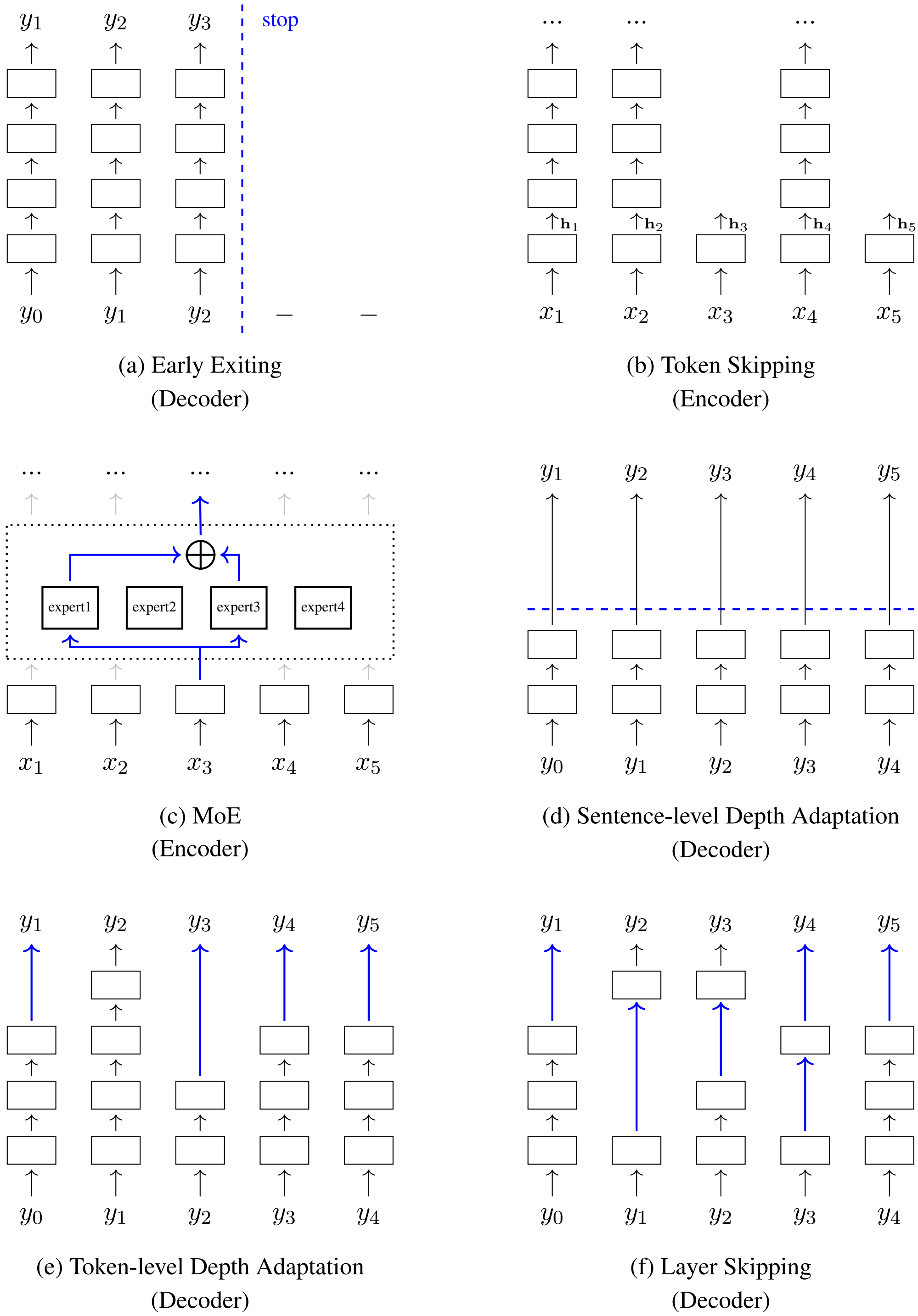


Figure 6.16: Methods of conditional computation, including early exiting, token skipping, MoE, sentence-level depth adaptation, token-level depth adaptation, and layer skipping. While these methods are illustrated using either the encoding or decoding process, most of them can be applied to both Transformer encoders and decoders.

(e)). To do this, we can either determine the appropriate depth for the entire sequence (call it a **sentence-level depth-adaptive model**), or use an adaptive depth for each token (call it a **token-level depth-adaptive model**). Here we consider token-level depth-adaptive models but the methods can be easily extended to sequence-level depth-adaptive models.

Suppose there are L stacked layers at position i[24]. We would ideally like to find a layer in the stack, which can be used as the last hidden layer for making predictions, and whose depth is as low as possible. However, we cannot simply use the L-th layer of the stack as the oracle for this problem, because we never know in advance what the last layer generates during inference. Instead, we need to determine whether the network should stop growing at depth i, considering the layers generated so far.

Now suppose we have a Transformer decoder which produces a distribution over a vocabulary V at each step. As usual, we denote the output of the l-th layer at step i by $\mathbf{s}_i^l$. For each $\mathbf{s}_i^l$, we create an output layer that produces a distribution $\mathbf{p}_i^l$ over the vocabulary (call it an **early exit classifier**), given by

$$\mathbf{p}_i^l = \mathrm{Softmax}(\mathbf{s}_i^l \cdot \mathbf{W}_{\mathrm{o}}^l) \tag{6.184}$$

where $\mathbf{W}_{\mathrm{o}}^l \in \mathbb{R}^{d\times|V|}$ is the parameter matrix. Hence we have $L-1$ additional output layers, each corresponding to a hidden layer from depth 1 to $L-1$. At training time, we consider the cross-entropy losses of $\{\mathbf{p}_i^1, ..., \mathbf{p}_i^{L-1}\}$, and train these layers together with the Transformer model. At test time, the depth of the network grows as usual, and we use $\{\mathbf{p}_i^1, ..., \mathbf{p}_i^l\}$ and/or $\{\mathbf{s}_i^1, ..., \mathbf{s}_i^l\}$ to determine whether we should exit at the l-th layer. There are several exit criteria, for example,

- Common criteria are based on measures of the confidence of predictions. A simple method is to compute the entropy of $\mathbf{p}_i^l$, and exit if this entropy is above a pre-defined value.
- Alternatively, one can view the maximum probability of the entries of $\mathbf{p}_i^l$ as the confidence of the prediction.
- Instead of considering the output of a single layer, we can also examine the change in the outputs or hidden states over a number of layers. For example, we can measure the similarity between $\mathbf{p}_i^{l-1}$ and $\mathbf{p}_i^l$ or between $\mathbf{s}_i^{l-1}$ and $\mathbf{s}_i^l$. If the similarity is above a given threshold, then we say that the output of the neural network tends to converge and the number of layers can stop growing.
- The above methods can be extended to examine the change in the predictions made by the classifiers associated with the layers. For example, the model can choose to exit if the predictions made by the classifiers remain unchanged for a number of layers.

Discussions of these criteria can be found in the related papers [Xin et al., 2020; Zhou et al., 2020; Schuster et al., 2022]. There are a variety of ways to improve these early exit methods. One is to explore other forms of the prediction for each layer. For example, we can develop a model that directly predicts how many layers we need to model the input [Elbayad

[24] A layer is a standard Transformer block consisting of a few sub-layers.

et al., 2020]. Another line of research on early exit focuses on better training for these models, for example, we can consider various loss functions for training the classifiers [Schwartz et al., 2020; Schuster et al., 2022]. In addition, there is also interest in learning the combination of the outputs of multiple layers so that we can make predictions by using multiple levels of representation [Zhou et al., 2020; Liao et al., 2021].

A problem with token-level adaptive-depth models is that the representations at certain depths may be absent in the previous steps. In this case, standard self-attention is not directly applicable, because we may not attend to the previous tokens in the same level of representation. For training, this can be addressed by using all the L layers of the full model. For inference, we can either duplicate the layer from which we exit to fill up the layer stack, or modify the self-attention model to enable it to attend to the representations of the previous tokens at different depths.

It is also possible to select any subset of the layers for constructing a shallow network. The adaptive models therefore can be generalized to skipping models (see Figure 6.16 (f)). As with the early exit problem, the skipping problem can be framed as a learning task, in which a classifier is trained to decide whether a layer should be dropped. The learning-to-skip problem has been studied in the field of computer vision [Wang et al., 2018b; Wu et al., 2018b]. However, learning a skipping model for large-scale, deep neural networks is difficult. For practical systems, it still seems reasonable to use heuristics or cheap models to obtain a neural network having skipped layers, which has been discussed in recent pre-trained NLP models [Wang et al., 2022c; Del Corro et al., 2023].

6.4.7 Model Transfer and Pruning

Many large Transformer models have been successfully developed to address NLP problems. A common question is: can we transform a large, well-trained model into a smaller one that allows for more efficient inference? At a high level, this can be thought of as a **transfer learning** problem in which the knowledge is transferred from one model to another. However, we will not discuss this general topic, which spans a broad range of issues and models, many of which fall outside the scope of this chapter. Instead, we narrow our discussion to two approaches that are widely used for training smaller neural networks from larger ones.

1. Knowledge Distillation

Knowledge distillation is a process of compressing the knowledge in a large neural network (or an ensemble of neural networks) into a smaller neural network [Hinton et al., 2015]. In the supervised learning of neural networks, objective functions are generally designed to quantify the error between the predicted and true answers. Hence, by minimizing this loss, models are trained to output the correct answers. While models are typically optimized on the training data in this manner, what we really want is to generalize to unseen data. This is, however, difficult because training only with ground-truth labels provides no direct supervision signal regarding generalization. In knowledge distillation, instead of forcing a model to match the ground-truth output, we train it to mimic the behavior (or output distributions) of a larger model. To achieve this, we directly transfer the knowledge of a pre-trained model to the model we wish to train.

A frequently used approach to knowledge distillation is **teacher-student training**. A teacher model is typically a relatively large neural network that has already been trained and can generalize well. A student model is a smaller neural network, such as one with fewer layers, to which we transfer the knowledge. A simple way to distill the knowledge from the teacher model into the student model is to use the output of the teacher model as the "correct" answer for training the student model. Suppose we have a teacher Transformer model that can generate a sequence of distributions $\{\Pr(\cdot|y_0,\mathbf{x}),...,\Pr(\cdot|y_0...y_{n-1},\mathbf{x})\}$ for the input $\mathbf{x}$. To keep the notation simple, we denote the distribution $\Pr(\cdot|y_0...y_{i-1},\mathbf{x})$ as $\widetilde{\mathbf{p}}_i$. Similarly, we denote the output of the student Transformer model for the same input as $\mathbf{p}_i$. As usual, we consider a loss function $\mathrm{Loss}(\widetilde{\mathbf{p}}_i,\mathbf{p}_i)$ (such as the cross-entropy function) for computing the distance between $\widetilde{\mathbf{p}}_i$ and $\mathbf{p}_i$. Then, we can define the loss over the entire sequence as

$$\mathcal{L}(\mathbf{x},\theta) = \frac{1}{n}\sum_{i=1}^{n}\mathrm{Loss}(\widetilde{\mathbf{p}}_i,\mathbf{p}_i) \tag{6.185}$$

where θ denotes the parameters of the student model[25]. Using this loss, we can optimize θ for any given set of source sequences $\{\mathbf{x}_1,...,\mathbf{x}_K\}$ by minimizing the objective $\sum_{k=1}^{K}\mathcal{L}(\mathbf{x}_k,\theta)$.

Several extensions to this basic method have been developed to model the problem of knowledge transfer between two models. A simple way is to use the hidden states instead of the output probabilities as the training targets [Romero et al., 2014]. In this case, the objective is to minimize the difference between some hidden states of the teacher model and the corresponding states of the student model. Rather than using the outputs of various layers as the targets for training the student model, another technique is to model the relations between samples and train the student model by minimizing the difference between the relation encodings of the teacher and student models [Park et al., 2019; Peng et al., 2019]. For example, we can develop a relation encoding model based on the Transformer architecture. The goal is then to optimize the student model so that its corresponding relation encoding of a group of samples is as close as possible to that of the teacher model.

For sequence generation problems, a special case of knowledge distillation, which can be viewed as a means of **data augmentation**, is often used for developing lightweight models [Kim and Rush, 2016]. For example, consider the problem of transferring the translation ability of a well-developed machine translation model (i.e., the teacher model) to a new model (i.e., the student model). Given a set of source-side sentences $\{\mathbf{x}_1,...,\mathbf{x}_K\}$, we can use the teacher model to translate each $\mathbf{x}_k$ to a target-side sentence $\widetilde{\mathbf{y}}_k$. Then, by treating $\mathbf{x}_k$ and $\widetilde{\mathbf{y}}_k$ as paired sentences, we obtain a bilingual dataset consisting of $\{(\mathbf{x}_1,\widetilde{\mathbf{y}}_1),...,(\mathbf{x}_K,\widetilde{\mathbf{y}}_K)\}$. We can use this bilingual dataset as the labeled dataset to train the student model as usual. One advantage of this data augmentation method is that it is architecture-agnostic. It does not require access to the internal architectures of either the teacher or the student models. Hence, it can be applied even if the teacher model is used as a black-box model. More detailed discussions on knowledge distillation can be found in the surveys by Gou et al. [2021] and Wang and Yoon [2021].

[25]We omit the parameters of the teacher model because they are fixed throughout the training process.

2. Structured Pruning

Pruning is among the most popular model compression methods and has been applied to a broad range of systems. One common approach is **unstructured pruning**, in which we retain only a sparse subset of connections between neurons. However, as with most sparse models, networks pruned in this way typically require specialized implementations and hardware support, which in turn limits their efficiency gains in certain applications. A simpler yet more aggressive alternative is **structured pruning**. In deep learning, structured pruning is a technique that removes entire groups of neurons or connections simultaneously. For example, we can remove an entire layer of neurons from a neural network to obtain a shallower model. Given their multi-layer and multi-head architecture, Transformers are naturally suited to structured pruning, and we can prune a Transformer network in several different ways. For instance, we can prune specific heads in the multi-head attention mechanism [Voita et al., 2019; Michel et al., 2019], or entire layers within the stack [Hou et al., 2020; Kim and Awadalla, 2020].

Formally, we can represent a neural network as a set of parameter groups $\{\theta_1, ..., \theta_R\}$, each corresponding to a specific component or sub-module of the network. Our goal is to find a subset of $\{\theta_1, ..., \theta_R\}$ that allows us to build a model yielding strong performance while maintaining significantly lower computational complexity. However, an exhaustive search for such an optimal subset is infeasible because there is a combinatorial number of possible candidates, and evaluating all of them is computationally prohibitive.

One approach to structured pruning is to randomly discard components of a model. We can repeat this random pruning process multiple times to generate a pool of model candidates, and select the best-performing one. Another approach relies on heuristics to determine which components are redundant and can be safely removed. Common measures for evaluating the importance of a parameter group θ_r include various metrics based on the norms of its weights or gradients [Santacroce et al., 2023]. We can then prune θ_r if these metrics fall below or exceed predefined thresholds. A third approach frames the pruning problem as a continuous optimization task by introducing trainable gates that indicate the presence or absence of different components [McCarley et al., 2019; Wang et al., 2020d; Lagunas et al., 2021]. The final pruned model can be directly derived using these trained gates. It is worth noting that, in many cases, pruning is no longer merely a post-processing step for a fully trained model, but rather an integral part of the training process itself.

6.4.8 Sequence Compression

In sequence modeling and generation problems, time and space complexities are strongly influenced by the length of the input or output sequences. Therefore, shorter sequences are generally preferred. This is particularly important for Transformer models, as their time and space complexities scale quadratically with sequence length, making memory footprint and latency significant computational bottlenecks for very long sequences. While previous subsections discussed modifications to the Transformer architecture to handle long sequences, in this section, we instead explore methods for compressing sequences to more manageable lengths.

One simple approach is to map the input sequence to a fixed-size representation. For example, using the recurrent models discussed in Section 6.4.2, we can encode a sequence of vectors into a single vector. This method can be easily extended to generate a larger representation so that this representation can retain more information from the original input. For example, we can select a fixed number of the hidden states over the sequence to form a new sequence of fixed length. Another way to represent a variable-length sequence as a fixed-length sequence is to perform cross-attention between the input vectors and a set of learnable hidden representations. In the work of Jaegle et al. [2021], this is done by introducing r hidden representations $\{\mathbf{u}_1, ..., \mathbf{u}_r\}$, and then applying cross-attention between the input vectors $\mathbf{x}_1, ..., \mathbf{x}_m$ and these hidden representations. The attention model can be a standard QKV attention model in which we view $\{\mathbf{u}_1, ..., \mathbf{u}_r\}$ as queries and $\{\mathbf{x}_1, ..., \mathbf{x}_m\}$ as keys and values. The output of this model is a sequence of r vectors, which can be used as fixed-length input to downstream systems.

A second approach is to use downsampling to compress the sequence into a shorter one. A typical method of downsampling is strided convolution, which has been widely used in computer vision and speech processing. For example, suppose there is a sequence of m vectors in $\mathbb{R}^d$. We can define a filter with a width of 2 and a stride of 2. By taking the sequence as input, the filter produces a sequence of $\frac{m}{2}$ new vectors in $\mathbb{R}^d$, and so we have a reduction factor of 2. Also, we can stack multiple convolutional layers or pooling layers to achieve a desired level of length reduction, called **progressive downsampling**. However, it seems inevitable that downsampling will lead to information loss [Han et al., 2020; Burchi and Vielzeuf, 2021]. We need to consider a trade-off between the compressed sequence length and the performance of downstream systems [Xu et al., 2023b].

In NLP, the problem of sequence compression is also closely related to the problem of tokenizing input strings. Therefore, tokenization is a practical approach that can be taken to address the length issue. Segmenting a string into small tokens (such as characters) generally reduces the sparsity of the data, which makes it easier to learn the embeddings of these tokens, but such approaches often lead to a long sequence. By contrast, we will have a shorter sequence if we segment the input string into larger units, but this may suffer from data sparsity. In deterministic tokenization methods, which produce tokenization results using statistics collected from the entire dataset, the sequence length can be controlled to some extent by adjusting specific hyperparameters, for example, in byte pair encoding [Sennrich et al., 2016b], increasing the size of the vocabulary generally reduces the number of resulting tokens. Another way to obtain an appropriate sequence of tokens is to use a model for choosing among tokenization candidates [Kudo, 2018; Provilkov et al., 2020]. As with many probabilistic models for text generation, in this case, we can incorporate priors to the criterion for tokenization selection so that we can express a preference for shorter sequences over longer sequences.

A fourth approach to sequence compression is to drop some of the tokens in the sequence. For instance, in many practical applications, sequences are simply truncated when their lengths exceed a predefined threshold. We can relate this to the early exiting and skipping approaches in conditional computation. Thus the methods discussed in Section 6.4.6 can be directly applied. The token dropping methods can also be viewed as pruning methods, called **token**

pruning. By discarding tokens that are less important for representing the entire sequence, token pruning can significantly reduce the sequence length while maintaining the performance of NLP systems on downstream tasks [Kim et al., 2023].

6.4.9 High Performance Computing Methods

So far in this section, we have discussed efficient Transformer models from the perspective of deep learning. However, we have not considered their hardware efficiency. As modern hardware provides a variety of execution paradigms, the practical time and memory footprint savings generally depend on the specifications of the hardware systems. One line of research on the efficient use of computing resources explores parallel computing methods. There have been many attempts to develop large-scale Transformer models by using a cluster of machines. Typically, scaling Transformers to models with billions or even tens of billions of parameters requires a careful design of parallelism strategies for sharding the big networks. More efficient implementations of such systems also require consideration of communication overhead in the cluster, as well as the utilization of sparse models that activate only a small subset of the parameters for each sample, thereby enabling the use of very large models. Most of these methods have been extensively studied in the literature [Lepikhin et al., 2021; Barham et al., 2022; Fedus et al., 2022b]. The results of these studies are foundational to many follow-on works investigating the **scaling laws** for large language models [Brown et al., 2020; Chowdhery et al., 2022]. Since large-scale distributed models are generic and not specialized to the case of Transformers, we skip their discussion here. Interested readers can refer to the above papers for more detailed discussions.

In this subsection, we consider hardware-aware methods to seek greater computational efficiency for Transformer models. We first consider a simple but widely used method that aims to represent and execute neural networks using lower- or mixed-precision formats [Gholami et al., 2022]. Conventional neural networks are typically based on single-precision and/or double-precision floating-point representations of data. While single-precision floating-point data types provide a sufficiently precise way to represent parameters and intermediate states in most cases, in some applications, they are not essential. As an alternative, one can use half-precision (or even lower-precision) formats for representing floating-point numbers in neural networks. The size of the resulting model is thus half the size of the original model. One advantage of using half-precision floating-point representations is that, although processing such data types requires new APIs for linear algebra operations and hardware support, it does not change the model architecture, requiring only slight modifications to the system. For example, half-precision floating-point representations can be applied to either the training or inference of Transformers, or both.

Recently, the deployment of large Transformer models has been further improved by quantizing these models. In signal processing, quantization is a process of mapping continuous values (i.e., floating-point representations) to a set of discrete values (i.e., fixed-point representations). This process is generally implemented using a system called a quantizer. In the context of neural networks, a quantizer involves two functions — the quantization function and the de-quantization function. The quantization function maps a floating-point number to a (lower-bit) integer. A simple quantization function is given by

$$Q(x) = \lfloor \frac{x}{s} \rceil \tag{6.186}$$

where $\lfloor \cdot \rceil$ is a rounding function[26], x is the real-valued input, and s is the quantization step size that controls the level of quantization. The quantization function is coupled with a de-quantization function

$$D(r) = s \cdot r \tag{6.187}$$

With this notation, the quantizer can be expressed as $D(Q(x)) = s \cdot \lfloor \frac{x}{s} \rceil$. The difference between $D(Q(x))$ and x is called the quantization error. A smaller value of s typically reduces the quantization error. In practice, however, we wish to choose an appropriate value of s in order to spread the possible values of $Q(x)$ evenly across the integer representation range. For example, $s = \frac{\max\{D(r)\}}{2^p - 1}$, where p is the number of bits used to represent an integer and $\max\{D(r)\}$ is the maximum representable value after de-quantization. The equations above show one of the simplest cases of quantization. More general discussions of quantization can be found in books on digital signal processing and related surveys [Oppenheim and Schafer, 1975; Rabiner and Gold, 1975; Gray, 1998].

Applying quantization to Transformers is relatively straightforward. The idea is that we quantize the inputs and model parameters using $Q(x)$ and feed them to a quantized Transformer model in which all layers operate on integer-valued tensors. In other words, we implement the model using integer-only arithmetic. However, the price to be paid for this compressed model, as with many approximation approaches to deep learning, is that its predictions are less accurate than those of the standard Transformer. Using integer operations to approximate continuous-valued operations generally leads to approximation errors. These errors will accumulate if the quantized neural network is deep. Furthermore, Transformer models involve components (such as self-attention sub-layers) that require relatively complex linear algebra operations. Simply applying quantization to these modules will lead to a significant loss in accuracy. One solution is to simplify the model architecture and develop new sub-models that are more suited for quantization. Alternatively, a more common paradigm in quantized neural networks is to add de-quantization functions to the neural networks so that the outputs of a layer are floating-point tensors and can be used as usual in the following steps. Consider a simple example where we multiply a real-valued input matrix $\mathbf{a}$ with a real-valued parameter matrix $\mathbf{A}$. We first quantize $\mathbf{a}$ and $\mathbf{A}$, and multiply them using integer-based matrix multiplication. The result is then de-quantized to a real-valued matrix. In this way, we obtain an approximation $D(Q(\mathbf{a}) \cdot Q(\mathbf{A}))$ to $\mathbf{a} \cdot \mathbf{A}$ at a very low computational cost.

However, sandwiching each layer between $Q(\cdot)$ and $D(\cdot)$ will lead to the additional cost of running $Q(\cdot)$ and $D(\cdot)$. In some practical applications, the computational overhead introduced by $Q(\cdot)$ and $D(\cdot)$ is even larger than the time savings of performing integer-based operations. In general, the benefits of quantizing neural networks are greater than the costs only if the

[26] $\lfloor a \rceil$ returns the integer closest to a.

neural networks are sufficiently large. Therefore, in practice, it is common to perform quantized computation only for operations whose computational costs are high. For example, in recent large language models, quantization is primarily applied to the multiplication of large matrices, yielding significant time and memory savings.

While quantization approaches can be used in both training and inference, a widely used approach is to quantize Transformer models after training (called **post-training quantization**). In this approach, quantization is performed on well-trained floating-point-based neural networks, resulting in fewer quantization-related errors. However, these errors cannot be easily compensated for because they are introduced after training is complete. A more promising approach is to incorporate quantization during training so that the model can learn to compensate for quantization-related errors [Jacob et al., 2018; Nagel et al., 2021]. There have been several attempts to apply quantization-aware training to Transformers [Bondarenko et al., 2021; Stock et al., 2021; Yang et al., 2023b]. In addition to computational efficiency, another important consideration for high-performance systems is the constraints of the memory hierarchy. In general, better system design requires considering the speeds and capacities of different memory levels. The problem is even more complicated when training large Transformer models on modern hardware where both GPUs and CPUs are utilized. A general principle of system design is that memory transfers between different memory levels should be minimized. While we would ideally like to have a large, high-level memory on which we can store all the data we need to process, in many practical situations, the size of fast, on-chip memory is orders of magnitude smaller than the total size of the data. In this case, we can re-order the memory access in the algorithms so that data used in adjacent computation steps can be loaded into high-speed memory all at once. This idea motivates the development of many fast linear algebra libraries. For example, there are matrix multiplication algorithms that are highly optimized for different shapes of input matrices.

It is relatively straightforward to use these optimized linear algebra algorithms to build a Transformer-based system, but the modules of this system are not jointly optimized for efficiency. For example, a self-attention sub-layer involves a series of scaling, normalization, and matrix multiplication operations. Although each of these operations is implemented in supported and efficient linear algebra libraries, successive calls to them still require multiple memory transfers when switching from one operation to another. In practice, a better approach would be to keep some of the intermediate states in the on-chip memory and reuse them in subsequent computation steps instead of fetching them again from slower memory. For example, on modern GPUs, a simple way to achieve this is to merge multiple operations into a single operation, a technique known as **kernel fusion**. For Transformer models, a general idea is to design data partitioning and layout strategies that maximize the computation on each data block loaded into high-performance memory while simultaneously minimizing memory transfers. There have been several attempts to use these strategies to improve the attention models in Transformers [Ivanov et al., 2021; Pope et al., 2023]. Some of these methods, such as FlashAttention and PagedAttention, have been successfully incorporated into recent large language models [Dao et al., 2022; Kwon et al., 2023].

6.5 Applications

Transformers have a wide range of applications across NLP tasks. While the Transformer model introduced by Vaswani et al. [2017] is based on a standard encoder-decoder architecture, it is primarily used in three different configurations:

- **Decoder-only Models**. If the cross-attention sub-layers are removed, the decoder becomes a standard language model. Hence, this decoder-only model can be applied to text generation problems. For example, given a sequence of left-context tokens, we use the model to predict subsequent tokens.
- **Encoder-only Models**. Transformer encoders can be treated as sequence models that take a sequence of tokens as input and produce a sequence of representations, each corresponding to an input token. These representations can be viewed as a form of encoding of the input sequence, and are often taken as input to a prediction model. This encoder-predictor architecture forms the basis of many NLP systems, such as systems for sentence classification and sequence labeling. Pre-trained Transformer encoders can also be used to map texts into the same vector space so that we can compute the distance or similarity between any two texts.
- **Encoder-Decoder Models**. Encoder-decoder models are typically used to model sequence-to-sequence problems. Applications include many tasks in NLP and related fields, such as machine translation and image captioning.

Note that while most Transformer-based systems can be categorized into the above three types, the same NLP problem can generally be addressed using different types of models. For example, recent decoder-only models have demonstrated strong performance on a broad range of problems by framing them as text generation tasks, even though many of these problems were historically addressed using encoder-decoder or encoder-only models. To illustrate how the above models are applied, this section considers a few applications where Transformers are chosen as backbone models.

6.5.1 Language Modeling

Language modeling is an NLP task in which we predict the next token given its preceding tokens. This is generally formulated as a problem of estimating the distribution of tokens at position $i+1$ given tokens at positions $0 \sim i$. Here we denote this distribution by $\Pr(\cdot \mid x_0, \dots, x_i)$, where $x_0, \dots, x_i$ are the observed tokens. The predicted token is obtained by maximizing the probability:

$$\hat{x}_{i+1} = \underset{x_{i+1} \in V}{\arg\max} \Pr(x_{i+1} | x_0, ..., x_i) \tag{6.188}$$

where V is the vocabulary. The prediction can be extended to the tokens following $\hat{x}_{i+1}$

$$\hat{x}_{k+1} = \underset{x_{k+1} \in V}{\arg\max} \Pr(x_{k+1} | x_0, ..., x_i, \hat{x}_{i+1}, ..., \hat{x}_k) \tag{6.189}$$

This model forms the basis of many systems for text generation: given the context tokens $x_0...x_i$, we generate the remaining tokens $\hat{x}_{i+1}...\hat{x}_{k+1}$ to make the sequence complete and coherent.

As discussed in Section 6.1.1, Transformer decoders are essentially language models. A key difference between the problem of decoding in an encoder-decoder Transformer and the problem of language modeling is that the Transformer decoder makes predictions conditioned on the context tokens on both the encoder and decoder sides, rather than being conditioned only on preceding tokens. To modify the Transformer decoder to implement a standard language model, the cross-attention sub-layers are simply removed and a Transformer decoding block can be expressed as

$$\mathbf{S}^l = \text{Layer}_{\text{ffn}}(\mathbf{S}^l_{\text{self}}) \tag{6.190}$$

$$\mathbf{S}^l_{\text{self}} = \text{Layer}_{\text{self}}(\mathbf{S}^{l-1}) \tag{6.191}$$

Here, $\mathbf{S}^l$ denotes the output of the block at depth l. $\text{Layer}_{\text{self}}(\cdot)$ denotes the self-attention sub-layer, and $\text{Layer}_{\text{ffn}}(\cdot)$ denotes the FFN sub-layer. We see that this decoding block has the same form as an encoding block. The difference between the decoding and encoding blocks arises from the masking strategies adopted during training: the former masks the attention from a position i to any right-context position $k > i$, whereas the latter has no such restriction. A Softmax layer is stacked on top of the last block and is used to produce the distribution over the vocabulary at each position. For inference, the Transformer decoder works in an autoregressive manner, as described in Eq. (6.189).

Training these models follows a standard paradigm. We learn the model by repeatedly updating the parameters, based on the gradients of the loss on the training samples. This paradigm can be extended to the training of large Transformer-based language models, which have been widely applied in generative AI. However, training Transformer models at scale, including decoder-only, encoder-only, and encoder-decoder models, may lead to new difficulties, such as training instabilities. We will discuss these issues further in the following chapters, where large-scale pre-training is the primary focus.

6.5.2 Text Encoding

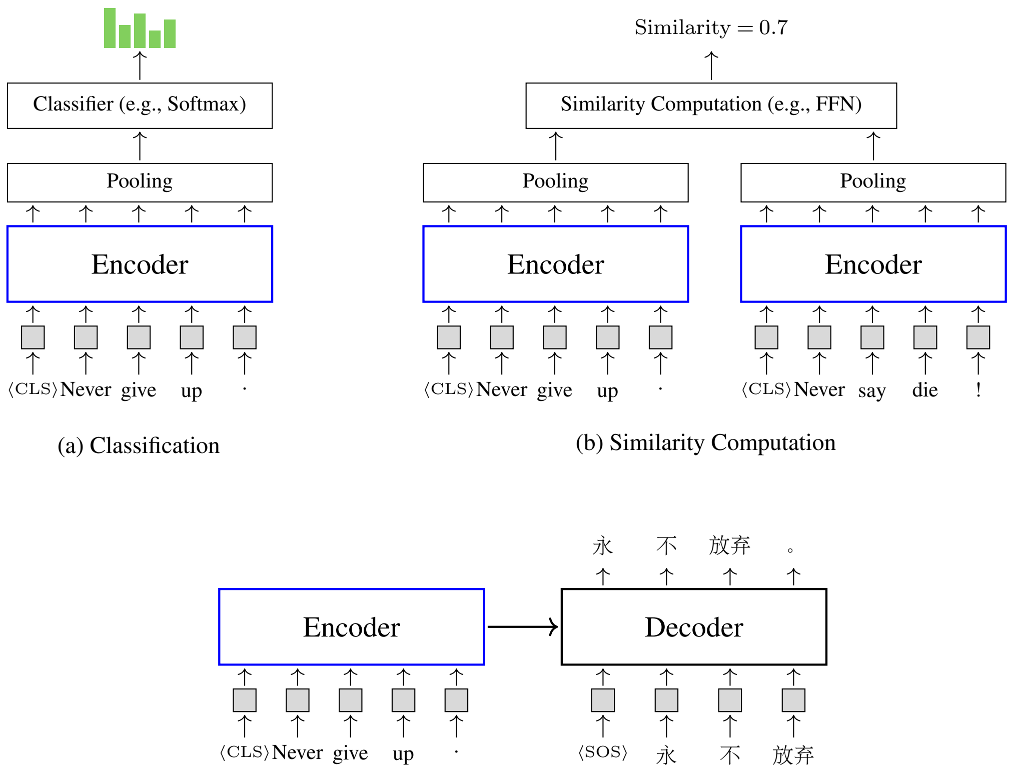
For many NLP problems, a widely used paradigm is to first learn a representation of an input sequence, and then make predictions for downstream tasks based on this representation. As a result, we separate sequence representation from specific NLP tasks. One of the advantages of this paradigm is that we can train a sequence model that is not specialized to particular problems, thereby enabling it to generalize well.

Clearly, Transformer encoders are a type of sequence model and can be used as text encoders. Consider a Transformer encoder with L encoding layers. The output of the last encoding layer can be viewed as the sequence representation. Here, we prepend a special token x_0 to each sequence, indicating the beginning of the sequence (often written as $\langle\text{SOS}\rangle$ or $[\text{CLS}]$). Given a sequence of $m+1$ input tokens $x_0, x_1, ..., x_m$, the output of the encoder will be a sequence of $m+1$ vectors $\mathbf{h}^L_0, \mathbf{h}^L_1, ..., \mathbf{h}^L_m$. Since x_0 does not correspond to an actual word and has a fixed positional embedding, it serves as a global representation token for collecting information from other positions via the self-attention mechanism. Hence, $\mathbf{h}_0^L$ acts as a representation of the entire sequence, without being biased toward any specific tokens or positions. In many cases, we require a single representation of a sequence to serve as input for downstream components of the system. For example, we can construct a sentence classification system based on a single vector generated from $\{\mathbf{h}_0^L, \ldots, \mathbf{h}_m^L\}$. In this scenario, we can simply use $\mathbf{h}_0^L$ as the sequence representation. A more general approach is to add a pooling layer to the encoder, which allows us to explore various pooling methods to generate the sequence embedding from $\{\mathbf{h}_0^L, ..., \mathbf{h}_m^L\}$.

In text encoding, token sequences are represented by real-valued vectors, often referred to as sentence representations or sentence embeddings, which can be interpreted as points in a multi-dimensional space [Hill et al., 2016]. Consequently, another use of text encoding is to compute the semantic or syntactic similarity of token sequences based on their relative proximity in this space. A straightforward method for this is to compute the Euclidean distance or cosine similarity between sequence embeddings. The shorter the distance between two sequences, the more similar they are considered to be. There are many distance metrics available, and it is possible to combine them to obtain a more robust measure of sequence similarity. Such similarity computations are applied in areas such as textual entailment, information retrieval, and translation evaluation [Cer et al., 2018; Reimers and Gurevych, 2019]. Additionally, they are frequently used to assess the inherent quality of text encoding models.

Text encoding is also a crucial component of sequence-to-sequence models. In sequence-to-sequence tasks, we can develop a separate Transformer encoder for source-side sequence modeling within an encoder-decoder system (see Figure 6.17). For instance, we can pre-train a Transformer encoder on a large-scale corpus of source-side texts and subsequently use it as the encoder in a downstream encoder-decoder model. It is worth noting that while the encoder is based on the Transformer architecture, the decoder is not restricted to this architecture. This flexibility enables us to incorporate pre-trained Transformer encoders into hybrid sequence-to-sequence architectures, such as systems that combine a Transformer encoder with an LSTM decoder.

In supervised learning scenarios, training a Transformer encoder is straightforward. We can treat it as a regular component of the target model and train this model on task-specific labeled data. However, such a method requires the encoder to be optimized on each task, and the resulting encoder might not always generalize well to other tasks, especially given that labeled data is scarce in most cases. A more prevalent approach is to frame the training of text encoders as an independent task in which supervision signals are derived solely from raw text. This led researchers to develop self-supervised Transformer encoders, such as BERT, which make use of large-scale unlabeled text, and these encoders were found to generalize well across many downstream tasks. Further discussions of pre-trained Transformer encoders can be found in Chapter 7.



(a) Classification

(b) Similarity Computation

(c) Sequence-to-Sequence Modeling

Figure 6.17: Integrating Transformer encoders as components of different systems. A common approach is to feed the output of the encoder (with pooling) into a classifier to obtain a sequence classification system. Another way to utilize Transformer encoders is to compute the similarity between two sequences. We use the same encoder to represent the two sequences, and then build a neural network over the two representations to produce a similarity score between them. Furthermore, Transformer encoders can also be used in encoder-decoder systems to model sequence-to-sequence problems.

6.5.3 Speech Translation

As illustrated in Section 6.1, the standard encoder-decoder Transformer model was proposed to model sequence-to-sequence problems. Here we consider the problem of translating speech in one language to text in another language — a problem that is conventionally addressed using both automatic speech recognition (ASR) and machine translation techniques. Instead of cascading an automatic speech recognition system and a machine translation system, we can use Transformer models to build an end-to-end speech-to-text (S2T) translation system to directly translate the input speech to the output text.

To simplify the discussion, we assume that the input to an S2T translation system is a sequence of source-side acoustic feature vectors, denoted by $\mathbf{a}_1...\mathbf{a}_m$, and the output of the system is a sequence of target-side tokens, denoted by $y_1...y_n$.[27] Mapping $\mathbf{a}_1...\mathbf{a}_m$ to $y_1...y_n$

[27] In order to obtain the input sequence to the system, we need to discretize continuous speech into signals represented by feature vectors. This process is typically nontrivial, requiring either a feature extractor based on a

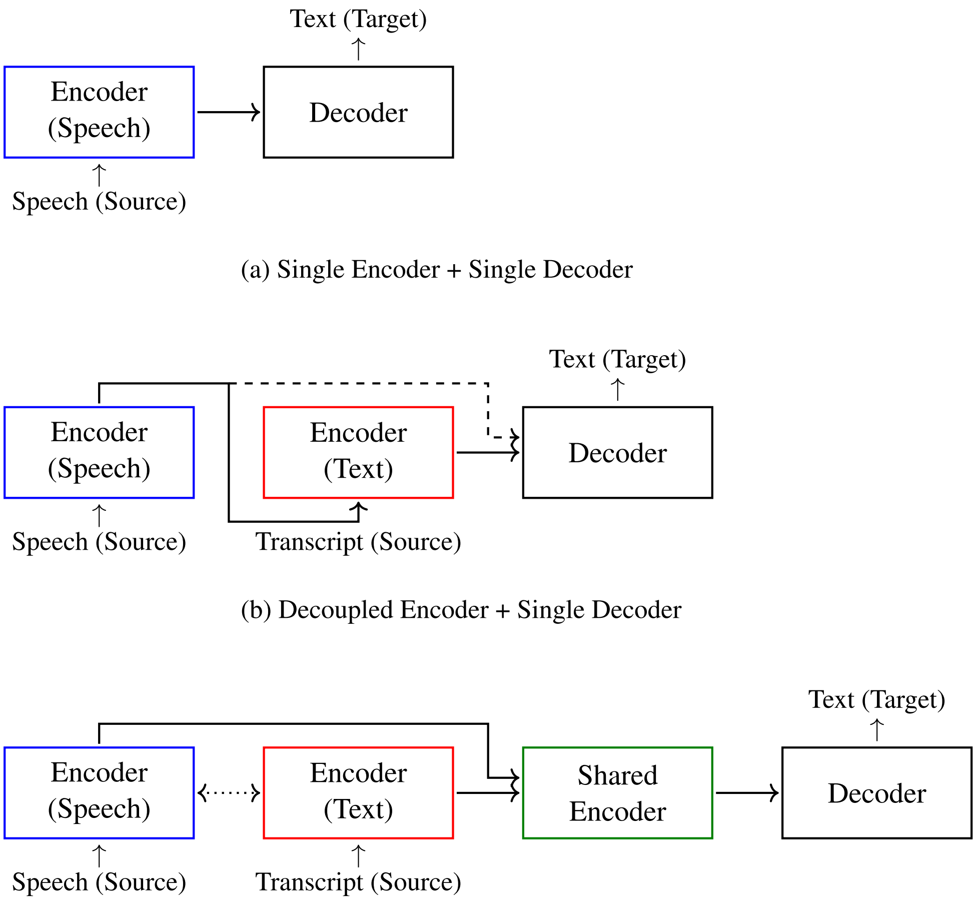
is a sequence-to-sequence problem. Thus it is straightforward to model the problem using an encoder-decoder Transformer model, and the training and inference of this model are standard, as in neural machine translation.

In S2T translation, however, we have to deal with sequence mappings across both modalities and languages. This poses new challenges compared to conventional machine translation problems and influences the design of S2T translation models. There have been several improvements to Transformers to adapt them better for S2T translation tasks. Some improvements focus on the design of Transformer layers [Di Gangi et al., 2019]. For example, in Gulati et al. [2020]'s system, a CNN sub-layer and relative positional embeddings are integrated into each Transformer layer, enabling the model to efficiently capture both local and global features.

Another line of research on S2T translation focuses on improving the encoder-decoder architecture. This involves modifying the encoder, the decoder, or both. To illustrate, Figure 6.18 shows the architectures of three S2T translation models. All of them are based on Transformers but have different encoder architectures. As shown in the figure, the standard encoder-decoder architecture has one Transformer encoder for reading the source-side input $\mathbf{a}_1, ..., \mathbf{a}_m$ and one Transformer decoder for producing the target-side output $y_1, ..., y_n$. By contrast, the decoupled encoder model decomposes the encoder into two stacked encoders — one for acoustic modeling (referred to as the **speech encoder**) and one for textual modeling (referred to as the **text encoder**) [Liu et al., 2020d; Xu et al., 2021a]. This design reflects a modeling hierarchy in which representations at different levels of the network are concerned with different aspects of the problem. For example, the speech encoder models low-level features when mapping acoustic embeddings into larger language units, and the text encoder models the semantic or syntactic features when representing the entire input sequence. An advantage of separating out the text encoder is that the encoding process follows our prior knowledge: we need to first transcribe the speech input and then translate the transcript into the target language. Therefore, we can train the speech encoder in a manner similar to how we train an ASR system. This enables us to pre-train the speech encoder and the text encoder on unlabeled data and subsequently incorporate the pre-trained encoders into S2T translation systems.

An alternative encoding architecture is the two-stream architecture, as shown in Figure 6.18(c). Like the decoupled encoder architecture, this architecture has a speech encoder and a text encoder, but the two encoders work in parallel rather than in sequence [Ye et al., 2021]. The speech encoder takes acoustic features as input, and the text encoder takes tokens (or their embeddings) as input. A third encoder, called the **shared encoder**, integrates the outputs from both the speech and text encoders, merging the representations from the two modalities. This two-stream architecture is flexible because it provides multiple ways to train S2T translation models. A common approach is to train each branch individually. For example, if we mask the speech encoder, the model will transform into a machine translation model which can be trained using bilingual texts. Conversely, if we mask the text encoder, we can train the model

variety of signal processing operations or a neural network that learns feature mappings in an end-to-end manner. But we will not dive into the details of these methods and simply treat the input feature extractor as an upstream system.



(a) Single Encoder + Single Decoder

(b) Decoupled Encoder + Single Decoder

(c) Two-stream Encoder + Single Decoder

Figure 6.18: Architectures of speech-to-text translation models based on Transformers. In addition to the standard encoder-decoder architecture, we can explicitly model the acoustic and textual (semantic) information using two separate encoders, called the speech encoder and the text encoder. In the decoupled encoder architecture, the two encoders are stacked, that is, text encoding is a subsequent process after speech encoding. In the two-stream encoder architecture, the two encoders work in parallel, and their outputs are merged using an additional encoder, called the shared encoder. The dotted line indicates the potential for interaction between the two encoders. For example, we could define a loss function to minimize the difference between their outputs, thereby guiding the model towards more aligned representations.

as a standard S2T translation model. For inference, the text encoder can be dropped, and the speech input is modeled using the speech encoder and the shared encoder.

In deep learning, training is often intertwined with architecture design. Because we are dealing with data in two modalities and two languages, we can develop multiple supervision signals for the multi-task learning of S2T translation models. A widely used method is to introduce an ASR-related loss into the training of speech encoders. For example, in the decoupled encoder model, a classifier can be constructed based on the output from the speech encoder. By minimizing the connectionist temporal classification (CTC) loss for this classifier, the speech encoder can be optimized in a manner similar to ASR. In general, training S2T translation models is challenging because speech-to-text aligned data is scarce. Typical responses to this challenge include data augmentation, pre-training, knowledge distillation with machine translation, and so on. However, an in-depth discussion of these methods goes beyond the scope of this section on Transformers. Interested readers can refer to a recent survey on speech translation for more detailed information [Xu et al., 2023a].

6.5.4 Vision Models

While Transformers were first used in NLP, their application to other domains has been a prominent research topic. In computer vision, for instance, there is a notable trend of shifting from CNNs to Transformers as backbone models. In this subsection, we consider the **Vision Transformer** (**ViT**) — an interesting application of Transformers to image classification [Dosovitskiy et al., 2021]. The Vision Transformer is a milestone model that opened the door to purely Transformer-based vision models. Here, we consider the basic structure of the Vision Transformer to keep this section focused and coherent, although extensive literature exists on the Vision Transformer and its variants. More detailed discussions of vision models can be found in recent surveys [Han et al., 2022; Liu et al., 2023e].

The core idea behind the Vision Transformer is to transform an image into a sequence of visual tokens, and input this sequence into a Transformer encoder to generate a representation of the image. The Transformer encoder is standard, and so we will not discuss it here, given the introduction to Transformers we have presented so far in this chapter. Mapping a 2D image into a sequence of tokens requires additional processing. Suppose we have an image represented as an $H \times W \times C$ feature map, where H is the height, W is the width, and C is the number of channels. The first step is to partition this image into a set of **patches**. Suppose all patches are squares of side length P. Then, the resulting patches can be represented by feature maps of shape $P \times P \times C$. By ordering these patches in a specific manner, we obtain a sequence of $\frac{HW}{P^2}$ patches, with each patch being treated as a "token."

Given this patch sequence, the subsequent steps are straightforward. For the patch at each position, we obtain a d-dimensional embedding via a linear transformation of the flattened patch. The input to the Transformer encoder is a sequence of d-dimensional vectors, each of which is the sum of the corresponding patch embedding and positional embedding. Figure 6.19 illustrates the patching and embedding steps in the Vision Transformer.

Once we have a sequence of vectors for representing the image, we can employ the Transformer encoder to encode the sequence. The encoding process is exactly the same as that in text encoding, as discussed in Section 6.5.2. For classification problems, we need only a single representation of the input. It is convenient to take the output of the encoder at position 0 (denoted by $\mathbf{h}_0^L$) and feed it into a classifier. Given that the first token ([CLS]) serves as a special global token that is attended to by all other tokens, $\mathbf{h}_0^L$ can be viewed as an aggregated representation of the entire sequence.

A standard way to train the Vision Transformer is to minimize a task-specific loss on labeled data, such as ImageNet. More recently, inspired by self-supervised learning in BERT-

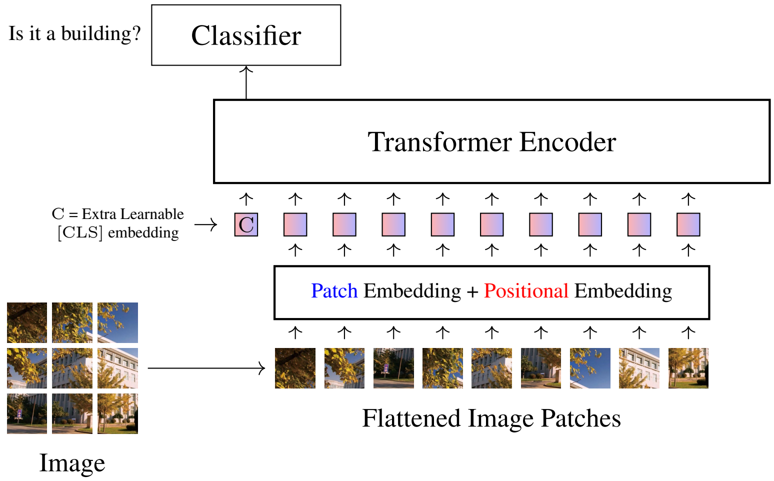


Figure 6.19: Illustration of the Vision Transformer for image classification [Dosovitskiy et al., 2021]. There are three steps. In the first step, the input image is partitioned into patches, which are then flattened and mapped into embeddings. In the second step, a Transformer encoder is employed to process the sequence of embeddings, representing the image as a real-valued vector (e.g., the output of the encoder at the first position). In the last step, a classifier is built on top of this image representation.

like models, there have been successful attempts to train Transformer-based image encoders on large-scale unlabeled data [Caron et al., 2021; Bao et al., 2021; He et al., 2022]. Note that one of the most significant contributions of the Vision Transformer is that it unifies the representation models for different modalities. This suggests that if an object, whether an image or text, is represented as a sequence of embeddings, it can be easily modeled using the Transformer architecture.

6.5.5 Multimodal Models

The above discussion of the Vision Transformer offers the possibility of unifying the representations from multiple modalities using the same Transformer architecture. In fact, many recent multimodal systems draw heavy inspiration from Transformers [Xu et al., 2023c]. Such systems convert objects from different modalities into vector sequences and feed these vectors into a single Transformer model. The output is a fused representation of all inputs, which can then be used in downstream systems.

As a simple example, consider the task of encoding a pair consisting of text and its corresponding image. First, we represent both the text and the image as sequences of embeddings that have the same dimensionality. This is a common step in sequence modeling, which we have encountered many times so far. We can do this by using either a simple embedding model (e.g., a word or patch embedding model) or a pre-trained model (e.g., a vision model). Then, these two sequences are concatenated into a long sequence comprising both textual and visual embeddings. The subsequent step is standard: a Transformer encoder takes the concatenated sequence of embeddings as input and produces representations of the text and image as output. Note that concatenating textual and visual sequences is one of the simplest methods for vision-text modeling. There are several alternative ways to merge information from different modalities. For example, we can feed visual representations into the attention layers of a text encoder or decoder [Li et al., 2022e; Alayrac et al., 2022].

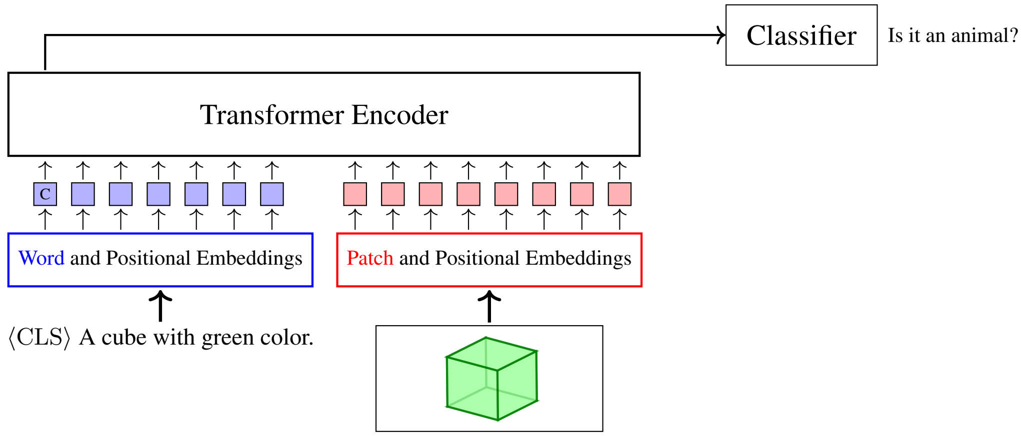
The above multimodal encoder can be used in both encoder-only and encoder-decoder systems. For encoder-only systems, consider an example where, given an image and its description, we predict the class of the image using a classifier built on top of the encoder [Kim et al., 2021]. For encoder-decoder systems, we pair the encoder with a decoder, as in sequence-to-sequence modeling [Cho et al., 2021]. For example, we might employ a Transformer decoder to generate text based on the output of the encoder. A common application of this architecture is **visual question answering** (**VQA**), where an image and a question about the image are provided, and the system is tasked with generating an answer [Antol et al., 2015]. The architectures of these models are illustrated in Figure 6.20 (a-b).

Large language models have also been extended to handle both textual and other modalities, such as images, video, and audio, leading to new advances in multimodal processing [Liu et al., 2023a; Yin et al., 2023]. By representing all inputs as a sequence of token embeddings, the problem becomes straightforward: we predict the next token given its context. This can be done using decoder-only systems, as shown in Figure 6.20 (c).

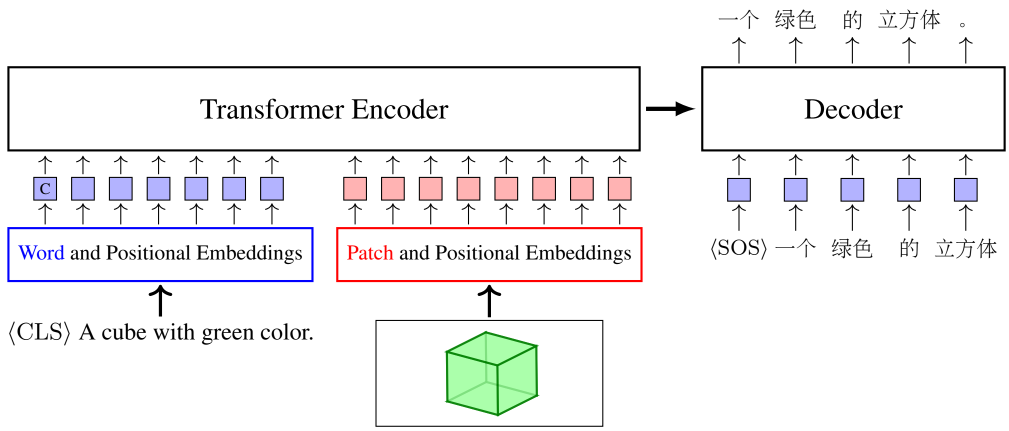
6.6 Summary

Transformer models have been widely adopted since the architecture was first proposed by Vaswani et al. [2017]. This adoption has accelerated the development of a wide range of algorithms, systems, and concepts. A thorough discussion of Transformers requires broad coverage, and so it is impossible to address all topics or provide an exhaustive list of references. Figure 6.21 provides a high-level overview of the Transformer landscape. Note that these models and related techniques can be categorized in various ways, and we present one such perspective. To summarize, we highlight the following points:

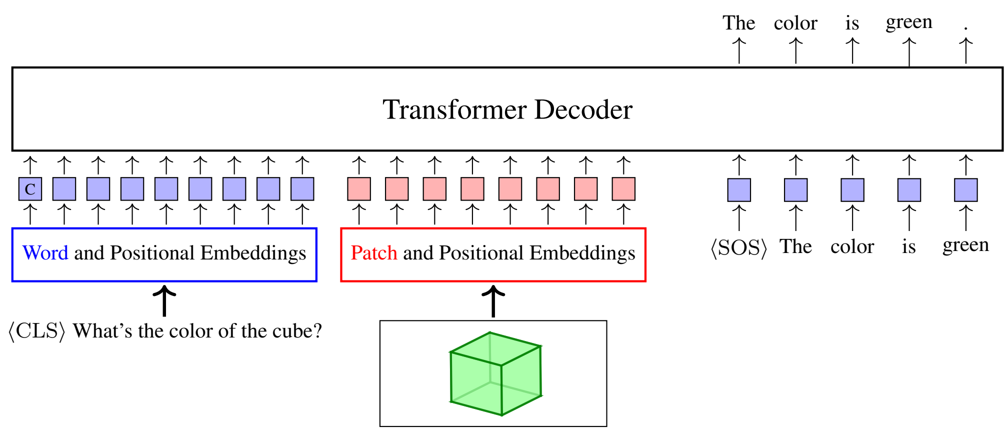
- **Foundations of Transformers**. Although the impact of Transformers has been revolutionary, they are not completely "new" models. From a deep learning perspective, Transformers are composed of common building blocks, including word and positional embeddings [Bengio et al., 2003a; Mikolov et al., 2013c; Gehring et al., 2017b], attention mechanisms [Bahdanau et al., 2014; Luong et al., 2015], residual connections [He et al., 2016b], layer normalization [Ba et al., 2016], and so on. Many of these components were presented in earlier systems, for example, similar ideas with QKV attention can be found in memory networks [Sukhbaatar et al., 2015] and hierarchical attention networks [Yang et al., 2016]. Transformers offer a novel approach to integrating these components, resulting in a unique architecture. For example, in Transformers, the combination of multi-head attention and dot-product QKV attention, along with the incorporation of layer normalization and residual connections, gives rise to a distinctive neural network block, specifically a self-attention sub-layer. This design has since become a de facto standard in many follow-on sequence modeling systems.



(a) Multi-modal Encoder + Classifier



(b) Multi-modal Encoder + Text Decoder (Translation)



(c) Multi-modal Decoder (Language Modeling)

Figure 6.20: Vision-text models. Blue boxes represent word+position embeddings, and red boxes represent image patch+position embeddings.

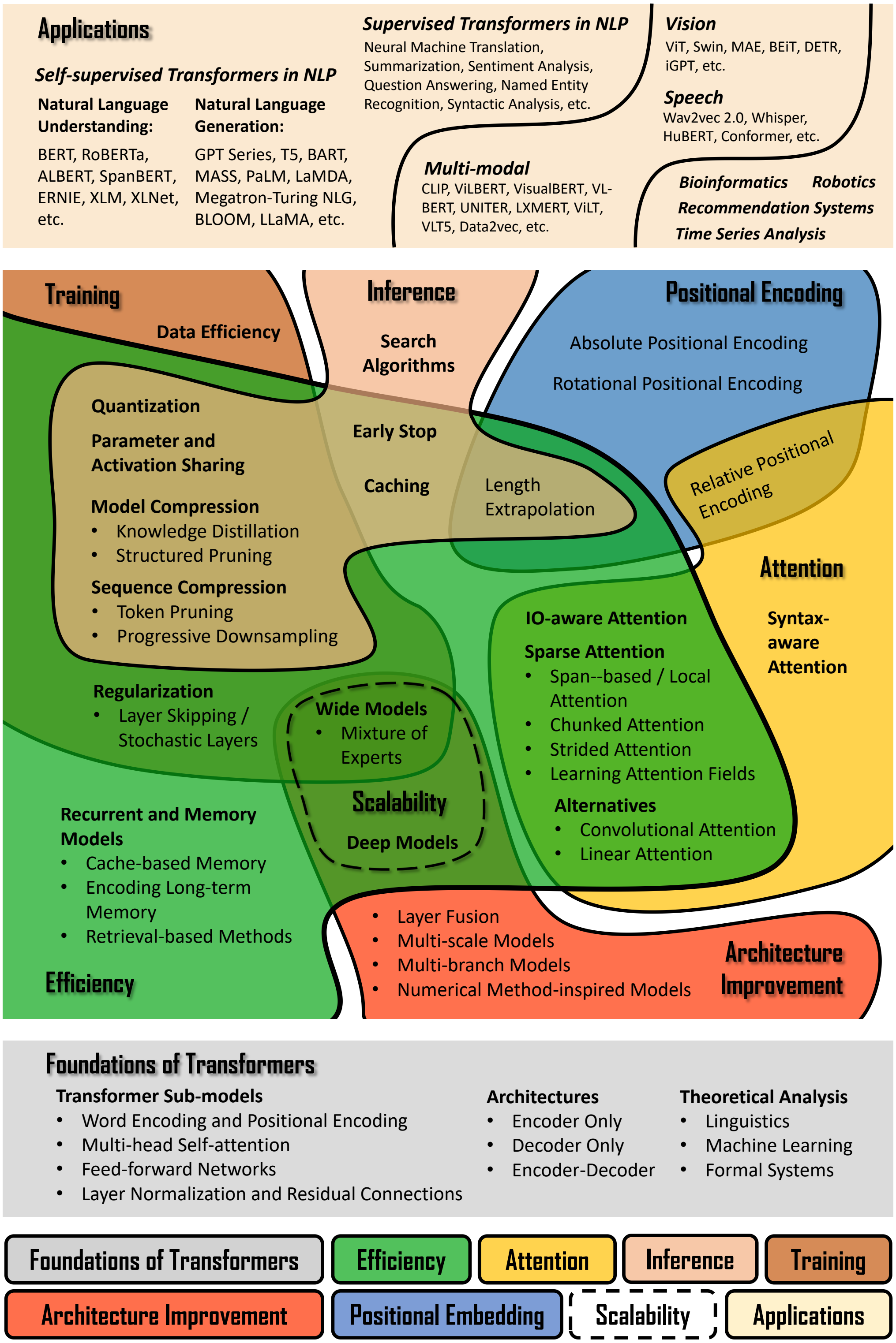


Figure 6.21: An overview of Transformers.

- **Attention Models**. The success of Transformers on NLP tasks has largely been attributed to the use of multi-head self-attention for sequence modeling. This has led to a surge of interest in enhancing the attention mechanisms within Transformers. While it is impossible to detail every attention model, there are several notable research directions. One prominent direction involves modifying the forms of QKV attention and multi-head attention for improved performance. The scope of this direction is vast, as there are numerous aspects to consider when enhancing Transformers [Lin et al., 2022a]. For example, one may add new components to self-attention sub-layers to adapt them to specific tasks, resulting in various Transformer variants. A second direction is to incorporate prior knowledge into the design of attention models. This makes sense, because much of the emphasis in traditional NLP has been on using linguistic insights to guide system design, and we generally want NLP systems to be linguistically explainable. For example, many Transformer-based systems take syntactic parses as input in various forms and make use of syntax in sequence modeling. A third direction is to develop efficient attention models [Tay et al., 2020b]. Self-attention has long been criticized for its quadratic time complexity and dependency on all previous tokens for each new token. In response, many researchers have focused on simplifying the structure of self-attention, or on approximating it using sparse or recurrent models. This concern for efficiency also motivates the development of alternatives to self-attention, such as attention models with linear time complexity. In addition to exploring stronger and more efficient attention models, it is natural to examine what knowledge is learned by such models. Interestingly, researchers have found that the underlying structure of languages can be learned by multi-head self-attention models, although these models are not trained to represent such knowledge [Manning et al., 2020].
- **Word and Positional Embeddings**. Transformers represent each input word (or token) through a combination of word and positional embeddings. Learning these word embeddings is not unique to Transformers; one can either utilize pre-trained embeddings, such as Word2Vec or GloVe, or treat them as learnable parameters within the model. A closely related issue is the tokenization of input sequences. Generally, tokenization determines the number of resulting tokens and affects the difficulty of learning their corresponding embeddings. Therefore, selecting an appropriate tokenization method is crucial for many applications. Furthermore, positional embeddings play a vital role in Transformers, as the underlying attention mechanisms are order-insensitive by design [Dufter et al., 2022]. While positional encoding is a general research topic, much effort has been dedicated to tailoring it for Transformers, leading to various architectural modifications [Shaw et al., 2018; Huang et al., 2018]. Additionally, studies show that for sequences significantly longer than those encountered during training, better extrapolation can be achieved by replacing sinusoidal embeddings with rotary positional embeddings (RoPE) or by scaling attention weights with positional scalars [Raffel et al., 2020; Su et al., 2021; Press et al., 2021].
- **Training and Model Scaling**. In the era of deep learning, high-performance systems are

typically built using large-scale neural networks. A straightforward approach to increasing Transformer capacity is to stack more layers or enlarge the hidden representation size. Indeed, deep and wide Transformer models consistently outperform their smaller counterparts. However, training extremely large models poses significant challenges, particularly when applying gradient descent over vast datasets, which demands substantial computational resources. Engineering solutions involve distributing training across computer clusters [Lepikhin et al., 2021; Chowdhery et al., 2022]. Although distributed training is a general paradigm not restricted to Transformers, it heavily influences architectural design; for instance, sparse mixture-of-experts (MoE) models can facilitate training with massive parameter sets and serve as the foundation for many expansive systems. Scaling also enables the study of scaling laws — quantifying how performance scales with model size, data volume, and compute budget [Hestness et al., 2017; Kaplan et al., 2020]. This scaling is occasionally accompanied by "emergence" [Wei et al., 2022b], where models manifest unpredictable new abilities. In recent research, the acquisition of such emergent abilities has been viewed as a hallmark of powerful large language models.

- **Efficient Models**. There are different goals for efficiency. A system may prioritize memory efficiency for memory-bound problems or low latency when inference speed is critical. Balancing these requirements leads to diverse optimization strategies. In Transformers, many optimizations involve modifying attention mechanisms; for instance, several variants reduce the memory footprint when processing long sequences [Tay et al., 2020b], while others minimize computation to decrease latency. Furthermore, Transformers can benefit from architecture-independent optimizations typical of neural networks, such as conditional computation, knowledge distillation, structured pruning, and sequence compression. Efficiency can also be addressed from a computer architecture perspective [Kim et al., 2023]. In sequence-to-sequence tasks, for example, the encoding process is typically compute-intensive, whereas the autoregressive decoding process is often IO-intensive. Therefore, we can employ different optimization methods for different components of Transformers.
- **Inference**. The inference problem is commonly discussed in sequence generation. In NLP, we often need to find the "best" hypothesis in a space involving sequences of hundreds or even thousands of tokens over a vocabulary. As a classic search problem in artificial intelligence, various algorithms are applicable, including breadth-first, depth-first, and A* search. In many practical applications of NLP, the efficiency of the search systems is an important consideration. As a result, optimized search algorithms are required. Most of these algorithms have been explored in machine translation and ASR, and are directly applicable to neural text generation models like Transformers. Beyond traditional search, recent optimizations such as speculative decoding are specifically tailored to the Transformer architecture [Leviathan et al., 2023]. Moreover, the efficient attention models mentioned previously are now widely deployed to accelerate large language models and neural machine translation systems [Heafield et al., 2021; Dao

et al., 2023].

- **Applications**. Applications of Transformers cover a wide variety of NLP problems. During the development of Transformers, they were at first used to build supervised models that perform particular tasks. Later, a greater success was achieved by using them as backbone networks for large scale self-supervised learning of foundation models [Bommasani et al., 2021]. This shift markedly changed the NLP paradigm: a model is first pre-trained on vast amounts of text to acquire general linguistic knowledge, and then adapted to downstream tasks through efficient methods such as fine-tuning or prompting. Beyond NLP, we have witnessed an explosion of Transformer applications in fields like computer vision, speech processing, and bioinformatics. The core idea is to represent diverse data types as sequences of tokens, extending Transformers into general-purpose representation models capable of handling multi-modal data.
- **Large Language Models as Foundation Models**. Transformers form the basis of recent large language models, such as the GPT series, which show surprising breakthroughs in NLP [Bubeck et al., 2023; Yang et al., 2023a]. Much of the research in large language models is more or less related to Transformers. As discussed in Section 6.5.1, training these language models is often synonymous with training large-scale Transformer decoders, and architectural refinements to the latter directly benefit the former. On the other hand, the rapid development of large language models has spurred further innovation in Transformer-related techniques, particularly in developing efficient and low-cost adaptation methods for massive models.
- **Theoretical Analysis**. While Transformers dominate empirically, their theoretical foundations have historically received less focus compared to engineering improvements — an empirical-first trend common across the broader machine learning community. In response, researchers have begun to analyze Transformers more deeply. One approach is to view Transformers as deep neural networks and interpret them via mathematical tools. For instance, the residual connections within Transformers can be viewed as Euler discretizations of ODEs. This equivalence suggests that we can leverage insights from numerical ODE methods to inform model design. Another promising avenue aims to develop a theoretical understanding of the self-attention mechanism, which distinguishes Transformers from earlier deep learning models. Studies have interpreted self-attention through various theoretical lenses, including data compression [Yu et al., 2023a], optimization [Li et al., 2022c], and function approximation [Yun et al., 2019]. Moreover, Transformers can be mapped to formal systems, such as Turing machines [Pérez et al., 2018], counter machines [Bhattamishra et al., 2020], regular and context-free languages [Hahn, 2020], Boolean circuits [Hao et al., 2022; Merrill et al., 2022], programming languages [Weiss et al., 2021], and first-order logic [Chiang et al., 2023a]. These connections provide theoretical tools to rigorously study the expressivity of Transformers. It is worth noting, however, that while we can analyze Transformers through these diverse perspectives, no unified theory yet explains the fundamental nature of these models. Addressing this theoretical gap remains a central challenge in the

field, yet it is essential for developing complex neural systems with explainable and predictable behaviors.

III Large Language Models

Chapter 7

Pre-training

The development of neural sequence models, such as Transformers, along with the improvements in large-scale self-supervised learning, has opened the door to universal language understanding and generation. This achievement is largely driven by pre-training: we extract common components from many neural network-based systems and train them on massive amounts of unlabeled data using self-supervision. These pre-trained models serve as foundation models that can be easily adapted to different tasks via fine-tuning or prompting. As a result, the paradigm of NLP has been fundamentally transformed. In many cases, large-scale supervised learning for specific tasks is no longer required, and instead, we only need to adapt pre-trained foundation models.

While pre-training has gained popularity in recent NLP research, this concept dates back decades to the early days of deep learning. For example, early attempts to pre-train deep learning systems include unsupervised learning for RNNs, deep feedforward networks, autoencoders, and others [Schmidhuber, 2015]. In the modern era of deep learning, we experienced a resurgence of pre-training, sparked in part by the large-scale unsupervised training of various word embedding models [Mikolov et al., 2013c; Pennington et al., 2014]. During the same period, pre-training also attracted significant interest in computer vision, where the backbone models were trained on relatively large labeled datasets such as ImageNet, and then applied to different downstream tasks [He et al., 2019; Zoph et al., 2020]. Large-scale research on pre-training in NLP began with the development of language models using self-supervised learning. This family of models covers several well-known examples like **BERT** [Devlin et al., 2019] and **GPT** [Brown et al., 2020], all with a similar idea that general language understanding and generation can be achieved by training models to predict masked or future words in a huge amount of text. Despite the simplicity of this approach, the resulting models show remarkable capabilities in capturing linguistic structures, even without being explicitly trained to do so. The generality of the pre-training tasks leads to systems that exhibit strong performance in a large variety of NLP problems, even outperforming previously state-of-the-art supervised systems. More recently, pre-trained large language models have achieved greater success, showing the exciting prospects for more general artificial intelligence [Bubeck et al., 2023].

This chapter discusses the concept of pre-training in the context of NLP. It begins with a general introduction to pre-training methods and their applications. BERT is then used as an example to illustrate how a sequence model is trained via a self-supervised task, called **masked language modeling**. This is followed by a discussion of methods for adapting pre-trained sequence models to various NLP tasks. As this chapter focuses primarily on the pre-training paradigm, we will defer the detailed discussion of generative large language models to subsequent chapters.

7.1 Pre-training in NLP

Pre-training in NLP typically involves two types of tasks: sequence modeling (or sequence encoding) and sequence generation. While these tasks take different forms, for simplicity, we describe them using a single unified model defined as follows:

$$\begin{aligned} \mathbf{o} &= g(x_0, x_1, \cdots, x_m; \theta) \\ &= g_\theta(x_0, x_1, \cdots, x_m), \end{aligned} \tag{7.1}$$

where $\{x_0, x_1, \ldots, x_m\}$ denotes a sequence of input tokens[1], x_0 denotes a special symbol ($\langle s \rangle$ or $[\text{CLS}]$) attached to the beginning of the sequence, $g(\cdot;\theta)$ (also written as $g_\theta(\cdot)$) denotes a neural network with parameters θ, and $\mathbf{o}$ denotes the output of the neural network. The form of the output $\mathbf{o}$ varies depending on the specific problem. For example, in token prediction tasks (such as language modeling), $\mathbf{o}$ is a distribution over a vocabulary; in sequence encoding tasks, $\mathbf{o}$ is a representation of the input sequence, often expressed as a sequence of real-valued vectors.

There are two fundamental aspects to consider here:

- Optimizing θ for a pre-training task. Unlike standard learning problems in NLP, pre-training does not assume specific downstream tasks to which the model will be applied. Instead, the goal is to train a model that can generalize across various tasks.
- Applying the pre-trained model $g_{\hat{\theta}}(\cdot)$ to downstream tasks. To adapt the model to these specific tasks, we need to fine-tune the parameters $\hat{\theta}$ using labeled data or prompt the model with task descriptions.

In this section, we discuss the basic concepts for addressing these two aspects.

7.1.1 Unsupervised, Supervised and Self-supervised Pre-training

In deep learning, pre-training refers to the process of optimizing a neural network before it is further fine-tuned and applied to downstream tasks. This approach is based on the assumption that a model pre-trained on one task can be effectively adapted to perform another. As a result, we do not need to train a deep, complex neural network from scratch on tasks with limited labeled data. Instead, we can leverage tasks where supervision signals are easier to obtain.

[1] Here we assume that tokens are the basic units of text produced by tokenization. Sometimes, we use the terms *token* and *word* interchangeably, though they have closely related but slightly different meanings in NLP.

This reduces the reliance on task-specific human annotations, enabling the development of more general models that are not confined to particular problems.

During the resurgence of neural networks driven by deep learning, many early attempts at pre-training were focused on **unsupervised learning**. In these methods, the parameters of a neural network are optimized using a criterion that is not directly related to specific target tasks. For example, a model can be trained to minimize the reconstruction cross-entropy of the input vector for each layer [Bengio et al., 2006]. Unsupervised pre-training is commonly employed as a preliminary step before supervised learning, offering several advantages, such as aiding in the discovery of better local minima and serving as a regularizer during the training process [Erhan et al., 2010]. These benefits make the subsequent supervised learning phase easier and more stable.

A second approach involves pre-training a neural network on **supervised learning** tasks. For example, consider a sequence model designed to encode input sequences into vector representations. During pre-training, this model is combined with a classification layer and trained on an initial supervised task, such as sentiment analysis (e.g., classifying whether a sentence is positive or negative). To adapt the pre-trained sequence model to a downstream task (e.g., determining if a sequence is subjective or objective), we replace the initial classification layer with a new one for the target task. Typically, we need to fine-tune the parameters of this combined model using task-specific labeled data to optimize its performance. The fine-tuned model is then employed to classify new sequences. A major advantage of supervised pre-training is its straightforward alignment with standard supervised learning paradigms. However, as model complexity increases, the demand for labeled data grows accordingly, making this pre-training approach challenging when large-scale labeled datasets are unavailable.

A third approach is **self-supervised learning**. In this paradigm, a neural network is trained using supervision signals derived from the data itself, rather than from human annotations. This is generally achieved by constructing training tasks directly from unlabeled data, such as predicting missing parts of the input. While self-supervised learning has recently become the dominant method in NLP, the underlying philosophy shares similarities with older machine learning concepts like **self-training**. In self-training, a model requires a small amount of seed data to build an initial classifier, which then generates pseudo-labels for unlabeled data. These pseudo-labels are subsequently used to iteratively refine and bootstrap the model. This method has been successfully applied to NLP areas like word sense disambiguation [Yarowsky, 1995] and document classification [Blum and Mitchell, 1998]. However, unlike standard self-training, self-supervised pre-training in NLP does not rely on an initial model to assign discrete pseudo-labels. Instead, supervision signals are derived directly from the text, and the entire model is trained from scratch. A well-known example is training sequence models by masking tokens in a text and requiring the model to predict them based on the surrounding context. This enables large-scale self-supervised learning for deep neural networks, leading to the success of pre-training in many understanding, generation, and reasoning tasks.

Figure 7.1 shows a comparison of the above three pre-training approaches. Self-supervised pre-training is so successful that most current state-of-the-art NLP models are based on this paradigm. Therefore, throughout this chapter and the remainder of the book, we will focus

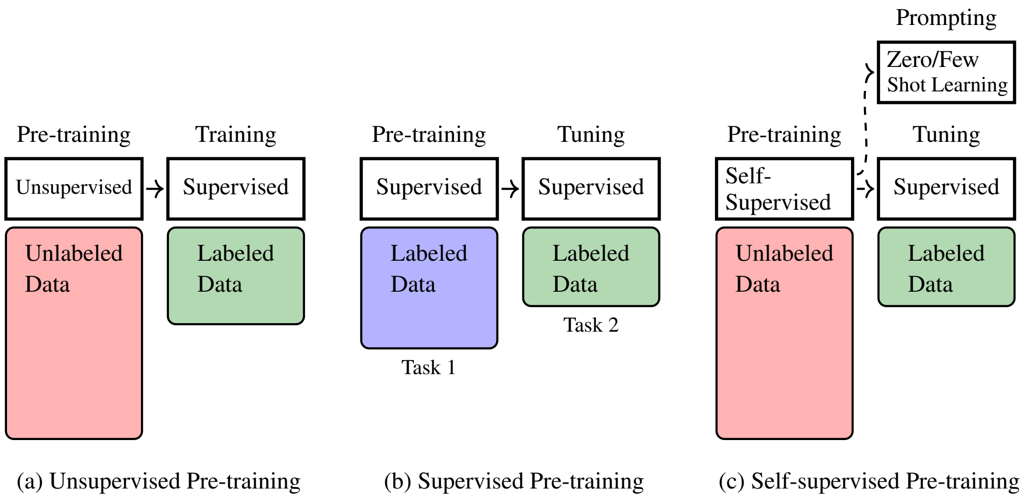


Figure 7.1: Illustration of unsupervised, supervised, and self-supervised pre-training. In unsupervised pre-training, pre-training is performed on large-scale unlabeled data. It can be viewed as a preliminary step to have a good starting point for the subsequent optimization process, though considerable effort is still required to further train the model with labeled data after pre-training. In supervised pre-training, the underlying assumption is that different (supervised) learning tasks are related. So we can first train the model on one task, and transfer the resulting model to another task with some training or tuning effort. In self-supervised pre-training, a model is pre-trained on large-scale unlabeled data via self-supervision. The model can be effectively trained in this way, and we can efficiently adapt it to new tasks through fine-tuning or prompting.

on self-supervised pre-training. We will show how sequence models are pre-trained via self-supervision and how the pre-trained models are applied.

7.1.2 Adapting Pre-trained Models

As mentioned above, two major types of models are widely used in NLP pre-training.

- **Sequence Encoding Models**. Given a sequence of words or tokens, a sequence encoding model represents this sequence as either a real-valued vector or a sequence of vectors, and obtains a representation of the sequence. This representation is typically used as input to another model, such as a sentence classification system.
- **Sequence Generation Models**. In NLP, sequence generation generally refers to the problem of generating a sequence of tokens based on a given context. The term *context* has different meanings across applications. For example, it refers to the preceding tokens in language modeling, and refers to the source-language sequence in machine translation[2].

We need different techniques for applying these models to downstream tasks after pre-

[2] More precisely, in autoregressive decoding of machine translation, each target-language token is generated based on both its preceding tokens and source-language sequence.

training. Here we are interested in the following two methods.

1. Fine-tuning of Pre-trained Models

For sequence encoding pre-training, a common method of adapting pre-trained models is fine-tuning. Let $\mathrm{Encode}_\theta(\cdot)$ denote an encoder with parameters θ, for example, $\mathrm{Encode}_\theta(\cdot)$ can be a standard Transformer encoder. Provided we have pre-trained this model in some way and obtained the optimal parameters $\hat{\theta}$, we can employ it to encode an input sequence, as follows

$$\mathbf{H} = \mathrm{Encode}_{\hat{\theta}}(\mathbf{x}), \tag{7.2}$$

where $\mathbf{x}$ is the input sequence $\{x_0, \cdots, x_m\}$, and $\mathbf{H}$ is the output representation which is a sequence of real-valued vectors $\{\mathbf{h}_0, \mathbf{h}_1, \cdots, \mathbf{h}_m\}$. Because the encoder does not work as a standalone NLP system, it is often integrated as a component into a larger system. Consider, for example, a text classification problem in which we identify the polarity (e.g., positive, negative, or neutral) of a given text. We can build a text classification system by placing a classifier on top of the encoder. Let $\mathrm{Classify}_\omega(\cdot)$ be a neural network with parameters ω. Then, the text classification model can be expressed in the form

$$\begin{aligned} \mathrm{Pr}_{\omega,\hat{\theta}}(\cdot|\mathbf{x}) &= \mathrm{Classify}_\omega(\mathbf{H}) \\ &= \mathrm{Classify}_\omega(\mathrm{Encode}_{\hat{\theta}}(\mathbf{x})). \end{aligned} \tag{7.3}$$

Here $\mathrm{Pr}_{\omega,\hat{\theta}}(\cdot|\mathbf{x})$ is a probability distribution over the label set $\{\mathrm{positive}, \mathrm{negative}, \mathrm{neutral}\}$, and the label with the highest probability in this distribution is selected as the output. To keep the notation uncluttered, we will use $F_{\omega,\hat{\theta}}(\cdot)$ to denote $\mathrm{Classify}_\omega(\mathrm{Encode}_{\hat{\theta}}(\cdot))$.

Because the model parameters ω and $\hat{\theta}$ are not explicitly optimized for the target classification task, we cannot directly use this combined model. Instead, we must use a modified version of the model that is adapted to the task. A typical approach is to fine-tune the model by providing explicit labels from downstream tasks. We can train $F_{\omega,\hat{\theta}}(\cdot)$ on a labeled dataset, treating it as a standard supervised learning task. This fine-tuning process yields further optimized parameters, $\tilde{\omega}$ and $\tilde{\theta}$. Alternatively, we can freeze the encoder parameters $\hat{\theta}$ to maintain their pre-trained state and focus solely on optimizing ω. This allows the classifier to be efficiently adapted to work effectively with the pre-trained encoder.

Once we have obtained a fine-tuned model, we can use it to classify a new text. For example, suppose we have a comment posted on a travel website:

I love the food here. It's amazing!

We first tokenize this text into tokens[3], and then feed the token sequence $\mathbf{x}_{\mathrm{new}}$ into the fine-

[3] The text can be tokenized in many different ways. One of the simplest is to segment the text into English words and punctuation $\{\mathrm{I}, \mathrm{love}, \mathrm{the}, \mathrm{food}, \mathrm{here}, ., \mathrm{It}, \text{'s}, \mathrm{amazing}, !\}$

tuned model $F_{\tilde{\omega},\tilde{\theta}}(\cdot)$. The model generates a distribution over classes by

$$F_{\tilde{\omega},\tilde{\theta}}(\mathbf{x}_{\text{new}}) = \begin{bmatrix} \Pr(\text{positive}|\mathbf{x}_{\text{new}}) & \Pr(\text{negative}|\mathbf{x}_{\text{new}}) & \Pr(\text{neutral}|\mathbf{x}_{\text{new}}) \end{bmatrix}. \quad (7.4)$$

We then select the label with the highest probability as the output. In this example, it is positive.

In general, the amount of labeled data used in fine-tuning is small compared to that of the pre-training data, and so fine-tuning is less computationally expensive. This makes the adaptation of pre-trained models very efficient in practice: given a pre-trained model and a downstream task, we just need to collect some labeled data, and slightly adjust the model parameters on this data. A more detailed discussion of fine-tuning can be found in Section 7.4.

2. Prompting of Pre-trained Models

Unlike sequence encoding models, sequence generation models are often employed independently to address language generation problems, such as question answering and machine translation, without the need for additional modules. It is therefore straightforward to fine-tune these models as complete systems on downstream tasks. For example, we can fine-tune a pre-trained encoder-decoder multilingual model on some bilingual data to improve its performance on a specific translation task.

Among various sequence generation models, a notable example is large language models trained on very large amounts of data. These models are trained simply to predict the next token given its preceding tokens. Although token prediction is a simple task that has traditionally been used for language modeling, performing it at scale enables models to acquire broad linguistic and world knowledge. As a result, pre-trained LLMs exhibit remarkable token prediction capabilities, making it possible to frame numerous NLP tasks as simple text generation problems via prompting. For example, we can frame the above text classification problem as a text generation task.

I love the food here. It's amazing! I'm ______

Here __ indicates the word or phrase we want to predict (call it the **completion**). If the predicted word is *happy*, or *glad*, or *satisfied*, or a related positive word, we can classify the text as positive. This example shows a simple prompting method in which we concatenate the input text with *I'm* to form a prompt. Then, the completion helps decide which label is assigned to the original text.

Given the strong language understanding and generation capabilities of large language models, a prompt can instruct these models to perform more complex tasks. The following example demonstrates how to instruct an LLM to perform polarity classification:

> Assume that the polarity of a text is a label chosen from {positive, negative, neutral}. Identify the polarity of the input.
>
> Input: I love the food here. It's amazing!
>
> Polarity: ______

The first two sentences are a description of the task. Input and Polarity are indicators of the input and output, respectively. We expect the model to complete the text and at the same time give the correct polarity label. By using instruction-based prompts, we can adapt large language models to solve NLP problems without the need for additional training.

This example also demonstrates the zero-shot learning capability of large language models, which can perform tasks that were not seen during training during. Another method for eliciting new capabilities from a neural network is few-shot learning. This is typically achieved through **in-context learning** (**ICL**). More specifically, we add some samples that demonstrate how an input corresponds to an output. These samples, known as **demonstrations**, are used to teach large language models how to perform the task. Below is an example with demonstrations

> Assume that the polarity of a text is a label chosen from {positive, negative, neutral}. Identify the polarity of the input.
>
> Input: The traffic is terrible during rush hours, making it difficult to reach the airport on time.
>
> Polarity: Negative
>
> Input: The weather here is wonderful.
>
> Polarity: Positive
>
> Input: I love the food here. It's amazing!
>
> Polarity: ______

Prompting and in-context learning play important roles in the recent rise of large language models. We will discuss these techniques in greater detail in Chapter 9. However, it is worth noting that while prompting is a powerful way to adapt large language models, some tuning efforts are still needed to ensure the models can follow instructions accurately. Additionally, the fine-tuning process is crucial for aligning the values of these models with human values. More detailed discussions of fine-tuning can be found in Chapter 10.

7.2 Self-supervised Pre-training Tasks

In this section, we consider self-supervised pre-training approaches for different neural architectures, including decoder-only, encoder-only, and encoder-decoder architectures. We restrict our discussion to Transformers since they form the basis of most pre-trained models in NLP. However, pre-training is a broad concept, and so we just give a brief introduction to basic approaches in order to make this section concise.

7.2.1 Decoder-only Pre-training

The decoder-only architecture has been widely used in developing language models [Radford et al., 2018]. For example, we can use a Transformer decoder as a language model by simply removing its cross-attention sub-layers. Such a model predicts the probability distribution over the vocabulary for the next token based on its preceding tokens, and the predicted output is the token with the highest probability. The standard way to train this model, as in traditional language modeling, is to minimize a loss function over a collection of token sequences.

Let $\mathrm{Decoder}_\theta(\cdot)$ denote a decoder with parameters θ. At each position i, the decoder generates a distribution over the next token based on the preceding context $\{x_0, \ldots, x_i\}$, denoted by $\Pr_\theta(\cdot|x_0, \ldots, x_i)$ (or $\mathbf{p}^\theta_{i+1}$ for short). Suppose we have the gold-standard target distribution at the same position, denoted by $\mathbf{p}^{\mathrm{gold}}_{i+1}$. For language modeling, we can think of $\mathbf{p}^{\mathrm{gold}}_{i+1}$ as a one-hot representation of the correct next word. We then define a loss function $\mathcal{L}(\mathbf{p}^\theta_{i+1}, \mathbf{p}^{\mathrm{gold}}_{i+1})$ to measure the discrepancy between the model prediction and the target distribution. In NLP, the cross-entropy loss is typically used.

Given a token sequence of $\mathbf{x} = \{x_0, \cdots, x_m\}$, the loss on this sequence is the sum of the loss over the positions $\{0, \cdots, m-1\}$, given by

$$\begin{aligned} \mathrm{Loss}_\theta(x_0, \cdots, x_m) &= \sum_{i=0}^{m-1} \mathcal{L}(\mathbf{p}^\theta_{i+1}, \mathbf{p}^{\mathrm{gold}}_{i+1}) \\ &= \sum_{i=0}^{m-1} \mathrm{CrossEntropy}(\mathbf{p}^\theta_{i+1}, \mathbf{p}^{\mathrm{gold}}_{i+1}), \end{aligned} \tag{7.5}$$

where $\mathrm{CrossEntropy}(\cdot)$ denotes the standard cross-entropy loss, and $\mathbf{p}^{\mathrm{gold}}_{i+1}$ is the one-hot representation of x_{i+1}.

This loss function can be extended to a set of sequences $\mathcal{D}$. In this case, the objective of pre-training is to find the optimal parameters that minimize the loss on $\mathcal{D}$

$$\hat{\theta} = \arg\min_\theta \sum_{\mathbf{x} \in \mathcal{D}} \mathrm{Loss}_\theta(\mathbf{x}). \tag{7.6}$$

Note that this objective is mathematically equivalent to maximum likelihood estimation, and can be re-expressed as:

$$\begin{aligned} \hat{\theta} &= \arg\max_\theta \sum_{\mathbf{x} \in \mathcal{D}} \log \Pr_\theta(\mathbf{x}) \\ &= \arg\max_\theta \sum_{\mathbf{x} \in \mathcal{D}} \sum_{i=0}^{|\mathbf{x}|-2} \log \Pr_\theta(x_{i+1}|x_0, \ldots, x_i). \end{aligned} \tag{7.7}$$

With these optimized parameters $\hat{\theta}$, we can use the pre-trained language model $\mathrm{Decoder}_{\hat{\theta}}(\cdot)$ to compute the probability $\mathrm{Pr}_{\hat{\theta}}(x_{i+1}|x_0, \cdots, x_i)$ at each position of a given sequence.

7.2.2 Encoder-only Pre-training

As defined in Section 7.1.2, an encoder $\mathrm{Encode}_\theta(\cdot)$ is a function that reads a sequence of tokens $\mathbf{x} = \{x_0, \dots, x_m\}$ and produces a sequence of vectors $\mathbf{H} = \{\mathbf{h}_0, \dots, \mathbf{h}_m\}$[4]. Training this model is not straightforward, as we do not have gold-standard data for measuring the quality of the continuous representations produced by this function. A typical approach to encoder pre-training is to combine the encoder with a task-specific output layer to obtain supervision signals that are readily available. Figure 7.2 shows a common architecture for pre-training Transformer encoders, where we add a softmax layer on top of the encoder. This architecture is essentially the same as that of the decoder-based language model, and the output is a sequence of probability distributions:

$$\begin{bmatrix} \mathbf{p}_1^{\mathbf{W},\theta} \\ \vdots \\ \mathbf{p}_m^{\mathbf{W},\theta} \end{bmatrix} = \mathrm{Softmax}_{\mathbf{W}}(\mathrm{Encode}_\theta(\mathbf{x})). \tag{7.8}$$

Here $\mathbf{p}_i^{\mathbf{W},\theta}$ is the output distribution $\mathrm{Pr}(\cdot|\mathbf{x})$ at position i. We use $\mathrm{Softmax}_{\mathbf{W}}(\cdot)$ to denote that the softmax layer is parameterized by $\mathbf{W}$, that is, $\mathrm{Softmax}_{\mathbf{W}}(\mathbf{H}) = \mathrm{Softmax}(\mathbf{H} \cdot \mathbf{W})$. For notational simplicity, we will sometimes drop the superscripts $\mathbf{W}$ and θ affixed to each probability distribution.

The difference between this model and standard language models is that the output $\mathbf{p}_i$ has different meanings in encoder pre-training and language modeling. In language modeling, $\mathbf{p}_i$ is the probability distribution of predicting the next token. This follows an autoregressive decoding process: a language model only observes the words up to position i and predicts the next. By contrast, in encoder pre-training, the entire sequence is observed at once, and thus it is trivial to predict a token when the model has already seen it.

1. Masked Language Modeling

One of the most popular methods of encoder pre-training is **masked language modeling**, which forms the basis of the well-known BERT model [Devlin et al., 2019]. The idea of masked language modeling is to create prediction challenges by masking out some of the tokens in the input sequence and training a model to predict the masked tokens. In this sense, the conventional language modeling problem, which is sometimes called **causal language modeling**, is a special case of masked language modeling: at each position, we mask the

[4] If we view $\mathbf{h}_i$ as a row vector, $\mathbf{H}$ can be written as

$$\mathbf{H} = \begin{bmatrix} \mathbf{h}_0 \\ \vdots \\ \mathbf{h}_m \end{bmatrix}.$$

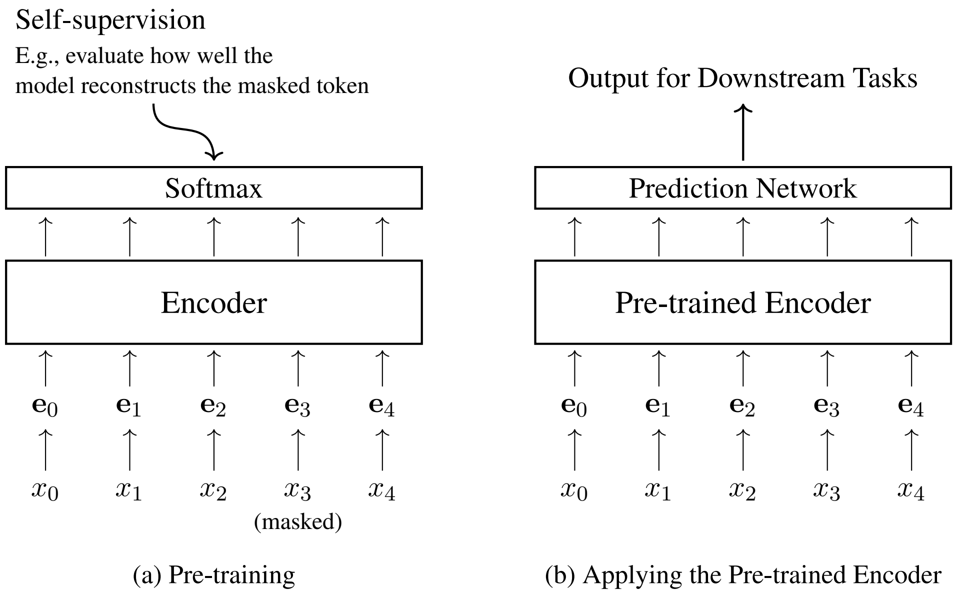


Figure 7.2: Pre-training a Transformer encoder (left) and then applying the pre-trained encoder (right). In the pre-training phase, the encoder, together with a softmax layer, is trained via self-supervision. In the application phase, the softmax layer is removed, and the pre-trained encoder is combined with a prediction network to address specific problems. In general, for better adaptation to these tasks, the system is fine-tuned using labeled data.

tokens in the right context, and predict the token at this position using its left context. However, in causal language modeling, only the left context is used for token prediction, and the model does not have access to tokens in the right context. By contrast, in masked language modeling, all the unmasked tokens are used for word prediction, leading to a bidirectional model that makes predictions based on both left and right contexts.

More formally, for an input sequence $\mathbf{x} = x_0 \cdots x_m$, suppose that we mask the tokens at positions $\mathcal{A}(\mathbf{x}) = \{i_1, \cdots, i_u\}$. Hence we obtain a masked token sequence $\bar{\mathbf{x}}$ where the token at each position in $\mathcal{A}(\mathbf{x})$ is replaced with a special symbol $[\mathrm{MASK}]$. For example, for the following sequence

$$\text{The early bird catches the worm}$$

we may have a masked token sequence like this

$$\text{The } [\mathrm{MASK}] \text{ bird catches the } [\mathrm{MASK}]$$

where we mask the tokens *early* and *worm* (i.e., $i_1 = 2$ and $i_2 = 6$).

Now we have two sequences $\mathbf{x}$ and $\bar{\mathbf{x}}$. The model is then optimized so that we can correctly predict $\mathbf{x}$ based on $\bar{\mathbf{x}}$. This can be thought of as an autoencoding-like process, and the training objective is to maximize the reconstruction probability $\Pr(\mathbf{x}|\bar{\mathbf{x}})$. Note that there is a simple position-wise alignment between $\mathbf{x}$ and $\bar{\mathbf{x}}$. Because an unmasked token in $\bar{\mathbf{x}}$ is the same as the token in $\mathbf{x}$ at the same position, there is no need to consider the prediction for this unmasked token. This leads to a simplified training objective which only maximizes the probabilities for masked tokens. We can express this objective in a maximum likelihood estimation fashion

$$(\widehat{\mathbf{W}}, \hat{\theta}) = \arg\max_{\mathbf{W},\theta} \sum_{\mathbf{x}\in\mathcal{D}} \sum_{i\in\mathcal{A}(\mathbf{x})} \log \Pr_i^{\mathbf{W},\theta}(x_i|\bar{\mathbf{x}}), \tag{7.9}$$

or alternatively express it using the cross-entropy loss

$$(\widehat{\mathbf{W}}, \hat{\theta}) = \arg\min_{\mathbf{W},\theta} \sum_{\mathbf{x}\in\mathcal{D}} \sum_{i\in\mathcal{A}(\mathbf{x})} \mathrm{CrossEntropy}(\mathbf{p}_i^{\mathbf{W},\theta}, \mathbf{p}_i^{\mathrm{gold}}), \tag{7.10}$$

where $\Pr_k^{\mathbf{W},\theta}(x_k|\bar{\mathbf{x}})$ is the probability of the true token x_k at position k given the corrupted input $\bar{\mathbf{x}}$, and $\mathbf{p}_k^{\mathbf{W},\theta}$ is the probability distribution at position k given the corrupted input $\bar{\mathbf{x}}$. To illustrate, consider the above example where two tokens of the sequence "*the early bird catches the worm*" are masked. For this example, the objective is to maximize the sum of log-probabilities

$$\begin{aligned} \mathrm{Loss} = \; & \log \Pr(x_2 = \textit{early}|\bar{\mathbf{x}} = [\mathrm{CLS}]\ \textit{The}\ \underbrace{[\mathrm{MASK}]}_{\bar{x}_2}\ \textit{bird catches the}\ \underbrace{[\mathrm{MASK}]}_{\bar{x}_6}) + \\ & \log \Pr(x_6 = \textit{worm}|\bar{\mathbf{x}} = [\mathrm{CLS}]\ \textit{The}\ \underbrace{[\mathrm{MASK}]}_{\bar{x}_2}\ \textit{bird catches the}\ \underbrace{[\mathrm{MASK}]}_{\bar{x}_6}). \end{aligned} \tag{7.11}$$

Once we obtain the optimized parameters $\widehat{\mathbf{W}}$ and $\hat{\theta}$, we can drop $\widehat{\mathbf{W}}$. Then, we can further fine-tune the pre-trained encoder $\mathrm{Encode}_{\hat{\theta}}(\cdot)$ or directly apply it to downstream tasks.

2. Permuted Language Modeling

While masked language modeling is simple and widely applied, it introduces new issues. One drawback is the use of a special token, [MASK], which is employed only during training but not at test time. This leads to a discrepancy between training and inference. Moreover, the auto-encoding process overlooks the dependencies between masked tokens. For example, in the previously discussed example, the prediction of x_2 (i.e., the first masked token) is made independently of x_6 (i.e., the second masked token), even though x_6 and x_2 are interdependent in the context of the sentence.

These issues can be addressed using the **permuted language modeling** approach to pre-training [Yang et al., 2019]. Similar to causal language modeling, permuted language modeling involves making sequential predictions of tokens. However, unlike causal modeling where predictions follow the natural sequence of the text (like left-to-right or right-to-left), permuted language modeling allows for predictions in any order. The approach is straightforward: we determine an order for token predictions and then train the model in a standard language modeling manner, as described in Section 7.2.1. Note that in this approach, the actual order of tokens in the text remains unchanged, and only the order in which we predict these tokens differs from standard language modeling. For example, consider a sequence of 5 tokens $x_0x_1x_2x_3x_4$. Let $\mathbf{e}_i$ represent the embedding of x_i (i.e., combination of the token embedding and positional embedding). In standard language modeling, we would generate this sequence in the order of $x_0 \to x_1 \to x_2 \to x_3 \to x_4$. The probability of the sequence can be modeled via a generation process.

$$\begin{aligned}\Pr(\mathbf{x}) \;=\; & \Pr(x_0)\cdot\Pr(x_1|x_0)\cdot\Pr(x_2|x_0,x_1)\cdot\Pr(x_3|x_0,x_1,x_2)\cdot \\ & \Pr(x_4|x_0,x_1,x_2,x_3). \qquad (7.12)\\ & \qquad (7.13)\end{aligned}$$

In practice, the model conditions on token embeddings rather than the tokens themselves. Therefore, we can rewrite the above expression as

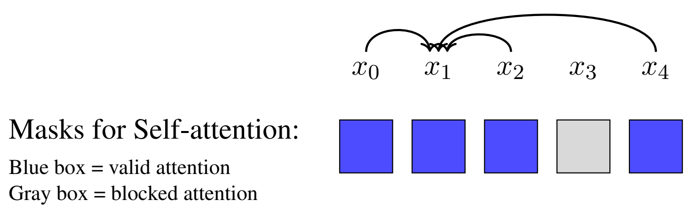
$$\begin{aligned}\Pr(\mathbf{x}) \;=\; & \Pr(x_0)\cdot\Pr(x_1|\mathbf{e}_0)\cdot\Pr(x_2|\mathbf{e}_0,\mathbf{e}_1)\cdot\Pr(x_3|\mathbf{e}_0,\mathbf{e}_1,\mathbf{e}_2)\cdot \\ & \Pr(x_4|\mathbf{e}_0,\mathbf{e}_1,\mathbf{e}_2,\mathbf{e}_3). \end{aligned} \qquad (7.14)$$

Now, let us consider a different order for token prediction: $x_0 \to x_4 \to x_2 \to x_1 \to x_3$. The sequence generation process can then be expressed as follows:

$$\begin{aligned}\Pr(\mathbf{x}) \;=\; & \Pr(x_0)\cdot\Pr(x_4|\mathbf{e}_0)\cdot\Pr(x_2|\mathbf{e}_0,\mathbf{e}_4)\cdot\Pr(x_1|\mathbf{e}_0,\mathbf{e}_4,\mathbf{e}_2)\cdot \\ & \Pr(x_3|\mathbf{e}_0,\mathbf{e}_4,\mathbf{e}_2,\mathbf{e}_1). \end{aligned} \qquad (7.15)$$

This new prediction order allows some tokens to be generated based on a broader context, rather than being limited to just the preceding tokens as in standard language models. For example, in generating x_3, the model considers both its left context (i.e., $\{\mathbf{e}_0,\mathbf{e}_1,\mathbf{e}_2\}$) and right context (i.e., $\{\mathbf{e}_4\}$). The embeddings $\mathbf{e}_0,\mathbf{e}_1,\mathbf{e}_2,\mathbf{e}_4$ incorporate the positional information of x_0,x_1,x_2,x_4, preserving the original order of the tokens. As a result, this approach is somewhat akin to masked language modeling: we mask out x_3 and use its surrounding tokens x_0,x_1,x_2,x_4 to predict this token.

The implementation of permuted language models is relatively easy for Transformers. Because self-attention itself is permutation-invariant, and positional information is encoded separately using positional embeddings, we do not need to explicitly reorder the input sequence to have a factorization like Eq. (7.15). Instead, permutation can be done by setting appropriate masks for self-attention. For example, consider the step of computing $\Pr(x_1|\mathbf{e}_0,\mathbf{e}_4,\mathbf{e}_2)$. We can keep x_0,x_1,x_2,x_3,x_4 in their original positions and simply prevent x_1 from attending to x_3 during the self-attention computation, as illustrated below



For a clearer illustration, we compare the self-attention masking results of causal language modeling, masked language modeling and permuted language modeling in Figure 7.3.

3. Pre-training Encoders as Classifiers

Another common approach to training an encoder is to use classification tasks. In self-supervised learning, this is typically done by creating new classification challenges from the unlabeled text. There are many different ways to design the classification tasks. Here we present two popular tasks.

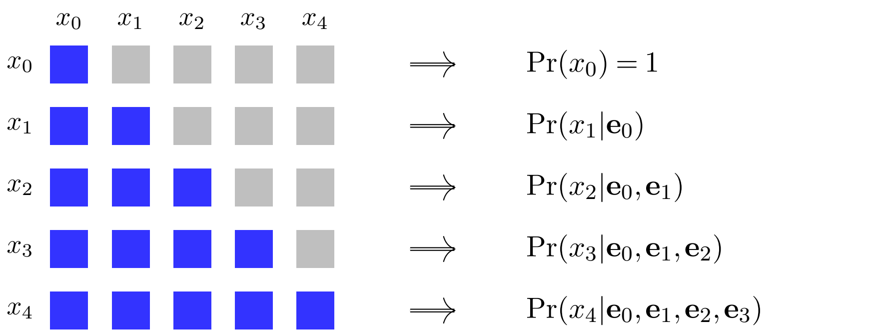
A simple method, called **next sentence prediction** (**NSP**), is presented in BERT's original paper [Devlin et al., 2019]. NSP is based on the assumption that a good text encoder should capture the relationship between two sentences. To model this relationship, NSP uses the encoder output of two consecutive sentences Sent_A and Sent_B to determine whether Sent_B is the next sentence following Sent_A. For example, suppose $\mathrm{Sent}_A =$ *'It is raining .'* and $\mathrm{Sent}_B =$ *'I need an umbrella .'*. The input sequence of the encoder could be

$$\text{[CLS] It is raining . [SEP] I need an umbrella . [SEP]}$$

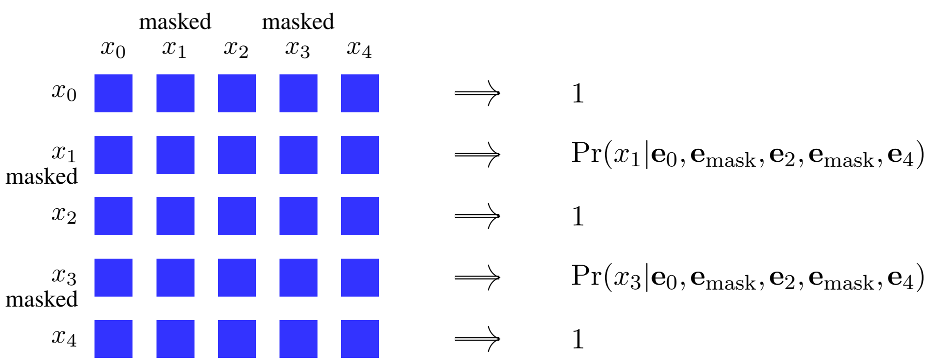
where [CLS] is the start symbol (i.e., x_0) which is commonly used in encoder pre-training, and [SEP] is a separator that separates the two sentences. The processing of this sequence follows a standard procedure of Transformer encoding: we first represent each token x_i as its corresponding embedding $\mathbf{e}_i$, and then feed the embedding sequence $\{\mathbf{e}_0, \cdots, \mathbf{e}_m\}$ into the encoder to obtain the output sequence $\{\mathbf{h}_0, \cdots, \mathbf{h}_m\}$. Since $\mathbf{h}_0$ is generally considered as the representation of the entire sequence, we add a softmax layer on top of it to construct a binary classification system. This process is illustrated as follows:

token:	[CLS]	It	is	raining	.	[SEP]	I	need	an	umbrella	.	[SEP]
embedding:	$\mathbf{e}_0$	$\mathbf{e}_1$	$\mathbf{e}_2$	$\mathbf{e}_3$	$\mathbf{e}_4$	$\mathbf{e}_5$	$\mathbf{e}_6$	$\mathbf{e}_7$	$\mathbf{e}_8$	$\mathbf{e}_9$	$\mathbf{e}_{10}$	$\mathbf{e}_{11}$
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
	Encoder											
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
encoding:	$\mathbf{h}_0$	$\mathbf{h}_1$	$\mathbf{h}_2$	$\mathbf{h}_3$	$\mathbf{h}_4$	$\mathbf{h}_5$	$\mathbf{h}_6$	$\mathbf{h}_7$	$\mathbf{h}_8$	$\mathbf{h}_9$	$\mathbf{h}_{10}$	$\mathbf{h}_{11}$
	$\downarrow$											
	Softmax											
	$\downarrow$											
	Is Next or Not?											

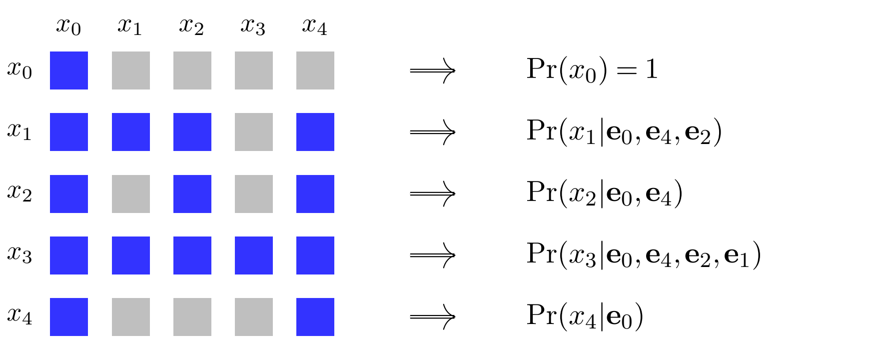
To generate training samples, we extract sentence pairs from a text corpus. A positive sample is created by selecting two actual consecutive sentences, while a negative sample is formed by pairing a sentence with a randomly chosen sentence from the corpus. As a result,



(a) Causal Language Modeling (order: $x_0 \to x_1 \to x_2 \to x_3 \to x_4$)



(b) Masked Language Modeling (order: $x_0, [\text{MASK}], x_2, [\text{MASK}], x_4 \to x_1, x_3$)



(c) Permuted Language Modeling (order: $x_0 \to x_4 \to x_2 \to x_1 \to x_3$)

Figure 7.3: Comparison of self-attention masking results of causal language modeling, masked language modeling and permuted language modeling. The gray cell denotes that the token at position j does not attend to the token at position i. The blue cell (i, j) denotes that the token at position j attends to the token at position i. $\mathbf{e}_{\text{mask}}$ represents the embedding of the symbol $[\text{MASK}]$, which is a combination of the token embedding and the positional embedding.

training this model is the same as training a classifier. Typically, NSP is used as an additional training loss function for pre-training based on masked language modeling.

A second approach to training Transformer encoders as classifiers applies classification-based supervision signals at the token level. For example, Clark et al. [2019b] propose, in their ELECTRA model, training a Transformer encoder to identify whether each input token is identical to the original token or has been altered in some manner. The first step of this method is to generate a new sequence from a given sequence of tokens, where some of the tokens are altered. To do this, a small masked language model (call it the generator) is applied: we randomly mask some of the tokens, and train this model to predict the masked tokens. For each training sample, this masked language model outputs a token at each masked position, which may differ from the original token. At the same time, we train another Transformer encoder (call it the discriminator) to determine whether each predicted token is the same as the original token or altered. More specifically, we use the generator to generate a sequence where some of the tokens are replaced. Below is an illustration.

original:	[CLS]	The	boy	spent	hours	working	on	toys	.
	↓	↓	↓	↓	↓	↓	↓	↓	↓
masked:	[CLS]	The	boy	spent	[MASK]	working	on	[MASK]	.
	↓	↓	↓	↓	↓	↓	↓	↓	↓
	Generator (small masked language model)								
	↓	↓	↓	↓	↓	↓	↓	↓	↓
replaced:	[CLS]	The	boy	spent	decades	working	on	toys	.

Then, we use the discriminator to label each of these tokens as original or replaced, as follows

replaced:	[CLS]	The	boy	spent	decades	working	on	toys	.
	↓	↓	↓	↓	↓	↓	↓	↓	↓
	Discriminator (the model we want)								
	↓	↓	↓	↓	↓	↓	↓	↓	↓
label:	original	original	original	original	replaced	original	original	original	original

For training, the generator is optimized as a masked language model with maximum likelihood estimation, and the discriminator is optimized as a classifier using a classification-based loss. In ELECTRA, the maximum likelihood-based loss and the classification-based loss are combined for jointly training both the generator and discriminator. An alternative approach is to use generative adversarial networks (GANs), that is, the generator is trained to fool the discriminator, and the discriminator is trained to distinguish the output of the generator from the true distribution. However, GAN-style training complicates the training task and is more difficult to scale up. Nevertheless, once training is complete, the generator is discarded, and the encoding part of the discriminator is applied as the pre-trained model for downstream tasks.

7.2.3 Encoder-Decoder Pre-training

In NLP, encoder-decoder architectures are often used to model sequence-to-sequence problems, such as machine translation and question answering. In addition to these typical sequence-to-sequence problems in NLP, encoder-decoder models can be extended to deal with many other problems. A simple idea is to treat text as both the input and output of a problem, and so we can directly apply encoder-decoder models. For example, given an input text, we can instruct a model to generate text describing its sentiment (e.g., *positive*, *negative*, or *neutral*).

Such an idea allows us to develop a single text-to-text system to address any NLP problem. We can formulate different problems into the same text-to-text format. We first train an encoder-decoder model to gain general-purpose knowledge of language via self-supervision. This model is then fine-tuned for specific downstream tasks using targeted text-to-text data.

1. Masked Encoder-Decoder Pre-training

In Raffel et al. [2020]'s **T5** model, many different tasks are framed as the same text-to-text task. Each sample in T5 follows the format

$$\text{Source Text} \quad \rightarrow \quad \text{Target Text}$$

Here $\rightarrow$ separates the source text, which consists of a task description or instruction and the input given to the system, from the target text, which is the response to the input task. As an example, consider a task of translating from Chinese to English. A training sample can be expressed as:

$$[\text{CLS}]\ \text{Translate from Chinese to English: 你好！} \quad \rightarrow \quad \langle s \rangle\ \text{Hello!}$$

where $[\text{CLS}]$ and $\langle s \rangle$ are the start symbols on the source and target sides, respectively[5].

[5] We could use the same start symbol for different sequences. Here we use different symbols to distinguish between the sequences on the encoder and decoder-sides.

Likewise, we can express other tasks in the same way. For example

[CLS] Answer: when was Albert Einstein born?
$\rightarrow$ $\langle s \rangle$ He was born on March 14, 1879.

[CLS] Simplify: the professor, who has published numerous papers in his field, will be giving a lecture on the topic next week.
$\rightarrow$ $\langle s \rangle$ The experienced professor will give a lecture next week.

[CLS] Text: John bought a new car. Hypothesis: John has a car.
$\rightarrow$ $\langle s \rangle$ Entailment

[CLS] Score the translation from English to Chinese. English: when in Rome, do as the Romans do. Chinese: 人 在 罗马 就 像 罗马 人 一样 做事 。
$\rightarrow$ $\langle s \rangle$ 0.81

where instructions are highlighted in gray. Interestingly, in the last example, we frame a continuous scoring problem as a text generation task. Our goal is to generate the text string representing the number 0.81, rather than outputting a continuous numerical value.

The approach described above provides a new framework of universal language understanding and generation. Both the task instructions and the problem inputs are provided to the system in text form. The system then follows the instructions to complete the task. This method puts different problems together, with the benefit of training a single model that can perform many tasks simultaneously.

In general, fine-tuning is necessary for adapting the pre-trained model to a specific downstream task. In this process, one can use different ways to instruct the model for the task, such as using a short name of the task as the prefix to the actual input sequence or providing a detailed description of the task. Because task instructions are provided as text and processed as part of the input, general instruction-following capabilities can be acquired during the pre-training phase. This greatly facilitates zero-shot learning. For instance, pre-trained models can often generalize to solve new problems where task-specific instructions have never been encountered during training.

There have been several powerful methods of self-supervised learning for either Transformer encoders or decoders. Applying these methods to pre-train encoder-decoder models is relatively straightforward. One common choice is to train encoder-decoder models as language models. For example, the encoder receives a sequence prefix, while the decoder generates the remaining sequence. However, this differs from standard causal language modeling, where the entire sequence is autoregressively generated from the first token. In our case, the encoder processes the prefix at once, and then the decoder predicts subsequent tokens in the manner of causal language modeling. Put more precisely, this is a **prefix language modeling** problem: a language model predicts the subsequent sequence given a prefix, which serves as the context for prediction.

Consider the following example

$$\underbrace{\text{[CLS] The puppies are frolicking}}_{\text{Prefix}} \rightarrow \underbrace{\langle s \rangle \text{ outside the house .}}_{\text{Subsequent Sequence}}$$

We can directly train an encoder-decoder model using examples like this. Then, the encoder learns to understand the prefix, and the decoder learns to continue writing based on this understanding. For large-scale pre-training, it is easy to create a large number of training examples from unlabeled text.

It is worth noting that for pre-trained encoder-decoder models to be effective in multilingual and cross-lingual tasks (e.g., machine translation), they should be trained on multilingual data. This typically requires that the vocabulary includes tokens from multiple languages. By doing so, the models can learn shared representations across different languages, thereby enabling capabilities in both language understanding and generation in a multilingual and cross-lingual context.

A second approach to pre-training encoder-decoder models is masked language modeling. In this approach, as discussed in Section 7.2.2, tokens in a sequence are randomly replaced with a mask symbol, and the model is then trained to predict these masked tokens based on the corrupted sequence.

As an illustration, consider the task of masking and reconstructing the sentence

$$\text{The puppies are frolicking outside the house .}$$

By masking two tokens (say, *frolicking* and *the*), we have the BERT-style input and output of the model, as follows

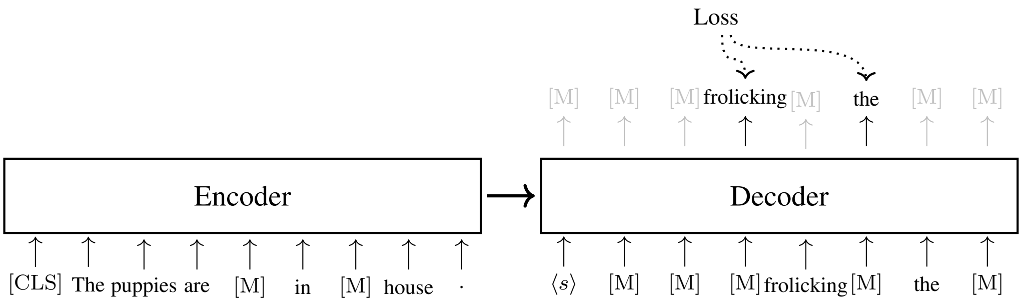
$$\begin{aligned} & \text{[CLS] The puppies are [MASK] outside [MASK] house .} \\ \rightarrow\ & \langle s \rangle \ \_\_\ \_\_\ \_\_\ \text{ frolicking } \_\_\ \text{ the } \_\_\ \_\_ \end{aligned}$$

Here __ denotes the masked position at which we do not make token predictions. By varying the percentage of the tokens in the text, this approach can be generalized towards either BERT-style training or language modeling-style training [Song et al., 2019]. For example, if we mask out all the tokens, then the model is trained to generate the entire sequence

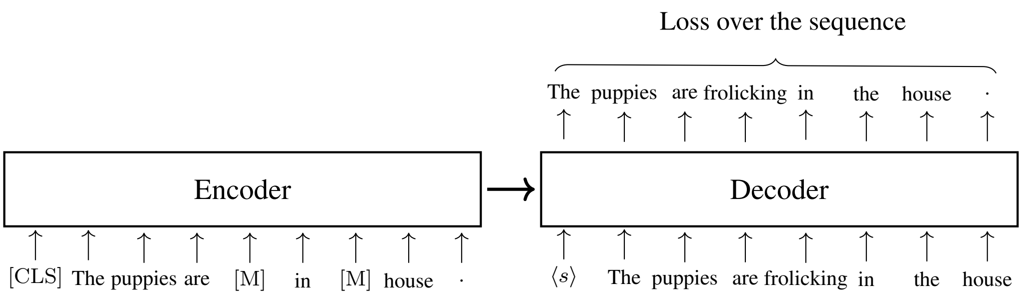
$$\begin{aligned} & \text{[CLS] [MASK] [MASK] [MASK] [MASK] [MASK] [MASK] [MASK] [MASK]} \\ \rightarrow\ & \langle s \rangle \text{ The puppies are frolicking outside the house .} \end{aligned}$$

In this case, we train the decoder as a language model.

Note that, in the context of the encoder-decoder architecture, we can use the encoder to read the masked sequence, and use the decoder to predict the original sequence. With this objective, we essentially have a denoising autoencoder: the encoder transforms a corrupted



(a) Training an encoder-decoder model with BERT-style masked language modeling



(b) Training an encoder-decoder model with denoising autoencoding

Figure 7.4: Training an encoder-decoder model using BERT-style and denoising autoencoding methods. In both methods, the input to the encoder is a corrupted token sequence where some tokens are masked and replaced with $[\mathrm{MASK}]$ (or $[\mathrm{M}]$ for short). The decoder predicts these masked tokens, but in different ways. In BERT-style training, the decoder only needs to compute the loss for the masked tokens, while the remaining tokens in the sequence can be simply treated as $[\mathrm{MASK}]$ tokens. In denoising autoencoding, the decoder predicts the sequence of all tokens in an autoregressive manner. As a result, the loss is obtained by accumulating the losses of all these tokens, as in standard language modeling.

input into some hidden representation, and the decoder reconstructs the uncorrupted input from this hidden representation. Here is an example of input and output for denoising training.

$$\begin{array}{ll} & [\mathrm{CLS}]\ \text{The puppies are}\ [\mathrm{MASK}]\ \text{outside}\ [\mathrm{MASK}]\ \text{house .} \\ \rightarrow & \langle s\rangle\ \text{The puppies are frolicking outside the house .} \end{array}$$

By learning to map from this corrupted sequence to its uncorrupted counterpart, the model gains the ability to understand on the encoder side and to generate on the decoder side. See Figure 7.4 for an illustration of how an encoder-decoder model is trained with BERT-style and denoising autoencoding objectives.

As we randomly select tokens for masking, we can mask consecutive tokens [Joshi et al., 2020]. Here is an example.

$$\begin{array}{rl} & \text{[CLS] The puppies are [MASK] outside [MASK] [MASK] .} \\ \rightarrow & \langle s \rangle \text{ The puppies are frolicking outside the house .} \end{array}$$

Another way to consider consecutive masked tokens is to represent them as spans. Here we follow Raffel et al. [2020]'s work, and use [X], [Y] and [Z] to denote sentinel tokens that cover one or more consecutive masked tokens. Using this notation, we can re-express the above training example as:

$$\begin{array}{rl} & \text{[CLS] The puppies are [X] outside [Y] .} \\ \rightarrow & \langle s \rangle \text{ [X] frolicking [Y] the house [Z]} \end{array}$$

The idea is that we represent the corrupted sequence as a sequence containing placeholder slots. The training task is to fill these slots with the correct tokens using the surrounding context. An advantage of this approach is that the sequences used in training would be shorter, making the training more efficient. Note that masked language modeling provides a very general framework for training encoder-decoder models. Various settings can be adjusted to have different training versions, such as altering the percentage of tokens masked and the maximum length of the masked spans.

2. Denoising Training

If we view encoder-decoder pre-training as a denoising autoencoding problem, there are many ways to corrupt the input. For instance, beyond random token masking, we can also modify tokens or rearrange their order.

Suppose we have an encoder-decoder model that can map an input sequence $\mathbf{x}$ to an output sequence $\mathbf{y}$

$$\begin{aligned} \mathbf{y} &= \mathrm{Decode}_{\omega}(\mathrm{Encode}_{\theta}(\mathbf{x})) \\ &= \mathrm{Model}_{\theta,\omega}(\mathbf{x}), \end{aligned} \tag{7.16}$$

where θ and ω are the parameters of the encoder and the decoder, respectively. In denoising autoencoding, we add some noise to $\mathbf{x}$ to obtain a noisy, corrupted input $\mathbf{x}_{\mathrm{noise}}$. By feeding $\mathbf{x}_{\mathrm{noise}}$ into the encoder, the decoder is trained to recontruct the original input. The training objective can be defined as:

$$(\hat{\theta}, \hat{\omega}) = \arg\min_{\theta,\omega} \mathrm{Loss}(\mathrm{Model}_{\theta,\omega}(\mathbf{x}_{\mathrm{noise}}), \mathbf{x}), \tag{7.17}$$

Here the loss function $\mathrm{Loss}(\mathrm{Model}_{\theta,\omega}(\mathbf{x}_{\mathrm{noise}}), \mathbf{x})$ evaluates how well the model $\mathrm{Model}_{\theta,\omega}(\mathbf{x}_{\mathrm{noise}})$ reconstructs the original input $\mathbf{x}$. We can choose the cross-entropy loss as usual.

Given the model architecture and training framework, the remaining issue is how to corrupt the input. Lewis et al. [2020a], in their **BART** model, propose several ways to corrupt the input sequence.

- **Token Masking**. This is the same masking method that we used in masked language modeling. Tokens are randomly selected and replaced with [MASK].
- **Token Deletion**. This method is similar to token masking. However, rather than replacing the selected tokens with a special symbol [MASK], these tokens are removed from the sequence. See the following example for a comparison of the token masking and token deletion methods.

  Original ($\mathbf{x}$): The puppies are $\underline{\text{frolicking}}$ outside $\underline{\text{the}}$ house .
  Token Masking ($\mathbf{x}_{\text{noise}}$): The puppies are [MASK] outside [MASK] house .
  Token Deletion ($\mathbf{x}_{\text{noise}}$): The puppies are ~~frolicking~~ outside ~~the~~ house .

  where the underlined tokens in the original sequence are masked or deleted.
- **Span Masking**. Non-overlapping spans are randomly sampled over the sequence. Each span is masked by [MASK]. We also allow spans of length 0, and, in such cases, [MASK] is simply inserted at a position in the sequence. For example, we can use span masking to corrupt the above sequence as

  Original ($\mathbf{x}$): The $\underline{0}$ puppies are $\underline{\text{frolicking outside the}}$ house .
  Span Masking ($\mathbf{x}_{\text{noise}}$): The [MASK] puppies are [MASK] house .

  Here the span *frolicking outside the* is replaced with a single [MASK]. $\underline{0}$ indicates a length-0 span, and so we insert an [MASK] between *The* and *puppies*. Span masking introduces new prediction challenges in which the model needs to know how many tokens are generated from a span. This problem is very similar to fertility modeling in machine translation [Brown et al., 1993].

If we consider a sequence consisting of multiple sentences, additional methods of corruption can be applied. In the BART model, there are two such methods.

- **Sentence Reordering**. This method randomly permutes the sentences so that the model can learn to reorder sentences in a document. Consider, for example, two consecutive sentences

  Hard work leads to success . Success brings happiness .

  We can reorder the two sentences to have a corrupted input sequence

  Success brings happiness . Hard work leads to success .

- **Document Rotation**. The goal of this task is to identify the start token of the sequence. First, a token is randomly selected from the sequence. Then, the sequence is rotated so that the selected token is the first token. For example, suppose we select the token *leads* from the above sequence. The rotated sequence is

selected
↓
~~Hard work~~ leads to success . Success brings happiness . Hard work

where the subsequence *Hard work* before *leads* is appended to the end of the sequence.

For pre-training, we can apply multiple corruption methods to learn robust models, for example, we randomly choose one of them for each training sample. In practice, the outcome of encoder-decoder pre-training depends heavily on the input corruption methods used, and so we typically need to choose appropriate training objectives through careful experimentation.

7.2.4 Comparison of Pre-training Tasks

So far, we have discussed a variety of pre-training tasks. Because the same training objective can often be applied to different architectures (e.g., using masked language modeling for both encoder-only and encoder-decoder pre-training), categorizing these tasks solely based on model architecture is insufficient. Instead, we summarize them based on their training objectives.

- **Language Modeling**. Typically, this approach refers to an autoregressive process of sequence generation. At each step, the model predicts the next token based exclusively on its preceding context.
- **Masked Language Modeling**. Masked language modeling follows a general mask-predict framework. It randomly masks tokens in a sequence and predicts these tokens using the entire masked sequence.
- **Permuted Language Modeling**. Permuted language modeling shares similarities with masked language modeling but removes the fixed left-to-right restriction by permuting the factorization order. It keeps the input sequence fixed but reorders the factorization (prediction) order. Each prediction is based on some context tokens that are randomly selected.
- **Discriminative Training**. In discriminative training, supervision signals are created from classification tasks. Pre-trained models are integrated into classifiers and trained together with the remaining parts of the classifiers to enhance their classification performance.
- **Denoising Autoencoding**. This approach is applied to the pre-training of encoder-decoder models. The input is a corrupted sequence (with noise such as masking, deletion, or permutation), and the model is trained to reconstruct the original sequence.

Table 7.1 illustrates these methods and their variants using examples. While the input/output formats in the examples are generally model-agnostic, we explicitly indicate the compatible model architectures for each pre-training task. In each example, the input consists of a token sequence, and the output is either a token sequence or some probabilities. For generation tasks, such as language modeling, superscripts are used to indicate the generation

Method	Enc	Dec	E-D	Input	Output
Causal LM		•	•		The^{1} kitten^{2} is^{3} chasing^{4} the^{5} ball^{6} .^{7}
Prefix LM		•	•	[C] The kitten is	chasing^{1} the^{2} ball^{3} .^{4}
Masked LM	•		•	[C] The kitten [M] chasing the [M] .	_ _ is _ _ ball _
MASS-style	•		•	[C] The kitten [M] [M] [M] ball .	_ _ is chasing the _ _
BERT-style	•		•	[C] The kitten [M] playing the [M] .	_ kitten is chasing _ ball _
Permuted LM	•			[C] The kitten is chasing the ball .	The^{5} kitten^{7} is^{6} chasing^{1} the^{4} ball^{2} .^{3}
Next Sentence Prediction	•			[C] The kitten is chasing the ball . Birds eat worms .	Pr(IsNext | representation-of-[C])
Sentence Comparison	•			Sent$_a$: The kitten is chasing the ball . Sent$_b$: A cat is running after a toy .	Score(Sent$_a$, Sent$_b$)
Token Classification	•			[C] The kitten is chasing the ball .	Pr(·|The) Pr(·|kitten) ... Pr(·|.)
Token Reordering			•	[C] . kitten the chasing The is ball	The^{1} kitten^{2} is^{3} chasing^{4} the^{5} ball^{6} .^{7}
Token Deletion			•	[C] The kitten is ~~chasing~~ the ball .	The^{1} kitten^{2} is^{3} chasing^{4} the^{5} ball^{6} .^{7}
Span Masking			•	[C] The kitten [M] is [M] .	The^{1} kitten^{2} is^{3} chasing^{4} the^{5} ball^{6} .^{7}
Sentinel Masking			•	[C] The kitten [X] the [Y]	[X]^{1} is^{2} chasing^{3} [Y]^{4} ball^{5} .^{6}
Sentence Reordering			•	[C] The ball rolls away swiftly . The kitten is chasing the ball .	The^{1} kitten^{2} is^{3} chasing^{4} the^{5} ball^{6} .^{7} The^{8} ball^{9} rolls^{10} away^{11} swiftly^{12} .^{13}
Document Rotation			•	[C] chasing the ball . The ball rolls away swiftly . The kitten is	The^{1} kitten^{2} is^{3} chasing^{4} the^{5} ball^{6} .^{7} The^{8} ball^{9} rolls^{10} away^{11} swiftly^{12} .^{13}

Table 7.1: Comparison of pre-training tasks, including language modeling, masked language modeling, permuted language modeling, discriminative training, and denoising autoencoding. [C] = [CLS], [M] = [MASK], [X], [Y] = sentinel tokens. Enc, Dec and E-D indicate whether the approach can be applied to encoder-only, decoder-only, encoder-decoder models, respectively. For generation tasks, superscripts are used to represent the order of the tokens.

order on the target side. If the superscripts are omitted, it implies that the output tokens can be predicted either autoregressively or in parallel (non-autoregressively). On the source side, we assume that the sequence undergoes a standard Transformer encoding process, meaning that each token can see the entire sequence in self-attention. The only exception is in permuted language modeling, where an autoregressive generation process is implemented by setting attention masks on the encoder side. To simplify the discussion, we remove the token $\langle s \rangle$ from the target-side of each example.

While these pre-training tasks are different, it is possible to compare them in the same framework and experimental setup [Dong et al., 2019; Raffel et al., 2020; Lewis et al., 2020a]. Note that we cannot list all the pre-training tasks here as there are many of them. For more discussions on pre-training tasks, the interested reader may refer to some surveys on this topic [Qiu et al., 2020b; Han et al., 2021a].

7.3 Example: BERT

In this section, we introduce BERT, one of the most widely used pre-trained sequence encoding models in NLP.

7.3.1 The Standard Model

The standard BERT model, introduced by Devlin et al. [2019], is a bidirectional Transformer encoder trained using two self-supervised tasks: masked language modeling and next sentence prediction. The overall training loss is simply the sum of the losses from these two tasks:

$$\mathrm{Loss}_{\mathrm{BERT}} = \mathrm{Loss}_{\mathrm{mlm}} + \mathrm{Loss}_{\mathrm{nsp}}. \tag{7.18}$$

As is standard in deep learning, we optimize the model parameters by minimizing this loss. During training, we sample mini-batches from a text corpus, compute the $\mathrm{Loss}_{\mathrm{BERT}}$ over each batch, and iteratively update the model parameters using gradient descent or its variants. This process repeats until a stopping criterion is met, such as the convergence of the training loss or reaching a predefined maximum number of steps.

1. Loss Functions

In general, BERT models are used to represent a single sentence or a pair of sentences, and thus can handle various downstream language understanding problems. In this section we assume that the input representation is a sequence containing two sentences Sent_A and Sent_B, given by

$$[\mathrm{CLS}]\ \ \mathrm{Sent}_A\ \ [\mathrm{SEP}]\ \ \mathrm{Sent}_B\ \ [\mathrm{SEP}]$$

Here we follow the notation of [Devlin et al., 2019] and use [SEP] to denote the separator.

Given this sequence, we can obtain $\mathrm{Loss}_{\mathrm{mlm}}$ and $\mathrm{Loss}_{\mathrm{nsp}}$ separately. For masked language modeling, a subset of tokens (typically 15%) is randomly selected for prediction. Then the sequence is modified in three ways:

- **Token Masking**. 80% of the selected tokens are masked and replaced with the symbol [MASK]. For example

Original:	[CLS] It is <u>raining</u> . [SEP] <u>I</u> need <u>an</u> <u>umbrella</u> . [SEP]
Masked:	[CLS] It is [MASK] . [SEP] I need [MASK] umbrella . [SEP]

  where the selected tokens are underlined. Predicting masked tokens encourages the model to learn representations from surrounding context.

- **Random Replacement**. 10% of the selected tokens are changed to a random token. For example

Original:	[CLS] It is <u>raining</u> . [SEP] <u>I</u> need <u>an</u> <u>umbrella</u> . [SEP]
Random Token:	[CLS] It is raining . [SEP] I need a hat . [SEP]

This helps the model learn to recover a token from a corrupted input.

- **Unchanged**. 10% of the selected tokens are kept unchanged. For example,

Original:	[CLS] It is raining . [SEP] I need an umbrella . [SEP]
Unchanged Token:	[CLS] It is raining . [SEP] I need an umbrella . [SEP]

Although predicting an unchanged token is trivial, this strategy is crucial. It encourages the model to maintain a meaningful representation of the actual observed word, thereby helping to mitigate the discrepancy between pre-training (which uses [MASK]) and fine-tuning (which does not).

Let $\mathcal{A}(\mathbf{x})$ denote the set of selected positions in a token sequence $\mathbf{x}$, and $\bar{\mathbf{x}}$ denote the modified sequence of $\mathbf{x}$. The loss function of masked language modeling can be defined as:

$$\mathrm{Loss}_{\mathrm{mlm}} = -\sum_{i \in \mathcal{A}(\mathbf{x})} \log \Pr_i(x_i|\bar{\mathbf{x}}), \tag{7.19}$$

where $\Pr_i(x_i|\bar{\mathbf{x}})$ is the probability of predicting x_i at the position i given $\bar{\mathbf{x}}$. Figure 7.5 shows a running example of computing $\mathrm{Loss}_{\mathrm{mlm}}$.

For next sentence prediction, we follow the method described in Section 7.2.2. Each training sample is assigned a label from the set $\{\mathrm{IsNext}, \mathrm{NotNext}\}$, for example,

Sequence:	[CLS] It is raining . [SEP] I need an umbrella . [SEP]
Label:	IsNext
Sequence:	[CLS] The cat sleeps on the windowsill . [SEP] Apples grow on trees . [SEP]
Label:	NotNext

The output vector of the encoder for the first token [CLS] is viewed as the sequence representation, denoted by $\mathbf{h}_{\mathrm{cls}}$ (or $\mathbf{h}_0$). A classifier is built on top of $\mathbf{h}_{\mathrm{cls}}$. Then, we can compute the probability of a label c given $\mathbf{h}_{\mathrm{cls}}$, i.e., $\Pr(c|\mathbf{h}_{\mathrm{cls}})$. There are many loss functions one can choose for classification problems. For example, in maximum likelihood training, we can define $\mathrm{Loss}_{\mathrm{nsp}}$ as

$$\mathrm{Loss}_{\mathrm{nsp}} = -\log \Pr(c_{\mathrm{gold}}|\mathbf{h}_{\mathrm{cls}}), \tag{7.20}$$

where c_{gold} is the correct label for this sample.

2. Model Setup

As shown in Figure 7.6, BERT models are based on the standard Transformer encoder architecture. The input is a sequence of embeddings, each of which is the sum of the token embedding,

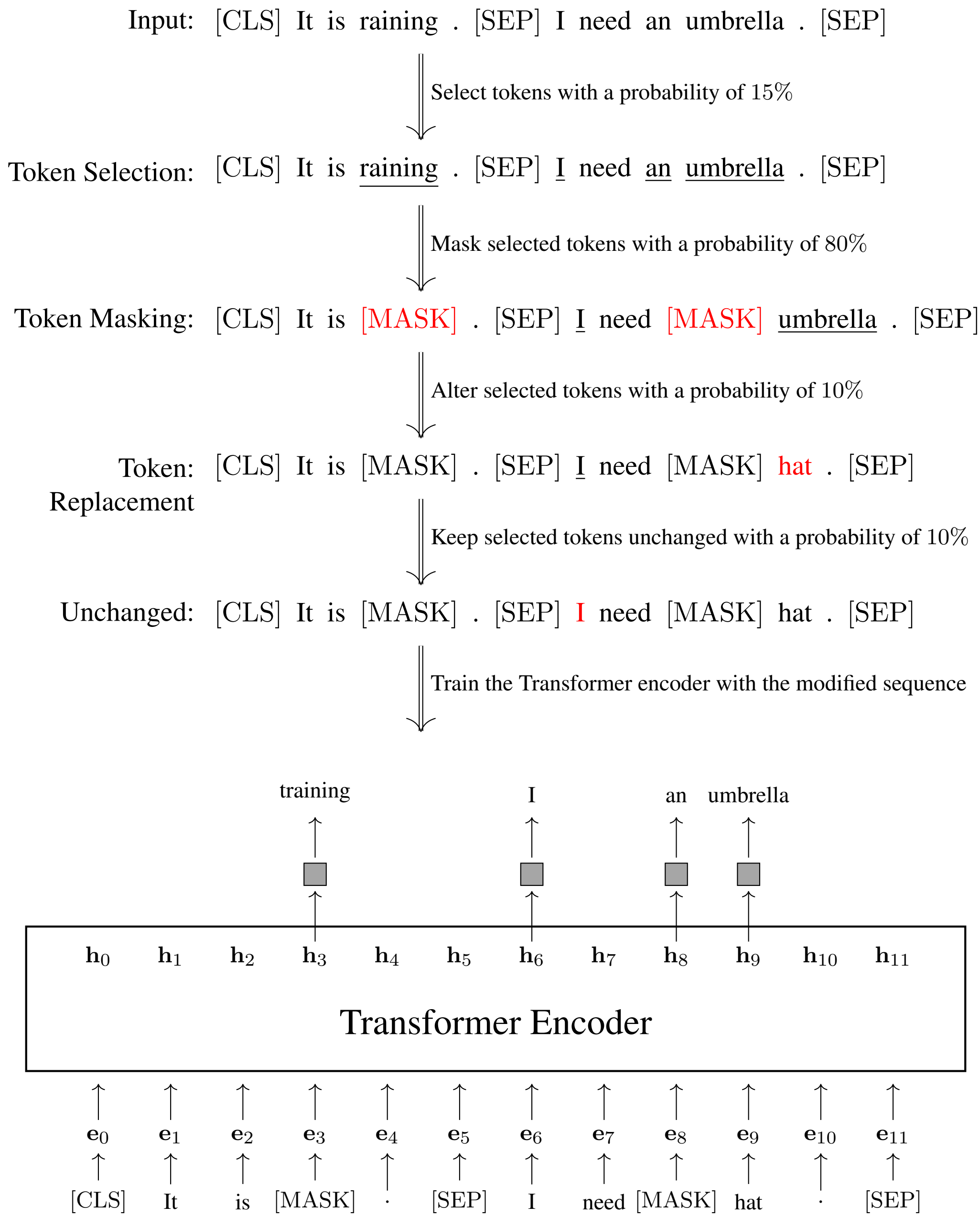


Figure 7.5: A running example of BERT-style masked language modeling. First, 15% of the tokens are randomly selected. These selected tokens are then processed in one of three ways: replaced with a [MASK] token (80% of the time), replaced with a random token (10% of the time), or kept unchanged (10% of the time). The model is trained to predict these selected tokens based on the modified sequence. $\mathbf{e}_i$ represents the embedding of the token at the position i. Gray boxes represent the softmax layers.

the positional embedding, and the segment embedding.

$$\mathbf{e} = \mathbf{x} + \mathbf{e}_{\text{pos}} + \mathbf{e}_{\text{seg}}. \tag{7.21}$$

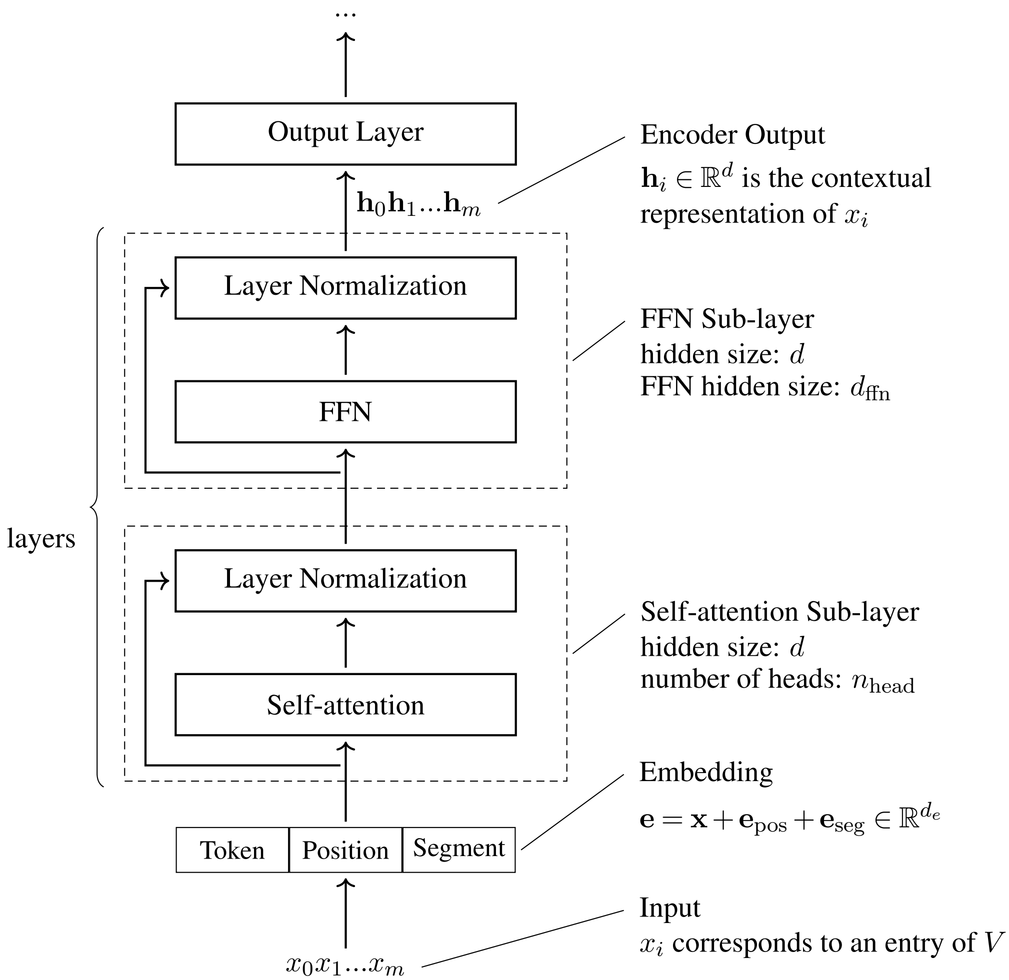


Figure 7.6: The model architecture of BERT (Transformer encoder). The input tokens are first represented as embeddings, each of which is the sum of the corresponding token embedding, positional embedding, and segment embedding. Then, the embedding sequence is processed by a stack of Transformer layers. Each layer in this stack includes a self-attention sub-layer and an FFN sub-layer. The output of the BERT model is a sequence of vectors produced by the final Transformer layer.

Both the token embedding ($\mathbf{x}$) and the positional embedding ($\mathbf{e}_{\text{pos}}$) are identical to those used in the original Transformer. The segment embedding ($\mathbf{e}_{\text{seg}}$) is a new addition that indicates whether a token belongs to Sent_A or Sent_B. This is illustrated by the following example:

Token	[CLS]	It	is	raining	.	[SEP]	I	need	an	umbrella	.	[SEP]
$\mathbf{x}$	$\mathbf{x}_0$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$	$\mathbf{x}_6$	$\mathbf{x}_7$	$\mathbf{x}_8$	$\mathbf{x}_9$	$\mathbf{x}_{10}$	$\mathbf{x}_{11}$
$\mathbf{e}_{\text{pos}}$	PE(0)	PE(1)	PE(2)	PE(3)	PE(4)	PE(5)	PE(6)	PE(7)	PE(8)	PE(9)	PE(10)	PE(11)
$\mathbf{e}_{\text{seg}}$	$\mathbf{e}_A$	$\mathbf{e}_A$	$\mathbf{e}_A$	$\mathbf{e}_A$	$\mathbf{e}_A$	$\mathbf{e}_A$	$\mathbf{e}_B$	$\mathbf{e}_B$	$\mathbf{e}_B$	$\mathbf{e}_B$	$\mathbf{e}_B$	$\mathbf{e}_B$

The main part of BERT models is a multi-layer Transformer network. A Transformer layer consists of a self-attention sub-layer and an FFN sub-layer. Both sub-layers follow the post-norm architecture: $\mathrm{output} = \mathrm{LNorm}(F(\mathrm{input}) + \mathrm{input})$, where $F(\cdot)$ is the core function of the sub-layer (either a self-attention model or an FFN), and $\mathrm{LNorm}(\cdot)$ is the layer normalization unit. Typically, a number of Transformer layers are stacked to form a deep network. For each position in the sequence, the final output representation is the real-valued vector produced by the uppermost layer.

There are several aspects one may consider in developing BERT models.

- **Vocabulary Size** ($|V|$). In Transformers, each token corresponds to an entry in a vocabulary V. Large vocabularies can cover more surface form variants of words, but may lead to increased storage requirements.
- **Embedding Size** (d_e). Every token is represented as a d_e-dimensional real-valued vector. As presented above, this vector is the sum of the token embedding, positional embedding, and segment embedding, all of which are also d_e-dimensional real-valued vectors.
- **Hidden Size** (d). The input and output of a sub-layer are of d dimensions. Besides, most of the hidden states of a sub-layer are d-dimensional vectors. In general, d can be roughly viewed as the width of the network.
- **Number of Heads** (n_{head}). In self-attention sub-layers, one needs to specify the number of heads used in multi-head self-attention. A larger number of heads allows the model to jointly attend to information from a greater number of representation subspaces. In practical systems, we often set $n_{\mathrm{head}} \geq 4$.
- **FFN Hidden Size** (d_{ffn}). The size of the hidden layer of the FFNs used in Transformers is typically larger than d. For example, a typical setting is $d_{\mathrm{ffn}} = 4d$. For larger Transformers, such as recent large models, d_{ffn} may be set to a very large value.
- **Model Depth** (L). Using deep networks is an effective way to improve the expressive power of Transformers. For BERT models, L is typically set to 12 or 24. However, networks with even greater depth are also feasible and can be applied for further enhancements.

Different settings of these hyper-parameters lead to different model sizes. Two widely used configurations are $\mathrm{BERT}_{\mathrm{base}}$ and $\mathrm{BERT}_{\mathrm{large}}$:

- $\mathrm{BERT}_{\mathrm{base}}$: $d = 768$, $L = 12$, $n_{\mathrm{head}} = 12$, total number of parameters $= 110\mathrm{M}$.
- $\mathrm{BERT}_{\mathrm{large}}$: $d = 1{,}024$, $L = 24$, $n_{\mathrm{head}} = 16$, total number of parameters $= 340\mathrm{M}$.

The training of BERT models follows the standard procedure for Transformers. Training larger models such as $\mathrm{BERT}_{\mathrm{large}}$ requires more training effort and time. This is a common problem for pre-training, especially when a model is trained on a very large amount of data. In practice, training efficiency is often a major concern. A common practice is to first train a BERT model on relatively short sequences for a large number of training steps, and then continue training it on full-length sequences for the remaining training steps.

7.3.2 More Training and Larger Models

BERT is a milestone in NLP and has sparked extensive subsequent work. One direction is to scale up pre-training, which generally involves two approaches: increasing the amount of training data and increasing the number of model parameters.

RoBERTa, an extension of the standard BERT model, is an example of such efforts [Liu et al., 2019]. It introduces two major improvements. First, simply using more training data and more compute can improve BERT models without the need to change the model architecture. Second, removing the NSP loss does not degrade performance on downstream tasks if the training is scaled up. These findings suggest exploring a general direction of pre-training: we can continue to improve pre-training by scaling it up on simple pre-training tasks.

A second approach to improving BERT models is to increase the number of model parameters. For example, in He et al. [2021]'s work, a 1.5 billion-parameter BERT-like model is built by increasing both the model depth and hidden size. However, scaling up BERT and various other pre-trained models introduces new challenges in training, for example, training very large models often becomes unstable and difficult to converge. This makes the problem more complicated, and requires careful consideration of various aspects, such as model architecture, parallel computation, and parameter initialization. In another example, Shoeybi et al. [2019] successfully trained a 3.9 billion-parameter BERT-like model, where hundreds of GPUs were used to manage the increased computational demands.

7.3.3 More Efficient Models

Compared to its predecessors, BERT is a relatively large model for the time it was proposed. This increase in model size results in larger memory requirements and slower inference. Developing smaller and faster BERT models is part of the broader challenge of building efficient Transformers, which has been extensively discussed in Chapter 6. However, a deeper exploration of this topic is beyond the scope of this section. Here, we instead focus on a few efficient variants of BERT.

Several research threads are of particular interest for developing efficient BERT models. First, work on knowledge distillation, such as training a student model using the outputs of a well-trained teacher model, demonstrates that more compact models can be obtained by transferring knowledge from larger ones. Given that BERT models are multi-layer networks with several different types of layers, knowledge distillation can be applied at different levels of representation. For example, beyond distilling knowledge from the output layers, it is also possible to incorporate a training loss that measures the difference between the outputs of hidden layers for the teacher and student models [Sun et al., 2020b; Jiao et al., 2020]. Indeed, knowledge distillation has been one of the most widely used techniques for learning small pre-trained models.

Second, conventional model compression methods can be directly applied to BERT. One common approach is to apply general-purpose pruning methods to Transformer encoders [Gale et al., 2019]. This generally involves removing entire layers [Fan et al., 2019] or a certain percentage of parameters in the networks [Sanh et al., 2020; Chen et al., 2020b]. Pruning is also applicable to multi-head attention models. For example, Michel et al. [2019] show that removing some of the heads does not significantly degrade performance of BERT models, but speeds up the inference of these models. Another approach to compressing BERT models is quantization [Shen et al., 2020b]. By representing model parameters as low-precision numbers, we can compress the models greatly. While this method is not specific to BERT models, it proves effective for large Transformer-based architectures.

Third, considering that BERT models are relatively deep and large networks, another thread of research uses dynamic networks to adapt these models for efficient inference. An idea in this paradigm is to dynamically choose the layers for processing a token, for example, in depth-adaptive models we exit at some optimal depth and thus skip the rest of the layers in the layer stack [Xin et al., 2020; Zhou et al., 2020]. Similarly, we can develop length-adaptive models in which the length of the input sequence is dynamically adjusted. For example, we can skip some of the tokens in the input sequence so that the model can reduce computational load on less important tokens, enhancing overall efficiency.

Fourth, it is also possible to share parameters across layers to reduce the size of BERT models. A simple way to do this is to share the parameters of a whole Transformer layer across the layer stack [Dehghani et al., 2018; Lan et al., 2020]. In addition to the reduced number of parameters, this enables reuse of the same layer in a multi-layer Transformer network, leading to savings of memory footprint at test time.

7.3.4 Multilingual Models

The initial BERT model was primarily focused on English. Soon after this model was proposed, it was extended to many languages. One simple way to do this is to develop a separate model for each language. Another approach, which has become more popular in recent work on large language models, is to train multilingual models directly on data from all the languages. Accordingly, **multilingual BERT** (**mBERT**) models were developed by training on text from 104 languages [6]. The primary difference from monolingual BERT models is that mBERT models use larger vocabularies to cover tokens from multiple languages. As a result, the representations of tokens from different languages are mapped into the same space, allowing for the sharing of knowledge across languages.

One of the most important applications of multilingual pre-trained models is cross-lingual transfer learning. In a zero-shot cross-lingual setting, a model is fine-tuned on a task using data from a source language and then applied to the same task in a target language without any target-language training data. In cross-lingual text classification, for example, we can fine-tune a multilingual pre-trained model on annotated English documents and then directly use the fine-tuned model to classify Chinese documents.

An improvement to multilingual pre-trained models like mBERT is to introduce bilingual data into pre-training. Rather than training solely on monolingual data from multiple languages, bilingual training explicitly models the relationship between tokens in two languages. The resulting model will have cross-lingual transfer abilities, and thus can be easily adapted to different languages. Lampple and Conneau [2019] propose an approach to pre-training **cross-**

[6] https://github.com/google-research/bert/

lingual language models (**XLMs**). In their work, a cross-lingual language model can be trained using either causal language modeling or masked language modeling. For masked language modeling pre-training, the model is treated as an encoder. The training objective is the same as BERT: we maximize the probabilities of some randomly selected tokens which are either masked, replaced with random tokens, or kept unchanged in the input. If we consider bilingual data in pre-training, we sample a pair of aligned sentences each time. Then, the two sentences are packed together to form a single sequence used for training. For example, consider an English-Chinese sentence pair

鲸鱼 是 哺乳 动物 。 $\leftrightarrow$ Whales are mammals .

We can pack them to obtain a sequence, like this

[CLS] 鲸鱼 是 哺乳 动物 。 [SEP] Whales are mammals . [SEP]

We then select a certain percentage of the tokens and replace them with [MASK].

[CLS] [MASK] 是 [MASK] 动物 。 [SEP] Whales [MASK] [MASK] . [SEP]

The goal of pre-training is to maximize the product of the probabilities of the masked tokens given the above sequence. By performing training in this way, the model can learn to represent both the English and Chinese sequences, as well as to capture the correspondences between tokens in the two languages. For example, predicting the Chinese token 鲸鱼 may require the information from the English token *Whales*. Aligning the representations of the two languages essentially transforms the model into a "translation" model. This training objective is therefore also called **translation language modeling**. Figure 7.7 shows an illustration of this approach.

A benefit of multilingual pre-trained models is their inherent capability of handling code-switching. In NLP and linguistics, code-switching refers to switching among languages in a text. For example, the following is a mixed language text containing both Chinese and English:

周末 我们 打算 去 做 hiking ， 你 想 一起 来 吗 ？
(We plan to go hiking this weekend, would you like to join us?)

For multilingual pre-trained models, we do not need to identify whether a token is Chinese or English. Instead, every token is just an entry of the shared vocabulary. This can be imagined as creating a "new" language that encompasses all the languages we want to process.

The result of multilingual pre-training is influenced by several factors. Given that the model architecture is fixed, one needs to specify many configurations, such as the size of the shared vocabulary, the number (or percentage) of samples in each language, and the size of the model. Conneau et al. [2020] point out several interesting issues regarding large-scale multilingual pre-training for XLM-like models. First, as the number of supported languages increases,

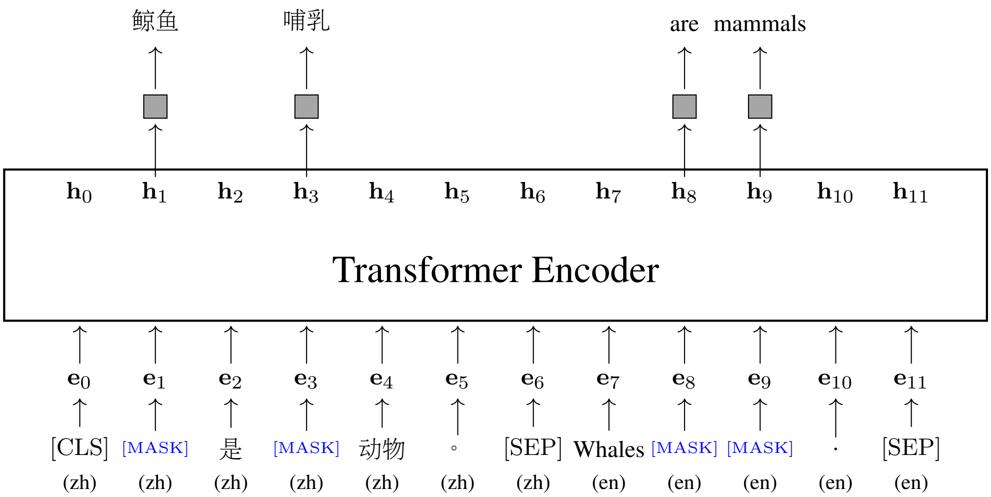


Figure 7.7: An illustration of translation language modeling (TLM). For ease of understanding, we present a simplified example where all selected tokens are masked. The model is trained to predict these masked tokens. Because the input sequence contains parallel tokens from two languages, predicting a masked token in one language allows the model to attend to context in the other language, thereby explicitly forcing cross-lingual alignment. In the original XLM formulation by Lample and Conneau [2019], the input embedding (i.e., $\mathbf{e}_i$) is the sum of the token embedding, positional embedding, and a specific language embedding. This requires that each token be assigned a language label, allowing the model to distinguish between languages. However, in modern multilingual pre-training architectures (like XLM-R), explicitly specifying the language is unnecessary, as a shared vocabulary is sufficient. Furthermore, relying on language embeddings makes it difficult to handle code-switching. Therefore, we assume here that all token representations are language-agnostic.

a larger model is needed to handle these languages. Second, a larger shared vocabulary is helpful for modeling the increased diversity in languages. Third, low-resource languages significantly benefit from cross-lingual transfer, particularly when linguistically similar high-resource languages are included in the pre-training corpus. However, a phenomenon known as the **curse of multilinguality** (or capacity interference) emerges when a model of fixed capacity is trained on too many languages: the addition of more languages eventually forces parameter competition, degrading the performance on high-resource languages. Therefore, in practical systems, there is a trade-off between the number of supported languages and the model capacity.

7.4 Applying BERT Models

Once a BERT model is pre-trained, it can then be used to solve NLP problems. But BERT models are not immediately ready for downstream tasks. In general, additional fine-tuning is required to adapt them. As a first step, we need a predictor to align the model output with the task of interest. Let $\mathrm{BERT}_{\hat{\theta}}(\cdot)$ be a BERT model with pre-trained parameters $\hat{\theta}$, and

$\mathrm{Predict}_{\omega}(\cdot)$ be a prediction network with parameters ω. By integrating the prediction network with the output of the BERT model, we develop a model to tackle the downstream tasks. This model can be expressed as:

$$\mathbf{y} = \mathrm{Predict}_{\omega}(\mathrm{BERT}_{\hat{\theta}}(\mathbf{x})), \tag{7.22}$$

where $\mathbf{x}$ is the input and $\mathbf{y}$ is the output that fits the problem. For example, in classification problems, the model outputs a probability distribution over labels.

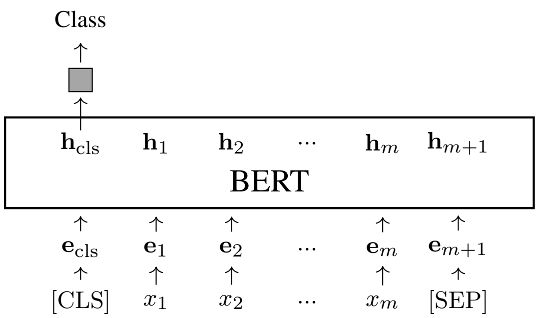
Then, we collect a set of labeled samples $\mathcal{D}$, and fine-tune the model by

$$(\tilde{\omega}, \tilde{\theta}) = \underset{\omega, \hat{\theta}^{+}}{\arg\min} \sum_{(\mathbf{x}, \mathbf{y}_{\mathrm{gold}}) \in \mathcal{D}} \mathrm{Loss}(\mathbf{y}_{\omega, \hat{\theta}^{+}}, \mathbf{y}_{\mathrm{gold}}), \tag{7.23}$$

where $(\mathbf{x}, \mathbf{y}_{\mathrm{gold}})$ represents a tuple of an input and its corresponding output. The notation of this equation seems a bit complicated, but the training/tuning process is standard. We optimize the model by minimizing the loss over the training data. The outcome is the optimized parameters $\tilde{\omega}$ and $\tilde{\theta}$. The optimization starts with the pre-trained parameters $\hat{\theta}$. Here we use $\hat{\theta}^{+}$ to indicate that the parameters are initialized with $\hat{\theta}$, and use $\mathbf{y}_{\omega, \hat{\theta}^{+}}$ to denote the model output computed using the parameters ω and $\hat{\theta}^{+}$.

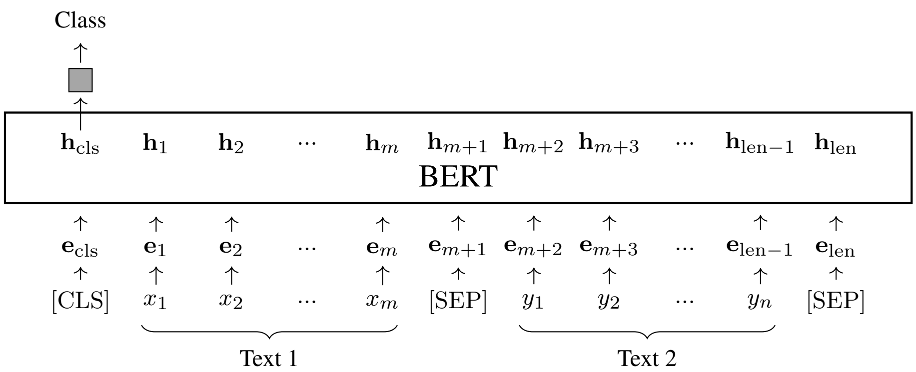
With the fine-tuned parameters $\tilde{\omega}$ and $\tilde{\theta}$, we can apply the model $\mathrm{Predict}_{\tilde{\omega}}(\mathrm{BERT}_{\tilde{\theta}}(\cdot))$ to new data of the same tasks for which the model was fine-tuned. The form of the downstream tasks determines the input and output formats of the model, as well as the architecture of the prediction network. In the following we list some tasks for which BERT models are well suited.

- **Classification** (Single Text). One of the most widely used applications of BERT models is text classification. In this task, a BERT model receives a sequence of tokens and encodes it as a sequence of vectors. The first output vector $\mathbf{h}_{\mathrm{cls}}$ (or $\mathbf{h}_0$) is typically used as the representation of the entire text. The prediction network takes $\mathbf{h}_{\mathrm{cls}}$ as input to produce a distribution of labels. Let $[\mathrm{CLS}]x_1 x_2 \cdots x_m$ be the input sequence. See below for an illustration of BERT-based text classification.



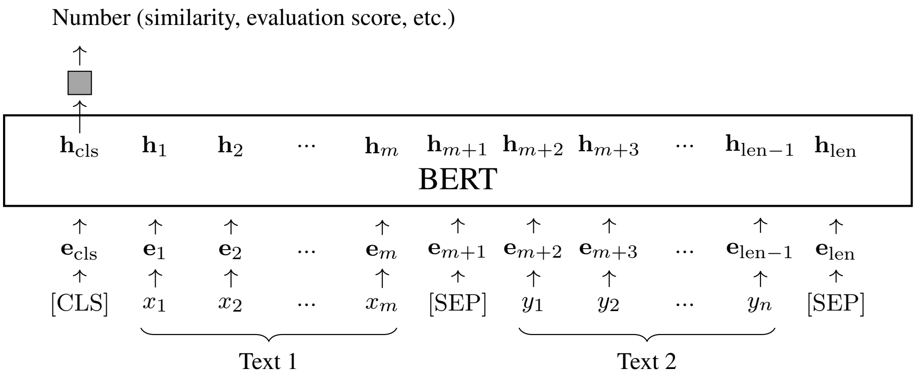
Here the gray box denotes the prediction network. Many NLP problems can be categorized as text classification tasks, and there have been several text classification benchmarks for evaluating pre-trained models. For example, we can classify texts by their grammatical correctness (grammaticality) or emotional tone (sentiment) [Socher et al., 2013; Warstadt et al., 2019]. Note that the prediction network could be any classification model, such as a deep neural network or a more traditional classification model. The entire model can then be trained or fine-tuned in the manner of a standard classification model. For example, the prediction network can be simply a softmax layer and the model parameters can be optimized by maximizing the probabilities of the correct labels.

- **Classification** (Pair of Texts). Classification can also be performed on a pair of texts. Suppose we have two texts, $x_1 \cdots x_m$ and $y_1 \cdots y_n$. We can concatenate these texts to form a single sequence with length len. Then, we predict a label for this combined text sequence based on the $\mathbf{h}_{\text{cls}}$ vector, as follows:



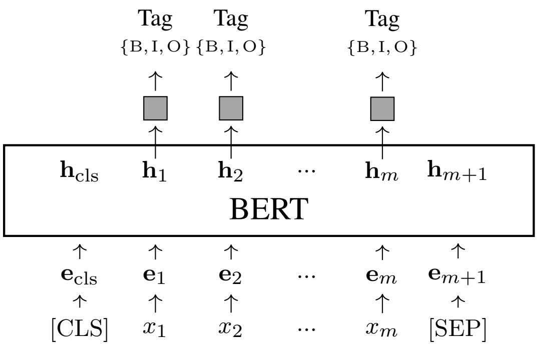
where $\text{len} = n + m + 2$. Text pair classification covers several problems, including semantic equivalence judgement (determine whether two texts are semantically equivalent) [Dolan and Brockett, 2005], text entailment judgement (determine whether a hypothesis can be logically inferred or entailed from a premise) [Bentivogli and Giampiccolo, 2011; Williams et al., 2018], grounded commonsense inference (determine whether an event is likely to happen given its context) [Zellers et al., 2018], and question-answering inference (determine whether an answer corresponds to a given question).

- **Regression**. Instead of generating a label distribution, we can have the prediction network output a real-valued score. For example, by adding a sigmoid layer to the prediction network, the system can be employed to compute the similarity between two given sentences. The architecture is the same as that of BERT-based classification systems, with only the output layer changed.



For training or fine-tuning, we can minimize the regression loss of the model output as usual.

- **Sequence Labeling**. Sequence labeling is a machine learning approach applicable to a wide range of NLP problems. This approach assigns a label to each token in an input sequence, and some linguistic annotations can then be derived from this sequence of labels. An example of sequence labeling in NLP is part-of-speech (POS) tagging. We label each word in a sentence with its corresponding POS tag. Another example is named entity recognition (NER) in which we label each word with an NER tag, and named entities are identified using these tags. See below for an illustration of the model architecture for NER.



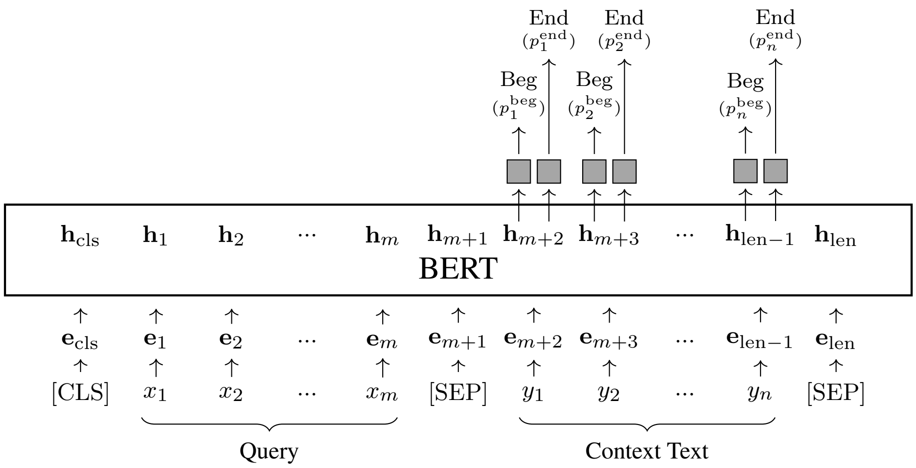
Here $\{\mathrm{B,I,O}\}$ is the tag set of NER. For example, B-ORG means the beginning of an organization, I-ORG means the word is inside an organization, and O means the word does not belong to any named entity. This NER model can output a distribution over the tag set at each position, denoted as $\mathbf{p}_i$. The training or fine-tuning of the model can be performed over these distributions $\{\mathbf{p}_1, \cdots, \mathbf{p}_m\}$. For example, suppose $p_i(\mathrm{tag}_i)$ is the probability of the correct tag at position i. The training loss can be defined to be the negative log-likelihood

$$\mathrm{Loss} = -\frac{1}{m}\sum_{i=1}^{m} \log p_i(\mathrm{tag}_i). \tag{7.24}$$

Finding the best label sequence given a trained NER model is a well-studied problem in

NLP. This is often achieved via dynamic programming, which, in the context of path finding over a lattice, has linear complexity [Huang, 2009].

- **Span Prediction**. Some NLP tasks require predicting a span in a text. A common example is reading comprehension. In this task, we are given a query $x_1 \cdots x_m$ and a context text $y_1 \cdots y_n$. The goal is to identify a continuous span in $y_1 \cdots y_n$ that best answers the query. This problem can be framed as a sequence labeling-like task in which we predict a label for each y_j to indicate the beginning or ending of the span. Following Seo et al. [2017], we add two networks on top of the BERT output for y_j: one for generating the probability of y_j being the beginning of the span (denoted by p_j^{beg}), and one for generating the probability of y_j being the ending of the span (denoted by p_j^{end}). The resulting model architecture is shown as follows:



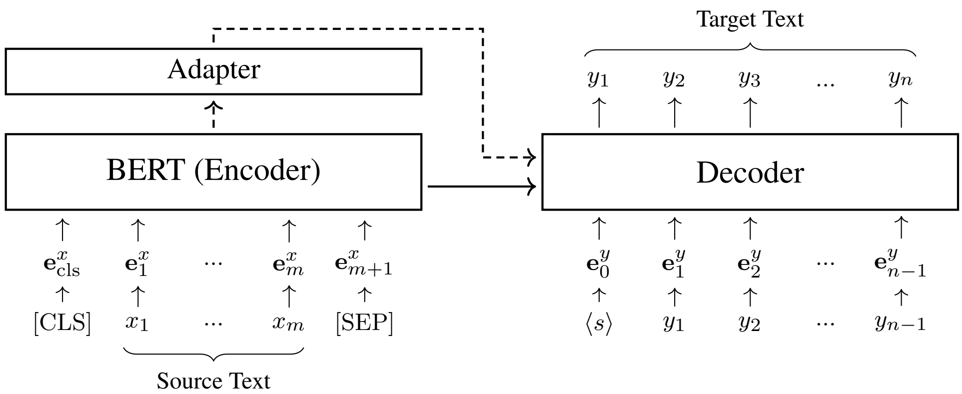
We pack the query and context text together to obtain the input sequence. The prediction networks are only applied to outputs for the context text, generating the probabilities p_j^{beg} and p_j^{end} at each position. The loss can be computed by summing the negative log likelihoods of the two models across the entire context text.

$$\text{Loss} \quad = \quad -\frac{1}{n}\sum_{j=1}^{n}\left(\log p_j^{\text{beg}} + \log p_j^{\text{end}}\right). \tag{7.25}$$

At test time, we search for the best span by

$$(\hat{j}_1, \hat{j}_2) \quad = \quad \underset{1 \le j_1 \le j_2 \le n}{\arg\max} \left(\log p_{j_1}^{\text{beg}} + \log p_{j_2}^{\text{end}}\right). \tag{7.26}$$

- **Encoding for Encoder-Decoder Models**. While our focus in this section has been primarily on language understanding problems, it is worth noting that BERT models can be applied to a broader range of NLP tasks. In fact, BERT models can be used in all the scenarios where we need to encode a piece of text. One application that we have not mentioned is text generation which includes a range of tasks such as machine translation, summarization, question answering, and dialogue generation. These tasks can be formulated as sequence-to-sequence problems: we use an encoder to represent the source text, and a decoder to generate the corresponding target text. A straightforward method to apply BERT models is to consider them as encoders. Before fine-tuning, we can initialize the parameters of the encoder with those from a pre-trained BERT model. Then, the encoder-decoder model can be fine-tuned on pairs of texts as usual. The following shows the architecture of a neural machine translation system where a BERT model is applied on the source side.



Here $x_1 \cdots x_m$ denotes the source sequence, $y_1 \cdots y_n$ denotes the target sequence, $\mathbf{e}^x_1 \cdots \mathbf{e}^x_m$ denotes the embedding sequence of $x_1 \cdots x_m$, and $\mathbf{e}^y_1 \cdots \mathbf{e}^y_n$ denotes the embedding sequence of $y_1 \cdots y_n$. The adapter, which is optional, maps the output of the BERT model to the form that is better suited to the decoder.

Fine-tuning BERT models is a complicated engineering problem, influenced by many factors, such as the amount of fine-tuning data, the model size, and the optimizer used in fine-tuning. In general, we wish to fine-tune these models sufficiently so that they can perform well in the downstream tasks. However, fine-tuning BERT models for specific tasks may lead to overfitting, which in turn reduces their ability to generalize to other tasks. For example, suppose we have a BERT model that performs well on a particular task. If we then fine-tune it for new tasks, this may decrease its performance on the original task. This problem is related to the **catastrophic forgetting** problem in continual training, where a neural network forgets previously learned information when updated on new samples. In practical applications, a common way to alleviate catastrophic forgetting is to add some old data into fine-tuning and train the model with more diverse data. Also, one may use methods specialized to catastrophic forgetting, such as experience replay [Rolnick et al., 2019] and elastic weight consolidation [Kirkpatrick et al., 2017]. The interested reader can refer to some surveys for more detailed discussions of this issue in continual learning [Parisi et al., 2019; Wang et al., 2023a;f].

7.5 Summary

In this chapter we have introduced the general concept of pre-training in NLP. In particular, we have discussed self-supervised pre-training and its application to encoder-only, decoder-only, and encoder-decoder architectures. Moreover, we have presented and compared a variety of pre-training tasks for these architectures. As an example, we used BERT to illustrate how sequence models are pre-trained via masked language modeling and applied to different downstream tasks.

Recent years have shown remarkable progress in NLP, led by the large-scale use of self-supervised pre-training. Significant advances have been made across many tasks, not only in NLP but also in computer vision and other areas of AI. One idea behind these advances is that a significant amount of knowledge about the world can be learned by simply training these AI systems on huge amounts of unlabeled data. For example, a language model can learn some general knowledge of a language by repeatedly predicting masked words in large-scale text. As a result, this pre-trained language model can serve as a foundation model, which can be easily adapted to address specific downstream NLP tasks. This paradigm shift in NLP has enabled the development of powerful systems for language understanding, generation, and reasoning [Manning, 2022]. However, it is important to recognize that we are still in the early stages of creating truly intelligent systems, and there is a long way to go. Nevertheless, large-scale pre-training has opened a door to intelligent systems that researchers have long aspired to develop, though several key research areas remain open for exploration, such as learning efficiently from limited data and acquiring complex reasoning and planning abilities.

Note that this chapter is mostly introductory and cannot cover all aspects of pre-training. For example, there are many methods to fine-tune a pre-trained model, offering different ways to adapt the model to diverse scenarios. Moreover, large language models, which are considered one of the most significant achievements in AI in recent years, are not covered in this chapter. We leave the discussion of these topics to the following chapters.

Chapter 8

Generative Models

One of the most significant advances in NLP in recent years is the development of large language models (LLMs). This breakthrough has facilitated the creation of systems that can understand and generate natural language at a near-human level. These systems have even demonstrated reasoning capabilities, a long-standing challenge in AI. With these achievements, NLP has made substantial progress and entered a new era of research in which increasingly complex problems are being addressed, such as building conversational agents that can interact with humans fluently.

The concept of language modeling dates back to early experiments conducted by Shannon [1951]. In his work, Shannon proposed a language model to estimate how predictable English text is — *how well the next letter of a text can be predicted when the preceding N letters are known*. Although Shannon's experiments were preliminary, the fundamental goals and methods of language modeling have remained largely unchanged over the decades. For many years, particularly before 2010, the dominant approach to language modeling was the n-gram approach [Jurafsky and Martin, 2008]. In n-gram language modeling, we estimate the probability of a word given its preceding $n-1$ words, and thus the probability of a sequence can be approximated by the product of n-gram probabilities. These probabilities are typically estimated from smoothed relative counts of n-grams in text. Despite its simplicity, it has been widely used in NLP. For example, the success of modern statistical speech recognition and machine translation systems has largely depended on the use of n-gram language models [Jelinek, 1998; Koehn, 2010].

Applying neural networks to language modeling has long been an active area of research, but a major breakthrough came with the advancement of deep learning techniques. A widely cited study is the work of Bengio et al. [2003a], in which n-gram probabilities are modeled using a feed-forward network trained end-to-end. An important outcome of this model is the distributed representations of words, known as word embeddings. Rather than representing words as discrete variables, word embeddings map words to low-dimensional real-valued vectors, making it possible to capture semantic relationships among words and word n-grams in a continuous representation space. As a result, language models are no longer burdened by the curse of dimensionality, but can represent an exponential number of n-grams using a compact neural model.

The idea of learning word representations through neural language models inspired subsequent research on representation learning in NLP. However, this approach did not attract significant interest from the NLP community in the first few years after its proposal. Around 2012, however, significant progress was made in learning word embeddings from large-scale text via simple word prediction tasks. Several methods, including Word2Vec, were proposed to effectively learn such embeddings, which were then successfully applied in a variety of NLP systems [Mikolov et al., 2013a;c]. As a result of these advances, researchers began exploring sequence representation learning using more powerful language models, such as LSTM-based models [Sutskever et al., 2014; Peters et al., 2018]. Subsequently, progress and interest in sequence representation grew rapidly after the Transformer architecture was proposed. With the rise of the Transformer architecture, the concept of language modeling was generalized to encompass models that learn to predict words under different paradigms. Many powerful Transformer-based models were pre-trained using these word prediction tasks and were successfully applied to a variety of downstream tasks [Devlin et al., 2019].

Indeed, training language models on large-scale data has led NLP research to exciting times. While language modeling has long been seen as a foundational technique with no direct link to the broader goals of artificial intelligence, it has led to the emergence of intelligent systems that can acquire general knowledge through repeated word prediction. Recent research demonstrates that a single, well-trained LLM can perform a wide range of tasks and generalize to new tasks with minimal adaptation [Bubeck et al., 2023]. This represents a step toward more advanced forms of artificial intelligence and inspires further exploration into developing more powerful language models as a class of foundation models.

In this chapter, we discuss the basic concepts of generative LLMs. For simplicity, we use LLMs to refer to generative models such as GPT, although in a broader sense the term may also include non-generative models such as BERT. We begin with a general introduction to LLMs, including the key steps involved in building such models. We then discuss two key scaling challenges for LLMs: how LLMs are trained at scale and how they can be improved to handle long-context inputs. Finally, we conclude with a brief summary.

8.1 A Brief Introduction to LLMs

In this section, we introduce the basic concepts of LLMs required for the remainder of this chapter and subsequent chapters. We will use the terms *word* and *token* interchangeably. Both refer to the basic units used in language modeling, though their original meanings differ slightly.

Before presenting the technical details, let us first consider how language models work. The core objective of language modeling is to estimate the probability of a token sequence. Let $\{x_0, x_1, \cdots, x_m\}$ be a sequence of tokens, where x_0 is the start symbol $\langle s \rangle$ (or $\langle \text{SOS} \rangle$)[1].

[1] In some models such as BERT, a special token like [CLS] is prepended, although it serves a different purpose from the start symbol in autoregressive models.

The probability of this sequence can be factorized using the chain rule

$$\begin{aligned}
\Pr(x_0,\cdots,x_m) &= \Pr(x_0)\cdot\Pr(x_1|x_0)\cdot\Pr(x_2|x_0,x_1)\cdots\Pr(x_m|x_0,\cdots,x_{m-1}) \\
&= \prod_{i=0}^{m}\Pr(x_i|x_0,\cdots,x_{i-1}),, \qquad (8.1)
\end{aligned}$$

or alternatively in a logarithmic form

$$\log\Pr(x_0,\cdots,x_m) = \sum_{i=0}^{m}\log\Pr(x_i|x_0,\cdots,x_{i-1}).. \qquad (8.2)$$

Here, $\Pr(x_i|x_0,\cdots,x_{i-1})$ is the probability of the token x_i given all its preceding tokens $\{x_0,\cdots,x_{i-1}\}$ [2]. In the era of deep learning, a typical approach to language modeling is to estimate these probabilities using deep neural networks. Neural networks trained for this task receive a context sequence $x_0,\cdots,x_{i-1}$ and output a probability distribution over the vocabulary V (denoted by $\Pr(\cdot|x_0,\cdots,x_{i-1})$). The probability $\Pr(x_i|x_0,\cdots,x_{i-1})$ corresponds to the entry associated with token x_i in this distribution.

When applying a trained language model, a common task is to find the most likely next token given its context. This can be formulated as

$$\hat{x}_i = \arg\max_{x_i\in V}\Pr(x_i|x_0,\cdots,x_{i-1}). \qquad (8.3)$$

We can perform this token prediction iteratively to generate text sequentially: at each step, we predict the most likely token $\hat{x}_i$, and then append it to the context to predict the subsequent token $\hat{x}_{i+1}$. This results in an autoregressive, left-to-right generation process that implements Eqs. (8.1) and (8.2). To illustrate, consider the generation of three tokens given the prefix "$\langle s\rangle\ a$", as shown in Table 8.1. Next, we will discuss how LLMs are constructed, trained, and applied in practice.

8.1.1 Decoder-only Transformers

As is standard practice, the input to a language model is a sequence of tokens (denoted by $\{x_0,\cdots,x_{m-1}\}$). At each step, a new token is generated and appended to the sequence for the next prediction. To do this, the language model outputs a probability distribution $\Pr(\cdot|x_0,\cdots,x_{i-1})$ at each position i, and the next token is generated according to this distribution (e.g., by sampling or greedy decoding). The model is trained by maximizing the log-likelihood $\sum_{i=1}^{m}\log\Pr(x_i|x_0,\cdots,x_{i-1})$[3].

Here, we focus on the decoder-only Transformer architecture, as it is one of the most widely adopted model architectures for LLMs. The input sequence of tokens is represented by a sequence of d_e-dimensional vectors $\{\mathbf{e}_0,\cdots,\mathbf{e}_{m-1}\}$. Each vector $\mathbf{e}_i$ is the sum of the

[2] We assume that when $i=0$, $\Pr(x_i|x_0,\cdots,x_{i-1}) = \Pr(x_0) = 1$. Hence, $\Pr(x_0,\cdots,x_m) = \Pr(x_0)\Pr(x_1,\cdots,x_m|x_0) = \Pr(x_1,\cdots,x_m|x_0)$.

[3] Note that $\sum_{i=1}^{m}\log\Pr(x_i|x_0,\cdots,x_{i-1}) = \sum_{i=0}^{m}\log\Pr(x_i|x_0,\cdots,x_{i-1})$ since $\log\Pr(x_0)=0$.

Context	Predict	Decision Rule	Sequence Probability
$\langle s \rangle\ \ a$	b	$\arg\max_{x_2 \in V} \Pr(x_2 \| \langle s \rangle\ a)$	$\Pr(\langle s \rangle) \cdot \Pr(a \| \langle s \rangle) \cdot$ $\Pr(b \| \langle s \rangle\ a)$
$\langle s \rangle\ \ a\ \ b$	c	$\arg\max_{x_3 \in V} \Pr(x_3 \| \langle s \rangle\ a\ b)$	$\Pr(\langle s \rangle) \cdot \Pr(a \| \langle s \rangle) \cdot \Pr(b \| \langle s \rangle\ a) \cdot$ $\Pr(c \| \langle s \rangle\ a\ b)$
$\langle s \rangle\ \ a\ \ b\ \ c$	d	$\arg\max_{x_4 \in V} \Pr(x_4 \| \langle s \rangle\ a\ b\ c)$	$\Pr(\langle s \rangle) \cdot \Pr(a \| \langle s \rangle) \cdot \Pr(b \| \langle s \rangle\ a) \cdot$ $\Pr(c \| \langle s \rangle\ a\ b) \cdot$ $\Pr(d \| \langle s \rangle\ a\ b\ c)$

Table 8.1: Illustration of generating the three tokens $b\ \ c\ \ d$ given the prefix $\langle s \rangle\ \ a$ via a language model. In each step, the model picks a token x_i from V so that $\Pr(x_i | x_0, \cdots, x_{i-1})$ is maximized. This token is then appended to the end of the context sequence. In the next step, we repeat the same process, but based on the new context.

token embedding for x_i and the positional embedding for position i. The core of the model consists of a stack of Transformer blocks (or layers). Each Transformer block consists of two sub-layers: a self-attention sub-layer and an FFN sub-layer. These sub-layers can be defined using the post-norm architecture

$$\text{output} \quad = \quad \text{LNorm}(F(\text{input}) + \text{input}), \tag{8.4}$$

or the pre-norm architecture

$$\text{output} \quad = \quad F(\text{LNorm}(\text{input})) + \text{input}, \tag{8.5}$$

where input and output denote the input and output, both being $m \times d$ matrices. The i-th rows of both input and output can be interpreted as the contextual representations of the i-th token in the sequence.

$F(\cdot)$ is the core function of a sub-layer. For FFN sub-layers, $F(\cdot)$ is a 2-layer FFN. For self-attention sub-layers, $F(\cdot)$ is a multi-head self-attention function. In general, self-attention is expressed in the form of QKV attention

$$\text{Att}_{\text{qkv}}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \quad = \quad \text{Softmax}(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}} + \mathbf{Mask})\mathbf{V}, \tag{8.6}$$

where $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V} \in \mathbb{R}^{m \times d}$ are the queries, keys, and values, respectively. It is important to note that only previous tokens are considered when predicting a token. So a masking variable $\mathbf{Mask} \in \mathbb{R}^{m \times m}$ is incorporated into self-attention to achieve this. The entry (i, k) of $\mathbf{Mask}$ has a value of 0 if $i \geq k$, and a value of $-\infty$ otherwise.

Given a representation $\mathbf{H} \in \mathbb{R}^{m \times d}$, the multi-head self-attention function can be defined as

$$F(\mathbf{H}) \quad = \quad \text{Merge}(\text{head}_1, \cdots, \text{head}_\tau)\mathbf{W}^{\text{head}}, \tag{8.7}$$

where $\text{Merge}(\cdot)$ represents a concatenation of its inputs, and $\mathbf{W}^{\text{head}} \in \mathbb{R}^{d \times d}$ represents a

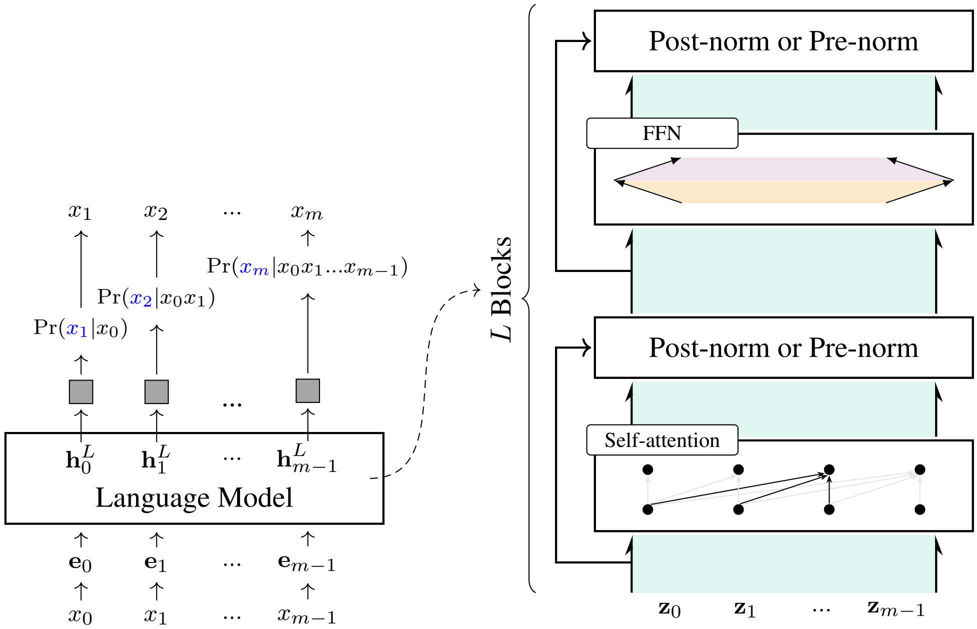


Figure 8.1: The Transformer-decoder architecture for language modeling. The central components are L stacked Transformer blocks, each comprising a self-attention sub-layer and an FFN sub-layer. To prevent the model from accessing the right-context, a masking variable is incorporated into self-attention. The output layer uses a Softmax function to generate a probability distribution for the next token, given the sequence of previous tokens. During inference, the model takes the previously predicted token to predict the next one, repeating this process until the end of the sequence is reached. $\{\mathbf{z}_0, \cdots, \mathbf{z}_{m-1}\}$ denote the inputs of a Transformer block, and $\{\mathbf{h}_0^L, \cdots, \mathbf{h}_{m-1}^L\}$ denote the outputs of the last Transformer block.

parameter matrix. head_j is the output of QKV attention on a subspace of the representation.

$$\mathrm{head}_j = \mathrm{Att}_{\mathrm{qkv}}(\mathbf{Q}^{[j]}, \mathbf{K}^{[j]}, \mathbf{V}^{[j]}). \tag{8.8}$$

Here, $\mathbf{Q}^{[j]}$, $\mathbf{K}^{[j]}$, and $\mathbf{V}^{[j]}$ are the queries, keys, and values projected onto the j-th subspace via linear transformations

$$\mathbf{Q}^{[j]} = \mathbf{H}\mathbf{W}_j^q \tag{8.9}$$

$$\mathbf{K}^{[j]} = \mathbf{H}\mathbf{W}_j^k \tag{8.10}$$

$$\mathbf{V}^{[j]} = \mathbf{H}\mathbf{W}_j^v, \tag{8.11}$$

where $\mathbf{W}_j^q$, $\mathbf{W}_j^k$, and $\mathbf{W}_j^v \in \mathbb{R}^{d \times \frac{d}{\tau}}$ are the parameter matrices of the transformations.

Suppose we have L Transformer blocks. A Softmax layer is built on top of the output of the last block. The Softmax layer outputs a sequence of m distributions over the vocabulary, like this

$$\begin{bmatrix} \Pr(\cdot|x_0) \\ \Pr(\cdot|x_0, x_1) \\ \vdots \\ \Pr(\cdot|x_0, \cdots, x_{m-1}) \end{bmatrix} = \mathrm{Softmax}(\mathbf{H}^L \mathbf{W}^o), \tag{8.12}$$

where $\mathbf{H}^L$ is the output of the last Transformer block, and $\mathbf{W}^o \in \mathbb{R}^{d\times|V|}$ is the parameter matrix.

Figure 8.1 shows the Transformer architecture for language modeling. Inference with this model follows an autoregressive process. Each time the language model takes the prefix $x_0, \cdots, x_{i-1}$ as input and predicts a token x_i that maximizes the probability $\Pr(x_i|x_0, \cdots, x_{i-1})$. It is important to note that, despite different implementation details, many LLMs share the same architecture described above. These models are called large because both their depth and width are significant. Table 8.2 shows the model sizes for a few LLMs, as well as their model configurations.

8.1.2 Training LLMs

Now suppose we are given a training set $\mathcal{D}$ comprising K sequences. The log-likelihood of each sequence $\mathbf{x} = x_0 \cdots x_m$ in $\mathcal{D}$ can be computed using a language model

$$\mathcal{L}_\theta(\mathbf{x}) = \sum_{i=1}^{m} \log \Pr_\theta(x_i|x_0, \cdots, x_{i-1}). \tag{8.13}$$

where the subscript θ in $\mathcal{L}_\theta(\cdot)$ and $\Pr_\theta(\cdot)$ denotes the parameters of the language model. Then, the objective of maximum likelihood training is defined as

$$\hat{\theta} = \arg\max_{\theta} \sum_{\mathbf{x}\in\mathcal{D}} \mathcal{L}_\theta(\mathbf{x}). \tag{8.14}$$

Training Transformer-based language models with the above objective is commonly viewed as a standard optimization process for neural networks. This can be achieved using gradient descent algorithms, which are widely supported by off-the-shelf deep learning toolkits. Somewhat surprisingly, model performance has continued to improve as language models evolved into more computationally intensive architectures and were trained on larger datasets [Kaplan et al., 2020]. These successes have motivated NLP researchers to continually scale up both training data and model size to build increasingly powerful language models.

However, as language models become larger, we confront new training challenges, which significantly change the problem compared to training relatively small models. One of these challenges arises from the need for large-scale distributed systems to manage the data, model parameters, and training routines. Developing and maintaining such systems requires substantial effort in both software and hardware engineering, alongside deep learning expertise. A related issue is that scaling up training demands massive computational resources to com-

LLM	# of Parameters	Depth L	Width d	# of Heads (Q/KV)
GPT-1 [Radford et al., 2018]	0.117B	12	768	12/12
GPT-2 [Radford et al., 2019]	1.5B	48	1,600	25/25
GPT-3 [Brown et al., 2020]	175B	96	12,288	96/96
LLaMA2 [Touvron et al., 2023b]	7B	32	4,096	32/32
	13B	40	5,120	40/40
	70B	80	8,192	64/64
LLaMA3/3.1 [Dubey et al., 2024]	8B	32	4,096	32/8
	70B	80	8,192	64/8
	405B	126	16,384	128/8
Gemma2 [Team et al., 2024]	2B	26	2,304	8/4
	9B	42	3,584	16/8
	37B	46	4,608	32/16
Qwen2.5 [Yang et al., 2024]	0.5B	24	896	14/2
	7B	28	3,584	28/4
	72B	80	8,192	64/8
DeepSeek-V3 [Liu et al., 2024a]	671B	61	7,168	128/128
Falcon [Penedo et al., 2023]	7B	32	4,544	71/71
	40B	60	8,192	128/128
	180B	80	14,848	232/232
Mistral [Jiang et al., 2023a]	7B	32	4,096	32/32

Table 8.2: Comparison of some LLMs in terms of model size, model depth, model width, and number of heads (a/b means a heads for queries and b heads for both keys and values).

plete the process within a reasonable timeframe. For instance, training an LLM with tens of billions of parameters from scratch generally requires hundreds or thousands of GPUs. This requirement drastically increases training costs, especially considering the many experimental iterations needed during model development. Furthermore, from an optimization perspective, the training process can become highly unstable when scaling up network depth and parameter count. As a result, architectural modifications are often required to adapt LLMs for large-scale training. In Section 8.2 we will discuss these issues in more detail.

8.1.3 Fine-tuning LLMs

Once we have pre-trained an LLM, we can apply it to perform various NLP tasks. Traditionally, language models were used as components of other systems. For example, they were widely applied to score translations in statistical machine translation systems. By contrast, in generative AI, LLMs are considered complete systems and are employed to address NLP problems by leveraging their generative nature. A common approach is to describe the target task in text and then prompt the LLM to generate text based on this description. This is a standard text generation task where the model continues or completes the text starting from a given context.

More formally, let $\mathbf{x} = x_0 \cdots x_m$ denote a sequence of tokens representing the context provided by users, and $\mathbf{y} = y_1 \cdots y_n$ denote a token sequence following the context. The inference process of an LLM can be defined as finding the most likely sequence $\mathbf{y}$ given $\mathbf{x}$:

$$\begin{aligned} \hat{\mathbf{y}} &= \arg\max_{\mathbf{y}} \log \Pr(\mathbf{y}|\mathbf{x}) \\ &= \arg\max_{\mathbf{y}} \sum_{i=1}^{n} \log \Pr(y_i|x_0, \cdots, x_m, y_1, \cdots, y_{i-1}). \end{aligned} \tag{8.15}$$

Here, $\sum_{i=1}^{n} \log \Pr(y_i|x_0, \cdots, x_m, y_1, \cdots, y_{i-1})$ is mathematically equivalent to the right-hand side of Eq. (8.2). It models the log probability of predicting tokens starting from position $m+1$, rather than position 0. Throughout this chapter and subsequent ones, we will employ separate variables $\mathbf{x}$ and $\mathbf{y}$ to distinguish the input and output of an LLM, though they can be viewed as sub-sequences belonging to the same continuous sequence. By adopting this notation, we can see that the above equation closely resembles the formulations used in other NLP text generation models, such as neural machine translation architectures.

To illustrate how LLMs are applied, consider the problem of determining the grammaticality of a given sentence. We can define a template like this

> {*sentence*}
> Question: Is this sentence grammatically correct?
> Answer: ____

Here __ represents the text we intend to generate. {*sentence*} is a placeholder variable that will be replaced by the actual sentence provided by users. For example, suppose we have a sentence "*John seems happy today.*". We can replace the {*sentence*} in the template with this sentence to construct an input to the language model

> John seems happy today.
> Question: Is this sentence grammatically correct?
> Answer: ____

To perform the task, the language model is given the context $\mathbf{x}$ ="John seems happy today .\n Question : Is this sentence grammatically correct?\n Answer :"[4]. It then generates the following text as the answer, based on the context. For example, the language model may output "Yes" (i.e., $\mathbf{y}$ = "Yes") if this text is the one with the highest probability given this context.

[4]\n is a special character used for line breaks.

Likewise, we can define more templates to address other tasks. For example, we can translate an English sentence into Chinese using the following template

> {*sentence*}
> Question: What is the Chinese translation of this English sentence?
> Answer: ____

or using an instruction-like template

> {*sentence*}
> Translate this sentence from English into Chinese.
> ____

or using a code-like template.

> [src-lang] = English [tgt-lang] = Chinese [input] = {*sentence*}
> [output] = ____

The above templates provide a simple but effective method to "prompt" a single LLM to perform various tasks without adapting the structure of the model. However, this approach requires that the LLM can recognize and follow the instructions or questions. One way to do this is to incorporate training samples with instructions and their corresponding responses into the pre-training dataset. While this method is straightforward, building and training LLMs from scratch is computationally expensive. Moreover, making instruction-following data effective for pre-training requires a significant amount of such data, but collecting large-scale labeled data for all tasks of interest is very difficult.

A second method, which has been a de facto standard in recent research, is to adapt LLMs via fine-tuning. As such, the token prediction ability learned in the pre-training phase can be generalized to accomplish new tasks. The idea behind fine-tuning is that some general knowledge of language has been acquired during pre-training, but we need a mechanism to activate this knowledge for applying it to new tasks. To achieve this, we can slightly fine-tune the model parameters using instruction-following data. This approach is called **instruction fine-tuning**.

An instruction fine-tuning sample, which is represented by a sequence of tokens, can be seen as a tuple consisting of an input and the desired output. Here, the input includes instructions, system information (or system prefix), and any other user-provided information[5]. To illustrate, consider the following examples (blue text = input and underlined text = output).

[5] System information refers to a sequence of tokens added at the beginning of an input in order to guide the behavior of an LLM, such as, *you are a helpful assistant and should not output toxic content*.

> Read the following sentence, and answer whether it is grammatically correct.
> LLMs are powerful models but are expensive to build.
>
> Yes

> LLMs are powerful models but are expensive to build.
> Does this sentence make sense grammatically?
> Answer Option
> - Yes
> - No
>
> Yes

> Yes or No? Yes = the following sentence is grammatically correct. No = it contains grammatical errors.
> LLMs are powerful models but are expensive to build.
>
> Yes

All these samples describe the same binary classification task, but with different instructions. To increase the diversity in the fine-tuning data and make LLMs generalize, we can define more tasks using instructions.

> Write a short story about three characters.
> Character 1: A boy whose name is Andy.
> Character 2: Andy's dog.
> Character 3: Andy's friend, Jane.
>
> In the heart of a bustling city, Andy, his loyal dog Rocket, and his friend Jane discovered a hidden garden beneath the glow of twilight. Surrounded by ancient trees and whispering winds, they shared dreams and secrets, forging an unbreakable bond. As the night deepened, they made a pact under the stars to always find magic in the mundane. Together, they realized that true adventure lay in the moments spent with friends, and every day was a treasure waiting to be discovered.

Show me a recipe for making ice cream.

Ingredients:
2 cups heavy cream; 1 cup milk; 3/4 cup sugar; 2 tsp vanilla extract;
Pinch of salt.
Steps:
1. Whisk all ingredients until sugar dissolves.
2. Chill mixture for 2 hours in the fridge.
3. Freeze in a shallow dish, stirring every 30 minutes, 3-4 times.
4. Freeze solid for 2 hours.
5. Serve your homemade ice cream.

If you buy 5 apples and each apple costs $1.20, how much do you spend in total?

$6.00

Write a Python program to calculate the sum of squares of the following numbers.
1 , 2 , 10 , -9 , 78

```
numbers = [1,2,10,-9,78]
sum_of_squares = sum(x**2 for x in numbers)
print(sum_of_squares)
```

To acquire instruction-following abilities, a certain amount of fine-tuning data is required. This data may include diverse instructions and possible responses. It has been found that scaling the number of fine-tuning tasks is beneficial for improving the performance of LLMs [Chung et al., 2022]. Note that although more fine-tuning data is favorable, the amount of this data is generally orders of magnitude smaller than that of the pre-training data. For example, LLMs can be fine-tuned with tens or hundreds of thousands of samples, or even fewer if these samples are of high quality [Zhou et al., 2023a; Chen et al., 2023b], whereas pre-training such models may require billions or trillions of tokens, resulting in significantly larger computational demands and longer training times [Touvron et al., 2023a].

It is also worth noting that we should not expect the fine-tuning data to cover all the downstream tasks to which we intend to apply LLMs. A common understanding of how the pre-training + fine-tuning approach works is that LLMs have acquired knowledge for understanding instructions and generating responses in the pre-training phase. However, these abilities are not fully activated until we introduce some form of supervision. The general instruction-following behavior emerges as we fine-tune the models with a relatively small amount of labeled data. As a result, we can achieve some level of **zero-shot learning**: the fine-tuned models can handle new tasks that they have not been explicitly trained or fine-tuned for [Sanh et al., 2022; Wei et al., 2022a]. This zero-shot learning ability distinguishes generative LLMs from earlier pre-trained models like BERT, which are primarily fine-tuned for specific tasks.

Once an instruction dataset is curated, the fine-tuning process itself is relatively straightforward. It can be viewed as standard neural network training akin to pre-training, but executed on a much smaller training dataset. Let $\mathcal{D}_{\text{tune}}$ be the fine-tuning dataset and $\hat{\theta}$ be the model parameters optimized via pre-training. We can modify Eq. (8.14) to obtain the objective of fine-tuning

$$\tilde{\theta} = \arg\max_{\hat{\theta}+} \sum_{\text{sample} \in \mathcal{D}_{\text{tune}}} \mathcal{L}_{\hat{\theta}+}(\text{sample}). \tag{8.16}$$

Here $\tilde{\theta}$ denotes the optimal parameters. The use of notation $\hat{\theta}^{+}$ means that the fine-tuning starts with the pre-trained parameters $\hat{\theta}$.

For each sample $\in \mathcal{D}_{\text{tune}}$, we divide it into an input segment $\mathbf{x}_{\text{sample}}$ and an output segment $\mathbf{y}_{\text{sample}}$, that is,

$$\text{sample} = [\mathbf{x}_{\text{sample}}, \mathbf{y}_{\text{sample}}]. \tag{8.17}$$

We then define the loss function to be

$$\mathcal{L}_{\hat{\theta}+}(\text{sample}) = -\log \Pr_{\hat{\theta}+}(\mathbf{y}_{\text{sample}} | \mathbf{x}_{\text{sample}}). \tag{8.18}$$

In other words, we compute the loss over the sub-sequence $\mathbf{y}_{\text{sample}}$, rather than the entire sequence. In a practical implementation of back-propagation for this equation, the sequence $[\mathbf{x}_{\text{sample}}, \mathbf{y}_{\text{sample}}]$ is constructed in the forward pass as usual. However, in the backward pass, error gradients are propagated back only through the parts of the network that correspond to $\mathbf{y}_{\text{sample}}$, leaving the rest of the network unchanged. As an example, consider a sequence

$$\underbrace{\langle s \rangle \text{ Square this number . 2 .}}_{\text{Context (Input)}} \ \underbrace{\text{The result is 4 .}}_{\text{Prediction (Output)}}$$

The loss is calculated and back propagated only for The result is 4 ..

Instruction fine-tuning also requires substantial engineering work. In order to achieve satisfactory results, one may experiment with different settings of the learning rate, batch size, number of fine-tuning steps, and so on. This typically requires many fine-tuning runs and evaluations. The cost and experimental effort of fine-tuning remain critical and should not be overlooked, though they are much lower than those of the pre-training phase.

While we focus on instruction fine-tuning for an illustrative example here, fine-tuning techniques play an important role in developing various LLMs and are more widely used. Examples include fine-tuning LLMs as chatbots using dialog data, and adapting these models to handle very long sequences. The wide application of fine-tuning has led researchers to improve these techniques, such as designing more efficient fine-tuning algorithms. While the research on fine-tuning is fruitful, in this section we just provide an overview of the key steps. We will see more detailed discussions on this topic in the following chapters.

8.1.4 Aligning LLMs with the World

Instruction fine-tuning provides a simple way to adapt LLMs to tasks that can be well defined. This can broadly be viewed as an **alignment** problem. Here, alignment refers to the process of guiding LLMs to behave in ways that accord with human values and goals. The guidance can come from labeled data, human feedback, or any other form of human preferences. For example, we want LLMs not only to be accurate in following instructions, but also to be unbiased, truthful, and harmless. So we need to supervise the models towards human values and expectations. A typical example is that when we ask an LLM how to build a weapon, it may provide a list of key steps to do so if it is not carefully aligned. However, a responsible model should recognize and avoid responding to requests for harmful or illegal information. Alignment in this case is crucial for ensuring that LLMs act responsibly and adhere to ethical guidelines.

A concept closely related to alignment is AI safety. An ultimate goal of AI research is to build intelligent systems that are safe and socially beneficial. To achieve this, we must ensure these systems remain robust, secure, and controllable under a wide range of real-world conditions, including instances of misuse or adversarial attacks. For LLMs, safety can be significantly enhanced through proper alignment with human guidance, achieved via human-annotated data and dynamic interactions with users during deployment.

Alignment is difficult as human values and expectations are diverse and shifting. Sometimes, it is hard to describe precisely what humans want, unless we see the response of LLMs to user requests. This makes alignment no longer a problem of tuning LLMs on predefined tasks, but a bigger problem of training them through interactions with the real world.

As a result of the concerns with controlling AI systems, there has been a surge in research on the alignment issue for LLMs. Typically, two alignment steps are employed after LLMs are pre-trained on large-scale unlabeled data.

- **Supervised Fine-tuning** (**SFT**). This involves continuing the training of pre-trained LLMs on new, task-oriented, labelled data. A commonly used SFT technique is instruction fine-tuning. As described in the previous subsection, by learning from instruction-response annotated data, LLMs can align with the intended behaviors for following instructions, thereby becoming capable of performing various instruction-described tasks. Supervised fine-tuning can be seen as following the pre-training + fine-tuning paradigm, and offers a relatively straightforward method to adapt LLMs.
- **Learning from Human Feedback**. After pre-training and supervised fine-tuning, an LLM can respond to user requests when prompted appropriately. However, the model may still generate content that is non-factual, biased, or harmful. To better align the LLM with human values, an effective approach is to learn directly from human feedback. For example, given an instruction and the model's generated response, human experts

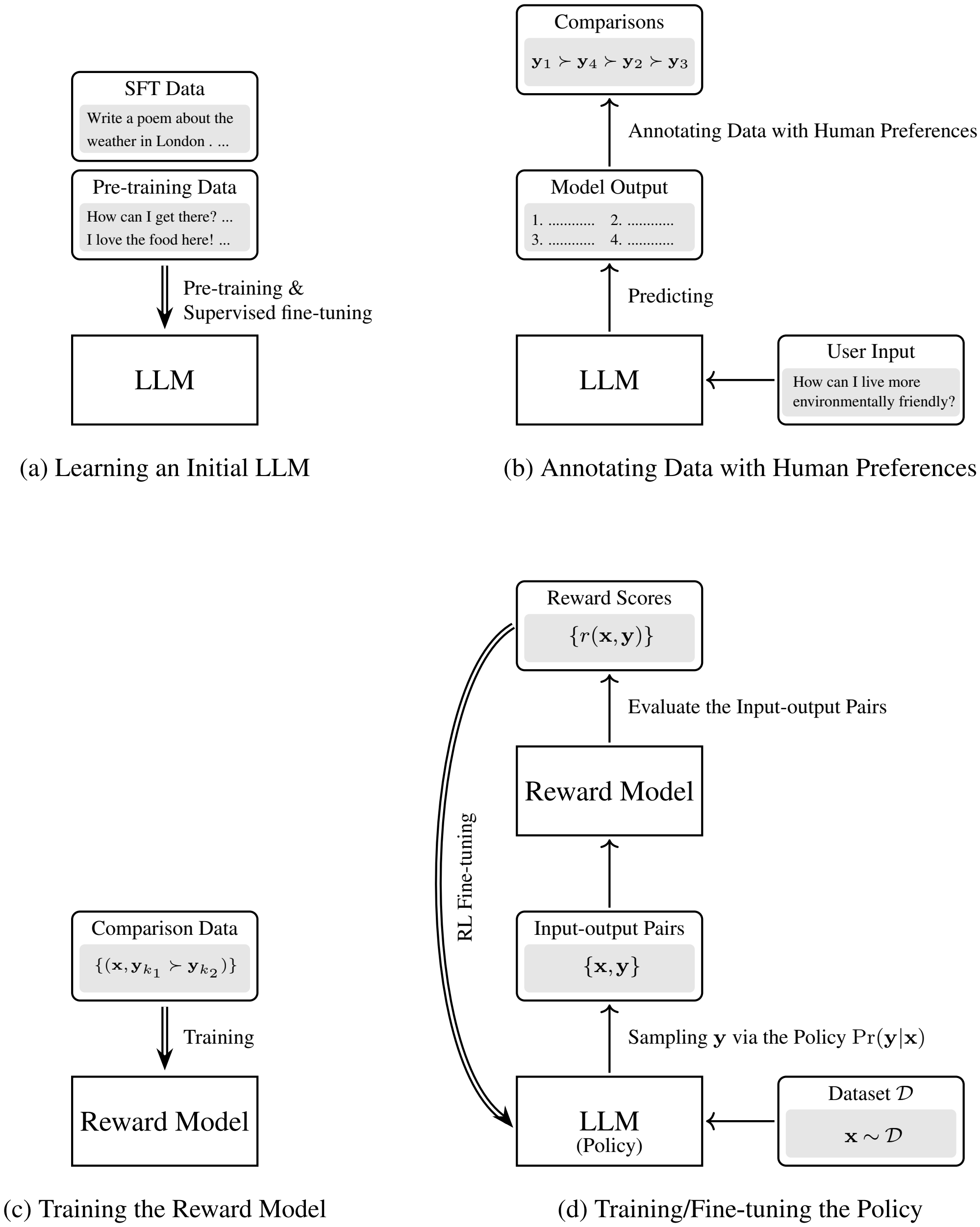


Figure 8.2: An overview of RLHF. There are 4 key steps involved: a) training an initial LLM (i.e., policy) using pre-training and supervised fine-tuning; b) collecting human preference data by ranking the outputs of the LLM; c) training a reward model using the ranking results; d) RL fine-tuning of the policy based on the reward model. Double line arrows mean training or fine-tuning.

are asked to evaluate how well the response aligns with their preferences and safety guidelines. This feedback is then used to further train the LLM for superior alignment.

A typical method for learning from human feedback is to consider it as a reinforcement learning (RL) problem, known as **reinforcement learning from human feedback** (**RLHF**) [Ouyang et al., 2022]. The RLHF method was initially proposed to address general sequential decision-making problems [Christiano et al., 2017], and was later successfully employed in the development of the GPT series models [Stiennon et al., 2020]. As a reinforcement learning approach, the goal of RLHF is to learn a policy by maximizing some reward from the environment. Specifically, RLHF consists of two main components:

- **Agent**. The agent, also referred to as an LM agent, is the LLM we intend to optimize. This agent operates by interacting with its environment: it receives text from the environment and produces text responses. The policy of this agent is the conditional probability distribution defined by the LLM, denoted as $\Pr(\mathbf{y}|\mathbf{x})$.
- **Reward Model**. The reward model acts as a proxy for the human environment. Every time the agent produces an output sequence, the reward model evaluates it and assigns a numerical score (i.e., the reward). This score explicitly informs the agent about the quality and alignment of its output.

In RLHF, two distinct learning phases are executed: 1) reward model learning, which involves training the reward model using human feedback over the outputs of the agent, and 2) policy learning, which uses reinforcement learning algorithms to optimize the policy under the guidance of the trained reward model. The key steps involved in RLHF can be outlined as follows:

- Build an initial policy using pre-training and instruction fine-tuning.
- Use the policy to generate multiple outputs for each input, and then collect human feedback on these outputs (e.g., comparisons of the outputs).
- Learn a reward model from the human feedback.
- Fine-tune the policy with the supervision from the reward model.

Figure 8.2 shows an overview of RLHF. Given that this section serves only as a brief introduction to concepts of LLMs, a detailed discussion of RLHF techniques will not be included. We instead illustrate the basic ideas behind RLHF using a simple example.

Suppose we have trained an LLM via pre-training and instruction fine-tuning. This LLM is deployed to respond to requests from users. For example, a user may input

> How can I live a more environmentally friendly life?

We use the LLM to generate 4 different outputs (denoted by $\{\mathbf{y}_1, \cdots, \mathbf{y}_4\}$) by sampling from the output distribution

Output 1 ($\mathbf{y}_1$):	Consider switching to an electric vehicle or bicycle instead of traditional cars to reduce carbon emissions and protect our planet.
Output 2 ($\mathbf{y}_2$):	Adopt a minimalist lifestyle. Own fewer possessions to reduce consumption and the environmental impact of manufacturing and disposal.
Output 3 ($\mathbf{y}_3$):	Go off-grid. Generate your own renewable energy and collect rainwater to become completely self-sufficient and reduce reliance on non-renewable resources.
Output 4 ($\mathbf{y}_4$):	Support local farm products to reduce the carbon footprint of transporting food, while enjoying fresh, healthy food.

We then ask annotators to evaluate these outputs. One straightforward way is to assign a rating score to each output. In this case, the reward model learning problem can be framed as a task of training a regression model. But giving numerical scores to LLM outputs is not an easy task for annotators. It is usually difficult to design an annotation standard that all annotators can agree on and easily follow. An alternative method, which is more popular in the development of LLMs, is to rank these outputs. For example, a possible ranking of the above outputs is

$$\mathbf{y}_1 \succ \mathbf{y}_4 \succ \mathbf{y}_2 \succ \mathbf{y}_3$$

A reward model is then trained using this ranking result. In general, a reward model in RLHF is a language model that shares the same architecture as the target LLM, but with a smaller model size. Given the input $\mathbf{x}$ and output $\mathbf{y}_k$, we concatenate them to form a sequence $\mathrm{seq}_k = [\mathbf{x}, \mathbf{y}_k]$. This sequence is processed from left to right using forced decoding. Since each position can only access its left context in language modeling, the output of the top-most Transformer layer at the first position cannot be used as the representation of the sequence. Instead, a special symbol (e.g., $\langle /s \rangle$) is added to the end of the sequence, and the corresponding output of the Transformer layer stack is considered as the representation of the entire sequence. An output layer, such as a linear transformation layer, is built on top of this representation to generate the reward, denoted by $R(\mathrm{seq}_k)$ or $R(\mathbf{x}, \mathbf{y}_k)$.

We train this reward model using ranking loss. For example, a pair-wise ranking loss function can be written in the form

$$\mathrm{Loss}_\omega(\mathcal{D}_r) \quad = \quad -\mathbb{E}_{(\mathbf{x},\mathbf{y}_{k_1},\mathbf{y}_{k_2})\sim\mathcal{D}_r} \log(\mathrm{Sigmoid}(R_\omega(\mathbf{x},\mathbf{y}_{k_1}) - R_\omega(\mathbf{x},\mathbf{y}_{k_2}))), \quad (8.19)$$

where ω represents the parameters of the reward model, and $\mathcal{D}_r$ represents a set of tuples of an input and a pair of outputs. $(\mathbf{x}, \mathbf{y}_{k_1}, \mathbf{y}_{k_2}) \sim \mathcal{D}_r$ is a sampling operation which draws a sample $(\mathbf{x}, \mathbf{y}_{k_1}, \mathbf{y}_{k_2})$ from $\mathcal{D}_r$ with some probability. As an example, suppose we first draw a model input $\mathbf{x}$ with a uniform distribution and then draw a pair of model outputs with a probability of

$\mathbf{y}_{k_1} \succ \mathbf{y}_{k_2}$ given $\mathbf{x}$ (denoted by $\Pr(\mathbf{y}_{k_1} \succ \mathbf{y}_{k_2}|\mathbf{x})$). The corresponding loss function is given by

$$\begin{aligned} & \mathrm{Loss}_{\omega}(\mathcal{D}_r) \\ = & -\sum \Pr(\mathbf{x}) \cdot \Pr(\mathbf{y}_{k_1} \succ \mathbf{y}_{k_2}|\mathbf{x}) \cdot \log(\mathrm{Sigmoid}(R_{\omega}(\mathbf{x},\mathbf{y}_{k_1}) - R_{\omega}(\mathbf{x},\mathbf{y}_{k_2}))) \\ = & -\frac{1}{K}\sum \Pr(\mathbf{y}_{k_1} \succ \mathbf{y}_{k_2}|\mathbf{x}) \cdot \log(\mathrm{Sigmoid}(R_{\omega}(\mathbf{x},\mathbf{y}_{k_1}) - R_{\omega}(\mathbf{x},\mathbf{y}_{k_2}))), \end{aligned} \tag{8.20}$$

where K represents the number of model inputs involved in sampling. While the form of these functions may seem complex, their idea is simple: we penalize the model if the predicted ranking of two outputs differs from the human-labeled ranking. By contrast, the model receives a bonus, if the predicted ranking matches the human-labeled ranking.

We can train the reward model by minimizing the above ranking loss

$$\hat{\omega} = \arg\min_{\omega} \mathrm{Loss}_{\omega}(\mathcal{D}_r). \tag{8.21}$$

The resulting model $R_{\hat{\omega}}(\cdot)$ can be employed to evaluate any given pair of input and output. Note that although the reward model is trained using a ranking-based objective, it is used for scoring. This allows it to provide continuous supervision signals, which is very beneficial for training other models.

We now turn to the policy learning problem. A commonly adopted objective is to maximize the reward on a set of input-output pairs. Following an analogous form of Eq. (8.16), we obtain a simple training objective for RL fine-tuning

$$\tilde{\theta} = \arg\max_{\hat{\theta}+} \mathbb{E}_{(\mathbf{x},\mathbf{y}_{\hat{\theta}+})\sim\mathcal{D}_{\mathrm{rlft}}} R_{\hat{\omega}}(\mathbf{x},\mathbf{y}_{\hat{\theta}+}), \tag{8.22}$$

where the optimal parameters $\tilde{\theta}$ are obtained by fine-tuning the pre-trained parameters $\hat{\theta}$. $\mathcal{D}_{\mathrm{rlft}}$ is the RL fine-tuning dataset. For each sample $(\mathbf{x},\mathbf{y}_{\hat{\theta}+})$, $\mathbf{x}$ is sampled from a prepared dataset of input sequences, and $\mathbf{y}_{\hat{\theta}+}$ is sampled from the distribution $\Pr_{\hat{\theta}+}(\mathbf{y}|\mathbf{x})$ given by the policy.

In practice, more advanced reinforcement learning algorithms, such as **proximal policy optimization** (**PPO**), are often used for achieving more stable training, as well as better performance. We leave the detailed discussion of reinforcement learning algorithms to the following parts of this book where RLHF is extensively used for alignment.

An interesting question arises here: why not consider learning from human preferences as a standard supervised learning problem? This question is closely related to our aforementioned discussion on the difficulty of data annotation. Often, describing human values and goals is challenging, and it is even more difficult for humans to provide outputs that are well aligned. As an alternative, annotating the preferences of a given list of model outputs offers a simpler task. By doing so, we can create a model that understands human preferences, which can then be used as a reward model for training policies. From the perspective of machine learning, RLHF is particularly useful for scenarios where the desired behavior of an agent is difficult to demonstrate but can be easily recognized by humans. Another advantage of RLHF is its ability to explore the sample space. By employing sampling techniques, models trained with reinforcement learning can venture beyond the annotated data set to explore additional samples. This exploratory ability allows RLHF to discover potentially beneficial policies that are not immediately apparent from the labeled data alone.

8.1.5 Prompting LLMs

We have so far shown that LLMs can be used to perform various tasks by giving them appropriate prompts. There are no restrictions on these prompts, which can include any information we wish to provide to LLMs, such as natural language instructions and conversational context. Since this approach requires no additional training or tuning, adapting LLMs becomes highly efficient once they are developed. This fundamentally shifts the paradigm of NLP: we no longer need to develop task-specific systems but can instead use a single, well-trained LLM to perform different tasks simply by prompting it. As a result, an appealing aspect of LLMs emerges: users can easily create customized systems by designing their own prompts for LLMs. Given the important role played by prompting in LLMs, **prompt engineering** has become a very active area of research in NLP.

The term *prompt* is used in many different ways in the literature. In this chapter, this term refers to the entire input to LLMs, and so we use the terms *prompt* and *model input* interchangeably. Before discussing prompting further, let us first see a few examples where the prompts are more complex than those presented in the previous subsections. Note that this subsection is not aimed at writing high-quality prompts but rather at highlighting some interesting issues in prompting LLMs.

One of the popular ways to use LLMs is to assign them a "role" in generating responses. For example, LLMs can act as a psychologist when answering questions.

> Please explain what delayed gratification is.
>
> Note, you are a researcher with a deep background in psychology. Your own field of study is developmental psychology in children, with a focus on aspects of control and intelligence. You are very familiar with the concept of delayed gratification and have applied it to your own research. You have published papers on delayed gratification and conducted related experiments.
>
> Delayed gratification is the process of resisting an immediate reward in anticipation of receiving a more valuable reward in the future. It is a significant concept in developmental psychology, especially in relation to how children learn self-control and develop cognitive abilities. Research shows that the ability to delay gratification is associated with positive outcomes in academic performance, social competence, and emotional well-being. Studies and experiments in this area highlight the importance of cultivating this skill early in life to support long-term success and resilience.

Another example is the use of LLMs in detecting and correcting errors such as syntactic or semantic mistakes in text. For an LLM which is trained on both code and natural language data, we may use it for code debugging[6].

Fix the bugs in this C language program.

```
#include <stdio.h>
int main() {
    printg("Hello, World!")
    return 0;
}
```

1. The function name 'printg' is incorrect. It should be 'printf'.
2. There is a missing semicolon at the end of the 'printf' function call.

Here is the corrected version of the program:

```
#include <stdio.h>
int main() {
    printf("Hello, World!");
    return 0;
}
```

Also, LLMs can be fine-tuned to act as a conversation partner [7].

Hi! I'm an assistant. How can I help you?

Who won the FIFA World Cup 2022?

Argentina won the FIFA World Cup 2022.

Where was it held?

The 2022 FIFA World Cup was held in Qatar.

How many times has Argentina won the World Cup?

Argentina has won the FIFA World Cup three times.

...

These examples and previous ones have shown that appropriate responses can be generated via prompts involving clear instructions and questions. However, when problem solving requires knowledge that is not explicitly specified, LLMs may make mistakes, even when the instructions are clear and precise. A family of challenging tasks for LLMs involves arithmetic reasoning and commonsense reasoning. For example, we can ask an LLM to solve primary school math problems presented in natural language.

[6] In this example, the code is not tokenized for easier reading.

[7] To fine-tune an LLM for multi-turn dialogue, one needs to consider conversation history in the context for predicting the response in the current round of conversation. This makes the actual prompt used in response generation relatively longer than that used in single-turn dialogue.

> Jack has 7 apples. He ate 2 of them for dinner, but then his mom gave him 5 more apples. The next day, Jack gave 3 apples to his friend John. How many apples does Jack have left in the end?
>
> The answer is 10.

The correct answer should be 7, so the model output is incorrect.

One approach to addressing such issues is to incorporate learning into prompts, called **in-context learning** or (**ICL**). The idea of ICL is to demonstrate the ways to solve problems in prompts, and condition predictions on these demonstrations. Here is an example where a similar problem and the corresponding answer are presented in the prompt (green = demonstrations).

> Tom has 12 marbles. He wins 7 more marbles in a game with his friend but then loses 5 marbles the next day. His brother gives him another 3 marbles as a gift. How many marbles does Tom have now?
>
> The answer is 17.
>
> Jack has 7 apples. He ate 2 of them for dinner, but then his mom gave him 5 more apples. The next day, Jack gave 3 apples to his friend John. How many apples does Jack have left in the end?
>
> The answer is 12.

But the LLM still made mistakes this time. A reason for this might be that solving math problems does not only involve problem-answer mappings but also, to a larger extent, the underlying logical inference in multiple steps. A method to improve the inference abilities of LLMs is **chain-of-thought prompting** (**COT prompting**) [Wei et al., 2022c]. In COT prompting, we decompose complex reasoning problems into multiple problem-solving intermediate steps. These steps are demonstrated in prompts so that LLMs can be prompted to learn to mimic the underlying thought processes in solving the problems. COT prompting has been found to be very useful for achieving good results on challenging tasks, such as the GSM8K mathematical reasoning benchmark.

Consider the above demonstration example *Tom has 12 marbles. He wins 7 more marbles in a game with his friend but then loses 5 marbles the next day. His brother gives him another 3 marbles as a gift. How many marbles does Tom have now?* The corresponding logical reasoning steps can be structured:

1. Tom starts with 12 marbles.
2. He wins 7 more marbles in a game, so we add those to his initial count: 12 marbles + 7 marbles = 19 marbles.
3. The next day, Tom loses 5 marbles, so we subtract those from his current total: 19 marbles - 5 marbles = 14 marbles.

4. His brother gives him 3 more marbles, so we add those to his current total: 14 marbles + 3 marbles = 17 marbles.
5. After all these events, Tom has 17 marbles.

We can add these reasoning steps into the prompt, and get the correct answer.

Tom has 12 marbles. He wins 7 more marbles in a game with his friend but then loses 5 marbles the next day. His brother gives him another 3 marbles as a gift. How many marbles does Tom have now?

Tom initially has 12 marbles. After winning a game, he gains 7 more, increasing his total to 19 marbles. He loses 5 marbles the next day, bringing his total down to 14 marbles. His brother gifts him 3 more marbles, increasing his total to 17 marbles. Therefore, Tom now has 17 marbles. So the answer is 17.

Jack has 7 apples. He ate 2 of them for dinner, but then his mom gave him 5 more apples. The next day, Jack gave 3 apples to his friend John. How many apples does Jack have left in the end?

<u>Jack starts with 7 apples. He eats 2 apples for dinner, so we subtract 2 from 7, leaving him with 5 apples. His mom gives him 5 more apples, adding 5 to his current total. Now he has 5 + 5 = 10 apples. The next day, Jack gives 3 apples to his friend John. Subtracting 3 from 10, Jack is left with 7 apples. Therefore, Jack has 7 apples left. So the answer is 7.</u>

This approach is also called **one-shot COT prompting**. Likewise, we can add more demonstrations to have a **few-shot COT prompting** approach.

Furthermore, a specialized variant called **zero-shot COT** introduces trigger phrases that encourage LLMs to think step-by-step, without explicitly providing any intermediate reasoning examples in the prompt [Kojima et al., 2022]. Interestingly, simply appending the sentence "Let's think step by step." to the end of a prompt is often sufficient to induce the LLM to generate internal reasoning steps and arrive at the correct final answer.

Jack has 7 apples. He ate 2 of them for dinner, but then his mom gave him 5 more apples. The next day, Jack gave 3 apples to his friend John. How many apples does Jack have left in the end?
Let's think step by step.

1. Initial Quantity: Jack starts with 7 apples.
2. After Dinner: He eats 2 apples, so $7-2=5$ apples remain.
3. His Mom Gives More: His mom gives him 5 more apples, so $5+5=10$ apples are now with him.
4. Giving to John: The next day, Jack gives 3 apples to his friend John, so $10-3=7$ apples are left.

In the end, Jack has 7 apples left.

Zero-shot, one-shot, and few-shot learning are common concepts in the area of in-context learning for LLMs and are not restricted to COT prompting. Broadly speaking, any prompting that involves only simple instructions without any demonstrations can be considered a form of zero-shot learning. This zero-shot learning ability emerges as LLMs are pre-trained and/or fine-tuned. Also, one-shot and few-shot learning methods are more often considered when LLMs do not acquire the corresponding zero-shot learning ability. These methods are therefore important for in-context learning when addressing new tasks. Examples include those for performing various NLP tasks by demonstrating task-formatted samples. See the following examples for sentiment sentence classification and phrase translation via few-shot learning.

Given the following text snippets, classify their sentiment as Positive, Negative, or Neutral.

Example 1: "I had an amazing day at the park!"
Sentiment: Positive

Example 2: "The service at the restaurant was terrible."
Sentiment: Negative

Example 3: "I think it's going to rain today."
Sentiment: Neutral

Text: "This movie was a fantastic journey through imagination."
Sentiment: Positive

Translate the following Chinese phrases into English.

Example 1: “你好”
Translation: “Hello”

Example 2: “谢谢你”
Translation: “Thank you”

Phrase to translate: “早上好”
Translation: “Good Morning”

Above, we have presented examples to illustrate the fundamental in-context learning capabilities of prompting LLMs. This section, however, does not include more advanced prompting techniques in order to keep the content concise and compact. More discussions on prompting can be found in Chapter 9.

8.2 Training at Scale

As a first step in developing LLMs, we need to train these models on large amounts of data. The training task is standard: the objective is to maximize the log-likelihood, which can be achieved via gradient descent. However, as we scale up both the model size and the amount of data, the optimization problem becomes highly challenging. For example, large models generally make the training process unstable. In this section, we discuss several issues inherent to large-scale training for LLMs, including data preparation, model modification, and distributed training. We also discuss the scaling laws for LLMs, which help us understand their training efficiency and effectiveness.

8.2.1 Data Preparation

The importance of data cannot be overstated in NLP. As larger neural networks are developed, the demand for data continues to increase. For example, developing LLMs may require trillions of tokens for pre-training (see Table 8.3), orders of magnitude larger than those used in training conventional NLP models. In general, we may want to gather as much training data as possible. However, larger training datasets do not mean better training results, and the development of LLMs raises new issues in creating or collecting these datasets.

A first issue is the quality of data. High-quality data has long been seen as crucial for training data-driven NLP systems. Directly feeding raw text from various sources into a model is generally undesirable. For instance, a significant portion of the data used to train recent LLMs originates from web scraping, which often contains errors and inappropriate content, such as toxic language and fabricated facts. In addition, the internet is increasingly flooded with machine-generated content due to the widespread use of AI, presenting further challenges for filtering web-scraped corpora. Researchers have demonstrated that training LLMs on unfiltered data can be harmful [Raffel et al., 2020]. Improving data quality typically involves integrating rigorous filtering and cleaning steps into the data processing workflow. For example, Penedo

LLM	# of Tokens	Data
GPT3-175B [Brown et al., 2020]	0.5T	Webpages, Books, Wikipedia
Falcon-180B [Almazrouei et al., 2023]	3.5T	Webpages, Books, Conversations, Code, Technical Articles
LLaMA2-65B [Touvron et al., 2023a]	1.0T $\sim$ 1.4T	Webpages, Code, Wikipedia, Books, Papers, Q&As
PaLM-450B [Chowdhery et al., 2022]	0.78T	Webpages, Books, Conversations, Code, Wikipedia, News
Gemma-7B [Gemma Team, 2024]	6T	Webpages, Mathematics, Code

Table 8.3: Amounts of training data used in some LLMs in terms of the number of tokens.

et al. [2023] showed that by adopting a series of heuristic processing techniques, 90% of their web-scraped data could be discarded prior to LLM training. In addition to large-scale web-scraped text, LLM training data often includes books, academic papers, user-generated content from social media, and other diverse corpora. Most of the latest LLMs are trained on such combined datasets, which are found to be important for the strong performance of the resulting models.

A second issue is the diversity of data. We want the training data to cover as many data types as possible, so that the trained models can adapt to different downstream tasks easily. It has been widely recognized that the quality and diversity of training data both play very important roles in LLMs. An interesting example is that incorporating programming code into training data has been found to be beneficial for LLMs. The benefits are demonstrated not only in enhancing the programming abilities of LLMs, but also in improving reasoning for complex problems, especially those requiring COT prompting. The concept "diversity" can be extended to include language diversity as well. For example, many LLMs are trained on multilingual data, and therefore we can handle multiple languages using a single model. While this approach shows strong abilities in multilingual and cross-lingual tasks, its performance on specific languages largely depends on the volume and quality of the data for those languages. It has been shown in some cases to provide poor results for low-resource languages.

A third issue is bias within the training data. This is not a problem unique to LLMs but exists across many NLP systems. A common example is gender bias, where LLMs exhibit unwarranted preferences or stereotypes. This can partly be attributed to class imbalances in the training corpora. For example, the term *nurse* is disproportionately associated with women in historical text. To debias the data, a common practice is to balance the representation of different demographic categories, such as gender, ethnicity, and dialects. Data bias is intrinsically linked to the diversity issue mentioned above. Because many LLMs are trained and aligned using English-centric data, they inherently lean towards the cultural values and perspectives prevalent among English-speaking populations. Increasing language and cultural diversity in the training data can partially mitigate these biases.

Another issue in large-scale data collection is privacy. If LLMs are trained on data from extensive sources, this potentially leads to risks regarding the exposure of sensitive information, such as intellectual property and personal data. This is particularly concerning given the capacity of LLMs to represent patterns from the data they are trained on, which might inadvertently involve memorizing and reproducing specific details. A simple approach to privacy protection is to remove or anonymize sensitive information. For example, anonymization techniques can be applied to remove personally identifiable information from training data to prevent LLMs from learning from such data. However, in practice, erasing or redacting all sensitive data is difficult. Therefore, many LLMs, particularly those launched for public service, typically work with systems that can detect the potential exposure of sensitive data, or are fine-tuned to reject certain requests that could lead to information leakage.

8.2.2 Model Modifications

Training LLMs is challenging. A commonly encountered issue is that the optimization process becomes increasingly unstable as models scale up in size. For example, one needs to choose a small learning rate to achieve stable training with gradient descent, but this in turn drastically increases training time. Sometimes, even when the training configuration is carefully designed, training may diverge at certain points during optimization. The training of LLMs is generally influenced by many factors, such as parameter initialization, batching, and regularization. Here, we focus on common modifications and improvements to the standard Transformer architecture, which are important for stable LLM training.

1. Layer Normalization with Residual Connections

Layer normalization is used to stabilize training for deep neural networks. It is a process of subtracting the mean and dividing by the standard deviation. By normalizing the layer output in this manner, we can effectively mitigate internal covariate shift and enhance training stability. In Transformers, layer normalization is typically employed alongside residual connections. As described in Section 8.1.1, a sub-layer can be based on either the post-norm architecture, in which layer normalization is performed right after a residual block, or the pre-norm architecture, in which layer normalization is performed inside a residual block. While both architectures are widely used in Transformer-based systems [Wang et al., 2019a], the pre-norm architecture has proven to be especially useful in training deep Transformers. Given this, most LLMs are based on the pre-norm architecture, expressed as $\mathrm{output} = F(\mathrm{LNorm}(\mathrm{input})) + \mathrm{input}$.

A widely-used form of the layer normalization function is given by

$$\mathrm{LNorm}(\mathbf{h}) \quad = \quad \boldsymbol{\alpha} \odot \frac{\mathbf{h} - \mu}{\sigma + \epsilon} + \boldsymbol{\beta}, \tag{8.23}$$

where $\mathbf{h}$ is a d-dimensional real-valued vector, μ is the mean of all the entries of $\mathbf{h}$, and σ is the corresponding standard deviation. ϵ is introduced for the sake of numerical stability. $\boldsymbol{\alpha} \in \mathbb{R}^d$ and $\boldsymbol{\beta} \in \mathbb{R}^d$ are the gain and bias terms.

A variant of layer normalization, called root mean square (RMS) layer normalization, only rescales the input vector but does not re-center it [Zhang and Sennrich, 2019]. The RMS layer normalization function is given by

$$\mathrm{RMSNorm}(\mathbf{h}) = \boldsymbol{\alpha} \odot \frac{\mathbf{h}}{\sigma_{\mathrm{rms}} + \epsilon}, \tag{8.24}$$

where σ_{rms} is the root mean square of $\mathbf{h}$, i.e., $\sigma_{\mathrm{rms}} = (\frac{1}{d}\sum_{k=1}^{d} h_k^2)^{\frac{1}{2}}$. This layer normalization function is used in LLMs like the LLaMA series.

2. Activation Functions in FFNs

In Transformers, FFN sub-layers are designed to introduce non-linearity into representation learning, and have been found to be useful for preventing the representations learned by self-attention from degeneration[8] [Dong et al., 2021]. A standard form of the FFN used in these sub-layers can be expressed as

$$\mathrm{FFN}(\mathbf{h}) = \sigma(\mathbf{h}\mathbf{W}_h + \mathbf{b}_h)\mathbf{W}_f + \mathbf{b}_f, \tag{8.25}$$

where $\mathbf{W}_h \in \mathbb{R}^{d \times d_h}$, $\mathbf{b}_h \in \mathbb{R}^{d_h}$, $\mathbf{W}_f \in \mathbb{R}^{d_h \times d}$, and $\mathbf{b}_f \in \mathbb{R}^{d}$ are the parameters, and d_h is the hidden size. The function $\sigma(\cdot)$ is the activation function of the hidden layer. A common choice for $\sigma(\cdot)$ is the **rectified linear unit** (**ReLU**), given by

$$\sigma_{\mathrm{relu}}(\mathbf{h}) = \max(\mathbf{0}, \mathbf{h}). \tag{8.26}$$

In practical implementations, increasing d_h is beneficial, and thus it is typically set to a much larger number than d in LLMs. However, an excessively large hidden size poses challenges for both training and deployment. In such cases, the choice of activation function plays a more important role in wide FFNs.

Several effective alternatives to the ReLU have been proposed in modern LLMs. One of the most widely used is the **gaussian error linear unit** (**GeLU**), which can be viewed as a smooth approximation to the ReLU. Instead of applying a hard threshold based on the sign of the input, the GeLU function weights each input by the cumulative distribution function of a standard normal distribution. Specifically, the GeLU function is defined element-wise as

$$\sigma_{\mathrm{gelu}}(\mathbf{h}) = \mathbf{h} \odot \Phi(\mathbf{h}), \tag{8.27}$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{1})$. The GeLU function has been adopted in several LLMs, such as BERT, GPT-3, and BLOOM.

Another popular family of activation functions is the **gated linear unit** (**GLU**) family. The basic form of GLUs is given by

$$\sigma_{\mathrm{glu}}(\mathbf{h}) = \sigma(\mathbf{h}\mathbf{W}_1 + \mathbf{b}_1) \odot (\mathbf{h}\mathbf{W}_2 + \mathbf{b}_2), \tag{8.28}$$

[8] Here, degeneration refers to the phenomenon where the rank of the representation matrix is significantly reduced after processing.

where $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{d \times d_h}$ and $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^{d_h}$ are model parameters. Note that the GLU mechanism effectively replaces the single linear projection and activation step in a standard FFN, mapping the input $\mathbf{h}$ directly to the expanded hidden dimension d_h. Different choices for the underlying activation $\sigma(\cdot)$ yield different variants of GLU. For instance, if $\sigma(\cdot)$ is defined as the GELU function, we obtain the GeGLU function

$$\sigma_{\text{geglu}}(\mathbf{h}) \quad = \quad \sigma_{\text{gelu}}(\mathbf{h}\mathbf{W}_1 + \mathbf{b}_1) \odot (\mathbf{h}\mathbf{W}_2 + \mathbf{b}_2). \tag{8.29}$$

This activation function has been successfully applied in LLMs like Gemma.

Alternatively, if we define $\sigma(\cdot)$ to be the Swish function, $\sigma_{\text{swish}}(\mathbf{h}) = \mathbf{h} \odot \text{Sigmoid}(c\mathbf{h})$ [Ramachandran et al., 2017], we arrive at the SwiGLU function

$$\sigma_{\text{swiglu}}(\mathbf{h}) \quad = \quad \sigma_{\text{swish}}(\mathbf{h}\mathbf{W}_1 + \mathbf{b}_1) \odot (\mathbf{h}\mathbf{W}_2 + \mathbf{b}_2). \tag{8.30}$$

Both the PaLM and LLaMA model series rely on the SwiGLU function for their FFN layers. For a more comprehensive discussion on GLU variants and their impacts, readers are referred to the work of Shazeer [2020].

3. Removing Bias Terms

Another popular model design is to remove the bias terms from the affine transformations used in LLMs. This simplification can be applied to layer normalization, input projections for QKV attention, and the FFN sub-layers. For example, we can modify Eq. (8.25) to obtain an FFN with no bias terms

$$\text{FFN}(\mathbf{h}) \quad = \quad \sigma(\mathbf{h}\mathbf{W}_h)\mathbf{W}_f. \tag{8.31}$$

Chowdhery et al. [2022] report that removing bias terms helps improve the training stability of LLMs. This method has been used in several LLMs, such as LLaMA and Gemma.

4. Other Issues

Many LLMs also involve modifications to their positional embedding models. For example, one can replace sinusoidal positional encodings with rotary position embeddings. These models will be discussed in Section 8.3.

Note that while model modifications are common in LLM development, the training stability can be improved in many different ways. For example, increasing the batch size as the training process proceeds has been found to be useful for some LLMs. In general, achieving stable and efficient large-scale LLM training requires carefully designed setups, such as learning rate scheduling, optimizer choices, parallel training strategies, and mixed-precision training methods. Some of these aspects are highly engineering-intensive, and therefore, we typically need a number of training runs to obtain satisfactory LLMs.

8.2.3 Distributed Training

Training LLMs requires significant computational resources. A common approach to improving training efficiency is to use large-scale distributed systems. Fortunately, alongside the rise of neural networks in AI, deep learning software and hardware have been developed, making it easier to implement LLMs and perform computations. For example, one can now easily fine-tune an LLM using deep learning software frameworks and a machine with multiple GPUs. However, scaling up the training of LLMs is still challenging, and requires significant effort to develop hardware and software systems for stable and efficient distributed training.

An important consideration in distributed training is parallelism. There are several forms of parallelism: data parallelism, model parallelism, tensor parallelism, and pipeline parallelism. Despite the different ways to distribute computations across devices, these parallelism methods are based on a similar idea: the training problem can be divided into smaller tasks that can be executed simultaneously. The issue of parallelism in training LLMs has been extensively studied [Narayanan et al., 2021; Fedus et al., 2022b]. Here we sketch the basic concepts.

- **Data Parallelism**. This method is one of the most widely used parallelism methods for training neural networks. To illustrate, consider the simplest case where the standard delta rule is used in gradient descent

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\partial L_{\theta_t}(\mathcal{D}_{\text{mini}})}{\partial \theta_t}, \tag{8.32}$$

  where the new parameters θ_{t+1} are obtained by updating the latest parameters θ_t with a small step η in the direction of the negative loss gradient. $\frac{\partial L_{\theta_t}(\mathcal{D}_{\text{mini}})}{\partial \theta_t}$ is the gradient of the loss with respect to the parameters θ_t, and is computed on a minibatch of training samples $\mathcal{D}_{\text{mini}}$. In data parallelism, we divide $\mathcal{D}_{\text{mini}}$ into N smaller batches, denoted by $\{\mathcal{D}^1, \cdots, \mathcal{D}^N\}$. Then, we distribute these batches to N workers, each with a corresponding batch. Once the data is distributed, these workers can work at the same time. The gradient of the entire minibatch is obtained by aggregating the gradients computed by the workers, as follows

$$\frac{\partial L_{\theta_t}(\mathcal{D}_{\text{mini}})}{\partial \theta_t} = \underbrace{\frac{\partial L_{\theta_t}(\mathcal{D}^1)}{\partial \theta_t}}_{\text{worker 1}} + \underbrace{\frac{\partial L_{\theta_t}(\mathcal{D}^2)}{\partial \theta_t}}_{\text{worker 2}} + \cdots + \underbrace{\frac{\partial L_{\theta_t}(\mathcal{D}^N)}{\partial \theta_t}}_{\text{worker } N}. \tag{8.33}$$

  Under ideal conditions where the workers coordinate well and the communication overhead is small, data parallelism can achieve nearly an N-fold speed-up for training.

- **Model Parallelism**. Although data parallelism is simple and effective, it requires each worker to run the entire LLM and perform the complete forward and backward passes. As LLMs grow larger, it sometimes becomes infeasible to load and execute an LLM on a single device. In this case, we can decouple the LLM into smaller components and run these components on different devices. One simple way to do this is to group consecutive layers in the layer stack and assign each group to a worker, called naive model parallelism. The workers operate sequentially following the layer order in the stack, that is, in the forward pass we process the input from lower-level to upper-level layers, and in the backward pass we propagate the error gradients from upper-level to lower-level layers. Consider, for example, a Transformer decoder with L stacked blocks. To distribute the computation load, each block is assigned to a worker. See the following illustration for a single run of the forward and backward passes of this model.

Worker L				B_L ($\uparrow$)	B_L ($\downarrow$)			
...			...			...		
Worker 2		B_2 ($\uparrow$)					B_2 ($\downarrow$)	
Worker 1	B_1 ($\uparrow$)							B_1 ($\downarrow$)

Here B_l denotes the computation of block l, and the symbols $\uparrow$ and $\downarrow$ denote the forward and backward passes, respectively. Note that this parallelism method forces the workers to run in sequence, so a worker has to wait for the previous worker to finish their job. This results in the devices being idle for most of the time. In practical systems, model parallelism is generally used together with other parallelism mechanisms to maximize the use of devices.

- **Tensor Parallelism**. Parallelism can also be performed in a single computation step. A common approach is splitting a large parameter matrix into chunks, multiplying an input tensor with each of these chunks separately, and then concatenating the results of these multiplications to form the output. For example, consider the multiplication of the representation $\mathbf{h} \in \mathbb{R}^d$ with the parameter matrix $\mathbf{W}_h \in \mathbb{R}^{d \times d_h}$ in an FFN sub-layer (see Eq. (8.25)). We can slice the matrix $\mathbf{W}_h \in \mathbb{R}^{d \times d_h}$ vertically into M sub-matrices

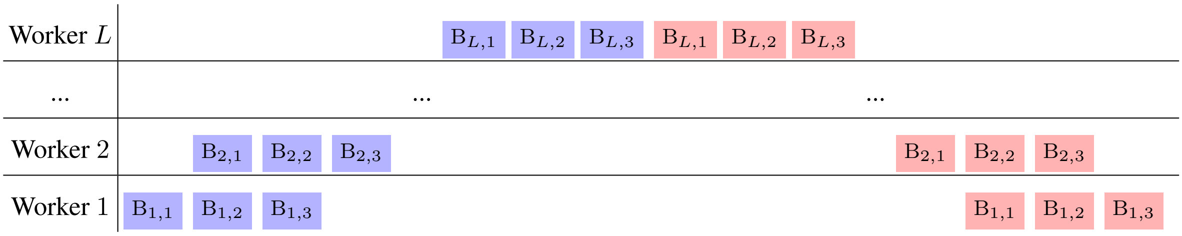
$$\mathbf{W}_h = \begin{bmatrix} \mathbf{W}_h^1 & \mathbf{W}_h^2 & \cdots & \mathbf{W}_h^M \end{bmatrix}, \tag{8.34}$$

where each sub-matrix $\mathbf{W}_h^k$ is of shape $d \times \frac{d_h}{M}$. The multiplication of $\mathbf{h}$ with $\mathbf{W}_h$ can be expressed as

$$\begin{aligned} \mathbf{h}\mathbf{W}_h &= \mathbf{h} \begin{bmatrix} \mathbf{W}_h^1 & \mathbf{W}_h^2 & \cdots & \mathbf{W}_h^M \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{h}\mathbf{W}_h^1 & \mathbf{h}\mathbf{W}_h^2 & \cdots & \mathbf{h}\mathbf{W}_h^M \end{bmatrix}. \end{aligned} \tag{8.35}$$

We can perform matrix multiplications $\{\mathbf{h}\mathbf{W}_h^1, \mathbf{h}\mathbf{W}_h^2, \cdots, \mathbf{h}\mathbf{W}_h^M\}$ on M devices separately. As a result, we distribute a large matrix multiplication across multiple devices, each of which may have relatively small memory. From the perspective of modern GPU design, tensor parallelism over GPUs provides a two-level, tile-based approach to parallel computing. First, at a higher level, we decompose a matrix multiplication into sub-matrix multiplications that can directly fit into the memory of GPUs. Then, at a lower level, we execute these sub-matrix multiplications on GPUs using tile-based parallel algorithms that are specifically optimized for GPUs.

- **Pipeline Parallelism**. Above, in model parallelism, we have described a simple approach to distributing groups of model components across multiple devices. But this method is inefficient because only one device is activated at a time during processing. Pipeline parallelism addresses this issue by introducing overlaps between computations on different devices [Harlap et al., 2018; Huang et al., 2019]. To do this, a batch of samples is divided into a number of micro-batches, and then these micro-batches are processed by each worker as usual. Once a micro-batch is processed by a worker and passed to the next one, the following micro-batch immediately takes its place on the same worker. In other words, we create a pipeline in which different computation steps can overlap if multiple jobs are fed into the pipeline. The following shows an illustration of pipeline parallelism for processing 3 micro-batches.



Here $\mathrm{B}_{l,k}$ represents the processing of the k-th micro-batch by the l-th worker. Ideally we would like to maximize the number of micro-batches, and thus minimize the idle time of the workers. However, in practice, using small micro-batches often reduces GPU utilization and increases task-switching costs. This may, in turn, decrease the overall system throughput.

The ultimate goal of parallel processing is to achieve linear growth in efficiency, that is, the number of samples that can be processed per unit of time increases linearly with the number of devices. However, distributed training is complicated, and influenced by many factors in addition to the parallelism method we choose. One problem, which is often associated with distributed systems, is the cost of communication. We can think of a distributed system as a group of networked nodes. Each of these nodes can perform local computation or pass data to other nodes. If there is a large number of such nodes, it will be expensive to distribute and collect data across them. Sometimes, the time savings brought about by parallelism are offset by the communication overhead of a large network. Another problem with large-scale distributed systems is that the synchronization of nodes introduces additional costs. As is often the case, some nodes may take longer to complete their tasks, causing others to wait for the slowest ones. While we can use asynchronous training to handle heterogeneity in computational resources, this may lead to stale gradients and non-guaranteed convergence. Moreover, as more nodes are added to the network, there is a higher chance of having crashed nodes during training. In this case, we need to ensure that the whole system is fault-tolerant. In many practical settings, to increase scalability, one needs to take into account additional issues, including architecture design, data transfer and computation overlap, load balancing, memory bandwidth, and so on.

Training LLMs is so computationally expensive that, even though distributed training is already in use, researchers and engineers often still employ various model compression and speed-up methods to improve training efficiency [Weng, 2021]. One example is mixed precision training, in which low-precision data (such as FP16 and FP8 data) is used for gradient computation on each node, and higher-precision data (e.g., FP32/FP64) is used for updating the model [Micikevicius et al., 2018]. A key operation in this approach is gradient accumulation, where gradients need to be accumulated and synchronized across nodes. However, due to the non-associativity of floating-point addition, this can lead to slight numerical differences in accumulated gradients on different nodes, which may affect model convergence and final performance. This problem becomes more pronounced when there is a large number of nodes involved in distributed training, especially given that low-precision numerical computations may encounter overflow and underflow issues, as well as inconsistencies across different hardware devices. Therefore, the design of distributed systems needs to consider these numerical computation issues to ensure satisfactory results and convergence.

8.2.4 Scaling Laws

The success of LLMs reveals that training larger language models using more resources can lead to improved model performance. Researchers have described this phenomenon as the **scaling laws** of LLMs. More specifically, scaling laws describe the relationships between the performance of LLMs and the factors involved in training, such as the model size, the amount of computation used for training, and the amount of training data. For example, Hestness et al. [2017] show that the performance of deep neural networks is a power-law function of the training data size. In the beginning, when the amount of training data is relatively small, the performance of the model improves slowly. Afterward, when more training data is used, the model enters a phase of rapid performance improvement, and the performance curve follows a power-law trend. Ultimately, the improvement in performance becomes slow again, and more data does not lead to significant gains. Figure 8.3 shows an example of such curves.

In NLP, a traditional view holds that the performance gains will disappear at a certain point as the training is scaled up. However, recent results show that, if we consider the problem at larger scales, scaling up training is still a very effective method for obtaining stronger LLMs. For example, both closed-source and open-source LLMs can benefit from more data, even though trillions of tokens have already been used for training.

With the increase in the scale of model training, LLMs exhibit new capabilities, known as the **emergent abilities** of LLMs. For example, Wei et al. [2022b] studied the scaling properties of LLMs across different model sizes and amounts of computational resources. Their work shows that some abilities emerge when we scale the model size to a certain level. The observation of emergent abilities has demonstrated the role of large-scale training in enhancing the performance of LLMs, and it has also, to some extent, motivated researchers to continuously attempt to train larger models. As larger and stronger LMs continue to appear, our understanding of scaling laws continues to mature. This understanding helps researchers predict the performance of LLMs during training and estimate the computational resources

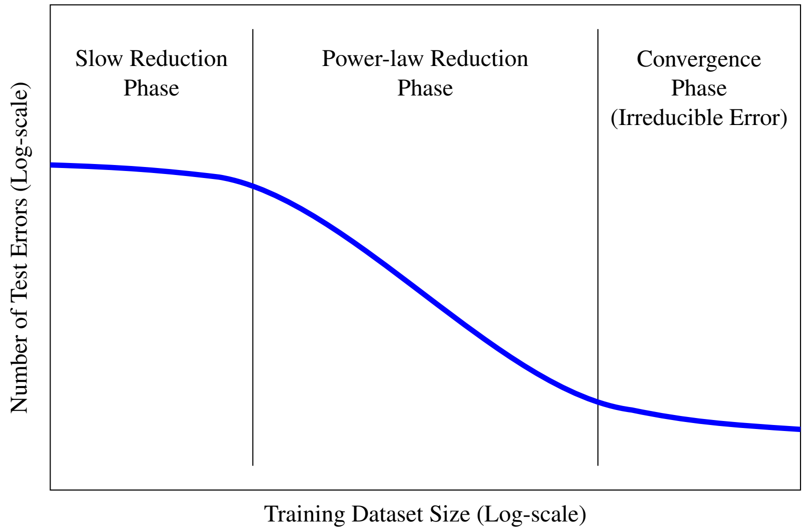


Figure 8.3: A scaling law of test error against a variable of interest (e.g., training dataset size) [Hestness et al., 2017]. The curve of the scaling law can be divided into three phases. Initially, the test error decreases slowly when more training data is used, but this only lasts for a short period. In the second phase, the test error decreases drastically, and the curve becomes a power-law curve. After that, the error reduction slows down again in the third phase. Note that there are irreducible errors that cannot be eliminated, regardless of the amount of training data.

required to achieve a given level of performance.

To understand how model performance scales with various factors involved in training, it is common to express model performance as a function of these factors. For example, in the simplest case, we can express the loss or error of an LLM as a function of a single variable of interest. However, there are no universal scaling laws that can describe this relationship. Instead, different functions are proposed to fit the learning curves of LLMs.

Let x be the variable of interest (such as the number of model parameters) and $\mathcal{L}(x)$ be the loss of the model given x (such as the cross-entropy loss on test data). The simplest form of $\mathcal{L}(x)$ is a power law

$$\mathcal{L}(x) = ax^b, \tag{8.36}$$

where a and b are parameters that are estimated empirically. Despite its simplicity, this function has successfully interpreted the scaling behavior of language models and machine translation systems in terms of model size (denoted by N) and training dataset size (denoted by D) [Gordon et al., 2021; Hestness et al., 2017]. For example, Kaplan et al. [2020] found that the performance of their language model improves as a power law of either N or D after an initial transient period, and expressed these relationships using $\mathcal{L}(N) = \left(\frac{N}{8.8\times10^{13}}\right)^{-0.076}$ and $\mathcal{L}(D) = \left(\frac{D}{5.4\times10^{13}}\right)^{-0.095}$ (see Figure 8.4).

An improvement to this scaling law is to add an **irreducible error** term to the power law.

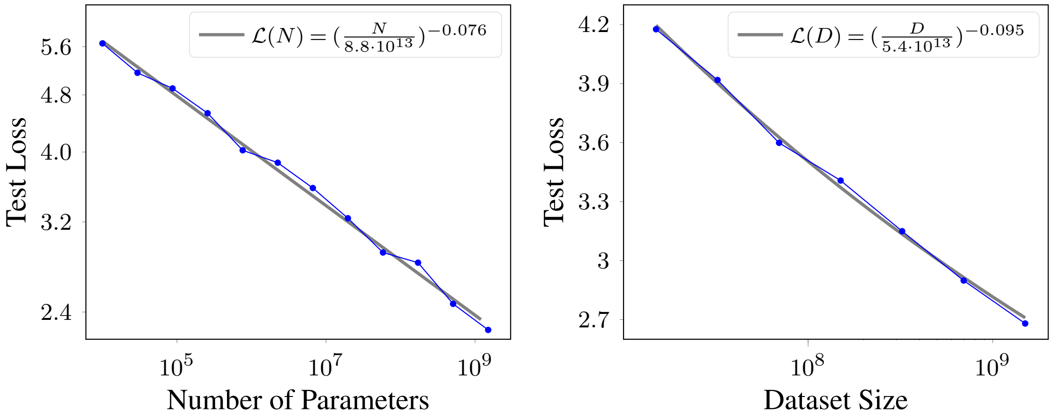


Figure 8.4: Test loss against model size (N) and training dataset size (D) (data points are plotted for illustrative purposes). We plot test loss as a function of N, which is defined as $\mathcal{L}(N) = \left(\frac{N}{8.8 \times 10^{13}}\right)^{-0.076}$, and a function of D, which is defined as $\mathcal{L}(D) = \left(\frac{D}{5.4 \times 10^{13}}\right)^{-0.095}$ [Kaplan et al., 2020].

The form of $\mathcal{L}(x)$ is then given by

$$\mathcal{L}(x) \quad = \quad ax^b + \epsilon_\infty, \tag{8.37}$$

where ϵ_∞ is the irreducible error that accounts for the error due to unknown variables, which is present even as $x \to \infty$. Eq. (8.37) is one of the most widely used forms for modeling scaling laws of LLMs. For example, Rosenfeld et al. [2020] developed a scaling law that involves both model scaling and dataset scaling, as follows

$$\mathcal{L}(N, D) \quad = \quad aN^b + cD^d + \epsilon_\infty. \tag{8.38}$$

An example of such a formulation is the Chinchilla scaling law. It states that the test loss per token is the sum of the inverse power-law functions of N and D, with an additional irreducible error term. Hoffmann et al. [2022] express this scaling law as

$$\mathcal{L}(N, D) \quad = \quad \underbrace{\frac{406.4}{N^{0.34}}}_{\text{model scaling}} + \underbrace{\frac{410.7}{D^{0.28}}}_{\text{dataset scaling}} + \underbrace{1.69}_{\text{irreducible error}}. \tag{8.39}$$

All the scaling laws mentioned above are based on monotonic functions. So they cannot cover functions with inflection points, such as double descent curves. In response, researchers have explored more sophisticated functions to fit the learning curves. Examples of such functions can be found in Alabdulmohsin et al. [2022] and Caballero et al. [2023]'s work.

The significance of scaling laws lies in providing directional guidance for LLM research: if we are still in the region of the power-law curve, using more resources to train larger models is a very promising direction. While this result compels large research groups and companies to invest heavily in computational resources to train larger models, which is very expensive, scaling laws continuously push the boundaries of AI forward. On the other hand, understanding scaling laws helps researchers make informed decisions when training LLMs. For example, given the computational resources at hand, the ultimate performance of an LLM can be estimated.

As a final note on scaling laws in this section, for LLMs, a lower test loss does not always imply better performance on all downstream tasks. To adapt LLMs, there are several steps, such as fine-tuning and prompting, that may influence the final results. Therefore, the scaling laws for different downstream tasks might vary in practice.

8.3 Long Sequence Modeling

We have already seen that, in large-scale training, more powerful language models can be developed by scaling up both data and computational resources. However, scaling can also occur along other dimensions. For instance, in many modern applications, LLMs often need to be adapted to process very long sequences. A common scenario involves pre-training an LLM on texts of standard length, and subsequently applying it to sequences that far exceed those seen during pre-training. Here, we use $\Pr(\mathbf{y}|\mathbf{x})$ to denote the conditional probability of generating text, where $\mathbf{x}$ is the context and $\mathbf{y}$ is the generated text. Broadly speaking, long sequence modeling problems can be categorized into three types:

- **Long-context text generation** (i.e., where $\mathbf{x}$ is a long sequence). For example, generating a short summary for a lengthy document.
- **Long-text generation** (i.e., where $\mathbf{y}$ is a long sequence). For example, generating a comprehensive story based on a few keywords.
- **Long-context and long-text generation** (i.e., where both $\mathbf{x}$ and $\mathbf{y}$ are long sequences). For example, generating a translation of a long document from Chinese to English.

Recently, NLP researchers have been more interested in applying and evaluating LLMs on tasks where extremely long input texts are involved. Imagine an LLM that reads a C++ source file containing tens of thousands of lines, and outlines the functionality of the program corresponding to the source file. Such models, capable of handling extensive textual contexts, are typically called **long-context LLMs**. In this section we will restrict ourselves to long-context LLMs, but the methods discussed here can be applied to other problems.

For Transformers, dealing with long sequences is computationally expensive, as the computational cost of self-attention grows quadratically with the sequence length. This makes it infeasible to train and deploy such models for very long inputs. Two strands of research have tried to adapt Transformers to long-context language modeling.

- The first explores efficient training methods and model architectures to learn self-attention models from long-sequence data.
- The other adapts pre-trained LLMs to handle long sequences with modest or no fine-tuning.

Here, we will discuss the former briefly since Chapter 6 extensively covers many methods in this strand. We will focus on the latter, highlighting popular methods in recent LLMs. We will also discuss the strengths and limitations of these long-sequence models.

8.3.1 Optimization from an HPC Perspective

We begin our discussion by considering improvements to standard Transformer models from the perspective of high-performance computing. Most of these improvements, though not specifically designed for LLMs, have been widely applied across various deep learning models [Kim et al., 2023]. A commonly used approach is to adopt reduced-precision data formats. For example, rather than using standard 32-bit floating-point (FP32) arithmetic, models can employ 16-bit floating-point (FP16/BF16) or even 8-bit integer (INT8) quantization. Using these low-precision data types can improve computational efficiency and memory throughput, so that longer sequences can be processed more easily. An alternative approach is to improve Transformers by using hardware-aware techniques. For example, on modern GPUs, the efficiency of Transformers can be improved by using IO-aware implementations of the self-attention function [Dao et al., 2022; Kwon et al., 2023].

Another way to handle long sequences is through sequence parallelism [Li et al., 2023b; Korthikanti et al., 2023]. Specifically, consider the general problem of attending the query $\mathbf{q}_i$ at the position i to the keys $\mathbf{K}$ and values $\mathbf{V}$. We can divide $\mathbf{K}$ along the sequence dimension (i.e., by rows) and obtain a set of sub-matrices $\{\mathbf{K}^{[1]}, \cdots, \mathbf{K}^{[n_u]}\}$, each corresponding to a segment of the sequence. Similarly, we can obtain the sub-matrices of $\mathbf{V}$, denoted by $\{\mathbf{V}^{[1]}, \cdots, \mathbf{V}^{[n_u]}\}$. Then, we assign each pair of $\mathbf{K}^{[u]}$ and $\mathbf{V}^{[u]}$ to a compute node (e.g., a GPU of a GPU cluster). The assigned nodes can run in parallel, thereby parallelizing the attention operation.

Recall that the output of the self-attention model can be written as

$$\mathrm{Att}_{\mathrm{qkv}}(\mathbf{q}_i, \mathbf{K}, \mathbf{V}) = \sum_{j=0}^{m-1} \alpha_{i,j} \mathbf{v}_j, \tag{8.40}$$

where $\alpha_{i,j}$ is the attention weight between positions i and j. In Transformers, $\alpha_{i,j}$ is obtained by normalizing the scaled dot product between $\mathbf{q}_i$ and $\mathbf{k}_j$. Let $\beta_{i,j}$ denote the attention score between $\mathbf{q}_i$ and $\mathbf{k}_j$. We have

$$\beta_{i,j} = \frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{d}} + \mathrm{Mask}(i,j), \tag{8.41}$$

where $\mathrm{Mask}(i,j)$ is the causal mask value for the position pair (i,j). Then, we define the attention weight $\alpha_{i,j}$ to be

$$\begin{aligned} \alpha_{i,j} &= \mathrm{Softmax}(\beta_{i,j}) \\ &= \frac{\exp(\beta_{i,j})}{\sum_{j'} \exp(\beta_{i,j'})}. \end{aligned} \tag{8.42}$$

On each compute node, we need to implement these equations. Given the keys and values assigned to this node, computing the numerator of the right-hand side of Eq. (8.42) (i.e., $\exp(\beta_{i,j})$) is straightforward, as all the required information is stored on the node. However, computing the denominator of the right-hand side of Eq. (8.42) involves a sum of $\exp(\beta_{i,j'})$ over all j', which requires transferring data to and from other nodes. To illustrate, suppose that $\mathbf{v}_j$ and $\mathbf{k}_j$ are placed on node u. We can rewrite Eq. (8.42) as

$$\alpha_{i,j} = \frac{\overbrace{\exp(\beta_{i,j})}^{\text{node } u}}{\underbrace{\sum_{\mathbf{k}_{j'} \in \mathbf{K}^{[1]}} \exp(\beta_{i,j'})}_{\text{node } 1} + \cdots + \underbrace{\sum_{\mathbf{k}_{j'} \in \mathbf{K}^{[u]}} \exp(\beta_{i,j'})}_{\text{node } u} + \cdots + \underbrace{\sum_{\mathbf{k}_{j'} \in \mathbf{K}^{[n_u]}} \exp(\beta_{i,j'})}_{\text{node } n_u}}, \tag{8.43}$$

where the notation $\mathbf{k}_{j'} \in \mathbf{K}^{[u]}$ indicates that $\mathbf{k}_{j'}$ is a row vector of $\mathbf{K}^{[u]}$. In a straightforward implementation, we first perform the summations $\{\sum_{\mathbf{k}_{j'} \in \mathbf{K}^{[u]}} \exp(\beta_{i,j'})\}$ separately on the corresponding nodes. Then, we collect these summation results from different nodes to combine them into a final result. This corresponds to a collective operation in the context of parallel processing. There are many efficient implementations of such operations, such as all-reduce operations. Hence the sum of all $\exp(\beta_{i,j})$ values can be computed using optimized routines in collective communication toolkits.

Given the attention weights $\{\alpha_{i,j}\}$, we then compute the attention results using Eq. (8.40). The problem can be re-expressed as

$$\mathrm{Att}_{\mathrm{qkv}}(\mathbf{q}_i, \mathbf{K}, \mathbf{V}) = \underbrace{\sum_{\mathbf{v}_{j'} \in \mathbf{V}^{[1]}} \alpha_{i,j'} \mathbf{v}_{j'}}_{\text{node } 1} + \cdots + \underbrace{\sum_{\mathbf{v}_{j'} \in \mathbf{V}^{[u]}} \alpha_{i,j'} \mathbf{v}_{j'}}_{\text{node } u} + \cdots + \underbrace{\sum_{\mathbf{v}_{j'} \in \mathbf{V}^{[n_u]}} \alpha_{i,j'} \mathbf{v}_{j'}}_{\text{node } n_u}. \tag{8.44}$$

Like Eq. (8.43), Eq. (8.44) can be implemented as a reduction operation in parallel processing. First, we perform the weighted summations of values on different nodes simultaneously. Then, we collect the results from these nodes via collective operations.

Note that, although this section primarily focuses on long sequence modeling, much of the motivation for sequence parallelism comes from the distributed training methods of deep networks, as discussed in Section 8.2.3. As a result, the implementation of these methods can be built on the same parallel processing library.

8.3.2 Efficient Architectures

One difficulty of applying Transformers to long sequences is that standard self-attention suffers from quadratic time and memory complexity with respect to the sequence length. Moreover, autoregressive generation requires maintaining a **key-value cache** (or **KV cache**), whose size increases as more tokens are processed. Although the KV cache footprint grows linearly with sequence length, for extremely long contexts, this memory requirement becomes prohibitive, making standard LLM deployment impractical. As a result, the model architecture of long-context LLMs generally moves away from the standard Transformer, turning instead to the development of more efficient variants and alternatives.

One approach is to replace standard self-attention with sparse attention. This family of models is motivated by the observation that attention distributions in long sequences are often highly skewed. Thus, only a small subset of tokens significantly impacts the context of any given token. Consequently, we can prune the vast majority of attention weights, representing the mechanism in a highly compressed form. To illustrate, consider the standard self-attention formulation:

$$\mathrm{Att}_{\mathrm{qkv}}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \alpha(\mathbf{Q},\mathbf{K})\mathbf{V}, \tag{8.45}$$

where the attention weight matrix $\alpha(\mathbf{Q},\mathbf{K}) \in \mathbb{R}^{m\times m}$ is obtained by

$$\begin{aligned}\alpha(\mathbf{Q},\mathbf{K}) &= \mathrm{Softmax}(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}} + \mathbf{Mask}) \\ &= \begin{bmatrix} \alpha_{0,0} & 0 & 0 & \cdots & 0 \\ \alpha_{1,0} & \alpha_{1,1} & 0 & \cdots & 0 \\ \alpha_{2,0} & \alpha_{2,1} & \alpha_{2,2} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha_{m-1,0} & \alpha_{m-1,1} & \alpha_{m-1,2} & \cdots & \alpha_{m-1,m-1} \end{bmatrix}.\end{aligned} \tag{8.46}$$

Each row vector $\begin{bmatrix}\alpha_{i,0} & \cdots & \alpha_{i,i} & 0 & \cdots & 0\end{bmatrix}$ corresponds to a distribution over tokens indicating how the i-th token attends to others. Since language models predict the next token only based on their left-context, we normally write the output of the attention model at position i as

$$\begin{aligned}\mathrm{Att}_{\mathrm{qkv}}(\mathbf{q}_i,\mathbf{K}_{\leq i},\mathbf{V}_{\leq i}) &= \begin{bmatrix}\alpha_{i,0} & \cdots & \alpha_{i,i}\end{bmatrix}\begin{bmatrix}\mathbf{v}_0 \\ \vdots \\ \mathbf{v}_i\end{bmatrix} \\ &= \sum_{j=0}^{i}\alpha_{i,j}\mathbf{v}_j,\end{aligned} \tag{8.47}$$

where $\mathbf{K}_{\leq i} = \begin{bmatrix}\mathbf{k}_0 \\ \vdots \\ \mathbf{k}_i\end{bmatrix}$ and $\mathbf{V}_{\leq i} = \begin{bmatrix}\mathbf{v}_0 \\ \vdots \\ \mathbf{v}_i\end{bmatrix}$ are the keys and values up to position i.

In standard self-attention, the vector $\begin{bmatrix}\alpha_{i,0} & \cdots & \alpha_{i,i}\end{bmatrix}$ is assumed to be dense, meaning almost all valid entries are non-zero. In sparse attention, only a predefined subset of these entries is computed, while the remainder are explicitly ignored (forced to zero). Let $G \subseteq \{0,\cdots,i\}$ denote the index set of the active (non-zero) entries. The output of the sparse attention model at position i then becomes:

$$\text{Att}_{\text{sparse}}(\mathbf{q}_i, \mathbf{K}_{\leq i}, \mathbf{V}_{\leq i}) = \sum_{j \in G} \alpha'_{i,j} \mathbf{v}_j. \tag{8.48}$$

Here $\{\alpha'_{i,j}\}$ are normalized over G. Hence their values are different from the original attention weights (in fact we have $\alpha'_{i,j} > \alpha_{i,j}$). The sparsity of the model is determined by the size of G. Sparse attention models differ in the way we define G. One simple approach is to define G based on heuristically designed patterns. For example, a widely-used pattern involves having G cover a window of tokens located near position i [Parmar et al., 2018].

While sparse attention reduces the computation through the use of sparse operations, such models still have significant limitations as we need to keep the entire KV cache (i.e., $\mathbf{K}_{\leq i}$ and $\mathbf{V}_{\leq i}$) during inference. If the sequence is very long, storing this cache will become highly memory-intensive. To address this, we can consider a different class of attention models where the KV cache is not explicitly retained. Linear attention is one such approach [Katharopoulos et al., 2020]. It uses a kernel function $\phi(\cdot)$ to map each query and key into a feature space, yielding $\mathbf{q}'_i = \phi(\mathbf{q}_i)$ and $\mathbf{k}'_i = \phi(\mathbf{k}_i)$, respectively. By replacing the Softmax normalization with a linear formulation[9], the form of the resulting attention model is given by

$$\begin{aligned} \text{Att}_{\text{qkv}}(\mathbf{q}_i, \mathbf{K}_{\leq i}, \mathbf{V}_{\leq i}) &\approx \text{Att}_{\text{linear}}(\mathbf{q}'_i, \mathbf{K}'_{\leq i}, \mathbf{V}_{\leq i}) \\ &= \frac{\mathbf{q}'_i \mu_i}{\mathbf{q}'_i \nu_i}, \end{aligned} \tag{8.49}$$

where μ_i and ν_i are variables that are computed in a recurrent manner

$$\mu_i = \mu_{i-1} + \mathbf{k}'^{\top}_i \mathbf{v}_i \tag{8.50}$$

$$\nu_i = \nu_{i-1} + \mathbf{k}'^{\top}_i. \tag{8.51}$$

μ_i and ν_i can be seen as summaries of the history up to position i. A benefit of this model is that we do not need to explicitly store all past keys and values. Instead only the latest representations μ_i and ν_i are used. So the computational cost of each step is a constant, and the model can be easily extended to deal with long sequences.

In fact, this sequential approach to long sequence modeling arises naturally when we adopt the perspective of recurrent models. Such models read one token (or a small number of tokens) at a time, update the recurrent state using these inputs, and then discard them before the next token arrives. The output at each step is generated based only on the recurrent state, rather than on all the previous states. The memory footprint is determined by the recurrent state which has a fixed size. Recurrent models can be used in real-time learning scenarios where data arrives in a stream and predictions can be made at any time step. In NLP, applying recurrent models

[9] In the new space after this transformation, the Softmax normalization can be transformed into the simple scaling normalization.

to language modeling is one of the earliest successful attempts to learn representations of sequences. Although Transformers have been used as the foundational architecture in LLMs, recurrent models are still powerful models, especially for developing efficient LLMs. For example, recurrent models, particularly modern state space models like Mamba [Gu and Dao, 2023], have seen a resurgence in language modeling and have been reconsidered as a promising alternative to Transformers.

Figure 8.5 shows a comparison of the models discussed in this subsection. Since these models, along with others not mentioned here, have been extensively discussed in Chapter 6 and in related surveys [Tay et al., 2020b], a more exhaustive discussion is omitted here.

8.3.3 Cache and Memory

LLMs based on the standard Transformer architecture employ global (dense) attention mechanisms. During autoregressive inference, these models require storing the entire left-context to make predictions for future tokens. This requires a KV cache, where the representations (i.e., keys and values) of all previously generated tokens are retained. As a result, the memory footprint of this cache grows linearly as inference proceeds. In the preceding sections, we discussed methods for mitigating this context bottleneck via efficient attention algorithms, such as sparse and linear attention. A related alternative approach is to explicitly encode and compress the context via an auxiliary memory module.

1. Fixed-size KV Cache

A straightforward approach is to represent the keys and values using a fixed-size memory model. Suppose we have a memory Mem which retains the contextual information. We can write the attention operation at position i in a general form

$$\mathrm{Att}(\mathbf{q}_i, \mathrm{Mem}) \quad = \quad \mathrm{Att}_{\mathrm{qkv}}(\mathbf{q}_i, \mathbf{K}_{\leq i}, \mathbf{V}_{\leq i}). \tag{8.52}$$

In this model, Mem is simply the KV cache, i.e., $\mathrm{Mem} = (\mathbf{K}_{\leq i}, \mathbf{V}_{\leq i})$. Thus the size of Mem is determined by i. If Mem is defined to have a fixed size, then the cost of performing $\mathrm{Att}(\mathbf{q}_i, \mathrm{Mem})$ will be fixed. There are several alternative ways to design Mem.

- One of the simplest methods is to consider a fixed-size window of previous keys and values. Mem is therefore given by

$$\mathrm{Mem} \quad = \quad (\mathbf{K}_{[i-n_c+1,i]}, \mathbf{V}_{[i-n_c+1,i]}), \tag{8.53}$$

  where n_c denotes the size of the window. The notation $\mathbf{K}_{[i-n_c+1,i]}$ and $\mathbf{V}_{[i-n_c+1,i]}$ denote the keys and values over positions from $i - n_c + 1$ to i.[10] This model can be seen as a type of local attention model.

[10] More formally, we write $\mathbf{K}_{[i-n_c+1,i]} = \begin{bmatrix} \mathbf{k}_{i-n_c+1} \\ \vdots \\ \mathbf{k}_i \end{bmatrix}$ and $\mathbf{V}_{[i-n_c+1,i]} = \begin{bmatrix} \mathbf{v}_{i-n_c+1} \\ \vdots \\ \mathbf{v}_i \end{bmatrix}$. Sometimes we denote $\mathbf{K}_{[i-n_c+1,i]}$ by $\{\mathbf{k}_{i-n_c+1}, \cdots, \mathbf{k}_i\}$ and $\mathbf{V}_{[i-n_c+1,i]}$ by $\{\mathbf{v}_{i-n_c+1}, \cdots, \mathbf{v}_i\}$ for notational simplicity.

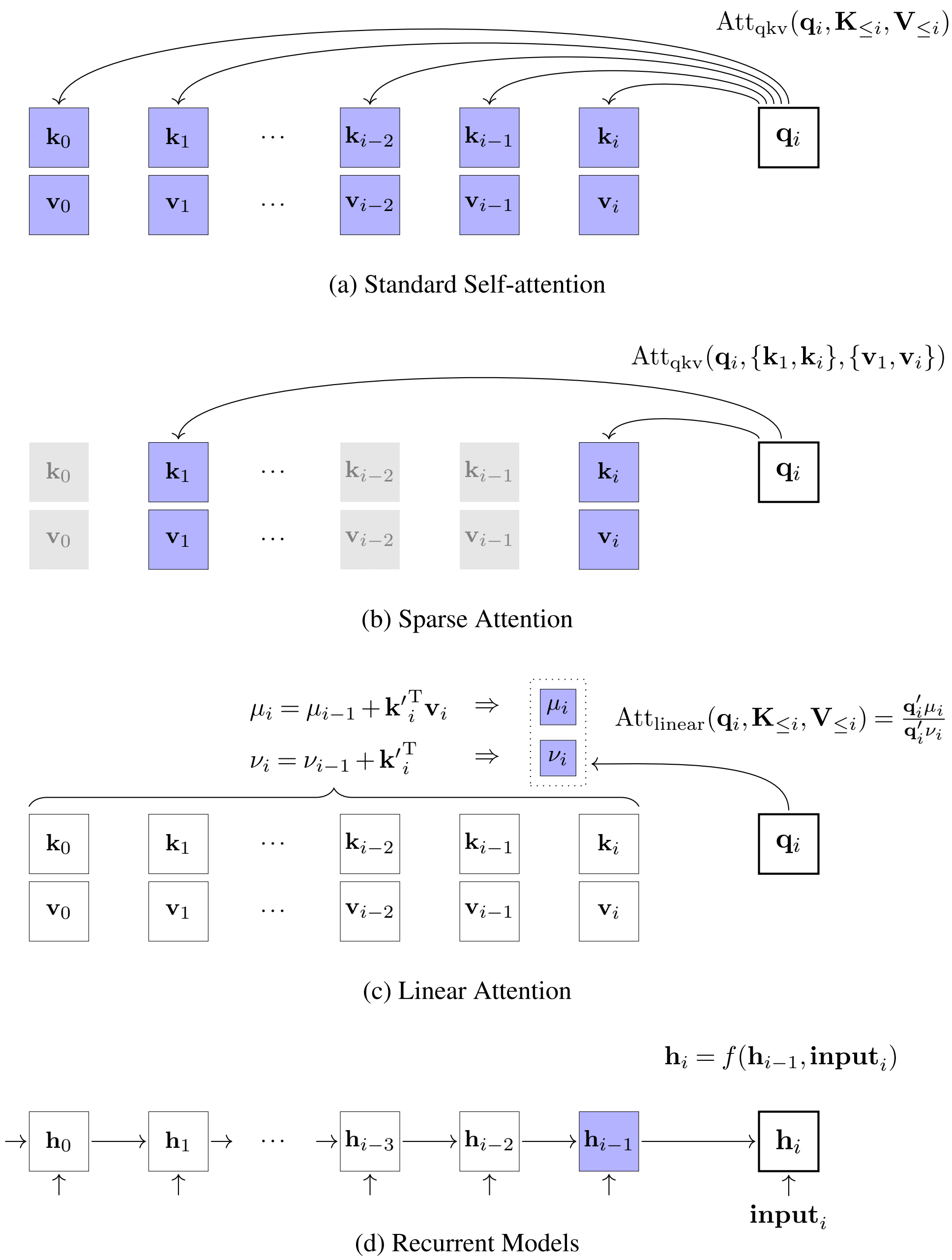


Figure 8.5: Illustrations of self-attention, sparse attention, linear attention and recurrent models. Blue boxes = cached states for producing the output at position i. $f(\cdot)$ = a recurrent cell.

- It is also possible to define Mem as a pair of summary vectors, which leads to a more compressed representation of the history. A simple way to summarize the previous keys and values is to use the moving averages of these vectors. For example, Mem can be defined as the unweighted moving average of the previous n_c keys and values

$$\mathrm{Mem} = \left(\frac{\sum_{j=i-n_c+1}^{i}\mathbf{k}_j}{n_c}, \frac{\sum_{j=i-n_c+1}^{i}\mathbf{v}_j}{n_c}\right). \tag{8.54}$$

Alternatively, we can use a weighted moving average

$$\mathrm{Mem} = \left(\frac{\sum_{j=i-n_c+1}^{i}\beta_{j-i+n_c}\mathbf{k}_j}{\sum_{j=1}^{n_c}\beta_j}, \frac{\sum_{j=i-n_c+1}^{i}\beta_{j-i+n_c}\mathbf{v}_j}{\sum_{j=1}^{n_c}\beta_j}\right). \tag{8.55}$$

Here $\{\beta_1, \cdots, \beta_{n_c}\}$ are the coefficients, which can either be learned as model parameters or determined via heuristics. For example, they can be set to increasing coefficients (i.e., $\beta_1 < \beta_2 < \cdots < \beta_{n_c-1} < \beta_{n_c}$) in order to give larger weight to positions that are closer to i. We can extend the moving average to include all the positions up to i. This leads to the cumulative average of the keys and values, given in the form

$$\mathrm{Mem} = \left(\frac{\sum_{j=0}^{i}\mathbf{k}_j}{i+1}, \frac{\sum_{j=0}^{i}\mathbf{v}_j}{i+1}\right). \tag{8.56}$$

In general, the cumulative average can be written in recursive form

$$\mathrm{Mem}_i = \frac{(\mathbf{k}_i, \mathbf{v}_i) + i \cdot \mathrm{Mem}_{i-1}}{i+1}, \tag{8.57}$$

where Mem_i and Mem_{i-1} denote the cumulative averages of the current and previous positions, respectively. An advantage of this model is that we only need to store a single key-value pair during inference, rather than storing all the key-value pairs. Note that the above memory models are related to recurrent models, and more advanced techniques have been used to develop alternatives to self-attention mechanisms in Transformers [Ma et al., 2023].

- The memory Mem can also be a neural network. At each step, it takes both the previous output of the memory and the current states of the model as input, and produces the new output of the memory. This neural network can be formulated as the function

$$\mathrm{Mem} = \mathrm{Update}(S_{\mathrm{kv}}, \mathrm{Mem}_{\mathrm{pre}}). \tag{8.58}$$

Here Mem and $\mathrm{Mem}_{\mathrm{pre}}$ represent the outputs of the memory at the current step and the previous step, respectively. S_{kv} is a set of key-value pairs, representing the recent states of the model. This formulation is general and allows us to develop various memory models by selecting different $\mathrm{Update}(\cdot)$ and S_{kv} configurations. For example, if S_{kv} only contains the latest key-value pair $(\mathbf{k}_i, \mathbf{v}_i)$ and $\mathrm{Update}(\cdot)$ is defined as a recurrent cell, then Eq. (8.58) can be expressed as an RNN-like model

$$\mathrm{Mem} = f((\mathbf{k}_i, \mathbf{v}_i), \mathrm{Mem}_{\mathrm{pre}}), \tag{8.59}$$

where $f(\cdot)$ is a recurrent cell. Recurrence can also be applied to segment-level modeling for efficiency considerations. A simple approach is to divide the sequence into segments, and treat S_{kv} as a segment. Applying recurrent models to $\text{Update}(\cdot)$ will result in memory models that operate on segments. A special case is to define $\text{Update}(\cdot)$ as an FIFO function that adds S_{kv} into the memory and removes the oldest key-value segment from the memory, given by

$$\text{Mem} = \text{FIFO}(S_{\text{kv}}, \text{Mem}_{\text{pre}}). \tag{8.60}$$

Consider a memory which includes two segments, one for current segment, and one for the previous segment. In the attention operation, each position can access the history key-value pairs in the two most recent consecutive segments. While conceptually simple, this sliding-chunk local memory and its variants have been widely adopted in segment-level recurrent architectures [Dai et al., 2019; Hutchins et al., 2022; Bulatov et al., 2022].

- The above memory models can be extended to involve multiple memories. An example of this approach is compressive Transformer [Rae et al., 2019b]. It employs two distinct fixed-size memories: one for modeling local context (denoted by Mem), and the other for modeling and compressing long-term history (denoted by CMem). The KV cache in this model is the combination of Mem and CMem. The attention function can be written as

$$\text{Att}_{\text{com}}(\mathbf{q}_i, \text{Mem}, \text{CMem}) = \text{Att}_{\text{qkv}}(\mathbf{q}_i, [\text{Mem}, \text{CMem}]), \tag{8.61}$$

where $[\text{Mem}, \text{CMem}]$ is a combined memory of Mem and CMem. Similar to standard segment-level models, the compressive Transformer processes the sequence in blocks. Each segment comprises n_s consecutive tokens, and S_{kv}^k denotes the raw key-value pairs generated from the k-th segment. Upon the arrival of a new segment, Mem is updated via a FIFO queue: it appends the n_s new key-value pairs from S_{kv}^k and simultaneously removes the n_s oldest pairs:

$$\text{Mem} = \text{FIFO}(S_{\text{kv}}^k, \text{Mem}_{\text{pre}}). \tag{8.62}$$

Rather than discarding these ejected pairs, they are passed through a compression network. The n_s raw pairs are compressed into a smaller set of $\frac{n_s}{c}$ condensed key-value pairs. Then CMem serves as a secondary FIFO queue, appending these $\frac{n_s}{c}$ compressed pairs and discarding its oldest $\frac{n_s}{c}$ entries to maintain fixed size. It is given by

$$\text{CMem} = \text{FIFO}(C_{\text{kv}}^k, \text{CMem}_{\text{pre}}), \tag{8.63}$$

where C_{kv}^k represents the set of compressed key-value pairs. The compressive Transformer implicitly assumes that local context should be represented explicitly with minimal information loss, while long-range context can be more compressed.

- We have already seen that both global and local contexts are useful and can be modeled using attention models. This view motivates the extension to attention models for combining both local and long-term memories [Ainslie et al., 2020; Zaheer et al., 2020; Gupta and Berant, 2020]. A simple but widely-used approach is to involve the first few tokens of the sequence in attention, which serves as global tokens. This approach is usually applied along with other sparse attention models. An advantage of incorporating global tokens is that it serves as an attention sink that absorbs excessive attention scores, and thus stabilizes model performance when the context size is very large [Xiao et al., 2024]. One drawback, however, is that using a fixed-size global memory may result in information loss. When dealing with long sequences, we need to enlarge the KV cache for sufficient representations of the context, but this in turn increases the computational cost.

Figure 8.6 shows illustrations of the above approaches. Note that, while we focus on optimization of the KV cache here, this issue is closely related to those discussed in the previous section. All of the methods we have mentioned so far can broadly be categorized as efficient attention approaches, which are widely used in various Transformer variants.

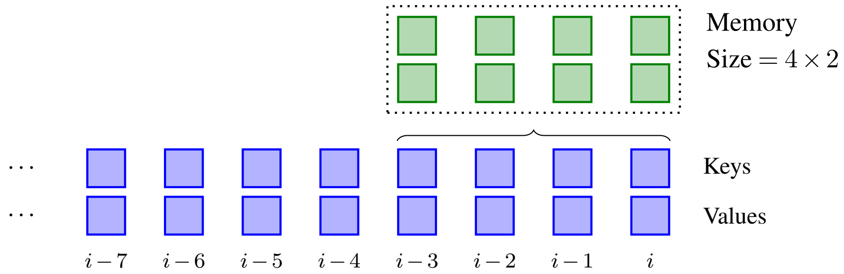
2. Memory-based Models

The memory modeling discussed above is based on updates to the KV cache, and the resulting models are typically referred to as **internal memories**. We now consider another family of models, called **external memories**, which operate independently to access large-scale contexts for LLMs. Many of these approaches are rooted in **memory-based methods**, a concept extensively studied in classical machine learning [Bishop, 2006]. A common example is nearest-neighbor algorithms: we store context representations in a datastore and query it to find the most similar stored representations. These retrieved context representations are then incorporated to enhance the attention mechanism for the current query.

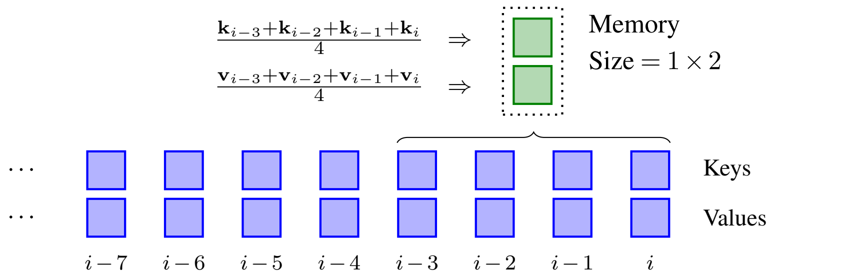
Here, we consider the k**-nearest neighbors** (k**-NN**) method which is one of the most popular memory-based methods. Since our focus is language modeling in this section, we define a sample in the datastore as a key-value pair corresponding to some context state. Note that "context" is a broad concept here, not just a sequence prefix in text generation. One might, for example, view the entire dataset as the context for predicting tokens. This allows us to retrieve the most similar context in a set of sequences, rather than a given sequence prefix. Although we will restrict ourselves to context modeling for a single sequence, in this subsection, we discuss a more general case.

Suppose we have a set of keys $\{\mathbf{k}_j\}$ with corresponding values $\{\mathbf{v}_j\}$, and suppose we store these key-value pairs in a vector database[11]. For each query $\mathbf{q}_i$, we find its k nearest neighbors by growing the radius of the sphere centered at $\mathbf{q}_i$ until it contains k data points in $\{\mathbf{k}_j\}$. This results in a set of k keys along with their corresponding values, denoted by Mem_{knn}. As before, we denote Mem as the local memory for the query, such as the KV cache of neighboring tokens. Our goal is to allow the query $\mathbf{q}_i$ to attend to both the local memory

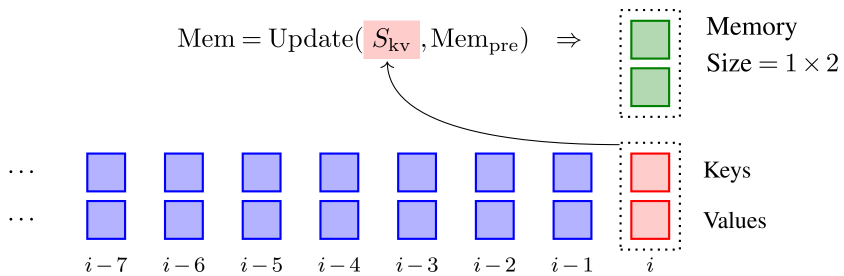
[11] A vector database, or vector store, is a database that provides highly optimized retrieval interfaces for finding stored vectors that closely match a query vector.



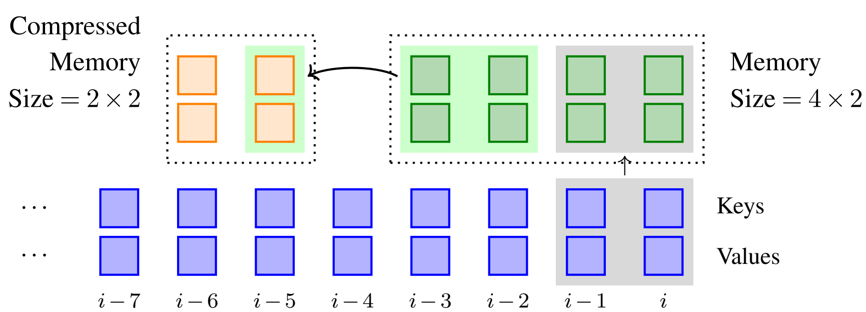
(a) Window-based Cache



(b) Moving Average-based Cache



(c) Recurrent Network as Cache



(d) Hybrid Cache (Compressed Memory + Local Memory)

Figure 8.6: Illustrations of fixed-size KV caches in LLMs. Blue boxes represent the keys and values generated during LLM inference, green boxes represent the keys and values stored or encoded in the primary memory, and orange boxes represent the keys and values stored or encoded in the compressed memory.

Mem and the long-term memory Mem_{knn}. There are, of course, several ways to incorporate Mem and Mem_{knn} into the attention model. For example, we might simply combine them to form a single KV cache $[\mathrm{Mem}, \mathrm{Mem}_{knn}]$, and attend $\mathbf{q}_i$ to $[\mathrm{Mem}, \mathrm{Mem}_{knn}]$ via standard QKV attention. Or we might use Mem and Mem_{knn} in separate attention steps. An example of such an approach is the model developed by Wu et al. [2021]. It linearly combines the two types of attention, given by

$$\begin{aligned}
\mathrm{Att}(\mathbf{q}_i, \mathrm{Mem}, \mathrm{Mem}_{knn}) &= \mathbf{g} \odot \mathrm{Att}_{\mathrm{local}} + (1 - \mathbf{g}) \odot \mathrm{Att}_{knn} & (8.64) \\
\mathrm{Att}_{\mathrm{local}} &= \mathrm{Att}(\mathbf{q}_i, \mathrm{Mem}) & (8.65) \\
\mathrm{Att}_{knn} &= \mathrm{Att}(\mathbf{q}_i, \mathrm{Mem}_{knn}). & (8.66)
\end{aligned}$$

Here $\mathbf{g} \in \mathbb{R}^d$ is the coefficient vector, which can be the output of a learned gate.

Given the k-NN-based memory model described above, the remaining task is to determine which key-value pairs are retained in the datastore. For standard language modeling tasks, we consider the previously seen tokens in a sequence as the context, so we can add the keys and values of all these tokens into the datastore. In this case, the resulting k-NN-based attention model is essentially equivalent to a sparse attention model [Gupta et al., 2021].

Alternatively, we can extend the context from one sequence to a collection of sequences. For example, we might collect all key-value pairs across the sequences in a training dataset and add them to the datastore to model a larger context. Thus, LLMs can predict tokens based on a generalized context. A problem with this approach is that the computational cost can be large if many sequences are involved. Since these sequences are part of our training data, we can build and optimize an index for the vectors in the datastore before running the LLMs. As a result, the retrieval of similar vectors can be very efficient, as commonly implemented in modern vector databases.

In fact, all the aforementioned methods represent instances of the broader retrieval-augmented paradigm. Beyond directly injecting retrieved vectors into the attention mechanism, retrieval can be applied at the output distribution level. A typical application is **k-NN language modeling** (**k-NN LM**) [Khandelwal et al., 2020]. The intuition is that similar hidden states within a Transformer network are highly predictive of similar next tokens across different contexts. In a k-NN LM, each item in the datastore is stored as a tuple $(\mathbf{z}, w)$, where $\mathbf{z}$ represents the final hidden state of the LLM at a specific position, and w represents the actual next token observed at that position. Typically, this datastore is constructed offline by passing the entire training dataset through the LLM.

During inference, given a sequence prefix, the LLM generates a current hidden state $\mathbf{h}_i$. We query the datastore using $\mathbf{h}_i$ to retrieve the k closest matching states, yielding a set of tuples $\{(\mathbf{z}_1, w_1), \cdots, (\mathbf{z}_k, w_k)\}$. The target tokens $\{w_1, \cdots, w_k\}$ serve as reference outputs. To utilize these references, we convert the retrieval distances into a probability distribution over the vocabulary V. Let $d(\mathbf{h}_i, \mathbf{z}_j)$ be the distance metric. The probability mass assigned to a token $v \in V$ by the k-NN retrieval is computed by aggregating the softmax-normalized similarities of the retrieved neighbors:

$$\Pr_{k\text{nn}}(v|\mathbf{h}_i) \quad \propto \sum_{\substack{(\mathbf{z}_j, w_j) \in \text{top-}k, \\ w_j = v}} \exp\left(-d(\mathbf{h}_i, \mathbf{z}_j)\right). \tag{8.67}$$

If a token v does not appear among the k retrieved neighbors, its probability $\Pr_{k\text{nn}}(v|\mathbf{h}_i)$ is zero. Finally, we use a tunable hyperparameter $\lambda \in [0,1]$ to linearly interpolate between this retrieval-based distribution and the standard parametric distribution $\Pr_{\text{lm}}(\cdot|\mathbf{h}_i)$ generated by the LLM's classification head:

$$\Pr(\cdot|\mathbf{h}_i) \quad = \quad \lambda \cdot \Pr_{k\text{nn}}(\cdot|\mathbf{h}_i) + (1-\lambda) \cdot \Pr_{\text{lm}}(\cdot|\mathbf{h}_i). \tag{8.68}$$

Then, as usual, we can select the next token y by maximizing the probability $\Pr(y|\mathbf{h}_i)$.

As with information retrieval (IR) systems, the datastore can store and retrieve textual data. For example, we can store a collection of text documents in a search engine with full-text indexing, and then search it for documents that match a given text-based query. Applying IR techniques to LLMs leads to a general framework called **retrieval-augmented generation** (**RAG**). The RAG framework works as follows. We use the context $\mathbf{x}$ as the query and find the k most relevant document chunks $\{\mathbf{c}_1, \cdots, \mathbf{c}_k\}$ from the datastore via efficient IR techniques[12]. These search results are combined with the original context via a prompting template $g(\cdot)$[13], resulting in an augmented input for the LLM

$$\mathbf{x}' \quad = \quad g(\mathbf{c}_1, \cdots, \mathbf{c}_k, \mathbf{x}). \tag{8.69}$$

Then, we use $\mathbf{x}'$ as the context and predict the following text using the model $\Pr(\mathbf{y}|\mathbf{x}')$. One advantage of RAG is that we need not modify the architecture of LLMs, but instead augment the input to LLMs via an additional IR system. Figure 8.7 shows a comparison of the use of different external memories in LLMs.

3. Memory Capacity

A memory model in LLMs, whether in the form of a simple key-value cache or an external datastore, can broadly be viewed as an encoder of contextual information. Ideally, before we say that a memory model is representative of the entire context in token prediction, we must ensure that the model can accurately reconstruct any part of that context. The standard KV cache is one such model, as it retains the complete sequence history. In this case, the model is said to possess sufficient (or unbounded) capacity for memorizing the context. In many

[12] In practical applications, queries are typically generated using a query generation system, which may expand it with variations of tokens and query intent.

[13] For example, the template could be:

> message = {*$\mathbf{c}_1$*} $\cdots$ {*$\mathbf{c}_k$*}
> input: {*$\mathbf{x}$*}
> output: ____

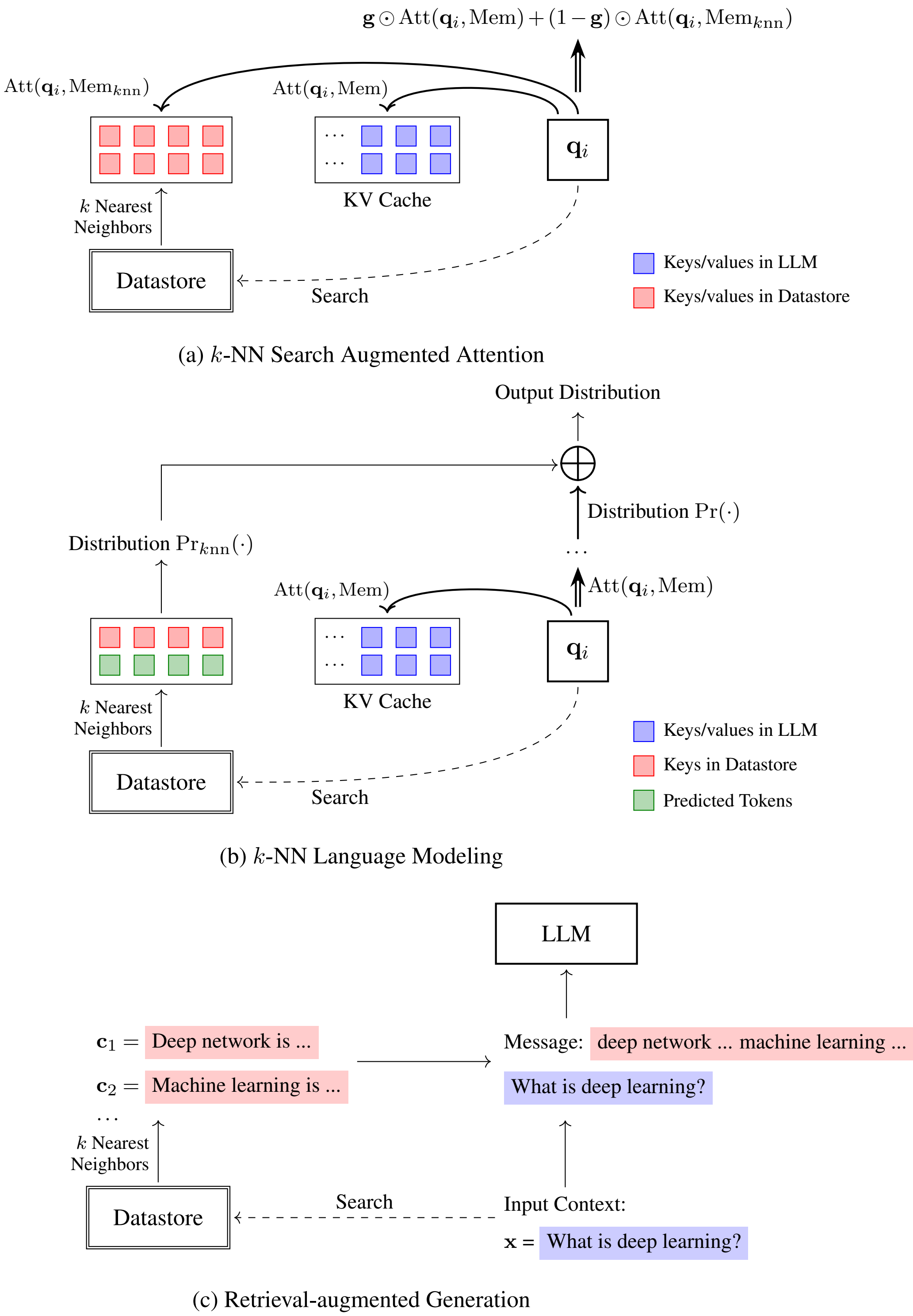


Figure 8.7: Illustrations of external memories (or datastores) for language modeling.

practical applications, however, complete memorization is unnecessary. Instead, the primary goal is to enable LLMs to access salient contextual information. As a result, efficient and compressed memory models have been developed, as described in this section. Note that the longer the sequence, the more difficult it becomes for a low-capacity memory model to capture crucial contextual information. It is therefore common practice to simply scale up the memory capacity when processing long contexts.

While high-capacity memory models are generally desirable, they are computationally expensive to train and deploy. A challenging scenario occurs when tokens arrive in a continuous stream, causing the context to grow indefinitely. Developing LLMs for such streaming tasks is difficult, as it requires training Transformers on extremely long sequences. One potential solution is to use non-parametric approaches, such as retrieval-based methods. For instance, as discussed previously, we can employ a vector database to store all generated key-value pairs, and thus effectively represent the context via an external memory model. Although this approach sidesteps the inherent context-length limitations of Transformers, the construction and maintenance of external memory models lead to significant computational overhead. Furthermore, these models are typically applied to problems where the context is provided in advance and remains static during inference, and are hence unsuitable for dynamic, streaming context modeling.

In cases where the size of the context continuously grows, applying fixed-size memory models is a commonly used approach. For example, in recurrent models, a sequence of arbitrary length can be summarized into a set of hidden states, resulting in a fixed computational cost per step. While recurrent models were initially found to be not very good at handling long-distance dependencies in sequence modeling in early applications of deep learning to NLP, recent advancements have shown that their variants are now effective in modeling extremely long sequences. [Bulatov et al., 2022; Hutchins et al., 2022; Munkhdalai et al., 2024; Ma et al., 2024].

There is no general definition of memory capacity in LLMs. A simple approach might consider how much storage is used to retain contextual information. For example, memory capacity could be defined by the size of the KV cache in Transformers or the vector database used in retrieval-based methods. A related concept is model complexity. In machine learning, there are several ways to define the model complexity. One of the simplest methods is by counting the number of parameters. However, it should be emphasized that the memory models discussed here primarily serve to store information, rather than add trainable parameters. Therefore, a model with a large memory capacity is not necessarily more complex. Nevertheless, in practice determining the capacity of a memory model is not straightforward. In general, we need to consider the trade-off between maximizing the performance and controlling the memory footprint.

8.3.4 Sharing across Heads and Layers

In Transformers, the KV cache is a data structure that can be dynamically adjusted along multiple dimensions, such as heads, layers, and sequence length. For example, consider an LLM with L layers. Each layer has τ attention heads, and each head produces a d_h-

dimensional output. During inference, we store the keys and values for up to m tokens. The space complexity of this caching mechanism is $O(L \cdot \tau \cdot d_h \cdot m)$. As we have seen previously, this complexity can be reduced by caching the keys and values for fewer tokens. For example, in sliding window attention, a fixed-size window is used to cache the keys and values in local context. The sliding-window model has a space complexity of $O(L \cdot \tau \cdot d_h \cdot m_w)$, with m_w being the size of the window.

In addition to reducing m, we can also decrease the size of the KV cache along other dimensions. A widely-used approach is to enable sharing across heads in multi-head self-attention. Recall from Section 8.1.1 that multi-head self-attention uses multiple sets of queries, keys, and values (each set is called a head), each performing the QKV attention mechanism as usual. This can be expressed as

$$\text{Output} \quad = \quad \text{Merge}(\text{head}_1, \cdots, \text{head}_\tau)\mathbf{W}^{\text{head}}, \tag{8.70}$$

where $\text{head}_j \in \mathbb{R}^{d_h}$ is the output of the j-th head, computed using the standard QKV attention function

$$\text{head}_j = \text{Att}_{\text{qkv}}(\mathbf{q}_i^{[j]}, \mathbf{K}_{\leq i}^{[j]}, \mathbf{V}_{\leq i}^{[j]}). \tag{8.71}$$

Here, $\mathbf{q}_i^{[j]}$ denotes the query vector at position i, while $\mathbf{K}_{\leq i}^{[j]}$ and $\mathbf{V}_{\leq i}^{[j]}$ denote the corresponding key and value matrices up to position i, with each vector projected into the j-th feature subspace. So this multi-head attention model can be interpreted as performing attention on a group of feature subspaces in parallel (see Figure 8.8 (b)). The KV cache needs to retain the keys and values for all these heads, that is, $\{(\mathbf{K}_{\leq i}^{[1]}, \mathbf{V}_{\leq i}^{[1]}), \cdots, (\mathbf{K}_{\leq i}^{[\tau]}, \mathbf{V}_{\leq i}^{[\tau]})\}$.

One refinement of the multi-head attention model, called **multi-query attention** (**MQA**), is to share keys and values across heads, while maintaining distinct queries for each head [Shazeer, 2019]. In MQA, there is a single set of keys and values $(\mathbf{K}_{\leq i}, \mathbf{V}_{\leq i})$. In addition, there are τ queries $\{\mathbf{q}_i^{[1]}, \cdots, \mathbf{q}_i^{[\tau]}\}$, each corresponding to a different head. For each head, we have

$$\text{head}_j = \text{Att}_{\text{qkv}}(\mathbf{q}_i^{[j]}, \mathbf{K}_{\leq i}, \mathbf{V}_{\leq i}). \tag{8.72}$$

Figure 8.8 (c) illustrates this model. By sharing keys and values, the size of the KV cache would be $O(L \cdot d_h \cdot m)$.

Grouped query attention (**GQA**) is a natural extension to multi-head attention and MQA [Ainslie et al., 2023]. In GQA, heads are divided into n_g groups, each corresponding to a shared set of keys and values. Hence we have n_g sets of keys and values $\{(\mathbf{K}_{\leq i}^{[1]}, \mathbf{V}_{\leq i}^{[1]}), \cdots, (\mathbf{K}_{\leq i}^{[n_g]}, \mathbf{V}_{\leq i}^{[n_g]})\}$. See Figure 8.8 (d) for an illustration. Let $g(j)$ be the group id for the j-th head. The GQA model can be expressed as

$$\text{head}_j = \text{Att}_{\text{qkv}}(\mathbf{q}_i^{[j]}, \mathbf{K}_{\leq i}^{[g(j)]}, \mathbf{V}_{\leq i}^{[g(j)]}). \tag{8.73}$$

The size of the KV cache of GQA is $O(L \cdot n_g \cdot d_h \cdot m)$. One benefit of GQA is that we can trade off between computational efficiency and model expressiveness by adjusting n_g. When $n_g = \tau$,

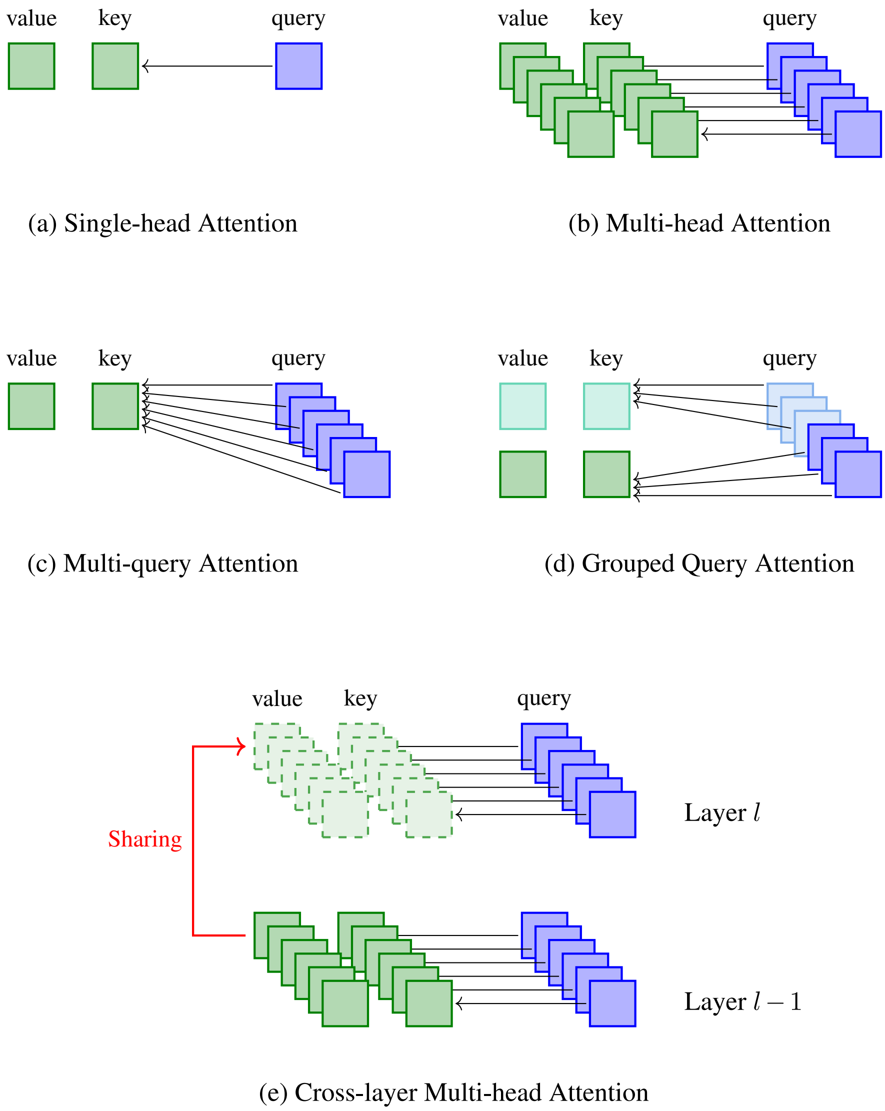


Figure 8.8: Illustration of QKV attention based on different multi-head and sharing mechanisms. (a) = single-head attention, and (b-e) = attention with multiple heads.

the model becomes the standard multi-head attention model. By contrast, when $n_g = 1$, it becomes the MQA model.

Sharing can also be performed across layers. Such a method falls into the family of shared weight and shared activation methods, which have been extensively used in Transformers [Dehghani et al., 2018; Lan et al., 2020]. For example, one can share KV activations or attention weights across layers to reduce both computation and memory footprints [Xiao et al., 2019; Brandon et al., 2024]. Figure 8.8 (e) shows an illustration of this method, where a query in a layer directly accesses the KV cache of a lower-level layer.

8.3.5 Position Extrapolation and Interpolation

Since Transformer layers are order-invariant with respect to input, we need some way to encode positional information in the input tokens. To do this, it is common to add positional embeddings to token embeddings, and then feed these combined embeddings into the Transformer layer stack. In this case, the embedding at position i can be expressed as

$$\mathbf{e}_i = \mathbf{x}_i + \mathrm{PE}(i), \tag{8.74}$$

where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the token embedding, and $\mathrm{PE}(i) \in \mathbb{R}^d$ denotes the positional embedding at position i. Because the token embedding $\mathbf{x}_i$ is position-agnostic, the positional embedding $\mathrm{PE}(i)$ encodes the absolute or relative context of the token. A straightforward implementation is to treat $\mathrm{PE}(i)$ as a set of learnable variables, trained jointly with the rest of the model parameters. This allows the model to learn a unique vector representation for every discrete position, and thus distinguish tokens based on where they appear in a sequence.

Using learned vectors to represent absolute positions works reasonably well when the sequence lengths encountered during inference closely match those seen during training. In practical LLM pre-training, however, we must impose strict maximum length limits (e.g., m_l) on training sequences to maintain computational tractability, yet we often desire to apply the trained model to significantly longer sequences during downstream deployment. In such scenarios, using learned positional embeddings has obvious drawbacks, as there are no trained embeddings for positions that are not observed in the training phase.

An alternative approach to modeling positional information is to develop positional embeddings that can generalize: once trained, the embedding model can be used to handle longer sequences. Suppose that we train a positional embedding model on sequences with a maximum length of m_l, and we wish to apply the trained model to a sequence of length m ($m \gg m_l$). If the model is only trained on a limited range of positions, then this model will simply fail to deal with new data outside that range. See Figure 8.9 (a) for an illustration where the learned embedding model cannot model data points outside the training domain if it lacks the ability to extrapolate.

There are several approaches to making positional embedding models generalize. They can be grouped into two classes.

- **Extrapolation**. The model learned on observed data points (i.e., positions) can be directly employed to assign meaningful values to data points beyond the original range. For instance, if a model learns a continuous functional relationship for positions $\{1, \cdots, 10\}$, it can theoretically evaluate that same function at position 15, maintaining the correct relative distance and geometric properties with respect to earlier tokens. Figure 8.9 (b) illustrates this concept, where a function fitted to the training domain smoothly extends to estimate values for out-of-distribution coordinates.
- **Interpolation**. This approach maps a larger range of data points into the original observation range. For example, suppose we have a model designed for numbers in the range $[1, 10]$. When given a new range of $[1, 20]$, we can scale this down by dividing

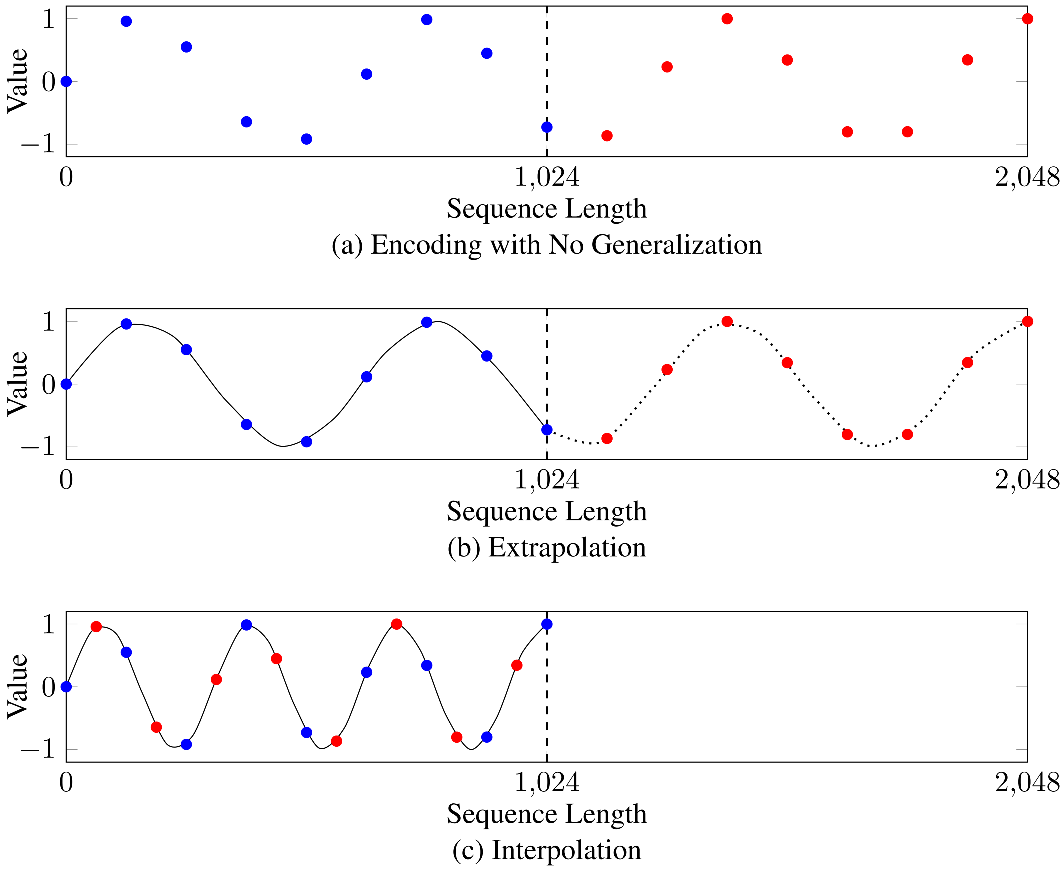


Figure 8.9: Illustrations of different positional embedding behaviors across a sequence length continuum. Blue points denote positions explicitly observed during training, while red points denote unseen positions encountered at inference time. In sub-figure (a), the embedding model memorizes the training domain and cannot generalize. In sub-figures (b) and (c), the model successfully generalizes to unseen lengths via extrapolation and interpolation, respectively.

every number by 2, thereby fitting all numbers into $[1,10]$. This scaling allows us to use the model trained on the range $[1,10]$ to describe data points in the expanded range of $[1,20]$. See Figure 8.9 (c) for an illustration of this approach.

In fact, positional embeddings in many systems have achieved some level of generalization. For example, sinusoidal encoding, the most common positional embedding method, employs sine and cosine functions that can naturally extend to sequences of any length. Although this approach seems direct and simple, it does not perform well when we significantly extend the sequences for processing. In this subsection, we will discuss several alternative methods based on either extrapolation or interpolation.

1. Attention with Learnable Biases

One problem with Eq. (8.74) is that the embedding model treats each token independently and therefore ignores the distance between different tokens. A common improvement to this model, known as relative positional embedding, is to consider the pairwise relationship between tokens [Shaw et al., 2018]. The general idea behind this is to compute the offset between any pair of positions and incorporate it into the self-attention model. One of the simplest forms of self-attention with relative positional embedding is given by

$$\mathrm{Att}_{\mathrm{qkv}}(\mathbf{q}_i, \mathbf{K}_{\leq i}, \mathbf{V}_{\leq i}) = \sum_{j=0}^{i} \alpha(i,j)\mathbf{v}_j \tag{8.75}$$

$$\alpha(i,j) = \mathrm{Softmax}\left(\frac{\mathbf{q}_i \mathbf{k}_j^{\top} + \mathrm{PE}(i,j)}{\sqrt{d}} + \mathrm{Mask}(i,j)\right). \tag{8.76}$$

The only difference between this model and the original self-attention model is that a bias term $\mathrm{PE}(i,j)$ is added to the query-key product in this new model. Intuitively, $\mathrm{PE}(i,j)$ can be interpreted as a distance penalty for the pair of positions i and j. As the distance between i and j increases, $\mathrm{PE}(i,j)$ decreases.

$\mathrm{PE}(i,j)$ can be defined in several different ways. Here, we consider the T5 version of relative positional embedding, called the T5 bias [Raffel et al., 2020]. For each pair of query $\mathbf{q}_i$ and key $\mathbf{k}_j$, the offset between them is defined to be[14]

$$d(i,j) = i - j. \tag{8.77}$$

A naive formulation for the bias $\mathrm{PE}(i,j)$ would assign a distinct learnable scalar for every possible discrete offset, i.e., $\mathrm{PE}(i,j) = u_{i-j}$, where u_{i-j} is the parameter tied to the exact distance $i-j$. However, simply assigning a unique value to each offset will restrict the model to offsets observed during training. When $i-j$ is larger than the maximum trained offset, the model cannot generalize.

The T5 bias introduces a generalized version of this approach. Rather than allocating a unique parameter to every possible integer offset, it groups these positional offsets into distinct "buckets," where all offsets mapping to the same bucket share a single learnable parameter. Specifically, given n_b buckets, the allocation strategy is defined as follows:

- For the first half of the buckets (indices 0 to $\lfloor \frac{n_b}{2} \rfloor - 1$), the mapping is one-to-one. Each offset is mapped to a unique bucket: bucket $0 \leftrightarrow$ offset 0, bucket $1 \leftrightarrow$ offset 1, and so on. We can express this linear mapping as $b(i-j) = i-j$.
- For the remaining half of the buckets (indices $\lfloor \frac{n_b}{2} \rfloor$ to $n_b - 1$), the bucket size increases logarithmically. As a result, larger offsets are grouped into progressively wider bins. For an offset $i - j \geq \lfloor \frac{n_b}{2} \rfloor$, the bucket index is computed as:

$$b(i-j) = \lfloor \frac{n_b}{2} \rfloor + \left\lfloor \frac{\log(i-j) - \log(\lfloor \frac{n_b}{2} \rfloor)}{\log(\mathrm{dist}_{\max}) - \log(\lfloor \frac{n_b}{2} \rfloor)} \cdot \left(n_b - \lfloor \frac{n_b}{2} \rfloor - 1\right) \right\rfloor, \tag{8.78}$$

[14]For language modeling, a query is only allowed to attend to its left-context, and so we have $i - j \geq 0$. In the more general case of self-attention, where a token can attend to all tokens in the sequence, we may have negative offsets when $i < j$.

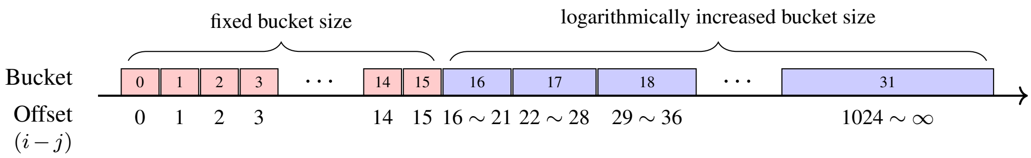


Figure 8.10: Illustration of distributing query-key offsets into buckets in the T5 model ($n_b = 32$ and $\text{dist}_{\text{max}} = 1024$). Boxes represent buckets. In the first half of the buckets, we use a fixed bucket size. In the second half of the buckets, we increase the bucket size logarithmically. The last bucket contains all the query-key offsets that are not covered by previous buckets.

where dist_{max} is a hyperparameter indicating the maximum expected offset threshold.

- When $i - j \geq \text{dist}_{\text{max}}$, the offset is clipped and placed into the final bucket $n_b - 1$. In other words, this bucket serves as a catch-all for any arbitrarily large distances extending beyond the expected sequence length.

Together, the bucketing function is defined as

$$\begin{aligned} & b(i-j) \\ = \; & \begin{cases} i-j & \text{if } 0 \leq i-j < \lfloor \frac{n_b}{2} \rfloor, \\ \min\left(n_b - 1, \lfloor \frac{n_b}{2} \rfloor + \left\lfloor \frac{\log(i-j) - \log(\lfloor \frac{n_b}{2} \rfloor)}{\log(\text{dist}_{\text{max}}) - \log(\lfloor \frac{n_b}{2} \rfloor)} \cdot \left(n_b - \lfloor \frac{n_b}{2} \rfloor - 1\right) \right\rfloor \right) & \text{if } i-j \geq \lfloor \frac{n_b}{2} \rfloor. \end{cases} \end{aligned} \tag{8.79}$$

Figure 8.10 shows an illustration of these buckets. We see that in the first half of the buckets, each bucket is associated with only one value of $i - j$, while in the second half, the bucket size increases as $i - j$ grows. The last bucket is designed to handle sequences of arbitrary length.

All relative distances mapping to the same bucket $b(i - j)$ share an identical learnable bias scalar $u_{b(i-j)}$. Substituting $\text{PE}(i, j) = u_{b(i-j)}$ back into Eq. (8.76), the final attention weight computation becomes[15]

$$\alpha(i,j) \quad = \quad \text{Softmax}\left(\frac{\mathbf{q}_i \mathbf{k}_j^{\top} + u_{b(i-j)}}{\sqrt{d}} + \text{Mask}(i,j)\right), \tag{8.81}$$

The parameters $\{u_0, \cdots, u_{n_b - 1}\}$ are learned jointly with other parameters of the model. It should be emphasized that this model can generalize to long sequences. This is because $\text{PE}(i, j)$s with similar query-key offsets share the same parameter, and this sharing strategy is particularly important for achieving good generalization, given that large query-key offsets are

[15]Note that, in Raffel et al. [2020]'s T5 model, the rescaling operation for the query-key product is removed. The attention weight $\alpha(i, j)$ is then given by

$$\alpha(i,j) \quad = \quad \text{Softmax}(\mathbf{q}_i \mathbf{k}_j^{\top} + u_{b(i-j)} + \text{Mask}(i,j)). \tag{8.80}$$

rare in training. In practice, we often set n_b to a moderate number, and thus it can help control the overfitting of positional embedding models.

2. Attention with Fixed Biases

Relative positional embedding models are based on a set of learned biases added to the query-key product in self-attention. An alternative approach is to assign fixed values to these biases via predefined heuristics, rather than learning them from data. One benefit of this heuristic-based approach is that it does not rely on training and thus can be directly applied to any sequence once the biases are set.

One example of such an approach is Press et al. [2022]'s approach, called **attention with linear biases** or **ALiBi** for short. In the ALiBi approach, the bias term is defined as the negative scaled query-key offset

$$\begin{aligned} \mathrm{PE}(i,j) &= -\beta \cdot (i-j) \\ &= \beta \cdot (j-i), \end{aligned} \tag{8.82}$$

where β is the scaling factor. Adding this term to the query-key product, we obtain a new form of attention weights

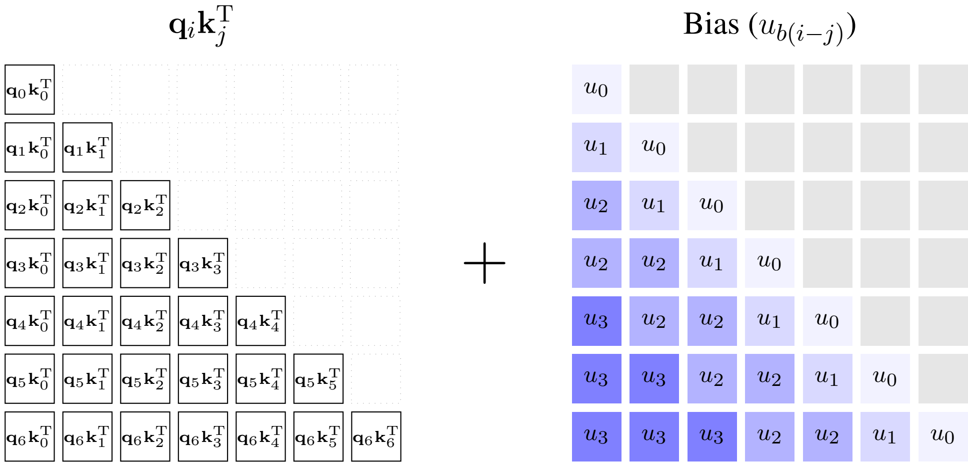
$$\alpha(i,j) = \mathrm{Softmax}\left(\frac{\mathbf{q}_i\mathbf{k}_j^{\top} + \beta \cdot (j-i)}{\sqrt{d}} + \mathrm{Mask}(i,j)\right). \tag{8.83}$$

This model can be interpreted as imposing a linear penalty on the attention score $\mathbf{q}_i\mathbf{k}_j^{\top}$ for every step the key j moves away from the query i. See Figure 8.11 for a comparison of the T5 bias and the ALiBi bias.

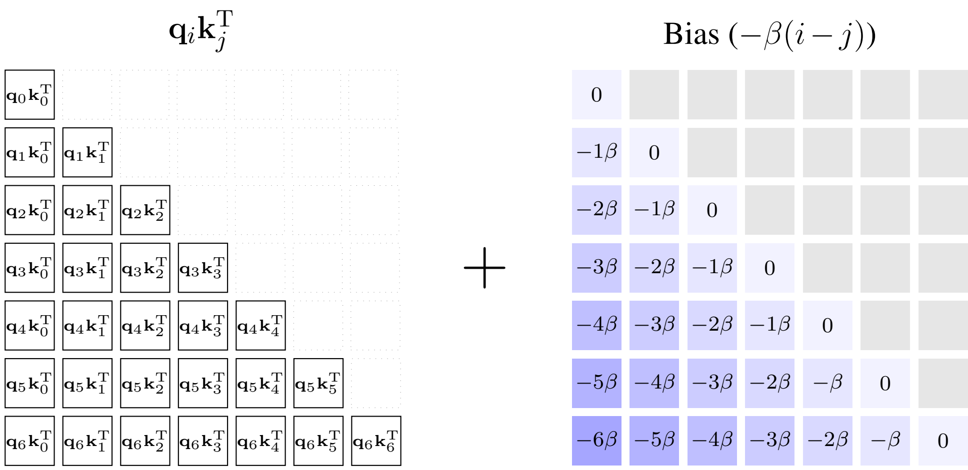
In principle, the scalar hyperparameter β could be tuned via grid search on a validation dataset. However, Press et al. [2022] demonstrated that setting β systematically across attention heads yields strong performance across a variety of tasks without any tuning. Specifically, for a multi-head self-attention layer comprising n_{head} heads, the β values decrease geometrically. The scalar for the k-th head is defined as

$$\beta_k = 2^{-\frac{8k}{n_{\text{head}}}}. \tag{8.84}$$

The ALiBi approach provides a simple form of relative positional embeddings. There are other similar methods for designing query-key biases using the offset $i-j$. Table 8.4 shows a comparison of such biases. As an aside, it is worth noting that the form of the right-hand side of Eq. (8.82) is similar to distance-based features used in conventional feature-based systems. For example, in statistical machine translation systems, such features are widely used to model word reordering problems, resulting in models that can generalize well across different translation tasks [Koehn, 2010].



(a) The T5 bias ($n_b = 3$ and $\mathrm{dist}_{\max} = 5$)



(b) The ALiBi bias

Figure 8.11: Query-key products with biases (above = the T5 bias and below = the ALiBi bias). The color scale of the biases ranges from light blue denoting small absolute values to deep blue denoting large absolute values.

3. Rotary Positional Embedding

Similar to sinusoidal embeddings, rotary positional embeddings use predefined frequency scales [Su et al., 2024]. Recall that in the sinusoidal embedding model, absolute positions are represented as fixed vectors composed of sine and cosine functions with varying frequencies, which are then additively combined with token embeddings. Rotary positional embeddings, instead, model positional context by applying geometrically meaningful rotations to token embeddings in the complex domain. This yields a formulation based on multiplicative

Entry	Query-Key Bias (PE(i,j))
T5 [Raffel et al., 2020]	$u_{b(i-j)}$
ALiBi [Press et al., 2022]	$-\beta \cdot (\ i-j\ )$
Kerple [Chi et al., 2022]	$-\beta_1 (\ i-j\ )^{\beta_2}$ (power)
	$-\beta_1 \log(1+\beta_2(\ i-j\ ))$ (logarithmic)
Sandwich [Chi et al., 2023]	$\sum_{k=1}^{\bar{d}/2} \cos\big((\ i-j\ )/10000^{2k/\bar{d}}\big)$
FIRE [Li et al., 2024b]	$f\big(\psi(\ i-j\ )/\psi(\max(m_{\text{len}}, i))\big)$

Table 8.4: Query-key biases as relative positional embeddings. β, β_1, β_2, $\bar{d}$, and m_{len} are hyper-parameters. In the T5 model, $b(i-j)$ denotes the bucket assigned to $i-j$. In the FIRE model, $\psi(\cdot)$ represents a monotonically increasing bounding function, such as $\psi(x) = \log(cx+1)$, and $f(\cdot)$ is an FFN.

interactions:

$$\mathbf{e}_i = \mathbf{x}_i R(i), \tag{8.85}$$

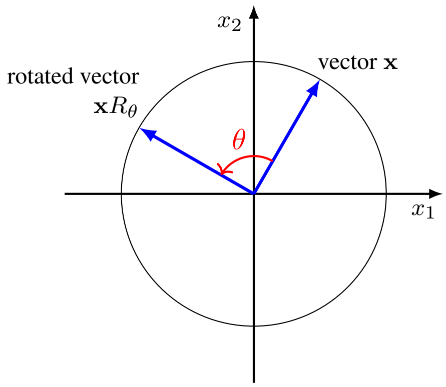
where $R(i) \in \mathbb{R}^{d\times d}$ is the rotation matrix representing the rotations performed on the token embedding $\mathbf{x}_i \in \mathbb{R}^d$.

For simplicity, we will first consider embeddings with only two dimensions and return to a discussion of the more general formulation later. Suppose we have a 2-dimensional token embedding represented as a row vector $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$. We can represent it as a vector in a plane, originating at the origin $(0,0)$ and terminating at (x_1, x_2). A counterclockwise rotation of this vector refers to an operation of moving the vector around the origin while maintaining its magnitude, as shown in Figure 8.12 (a). The degree of rotation is usually defined by a specific angle, denoted by θ. The rotation can be expressed mathematically in the form

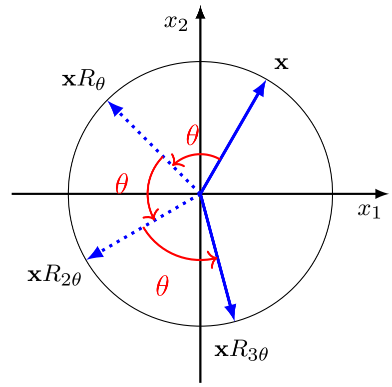
$$\begin{aligned} \text{Ro}(\mathbf{x}, \theta) &= \mathbf{x} R_\theta \\ &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta \cdot x_1 - \sin\theta \cdot x_2 & \sin\theta \cdot x_1 + \cos\theta \cdot x_2 \end{bmatrix}. \end{aligned} \tag{8.86}$$

where $R_\theta = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ is the rotation matrix. If two or more rotations are performed on the same vector, we can rotate the vector further. This follows from the fact that the composition of successive rotations is itself a rotation. More formally, rotating a vector by an angle θ, repeated t times, can be expressed as

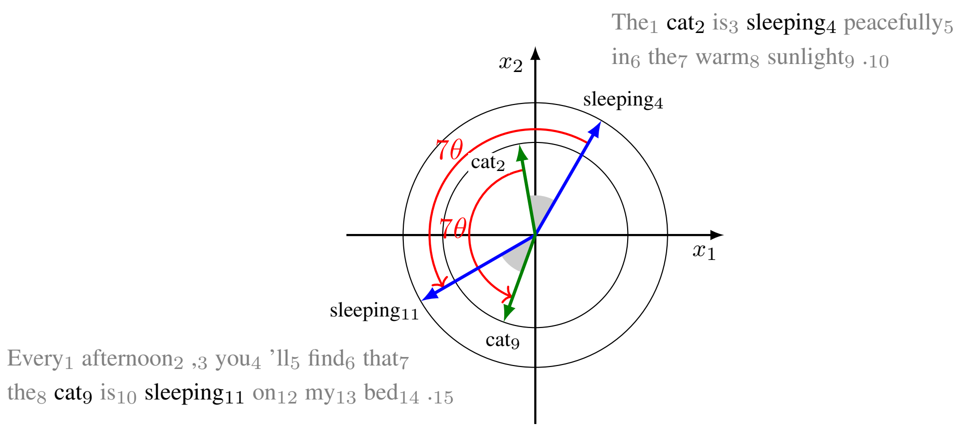
$$\begin{aligned} \text{Ro}(\mathbf{x}, t\theta) &= \mathbf{x} R_{t\theta} \\ &= \begin{bmatrix} \cos t\theta \cdot x_1 - \sin t\theta \cdot x_2 & \sin t\theta \cdot x_1 + \cos t\theta \cdot x_2 \end{bmatrix}. \end{aligned} \tag{8.87}$$



(a) Single-step Rotation



(b) Multi-step Rotation



(c) Angles between embeddings of two tokens at different positions

Figure 8.12: Illustrations of vector rotations in a plane. Sub-figures (a) and (b) show rotations of a vector in a single step and multiple steps, respectively. Sub-figure (c) shows the embeddings of tokens *cat* and *sleeping* in two different sentences. We show these sentences with a subscript affixed to each token to indicate its position. If we represent tokens as vectors, we can add positional information by rotating these vectors. This rotation preserves the "distances" between the vectors. For example, given that the distance between *cat* and *sleeping* is the same in both sentences, the angle between their embeddings also remains the same during rotation.

If we interpret t as the position of a token represented by $\mathbf{x}$ in a sequence, then we will find that the above equation defines a simple positional embedding model. As shown in Figure 8.12 (b), we start moving the token from position 0. Each time we move one step forward, the vector is rotated by the angle θ. Upon arriving at the position t, the representation of the token with positional context is given by $\mathrm{Ro}(\mathbf{x}, t\theta)$. As the rotations do not change the magnitude of the embedding, the original "meaning" of the token is retained. The positional information is injected into the embedding when it gets rotated.

A common way to understand vector rotation is via complex numbers. It is easy to transform each vector $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$ in the 2D Euclidean space $\mathbb{R}^2$ to a complex number $\mathbf{x}' = x_1 + \mathrm{i}x_2$ in the complex space $\mathbb{C}$ via a bijective linear map. Then, the rotation of $\mathbf{x}$ with the angle $t\theta$ corresponds to the multiplication by $e^{\mathrm{i}t\theta}$. Given that $e^{\mathrm{i}t\theta} = \cos t\theta + \mathrm{i}\sin t\theta$, the rotation operation can be re-expressed in the form

$$\begin{aligned} \mathbf{x}R_{t\theta} &\mapsto \mathbf{x}'e^{\mathrm{i}t\theta} \\ &= (x_1 + \mathrm{i}x_2)(\cos t\theta + \mathrm{i}\sin t\theta) \\ &= \cos t\theta \cdot x_1 - \sin t\theta \cdot x_2 + \mathrm{i}(\sin t\theta \cdot x_1 + \cos t\theta \cdot x_2). \end{aligned} \tag{8.88}$$

Here we denote the token representation $\mathbf{x}'e^{\mathrm{i}t\theta}$ by $C(\mathbf{x}, t\theta)$. The inner product of the representations of the tokens at positions t and s can be written as

$$\langle C(\mathbf{x}, t\theta), C(\mathbf{y}, s\theta) \rangle = (\mathbf{x}'\overline{\mathbf{y}'})e^{\mathrm{i}(t-s)\theta}, \tag{8.89}$$

where $\overline{\mathbf{y}'}$ is the complex conjugate of $\mathbf{y}'$. The result of this inner product involves a term $t - s$, and so it can model the offset between the two tokens.

Now we go back to representations in the 2D Euclidean space. The dot-product of $\mathrm{Ro}(\mathbf{x}, t\theta)$ and $\mathrm{Ro}(\mathbf{y}, s\theta)$ can be written as a function of $(t - s)\theta$

$$\begin{aligned} \mathrm{Ro}(\mathbf{x}, t\theta)[\mathrm{Ro}(\mathbf{y}, s\theta)]^{\top} &= \mathbf{x}R_{t\theta}[\mathbf{y}R_{s\theta}]^{\top} \\ &= \mathbf{x}R_{t\theta}[R_{s\theta}]^{\top}\mathbf{y}^{\top} \\ &= \mathbf{x}R_{(t-s)\theta}\mathbf{y}^{\top}. \end{aligned} \tag{8.90}$$

Given this result, if we consider $\mathrm{Ro}(\mathbf{x}, t\theta)$ and $\mathrm{Ro}(\mathbf{y}, s\theta)$ as the query and the key, then the self-attention operation will implicitly involve the modeling of relative positional context.

This rotary positional embedding can be extended to multi-dimensional embeddings. For a d-dimensional token embedding $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_d \end{bmatrix}$, we can treat it as a $\frac{d}{2}$-dimensional complex vector $\mathbf{x}' = \begin{bmatrix} x'_1 & x'_2 & \cdots & x'_{d/2} \end{bmatrix} = \begin{bmatrix} x_1 + \mathrm{i}x_2 & x_3 + \mathrm{i}x_4 & \cdots & x_{d-1} + \mathrm{i}x_d \end{bmatrix}$, where each consecutive pair of items forms a complex number. Then, the rotary positional embedding in the complex space is given by

$$C(\mathbf{x}, t\theta) = \sum_{k=1}^{d/2} x'_k e^{\mathrm{i}t\theta_k} \vec{e}_k, \tag{8.91}$$

where $\vec{e}_k$ is the standard basis vector with a single non-zero value in the k-th coordinate and 0's elsewhere [Biderman et al., 2021].

Although this formula involves a complicated expression, its equivalent form in the d-

dimensional Euclidean space is relatively easy to understand. We can write it as

$$\mathrm{Ro}(\mathbf{x}, t\theta) = \begin{bmatrix} x_1 & x_2 & \cdots & x_d \end{bmatrix} \begin{bmatrix} R_{t\theta_1} & & & \\ & R_{t\theta_2} & & \\ & & \ddots & \\ & & & R_{t\theta_{d/2}} \end{bmatrix}, \tag{8.92}$$

where $R_{t\theta_k} = \begin{bmatrix} \cos t\theta_k & \sin t\theta_k \\ -\sin t\theta_k & \cos t\theta_k \end{bmatrix}$. $\theta = \begin{bmatrix} \theta_1, \cdots, \theta_{d/2} \end{bmatrix}$ are the parameters for controlling the angles of rotations in different dimensions. Typically, θ_k is set to $10000^{-\frac{2(k-1)}{d}}$, which is analogous to the setting in sinusoidal embeddings.

In a practical implementation, Eq. (8.92) can be rewritten into a form that relies solely on the element-wise product and addition of vectors:

$$\mathrm{Ro}(\mathbf{x}, t\theta) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{d-1} \\ x_d \end{bmatrix}^{\top} \odot \begin{bmatrix} \cos t\theta_1 \\ \cos t\theta_1 \\ \vdots \\ \cos t\theta_{d/2} \\ \cos t\theta_{d/2} \end{bmatrix}^{\top} + \begin{bmatrix} -x_2 \\ x_1 \\ \vdots \\ -x_d \\ x_{d-1} \end{bmatrix}^{\top} \odot \begin{bmatrix} \sin t\theta_1 \\ \sin t\theta_1 \\ \vdots \\ \sin t\theta_{d/2} \\ \sin t\theta_{d/2} \end{bmatrix}^{\top}. \tag{8.93}$$

Finally, we rewrite Eq. (8.85) to obtain the form of the embedding at position i

$$\mathbf{e}_i = \mathrm{Ro}(\mathbf{x}_i, i\theta). \tag{8.94}$$

4. Position Interpolation

In position interpolation, our goal is to map the positions in the new sequence to match the observed range in training. Suppose the model is trained on position indices in $[0, m_l]$. When inference requires processing a length $m > m_l$, we compress the positions in the range $[0, m]$ such that their representations fit into the original domain $[0, m_l]$.

To illustrate, consider the rotary positional embedding model described above. The embedding of each token is given by the transformation $\mathrm{Ro}(\mathbf{x}_i, i\theta)$, where the parameter vector is $\theta = \begin{bmatrix} \theta_1, \cdots, \theta_{d/2} \end{bmatrix}$. As shown in Eq. (8.93), $\mathrm{Ro}(\mathbf{x}_i, i\theta)$ relies on a combination of periodic functions across different feature dimensions

$$\cos i\theta = \begin{bmatrix} \cos i\theta_1 & \cdots & \cos i\theta_{d/2} \end{bmatrix} \tag{8.95}$$

$$\sin i\theta = \begin{bmatrix} \sin i\theta_1 & \cdots & \sin i\theta_{d/2} \end{bmatrix}, \tag{8.96}$$

θ_k follows an exponential form

$$\theta_k = b^{-\frac{2(k-1)}{d}}, \tag{8.97}$$

where b is the base. The period of $\cos i\theta_k$ and $\sin i\theta_k$ is

$$T_k = 2\pi \cdot b^{\frac{2(k-1)}{d}}. \tag{8.98}$$

The key intuition behind linear position interpolation is to adjust these periods so that the new extended positions can be encoded within the pre-trained boundaries $[0, m_l]$. The most straightforward way to achieve this is to linearly scale up T_k by the extrapolation ratio $\frac{m}{m_l}$, yielding:

$$T'_k = \frac{m}{m_l} \cdot 2\pi \cdot b^{\frac{2(k-1)}{d}}. \tag{8.99}$$

By doing so, all position indices in $[0, m]$ are uniformly compressed into $[0, m_l]$. This linear scaling can be implemented by dividing the position index in the embedding model by the scaling factor [Chen et al., 2023c]. The modified model with linear positional interpolation becomes

$$\mathrm{Ro}'(\mathbf{x}_i, i\theta) = \mathrm{Ro}\left(\mathbf{x}_i, \frac{m_l}{m} i\theta\right). \tag{8.100}$$

Another method of positional interpolation is to scale the base b, a technique known as **NTK-aware scaled RoPE**[16]. Suppose the base b is scaled by a factor λ. The goal is to enforce the period of this new model in the highest frequency dimension (i.e., the last dimension where $k = \frac{d}{2}$) to match that of the linear positional interpolation model. Substituting $k = \frac{d}{2}$ into the exponent yields $\frac{2(d/2-1)}{d} = \frac{d-2}{d}$. Equating the periods gives:

$$2\pi \cdot (\lambda b)^{\frac{d-2}{d}} = \frac{m}{m_l} \cdot 2\pi \cdot b^{\frac{d-2}{d}}. \tag{8.101}$$

By solving this equation, we can isolate λ:

$$\lambda^{\frac{d-2}{d}} = \frac{m}{m_l}. \tag{8.102}$$

Then, we have

$$\lambda = \left(\frac{m}{m_l}\right)^{\frac{d}{d-2}}. \tag{8.103}$$

This approach yields a modified embedding model:

$$\mathrm{Ro}'(\mathbf{x}_i, i\theta) = \mathrm{Ro}(\mathbf{x}_i, i\theta'), \tag{8.104}$$

where

$$\theta' = \left[(\lambda b)^{-\frac{0}{d}}, (\lambda b)^{-\frac{2}{d}}, \cdots, (\lambda b)^{-\frac{d-2}{d}}\right]. \tag{8.105}$$

[16] This method was first proposed in an online forum: https://www.reddit.com/r/LocalLLaMA/comments/14lz7j5/ntkaware_scaled_rope_allows_llama_models_to_have/

Note that scaling the base provides a non-uniform method for scaling the periods across different dimensions of θ. This method has been found to be helpful for extending LLMs to longer sequences, and several improvements have been developed [Peng et al., 2024; Ding et al., 2024].

8.3.6 Remarks

In this section, we have presented a variety of methods for long-context language modeling. We close this section by discussing some interesting issues related to these methods.

1. Need for Long Context

One of the ultimate goals of long-context LLMs is to enable these models to precisely encode infinite context. Here, the term *infinite context* refers to the ability of an LLM to continuously process incoming tokens. This motivates the development of LLMs that can handle very long context or streaming data. As discussed in Section 8.3.3, it is common to use fixed-size memory models to process continuously expanding context. Many such systems are based on recurrent architectures or their variants, because they are inherently suited to model sequential data where the effects of past inputs continue indefinitely. Another way to achieve infinite memory is to develop alternatives to self-attention models, for example, one can use continuous-space attention models to encode context, which removes the dependency on context length [Martins et al., 2022].

When studying long-context LLMs, it is natural to wonder what mechanisms may explain the use of long context in language modeling. Can we compress the representation of infinite context into a relatively small model? Are all context tokens useful for predicting the next token? Can we know in advance which contextual information is critical for prediction? General answers to all these questions are not obvious, but they inspire follow-on research on explainable models, and some interesting results have been found. For example, Deletang et al. [2024] conducted extensive experiments to show that LLMs are powerful in-context compressors. Although viewing predictive models as compression models has long been studied in machine learning, it also provides insights into our understanding of LLM scaling laws. Furthermore, Pal et al. [2023] and Wu et al. [2024] investigated whether the internal features learned up to the current computational step, even if not explicitly optimized for it, are already sufficient to accurately predict tokens multiple steps into the future. Note that the need for long-context in language modeling is highly dependent on the problem that we address. A related issue is where to apply LLMs and how to evaluate them. For example, in summarization tasks we may only need to distill and focus on a few key aspects of the text, while in retrieval-like tasks we need to "memorize" the entire context so that the relevant information can be accessed. We will discuss the evaluation issue later in this subsection.

2. Pre-training or Adapting LLMs?

The training of LLMs requires significant computational resources. Although it is straightforward to train LLMs on long sequences, the training becomes computationally prohibitive for large datasets. It is common practice to pre-train LLMs on general datasets, and then adapt them with modest fine-tuning effort. For example, LLMs with relative or rotary positional embeddings can be directly trained on large-scale data in the pre-training phase. While the resulting models may exhibit some length extrapolation capabilities in the inference phase, it may be more effective to fine-tune them on longer sequences.

Ideally, we would like to pre-train LLMs with standard Transformer architectures and adapt them to new tasks. This allows us to use many off-the-shelf LLMs and efficiently adapt them to handle long sequences. However, when new architectures are adopted, it seems inevitable that we need to train these models from scratch. This poses practical difficulties for developing long-context LLMs, as we cannot leverage well-developed, pre-trained models and must instead train them ourselves. On the other hand, fine-tuning is still an effective way to adapt LLMs with certain architectures that are different from those in pre-training. An example is models augmented with external memories. In these models, the pre-trained LLMs are fixed, and the focus is on how LLMs interact with external memory modules. In RAG, for instance, it is common to fine-tune LLMs to improve their use of retrieval-augmented inputs. Another example is to train an LLM with full attention and then replace it with sparse attention in the fine-tuning phase. The pre-trained LLM provides an initialization for the parameters of the new model, and this model is then fine-tuned as usual.

3. Evaluating Long-context LLMs

Evaluating long-context LLMs is important, but it is a relatively new problem in NLP. The general idea is that, if we input a long context to an LLM, then we can examine the output of the LLM to determine whether it understands the entire context and makes use of it in predicting subsequent tokens. In conventional NLP research, such evaluations are often aimed at examining the ability of NLP models in handling long-range dependencies. However, the length of contexts used in recent LLMs is much larger than those used in NLP systems a few years ago. This motivates researchers to develop new evaluation benchmarks and metrics for long-context LLMs.

One approach is to use the perplexity metric. However, despite its simplicity, this method tends to reflect the ability of LLMs to make use of local context rather than global context. It is therefore tempting to develop evaluation methods that are specific to long-context LLMs. Popular methods include various synthetic tasks where artificially generated or modified data is used to evaluate specific capabilities of long-context LLMs. In needle-in-a-haystack[17] and passkey retrieval tasks [Mohtashami and Jaggi, 2024; Chen et al., 2023c], for instance, LLMs are required to identify and extract a small, relevant piece of information from a large volume of given text. The assumption here is that an LLM with sufficient memory should remember earlier parts of the text as it processes new information. This LLM can thus pick out from the text the relevant details, which might be sparse and hidden among much irrelevant information. Alternatively, in copy memory tasks (or copy tasks for short), LLMs are required to repeat the input text or a specific segment multiple times. These tasks were initially proposed to test the extent to which recurrent models can retain and recall previously seen tokens [Hochreiter

[17] https://github.com/gkamradt/LLMTest_NeedleInAHaystack

and Schmidhuber, 1997; Arjovsky et al., 2016], and have been adopted in evaluating recent LLMs [Bulatov et al., 2022; Gu and Dao, 2023].

Another approach to evaluating long-context LLMs is to test them on NLP tasks that involve very long input sequences. Examples include long-document or multi-document summarization, long-document question answering, code completion, and so on. A benefit of this approach is that it can align evaluations with user expectations.

Although many methods have been developed, there is still no general way to evaluate long-context LLMs [Liu et al., 2024c]. One problem is that most of these methods focus on specific aspects of LLMs, rather than their fundamental ability to model very long contexts. Even though an LLM can pick out the appropriate piece of text from the input, we cannot say that it truly understands the entire context. Instead, it might just remember some important parts of the context, or even simply recall the answer via the model learned in pre-training. Moreover, the data used in many tasks is small-scale and relatively preliminary, leading to discrepancies between evaluation results and actual application performance. A more interesting issue is that the results of LLMs are influenced by many other factors and experimental setups, for example, using different prompts can lead to very different outcomes. This makes evaluation even more challenging because improvements may not solely result from better modeling of long contexts, and there is a risk of overclaiming performance gains. Nevertheless, many open questions remain in the development and evaluation of long-context LLMs. For example, these models still suffer from limitations such as restricted context length and high latency. Studying these issues is likely to prove a valuable future direction.

8.4 Summary

In this chapter, we have discussed the concept of LLMs and related techniques. This can be considered a general, though not comprehensive, introduction to LLMs, laying the foundation for further discussions on more advanced topics in subsequent chapters. Furthermore, we have explored two ways to scale up LLMs. The first focuses on the large-scale pre-training of LLMs, which is crucial for developing state-of-the-art models. The second focuses on methods for adapting LLMs to long inputs, including optimizing attention models, designing more efficient and compressed KV caches, incorporating memory models, and exploring better positional embeddings.

The strength of LLMs lies in their ability to break the constraints of training NLP models for a limited number of specific tasks. Instead, LLMs learn from large amounts of text through the simple task of token prediction — predicting the next token in a sequence given its preceding context. A general view is that, by repeating this token prediction task a large number of times, LLMs can acquire some knowledge of the world and language, which can then be applied to new tasks. As a result, LLMs can be prompted to perform any task by framing it as a task of predicting subsequent tokens given prompts. This emergent ability in language models comes from several dimensions, such as scaling up training, model size, and context size. It is undeniable that scaling laws are currently the fundamental principle adopted in developing large language models, although simply increasing model size has yet to prove sufficient for achieving AGI. These continuously scaled LLMs have been found to show capabilities in general-purpose language understanding, generation, and reasoning. More recently, it has been found that scaling compute at inference time (i.e., test-time compute) can also lead to significant improvements in complex reasoning tasks [OpenAI, 2024].

Given their amazing power, LLMs have attracted considerable interest, both in terms of techniques and applications. As a result, the explosion of research interest in LLMs has also led to a vast number of new techniques and models. However, we do not attempt to provide a comprehensive literature review on all aspects of LLMs, given the rapid evolution of the field. Nevertheless, one can still gain knowledge about LLMs from general reviews [Zhao et al., 2023; Minaee et al., 2024] or more focused discussions on specific topics [Ruan et al., 2024].

Chapter 9

Prompting

prompting refers to the process of providing an LLM with a specific input or cue to generate a desired output or perform a task. For example, if we want the LLM to translate a sentence from English to Chinese, we can prompt it as follows

> Translate the text from English to Chinese.
>
> Text: The early bird catches the worm.
>
> Translation: ____

Prompting is crucial for LLMs because it directly influences how effectively these models understand and respond to user queries. A well-crafted prompt can guide an LLM to generate more accurate, relevant, and contextually appropriate responses. Furthermore, this process can be iteratively refined. By analyzing the responses of the LLM, users can adjust their prompts to align more closely with their specific needs. Given the importance of prompting in applying LLMs, prompt design has become an essential skill for users and developers working with LLMs. The growing importance of prompt design has given rise to an active research field known as **prompt engineering**, in which we design effective prompts to make better use of LLMs and enhance their practical utility in real-world applications.

An important concept related to prompting is **in-context learning**. When prompting an LLM, we can add new information to the context, such as demonstrations of problem-solving. This allows the LLM to learn from this context how to solve the problem. Here is an example of prompting LLMs with a few demonstrations of how to classify text based on sentiment polarity.

> Here are some examples of text classification.
>
> Example 1: We had a delightful dinner together. $\rightarrow$ Label: Positive
>
> Example 2: I'm frustrated with the delays. $\rightarrow$ Label: Negative
>
> What is the label for "That comment was quite hurtful."?
>
> Label: ____

In-context learning is often seen as an emergent ability of LLMs that arises after pre-training. Though LLMs can be trained or tuned to perform new tasks, in-context learning provides a very efficient way to adapt these models without any training or tuning effort. Perhaps this is one of the most notable features of LLMs: they indeed learn general knowledge about the world and language during pre-training, which we can easily apply to new challenges. Moreover, in-context learning reflects the broader trend of making AI systems more generalizable and user-friendly. Instead of requiring specialized engineers to fine-tune models for every unique task, users can interact with LLMs in a more intuitive way, simply providing examples or adjusting the context as needed.

In this chapter, we focus on prompting techniques in LLMs. We begin by considering several interesting prompt designs commonly used in prompt engineering. Then, we discuss a series of refinements to these methods. Finally, we explore approaches for automating prompt design.

9.1 General Prompt Design

This section presents basic concepts in prompt design, along with examples of how to prompt LLMs for various NLP tasks. Since the effectiveness of prompting depends heavily on the LLMs being used, prompts often vary across different LLMs, which makes it difficult to provide a comprehensive list of prompts for all LLMs and downstream tasks. Therefore, this discussion does not focus on any specific LLM. Instead, the goal is to provide guiding principles for prompt design.

9.1.1 Basics

The term *prompt* is used in many different ways. In this chapter, we define a prompt as the input text to an LLM, denoted by $\mathbf{x}$. The LLM generates an output $\mathbf{y}$ according to the conditional probability $\Pr(\mathbf{y}|\mathbf{x})$. In this generation process, the prompt serves as the condition upon which predictions are made, and it can contain any information that helps describe and solve the problem.

A prompt can be obtained using a prompt template (or template for short) [Liu et al., 2023b]. A template is a piece of text containing placeholders or variables, where each placeholder can be filled with specific information. The following are two templates for asking the LLM for weekend suggestions.

> Please give me some suggestions for a fun weekend.
>
> ____

> If $\{*\text{premise}*\}$, what are your suggestions for a fun weekend.
>
> ____

In the first template, we simply instruct the LLM to return some suggestions. So the template is just a piece of text with no variables. In the second template, the variable $\{*\text{premise}*\}$ needs to be specified by the user to provide a premise for making suggestions. For example, if we input

$$\text{premise} \quad = \quad \text{the weather is nice this weekend,}$$

then we can generate a prompt

> If the weather is nice this weekend,
> what would you suggest for a fun weekend?
>
> ____

We can also design a template with multiple variables. Here is an example in which we compare the semantic similarity between two sentences.

> Here is a sentence
> $\{*\text{sentence1}*\}$
> Here is another sentence
> $\{*\text{sentence2}*\}$
> Compute the semantic similarity between the two sentences
>
> ____

A popular way to format prompts is to write each input or output in a "name: content" style. For example, we can describe a conversation between two people, named John and David, and use the LLM to continue the conversation. The following is a template for such prompts.

> John: $\{*\text{utterance1}*\}$
> David: $\{*\text{utterance2}*\}$
> John: $\{*\text{utterance3}*\}$
> David: $\{*\text{utterance4}*\}$
> John: $\{*\text{utterance5}*\}$
> David: $\{*\text{utterance6}*\}$
> John: $\{*\text{utterance7}*\}$
> David: ____

The "name: content" format can be used to define the task that we want the LLM to perform. For example, given that "Q" and "A" are commonly used abbreviations for "Question" and "Answer", respectively, we can use the following template to do question-answering.

> Q: $\{*\text{question}*\}$
> A: ____

This format can be used to describe more complex tasks. For instance, the following is an example of providing a specification for a translation task:

> Task: Translation
> Source language: English
> Target language: Chinese
> Style: Formal text
> Template: Translate the following sentence: $\{*\text{sentence}*\}$
> ____

In practical systems, it is common to represent and store such data in key-value pairs, such as JSON[1].

When the problem is difficult to describe in an attribute-based manner, it is more common to instruct the LLM using a clear and detailed description. There are many ways to do this.

[1] The JSON representation is

```
{
    "Task": "Translation",
    "Source language": "English",
    "Target language": "Chinese",
    "Style": "Formal text",
    "Template": "Translate the following sentence: {*sentence*}",
}
```

One example is to assign a role to the LLM and provide sufficient context. The following is a template that instructs an LLM to act as an expert and answer questions from children.

> You are a computer scientist with extensive knowledge in the field of deep learning.
>
> Please explain the following computer-related concept to a child around 10 years old, using simple examples whenever possible.
>
> $\{*\text{concept}*\}$
>
> ____

The text "You are a computer scientist ... deep learning. " is sometimes referred to as a system prompt (or system information), and is provided to help the LLM understand the context or constraints of the task it is being asked to perform.

9.1.2 In-context Learning

Learning can occur during inference in large language models. In-context learning is one such method, where prompts involve demonstrations of problem-solving, and LLMs can learn how to solve new problems from these demonstrations. Since we do not update model parameters in this process, in-context learning can be viewed as a way to efficiently activate the knowledge learned in pre-training without additional training or fine-tuning. This enables LLMs to quickly adapt to new problems and extends the capabilities of pre-trained models without task-specific adjustments.

In-context learning can be illustrated by comparing three methods: zero-shot learning, one-shot learning, and few-shot learning. Zero-shot learning, as its name implies, does not involve a traditional "learning" process. Instead, it directly uses LLMs to address new problems that were not observed during training. In practice, we can iteratively adjust prompts to guide the LLMs in generating better responses, without demonstrating problem-solving steps or providing examples. Consider the following example. Suppose we want to use an LLM as an assistant that can help correct English sentences. A zero-shot learning prompt is given by

SYSTEM	You are a helpful assistant, and are great at grammar correction.
USER	You will be provided with a sentence in English. The task is to output the correct sentence. Input: She don't like going to the park. Output: ____

Here the gray words are used to indicate different fields of the prompt.

In one-shot learning, we extend this prompt by adding a demonstration of how to correct sentences, thereby allowing the LLM to learn from this newly added experience.

SYSTEM	You are a helpful assistant, and are great at grammar correction.
DEMO	You will be provided with a sentence in English. The task is to output the correct sentence. Input: There is many reasons to celebrate. Output: There are many reasons to celebrate.
USER	You will be provided with a sentence in English. The task is to output the correct sentence. Input: She don't like going to the park. Output: ____

Furthermore, we can add more demonstrations to enable few-shot learning.

SYSTEM	You are a helpful assistant, and are great at grammar correction.
DEMO1	You will be provided with a sentence in English. The task is to output the correct sentence. Input: There is many reasons to celebrate. Output: There are many reasons to celebrate.
DEMO2	You will be provided with a sentence in English. The task is to output the correct sentence. Input: Me and my friend goes to the gym every day. Output: My friend and I go to the gym every day.
USER	You will be provided with a sentence in English. The task is to output the correct sentence. Input: She don't like going to the park. Output: ____

In few-shot learning, we essentially provide a pattern that maps some inputs to the corresponding outputs. The LLM attempts to follow this pattern in making predictions, provided that the prompt includes a sufficient (though typically small) number of demonstrations. It is also possible to use simpler patterns to achieve this. For example, one can use the following few-shot learning prompt for translating words from Chinese to English.

DEMO	现在	→	now
	来	→	come
	去	→	go
	男孩	→	boy
USER	女孩	→	____

If the LLM is powerful enough, few-shot learning can enable it to address complex problems, such as mathematical reasoning. For example, consider the following task of summing two numbers and then dividing the sum by their product.

DEMO	12 5	→	$(12+5)/(12\times5)=0.283$
	3 1	→	$(3+1)/(3\times1)=1.33$
	−9 4	→	$(-9+4)/(-9\times4)=0.138$
	15 15	→	$(15+15)/(15\times15)=0.133$
USER	19 73	→	____

In many practical applications, the effectiveness of in-context learning relies heavily on the quality of prompts and the fundamental abilities of pre-trained LLMs. On one hand, significant prompt engineering effort is required to develop appropriate prompts that help LLMs learn more effectively from demonstrations. On the other hand, stronger LLMs can make better use of in-context learning for performing new tasks. For example, suppose we wish to use an LLM to translate words from Inuktitut to English. If the LLM lacks pre-training on Inuktitut data, its understanding of Inuktitut will be weak, and it will be difficult for the model to perform well in translation regardless of how we prompt it. In this case, we need to continue training the LLM with more Inuktitut data, rather than trying to find better prompts.

It is also interesting to explore how in-context learning emerges during pre-training and why it works during inference. One intuitive explanation is that while LLMs acquire problem-solving knowledge during pre-training, the large output space makes it difficult to select the correct prediction for novel tasks. Providing demonstrations guides the LLMs along the "correct" paths. Furthermore, some researchers have tried to interpret in-context learning from several different perspectives, including Bayesian inference [Xie et al., 2022], gradient descent [Dai et al., 2023; Von Oswald et al., 2023], linear regression [Akyürek et al., 2023], meta-learning [Garg et al., 2022], and so on.

9.1.3 Prompt Engineering Strategies

Designing prompts is highly empirical. In general, there are many ways to prompt an LLM to perform the same task, and we often need to conduct several trial-and-error iterations to find a satisfactory prompt. To write good prompts more efficiently, one can follow certain strategies. Examples of common prompting principles include:

- **Describing the task as clearly as possible**. When we apply an LLM to solve a problem, we need to provide a precise, specific, and clear description of the problem and instruct the LLM to behave as intended. This is particularly important when we want the output of the LLM to meet certain expectations. For example, suppose we are curious about climate change. A simple prompt for asking the LLM to provide some information is

  > Tell me about climate change.
  >
  > ____

  Since this instruction is too general, the LLM may generate a response that addresses any aspect of climate change, which may not align with our specific interests. In this case, we can instead use prompts that are specific and detailed. One such example is

  > Provide a detailed explanation of the causes and effects of climate change, including the impact on global temperatures, weather patterns, and sea levels. Also, discuss possible solutions and actions being taken to mitigate these effects.
  >
  > ____

  Now suppose we intend to explain climate change to a 10-year-old child. We can adjust the above prompt further.

  > Explain the causes and effects of climate change to a 10-year-old child. Talk about how it affects the weather, sea levels, and temperatures. Also, mention some things people are doing to help. Try to explain in simple terms and do not exceed 500 words.
  >
  > ____

- **Guiding LLMs to think**. LLMs have exhibited a surprisingly strong ability to "think". A common example is that well-developed LLMs have achieved impressive performance in mathematical reasoning tasks, which are considered challenging. In prompt engineering, the thinking ability of LLMs needs to be activated through appropriate prompting, especially for problems that require significant reasoning efforts. In many cases, an LLM that is instructed to "think" can produce completely different results compared with instructing the same model to perform the task directly. For example, Kojima et al. [2022] found that simply appending "Let's think step by step" to the end of each prompt can improve the performance of LLMs on several reasoning tasks. LLMs can be prompted to "think" in a number of ways. One method is to instruct LLMs to generate steps for reasoning about the problem before reaching the final answer. For example, consider a task of solving mathematical problems. The following is a simple prompt for this task.

> You are a mathematician. You will be provided with a math problem. Please solve the problem.
>
> ——

Since solving math problems requires a detailed reasoning process, LLMs would probably make mistakes if they attempted to work out the answer directly. So we can explicitly ask LLMs to follow a given reasoning process before coming to a conclusion.

> You are a mathematician. You will follow these detailed reasoning steps when solving math problems.
>
> Step 1: Problem Interpretation.
> The mathematician carefully listens to your query and understands the intricate details of the mathematical challenge you have presented.
>
> Step 2: Strategy Formulation.
> Drawing upon their extensive knowledge, the mathematician chooses the most effective strategy tailored to the type of math problem, whether it is algebra, calculus, or geometry.
>
> Step 3: Detailed Calculation.
> With precision and expertise, the mathematician performs the necessary calculations step by step, adhering to all mathematical principles.
>
> Step 4: Solution Review.
> Before providing the final answer, the mathematician meticulously checks the calculations for accuracy and offers a concise explanation or rationale for the solution.
>
> You will be provided with a math problem. Please solve the problem.
>
> $\{*\text{problem}*\}$
>
> ——

Another method to guide LLMs to "think" is through multiple rounds of interaction with LLMs. For example, as a first step, we can instruct LLMs to solve the problem directly

> You will be provided with a math problem. Please solve the problem.
>
> {∗problem∗}
>
> ——

Now we have an initial answer to the problem. As a second step, we prompt LLMs to evaluate the correctness of the answer and, if necessary, rework it to find a better solution.

> You will be provided with a math problem, along with a solution. Evaluate the correctness of this solution, and identify any errors if present. Then, work out your own solution.
>
> Problem: {∗problem∗}
>
> Solution: {∗solution∗}
>
> ——

The prompts presented here are closely related to a long line of research on reasoning problems in LLMs. It is impossible to provide a complete discussion of all related issues because this topic covers a large family of methods. But we will see a relatively more detailed discussion on how to improve prompting through more reasoning in Section 9.2.

- **Providing reference information**. As discussed in the previous section, we can include demonstrations in prompts and allow LLMs to learn in-context from these demonstrations how to perform the task. In fact, given the remarkable language understanding abilities of LLMs, we can add any type of text into the prompts and so these models can predict based on enriched contexts. In many applications, we have a variety of information that is relevant to user queries. Instead of using LLMs to make unconstrained predictions, we often want LLMs to produce outputs that are confined to the relevant text. One such example is RAG, where the relevant text for the user query is retrieved using an IR system, and we prompt LLMs to generate responses based on this provided relevant text. The following prompt shows an example.

> You are an expert that can generate answers to input queries. You have now been provided with a query and the corresponding context information. Please generate an answer based on this context information. Note that you need to provide the answer in your own words, not just copy from the context provided.
>
> Context information: $\{*\text{IR-result}*\}$
>
> Query: $\{*\text{query}*\}$
>
> ——

If the context information is highly reliable, we can even restrict LLMs to answering using only the provided text. An example prompt is shown as follows

> You are an expert tasked with generating answers from input queries. You have been provided with a query and corresponding context information, organized in a table where each row represents a useful record. Please generate an answer using only this context information. Ensure that you provide the answer in your own words.
>
> Context information: $\{*\text{table}*\}$
>
> Query: $\{*\text{query}*\}$
>
> ——

When dealing with real-world problems, we often have prior knowledge and additional information about the problems that help produce better answers. Considering such information in prompting is generally helpful in improving the result.

- **Paying attention to prompt formats**. In general, the performance of LLMs is highly sensitive to the prompts we input. Sometimes a small modification to a prompt can lead to a big change in model output. An interesting example is that changing the order of sentences in a prompt may cause LLMs to generate different results. To make prompts easy to read and reduce ambiguity, it is common to format them in a way that ensures clarity. One example is that we define several fields for prompts and fill different information in each field. Another example is that we can use code-style prompts for LLMs which can understand and generate both natural language and code. See the following for a code-style prompt that performs translation where one demonstration is presented.

> [English] = [I have an apple.]
> [German] = [Ich habe einen Apfel.]
> [English] = [I have an orange.]
> [German] = ____

LLMs can receive text in various formats. This allows us to use control characters, XML tags, and specific formatting to represent complex data. Furthermore, it is useful to specify how the input and output should be formatted or structured. For example, we can delimit sections of text using quotes and prompt LLMs accordingly (e.g., adding a sentence like "the input text is delimited by double quotes" to the prompt).

Above, we have discussed only a few strategies for writing good prompts. There are, of course, many such methods, and one needs to develop their own through practice. Interested readers can refer to various online documents for more information, such as OpenAI's manual on the GPT series models[2].

9.1.4 More Examples

In this subsection, we consider more examples of prompting LLMs to perform various NLP tasks. The motivation here is not to give standard prompts for these tasks, but rather to use simple examples to illustrate how LLMs can be prompted to address NLP problems.

1. Text Classification

Text classification is perhaps one of the most common problems in NLP. Many tasks can be broadly categorized as assigning pre-defined labels to a given text. Here, we consider the polarity classification problem in sentiment analysis. We choose polarity classification for illustration because it is one of the most popular and well-defined text classification tasks. In a general setup for polarity classification, the goal is to categorize a given text into one of three categories: negative, positive, or neutral. Below is a simple prompt for this task (for easy reading, we highlight the task description in the prompt).

> Analyze the polarity of the following text and classify it as positive, negative, or neutral.
>
> Text:
> The service at the restaurant was slower than expected, which was a bit frustrating.
>
> The polarity of the text can be classified as negative.

[2]See https://platform.openai.com/docs/guides/prompt-engineering/six-strategies-for-getting-better-results.

For completeness, we show the response generated by the LLM (shown in underlined text).

Although the answer is correct, the LLM produces a descriptive sentence rather than a discrete label. The challenge arises because LLMs are inherently designed to generate text rather than assign discrete labels. Thus, classification is treated as a text generation problem. As a result, we need another system to map the LLM's output to the label space (a process referred to as **label mapping**). For instance, we extract "negative" from "The polarity of the text can be classified as negative". This is trivial in most cases because we can identify label words via simple heuristics. But occasionally, LLMs may not express the classification results using these label words. In this case, the problem becomes more complicated, as we need a mapping from generated text to predefined labels.

One method to induce output labels from LLMs is to reframe the problem as a cloze task. For example, the following shows a cloze-like prompt for polarity classification.

> Analyze the polarity of the following text and classify it as positive, negative, or neutral.
>
> Text:
> The service at the restaurant was slower than expected, which was a bit frustrating.
>
> The polarity of the text is negative

We can use LLMs to complete the text and fill in the blank with the most appropriate word. Ideally, the filled word should be $\mathrm{positive}$, $\mathrm{negative}$, or $\mathrm{neutral}$. However, LLMs are not guaranteed to generate these label words. One method to address this problem is to constrain the output space to the set of label words and select the one with the highest probability. Then, the output label is given by

$$\mathrm{label} = \arg\max_{y \in Y} \Pr(y|\mathbf{x}), \tag{9.1}$$

where y denotes the token (or word) filled in the blank, and Y denotes the set of label words $\{\mathrm{positive}, \mathrm{negative}, \mathrm{neutral}\}$.

Another approach to generating labels with LLMs is to constrain the output with prompts. For example, we can prompt LLMs to predict from a predefined label set. Here is an example.

> Analyze the polarity of the following text and classify it as positive, negative, or neutral.
>
> Text:
> The service at the restaurant was slower than expected, which was a bit frustrating.
>
> What is the polarity of the text?
>
> Just answer: positive, negative, or neutral.
>
> Negative

Sentiment analysis is a common NLP problem that has generally been well understood by LLMs through pre-training or fine-tuning. Thus we can prompt LLMs using simple instructions to perform the task. However, for new classification problems, it may be necessary to provide additional details about the task, such as the classification standards, so that the LLMs can perform correctly. To do this, we can add a more detailed description of the task and/or demonstrate classification examples in the prompts. To illustrate, consider the following example.

> Analyze the polarity of the following text and classify it as positive, negative, or neutral. Here's what each category represents:
>
> Positive: This indicates that the text conveys a positive emotion or attitude. For example, texts expressing happiness, satisfaction, excitement, or admiration are considered positive.
>
> Negative: This refers to a text that expresses a negative emotion or attitude. It encompasses feelings of sadness, anger, frustration, or criticism.
>
> Neutral: Neutral sentiment is used to describe texts that do not exhibit clear positive or negative emotions but instead convey informational, factual, or indifferent tones.
>
> Text:
> The service at the restaurant was slower than expected, which was a bit frustrating.
>
> What is the polarity of the text?
>
> Negative

While it seems straightforward to use LLMs for classification problems, there are still issues that have not been well addressed. For example, when dealing with a large number of categories, it remains challenging to effectively prompt LLMs. Note that if we face a very difficult classification problem and have a certain amount of labeled data, fine-tuning LLMs or adopting "BERT + classifier"-like architectures is also desirable.

2. Information Extraction

Many NLP tasks can be framed as information extraction problems, which involve identifying and extracting specific pieces of information from unstructured text. Such information includes named entities, relationships, events, and other relevant data points. The goal of information extraction is to transform raw data into a structured format that can be easily analyzed and used in various downstream applications.

As information extraction covers a wide range of problems, we cannot discuss them all here. Instead, we start with the task of named entity recognition — a task that has long been a central focus in NLP. Named entity recognition is the task of identifying and classifying named entities in text. These entities often include proper nouns and other expressions such as dates, monetary values, and percentages. First, consider a simple example of extracting person names from a given text[3].

> Identify all person names in the provided text.
>
> Text:
> Is the UK really doing that badly or have travel trends shifted, post-pandemic? For Tom Jenkins, CEO of the European Tourism Organisation, it's the latter. "I think the UK is doing perfectly well but we'll see more people going to Europe," he says of 2024, adding that there's "allegedly a slight plateau of American demand for the UK."
>
> The person name identified in the provided text is:
>
> Tom Jenkins

Next, we can expand the scope to extract all named entities.

[3] The text is from https://edition.cnn.com/travel

Identify and classify all named entities in the provided text into categories such as person names, locations, dates, and organizations. List each entity with its type on one line.

Text:
Is the UK really doing that badly or have travel trends shifted, post-pandemic? For Tom Jenkins, CEO of the European Tourism Organisation, it's the latter. "I think the UK is doing perfectly well but we'll see more people going to Europe," he says of 2024, adding that there's "allegedly a slight plateau of American demand for the UK."

1. Tom Jenkins - Person Name

2. European Tourism Organisation - Organization

3. UK - Location

4. Europe - Location

5. 2024 - Date

Given these identified named entities, we can apply other information extraction tasks to the text. For example, we can identify the relationships between these named entities (call it **relation extraction**). The corresponding prompt is given below.

Given a text and a list of named entities identified within it, analyze and describe the relationships between these entities. Explain how each entity is contextually related to others.

Text:
Is the UK really doing that badly or have travel trends shifted, post-pandemic? For Tom Jenkins, CEO of the European Tourism Organisation, it's the latter. "I think the UK is doing perfectly well but we'll see more people going to Europe," he says of 2024, adding that there's "allegedly a slight plateau of American demand for the UK."

Named Entity:
Tom Jenkins, European Tourism Organisation, UK, Europe, 2024

Tom Jenkins - European Tourism Organisation: Tom Jenkins is the CEO of the European Tourism Organisation. This indicates a professional association where Tom Jenkins likely has a significant role in directing the organization's activities, especially those related to tourism in Europe.

...

If an LLM has been fine-tuned to follow instructions, adapting it to various information extraction tasks is relatively straightforward. For example, the following is a general prompt template for information extraction.

> You will be provided with a text. Your task is to {*task-description*}
>
> Text: {*text*}
>
> ____

Here task-description is the instruction for the task. Examples of such instructions include

extract the keywords from the text.

extract the key events from the text.

detect the coreference and anaphoric links between entities from the text.

Typically, providing a more detailed task description will yield more accurate responses from the LLM.

3. Text Generation

In NLP, the term *text generation* is used broadly to cover a wide range of problems, many of which focus on generating new content based on an input. These problems can be roughly categorized into two classes: 1) continuing from an initial input text (call it **text completion**), and 2) converting the input text into another text (call it **text transformation**). Here, we present examples of prompts for the former and discuss the latter later.

A simple example of text completion is continuing to write a story starting from a few initial sentences.

> You are a writer who is great at writing stories. You are provided with a few sentences to start with. Please continue the story from the provided sentences.
>
> Start of the story:
> One sunny afternoon, Lily discovered a tiny, golden key hidden under a rock in her backyard. Curious, she picked it up and wondered what it might unlock. Just then, her cat, Whiskers, began to meow excitedly, leading her towards the old oak tree at the edge of the garden.
>
> Lily followed Whiskers, her heart pounding with excitement and a touch of nervousness. The old oak tree had always seemed mysterious to her, its gnarled branches casting long shadows even on the brightest days. As they reached the tree, Whiskers darted ahead and started pawing at something buried beneath a pile of autumn leaves ...

Likewise, we can complete a conversation given the conversational context.

> You are provided with a conversation between a tourist and a taxi driver. Please complete the conversation.
>
> Tourist: Could you take me to the downtown museum, please?
>
> Taxi Driver: Of course! First time in the city?
>
> Tourist: Yes, any must-see places around?
>
> Taxi Driver: Definitely check out the botanical gardens after the museum. They're nearby and quite beautiful.
>
> Tourist: Great, thanks for the tip!
>
> Taxi Driver: No problem. Enjoy your visit!

Beyond simple continuation, this paradigm can be extended to instruction-driven generation tasks, where the model "completes" a response based on user requirements. These requirements can dictate style, tone, length, and specific content. For instance, users may request a formal report, a creative story, or programming code. Below is a prompt for generating a traditional five-character regulated Chinese poem based on predefined constraints.

> You are a poet. Please write a traditional Chinese poem. Here are the requirements.
>
> Theme: Spring's rejuvenation
>
> Structure: Five-character regulated poem
>
> Emotion: Convey a sense of happiness and renewal through vivid spring imagery.
>
> 春意盎然
>
> 翠柳风中舞，
> 红花雨后新。
> 山明水又绿，
> 天宽鸟自频。

Furthermore, when an LLM is pre-trained or fine-tuned on a corpus encompassing both natural language and programming languages, we can prompt it to perform text-to-code generation tasks. In this case, the model simply translates natural language instructions into functional code. The following example shows how to prompt the LLM to write a Python function.

Please write a Python function to calculate the average of a list of numbers.

```
def calculate_average(numbers):
    if numbers:    # Check if the list is not empty
        return sum(numbers) / len(numbers)
    else:
        return 0   # Return 0 if the list is empty
```

4. Text Transformation

Typical text transformation tasks include machine translation, summarization, text style transfer, and so on. The prompts used to perform these tasks are not significantly different from the ones presented previously. Here, we present a few examples to illustrate these prompts.

Translate the following text from English to Spanish.

Text:

The quick brown fox jumps over the lazy dog.

El rápido zorro marrón salta sobre el perro perezoso.

Summarize the following article in no more than 50 words:

Article:

In recent years, urban areas have been facing significant challenges related to sustainability and environmental management. Cities are experiencing higher rates of pollution, increased traffic congestion, and greater demands on infrastructure. This growth has led to numerous environmental issues, including elevated levels of air and water pollution, increased waste production, and strained public services ...

Urban areas are grappling with sustainability challenges, such as rising pollution, traffic congestion, and infrastructure demands ...

> Rewrite this text in a formal tone.
>
> Text:
>
> Hey, what's up? Long time no see!
>
> Hello, how have you been? It has been quite some time since we last met!

5. Question Answering

In question answering, a system is given a question and is expected to produce an appropriate answer. Prompt templates for general-purpose question answering can be:

> $\{*\text{question}*\}$
>
> ____

> Question: $\{*\text{question}*\}$
>
> Answer: ____

> Q: $\{*\text{question}*\}$
>
> A: ____

Question answering is important in NLP because many problems can be framed as question-answering tasks. In particular, many recent reasoning tasks are defined in the form of question answering. For example, in the MMLU benchmark [Hendrycks et al., 2021], each example consists of a multiple-choice question, and LLMs are required to select the correct answer. The following shows an example prompt for answering a question in this dataset.

> $(\mathbb{Z}, *)$ is a group with $a * b = a + b + 1$ for all a, b in $\mathbb{Z}$. The inverse of a is
>
> (A) 0
>
> (B) -2
>
> (C) $a-2$
>
> (D) $(2+a)*-1$
>
> D

Another widely-used benchmark is the GSM8K dataset [Cobbe et al., 2021]. It consists of thousands of grade school math word problems. For each problem, we use LLMs to generate a solution in natural language. For example, the following is a prompt used to solve problems in GSM8K.

DEMO Q: Jerry's two daughters play softball on different teams. They each have 8 games this season. Each team practices 4 hours for every game they play. If each game lasts for 2 hours, how many hours will Jerry spend at the field watching his daughters play and practice altogether?

A: Jerry will spend 8 games * 2 hours per game = $\ll 8*2=16 \gg$ 16 hours watching one daughter play her games. He will spend $16*2 = \ll 16*2=32 \gg$ 32 hours watching both daughters play their games. He will spend 8 games * 4 hours of practice = $\ll 8*4=32 \gg$ 32 hours watching one daughter practice. He will spend $32*2 = \ll 32*2=64 \gg$ 64 hours watching both daughters practice. He will spend a total of 32 hours watching games + 64 hours watching practice $= \ll 32+64=96 \gg$ 96 hours. #### 96

Q: Mary bought six apples from the store. From the apples she bought, for each that Mary ate, she planted two trees from the remaining ones. How many apples did Mary eat?

A: She planted eight trees. This means she used half of that amount of apples, which is 8 trees /2 trees/apple $= \ll 8/2=4 \gg$ 4 apples. That means that she planted four of the six apples she bought, leaving only 6 apples $-$ 4 apples $= \ll 6-4=2 \gg$ 2 apples to be eaten. #### 2

USER Q: Boris has 100 apples. Beck has 23 fewer apples than Boris. If Boris gives Beck 10 apples, how many fewer apples does Beck have than Boris now?

A: <u>Boris starts with 100 apples and gives Beck 10 apples, leaving him with 100 apples -10 apples $= \ll 100-10=90 \gg$ 90 apples. Initially, Beck has 23 fewer apples than Boris, giving him a starting count of $100-23 = \ll 100-23=77 \gg$ 77 apples. After receiving 10 apples from Boris, Beck's total comes to 77 apples $+10$ apples $= \ll 77+10=87 \gg$ 87 apples. Consequently, Beck now has $90-87 = \ll 90-87=3 \gg$ 3 fewer apples than Boris. #### 3</u>

Here a few-shot prompt is adopted. The LLM learns from these demonstrations of problem-solution pairs not only how to solve problems but also how to format the output. For example, the final result of the calculation appears after the #### token, and $\ll \cdots \gg$ annotates the detailed calculation steps (called **calculation annotation**)[4].

[4] During prediction, a calculator is used when we see $\ll \cdots \gg$. More specifically, once the LLM encounters

9.2 Advanced Prompting Methods

So far in this chapter, we have introduced the basic concepts related to LLM prompting and presented a number of prompts for NLP tasks. We now consider several techniques for enhancing the effectiveness of prompting.

9.2.1 Chain of Thought

We have encountered the concept of **chain of thought** (**CoT**) several times in this chapter and previous ones [Wei et al., 2022c; Chowdhery et al., 2022]. CoT methods provide a simple way to prompt LLMs to generate step-by-step reasoning for complex problems, thereby approaching tasks in a more human-like manner. Rather than coming to a conclusion directly, the CoT methods instruct LLMs to generate reasoning steps or to learn from demonstrations of detailed reasoning processes. To illustrate CoT, we consider the problem of algebraic calculation, as commonly described in the literature. Suppose we are given an algebraic problem

Calculate the average of the numbers 2, 4, and 9.

We can pose this as a question to prompt an LLM.

> Q: Please calculate the average of the numbers 2, 4, and 9.
>
> A: <u>The answer is 6.</u>

For nontrivial reasoning tasks, directly prompting an LLM may not always yield the correct answer. A simple improvement is to add demonstrations of similar problems in the prompt, and thus the LLM can learn from these demonstrations.

> Q: Please calculate the average of the numbers 1, 3, 5, and 7.
>
> A: The answer is 4.
>
> Q: Please calculate the average of the numbers 2, 4, and 9.
>
> A: <u>The answer is 7.</u>

The problem here is that, although we have shown a similar question-answer pair, it remains difficult for the LLM to reason out the correct answer. In CoT, LLMs not only learn from the correspondence between questions and answers but also benefit from the detailed problem-solving steps that are used to derive the answers. To do this, we can incorporate some reasoning steps into the prompt to obtain a CoT prompt.

"=" in a $\ll \cdots \gg$, then the calculator calculates the expression on the left-hand side of "=". This method helps reduce the calculation errors made by LLMs.

Q: Please calculate the mean square of the numbers 1, 3, 5, and 7.

A: Calculate the square of each number: $1^2 = 1, 3^2 = 9, 5^2 = 25$, and $7^2 = 49$. Sum the squares, $1+9+25+49 = 84$. There are 4 numbers in total. Divide the sum by the number of items, $84/4 = 21$. The answer is 21.

Q: Please calculate the average of the numbers 2, 4, and 9.

A: Calculate $2+4+9$, which equals 15. There are three numbers. Divide the total sum by the count, resulting in $15/3 = 5$. The answer is 5.

Here we highlight the reasoning steps in green. By providing detailed reasoning demonstrations, the LLM is encouraged to generate intermediate steps that lead to the correct answer.

There are several benefits to using CoT prompting. First, CoT allows LLMs to decompose complex problems into smaller, sequential reasoning steps. This resembles how humans solve complex problems step by step, making it particularly effective for tasks requiring detailed, multi-step reasoning. Second, CoT makes the reasoning process more transparent and interpretable. Since the intermediate reasoning steps are visible, we can better understand how a conclusion was reached. Third, if users can follow the logic behind the reasoning process, they will be more likely to trust the predictions of an LLM. This is particularly important when applying LLMs in fields like medicine, education, and finance. Fourth, CoT is an in-context learning approach, and thus, it is applicable to most well-trained, off-the-shelf LLMs. Moreover, CoT provides efficient ways to adapt LLMs to different types of problems. It can even inspire more creative solutions by exploring various alternative reasoning paths, which might not be obvious when arriving at a conclusion directly.

The method described above requires providing one or more examples of CoT reasoning, typically called the few-shot CoT method. By contrast, the zero-shot CoT method does not require such examples. It instead prompts LLMs to reason step-by-step by incorporating specific instructions into prompts. For example, below is a zero-shot CoT prompt.

Q: Please calculate the average of the numbers 2, 4, and 9.

A: Let's think step-by-step.

We have three numbers: 2, 4, and 9. Add these numbers together, $2+4+9 = 15$. Determine how many numbers there are, which in this case is three. The average is calculated by dividing the total sum by the number of elements. Completing the division gives $15/3 = 5$. So the answer is 5.

Following the instruction "Let's think step by step", the LLM is prompted to generate detailed reasoning steps. As discussed in Kojima et al. [2022]'s work, prompting with such instructions may result in LLMs generating only the reasoning steps without a clear conclusion. In this case, a second round of prompting can be used to extract the answer from these reasoning steps. For example, Kojima et al. [2022] create a second prompt which combines both the input and output from the first round of prompting. Using this combined input, the LLM can continue its reasoning process and then generate the correct answer. Furthermore, it is possible to prompt LLMs to reason using instructions other than "Let's think step by step", such as "Let's think logically" and "Please show me your thinking steps first".

While we have illustrated CoT methods using an algebraic reasoning problem, these methods can be applied to a variety of problems. Typical applications of CoT include mathematical reasoning, logical reasoning, commonsense reasoning, symbolic reasoning, code generation, etc. See Figure 9.1 for more examples of applying CoT in various tasks.

CoT is one of the most active research topics in prompt engineering. This has not only improved the performance of LLM prompting, but has also motivated a wide range of research on evaluating and verifying LLM reasoning capabilities. Although we have focused on the basic idea of CoT in this section, it can be improved in several ways. For example, we can consider the reasoning process as a problem of searching through many possible paths, each of which may consist of multiple intermediate states (i.e., reasoning steps). In general, we wish the search space to be well-defined and sufficiently large, so that we are more likely to find the optimal result. For this reason, an area of current LLM research is aimed at designing better structures for representing reasoning processes, allowing LLMs to tackle more complex reasoning challenges. These structures include tree-based structures [Yao et al., 2024], graph-based structures [Besta et al., 2024], etc. By using these compact representations of reasoning paths, LLMs can explore a wider range of decision-making paths, analogous to System 2 thinking[5]. Another line of research focuses on prompting LLMs with multi-round interactions. This involves decomposing complex problems into sub-problems, verifying and refining model outputs, employing model ensembling, and so on. Note that these methods and the issues involved are not limited to CoT. In fact, they are often used as more general approaches to improving LLMs, while CoT can be seen as a way to test the capabilities of LLMs. We will see discussions of some of these issues in the following subsections.

Before leaving our discussion of CoT, we should consider its practical limitations. One major limitation is the need for detailed, multi-step reasoning demonstrations in few-shot CoT scenarios, which may be difficult to obtain, either automatically or manually. Also, there is no standard method for breaking down complex problems into simpler problem-solving steps. This often heavily depends on the user's experience. In addition, errors in intermediate steps can also affect the accuracy of the final conclusion. For further discussion on the pros and cons of CoT, the interested reader can refer to recent surveys on this topic [Chu et al., 2023; Yu et al., 2023b; Zhang et al., 2023a].

9.2.2 Problem Decomposition

We have seen that LLMs can benefit from solving a complex problem by breaking it down into simpler sub-problems. Such an approach can be seen as an example of a broader paradigm

[5] System 1 and System 2 thinking, as described by Kahneman [2011], represent two different modes of cognitive processing. System 1 is fast, automatic, intuitive, and emotional. This mode of thinking operates effortlessly and quickly, and is often what guides our daily decisions, judgments, and impressions. System 2 is slow, deliberate, and analytical. It is activated when we need to perform complex computations.

Q: What home entertainment equipment requires cable?
Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: The answer must require cable. Of the above choices, only television requires cable. So the answer is (c).

Q: What type of person typically contracts illness?
Answer Choices: (a) hospital (b) head (c) sick person (d) elderly person (e) doctor's office

A: The answer must be a person susceptible to illness. Of the choices, "elderly per son" is correct because they typically have weaker immune systems.
So the answer is (d).

(a) CSQA

Q: Do hamsters provide food for any animals?

A: Hamsters are prey animals. Prey are food for predators. Thus, hamsters provide food for some animals. So the answer is yes.

Q: Can you use Microsoft Office without internet?

A: Yes, you can use Microsoft Office applications, although features like cloud storage and online collaboration require internet. So the answer is yes.

(b) StrategyQA

Q: Complete the rest of the sequence, making sure that the parentheses are closed properly. Input: [{

A: Let's think step by step.
0: empty stack
1: [; stack: [
2: { ; stack: [{
So the answer is }].

Q: Complete the rest of the sequence, making sure that the parentheses are closed properly. Input: $<$ [[

A: Let's think step by step.
0: empty stack
1: $<$; stack: $<$
2: [; stack: $<$ [
3: [; stack: $<$ [[
So the answer is]] $>$.

(c) Dyck languages

Q: Take the last letters of the words in "Elon Musk" and concatenate them.

A: The last letter of "Elon" is "n". The last letter of "Musk" is "k". Concatenating them is "nk". The answer is nk.

Q: Take the last letters of each word in "Renata Mariela Mona Kristin" and concatenate them.

A: The last letter of "Renata" is "a". The last letter of "Mariela" is "a". The last letter of "Mona" is "a". The last letter of "Kristin" is "n". Concatenating them is "aaan". The answer is aaan.

(d) Last Letter Concatenation

Figure 9.1: CoT in four different reasoning tasks, including CSQA, StrategyQA, Dyck languages, and Last Letter Concatenation. The CoT parts are highlighted in green.

known as **problem decomposition**, which has been extensively explored and discussed in psychology and computer science. From a psychological perspective, complex problem-solving refers to a process of addressing a problem using knowledge that helps overcome the barriers of the problem[6]. There is often no clear path to solving a complex problem. However, it is often advantageous to employ strategies that decompose the problem, thereby making it easier to tackle the corresponding sub-problems with less effort. For example, consider writing a blog about the risks of AI. If we simply prompt an LLM with the instruction "Please write a blog about the risks of AI", the LLM may generate a blog with arbitrary structures and writing styles. A better method, instead, could be to outline the blog and provide more detailed information about each section. Consider the following prompt

> You are a blog writer. Please follow the provided outline below to write a blog about the risks of AI.
>
> - Introduction
>   Introduce AI, its relevance, and the importance of understanding its risks for youth.
> - Privacy Concerns
>   Discuss how AI might compromise personal privacy through interactions online.
> - Misinformation
>   Explore AI's role in spreading misinformation and influencing young people's decisions.
> - Cyberbullying
>   Highlight how AI tools can be utilized in cyberbullying and the impact on mental health.
> - Tips for Safe AI Use
>   Offer guidelines for responsible AI usage and promote critical thinking.
> - Conclusion
>   Recap main points and encourage proactive engagement with AI ethics.
>
> ____

Here we give the title and major points for each section. Then, the LLM can use this structure to break down the writing task by filling in content for these sections. Note that the way to structure the blog can be provided by humans or even generated automatically. For example, we can use the LLM to first generate the outline, and then ask it to follow this outline to complete the writing.

In computer science, decomposing complex problems is a commonly used strategy in software and hardware system design. A well-known example is the divide-and-conquer paradigm, which is often used to design algorithms for computation problems that can be reduced to simpler, more manageable problems. For example, consider the problem of determining whether a document discusses the risks of AI. We can instruct the LLM with the following prompt.

[6]A relatively formal definition can be found in Frensch and Funke [2014]'s book: *complex problem-solving occurs to overcome barriers between a given state and a desired goal state by means of behavioral and/or cognitive, multi-step activities*.

> You are provided with a text. Please determine whether it discusses the risks of AI.
>
> {∗document∗}
>
> ——

If the document is long, inference can become computationally expensive. Alternatively, we can divide the document into relatively short segments and perform the same task on each segment. These segments can be processed in parallel to further reduce the computational cost. Next, we determine the relevance of each segment to the topic of AI risks. The final output is then generated using another prompt.

> Your task is to determine whether a text discusses the risks of AI. This text has been divided into segments, and you have obtained the relevance of each segment to the topic of AI risks. Based on this, please provide your final result.
>
> Segment 1: {∗relevance-to-the-topic1∗}
>
> Segment 2: {∗relevance-to-the-topic2∗}
>
> Segment 3: {∗relevance-to-the-topic3∗}
>
> ...
>
> ——

Now let us return to a more general discussion of problem decomposition in prompting. While problem decomposition can be applied to various NLP problems, it has been more extensively discussed and tested in reasoning tasks recently. For complex reasoning tasks, we often need a multi-step reasoning path to reach a correct conclusion. We can use LLMs to perform such reasoning in three different ways. First, LLMs can directly reach the conclusion. In other words, they can predict without explicit reasoning processes, and the underlying reasoning process remains implicit and uninterpretable. Second, LLMs are prompted to generate a multi-step reasoning path that leads to the conclusion, like CoT. However, we run LLMs just once, and all intermediate steps in reasoning are generated in a single prediction. Third, we break down the original problem into a number of sub-problems, which are either addressed in separate runs of LLMs or tackled using other systems. Here we focus our attention on the third approach, which is closely related to problem decomposition. Note, however, that a more comprehensive discussion could cover all these approaches, while the first two have been discussed to some extent in this chapter.

A general framework for problem decomposition involves two elements.

- **Sub-problem Generation**. This involves decomposing the input problem into a number of sub-problems.

- **Sub-problem Solving**. This involves solving each sub-problem and deriving intermediate and final conclusions through reasoning.

These two issues can be modeled in different ways, leading to various problem decomposition methods. One approach is to treat them as separate steps in a two-step process. For example, consider the blog writing task described at the beginning of this subsection. In the first step, we decompose the entire problem into sub-problems all at once (i.e., outline the blog). In the second step, we solve the sub-problems either sequentially or in another order (i.e., fill in content for each section as needed). The final output of this process combines the results from solving each sub-problem. While this method is simple and straightforward, it assumes that the problem is compositional, making it more suitable for tasks like writing and code generation.

However, many real-world problems require complex reasoning. A key characteristic of such problems is that their reasoning paths are not fixed in advance and may depend on intermediate results obtained during reasoning. In such cases, it is undesirable to use fixed sub-problem generation in advance. Instead, sub-problems should be generated dynamically based on the input problem, and, if possible, generated on the fly during the reasoning process. This makes problem decomposition more challenging compared with designing divide-and-conquer algorithms. Ideally, we would like to jointly design both the systems for sub-problem generation and sub-problem solving. But a more practical and widely used approach is to adopt separate models for these tasks. A straightforward way to achieve this is to adapt an LLM for these tasks by either prompting or tuning the model.

Here we consider a method based on the above idea, called **least-to-most prompting** [Zhou et al., 2023b]. The motivation for this method arises from the challenges of solving difficult reasoning problems — those that cannot be addressed by simply generalizing from a few examples. For these problems, a more effective problem-solving strategy is to follow a progressive sequence of sub-problems that systematically lead to the conclusion. More specifically, in the least-to-most prompting method, sub-problem generation is performed by prompting an LLM with instructions and/or demonstrations. For example, below is a 2-shot prompt for sub-problem generation in least-to-most prompting.

TASK Your task is to decompose a problem into several sub-problems. You will be given a few examples to illustrate how to achieve this.

DEMO Q: In a community, 5% of the population are infants, 15% are children, 40% are adults, and 40% are seniors. Which group makes up the largest portion of the population?

A: To answer the question "Which group makes up the largest portion of the population?", we need to know: "How many percent are infants?", "How many percent are children?", "How many percent are adults?", "How many percent are seniors?".

Q: Alice, Bob, and Charlie brought beads for their group project in their craft class. Alice has twice as many beads as Bob, and Bob has five times as many beads as Charlie. If Charlie has 6 beads, how many beads can they use for their craft project?

A: To answer the question "How many beads can they use for their craft project?", we need to know: "How many beads does Bob have?", "How many beads does Alice have?".

USER Q: The environmental study conducted from 2015 to 2020 revealed that the average temperature in the region increased by 2.3 degrees Celsius. What was the duration of the environmental study?

A: To answer the question "What was the duration of the environmental study?", we need to know: "When did the environmental study start?", "When did the environmental study end?".

By learning from the examples, the LLM can generate two sub-problems for answering the new problem "What was the duration of the environmental study?" (highlighted in blue and orange). Given these sub-problems, we solve them sequentially. For each sub-problem, we take all previously-generated QA pairs as context, and then produce the answer. For the example above, we need to answer the first sub-problem by prompting the LLM, like this

The environmental study conducted from 2015 to 2020 revealed that the average temperature in the region increased by 2.3 degrees Celsius.

SUB-PROB1 Q: When did the environmental study start?

A: The environmental study started in 2015.

Once we have the answer to the first sub-problem, we proceed to the second one. This time, we include both the first sub-problem and its corresponding answer in the input.

> The environmental study conducted from 2015 to 2020 revealed that the average temperature in the region increased by 2.3 degrees Celsius.
>
> SUB-PROB1 Q: When did the environmental study start?
> A: The environmental study started in 2015.
>
> SUB-PROB2 Q: When did the environmental study end?
> A: The environmental study ended in 2020.

Finally, we use the LLM to solve the original problem given the answers to all the sub-problems.

> The environmental study conducted from 2015 to 2020 revealed that the average temperature in the region increased by 2.3 degrees Celsius.
>
> SUB-PROB1 Q: When did the environmental study start?
> A: The environmental study started in 2015.
>
> SUB-PROB2 Q: When did the environmental study end?
> A: The environmental study ended in 2020.
>
> FINAL Q: What was the duration of the environmental study?
> A: The duration of the environmental study was 5 years.

The least-to-most method offers a basic approach to prompting LLMs to generate and solve sub-problems separately. We can improve it in several ways. One simple improvement is to apply various advanced prompting techniques, which do not require changes to the problem decomposition framework. For example, we can incorporate CoT into the prompting to enhance the reasoning performance of sub-problem generation and solving.

Another improvement is to explore methods for better decomposing problems and organizing problem-solving paths. To describe these approaches, we will use the symbol p_0 to denote the input problem, and use the symbols $\{p_1, \cdots, p_n\}$ to denote the sub-problems corresponding to p_0. For least-to-most prompting, we decompose p_0 into $\{p_1, \cdots, p_n\}$, given by

$$\{p_1, \cdots, p_n\} = G(p_0), \tag{9.2}$$

where $G(\cdot)$ denotes the function of sub-problem generation. Then, we solve the sub-problems $\{p_1, \cdots, p_n\}$ sequentially, yielding a sequence of answers $\{a_1, \cdots, a_n\}$. For answering the i-th sub-problem p_i, we include both the original problem p_0 and all previously generated problem-answer pairs in the context for prediction. The answer a_i is given by

$$a_i = S_i(p_i, \{p_0, p_{<i}, a_{<i}\}), \tag{9.3}$$

where $p_{<i} = \{p_1, \cdots, p_{i-1}\}$ and $a_{<i} = \{a_1, \cdots, a_{i-1}\}$. $S_i(\cdot)$ denotes the function that solves the sub-problem p_i given the context $\{p_0, p_{<i}, a_{<i}\}$. The last step is to generate the answer to the original problem p_0, which can be expressed in a similar manner to Eq. (9.3).

$$a_0 = S_0(p_0, \{p_{\leq n}, a_{\leq n}\}). \tag{9.4}$$

One way to refine this model is to modify the $G(\cdot)$ function so that the model can dynamically generate answers. Instead of generating all sub-problems at one time, we can generate each of them during problem-solving [Dua et al., 2022]. To do this, we can replace Eq. (9.2) with

$$p_i = G_i(p_0, \{p_{<i}, a_{<i}\}). \tag{9.5}$$

Hence we obtain a sub-problem generation model that operates in a step-by-step manner. At step i, we first prompt an LLM with the original problem p_0 together with the previous reasoning history $\{p_{<i}, a_{<i}\}$ to generate a new sub-problem p_i. We then solve this sub-problem using the same or another LLM conditioned on the same context (see Eq. (9.3)). This method effectively extends the reasoning capacity of LLMs by allowing them to dynamically generate and solve sub-problems in intermediate reasoning steps. As a result, the reasoning paths are not fixed in advance, and the models can choose and adapt their reasoning strategies during problem-solving.

Another way to improve the above model is to focus on developing better sub-problem solvers. In our previous discussion, we restricted $S_i(\cdot)$ to LLMs that are prompted to solve the sub-problem p_i. In fact, we can expand this function to any system that is capable of addressing the sub-problem. For example, $S_i(\cdot)$ could make calls to IR systems, thereby allowing us to access a broader range of data for problem-solving. Another example is using $S_i(\cdot)$ as a calculator to accurately compute results in mathematical problem-solving. If the sub-problem p_i is complex and requires multiple intermediate problem-solving steps, it is also possible to further decompose p_i into smaller sub-problems. For example, $S_i(\cdot)$ can be defined as a recursive program that generates and solves sub-problems. This incorporates recursion into problem-solving and allows us to address problems by iteratively decomposing them. As a result, we can define a hierarchical structure for problem-solving [Khot et al., 2023].

A further generalization of this formulation is to view it as a reinforcement learning problem. A typical method is to model a problem-solving process as a decision making process. In each step of this process, an action is taken based on the current state. These actions can include all functions for sub-problem generation and solving (i.e., $G_i(\cdot)$ and $S_i(\cdot)$). Thus, the action sequence corresponds to a problem-solving path. Since the discussion of reinforcement learning problems is beyond the scope of this chapter, we skip the precise description of this learning task. Nevertheless, developing an agent or controller to determine when and how to generate and solve a sub-problem is also a natural choice.

In NLP, problem decomposition is related to a long line of research on multi-hop question answering [Mavi et al., 2024]. This task requires the system to gather and combine information from multiple pieces of text to provide an accurate answer to a complex question. For example, to answer the question "What is the capital of the country where Albert Einstein was born?", we need to know "Where was Albert Einstein born?" and "What is the capital of Germany?". Earlier work in this area and related ones has investigated the issue of problem decomposition, though the methods might not be based on LLMs. For example, a popular method is to develop an additional neural model to generate simpler questions that address different aspects of the original question [Andreas et al., 2016; Talmor and Berant, 2018; Min et al., 2019]. This question generator can create questions in a batch or sequential manner.

Broadly speaking, problem decomposition is also related to the compositionality issue in NLP [Drozdov et al., 2022; Press et al., 2023]. For example, in semantic parsing, we map natural language sentences into structured meaning representations by breaking them down into constituent parts and understanding the sentences based on the meanings of these parts and the rules used to combine them. In early studies of this field, highly compositional sentences were considered easier test cases for evaluating systems, as it is relatively straightforward to decompose such sentences and compose the meanings of their parts. However, the task becomes much more difficult when more generalization is required for modeling compositionality in new data. In this case, we want systems to have improved abilities of **compositional generalization**. In more recent research on LLMs, this issue has been frequently discussed in compositional reasoning tasks, such as SCAN[7], as it is considered an important aspect of testing the language understanding and reasoning abilities of LLMs. This also presents new tasks for developing and examining problem decomposition methods.

In LLMs, one interesting application of problem decomposition is tool use. In some cases, it is necessary to integrate external tools into LLMs to access accurate data not available during training or fine-tuning. For example, LLMs can integrate with APIs to fetch real-time data such as weather updates, stock market prices, or news feeds, enabling them to provide up-to-date responses to user queries. When using tools, LLM predictions might include markers that indicate where and how to call external APIs. This requires decomposing the problem into sub-problems, with some handled by the LLMs and others by external tools. More detailed discussions on this topic will be presented in Section 9.2.5.

9.2.3 Self-refinement

In many cases, LLM predictions can be inaccurate or flawed. Given that current LLMs are capable of performing tasks such as refinement and correction, it is valuable to explore methods that enable these models to self-refine their outputs. Self-refinement is a common phenomenon in human cognitive processes and daily behavior. For example, when designing a product, a

[7] The SCAN tasks (Simplified versions of the CommAI Navigation tasks) are designed to evaluate the ability of LLMs to perform compositional generalization [Lake and Baroni, 2018]. They involve translating natural language commands into a sequence of actions. For example, a command "jump opposite left and walk thrice" can be translated into the action sequence "LTURN LTURN JUMP WALK WALK WALK".

designer might first create a basic prototype and then refine the design after evaluation and testing to enhance user experience and functionality. This refinement process can be iterated several times until the design is satisfactory. The predict-then-refine paradigm has a long history in NLP. One early example is Brill's tagger [Brill, 1992], where an initial POS tagging result of a sentence can be iteratively refined using a rule-based system. In the era of deep learning, a good deal of work on sequence-to-sequence problems, such as grammar correction and text rewriting, can also be viewed as examples of this paradigm.

We can prompt LLMs to do self-refinement. Consider a simple example where we first use an LLM to translate a text from Chinese to English and then use the same model to refine the translation. Here is how we can prompt the model to translate:

> Translate the following text from Chinese to English:
>
> Chinese:
> 一系列考古发现奠定红山文化在中华文明起源研究中的重要地位。最新公布的研究成果认为，大约从距今5800年开始，中华大地上各个区域相继出现较为明显的社会分化，进入文明起源的加速阶段。
>
> English:
> A series of discoveries have cemented the significant role of the Hongshan culture in studies on the origins of Chinese civilization. The latest research findings suggest that, starting from around 5800 years ago, many regions across China began to experience noticeable social differentiations, entering an accelerated phase in the origin of civilization.

Then, we can prompt it to refine the translation:

You are provided with a Chinese text along with its English translation. Please refine the translation to improve its accuracy and fluency if necessary.

Chinese:
一系列考古发现奠定红山文化在中华文明起源研究中的重要地位。最新公布的研究成果认为，大约从距今5800年开始，中华大地上各个区域相继出现较为明显的社会分化，进入文明起源的加速阶段。

English:
A series of discoveries have cemented the significant role of the Hongshan culture in studies on the origins of Chinese civilization. The latest research findings suggest that, starting from around 5800 years ago, many regions across China began to experience noticeable social differentiations, entering an accelerated phase in the origin of civilization.

Refinement:
A series of archaeological discoveries have cemented the significant role of the Hongshan culture in studies on the origins of Chinese civilization. The latest research findings suggest that, starting from around 5800 years ago, various regions across China began to experience noticeable social differentiations, entering an accelerated phase in the origin of civilization.

Here modifications to the input translation are highlighted in green. This process can be iterative: at each step, we use the translation from the previous step as input, and prompt the LLM to produce a potentially better translation.

In the above example, the refinement of the initial model output was achieved by prompting the LLM with a simple instruction like "Please refine it!". However, the refinement is based solely on the instruction-following ability of the LLM, and there is no guidance or supervision on how and where to improve the model output. A more effective approach would be to consider feedback on specific aspects that need refinement. For example, we can prompt the LLM with "Please correct all the grammatical errors in the translation", so that the model can focus more on grammatical error correction during refinement.

A general framework of self-refinement with LLMs involves three steps [Madaan et al., 2024].

- **Prediction**. We use an LLM to produce the initial model output.
- **Feedback Collection**. We obtain feedback on the model output.
- **Refinement**. We use the LLM to refine the model output based on the feedback.

The last two steps can be repeated multiple times, which leads to an iterative self-refinement process. In this framework, receiving accurate and detailed feedback is critical as it guides the subsequent refinement step. In general, feedback can be obtained in various ways [Fernandes et al., 2023; Pan et al., 2024]. We can manually examine the model's predictions to identify issues. Also, we can use a feedback model to automatically generate feedback. For instance, a reward model trained on some labeled data can help evaluate the model output and return a score indicating its quality.

Since the focus of this chapter is on LLM prompting, we consider using LLMs to provide feedback. To illustrate, let us consider an example of generating good responses to user questions:

- First, we prompt an LLM to answer the input question, as usual.

> Generate a response to the question: "What are some major environmental concerns today?"
>
> Response:
>
> <u>Major environmental concerns today include climate change, air and water pollution, deforestation, loss of biodiversity, and ocean degradation.</u>

- Then, using the following prompt, we prompt the same LLM again to get feedback on the generated response.

> You will be provided with a question and its corresponding response. Evaluate the accuracy and depth of this response, and identify any factual inaccuracies or areas where additional details could enhance understanding.
>
> Question:
>
> What are some major environmental concerns today?
>
> Response:
>
> Major environmental concerns today include climate change, air and water pollution, deforestation, loss of biodiversity, and ocean degradation.
>
> Feedback:
>
> <u>The response provided is accurate and highlights key environmental concerns. However, it would benefit from more specific and detailed descriptions of these concerns. Here are suggestions for improvement:</u>
>
> 1. <u>Include specific examples of environmental issues.</u>
> 2. <u>Add information about major pollutants.</u>
> 3. <u>Specify the regions most affected.</u>

- Given the feedback the LLM produces, we prompt it to refine the response.

> You will be provided with a question, its corresponding response, and some feedback on this response.
>
> Question:
> What are some major environmental concerns today?
>
> Response:
> Major environmental concerns today include climate change, air and water pollution, deforestation, loss of biodiversity, and ocean degradation.
>
> Feedback:
> The response provided is accurate and highlights key environmental concerns. However, it would benefit from more specific and detailed descriptions of these concerns. Here are suggestions for improvement:
> 1. Include specific examples of environmental issues.
> 2. Add information about major pollutants.
> 3. Specify the regions most affected.
>
> Now refine the response based on the provided feedback.
>
> Refined Response:
> Key environmental concerns include climate change, evidenced by rising global temperatures and severe weather patterns; significant air and water pollution in urban and industrial areas; rampant deforestation in regions like the Amazon, causing biodiversity loss; and ocean degradation, highlighted by coral reef bleaching and widespread overfishing.

Ideally, if a strong LLM is adopted, we would like to have it perform all three steps without extra training. On the other hand, if we have enough labeled data for the task of interest, we can enhance the performance of the LLM using supervised learning. For example, we can fine-tune the LLM to better adapt it to refinement tasks, or alternatively, use task-specific models, which may not necessarily be based on LLMs [Welleck et al., 2023; Schick et al., 2023]. In a broader sense, improving LLMs for self-refinement tasks can be seen as an alignment issue. For example, it has been found that some self-correction abilities can be activated through RLHF [Ganguli et al., 2023]. However, discussing these issues is beyond the scope of this chapter. Further discussion can be found in Chapter 10.

In the context of LLMs, self-refinement relates to several concepts that mimic advanced reasoning capabilities, such as the ability to self-reflect. One perspective is that endowing LLMs with self-reflection enables them to generate more accurate predictions and exhibit self-correction. This reflective capacity can be activated in various ways, such as by prompting the LLMs to engage in deliberate, step-by-step reasoning, or by providing in-context examples from which the models can learn and adjust. To illustrate, we examine the **deliberate-then-generate** (**DTG**) method proposed by Li et al. [2023a], which explicitly prompts LLMs to deliberate. In the DTG framework, given an initial, potentially flawed model output, the LLM is prompted to identify the specific error types before providing an improved revision. Below is a DTG prompt template for Chinese-to-English translation tasks.

> Given the Chinese sentence: $\{*\text{source}*\}$
>
> The English translation is: $\{*\text{target}*\}$
>
> Please first detect the type of error, and then refine the translation.
>
> Error Type: ______
>
> ______

We aim to first predict the error type (red), and then produce a refined translation (blue). This process of deliberation is guided by the instruction "Please first detect the type of error, and then refine the translation". It encourages LLMs to initially engage in thoughtful analysis and then give better results. Since error type prediction and refinement are performed in a single run of the LLM, this method incorporates both steps of feedback and refinement into one process.

In the above prompts, we assume that the LLM we use is able to review the input translation and correctly identify its error types. However, this raises new difficulties as the model may not be good at finding errors in translations. This will in turn result in extra fine-tuning or prompt engineering efforts. So a simpler method is to reduce the burden of error identification and use LLMs for deliberation only. To do this, we can replace the input translation with a random translation and assign a default error type. An example of such a prompt is shown below.

> Given the Chinese sentence:
>
> 一系列考古发现奠定红山文化在中华文明起源研究中的重要地位。
>
> The English translation is:
>
> A variety of innovative techniques have redefined the importance of modern art in contemporary cultural studies.
>
> Please first detect the type of error, and then refine the translation.
>
> Error Type: Incorrect Translation
>
> ______

In this example, the input translation is not generated by the LLM but is instead randomly sampled from the dataset. Thus, it is simply an incorrect translation for the source sentence, and we can set the error type accordingly. The LLM then generates a new translation by taking both the source sentence and the incorrect translation as input. The design of this prompt can also be considered as activating the learning capabilities of LLMs through "negative evidence" [Marcus, 1993], thereby enabling them to reflect and produce better outcomes through contrastive analysis. Nevertheless, this method does not rely on any feedback and can enhance the performance of a single LLM prediction via simple prompting.

Note that while DTG is non-iterative, iterative learning and refinement are commonly used in NLP. An advantage of these iterative approaches is that they mimic human learning and problem-solving, where continuous feedback and adjustments lead to progressively improved outcomes. Iterative methods can be applied to a range of LLM prompting problems. For example, in problem decomposition, one can incorporate new sub-problems and their solutions into the context at each step, and thus LLMs can progressively approach the solution of the original problem. On the other hand, iterative methods raise several issues that are absent in non-iterative methods. For example, errors in earlier steps may negatively impact subsequent problem-solving, and determining when to stop iterating often requires additional engineering effort.

9.2.4 Ensembling

Model ensembling for text generation has been extensively discussed in the NLP literature. The idea is to combine the predictions of two or more models to generate a superior final output. This technique is directly applicable to LLMs. For example, we can collect a set of LLMs and run each of them on the same input. The final output is obtained by combining the predictions of these models.

For LLM prompting, it is also possible to improve performance by combining predictions based on different prompts. Suppose we have an LLM and a collection of prompts that address the same task. We can run this LLM with each of the prompts and then combine the predictions. For example, below are three different prompt templates for text simplification.

> Make this text simpler.
> {*text*}
> ____

> Condense and simplify this text.
> {*text*}
> ____

> Rewrite for easy reading.
> {*text*}
> ____

Each of these prompts will lead to a different prediction, and we can combine these three predictions to produce the final output.

Formally, let $\{\mathbf{x}_1, \cdots, \mathbf{x}_K\}$ be K prompts for performing the same task. Given an LLM $\Pr(\cdot|\cdot)$, we can find the most likely prediction for each $\mathbf{x}_i$ using $\hat{\mathbf{y}}_i = \arg\max_{\mathbf{y}_i} \Pr(\mathbf{y}_i|\mathbf{x}_i)$. These predictions can be combined to form a "new" prediction:

$$\hat{\mathbf{y}} = \mathrm{Combine}(\hat{\mathbf{y}}_1, \cdots, \hat{\mathbf{y}}_K). \tag{9.6}$$

Here $\mathrm{Combine}(\cdot)$ is the combination model, which can be designed in several different ways. For example, we can select the best prediction by voting or by identifying the one that overlaps the most with others. Another approach is to combine token-level probability scores during decoding. Let $\hat{y}_j$ be the predicted token at the j-th step for model combination. The predicted token $\hat{y}_j$ is given by

$$\hat{y}_j = \arg\max_{y_j} \sum_{k=1}^{K} \log \Pr(y_j|\mathbf{x}_k, \hat{y}_1, \cdots, \hat{y}_{j-1}). \tag{9.7}$$

The interested reader can refer to Chapter 5 for more details of these methods.

In ensembling for LLM prompting, it is generally advantageous to use diverse prompts so that the combination can capture a broader range of potential responses. This practice is common in ensemble learning, as diversity helps average out biases and errors that may be specific to any single model or configuration. From the Bayesian viewpoint, we can treat the prompt $\mathbf{x}$ as a latent variable, given the problem of interest, p. This allows the predictive distribution of $\mathbf{y}$ given p to be written as the distribution $\Pr(\mathbf{y}|\mathbf{x})$ marginalized over all possible prompts

$$\Pr(\mathbf{y}|p) = \int \Pr(\mathbf{y}|\mathbf{x}) \Pr(\mathbf{x}|p) d\mathbf{x}. \tag{9.8}$$

The integral computes the total probability of $\mathbf{y}$ by considering all possible values of $\mathbf{x}$, weighted by their probabilities given p. Here $\Pr(\mathbf{y}|\mathbf{x})$ is given by the LLM, and $\Pr(\mathbf{x}|p)$ is the prior distribution of prompts for the problem. This formulation is advantageous because the integral accounts for uncertainty in the choice of $\mathbf{x}$. As a result, the final predictive distribution $\Pr(\mathbf{y}|p)$ becomes more robust to variations and biases in individual prompts. However, computing this integral directly can be computationally infeasible due to the potentially infinite space of $\mathbf{x}$. One approach to addressing this issue is to employ methods like Monte Carlo sampling, which approximate the integral using a manageable, finite number of prompts.

While the Bayesian treatment is mathematically well-defined, it is common practice in NLP to assume a non-informative or uniform prior and focus instead on constructing a set of diverse prompts. Consequently, the output can be computed using a straightforward combination model, as described in Eq. (9.6). The issue of creating high-quality, diverse prompts has been studied in CoT and other in-context learning areas. Most of the research focuses on incorporating a variety of demonstration examples across different prompts. Here, we list some of these methods.

- Given a problem, we manually create a number of demonstrations and use different ones for different prompts.
- Given a problem, we use LLMs to automatically generate demonstrations and prompts.
- Given a prompt, we create different prompts by changing the order of demonstrations in the prompt.
- Given a prompt, we use LLMs to generate a number of similar prompts.
- Given a prompt, we transform it into other forms, e.g., translating it into other languages.

Of course, in practice, we can combine these methods to achieve greater diversity. An underlying assumption here is that diverse prompts can lead to diverse model outputs. This is particularly true when dealing with relatively novel and complex problems. For stronger and more robust LLMs, the variance in output for similar prompts might not be large. In this case, the benefit of using multiple prompts can be modest.

In addition to providing diverse prompts for LLMs, another approach is to make use of the inherent variance in the outputs of LLMs. One simple way is to sample multiple outputs from the hypothesis space. This is straightforward for LLMs because their search algorithms are naturally suited to producing multiple results. For example, if beam search is used, we can easily collect all completed hypotheses in the beam to generate multiple output candidates. Also, we can modify the search algorithms to implement more powerful sampling methods and explore a wider range of high-quality hypotheses.

The next question is how to combine diverse model predictions. This leads us back to the model combination issue as described in Eq. (9.6). Here we consider the **self-consistency** method, which outputs not necessarily the highest-probability prediction, but rather the one that is most consistent with the other predictions [Wang et al., 2022d; 2023c]. First, an LLM is prompted with CoT as usual and generates multiple reasoning paths by sampling. Then, the frequency of each answer is counted across these reasoning paths. The final output is the answer that appears most frequently. Below are three different predictions generated by prompting an LLM with CoT.

Prediction 1 (Correct):

> Q: Three friends play a game where they flip a fair coin. Each friend flips the coin once. What is the probability that exactly one of them flips heads?
>
> A: Let's think step by step.
> <u>Each coin flip is an independent event with a 50% chance of landing heads (H) and 50% tails (T). Consider the possible outcomes for three flips: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT. Out of these, only HHH, HHT, HTH, and THH are relevant. We are only interested in scenarios with exactly one head: HTT, THT, and TTH. There are 3 such outcomes out of 8 total, giving a probability of 3/8.</u>

Prediction 2 (Correct):

> Q: Three friends play a game where they flip a fair coin. Each friend flips the coin once. What is the probability that exactly one of them flips heads?
>
> A: Let's think step by step.
>
> Using a binomial probability formula, where p (success or heads) is 0.5, and each of the 3 trials (flips) is independent, we look for exactly one success (head). The formula for exactly k successes in n trials is given by the binomial coefficient formula $\binom{n}{k}p^k(1-p)^{n-k}$. Plugging in the values, we get $\binom{3}{1}(0.5)^1(0.5)^{3-1} = 3 \cdot 0.5 \cdot 0.25 = 0.375$.

Prediction 3 (Wrong):

> Q: Three friends play a game where they flip a fair coin. Each friend flips the coin once. What is the probability that exactly one of them flips heads?
>
> A: Let's think step by step.
>
> Since each coin flip has a 50% chance of landing heads and a 50% chance of landing tails, and we are looking for the case where only one of the three flips results in heads, we might initially think this is simply a matter of picking which flip is the heads. There are three flips, so one might incorrectly assume that the chance of picking one specific outcome like this would be 1 out of 3. Thus, they might conclude that the probability of exactly one head is $1/3 = 33.3\%$.

Predictions 1 and 2 correctly identify the three cases where exactly one head is flipped, both obtaining a probability of 37.5%. The reasoning in Prediction 3 fails to account for the total number of outcomes possible with three coin flips, thus giving a wrong answer of 33.3%. Therefore, we select 37.5% as the final answer because it is the consensus.

Self-consistency provides a criterion for determining the best prediction in a pool of candidates. Since the prompt and the model are fixed in this method, it is not strictly a prompt ensembling method. Instead, it can be seen as an instance of output ensembling methods, also known as hypothesis selection methods, which have long been explored in NLP, particularly for text generation problems [Xiao et al., 2013]. In these methods, multiple outputs are generated by varying model architectures or parameters. Each output is then assigned a score by some criterion, and the outputs are re-ranked based on these scores. There are various ways to define the scoring function, such as measuring the agreement between an output and others, and using a stronger model to rescore each output[8]. Figure 9.2 shows a comparison of different

[8]An interpretation of self-consistency is to view it as a minimum Bayes risk search process. It searches for the best output by minimizing the Bayes risk. More specifically, a risk function $R(\mathbf{y}, \mathbf{y}_r)$ is defined on each pair of outputs (denoted by $(\mathbf{y}, \mathbf{y}_r)$), representing the cost of replacing $\mathbf{y}$ with $\mathbf{y}_r$. Given a set of outputs Ω, the risk of an

ensembling methods for LLMs.

Now, let us briefly review the methods we have discussed so far in this section, such as problem decomposition and self-refinement. It is apparent that these methods enhance decision-making by introducing more "choices" into the reasoning process. To some extent, they all involve evaluating and providing feedback on the results of LLMs. For example, in self-refinement, we need to offer suggestions for improving the prediction of LLMs, and in output ensembling, we select the optimal output from a pool of candidates. In this sense, these methods fall under the broader category of predict-then-verify approaches, where predictions are initially made, then verified and refined. The fundamental problem here involves verifying and evaluating the reasoning results or intermediate steps. This issue is somewhat related to the problem of training reward models in RLHF, although RLHF addresses a different aspect. In fact, the development of verifiers has been explored and implemented in reasoning with LLMs. Most work, rather than developing heuristic-based inference-time algorithms, focuses on learning verifiers in a supervised manner. A straightforward method is to train verifiers as binary classifiers, such as classifying an answer as correct or incorrect, although these verifiers are typically used as scoring models. Given a reasoning path for a problem, the verifiers can be used to score either the entire path (called outcome-based approaches) [Cobbe et al., 2021], or each individual reasoning step (called process-based approaches) [Uesato et al., 2022; Lightman et al., 2024].

9.2.5 RAG and Tool Use

RAG is generally employed when standard LLMs lack sufficient factual accuracy or access to up-to-date knowledge. By drawing from external databases and documents, RAG can significantly improve contextual relevance and factual grounding of responses. This approach is particularly useful in scenarios that require high factual accuracy and up-to-date information, such as complex question answering.

While RAG has been introduced in previous chapters, we briefly outline its core steps here for completeness.

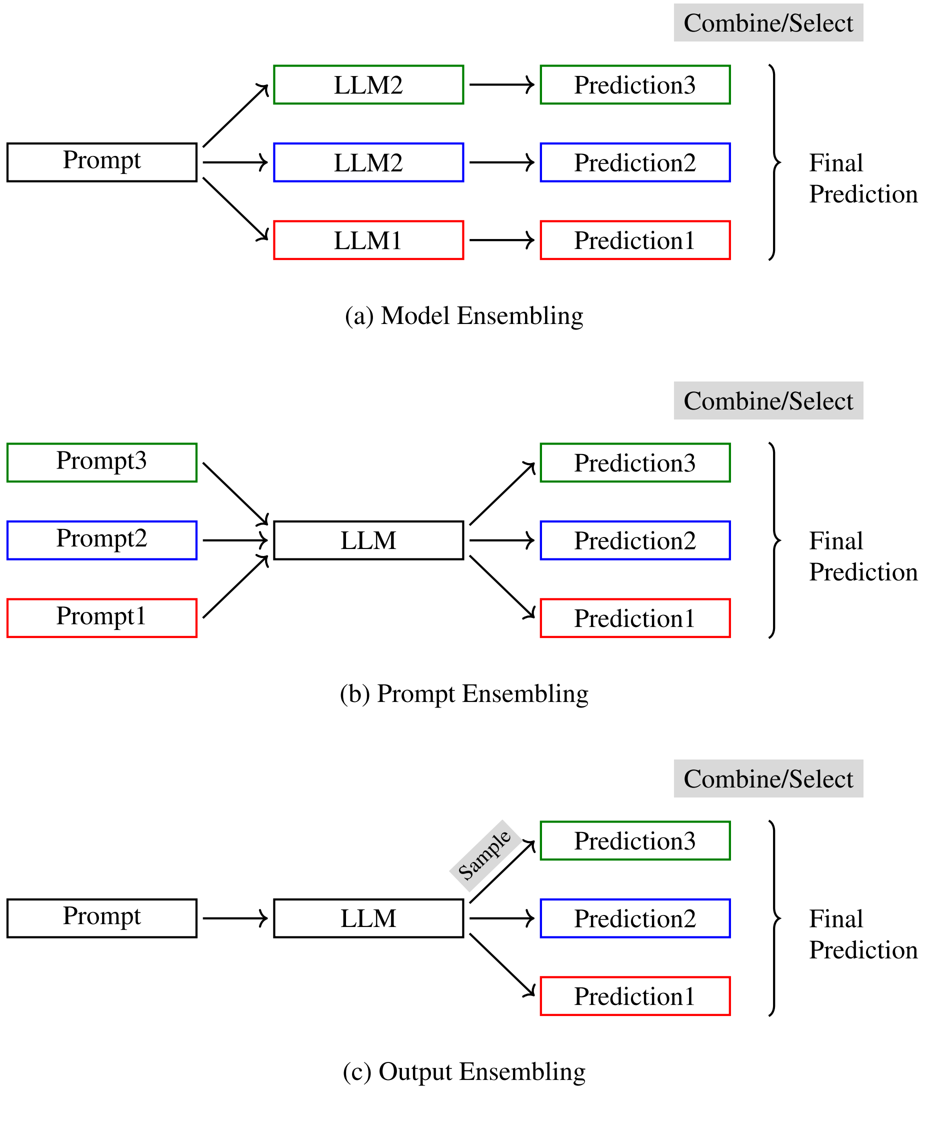
- We prepare a collection of texts that serves as an external knowledge source.
- We retrieve relevant texts for a given query.
- We input both the retrieved texts and the query into an LLM, which is then prompted to produce the final prediction.

Steps 1 and 2 can be implemented by using an external information retrieval system. For example, we can store the collection of texts in a vector database and then retrieve relevant texts using vector-based search techniques. Since information retrieval is not the focus of this chapter, we will assume that such systems are available off-the-shelf and use them directly.

---

output $\mathbf{y} \in \Omega$ is given by

$$\begin{aligned}\mathrm{Risk}(\mathbf{y}) &= \mathbb{E}_{\mathbf{y}_r \sim \Pr(\mathbf{y}_r|\mathbf{x})} R(\mathbf{y}, \mathbf{y}_r) \\ &= \sum_{\mathbf{y}_r \in \Omega} R(\mathbf{y}, \mathbf{y}_r) \cdot \Pr(\mathbf{y}_r|\mathbf{x}).\end{aligned} \quad (9.9)$$



(a) Model Ensembling

(b) Prompt Ensembling

(c) Output Ensembling

Figure 9.2: Ensembling methods for LLMs. In standard model ensembling (a), multiple LLMs varying in architectures or parameters are used. Each LLM receives the same prompt and produces a prediction. These predictions are combined to generate the final prediction. In prompt ensembling (b), we have one LLM and multiple prompts. The LLM produces a prediction for each prompt, and these predictions are combined as usual. In output ensembling (c), the LLM samples multiple predictions over the prediction space given a prompt. It can be seen as a method to boost the performance of the LLM itself. Note that these ensembling methods can be combined to increase the diversity of predictions. For example, we can use both prompt ensembling and output ensembling to obtain more diverse predictions.

Here we present how to prompt LLMs to make use of retrieved texts. To illustrate, consider an example of using LLMs to answer the following question.

Where will the 2028 Olympics be held?

We can simply input this question into an online search engine. It will then return the relevant pieces of text found on the internet, for example,

> (Wikipedia)
> *The 2028 Summer Olympics, officially the Games of the XXXIV Olympiad and commonly known as Los Angeles 2028 or LA28, is an upcoming international multi-sport event scheduled to take place from July 14-30, 2028, in the United States. ...*
>
> (The Sporting News)
> *In 2028, Los Angeles will become the third city, following London and Paris respectively, to host three Olympics after hosting the Summer Games in 1932 and 1984. It will also be the first time the United States has hosted an Olympic Games since the 2002 Winter Games in Salt Lake City. ...*
>
> ...

We can use these retrieved texts as additional context, and prompt an LLM to generate a response based on these texts. Below is an example RAG prompt.

> Your task is to answer the following question. To help you with this, relevant texts are provided. Please base your answer on these texts.
>
> Question:
> Where will the 2028 Olympics be held?
>
> Relevant Text 1:
> The 2028 Summer Olympics, officially the Games of the XXXIV Olympiad and commonly known as Los Angeles 2028 or LA28 ...
>
> Relevant Text 2:
> In 2028, Los Angeles will become the third city, following London and Paris respectively, to host three Olympics after ...
>
> ...
>
> The 2028 Olympics will be held in Los Angeles.

This prompt assumes that the provided texts are relevant to the question and expects the LLM to generate a faithful response using these texts. However, the information retrieval system may sometimes provide irrelevant or incorrect texts, which may lead the LLM to produce an incorrect answer. One straightforward way to address this issue is to improve the accuracy of the information retrieval system. Nevertheless, as with most AI systems, errors may still occur. Therefore, it is also necessary to enhance the robustness of the LLM, so that it can make reasonable predictions even when the input is inaccurate. Below is a new prompt that enables the LLM to be more faithful to the facts, and allows it to choose not to answer questions when the information provided is inaccurate.

> Your task is to answer the following question. To help you with this, relevant texts are provided. Please base your answer on these texts.
>
> Please note that your answers need to be as accurate as possible and faithful to the facts. If the information provided is insufficient for an accurate response, you may simply output "No answer!".
>
> Question:
>
> Where will the 2028 Olympics be held?
>
> Relevant Text 1:
>
> The 2024 Summer Olympics, officially the Games of the XXXIII Olympiad and branded as Paris 2024, were an international multi-sport event ...
>
> ...
>
> No answer!

In this example, the LLM refuses to answer because the provided information is insufficient and irrelevant to the question.

Both RAG and fine-tuning are common methods for adapting LLMs using task-specific data. Standard RAG is training-free and can be directly applied to LLMs. To further improve RAG, it is also possible to fine-tune LLMs, though this will require some training effort. For example, we can fine-tune LLMs using human-labelled data to supervise them in learning to refuse to answer. Note that, while the examples shown above seem simple, RAG is not trivial. From the prompt engineering perspective, different use cases may require different prompts, though our broader goal is to develop prompting strategies that generalize across tasks. In many cases, we need to control how much we depend on the retrieved context to make predictions. Sometimes, LLMs must generate responses strictly based on the provided texts, while at other times, they may need to generate responses using their existing knowledge if the provided texts are insufficient. There are many aspects of RAG, such as improvements to the retrieval systems, that cannot be covered in this chapter. Interested readers can refer to surveys of RAG techniques for more information [Li et al., 2022d; Gao et al., 2023c].

One reason we discuss RAG here is that it can be broadly regarded as an instance of the general problem decomposition framework (see Section 9.2.2). RAG divides problem-solving into two steps. In the first step, we collect relevant and supporting information for a given query from various knowledge sources. In the second step, we use LLMs to generate responses based on the collected information. If we extend the concept of problem decomposition further, we will find that many tasks requiring the use of external systems or tools can be treated as similar problems. One such example is tool use in LLMs. In many applications, LLMs need to employ external databases, APIs, and even simulation tools to generate accurate responses. For example, LLMs can access real-time data from financial markets to provide financial information or access medical knowledge bases to offer personalized medical information. This integration extends the capabilities of LLMs by allowing them to interact with, and in some contexts, influence or control external systems. Consequently, LLMs can function as simple autonomous agents rather than merely text generators. [Franklin and Graesser, 1996].

Tool use is a broad research area. Here we narrow our discussion to tasks that can be facilitated by calling external APIs to solve some of the sub-problems [Parisi et al., 2022; Gao et al., 2023b]. Consider again the example of asking an LLM to answer "Where will the 2028 Olympics be held?". Suppose the LLM can access a web search tool. We can then prompt the LLM to answer the question using web search, like this:

> Your task is to answer the following question. You may use external tools, such as web search, to assist you.
>
> Question:
>
> Where will the 2028 Olympics be held?
>
> The information regarding this question is given as follows:
>
> `{tool: web-search, query: "2028 Olympics"}`
>
> So the answer is: Los Angeles

Here `tool: web-search, query: "2028 Olympics"` indicates a request to the web search system using the query "2028 Olympics". The generated tool command is intercepted by an external system, which executes the corresponding tool and appends the returned result to the context. The updated context is then fed back into the LLM for subsequent prediction, and thus the model can generate the correct answer using the retrieved information.

Consider another example where we ask the LLM to solve a mathematical problem.

> Problem:
>
> A swimming pool needs to be filled with water. The pool measures 10 meters in length, 4 meters in width, and 2 meters in depth. Calculate the volume of the pool in cubic meters and then determine how many liters of water are needed to fill it (considering 1 cubic meter equals 1000 liters).
>
> Solution:
>
> To solve this problem, the LLM needs to first calculate the volume of the pool by using the formula for the volume of a rectangular prism: Length$\times$ Width $\times$ Depth. Therefore, the volume is $10\,\mathrm{m} \times 4\,\mathrm{m} \times 2\,\mathrm{m} =$ `{tool: calculator, expression: 10 * 4 * 2}` m^3. Next, to find out how many liters of water are needed, the LLM multiplies the volume in cubic meters by 1000 (since 1 cubic meter equals 1000 liters). Thus, 80 $\times 1000$ $=$ `{tool: calculator, expression: 80*1000}` liters.

Here the string `{tool: calculator, expression: 10 * 4 * 2}` triggers the invocation of a mathematical interpreter to calculate the result of the expression. Note that the result (i.e., 80) will replace `{tool: calculator, expression: 10 * 4 * 2}` and can be referred to in the following token predictions. For example, in the last step of problem-solving, 80 is used instead of `{tool: calculator, expression: 10 * 4 * 2}`.

A key difference between the tool use examples here and the previously discussed RAG examples is that in tool use, external functions can be called during inference. In contrast, in RAG, the retrieved texts are provided before the prediction process begins. However, from the language modeling perspective, they are similar: before generating the final result, we use external tools, either manually or automatically, to obtain sufficient and relevant context. A high-level interpretation of these approaches is that they both rely on an "agent" that can determine where and how to call external functions to generate the context necessary for prediction.

An issue with tool use is that the original LLM is not trained to generate the necessary markers for tool use. Therefore, we need to fine-tune the LLM to adapt it for these tasks [Schick et al., 2024]. As this chapter focuses on prompting, we will not present the details of this fine-tuning process. To put it simply, we first need to annotate data. For each fine-tuning example, we replace parts of the output that require the use of external tools with predefined commands or markers. Then, we use this labeled data to fine-tune the parameters of the LLM as usual. As a result, the LLM can gain the ability to generate commands for calling external tools. During inference, we can execute these tool use commands in the model outputs to get assistance from external tools.

9.3 Learning to Prompt

So far in this chapter, we have considered several basic prompting strategies and various refinements to them. However, all the prompts we have discussed were designed manually. This leads to a number of problems. First, designing high-quality prompts is difficult and often requires extensive experimentation to identify effective prompts. Since different LLMs may vary in their sensitivity to prompting styles, developing prompts that generalize well across models can be resource-intensive. Second, manual prompt design relies heavily on human expertise, which can limit the diversity of approaches and overlook potentially effective prompts that are not immediately obvious to humans. Third, prompts created by humans can be complex and verbose, leading to longer inputs for LLMs and higher computational costs.

In this section, we discuss techniques for automated prompting. These methods aim to automatically create, optimize, and represent prompts so that the downstream tasks can be addressed more effectively and efficiently. In particular, we consider three issues here.

- How can we automate the process of designing and optimizing prompts for LLMs?
- Are there representations of prompts beyond plain text strings, and how can we learn such representations?
- How can we make prompts more concise and compact, thereby reducing their complexity

and length?

Note that there are many settings in which we can investigate these issues. For example, prompts may be optimized for a specific LLM or designed to generalize across different LLMs. These settings can lead to different methods and application scenarios, but these methods may overlap in some ways. In the following discussion, we will cover several different scenarios and discuss the connections between various methods.

9.3.1 Prompt Optimization

Because prompt design is labor-intensive, it is desirable to use machine learning models to discover the optimal prompt for a specific task (call it **automatic prompt design** or **prompt optimization**). This approach can broadly be regarded as an instance of **automated machine learning** (**AutoML**), which aims to reduce or eliminate the need for expert-driven manual design of machine learning models. Although our focus here is on the design of prompts, prompts themselves are discrete structures. Therefore, prompt optimization is similar to the search over discrete model architectures in machine learning. One closely related area is **neural architecture search** (**NAS**), where the optimal neural networks are identified by exploring a space of possible neural networks [Zoph and Le, 2016; Elsken et al., 2019a]. If we consider prompt optimization as a search process, then we can describe a general prompt optimization framework involving the following components:

- **Prompt Search Space**. This defines all possible prompts that the algorithms can explore. For example, one can edit some seed prompts to generate a set of diverse candidate prompts.
- **Performance Estimation**. Once a prompt is chosen, it needs to be evaluated. For example, a straightforward way is to input it to an LLM and measure its performance on a validation set.
- **Search Strategy**. The search process is generally the same as that used in many AI systems. At each step, the system explores a set of promising prompts in the search space and evaluates them. This process continues as more prompts are explored. The outcome of the search is the best-performing prompt observed until the search stops.

This is a very general framework, and different prompt optimization systems can vary in their design of each component. A widely used approach is to use LLMs as the basis to develop these components. Initially, a few prompts are provided. Then, the following process is iterated until a stopping criterion is met: 1) the prompts are evaluated on a validation set; 2) a candidate pool is maintained by keeping only the most promising prompts; and 3) new prompts are created by employing LLMs to infer similar prompts from this candidate pool. One benefit of this approach is that it allows us to use off-the-shelf LLMs to perform the tasks mentioned above without the need for substantial system development. To achieve this, we can prompt or fine-tune LLMs to adapt them to these tasks. Here we consider Zhou et al. [2023c]'s method for illustrating LLM-based prompt optimization. It involves the following steps.

- **Initialization**. Let C represent the pool of candidate prompts to be explored. The first step is to add initial prompts into C. We can do this in several ways. A simple method is to create such prompts by hand for a given task. However, in many cases where humans have limited knowledge about how to write effective prompts for the task, developing prompts becomes challenging. In these cases, it is desirable to use LLMs to generate prompts. For example, we can directly instruct LLMs to produce prompts, providing them with a description of the task.

> You are given a task to complete using LLMs. Please write a prompt to guide the LLMs.
>
> $\{*\text{task-description}*\}$
>
> ____

This method is straightforward, but it still requires a human-provided description of the task. An alternative method is to use LLMs to generate prompts given examples of the input and output of the task. Here is a prompt template.

> You are provided with several input-output pairs for a task. Please write an instruction for performing this task.
>
> Input: $\{*\text{input1}*\}$ Output: $\{*\text{output1}*\}$
>
> Input: $\{*\text{input2}*\}$ Output: $\{*\text{output2}*\}$
>
> ...
>
> ____

In this way, LLMs can infer the corresponding instruction for the task from the provided inputs and outputs.

- **Evaluation**. Once we obtain the candidate pool C, we need to evaluate the prompts in C. One method is to feed each prompt into an LLM and assess the results on the downstream task. For example, we can evaluate the output of the LLM given an input using a pre-defined metric, or alternatively, use the log-likelihood of the output as a measure of the quality of the prompt.
- **Pruning**. If C contains a large number of prompts, it is reasonable to prune the unpromising prompts within it, thus reducing the computational burden in subsequent steps. This is a standard pruning problem. Given the evaluation score for each prompt, a simple method is to keep only a certain percentage of the prompts and discard the rest.
- **Expansion**. Expansion is a key operation in search algorithms used to explore different states in the search space. The expansion operation here can be defined as a function:

$$C' = \text{Expand}(C, f), \tag{9.10}$$

where C' is the set of new prompts generated from C using the model f. If we consider f as an LLM, we can perform the expansion operation by instructing f to generate new and relevant prompts based on C. Below is an example.

> Below is a prompt for an LLM. Please provide some new prompts to perform the same task.
>
> Input: $\{*\text{prompt}*\}$
>
> ____

Then, we replace C with C'. The steps of evaluation, pruning and expansion can be repeated, and so we can gradually explore a wider range of prompts.

In prompt optimization, the expansion step plays a key role, as it defines how we explore the search space, and our goal is to find optimal results with minimal effort. One improvement to this step is to treat the problem as a paraphrasing task. A simple method is to apply off-the-shelf paraphrasing systems, either based on LLMs or other models, to transform input prompts into semantically equivalent forms [Jiang et al., 2020]. Alternatively, we can define specific edit operations, such as insertions and modifications, for each token. A given prompt can be edited into new prompts by applying these operations [Prasad et al., 2023]. Also, further evaluation and pruning can be applied to filter out low-quality prompts. In addition to framing prompt generation as a paraphrasing problem, we can improve the quality of prompts during expansion by learning from feedback [Pryzant et al., 2023]. This approach is somewhat related to the self-refinement issue discussed in Section 9.2.3. An LLM can be used to generate feedback on an input prompt, which is then revised based on this feedback. This feedback-and-revision cycle can be repeated multiple times until the result converges or the desired outcome is achieved.

Another approach to prompt optimization is to apply classic optimization techniques. For example, the problem can be framed as an evolutionary computation problem, where prompts are treated as candidates that evolve generation by generation as the optimization progresses [Guo et al., 2024]. Since many powerful optimization algorithms have been developed in related fields, they can be directly applied to this problem.

In practice, we might be tempted to use existing LLM APIs to implement the steps described above. Such an approach, however, depends heavily on the inference and in-context learning abilities of the LLMs. If these LLMs are not strong and lack adaptation to the tasks, they may introduce errors into search, for example, generating incorrect prompts during expansion. In such cases, it is preferable to train models that are better suited to the tasks. One approach in this research direction employs reinforcement learning, which has been widely used in solving discrete decision making and optimization problems. For example, Deng et al. [2022] developed a prompt generator by integrating an FFN-based adaptor into an LLM. The prompt generator is trained as a typical policy network, but only the parameters of the adaptor are updated while the remaining parameters of the model are kept unchanged. During training, the reward is obtained by testing the generated prompts using another LLM, similar to the evaluation method as discussed above. Once the training is complete, the prompt generator is then employed to generate new prompts.

Note that, in our discussion here, prompts are simply seen as sequences of tokens, and the output of prompt optimization is such a sequence. However, in a strict sense, prompts often contain multiple components, such as instructions, demonstrations, and user inputs. While our discussed approaches are mostly general, much work in prompt optimization has focused on learning better instructions for prompting. Specifically, the goal is to generate instructions that effectively guide LLMs based on a given task. Of course, the concept of prompt optimization can also be extended to learning other parts of prompts. For example, there has been substantial research interest in learning to select or generate demonstrations in CoT [Liu et al., 2022; Rubin et al., 2022; Zhang et al., 2023b]. One of the differences between learning instructions and learning demonstrations is that generating high-quality demonstrations using LLMs is relatively easy and the focus of learning demonstrations is typically on how to sample appropriate demonstrations from a pool of candidates. In contrast, the difficulty in learning instructions is partly because pre-trained LLMs are not suited to predict the quality of instructions, and testing these instructions on downstream tasks is computationally expensive. This makes the optimization methods costly to apply, and exploring a wide variety of instructions poses significant challenges.

9.3.2 Soft Prompts

Although developing natural language prompts, either manually or automatically, is a straightforward and widely applied approach, it presents some problems. One problem is that natural language prompts can be complex and lengthy, resulting in significant computational burdens when processed by LLMs. In many applications, users may need to perform a task repeatedly, and inputting the same long prompt into the LLMs a large number of times is clearly inefficient. Another problem is that while prompts are typically represented as discrete token sequences (call them **hard prompts**) in regular LLM input, the LLMs encode them as continuous vectors. This raises the question of whether there are more compact and efficient ways to represent prompts.

In this subsection, we introduce the concept of **soft prompts**, which can be viewed as hidden, distributed representations of prompts. When prompting LLMs, we are concerned with communicating tasks or questions to elicit the desired responses. We can define hard prompts as explicit, predefined text sequences that users input directly into LLMs to guide the responses. In contrast, we can think of soft prompts as implicit, adaptable prompting patterns represented in the continuous hidden space of LLMs. Unlike hard prompts, which are expressed in natural language and should be understandable for humans, soft prompts are encoded in a format that is more comprehensible to the model rather than to humans. To illustrate, consider a simple prompt

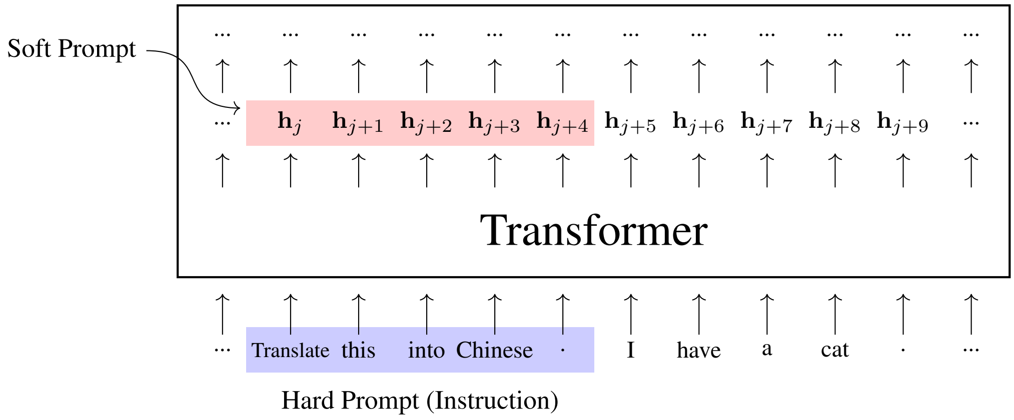


Figure 9.3: Illustration of hard and soft prompts. Here the hard prompt is the instruction we input to the LLM for performing the task. The LLM encodes this instruction as usual, and the intermediate representations corresponding to the instruction can be viewed as some sort of soft prompt.

> Translate the sentence into Chinese.
>
> Consider it done!
>
> ____

Here, the instruction "Translate the sentence into Chinese" can be seen as a hard prompt, denoted by the token sequence $c_1 \cdots c_5$. By feeding these tokens into an LLM, they are transformed into a sequence of real-valued vectors $\mathbf{h}_1 \cdots \mathbf{h}_5$, each corresponding to a token. We can view $\mathbf{h}_1 \cdots \mathbf{h}_5$ as an intuitive illustration of soft prompts, as shown in Figure 9.3.

While the above example shows that soft prompts can be generated by transforming hard prompts, there is not necessarily a direct correspondence between them. In fact, we do not even need to interpret soft prompts using meaningful text. Instead, soft prompts are represented as continuous vectors in the hidden space of LLMs and can be optimized directly as learnable parameters. Such a treatment allows us to explore prompting methods beyond text. As another benefit, soft prompts provide dense, learnable representations for encoding how we guide LLMs to generate specific outputs. The training and application of these representations require significantly lower computational costs than those required for processing long hard prompts. This approach would be of great practical value in LLM inference applications where the same prompt is repeatedly used.

1. Adapting LLMs to Simplified Prompts

One obvious way to adapt an LLM for a particular task is to simply fine-tune the model using labeled data. This leads to a variety of LLM alignment methods, such as supervised fine-tuning, which update the model parameters by aligning the responses to given prompts with supervision signals. Fine-tuned LLMs embed task-related information in model parameters, and thus these models can respond correctly when dealing with similar prompts to those in fine-tuning.

If we take this idea further, we can expect LLMs to absorb the knowledge about prompting of a task as much as possible during fine-tuning. Consequently, the prompting information is partially captured in the model parameters, and the fine-tuned LLMs can perform the task with less prompting. Here we consider a simple form of prompt, where only an instruction (denoted by $\mathbf{c}$) and a user input (denoted by $\mathbf{z}$) are included. A prompt can be expressed using the following tuple:

$$\mathbf{x} \quad = \quad (\mathbf{c}, \mathbf{z}). \tag{9.11}$$

Given a set of prompt-response pairs $\mathcal{D} = \{(\mathbf{x}, \mathbf{y})\}$, the objective of fine-tuning is to minimize the total loss incurred over this set. A popular method is to minimize the negative log-likelihood (i.e., maximize the log-likelihood) with respect to the model parameters θ:

$$\begin{aligned} \hat{\theta} &= \arg\max_{\theta} \sum_{(\mathbf{x},\mathbf{y}) \in \mathcal{D}} \log \Pr_{\theta}(\mathbf{y}|\mathbf{x}) \\ &= \arg\max_{\theta} \sum_{(\mathbf{x},\mathbf{y}) \in \mathcal{D}} \log \Pr_{\theta}(\mathbf{y}|\mathbf{c}, \mathbf{z}), \end{aligned} \tag{9.12}$$

where $\Pr_{\theta}(\cdot|\cdot)$ is the probability predicted by an LLM with the parameters θ[9].

In general, the instruction in each fine-tuning example should follow the guideline of prompt design, for example, a good instruction should be as clear as possible and provide a detailed description of the task. However, the method described in the above equation does not restrict the instruction to any particular form. This flexibility allows us to instruct LLMs in any way that we want. Consider an example where we intend to instruct LLMs to translate an English sentence into Chinese. Of course, as mentioned earlier in this chapter, we can prompt LLMs using the instruction

Translate the following sentence from English to Chinese.

If we want the instruction to be simpler, we may rephrase it into a simpler form

Translate this into Chinese.

We can even define the instruction as a single phrase

Translate!

With fine-tuning, we can adapt LLMs to follow any of these instructions. From an efficient prompting perspective, there are computational advantages in simplifying instructions in prompting. For example, we can use simple instructions like “Translate!” to perform tasks that

[9]In practice, we initialize θ with the parameters obtained from pre-training, and then adjust θ moderately to ensure that the results after fine-tuning do not deviate too much from the pre-trained results.

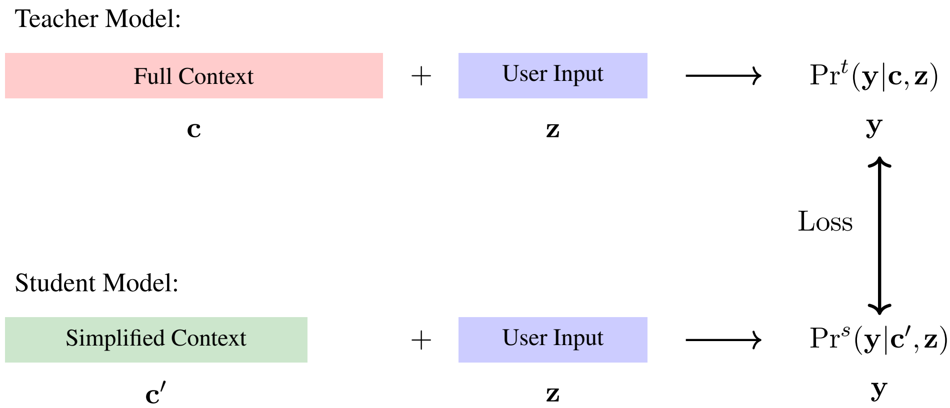


Figure 9.4: Illustration of context distillation [Snell et al., 2022]. The teacher model is a standard LLM, which takes both the context and the user input as model input and produces a prediction as model output. Then, we simplify the context (e.g., simplifying the instruction in prompting) and use the student model to make predictions based on the simplified context and the user input. The student model is trained by minimizing the loss between the predictions produced by the two models.

would typically require more complex and detailed instructions. This can make subsequent prompting during inference much easier. On the other hand, fine-tuning LLMs with overly simplified instructions may be harmful to the generalization of the models. Since simplified instructions can lead to a loss of information, it is more likely that the LLMs will overfit the fine-tuning data and fail to generalize beyond those instructions. This issue becomes more pronounced when both complex and simplified instructions are included during fine-tuning, since the available labeled data is often insufficient to cover a wide range of instruction variations.

An alternative way to adapt LLMs for simplified instructions is through knowledge distillation. As an example, we consider the context distillation method [Snell et al., 2022]. The goal of this method is to learn a student model that can make use of simplified instructions from a well-trained instruction-following teacher model. Figure 9.4 shows an illustration of this approach. The teacher model is built using a standard fine-tuning process: we first collect a certain amount of data that includes instructions, user inputs, and correct responses, and then we continue to train a pre-trained model with this dataset. For building the student model, we need to construct a new dataset $\mathcal{D}'$ where each sample is a tuple consisting of an instruction, a corresponding simplified instruction, and a user input, denoted by $\mathbf{x}' = (\mathbf{c}, \mathbf{c}', \mathbf{z})$. Knowledge distillation is performed by minimizing a loss function defined on the outputs of the teacher and student models:

$$\hat{\theta} \quad = \quad \arg\min_{\theta} \sum_{\mathbf{x}' \in \mathcal{D}'} \mathrm{Loss}(\Pr^t(\cdot|\cdot),\ \Pr^s_{\theta}(\cdot|\cdot),\ \mathbf{x}'), \tag{9.13}$$

where $\Pr^t(\cdot|\cdot)$ denotes the pre-trained teacher model, and $\Pr^s_{\theta}(\cdot|\cdot)$ denotes the student model with the parameters θ. To keep the notation simple we will write $\mathrm{Loss}(\mathrm{Pr}^t(\cdot|\cdot),\ \mathrm{Pr}^s_\theta(\cdot|\cdot),\ \mathbf{x}')$ as Loss for short. A commonly used loss is the sequence-level loss, which has the basic form:

$$\mathrm{Loss} = -\sum_{\mathbf{y}} \mathrm{Pr}^t(\mathbf{y}|\mathbf{c},\mathbf{z}) \log \mathrm{Pr}^s_\theta(\mathbf{y}|\mathbf{c}',\mathbf{z}). \tag{9.14}$$

But this function is computationally infeasible because it requires summing over an exponentially large number of possible output sequences. A variant of this method is to train the student model using outputs generated by the teacher model. For each sample, we use the teacher model to produce an output $\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \log \mathrm{Pr}^t(\mathbf{y}|\mathbf{c},\mathbf{z})$. Then we consider $\hat{\mathbf{y}}$ as the target for learning, and the loss function is given by

$$\mathrm{Loss} = -\log \mathrm{Pr}^s_\theta(\hat{\mathbf{y}}|\mathbf{c}',\mathbf{z}). \tag{9.15}$$

Alternatively, we can minimize the distances between the probability distributions generated by the two models [Askell et al., 2021]. For example, the loss function can be defined as the KL divergence between the two output distributions:

$$\mathrm{Loss} = \mathrm{D}_{\mathrm{KL}}(\mathrm{P}^t \,\|\, \mathrm{P}^s_\theta), \tag{9.16}$$

where

$$\mathrm{P}^t = \mathrm{Pr}^t(\cdot|\mathbf{c},\mathbf{z}), \tag{9.17}$$

$$\mathrm{P}^s_\theta = \mathrm{Pr}^s_\theta(\cdot|\mathbf{c}',\mathbf{z}). \tag{9.18}$$

Simplifying prompts through fine-tuning or distillation does not necessarily produce explicit soft prompts in the conventional sense. However, these approaches share a common intuition: task-specific prompting behavior can gradually be absorbed into compact continuous representations within the model.

Although we have restricted ourselves to knowledge distillation for instructions, the approaches discussed here are general. By learning from the outputs of the teacher model, the knowledge in prompting can be distilled into the parameters of the student model. Therefore, the distilled model can be viewed as implicitly encoding task-specific prompting behavior in its parameters. This method can be applied to many other problems in prompt learning, such as compressing long contexts and learning soft prompts as specific components of LLMs.

2. Learning Soft Prompts for Parameter-efficient Fine-tuning

Full fine-tuning, which updates all model parameters, is a common method for adapting LLMs to downstream tasks. Although fine-tuning is much less computationally expensive than pre-training, it is still costly to apply in practice. This issue motivates the development of parameter-efficient fine-tuning methods, which aim to minimize the number of parameters that need to be updated.

One approach, known as **prefix fine-tuning**, prepends a sequence of trainable vectors, or prefixes, to the input of each Transformer layer [Li and Liang, 2021]. These prefixes can be thought of as soft prompts that serve as additional context to guide the behavior of the model under specific tasks. During fine-tuning, we need only to learn the prefixes for embedding task-specific knowledge. Thus, this method is efficient because it only modifies a small part of the model rather than adjusting the entire set of model parameters.

Specifically, let the input of a layer at depth l be denoted by $\mathbf{H}^l = \mathbf{h}_0^l \mathbf{h}_1^l \cdots \mathbf{h}_m^l$. The output of the layer can be expressed as:

$$\mathbf{H}^{l+1} = \text{Layer}(\mathbf{H}^l). \tag{9.19}$$

In prefix fine-tuning, we extend the sequence $\mathbf{h}_0^l \mathbf{h}_1^l \cdots \mathbf{h}_m^l$ by adding a few vectors at the beginning, which we denote as $\mathbf{p}_0^l \mathbf{p}_1^l \cdots \mathbf{p}_n^l$. Hence $\mathbf{H}^l$ can be written in the form:

$$\mathbf{H}^l = \underbrace{\mathbf{p}_0^l\ \mathbf{p}_1^l\ \cdots\ \mathbf{p}_n^l}_{\text{trainable}}\ \underbrace{\mathbf{h}_0^l\ \mathbf{h}_1^l\ \cdots\ \mathbf{h}_m^l}_{\text{previous layer output}}. \tag{9.20}$$

We retain only the last $m+1$ representations as the hidden states passed to the next layer.

$$\begin{aligned} \overline{\mathbf{H}}^{l+1} &= \text{Layer}(\mathbf{H}^l)[-m-1:] \\ &= \mathbf{h}_0^{l+1}\mathbf{h}_1^{l+1}\cdots\ \mathbf{h}_m^{l+1}, \end{aligned} \tag{9.21}$$

where $[-m-1:]$ denotes the slicing operation that extracts the last $m+1$ elements of a sequence. Given $\overline{\mathbf{H}}^{l+1}$, the input of the next layer can be expressed in the same form as Eq. (9.20):

$$\begin{aligned} \mathbf{H}^{l+1} &= \mathbf{p}_0^{l+1}\ \mathbf{p}_1^{l+1}\ \cdots\ \mathbf{p}_n^{l+1}\ \overline{\mathbf{H}}^{l+1} \\ &= \mathbf{p}_0^{l+1}\ \mathbf{p}_1^{l+1}\ \cdots\ \mathbf{p}_n^{l+1}\mathbf{h}_0^{l+1}\mathbf{h}_1^{l+1}\cdots\mathbf{h}_m^{l+1}. \end{aligned} \tag{9.22}$$

Here each $\mathbf{p}_i \in \mathbb{R}^d$ can be seen as a learnable parameter. During training, $\mathbf{p}_0^l \mathbf{p}_1^l \cdots \mathbf{p}_n^l$ are trained as usual, and the parameters of the original Transformer model are kept fixed.

Figure 9.5 shows an illustration of prefix fine-tuning for a translation task. Here, only the prefix vectors $\mathbf{p}_0^l$ and $\mathbf{p}_1^l$ are updated via gradients propagated from the output layer (i.e., the Chinese translation). By adjusting these vectors for the translation task, the model adapts accordingly. This makes $\mathbf{p}_0^l$ and $\mathbf{p}_1^l$ serve as prompts which activate the LLM to perform the task without needing explicit input prompts like "Translate the following sentence from English to Chinese". At test time, we prepend the optimized $\mathbf{p}_0^l$ and $\mathbf{p}_1^l$ to the layer, and the LLM will then translate the input sentence. Note that prefix fine-tuning introduces additional $L \times n \times d$ parameters, where L is the number of layers, n is the number of prefixes, and d is the dimensionality of each prefix. However, this number is much smaller compared to the total number of parameters in the LLM, making the fine-tuning process highly efficient.

While prefix fine-tuning is simple, it still requires modifications to LLMs. Alternatively, separating soft prompts from the LLMs allows us to preserve the original model architecture, making it more efficient for deployment across different tasks without the need to adjust the core model. One such method is prompt tuning [Lester et al., 2021]. Like prefix fine-tuning,

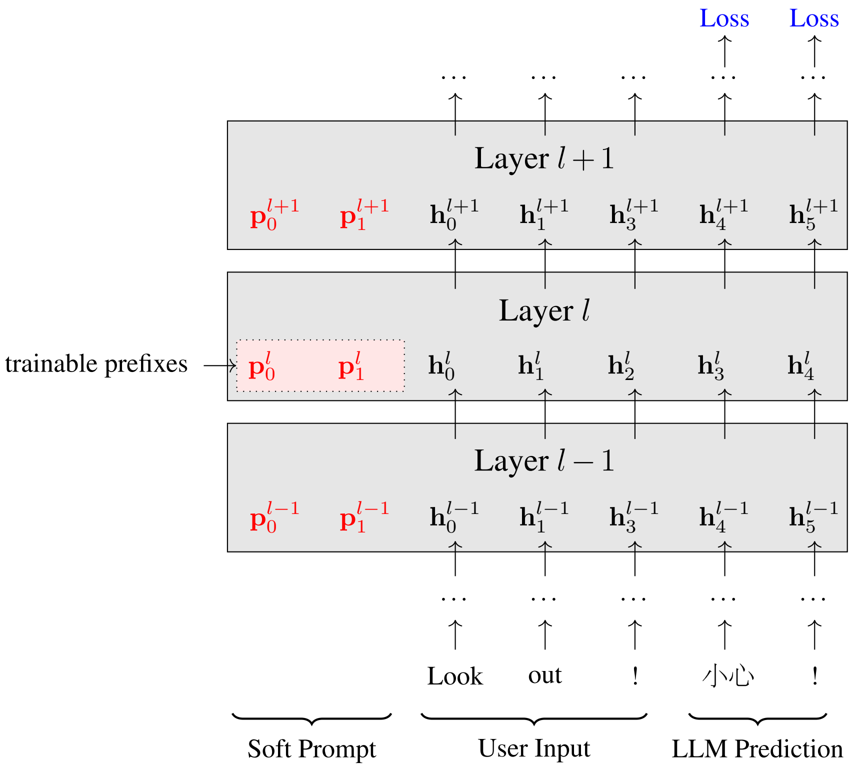


Figure 9.5: Illustration of prefix fine-tuning for a translation task (Look out! $\rightarrow$ 小心!). For each layer, we add two prefixes $\mathbf{p}_0^l$ and $\mathbf{p}_1^l$ at the beginning. The LLM is trained to minimize the loss on the predictions given the input. During this process, only the prefixes are optimized while the rest of the parameters remain fixed. Therefore, the model can adapt to the task with only a small number of trainable parameters. At inference time, the LLM works with optimized prefixes, and can perform the task without the need for explicit hard prompts.

prompt tuning incorporates trainable vectors so that LLMs can adapt to given tasks by adjusting these vectors. However, prompt tuning differs in that it modifies only the embedding layer.

Recall that in LLMs each input token z_i is represented by an embedding $\mathbf{e}_i$. These embeddings are generally learned through a token embedding model and are then used as the real inputs to the LLMs, replacing the symbolically represented tokens. In prompt tuning, a number of pseudo embeddings $\mathbf{p}_0 \cdots \mathbf{p}_n$ are added at the beginning of the token embedding sequence. So the actual input to the LLMs can be expressed as:

$$\underbrace{\mathbf{p}_0\ \mathbf{p}_1\ \cdots\ \mathbf{p}_n}_{\text{trainable}}\ \underbrace{\mathbf{e}_0\ \mathbf{e}_1\ \cdots\ \mathbf{e}_m}_{\text{token embeddings}}. \tag{9.23}$$

Note that a pseudo embedding does not need to correspond to any token in natural language. Instead these embeddings can be seen as "soft prompt embeddings" that serve to condition the LLMs. By training soft prompt embeddings on task-specific data, they learn to interact adaptively with the token embeddings $\mathbf{e}_0 \cdots \mathbf{e}_m$ and guide the behavior of LLMs. Since prompt

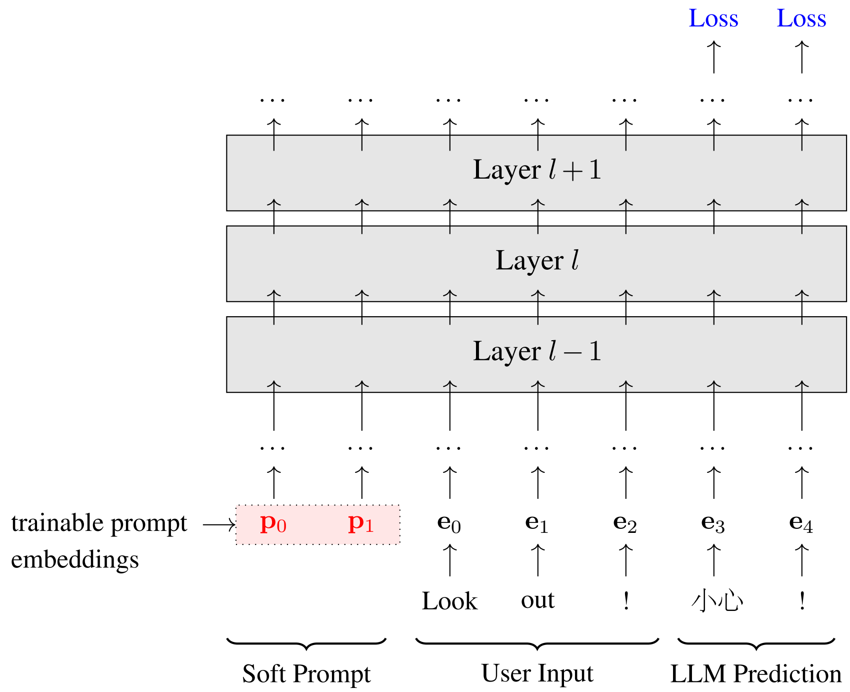
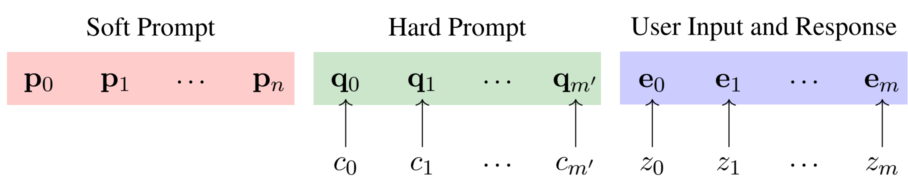


Figure 9.6: Illustration of prompt tuning for a translation task (Look out! → 小心!). Instead of using fixed textual prompts, soft prompts are learnable embeddings that are added at the beginning of the embedding sequence. During fine-tuning, only these prompt embeddings are optimized to efficiently adapt the LLM to the given task. Once optimized, the prompt embeddings are used to instruct the LLM to perform the task as new data arrives.

tuning does not change the underlying parameters of pre-trained LLMs, it is considered a lightweight and efficient method of fine-tuning, improving task-specific performance while maintaining their generalization capabilities. See Figure 9.6 for an illustration of prompt tuning.

Since $\mathbf{p}_0\ \mathbf{p}_1\ \cdots\ \mathbf{p}_n$ is itself a sequence, we can employ sequence models to better represent it. For example, a Transformer model can encode this sequence, and the resulting representation can then be used as the input to the LLM. In other words, we can develop an additional model for encoding soft prompts. Another way to improve prompting is by combining soft and hard prompts, thereby taking advantage of both types [Liu et al., 2023c]. In the embedding sequence, we can arrange or intersperse these prompts. This would result in different prompt patterns. For example, a simple pattern that uses both types of prompt is:



where $c_0 \cdots c_{m'}$ denotes the hard prompt and $\mathbf{q}_0 \cdots \mathbf{q}_{m'}$ denotes the corresponding embedding sequence.

Here we have considered methods for inserting soft prompts in LLMs. But we skip the details of training these soft prompts and assume that the reader is familiar with the standard supervised learning process, that is, maximizing the likelihood of the correct model output given the model input. In fact, learning soft prompts can be related to many issues in LLM fine-tuning. For example, if we consider it as a context compression problem, we can apply the knowledge distillation methods described previously. In Mu et al. [2024]'s work, prompts are compressed and represented as a few pseudo tokens, which are appended to each input sequence. The embeddings of these pseudo tokens are optimized to mimic the predictions of a standard-prompted model. In other words, the prompting knowledge is distilled from a teacher model into the pseudo tokens.

Broadly speaking, many parameter-efficient fine-tuning methods can be thought of as learning some sort of soft prompt [Lialin et al., 2023]. When we fine-tune a part of an LLM for a task, this process can essentially be seen as injecting task-related prompting information into that specific part of the model. Another widely-used approach to parameter-efficient fine-tuning is to add an adaptor layer between the existing model layers. This approach allows us to fine-tune only the adaptor layer on specific tasks without altering the underlying architecture or retraining the entire model. In this sense, adaptor layers can be viewed as soft prompts that encode prompting and task-related information and interact with the original LLM to help it adapt. To summarize, Figure 9.7 shows a comparison of different methods of using soft prompts in LLMs.

3. Learning Soft Prompts with Compression

Another approach to learning soft prompts is to frame it as a compression problem. As a simple example, consider the problem of approximating a long context using a continuous representation [Wingate et al., 2022]. Suppose we have a user input $\mathbf{z}$ and its context $\mathbf{c}$ (such as long instructions and demonstrations). Now we want to learn a compressed representation of the context, denoted by σ, such that the prediction conditioned on $\mathbf{z}$ and σ is as close as possible to that conditioned on $\mathbf{z}$ and $\mathbf{c}$. This goal can be loosely formulated as

$$\hat{\sigma} \quad = \quad \arg\min_{\sigma} s(\hat{\mathbf{y}}, \hat{\mathbf{y}}_{\sigma}), \tag{9.24}$$

where $\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \Pr(\mathbf{y}|\mathbf{c}, \mathbf{z})$ and $\hat{\mathbf{y}}_{\sigma} = \arg\max_{\mathbf{y}} \Pr(\mathbf{y}|\sigma, \mathbf{z})$ are the LLM predictions given the full context and the compressed context, respectively. The function $s(\cdot, \cdot)$ measures the discrepancy between the two predictions.

One general framework for achieving this is knowledge distillation, where $\hat{\mathbf{y}}$ and $\hat{\mathbf{y}}_{\sigma}$ can be seen as the predictions of the teacher model and the student model, respectively. This formalization links our discussion to the context distillation problem discussed earlier. The training objective can be defined analogously to Eqs. (9.15) and (9.16). For instance, a

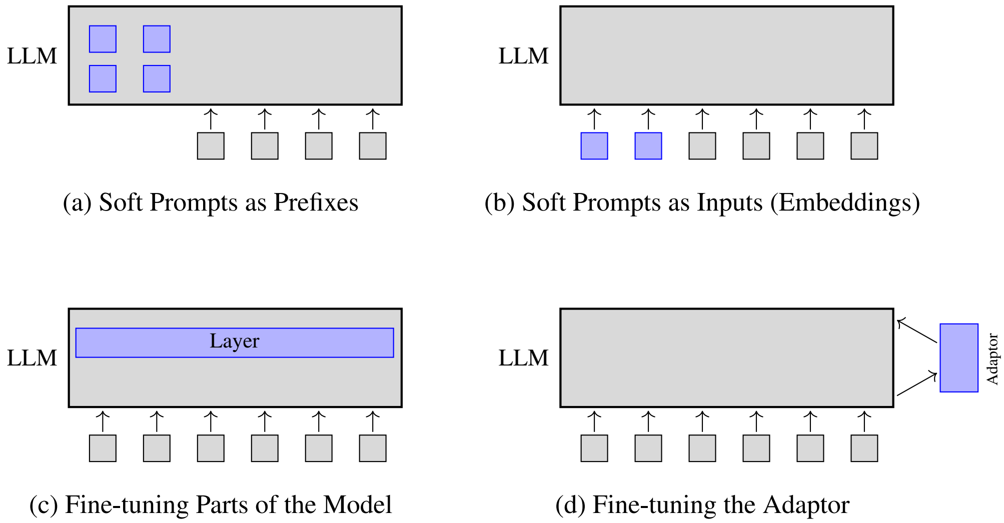


Figure 9.7: Illustrations of using soft prompts in LLMs. Here tunable soft prompts are shown in blue, and components whose parameters are fixed during fine-tuning are shown in gray. In sub-figure (a), soft prompts are prefixes appended to each layer of the LLM. In sub-figure (b), soft prompts are used as input embeddings for the LLM. In sub-figures (c) and (d), soft prompts are broadly treated as components of the model that are fine-tuned for task adaptation.

straightforward maximum likelihood objective is given by

$$\hat{\sigma} \quad = \quad \arg\max_{\sigma} \log \Pr(\hat{\mathbf{y}}|\sigma, \mathbf{z}). \tag{9.25}$$

Alternatively, we can minimize the KL divergence between the output distributions, giving

$$\hat{\sigma} \quad = \quad \arg\min_{\sigma} \mathrm{D}_{\mathrm{KL}}(\Pr(\cdot|\mathbf{c}, \mathbf{z}) \,||\, \Pr(\cdot|\sigma, \mathbf{z})). \tag{9.26}$$

The difference with the models in Eqs. (9.15) and (9.16) is that here the compressed context is represented as real-valued vectors (call them **prompt embeddings**), rather than as normal tokens. By applying the above methods, we distill the context from the token sequence $\mathbf{c}$ into the embeddings σ. Note that the teacher model $\Pr(\cdot|\mathbf{c}, \mathbf{z})$ and the student model $\Pr(\cdot|\sigma, \mathbf{z})$ may not share the same architecture or model settings. In practice, we generally wish for the teacher model to be stronger, while the student model should be smaller and more efficient.

While compressing full context into continuous representations is a straightforward approach to learning soft prompts, it requires a teacher model that can deal with long input sequences. In many cases, however, the context is so long that applying an LLM is too costly or infeasible. Modeling long input sequences can fall under the broad family of efficient methods for long-context LLMs. Many techniques have been developed to address this issue. For example, one can use a fixed-size KV cache to store the past information at each step during inference. Efficient Transformer architectures and long-context LLMs have been extensively

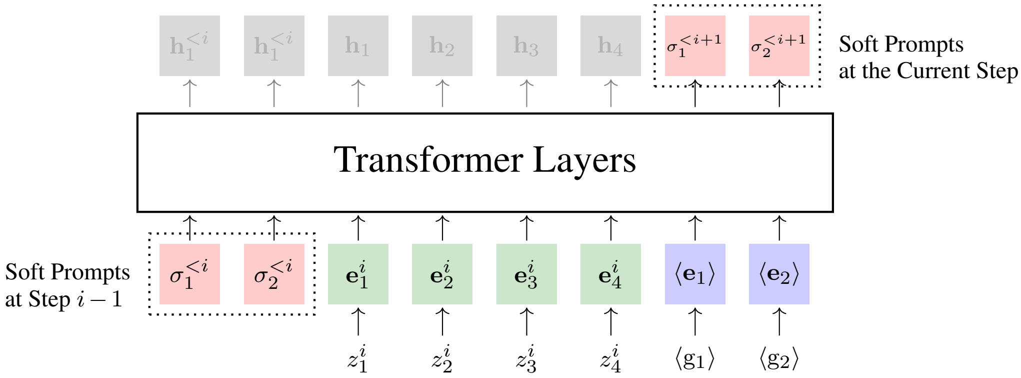


Figure 9.8: Illustration of compressing a context segment into soft prompts ($\kappa = 2$ and $m_i = 4$). The input to the LLM includes the soft prompts from the previous step ($\sigma_1^{<i}$ and $\sigma_2^{<i}$), the tokens of the segment (z_1, z_2, z_3, and z_4), and the summary tokens ($\langle \mathrm{g}_1 \rangle$ and $\langle \mathrm{g}_2 \rangle$). Given these, the LLM operates as usual. We then extract the outputs at the last Transformer layer that correspond to the summary tokens. These outputs can be viewed as the soft prompts that accumulated up to this segment.

discussed in this book. For more detailed discussions of these topics, interested readers can refer to Chapters 6 and 8.

There are also methods specifically designed to compress long sequences into soft prompts. Here we consider Chevalier et al. [2023]'s method as an example. The basic idea is to gradually learn soft prompts by compressing the input sequence into a fixed-size representation. For simplicity, we use $\mathbf{z}$ to denote the entire input sequence to be compressed. Given a long input sequence, we first divide it into a number of segments $\{\mathbf{z}^1, \cdots, \mathbf{z}^K\}$. We then process these segments in sequence, each time generating a representation of the input we have processed so far, denoted by $\sigma^{<i+1}$. To do this, a few summary tokens $\langle \mathrm{g}_1 \rangle, \cdots, \langle \mathrm{g}_\kappa \rangle$ are introduced. At each step, we take a segment $\mathbf{z}^i = z_1^i \cdots z_{m_i}^i$, along with the previous compressed representation $\sigma^{<i}$ and the summary tokens $\langle \mathrm{g}_1 \rangle, \cdots, \langle \mathrm{g}_\kappa \rangle$ as input, and use an LLM to produce the corresponding hidden representation sequence at the last Transformer layer. An example of this process is illustrated in Figure 9.8.

Here $\sigma^{<i}$ is essentially a memory. The model operates in an RNN fashion. Each time we take a segment and update this memory by encoding both the previous memory state and the segment. Therefore, the $\sigma^{<i}$ produced at the last segment is a representation of the entire context sequence. The Transformer model for learning these representations can be a standard LLM but we need to fine-tune it to adapt to this context representation task.

Note that here we simply consider *prompt* and *context* as similar terms, even though they are not the same. Although we use the term *prompt* broadly to include contextual information supplied to the model., we can often view it as a type of context. From this perspective, the methods discussed here can be applied to general text compression problems.

9.3.3 Prompt Length Reduction

While soft prompts provide dense hidden representations, they are generally not directly interpretable. This lack of interpretability makes it difficult for users to understand how prompts influence LLM outputs. Moreover, modifying soft prompts often requires additional fine-tuning, which limits their utility in dynamic environments where prompts frequently change.

One alternative approach to improving prompt efficiency is to simplify the prompt text itself. Unlike soft prompts, simplified textual prompts remain interpretable and can be modified directly by users. For example, below is a prompt for answering questions on healthcare and finance.

> The task involves developing a language model capable of understanding and responding to user inquiries across various domains, with a particular emphasis on healthcare and finance. Considering the broad range of potential queries, from the specifics of medical diagnoses to the nuances of financial regulations, the model must ensure a comprehensive understanding and accurate responses.
>
> Question:
>
> What are the best practices for using artificial intelligence in diagnosing cardiovascular diseases?
>
> ____

We can simplify the task description by deleting the unimportant parts.

> *The task involves developing a language model capable of understanding and responding to user inquiries ~~across various domains, with a particular emphasis~~ on healthcare and finance. ~~Considering the broad range of potential queries, from the specifics of medical diagnoses to the nuances of financial regulations,~~ The model must ensure a comprehensive understanding and accurate responses.*

We can also paraphrase it into a shorter text.

> *The task involves developing a language model focused on healthcare and finance, capable of understanding and accurately responding to a wide range of user inquiries.*

This problem can be viewed as a standard NLP task — text simplification. Therefore, the methods used for text simplification are generally applicable and not limited to prompt simplification. There are many ways to achieve this. One simple method is to define some heuristics and identify redundant words that can be eliminated without losing essential information. For example, we can examine each token in a sequence in terms of its contribution to the overall meaning and remove those that provide minimal value [Li et al., 2023c; Jiang et al., 2023b]. Another method involves framing the problem as a sequence-to-sequence task. Given labeled data for text simplification, we can train an encoder-decoder model to transform each input text into its simplified form. In addition, given that many LLMs have been fine-tuned and aligned to perform text simplification tasks, it is straightforward to use these models to simplify prompts. For example, we can prompt an LLM to simplify a prompt while satisfying constraints such as length limits.

9.4 Summary

In this chapter, we have discussed a variety of topics related to LLM prompting. Our discussion has focused mainly on two aspects:

- How to design basic prompts to guide the predictions of LLMs and refine these prompts for more effective and efficient problem-solving?
- How to automate the design and representation of prompts?

Solutions to these issues involve both general prompt designs and more advanced techniques, such as CoT and prompt learning, which have been explored extensively in recent research.

In NLP, prompting can be viewed as a technology that has evolved along with LLMs, and in a sense, it has opened the door to the practical application of these models in an impressive range of problem domains. In fact, if we expand the concept of prompts to some extent, it can be traced back to the early days of machine learning and NLP. For example, many NLP systems use hand-crafted features and templates to “prompt” specific tasks. Imagine developing a feature to indicate whether a text is formal or informal. We can feed this feature into a machine translation system to condition the translation on the type of the input text.

The widespread use of the modern concept of prompts began with the rise of large pre-trained models in the field of NLP. Initially, these models, such as BERT, were adapted to specific downstream tasks mainly through fine-tuning. However, researchers soon discovered that by designing specific “prompts” — adding certain words or sentences to the input — the models could be triggered to respond to specific tasks without extensive fine-tuning. This motivated the NLP community to develop and apply universal foundation models that can be prompted to address various tasks without changing the underlying architecture and the pre-training procedure.

Prompting approaches were first experimented with smaller models and later demonstrated impressive capabilities with large models like GPT-3, which could generate high-quality text in response to simple prompts across various tasks. As prompting technology evolved, prompt engineering emerged as a critical area of research. As discussed in this chapter, it broadly involves designing effective prompts to maximize model performance, encompassing both hand-crafted and automatically generated prompts. More recent research has explored how to enhance the effectiveness of prompting through techniques like few-shot learning, zero-shot learning, and CoT reasoning, enabling LLMs to work effectively across a wide range of scenarios. A general discussion of prompting can be very broad, and we cannot cover all details in this chapter. For more advanced techniques of prompting, the reader can refer to recent surveys. Topics include in-context learning [Li, 2023; Dong et al., 2022], CoT [Chu et al., 2023; Yu et al., 2023b; Zhang et al., 2023a], efficient prompting [Chang et al., 2024], and general prompt engineering [Liu et al., 2023d; Chen et al., 2023a].

Note that although we would ideally like to develop general prompting methods without adjusting model architectures and parameters, the results of prompting generally depend heavily on the quality and size of the given LLMs. For stronger models, such as commercialized online LLMs, simple prompts may be sufficient to instruct these models to perform tasks correctly. In this case, prompt engineering is relatively easy, though careful prompt refinement remains necessary. By contrast, for less capable or smaller-scale models, we may need to carefully design the prompts to achieve the desired results. In many cases, fine-tuning is still necessary to adapt the models to sophisticated prompting strategies.

Chapter 10

Alignment

Alignment is not a new concept in NLP, but its meaning varies across different domains and over time. In traditional NLP, the term *alignment* typically refers to the tasks that link corresponding elements in two sets, such as aligning words between a Chinese sentence and an English sentence. As LLMs become increasingly important in NLP research, this term is more broadly used to refer to aligning model outputs with human expectations. The problem that alignment addresses is that the output of a model may not align with the specific goals or contexts intended by users. For example, pre-trained LLMs may not be able to follow user instructions because they were not trained to do so. Another example is that LLMs may generate harmful content or perpetuate biases inherent in their training data. This poses new challenges in ensuring that LLM outputs are not only accurate and relevant, but also ethically sound and non-discriminatory.

Simply pre-training LLMs can result in a variety of alignment problems. Our ultimate goal is to resolve or mitigate all these problems to ensure LLMs are both accurate and safe. There is an interesting issue here: since large language models are trained on vast amounts of data, we have reason to believe that if we have sufficient data covering a variety of tasks and aligned with human preferences, pre-training could make LLMs accurate and safe enough, perhaps even eliminating the need for alignment. However, the reality is that it is nearly impossible to gather data that encompasses all tasks or adequately represents human preferences. This makes it difficult to achieve model alignment through pre-training alone, or at least, at this stage, alignment remains a very necessary and critical step in the development of LLMs.

In this chapter, we will focus on alignment methods for LLMs. We will begin by discussing the general alignment tasks. Then we will consider two widely-used approaches, known as **instruction alignment** and **human preference alignment**, respectively. The former resorts to supervised fine-tuning techniques and guides the LLMs to generate outputs that adhere closely to user instructions. On the other hand, the latter typically relies on reinforcement learning techniques, where the LLMs are trained based on feedback from humans. While these methods are motivated by different goals, they are commonly used together to develop well-aligned LLMs.

10.1 An Overview of LLM Alignment

Alignment can be achieved in several different ways. We need different methods for LLM alignment because this problem is itself complicated and multifaceted, requiring a blend of technical considerations. Here we consider three widely-used approaches to aligning LLMs.

The first approach is to fine-tune LLMs with labeled data. This approach is straightforward as it simply extends the pre-existing training of a pre-trained LLM to adapt it to specific tasks. An example of this is **supervised fine-tuning** (**SFT**), in which the LLM is further trained on a dataset comprising task-specific instructions paired with their expected outputs. The SFT dataset is generally much smaller compared to the original training set, but this data is highly specialized. The result of SFT is that the LLM can learn to execute tasks based on user instructions. These tasks can either be ones previously encountered in SFT, or new tasks similar to those. For example, by fine-tuning the LLM with a set of question-answer pairs, the model can respond to specific questions, even if not directly covered in the SFT dataset. This method proves particularly useful when it is relatively easy to describe the input-output relationships and straightforward to annotate the data.

The second approach is to fine-tune LLMs using reward models. One difficulty in alignment is that human values and expectations are complex and hard to describe. In many cases, even for humans themselves, articulating what is ethically correct or culturally appropriate can be challenging. As a result, collecting or annotating fine-tuning data is not as straightforward as it is with SFT. Moreover, aligning LLMs is not just a task of fitting data, or in other words, the limited samples annotated by humans are often insufficient to comprehensively describe these behaviors. What we really need here is to teach the model how to determine which outputs are more in line with human preferences, for example, we not only want the outputs to be technically accurate but also to align with human expectations and values. One idea is to develop a reward model analogous to a human expert. This reward model would work by rewarding the LLM whenever it generates responses that align more closely with human preferences, much like how a teacher provides feedback to a student. To obtain such a reward model, we can train a scoring function from human preference data. The trained reward model is then used as a guide to adjust and refine the LLM. This frames the LLM alignment task as a reinforcement learning task. The resulting methods, such as **reinforcement learning from human feedback** (**RLHF**), have been demonstrated to be particularly successful in adapting LLMs to follow the subtleties of human behavior and social norms.

The third approach is to perform alignment during inference rather than during training or fine-tuning. From this perspective, prompting in LLMs can also be seen as a form of alignment, but it does not involve training or fine-tuning. So we can dynamically adapt an LLM to various tasks at minimal cost. Another method to do alignment at inference time is to rescore the outputs of an LLM. For example, we could develop a scoring system to simulate human feedback on the outputs of the LLM (like a reward model) and prioritize those that receive more positive feedback.

The three methods mentioned above are typically used in sequence once the pre-training is complete: we first perform SFT, then RLHF, and then prompt the LLM in some way

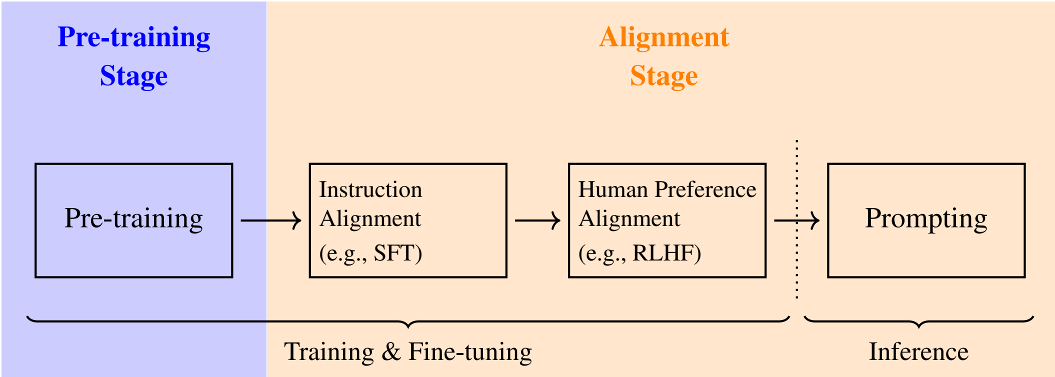


Figure 10.1: Schematic illustration of the pre-train-then-align method for developing LLMs. In the pre-training stage, we train an LLM on vast amounts of data using next token prediction. Then, in the alignment stage, we align the LLM to user instructions, intents, and preferences. This includes instruction alignment, human preference alignment, and prompting.

during inference. This roughly divides the development of LLMs into two stages — the pre-training stage and the alignment stage. Figure 10.1 shows an illustration of this. Since prompting techniques have been intensively discussed in the previous chapter, we will focus on fine-tuning-based alignment methods in the rest of this chapter.

10.2 Instruction Alignment

One feature of LLMs is that they can follow the prompts provided by users to perform various tasks. In many applications, a prompt consists of a simple instruction and user input, and we want the LLM to follow this instruction to perform the task correctly. This ability of LLMs is also called the instruction-following ability. For example, below is a prompt where we want the LLM to extract key points and provide a concise summary for a lengthy article.

Instruction	Summarize this text in three sentences.
Input	Daylight Savings Time (DST) - the process of moving clocks forward by one hour in the summer - was started in Germany in 1916. During World War One it was a way to save ...
Output	____

This task requires the LLM to understand the instruction "Summarize this text in three sentences" and perform the summarization accordingly. However, LLMs are typically trained for next-token prediction rather than for generating outputs that follow instructions. Applying a pre-trained LLM to the above example would likely result in the model continuing to write the input article instead of summarizing the main points. The goal of instruction alignment (or **instruction fine-tuning**) is to tune the LLM to accurately respond to user instructions and intentions. The rest of this section will discuss some issues related to instruction alignment, including fine-tuning LLMs to follow instructions, generating or collecting instruction data, and generalizing instruction alignment.

10.2.1 Supervised Fine-tuning

One straightforward approach to adapting LLMs to follow instructions is to fine-tune these models using annotated input-output pairs [Ouyang et al., 2022; Wei et al., 2022a]. Unlike standard language model training, here we do not wish to maximize the probability of generating a complete sequence, but rather maximize the probability of generating the rest of the sequence given its prefix (i.e., generating the output given the input). This approach makes instruction fine-tuning a bit different from pre-training. Let $\mathbf{x} = x_0...x_m$ be an input sequence (e.g., instruction + user input) and $\mathbf{y} = y_1...y_n$ be the corresponding output sequence. The SFT data is a collection of such input-output pairs (denoted by S), where each output is the correct response for the corresponding input instruction. For example, below is an SFT dataset

$\mathbf{x}$ (instruction + user input)	$\mathbf{y}$ (output)
Summarize the following article. Article: In recent years, solar energy has seen unprecedented growth, becoming the fastest-growing ...	$\{*$summary$*\}$
Analyze the sentiment of the following review. Review: I absolutely loved the new dining experience. The food was divine and the service was impeccable.	Positive
Translate the following sentence into French. Sentence: practice indeed helps.	La pratique aide effectivement.
Extract the main financial figures from the following earnings report. Report: The company reported a revenue of $10 million in the first quarter with a profit margin of 15% ...	Revenue: $10 million, Profit Margin: 15%
Classify the following email as spam or not spam. Text: Congratulations! You've won a $500 gift card. Click here to claim now.	Spam
Provide a solution to the following technical issue. Issue: my computer is running slow and often freezes.	First, check for ...

where the instructions are highlighted. This dataset contains instructions and the corresponding outputs for several different NLP problems, and so we can fine-tune an LLM to handle multiple tasks simultaneously.

In SFT, we aim to maximize the probability of the correct output given the input. Consider an LLM with pre-trained parameters $\hat{\theta}$. The fine-tuning objective can then be formulated as:

$$\tilde{\theta} = \arg\max_{\hat{\theta}+} \sum_{(\mathbf{x},\mathbf{y})\in\mathcal{D}} \log \Pr_{\hat{\theta}+}(\mathbf{y}|\mathbf{x}) \tag{10.1}$$

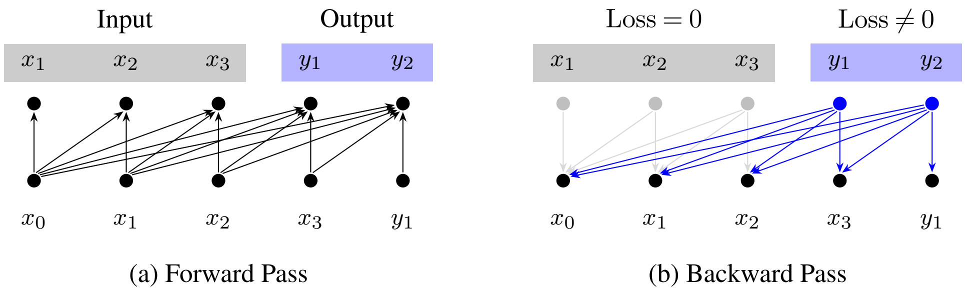


Figure 10.2: Illustration of supervised fine-tuning for LLMs. We concatenate the input and the output into a single sequence. During the forward pass, we run the LLM as usual. During the backward pass, we compute the loss only for the output part and simply set the loss for the input part to 0.

where $\tilde{\theta}$ denotes the parameters optimized via fine-tuning, and $\hat{\theta}^+$ represents an adjustment to $\hat{\theta}$. Here we will omit the superscript $+$ and use θ to represent $\hat{\theta}^+$ to keep the notation uncluttered. But the reader should keep in mind that the fine-tuning starts from the pre-trained parameters rather than randomly initialized parameters.

The objective function $\log \Pr_\theta(y_i|\mathbf{x}, \mathbf{y}_{<i})$ is computed by summing the log-probabilities of the tokens in $\mathbf{y}$, conditional on the input $\mathbf{x}$ and all the previous tokens $\mathbf{y}_{<i}$:

$$\log \Pr_\theta(\mathbf{y}|\mathbf{x}) = \sum_{i=1}^{n} \log \Pr_\theta(y_i|\mathbf{x}, \mathbf{y}_{<i}) \tag{10.2}$$

This formulation is equivalent to minimizing the cross-entropy loss.

Note that minimizing the conditional log-probability $\log \Pr_\theta(\mathbf{y}|\mathbf{x})$ is not a standard language model training problem. If we concatenate $\mathbf{x}$ and $\mathbf{y}$ as a single sequence, a more general form of language modeling is based on the joint log-probability $\log \Pr_\theta(\mathbf{x}, \mathbf{y})$, that is, we minimize the loss over all tokens of the sequence $\mathrm{seq}_{\mathbf{x},\mathbf{y}} = [\mathbf{x}, \mathbf{y}]$. We can write the probability of this sequence using the chain rule

$$\begin{aligned} \log \Pr_\theta(\mathrm{seq}_{\mathbf{x},\mathbf{y}}) &= \log \Pr_\theta(\mathbf{x}, \mathbf{y}) \\ &= \underbrace{\log \Pr_\theta(\mathbf{x})}_{\text{set to } 0} + \underbrace{\log \Pr_\theta(\mathbf{y}|\mathbf{x})}_{\text{loss computation}} \end{aligned} \tag{10.3}$$

There are two terms on the right-hand side of the equation. We can simply set the first term $\log \Pr_\theta(\mathbf{x})$ to 0, focusing solely on the second term $\log \Pr_\theta(\mathbf{y}|\mathbf{x})$ for loss computation. As a result, the training can be implemented using standard LLMs. For the sequence $\mathrm{seq}_{\mathbf{x},\mathbf{y}}$, we first run the forward pass as usual. Then, during the backward pass, we force the loss corresponding to $\mathbf{x}$ to be zero. Figure 10.2 shows an illustration of this process.

By taking $\log \Pr_\theta(\mathrm{seq}_{\mathbf{x},\mathbf{y}})$ as the objective function, we can describe SFT using a regular form of language model training:

$$\tilde{\theta} = \arg\max_{\theta} \sum_{(\mathbf{x},\mathbf{y}) \in \mathcal{D}} \log \Pr_{\theta}(\mathrm{seq}_{\mathbf{x},\mathbf{y}}) \tag{10.4}$$

The problem we considered above is fundamentally a **single-round prediction** problem, where the LLM generates a response based on a single input without any further interaction or feedback from the user. The input is processed, and the output is generated in one go. This is typical in scenarios where a single question is asked, and a single answer is provided, with no follow-up questions or clarifications. However, in practice, we sometimes have to handle multi-round prediction problems, for example, an LLM engages in a dialogue over multiple turns. In this setting, the LLM not only generates responses based on the initial input but also incorporates subsequent inputs that might refine or expand on earlier interactions. For example, we can use the LLM to act as a healthcare assistant chatbot and have a conversation with the user, like this

User	I've been feeling very tired lately.
Chatbot	I'm sorry to hear that. Besides feeling tired, have you noticed any other symptoms?
User	Yes, I'm also experiencing headaches frequently.
Chatbot	How long have these symptoms been going on?
User	About a week now.
Chatbot	It might be good to check in with a healthcare professional. Would you like help setting up an appointment?
User	Yes, please. Can it be after work hours?
Chatbot	Sure, I can arrange that. There are slots available next Wednesday and Thursday after 5 PM. Which day works better for you?
	...

In this task, there are several rounds of conversation, each involving the generation of a response based on the user's request or question and the conversational history. Suppose we have K rounds of conversation, denoted by $\{\mathbf{x}^1, \mathbf{y}^1, \mathbf{x}^2, \mathbf{y}^2, ..., \mathbf{x}^K, \mathbf{y}^K\}$. Here $\mathbf{x}^k$ and $\mathbf{y}^k$ denote the user request and the response, respectively, for each round k. The log-probability of generating the response can be written as $\log \Pr_{\theta}(\mathbf{y}^k | \mathbf{x}^1, \mathbf{y}^1, ..., \mathbf{x}^k)$. Our goal is then to maximize the sum of these log-probabilities

$$\tilde{\theta} = \arg\max_{\theta} \sum_{k=1}^{K} \log \Pr_{\theta}(\mathbf{y}^k | \mathbf{x}^1, \mathbf{y}^1, ..., \mathbf{x}^k) \tag{10.5}$$

A straightforward implementation of this involves calculating the conditional probability for each k. However, it requires running the LLM K times, each time with an increased conversational history to make predictions. A more efficient method is to perform loss computation of all responses in a single run of the LLM. To do this, we represent the conversation as a sequence $\mathrm{seq}_{\mathbf{x}^1,\mathbf{y}^1,...,\mathbf{x}^K,\mathbf{y}^K} = [\mathbf{x}^1, \mathbf{y}^1, ..., \mathbf{x}^K, \mathbf{y}^K]$ (or seq for short). The log-probability of this sequence is given by

$$\begin{aligned} \log \Pr_\theta(\mathrm{seq}) &= \log \Pr_\theta(\mathbf{x}^1, \mathbf{y}^1, ..., \mathbf{x}^K, \mathbf{y}^K) \\ &= \underbrace{\log \Pr_\theta(\mathbf{x}^1)}_{\text{set to 0}} + \underbrace{\log \Pr_\theta(\mathbf{y}^1|\mathbf{x}^1)}_{\text{loss computation}} + \cdots + \\ &\quad \underbrace{\log \Pr_\theta(\mathbf{x}^K|\mathbf{x}^1, \mathbf{y}^1, ..., \mathbf{y}^{K-1})}_{\text{set to 0}} + \\ &\quad \underbrace{\log \Pr_\theta(\mathbf{y}^K|\mathbf{x}^1, \mathbf{y}^1, ..., \mathbf{x}^K)}_{\text{loss computation}} \end{aligned} \tag{10.6}$$

The trick here is that we ignore the loss for generating user inputs (i.e., $\log \Pr_\theta(\mathbf{x}^1)$,..., $\log \Pr_\theta(\mathbf{x}^K|\mathbf{x}^1, \mathbf{y}^1, ..., \mathbf{y}^{K-1})$), as illustrated in Figure 10.3. Hence we only compute the probabilities of generating the responses given their conversational histories, in other words, the value on the right-hand side of Eq. (10.6) is actually equal to the value on the right-hand side of Eq. (10.5). As with Eq. (10.4), the training of this multi-round prediction model can be achieved by maximizing the log likelihood over a training dataset $\mathcal{D}$:

$$\tilde{\theta} = \arg\max_\theta \sum_{\mathrm{seq} \in \mathcal{D}} \log \Pr_\theta(\mathrm{seq}) \tag{10.7}$$

While implementing the SFT methods introduced above seems trivial as they are fundamentally the same as regular language model training, there are still issues that need to be considered in practice. For example,

- SFT requires labeled data. This makes SFT quite different from pre-training, where raw text is used as training data and is readily available. As in other supervised machine learning problems, data annotation and selection in SFT are not simple tasks. In general, we wish to develop SFT data that is both substantial in quantity and high in quality, and this data should be highly relevant to the tasks the LLM will perform. On the other hand, there is a need to fine-tune LLMs with less data to minimize computational and data construction costs. Often, the quality of LLMs is highly dependent on the data used in SFT. Thus, such data must be carefully developed and examined. As we will see in later subsections, SFT can be more efficient and effective through more advanced techniques for data construction.
- SFT is still computationally expensive for LLMs due to their large size. As a result, maintaining and updating such models is resource-intensive. For example, applying gradient updates to billions of parameters within an LLM requires significant computational power and memory. This often requires high-performance computing environments,

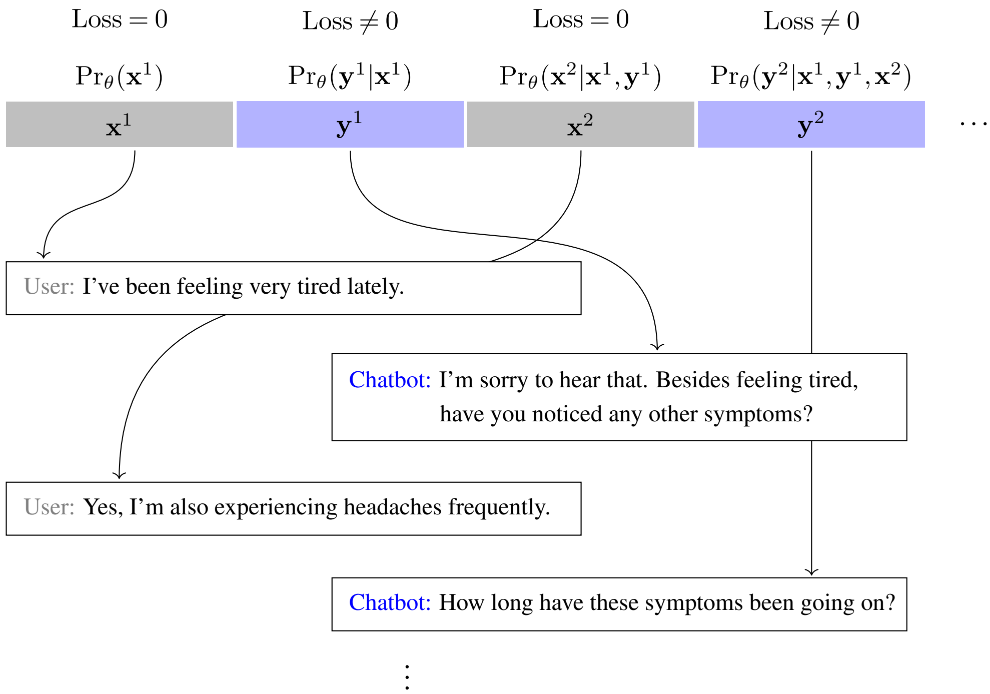


Figure 10.3: Illustration of supervised fine-tuning for conversational models. Here the LLM acts as a chatbot to respond to each request based on the conversational history. The conversation progresses by alternating between the user and the chatbot. In SFT, we treat the entire conversation as a sequence, just like in standard LLMs, but compute the loss only for the responses of the LLM.

which are costly to operate. To address these challenges, various optimization strategies, such as pruning, quantization, and the use of more efficient training algorithms, have been explored. In particular, there has been significant interest in parameter-efficient fine-tuning methods which are designed to maintain state-of-the-art performance without the need for extensive computational resources. We have seen in Chapter 9 that applying techniques like soft prompts can make the fine-tuning process more efficient. For further discussion on parameter-efficient methods, the reader can refer to related papers on this issue [Houlsby et al., 2019; Hu et al., 2022; Han et al., 2024].

- SFT can be regarded as a post-training step following pre-training. It is a separate training phase designed to preserve the advantages of the initial pre-training while incorporating new adjustments. This may seem paradoxical because updating a pre-trained LLM with further data potentially causes the model to forget some of its prior knowledge. Imagine a scenario where we have a large amount of SFT data and extensively fine-tune the LLM. In this case, the LLM could overfit the data, which in turn may reduce generalization performance or cause catastrophic forgetting. A common strategy to mitigate this issue is to employ regularization and early stopping techniques. Another practical approach is to use a smaller learning rate to gently adjust the weights of the LLM. In addition, fine-tuning with data from diverse sources and problem domains can also be beneficial. Nevertheless, in practice, the SFT step is often carefully examined and requires substantial engineering and experimental efforts to optimize.

10.2.2 Fine-tuning Data Acquisition

Fine-tuning data is so important that much recent work in LLM has focused on developing various datasets for instruction fine-tuning. As with most work in machine learning, there are generally two approaches to data acquisition — manual data generation and automatic data generation.

1. Manually Generated Data

One straightforward method is to recruit human annotators to create input-output pairs for the tasks of interest. Unlike data annotation in conventional NLP, such as text classification, where annotators simply assign labels to collected texts according to guidelines, creating fine-tuning data for LLMs requires more steps and effort, making it thus more challenging. Suppose we want to obtain fine-tuning data for the English-to-Chinese machine translation task. The first step is to write a prompt template to describe the task and format the problem clearly. For example,

Instruction	Translate the text from English to Chinese.
User Input	$\{*\text{text}*\}$
Output	$\underline{\{*\text{translation}*\}}$

Then, we collect pairs of source and target texts (i.e., Chinese texts and the corresponding translations), and replace the variables $\{*\text{text}*\}$ and $\{*\text{translation}*\}$ to generate the fine-tuning samples. For example, given a pair of English and Chinese sentences

$$\underbrace{\text{How's the weather today?}}_{\{*\text{text}*\}} \quad \rightarrow \quad \underbrace{\text{今天天气怎么样?}}_{\{*\text{translation}*\}}$$

we can generate a fine-tuning sample using the prompt template, like this

Instruction	Translate the text from English to Chinese.
User Input	How's the weather today?
Output	<u>今天天气怎么样?</u>

That is,

$$\begin{aligned}\mathbf{x} &= \text{Translate the text from English to Chinese.\textbackslash n How's the weather today?} \\ \mathbf{y} &= \text{今天天气怎么样?}\end{aligned}$$

We can use this $(\mathbf{x},\mathbf{y})$ pair to fine-tune the LLM, as described in the previous subsection.

One difficulty here is that there are many, many different ways to write prompt templates for the same task, and different people may produce prompt templates with varying qualities and complexities. Sometimes, we may write prompt templates with overly complex or verbose instructions. Sometimes, we may not even know exactly what the target task is and how to describe it. A widely-adopted strategy is to create prompt templates for existing NLP tasks, given that there have been so many well-established NLP problems and benchmarks [Bach et al., 2022; Wang et al., 2022e; Mishra et al., 2022]. In this case, annotators can be given the original task description and many examples. Then, they can use their own ways to express how to prompt the LLM to perform the tasks. Note that, while such a method can ease the process of creating and writing prompts, we still need annotation frameworks and crowdsourcing systems to manage the work and conduct quality control. For example, we generally need to design annotation guidelines and a unified format for writing prompt templates, especially when many annotators are contributing to the same task. One advantage of inducing prompts from existing NLP tasks is that, once the prompt templates have been developed, it is easy to generate prompts using the annotated samples in the original tasks. For example, given a bilingual dataset for English-to-Chinese translation, we can easily create a number of fine-tuning examples by filling the slots in the above template with the sentence pairs in this dataset.

Another approach is to directly use the naturally existing data available on the internet. A common example is by collecting question-and-answer pairs from QA websites to fine-tune LLMs for open-domain QA tasks [Joshi et al., 2017]. Many benchmarks in QA are built in this way because there are so many types of questions that it is impossible to think of them all by a small group of people. Instead, using data from those websites can ensure that the LLM fine-tuning data is at a good or acceptable level in terms of quantity and quality.

In addition to employing existing resources, another straightforward way to develop a fine-tuning dataset is to crowdsource the data. A simple approach is to allow users to input any question, after which responses are either manually given or automatically generated by an LLM and then manually annotated and corrected. It is thus possible to capture real user behavior and consequently gather inputs and outputs for a large number of "new" problems that traditional NLP tasks do not cover.

An issue related to the construction of the fine-tuning datasets is that we usually want the data to be as diverse as possible. Many studies have found that increasing the diversity of fine-tuning data can improve the robustness and generalization ability of LLMs. For this reason, there has been considerable interest in involving more diverse prompts and tasks in LLM fine-tunining datasets. We will provide further discussion on the generalization of fine-tuning in Section 10.2.4.

2. Automatically Generated Data

One limitation of manual data generation is that the quality and diversity largely depend on human experience and creativity. Therefore, if we want LLMs to handle a broad range of tasks, that is, to effectively execute any instruction, relying on human-annotated data for LLM fine-tuning is often inefficient. Moreover, the coverage of such data can be limited, and the data may even contain biases introduced by the annotators themselves. An alternative approach is to generate data automatically. For example, we can collect a number of questions through crowdsourcing, and employ a well-tuned LLM to generate answers to the questions. These question-answer pairs are then used as fine-tuning samples as usual. This method, though very simple, has been extensively applied to generate large-scale fine-tuning data for LLMs.

The above way of producing synthetic fine-tuning data is similar to those used in data augmentation for NLP. If we have an LLM, we can produce a prediction in response to any input. Repeating this process for different inputs allows us to create a sufficient number of fine-tuning samples. Such a method is particularly useful for fine-tuning new LLMs using a well-tuned LLM. However, one disadvantage of this approach is that it relies on human-crafted or collected inputs for data generation, which may turn out to be inappropriate for generalizing LLMs. In many LLM applications, a significant challenge arises from the broad range of users' questions and requests, many of which are not covered in existing NLP tasks and datasets. In these cases, it becomes necessary to generate not only the predictions but also the inputs themselves.

Here we consider **self-instruct** as an example to illustrate how to generate LLM fine-tuning samples [Wang et al., 2023e; Honovich et al., 2023]. The idea is that we can prompt an LLM to create a new instruction by learning from other instructions. Given this instruction, the LLM can then fill in other fields (such as the user input) and produce the predictions. Figure 10.4 shows a schematic illustration of self-instruct. Here we give a brief outline of the key steps involved.

- The self-instruct algorithm maintains a pool of tasks. Initially it contains a number of seed hand-crafted tasks, each with an instruction and input-output sample. As the algorithm proceeds, LLM-generated instructions and samples will be added to this pool.
- At each step, a small number of instructions are drawn from the instruction pool. For example, we can randomly select a few human-written instructions and a few LLM-generated instructions to ensure diversity.

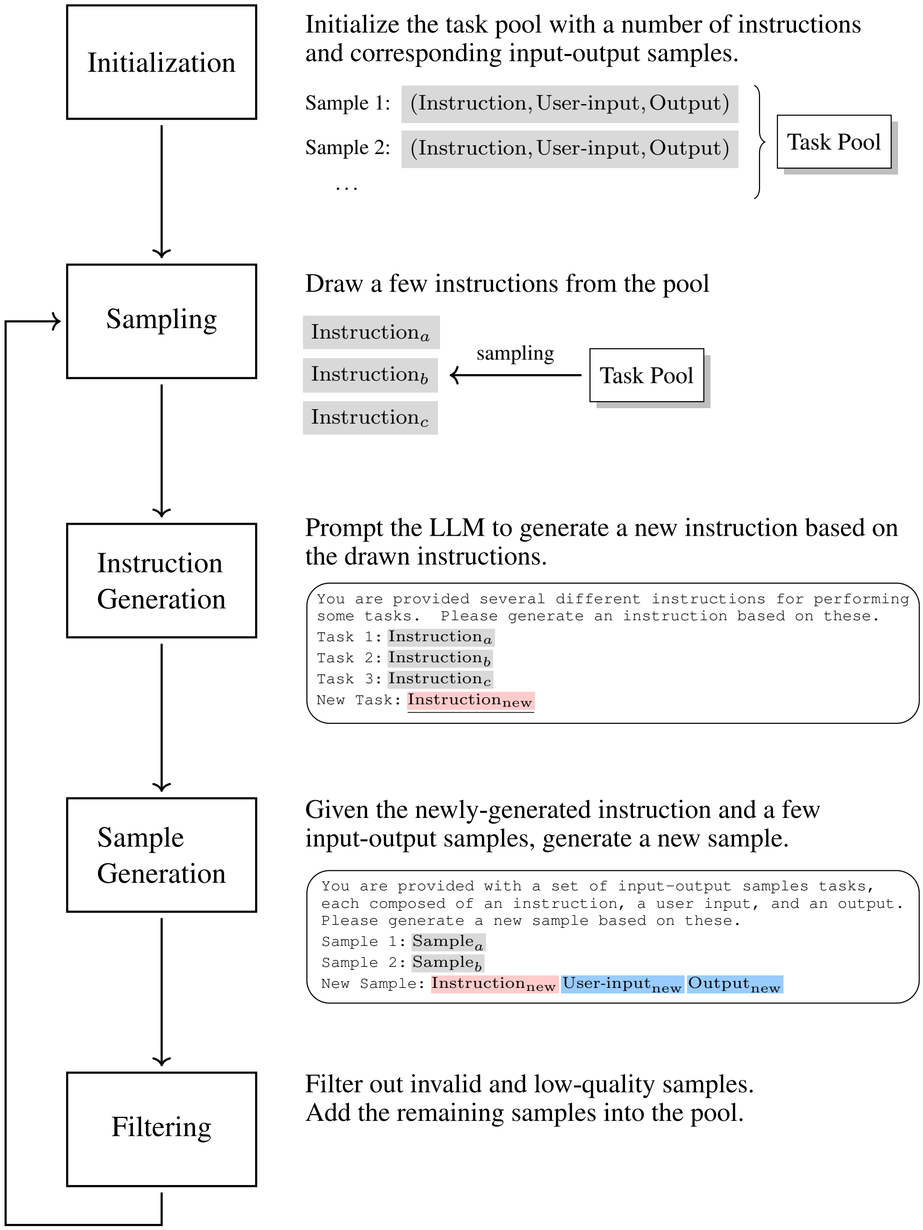


Figure 10.4: Illustration of self-instruct [Wang et al., 2023c]. This method maintains a pool of instructions and corresponding input-output samples. Initially, the pool contains a number of hand-crafted instructions and samples. Each time, we draw a few instructions from the pool. An LLM is then prompted to generate new instructions and samples based on those drawn. Finally, the newly-generated instructions and samples are filtered and added to the pool.

- The selected instructions are then used as demonstration examples. Thus, the LLM can in-context learn from these examples and produce a new instruction. Below is an example template for prompting the LLM.

> You are provided several different instructions for performing some tasks. Please generate an instruction based on these.
>
> Task 1: {instruction1}
>
> Task 2: {instruction2}
>
> Task 3: {instruction3}
>
> Task 4: {instruction4}
>
> New Task: ____

- Given the generated instruction, the LLM is then prompted to complete the sample by filling in the remaining input fields and generating the corresponding output. Below is a prompt template.

> You are provided with a set of input-output samples, each composed of an instruction, a user input, and an output. Please generate a new sample based on these.
>
> Sample 1: {instruction1}
> Input: {user-input1}
> Output: {output1}
>
> Sample 2: {instruction2}
> Input: {user-input2}
> Output: {output2}
>
> New Sample: {new-instruction}
>
> ____

- This newly-generated sample is examined by some heuristic rules (such as filtering out samples or instructions that are similar to those already in the pool). If it passes, the sample and instruction are added to the pool.

This generation process can be repeated many times to obtain a sufficient number of fine-tuning samples. Note that, above, we just show simple prompt templates for generating instruction and fine-tuning samples. Of course, we can develop better templates to generate more diverse and accurate instruction and fine-tuning samples. For example, for certain tasks like text classification, the LLM may tend to produce biased predictions, for example, most generated samples belong to a single class. In such cases, we can adjust the order of generation of different fields. More specifically, we can specify the output (i.e., the class) with some prior, and prompt the LLM to generate user input given both the instruction and the output. This method resembles **input inversion**, where the LLM generates the input based on the specified output [Longpre et al., 2023].

Using LLM-generated instructions and fine-tuning samples has been a common method for developing LLMs, especially given that manually developing such data is so expensive that most research groups cannot afford it. In several well-tuned LLMs, their fine-tuning datasets include a certain amount of synthetic data, which has proved useful [Ouyang et al., 2022; Taori et al., 2023; Chiang et al., 2023b]. There have been further studies on synthetic data generation for LLM fine-tuning. For example, one can generate more diverse instructions by introducing evolutionary algorithms [Xu et al., 2024], or use synthetic data as supervision signals in a more advanced fine-tuning process [Chen et al., 2024b]. More recently, there has also been considerable interest in using synthetic data in the pre-training stage [Gunasekar et al., 2023; Allal et al., 2024].

In many applications, a real-world scenario is that, given a task, we can collect or annotate a relatively small amount of fine-tuning data, for example, we can recruit experts to create questions for QA tasks in a specific domain. But the quantity and diversity of this data are in general not sufficient. In this case, we can use self-instruct techniques to generate more diverse question-answer pairs, and thus augment the fine-tuning data. This provides a way of bootstrapping the LLM starting from a seed set of fine-tuning samples. Note that using self-generated data is a common practice and has long been applied in NLP. For example, this approach has been successfully used in parsing and machine translation [Charniak, 1997; Sennrich et al., 2016a].

10.2.3 Fine-tuning with Less Data

With the increasing prominence of instruction fine-tuning, there has been a surge in demand for large-scale, high-quality fine-tuning data. For example, the FLAN fine-tuning dataset, which is compiled from 1,836 tasks, contains 15 million samples [Longpre et al., 2023]. Fine-tuning LLMs with such large datasets is typically a computationally expensive task, especially given that updating the large number of parameters in LLMs is resource-intensive. One approach for mitigating this issue is to explore efficient model training methods, for example, one can use parameter-efficient methods to update only a small portion of the model. However, many fine-tuning datasets contain a large amount of synthetic data, where errors and biases are still inevitable.

Another approach to efficient fine-tuning is to consider only the most relevant and impactful examples for fine-tuning. We can thus reduce the amount of data that needs to be processed while still maintaining the quality of the model updates. There are several methods to achieve this. For example, Zhou et al. [2023a] built an instruction-following dataset containing only 1,000 samples by carefully crafting the prompts and collecting samples from a variety of NLP tasks. They showed that the LLaMa 65B model fine-tuned with this dataset could be competitive with or even better than models with much more fine-tuning effort. This suggests that LLMs can be adapted to respond to diverse tasks without necessarily needing fine-tuning on all types of instruction-following data. Chen et al. [2024a] developed a system based on the GPT-3.5 model to assess the quality of each instruction-following sample. Therefore, they could select high-quality samples from existing datasets, showing better fine-tuning performance with fewer fine-tuning samples. Researchers have also developed methods to either select or filter out data using heuristics [Zhao et al., 2024; Ge et al., 2024], or to prioritize data that more significantly influences the fine-tuning process [Xia et al., 2024]. In fact, most of these methods can be seen as instances of larger families of data selection and filtering methods. And it is often the case that using higher quality (but maybe less) data is beneficial for training NLP models.

The discoveries in instruction fine-tuning somewhat differ from traditional views in NLP: the ability of models to handle complex problems can be activated with a small amount of annotated data, rather than requiring massive amounts of supervised data for extensive training. One possible explanation is that the ability of generating correct responses given instructions has been learned during pre-training, but such instruction-response mappings are not with high probabilities during inference. Fine-tuning can slightly adjust the models to get them to follow instructions, requiring significantly less training effort than pre-training. This is closely related to what is known as the **superficial alignment hypothesis**, which suggests that learning primarily occurs during pre-training, and the subsequent fine-tuning or alignment phase does not significantly contribute to the underlying knowledge base of an LLM [Zhou et al., 2023a]. Since the core abilities and knowledge of the model are already established from pre-training, effective fine-tuning for alignment with user needs can be achieved with relatively small training fine-tuning effort. This implies the possibility of fine-tuning LLMs with very little data. In another direction, it may not be necessary to restrict fine-tuning to paired instruction-response data. For example, Hewitt et al. [2024] found that instruction-following can be implicitly achieved by fine-tuning LLMs only on responses, without corresponding instructions.

A concept related to the discussion here is sample efficiency. A machine learning method is called **sample efficient** if it can learn effectively from a small number of training examples. In this sense, instruction fine-tuning is sample efficient compared with pre-training. From the perspective of machine learning, sample-efficient methods can be seen as efficient ways to sample the space of data, and are advantageous as they make optimal use of scarce data. Therefore, sampling-based learning techniques, such as many reinforcement learning algorithms, can benefit from these sample efficient approaches. For example, in human preference alignment, we can either efficiently sample preference data via reward models [Liu et al., 2024b] or improve the sampling efficiency in policy learning [Wang et al., 2024].

10.2.4 Instruction Generalization

In many machine learning and NLP problems, training a model to generalize is a fundamental goal. For example, in text classification, we expect our model to correctly classify new texts that were not seen during training. However, generalization poses additional challenges in instruction fine-tuning. We expect instruction-fine-tuned LLMs to not only generate appropriate responses for different inputs within a task but also to accurately perform various tasks as described by different instructions. To illustrate this issue, consider an LLM $\Pr(\mathbf{y}|\mathbf{c},\mathbf{z})$, where $\mathbf{c}$ is an instruction, $\mathbf{z}$ is a user input, and $\mathbf{y}$ is the corresponding model output (i.e., the response). Suppose that the performance of this model is evaluated in terms of a metric, written as $\mathrm{Performance}(\Pr(\mathbf{y}|\mathbf{c},\mathbf{z}))$ or $\mathrm{P}(\mathbf{c},\mathbf{z},\mathbf{y})$ for short. Informally, when we say this model can generalize within a given task (indicated by the instruction $\mathbf{c}^*$), we mean that there may be a value ϵ such that the average performance on new inputs is above this value:

$$\frac{1}{|\mathcal{Z}|}\sum_{\mathbf{z}'\in\mathcal{Z}}\mathrm{P}(\mathbf{c}^*,\mathbf{z}',\mathbf{y}') > \epsilon \tag{10.8}$$

where $\mathcal{Z}$ is the set of new inputs, and $\mathbf{z}'$ and $\mathbf{y}'$ are an input in this set and the corresponding output, respectively.

Likewise, we can say that this model can generalize across tasks if the average performance over all instruction-input pairs is above some ϵ:

$$\frac{1}{|\mathcal{D}|}\sum_{(\mathbf{c}',\mathbf{z}')\in\mathcal{D}}\mathrm{P}(\mathbf{c}',\mathbf{z}',\mathbf{y}') > \epsilon \tag{10.9}$$

where $\mathcal{D}$ is the set of new instruction-input pairs.

Here, we need to deal with variations in two dimensions: instruction and user input. This makes the generalization problem very complex, because, intuitively, a model needs to learn from a vast number of tasks and different input-output pairs associated with each task to achieve good generalization. As we have discussed several times in this book, achieving such generalization incurs much lower cost than pre-training. In general, fine-tuning LLMs with instruction-response data to some extent can lead to models yielding instruction following on new tasks. Nevertheless, it is typically believed that certain efforts are still needed to adapt LLMs to make them understand and execute instructions broadly.

One way to generalize instruction fine-tuning is to increase the diversity of the fine-tuning data. In earlier studies on instruction fine-tuning, researchers developed many datasets, covering a wide variety of NLP tasks and different instructions for each task [Wang et al., 2022e; Sanh et al., 2022; Longpre et al., 2023]. By transforming these tasks into a unified format, one can fine-tune an LLM with a sufficiently large number of samples, for example, there have been several instruction fine-tuning datasets that involve over 100 NLP tasks and 1M samples. However, these early datasets mostly focus on existing academic problems, but not those that users want to deal with in real-world applications. Much recent work has shifted focus to addressing new and more practical problems. For example, there has been considerable interest in constructing datasets that contain large and complicated demonstrations and responses from SOTA models to real user queries [Wang et al., 2023d; Teknium, 2023].

Perhaps the use of large and diverse fine-tuning datasets has its origins in attempts to scale LLMs in different dimensions. Indeed, scaling laws have been used broadly to motivate the development of a wide range of different instruction-fine-tuned LLMs. And it is reasonable to scale instruction fine-tuning to make an LLM follow broad instructions. From the perspective of LLM alignment, however, scaling instruction fine-tuning might not be efficient to achieve generalization.

One problem is that instruction fine-tuning relies on supervised learning that learns to generalize and perform tasks based on instruction-response mappings. However, such an approach does not capture subtle or complex human preferences (e.g., tone, style, or subjective quality) because these are hard to encode as explicit instruction-response data. Moreover, the generalization performance is bounded by the diversity and quality of the instruction-response dataset. Given these limitations, we would instead like to employ preference models as an additional fine-tuning step following instruction fine-tuning, so the LLMs can generalize further (see Section 10.3).

Another view is that some instruction-response mappings may already be learned during pre-training, and so the pre-trained LLMs have encoded such mappings. However, since we often do not know exactly what data is used in the pre-training, it is hard to judge whether we need to learn such mappings in the fine-tuning. A related question is whether out-of-distribution generalization is primarily achieved during pre-training or fine-tuning. While directly answering this question is beyond the scope of this chapter, it has been shown that pre-training on large and diverse datasets is effective in improving out-of-distribution performance [Hendrycks et al., 2020; Radford et al., 2021; Gunasekar et al., 2023]. This raises an interesting problem: if an LLM has been well pre-trained at scale, fine-tuning may not be as essential for out-of-distribution generalization, since the model may have already encountered sufficient distributional variation. This prompts researchers to fine-tune LLMs with modest effort or to explore new methods to achieve instruction-following. As discussed in the previous subsection, for example, instruction following can be yielded by fine-tuning on a small number of carefully selected instruction-response pairs [Zhou et al., 2023a], or even by using methods that are not explicitly designed to do so [Kung and Peng, 2023].

The above discussion provides two different strategies: one requires scaling up fine-tuning datasets for larger diversity, the other requires small but necessary fine-tuning datasets for efficient LLM adaptation. However, in practice, involving diverse instructions often helps. In many cases, we need to adapt our LLM for specific purposes. But the LLM, which has possibly encoded broad instruction-following mappings during pre-training, might tend to behave as a general-purpose instruction executor even with modest fine-tuning. An interesting phenomenon is that when fine-tuning on math data, the resulting LLM might not specialize in math outputs. Instead, this model might respond normally to general instructions, for example, it could generate poetry if instructed to do so [Hewitt, 2024]. This is not a bad thing, but it shows that LLMs may not easily change their nature of following general instructions. In this case, additional adaptations with more diverse data may help adjust the way the LLM follows instructions, particularly for those tasks we aim to address.

10.2.5 Using Weak Models to Improve Strong Models

So far we have explored a variety of instruction fine-tuning methods based on labeled data. One of the limitations of many such methods is that they require the data to be annotated by humans or generated by strong LLMs, which can provide accurate supervision signals in fine-tuning. However, in many cases, the LLM we have in hand is already strong (or at least is advantageous in specific aspects of problem solving), and thus it is not easy to find a superior model for supervision. Even for human experts, when the problem becomes complex, providing correct and detailed answers might be difficult, or sometimes infeasible. For example, when faced with an extremely long document, the experts would find it challenging to identify any inconsistencies, subtle biases, or missing key points without conducting an exhaustive and time-consuming review.

One may ask at this point: can we use weak LLMs to supervise strong LLMs? This seems to be a significant challenge, but it may reflect a future scenario where we need to supervise AI systems that are smarter than humans or any other AI systems [Burns et al., 2023b]. The problem of using smaller, less complex models to improve the training of larger, more complex models is also called the **weak-to-strong generalization** problem. While there have not been mature approaches to weak-to-strong generalization, using smaller models to assist stronger models has indeed proven useful in several areas of LLMs.

For instruction fine-tuning, one of the simplest ways of applying weak LLMs is to use these models to generate synthetic fine-tuning data. Suppose we have a collection of inputs X, where each input includes an instruction and a user input if necessary. For each $\mathbf{x} \in X$, we use a weak LLM $\Pr^{w}(\cdot)$ to generate a prediction $\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \Pr^{w}(\mathbf{y}|\mathbf{x})$. Then, the strong LLM $\Pr^{s}_{\theta}(\cdot)$ can be trained on these generated predictions (see Eq. (10.1)):

$$\tilde{\theta} = \arg\max_{\theta} \sum_{\mathbf{x} \in X} \log \Pr^{s}_{\theta}(\hat{\mathbf{y}}|\mathbf{x}) \tag{10.10}$$

where θ is the model parameters.

The above form transforms the fine-tuning problem into a knowledge distillation problem, in other words, we distill knowledge from the weak model to the strong model. Consequently, we can employ various knowledge distillation methods to achieve this goal. However, explaining weak-to-strong fine-tuning from the perspective of knowledge distillation is not straightforward. A major concern is that the strong model may merely imitate or overfit the errors of the weak model and fail to generalize. For example, the fine-tuned strong model still cannot solve difficult problems that the weak model cannot accurately predict. Fortunately, preliminary experiments in this line of research have shown positive and promising results. For example, Burns et al. [2023a] found that fine-tuning the strong pre-trained GPT-4 model with GPT-2-level supervision could improve generalization across several NLP tasks. To measure how the weak model improves the generalization of the strong model, we define the following terms:

- **Weak Performance** ($\mathrm{P}_{\mathrm{weak}}$). This is the test-set performance of the weak model, which can be regarded as the baseline performance.
- **Weak-to-strong Performance**($\mathrm{P}_{\mathrm{weak}\rightarrow\mathrm{strong}}$). This is the test-set performance of the strong model that is fine-tuned with the weak model.
- **Strong Ceiling Performance** ($\mathrm{P}_{\mathrm{ceiling}}$). This is the test-set performance of the strong model that is fine-tuned with ground truth data. For example, we fine-tune the strong model with human-annotated predictions and take the resulting model as a ceiling.

Then, the **performance gap recovered** (**PGR**) can be defined as

$$\mathrm{PGR} \quad = \quad \max\left\{0, \frac{\mathrm{P}_{\text{weak}\to\text{strong}} - \mathrm{P}_{\text{weak}}}{\mathrm{P}_{\text{ceiling}} - \mathrm{P}_{\text{weak}}}\right\} \tag{10.11}$$

This metric measures how much of the performance gap between the ceiling model and the weak model can be recovered by the weak-to-strong model. A PGR of 1 indicates that the weak-to-strong fine-tuning can completely closes the performance gap, whereas a PGR of 0 indicates no improvement. In Burns et al. [2023a]'s work, it is shown that PGR can be around 0.8 on 22 NLP classification tasks. It should be noted that, while the potential of weak-to-strong fine-tuning is promising, achieving substantial weak-to-strong generalization remains a challenging goal that needs further investigation [Aschenbrenner, 2024].

Fine-tuning LLMs with weak supervision is just one choice for using small models to improve large models. Although this section primarily focuses on fine-tuning LLMs, we also mention other methods here to give a more complete discussion (see Figure 10.5 for illustrations of these methods).

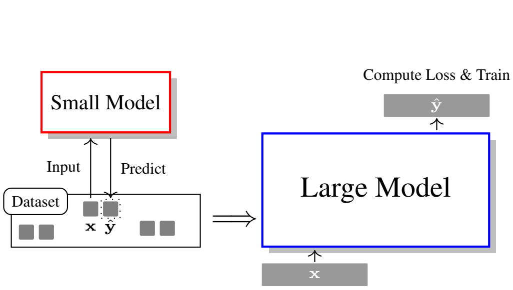
- Instead of using small models to generate synthetic data, it is also straightforward to incorporate knowledge distillation loss based on these models. For example, a simple loss function that measures the difference between the small and large models can be defined as:

  $$\mathrm{Loss}_{\mathrm{kd}} \quad = \quad \mathrm{KL}(\mathrm{Pr}^{w}(\cdot|\mathbf{x}) \;||\; \mathrm{Pr}^{s}_{\theta}(\cdot|\mathbf{x})) \tag{10.12}$$

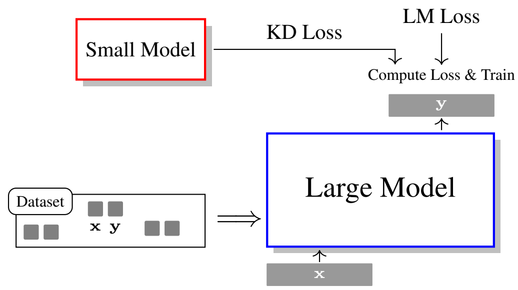
  Then, we can add this loss to the original loss of language modeling, and yield the following training objective

  $$\tilde{\theta} \quad = \quad \arg\max_{\theta} \sum_{(\mathbf{x},\mathbf{y})\in\mathcal{D}} \log \mathrm{Pr}^{s}_{\theta}(\mathbf{y}|\mathbf{x}) - \lambda \cdot \mathrm{Loss}_{\mathrm{kd}} \tag{10.13}$$

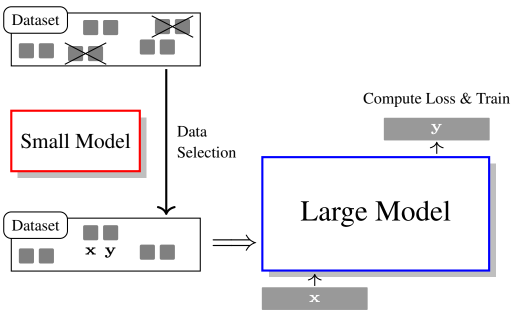
  where $\mathcal{D}$ is the set of input and output pairs, and λ is the coefficient of the interpolation. This method can be employed in either the pre-training or fine-tuning phase. We can adjust λ to control how much the small model influences the training. For example, we can gradually decrease λ to make the training rely more on the original language modeling loss as the large model becomes more capable.
- Another approach to involving small models in LLM pre-training and fine-tuning is to use them to do data selection or filtering. Given a sequence, we can compute the likelihood or cross-entropy using a small model. These quantities can then be used as criteria for selecting or filtering data. For example, sequences with low likelihood or high cross-entropy might be excluded from the training set, as they are less aligned with the small model's learned distribution. Conversely, sequences with high likelihood or low cross-entropy can be prioritized, ensuring that the training focuses on more relevant or high-quality data.
- Ensemble learning is a simple and effective way to build a strong model by combining



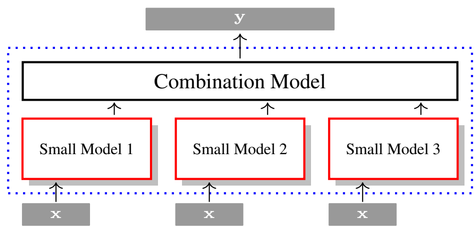
(a) Fine-tuning on data generated by a small model (weak-to-strong generalization)



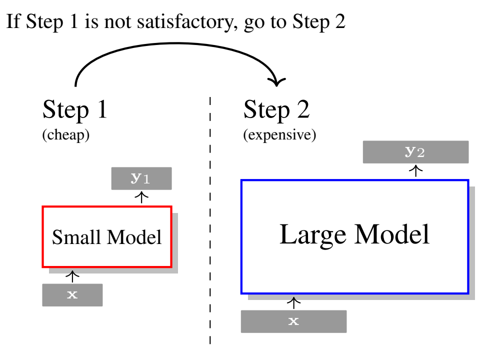
(b) Fine-tuning with KD Loss from a small model (weak-to-strong generalization)



(c) Data selection with a small model



(d) Ensemble of multiple small models



(e) Cascading (at inference time)

Figure 10.5: Illustrations of using small models to improve large models in LLMs. One approach involves using smaller models for the fine-tuning or pre-training of larger models. This includes generating synthetic data (a), incorporating auxiliary loss (b), and selecting appropriate data (c). Another approach involves combining small models and large models. This includes learning a strong model by aggregating multiple small models (d), and cascading small models with large models (e).

multiple weak models. Applying this technique to LLMs is straightforward. We can aggregate distributions predicted by multiple small models or specialized submodels, and derive the final prediction from the aggregated results. This aggregation can be done using methods such as majority voting, weighted averaging, or stacking.

- Small models can also be employed at inference time to improve overall efficiency. Suppose we have a large model that is slow but more accurate, and a small model that is fast but less accurate. In model cascading, the small model first processes the input data, quickly generating preliminary results. If these results meet certain pre-defined criteria, they can be directly used. However, if the initial results are not sufficiently good, the input is then passed to the larger, more accurate model to produce a better result. This approach significantly reduces computational costs and latency, as the small model can effectively handle many inputs without access to the large model.

10.3 Human Preference Alignment: RLHF

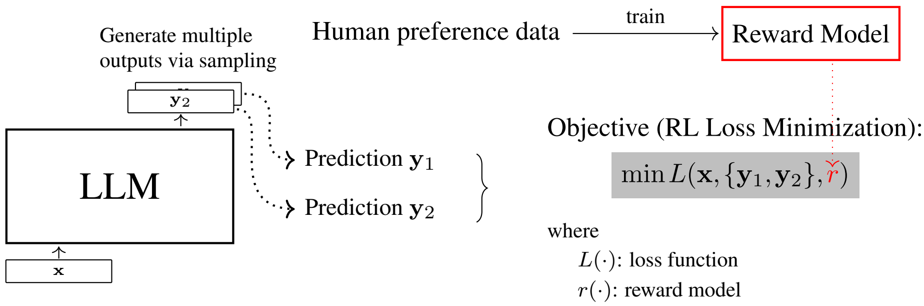
So far in this chapter, we have focused on fine-tuning LLMs using input-output paired labeled data. This approach allows us to adapt LLMs for instruction-following via supervised learning. In many applications, however, LLMs are required not only to follow instructions but also to act in ways that are more aligned with human values and preferences. Consider a scenario where a user asks an LLM how to hack into a computer system. If the LLM is not appropriately aligned, it may respond by providing details on how to perform this illegal activity. Instead, a more desirable response might be to advise the user against engaging in illegal activities and offer a general overview of the consequences of such actions. The difficulty in achieving this is that the ethical nuances and contextual considerations required for an LLM to respond appropriately in such scenarios are not always straightforward to encode into a fine-tuning dataset. What's even more challenging is that, often, humans themselves cannot precisely express their own preferences.

In this section, we discuss an alternative LLM fine-tuning method, called reinforcement learning from human feedback or RLHF for short [Christiano et al., 2017; Stiennon et al., 2020]. The basic idea behind RLHF is that LLMs can learn from comparisons of model outputs using reward models (see Figure 10.6). To do this, we can recruit human experts who indicate their preferences between pairs of outputs generated by the LLM. This preference data is used to train a reward model that can predict the perceived quality of LLM outputs. Once trained, the reward model provides feedback by assigning scores to new outputs that the LLM generates in response to the inputs. The LLM uses these scores to update its parameters through reinforcement learning algorithms. In the rest of this section, we will first introduce the basic knowledge of reinforcement learning to facilitate the discussion, and then discuss methods for training reward models and aligning LLMs with these models.

10.3.1 Basics of Reinforcement Learning

We begin by looking at some basic concepts of reinforcement learning. Note that the notation used here slightly differs from that used in the previous sections and chapters because we want to make our description more consistent with those in the reinforcement learning literature.

(a) Supervised fine-tuning (maximizing the prediction probability given the input)



(b) Reinforcement Learning from Human Feedback

Figure 10.6: Supervised fine-tuning vs. reinforcement learning from human feedback. In supervised fine-tuning, we optimize the LLM by maximizing the probability of the prediction given the input. In reinforcement learning from human feedback, we first train a reward model on human preference data (on each pair of predictions, evaluators are asked to choose which one they prefer). Then, we use this reward model to supervise the LLM during fine-tuning.

Nevertheless, we will show how this notation corresponds to the language modeling notation. The reader who is already familiar with reinforcement learning techniques may skip or skim this subsection.

A general reinforcement learning framework describes how an agent interacts with a dynamic environment. This interaction is modeled as a sequence of actions taken by the agent in response to the state of the environment. At each time step, the agent observes the current state, chooses an action based on its policy, performs the action, and then receives feedback from the environment in the form of a reward and a new state. This sequence of observe-act-receive feedback is repeated until the agent achieves its goal.

A reinforcement learning system involves several components:

- **Agent**. This is the learner or decision-maker in reinforcement learning. In the context of LLMs, it can be seen as the LLM itself.
- **Environment**. This includes everything external to the agent with which the agent interacts. But the environment in LLMs is less about a physical or virtual space and more about the framework within which the agent (e.g., an LLM) receives feedback and

learns.

- **State** (s). A state represents the current situation of the environment. Given a sequence of tokens for language modeling, a state at a time step can be viewed as the tokens we observed so far, that is, the context tokens we take to predict the next token. For example, we can define $(\mathbf{x}, \mathbf{y}_{<t})$ as the state when predicting the next token at the time step t.
- **Action** (a). Actions represent possible decisions the agent can make. We can see them as possible predicted tokens in the vocabulary.
- **Reward** (R). The reward is the feedback from the environment that evaluates the success of an action. For example, $r(s, a, s')$ denotes the reward the agent receives for taking the action a at the state s and moving to the next state s'. If the state-action sequence is given, we can denote the reward at the time step t as $r_t = r(s_t, a_t, s_{t+1})$. Also note that if the decision-making process is deterministic, we can omit s_{t+1} because it can be determined by s_t and a_t. In such cases, we can use $r(s_t, a_t)$ as shorthand for $r(s_t, a_t, s_{t+1})$.
- **Policy** (π). For an LLM, a policy is defined as the probability distribution over the tokens that the LLM predicts, given the preceding context tokens. Formally, this can be expressed as

$$\pi(a|s) = \Pr(y_t|\mathbf{x}, \mathbf{y}_{<t}) \tag{10.14}$$

where a corresponds to the token y_t, and s corresponds to the context $(\mathbf{x}, \mathbf{y}_{<t})$. Figure 10.7 illustrates how an LLM can be treated as a policy in the reinforcement learning framework.

- **Value Function** (V and Q). A **state-value function** (or value function, for short) assesses the expected discounted return (i.e., accumulated rewards) for an agent starting from a particular state s and following a specific policy π. It is defined as:

$$\begin{aligned} V(s) &= \mathbb{E}\Big[r(s_0, a_0, s_1) + \gamma r(s_1, a_1, s_2) + \gamma^2 r(s_2, a_2, s_3) + \cdots \;\Big|\; s_0 = s, \pi\Big] \\ &= \mathbb{E}\Big[r_0 + \gamma r_1 + \gamma^2 r_2 + \cdots \;\Big|\; s_0 = s, \pi\Big] \\ &= \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t r_t \;\Big|\; s_0 = s, \pi\Big] \end{aligned} \tag{10.15}$$

where $\gamma \in [0, 1]$ is the discount factor that adjusts the importance of future rewards, $s_0 = s$ indicates that the agent starts with the state s, and the expectation $\mathbb{E}$ is performed over all possible trajectories (i.e., state-action sequences). Similarly, an **action-value function** (or **Q-value function**) measures the expected return starting from a state s taking an action a and thereafter following a policy π, given by

$$Q(s, a) = \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t r_t \;\Big|\; s_0 = s, a_0 = a, \pi\Big] \tag{10.16}$$

where $a_0 = a$ indicates that the action taken at the initial state is a.

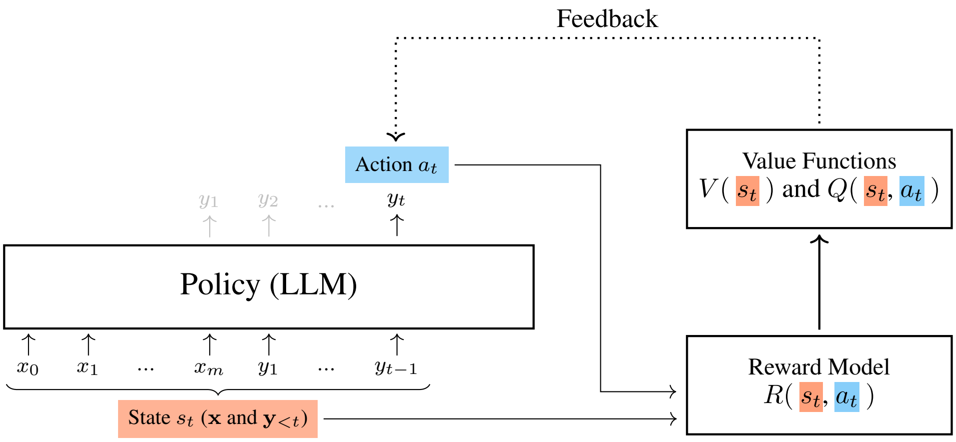


Figure 10.7: LLM as policy in reinforcement learning. At each step t, the LLM predicts a token y_t given the model input $\mathbf{x}$ and the previously-generated tokens $\mathbf{y}_{<t}$. This process can be framed as a reinforcement learning problem, where y_t serves as the action, $(\mathbf{x}, \mathbf{y}_{<t})$ as the state, and the predicted distribution $\Pr(y_t|\mathbf{x}, \mathbf{y}_{<t})$ as the policy. Once y_t is predicted, the LLM inputs both $(\mathbf{x}, \mathbf{y}_{<t})$ and y_t to the reward model, which evaluates how effectively the chosen token contributes to achieving the desired textual outcome. This evaluation generates reward scores which are used to compute the value functions $V(s_t)$ and $Q(s_t, a_t)$. These functions then provide feedback to the LLM and guide the policy training.

The goal of reinforcement learning is to learn a policy that maximizes the **cumulative reward** (or **return**) the agent receives over the long run. Given a state-action sequence $\tau = \{(s_1, a_1), ..., (s_T, a_T)\}$[1], the cumulative reward over this sequence can be written as

$$R(\tau) = \sum_{t=1}^{T} r_t \tag{10.17}$$

The expectation of this cumulative reward over a space of state-action sequences is given in the form

$$\begin{aligned} J(\theta) &= \mathop{\mathbb{E}}_{\tau \sim \mathcal{D}} \Big[R(\tau) \,\Big|\, \pi_\theta \Big] \\ &= \sum_{\tau \in \mathcal{D}} \Pr_\theta(\tau) R(\tau) \\ &= \sum_{\tau \in \mathcal{D}} \Pr_\theta(\tau) \sum_{t=1}^{T} r_t \end{aligned} \tag{10.18}$$

where $\tau \sim \mathcal{D}$ indicates that τ is drawn from the state-action sequence space $\mathcal{D}$, and the subscript

[1]We assume the state-action sequence begins with s_1 and a_1, rather than s_0 and a_0, to align with the notation commonly used in this chapter, where the prediction $\mathbf{y}$ typically starts from y_1. Of course, it is also common to denote a state-action sequence as $\{(s_0, a_0), ..., (s_T, a_T)\}$ or $\{(s_0, a_0), ..., (s_{T-1}, a_{T-1})\}$ in the literature. But this variation in notation does not affect the discussion of the models presented here.

θ indicates the parameters of the policy. $J(\theta)$ is also called the **performance function**.

Then the training objective is to maximize $J(\theta)$:

$$\tilde{\theta} = \arg\max_{\theta} J(\theta) \tag{10.19}$$

Now, we have a simple reinforcement learning approach: 1) we sample a number of state-action sequences; then, 2) we evaluate each sequence using the performance function; then, 3) we update the model to maximize this performance function. If we take Eq. (10.18) and use gradient descent to optimize the policy, this approach would constitutes a form of policy gradient methods [Williams, 1992].

Note that in many NLP problems, such as machine translation, rewards are typically sparse. For instance, a reward is only received at the end of a complete sentence. This means that $r_t = 0$ for all $t < T$, and r_t is non-zero only when $t = T$. Ideally, one might prefer feedback to be immediate and frequent (dense), and thus the training of the policy can be easier and more efficient. While several methods have been proposed to address sparse rewards, such as reward shaping, we will continue in our discussion to assume a sparse reward setup, where the reward is available only upon completing the prediction.

The model described in Eqs. (10.17-10.19) establishes a basic form of reinforcement learning, and many variants and improvements of this model have been developed. Before showing those more sophisticated models, let us take a moment to interpret the objective function $J(\theta)$ from the perspective of policy gradient. In gradient descent, we need to compute the gradient of $J(\theta)$ with respect to θ:

$$\begin{aligned}
\frac{\partial J(\theta)}{\partial \theta} &= \frac{\partial \sum_{\tau \in \mathcal{D}} \Pr_{\theta}(\tau) R(\tau)}{\partial \theta} \\
&= \sum_{\tau \in \mathcal{D}} \frac{\partial \Pr_{\theta}(\tau)}{\partial \theta} R(\tau) \\
&= \sum_{\tau \in \mathcal{D}} \Pr_{\theta}(\tau) \frac{\partial \Pr_{\theta}(\tau)/\partial \theta}{\Pr_{\theta}(\tau)} R(\tau) \\
&= \sum_{\tau \in \mathcal{D}} \Pr_{\theta}(\tau) \frac{\partial \log \Pr_{\theta}(\tau)}{\partial \theta} R(\tau)
\end{aligned} \tag{10.20}$$

In some cases, we will assume that every sequence in $\mathcal{D}$ is equally probable (i.e., $\Pr_{\theta}(\tau) = 1/|\mathcal{D}|$). In this case we can simplify Eq. (10.20) and need only consider the terms $\frac{\partial \log \Pr_{\theta}(\tau)}{\partial \theta}$ and $R(\tau)$:

$$\frac{\partial J(\theta)}{\partial \theta} = \frac{1}{m} \sum_{\tau \in \mathcal{D}} \frac{\partial \log \Pr_{\theta}(\tau)}{\partial \theta} R(\tau) \tag{10.21}$$

One advantage of this result is that $R(\tau)$ does not need to be differentiable, which means that we can use any type of reward function in reinforcement learning.

By treating the generation of the sequence τ as a Markov decision process, we can further derive $\frac{\partial \log \Pr_\theta(\tau)}{\partial \theta}$, and obtain:

$$\begin{aligned} \frac{\partial \log \Pr_\theta(\tau)}{\partial \theta} &= \frac{\partial}{\partial \theta} \log \prod_{t=1}^{T} \pi_\theta(a_t|s_t) \Pr(s_{t+1}|s_t, a_t) \\ &= \frac{\partial}{\partial \theta} \sum_{t=1}^{T} \underbrace{\log \pi_\theta(a_t|s_t)}_{\text{policy}} + \frac{\partial}{\partial \theta} \sum_{t=1}^{T} \underbrace{\log \Pr(s_{t+1}|s_t, a_t)}_{\text{dynamics}} \end{aligned} \quad (10.22)$$

where the gradient is decomposed into two parts: the policy gradient and the dynamics gradient. The policy component, $\log \pi_\theta(a_t|s_t)$, determines the log-probability of taking action a_t given state s_t, and it is parameterized by θ. The dynamics component, $\log \Pr(s_{t+1}|s_t, a_t)$, represents the log-probability of transitioning to state s_{t+1} from state s_t after taking action a_t. In typical reinforcement learning settings, the dynamics are not directly influenced by the policy parameters θ, and thus, their derivatives are often zero. In this case, therefore, Eq. (10.22) can be simplified to:

$$\frac{\partial \log \Pr_\theta(\tau)}{\partial \theta} = \frac{\partial}{\partial \theta} \sum_{t=1}^{T} \log \pi_\theta(a_t|s_t) \quad (10.23)$$

In other words, we only concentrate on optimizing the policy without concerning ourselves with the underlying dynamics.

Substituting Eq. (10.23) into Eq. (10.21), and expanding $R(\tau)$, we then obtain

$$\frac{\partial J(\theta)}{\partial \theta} = \frac{1}{|\mathcal{D}|} \sum_{\tau \in \mathcal{D}} \frac{\partial}{\partial \theta} \Big(\sum_{t=1}^{T} \log \pi_\theta(a_t|s_t) \sum_{t=1}^{T} r_t \Big) \quad (10.24)$$

While this policy gradient approach is straightforward, it suffers from the problem that the variance of the estimated gradients can be very high, making the learning process noisy and inefficient. One reason for this high variance problem is that rewards can vary greatly across different steps or scenarios. Imagine that in a sequence of action decisions, the reward model tends to assign small rewards to good actions (e.g., $R_t = 2$) and large penalties to poor actions (e.g., $R_t = -50$). Such varying reward scales for good and poor actions can result in a very low total reward for the entire sequence, even if it includes good actions.

One simple method for reducing the variance of the gradient is to set a baseline b and subtract it from $\sum_{t=1}^{T} r_t$, resulting in $\sum_{t=1}^{T} r_t - b$.[2] Here, the baseline can be interpreted as a reference point. By centering the rewards around this baseline, we remove systematic biases in

[2] In fact, the use of a baseline b does not change the variance of the total rewards $\sum_{t=1}^{T} r_t$. However, it is important to note that while introducing a baseline does not alter the overall variance of the rewards, it helps reduce the variance of the gradient estimates. This is because subtracting the baseline from the total rewards effectively reduces fluctuations around their mean, which makes the gradient estimates more stable. In general, the operation $\sum_{t=1}^{T} r_t - b$ centers the rewards around zero (e.g., b is defined as the expected value of $\sum_{t=1}^{T} r_t$), which can lead to reduced variance in the product $\sum_{t=1}^{T} \log \pi_\theta(a_t|s_t)(\sum_{t=1}^{T} r_t - b)$.

the reward signal, making the updates more stable and less sensitive to extreme fluctuations in individual rewards.

This policy gradient model with a baseline can be given by

$$\begin{aligned}\frac{\partial J(\theta)}{\partial \theta} &= \frac{1}{|\mathcal{D}|}\sum_{\tau\in\mathcal{D}}\frac{\partial}{\partial\theta}\Big(\sum_{t=1}^{T}\log\pi_\theta(a_t|s_t)\Big)\Big(\sum_{t=1}^{T}r_t-b\Big) \\ &= \frac{1}{|\mathcal{D}|}\sum_{\tau\in\mathcal{D}}\frac{\partial}{\partial\theta}\Big[\sum_{t=1}^{T}\log\pi_\theta(a_t|s_t)\Big(\sum_{k=1}^{T}r_k-b\Big)\Big] \\ &= \frac{1}{|\mathcal{D}|}\sum_{\tau\in\mathcal{D}}\frac{\partial}{\partial\theta}\Big[\sum_{t=1}^{T}\log\pi_\theta(a_t|s_t)\Big(\sum_{k=1}^{t-1}r_k+\sum_{k=t}^{T}r_k-b\Big)\Big] \qquad (10.25)\end{aligned}$$

Here we write $\sum_{k=1}^{T} r_k$ as the sum of two terms $\sum_{k=1}^{t-1} r_k$ and $\sum_{k=t}^{T} r_k$ to distinguish between the rewards accrued before and after the action at time step t. Note that in Markov decision processes, the future is independent of the past given the present. Therefore, the action taken at time step t cannot influence the rewards received before t, or in other words, the rewards prior to t are already "fixed" by the time the action at t is chosen. The term $\sum_{k=1}^{t-1} r_k$ does not contribute to the gradient and can be omitted, leading to a simplified version of Eq. (10.25)

$$\frac{\partial J(\theta)}{\partial \theta} = \frac{1}{|\mathcal{D}|}\sum_{\tau\in\mathcal{D}}\frac{\partial}{\partial\theta}\Big[\sum_{t=1}^{T}\log\pi_\theta(a_t|s_t)\Big(\sum_{k=t}^{T}r_k-b\Big)\Big] \qquad (10.26)$$

Also note that removing $\sum_{k=t}^{T} r_k$ can further reduce the variance of the gradient.

There are many ways to define the baseline b. Here we consider the value function of the state s_t, that is, the estimated value of being in state s_t: $V(s_t) = \mathbb{E}(r_t + r_{t+1} + \cdots + r_T)$. Hence we have

$$\begin{aligned}A(s_t,a_t) &= \sum_{k=t}^{T} r_k - b \\ &= \sum_{k=t}^{T} r_k - V(s_t) \qquad (10.27)\end{aligned}$$

where $\sum_{k=t}^{T} r_k$ represents the actual return received, and $V(s_t)$ represents the expected return. $A(s_t,a_t)$ (or A_t for short) is called the **advantage** at time step t, which quantifies the relative benefit of the action a_t compared to the expected value of following the policy from the state s_t onward.

By using the advantage function $A(s_t,a_t)$, the gradient of $J(\theta)$ can be written in the form

$$\frac{\partial J(\theta)}{\partial \theta} = \frac{1}{|\mathcal{D}|}\sum_{\tau\in\mathcal{D}}\frac{\partial}{\partial\theta}\Big(\sum_{t=1}^{T}\log\pi_\theta(a_t|s_t)A(s_t,a_t)\Big) \qquad (10.28)$$

This optimization objective corresponds to the **advantage actor-critic** (**A2C**) method in reinforcement learning [Mnih et al., 2016]. In this method, the actor aims at learning a policy. It updates the policy parameters using Eq. (10.28) to help focus more on actions that are likely to improve performance. The critic, on the other hand, updates its estimation of the value function, which is used to calculate the advantage function $A(s_t, a_t)$, thus serving as the evaluator of the policy being learned by the actor.

In the A2C method, $A(s_t, a_t)$ is typically expressed as the difference of the action-value function $Q(s_t, a_t)$ and the state-value function $V(s_t)$

$$A(s_t, a_t) \quad = \quad Q(s_t, a_t) - V(s_t) \tag{10.29}$$

At first glance, this model may seem challenging to develop because it requires two separate sub-models to calculate $Q(s_t, a_t)$ and $V(s_t)$ respectively. Fortunately, considering that $Q(s_t, a_t)$ can be defined as the return $r_t + V(s_{t+1})$, we can rewrite Eq. (10.29) as

$$A(s_t, a_t) \quad = \quad r_t + V(s_{t+1}) - V(s_t) \tag{10.30}$$

or alternatively, introduce the discount factor γ to obtain a more general form

$$A(s_t, a_t) \quad = \quad r_t + \gamma V(s_{t+1}) - V(s_t) \tag{10.31}$$

$A(s_t, a_t) = r_t + \gamma V(s_{t+1}) - V(s_t)$ is also called the **temporal difference** (**TD**) error. What we need is to train a critic network for the value function $V(s_t)$, and then use it to compute the advantage function[3].

Up to this point, we have spent considerable space discussing the basics of reinforcement learning, especially on how to derive the optimization objective for the A2C method. However, reinforcement learning is a vast field, and many technical details cannot be covered here. The interested reader can refer to reinforcement learning books for more details [Sutton and Barto, 2018; Szepesvári, 2010]. Nevertheless, we now have the necessary knowledge to further discuss RLHF. In the subsequent subsections, we will return to the discussion on LLM alignment, demonstrating how to use the A2C method for aligning with human preferences.

10.3.2 Training Reward Models

We have shown that reward models play a very important role in the general reinforcement learning framework and form the basis for computing value functions. We now consider the problem of training these reward models.

In RLHF, a reward model is a neural network that maps a pair of input and output token

[3]The training loss for the value network (or critic network) in A2C is generally formulated as the mean squared error between the computed return $r_t + \gamma V(s_{t+1})$ and the predicted state value $V(s_t)$. Suppose that the value network is parameterized by ω. The loss function is given by

$$\mathcal{L}_v(\omega) \quad = \quad \frac{1}{M} \sum \big(r_t + \gamma V_\omega(s_{t+1}) - V_\omega(s_t)\big)^2 \tag{10.32}$$

where M is the number of training samples, for example, for a sequence of T tokens, we can set $M = T$.

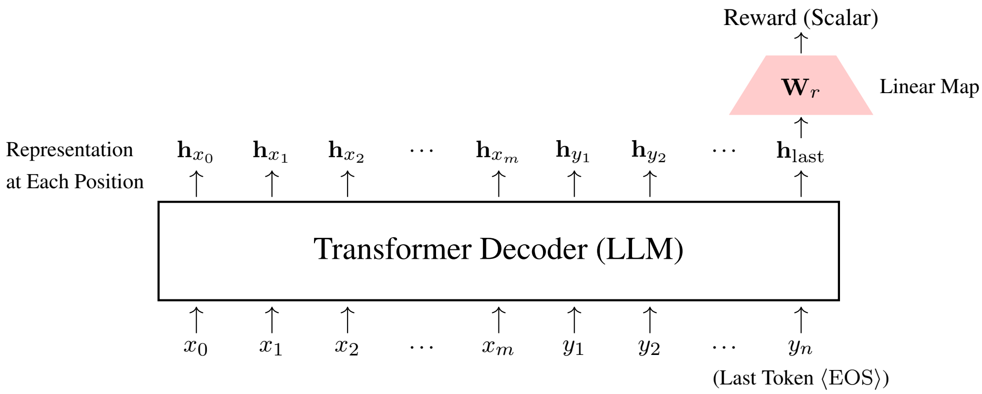


Figure 10.8: Architecture of the reward model based on Transformer. The main component of this model is still an LLM. We use the Transformer decoder as the sequence representation model. We extract the representation of the last position of the decoder as the representation of the entire sequence $[\mathbf{x}, \mathbf{y}]$. We then map this representation to a scalar through a linear transformation, which serves as the reward score for $[\mathbf{x}, \mathbf{y}]$.

sequences to a scalar. Given an input $\mathbf{x}$ and an output $\mathbf{y}$, the reward can be expressed as

$$r = \text{Reward}(\mathbf{x}, \mathbf{y}) \tag{10.33}$$

where $\text{Reward}(\cdot)$ is the reward model. r can be interpreted as a measure of how well the output $\mathbf{y}$ aligns with the desired behavior given the input $\mathbf{x}$. As discussed in the previous subsection, both $\mathbf{x}$ and $\mathbf{y}$ are assumed to complete texts. This means that the reward model evaluates the relationship between inputs and outputs that provide full semantic content. For example, when applying the reward model, it assigns a value of 0 (or another predetermined value) at each position t in the output sequence $\mathbf{y} = y_1...y_n$. Only at the final position, when $t = n$, does the reward model generate the actual reward score. To keep the notation uncluttered, we will use $r(\mathbf{x}, \mathbf{y})$ to denote the reward model $\text{Reward}(\mathbf{x}, \mathbf{y})$ from here on.

There are many ways to implement the reward model. One simple approach is to build the reward model based on a pre-trained LLM. More specifically, we can concatenate $\mathbf{x}$ and $\mathbf{y}$ to form a single token sequence $\text{seq}_{\mathbf{x},\mathbf{y}} = [\mathbf{x}, \mathbf{y}]$. We run a pre-trained LLM on this sequence, as usual, and at each position, we obtain a representation from the top-most Transformer layer. Then, we take the representation at the last position (denoted by $\mathbf{h}_{\text{last}}$) and map it to a scalar via linear transformation:

$$r(\mathbf{x}, \mathbf{y}) = \mathbf{h}_{\text{last}} \mathbf{W}_r \tag{10.34}$$

where $\mathbf{h}_{\text{last}}$ is a d-dimensional vector, and $\mathbf{W}r$ is a $d \times 1$ linear mapping matrix. This architecture of the reward model is illustrated in Figure 10.8.

To train the reward model, the first step is to collect human feedback on a set of generated outputs. Given an input $\mathbf{x}$, we use the LLM to produce multiple candidate outputs $\{\mathbf{y}_1, ..., \mathbf{y}_N\}$.

Human feedback can be obtained in several ways:

- **Pairwise Comparison** (**Pairwise Ranking**). Given two different outputs, human experts select which one is better.
- **Rating**. Human experts provide a score or rating to each output. This score is often a continuous or discrete numerical value, such as a score on a scale (e.g., 1-5 stars, or 1-10 points). In some cases, the rating might be binary, indicating a "yes/no" or "positive/negative" preference.
- **Listwise Ranking**. Human experts are asked to rank or order the given set of possible outputs.

Here we consider pairwise comparison feedback as it is one of the simplest and most common forms of human feedback used in RLHF. In this setting, each time, two outputs $(\mathbf{y}_a, \mathbf{y}_b)$ are randomly drawn from the candidate pool $\{\mathbf{y}_1, ..., \mathbf{y}_N\}$. Human experts are then presented with these pairs and asked to decide which output they prefer based on specific criteria, such as clarity, relevance, and accuracy. The human feedback can be encoded as a binary label, $\mathbf{y}_a \succ \mathbf{y}_b$ for a preference for $\mathbf{y}_a$, and $\mathbf{y}_b \succ \mathbf{y}_a$ for a preference for $\mathbf{y}_b$.

One simple and widely used model for describing such pairwise comparisons is the **Bradley-Terry model** [Bradley and Terry, 1952]. It is a probabilistic model that estimates the probability that one item is preferred over another. Adapting this model to the notation used here, we can write the probability that $\mathbf{y}_a$ is preferred over $\mathbf{y}_b$ in the form

$$\begin{aligned} \Pr(\mathbf{y}_a \succ \mathbf{y}_b | \mathbf{x}) &= \frac{e^{r(\mathbf{x}, \mathbf{y}_a)}}{e^{r(\mathbf{x}, \mathbf{y}_a)} + e^{r(\mathbf{x}, \mathbf{y}_b)}} \\ &= \frac{e^{r(\mathbf{x}, \mathbf{y}_a) - r(\mathbf{x}, \mathbf{y}_b)}}{e^{r(\mathbf{x}, \mathbf{y}_a) - r(\mathbf{x}, \mathbf{y}_b)} + 1} \\ &= \mathrm{Sigmoid}(r(\mathbf{x}, \mathbf{y}_a) - r(\mathbf{x}, \mathbf{y}_b)) \end{aligned} \tag{10.35}$$

When training the reward model, we want to maximize this preference probability. A loss function based on the Bradley-Terry model is given by

$$\mathcal{L}_r(\phi) = -\mathbb{E}_{(\mathbf{x}, \mathbf{y}_a, \mathbf{y}_b) \sim \mathcal{D}_r} \big[\log \Pr_\phi(\mathbf{y}_a \succ \mathbf{y}_b | \mathbf{x}) \big] \tag{10.36}$$

where $(\mathbf{x}, \mathbf{y}_a, \mathbf{y}_b)$ is drawn from a human-annotated dataset $\mathcal{D}_r$ consisting of preference pairs of outputs and their corresponding inputs. ϕ represents the parameters of the reward model, which includes both the parameters of the Transformer decoder and the linear mapping matrix $\mathbf{W}_r$. In practice, assuming $(\mathbf{x}, \mathbf{y}_a, \mathbf{y}_b)$ is uniformly sampled from $\mathcal{D}_r$, we can replace the expectation with a summation

$$\mathcal{L}_r(\phi) = -\frac{1}{|\mathcal{D}_r|} \sum_{(\mathbf{x}, \mathbf{y}_a, \mathbf{y}_b) \in \mathcal{D}_r} \log \Pr_\phi(\mathbf{y}_a \succ \mathbf{y}_b | \mathbf{x}) \tag{10.37}$$

The goal of training the reward model is to find the optimal parameters $\hat{\phi}$ that minimize this loss function, given by

$$\hat{\phi} = \arg\min_{\phi} \mathcal{L}_r(\phi) \tag{10.38}$$

Since the reward model itself is also an LLM, we can directly reuse the Transformer training procedure to optimize the reward model. The difference from training a standard LLM is that we only need to replace the cross-entropy loss with the pairwise comparison loss as described in Eq. (10.37). After the training of the reward model, we can apply the trained reward model $r_{\hat{\phi}}(\cdot)$ to supervise the target LLM for alignment.

It is worth noting that although we train the reward model to perform pairwise ranking, we apply it to score each input-output pair independently during the alignment process. The pairwise ranking objective ensures that the reward model is sensitive to subtle differences between outputs, but we rely on the continuous scores produced by the reward model to guide the optimization of the LLM. An advantage of this approach is that we can choose from or combine various ranking loss functions, and still apply the resulting reward models in the same way as we have done in this subsection. This consistency ensures a unified framework for aligning the LLM, regardless of the specific ranking loss used during reward model training.

10.3.3 Training LLMs

Having obtained the reward model, we then train the policy (i.e., the LLM) via the A2C method. Recall from Section 10.3.1 that a state-action sequence or trajectory τ can be evaluated by the utility function

$$U(\tau;\theta) = \sum_{t=1}^{T} \log \pi_\theta(a_t|s_t) A(s_t, a_t) \tag{10.39}$$

where $A(s_t, a_t)$ is the advantage of taking the action a_t given the state s_t. An estimate of $A(s_t, a_t)$ is defined as the TD error $r_t + \gamma V(s_{t+1}) - V(s_t)$, where the value function $V(s_t)$ is trained with the reward model.

Given this utility function, the A2C-based loss function can be written in the form

$$\begin{aligned} \mathcal{L}(\theta) &= -\mathbb{E}_{\tau\sim\mathcal{D}}\big[U(\tau;\theta)\big] \\ &= -\mathbb{E}_{\tau\sim\mathcal{D}}\big[\sum_{t=1}^{T} \log \pi_\theta(a_t|s_t) A(s_t, a_t)\big] \end{aligned} \tag{10.40}$$

where $\mathcal{D}$ is a space of state-action sequences. As usual, the goal of training the policy is to minimize this loss function

$$\tilde{\theta} = \arg\min_{\theta} \mathcal{L}(\theta) \tag{10.41}$$

If we map the problem back to the language modeling problem and adopt the notation from LLMs, the loss function can be written as:

$$\mathcal{L}(\theta) = -\mathbb{E}_{(\mathbf{x},\mathbf{y})\sim\mathcal{D}}\big[U(\mathbf{x},\mathbf{y};\theta)\big] \tag{10.42}$$

where

$$U(\mathbf{x},\mathbf{y};\theta) = \sum_{t=1}^{T}\log\pi_\theta(y_t|\mathbf{x},\mathbf{y}_{<t})A(\mathbf{x},\mathbf{y}_{<t},y_t) \tag{10.43}$$

Here $\pi_\theta(y_t|\mathbf{x},\mathbf{y}_{<t}) = \Pr_\theta(y_t|\mathbf{x},\mathbf{y}_{<t})$ is the LLM parameterized by θ.

In general, we do not have a human annotated input-output dataset $\mathcal{D}$ in RLHF, but a dataset containing inputs only. The outputs, in this case, are typically the predictions made by the LLM. The loss function is then defined as

$$\mathcal{L}(\theta) = -\mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\big[U(\mathbf{x},\mathbf{y};\theta)\big] \tag{10.44}$$

where $\mathcal{D}$ denotes the input-only dataset, and $\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})$ denotes that the output $\mathbf{y}$ is sampled by the policy $\pi_\theta(\cdot|\mathbf{x})$.

The above formulation provides a basic form of the A2C method for LLMs. Improved versions of this model are more commonly used in RLHF. In the following discussion, we will still use the reinforcement learning notation to simplify the presentation and will get back the language modeling notation later.

One common improvement of policy gradient methods is to use **importance sampling** to refine the estimation of $U(\tau;\theta)$. This can be written as

$$U(\tau;\theta) = \sum_{t=1}^{T}\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{ref}}}(a_t|s_t)}A(s_t,a_t) \tag{10.45}$$

Here we replace the log-probability $\log\pi_\theta(a_t|s_t)$ with the ratio $\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{ref}}}(a_t|s_t)}$. θ_{ref} denotes the parameters of the previous policy (such as an initial model from which we start the training). So $\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{ref}}}(a_t|s_t)}$, also called the **ratio function**, can be interpreted as the log-probability ratio between the current policy π_θ and the previous policy $\pi_{\theta_{\mathrm{ref}}}$ (call it the reference policy). By using the ratio function we reweight the observed rewards based on the likelihood of the actions under the current policy versus the reference policy. When $\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{ref}}}(a_t|s_t)} > 1$, the action a_t is more favored by the current policy compared to the reference policy. By contrast, when $\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{ref}}}(a_t|s_t)} < 1$, the action a_t is less favored by the current policy[4].

A problem with the model presented in Eq. (10.47) (as well as in Eq. (10.39)) is that the variance in the gradient estimates is often high, making the learning process unstable. To

[4]Consider a more general case where we wish to evaluate the policy using its expected reward (also see Eq. (10.18))

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\Big[R(\tau)\Big] \tag{10.46}$$

Here $\tau\sim\pi_\theta$ means that the sequence τ is generated by the policy π_θ. Alternatively, we can write $J(\theta)$ in another mitigate this issue, techniques such as clipping are often employed to bound the importance weights and prevent large updates. A clipped version of the utility function (also called the clipped surrogate objective function) is given by

$$U_{\text{clip}}(\tau;\theta) = \sum_{t=1}^{T} \text{Clip}\Big(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{ref}}}(a_t|s_t)}\Big) A(s_t,a_t) \tag{10.49}$$

$$\text{Clip}\Big(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{ref}}}(a_t|s_t)}\Big) = \min\Big(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{ref}}}(a_t|s_t)}, \text{bound}\big(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{ref}}}(a_t|s_t)}, 1-\epsilon, 1+\epsilon\big)\Big) \tag{10.50}$$

Here the function $\text{bound}(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{ref}}}(a_t|s_t)}, 1-\epsilon, 1+\epsilon)$ constrains the ratio function to the range $[1-\epsilon, 1+\epsilon]$.

A further improvement to the above model is to consider **trust regions** in optimization [Schulman et al., 2015]. In reinforcement learning, a large update to the policy can lead to instability, where the agent may start performing worse after an update. A reasonable idea is to optimize the model in the trust region, which refers to a region around the current parameter estimate where the model is well-behaved. One approach to incorporating trust regions is to impose a constraint on the size of the policy update, ensuring that the current policy does not deviate too significantly from the reference policy. This can be achieved by adding a penalty based on some form of divergence between the current and reference policies to the objective function. A simple form of such a penalty is given by the difference in the log-probability of the sequence τ under the current policy versus the reference policy:

$$\text{Penalty} = \log \pi_\theta(\tau) - \log \pi_{\theta_{\text{ref}}}(\tau) \tag{10.51}$$

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form

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_{\theta_{\text{ref}}}}\Big[\frac{\text{Pr}_\theta(\tau)}{\text{Pr}_{\theta_{\text{ref}}}(\tau)} R(\tau)\Big] \tag{10.47}$$

It is not difficult to find that the right-hand sides of these equations are essentially the same since $\mathbb{E}_{\tau\sim\pi_{\theta_{\text{ref}}}}\Big[\frac{\text{Pr}_\theta(\tau)}{\text{Pr}_{\theta_{\text{ref}}}(\tau)} R(\tau)\Big] = \sum_\tau \text{Pr}_{\theta_{\text{ref}}}(\tau) \frac{\text{Pr}_\theta(\tau)}{\text{Pr}_{\theta_{\text{ref}}}(\tau)} R(\tau) = \sum_\tau \text{Pr}_\theta(\tau) R(\tau) = \mathbb{E}_{\tau\sim\pi_\theta}\Big[R(\tau)\Big]$
Note that this equivalence holds only when the expectation is performed over the entire sequence space. In practice, however, we often only sample a relatively small number of sequences using a policy in policy learning. As a result, the sampling method itself matters. Eq. (10.47) offers an interesting manner to separate the sampling and reward computation processes: we first use a baseline policy (with θ_{ref}) to sample a number of sequences, and then use the target policy (with θ) to compute the expected reward. In this way, we separate the policy used for collecting the data, and the policy used for computing the gradient. This approach avoids the need to directly sample from the policy we are evaluating, which can be beneficial in cases where generating sequences from the target policy is expensive or difficult. In reinforcement learning, $\mathbb{E}_{\tau\sim\pi_{\theta_{\text{ref}}}}\Big[\frac{\text{Pr}_\theta(\tau)}{\text{Pr}_{\theta_{\text{ref}}}(\tau)} R(\tau)\Big]$ is often called a **surrogate objective**.

Eq. (10.47) can also be interpreted from a policy gradient perspective. For $\mathbb{E}_{\tau\sim\pi_{\theta_{\text{ref}}}}\Big[\frac{\text{Pr}_\theta(\tau)}{\text{Pr}_{\theta_{\text{ref}}}(\tau)} R(\tau)\Big]$, the gradient at $\theta = \theta_{\text{ref}}$ is given by

$$\frac{\partial}{\partial\theta} \mathbb{E}_{\tau\sim\pi_{\theta_{\text{ref}}}}\Big[\frac{\text{Pr}_\theta(\tau)}{\text{Pr}_{\theta_{\text{ref}}}(\tau)} R(\tau)\Big]\Big|_{\theta=\theta_{\text{ref}}} = \mathbb{E}_{\tau\sim\pi_{\theta_{\text{ref}}}}\Big[\frac{\partial \text{Pr}_\theta(\tau)|_{\theta=\theta_{\text{ref}}}}{\partial\theta} R(\tau)\Big] \tag{10.48}$$

The right-hand side is a standard form used in policy gradient methods, meaning that we compute the direction of the parameter update at the point $\theta = \theta_{\text{ref}}$ on the optimization surface.

In practice, this penalty can be approximated by considering only the policy probabilities and ignoring the dynamics. This gives

$$\text{Penalty} = \sum_{t=1}^{T} \log \pi_{\theta}(a_t|s_t) - \sum_{t=1}^{T} \log \pi_{\theta_{\text{ref}}}(a_t|s_t) \tag{10.52}$$

By including this penalty in the optimization objective, we encourage the current policy to remain close to the reference policy, limiting very large updates that could destabilize learning.

We can incorporate this penalty into the clipped surrogate objective function, and obtain

$$U_{\text{ppo-clip}}(\tau;\theta) = U_{\text{clip}}(\tau;\theta) - \beta \text{Penalty} \tag{10.53}$$

where β is the weight of the penalty. This training method is called **proximal policy optimization** (**PPO**), which is one of the most popular reinforcement learning methods used in LLMs and many other fields [Schulman et al., 2017].

Now we can write the objective of training LLMs in the form of PPO.

$$U(\mathbf{x},\mathbf{y};\theta) = U_{\text{ppo-clip}}(\mathbf{x},\mathbf{y};\theta) - \beta \text{Penalty} \tag{10.54}$$

where

$$U_{\text{ppo-clip}}(\mathbf{x},\mathbf{y};\theta) = \sum_{t=1}^{T} \text{Clip}\Big(\frac{\pi_{\theta}(y_t|\mathbf{x},\mathbf{y}_{<t})}{\pi_{\theta_{\text{ref}}}(y_t|\mathbf{x},\mathbf{y}_{<t})}\Big) A(\mathbf{x},\mathbf{y}_{<t},y_t) \tag{10.55}$$

$$\begin{aligned} \text{Penalty} &= \log \Pr_{\theta}(\mathbf{y}|\mathbf{x}) - \log \Pr_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x}) \\ &= \sum_{t=1}^{T} \log \Pr_{\theta}(y_t|\mathbf{x},\mathbf{y}_{<t}) - \sum_{t=1}^{T} \log \Pr_{\theta_{\text{ref}}}(y_t|\mathbf{x},\mathbf{y}_{<t}) \end{aligned} \tag{10.56}$$

Although the notation here appears a bit tedious, the idea of PPO is simple: we develop an objective by combining the clipped likelihood ratio of the target and reference policies with an advantage function, and then impose a penalty that ensures policy updates are not too large. The PPO-based RLHF is illustrated in Figure 10.9.

To summarize, implementing RLHF requires building four models, all based on the Transformer decoder architecture.

- **Reward Model** ($r_{\phi}(\cdot)$ where ϕ denotes the parameters). The reward model learns from human preference data to predict the reward for each pair of input and output token sequences. It is a Transformer decoder followed by a linear layer that maps a sequence (the concatenation of the input and output) to a real-valued reward score.
- **Value Model** or **Value Function** ($V_{\omega}(\cdot)$ where ω denotes the parameters). The value function receives reward scores from the reward model and is trained to predict the expected sum of rewards that can be obtained starting from a state. It is generally based on the same architecture as the reward model.

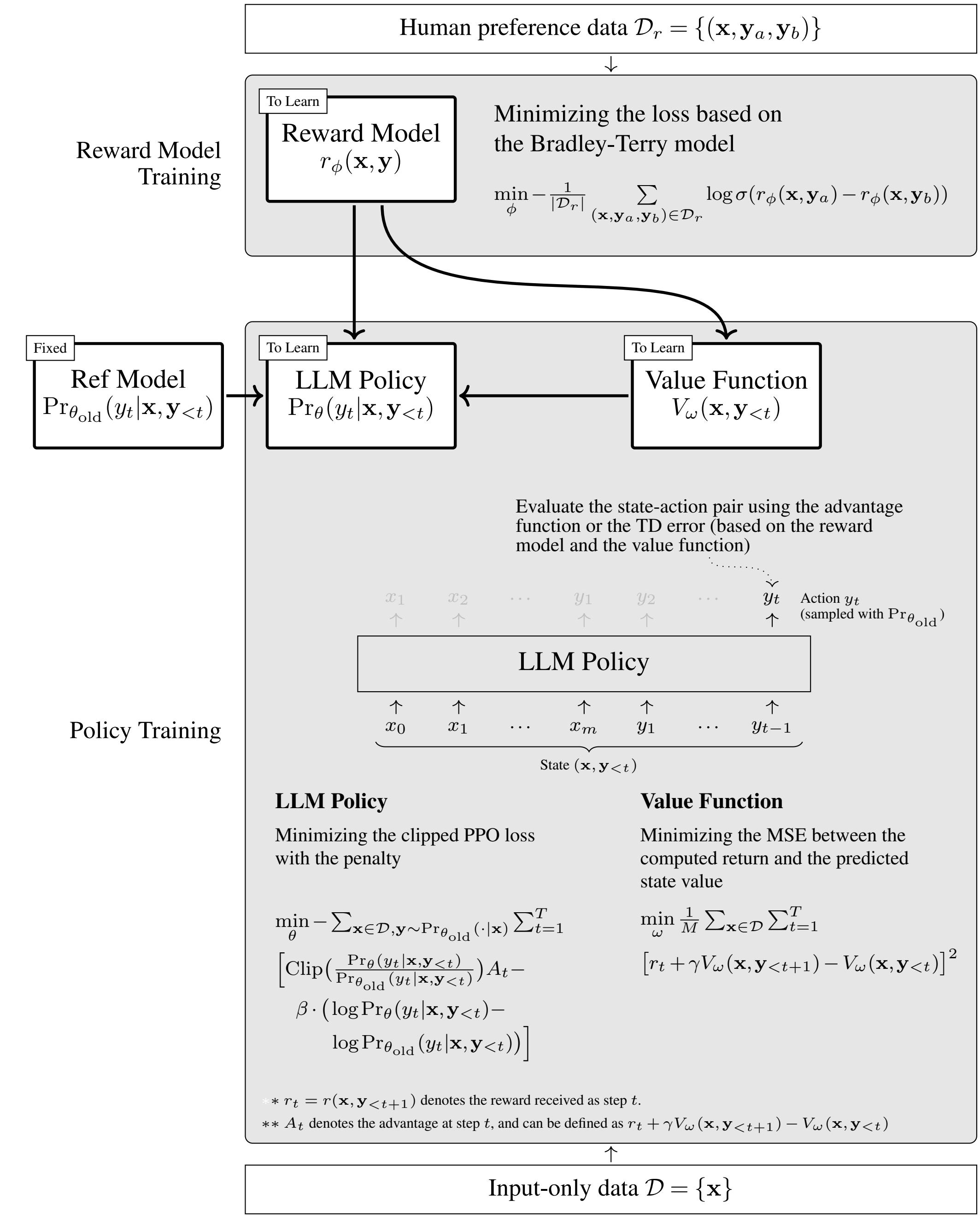


Figure 10.9: Illustration of RLHF. The first step is to collect human preference data and train the reward model using this data. Once the reward model is optimized, along with the reference model, we proceed to train both the policy and the value function. At each prediction step, we compute the sum of the PPO-based loss and update the parameters of the policy. This requires access to the reward model, the reference model, and the value function at hand. At the same time, we update the parameters of the value function by minimizing the MSE loss.

- **Reference Model** ($\pi_{\theta_{\mathrm{ref}}}(\cdot) = \mathrm{Pr}_{\theta_{\mathrm{ref}}}(\cdot)$ where θ_{ref} denotes the parameters). The reference model is the baseline LLM that serves as a starting point for policy training. In RLHF, it represents the previous version of the model or a model trained without human feedback. It is used to perform sampling over the space of outputs and contribute to the loss computation for policy training.
- **Target Model** or **Policy** ($\pi_{\theta}(\cdot) = \mathrm{Pr}_{\theta}(\cdot)$ where θ denotes the parameters). This policy governs how the LLM decides the most appropriate next token given its context. It is trained under the supervision of both the reward model and the value model.

In practice, these models need to be trained in a certain order. First, we need to initialize them using some other models. For example, the reward model and the value model can be initialized with a pre-trained LLM, while the reference model and the target model can be initialized with a model that has been instruction fine-tuned. Note that, at this point, the reference model is ready for use and will not be further updated. Second, we need to collect human preference data and train the reward model on this data. Third, both the value model and the policy are trained simultaneously using the reward model. At each position in an output token sequence, we update the value model by minimizing the MSE error of value prediction, and the policy is updated by minimizing the PPO loss.

10.4 Improved Human Preference Alignment

In the previous section, we reviewed the basic concepts of reinforcement learning and the general framework of RLHF. In this section, we will discuss some refinements of RLHF and alternative methods to achieve human preference alignment.

10.4.1 Better Reward Modeling

In Section 10.3.2, we highlighted the task of learning from human preferences as well as the use of pairwise ranking loss for training reward models. Here we consider more methods for reward modeling. Our discussion will be relatively general, and since the reward model is widely used in many reinforcement learning problems, it will be easy for us to apply the methods discussed here to RLHF and related applications.

1. Supervision Signals

The training of reward models can broadly be seen as a ranking problem, where the model learns to assign scores to outputs so that their order reflects the preferences indicated by humans. There are several methods to train a reward model from the perspective of ranking.

One approach is to extend pairwise ranking to listwise ranking. For each sample in a dataset, we can use the LLM to generate multiple outputs, and ask human experts to order these outputs. For example, given a set of four outputs $\{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4\}$, one possible order of them can be $\mathbf{y}_2 \succ \mathbf{y}_3 \succ \mathbf{y}_1 \succ \mathbf{y}_4$. A very simple method to model the ordering of the list is to accumulate the pairwise comparison loss. For example, we can define the listwise loss by accumulating the loss over all pairs of outputs:

$$\mathcal{L}_{\text{list}} = -\mathbb{E}_{(\mathbf{x},Y)\sim\mathcal{D}_r}\Big[\frac{1}{N(N-1)}\sum_{\substack{\mathbf{y}_a\in Y,\mathbf{y}_b\in Y\\ \mathbf{y}_a\neq\mathbf{y}_b}}\log\Pr(\mathbf{y}_a\succ\mathbf{y}_b|\mathbf{x})\Big] \tag{10.57}$$

where Y is a list of outputs, and N is the number of outputs in the list. $\Pr(\mathbf{y}_a\succ\mathbf{y}_b|\mathbf{x})$ can be defined using the Bradley-Terry model, that is, $\Pr(\mathbf{y}_a\succ\mathbf{y}_b|\mathbf{x}) = \text{Sigmoid}(r(\mathbf{x},\mathbf{y}_a)-r(\mathbf{x},\mathbf{y}_b))$. Here we omit the ϕ superscript on the $\Pr(\cdot)$ to keep the notation uncluttered.

An extension to the Bradley-Terry model for listwise ranking could involve a ranking mechanism that takes into account the entire list of outputs rather than just pairwise comparisons. One such model is the **Plackett-Luce model**, which generalizes the Bradley-Terry model to handle multiple items in a ranking [Plackett, 1975]. In the Plackett-Luce model, for each item in a list, we define a "worth" for this item that reflects its relative strength of being chosen over other items. For the reward modeling problem here, the worth of $\mathbf{y}$ in the list Y can be defined as

$$\alpha(\mathbf{y}) = \exp(r(\mathbf{x},\mathbf{y})) \tag{10.58}$$

Then the probability of selecting $\mathbf{y}$ from Y is given by

$$\begin{aligned}\Pr(\mathbf{y}\text{ is selected}|\mathbf{x},Y) &= \frac{\alpha(\mathbf{y})}{\sum_{\mathbf{y}'\in Y}\alpha(\mathbf{y}')}\\ &= \frac{\exp(r(\mathbf{x},\mathbf{y}))}{\sum_{\mathbf{y}'\in Y}\exp(r(\mathbf{x},\mathbf{y}'))}\end{aligned} \tag{10.59}$$

Suppose $\mathring{Y}$ is an ordered list $\mathbf{y}_{j_1}\succ\mathbf{y}_{j_2}\succ\cdots\succ\mathbf{y}_{j_N}$. The overall log-probability of this ordered list can be defined as the sum of the conditional log-probabilities at each stage of selection, given by

$$\begin{aligned}\log\Pr(\mathring{Y}|\mathbf{x}) &= \log\Pr(\mathbf{y}_{j_1}\succ\mathbf{y}_{j_2}\succ\cdots\succ\mathbf{y}_{j_N}|\mathbf{x})\\ &= \log\Pr(\mathbf{y}_{j_1}|\mathbf{x},\{\mathbf{y}_{j_1},\mathbf{y}_{j_2},...,\mathbf{y}_{j_N}\})+\\ &\quad\ \log\Pr(\mathbf{y}_{j_2}|\mathbf{x},\{\mathbf{y}_{j_2},...,\mathbf{y}_{j_N}\})+\\ &\quad\ \cdots+\\ &\quad\ \log\Pr(\mathbf{y}_{j_N}|\mathbf{x},\{\mathbf{y}_{j_N}\})\\ &= \sum_{k=1}^{N}\log\Pr(\mathbf{y}_{j_k}|\mathbf{x},\mathring{Y}_{\geq k})\end{aligned} \tag{10.60}$$

where $\mathring{Y}_{\geq k}$ represents the subset of the list of outputs that remain unselected at the k-th stage, i.e., $\mathring{Y}_{\geq k} = \{\mathbf{y}_{j_k},...,\mathbf{y}_{j_N}\}$. Given the log-probability $\log\Pr(\mathring{Y}|\mathbf{x})$, we can define the loss function based on the Plackett-Luce model by

$$\mathcal{L}_{\text{pl}} = -\mathbb{E}_{(\mathbf{x},\mathring{Y})\sim\mathcal{D}_r}\big[\log\Pr(\mathring{Y}|\mathbf{x})\big] \tag{10.61}$$

There are also many other pairwise and listwise methods for modeling rankings, such as RankNet [Burges et al., 2005] and ListNet [Cao et al., 2007]. All these methods can be categorized into a large family of learning-to-rank approaches, and most of them are applicable to the problem of modeling human preferences. However, discussing these methods is beyond the scope of this chapter. Interested readers can refer to books on this topic for more details [Liu, 2009; Li, 2011].

In addition to pairwise and listwise ranking, using pointwise methods to train reward models offers an alternative way to capture human preferences. Unlike methods that focus on the relative rankings between different outputs, pointwise methods treat each output independently. For example, human experts might assign a score to an individual output, such as a rating on a five-point scale. The objective is to adjust the reward model so that its outputs align with these scores. A simple way to achieve pointwise training is through regression techniques where the reward of each output is treated as a target variable. Let $\varphi(\mathbf{x},\mathbf{y})$ be the score assigned to $\mathbf{y}$ given $\mathbf{x}$ by humans. Pointwise reward models can be trained by minimizing a loss function, often based on mean squared error or other regression losses, between the predicted reward $r(\mathbf{x},\mathbf{y})$ and the actual human feedback $\varphi(\mathbf{x},\mathbf{y})$. For example, the loss function could be

$$\mathcal{L}_{\text{point}} = -\mathbb{E}\big[\varphi(\mathbf{x},\mathbf{y}) - r(\mathbf{x},\mathbf{y})\big]^2 \tag{10.62}$$

While pointwise methods are conceptually simpler and can directly guide the reward model to predict scores, they might not always be the best choice in RLHF. A problem is that these methods may struggle with high variance in human feedback, especially when different experts provide inconsistent scores for similar outputs. Because they focus on fitting to absolute scores rather than relative differences, inconsistencies in scoring can lead to poor model performance. Moreover, fitting to specific scored outputs might discourage generalization, particularly given that training data is often very limited in RLHF. In contrast, methods that consider relative preferences can promote the learning of more generalized patterns of success and failure. Nevertheless, there are scenarios where pointwise methods might still be suitable. For example, in tasks where training data is abundant and the costs of obtaining accurate, consistent annotations are low, pointwise methods can prove effective.

In fact, to make the supervision signal for training the reward model more robust, we can also introduce additional regularization terms into training. For example, if we consider the first term $U_{\text{ppo-clip}}(\mathbf{x},\mathbf{y};\theta)$ in Eq. (10.54) as a type of generalized reward, then the second term (i.e., the penalty term) can be viewed as a form of regularization for the reward model, except that here the goal is to train the policy rather than the reward model. Another example is that Eisenstein et al. [2023] develop a regularization term based on the squared sum of rewards, and add it to the pairwise comparison loss in RLHF:

$$\begin{aligned}
\mathcal{L}_{\text{reg}} &= \mathcal{L}_{\text{pair}} + \big(-\mathbb{E}_{(\mathbf{x},\mathbf{y}_a,\mathbf{y}_b)\sim\mathcal{D}_r}\big[r(\mathbf{x},\mathbf{y}_a) + r(\mathbf{x},\mathbf{y}_b)\big]^2\big) \\
&= -\mathbb{E}_{(\mathbf{x},\mathbf{y}_a,\mathbf{y}_b)\sim\mathcal{D}_r}\big[\log \Pr_\phi(\mathbf{y}_a \succ \mathbf{y}_b|\mathbf{x})\big] \\
&\quad -\mathbb{E}_{(\mathbf{x},\mathbf{y}_a,\mathbf{y}_b)\sim\mathcal{D}_r}\big[r(\mathbf{x},\mathbf{y}_a) + r(\mathbf{x},\mathbf{y}_b)\big]^2
\end{aligned} \tag{10.63}$$

Optimizing with this regularization term can help mitigate the underdetermination of reward models[5].

2. Sparse Rewards vs. Dense Rewards

As discussed in Section 10.3, the rewards in RLHF are very sparse: they are observed only at the end of sequences, rather than continuously throughout the generation process. Dealing with sparse rewards has long been a concern in reinforcement learning, and has been one of the challenges in many practical applications. For example, in robotics, it often needs to shape the reward function to ease optimization rather than relying solely on end-of-sequence rewards. Various methods have been developed to address this issue. One common approach is reward shaping, where the original function is modified to include intermediate rewards, thereby providing more immediate feedback. Also, one can adopt curriculum learning to sequentially structure tasks in a way that the complexity gradually increases. This can help models to master simpler tasks first, which prepares them for more complex challenges as their skills develop. There are many such methods that can mitigate the impact of sparse rewards, such as Monte Carlo methods and intrinsic motivation. Most of these methods are general and the discussion of them can be found in the broader literature on reinforcement learning, such as Sutton and Barto [2018]'s book.

Although we do not discuss methods for mitigating sparse rewards in detail here, an interesting question arises: why are sparse rewards so successful in RLHF? Recall from Section 10.3.1 that the supervision signal received at each time step t is not the reward for the current action, but rather some form of the accumulated rewards from t until the last time step. Such supervision signals are dense over the sequence, because the reward obtained at the end of the sequence can be transferred back to that time step, regardless of which time step it is. In other words, the sparse rewards are transformed into the dense supervision signals. Furthermore, from the perspective of reward shaping, Ng et al. [1999] show that the reward at t can be defined as

$$r'(s_t, a_t, s_{t+1}) = r(s_t, a_t, s_{t+1}) + f(s_t, a_t, s_{t+1}) \tag{10.64}$$

where $r'(\cdot)$ is the transformed reward function, $r(\cdot)$ is the original reward function, and $f(\cdot)$ is the shaping reward function. To ensure the optimality of the policy under the transformed reward function, the shaping reward function can be given in the form

$$f(s_t, a_t, s_{t+1}) = \gamma\Phi(s_{t+1}) - \Phi(s_t) \tag{10.65}$$

where $\Phi(s)$ is called the potential value of the state s. If we define $\Phi(s)$ as the common value function as in Eq. (10.15) and substitute Eq. (10.65) into Eq. (10.64), we obtain

$$r'(s_t, a_t, s_{t+1}) = r(s_t, a_t, s_{t+1}) + \gamma V(s_{t+1}) - V(s_t) \tag{10.66}$$

[5] A model is called underdetermined if there are multiple alternative sets of parameters that can achieve the same objective.

It is interesting to see that this function is exactly the same as the advantage function used in PPO. This relates advantage-based methods to reward shaping: the advantage is essentially a shaped reward.

On the other hand, one of the reasons for adopting end-of-sequence rewards lies in the nature of the RLHF tasks. Unlike traditional reinforcement learning environments where the agent interacts with a dynamic environment, RLHF tasks often involve complex decision-making based on linguistic or other high-level cognitive processes. These processes do not lend themselves easily to frequent and meaningful intermediate rewards because the quality and appropriateness of the actions can only be fully evaluated after observing their impact in the larger context of the entire sequence or task. In this case, the reward signals based on human feedback, though very sparse, are typically very informative and accurate. Consequently, this sparsity, together with the high informativeness and accuracy of the human feedback, can make the learning both robust and efficient.

3. Fine-grained Rewards

For many applications, our objective will be more complex than merely evaluating an entire text. For example, in sentiment analysis, we often do not just determine the sentiment of a text, but need to analyze the sentiment in more detail by associating it with specific aspects of a topic discussed in the text. Consider the sentence "The camera of the phone is excellent, but the battery life is disappointing." In this example, we would need to separately analyze the sentiments expressed about the camera and the battery. Such analysis, known as aspect-based sentiment analysis, helps provide a finer-grained understanding of the customer review compared to general sentiment analysis.

For the problem of reward modeling, we often need to model different parts of a sequence as well. A simple and straightforward way to do this is to divide a sequence into different segments and then compute the reward for each segment [Wu et al., 2023b]. Suppose that an output token sequence $\mathbf{y}$ can be divided into n_s segments $\{\bar{\mathbf{y}}_1, ..., \bar{\mathbf{y}}_{n_s}\}$ by some criterion. We can use the reward model to evaluate each of these segments. By taking $\mathbf{x}$, $\mathbf{y}$ and $\bar{\mathbf{y}}_k$ as input to the reward model, the reward score for the k-th segment is given by

$$r^k = r(\mathbf{x}, \mathbf{y}, \bar{\mathbf{y}}_k) \tag{10.67}$$

Then the reward score for the entire output sequence is given by

$$r(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^{n_s} r(\mathbf{x}, \mathbf{y}, \bar{\mathbf{y}}_k) \tag{10.68}$$

Here $r(\mathbf{x}, \mathbf{y})$ can be used to train the policy as usual.

A problem with this model is that training reward models at the segment level is not as straightforward as learning from human preferences on entire texts, as it is difficult to obtain segment-level human preference data. For rating-like problems (e.g., we rate a segment according to its level of misinformation), one simple approach is to assign a rating score to each segment and train the reward model using pointwise methods. For example, we can use a strong LLM to rate the sequences $\bar{\mathbf{y}}_1...\bar{\mathbf{y}}_{k-1}$ and $\bar{\mathbf{y}}_1...\bar{\mathbf{y}}_k$, and obtain the scores $s(\bar{\mathbf{y}}_1...\bar{\mathbf{y}}_{k-1})$ and $s(\bar{\mathbf{y}}_1...\bar{\mathbf{y}}_k)$. We can then define the score of the segment $\bar{\mathbf{y}}_k$ as the difference between $s(\bar{\mathbf{y}}_1...\bar{\mathbf{y}}_k)$ and $s(\bar{\mathbf{y}}_1...\bar{\mathbf{y}}_{k-1})$

$$s(\bar{\mathbf{y}}_k) = s(\bar{\mathbf{y}}_1...\bar{\mathbf{y}}_k) - s(\bar{\mathbf{y}}_1...\bar{\mathbf{y}}_{k-1}) \tag{10.69}$$

Using these segment-level scores, we can train the reward model with a regression loss function

$$\mathcal{L}_{\text{rating}} = -\mathbb{E}_{\bar{\mathbf{y}}_k}\left[s(\bar{\mathbf{y}}_k) - r(\mathbf{x},\mathbf{y},\bar{\mathbf{y}}_k)\right]^2 \tag{10.70}$$

Sometimes, alignment can be treated as a classification problem, for example, we assess whether a segment has ethical issues. In this case, the segment can be labeled as ethical or unethical, either by humans or using additional classifiers. Given the label of the segment, we can train the reward model using some classification loss function. For example, suppose that $r(\mathbf{x},\mathbf{y},\bar{\mathbf{y}}_k) = 1$ if the segment is classified as unethical, and $r(\mathbf{x},\mathbf{y},\bar{\mathbf{y}}_k) = -1$ otherwise[6]. The hinge loss of training binary classification models is given by

$$\mathcal{L}_{\text{hinge}} = \max(0, 1 - r(\mathbf{x},\mathbf{y},\bar{\mathbf{y}}_k) \cdot \hat{r}) \tag{10.71}$$

where $\hat{r} \in \{1, -1\}$ denotes the ground truth label.

The remaining issue here is how to split $\mathbf{y}$ into segments. One approach is to define a fixed-length segmentation, where $\mathbf{y}$ is divided into equal-length chunks. However, this may not always be ideal, as the content of the sequence may not align well with fixed boundaries. An alternative approach is to segment $\mathbf{y}$ based on specific linguistic or semantic cues, such as sentence boundaries, topic shifts, or other meaningful structures in the text. Such a segmentation can be achieved by using linguistic segmentation systems or prompting LLMs to identify natural breaks in the sequence. Another approach is to use dynamic segmentation methods based on the complexity of the sequence. For example, segments could be defined where there is a significant change in the reward score, which might correspond to shifts in the task being modeled.

4. Combination of Reward Models

A reward model can be viewed as a proxy for the environment. Since the true environment is often too complex or unknown, developing a perfect proxy for the environment is generally not possible. As a result, over-aligning LLMs with this imperfect proxy might lead to decreased performance, known as the **overoptimization problem** [Stiennon et al., 2020; Gao et al., 2023a][7]. We can also explain this through Goodhart's law, which states: *when a measure*

[6] To allow the reward model to output categories, we can replace the linear layer described in Section 10.3.2 with a Softmax layer.

[7] This problem is also called **reward hacking** or **reward gaming** [Krakovna et al., 2020; Skalse et al., 2022; Pan et al., 2022], which refers to the phenomenon where the agent attempts to trick the reward model but fails to align its actions with the true intended objectives of the task. Imagine a student who is assigned homework and is rewarded with points or praise for completing it. The student might then find ways to finish the homework

becomes a target, it ceases to be a good measure [Goodhart, 1984].

Addressing the overoptimization problem is not easy, and there is no mature solution yet. The ideal approach might be to develop an oracle reward model that can perfectly capture the true objectives of the task and prevent the agent from "tricking". However, creating such a model is extremely difficult due to the complexity of the real-world environment, as well as the challenge of defining all the relevant factors that contribute to the desired outcome. Instead, a more practical approach is to combine multiple reward models, thereby alleviating the misalignment between the training objective and the true objective that arises from using a single, specific reward model [Coste et al., 2024].

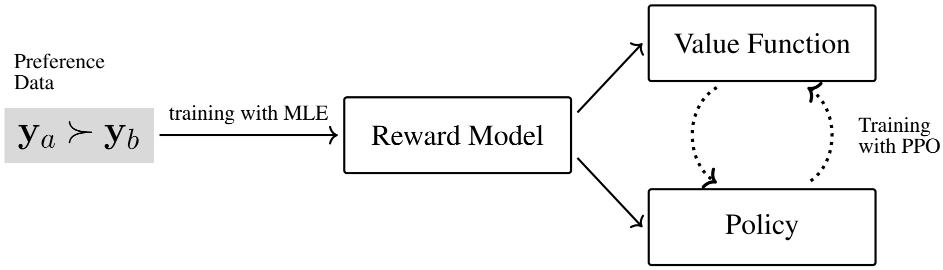
Given a set of reward models, combining them is straightforward, and in some cases, we can simply treat this problem as an ensemble learning problem. A simple yet common approach is to average the outputs of these models to obtain a more precise reward estimation:

$$r_{\text{combine}} = \frac{1}{K}\sum_{k=1}^{K} w_k \cdot r_k(\mathbf{x}, \mathbf{y}) \tag{10.72}$$

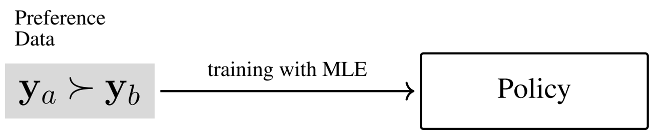
where $r_k(\cdot)$ is the k-th reward model in the ensemble, w_k is the weight of $r_k(\cdot)$, and K is the number of reward models. This combined reward can then be used to supervise the training of a policy. In fact, there are many ways to combine different models, for example, one can make predictions using Bayesian model averaging or develop a fusion network to learn to combine the predictions from different models. Alternatively, one can frame this task as a multi-objective optimization problem, and use multiple reward models to train the policy simultaneously. These methods have been intensively discussed in the literature on optimization and machine learning [Miettinen, 1999; Bishop, 2006].

In addition to model combination methods, another important issue is how to collect or construct multiple different reward models. One of the simplest approaches is to employ ensemble learning techniques, such as developing diverse reward models from different subsets of a given dataset or from various data sources. For RLHF, it is also possible to construct reward models based on considerations of different aspects of alignment. For example, we can develop a reward model to evaluate the factual accuracy of the output and another reward model to evaluate the completeness of the output. These two models are complementary to each other, and can be combined to improve the overall evaluation of the output. Another approach is to employ different off-the-shelf LLMs as reward models. This approach is simple and practical, as there have been a lot of well-developed LLMs and we just need to use them with no or little modification. An interesting issue, though not closely related to the discussion here, arises: can an LLM that aligns with other LLMs outperform those LLMs? Probably not at first glance. In part, this is because the target LLM merely imitates other LLMs based on limited supervision and thus cannot capture well the nuances of the behaviors of these supervisors. However, given the strong generalization ability of LLMs, this approach can, in fact, be quite beneficial. For example, using open-sourced or commercial LLMs as reward

with minimal effort to maximize the reward, such as copying and pasting solutions from the internet or previous assignments, rather than solving the problems themselves.



(a) RLHF (PPO)



(b) DPO

Figure 10.10: Standard RLHF (PPO) vs. DPO. In RLHF, the human preference data is used to train a reward model, which is then employed in training the policy as well as the value function. In DPO, the use of human preference data is more direct, and the policy is trained on this data without the need for reward model training.

models has demonstrated strong performance in aligning LLMs, even achieving state-of-the-art results on several popular tasks [Lambert et al., 2024].

10.4.2 Direct Preference Optimization

Although learning reward models is a standard step in reinforcement learning, it makes the entire training process much more complex than supervised training. Training a reliable reward model is itself not an easy task, and a poorly trained reward model can greatly affect the outcome of policy learning. We now consider an alternative alignment method, called **direct preference optimization** (**DPO**), which simplifies the training framework by eliminating the need to explicitly model rewards [Rafailov et al., 2024]. This method directly optimizes the policy based on user preferences, rather than developing a separate reward model. As a result, we can achieve human preference alignment in a supervised learning-like fashion. Figure 10.10 shows a comparison of the standard RLHF method and the DPO method.

Before deriving the DPO objective, let us first review the objective of policy training used in RLHF. As discussed in Section 10.3.3, the policy is typically trained by optimizing a loss function with a penalty term. The DPO method assumes a simple loss function where the quality of the output $\mathbf{y}$ given the input $\mathbf{x}$ is evaluated by the reward model $r(\mathbf{x}, \mathbf{y})$. The training objective is thus given by

$$\tilde{\theta} = \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\big[\underbrace{-r(\mathbf{x},\mathbf{y})}_{\text{loss}}+\beta\underbrace{\big(\log\pi_\theta(\mathbf{y}|\mathbf{x})-\log\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})\big)}_{\text{penalty}}\big] \quad (10.73)$$

Note that in this optimization problem, only the term $\pi_\theta(\mathbf{y}|\mathbf{x})$ depends on the target policy $\pi_\theta(\cdot)$. Both the reward model $r(\mathbf{x},\mathbf{y})$ and the reference model $\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})$ are assumed to be fixed given $\mathbf{x}$ and $\mathbf{y}$. This is a strong assumption compared with PPO, but as will be shown later, it simplifies the problem and crucial for deriving the DPO objective.

Since θ is the variable we want to optimize, we rearrange the right-hand side of Eq. (10.73) to isolate $\pi_\theta(\mathbf{y}|\mathbf{x})$ as an independent term:

$$\begin{aligned}
\tilde{\theta} &= \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\big[\beta\log\pi_\theta(\mathbf{y}|\mathbf{x})-\beta\log\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})-r(\mathbf{x},\mathbf{y})\big] \\
&= \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\Big[\log\pi_\theta(\mathbf{y}|\mathbf{x})-\Big(\log\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})+\frac{1}{\beta}r(\mathbf{x},\mathbf{y})\Big)\Big] \\
&= \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\Big[\underbrace{\log\pi_\theta(\mathbf{y}|\mathbf{x})}_{\text{dependent on }\theta}-\underbrace{\log\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})\exp\Big(\frac{1}{\beta}r(\mathbf{x},\mathbf{y})\Big)}_{\text{not dependent on }\theta}\Big] \quad (10.74)
\end{aligned}$$

This equation defines the objective function as the difference between the log-probability distribution function of y and another function of y. This form of the objective function seems not "ideal", as we usually prefer to see the difference between two distributions, so that we can interpret this difference as some kind of divergence between the distributions. A simple idea is to convert the second term (i.e., $\log\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})\exp(\frac{1}{\beta}r(\mathbf{x},\mathbf{y}))$) into a log-probability distribution over the domain of $\mathbf{y}$. If we treat $\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})\exp(\frac{1}{\beta}r(\mathbf{x},\mathbf{y}))$ as an unnormalized probability of y, we can convert it into a normalized probability by dividing it by a normalization factor:

$$Z(\mathbf{x}) = \sum_{\mathbf{y}}\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})\exp\Big(\frac{1}{\beta}r(\mathbf{x},\mathbf{y})\Big) \quad (10.75)$$

Hence we can define a probability distribution by

$$\pi^*(\mathbf{y}|\mathbf{x}) = \frac{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})\exp\big(\frac{1}{\beta}r(\mathbf{x},\mathbf{y})\big)}{Z(\mathbf{x})} \quad (10.76)$$

We then rewrite Eq. (10.74) as

$$\begin{aligned}
\tilde{\theta} &= \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\Big[\log\pi_\theta(\mathbf{y}|\mathbf{x}) - \log\frac{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})\exp\left(\frac{1}{\beta}r(\mathbf{x},\mathbf{y})\right)}{Z(\mathbf{x})} \\
&\qquad - \log Z(\mathbf{x})\Big] \\
&= \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\Big[\log\pi_\theta(\mathbf{y}|\mathbf{x}) - \log\pi^*(\mathbf{y}|\mathbf{x}) - \log Z(\mathbf{x})\Big] \\
&= \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\Big[\mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\Big[\log\pi_\theta(\mathbf{y}|\mathbf{x}) - \log\pi^*(\mathbf{y}|\mathbf{x})\Big] \\
&\qquad - \mathbb{E}_{\mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})}\big[\log Z(\mathbf{x})\big]\Big] \\
&= \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\Big[\underbrace{\mathrm{KL}\big(\pi_\theta(\cdot|\mathbf{x}) \mid\mid \pi^*(\cdot|\mathbf{x})\big)}_{\text{KL divergence}} - \underbrace{\log Z(\mathbf{x})}_{\text{constant wrt. }\theta}\Big] \qquad (10.77)
\end{aligned}$$

Since $\log Z(\mathbf{x})$ is independent of θ, it does not affect the result of the $\arg\min_\theta$ operation, and can be removed from the objective. Now we obtain a new training objective which finds the optimal policy π_θ by minimizing the KL divergence between $\pi_\theta(\cdot|\mathbf{x})$ and $\pi^*(\cdot|\mathbf{x})$

$$\tilde{\theta} = \arg\min_{\theta} \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}\Big[\mathrm{KL}\big(\pi_\theta(\cdot|\mathbf{x}) \mid\mid \pi^*(\cdot|\mathbf{x})\big)\Big] \qquad (10.78)$$

Clearly, the solution to this optimization problem is given by

$$\begin{aligned}
\pi_\theta(\mathbf{y}|\mathbf{x}) &= \pi^*(\mathbf{y}|\mathbf{x}) \\
&= \frac{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})\exp\left(\frac{1}{\beta}r(\mathbf{x},\mathbf{y})\right)}{Z(\mathbf{x})} \qquad (10.79)
\end{aligned}$$

Given this equation, we can express the reward $r(\mathbf{x},\mathbf{y})$ using the target model $\pi_\theta(\mathbf{y}|\mathbf{x})$, the reference model $\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})$, and the normalization factor $Z(\mathbf{x})$:

$$r(\mathbf{x},\mathbf{y}) = \beta\left(\log\frac{\pi_\theta(\mathbf{y}|\mathbf{x})}{\pi_{\theta_{\text{ref}}}(\mathbf{y}|\mathbf{x})} + \log Z(\mathbf{x})\right) \qquad (10.80)$$

This is interesting because we initially seek to learn the policy $\pi_\theta(\cdot)$ using the reward model $r(\mathbf{x},\mathbf{y})$, but eventually obtain a representation of the reward model based on the policy. Given the reward model defined in Eq. (10.80), we can apply it to the Bradley-Terry model to calculate the preference probability (also see Section 10.3.2):

$$\begin{aligned}
\Pr_\theta(\mathbf{y}_a \succ \mathbf{y}_b|\mathbf{x}) &= \mathrm{Sigmoid}(r(\mathbf{x},\mathbf{y}_a) - r(\mathbf{x},\mathbf{y}_b)) \\
&= \mathrm{Sigmoid}\Big(\beta\Big(\log\frac{\pi_\theta(\mathbf{y}_a|\mathbf{x})}{\pi_{\theta_{\mathrm{ref}}}(\mathbf{y}_a|\mathbf{x})} + \log Z(\mathbf{x})\Big) - \\
&\qquad \beta\Big(\log\frac{\pi_\theta(\mathbf{y}_b|\mathbf{x})}{\pi_{\theta_{\mathrm{ref}}}(\mathbf{y}_b|\mathbf{x})} + \log Z(\mathbf{x})\Big)\Big) \\
&= \mathrm{Sigmoid}\Big(\beta\log\frac{\pi_\theta(\mathbf{y}_a|\mathbf{x})}{\pi_{\theta_{\mathrm{ref}}}(\mathbf{y}_a|\mathbf{x})} - \beta\log\frac{\pi_\theta(\mathbf{y}_b|\mathbf{x})}{\pi_{\theta_{\mathrm{ref}}}(\mathbf{y}_b|\mathbf{x})}\Big)
\end{aligned} \tag{10.81}$$

This formula is elegant because it converts the difference in rewards into the difference in ratio functions, and we do not need to calculate the value of $Z(\mathbf{x})$. A direct result is that we no longer need a reward model, but only need the target policy and reference model to calculate the probability of preferences. Finally, we can train the target policy by minimizing the following DPO loss function

$$\mathcal{L}_{\mathrm{dpo}}(\theta) = -\mathbb{E}_{(\mathbf{x},\mathbf{y}_a,\mathbf{y}_b)\sim\mathcal{D}_r}\big[\log\Pr_\theta(\mathbf{y}_a \succ \mathbf{y}_b|\mathbf{x})\big] \tag{10.82}$$

The form of this loss function is very similar to that used in training reward models in RLHF (see Eq. (10.36)). But it should be noted that the loss function here depends on the parameters of the policy (i.e., θ) rather than the parameters of the reward model (i.e., ϕ).

The main advantage of DPO lies in its simplicity and efficiency. The DPO objective is very straightforward — it directly optimizes for preference-based feedback, rather than relying on separately developed reward models. Moreover, DPO is generally more sample-efficient, as it learns from a fixed dataset without the need for the computationally expensive sampling process used in PPO. This makes DPO a popular method for human preference alignment, especially when developing and applying reward models via reinforcement learning is challenging.

DPO can broadly be viewed as an **offline reinforcement learning** method, where the training data is pre-collected and fixed, and there is no exploration. In contrast, online reinforcement learning methods like PPO, which require exploring new states through interaction with the environment (using the reward model as a proxy), also have their unique advantages. One of the benefits of online reinforcement learning is that it allows the agent to continuously adapt to changes in the environment by learning from real-time feedback. This means that, unlike offline methods, online methods are not constrained by the static nature of pre-collected data and can discover new problem-solving strategies. In addition, exploration can help the agent cover a wider range of state-action pairs, thus improving generalization. This could be an important advantage for LLMs, as generalization is considered a critical aspect in applying such large models.

10.4.3 Automatic Preference Data Generation

Although learning from human preferences is an effective and popular method for aligning LLMs, annotating preference data is costly. Using human feedback does not only faces the problem of limited scalability, but it may also introduce bias because human feedback is inherently subjective. As a result, one can turn to AI feedback methods to address these scalability and consistency issues without the limitations associated with human annotators.

As with data generation for instruction fine-tuning, generating preference data using LLMs is straightforward. Given a set of inputs, we first use an LLM to generate pairs of outputs. Then, we prompt the LLM to label the preference between each pair of outputs, along with its corresponding input. Below is an example of prompting the LLM to generate a preference label for a pair of consumer service responses.

> Consider a customer service scenario where a customer poses a request. You will review two responses to this request. Please indicate which response is preferred. Note that a good response should be courteous, clear, and concise. It should address the customer's concern directly, provide helpful information or a solution, and maintain a positive tone.
>
> Request:
>
> *Hello, I noticed that my order hasn't arrived yet, though it was scheduled to arrive several days ago. Could you please update me on its status? Thank you!*
>
> Response A:
>
> *I'm very sorry for the delay and understand how disappointing this can be. We're doing our best to sort this out quickly for you.*
>
> Response B:
>
> *Hey, stuff happens! Your package will get there when it gets there, no need to stress.*
>
> <u>Response A is preferred.</u>

Once we collect such preference labels, we can use them, along with the output pair and input, to train the reward model. Of course, we can consider demonstrating a few examples or using advanced prompting techniques, such as CoT, to improve labeling performance. For example, we can include in the prompt an example showing how and why one of the two responses is preferred based on a CoT rationale.

In addition to preference labels, we can also obtain the probability associated with each label [Lee et al., 2023]. A simple method is to extract the probabilities for the label tokens, such as "A" and "B", from the probabilities output by the LLM. We can then use the Softmax function or other normalization techniques to re-normalize these probabilities into a distribution over the labels. These probabilities of preferred labels can serve as pointwise supervision signals for training the reward model, as discussed in Section 10.4.1.

For data generation, although it is easy to scale up, it is often necessary to ensure the data is accurate and diverse. Here, the data quality and diversity issues involve not only the labeling of preferences but also the inputs and outputs of the model. Therefore, we often need to use a variety of techniques to obtain large-scale, high-quality data. For example, one can generate diverse model outputs and annotations by using different LLMs, prompts, in-context demonstrations, and so on [Cui et al., 2024]. Dubois et al. [2024] report that the variability in pairwise preference data is important for training LLMs from either human or AI feedback.

While learning from AI feedback is highly scalable and generally objective, this method is more suited to well-defined tasks where objective performance metrics are available. By contrast, learning from human feedback is more advantageous when aligning AI systems with human values, preferences, and complex real-world tasks that require understanding of subtle or subjective context. These methods can be combined to train LLMs that benefit from both human insights and the scalability of AI feedback.

10.4.4 Step-by-step Alignment

So far, our discussion of alignment has primarily focused on the use of reward models for evaluating entire input-output sequence pairs. These methods can be easily adapted to scenarios where the correctness of an output can be examined by checking whether the desired result is included. For example, in the task of calculating a mathematical expression, a reward model can provide positive feedback if the answer is correct, and negative feedback if the answer is wrong. However, in many problems that require complex reasoning, simply examining the correctness of the final result is insufficient for learning. Imagine a student who is only given the final answer to a challenging math problem. Knowing whether the final answer is right or wrong does not help the student figure out where they went wrong and how to calculate the correct answer. A better approach would be to guide the student with a step-by-step breakdown of the problem-solving process and encourage understanding of the underlying concepts and logic behind these steps.

In Chapter 9, we studied CoT methods to prompt LLMs to explicitly write out intermediate steps or the reasoning process needed to reach a conclusion or solve a problem. We saw that breaking down a problem into smaller parts could make it easier to understand the solution path and increase the accuracy of the output. These methods can be naturally extended to the alignment of LLMs, that is, we supervise the model during the intermediate steps of reasoning. Consider a reasoning task where an LLM produces a sequence of reasoning steps $\mathbf{y} = \{\bar{\mathbf{y}}_1, ..., \bar{\mathbf{y}}_{n_s}\}$ for the given input. The result of the reasoning is assumed to be included in the last step $\bar{\mathbf{y}}_{n_s}$, and can be easily verified. For this reasoning problem, Uesato et al. [2022] categorize LLM fine-tuning approaches into two classes:

- **Outcome-based Approaches**. Supervision occurs only when the end result is verified. This is a standard method for learning from human feedback we have discussed in this chapter. For example, the LLM is optimized to maximize some form of the reward $r(\mathbf{x}, \mathbf{y})$.
- **Process-based Approaches**. Supervision is involved in all intermediate steps in addition to the last step. To do this, we need to develop a model to give a supervision signal at each step, and develop loss functions that can make use of such supervision signals.

Question (from The International Junior Math Olympiad):
Jessica has a lot of apps on her phone. The number of apps she has is divisible by 5. If she downloaded 10 more apps, she would still have less than 50 apps. What is the maximum original number of apps Jessica could have?

Response 1:

$\bar{\mathbf{y}}_1$: x must be a multiple of 5, i.e., $x \equiv 0 \pmod 5$.

$\bar{\mathbf{y}}_2$: After downloading 10 apps, the total total number of apps should still be less than 50: $x + 10 < 50$. Solving this inequality: $x < 40$.

$\bar{\mathbf{y}}_3$: Since x must be a multiple of 5 and less than 40, the largest value of x is 35.

$\bar{\mathbf{y}}_4$: Verify the result. If $x = 35$, then after downloading 10 apps, Jessica would have: $35 + 10 = 45$ which satisfy $45 < 50$.

$\bar{\mathbf{y}}_5$: The maximum original number of apps is $\underline{35}$.

correct

Response 2:

$\bar{\mathbf{y}}_1$: The number of apps is divisible by 5, so x must be a multiple of 5.

$\bar{\mathbf{y}}_2$: If she downloads 10 more apps, her new total will be $x + 10$, and this must be less than 50. Therefore, $x + 10 < 50$. Simplifying this gives $x < 40$.

$\bar{\mathbf{y}}_3$: x must be less than or equal to the largest multiple of 5, which is 40. problematic

$\bar{\mathbf{y}}_4$: But $x + 10$ should not be more than or equal to 50. So we need to subtract 5 from 40. problematic

$\bar{\mathbf{y}}_5$: Therefore, the final result is $\underline{35}$.

correct

Figure 10.11: Two LLM responses to a math problem. In response 1, both the final result and all the reasoning steps are correct. In response 2, the final result is correct, but there are mistakes in the reasoning process (highlighted in red). For outcome-based approaches, both responses are considered correct. For process-based approaches, the mistakes in response 2 can be considered in reward modeling.

Figure 10.11 shows two LLM outputs for an example math problem. Although the LLM gives the correct final answer in both cases, it makes mistakes during the problem-solving process in the second output. Outcome-based approaches overlook these mistakes and give positive feedback for the entire solution. By contrast, process-based approaches can take these mistakes into account and provide additional guidance on the detailed reasoning steps.

An important issue for process-based approaches is that we need to get step-level feedback during a (potentially) long reasoning path. We can collect or generate reasoning paths corresponding to problems from existing datasets. Human experts then annotate each step in these paths for correctness. These annotations can be used to directly train LLMs or as rewards in reward modeling. However, in practice, richer annotations are often introduced [Lightman et al., 2024]. In addition to the *correct* and *incorrect* labels, a step can also be labeled as *neutral* to indicate that while the step may be technically correct, it might still be problematic within the overall reasoning process. Furthermore, to improve the efficiency of data annotation, techniques such as active learning can be employed. Identifying obvious errors usually does not significantly contribute to learning from reasoning mistakes. Instead, annotating steps that the model confidently considers correct but are actually problematic is often more effective.

Given a set of step-level annotated reasoning paths and corresponding inputs, we can train a reward model to provide feedback for supervising policy learning. The reward model can be treated as a classification model, and so its architecture can be a Transformer decoder with a Softmax layer stacked on top. At step k, the reward model takes both the problem description (denoted by $\mathbf{x}$) and the reasoning steps generated so far (denoted by $\bar{\mathbf{y}}_{\leq k}$) as input and outputs a probability distribution over the label set $\{correct, incorrect\}$ or $\{correct, incorrect, neutral\}$. Then the learned reward model is used to evaluate reasoning paths by assessing the correctness of each step. A simple method to model correctness is to count the number of steps that are classified as *correct*, given by

$$r(\mathbf{x},\mathbf{y}) = \sum_{k=1}^{n_s} \delta(correct, C(\mathbf{x},\bar{\mathbf{y}}_{\leq k})) \tag{10.83}$$

where $C(\mathbf{x},\bar{\mathbf{y}}_{\leq k})$ denotes the label with the maximum probability. We can also use log-probabilities of classification to define the reward of the entire path

$$r(\mathbf{x},\mathbf{y}) = \sum_{k=1}^{n_s} \log \Pr(correct|\mathbf{x},\bar{\mathbf{y}}_{\leq k}) \tag{10.84}$$

where $\Pr(correct|\mathbf{x},\bar{\mathbf{y}}_{\leq k})$ denotes the probability of the *correct* label generated by the reward model. The reward score $r(\mathbf{x},\mathbf{y})$ can then be used to train the policy in RLHF as usual.

While we restrict our discussion to math problems, the approaches described here are general and can be applied to a wide variety of tasks that involve multi-step reasoning and decision-making. Moreover, we can consider various aspects when assessing the quality of a step, rather than just its correctness. For example, in dialogue systems, responses must not only be accurate but also contextually appropriate across multiple turns of conversation. If a model provides a correct response but fails to maintain coherence in the context of the ongoing dialogue, step-level feedback could help the model identify and correct such discrepancies. Also note that the process-based approaches are related to the fine-grained reward modeling approaches discussed in Section 10.4.1. All these approaches essentially aim to provide more detailed supervision to LLMs by breaking their outputs into smaller, more manageable steps. However, process-based feedback focuses more on evaluating the correctness of a step based on its preceding steps, while the approaches in Section 10.4.1 emphasize evaluating each step independently.

The idea of aligning LLMs step by step has great application potential, especially considering the recent shift towards more complex reasoning tasks in the use of LLMs. For example, both the GPT-o1 and GPT-o3 models are designed with more advanced reasoning techniques (such as long internal CoT) to solve challenging problems like scientific and mathematical reasoning [OpenAI, 2024]. These tasks often rely on long and complex reasoning paths, and therefore, it seems essential to introduce detailed supervision signals in the reasoning process. Moreover, from a practical perspective, effective supervision on long reasoning paths not only improves reasoning performance, but it also helps the model eliminate redundant or unnecessary reasoning steps, thereby reducing reasoning complexity and improving efficiency.

10.4.5 Inference-time Alignment

In this section we explored a variety of methods to align models with human preferences and annotations. However, one of the significant limitations of many such methods is that LLMs must be fine-tuned. For RLHF and its variants, training LLMs with reward models can be computationally expensive and unstable, leading to increased complexity and costs when applying these approaches. In this case, we can consider aligning models at inference time, thus avoiding the additional complexity and effort involved.

One simple way to achieve inference-time alignment is to use the reward model to select the best one from N alternative outputs generated by the LLM, a method known as **Best-of-N sampling** (**BoN sampling**). We can consider BoN sampling as a form of reranking. In fact, reranking methods have been widely used in NLP tasks, such as machine translation, for a long time. They are typically applied in situations where training complex models is costly. In such cases, directly reranking the outputs allows for the incorporation of these complex models at a very low cost[8].

In the BoN sampling process, the LLM takes the input sequence $\mathbf{x}$ and generates N different output sequences $\{\hat{\mathbf{y}}_1, ..., \hat{\mathbf{y}}_N\}$:

$$\{\hat{\mathbf{y}}_1, ..., \hat{\mathbf{y}}_N\} = \underset{\mathbf{y}}{\mathrm{argTopN}} \left[\Pr(\mathbf{y}|\mathbf{x})\right] \tag{10.85}$$

where the $\mathrm{argTopN}$ operation returns the top-N outputs that maximize the function $\Pr(\mathbf{y}|\mathbf{x})$. These outputs can be generated in a variety of ways, depending on the search algorithm used by the model (e.g., sampling or beam search). Once the N-best output candidates are generated, the reward model is used to evaluate and select the best one:

$$\hat{\mathbf{y}}_{\mathrm{best}} = \max\{r(\mathbf{x}, \hat{\mathbf{y}}_1), ..., r(\mathbf{x}, \hat{\mathbf{y}}_N)\} \tag{10.86}$$

It is worth noting that the result of BoN sampling is also influenced by the diversity of the N-best list. This is a common issue with most reranking methods. Typically, we wish the N-best output candidates to have relatively high quality but be sufficiently different from each other. In many text generation systems, the N-best outputs are very similar, often differing by

[8]Reranking methods can also help us explore what are known as model errors and search errors, although these issues are not often discussed in the context of LLMs. For example, suppose we have an old model and a new, more powerful model. We can use the new model to select the best output from the N-best list of the old model as the oracle output. The performance difference between the oracle output and the top-1 output of the original N-best list reflects the performance gain brought by the new model. If the performance gain is significant, we can say that the old model has more model errors. If the gain is small, it may indicate that the issue lies in search errors, as the best candidates were not found.

just one or two words. The diversity issue is even more challenging in LLMs, as the N-best outputs generated by an LLM can be different in their wordings, yet their semantic meanings are often quite similar. In practice, one can adjust the model hyperparameters and/or adopt different LLMs to generate more diverse output candidates for reranking. Nevertheless, as with many practical systems, we need to make a trade-off between selecting high-quality candidates and ensuring sufficient variation in the generated outputs.

BoN sampling can be used for training LLMs as well. A closely related method is **rejection sampling**. In this method, we first select the "best" outputs from the N-best lists via the reward model, and then take these selected outputs to fine-tune the LLM. In this way, we can introduce human preferences into the training of LLMs via a much simpler approach compared to RLHF. Many LLMs have adopted rejection sampling for fine-tuning [Nakano et al., 2021; Touvron et al., 2023b].

10.5 Summary

In this chapter, we have explored a range of techniques for aligning LLMs. In particular, we have discussed fine-tuning methods that enable LLMs to follow instructions and align them with human preferences. One of the benefits of fine-tuning LLMs is computation efficiency. Unlike pre-training based on large-scale neural network optimization, fine-tuning is a post-training step and so is less computationally expensive. Moreover, it is better suited to address problems that are not easily solved in pre-training, such as human value alignment. The widespread attention to the alignment issue has also led to a surge of research papers on this topic, which has posed challenges in writing this chapter, as it is difficult to cover all the latest techniques. However, we have tried to provide a relatively detailed introduction to the fundamental approaches to alignment, such as instruction fine-tuning and RLHF.

While we have focused on LLM alignment techniques in this chapter, the term *AI alignment* is a wide-ranging concept. It generally refers to the process of ensuring that the behavior of an AI system aligns with human values, goals, and expectations. The idea of AI alignment can be traced back to the early days of AI. A widely cited description of AI alignment comes from an article by the mathematician and computer scientist Norbert Wiener [Wiener, 1960]. The quote is as follows

> *If we use, to achieve our purposes, a mechanical agency with whose operation we cannot efficiently interfere ... we had better be quite sure that the purpose put into the machine is the purpose which we really desire.*

At that time, AI alignment was a distant concern for researchers. But today, it greatly influences the design of various AI systems. For example, in robotics, alignment is critical to ensuring that autonomous robots safely interact with humans and their environments. In autonomous driving, cars must not only follow traffic laws but also make complex, real-time decisions that prioritize human safety, avoid accidents, and navigate ethical dilemmas.

In current AI research, alignment is usually achieved by developing a surrogate objective that is analogous to the real goal and steering the AI system towards this objective. However, designing the objective of AI alignment is very difficult. One reason is that human values are diverse and often context-dependent, making it difficult to distill them into a single, universally applicable objective function. Also, the complexity of real-world environments, where values and goals often conflict or evolve over time, further complicates alignment efforts. Even if we could define an appropriate objective, AI systems may find unintended ways to achieve it, leading to “misaligned” outcomes that still technically satisfy the objective but in a harmful or counterproductive way.

These challenges have motivated and are motivating AI research towards more aligned systems, either through developing new mechanisms for perceiving the world or more efficient and generalizable methods to adapt these systems to given tasks. More importantly, as AI systems become more powerful and intelligent, especially given that recent advances in LLMs have shown remarkable capabilities in dealing with many challenging problems, the need for AI alignment has become more urgent. Researchers have started to be concerned with AI safety and warn the community that they need to develop and release AI systems with great caution to prevent these systems from being misaligned [Russell, 2019; Bengio et al., 2024].

Chapter 11

Inference

Once we have pre-trained and fine-tuned an LLM, we can apply it to make predictions on new data. This process is called inference, in which the LLM computes the probabilities of different possible outputs given an input, and selects the output that maximizes the probability. The inference problem is generally expressed in the following form:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \Pr(\mathbf{y}|\mathbf{x}), \tag{11.1}$$

where $\mathbf{x}$ is the input sequence, $\mathbf{y}$ is a possible output sequence, and $\hat{\mathbf{y}}$ is the best output sequence.

This is perhaps one of the most widely adopted formulas in NLP, and dates back to the early days of speech recognition and machine translation systems based on probabilistic models. Although for some applications, such as predicting a token using a very small language model, solving this optimization problem seems trivial, for most situations the computational challenges arise from both calculating $\Pr(\mathbf{y}|\mathbf{x})$ and performing the $\arg\max$ operation. The problems we therefore wish to address in this chapter involve: 1) computing the prediction probability efficiently given a trained LLM, and 2) devising an efficient (suboptimal) search for $\hat{\mathbf{y}}$.

At a high level, these are fundamental issues in artificial intelligence, which have been extensively studied. So many well-established techniques can be directly applied, for example, one can use greedy search algorithms to implement an efficient inference system. On the other hand, model-specific optimizations, such as efficient attention models for Transformers, can be considered to further improve efficiency. But, in many practical applications, we still need to make a trade-off between accuracy and efficiency, by carefully combining various techniques.

The importance of the inference problem in LLMs also lies in the fact that many application scenarios require processing extremely long sequences. Recent studies have found that injecting additional prompts and contextual information, such as long chain-of-thought prompts, during inference can significantly improve the performance of LLMs. This provides a new approach to scaling LLMs: better results can be achieved by increasing the compute at inference time. For instance, through inference-time scaling, OpenAI [2024]'s o1 and DeepSeek [2025]'s R1 systems have demonstrated impressive performance on complex reasoning and programming tasks. This, in turn, has encouraged the NLP field to focus more on the issue of efficient inference.

In this chapter, we will introduce basic concepts and algorithms of LLM inference, including prefilling-decoding frameworks, search (decoding) algorithms, and evaluation metrics of inference performance. We will then present methods for improving the efficiency of LLM inference, covering a range of techniques for speeding up the system and compressing the model. Finally, we will discuss inference-time scaling, which is considered an important application of inference optimization.

11.1 Prefilling and Decoding

In this section, we present the prefilling-decoding framework, the most common approach for interpreting and implementing LLM inference processes. We first introduce the notation and background knowledge, and then describe the details of the framework, such as the decoding algorithms for LLM inference.

11.1.1 Preliminaries

Although we have described LLMs many times in this book, we begin by briefly defining the notation to facilitate the subsequent discussion, and to make this chapter self-contained.

$\mathbf{x}$: The input token sequence. It is conceptually equivalent to a "prompt", which includes instructions, user inputs, and any additional context intended as input to the LLM. $\mathbf{x}$ comprises $m+1$ tokens, denoted by $x_0 \cdots x_m$, where x_0 is the start symbol $\langle\text{SOS}\rangle$.

$\mathbf{y}$: The output token sequence, also called the response to the input. $\mathbf{y}$ comprises n tokens, denoted by $y_1 \cdots y_n$.

$\mathbf{y}_{<i}$: The output tokens that precede position i, that is, $\mathbf{y}_{<i} = y_1 \cdots y_{i-1}$.

$\Pr(\mathbf{y}|\mathbf{x})$: The probability of generating $\mathbf{y}$ given $\mathbf{x}$ using the LLM. If the LLM is parameterized by θ, we can write it as $\Pr_\theta(\mathbf{y}|\mathbf{x})$.

$[\mathbf{x}, \mathbf{y}]$: The concatenated token sequence of $\mathbf{x}$ and $\mathbf{y}$. That is, $[\mathbf{x}, \mathbf{y}] = x_0 \cdots x_m y_1 \cdots y_n$. Occasionally, we use the notation $\text{seq}_{\mathbf{x},\mathbf{y}}$ to represent $[\mathbf{x}, \mathbf{y}]$.

$\Pr([\mathbf{x}, \mathbf{y}])$: The probability of generating the token sequence $[\mathbf{x}, \mathbf{y}]$ using the LLM.

As described in Eq. (11.1), the goal of LLM inference is to maximize $\Pr(\mathbf{y}|\mathbf{x})$. Modeling this conditional probability is common in NLP. At first glance, it seems to be a sequence-to-sequence problem, where we transform a sequence into another using encoding-decoding models. However, we are not discussing sequence-to-sequence problems or encoding-decoding architectures. Instead, as discussed in earlier chapters, this modeling problem can be addressed by using decoder-only models. To do this, we can interpret the log-scale probability $\log \Pr(\mathbf{y}|\mathbf{x})$ as the difference between $\log \Pr([\mathbf{x},\mathbf{y}])$ and $\log \Pr(\mathbf{x})$:

$$\log \Pr(\mathbf{y}|\mathbf{x}) \quad = \quad \log \Pr([\mathbf{x},\mathbf{y}]) - \log \Pr(\mathbf{x}), \tag{11.2}$$

where $\log \Pr([\mathbf{x},\mathbf{y}])$ and $\log \Pr(\mathbf{x})$ can be obtained by running the LLM on the sequences $[\mathbf{x},\mathbf{y}]$ and $\mathbf{x}$, respectively. For example, we can calculate the probability of generating $\mathbf{x}$ using the chain rule:

$$\begin{aligned} \log \Pr(\mathbf{x}) &= \log \Pr(x_0 \cdots x_m) \\ &= \log \big[\Pr(x_0)\Pr(x_1|x_0)\cdots\Pr(x_m|x_0\cdots x_{m-1})\big] \\ &= \underbrace{\log \Pr(x_0)}_{=0} + \sum_{j=1}^{m} \log \Pr(x_j|\mathbf{x}_{<j}) \\ &= \sum_{j=1}^{m} \log \Pr(x_j|\mathbf{x}_{<j}). \end{aligned} \tag{11.3}$$

In other words, we calculate the token prediction log-probability at each position of $\mathbf{x}$, and sum all these log-probabilities.

In common implementations of LLMs, however, we do not need to compute the log-probability of the input sequence, but use the LLM to directly compute the log-probability of the output sequence in the following form:

$$\log \Pr(\mathbf{y}|\mathbf{x}) \quad = \quad \sum_{i=1}^{n} \log \Pr(y_i|\mathbf{x},\mathbf{y}_{<i}), \tag{11.4}$$

where $[\mathbf{x},\mathbf{y}_{<i}]$ represents the context for predicting y_i. We use $\Pr(y_i|\mathbf{x},\mathbf{y}_{<i})$ to denote $\Pr(y_i|[\mathbf{x},\mathbf{y}_{<i}])$, following the commonly used notation in the literature.

Now, we have two sub-problems in addressing the inference issue described in Eq. (11.1):

- **Model Computation**: we model $\Pr(y_i|\mathbf{x},\mathbf{y}_{<i})$ and compute it in an efficient manner.
- **Search**: we find the optimal (or sub-optimal) output sequence in terms of $\log \Pr(\mathbf{y}|\mathbf{x})$.

The second sub-problem is a classic issue in NLP. We will show in Section 11.1.3 that there are several well-studied algorithms that can be applied to efficiently search the space of possible output sequences. The first sub-problem requires a language model to produce a distribution over a vocabulary V given a sequence of context tokens. We can do this by training a Transformer decoder, which outputs the distribution:

$$\Pr(\cdot|\mathbf{x},\mathbf{y}_{<i}) \quad = \quad \mathrm{softmax}(\mathbf{H}\mathbf{W}^{o})_{m+i} \tag{11.5}$$

$$\mathbf{H} \quad = \quad \mathrm{Dec}([\mathbf{x},\mathbf{y}_{<i}]). \tag{11.6}$$

Here $\mathrm{Dec}(\cdot)$ produces a sequence of representations, each corresponding to a position of the input sequence. So, if we input $[\mathbf{x},\mathbf{y}_{<i}]$ to the LLM, $\mathbf{H}$ is an $i' \times d$ matrix, where d is the

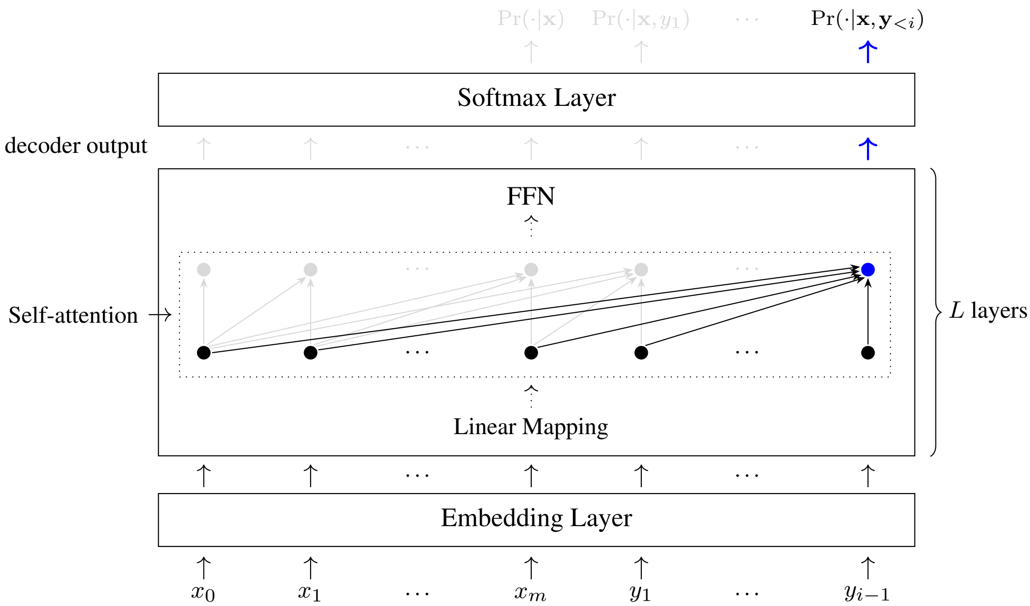


Figure 11.1: The decoder-only architecture for LLMs. The decoder consists of an embedding layer and a stack of Transformer layers. In each Transformer layer, the input passes through a linear mapping, a self-attention network, and an FFN. The output of the decoder is a sequence of representations that are taken as input to a softmax layer, which generates a distribution of tokens for each position.

dimensionality of each representation, and $i' = m + i$ is the number of context tokens. We can then use a softmax layer to transform these representations into distributions of tokens. $\mathbf{W}^o \in \mathbb{R}^{d \times |V|}$ is the linear mapping matrix of the softmax layer, and $\mathbf{H}\mathbf{W}^o$ transforms the d-dimensional representations in $\mathbf{H}$ into the $|V|$-dimensional representations. The use of the subscript $m+i$ indicates that the softmax function is performed only on the representation at position $m+i$. See Figure 11.1 for an illustration of this architecture.

$\mathrm{Dec}(\cdot)$ is a Transformer decoding network that consists of an embedding network and a number of stacked self-attention and FFN networks. We will not discuss Transformers in detail here, as readers can easily learn about these models from the literature. However, it is worth pointing out that the difficulty of inference is in part from the use of the self-attention mechanism in Transformers. Recall that a general form of single-head self-attention is given by:

$$\mathrm{Att}_{\mathrm{qkv}}(\mathbf{q}_{i'}, \mathbf{K}, \mathbf{V}) \quad = \quad \mathrm{softmax}(\frac{\mathbf{q}_{i'}\mathbf{K}^{\top}}{\sqrt{d}})\mathbf{V}, \tag{11.7}$$

where $\mathbf{q}_{i'} \in \mathbb{R}^d$ is the query at the position i' (i.e., position of y_i), and $\mathbf{K}$ and $\mathbf{V} \in \mathbb{R}^{i' \times d}$ are the keys and values up to i', respectively.

At each step during inference, we call the self-attention function $\mathrm{Att}_{\mathrm{qkv}}(\cdot)$, followed by an FFN, to generate a d-dimensional representation that integrates information from both the current token and its left context. This process is repeated through L layers of self-attention and FFN, forming a stack of Transformer layers. The output of the L-th layer in this stack is the final representation.

Each time, the model attends position i' to all previous positions, which results in $2i'$ vector products (i' times for $\mathbf{q}_{i'}\mathbf{K}^{\top}$ and i' times for the product of $\mathrm{softmax}(\frac{\mathbf{q}_{i'}\mathbf{K}^{\top}}{\sqrt{d}})$ and $\mathbf{V}$). Hence, generating a sequence of length len has a time complexity of $O(L \times d \times len^2)$ for the self-attention network. Clearly, the inference of this model is slow for long sequences due to its quadratic time complexity with respect to sequence length. Therefore, many improvements to Transformers and alternative models have focused on efficient methods that are faster than this quadratic time complexity, such as sparse attention mechanisms and linear-time models. A detailed discussion of efficient Transformers can be found in the previous chapters, and this section will focus on the standard Transformer architecture.

Note that in self-attention, the queries, keys, and values of a layer are linear mappings from the same input (i.e., the output of the previous layer). Once a new key-value pair is generated, it is repeatedly used in subsequent inference steps. Rather than regenerating these key-value pairs during inference, a more desirable way is to store them in a structure, called the **key-value cache**, or the **KV cache**. Thus, $(\mathbf{K},\mathbf{V})$ can straightforwardly be considered a KV cache. This cache is updated as follows:

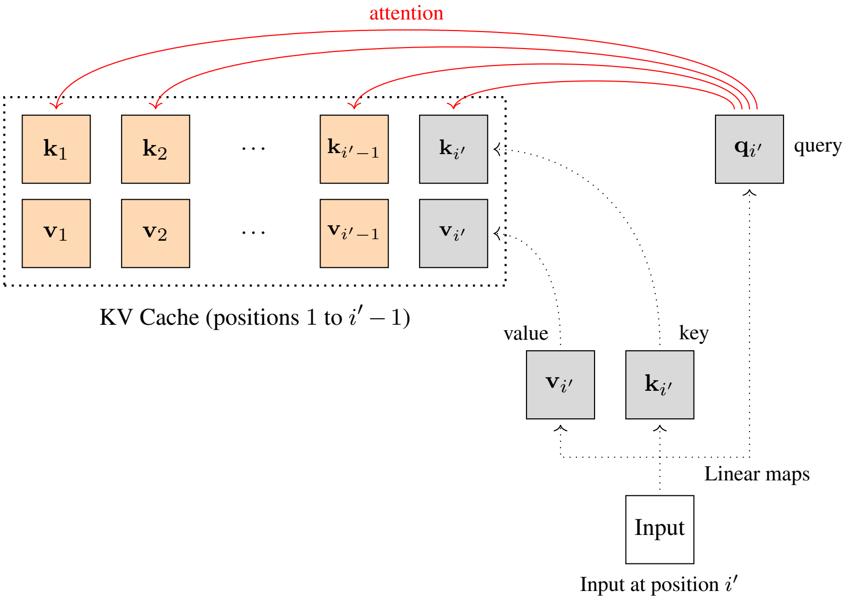
$$\mathbf{K} = \mathrm{Append}(\mathbf{K}, \mathbf{k}_{i'}), \tag{11.8}$$

$$\mathbf{V} = \mathrm{Append}(\mathbf{V}, \mathbf{v}_{i'}), \tag{11.9}$$

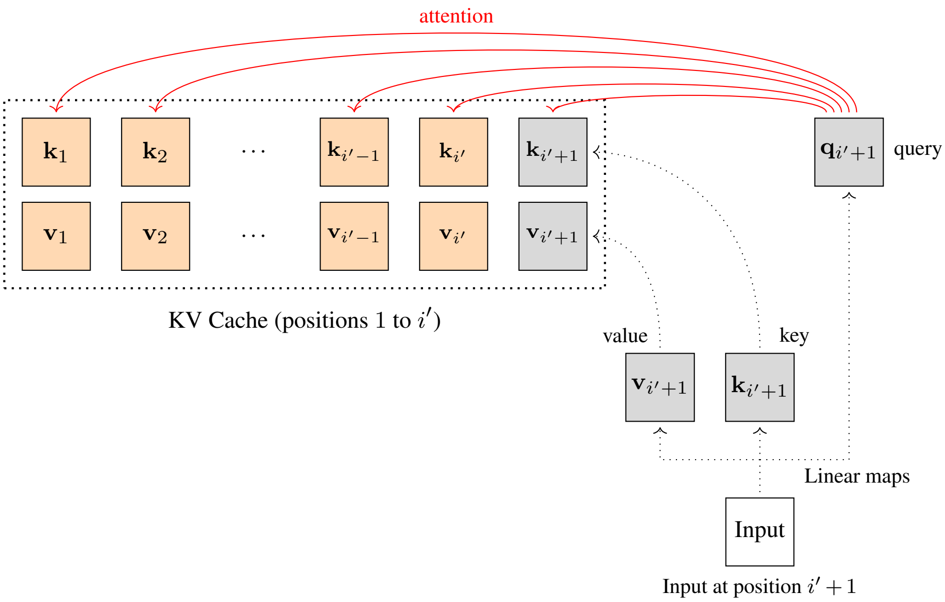
where $(\mathbf{k}_{i'}, \mathbf{v}_{i'})$ is the newly generated key-value pair at position i', and $\mathrm{Append}(\mathbf{a}, \mathbf{b})$ denotes a function that appends a row vector $\mathbf{b}$ to a matrix $\mathbf{a}$. Figure 11.2 shows how a Transformer decoder works with a KV cache.

Finally, the process of computing $\log \Pr(\mathbf{y}|\mathbf{x})$ is summarized as follows:

1. We concatenate $\mathbf{x}$ and $\mathbf{y}$ into a sequence $[\mathbf{x},\mathbf{y}]$. For each position i' of this sequence, we perform the following steps.
   (a) We compute the embedding of the token at position i', and feed the resulting embedding as an initial representation into the stack of Transformer layers.
   (b) In each Transformer layer, we pass the input representation through the self-attention network first and then through an FFN. In the self-attention network, the input representation is transformed into $\mathbf{q}_{i'}$, $\mathbf{k}_{i'}$, and $\mathbf{v}_{i'}$. Then, we update the KV cache $(\mathbf{K},\mathbf{V})$ using $\mathbf{k}_{i'}$ and $\mathbf{v}_{i'}$ (see Eqs. (11.8-11.9)). Then, we compute the output of the attention model by attending $\mathbf{q}_{i'}$ to $(\mathbf{K},\mathbf{V})$ (see Eq. (11.7)).
   (c) If $i' > m$ (i.e., $i = i' - m \geq 0$), we take the output of the Transformer stack and compute the token prediction probability $\Pr(y_i|\mathbf{x},\mathbf{y}_{<i})$ via the softmax layer (see Eq. (11.5)).
2. When reaching the end of the sequence, we obtain $\log \Pr(\mathbf{y}|\mathbf{x})$ by summing $\log \Pr(y_i|\mathbf{x},\mathbf{y}_{<i})$ over $i \in [1,n]$ (see Eq. (11.4)).



(a) Updating the KV Cache at Position i'



(b) Updating the KV Cache at Position $i'+1$

Figure 11.2: Illustration of the KV cache. We update the KV cache at a position, perform the attention operation, and then move to the next position to repeat the process.

11.1.2 A Two-phase Framework

As we have seen, language modeling is a standard autoregressive process, where each token is generated one at a time, conditioned on the previous tokens. For Transformers, this requires the model to maintain a KV cache that stores past representations, and attend the newly generated representation to this KV cache. If we think of the model $\Pr(\mathbf{y}|\mathbf{x})$ from the perspective of computing the KV cache, it is natural to divide inference into two phases:

- **Prefilling**. The prefilling phase computes the KV cache for the input sequence $\mathbf{x}$. It is called prefilling because the model prepares and stores the key-value pairs for each token in the input before the actual inference begins. The process of prefilling in an LLM can be expressed as:

$$\mathrm{cache} = \mathrm{Dec}_{\mathrm{kv}}(\mathbf{x}), \tag{11.10}$$

  where $\mathrm{Dec}_{\mathrm{kv}}(\cdot)$ is the decoding network (i.e., the same as $\mathrm{Dec}(\cdot)$), but it returns the KV cache in self-attention instead of the output representations. cache is a list, given by:

$$\mathrm{cache} = \{\mathrm{cache}^1, \cdots, \mathrm{cache}^L\}, \tag{11.11}$$

  where cache^l represents the key-value pairs for the l-th layer.

- **Decoding**. The decoding phase continues generating tokens based on the KV cache, as illustrated in Figure 11.2. When a new token is input into the decoder, we update the KV cache in each layer by adding the new key-value pair. The updated cache is then used for self-attention computation. The token generation stops when some stopping criterion is met, such as when the generated token is the end symbol. The goal of decoding is to find the best predicted sequence, which is given by:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} \Pr(\mathbf{y}|\mathrm{cache}). \tag{11.12}$$

  Here we use $\Pr(\mathbf{y}|\mathrm{cache})$ instead of $\Pr(\mathbf{y}|\mathbf{x})$ to emphasize that the decoding process actually relies on the KV cache rather than $\mathbf{x}$.

The prefilling and decoding processes are illustrated in Figure 11.3. Note that both these processes are autoregressive. However, as shown in Table 11.1, they differ in several aspects, which lead to very different implementations in practice.

In essence, while the underlying model of prefilling is based on token prediction, it can be considered an encoding process. This is because our goal is not to generate tokens, but to build a context representation (i.e., the KV cache) for the subsequent steps in the decoding phase. In this sense, it is similar to BERT, where we encode the input sequence into a sequence of contextualized token representations. On the other hand, unlike BERT which generates bidirectional sequence representations, prefilling is based on standard language modeling tasks, and is thus unidirectional. Note that, since the entire sequence $\mathbf{x}$ is input into the model all at once, all queries can be packed together and the self-attention operation is performed on $\mathbf{x}$

	Prefilling	Decoding
Goal	Set up initial context $\mathbf{x}$.	Continue generating tokens $\mathbf{y}$ after the initial input.
All-at-once Visibility	Tokens in $\mathbf{x}$ are presented all at once.	Tokens in $\mathbf{y}$ are presented sequentially, that is, predicting a token requires waiting for the previous tokens to be predicted first.
Context Use	Build the context or encoded representation of the input.	Use the cached key-value pairs (from prefilling) to generate further tokens.
Resource Limitation	Compute-bound	Memory-bound
Computational Cost	High	Very High

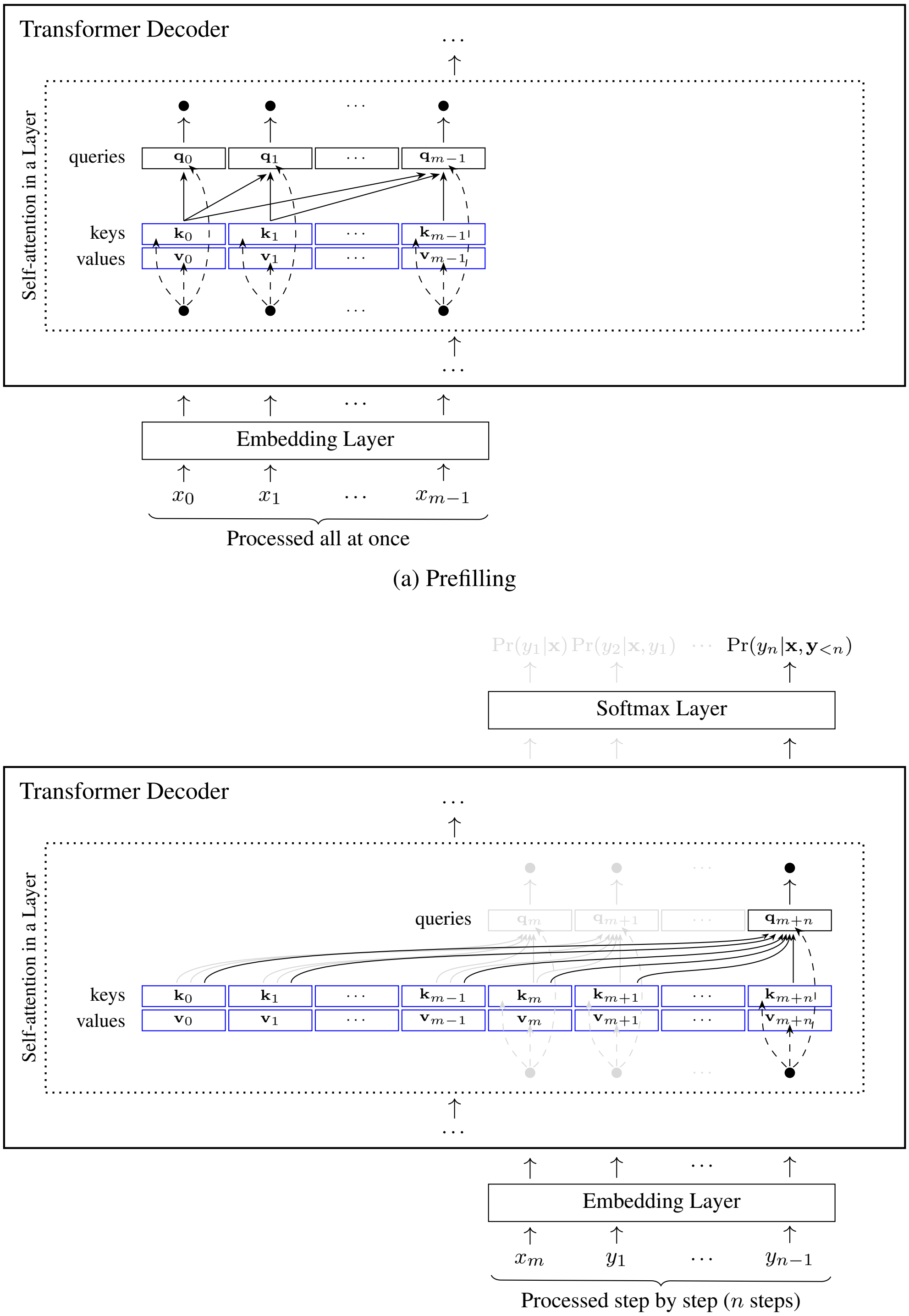
Table 11.1: Prefilling vs Decoding.

in parallel. Let $\mathbf{Q}$ be the queries that are packed into one matrix. The self-attention model in prefilling can be defined as:

$$\mathrm{Att}_{\mathrm{qkv}}(\mathbf{Q},\mathbf{K},\mathbf{V}) \quad = \quad \mathrm{softmax}(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d}}+\mathbf{Mask})\mathbf{V}, \tag{11.13}$$

where $\mathbf{Q},\mathbf{K},\mathbf{V} \in \mathbb{R}^{(m+1)\times d}$. $\mathbf{Mask} \in \mathbb{R}^{(m+1)\times(m+1)}$ is a mask that ensures that each token only attends to itself and the tokens that precede it in the sequence. It is represented by setting the values in the mask corresponding to future tokens to a large negative number, for example, for the query $\mathbf{q}_i$ and the key $\mathbf{k}_j$, we set the value of the entry (i,j) to $-\infty$ if $i<j$. One advantage of processing the sequence with a single self-attention computation is that we can make better use of the parallel computing capabilities of modern GPUs, and so speed up prefilling. In general, the prefilling process is considered compute-bound. This is because merging multiple computational operations into one operation reduces the number of data transfers and the performance bottleneck usually comes from the computational capacity rather than memory bandwidth.

Decoding is a standard left-to-right text generation process. The token sequence is generated autoregressively by predicting one token at a time based on the KV cache. Each time a new token is generated, we need to attend it to previous tokens, following Eq. (11.7). Therefore, the decoding process is memory-bound due to its frequent access to the KV cache. The cost of decoding grows significantly as more tokens are generated. In most cases, decoding is computationally more expensive than prefilling. Note that this is not just because, in decoding, the LLM generates tokens one by one and repeatedly updates the KV cache. As we will see in the following subsection, we may need to explore multiple different token sequences during decoding, which makes the problem more complex and increases its cost further.



(a) Prefilling

(b) Decoding (at the n-th step)

Figure 11.3: Illustration of the prefilling and decoding processes. In prefilling, the entire input sequence is processed together and the KV cache is filled. In decoding, the LLM generates the output sequence step by step based on the prefilled KV cache.

11.1.3 Decoding Algorithms

So far our discussion of LLM inference has primarily focused on the model computation problem, that is, how to compute $\Pr(\mathbf{y}|\mathbf{x})$. Now we turn to the discussion of the search problem. The problem can be stated as: given an LLM $\Pr(\mathbf{y}|\mathbf{x})$, how do we efficiently search for the best output sequence $\hat{\mathbf{y}}$ given the input sequence $\mathbf{x}$ (or the generated KV cache)? Naively, we can consider all of the output sequences, compute the prediction probability for each, and then select the output sequence having the highest probability. This method can guarantee the globally optimal solution, but direct exhaustive search is impractical for LLMs as the number of possible output sequences grows exponentially with the length of $\mathbf{y}$.

In practice, various heuristic search algorithms, such as greedy search and sampling-based search, are commonly employed to approximate the solution. Each of these methods offers trade-offs between search quality and computational efficiency. The search problem, therefore, becomes a balancing act between exploration and exploitation, where the goal is to find an efficient strategy that produces high-quality outputs without exploring the entire space.

Before giving a more detailed discussion of these methods, let us first informally define what a search space is and how it is represented. In LLM inference, we define a hypothesis as a tuple of input and output sequences. Since $\mathbf{x}$ is fixed during inference, we can simply consider each hypothesis as an output sequence. The search space, denoted by $\mathcal{Y}$, is then the set of all possible hypotheses (i.e., output sequences) that the model can generate. The search problem for LLM inference can be re-expressed as:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y} \in \mathcal{Y}} \Pr(\mathbf{y}|\mathbf{x}). \tag{11.14}$$

In NLP, $\mathcal{Y}$ is commonly represented in a tree data structure to facilitate search. Figure 11.4 shows an example of the search tree resulting from a small vocabulary. In this example, a node represents a prefix subsequence that can be shared by many sequences. The search starts with the root of the tree, which can be regarded as the beginning of all sequences that can be generated[1]. Each child node extends the prefix of its parent node by adding one token from the vocabulary to the sequence, along with the probability of predicting the token given the prefix. This process continues as each node further branches out into additional child nodes, each representing a new possible extension of the sequence with another token. The search tree thus grows deeper and wider, representing an ever-increasing number of potential sequences as more tokens are appended. This structure allows us to efficiently traverse through possible sequences, evaluating each in terms of the log-probability accumulated over the path from the root to that node. For example, in Figure 11.4, the path from the root to the node 17 corresponds to the output sequence "*Cats are playful.*". The prediction log-probability $\log \Pr(\mathbf{y}|\mathbf{x})$ is the sum of the log-probabilities of all the nodes on this path.

In general, the search tree is organized as levels, where each level consists of all nodes that are the same distance from the root node. Thus, a breadth-first search over the tree essentially performs left-to-right generation of tokens. Nodes in the same level correspond to sequences

[1] Here, since the predictions in LLMs are based on $\mathbf{x}$, we can think of the root as a representation of $\mathbf{x}$.

Path: node $0 \rightarrow$ node $3 \rightarrow$ node $9 \rightarrow$ node $11 \rightarrow$ node 17
Output: cats are playful.

Probability:
node $0 \rightarrow 0$
node $3 \rightarrow \log \Pr(\text{"cats"}|\mathbf{x})$
node $9 \rightarrow \log \Pr(\text{"are"}|\mathbf{x}, \text{"cats"})$
node $11 \rightarrow \log \Pr(\text{"playful"}|\mathbf{x}, \text{"cats are"})$
node $17 \rightarrow \log \Pr(\text{"."}|\mathbf{x}, \text{"cats are playful"})$

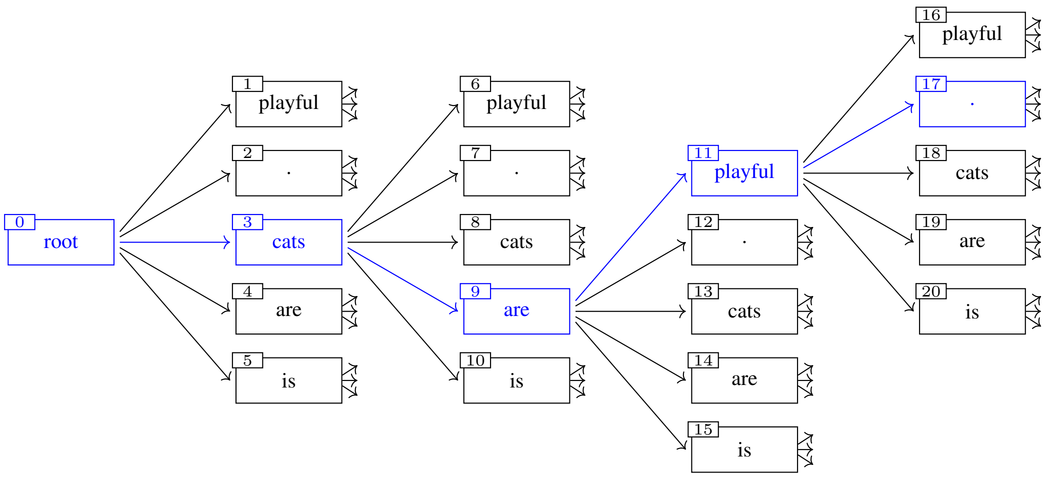


Figure 11.4: A search tree for decoding. At each node, we expand the tree by considering all possible tokens, each leading to a new node representing a potential continuation of the text. Here we highlight a path through nodes 0, 3, 9, 11, and 17. The path represents the output sequence "*cats are playful.*", whose log-probability can be computed by accumulating the log-probabilities of these nodes.

of the same length. As the search progresses, new tokens are appended to these sequences, expanding them incrementally.

Let Y_i be the set of the sequences that the LLM generates at step i. Y_i can be obtained by expanding each sequence in Y_{i-1} with all possible next tokens in the vocabulary V, given in the following recursive form:

$$Y_i = Y_{i-1} \times V, \tag{11.15}$$

where $Y_{i-1} \times V$ denotes the Cartesian product of Y_{i-1} and V (i.e., each sequence in Y_{i-1} is concatenated with each token in V). Note that if a sequence in Y_{i-1} is complete (e.g., ending with the $\langle \text{EOS} \rangle$ token), it will not be expanded any further. Let $\Psi(Y_i)$ be the set of all complete sequences in Y_i. Then, the search space can be expressed as:

$$\mathcal{Y} = \Psi(Y_1) \bigcup \Psi(Y_2) \bigcup \cdots \bigcup \Psi(Y_{n_{\max}}), \tag{11.16}$$

where $n_{\max}$ is the maximum length of a sequence.

Most decoding algorithms follow this level-by-level search process. However, $\mathcal{Y}$ consists of an exponentially large number of sequences, and a direct search in such a vast space is computationally infeasible. Therefore, practical decoding algorithms often rely on strategies to prune the search space and avoid exploring low-quality sequences. For example, at each decoding step, Y_i can be obtained in the following way:

$$Y_i = \mathrm{Prune}(Y_{i-1} \times V), \tag{11.17}$$

where $\mathrm{Prune}(\cdot)$ is a function that selectively removes sequences less likely to result in high-quality outcomes. In general, we expect that $|Y_i| << |Y_{i-1}| \cdot |V|$. Thus we can drastically reduce the number of sequences under consideration at each step, ensuring that the computational load does not grow exponentially with the sequence length.

Next, we will introduce these decoding algorithms. Some of them have already been discussed in sequence-to-sequence models (see Chapter 5), while others are more commonly used in LLMs.

1. Greedy Decoding

Greedy search (or greedy decoding) is one of the most widely used decoding methods in NLP, particularly in text generation tasks like machine translation. The idea behind greedy search is straightforward: at each step in generation, it selects the next token that has the highest prediction probability. For each sequence $\mathbf{y} = y_1 \cdots y_i \in Y_{i-1} \times V$, we can evaluate it using $\log \Pr(\mathbf{y}|\mathbf{x})$. This log-probability can be easily computed by noting that:

$$\begin{aligned}
\log \Pr(\mathbf{y}|\mathbf{x}) &= \log \Pr(y_1 \cdots y_i|\mathbf{x}) \\
&= \underbrace{\log \Pr(\mathbf{y}_{<i}|\mathbf{x})}_{\text{accumulated up to the parent node}} + \underbrace{\log \Pr(y_i|\mathbf{x}, \mathbf{y}_{<i})}_{\text{newly computed for the current node}},
\end{aligned} \tag{11.18}$$

Here the first term is the sum of the log-probabilities of the path from the root to the parent node, which has been computed in the previous decoding steps. At step i, we only need to compute the second term which is the standard token prediction log-probability produced by the LLM.

The "best" token at step i is then chosen as:

$$\begin{aligned}
y_i^{\text{top1}} &= \arg\max_{y_i \in V} \log \Pr(y_1 \cdots y_i|\mathbf{x}) \\
&= \arg\max_{y_i \in V} \big[\underbrace{\log \Pr(\mathbf{y}_{<i}|\mathbf{x})}_{\text{fixed wrt. } y_i} + \log \Pr(y_i|\mathbf{x}, \mathbf{y}_{<i})\big] \\
&= \arg\max_{y_i \in V} \log \Pr(y_i|\mathbf{x}, \mathbf{y}_{<i}).
\end{aligned} \tag{11.19}$$

Thus, the "best" sequence generated up to step i is given by:

$$\mathbf{y}^{\text{top1}} = y_1 \cdots y_{i-1} y_i^{\text{top1}}. \tag{11.20}$$

Finally, Y_i contains only this sequence:

$$Y_i = \{\mathbf{y}^{\mathrm{top1}}\}. \tag{11.21}$$

The greedy choice in one decoding step is illustrated in Figure 11.5 (a). Greedy search offers computational efficiency and simplicity in implementation for LLM inference. Its primary disadvantage, however, lies in its suboptimal nature — high-quality sequences are likely pruned at early stages of decoding. Therefore, greedy search is appealing for tasks that demand speed and simplicity. For tasks that require better search results, alternative strategies such as beam search, which explores multiple potential paths simultaneously, are preferable.

2. Beam Decoding

Beam search (or beam decoding) is a natural extension of greedy search. Instead of selecting the single most probable token at each step, beam search maintains a fixed number of the best candidates at each step, known as the "beam width". See Figure 11.5 (b) for an illustration of beam search.

Let K be the beam width. Given a parent node, which corresponds to the prefix $y_1 \cdots y_{i-1}$, we can select the top-K next tokens by:

$$\{y_i^{\mathrm{top1}}, \cdots, y_i^{\mathrm{top}K}\} = \underset{y_i \in V}{\mathrm{argTopK}} \ \Pr(y_i|\mathbf{x}, \mathbf{y}_{<i}), \tag{11.22}$$

where argTopK is a function that ranks the prediction probabilities of all possible next tokens and selects the top K candidates. Given these tokens, the top-K sequences for step i are given by:

$$\mathbf{y}^{\mathrm{top1}} = y_1 \cdots y_{i-1} y_i^{\mathrm{top1}} \tag{11.23}$$

$$\vdots$$

$$\mathbf{y}^{\mathrm{top}K} = y_1 \cdots y_{i-1} y_i^{\mathrm{top}K}. \tag{11.24}$$

Then, we can define Y_i as:

$$Y_i = \{\mathbf{y}^{\mathrm{top1}}, \cdots, \mathbf{y}^{\mathrm{top}K}\}. \tag{11.25}$$

We can adjust the beam width K to balance search efficiency and accuracy. But a very large beam width might not be helpful. In many practical applications, selecting a relatively small number for K, such as $K = 2$ or $K = 4$, is often sufficient to achieve satisfactory performance in LLM inference.

3. Sampling-based Decoding

Both greedy and beam search generate deterministic outputs, that is, given an LLM, the output of the model will always be the same every time it processes the same input. The deterministic nature of greedy and beam search ensures predictability and reliability in applications where

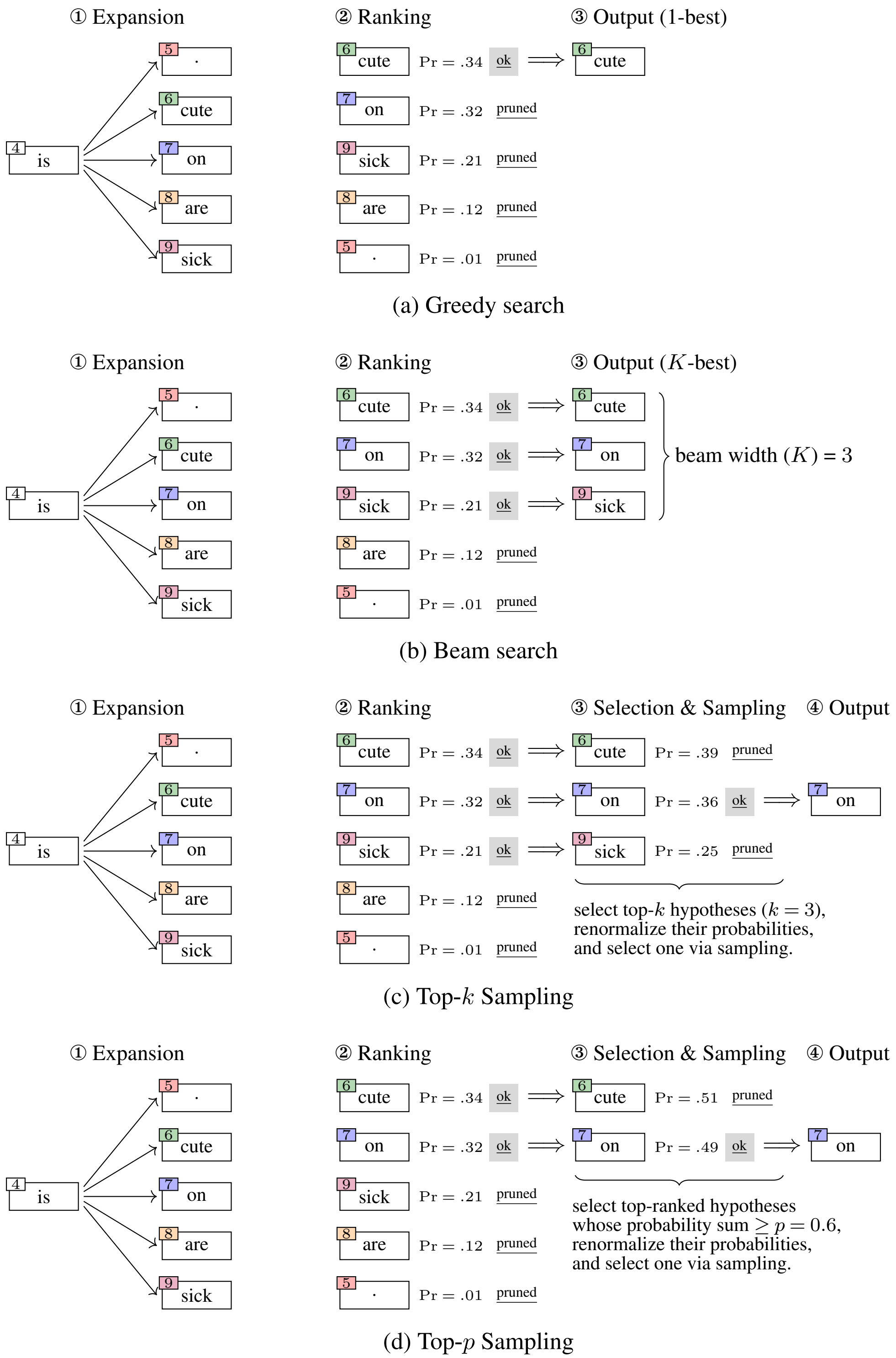


Figure 11.5: Illustrations of greedy decoding, beam decoding, top-k decoding and top-p decoding methods (in one decoding step).

consistent outcomes are critical, such as in formal document generation, where varying outputs could cause confusion or errors. On the other hand, one disadvantage of these methods is the lack of diversity and flexibility. For example, in creative tasks like story generation or conversational agents, generic or repetitive outputs generated by deterministic systems are often less engaging.

To add variation into LLM outputs, we can use sampling-based decoding methods. There are two commonly used methods.

- **Top-k Sampling**. This method selects the next token from the top-k most likely candidates at each step of the generation process [Fan et al., 2018]. Let $\overline{V}_i$ be the selection pool for top-k sampling. We can define it as:

$$\overline{V}_i = \{y_i^{\mathrm{top1}}, \cdots, y_i^{\mathrm{top}k}\}, \tag{11.26}$$

where $\{y_i^{\mathrm{top1}}, \cdots, y_i^{\mathrm{top}k}\}$ are the top-k tokens selected based on their prediction probabilities (see Eq. (11.22)). Once the selection pool is determined, we recompute the prediction probability distribution over $\overline{V}_i$. One of the simplest ways to do this is to renormalize their probabilities:

$$\overline{\Pr}(y_i|\mathbf{x}, \mathbf{y}_{<i}) = \frac{\Pr(y_i|\mathbf{x}, \mathbf{y}_{<i})}{\sum_{y_j \in \overline{V}_i} \Pr(y_j|\mathbf{x}, \mathbf{y}_{<i})}. \tag{11.27}$$

Alternatively, we can calculate the distribution by using the softmax function:

$$\overline{\Pr}(y_i|\mathbf{x}, \mathbf{y}_{<i}) = \frac{\exp(u_{y_i})}{\sum_{y_j \in \overline{V}_i} \exp(u_{y_j})}, \tag{11.28}$$

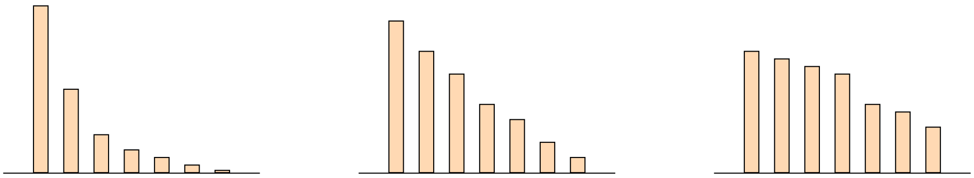
where u_{y_i} is the logit for token y_i. Then, we sample a token $\bar{y}_i$ from this distribution:

$$\bar{y}_i \sim \overline{\Pr}(y_i|\mathbf{x}, \mathbf{y}_{<i}). \tag{11.29}$$

The corresponding sequence is $\bar{\mathbf{y}} = y_1 \cdots y_{i-1}\bar{y}_i$, and Y_i is given by:

$$Y_i = \{\bar{\mathbf{y}}\}. \tag{11.30}$$

- **Top-p Sampling**. This sampling method, also known as **nucleus sampling**, follows a procedure similar to that of top-k sampling. Instead of drawing from a fixed size candidate pool, it selects the next token from the smallest set of tokens that together have a cumulative probability higher than a predefined threshold p [Holtzman et al., 2020b]. In this way we prevent the prediction from choosing from low-probability tokens in the long tail that could lead to incoherent or nonsensical outputs. To obtain the candidate pool in the top-p sampling method, we can sort all tokens by their predicted probabilities. Then, starting with the token with the highest probability, we continue to add tokens to the candidate pool until the cumulative probability of the tokens in the pool reaches or exceeds p (we denote the size of the candidate pool at this point as k_p). The candidate



(a) $\beta = 0.1$ (b) $\beta = 0.8$ (c) $\beta = 2.0$

Figure 11.6: Histogram estimates of the distributions generated by the softmax function with different values of the temperature parameter β.

pool can then be expressed as:

$$\overline{V}_i = \{y_i^{\text{top}1}, \cdots, y_i^{\text{top}k_p}\}. \tag{11.31}$$

The subsequent steps, such as the renormalization of the distribution and sampling, are the same as in the top-k sampling method (see Eqs.(11.27-11.30)).

See Figure 11.5 (c-d) for illustrations of the top-k and top-p sampling methods. By limiting the choices to a smaller set of high-probability tokens, these methods strike a balance between randomness and coherence. They allow for more diverse outputs while still maintaining a reasonable level of relevance and fluency. However, the value of k or p must be carefully chosen: if k or p is too small, the output may still be overly deterministic (more like greedy decoding), and if k or p is too large, the LLM might produce degenerate outputs.

In order to further control the randomness of the token selection process, the renormalized distribution $\overline{\Pr}(\cdot)$ is typically obtained by using the softmax function with the temperature parameter, given by:

$$\overline{\Pr}(y_i|\mathbf{x}, \mathbf{y}_{<i}) = \frac{\exp(u_{y_i}/\beta)}{\sum_{y_j \in \overline{V}_i} \exp(u_{y_j}/\beta)}, \tag{11.32}$$

Here β is a temperature parameter that controls the sharpness of the probability distribution derived from logits. In Figure 11.6, we show simple examples involving distributions generated by the above function with different temperatures. When the temperature is set to a higher value, the resulting probability distribution becomes more uniform, as the differences between the logits are diminished. This means that each token in the candidate pool has a more equal chance of being selected, leading to greater diversity in the generated output. By contrast, when the temperature is set to a lower value, the distribution becomes sharper, making the high-probability tokens even more likely to be chosen, which often results in more deterministic outputs. For example, if we set p to 1 and β to a very small number (approaching zero), the top-p sampling method will become equivalent to the greedy search method.

4. Decoding with Penalty Terms

One common improvement to decoding methods in text generation is to modify the search objective. For example, one can replace maximum a posteriori (MAP) decoding with minimum

Bayes risk (MBR) decoding [Kumar and Byrne, 2004b], where the focus shifts from selecting the single most probable output to choosing an output that minimizes the expected risk over a distribution of possible outputs. More details on MBR decoding can be found in Chapter 5. Here we explore methods that incorporate penalty terms into decoding. These methods offer a simple but effective way to make decoding more controllable.

Recall from Eq. (11.14) that the goal of decoding is to maximize the likelihood of the output sequence. With penalty terms, the objective is extended to include additional factors that penalize or reward certain behaviors in the generated text. A general form of the new objective is given by:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}\in\mathcal{Y}} \left[\Pr(\mathbf{y}|\mathbf{x}) - \lambda \cdot \mathrm{Penalty}(\mathbf{x},\mathbf{y})\right], \tag{11.33}$$

where $\mathrm{Penalty}(\mathbf{x},\mathbf{y})$ is a function that quantifies the degree to which the generated sequence $\mathbf{y}$ violates certain constraints or exhibits undesirable behaviors given the input $\mathbf{x}$. The design of $\mathrm{Penalty}(\cdot)$ is very flexible, thus allowing us to incorporate a wide range of constraints or prior knowledge into it. Below, we present some common types of penalty functions.

- **Repetition Penalty**. A repetition penalty discourages the model from generating repetitive or redundant text. The penalty function might measure the frequency of repeated tokens or phrases in the generated sequence and impose a penalty proportional to their occurrence.
- **Length Penalty**. A length penalty ensures that the generated sequence adheres to a desired length. For example, in text summarization tasks, the penalty function could penalize outputs that are too short or too long.
- **Diversity Penalty**. A diversity penalty promotes variation in the generated text. For example, in beam search, we can measure the similarity between generated hypotheses, and encourage the model to explore different hypotheses.
- **Constraint-based Penalty**. A constraint-based penalty enforces specific constraints related to the content or style of the generated text. For example, in machine translation, the penalty function could penalize outputs that deviate from a desired tone or terminology.

In general, we can consider $\mathrm{Penalty}(\mathbf{x},\mathbf{y})$ as a function that defines the cost of generating the surface form of the output sequence $\mathbf{y}$ given the input sequence $\mathbf{x}$. Alternatively, this function can be defined to assess the hidden states of an LLM when generating $\mathbf{y}$. For example, Su et al. [2022] develop a penalty term that calculates the maximum distance between the representation of the predicted token and the representations of the previously generated tokens. Therefore, the search objective will penalize degenerated outputs, such as texts with many repetitions.

The method described in Eq. (11.33) is general and can be easily adapted to different search algorithms. For example, in greedy search, we can keep the single sequence that maximizes $\Pr(\mathbf{y}|\mathbf{x}) - \lambda \cdot \mathrm{Penalty}(\mathbf{x},\mathbf{y})$ at each decoding step; in sampling-based search, we can rank and select the top-ranked sequences based on $\Pr(\mathbf{y}|\mathbf{x}) - \lambda \cdot \text{Penalty}(\mathbf{x}, \mathbf{y})$ to form the candidate pool.

5. Speculative Decoding

Speculative decoding stems from the concept of **speculative execution**, where a system makes educated guesses about future actions and performs them in advance. If the guess is correct, the results are immediately available, which speeds up processing. In the case of LLM inference, suppose we have two models. One is a smaller, faster model (called draft model), and the other is the full, more accurate model (called verification model). These two models represent two baselines in LLM inference: the draft model is efficient but not very accurate; the verification model is usually the one we want to run, but it is very slow. Given a prefix, we first use the draft model to speculatively predict a sequence of likely future tokens. This is a standard autoregressive decoding process, but it is still fast in practice due to the high efficiency of the draft model. Then, the verification model evaluates the speculated tokens in parallel. It checks whether the predicted tokens are correct or need to be adjusted. Note that, since we can deal with these tokens all at once, the verification can be done in a single step for all the tokens simultaneously, rather than in a token-by-token manner. If the speculated tokens are correct, they are accepted, and the process continues with the next set of tokens. If they are incorrect, the incorrect speculations are discarded, and the verification model is used to generate the correct tokens.

To be more specific, let us see the speculative decoding method presented in Leviathan et al. [2023]'s work. In this method, the draft model is a small language model, denoted by $\Pr_q(y_i|\mathbf{x}, \mathbf{y}_{<i})$, while the verification model is a normal LLM, denoted by $\Pr_p(y_i|\mathbf{x}, \mathbf{y}_{<i})$. The goal is that, given a prefix, we use the draft model to autoregressively predict up to τ tokens. The verification model is then employed to generate the last token at the point where errors begin to occur in the speculative predictions. Figure 11.7 illustrates one step in this decoding process.

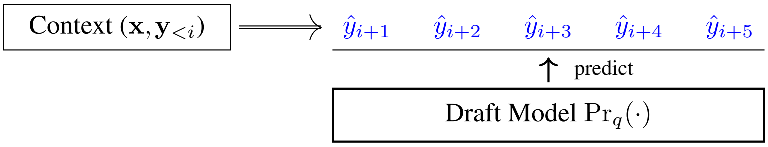
The speculative decoding algorithm can be summarized as follows.

- Given the prefix $[\mathbf{x}, \mathbf{y}_{\leq i}]$, we use the draft model to predict the next τ consecutive tokens, denoted by $\{\hat{y}_{i+1}, \cdots, \hat{y}_{i+\tau}\}$. This is a token-by-token generation process, given by:

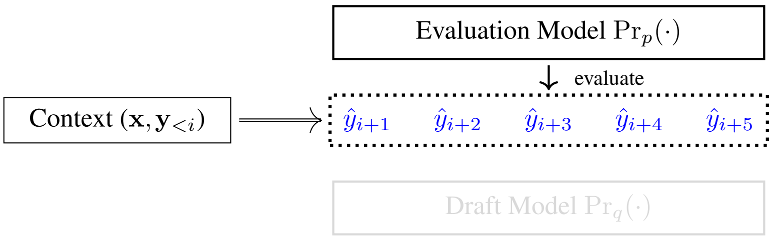
$$\hat{y}_{i+t} = \arg\max_{y_{i+t}} \Pr_q(y_{i+t}|\mathbf{x}, \mathbf{y}_{\leq i}, \hat{y}_{i+1} \cdots \hat{y}_{i+t-1}). \tag{11.34}$$

- We evaluate $\{\hat{y}_{i+1}, \cdots, \hat{y}_{i+\tau}\}$ using the verification model, that is, we compute $\{\Pr_p(\hat{y}_{i+1}|\mathbf{x}, \mathbf{y}_{\leq i}), \cdots, \Pr_p(\hat{y}_{i+\tau}|\mathbf{x}, \mathbf{y}_{\leq i}, \hat{y}_{i+1} \cdots \hat{y}_{i+\tau-1})\}$. Note that we can compute these probabilities in parallel, and so this verification step is efficient.

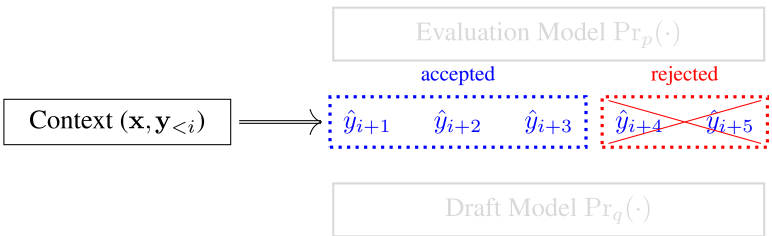
- We determine the maximum number of accepted speculated tokens. In order to keep the notation uncluttered, we denote $\Pr_q(\hat{y}_{i+t}|\mathbf{x}, \mathbf{y}_{\leq i}, \hat{y}_{i+1} \cdots \hat{y}_{i+t-1})$ and $\Pr_p(\hat{y}_{i+t}|\mathbf{x}, \mathbf{y}_{\leq i}, \hat{y}_{i+1} \cdots \hat{y}_{i+t-1})$ simply by $q(\hat{y}_{i+t})$ and $p(\hat{y}_{i+t})$, respectively. We then define that, if $q(\hat{y}_{i+t}) \leq p(\hat{y}_{i+t})$, then we accept this speculation. By contrast, if $q(\hat{y}_{i+t}) > p(\hat{y}_{i+t})$, we reject this speculation with probability $1 - \frac{p(\hat{y}_{i+t})}{q(\hat{y}_{i+t})}$. Starting from $\hat{y}_{i+1}$, the maximum number of accepted



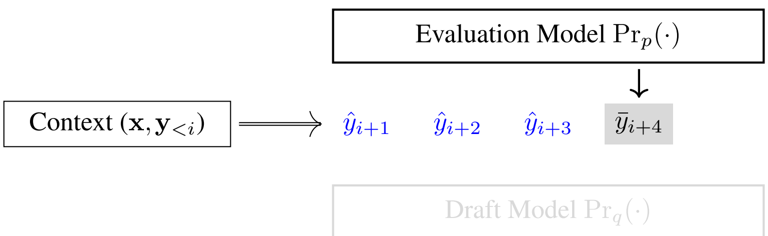
(a) Predict the next τ tokens given the context using the draft model ($\tau = 5$)



(b) Evaluate the predicted tokens using the evaluation model



(c) Determine the number of accepted tokens



(d) Predict a new token following the accepted tokens using the evaluation model

Figure 11.7: Illustration of one step of speculative decoding. The goal is to predict as many next tokens as possible using the draft model. There are four sub-steps. Given the context, we first use the draft model to predict the next τ tokens. Then, we evaluate these predictions in parallel using the evaluation model. Next, we determine the maximum number of predicted tokens that can be accepted. Finally, we use the evaluation model to predict a new token following these accepted tokens.

consecutive speculated tokens is defined as:

$$n_a = \min\left\{t-1 | 1 \le t \le \tau, r_t > \frac{p(\hat{y}_{i+t})}{q(\hat{y}_{i+t})}\right\}, \tag{11.35}$$

where r_t is a variable drawn from the uniform distribution $U(0,1)$.

- Given n_a, we keep the speculated tokens $\{\hat{y}_{i+1}, \cdots, \hat{y}_{i+n_a}\}$. We then use the verification model to make a new prediction at $i + n_a + 1$:

$$\bar{y}_{i+n_a+1} = \arg\max_{y_{i+n_a+1}} \Pr_p(y_{i+n_a+1}|\mathbf{x}, \mathbf{y}_{\le i}, \hat{y}_{i+1} \cdots \hat{y}_{i+n_s}). \tag{11.36}$$

- Above, we have described one step of speculative decoding. The result sequence (including both the context and predicted tokens) is illustrated as follows

$$\underbrace{[\mathbf{x},\ \mathbf{y}_{<i}]}_{\text{Context}}\ \underbrace{\hat{y}_{i+1} \ldots \hat{y}_{i+n_a}}_{\substack{n_a \text{ tokens} \\ \text{predicted using} \\ \text{the draft model}}}\ \underbrace{\bar{y}_{i+n_a+1}}_{\substack{\text{One token} \\ \text{predicted using} \\ \text{the verification model}}}$$

Once we have finished this step, we add the predicted tokens $\{\hat{y}_{i+1}, \cdots, \hat{y}_{i+n_a}, \bar{y}_{i+n_a+1}\}$ to the context, and repeat the above process.

In practice, we usually wish to use a smaller draft model so that predicting $\{\hat{y}_{i+1}, \cdots, \hat{y}_{i+n_a}\}$ would be computationally cheaper. But a very small draft model is less accurate and can result in smaller n_a. We therefore need to carefully select the draft model to make the trade-off between the computational efficiency and accuracy.

6. Stopping Criteria

Stopping criteria are a critical component of LLM inference. They typically involve rules or conditions that specify when the model should stop generating text during decoding. Most LLMs are trained to generate an end-of-sequence token (e.g., $\langle\text{EOS}\rangle$ or $\langle/s\rangle$) to signal the end of the generated text. So one of the simplest strategies is that the generation process stops when this token is produced. For beam search, which explores multiple hypotheses simultaneously, the process can continue until a given number of complete sequences have been generated.

In practical applications, it will generally be undesirable to generate very long sequences, and so we need to reduce the decoding cost and unnecessary verbosity. One commonly-used stopping criterion is the maximum length of the output. The model stops generating text once it has produced a predetermined number of tokens. Alternatively, we can stop the decoding based on the real cost, such as the computational resources or time constraints. For example, in real-time applications like chatbots, decoding may need to stop after a certain time limit to ensure responsiveness.

Another approach is to design stopping criteria based on the behavior of LLMs. For example, decoding can be stopped if the probability of predicting the next token falls below a certain threshold. In addition to probability-based stopping, a repetition detection module can be implemented to trigger the model to stop if it begins repeating tokens or phrases beyond a predefined limit. This helps prevent redundant or incoherent outputs.

11.1.4 Evaluation Metrics for LLM Inference

Evaluating the performance of LLMs during inference involves a variety of metrics to assess how well these models meet desired standards, such as accuracy, robustness, usability, and efficiency. As with most NLP systems, we can evaluate LLMs using accuracy-based metrics, such as perplexity and F1 score. We can also examine their robustness by testing how well they handle ambiguous or challenging inputs, including adversarial, perturbed, or out-of-distribution data. Additionally, usability can be assessed by measuring how well the generated outputs align with user expectations in terms of fluency, coherence, relevance, and diversity. Human evaluators can rate the naturalness of the text or assess whether the responses are contextually appropriate and logically consistent. Ethical and fairness metrics can also be included to ensure LLMs avoid perpetuating biases or generating harmful content.

All of the evaluation metrics mentioned above essentially focus on assessing the quality of the outputs. Given the high cost of deploying and applying LLMs, efficiency metrics are also very important for practitioners. Below are some commonly used efficiency metrics [Nvidia, 2025]:

- **Request Latency**. This metric measures the total time taken from when a request is sent to the LLM until the complete response is received. This includes the time taken for data transmission, processing by the model, and the return of the output to the user.
- **Throughput**. It refers to the number of tokens or requests the model can process per second.
- **Time to First Token** (TTFT). This metric measures the time it takes from the beginning of a request being sent to the generation of the first token of the response. If data transmission does not consume too much time, then TTFT is mainly the time for prefilling and predicting the first token.
- **Inter-token Latency** (ITL). This metric refers to the time taken to generate each subsequent token after the first one. It reflects the efficiency of the decoding process.
- **Tokens Per Second** (TPS). This metric quantifies the number of tokens that the model can generate per second.
- **Resource Utilization**. This involves measuring the computational resource usage (e.g., CPU and GPU utilization) and memory consumption of the model during inference.

In addition to these metrics, energy efficiency and cost efficiency are practical considerations for deploying LLMs at scale. Energy efficiency measures the amount of electrical power consumed by the model during inference. Cost efficiency, on the other hand, evaluates the total expenses related to deploying and maintaining the model.

In general, choosing the right evaluation metrics depends on the specific task and application. While quality-focused metrics are essential for assessing LLMs, efficiency metrics are equally crucial for their effective deployment in real-world applications. A comprehensive evaluation framework should include both sets of metrics to accurately estimate an LLM's performance and practicality.

11.2 Efficient Inference Techniques

In practical applications, we often wish a system to be as efficient as possible. For LLM inference, this typically involves two types of improvements: reducing memory requirements and accelerating the system. For example, we can modify the Transformer architecture to avoid memory explosion when processing very long input sequences. Another example is that we can compress input sequences to reduce computational overhead while preserving their semantic information. In addition, techniques like quantization and pruning can be employed to further optimize memory usage and inference speed.

Efficient inference is a wide-ranging topic that overlaps with several sub-fields of LLMs, such as architecture design and model compression. Most of these topics have been covered in previous chapters. For example, in Chapter 6, we discussed efficient Transformer architectures; in Chapter 8, we discussed long-context LLMs; and in Chapter 9, we discussed prompt compression methods for reducing prompt length. In this section, we focus on techniques that are commonly used in LLM deployment and serving.

11.2.1 More Caching

In real-world applications, it is common practice to store frequent requests and their corresponding responses in a cache. When a new request hits the cache, the system can retrieve the response directly from the cache instead of recomputing the result. One straightforward implementation is a key-value datastore (e.g., a hash table) that maps input sequences to their LLM-generated output sequences. In the simplest case, we can collect frequent queries, generate their responses using the LLM, and store these query-response pairs in the datastore. This creates a basic sequence-level caching mechanism that allows the system to bypass LLM computation when the input sequence exactly matches a cached query.

A straightforward extension of the caching mechanism is to cache prefixes and their corresponding hidden states. Given an input sequence $\mathbf{x}$ in a dataset $\mathcal{D}$, we can process it as in the standard prefilling phase. Thus, we obtain a sequence of prefixes and their corresponding KV cache states:

$$
\begin{aligned}
x_0 \ (\mathbf{x}_{<1}) \quad &\Rightarrow \quad \mathrm{cache}_{<1} \\
x_0 x_1 \ (\mathbf{x}_{<2}) \quad &\Rightarrow \quad \mathrm{cache}_{<2} \\
&\cdots \\
x_0 x_1 \cdots x_{m-1} \ (\mathbf{x}_{<m}) \quad &\Rightarrow \quad \mathrm{cache}_{<m}
\end{aligned}
$$

where $\mathrm{cache}_{<i}$ denotes the KV cache for the prefix $\mathbf{x}_{<i}$ (see also Eq. (11.10)). All these mappings can be stored in the prefix cache for efficient reuse.

When processing a new sequence that shares a common prefix with a previously seen sequence in $\mathcal{D}$, we can load the corresponding cached hidden states instead of recomputing them. Specifically, if a new input $\mathbf{x}'$ has $\mathbf{x}_{<k}$ (i.e., $\mathbf{x}'_{<k} = \mathbf{x}_{<k}$ for some $k \leq m$), we can initialize the KV cache with $\mathrm{cache}_{<k}$ and only compute the hidden states for the remaining tokens $\mathbf{x}'_{\geq k}$.

As usual, we can maintain a key-value datastore that maps frequently encountered prefixes to their precomputed KV caches. The lookup can be performed using a hash of the prefix tokens, allowing constant-time access to the cached states. Care must be taken to manage memory usage, as storing all possible prefixes may be infeasible for large datasets. Practical systems often employ least recently used (LRU) caching methods or other strategies to balance between computational savings and memory constraints.

11.2.2 Batching

Batching in LLM inference refers to the process of processing multiple input sequences simultaneously as a group (called a batch) rather than one at a time. Because modern GPUs excel at parallel processing, batching allows them to compute multiple sequences in a single forward pass, keeping the hardware fully occupied. Therefore, when serving LLMs at scale, batching is important for improving computational efficiency and maximizing hardware utilization[2].

To illustrate the idea of batching, Figure 11.8 (a-b) show simple examples with batch sizes of 1 and 4, respectively. When using a batch size of 1 (i.e., without batching), the GPU processes one input sequence at a time. Thus, the processing is sequential: the next sequence must wait for the current computation to finish. By contrast, when using a batch size of 4, the GPU can process four sequences simultaneously in a single forward pass. As the input sequences vary in length, we need to standardize their length using padding techniques. Here we use left padding, which adds dummy tokens to the beginnings of short sequences, so all the sequences in the batch would have the same length for prefilling. For decoding, tokens are generated simultaneously for all these sequences, and the generation process continues until the longest sequence reaches completion.

The above examples imply a trade-off between throughput and latency, which is a very important consideration in designing and implementing LLM inference systems. If we choose a smaller batch size, the latency would be lower, as fewer tokens need to be processed in a single run of inference. Imagine that we have only one sequence. The result becomes available immediately after generation completes, with no additional computational overhead. However, this low-latency advantage comes at the cost of underutilizing parallel computing resources, as the parallelism of GPUs remains largely idle during sequential processing. On the other hand, if we use a larger batch, we can make better use of the parallelism, as GPUs can be occupied by large-scale matrix computations. As a result, we can process more tokens in the same period of time and the throughput is improved. However, since the result is obtained only when the last token in the batch is predicted, the latency would be higher.

In practice, we usually prefer to use a slightly larger batch, but try to fill the batch with sequences of similar lengths to reduce the number of padding tokens and improve device utilization. For example, we can group the incoming user requests in a short period of time into

[2] See https://docs.nvidia.com/deeplearning/performance/dl-performance-gpu-background/index.html#understand-perf for a simple evaluation.

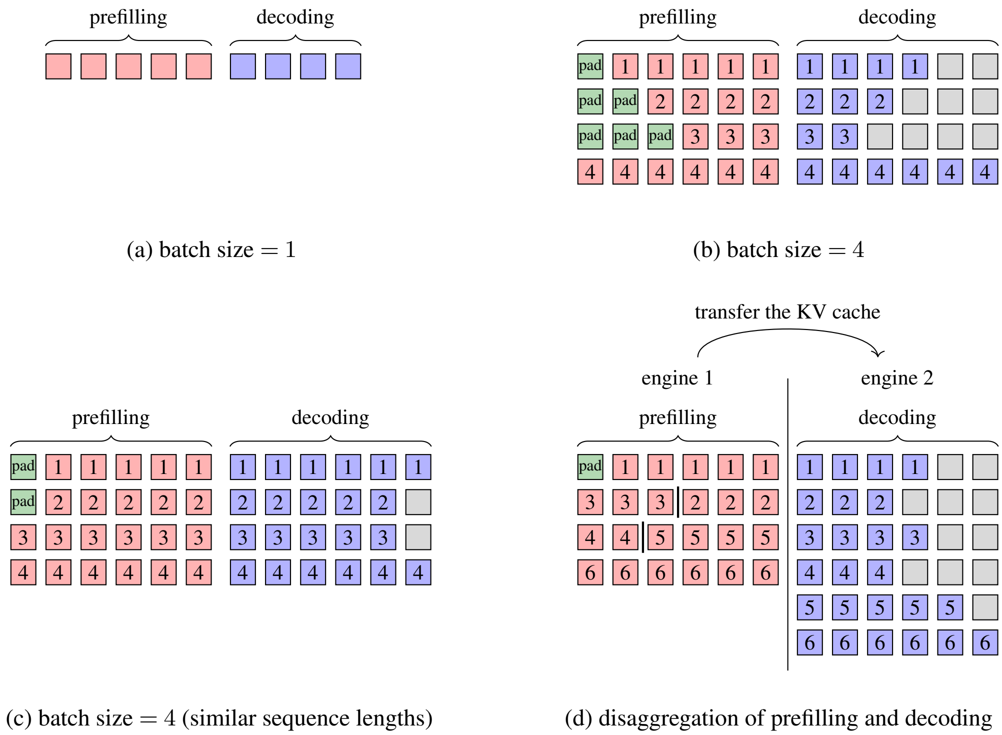


Figure 11.8: Illustrations of basic batching methods. We use a 2D layout to illustrate the batch, where each square represents a token. Red squares indicate tokens in the prefilling stage, blue squares represent tokens in the decoding stage, green squares denote padding tokens, and gray squares correspond to meaningless tokens. Subfigures (a) and (b) compare the cases where the batch size is 1 and 4, respectively. Subfigure (c) shows the strategy of grouping sequences with similar lengths into the same batch. Subfigure (d) illustrates the disaggregation of prefilling and decoding. In this approach, we can make better use of the parallelism of GPUs by concatenating multiple short sequences into a single long sequence for joint processing. This allows us to maximize the number of tokens processed in a batch while minimizing the number of padding tokens. However, as a trade-off, we need to copy the KV cache to the decoding engine and reorganize it after the prefilling phase, which introduces additional data transfer overhead.

buckets, each of which contains sequences with similar lengths. Then, we can fill the batch with sequences in the same bucket, so that we can minimize wasted computational resources, as illustrated in Figure 11.8 (c).

Another approach to implementing batching in LLMs is to disaggregate the prefilling and decoding processes [Wu et al., 2023a; Patel et al., 2024; Zhong et al., 2024]. For example, we can perform prefilling on one GPU, and perform decoding on another GPU. One advantage of disaggregation is that we can rearrange the input sequences in the batch to better fill it, because there is no interference between prefilling and decoding. For example, we can concatenate multiple short sequences into a longer one, thus ensuring that the lengths of sequences in the batch are as consistent as possible, as illustrated in Figure 11.8 (d). In this way, we can maximize the throughput of the prefilling phase. However, as a trade-off, we need to transfer the KV cache to the devices performing decoding, which also incurs extra communication overhead. Typically, this method requires a high-bandwidth, low-latency network to achieve optimal performance.

In this section, we will discuss several improvements to the above basic batching strategies. Most of them are based on an aggregated architecture, that is, decoding and prefilling can be considered as different stages of a model executed on the same device.

1. Scheduling

A practical LLM inference system typically consists of two components:

- **Scheduler**. Its primary role is to efficiently queue and dispatch tasks (i.e., input sequences) to the inference engine based on the current system load and task priorities. This often involves a variety of batching strategies that group certain requests together to maximize processing efficiency in some way.

- **Inference Engine**. It is responsible for the actual execution of the LLMs, processing the queued requests as they come in. As discussed previously, this engine involves both prefilling and decoding processes.

This architecture is illustrated in Figure 11.9. Incorporating scheduling into batch processing provides a flexible way to optimize both the system's throughput and latency, thereby achieving a better balance between them. For example, the batching methods shown in Figure 11.8 (a) and (b) can be considered one of the simplest scheduling strategies, called **request-level scheduling**. In this strategy, once a batch is filled and sent to the engine, the processing of the entire batch cannot be interrupted. The scheduler waits for this batch to be processed before handling the next batch [Timonin et al., 2022].

A more sophisticated scheduling strategy, called **iteration-based scheduling**, interacts with the inference engine at each token prediction step rather than at the sequence level. This approach allows dynamic batch adjustment during inference, as illustrated in Figure 11.10. Such fine-grained control lets the system prioritize critical tokens or sequences in real-time. For instance, if an urgent request arrives at some decoding step, the scheduler can add this request into the batch so that it can be processed as early as possible. In the following subsections, we will discuss batching methods based on iteration-based scheduling.

2. Continuous Batching

Continuous batching is an iteration-based scheduling method used in the Orca system [Yu et al., 2022]. In this method, an iteration refers to either the entire prefilling procedure or a single decoding step. For example, given an input sequence $\mathbf{x} = x_0 \cdots x_m$ and an output sequence $\mathbf{y} = y_1 \cdots y_n$, there are $n+1$ iterations in total: one for prefilling, and n for generating the output tokens (one per token). During scheduling, the batch can be adjusted between iterations. For example, we can either add a new input sequence to the batch, or remove a complete sequence from the batch at some iteration, even if the batch processing is not yet finished.

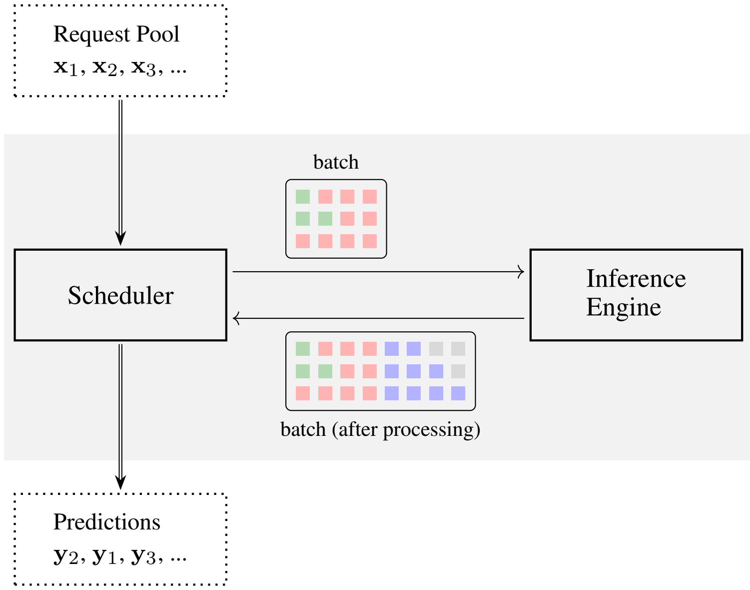
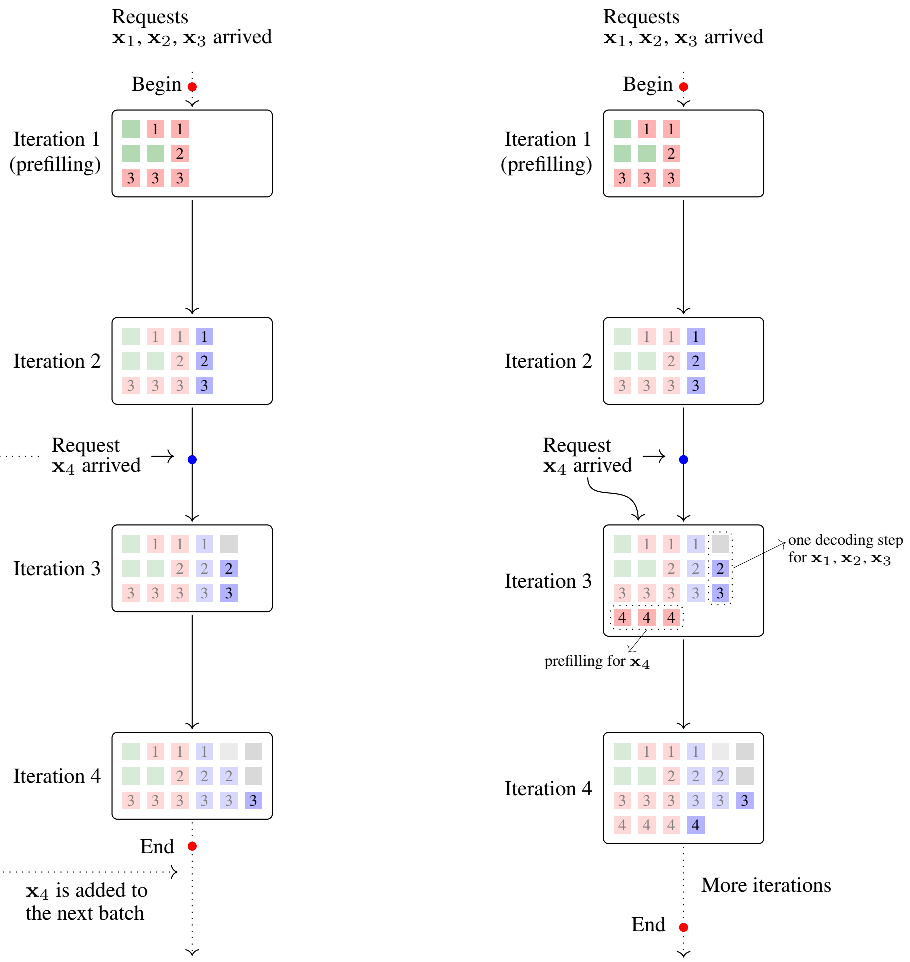


Figure 11.9: Illustration of the LLM inference architecture involving a scheduler and an inference engine. Each time, the scheduler selects a number of user requests to form a batch and sends it to the inference engine. The scheduler can interact with the inference engine and adjust the batch at certain points during inference, such as at the beginning of batch processing and at the start of each token prediction.

The general process of continuous batching includes the following steps:

- Initially, a batch is created with one or more input sequences, based on both the inference engine's processing capacity and the current user requests. The batch is then fed into the inference engine.
- The inference engine processes the batch iteration by iteration. After each iteration, the scheduler may adjust the batch in one of the following ways:
  - If a sequence in the batch completes generation (i.e., generates the end-of-sequence symbol), that sequence is removed from the batch.
  - If a new user request arrives and the inference engine has additional processing capacity, it is added to the batch.
  - If no sequences are added to or removed from the batch, the batch remains unchanged.
- The processing terminates only when all sequences have been completed and no new user requests arrive.

See Figure 11.11 for an example of continuous batching. In this example, we start with two user requests, $\mathbf{x}_1$ and $\mathbf{x}_2$. These two sequences are packed into a batch and sent to the inference engine for processing. After the engine completes two iterations, a new user request, $\mathbf{x}_3$, arrives. At this point, the scheduler adjusts the batch by adding $\mathbf{x}_3$ to it. The inference engine then continues processing the updated batch. Note that the inference engine now processes



(a) Request-level Scheduling (b) Iteration-level Scheduling

Figure 11.10: Illustrations of request-level scheduling and iteration-based scheduling. In request-level scheduling, once a batch is created and sent to the inference engine, we cannot adjust the batch. In other words, scheduling only occurs after the processing of a batch finishes. In iteration-level scheduling, we can perform scheduling during batch processing. For example, if a new request arrives at some point during inference, we can add it to the batch and continue processing.

different sequences in different ways: $\mathbf{x}_1$ and $\mathbf{x}_2$ proceed with the decoding process (i.e., predicting the next tokens), while $\mathbf{x}_3$ undergoes the prefilling process. After some time, the generation for $\mathbf{x}_2$ completes. As it happens, two more user requests, $\mathbf{x}_4$ and $\mathbf{x}_5$, arrive. The scheduler removes the completed sequence $\mathbf{x}_2$ from the batch and, considering the current load of the inference engine, adds $\mathbf{x}_4$ to the batch. However, $\mathbf{x}_5$ must wait until another sequence in the batch finishes before it can be added.

The idea behind continuous batching is to keep the inference engine fully utilized by processing as many sequences as possible, thereby maximizing computational resource usage.

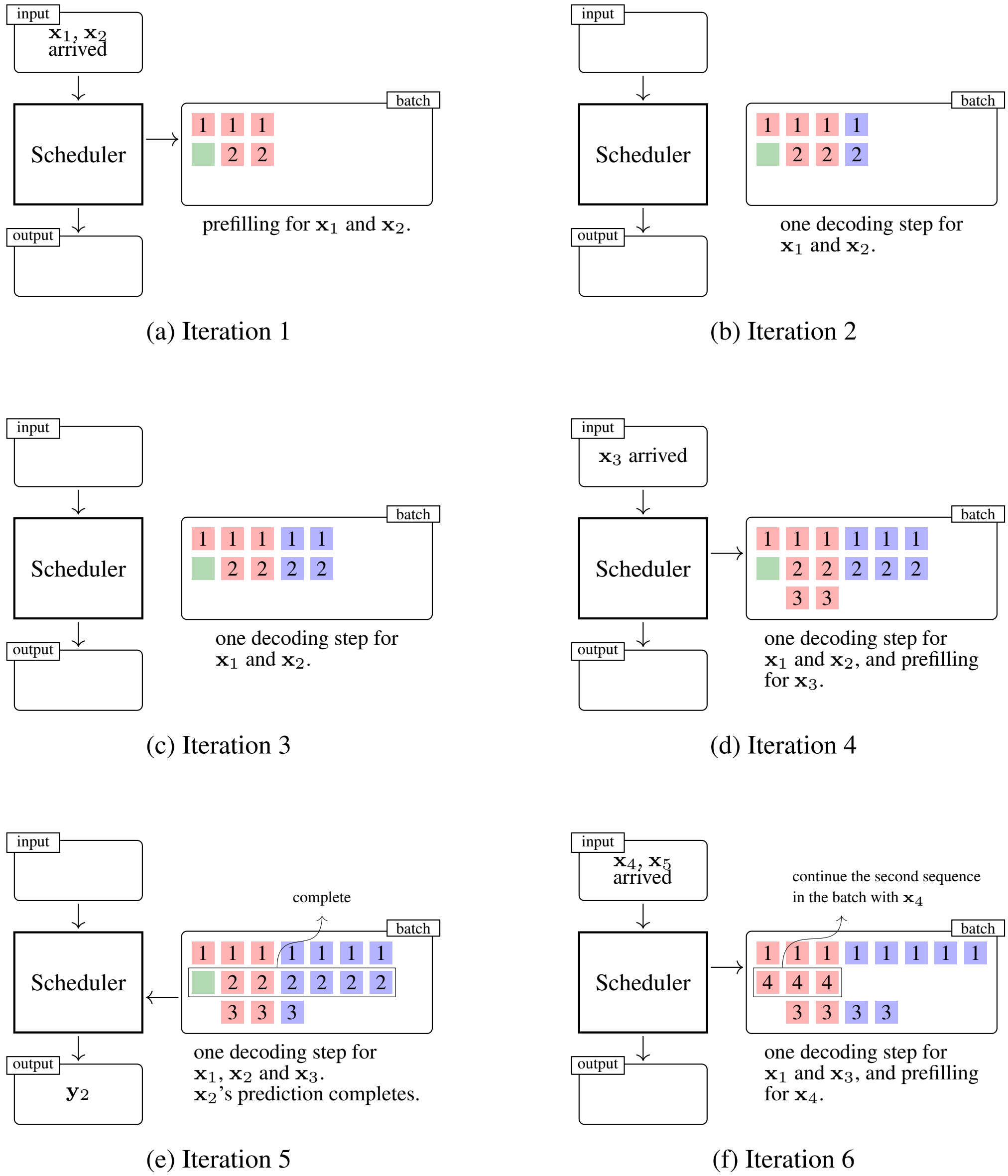


Figure 11.11: Illustration of batch adjustment in continuous batching. Instead of fixing a batch of input sequences and processing them to completion (as in request-level batching), continuous batching dynamically updates the batch during inference. The system continuously accepts and adds new requests (e.g., $\mathbf{x}_3$ and $\mathbf{x}_4$) into the current batch as long as there is available compute capacity.

A key difference between continuous batching and standard batching (see Figure 11.8) lies in the fact that, in continuous batching, prefilling and decoding can occur simultaneously across different sequences, whereas in standard batching, these two phases are performed sequentially for the entire batch. As discussed in Section 11.1.2, prefilling is considered a compute-bound process, while decoding is considered a memory-bound process. The intuition behind overlapping prefilling and decoding is to reduce idle times for both computation and data transfer. Consider two mini-batches: one for prefilling and one for decoding. While the prefilling mini-batch keeps the GPUs occupied, the decoding mini-batch can perform memory transfers concurrently.

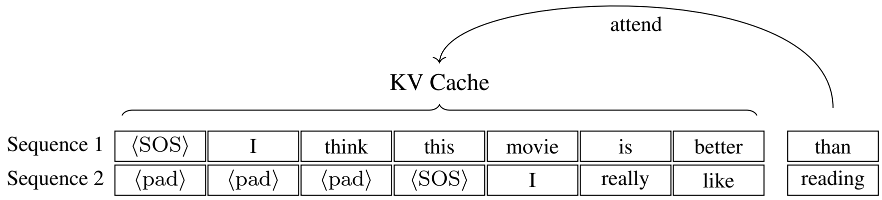
Another difference between continuous batching and standard batching is that continuous batching is prefilling-prioritized, while standard batching is decoding-prioritized [Agrawal et al., 2024]. In continuous batching, once the inference engine has spare computational resources, the scheduler will add new requests to the batch. In other words, these newly added requests will be processed for prefilling as early as possible. This approach improves system throughput, but at the cost of increased latency, as the newly added requests extend the processing time of earlier ones. In contrast, in standard batching, once the batch is created, we must wait for the last sequence in the batch to complete before processing new requests. This ensures relatively low latency, but results in lower device utilization and system throughput.

It is important to note that the cost of continuous batching is that we need to continuously reorganize the batches, which involves rearranging the data in memory. Each time a new request is added, the scheduler needs to reassess and optimize the current batch structure. This dynamic adjustment can incur additional memory and computational overhead, especially when the batches are frequently adjusted. Therefore, while this method can improve throughput, it may also lead to increased memory fragmentation and, in some cases, introduce additional latency.

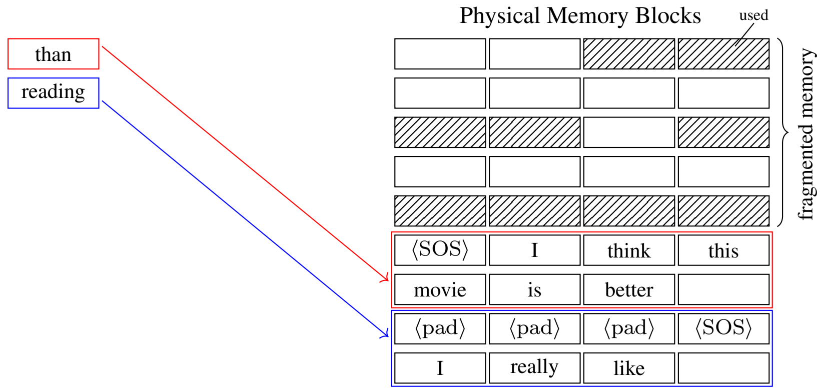
3. PagedAttention

PagedAttention (or paged KV caching) is a technique used in the vLLM system [Kwon et al., 2023]. Inspired by operating system paging, it optimizes memory usage during LLM inference — particularly for the KV cache — by addressing fragmented memory allocation in dynamic batching scenarios with variable-length sequences. The idea behind PagedAttention is to break down large memory requirements for KV caching into more manageable "pages" or chunks of memory. In this way, we do not need to store the KV cache of the full sequence in a continuous memory. Instead, the KV cache is divided into fixed-size blocks (analogous to memory pages in an operating system), which can be non-contiguously allocated in physical memory. One advantage of PagedAttention is that it enables flexible memory management, supporting dynamic sequence growth without requiring expensive reallocation or copying of large contiguous memory regions. Note that PagedAttention is not specifically designed for batching. But it indeed helps improve memory efficiency in batched inference scenarios, where memory management is more demanding and complicated.

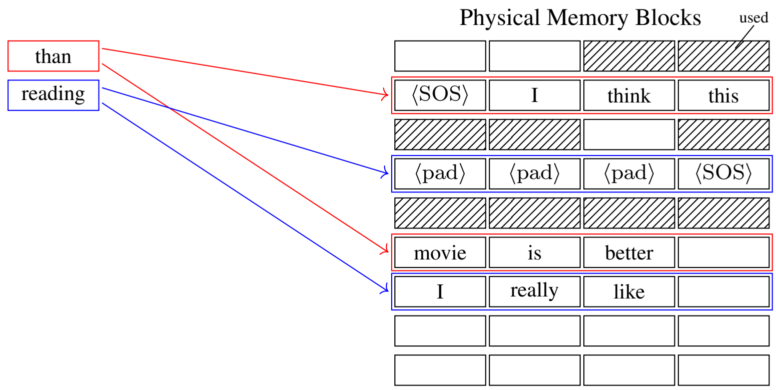
Consider a simple example of memory allocation in Figure 11.12 in which self-attention is performed for a batch consisting of two sequences. For each sequence, we need to attend the current token to the key-value pairs in the KV cache of this sequence, as required by self-attention. In the standard implementation of self-attention, the KV cache is stored in a contiguous block of memory, allowing us to efficiently access this continuous memory. However, in a paged KV caching system, the KV cache is divided into smaller, fixed-size memory blocks which are not necessarily contiguous. These smaller KV cache blocks can



(a) Two sequences in a batch



(b) Memory allocation for KV caching in standard self-attention



(c) Memory allocation for KV caching in PagedAttention

Figure 11.12: Illustration of memory allocation in PagedAttention. There are two sequences in the batch, as illustrated in sub-figure (a). Since the memory is fragmented, the KV cache is stored in a large unused block of memory in standard self-attention (see sub-figure (b)), but the fragmented memory is not used. By contrast, in PagedAttention (see sub-figure (c)), the KV cache is divided into smaller blocks and thus fits into fragmented memory.

be more effectively allocated to fragmented memory regions, thereby improving memory utilization. Another benefit of distributing chunks of the KV cache across different memory blocks is that it enables parallelization of the caching process. For example, if the input sequence is long and the memory bandwidth is sufficient, it would be beneficial to write and read the key and value vectors of different segments of the sequence in parallel across multiple memory blocks.

In general, storing contiguous data in non-contiguous regions can cause issues, for example, accessing fragmented data requires additional seek time, which reduces I/O efficiency. However, when handling large-scale data (e.g., performing multiplication on extremely large matrices), we typically do not process all the data at once but instead divide it into smaller blocks for block-level computation. From this perspective, it is also reasonable to partition the attention computation. If the paging strategy is well designed, the additional overhead in memory access can be minimal, while the improvement in memory utilization can be significant.

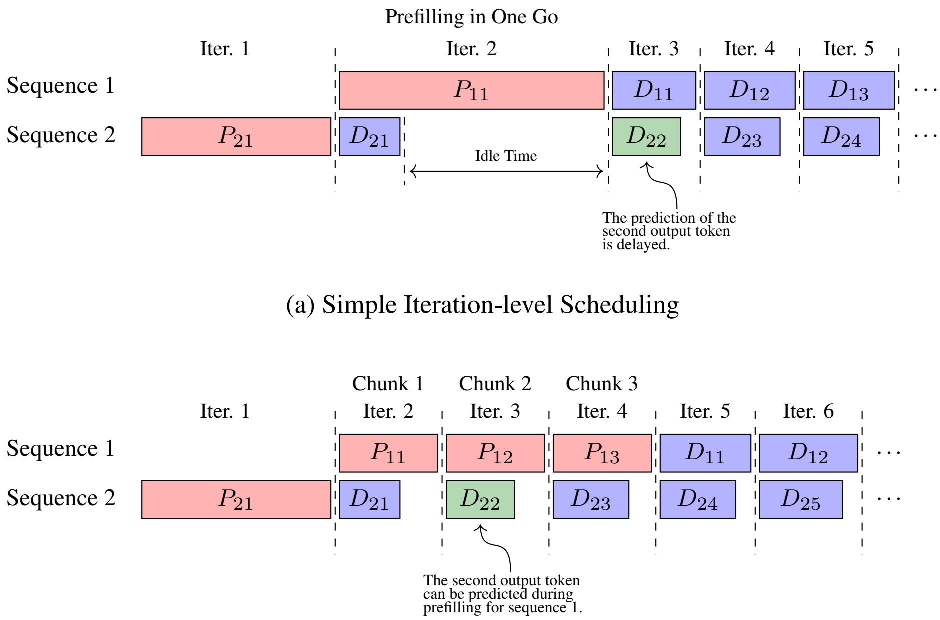
4. Chunked Prefilling

We have seen that, in iteration-level scheduling, prefilling and decoding for different sequences can occur simultaneously. This can be seen as a prefilling-prioritized strategy which can maximize the throughput. However, one such iteration can take a long time if the input sequence is very long and the prefilling process dominates the computation. In this case, decoding for other sequences has to wait until the prefilling completes, leading to increased latency for generating output tokens. Therefore, while prefilling-prioritized strategies are effective for maximizing hardware utilization, they may introduce significant variability in token generation latency, particularly when the system is handling a mix of long and short input sequences.

A simple way to reduce decoding latency is to make computations for different sequences in the batch comparable. One such method is to divide sequences into chunks and perform prefilling chunk by chunk. This approach, often referred to as chunked prefilling, processes smaller portions of each sequence at a time, allowing the system to better balance the computational load across sequences [Agrawal et al., 2023]. By choosing an appropriate chunk size, we can ensure that when prefilling and decoding overlap for two sequences, their processing within the same iteration tends to take a similar amount of time. As a result, decoding idle time is reduced and overall throughput is improved.

Figure 11.13 shows an illustration of chunked prefilling in a few iterations. In this example, the batch contains two sequences. The whole prefilling process of the first sequence is divided into three prefilling steps, giving rise to the chunks denoted P_{11}, P_{12} and P_{13}. Each chunk corresponds to one iteration and can thus overlap with one decoding step. In this way, during the prefilling of the first sequence, we can perform three decoding steps, rather than only a single decoding step, as is the case in standard iteration-level scheduling. As a result, the idle time of the decoding process is reduced, and the output tokens can be generated earlier.

Chunked Prefilling improves decoding efficiency by overlapping prefilling and decoding, but at the cost of additional memory overhead and scheduling complexity. In standard prefilling,



(a) Simple Iteration-level Scheduling

(b) Chunked Prefilling

Figure 11.13: Comparison of simple iteration-based scheduling and chunked prefilling. P_{xy} denotes the y-th prefilling step for sequence x, and D_{xy} denotes the y-th decoding step for sequence x. In simple iteration-based scheduling (or prefilling-prioritized scheduling), since prefilling is treated as a single iteration, D_{22} has to wait for the completion of the prefilling of sequence 1. In chunked prefilling, the prefilling process can be divided into multiple steps. Thus, D_{22} can execute during prefilling for sequence 1 (i.e., during P_{12}).

we process the whole input sequence once, building the KV cache in one go. By contrast, in chunked prefilling, each chunk needs a separate forward pass to compute its attention outputs and update the KV cache. As a result, we need to maintain the KV cache of early chunks while processing later chunks. This also compromises the parallelism of completing the prefilling for the entire sequence in a single pass. In practice, it is usually possible to balance throughput and latency by choosing an appropriate chunk size.

It is worth noting that the methods discussed in this subsection can broadly be categorized as priority-based scheduling methods. In these methods, we can give priority to certain requests, or to certain prefilling or decoding steps, so that system resources are allocated in a way that better aligns with specific performance goals. As presented above, for example, we may prioritize decoding over prefilling to minimize token generation latency, or prioritize prefilling over decoding to maximize overall throughput in batch-processing scenarios. Practitioners can design custom priority policies for specific needs and operational constraints in real-world applications, such as request deadlines and importance levels defined by users.

11.2.3 Parallelization

Parallelization is a widely used approach to scale up LLM inference, especially for large-scale deployments. In Chapter 7, we have discussed several common parallelization strategies to parallelize LLM pre-training, such as model parallelism, tensor parallelism, and pipeline parallelism. We have also discussed efficient architectures that are easy to deploy in distributed computing systems. For example, in MoE models, we assigns different experts to different devices[3]. Only the active experts for a given input are executed, which significantly improves computational efficiency while maintaining model quality. Many of these methods can be directly applied to LLM inference with minimal modifications.

However, applying these parallelization techniques to inference poses new challenges compared to pre-training. These issues become especially pronounced in real-time or low-latency inference scenarios, where load imbalance across devices and communication overhead can significantly impact performance. For example, unlike pre-training, where batches can be prepared in advance, inference must handle variable-length sequences in real time. This makes it harder to maintain optimal device utilization and complicates scheduling across heterogeneous computational resources. A related issue is load balancing. When a large number of requests arrive in a short period of time, the system must efficiently distribute workloads across available devices. For example, real-world requests typically exhibit highly variable computational demands due to differences in task types and prompt lengths. Such variability renders simple static load balancing approaches ineffective, and so we need to use finer-grained strategies that can adapt to runtime conditions. The problem becomes even more complicated when we deploy the system on heterogeneous hardware and there are strict latency constraints.

In the development of LLMs, parallelization is closely related to LLM serving. Generally, building a high-quality LLM serving system is not a simple task — it typically requires the combination of multiple techniques, such as architectural design, workload distribution, and LLM-specific hardware/software optimizations. As such, LLM serving constitutes an exceptionally broad subject that often demands substantial engineering expertise. Here, we will not go into the details of LLM serving. For related concepts and techniques, readers may refer to relevant open-source systems (such as vLLM[4], TensorRT-LLM[5] and TGI[6]) and papers [Pope et al., 2023; Li et al., 2024a].

11.2.4 Remarks

We have considered many methods for improving the efficiency of LLMs in this and previous chapters. Although these approaches address different issues, most of them essentially explore trade-offs between various performance factors. One important trade-off is between inference speed and accuracy. For example, techniques like quantization, pruning, and knowledge

[3] In LLMs, the experts are typically modular FFNs. So each expert is a part of the FFN component in the Transformer architecture.

[4] https://github.com/vllm-project/vllm

[5] https://github.com/NVIDIA/TensorRT-LLM

[6] https://github.com/huggingface/text-generation-inference

distillation can significantly reduce computational overhead and latency but may introduce minor degradations in model performance. Conversely, preserving full precision or using larger models enhances accuracy but at the cost of slower inference and higher resource demands.

Another important consideration in LLM inference is the memory-compute trade-off. As in computer system design, we need to consider the balance between memory usage and computation required to generate the output. In particular, storing intermediate results such as KV caches during inference can significantly reduce redundant computation, but at the cost of increased memory usage. In KV caching, storing past attention states avoids recomputation of self-attention over previous tokens, thereby reducing compute time per token. However, as the number of tokens grows, so does the memory footprint of the KV cache, especially when processing very long sequences or multiple sequences in parallel. In response, various techniques have been developed to reduce memory consumption by partially recomputing intermediate states. For instance, chunked or windowed attention limits the attention span to a recent subset of tokens, reducing KV cache size at the cost of reduced context or additional compute if past information must be reprocessed.

Note that considering the memory-compute trade-off is a very general principle. It can be extended beyond attention mechanisms and Transformers to other components in system design. An example is the choice of data precision. Using lower-precision formats such as FP16 or INT8 can reduce both memory usage and memory bandwidth requirements, effectively alleviating pressure on the memory subsystem. However, lower precision may lead to numerical instability or slight accuracy degradation, requiring careful calibration or retraining. Thus, this trade-off can also be seen as a memory-compute-accuracy triangle, where improvements in one dimension may come at the expense of another.

Beyond speed, accuracy, and memory, several other dimensions also influence LLM inference efficiency. Some of these dimensions have been discussed in this chapter, while others have not. Here we outline them as follows.

- **Throughput vs. Latency**: In large-scale multi-user LLM serving scenarios, we often aim to maximize system throughput. For example, as discussed in this section, we can batch multiple requests together to increase the number of tokens processed at the same time. However, batching increases waiting time and may lead to higher per-request latency, especially for short or interactive requests. By contrast, optimizing for low latency often requires serving requests individually or in smaller batches, which underutilizes hardware resources and reduces throughput. Achieving a good balance depends on the quality-of-service requirements and user interaction patterns.

- **Generalization vs. Specialization**: General-purpose LLMs are trained to perform a wide range of tasks with a single set of parameters. While flexible, they may be less efficient or accurate for specific tasks. Specialized models can yield better performance and lower inference costs for targeted applications. However, maintaining multiple specialized models increases system complexity and storage requirements. The trade-off between maintaining a single general model versus multiple specialized models is an important system-level design choice.

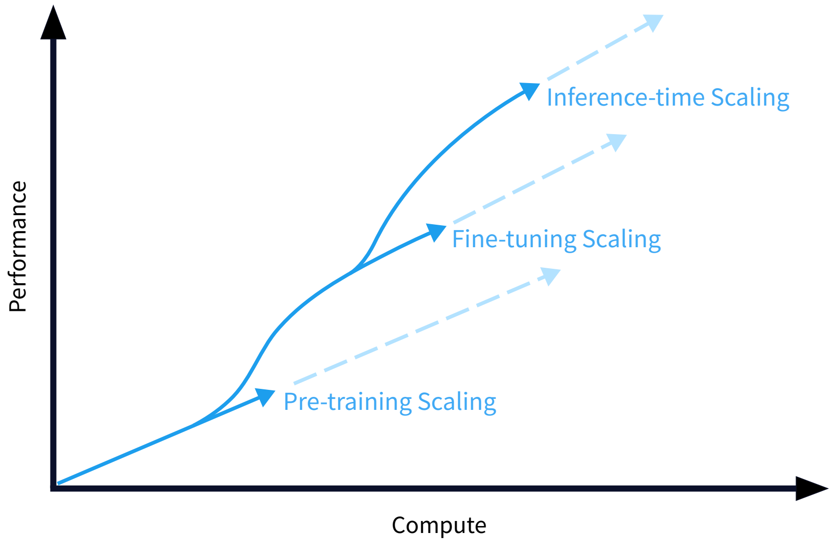


Figure 11.14: Scaling for pre-training, fine-tuning and inference stages [Briski, 2025].

- **Energy Efficiency vs. Performance**: High-performance inference often requires running large models at high throughput on powerful accelerators, which consumes considerable energy. This may be problematic for edge deployments or energy-sensitive environments. Techniques like model compression can improve energy efficiency, but usually with some degradation in output quality or increase in latency. Energy constraints thus introduce another important dimension in optimizing LLM inference.

11.3 Inference-time Scaling

Scaling laws can be considered one of the fundamental principles guiding the development of LLMs. In previous chapters, we discussed several times that scaling up training data, model size, and compute can effectively improve the performance of pretraining. In fact, scaling laws also apply to downstream stages such as fine-tuning and inference (see Figure 11.14). Here we consider **inference-time scaling**, which has been widely employed by recent LLMs to solve complex problems, such as complex math problems [Snell et al., 2025]. Unlike pre-training and fine-tuning scaling, which focuses on improving LLMs via parameter updates, inference-time scaling improves these models during inference without further training. This includes a large variety of methods which scale LLMs in different dimensions, such as ensembling multiple model outputs, increasing context length, adopting more aggressive decoding algorithms, and using external tools to extend model capabilities.

While inference-time scaling is wide-ranging, in this section we consider those methods that incorporate more compute into inference (called inference-time compute scaling). Here is a list of inference-time (test-time) compute scaling methods, organized by category:

- **Context Scaling**. It involves scaling the input or context to improve generation (or potentially scale the output).
- **Search Scaling**. It involves increasing computational effort during decoding.
- **Output Ensembling**. It involves combining multiple model outputs.
- **Generating and Verifying Thinking Paths**. It involves guiding LLMs to generate and verify thinking paths for solving complex reasoning problems.

We will describe these methods in the following subsections.

11.3.1 Context Scaling

Context scaling improves LLM performance by extending the input to the model. A straightforward approach is to incorporate more helpful context during inference, allowing the model to condition its predictions on more prior information. One example is few-shot prompting. It augments the context with multiple input-output examples, and so the model can learn task behavior implicitly from these examples without parameter updates. On top of few-shot prompting, we can use chain-of-thought prompting to encourage the model to produce intermediate reasoning steps before final answers. Note that chain-of-thought prompting is one of the most important methods in addressing reasoning problems. By explicitly providing intermediate steps in problem-solving, we can prompt the model to break down complex tasks into simpler sub-tasks, which is found to be very beneficial for generating accurate and interpretable outputs.

Beyond extending the prompt with examples or reasoning steps, another approach to context scaling involves dynamically incorporating external knowledge. This is often achieved through RAG. RAG systems first retrieve relevant document snippets from a large collection of documents or a database based on the current input. These retrieved pieces of information are then added to the context provided to the LLM. This essentially expands the context to include timely or specialized external knowledge. By doing so, the model grounds its responses in specific knowledge found in the external source. The LLM thus can generate responses that are not only relevant to the input but also factually accurate and up-to-date.

However, as the context grows, these methods often suffer from the constraints of finite context window length. While model architectures and techniques (like efficient attention models) are continually evolving to support longer contexts, processing extremely long inputs still poses challenges. Increased computational cost is one factor. More critically, when the context window becomes very large, the model might struggle to attend effectively to the most relevant information (e.g., the “lost in the middle” phenomenon). Therefore, effective context scaling is not just about adding more information, but also about strategically selecting, structuring, and presenting the most pertinent information within the model’s processing capabilities.

Here we omit the detailed discussion of these methods, as they have already been covered in previous chapters. See Chapters 8 and 9 for more details, including prompting, RAG, and long-sequence modeling methods.

11.3.2 Search Scaling

In LLMs, decoding is a search process that aims to efficiently find the best output sequence given the input sequence. Search scaling (or decoding scaling) typically involves two aspects: scaling the output length and scaling the search space.

Scaling the output length refers to increasing the number of tokens generated during inference. This is especially important in tasks that require long-form generation, such as story writing. More recently, generating outputs with long thinking paths has shown strong performance in math problem solving and code generation. For example, encouraging the model to generate long thinking paths before producing the final answers has been found to be very beneficial in performing complex reasoning. This idea has been widely used in developing recent LLMs for reasoning, such as OpenAI [2024]'s o1 and DeepSeek [2025]'s R1. We will discuss more about output length scaling in Section 11.3.4.

Scaling the search space, on the other hand, refers to expanding the set of candidate output sequences considered during search, so that higher-quality outputs can be found. As discussed in Section 11.1.3, a simple example is that in beam search we increase the beam width to allow more candidate sequences to be explored in parallel at each decoding step. This increases the chance of discovering better outputs, especially in tasks where the optimal solution is not immediately apparent from local decisions.

In addition to decoding algorithm adjustments, it is also possible to explore compact structures to encode a large number of outputs. For example, we can construct and navigate a tree or graph of reasoning steps [Yao et al., 2024]. In this paradigm, each node represents a partial solution or intermediate step, and edges represent transitions between reasoning states. Such structured search enables the model to consider multiple paths simultaneously. Another related direction is Monte Carlo tree search-inspired decoding, where the model stochastically explores and scores different paths based on learned heuristics or external reward models.

Search scaling is a very general idea, and it is often implicitly involved in the design of search procedures that exploit search structure, heuristics, and model uncertainty. Many of the above methods have been discussed previously, though they were not originally developed with scaling as their primary goal. However, search scaling inherently comes with computational costs. Increasing beam width, for instance, directly translates to higher memory usage and longer inference times. In practice, there is often a point of diminishing returns, where further expansion of the search space yields marginal improvements in output quality at a significant computational expense. Therefore, an effective strategy often involves finding an optimal balance between scaling and computational feasibility.

11.3.3 Output Ensembling

If we have multiple model outputs, it is often beneficial to combine them to mitigate the impact of individual model errors and synthesize a superior final output. Each model might capture different aspects of the underlying data distribution or possess unique strengths and weaknesses. By ensembling, we can average out the noise or random errors present in individual predictions, leading to a more stable and reliable outcome. In LLM ensembling, one of the simplest approaches is to average the probability distributions over the next token from each model, and select the best token using this averaged distribution. Or, if we regard the problem as a discrete decision-making task, majority voting can be employed. More sophisticated methods might involve re-ranking candidate outputs generated by different models based on a separate scoring function or even using a meta-learner to intelligently combine the predictions.

The "scaling" from output ensembling comes at the cost of running multiple models or sampling multiple outputs. This not only increases the latency of inference but also leads to the additional complexity of managing multiple models. But the quality of outputs does not continue to improve indefinitely as more models are added. In some cases, the benefits of output ensembling may diminish as the number of component models in the ensemble exceeds a certain threshold. Instead, the benefits of ensembling are generally greater when the individual models are diverse (i.e., they make different errors), even if there are a relatively small number of component models. Therefore, it is common practice to use a set of diverse LLMs which differ in their training data, model architectures, or fine-tuning objectives.

In LLMs, "scaling" often implies making things "bigger" for quality with more resources. However, in addition to scaling up the quality, scaling can mean more. It can also signify scaling up the robustness (making the system less prone to errors and more reliable) and exploration (covering a wider range of potential solutions). In output ensembling, these dimensions are naturally integrated. For instance, the very act of averaging or voting across different model outputs is a direct strategy to scale up robustness against individual model failures. Furthermore, by intentionally including varied models, ensembling increases the chances of discovering novel or superior solutions. In this sense, scaling is not limited to making models larger or running them longer — it also means strategies for making inference more robust, exploratory, and adaptive.

11.3.4 Generating and Verifying Thinking Paths

So far, we have viewed inference-time scaling as a general class of methods for scaling various aspects of inference, such as sequence length, model size, and/or search strategies. In fact, one successful application is the use of inference-time scaling to enhance the reasoning capabilities of LLMs. As we have seen, the reasoning performance of LLMs can be improved by using chain-of-thought methods. We can therefore make use of the chain-of-thought prompts to generate intermediate reasoning steps and reach a correct answer. However, reasoning problems are often so complicated that we cannot obtain high-quality solutions by providing simple chain-of-thought prompts. For example, when solving a math problem, we typically need to reason over a sequence of steps. At each step, we need to work out some intermediate result, verify it, and then determine what to do next. The reasoning path is not a fixed pattern but a dynamically generated thinking process that often involves trial-and-error, backtracking, and self-correction. This requires more sophisticated prompting strategies or search algorithms to navigate such complex reasoning. In this subsection, we focus on inference-scaling methods that go beyond simple chain-of-thought to address complex reasoning problems more effectively.

At a high level, methods for scaling the reasoning of LLMs can be categorized into two classes:

- **Training-free Methods**. These methods aim to improve reasoning capabilities without requiring any modification or retraining of the pre-trained parameters. Instead, they focus on techniques applied during inference, such as sophisticated prompting strategies (e.g., chain-of-thought) and algorithmic control over the reasoning process (e.g., search).
- **Training-based Methods**. These methods involve further training or fine-tuning the model parameters to explicitly improve reasoning abilities, such as supervised fine-tuning on datasets with reasoning examples (e.g., math problems with step-by-step solutions).

In the following, we first discuss training-free methods, and then training-based methods.

1. Solution-level Search with Verifiers

Given an input sequence (e.g., a math problem), there are many possible output sequences (e.g., solutions to the problem). If we have a model to evaluate or verify each solution, we can select the best one. This is the fundamental principle behind methods like best-of-N sampling, where multiple outputs are generated, and the optimal result is picked based on some selection mechanism. Such a selection process can be viewed as a search problem, which involves two components:

- **Search Algorithm**. This defines the strategy used to explore the space of possible output sequences (solutions) and generate a set of candidates. It can range from simple independent sampling to more sophisticated search techniques as discussed in Section 11.1.3.
- **Verifier**. This is a model or function responsible for evaluating the quality, correctness, or utility of each candidate solution generated by the search algorithm. It provides a score, a probability, or a judgment that allows the system to select the best among the candidates. The verifier can be another LLM, or even a set of predefined rules or heuristics.

Given an input problem $\mathbf{x}$, we define that an output solution $\mathbf{y}$ can be represented as a sequence of reasoning steps:

$$\mathbf{y} \quad = \quad (a_1, a_2, \cdots, a_{n_r}), \tag{11.37}$$

where a_i is the i-th reasoning step, and a_{n_r} is the last step which should contain the answer to the problem. See Figure 11.15 for an example of a multi-step reasoning path.

The search algorithm can efficiently generate a set of candidate solutions:

$$\mathcal{D}_c \quad = \quad \{\mathbf{y}_1, \cdots, \mathbf{y}_K\}. \tag{11.38}$$

Then, we can use a verifier, which evaluates each solution by the function $V(\mathbf{y})$, to score

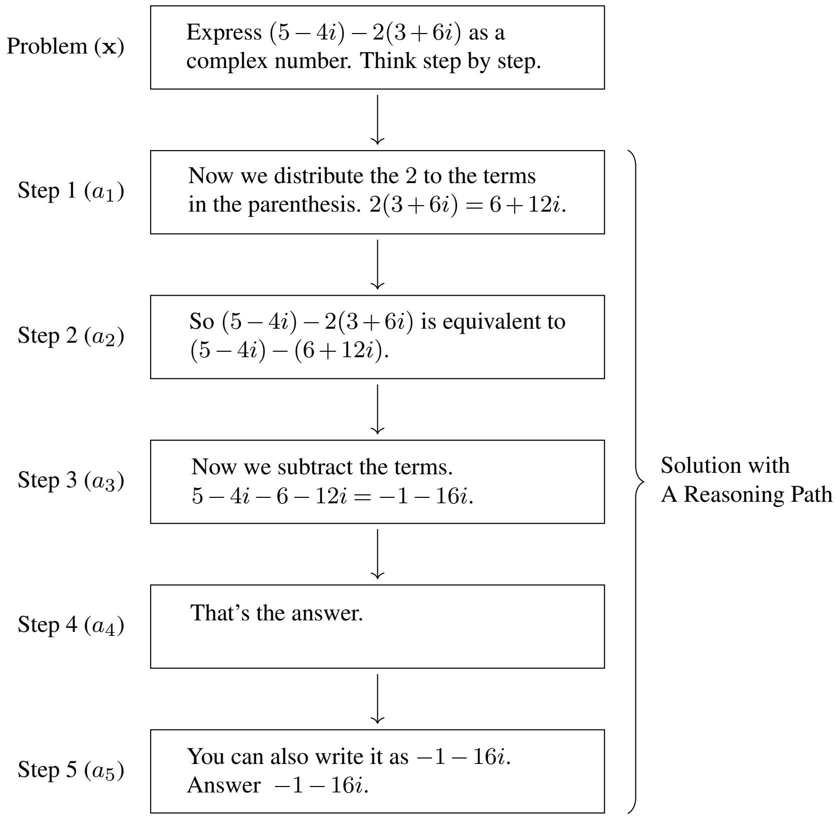


Figure 11.15: Illustration of multi-step reasoning. This example is from the PRM800K dataset [Lightman et al., 2024]. Given a math problem, the LLM is prompted to generate a thinking path (or reasoning path) consisting of several reasoning steps. Each step addresses a sub-problem based on the results of the previous steps. The answer to the original problem is contained in the last step.

the candidates in $\mathcal{D}_c$. The final output is the best candidate selected by the verifier:

$$\hat{\mathbf{y}} = \arg\max_{\mathbf{y}\in\mathcal{D}_c} V(\mathbf{y}). \tag{11.39}$$

Although verifying the entire reasoning path is possible, a simpler alternative is to verify only the final reasoning step. In this way the verifier function $V(\mathbf{y})$ is simplified to depend solely on the final answer contained within a_{n_r}. This can be achieved in various ways, depending on the nature of the problem and the expected answer format.

- For some math and coding problems, we can use off-the-shelf tools as verifiers. Examples include proof checkers for mathematical theorems, interpreters or compilers for code execution, and unit test systems for verifying program correctness against predefined test cases.
- If there is labeled data for evaluating the answer, such as human preference data, we can train a reward model on such data. The learned reward model is then used as the verifier

which assigns a scalar score to each candidate answer.

- If there are no existing systems or suitable reward models, we can use another LLM to act as the verifier. This LLM is prompted to assess the quality of the candidate answer. It could potentially be a more capable model, or the same LLM used with a specific "evaluator" prompt.
- Alternatively, simpler heuristic-based verifiers can be designed. A commonly used approach is to employ majority voting, where the most frequently occurring answer among a set of candidates is selected.

Based on these verifiers, we can search to obtain a set of candidate solutions for selection. One simple strategy, which is often referred to as **parallel scaling** [Brown et al., 2024; Snell et al., 2024], involves generating K candidate solutions by running the base LLM K times independently. In this process, we can adjust the temperature in sampling to control the diversity in the outputs. The verifier then assesses each of these K complete solutions, and the one with the highest score is selected as the final output. This is conceptually very similar to best-of-N sampling, which in previous chapters we primarily described as a method of selecting the best one from a set of sampled outputs using a reward model.

Another approach is **sequential scaling**, which builds a sequence of solutions incrementally [Gou et al., 2024; Zhang et al., 2024]. It starts with an initial solution generated by the LLM with prompting. Then, we use a verifier (often the same LLM) to evaluate the solution. This can be seen as a critique stage. The output of this stage is some form of feedback, such as textual critiques pinpointing errors or suggesting improvements, numerical scores reflecting solution quality, or even a revised plan or intermediate step to guide the next generation. This feedback, along with the original problem and the current solution, is then used to prompt the LLM to generate a potentially improved solution. This can be seen as a refine stage. This critique-refine cycle can be repeated, forming an iterative loop:

$$\mathbf{y}_{k+1} \quad = \quad \mathrm{Refine}(\mathbf{x}, \mathbf{y}_k, \mathrm{Feedback}(\mathbf{y}_k)), \tag{11.40}$$

where $\mathrm{Feedback}(\mathbf{y}_k)$ represents the feedback from the verifier. The $\mathrm{Refine}(\cdot)$ function generates the improved solution $\mathbf{y}_{k+1}$ by prompting the LLM with the original problem $\mathbf{x}$, the previous solution $\mathbf{y}_k$, and this feedback. The process can be iterated for K times, or until the solution quality, as assessed by the verifier, converges to a satisfactory level. This iterative framework, where a solution is progressively improved through cycles of generation, evaluation (critique), and revision, is precisely what constitutes self-refinement [Shinn et al., 2023; Madaan et al., 2024]. In such scenarios, the role of the verifier is not just to pick the best complete solution from a static set, but to actively guide the generation process itself.

See Figure 11.16 for illustrations of parallel scaling and sequential scaling. Note that there are other ways to perform search and obtain different sets of candidate solutions. One alternative method is to organize search as a tree structure. This approach, often referred to as tree search, provides a more structured way to explore the space of possible reasoning paths. In solution-level search, each node of the tree represents a complete solution. During search, we need to expand a node to a set of child nodes, representing new solutions that can be considered

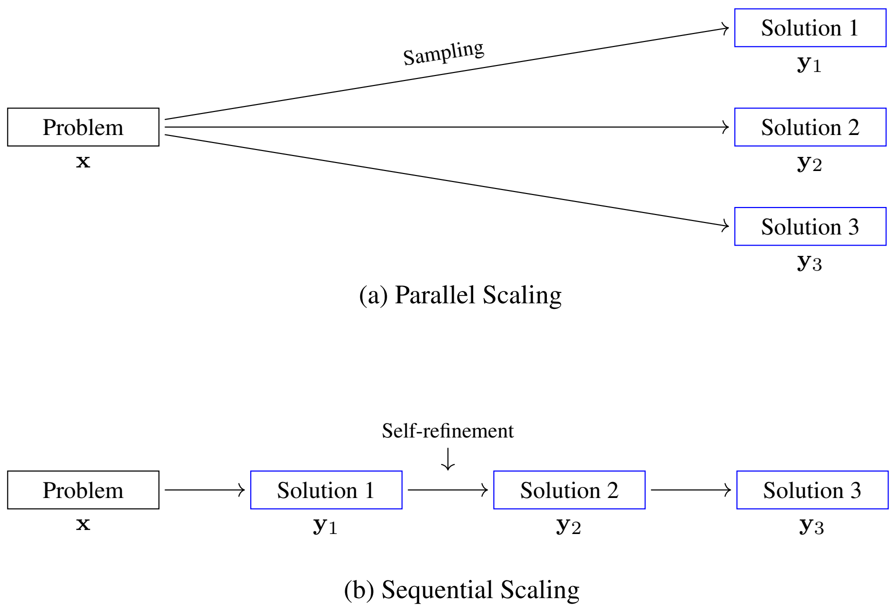


Figure 11.16: Illustrations of parallel scaling and sequential scaling. In parallel scaling, we obtain multiple solutions by running the LLM several times independently. In sequential scaling, the LLM generates an initial solution. Then, we use the LLM to refine it iteratively, with each refinement yielding a new, possibly better solution.

in verification. The expansion process typically involves taking an existing solution (the parent node) and using the LLM to generate variations or alternative solutions.

2. Step-level Search with Verifiers

While the methods discussed above primarily focus on generating complete solutions before final selection, the search process can also be integrated more deeply into the step-by-step generation of the reasoning path itself. This leads to approaches that perform step-level search with verifiers, where guidance or pruning occurs at intermediate reasoning steps $\{a_1, \cdots, a_{n_k}\}$ rather than only after a full solution $\mathbf{y}$ is formed.

Such fine-grained control is particularly beneficial for complex reasoning problems where a single incorrect intermediate step can render the entire subsequent reasoning chain invalid. By evaluating or guiding the generation at each intermediate step, the LLM can explore the reasoning space more effectively, potentially pruning unpromising paths early or allocating more resources to explore more plausible ones.

Step-level search with verifiers can also be modeled as a tree search problem. In this paradigm, each node (or state) corresponds to a partial reasoning path, $\mathbf{a}_{\leq i} = (a_1, \cdots, a_i)$, representing the sequence of i reasoning steps taken so far (i.e., a path from the root node to the current node). The objective of the search process is to explore the underlying state space, starting from an initial empty path, to find a complete path that constitutes a correct solution. Note that we use $\mathbf{a}_{\leq i}$ here to represent a partial reasoning path instead of $\mathbf{y}_{\leq i}$. While this makes notation a bit inconsistent with that used for representing complete solutions ($\mathbf{y}$) or full paths in solution-level search, it serves to highlight the focus on individual actions or steps.

The core components of step-level search with verifiers are:

- **Node Representation**. A node is a partial reasoning path $\mathbf{a}_{\leq i} = (a_1, \cdots, a_i)$. The root node is an empty path, and terminal nodes are complete reasoning paths.
- **Node Expansion**. Given a current partial path $\mathbf{a}_{\leq i}$, the LLM is used to generate one or more candidate next reasoning steps $\{a_{i+1}^{(1)}, \cdots, a_{i+1}^{(M)}\}$. Each candidate step, when appended to $\mathbf{a}_{\leq i}$, forms a new potential partial path $\mathbf{a}_{\leq i+1} = (a_1, \cdots, a_i, a_{i+1}^{(j)})$.
- **Verification**. The verifier $V(\cdot)$ evaluates the quality of a newly generated step in the context of the current partial path $\mathbf{a}_{\leq i} = (a_1, \cdots, a_i)$ and the original problem $\mathbf{x}$. As with solution-level verification, step-level verifiers might output a numerical score, a categorical label, and textual feedback.
- **Search**. This governs how the search space is explored. Based on the evaluations from the verifier, the search strategy decides which partial paths to extend further, which to prune, and the order of exploration.

This step-by-step verification allows for dynamic adjustments to the reasoning process. If a step a_{i+1} is deemed incorrect or unpromising by $V(\cdot)$, the search algorithm can backtrack and explore alternative steps from $\mathbf{a}_{\leq i}$, or even from an earlier node $\mathbf{a}_{\leq i'}$ (where $i' < i$). Conversely, if a step is highly rated, resources can be focused on extending that path. See Figure 11.17 for an illustration of step-level search with verifiers.

Clearly, this search framework is very similar to that used in decoding methods for LLMs, as discussed in Section 11.1.3. For example, beam search maintains a set of K most promising partial sequences at each generation step. This is a form of step-level search where the "verifier" is implicitly the LLM's own probability model, and the "search" is the pruning mechanism to maintain the beam size.

However, step-level search with explicit verifiers, as described here, presents differences from standard decoding. One of them is that the verifier can be a much more sophisticated component than just the raw output probabilities of the generative LLM. The design of step-level verifiers basically follows that of solution-level verification. A step-level verifier might be a language model that assesses the quality of an individual reasoning step within the context of the preceding path. This LLM can even be fine-tuned to enhance its verification capability. Alternatively, for domains with well-defined rules, it could be a symbolic engine or a set of programmatic checks. Furthermore, verifiers can be designed to predict the future utility or likelihood of success given the current partial path, drawing inspiration from value functions in reinforcement learning. Human expertise can also be incorporated to provide judgments on critical steps, especially in high-stakes scenarios.

One example of such a step-level verifier, particularly when using human feedback to assess intermediate progress, is the **process reward model** (**PRM**). A PRM is typically a separate language model trained to output a scalar reward for each reasoning step $a_{i'}$ within a partial path $\mathbf{a}_{\leq i}$. It provides a more direct and fine-grained supervisory signal compared to **outcome reward models** (**ORMs**) which only evaluate the final solution. However, the development

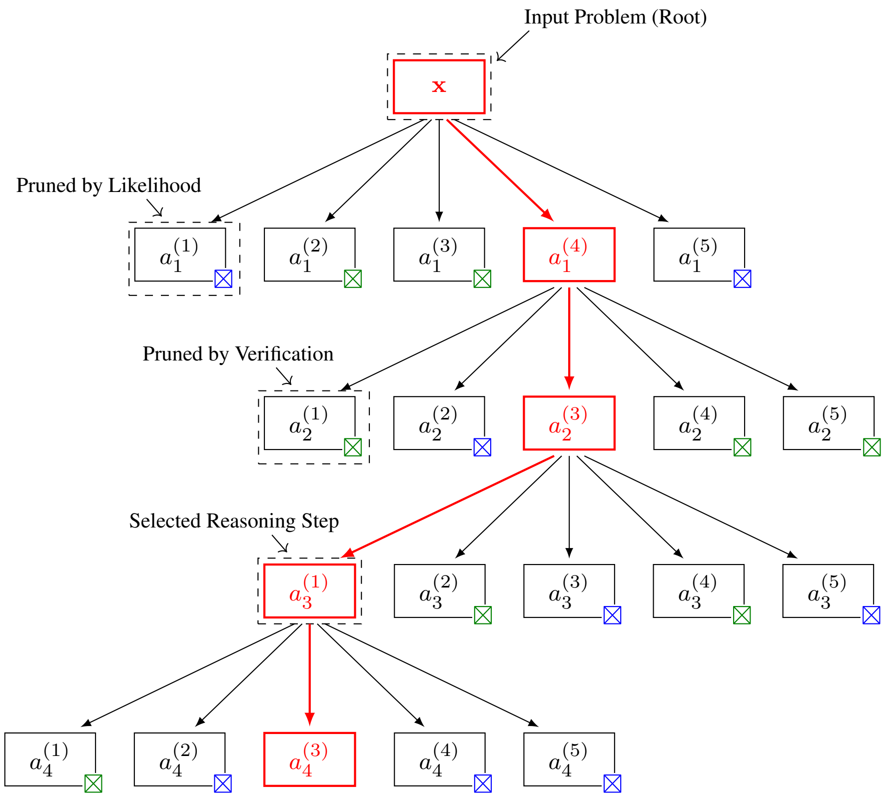


Figure 11.17: Illustration of step-level search with verifiers. $a_i^{(j)}$ = the j-th candidate for the i-th reasoning step, ⊠ = candidate pruned by the LLM's output probability, and ⊠ = candidate pruned by the verifier. Given the input problem as the root node, we expand the tree by generating multiple reasoning steps at each expansion. Each candidate can be pruned by either likelihood (as in standard decoding) or step-level verification. The unpruned candidates are then expanded to generate further reasoning steps. The process is iterated until a complete reasoning chain leading to a final answer is generated, or until a predefined search limit is reached.

of PRMs relies on step-level human annotations, such as preferences on different next steps. Collecting supervision for each intermediate step is considerably more labor-intensive and requires greater cognitive effort from human annotators than simply labeling final outcomes.

One alternative approach to developing training data for step-level verification is to use LLMs to generate such annotations automatically. For example, we can take a strong LLM, referred to as a teacher model, and prompt it to first generate a complete reasoning path for a given problem. Then, at each intermediate step within this path, we can prompt the same teacher LLM (or another capable LLM) to generate several alternative candidate next steps in addition to the one it originally chose. The teacher LLM can then be prompted again to evaluate these alternatives. These evaluation results (e.g., correct vs. incorrect) can then serve as data annotations. Alternatively, the generalization capabilities of PRMs can be leveraged. We can train a PRM on tasks where step-level verification is easier and then generalize this PRM to other tasks with little or no additional training.

Note that step-level verification also comes with its own problems. Frequent verification, especially if using an LLM as the verifier, can substantially increase computational costs and latency. The design of effective step-level verifiers is non-trivial itself. An inaccurate verifier might prematurely discard good reasoning paths or fail to identify flawed ones, thereby misleading the search. This makes the development of such systems more complex and difficult.

3. Encouraging Long Thinking

So far in this subsection, most of the methods are implicitly based on a simple idea: generating longer reasoning paths can help. In addition to CoT and search with verifications, we can consider alternative methods to achieve this. For example, we can prompt the LLM by explicitly asking for extended deliberation. Beyond direct prompting, we can also make modifications to the decoding process itself, such as adjusting token limits or applying penalties for short outputs. Another approach is to employ multi-stage generation schemes where the model incrementally builds upon its reasoning.

4. Training-based Scaling

As well as considering inference-time scaling methods without training, we also wish to consider methods that can improve intrinsic reasoning capabilities of LLMs by modifying their parameters through further training. While such training-based scaling methods typically require additional training cost and computational resources, they instill stronger reasoning skills directly into the model parameters, which in turn can lead to more effective and efficient reasoning performance. We can even combine them with training-free methods for better inference-time scaling results.

Although our discussion here is restricted to reasoning problems, methods for training-based scaling are common. Most of them have been discussed in Chapter 10. Here, we will briefly describe how these methods can be applied to improving inference-time scaling for reasoning problems.

- **Fine-tuning on Reasoning Data**. One of the most direct ways to enhance reasoning is by fine-tuning pre-trained LLMs on datasets specifically curated for reasoning tasks. These datasets can range from simple input-output pairs to more structured formats that include step-by-step reasoning processes. Typical examples include datasets of math word problems, logical deduction exercises, or code generation with explanations. By training on such data, the model learns from common reasoning patterns, and thus can generate detailed and coherent reasoning paths at test time.
- **Reinforcement Learning for Reasoning**. If we regard a verifier as a reward model, we can see that the methods discussed in the previous subsection are a direct application of the reward model to reasoning problems, though they are training-free. Of course, we can apply this reward model to LLM fine-tuning. This follows a standard paradigm of reinforcement learning. Given a reward model, the LLM, acting as a policy, is fine-tuned using reinforcement learning algorithms. The LLM generates reasoning steps

or full solutions, receives feedback (rewards) from the reward model, and updates its parameters to produce outputs that maximize these rewards. This process aligns the LLM output with notions of high-quality reasoning, thereby encouraging the LLM to generate more reliable reasoning paths. Another key issue is the training of the reward model. Generally, this reward model could be an outcome reward model that evaluates the correctness or quality of the final answer, or a process reward model that assesses the quality of each intermediate reasoning step, as discussed in the context of step-level verifiers. In some cases, we can even develop a reward model based on simple rules, such as giving bonuses to longer outputs.

- **Knowledge Distillation for Reasoning**. In this approach, a smaller, more efficient student LLM is trained to mimic the reasoning outputs or internal representations of a larger, more capable teacher LLM. The teacher model might generate detailed reasoning steps for a variety of problems. The student model then learns to reproduce these high-quality reasoning demonstrations. This strategy makes stronger reasoning capabilities more accessible by deploying them in smaller models that are less computationally expensive at inference time.
- **Iterative Refinement**. Training-based scaling can also involve iterative refinement. For example, an LLM can generate solutions to a set of problems. These solutions and their reasoning paths are then verified, either by humans or automatic verifiers. The correct reasoning paths are subsequently added to the training data, and the LLM is further fine-tuned on this augmented dataset. This creates a cycle where the LLM progressively improves its reasoning capabilities through repeated generation, critique, and learning.

The primary advantage of these training-based scaling methods is that they endow the LLM with stronger inherent reasoning skills. This directly contributes to improved inference-time scaling in several ways: it can lead to more efficient inference, as the LLM might require less extensive search or fewer generation samples to arrive at a correct solution. Moreover, the base quality of generated steps or solutions is higher. Therefore, a well-trained LLM might generalize its learned reasoning abilities to novel problems more effectively than an LLM relying solely on in-context learning or training-free inference schemes.

On the other hand, training-based approaches also present challenges, compared to the training-free counterparts. The creation of high-quality, large-scale training datasets for reasoning can be expensive and labor-intensive. The fine-tuning process itself, particularly for the largest LLMs or when using RL, can be computationally intensive and require substantial engineering effort. There is also the risk of the model overfitting to the specific types of problems or reasoning styles present in the training data, potentially limiting its performance on out-of-distribution tasks.

11.4 Summary

In this chapter, we have discussed the inference issue for LLMs. We have presented the prefilling-decoding framework and related decoding algorithms for LLM inference. Then, we have described several techniques for efficient inference. We have also discussed inference-time scaling, which has been considered one of the most important methods for improving LLM reasoning.

Inference over sequential data has long been a concern in AI [Wozengraft and Reiffen, 1961; Viterbi, 1967; Forney, 1972]. In the context of NLP, this line of work dates back to the very early days of speech recognition and statistical machine translation [Koehn, 2010], where researchers faced the challenge of efficiently searching vast hypothesis spaces to find the most probable output sequence. Techniques like beam search and various pruning strategies were developed then to make this computationally tractable. At that time, models were relatively weak, and much of the research focused on developing powerful search algorithms to reduce search errors. These foundational ideas continue to influence modern approaches.

As we enter the era dominated by deep learning methods, models based on deep neural networks have become extremely powerful. Even with very simple search algorithms, these models can achieve excellent results. In this context, inference no longer seems as "important" as it once was, and research attention has gradually shifted toward model architectures, training methods, and scaling up models.

However, history tends to repeat itself. With the rise of LLMs, inference has once again attracted significant attention. This renewed focus is primarily manifested in two aspects:

- The inference cost for LLMs is very high. For example, efficiently deploying LLMs in high-concurrency, low-latency scenarios remains a challenging problem, making inference efficiency critically important. In this context, efficient architecture designs, optimized search algorithms, and various inference optimization strategies hold substantial practical significance.
- Input and output sequence lengths have significantly increased in complex tasks. Especially in tasks like mathematical reasoning, the growth of sequence lengths further highlights the importance of inference efficiency. Moreover, scaling the inference process has recently proven to be an effective way to improve the reasoning capabilities of models. Therefore, achieving efficient inference scaling is emerging as a particularly promising research direction.

Inference is now a wide-ranging topic that encompasses many techniques. It involves not only the development of model architectures and decoding algorithms, but is increasingly shaped by the intricate engineering and sophisticated systems-level optimizations required to deploy LLMs effectively and efficiently. Many of these techniques are beyond the scope of NLP or a specific AI area. Instead, the frontier of LLM inference optimization now extends deeply into domains traditionally considered core computer science and engineering. This systemic perspective has brought many new ideas to the study of inference problems. Unfortunately, this chapter cannot cover all relevant techniques — indeed, that would be an almost impossible task in itself. Ultimately, the best way to better understand and master these techniques may still lie in hands-on practice.

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